Cell Biology

Introduction

Introduction

Course Outline

Atom→ Molecules→
Cell Structures→ Cells

Cell Functions

Propagation/Heredity

Expression Regulation

Cell Communication

Applied Cell Biology

Introduction

Course Objective

Understand the molecular logic of the "living state"

How a molecule's function arises from its structure,

How this information is conveyed to the rest of the cellular machinery

System Manipulation

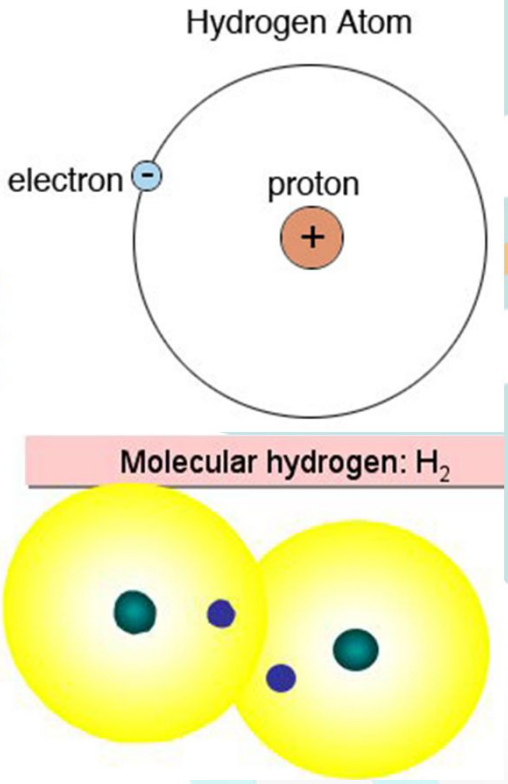
END

CELL BIOLOGY

Composition of Matter

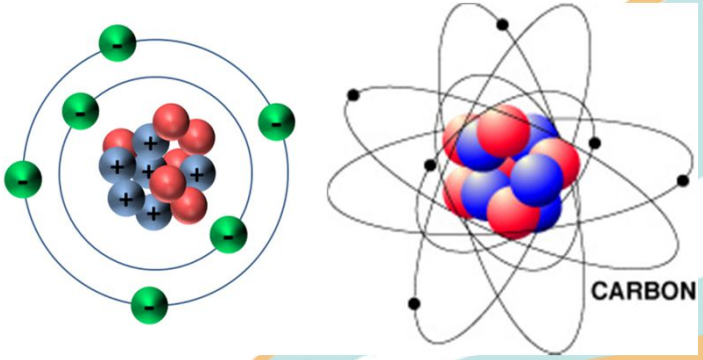
Composition of Matter

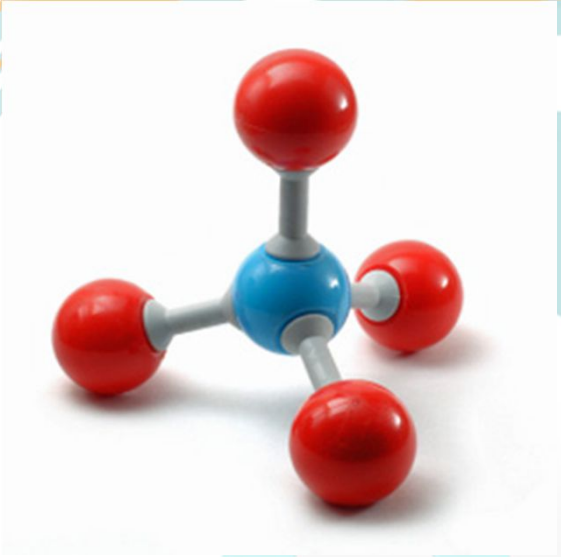
Atomic Structure



Composition of Matter

Carbon Atom

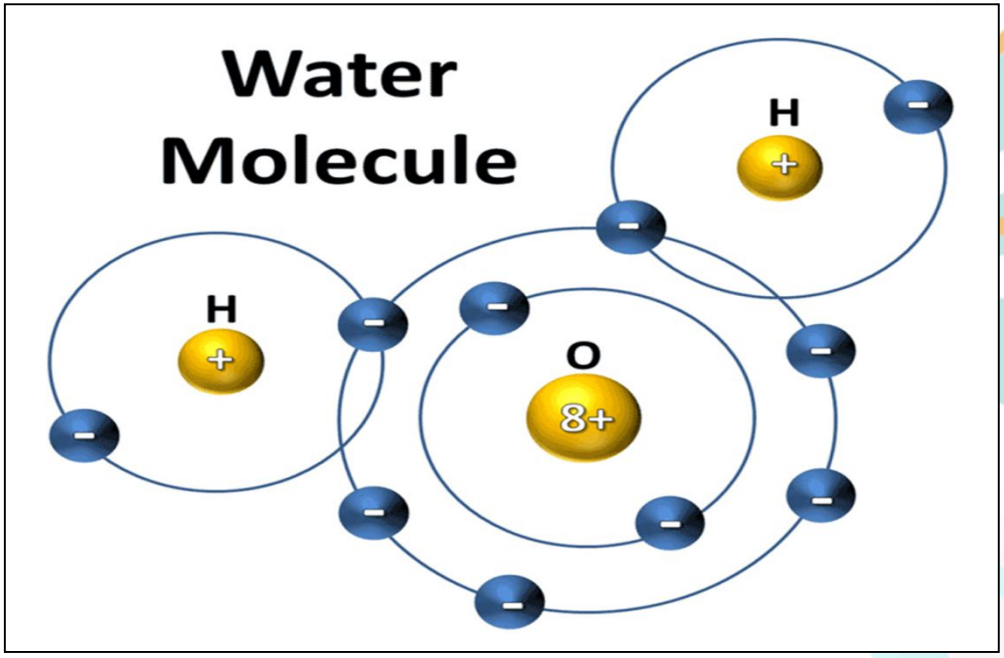




Composition of Matter

Electronegativity

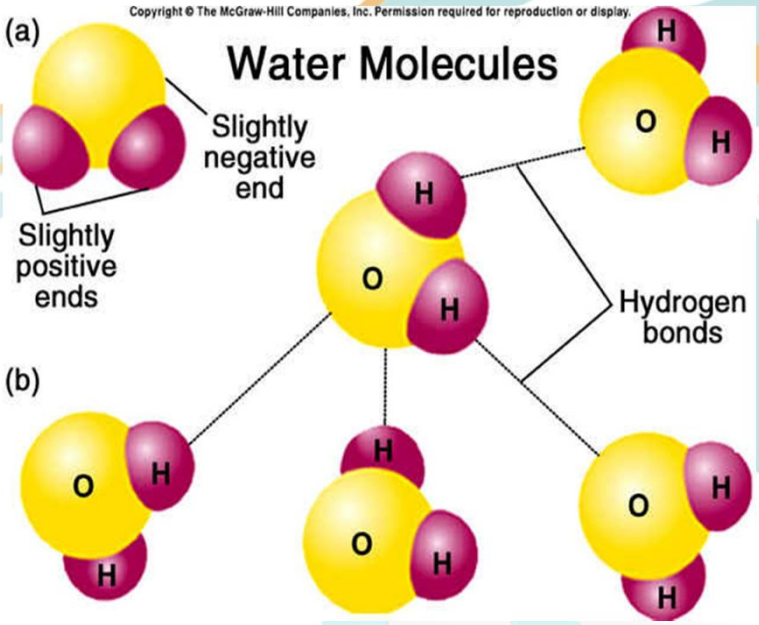
The tendency of an atom to attract electrons towards itself.



Composition of Matter

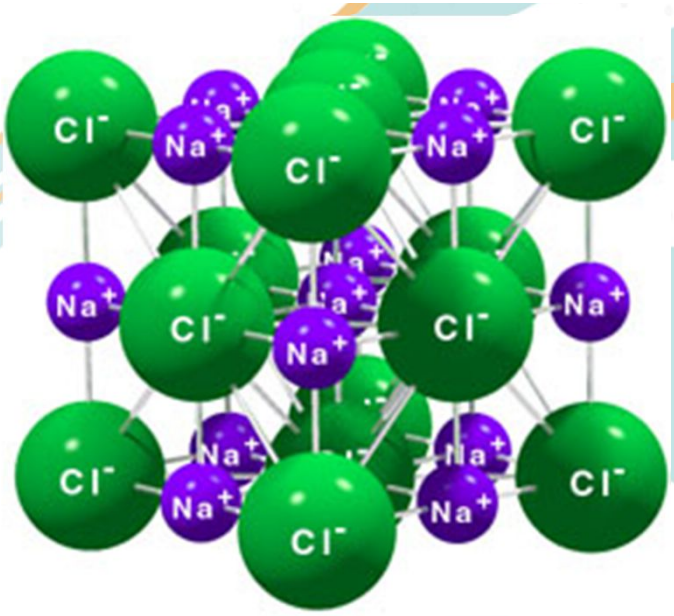
Polarity of H_2O

Results in Hydrogen Bonding



Composition of Matter

Ionic Bond

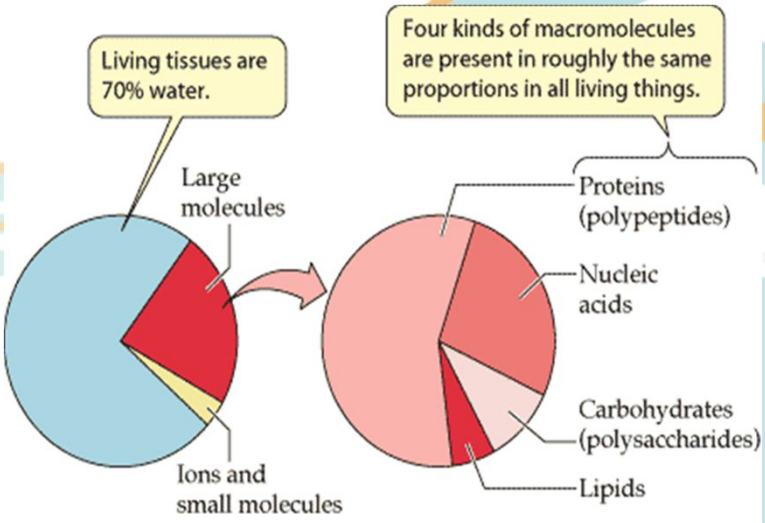


CELL BIOLOGY

Molecules of Life

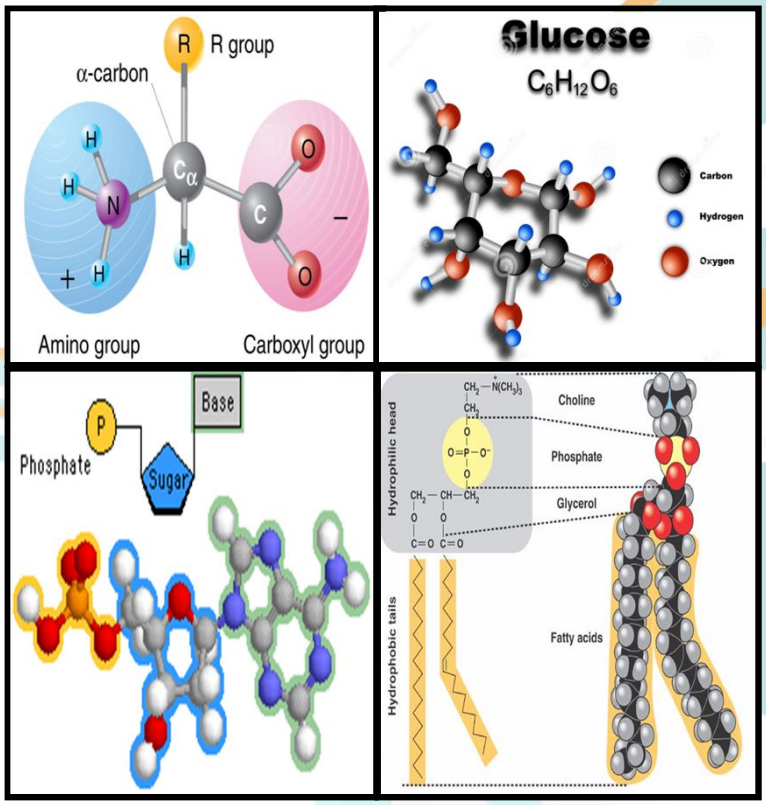
Molecules of Life

Composition of Life



Molecules of Life

Building Blocks of Life



Molecules of Life

Polymers & Monomers

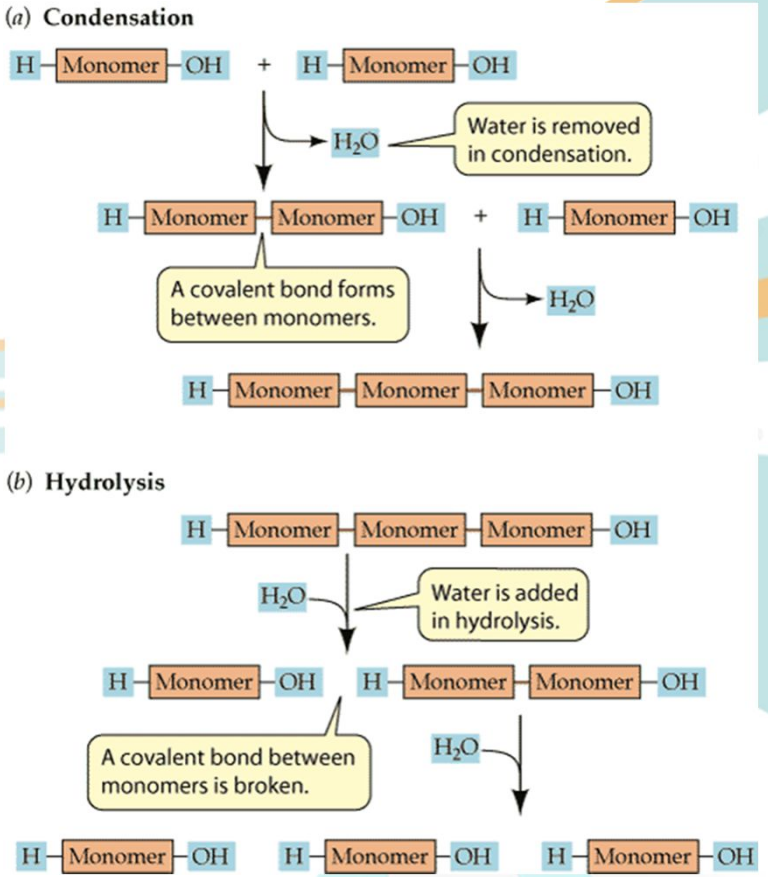
Macromolecules are polymers of smaller monomers molecules

Condensation reactions: Monomers join H_2O Removal

Hydrolysis reactions: Use water to break polymers into monomers.

Molecules of Life

Condensation & Hydrolysis



Proteins: Polymers of Amino Acids

Proteins: Polymers of Amino Acids

Protein Functions

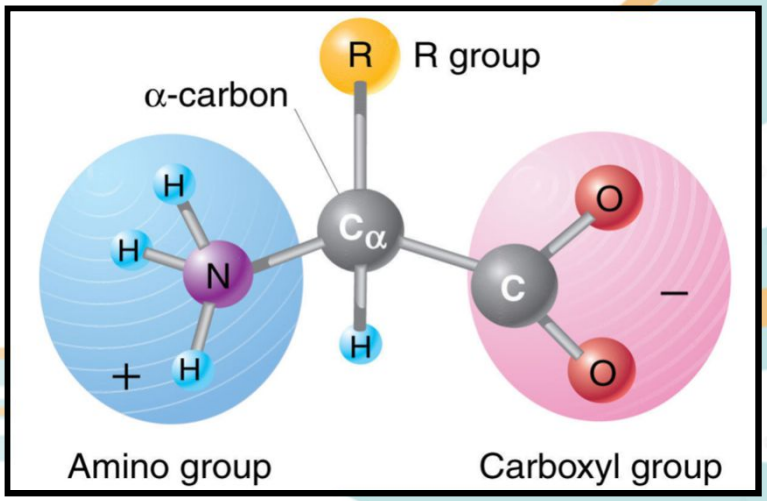
Most Abundant Polymer in the cell

Support, protection, catalysis, transport, defense, regulation & movement

Do not store genetic information

Proteins: Polymers of Amino Acids

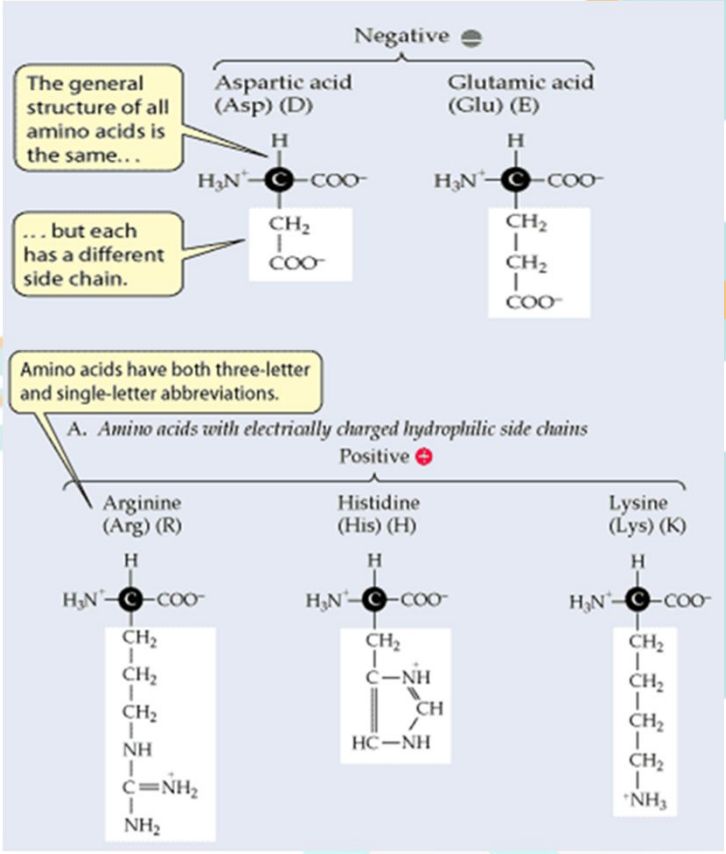
Amino Acid Structure



The side chains or R groups give different properties to each of the amino acids

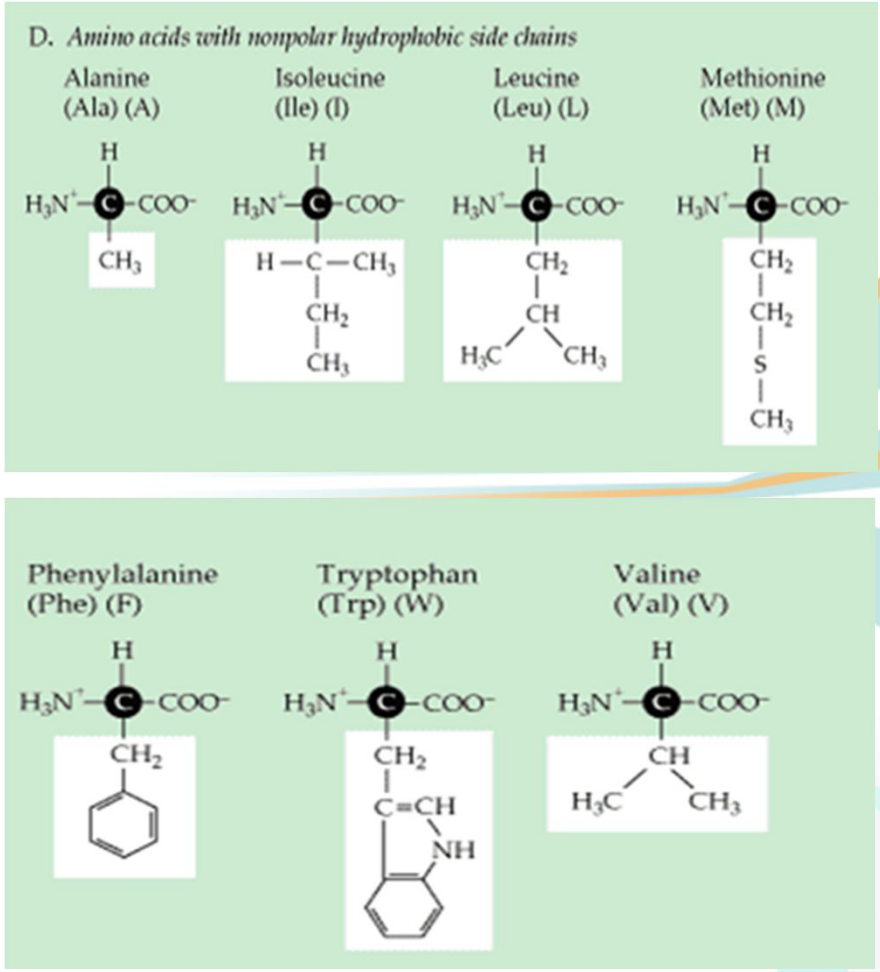
Proteins: Polymers of Amino Acids

Charged R Groups



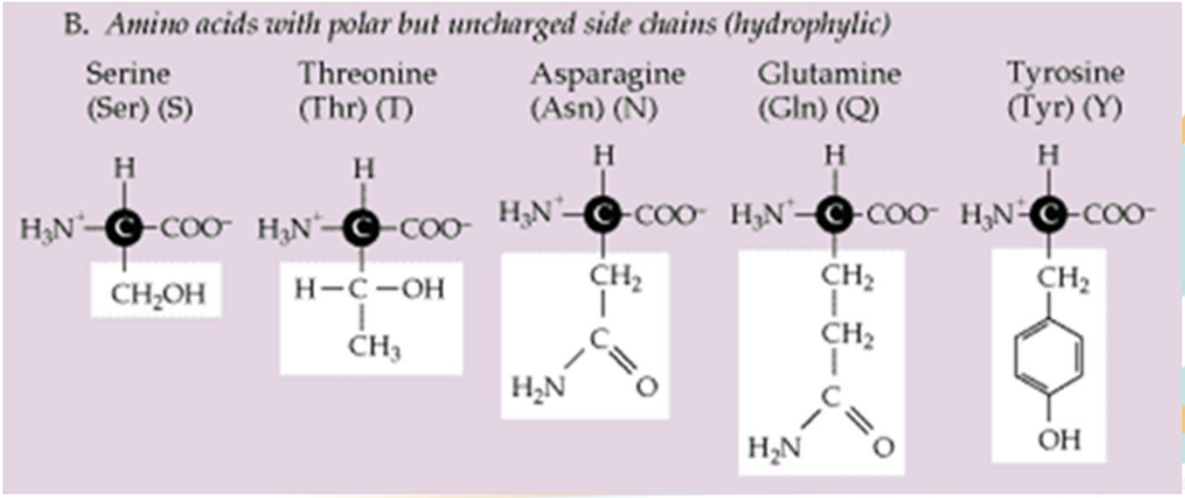
Proteins: Polymers of Amino Acids

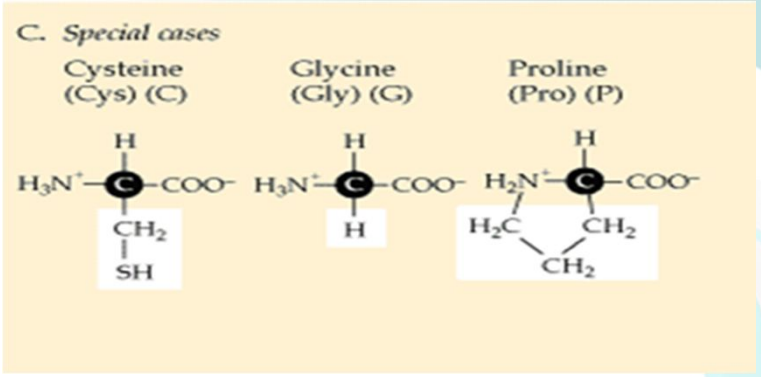
Nonpolar R Groups



Proteins: Polymers of Amino Acids

Polar & Special R Groups



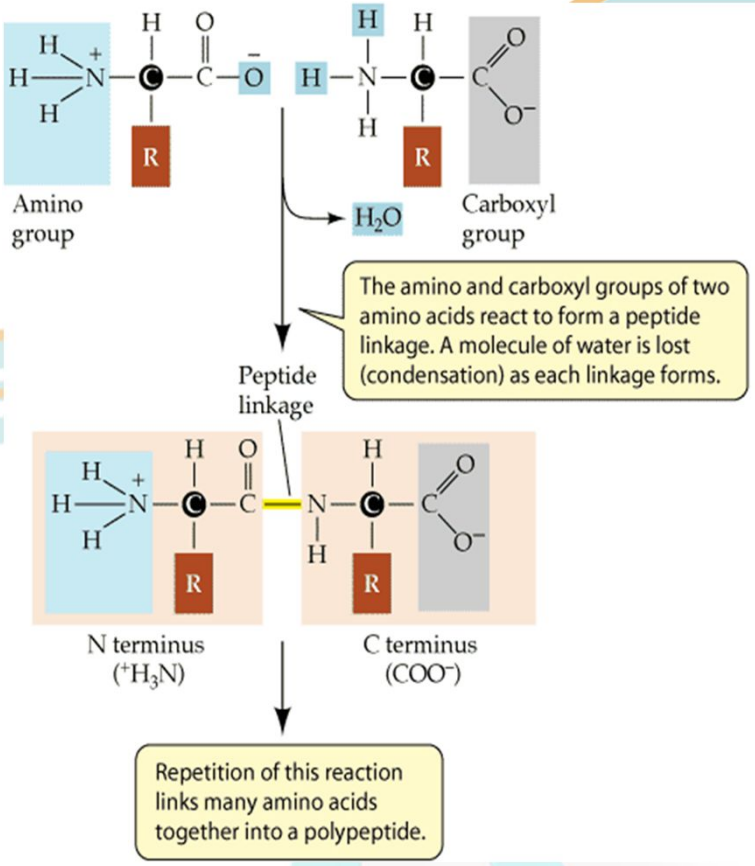


CELL BIOLOGY

Proteins: Structure

Proteins: Polymers of Amino Acids

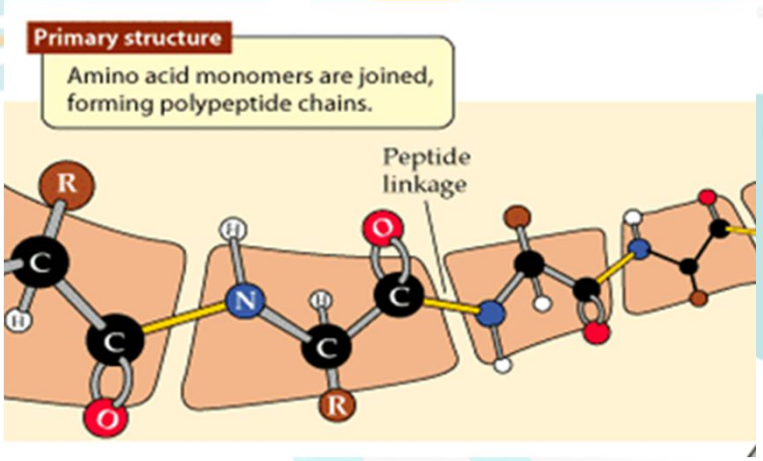
The Peptide Bond



Proteins: Structure

Primary Structure

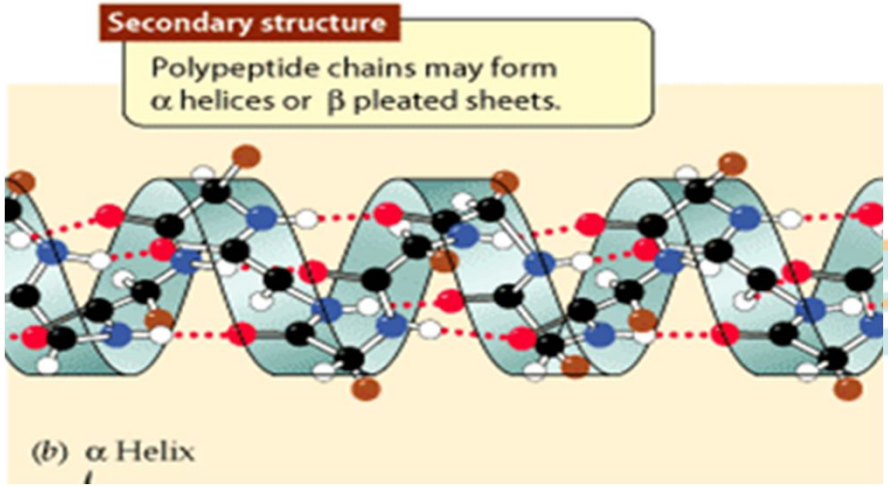
Sequence of amino acids bonded by peptide linkages (Diversity 20n)

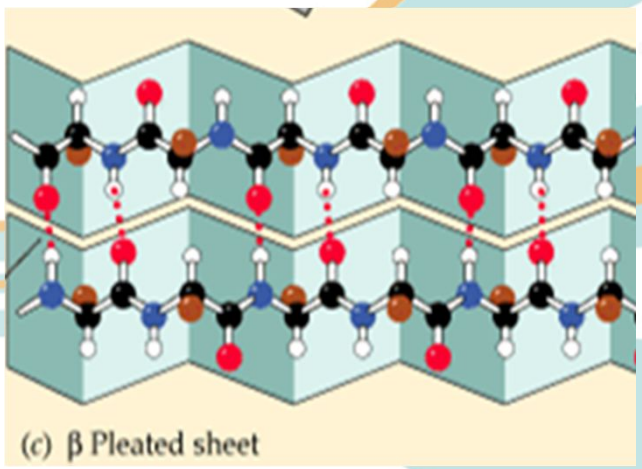


Proteins: Structure

Secondary Structure

α helices (hydrogen bonds between amino acid residues)

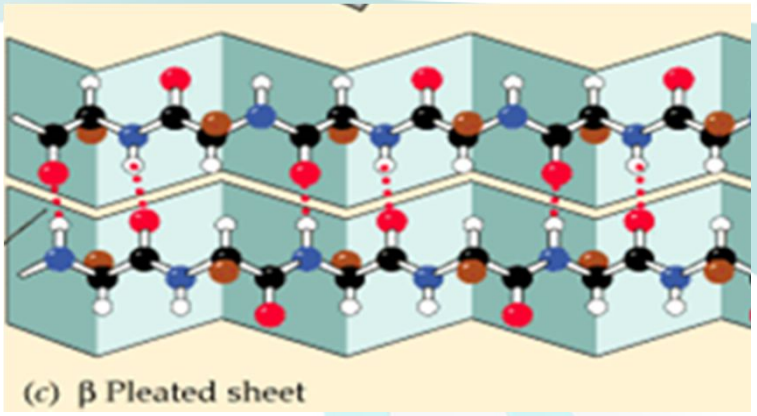




Proteins: Structure

Secondary Structure

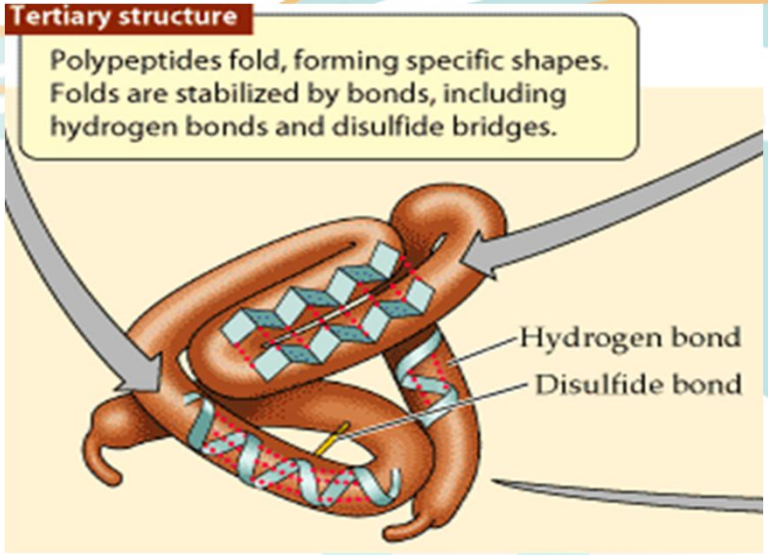
β pleated sheets (hydrogen bonds between amino acid residues)



Proteins: Structure

Tertiary Structure

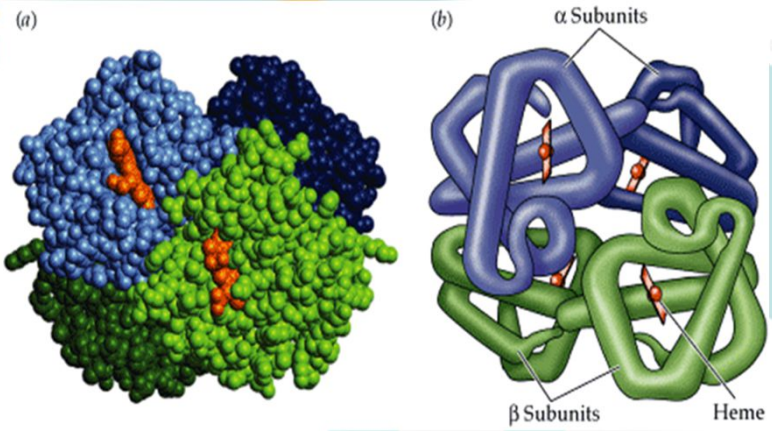
Generated by bending and folding of the polypeptide chain



Proteins: Structure

Quaternary structure

Arrangement of multiple folded protein molecules in a multi-subunit complex.



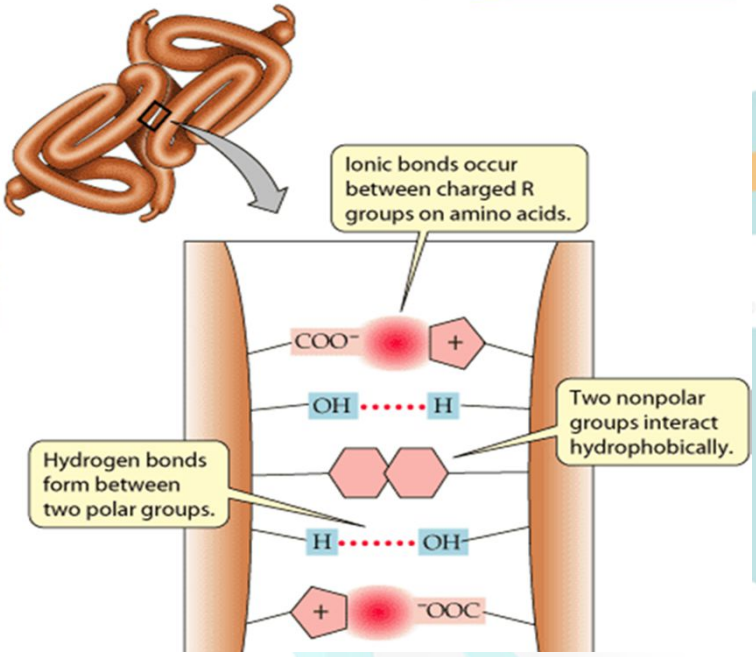
CELL BIOLOGY

Protein Interactions

Protein Interactions

Weak interactions

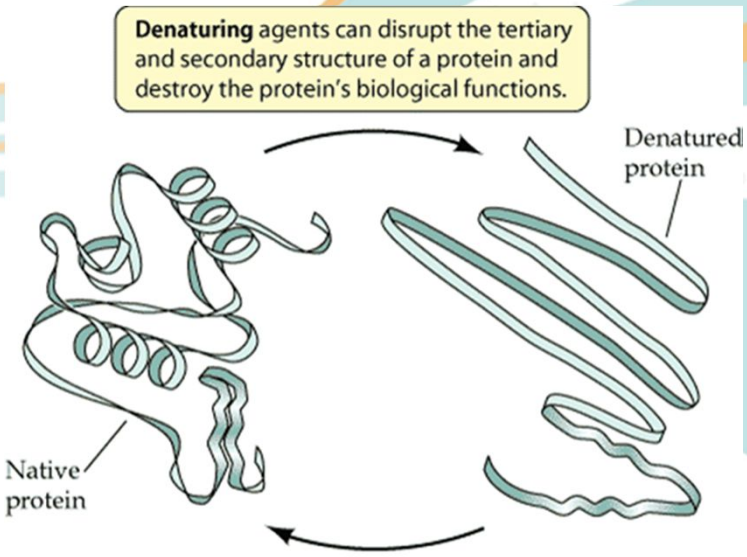
3D structure



Protein Interactions

Protein: Denaturation

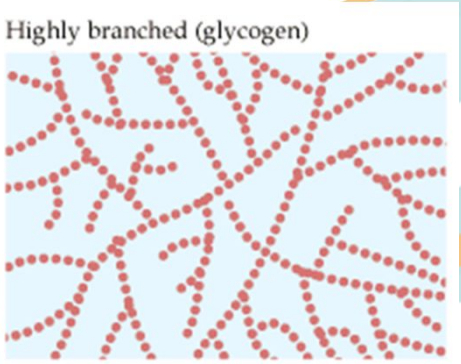
Heat, pH change: Tertiary and secondary structure as well as biological function.

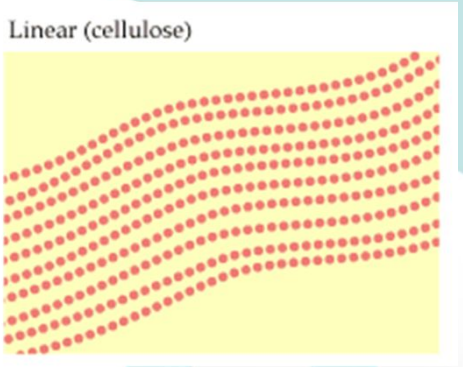


Glycosidic linkages

Carbohydrates

Disaccharide α or β orientation

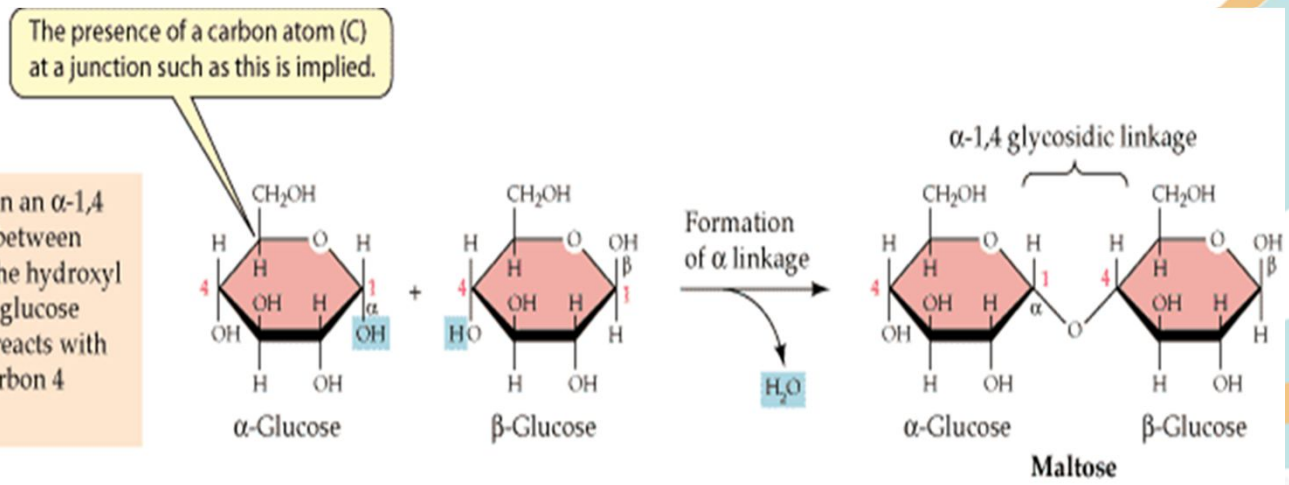




Glycosidic linkages

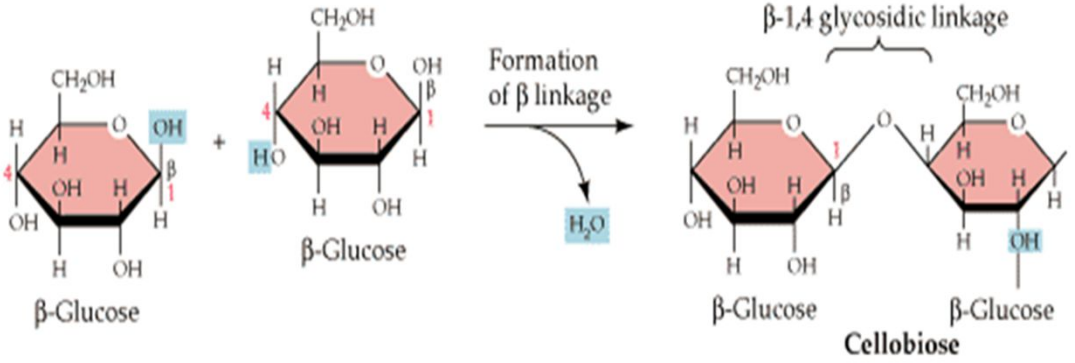
Disaccharide α or β orientation

Maltose is produced when an α-1,4 glycosidic linkage forms between two glucose molecules. The hydroxyl group on carbon 1 of one glucose in the α (down) position reacts with the hydroxyl group on carbon 4 of the other glucose.



Digestible

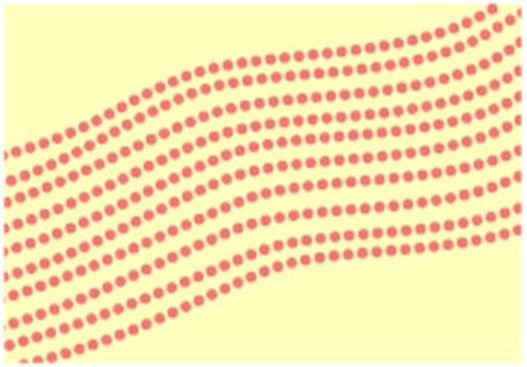
In **cellobiose**, two glucoses are linked by a β-1,4 glycosidic linkage.



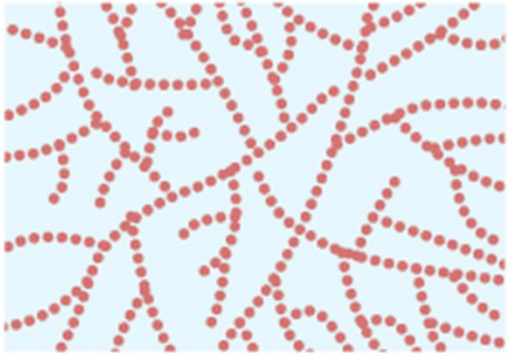
Glycosidic linkages

Cellulose & Glycogen

Linear (cellulose)

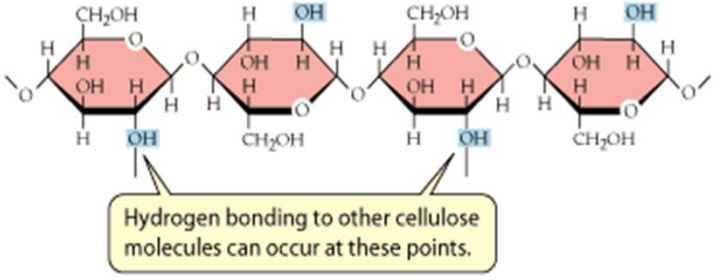


Highly branched (glycogen)



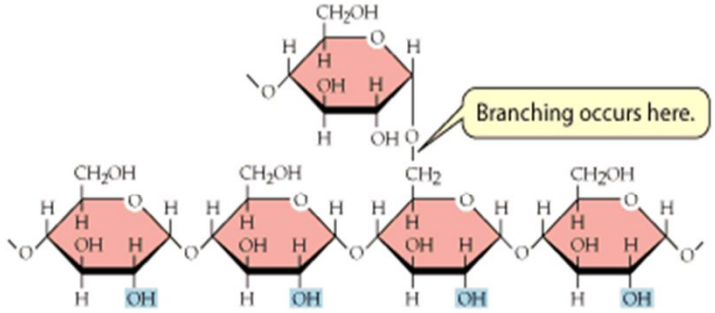
(a) Molecular structure

Cellulose



Cellulose is an unbranched polymer of glucose with β-1,4 glycosidic linkages that are chemically very stable.

Starch and glycogen



Glycogen and starch are polymers of glucose with α-1,4 glycosidic linkages. α-1,6 glycosidic linkages produce branching at carbon 6.

Glycosidic linkages

Cellulose & Starch

Lipids: Nonpolar Hydrocarbons

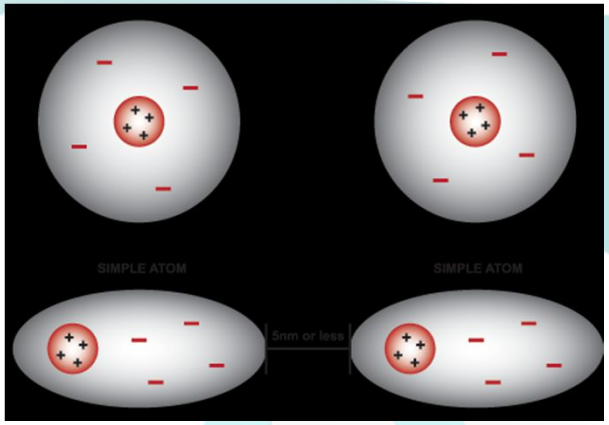
Lipids: Nonpolar Hydrocarbons

Week Bonds

Hydrophobic molecules

Water insoluble

Aggregate together (by hydrophobic and Van der Waals force)



Lipids: Nonpolar Hydrocarbons

Functions

Store energy

Phospholipids membranes

Carotenoids & Light

Hormones and vitamins

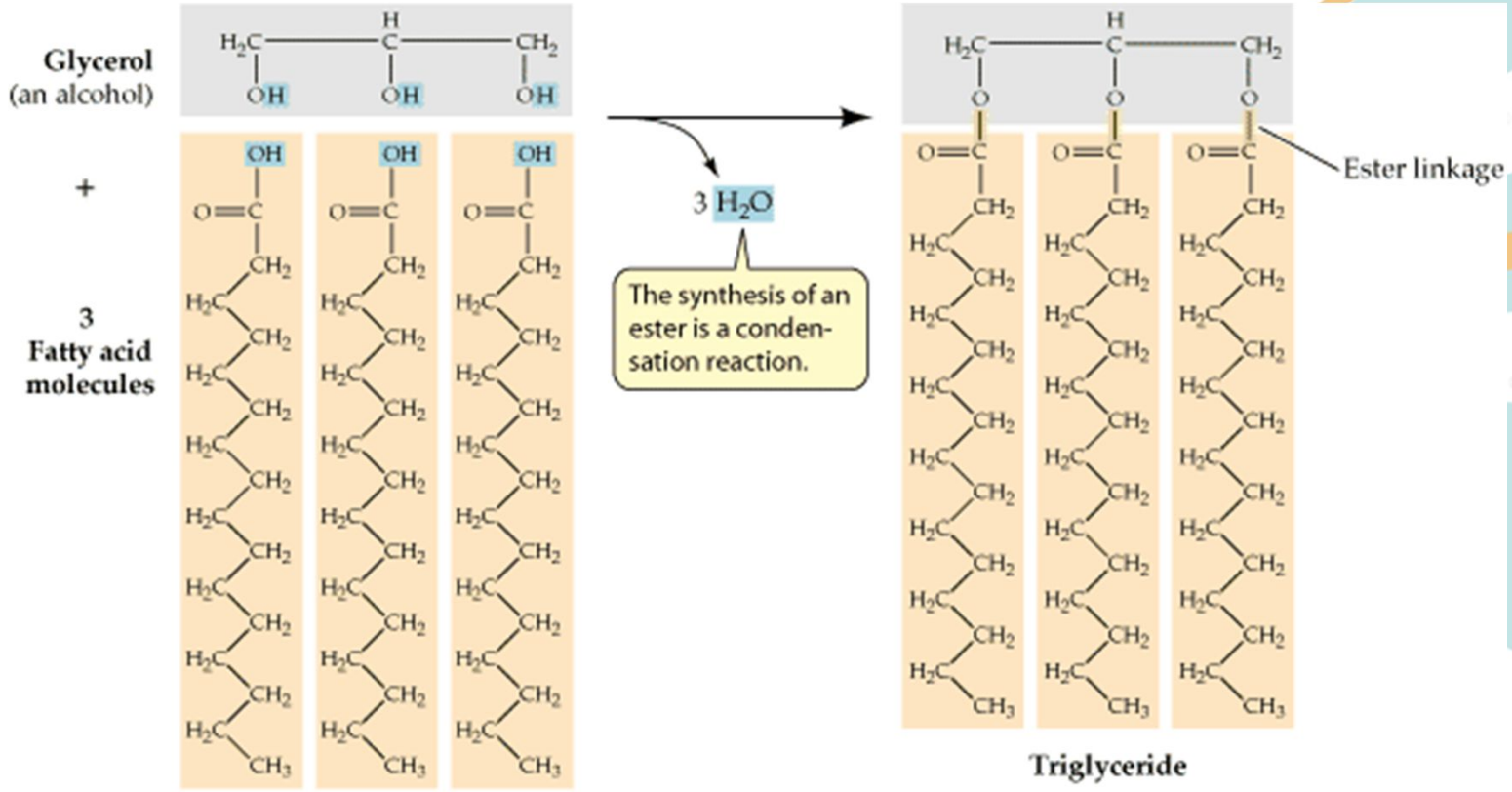
Animal fat is insulator

Insulation of nerves

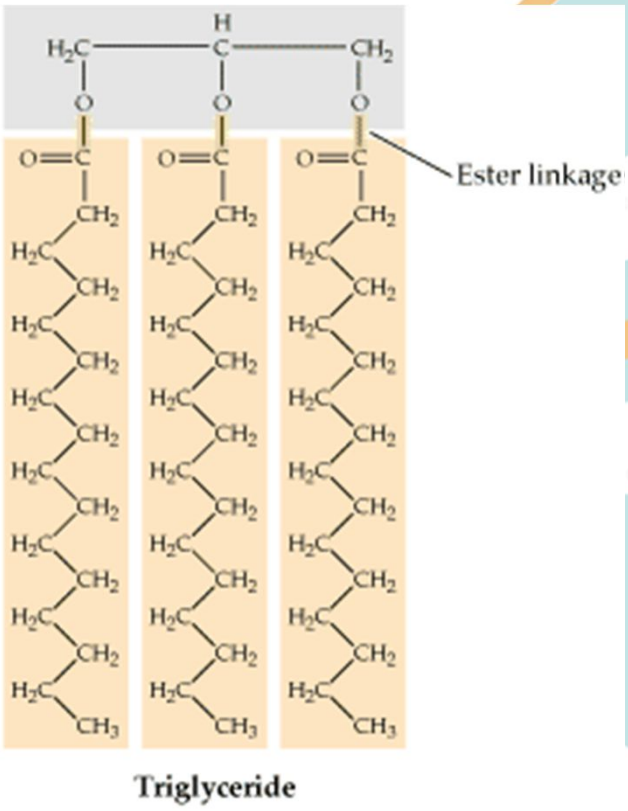
Water repellant for skin, fur and feathers

Lipids: Synthesis

Fats & oils: Triglycerides: (covalent ester linkage vs Hydrophobic bond)



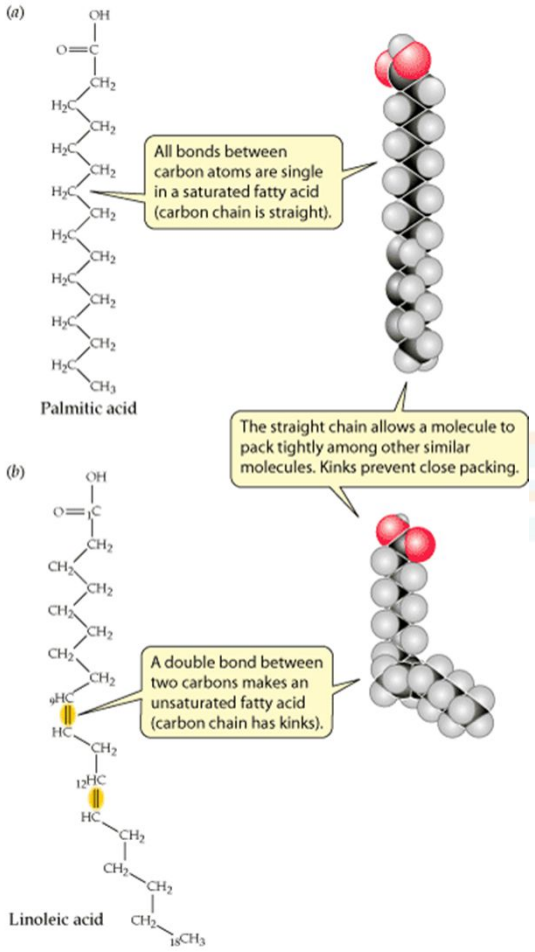
Lipids: Synthesis



CELL BIOLOGY

Phospholipids

Lipids: Saturated & unsaturated



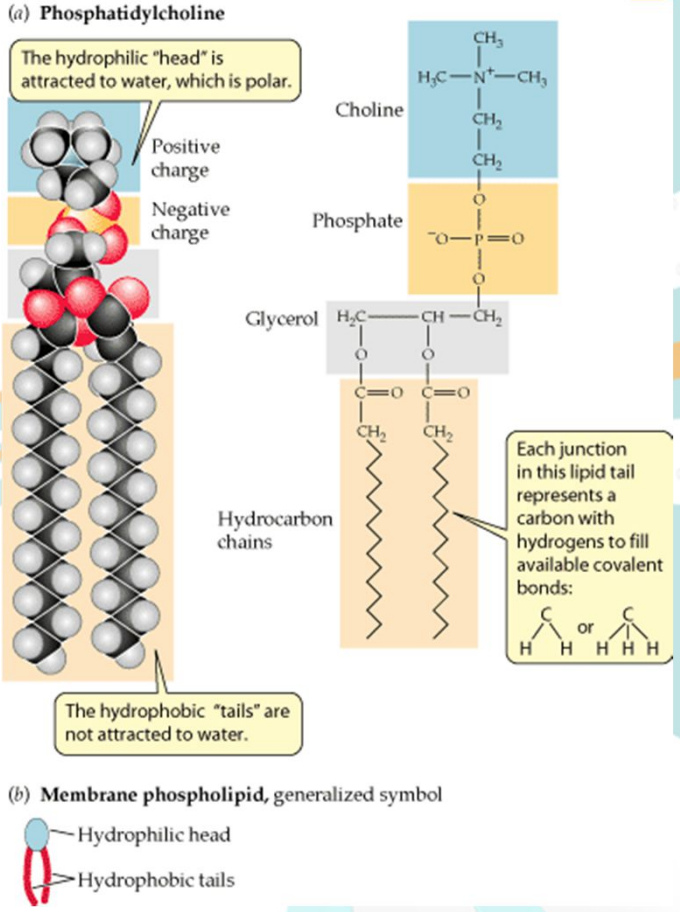
Fats (L) & Oils (S)

Saturated fatty acids have a hydrocarbon chain with no double bonds.

The hydrocarbon chains of unsaturated fatty acids have one or more double bonds that bend the chain, making close packing less possible.

Phospholipids

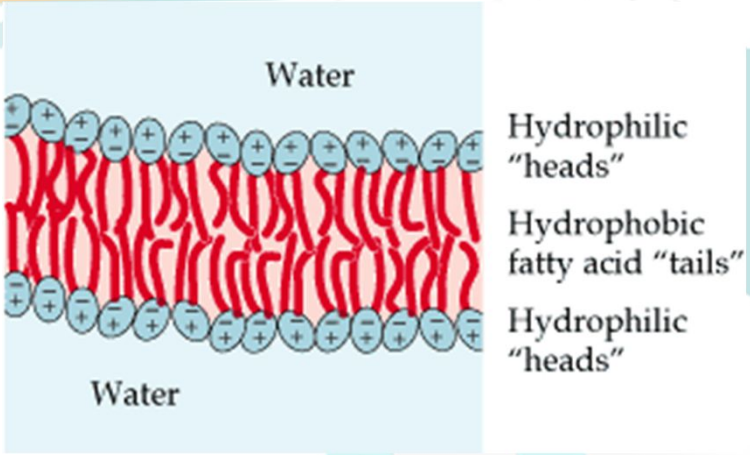
Amphiphatic Molecule



P-lipids: Aqueous Environment

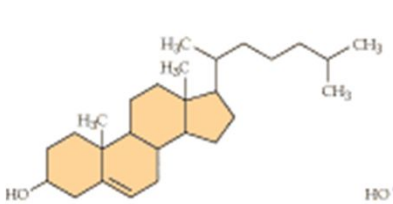
The Bilayer

Two molecules thick head groups directed outward, The tails are packed together in the interior of the bilayer.

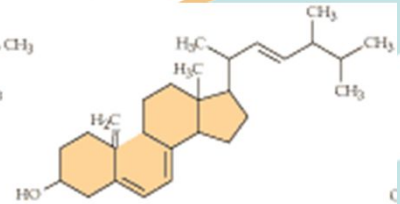


Lipids: Cholesterol

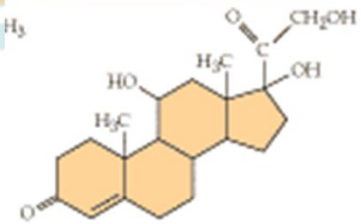
Vitamins & Signals



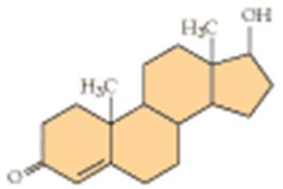
Cholesterol is a constituent of membranes and is the source of steroid hormones.



Vitamin D_2 can be produced in the skin by the action of light on a cholesterol derivative.



Cortisol is a hormone secreted by the adrenal glands.



Testosterone is a male sex hormone.

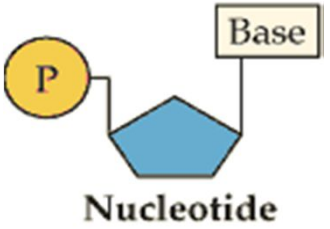
END

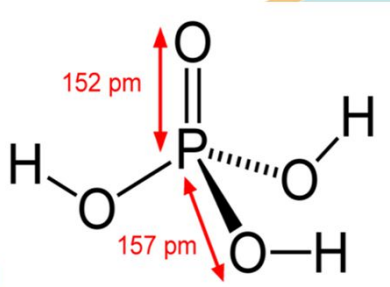
CELL BIOLOGY

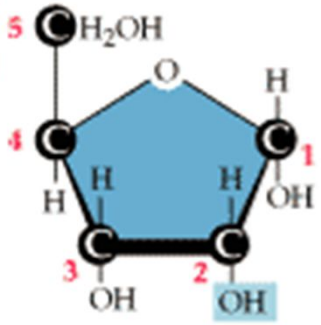
Nucleic Acids

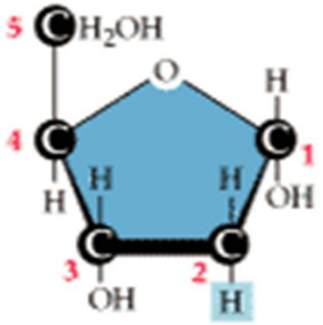
Nucleic Acids

DNA & RNA



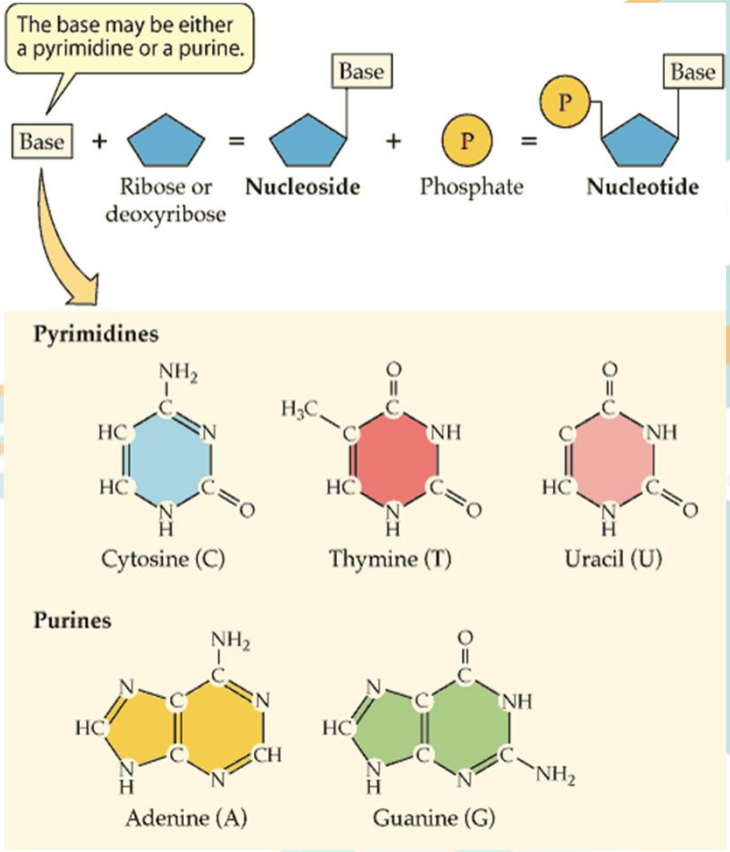






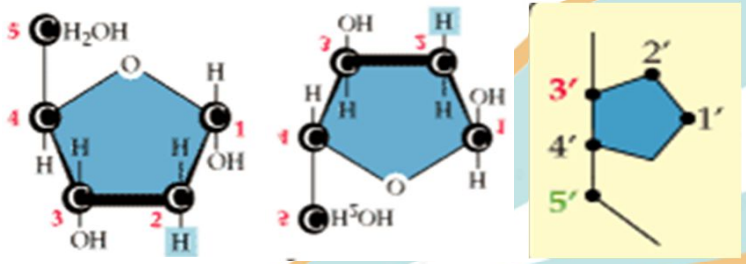
Nucleic Acids

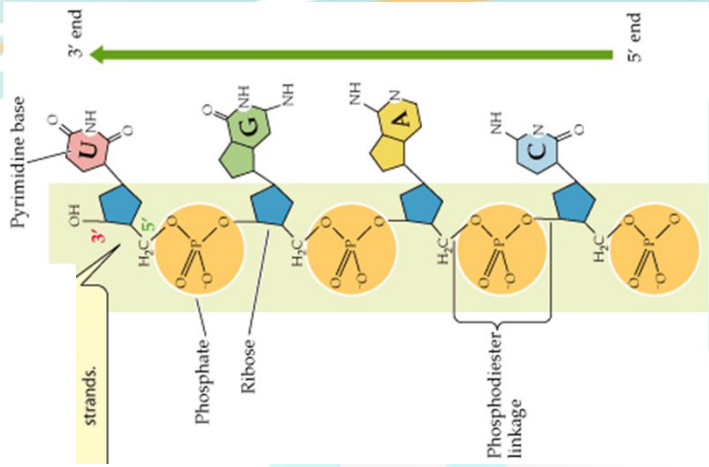
The Bases



Nucleic Acids

Prime Ends (3′& 5′)





END

CELL BIOLOGY

RNA & DNA

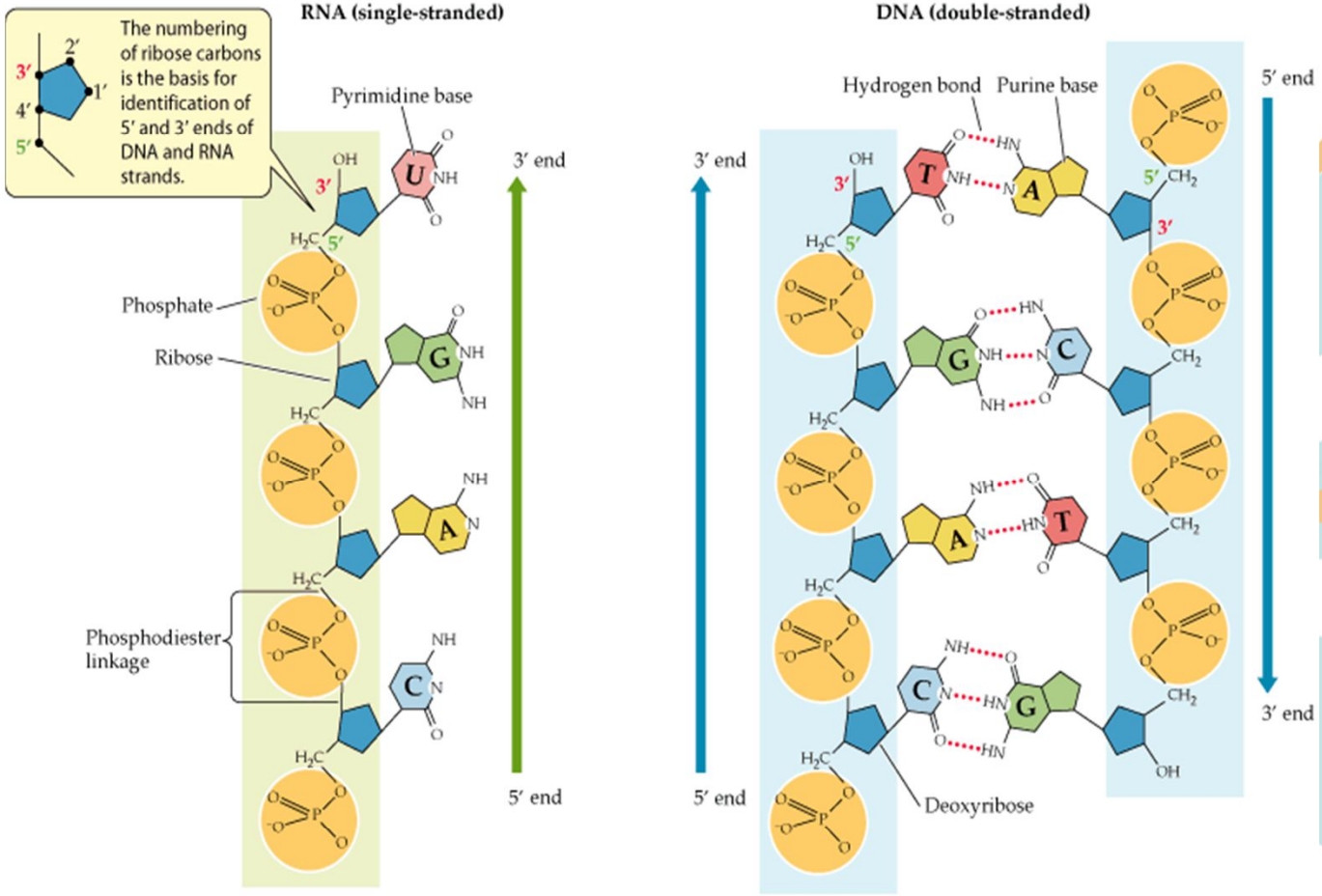
Nucleic Acids

RNA vs. DNA

3.3 Distinguishing RNA from DNA

NUCLEIC ACID	SUGAR	BASES
RNA	Ribose	Adenine Cytosine Guanine Uracil
DNA	Deoxyribose	Adenine Cytosine Guanine Thymine

Nucleic Acids



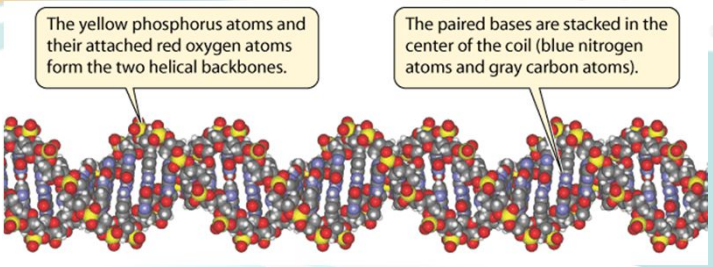
In RNA, the bases are attached to ribose. The bases in RNA are the purines adenine (A) and guanine (G) and the pyrimidines cytosine (C) and uracil (U).

In DNA, the bases are attached to deoxyribose, and the base thymine (T) is found instead of uracil. Hydrogen bonds between purines and pyrimidines hold the two strands of DNA together.

Nucleic Acids

DNA Uniform Width

Information in Sequence not Shape



Chicken vs. Egg: What Came 1st

Protein or DNA

Genetic information codes for enzymes

Enzymes are required to decipher genetic information and replicate it

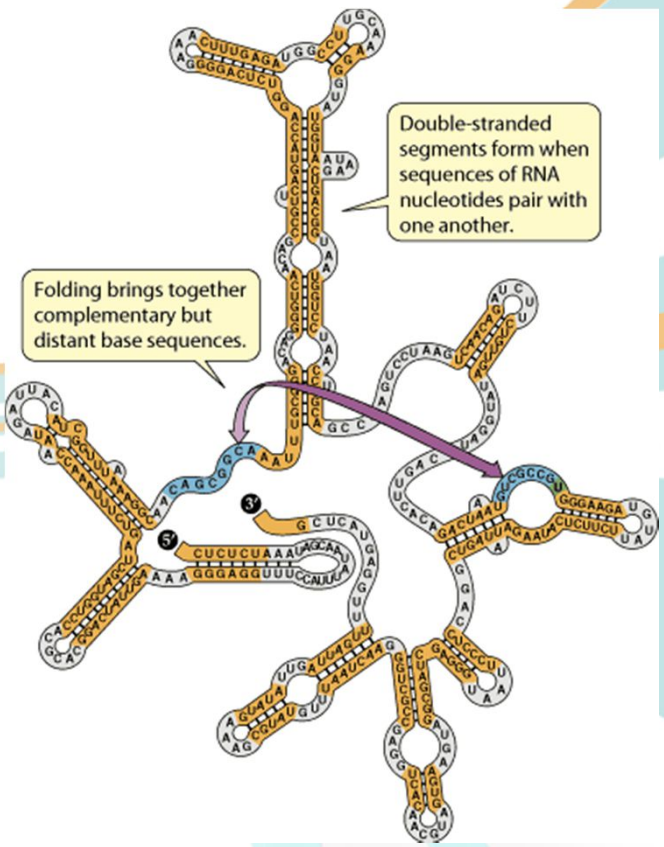
RNA Genetic Material & Enzyme

Many viruses use RNA as their hereditary material

RNAs can achieve chemical catalysis, like enzymes: ribosome the active site is composed entirely of RNA (peptide bond formation)

RNA: Information in Sequence and Shape

Why Then DNA?



END

CELL BIOLOGY

Unit of Life

Unit of Life

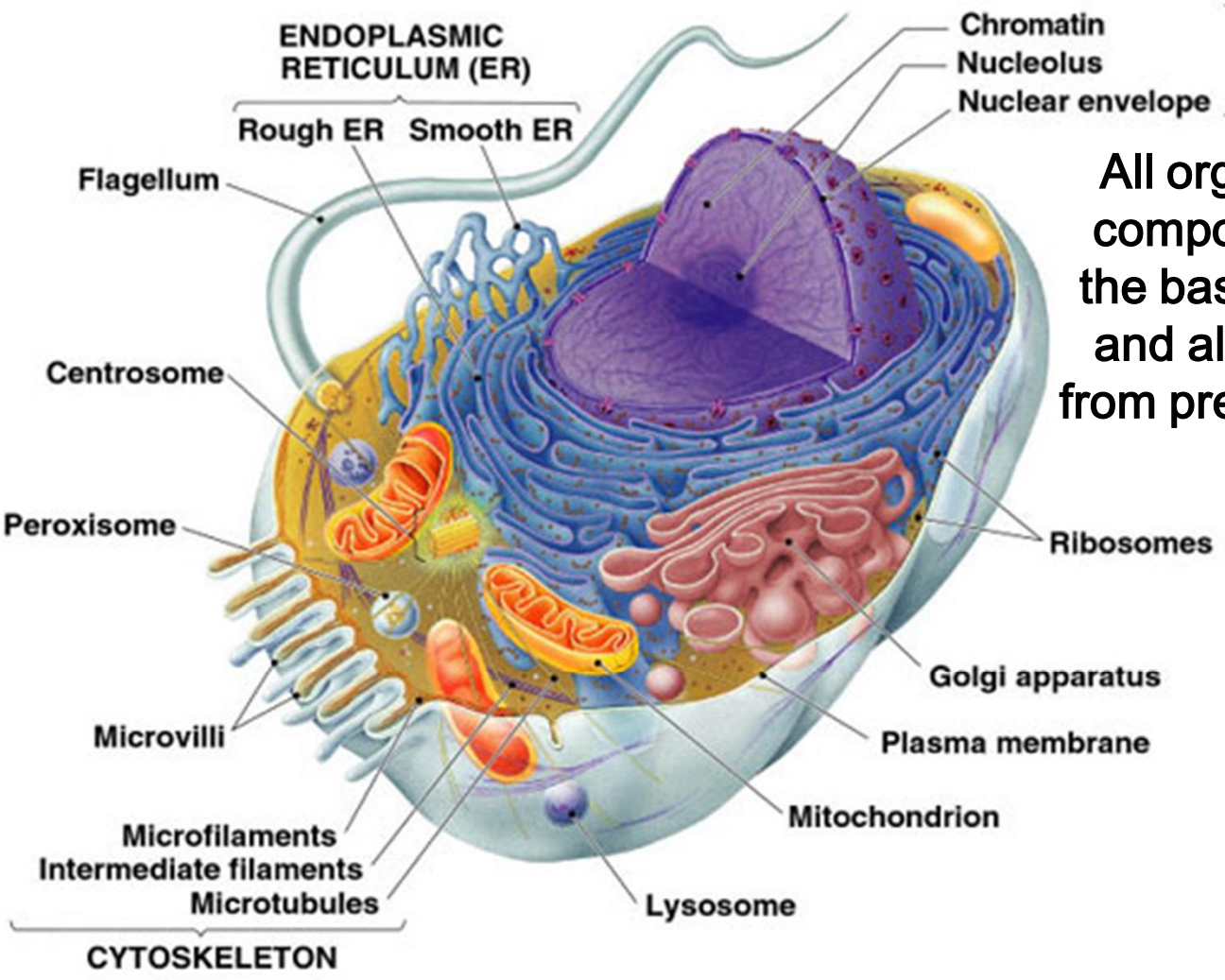
Cell Structure

Cellular compartments

Organelles

Structure & Function

Unit of Life



All organisms are composed of cells the basic unit of life and all cells come from preexisting cells

Not a random collection of molecules

Life

collection of macromolecules that can perform unique functions because the are enclosed in structural compartment that provides consistency (homeostasis).

CELL BIOLOGY

Journey into the Cell

Unit of Life

View animation

END

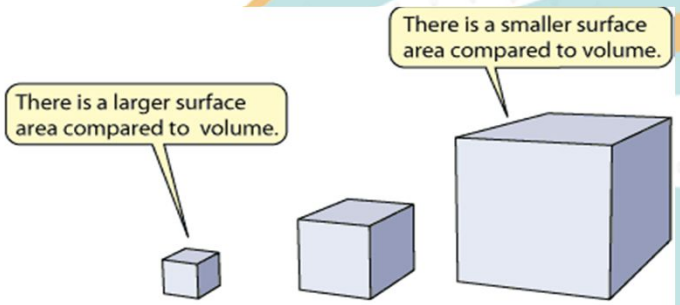
CELL BIOLOGY

Size Matters

Size Matters

Cell Structure

Cells are small to maintain large surface area to volume ratio



	1-mm cube	2-mm cube	4-mm cube
Surface area	6 sides × 1^2 = 6 mm^2	6 sides × 2^2 = 24 mm^2	6 sides × 4^2 = 96 mm^2
Volume	1^3 = 1 mm^3	2^3 = 8 mm^3	4^3 = 64 mm^3
Surface area-to-volume ratio	6/1	3/1	1.5/1

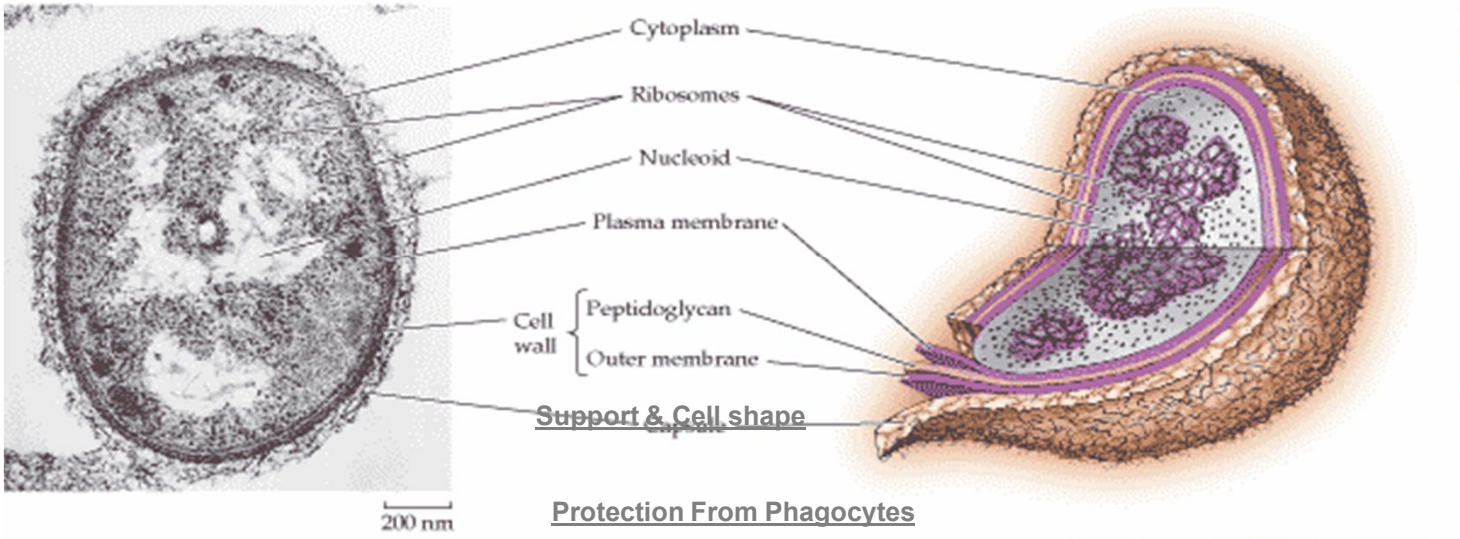
Prokaryotic Cells

No membrane enclosed internal compartments.

Plasma membrane regulates traffic (barrier).

Nucleoid region contains DNA.

Most have cell wall.



Support & Cell shape

Protection From Phagocytes

Prokaryotic Cells

Special Features

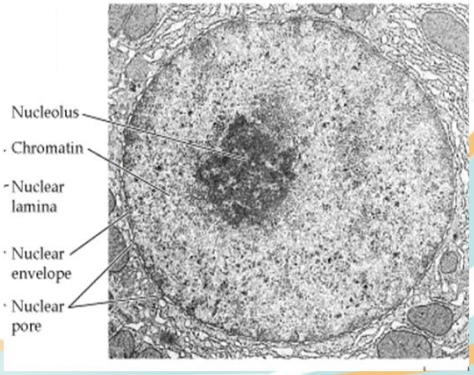
Cyanobacteria Chlorophyll containing have folds of plasma membrane, other have mesosomes (energy).

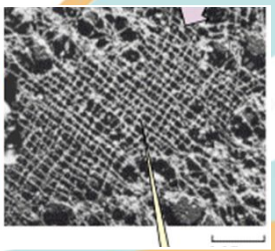
Some have actin like filaments and other have Flagella made-up of Flagellin.

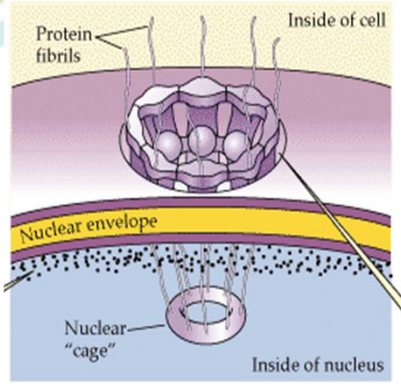
END

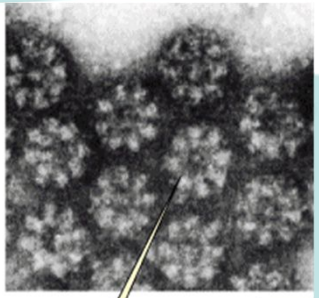
The Nucleus

Nuclear EMs









An octagon of protein complexes surrounds each **nuclear pore**. Protein fibrils on the nuclear side form a cagelike structure.

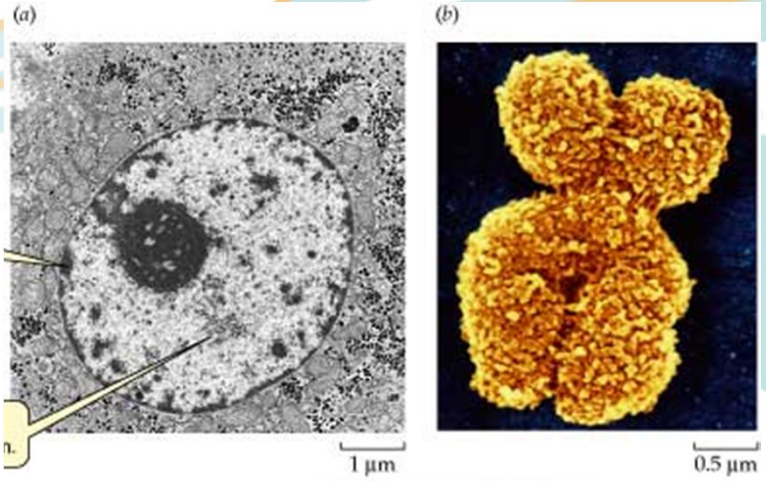
The Nucleus

Chromatin

Euchromatin

Hetrochromatin

Chromosome



END

CELL BIOLOGY

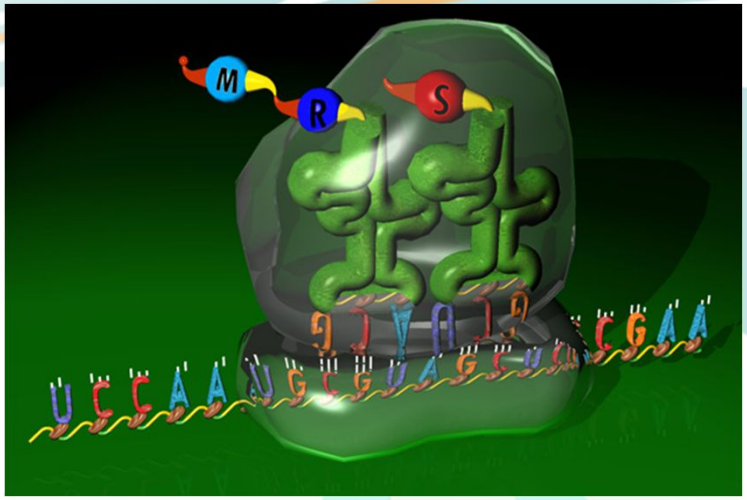
Ribosome

Ribosome

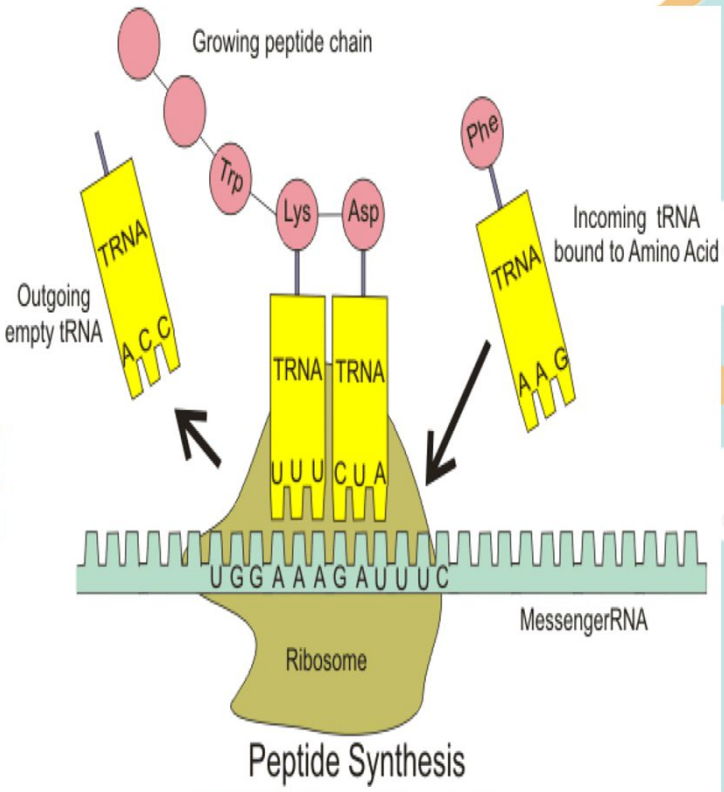
Protein & RNA

Free or attached to ER involved in protein synthesis.

Mitochondria & Chloroplasts.



Ribosome

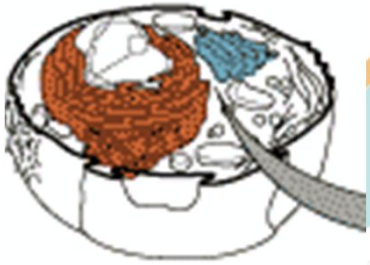


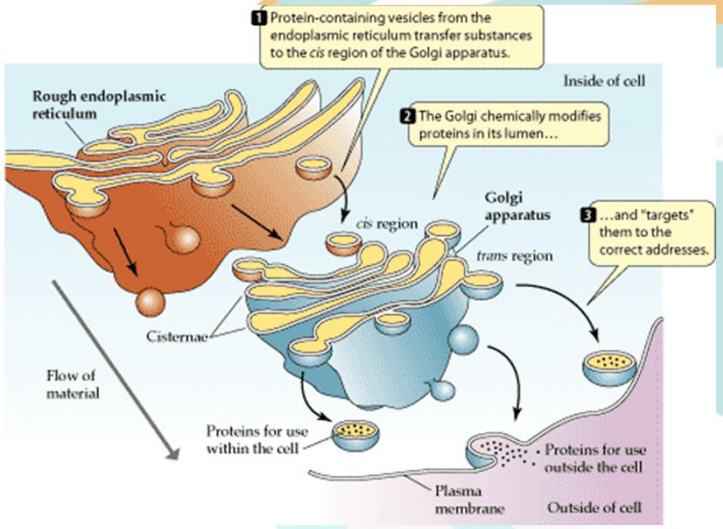
END

The Endoplasmic Reticulum

The Endoplasmic Reticulum

RER + SER

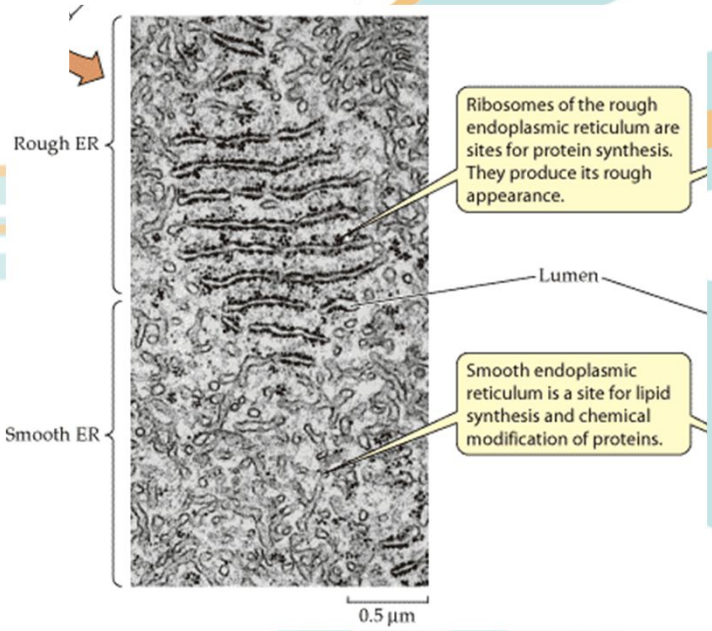




The Endoplasmic Reticulum

RER

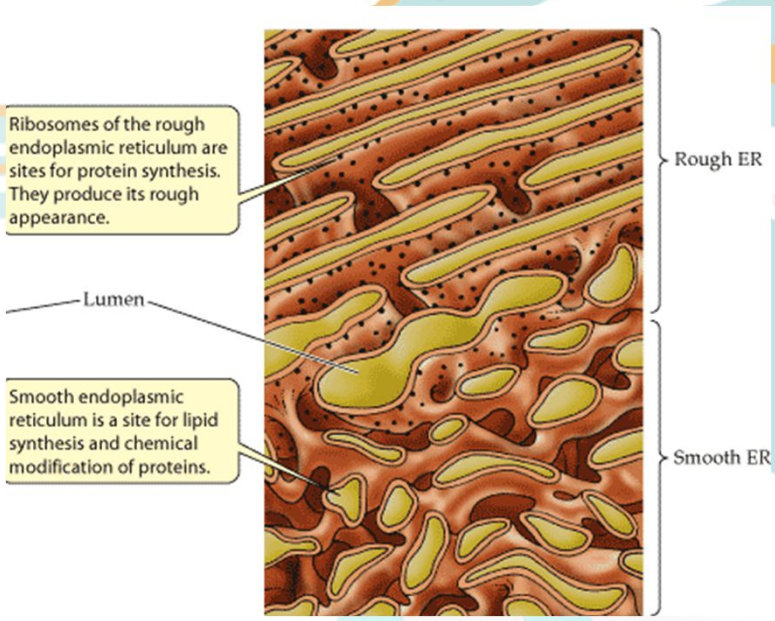
Segregates newly synthesized Proteins



The Endoplasmic Reticulum

RER

Chemically modifies proteins (Glycosylation address)

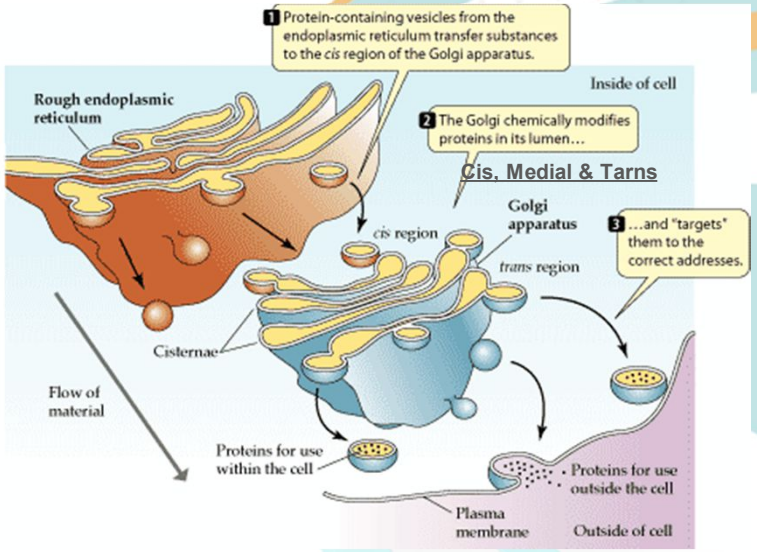


The Endoplasmic Reticulum

SER

Glycogen hydrolysis

Secreting cells have abundant SER



END

CELL BIOLOGY

Golgi & Lysosomes

Golgi

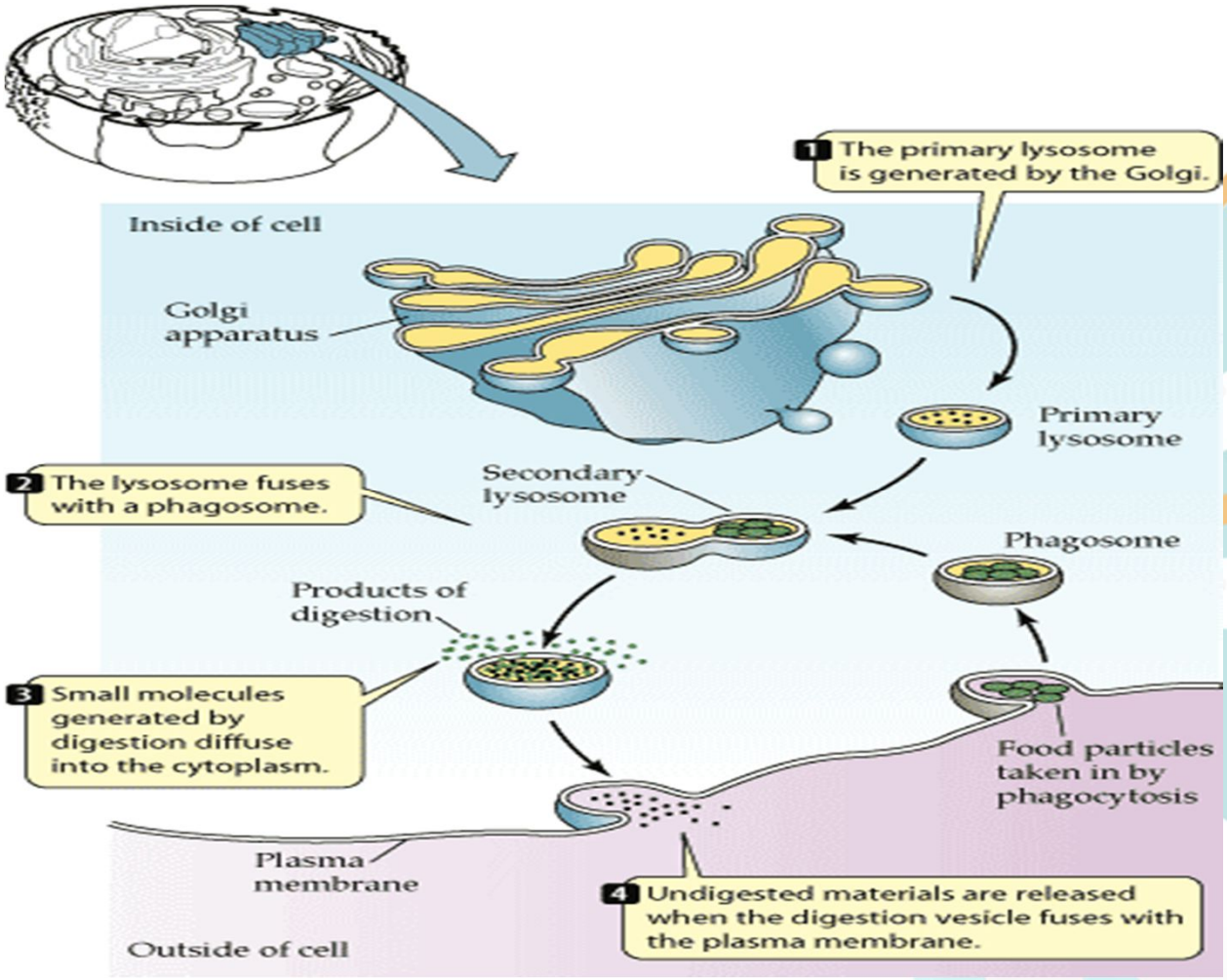
Golgi Apparatus

Receives materials from the rough ER and modifies them.

Concentrates sorts and packages proteins and sends them to other destinations.

Manufactures poly-saccharides for the plant wall

Golgi


1 The primary lysosome is generated by the Golgi.
Inside of cell
Golgi apparatus
Primary lysosome
2 The lysosome fuses with a phagosome.
Secondary lysosome
Phagosome
Products of digestion
3 Small molecules generated by digestion diffuse into the cytoplasm.
Food particles taken in by phagocytosis
Plasma membrane
4 Undigested materials are released when the digestion vesicle fuses with the plasma membrane.
Outside of cell

Lysosomes

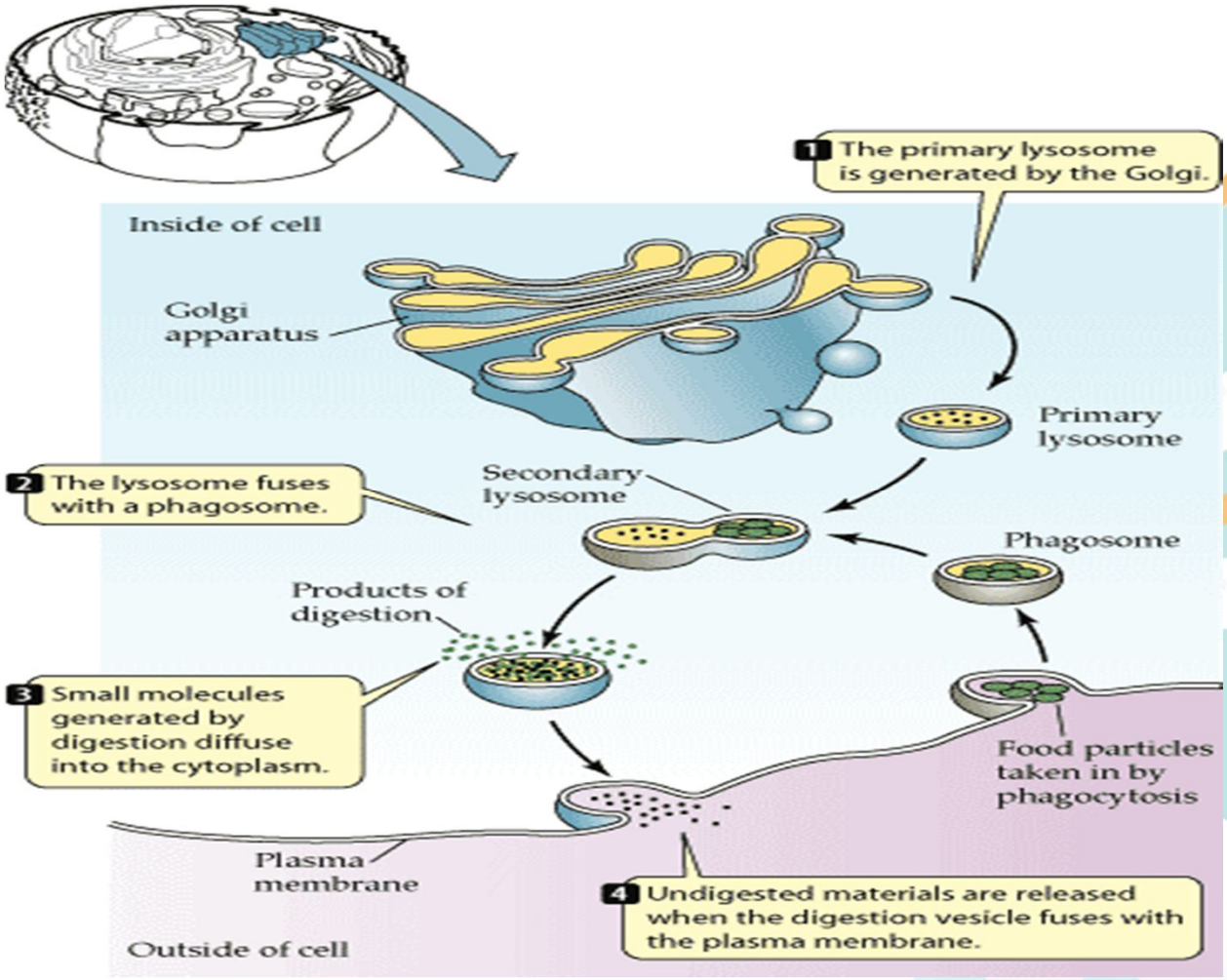
Garbage Disposal

Contain many digestive enzymes.

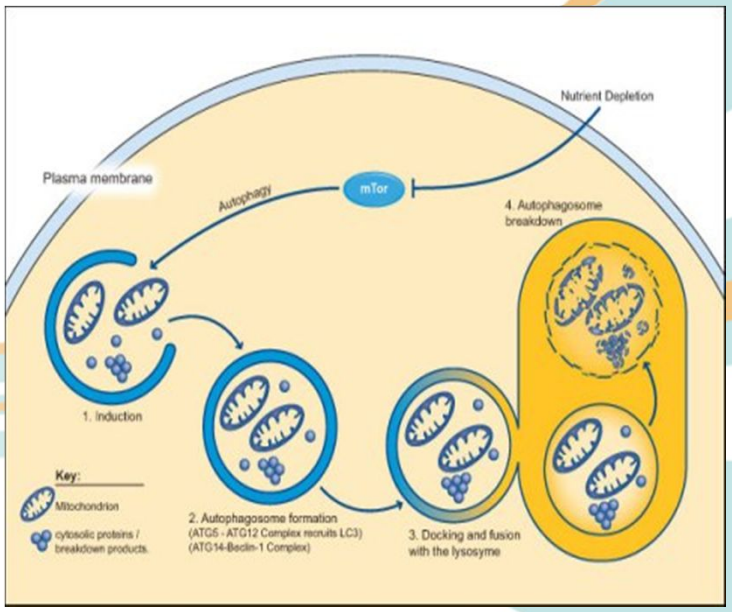
Fuse with phagosomes (autophagy).

Secondary lysosomes, where engulfed materials are digested phagocytosis.

Lysosomes


1 The primary lysosome is generated by the Golgi.
Inside of cell
Golgi apparatus
Primary lysosome
2 The lysosome fuses with a phagosome.
Secondary lysosome
Phagosome
Products of digestion
3 Small molecules generated by digestion diffuse into the cytoplasm.
Food particles taken in by phagocytosis
Plasma membrane
4 Undigested materials are released when the digestion vesicle fuses with the plasma membrane.
Outside of cell

Lysosomes



END

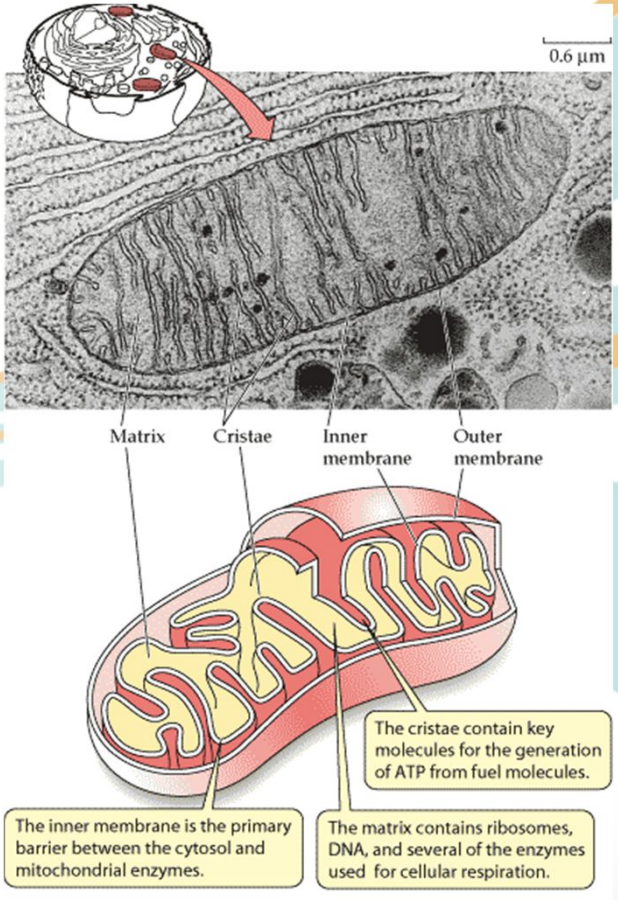
CELL BIOLOGY

Mitochondria

Mitochondria

Make ATP

Breakdown fuel



Mitochondria

Make ATP

Breakdown fuel

Some protists have one, liver more than a thousan and egg a few hundred thousand MT.

Mitochondria and chloroplasts contain their own DNA and ribosomes for making some of their own proteins

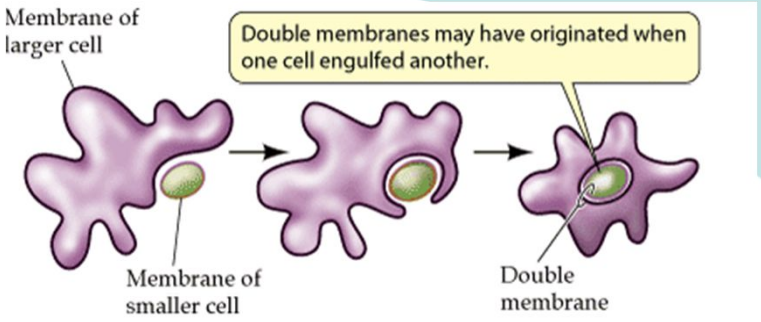
Mitochondria

Endosymbiosis

Mutual benefits permitted this symbiotic relationship

Photosynthesis

Detoxification of O_2



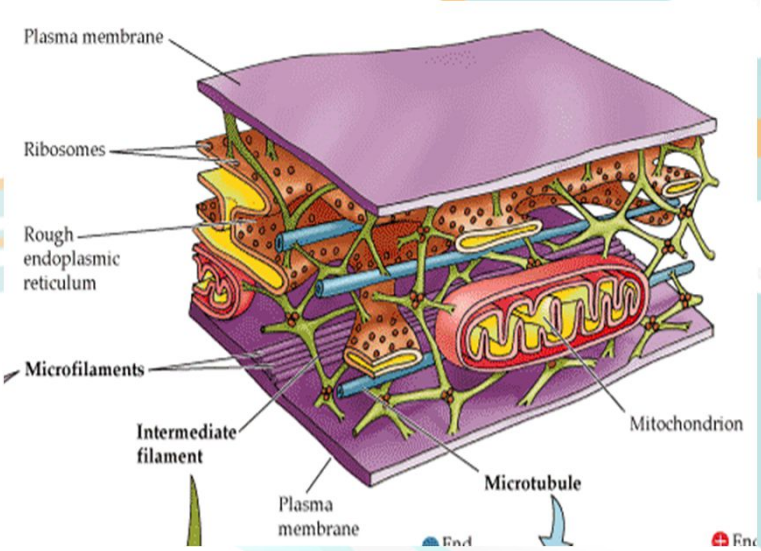
END

CELL BIOLOGY

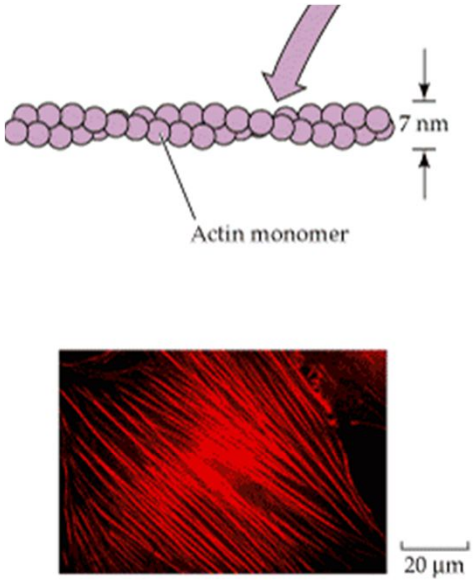
Cytoskeleton

Cytoskeleton

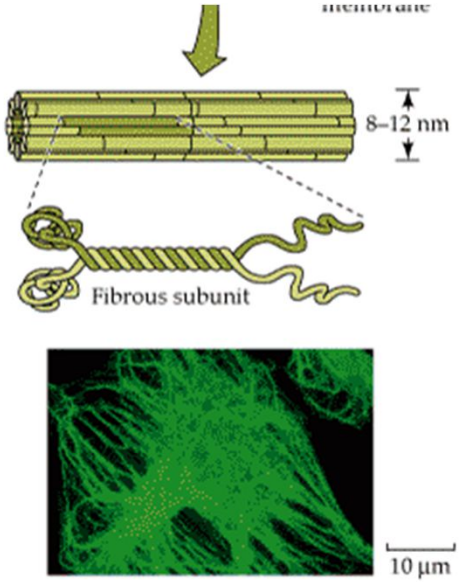
Shape & Function



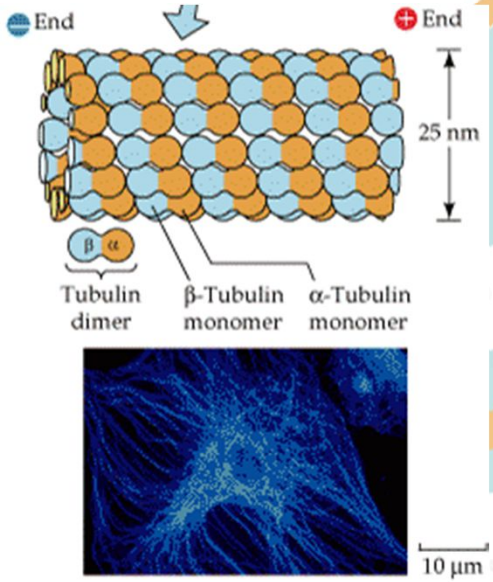
Cytoskeleton



Microfilaments are made up of strands of the protein actin and often interact with strands of other proteins. They change cell shape and drive cellular motion, including contraction, cytoplasmic streaming, and the "pinched" shape changes that occur during cell division. Microfilaments and myosin strands together drive muscle action.



Intermediate filaments are made up of fibrous proteins organized into tough, ropelike assemblages that stabilize a cell's structure and help maintain its shape. Some intermediate filaments help to hold neighboring cells together. Others make up the nuclear lamina.



Microtubules are long, hollow cylinders made up of many molecules of the protein tubulin. Tubulin consists of two subunits, α-tubulin and β-tubulin. Microtubules lengthen or shorten by adding or subtracting tubulin dimers. Microtubule shortening moves chromosomes. Interactions between microtubules drive the movement of cells. Microtubules serve as "tracks" for the movement of vesicles.

Cytoskeleton

Actin

Actin (cortical)
Stabilizes cell shape.

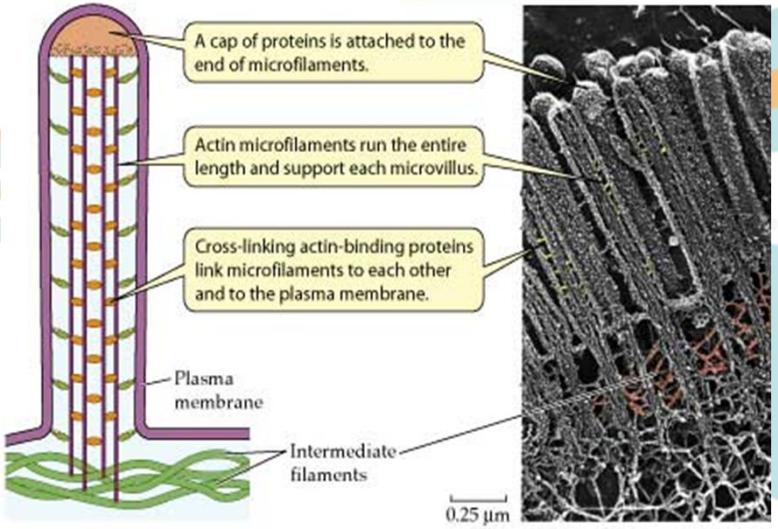
Generates movement local.

Muscle contraction.

Constriction ring in cell divission (Cytokinesis).
Cytoplasmic strreaming.

Cytoskeleton

Actin for Support



CELL BIOLOGY

Cytoskeleton 2

Cytoskeleton

IF

IF stabilize cell structure (hold organelles).

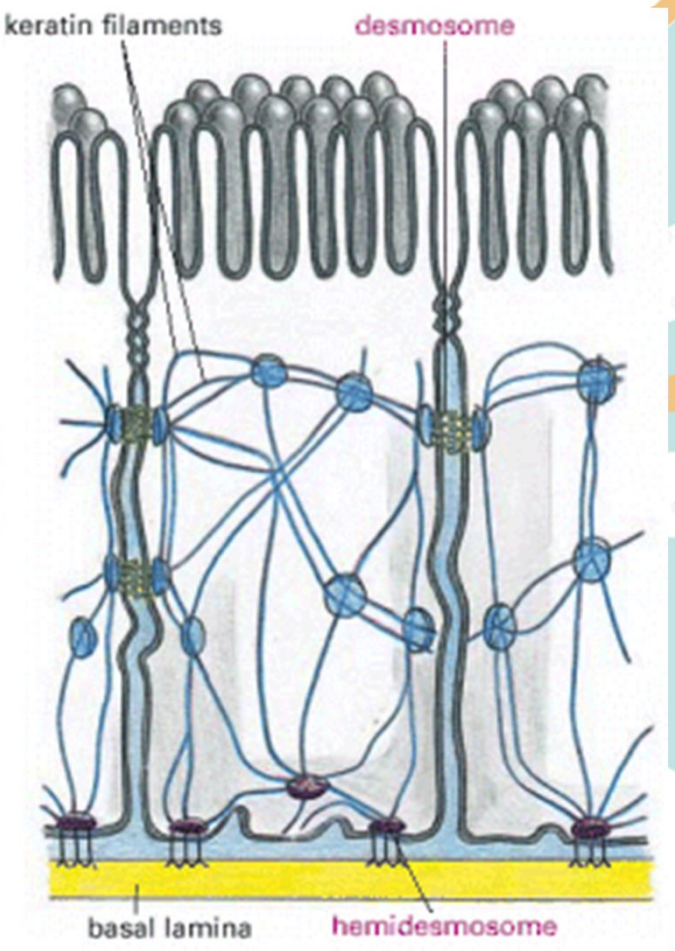
Stabilize and maintain tissue rigidity (desmosomes).

Lamins in nucleus.

Resist tention.

Cytoskeleton

IF



Cytoskeleton

Microtubules

Rigid internal skeleton resist compression.

Tracks along for motor proteins.

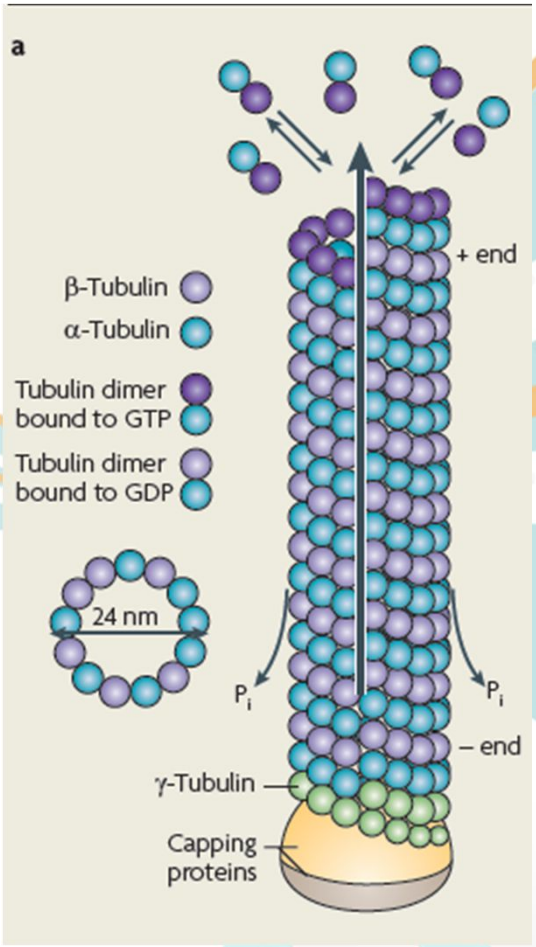
Tubulin is a dimer of

α-tubulin and β-tubulin.

The two ends different plus (+) end & minus (–) end (cat Tail & dynamic).

Cytoskeleton

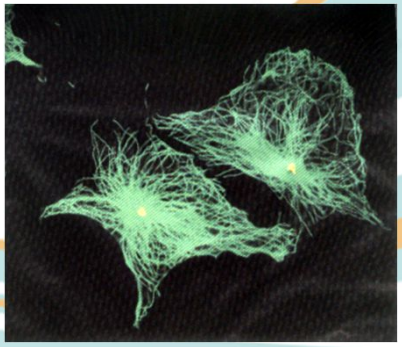
Microtubules 13

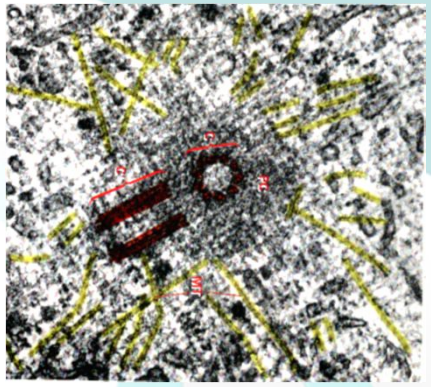


Cytoskeleton

Centrosome

Microtubule organizing center





END

CELL BIOLOGY

Motor Proteins

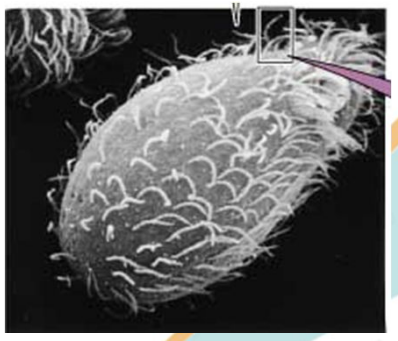
Motor Proteins

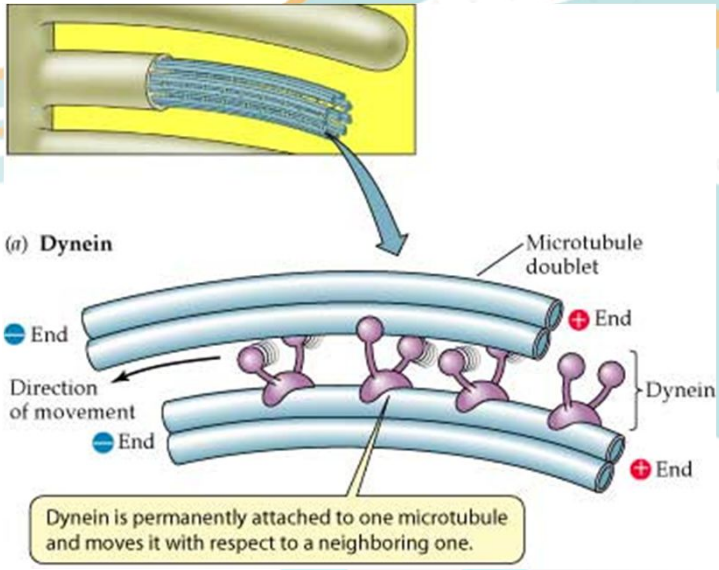
Dynein & Kinesin

Cargo & Motor Domain

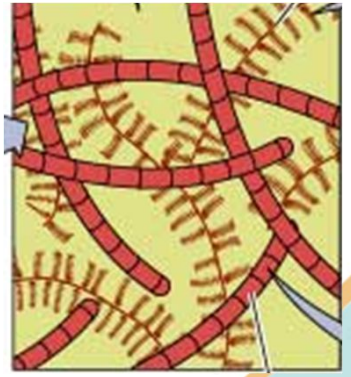
Motor Proteins

Cilia and Flagella





ECM



Holds cells in tissues.

Physical properties of tissues cartilage, skin.

Orients cell movements during development/injury

Chemical signaling from one cell to another.

END

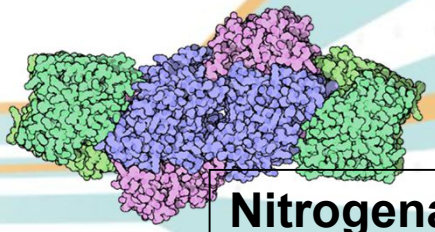
CELL BIOLOGY

The Miracle Workers

The Miracle Workers

The reaction is carried out under conditions of 150-250 atmospheres (atm), 450-500 °C; resulting in a yield of 10-20%

$$N_2 + 3H_2 \longrightarrow 2NH_3$$



Nitrogenase

$$N_2 + 8H^+ + 8e^- + 16ATP \longrightarrow 2NH_3 + H_2 + 16ADP + 16\ Pi$$

Energy Conversions & Enzymes

The Miracle Workers

One of biggest miracles of life is that it can perform slow reactions and the ones requiring extreme conditions, quickly and at physiological conditions

CELL BIOLOGY

Energy

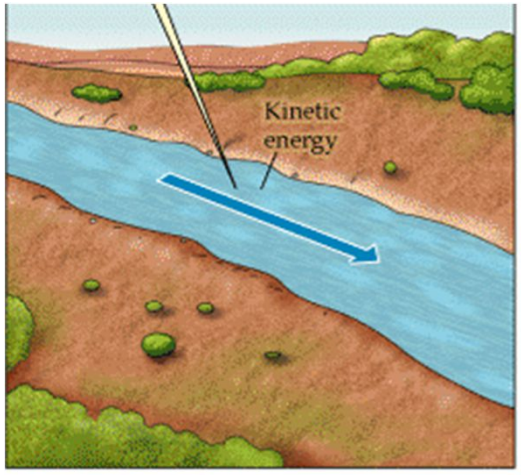
Energy

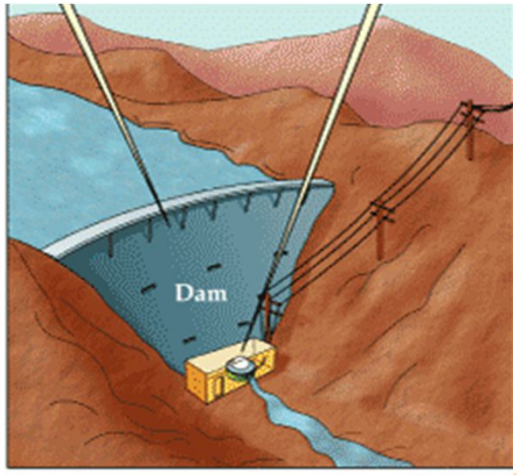
Capacity to do work

Potential energy is the energy of state or position; it includes the energy stored in chemical bonds.

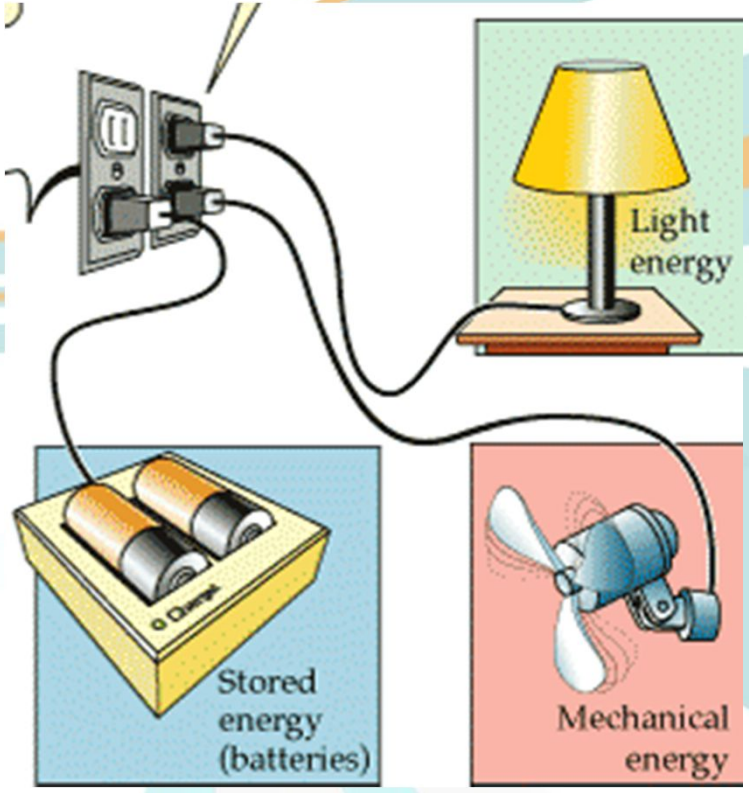
Kinetic energy is the energy of motion (and related forms such as electric energy, light, and heat).

Energy Conversions





Efficiency Not 100%

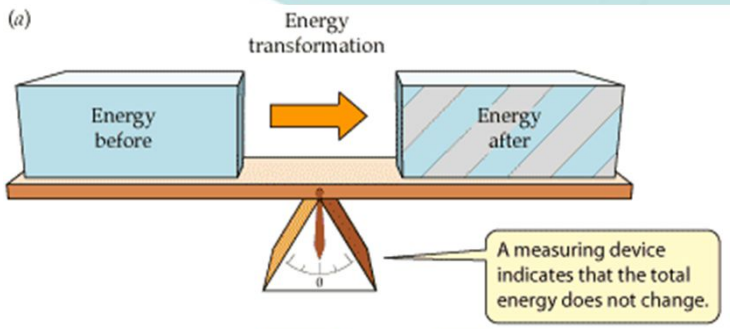


The Laws of Thermodynamics

The First Law

Energy cannot be created or destroyed.

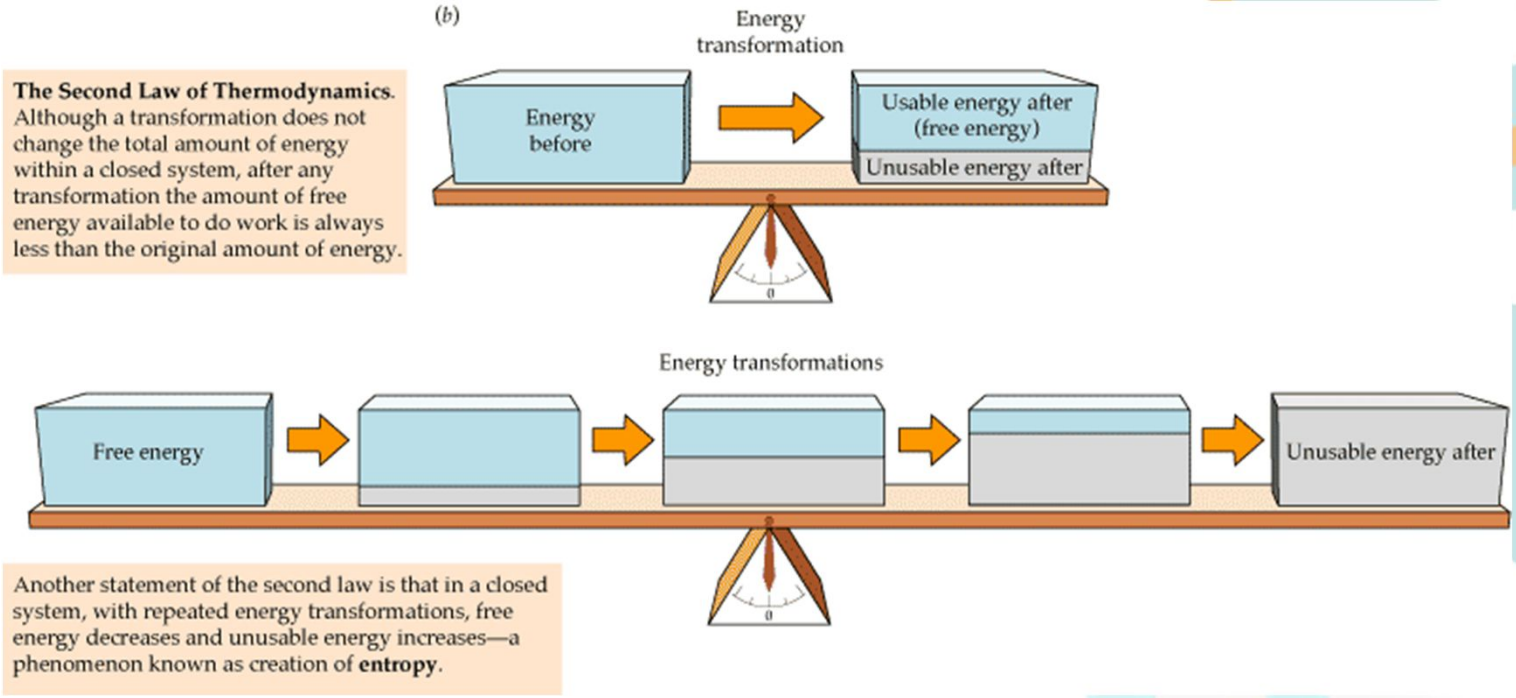
The First Law of Thermodynamics. The total amount of energy before a transformation equals the total amount after a transformation. No new energy is created, and no energy is lost.



The Laws of Thermodynamics

The Second Law

The quantity of energy available to do work (free energy) decreases and unusable energy (associated with entropy) increases.



The Second Law of Thermodynamics. Although a transformation does not change the total amount of energy within a closed system, after any transformation the amount of free energy available to do work is always less than the original amount of energy.

Another statement of the second law is that in a closed system, with repeated energy transformations, free energy decreases and unusable energy increases—a phenomenon known as creation of **entropy**.

Free Energy (ΔG)

The Second Law

total energy =

usable energy +
unusable energy

$H = G + TS$

(H = Enthalpy, G = free energy, T= absolute temperature, S = entropy)

$G = H - TS$

END

Free Energy (ΔG)

Free Energy (ΔG)

The Second Law

total energy =

usable energy +
unusable energy

$H = G + TS$

(H = Enthalpy, G = free energy, T= absolute temperature, S = entropy)

$G = H - TS$

Free Energy (ΔG)

Δ

Absolute G, H or S can not be measured

Changes in free energy, total energy, and entropy are related by the equation.

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G \text{ (reaction)} = G \text{ (products)} - G \text{ (reactants)}$$

$$\Delta H \text{ (reaction)} = H \text{ (products)} - H \text{ (reactants)}$$

$$\Delta S \text{ (reaction)} = S \text{ (products)} - S \text{ (reactants)}$$

ΔG of Combustion

Reactants		Products
$CH_4 + 2O_2$	⟶	$CO_2 + 2H_2O$
H = G + TS		H = G + TS
500 = 400 + 100		200 = 100 + 100

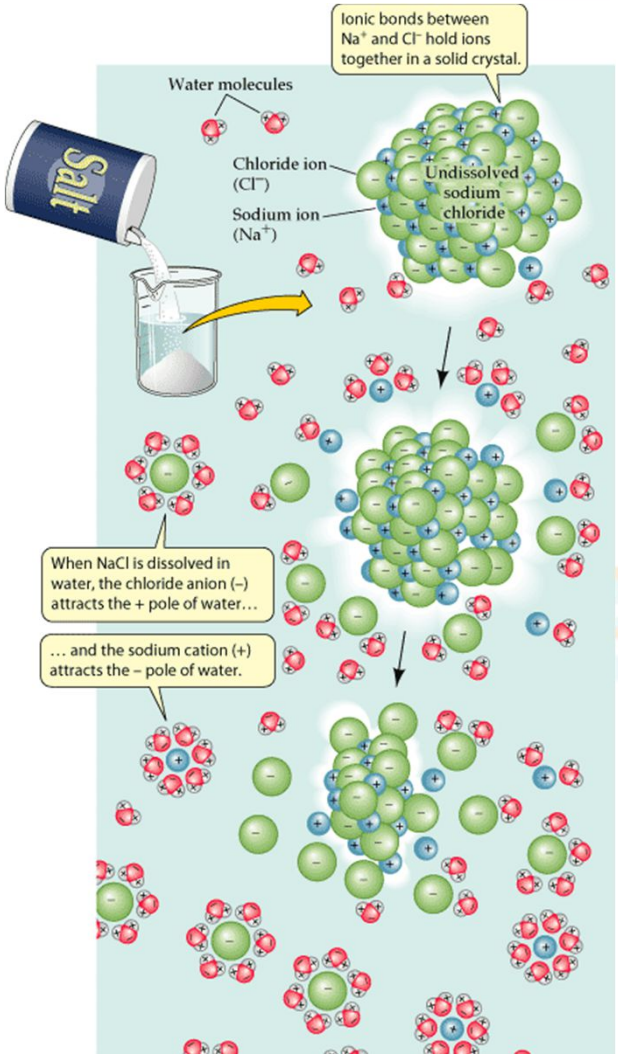
Reactants		Products
G = H – TS	⟶	G = H – TS
400 = 500 - 100		100 = 200 - 100

$$\Delta G_{(reaction)} = G_{(products)} - G_{(reactants)}$$

$$-300 = 100 - 400$$

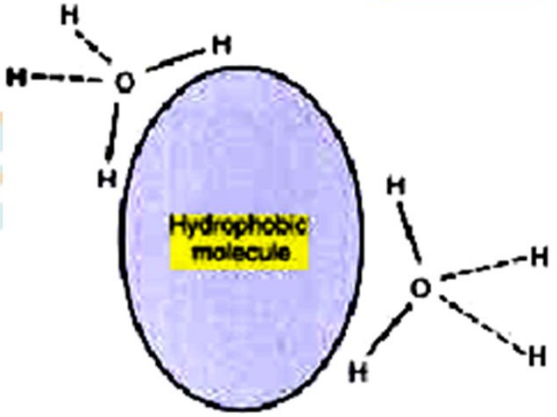
$$\Delta G / \Delta H = -300$$

Hydrophobicity Counterintuitive



????

H_2O adjacent to hydrophobic molecule are restricted to form hydrogen bonds with other water molecules.



Bonding (hydrogen) also reduces energy state.

Free Energy (ΔG)

$^{+}$ve or $^{-}$ve

If ΔG is $^{-}$ve energy is released

If ΔG is $^{+}$ve free energy is required

In living systems magnitude and sign of ΔG can depend significantly on changes in entropy ΔS

ΔS = +ve in hydrolysis

END

CELL BIOLOGY

Exergonic & Endergonic Reactions

Exergonic & Endergonic Reactions

ΔG $^{+}$ve or $^{-}$ve

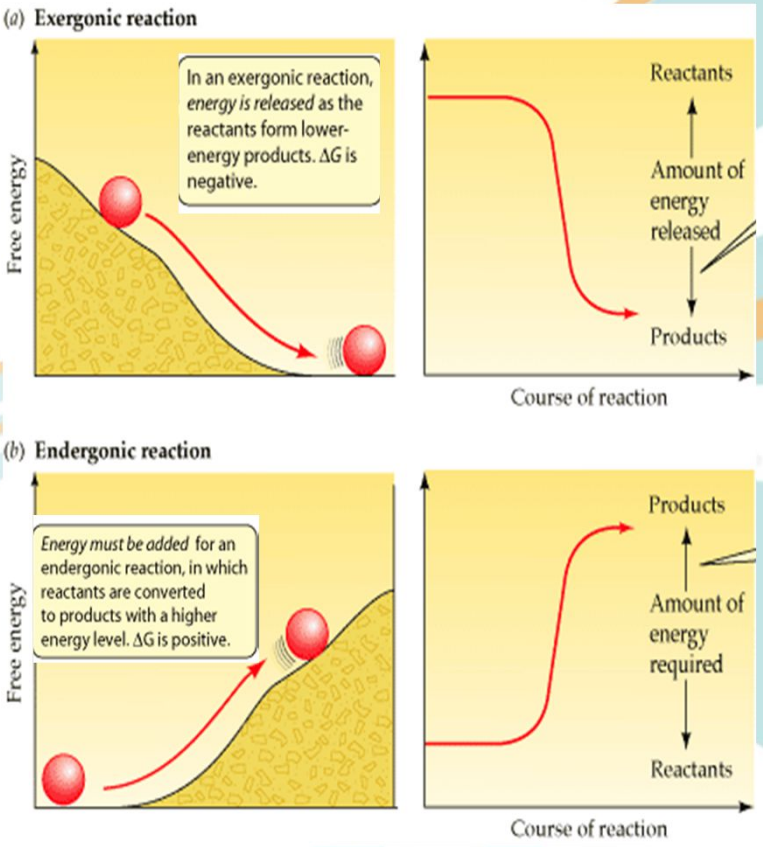
Endergonic reactions take up free energy and have a positive ΔG.

Endergonic reactions proceed only if free energy is provided

Exergonic reactions release free energy and have a negative ΔG.

Exergonic & Endergonic Reactions

ΔG $^{+}$ve or $^{-}$ve



Reversible Reactions

Equilibrium

Point at which the forward and reverse reactions proceed at the same rate.

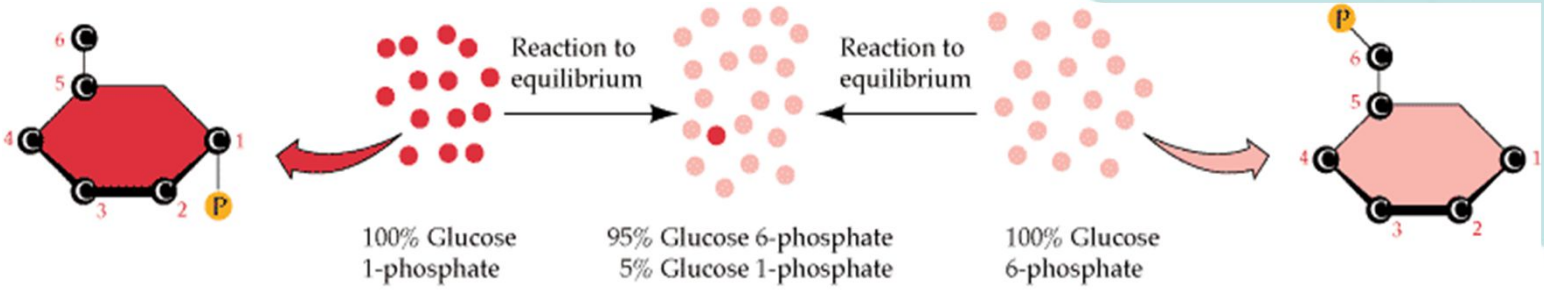
30%A
70%B
ΔG

Reversible Reactions

Equilibrium

The change in free energy (ΔG) of a reaction determines its point of chemical equilibrium,

For exergonic reactions, the equilibrium point lies toward completion (the conversion of all reactants into products).



ΔG = -1.7 kcal/mol

@ 0.02M, 25C & pH 7

Reversible Reactions

Equilibrium

At equilibrium
Amount reactant and product molecules stays constant
Although individual moles molecule keep changing their states
$\Delta G = 0$

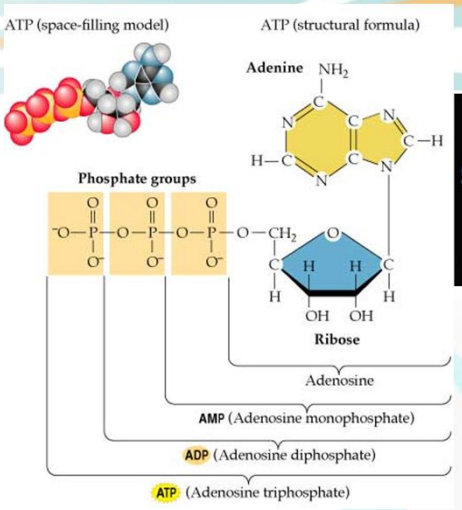
END

ATP Hydrolysis

ATP Hydrolysis

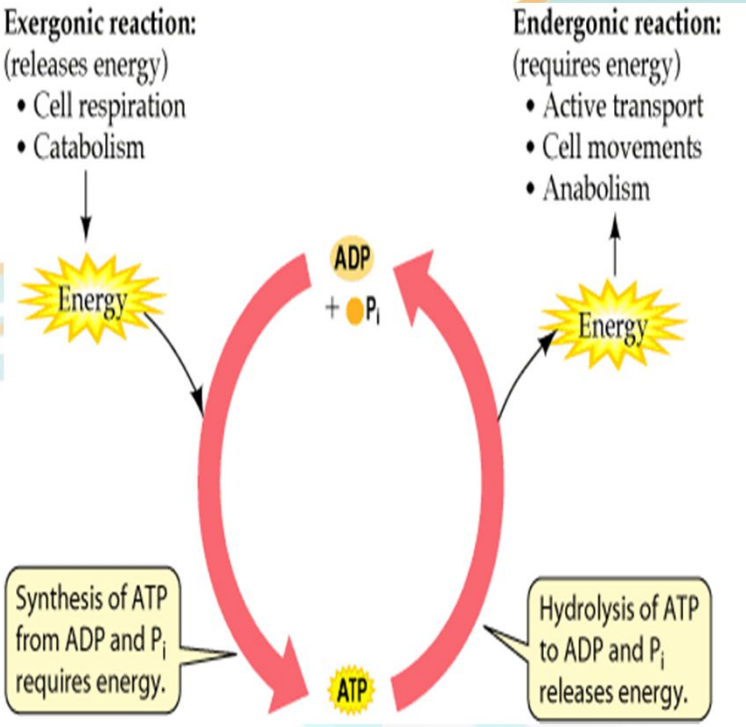
Energy Currency

Hydrolysis of ATP releases a relatively large amount of free energy (-12 kcal/mol).



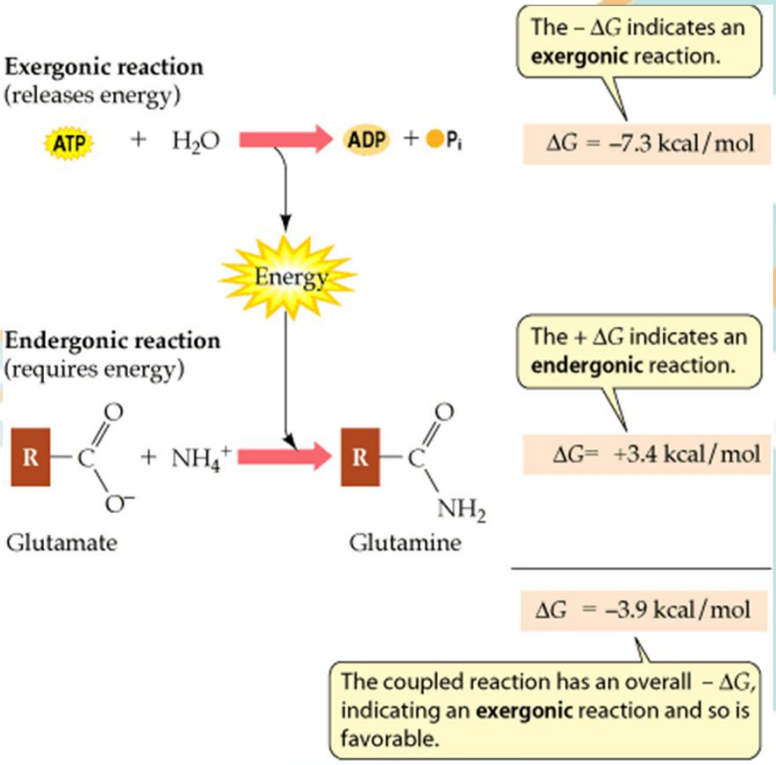
ATP Hydrolysis

Coupling



ATP Hydrolysis

Coupling



Speed of a Reaction

ΔG Independent

50% 600 poly arginine degradation in 7years Carboxypeptidase takes half a second

Reactants Need to cross an energy barrier small amount of added energy (Activation Energy E_a)

e.g. CH4 + O2

END

CELL BIOLOGY

Enzymes

Biological Catalysts

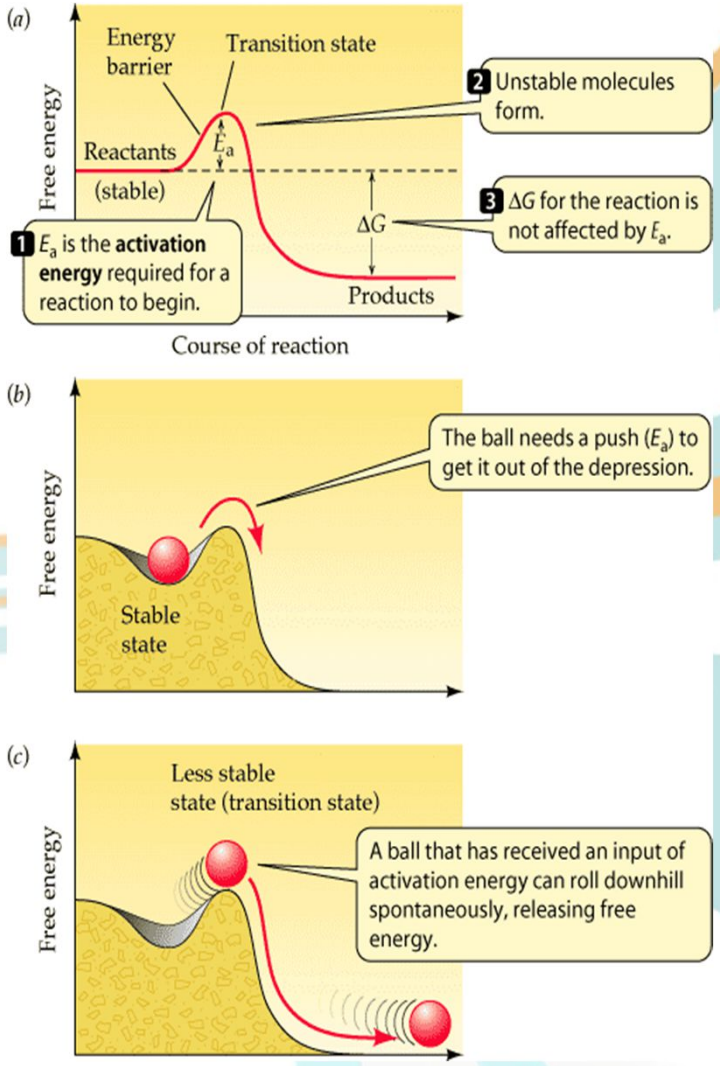
Enzymes

The more -ve ΔG is, the further the reaction proceeds to completion.

ΔG does not tell us the rate of a reaction (the speed at which it moves towards equilibrium).

Most Enzymes are proteins (ribozymes are RNA)

Activation Energy

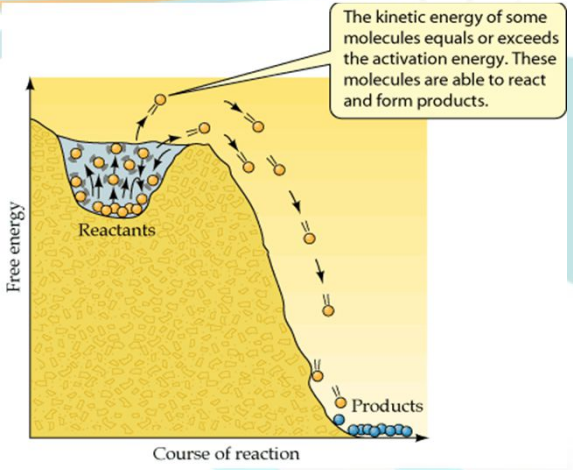


Activation Energy

Enzymes

At normal temperatures few molecules have enough kinetic energy for this transformation.

Enzymes lower the energy barrier.



END

CELL BIOLOGY

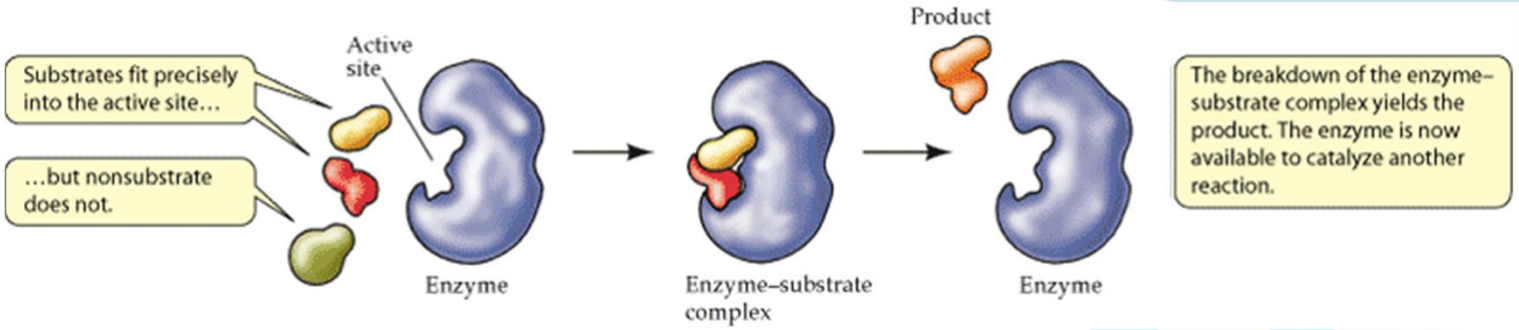
Enzymes
Mechanisms

Enzymes Specificity

Reactants (substrates) bind the a particular (active) site of the respective enzyme.

The active site = the catalytic + binding sites

The specificity of this binding is a function of three-dimensional shape, structure and environment of its active site.

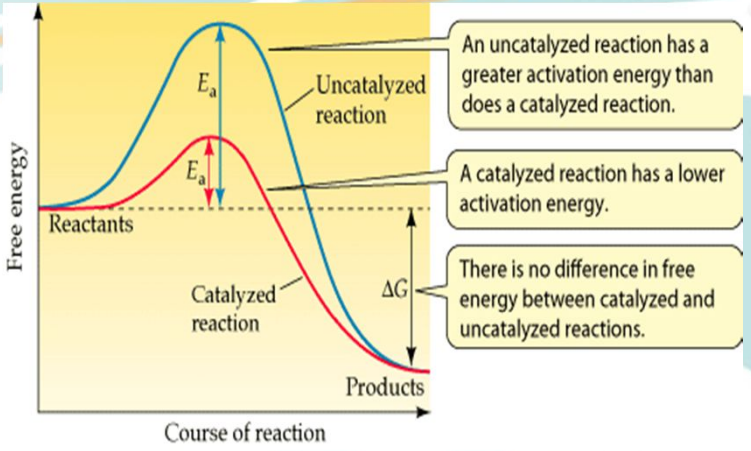


Enzymes Lower the Energy Barrier

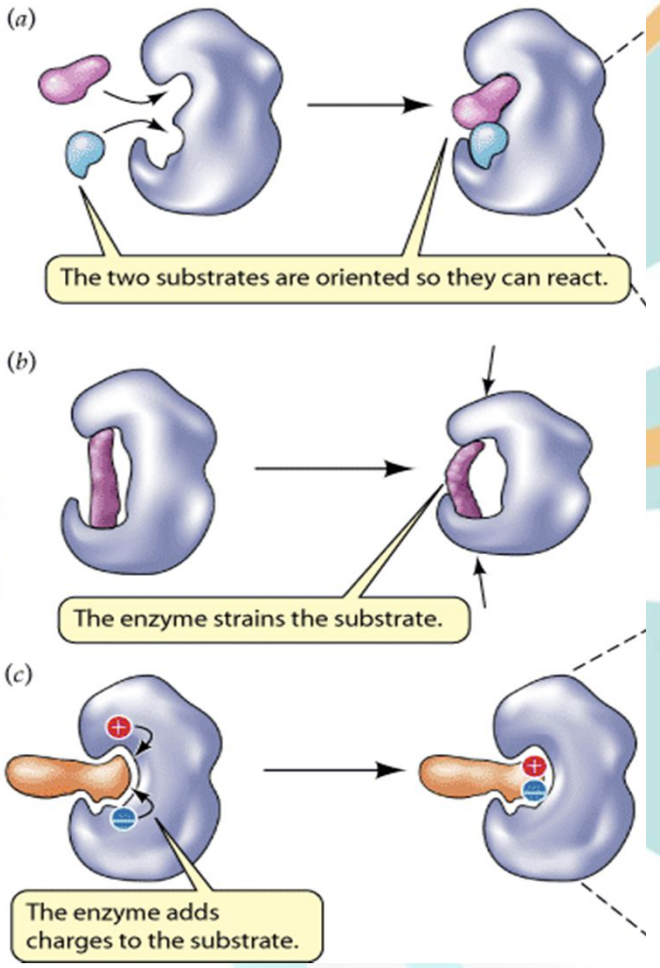
E_a

Enzymes Lower the Energy Barrier (forward and reverse reaction).

Equilibrium and ΔG are not changed



How do Enzymes Work



Induced Fit

Conformational Change

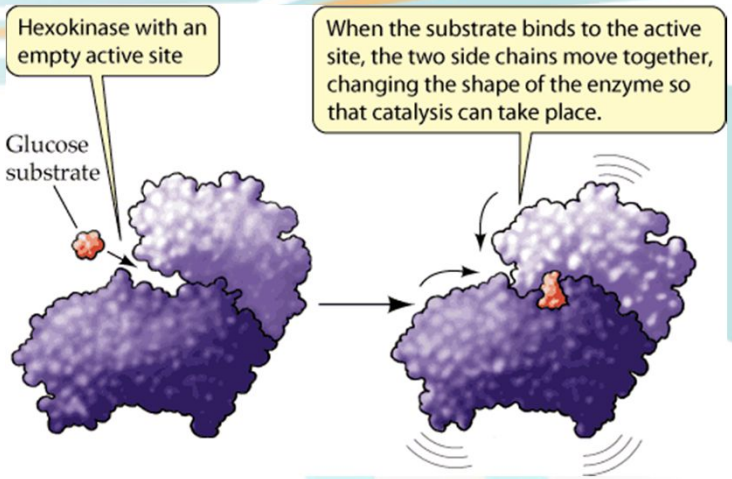
Upon binding to substrate, some enzymes change shape, facilitating catalysis.

The shape changes (Induced fitting) modifies active site and exposes/ aligns those regions that perform catalysis

Induced Fit

Conformational Change

Water exclusion at active site results in transfer of Pi from ATP to glucose preventing APT hydrolysis to ADP



CELL BIOLOGY

Enzyme Helpers

Enzyme Helpers

6.1 ***A Few Examples of Nonprotein "Partners" of Enzymes***

TYPE OF MOLECULE	ROLE IN CATALYZED REACTIONS
Cofactors	
Iron (Fe^{2+} or Fe^{3+})	Oxidation/reduction
Copper (Cu^{+} or Cu^{2+})	Oxidation/reduction
Zinc (Zn^{2+})	Helps bind NAD
Coenzymes	
Biotin	Carries $—COO^{-}$
Coenzyme A	Carries $—CH_2—CH_3$
NAD	Carries electrons
FAD	Carries electrons
ATP	Provides/extracts energy
Prosthetic groups	
Heme	Binds ions, O_2, and electrons; contains iron cofactor
Flavin	Binds electrons
Retinal	Converts light energy

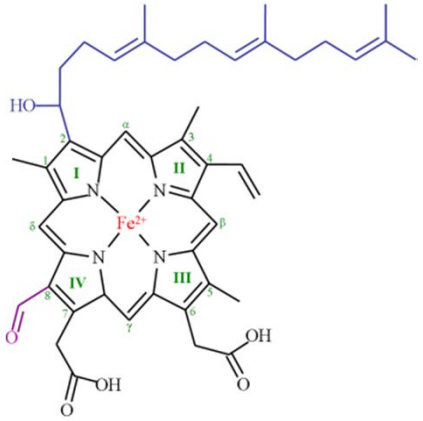
Enzyme Helpers

Cofactors: Inorganic ions (Cu, Zn & Fe) bound to some enzymes and essential to their function.

Coenzymes: Carbon containing molecules bind enzymes temporarily chemically changed during reaction (substrate).

Enzyme Helpers

Prosthetic groups: non-protein component permanently bound to enzymes (heme group), without its prosthetic group apoprotein vs. holoprotein.

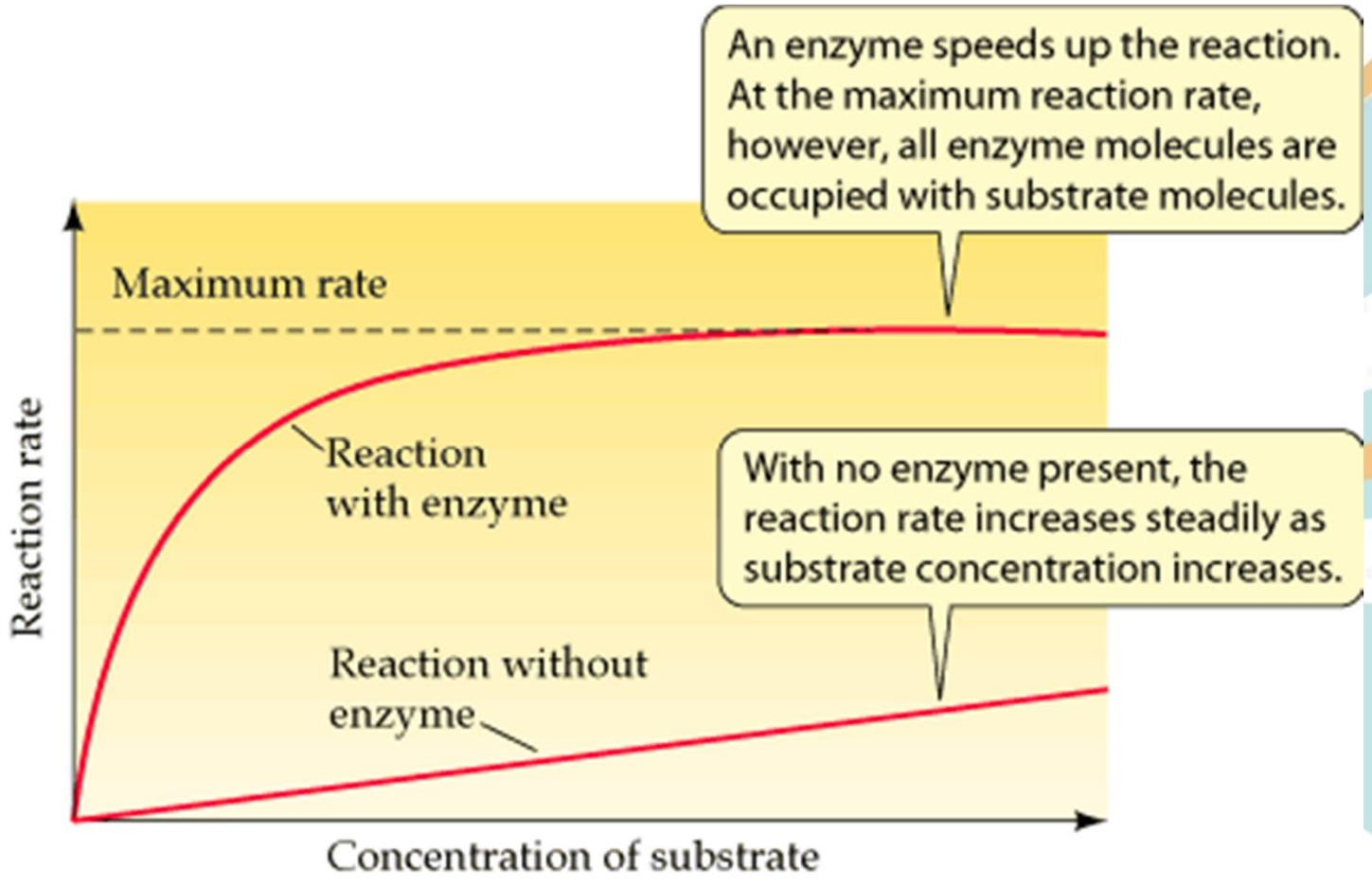


END

CELL BIOLOGY

Enzyme Regulation

Substrate Concentration



Enzyme Regulation

The Inhibitors

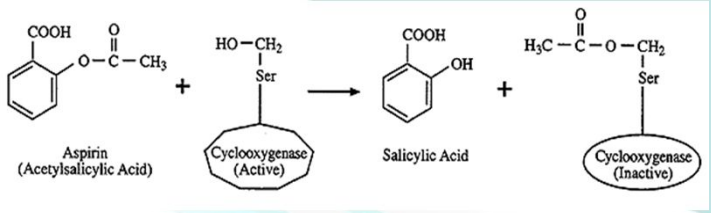
Some factors can activate enzymes and other can deactivate them (inhibit their function).

1. Irreversible inhibitors
2. Competitive inhibiters
3. Noncompetitive inhibiters

Enzyme Regulation

Irreversible inhibitors

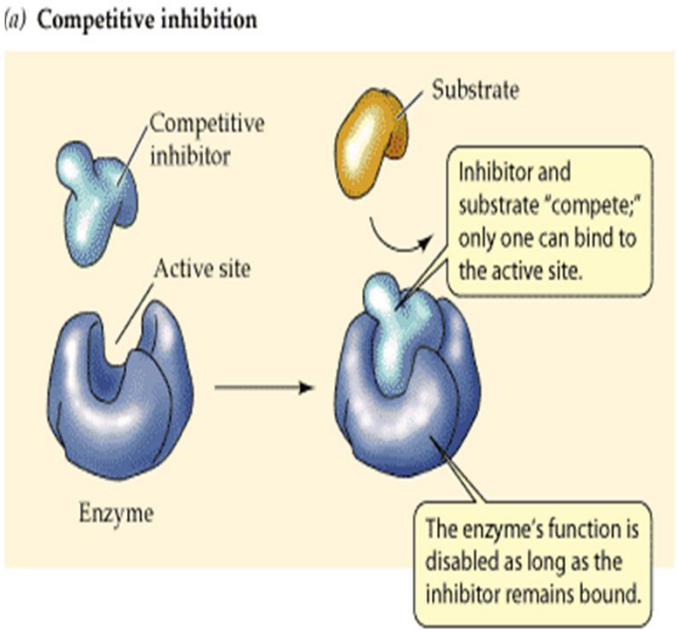
Permanently inhibit enzymes
function generally by modifying the active site

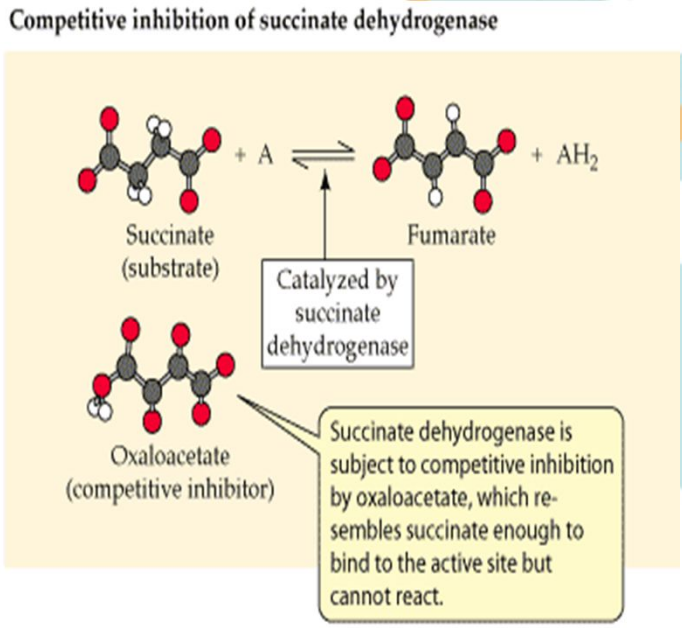


Enzyme Regulation

Reversible inhibitors

Competitive inhibiters are similar to the natural substrate (bind the active site) however they are different enough that no reaction is catalyzed.

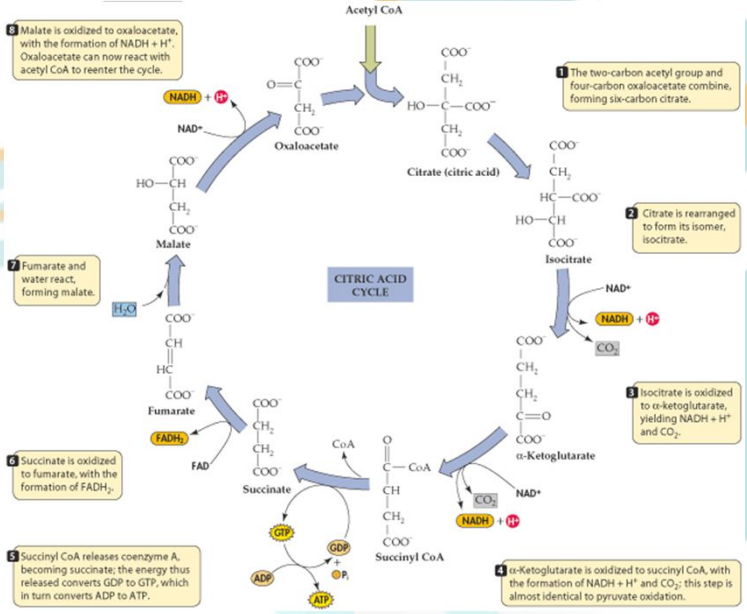




Enzyme Regulation

The Citric Acid Cycle

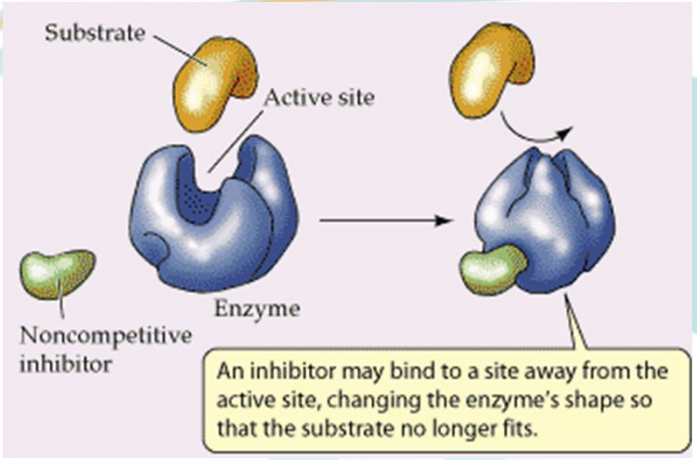
Feed Back/End Product Inhibition



Enzyme Regulation

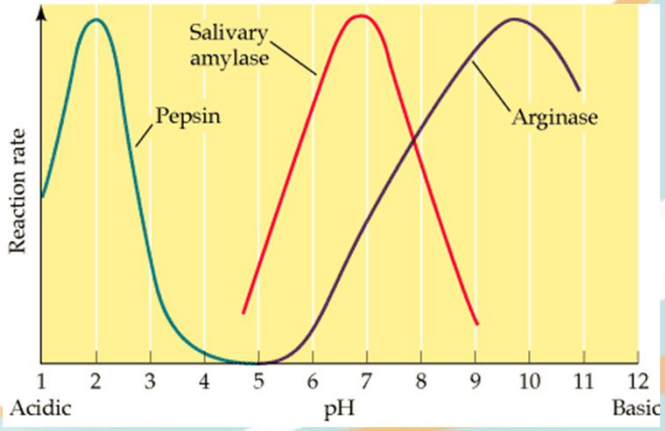
Noncompetitive inhibiters

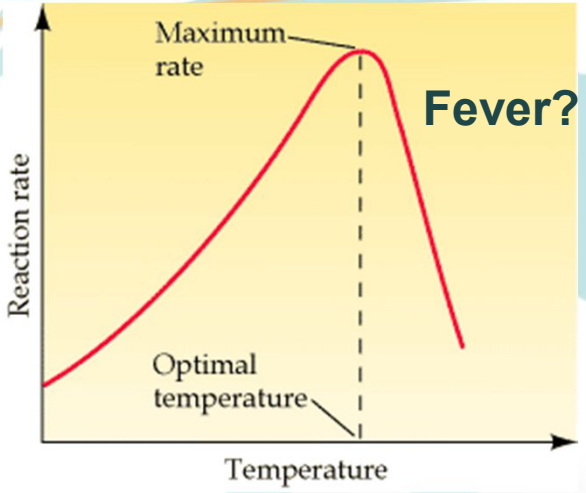
Bind the enzyme at nonactive site cause a conformational change prevents enzyme from binding its substrate.



Physical Factors

Influence Enzymes

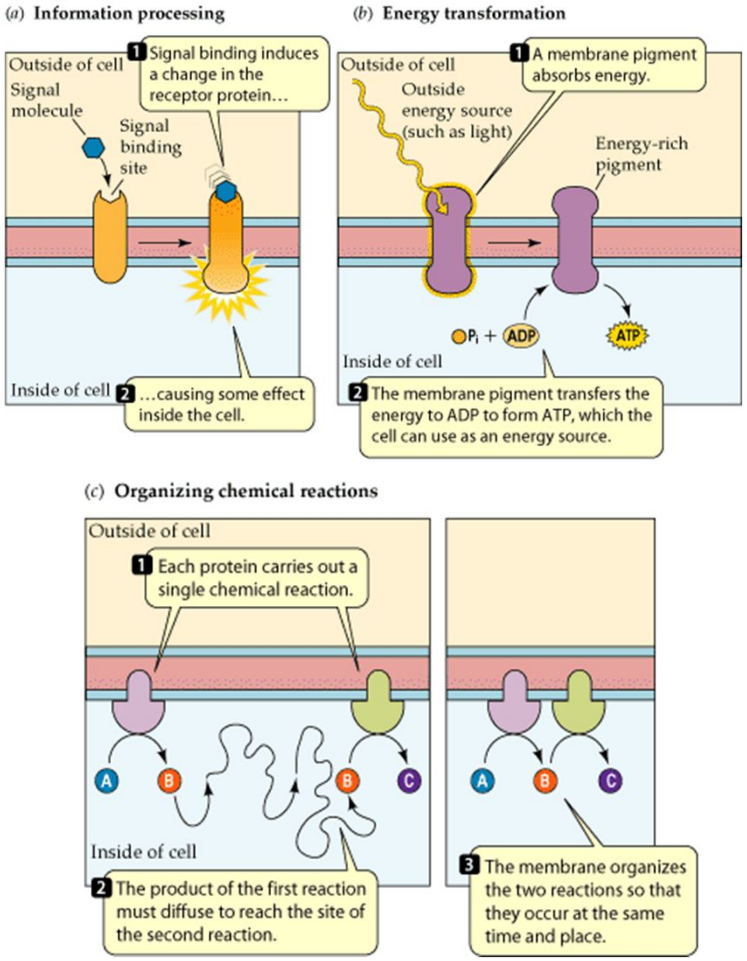


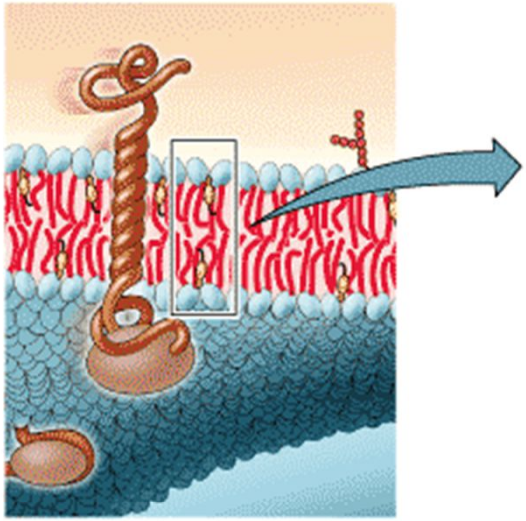


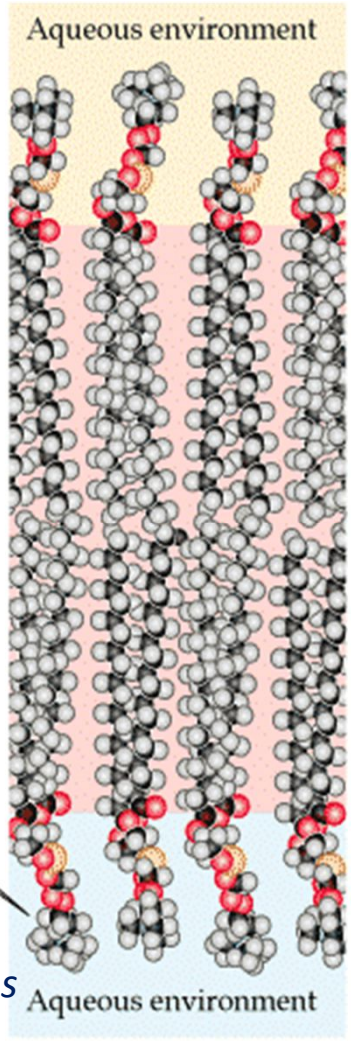
END

Membranes

More Than Barriers



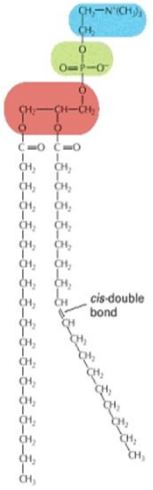


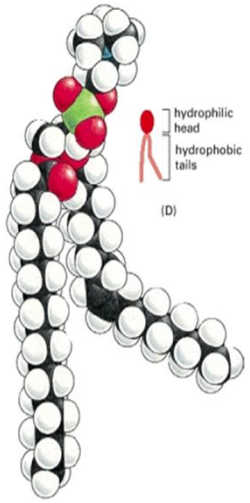


The nonpolar, hydrophobic fatty acid "tails" interact with one another in the interior of the bilayer.

The charged, or polar, hydrophilic "head" portions interact with polar water.

Amphipathic: *Hydrophilic regions + Hydrophobic regions*



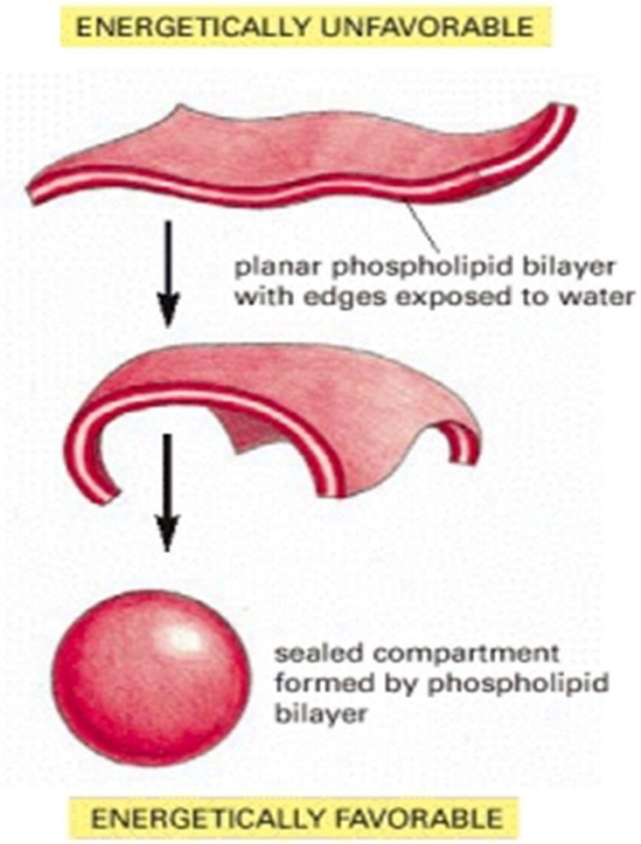


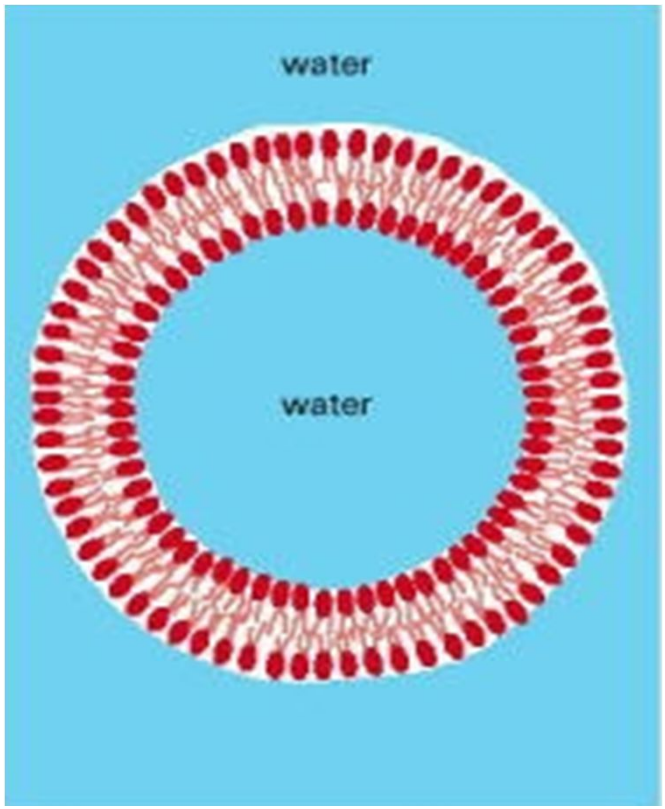
Fatty Acid Length: 14-24 Cs

One Tail Unsaturated Cis

Spontaneous Assembly

Free Energy Dictates





Features

Phospholipids coexist with water form a bilayer

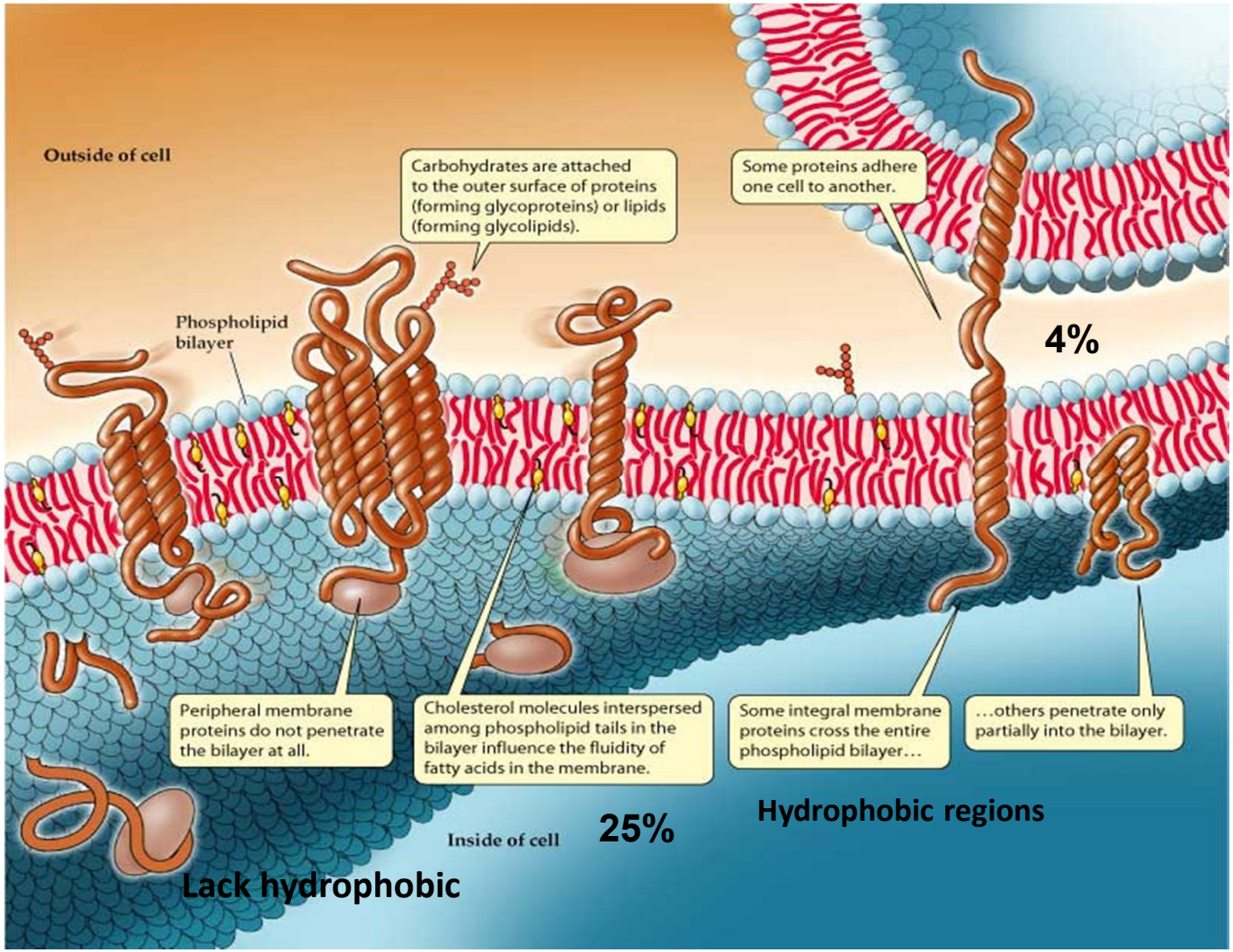
Dynamic structures: parts move & change.

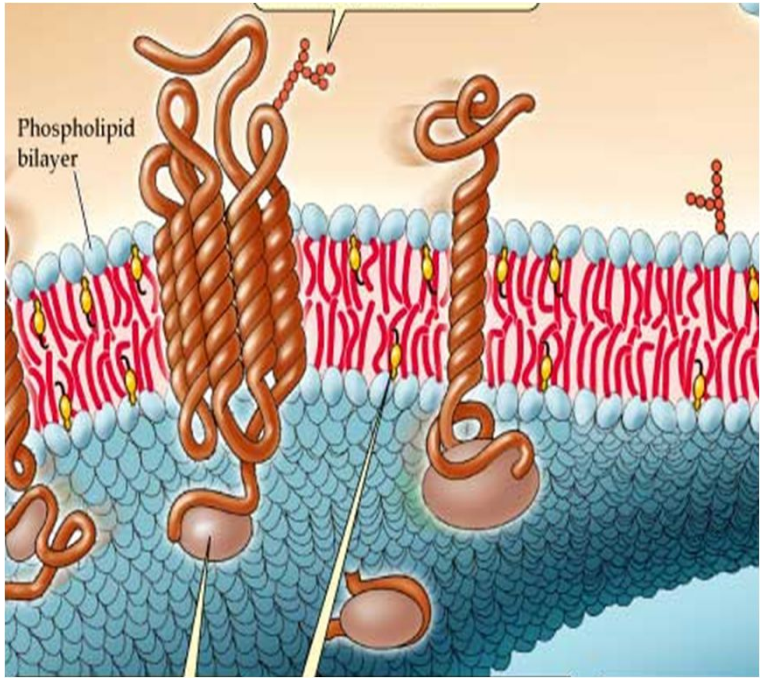
Vital physiological roles: cell cell & cell interaction with environmental

Physical integrity: lipids, create a barrier to hydrophilic materials such as water and ions

END

Membranes Components


Outside of cell
Carbohydrates are attached to the outer surface of proteins (forming glycoproteins) or lipids (forming glycolipids).
Some proteins adhere one cell to another.
Phospholipid bilayer
4%
Peripheral membrane proteins do not penetrate the bilayer at all.
Cholesterol molecules interspersed among phospholipid tails in the bilayer influence the fluidity of fatty acids in the membrane.
Some integral membrane proteins cross the entire phospholipid bilayer…
…others penetrate only partially into the bilayer.
Hydrophobic regions
Inside of cell
25%
Lack hydrophobic

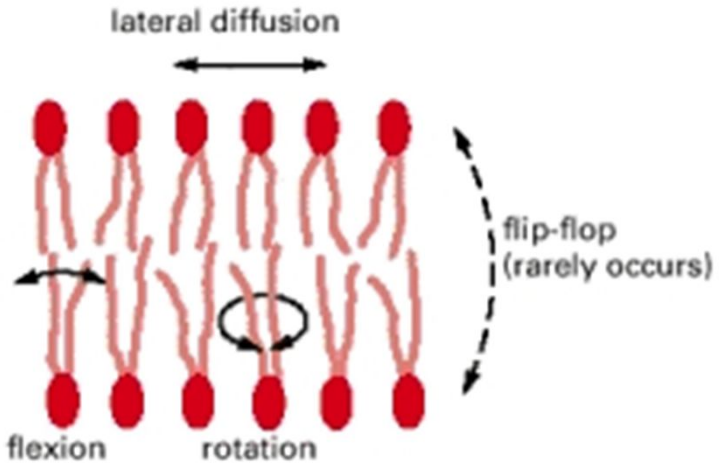


Fatty acids interact with each other & the polar regions face aqueous environment outside

Lipid “lake” in which a variety of proteins “float”

Each membrane has a set of proteins suitable to the specialized function of the cell or organelle it surrounds.

Carbohydrates attached either to lipids/protein molecules



Movement

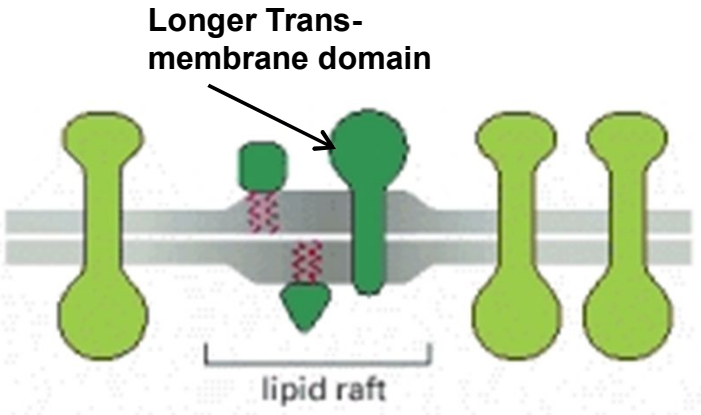
A phospholipid molecule travels from one end the other in about 1 second

Phospholipid molecule flip overs seldom (polar part of molecule hardly ever moves through the hydrophobic interior

Inner and outer halves of the bilayer have different phospholipids

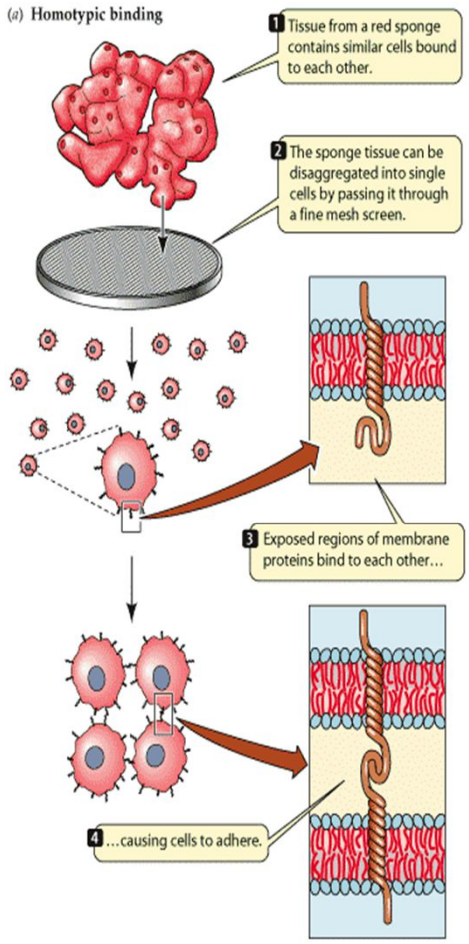
Lipid Rafts

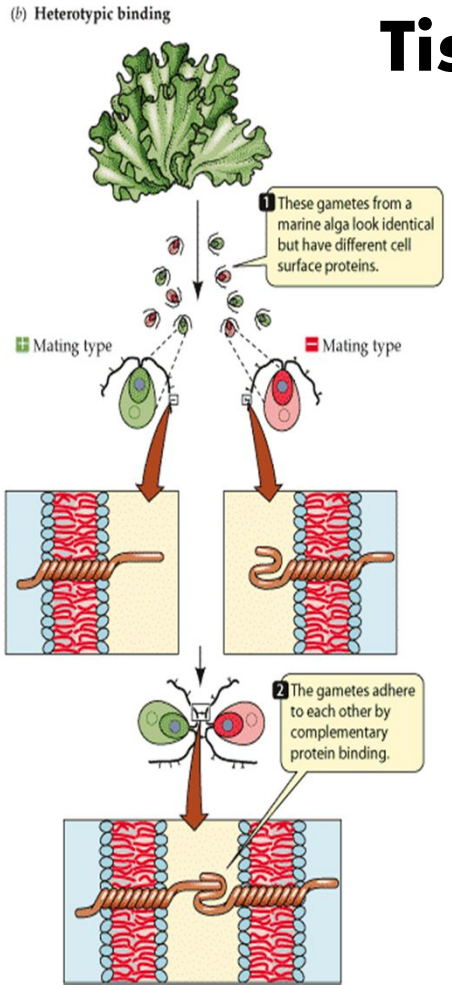
Some proteins migrate freely others “anchored” to specific semisolid region different lipid composition (very long fatty acid chains)



END

Recognition & Adhesion

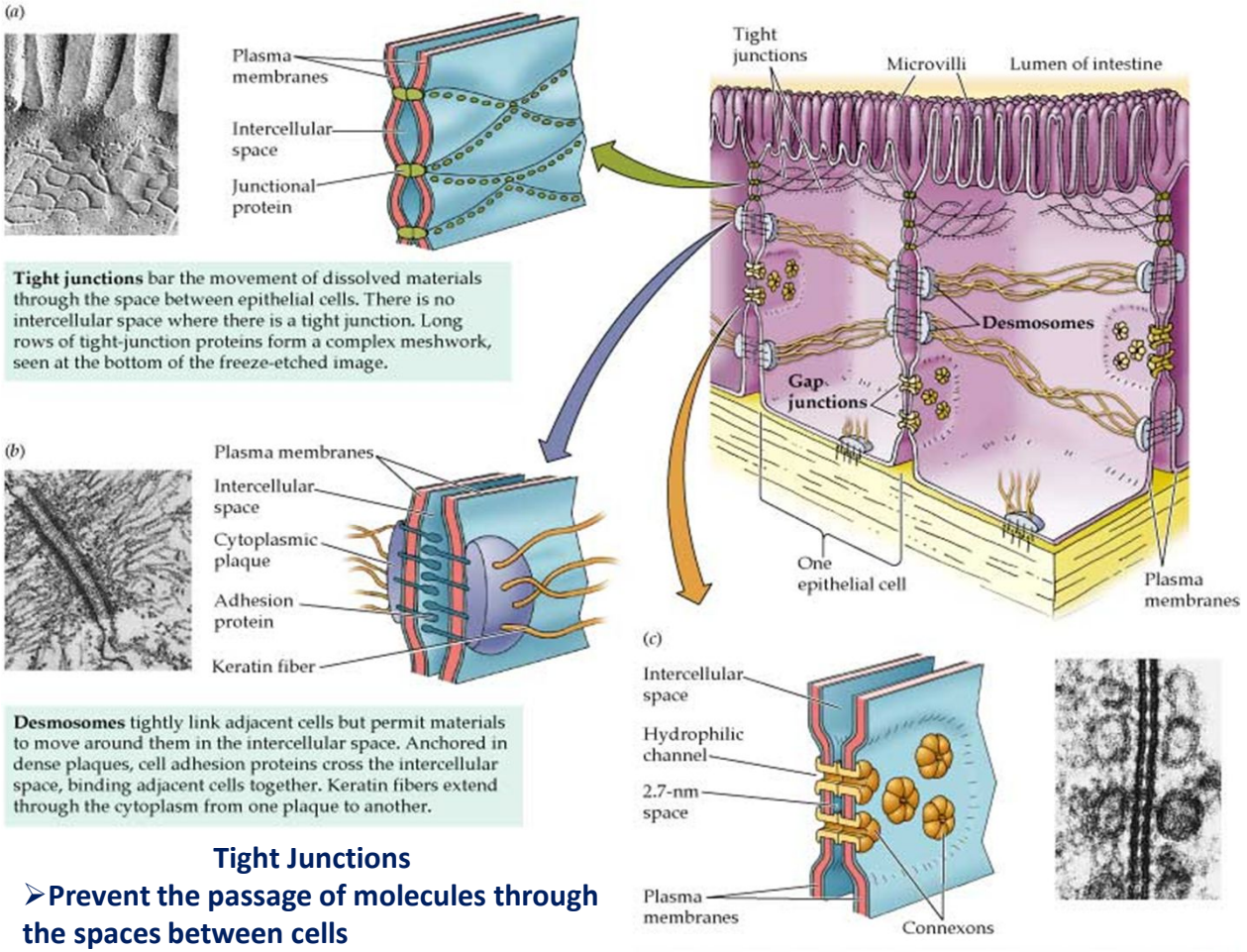




Tissue Formation

Cell recognition: One cell specifically binds to another cell

Cell adhesion: relationship between the two cells is “cemented”



Tight junctions bar the movement of dissolved materials through the space between epithelial cells. There is no intercellular space where there is a tight junction. Long rows of tight-junction proteins form a complex meshwork, seen at the bottom of the freeze-etched image.

Desmosomes tightly link adjacent cells but permit materials to move around them in the intercellular space. Anchored in dense plaques, cell adhesion proteins cross the intercellular space, binding adjacent cells together. Keratin fibers extend through the cytoplasm from one plaque to another.

Gap junctions let adjacent cells communicate. Dissolved molecules and electric signals may pass from one cell to the other through the channel formed by two connexons extending from adjacent cells.

Tight Junctions

- **Prevent the passage of molecules through the spaces between cells**
- **Restricting the migration of lipid molecules & membrane proteins**

Passive Transport

Does not require any input of energy to drive the process

Active Transport

Requires input of energy to drive the process

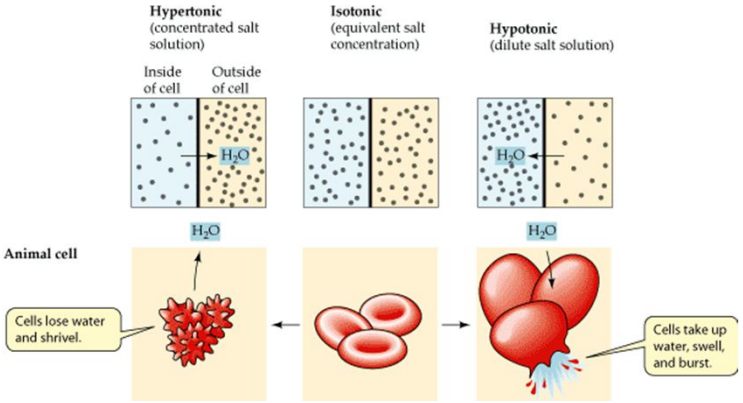
END

Transport Across Membrane

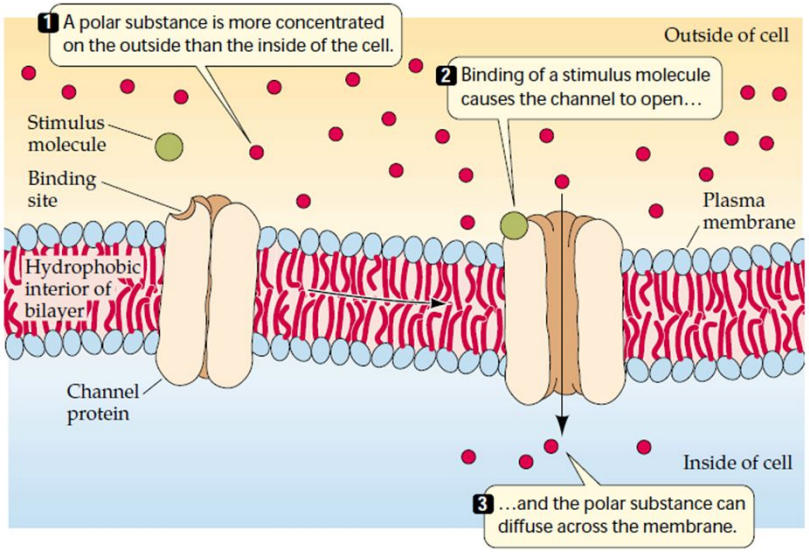
Aqua-porins:

Water molecules move through membranes by Osmosis (diffusion).

Number of solute particles not on the kinds of particles.



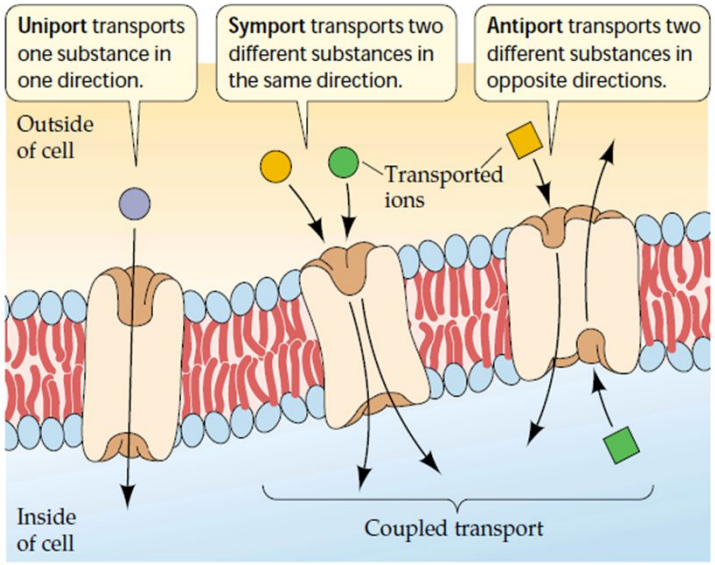
The Two Domains



Polar amino acids and water on the inside of the channel pore (to bind to the polar or charged substance and allow it to pass through)

Nonpolar amino acids on the outside of the macromolecule (to allow the channel protein to insert itself into the lipid bilayer)

Primary active transport: ATP required directly. Secondary active transport: ATP not used directly; ion conc gradient (PAT).



END

Cell Division

Life Span & Continuity

All living organisms have a limited life span.

The essence of life is to propagate.

Cells are the basic unit of life.

Cell division is the basic process that ensures the continuity of life.

The Necessity

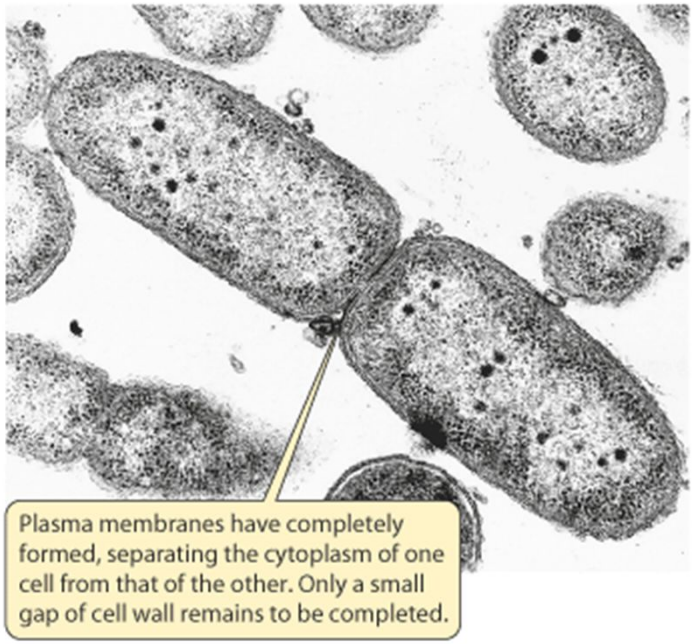
Unicellular organisms propagate by splitting into two cells.

Multicellular organisms: replace damaged or senescent cells.

Billions of cells die each day in humans, (blood epithelia)

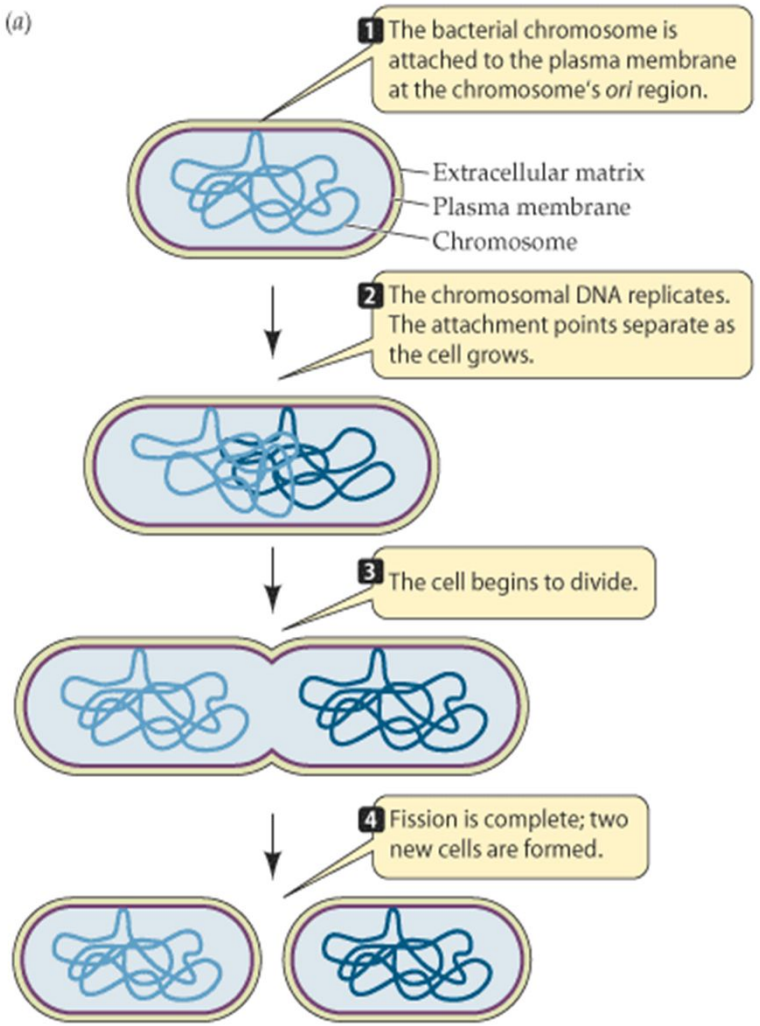
Meiosis: Organisms shuffle genes.

Even the Prokaryotes Do It



Favorable Conditions

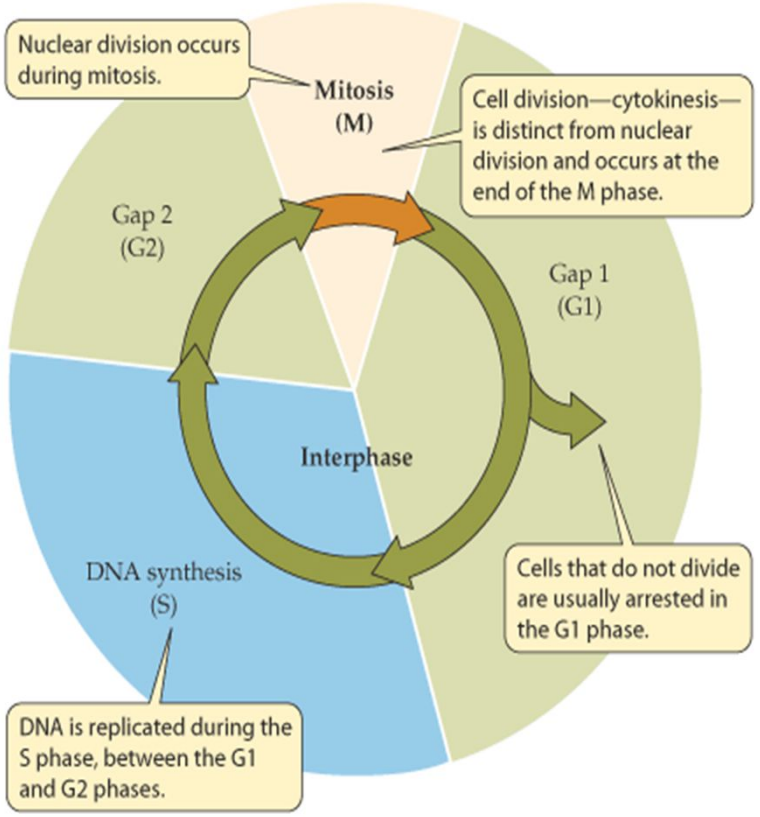
20 - 40 min

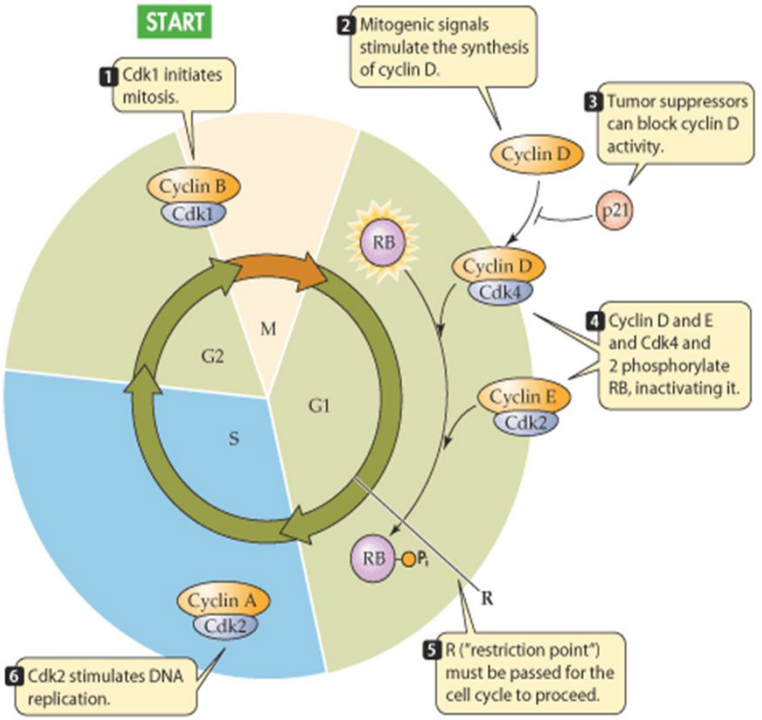

(a)
1 The bacterial chromosome is attached to the plasma membrane at the chromosome's *ori* region.
Extracellular matrix
Plasma membrane
Chromosome
2 The chromosomal DNA replicates. The attachment points separate as the cell grows.
3 The cell begins to divide.
4 Fission is complete; two new cells are formed.

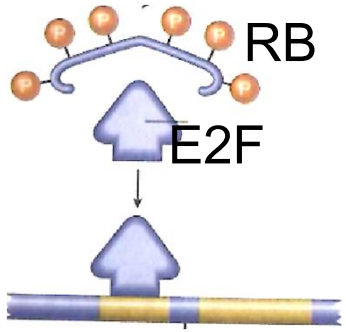
END

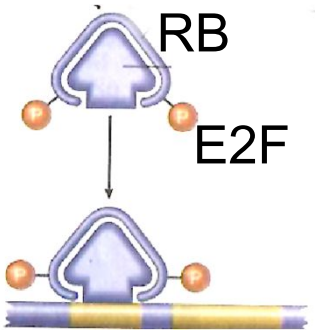
Cell Cycle

Cell Cycle


Nuclear division occurs during mitosis.
Mitosis (M)
Cell division—cytokinesis—is distinct from nuclear division and occurs at the end of the M phase.
Gap 2 (G2)
Gap 1 (G1)
Interphase
DNA synthesis (S)
Cells that do not divide are usually arrested in the G1 phase.
DNA is replicated during the S phase, between the G1 and G2 phases.


START
1 Cdk1 initiates mitosis.
2 Mitogenic signals stimulate the synthesis of cyclin D.
3 Tumor suppressors can block cyclin D activity.
Cyclin D
p21
Cyclin B
Cdk1
RB
Cyclin D
Cdk4
M
G2
G1
S
4 Cyclin D and E and Cdk4 and 2 phosphorylate RB, inactivating it.
Cyclin E
Cdk2
RB
Pi
R
Cyclin A
Cdk2
6 Cdk2 stimulates DNA replication.
5 R ("restriction point") must be passed for the cell cycle to proceed.

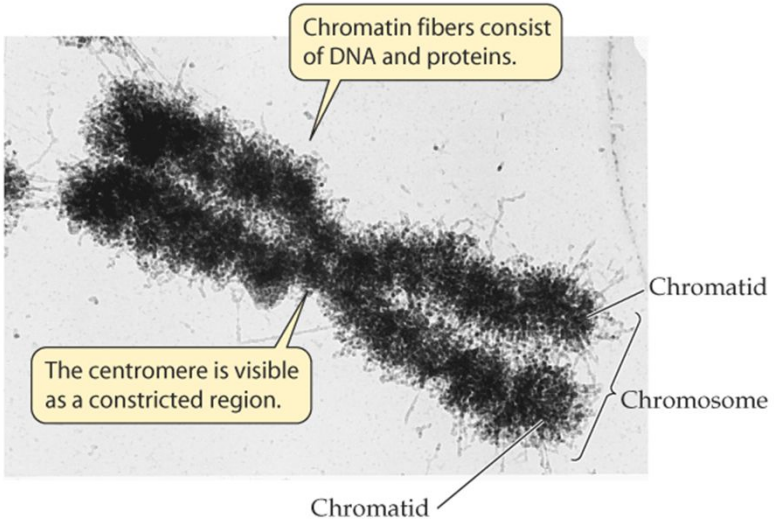

RB
E2F


RB
E2F

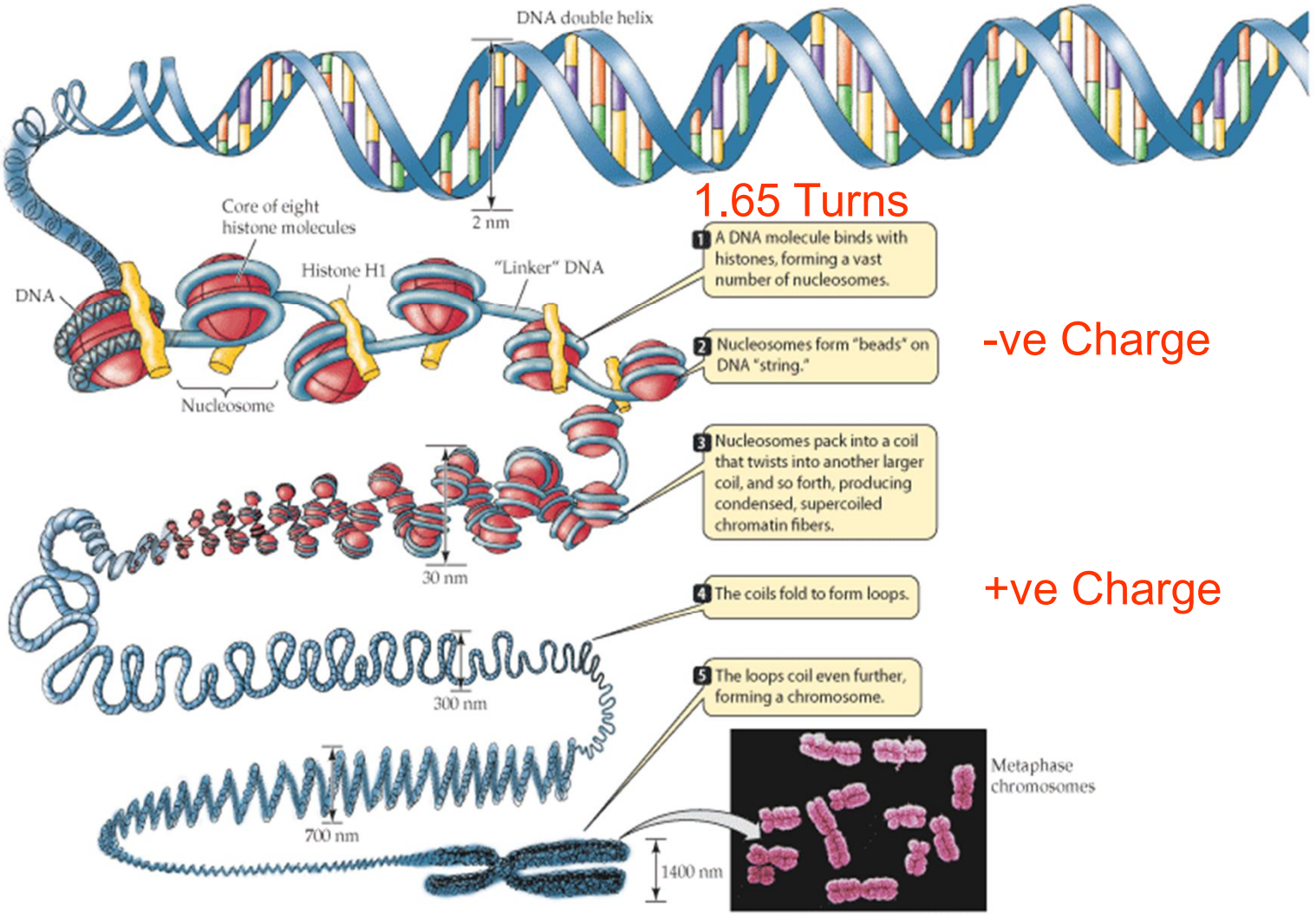
END

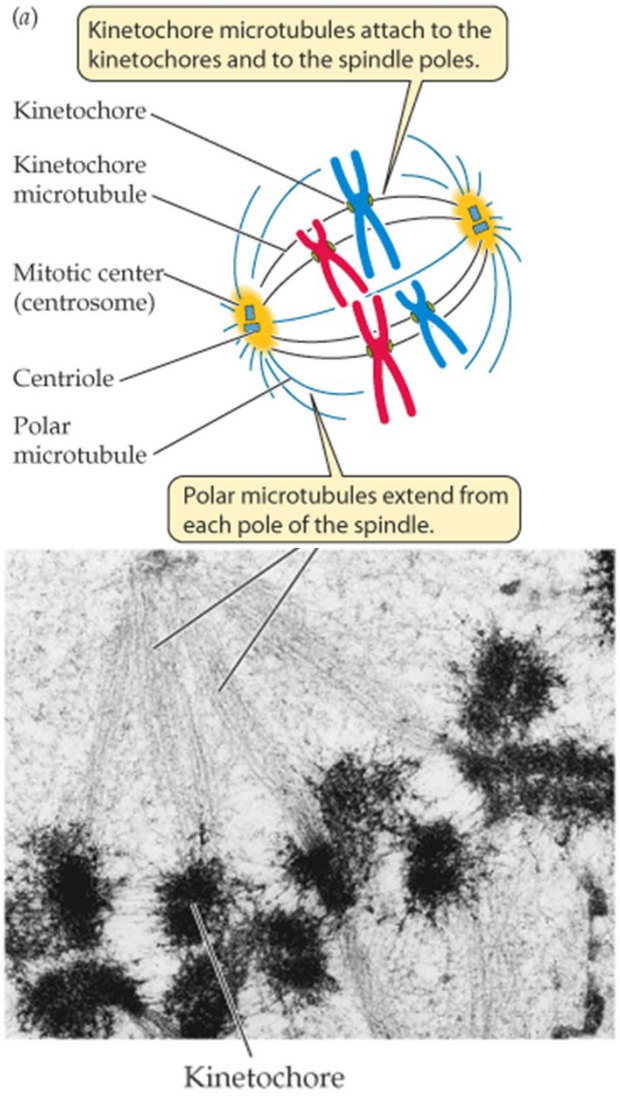
Chromosome

Anatomy



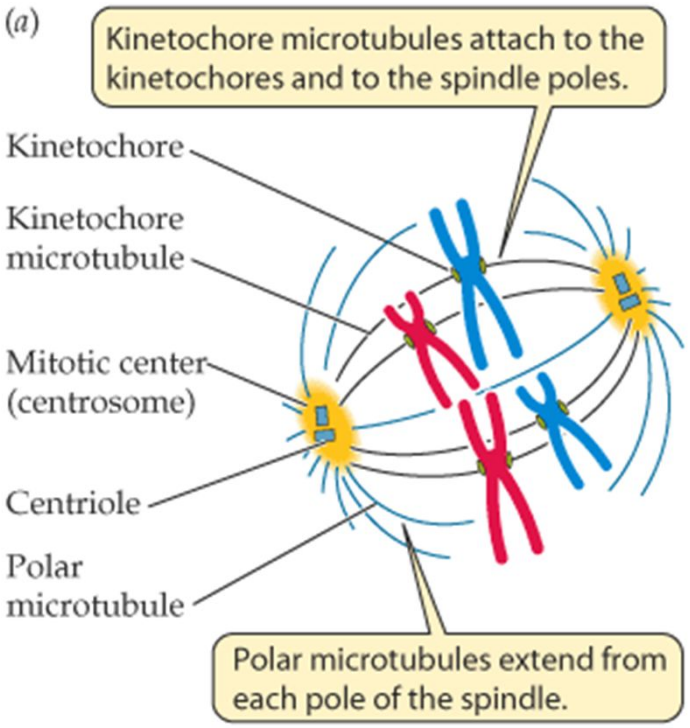
A chromosome is an organized structure of DNA and protein

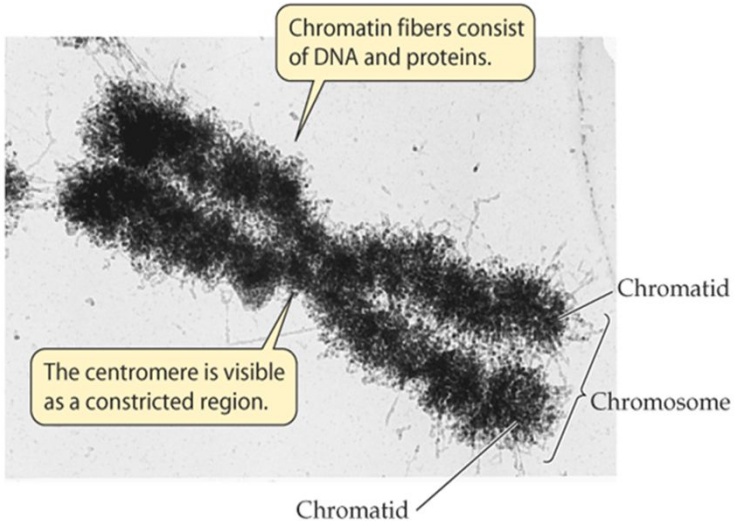

DNA double helix
2 nm
1.65 Turns
Core of eight histone molecules
Histone H1
"Linker" DNA
DNA
Nucleosome
1 A DNA molecule binds with histones, forming a vast number of nucleosomes.
2 Nucleosomes form "beads" on DNA "string."
-ve Charge
3 Nucleosomes pack into a coil that twists into another larger coil, and so forth, producing condensed, supercoiled chromatin fibers.
30 nm
+ve Charge
4 The coils fold to form loops.
300 nm
5 The loops coil even further, forming a chromosome.
700 nm
1400 nm
Metaphase chromosomes


(a)
Kinetochore microtubules attach to the kinetochores and to the spindle poles.
Kinetochore
Kinetochore microtubule
Mitotic center (centrosome)
Centriole
Polar microtubule
Polar microtubules extend from each pole of the spindle.
Kinetochore

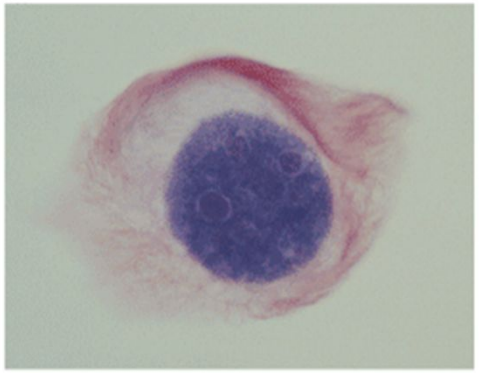
END

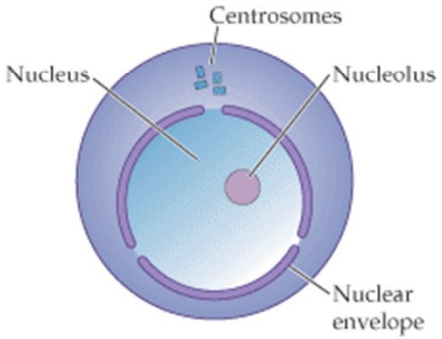
Mitosis


(a)
Kinetochore microtubules attach to the kinetochores and to the spindle poles.
Kinetochore
Kinetochore microtubule
Mitotic center (centrosome)
Centriole
Polar microtubule
Polar microtubules extend from each pole of the spindle.


Chromatin fibers consist of DNA and proteins.
The centromere is visible as a constricted region.
Chromatid
Chromosome
Chromatid

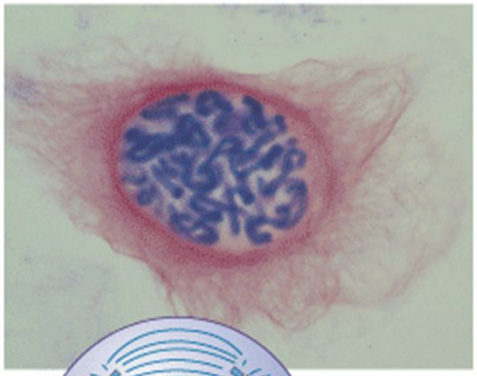
Interphase

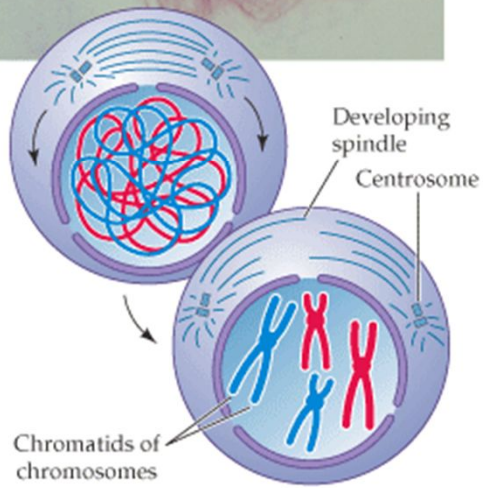




1 During the S phase of interphase, the nucleus replicates its DNA and centrosomes.

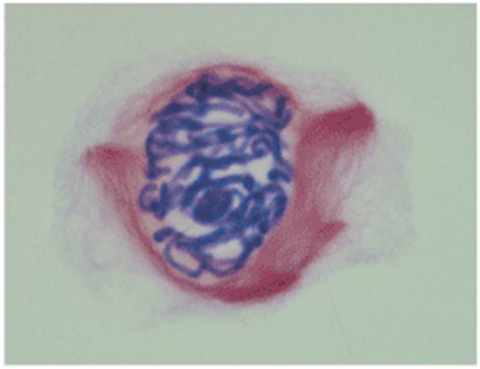
Prophase

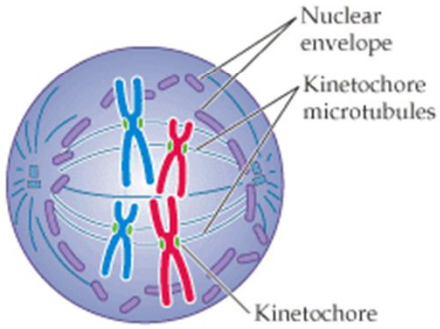




2 The chromatin coils and supercoils, becoming more and more compact and eventually condensing into visible chromosomes. The chromosomes consist of identical, paired sister chromatids.

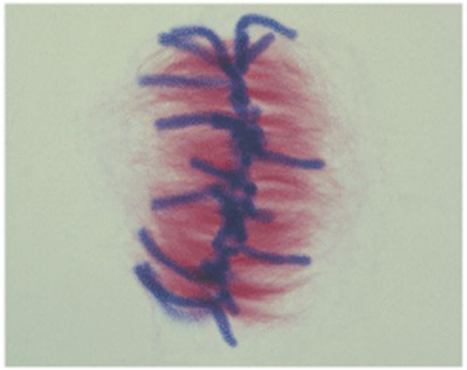
Prometaphase

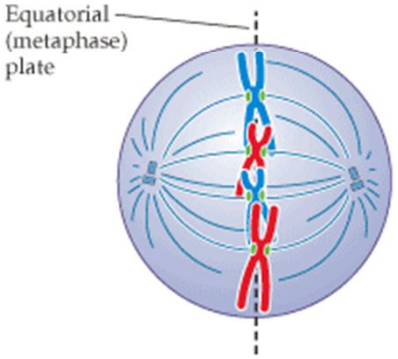




3 The nuclear envelope breaks down. Kinetochore microtubules appear and connect the kinetochores to the microtubule organizing centers.

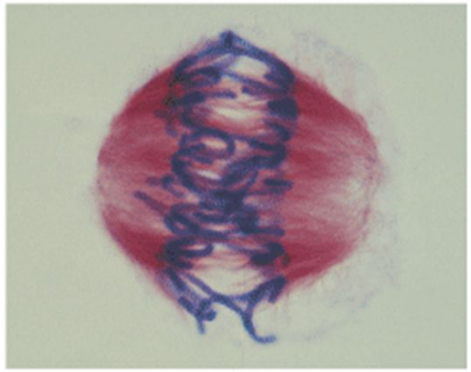
Metaphase

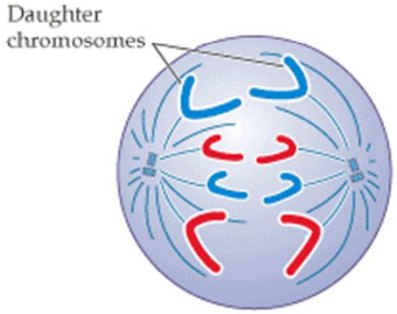




4 The centromeres (regions connecting paired chromatids) become aligned in a plane at the cell's equator.

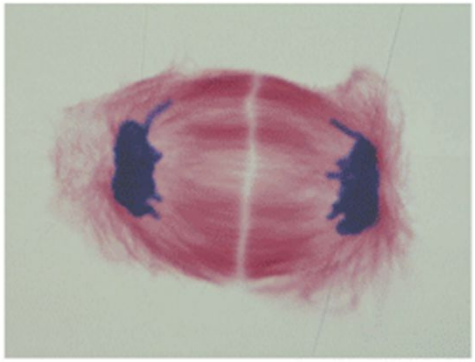
Anaphase

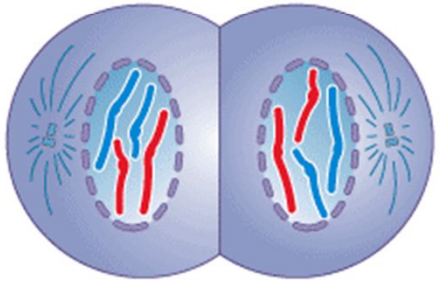




5 The paired sister chromatids separate, and the new daughter chromosomes begin to move toward the poles.

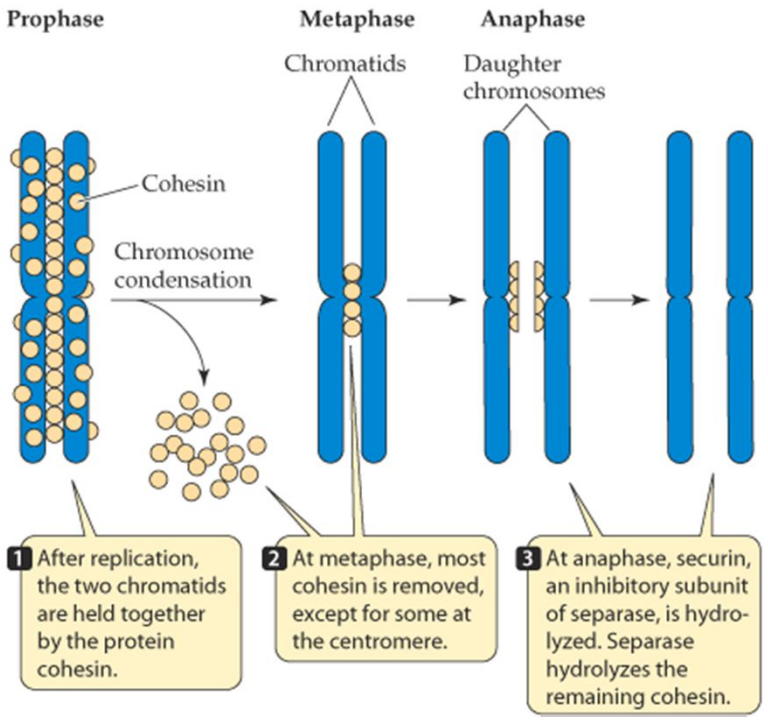
Telophase



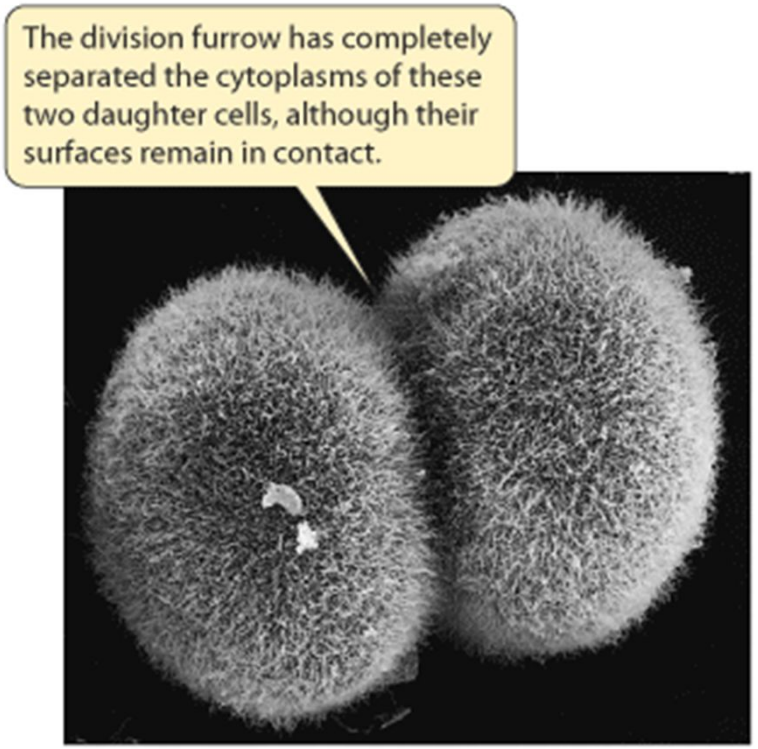


6 The daughter chromosomes reach the poles. Telophase passes into the next interphase as the nuclear envelopes and nucleoli re-form and the chromatin becomes diffuse.

The Glue


Prophase
Metaphase
Anaphase
Chromatids
Daughter chromosomes
Cohesin
Chromosome condensation
1 After replication, the two chromatids are held together by the protein cohesin.
2 At metaphase, most cohesin is removed, except for some at the centromere.
3 At anaphase, securin, an inhibitory subunit of separase, is hydrolyzed. Separase hydrolyzes the remaining cohesin.

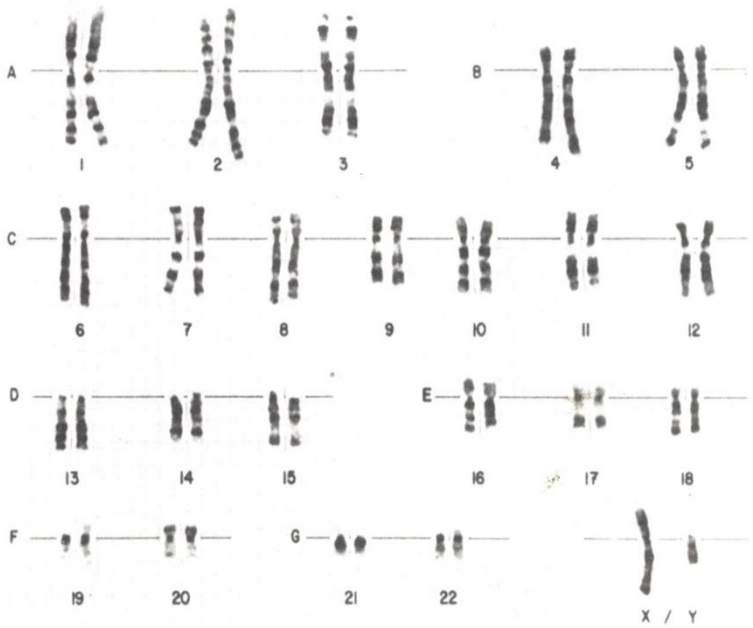
Cytokinesis



END

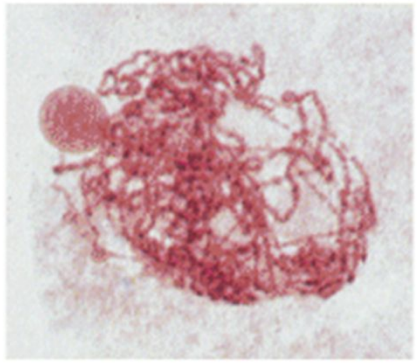
Meiosis I

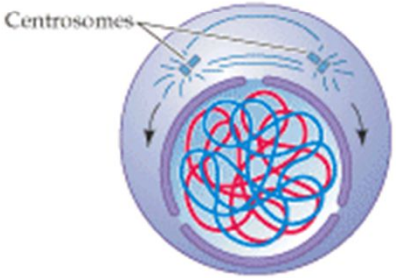
Karyotype



MEIOSIS I

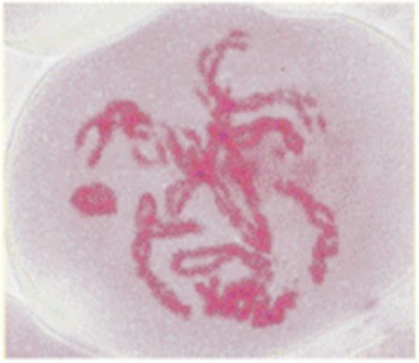
Early Prophase I

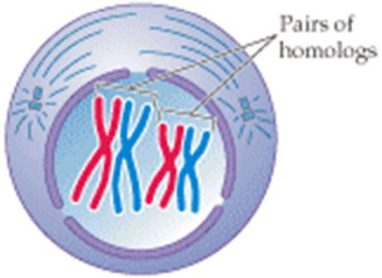




1 The chromatin begins to condense following interphase.

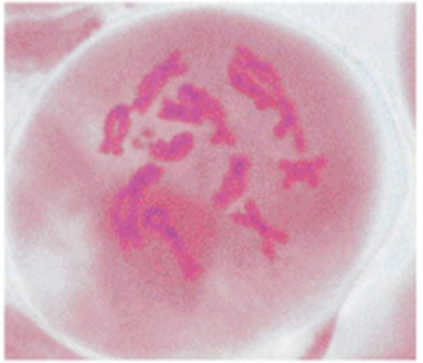
Mid-Prophase I





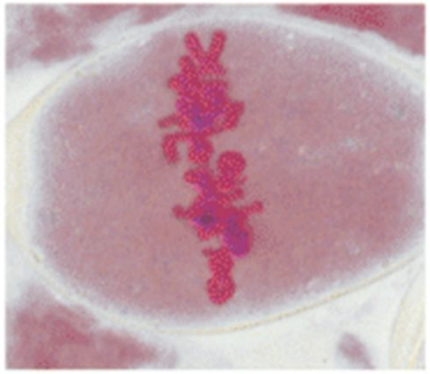
2 Synapsis aligns homologs, and chromosomes condense. Homologs are shown in different colors indicating those coming from each parent. In reality, their differences are very small, usually comprising different alleles of some genes.

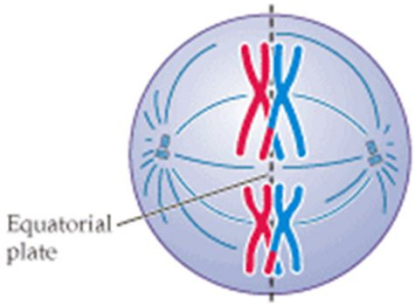
Late Prophase I–Prometaphase



3 The chromosomes continue to coil and shorten. Crossing over results in an exchange of genetic material. In prometaphase the nuclear envelope breaks down.

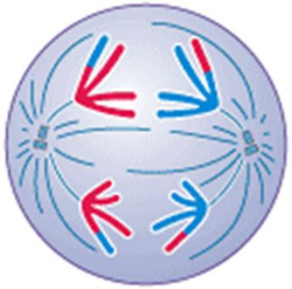
Metaphase I





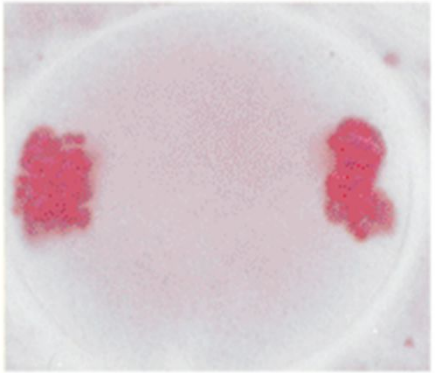
4 The homologous chromosomes line up on the equatorial (metaphase) plate.

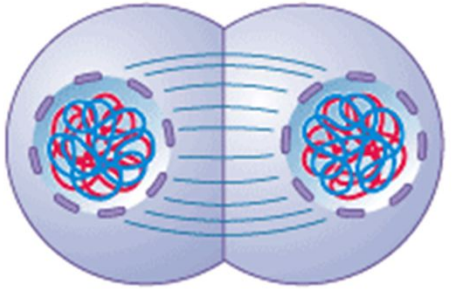
Anaphase I



5 The homologous chromosomes (each with two chromatids) move to opposite poles of the cell.

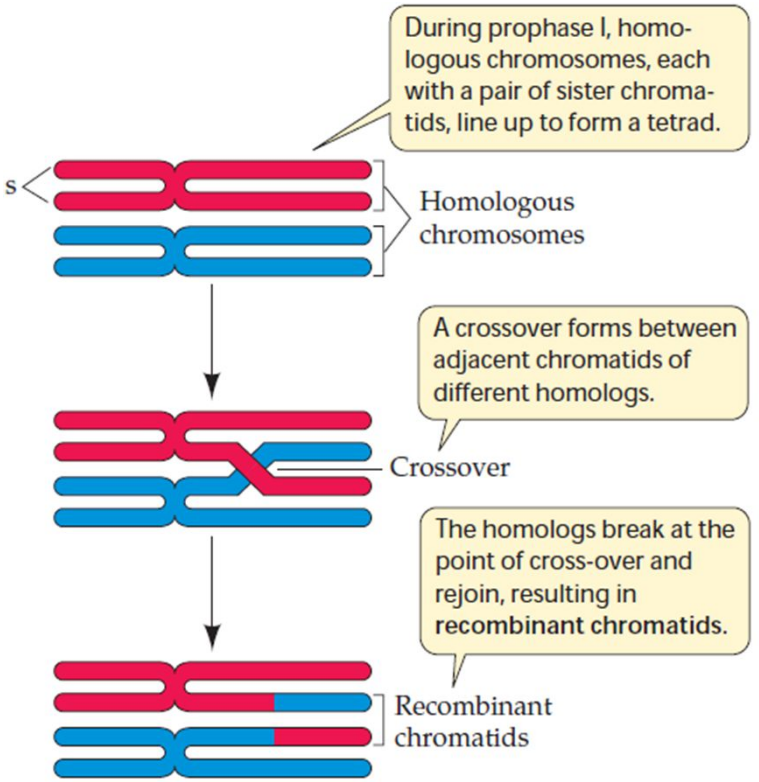
Telophase I





6 The chromosomes gather into nuclei, and the original cell divides.

Genetically Diverse Chromosomes

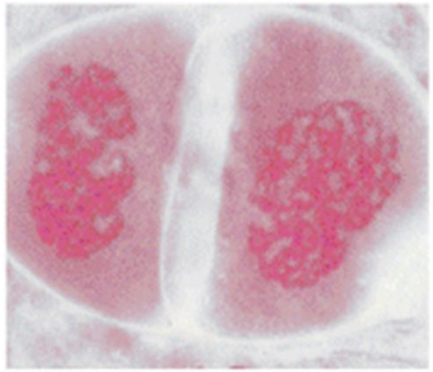


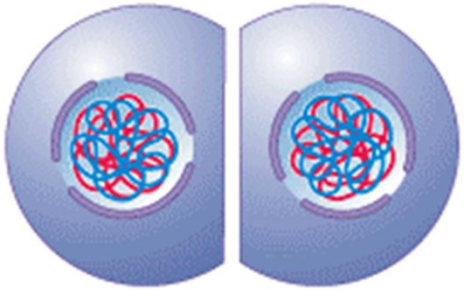
END

Meiosis II

MEIOSIS II

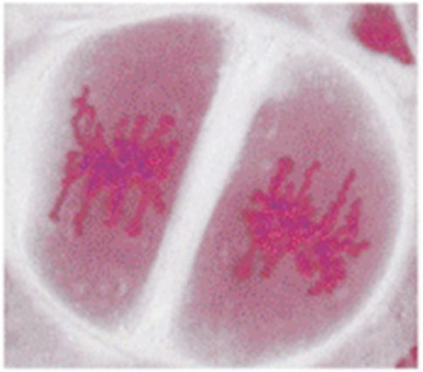
Prophase II

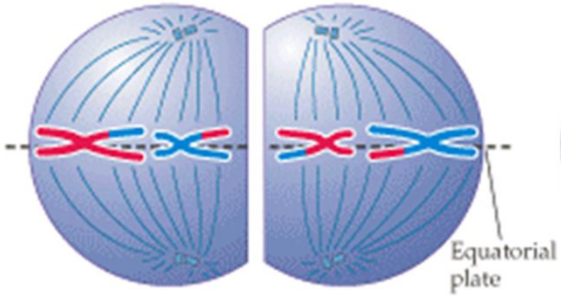




7 The chromosomes condense again, following a brief interphase (interkinesis) in which DNA does not replicate.

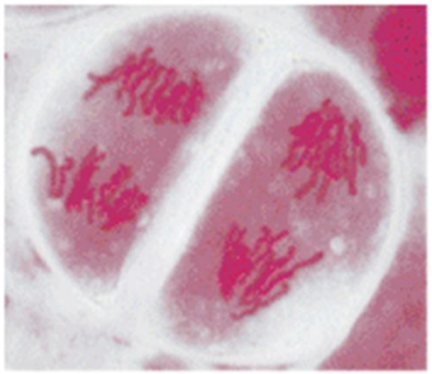
Metaphase II

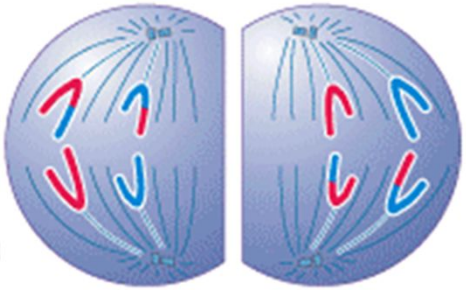




8 Kinetochores of the paired chromatids line up across the equatorial plates of each cell.

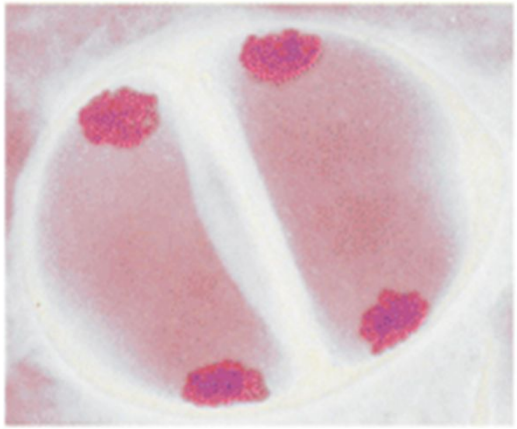
Anaphase II

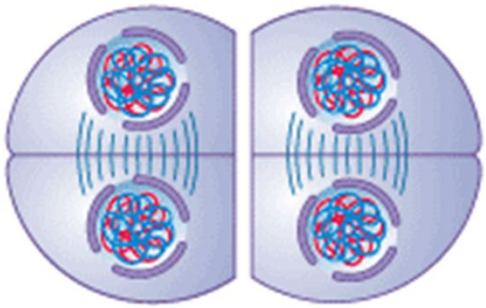




9 The chromatids finally separate, becoming chromosomes in their own right, and are pulled to opposite poles. Because of crossing over in prophase I, each new cell will have a different genetic makeup.

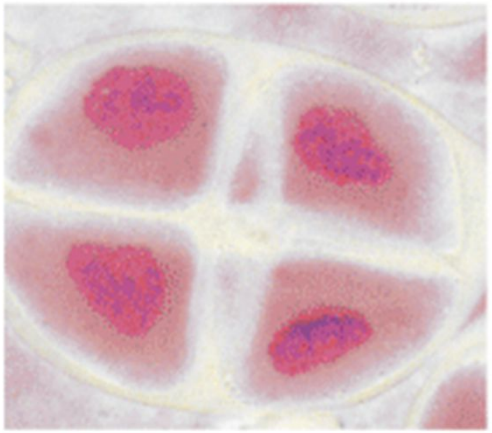
Telophase II

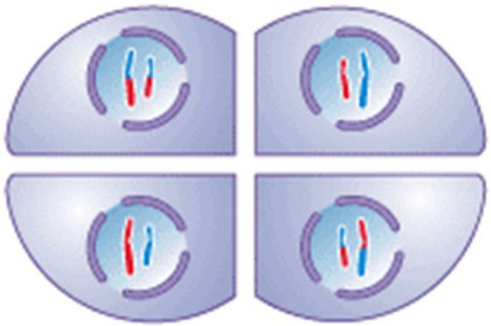




10 The chromosomes gather into nuclei, and the cells divide.

Products





11 Each of the four cells has a nucleus with a haploid number of chromosomes.

Features

Reduces chromosome number from diploid to haploid

Ensures each of the haploid products has a complete set of chromosomes

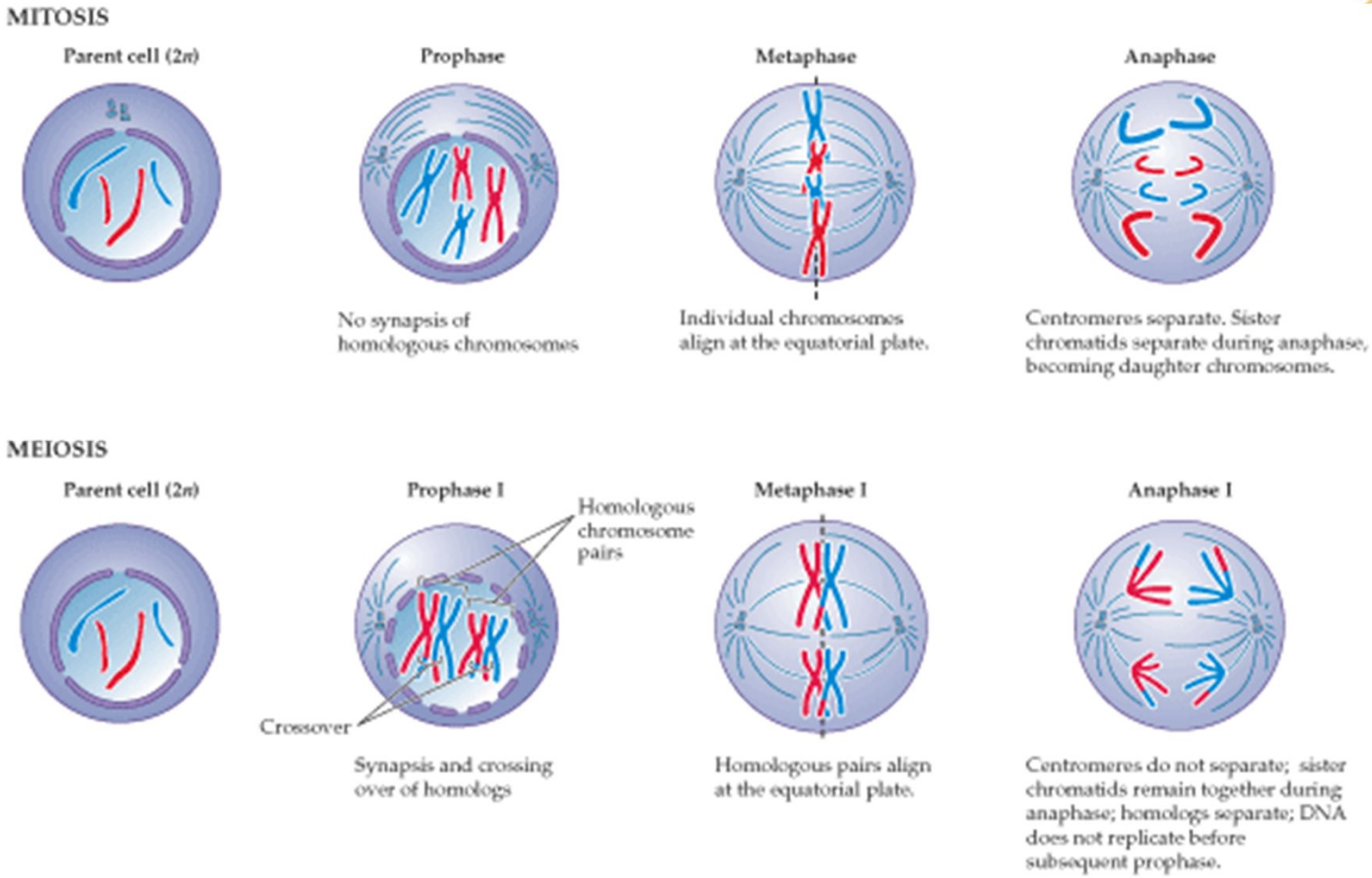
Promotes genetic diversity among the products.

END

Cell Biology

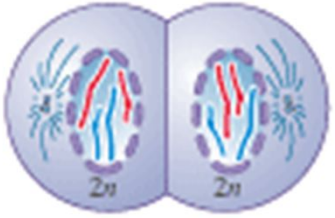
The Comparison

The Comparison



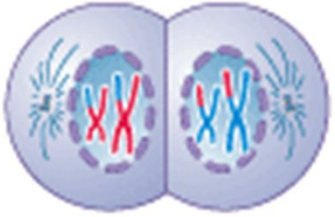
The Comparison

Two daughter cells (each 2n)

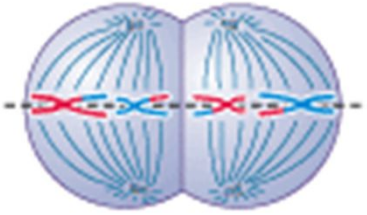


Mitosis is a mechanism for constancy: The parent nucleus produces two identical daughter nuclei.

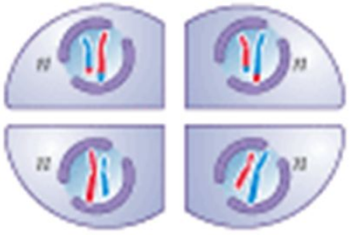
Telophase I



Metaphase II



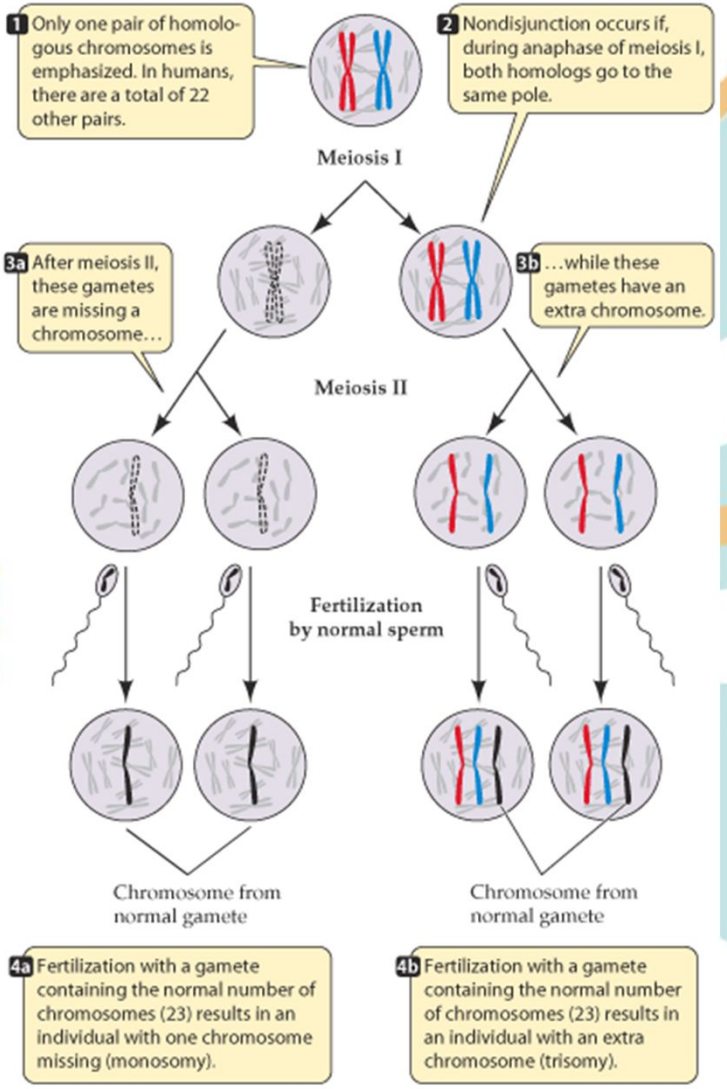
Four daughter cells (each n)



Chromatids separate.

Meiosis is a mechanism for diversity: The parent nucleus produces four different haploid daughter nuclei.

The Common Mistake



END

Genetics Introduction

Historic Background

Blending VS Particulate

Definitions

A character is an observable feature, such as flower color.

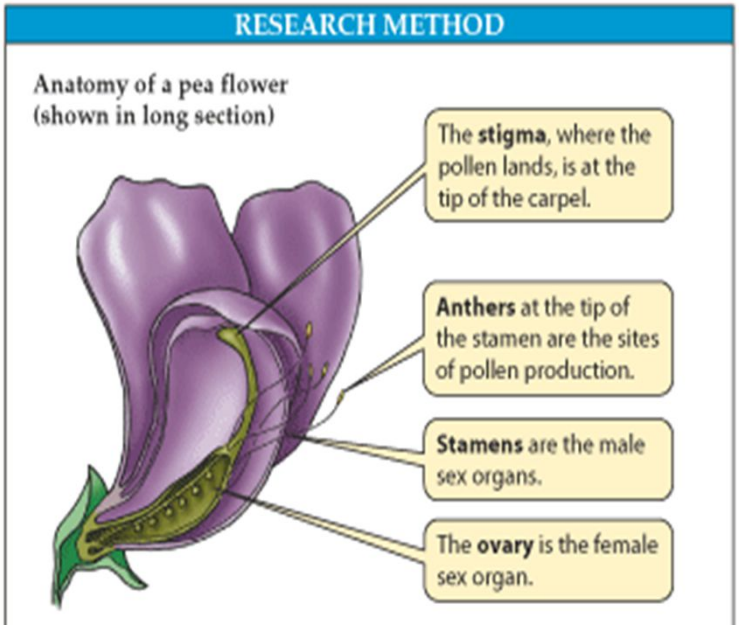
A trait is a particular form of a character, such as white flowers.

A heritable character trait is one that is passed from parent to offspring.

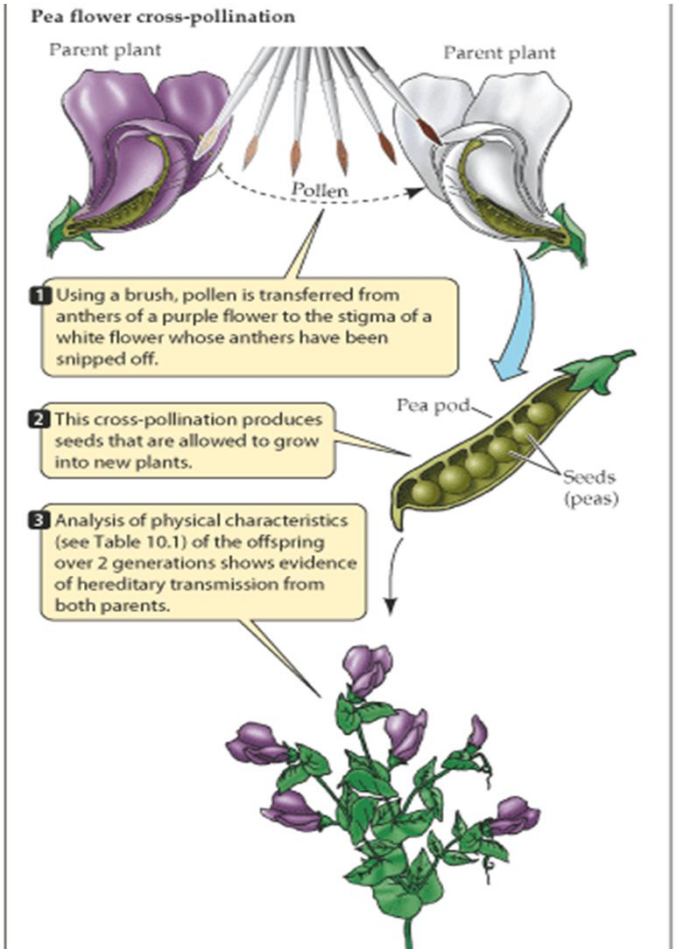
END

The Question

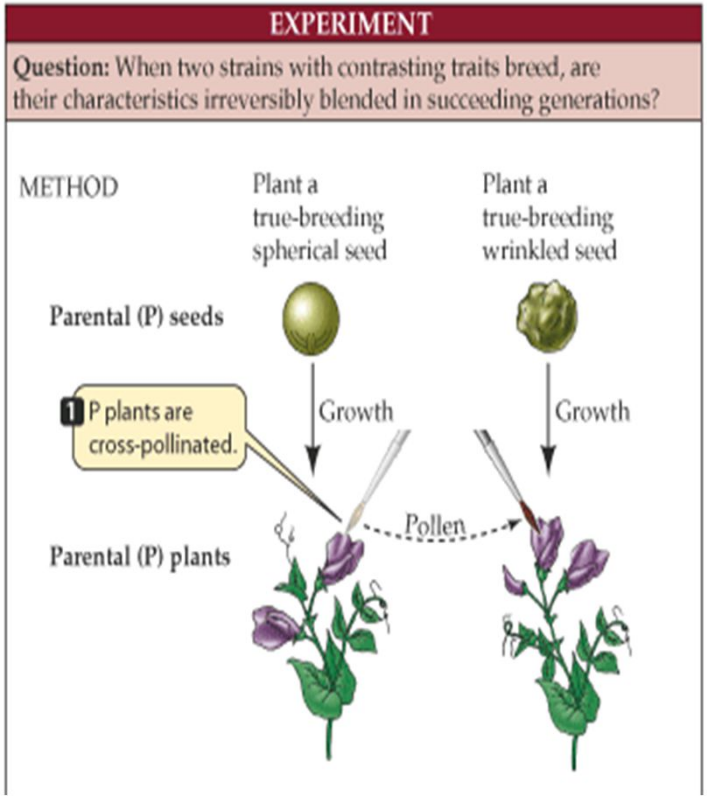
The Anatomy of the Subject



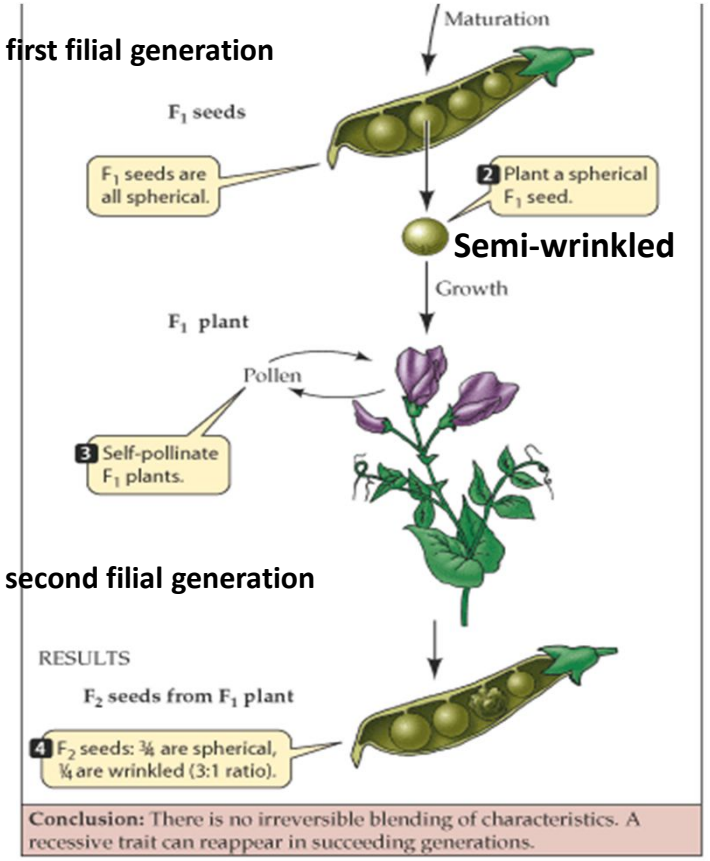
The Technique



The Monohybrid Cross

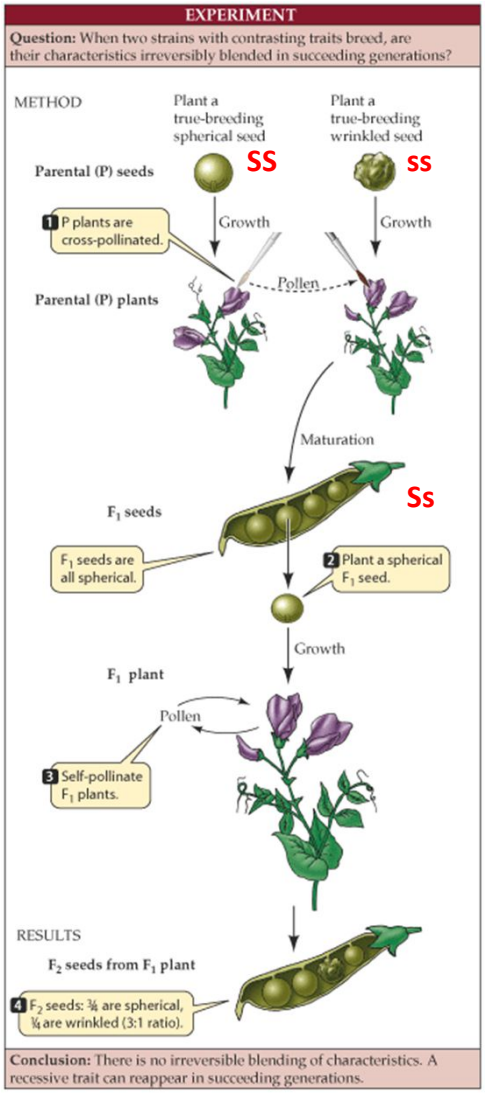


The Answer



END

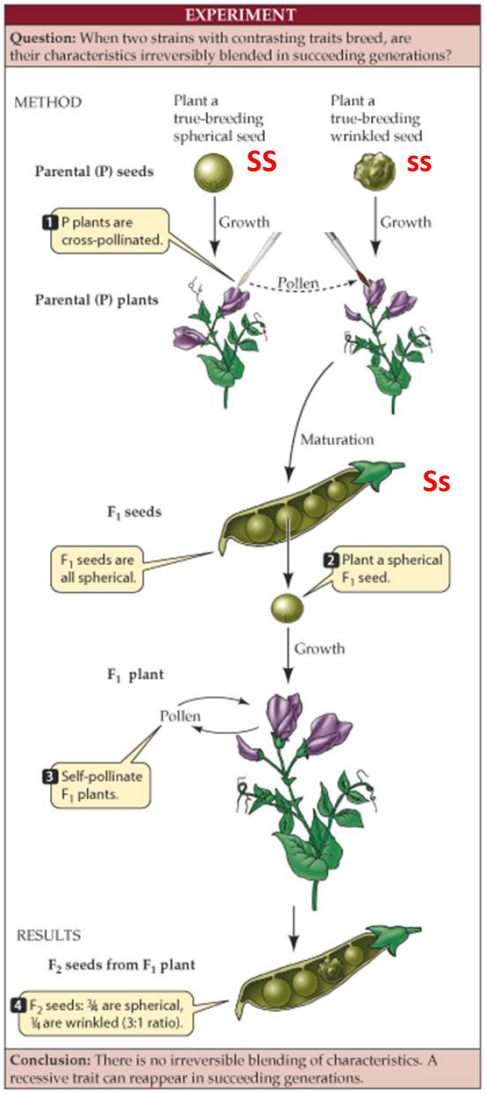
The Lessons Learned



ss trait not expressed in the F1 generation, (reappeared in F2) Mendal concluded SS trait domin-ant to the ss trait (recessive).

Mendel worked with his data mathematically & concluded each pea plant has two units of inheritance for each character, 1 from each parent.

During gamete production only 1 of these paired units is given to a gamete. Resulting zygote contains 2, (produced fusion of 2 gametes).

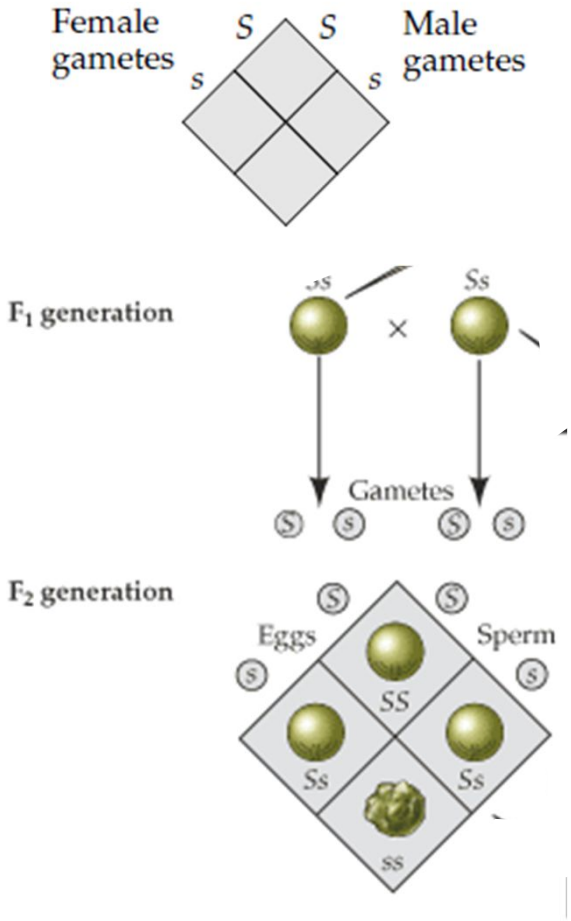


Mendel's unit of inheritance is now called a gene. Different forms of a gene (S & s) are called alleles.

Homozygous have two copies of the same allele (ss). Heterozygous have two different alleles of the gene in question (in this case, Ss).

The physical appearance of an organism is its phenotype. Phenotype to be the result of the genotype, or genetic constitution.

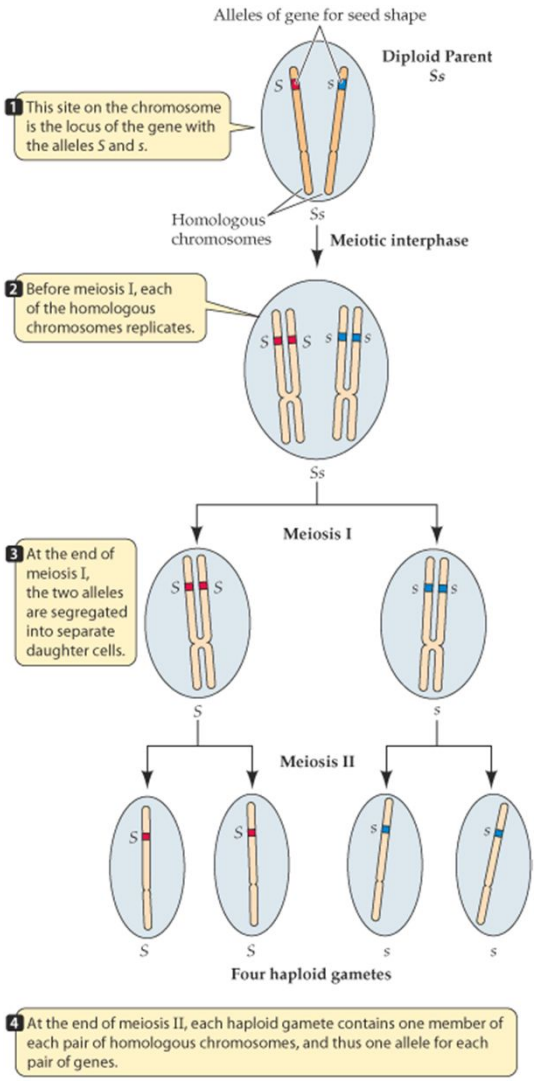
The Punnett Square

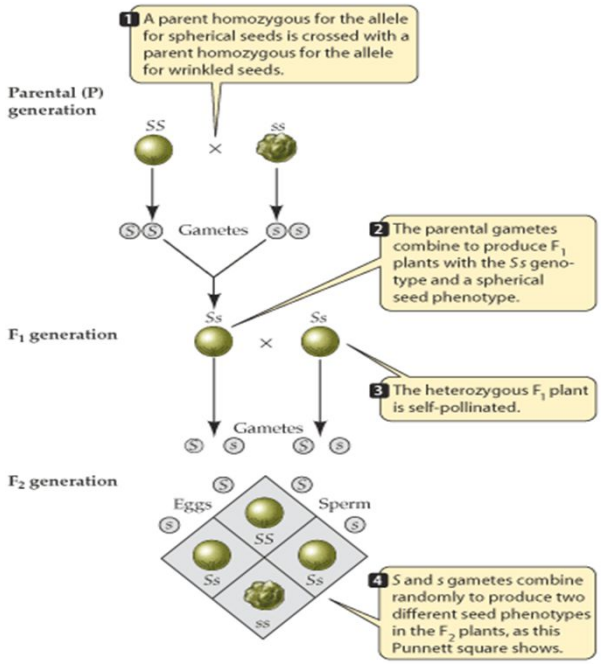


END

The First Law

When any individual produces gametes, the two alleles separate, so that each gamete receives only one member of the pair of alleles.



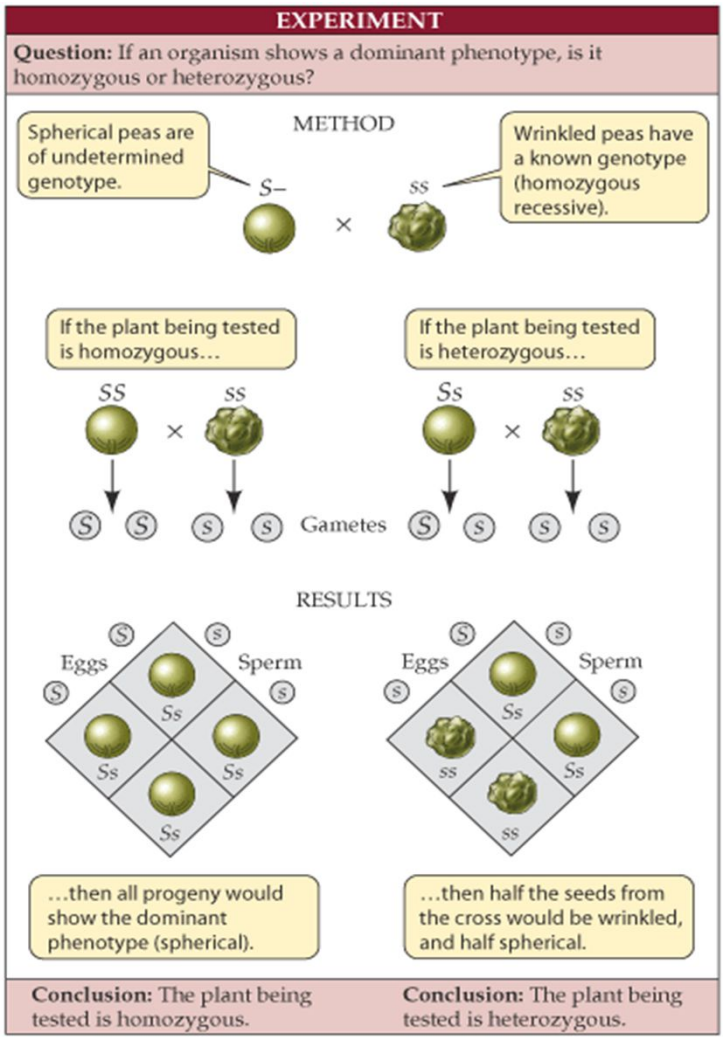


10.1 Mendel's Results from Monohybrid Crosses

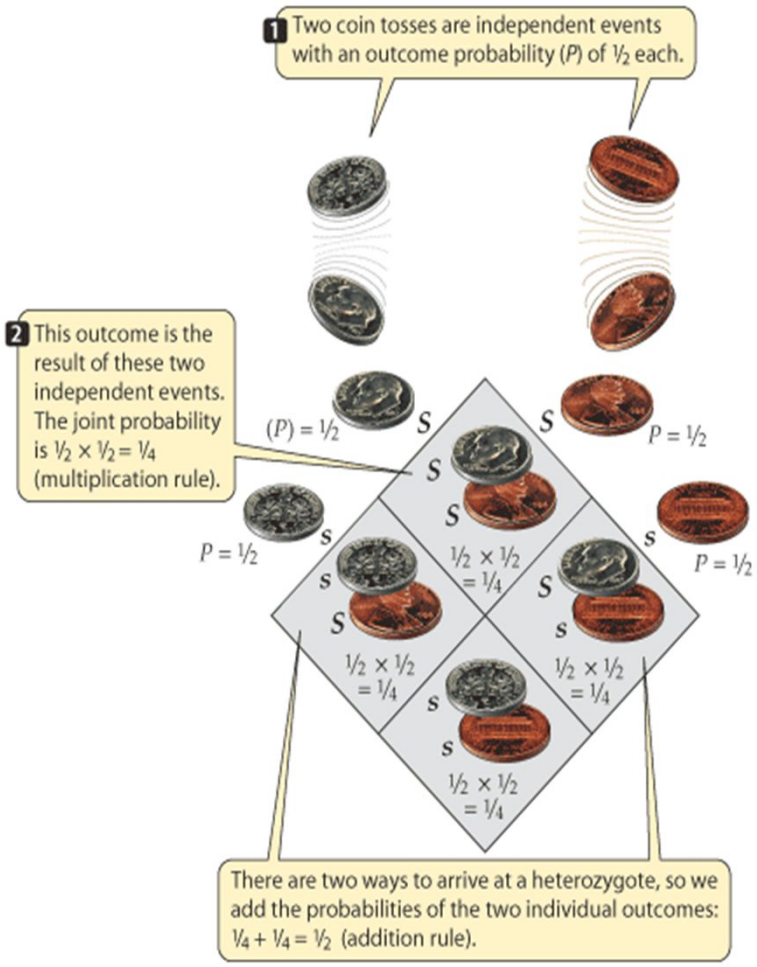
each member of the P generation is true-breeding

	PARENTAL GENERATION PHENOTYPES			F_2 GENERATION PHENOTYPES			
	DOMINANT		RECESSIVE	DOMINANT	RECESSIVE	TOTAL	RATIO
	Spherical seeds	×	Wrinkled seeds	5,474	1,850	7,324	2.96:1
	Yellow seeds	×	Green seeds	6,022	2,001	8,023	3.01:1
	Purple flowers	×	White flowers	705	224	929	3.15:1
	Inflated pods	×	Constricted pods	882	299	1,181	2.95:1
	Green pods	×	Yellow pods	428	152	580	2.82:1
	Axial flowers	×	Terminal flowers	651	207	858	3.14:1
	Tall stems (1 m)	×	Dwarf stems (0.3 m)	787	277	1,064	2.84:1

END


EXPERIMENT
Question: If an organism shows a dominant phenotype, is it homozygous or heterozygous?
METHOD
Spherical peas are of undetermined genotype.
S–
×
ss
Wrinkled peas have a known genotype (homozygous recessive).
If the plant being tested is homozygous…
SS
×
ss
S S s s
Gametes
If the plant being tested is heterozygous…
Ss
×
ss
S s s s
RESULTS
Eggs
Sperm
Ss
Ss
Ss
Ss
…then all progeny would show the dominant phenotype (spherical).
Eggs
Sperm
Ss
ss
Ss
ss
…then half the seeds from the cross would be wrinkled, and half spherical.
Conclusion: The plant being tested is homozygous.
Conclusion: The plant being tested is heterozygous.

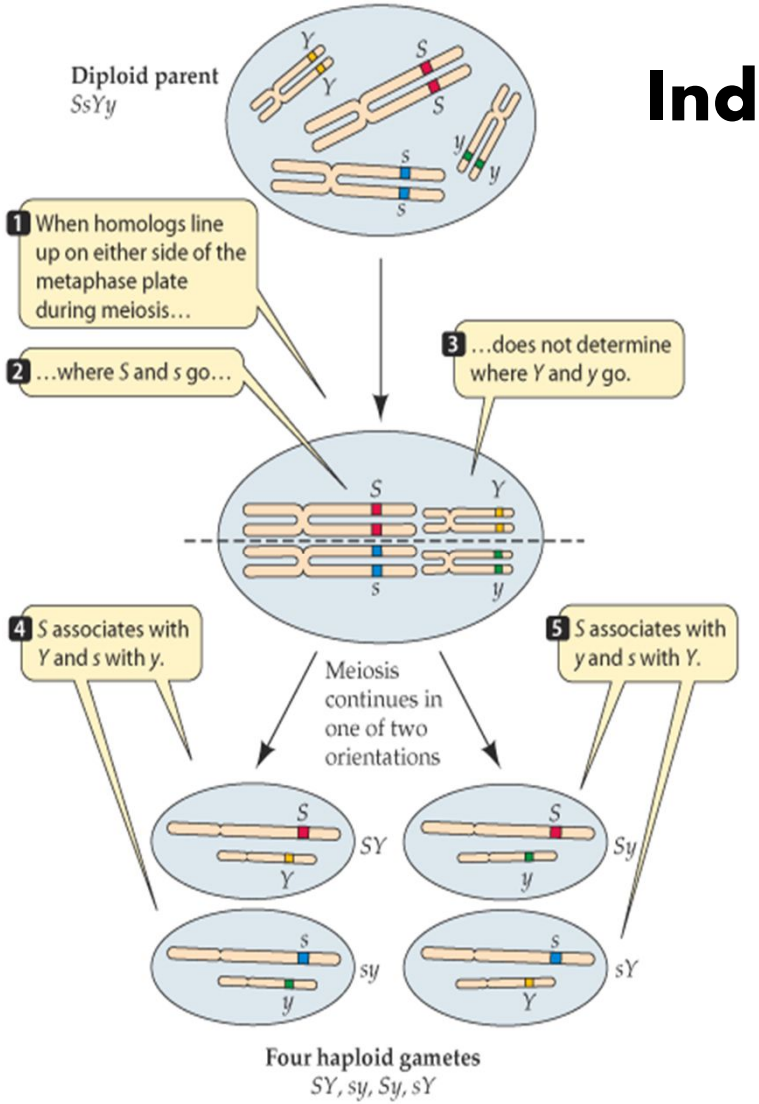
The Second Law



Probabilities

If an event is absolutely certain to happen, its probability is 1.

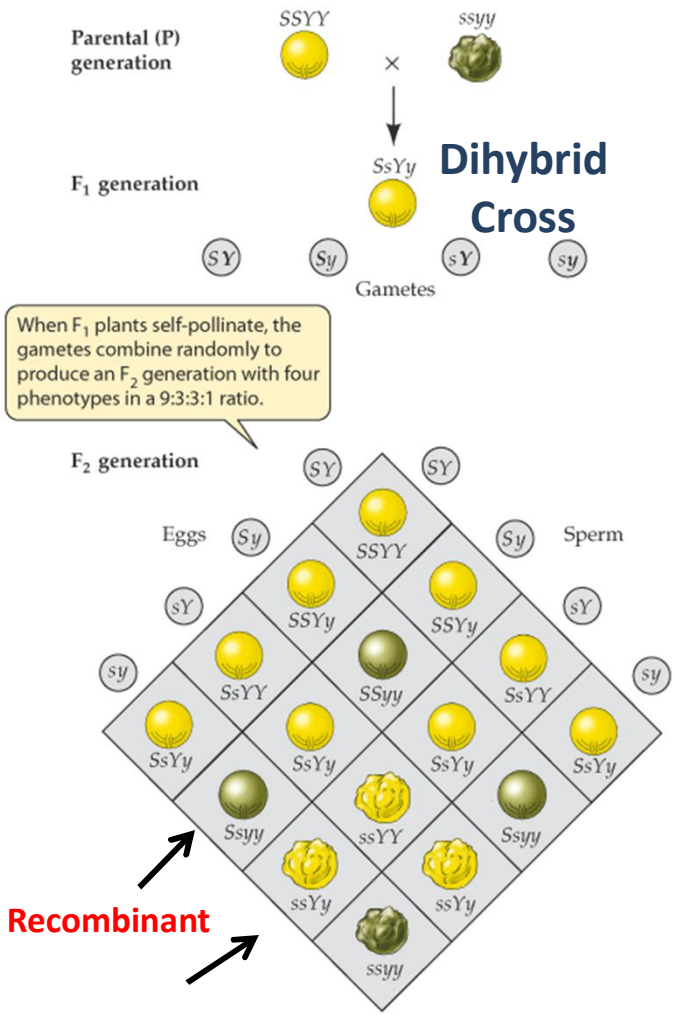
If it cannot possibly happen, its probability is 0. Otherwise, its probability lies between 0 and 1.



Independent Assortment

Alleles of different genes assort independently of one another during gamete formation.

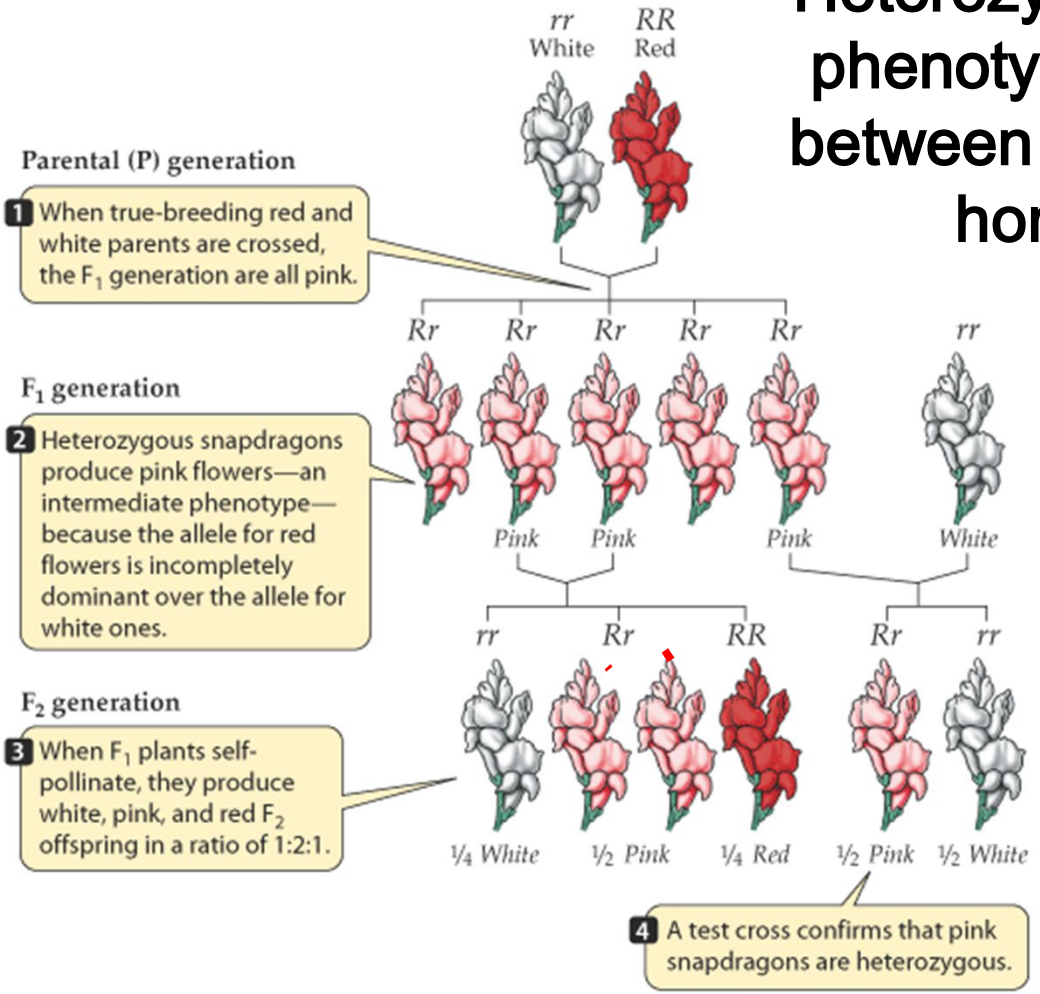
That is, the segregation of the alleles of gene A is independent of the segregation of alleles of gene B.


Parental (P) generation
SSYY
ssyy
×
F1 generation
SsYy
Dihybrid Cross
SY
Sy
sY
sy
Gametes
When F1 plants self-pollinate, the gametes combine randomly to produce an F2 generation with four phenotypes in a 9:3:3:1 ratio.
F2 generation
Eggs
Sperm
SSYY
SSYy
SSYy
SsYY
SSyy
SsYY
SsYy
SsYy
SsYy
SsYy
Ssyy
ssYY
Ssyy
ssYy
ssYy
ssyy
Recombinant

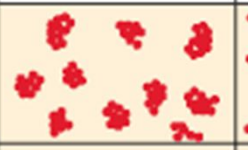
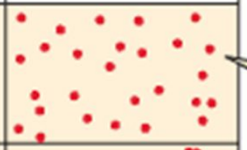
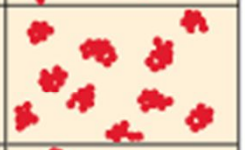
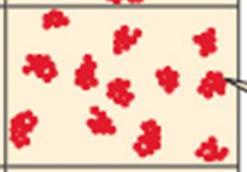
END

Aberrant Phenomenon

Heterozygotes show a phenotype intermediate between those of the two homozygotes



Both alleles appear in heterozygotes

Blood type of cells	Genotype	Antibodies made by body	Reaction to added antibodies	
			Anti-A	Anti-B
A	I^AI^A or I^Ai^O	Anti-B		
B	I^BI^B or I^Bi^O	Anti-A		
AB	I^AI^B	Neither anti-A nor anti-B		
O	i^Oi^O	Both anti-A and anti-B		

Red blood cells that do not react with antibody remain evenly dispersed.

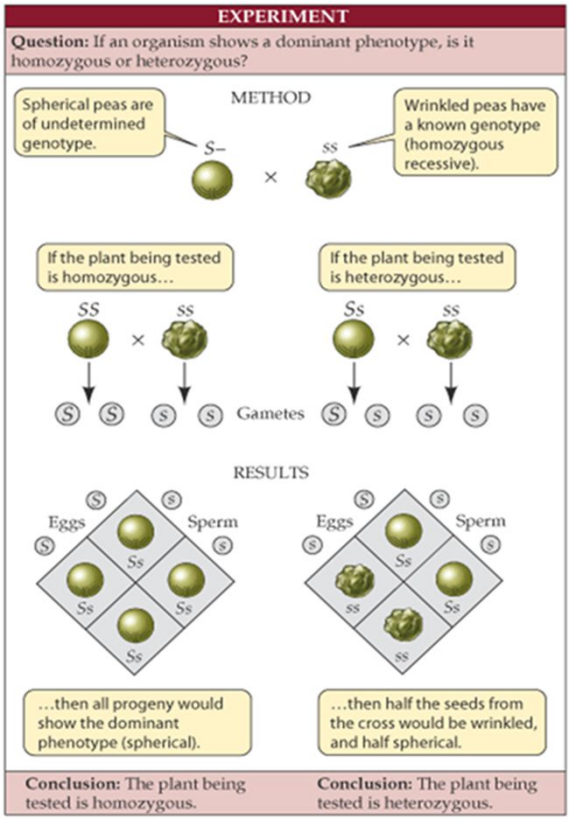
Red blood cells that react with antibody clump together (speckled appearance).

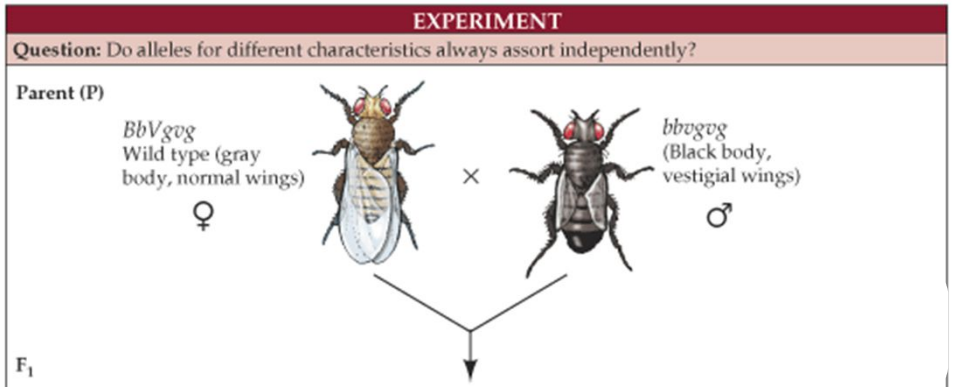
Effect of one gene dependent on the presence of one or more 'modifier genes'

END

Linkage & Recombination

Independent Assortment

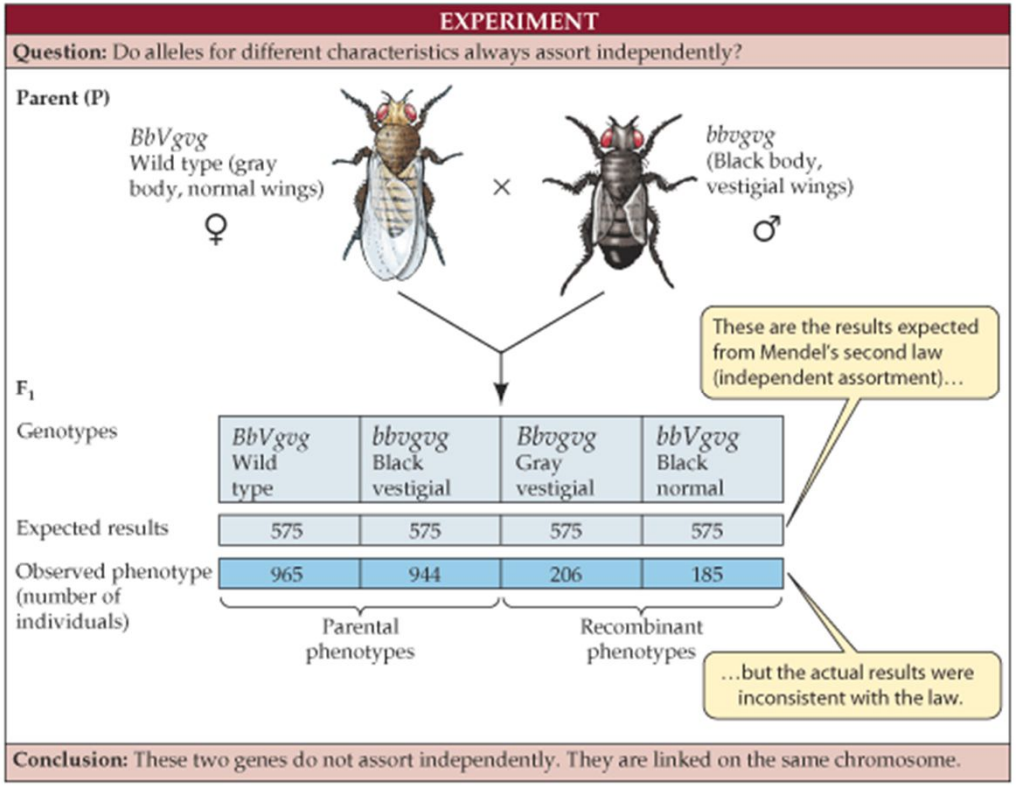


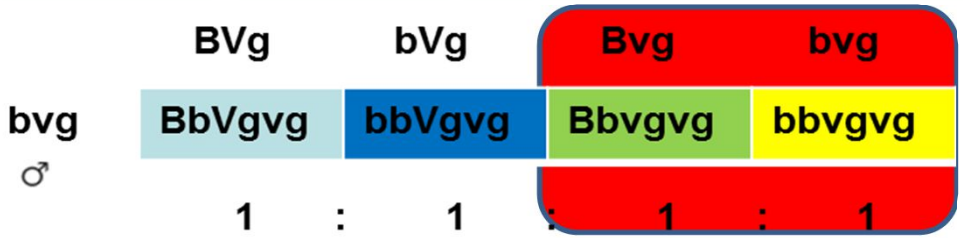


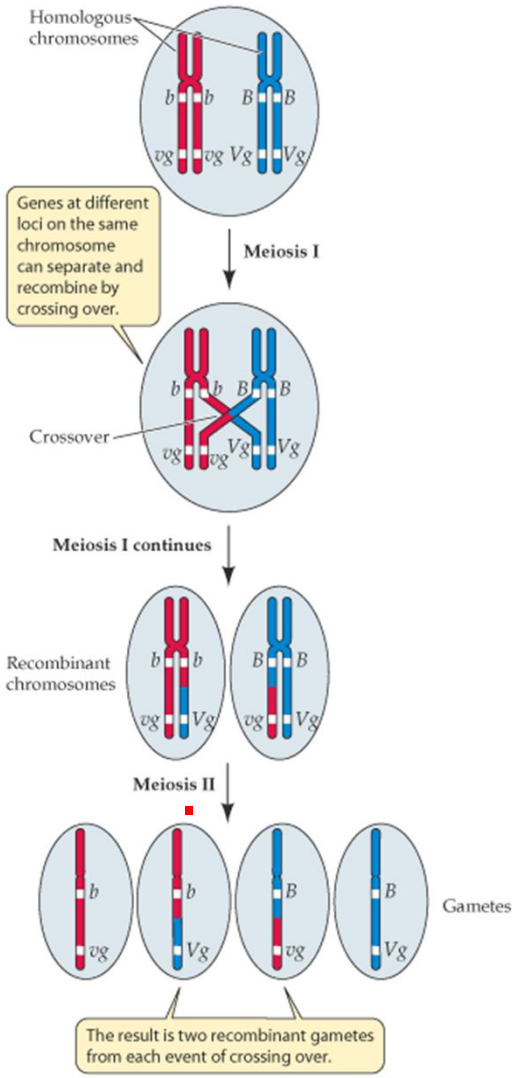
♂ \ ♀	BVg	bVg	Bvg	bvg
bvg	BbVgvg	bbVgvg	Bbvgvg	bbvgvg

1 : 1 : 1 : 1

No Independent Assortment

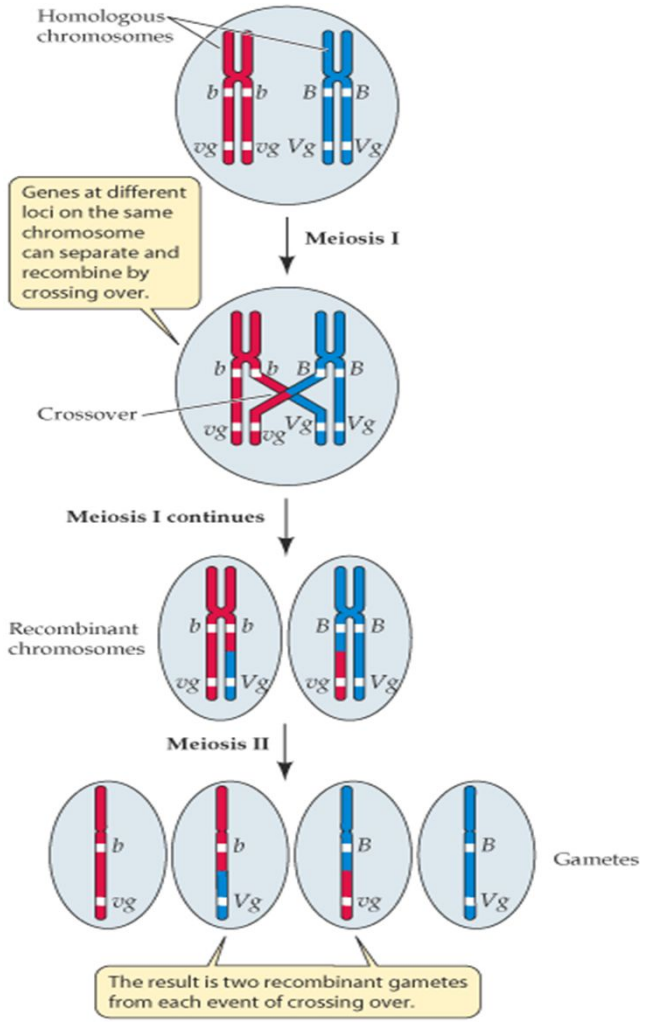






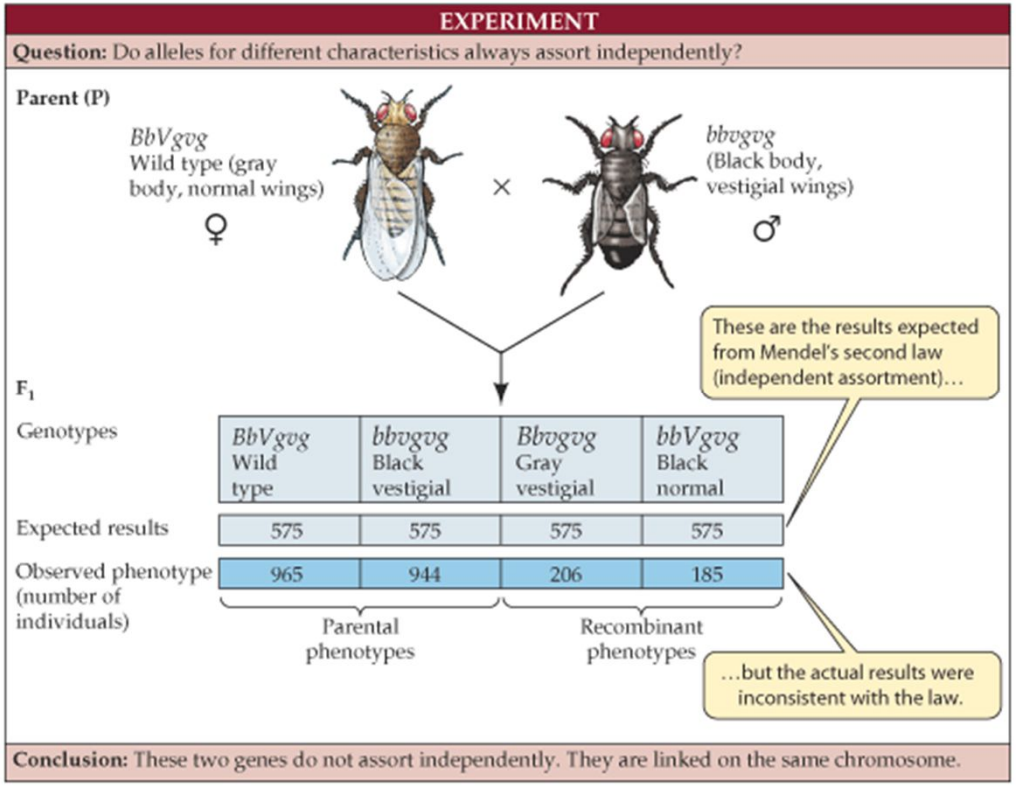
The number of linkage groups in a species equals the number of homologous chromosome pairs.

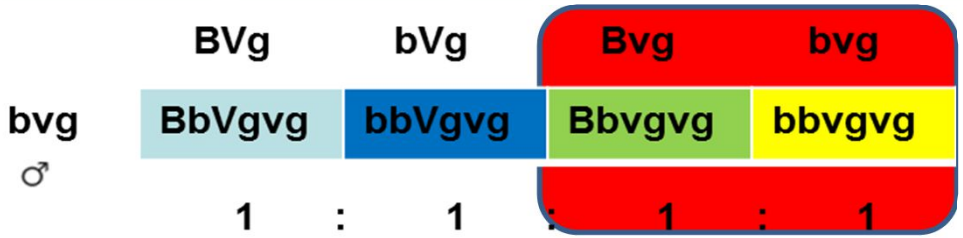
Number of genes in a cell far exceeds Number of chromosomes, (1 chromo must contain many genes)

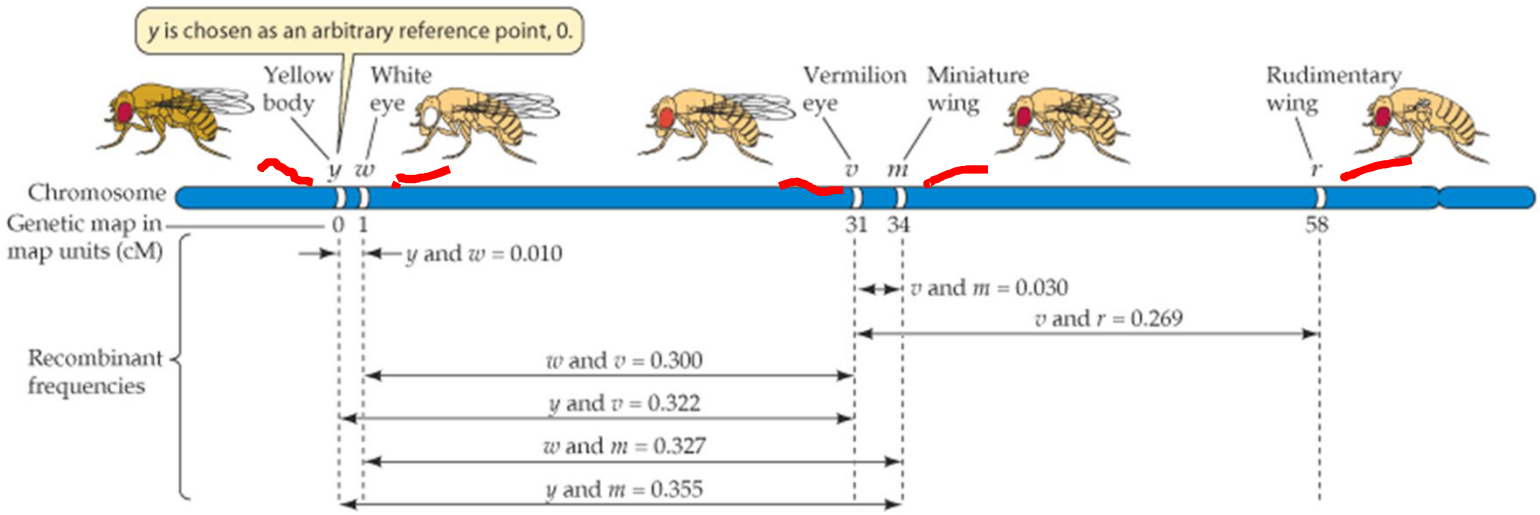

Homologous chromosomes
b b
vg vg
B B
Vg Vg
Genes at different loci on the same chromosome can separate and recombine by crossing over.
Meiosis I
Crossover
Meiosis I continues
Recombinant chromosomes
Meiosis II
Gametes
The result is two recombinant gametes from each event of crossing over.

Genetics Map

No Independent Assortment



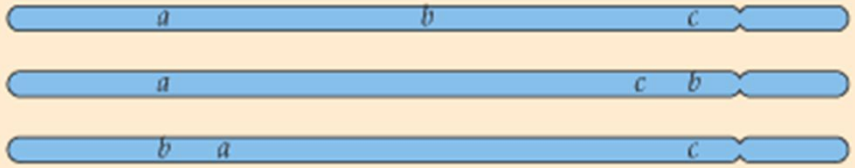




A map unit corresponds to a recombinant frequency of 0.01; it is also referred to as a *centimorgan* (cM)

If two loci are very close together on a chromosome, the odds of crossing over between them are small. In contrast, if two loci are far apart, crossing over could occur between them at many points.

1 At the outset, we have no idea of the individual distances, and there are several possible sequences (*a*-*b*-*c*, *a*-*c*-*b*, *b*-*a*-*c*).

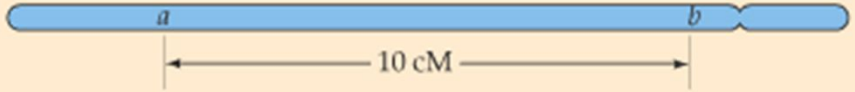


We make a cross *AABB* × *aabb*, and obtain an F_1 generation with a genotype *AaBb*. We test cross these *AaBb* individuals with *aabb*. Here are the genotypes of the first 1,000 progeny:

450 *AaBb*, 450 *aabb*, 50 *Aabb*, and 50 *aaBb*.

2 **How far apart are the *a* and *b* genes?** Well, what is the recombinant frequency? Which are the recombinant types, and which are the parental types?

Recombinant frequency (*a* to *b*) = (50 + 50)/1,000 = 0.1
So the map distance is
Map distance = 100 × recombinant frequency =
100 × 0.1 = 10 cM

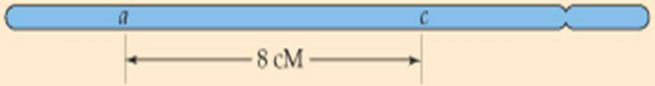


3 Now we make a cross *AACC* × *aacc*, obtain an F_1 generation, and test cross it, obtaining:

460 *AaCc*, 460 *aacc*, 40 *Aacc*, and 40 *aaCc*.

How far apart are the *a* and *c* genes?

Recombinant frequency (*a* to *c*) = (40 + 40)/1,000 = 0.08
Map distance = 100 × recombinant frequency =
100 × 0.08 = 8 cM



4 **How far apart are the *b* and *c* genes?**

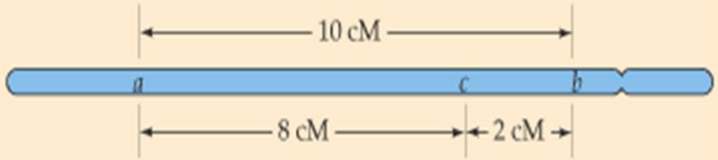
We make a cross *BBCC* × *bbcc*, obtain an F_1 generation, and test cross it, obtaining:

490 *BbCc*, 490 *bbcc*, 10 *Bbcc*, and 10 *bbCc*.

Determine the map distance between *b* and *c*.

Recombinant frequency (*b* to *c*) = (10 + 10)/1,000 = 0.02
Map distance = 100 × recombinant frequency =
100 × 0.02 = 2 cM

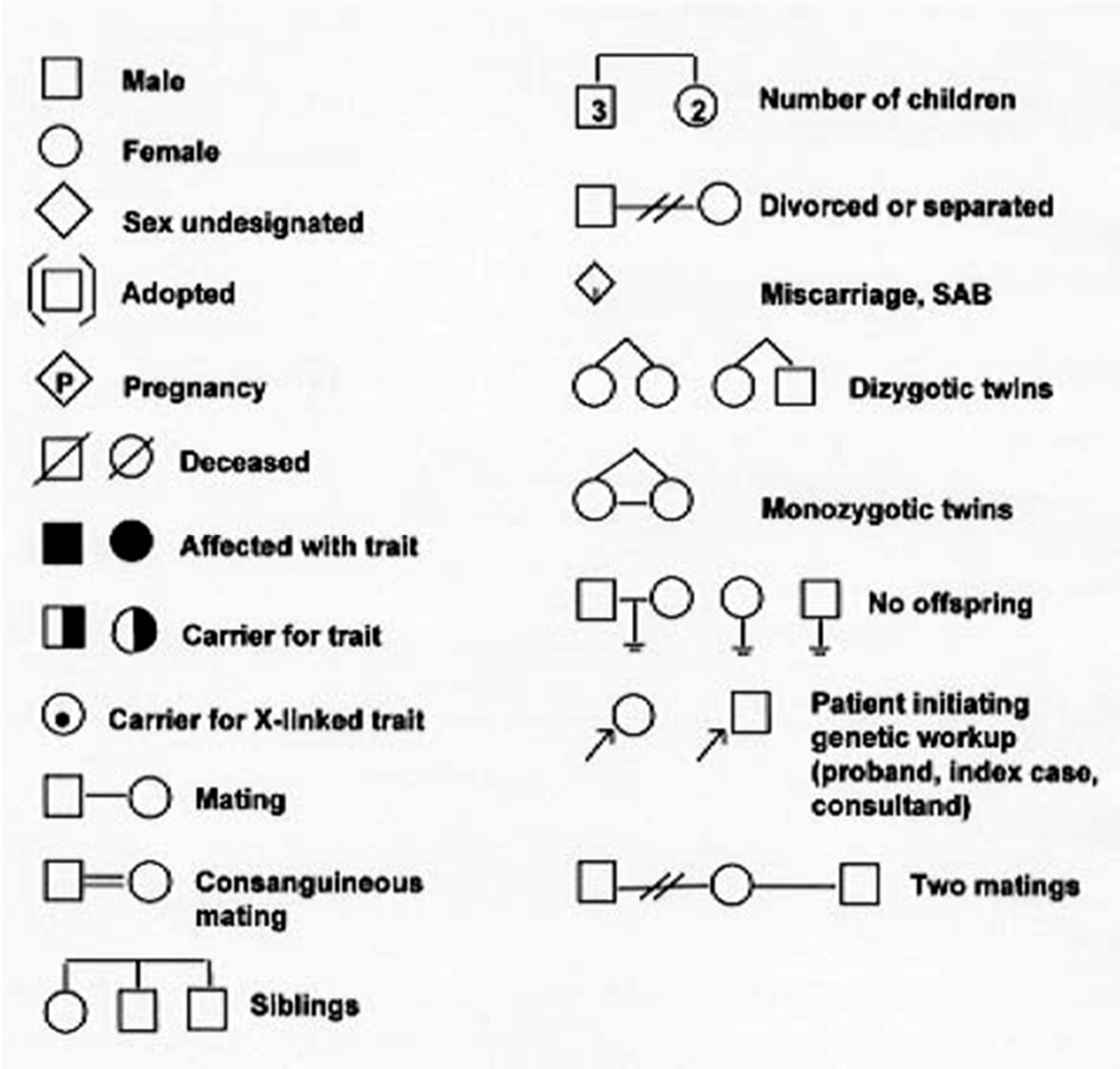
5 **Which of the three genes is between the other two?**
Because *a* and *b* are the farthest apart, *c* must be between them.



These numbers add up perfectly, but in most real cases they don't add up perfectly because of multiple crossovers.

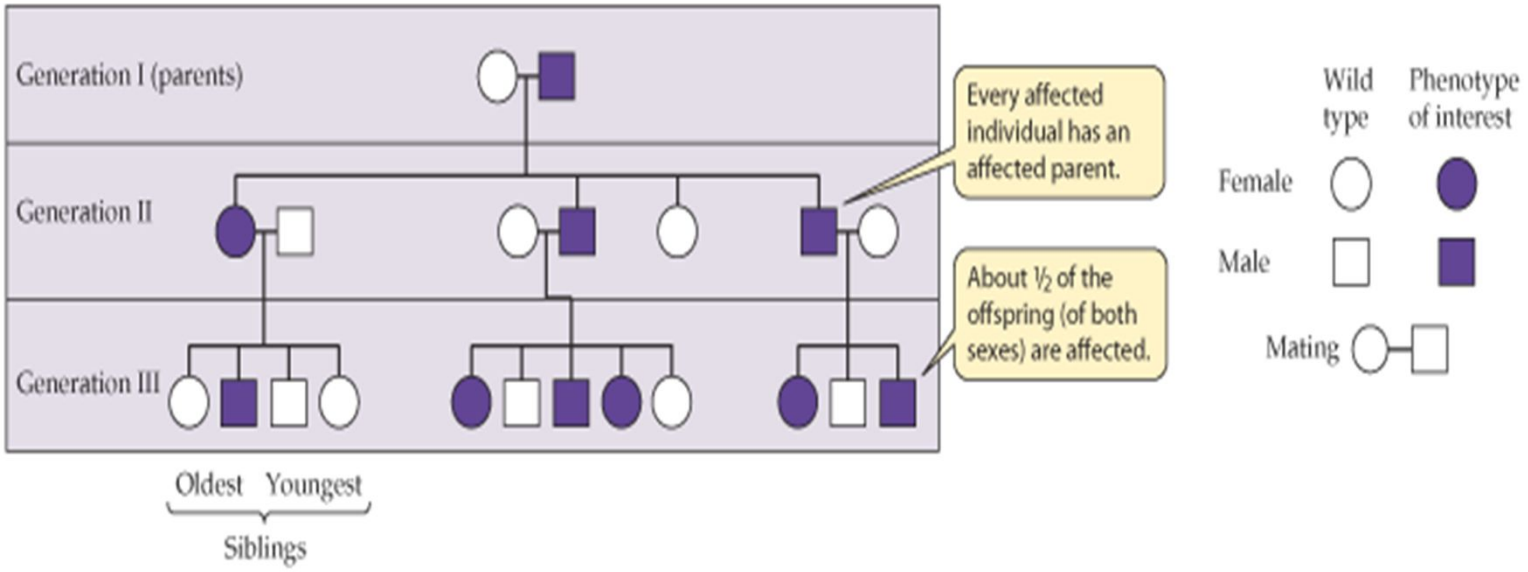
END

Clinical Genetics I



Pedigree Symbols

What Is My Mode of Inheritance



Features

Manifested in the heterozygous state

At least one parent of an index case is affected

Both males and females transmit the condition

When an effected person marries an unaffected child has one chance in two of having the disease

Reduce Penetrance: Some individuals inherit the mutant allele but are phenoltypically normal

Variable Expressivity: gene expression is variable e.g. neurofibromatosis type 1 range from brown spots to multiple tumors and skeletal deformities

Age of on set can delayed e.g. Huntington

Neurofibromatosis

Neurofibromatosis

Clinical Genetics II

Features

Manifested in the homozygous state

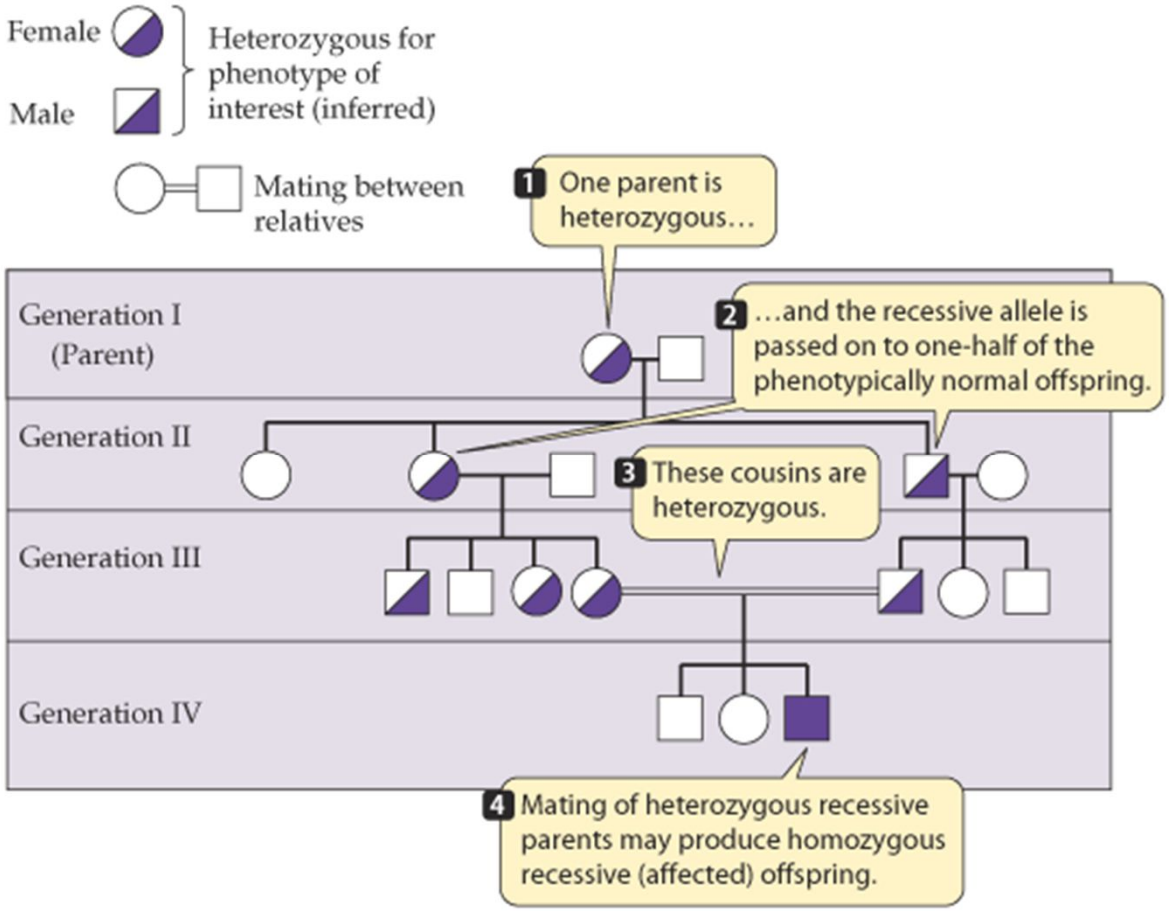
Parents usually not affected but the siblings may show the trait 1/4 (25%)

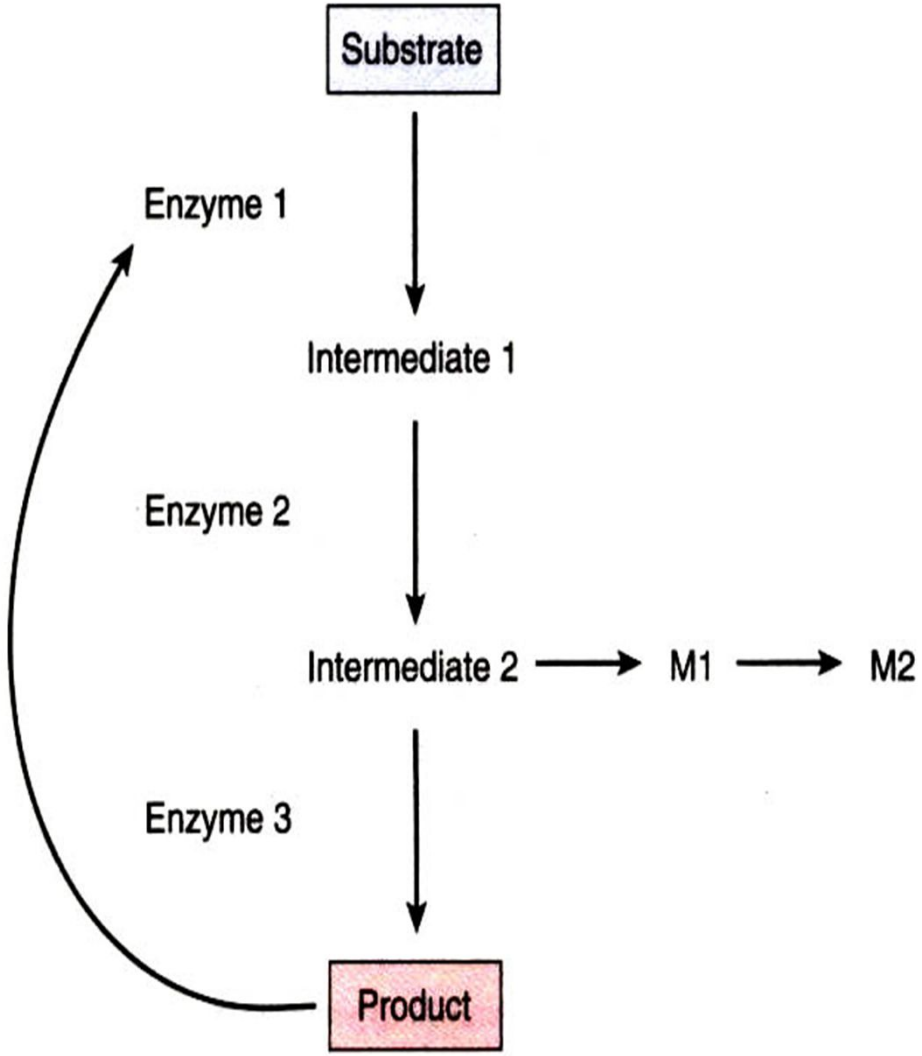
Affected individual may be product of consanguineous marriage (low frequency alleles)

The expression of the defect more uniform

Mostly enzyme proteins affected (half of normal enzymes ensures margin of safety)

Pedigree


Female
Male
Heterozygous for phenotype of interest (inferred)
Mating between relatives
1 One parent is heterozygous…
2 …and the recessive allele is passed on to one-half of the phenotypically normal offspring.
3 These cousins are heterozygous.
4 Mating of heterozygous recessive parents may produce homozygous recessive (affected) offspring.
Generation I (Parent)
Generation II
Generation III
Generation IV

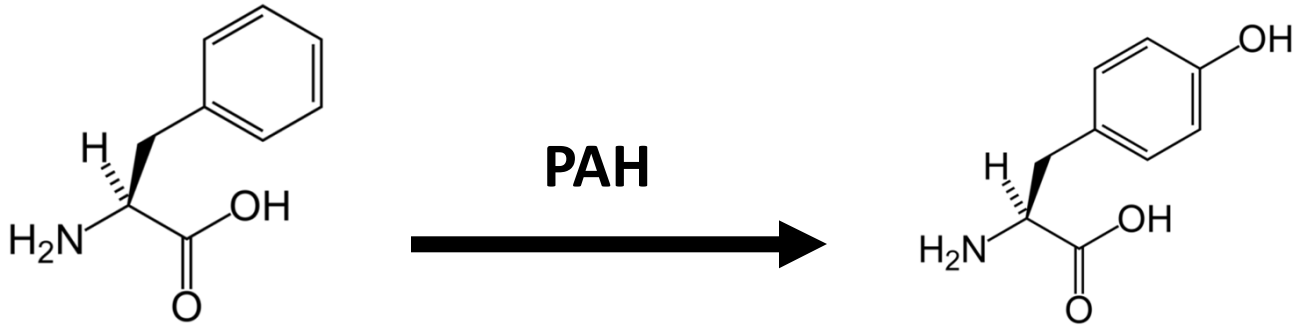

Substrate
Enzyme 1
Intermediate 1
Enzyme 2
Intermediate 2
M1
M2
Enzyme 3
Product

Mechanism

Phenylketonuria (PKU) is an autosomal recessive genetic disorder characterized by a deficiency in the enzyme phenylalanine hydroxylase (PAH)

This enzyme is necessary to metabolize the amino acid phenylalanine to the amino acid tyrosine

When PAH is deficient, phenylalanine accumulates and is converted into phenylketones, which are detected in the urine



Untreated PKU can cause problems with brain development, leading to progressive mental retardation and seizures

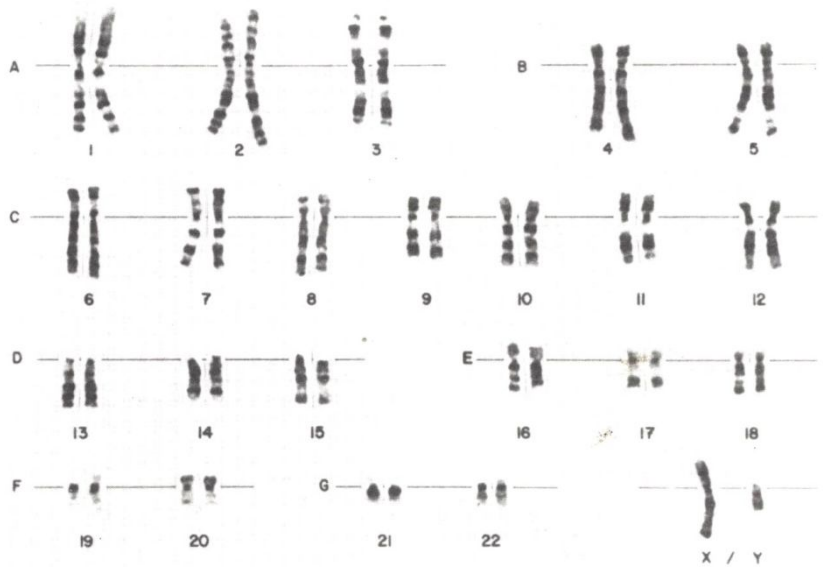
PKU is one of the few genetic diseases that can be controlled by diet

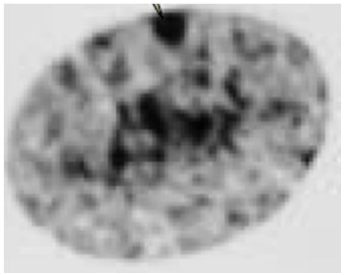
A diet low in phenylalanine and high in tyrosine can bring about a nearly total cure

the disease may present clinically with seizures, albinism (excessively fair hair and skin), and a "musty odor" to the baby's sweat and urine (due to phenylacetate, one of the ketones produced

H
H2N
OH
O
PAH
OH
H
H2N
OH
O

X-Linked Disorders





All sex-linked disorders are X-linked

No Y-linked diseases known

Transmitted by heterozygous female carriers to sons (hemizygous)

Hetrozygous female rarely express full phenotype

Partial expression due to X chromosome inactivation

Features

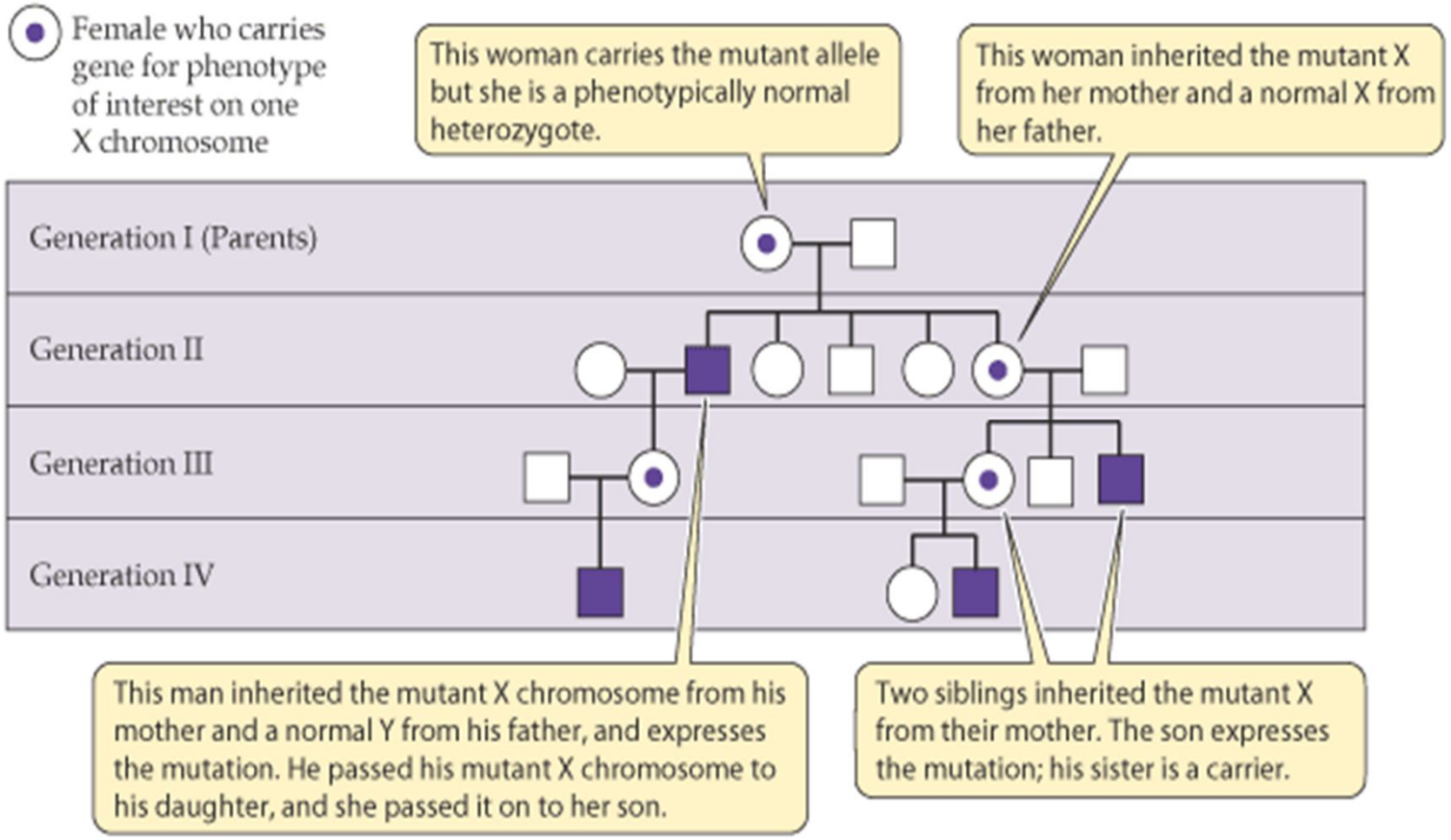
An affected male does not transmit the disorder to sons but all his daughters are carriers

Sons of heterozygous females have one chance in two of receiving mutant allele

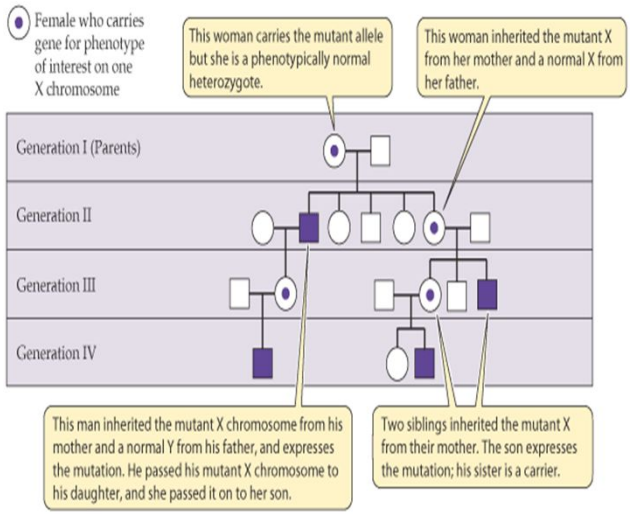
There are very few X-linked dominant diseases, inheritance pattern is characterized by transmission of the disease to 50% of the sons and daughters of an affected female

An affected male cannot transmit the disease to his sons but all daughters are affected VDRR

Pedigree

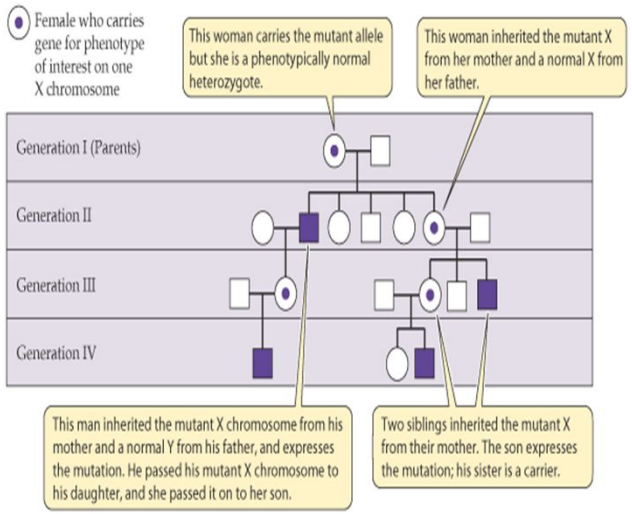

Female who carries gene for phenotype of interest on one X chromosome
This woman carries the mutant allele but she is a phenotypically normal heterozygote.
This woman inherited the mutant X from her mother and a normal X from her father.
Generation I (Parents)
Generation II
Generation III
Generation IV
This man inherited the mutant X chromosome from his mother and a normal Y from his father, and expresses the mutation. He passed his mutant X chromosome to his daughter, and she passed it on to her son.
Two siblings inherited the mutant X from their mother. The son expresses the mutation; his sister is a carrier.

Pedigree



END

Pedigree



END

Multifactorial Disorders

Features

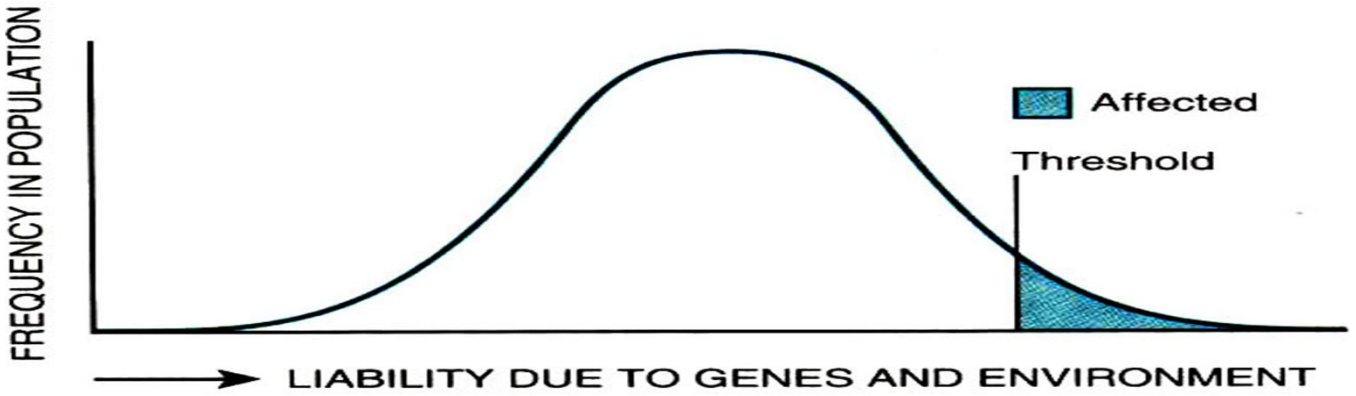
A multifactorial trait is governed by the additive effect of two or more genes of small effect but conditioned by environmental, nongenetic influences

Many physiologic characteristics of humans (e.g. diabetes, height, weight, blood pressure, hair color, gout, etc.)

Monozygous twins reared separately may achieve different heights because of nutritional/other environmental influences

Features

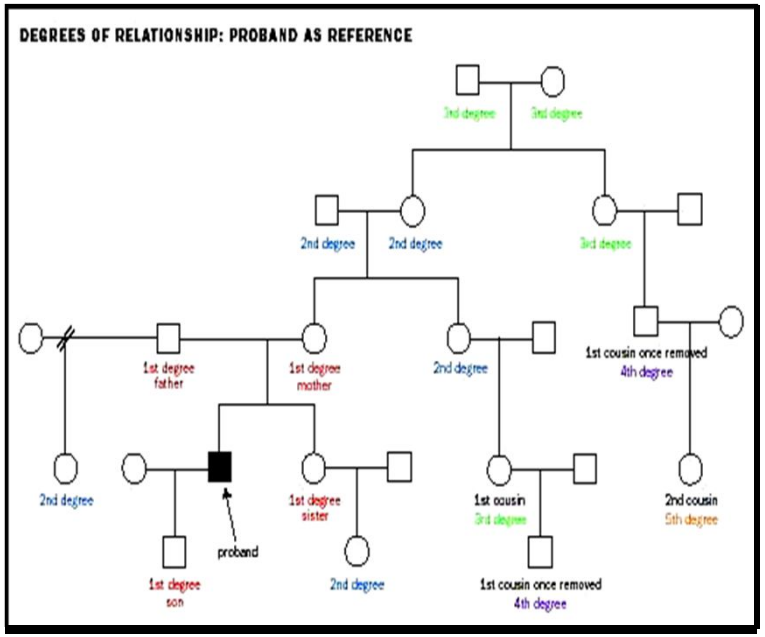
In a large population phenotypic attributes governed by multifactorial inheritance fall on a continuous Gaussian distribution



A disorder manifests itself only when a certain number of effector genes, as well as conditioning environmental influences are involved

this phenomenon is termed as threshold effect

Recurrence rate 2%-7% in for first degree relatives and 20%-40% for identical twins


DEGREES OF RELATIONSHIP: PROBAND AS REFERENCE
1st degree father
1st degree mother
2nd degree
1st cousin once removed
4th degree
2nd degree
proband
1st degree sister
1st cousin
3rd degree
2nd cousin
5th degree
1st degree son
2nd degree
1st cousin once removed
4th degree

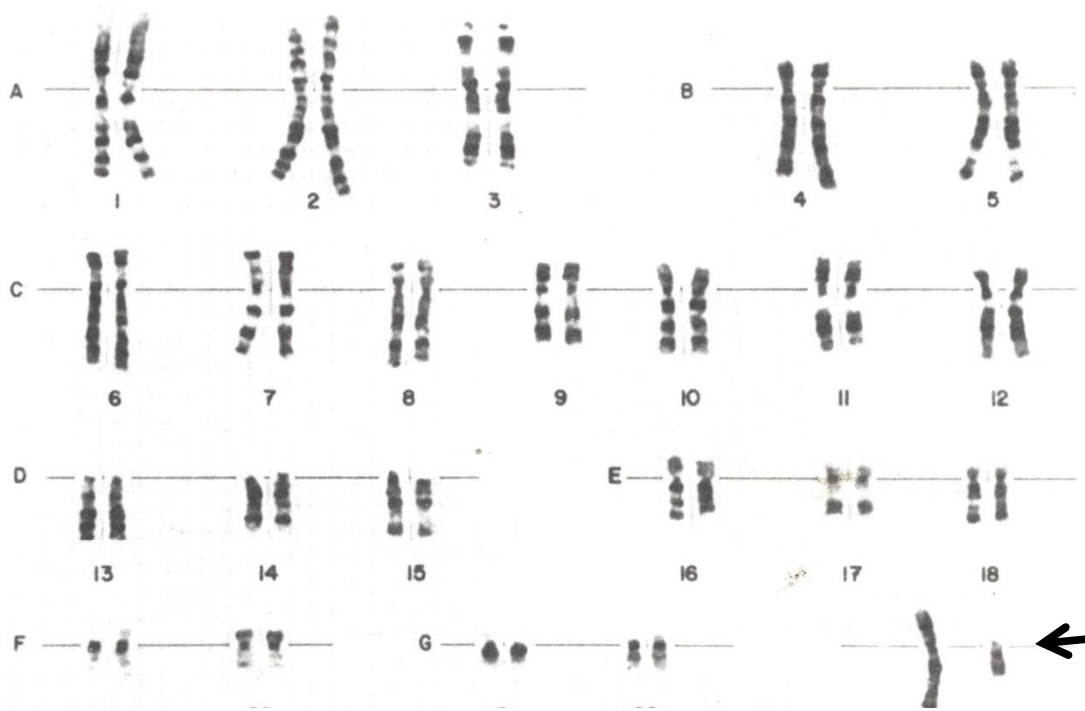
END

Cytogenetic Disorders

Karyotyping is used to diagnose Cytogenetic disorders

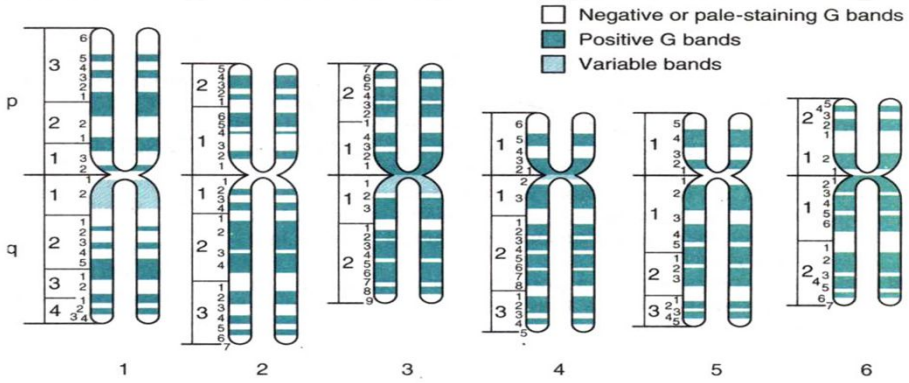
Karyotype is a photographic representation of a (G banded) stained metaphase chromosome spread arranged in order of decreasing length

1in 2000 newborns and 50% of first trimester abortions have chromosomal aberrations



Karyotype

Sex Chromosomes



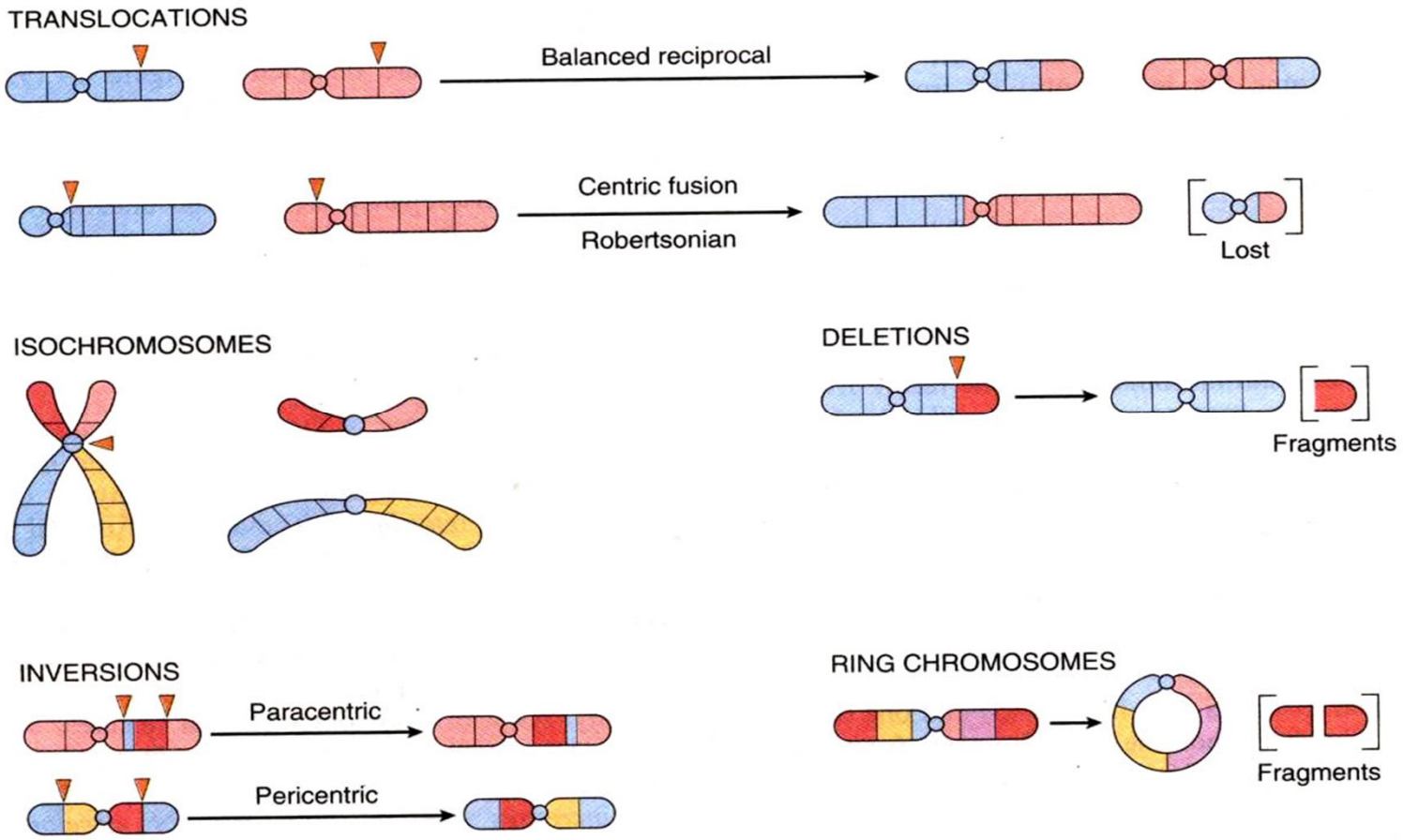
Chromosomal Nomenclature

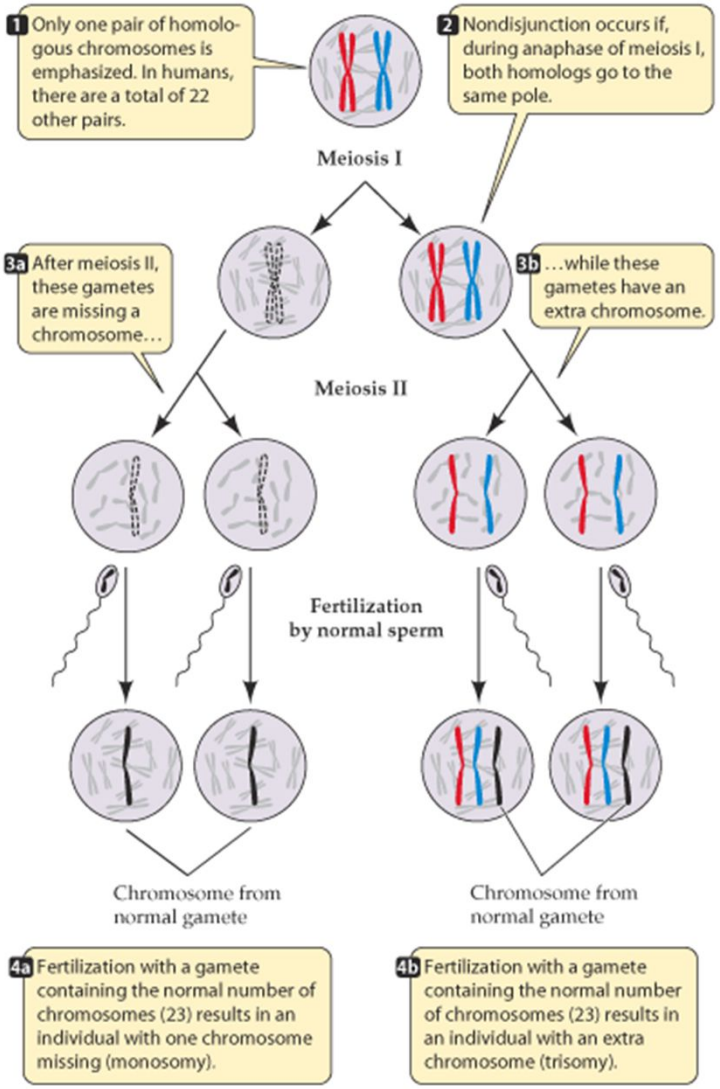
Any exact multiple of haploid (n) number of chromosomes (other than diploid 2n) are called Polyploid e.g. 3n, 4n etc

A chromosomal number that is not exact multiple of n is called aneuploid

Aneuploidy can result from non disjunction of homologous pair of chromosomes (first meiotic division) or failure of sister chromatids to separate (second meiotic division)

The result is monosomy (2n-1) or trisomy (2n+1)


TRANSLOCATIONS
Balanced reciprocal
Centric fusion
Robertsonian
Lost
ISOCHROMOSOMES
DELETIONS
Fragments
INVERSIONS
Paracentric
Pericentric
RING CHROMOSOMES
Fragments


1 Only one pair of homologous chromosomes is emphasized. In humans, there are a total of 22 other pairs.
2 Nondisjunction occurs if, during anaphase of meiosis I, both homologs go to the same pole.
Meiosis I
3a After meiosis II, these gametes are missing a chromosome…
3b …while these gametes have an extra chromosome.
Meiosis II
Fertilization by normal sperm
Chromosome from normal gamete
Chromosome from normal gamete
4a Fertilization with a gamete containing the normal number of chromosomes (23) results in an individual with one chromosome missing (monosomy).
4b Fertilization with a gamete containing the normal number of chromosomes (23) results in an individual with an extra chromosome (trisomy).

END

Autosomal Cytogenetic Disorders

Trisomy 21 (Down Syndrome)

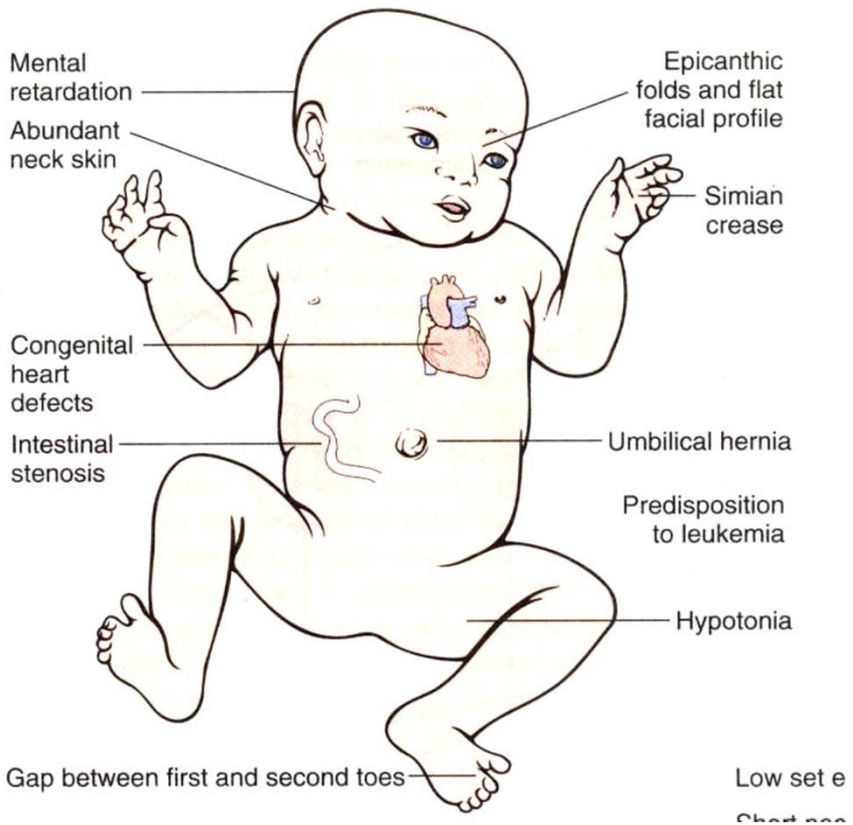
Arises from meiotic nondisjunction

Normal Parents 95% of time the extra chromosome comes from mother

Maternal age plays a significant role (1 in 1550 for women under 20 and 1 in 25 for women over 45)

4% of patients have 14:21 translocation

1% of patients are mosaic



Single palmar fold (also known as a simian crease), Epicanthic Folds, Flat Facial Profile, IQ 25-50, Cardiac Malformations

END

Cytogenetic Disorders of the Sex Chromosomes

Sex Chromosomes Disorders

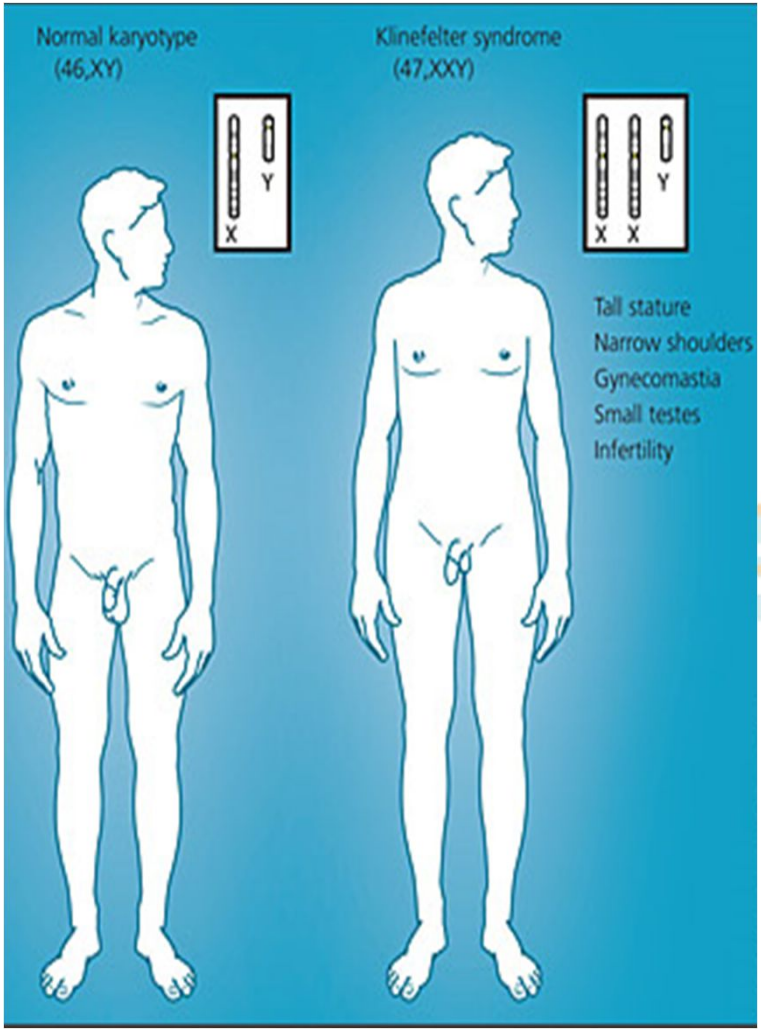
Trisomy 21 (Down Syndrome)

Numerical aberrations of sex chromosomal are relatively better tolerated than those caused by autosomes 46X-46XXXY

Only one X chromosome is active (females are mosaic) extra chromosome forms Barr body

People with extra Y chromosome are phenotypically normal

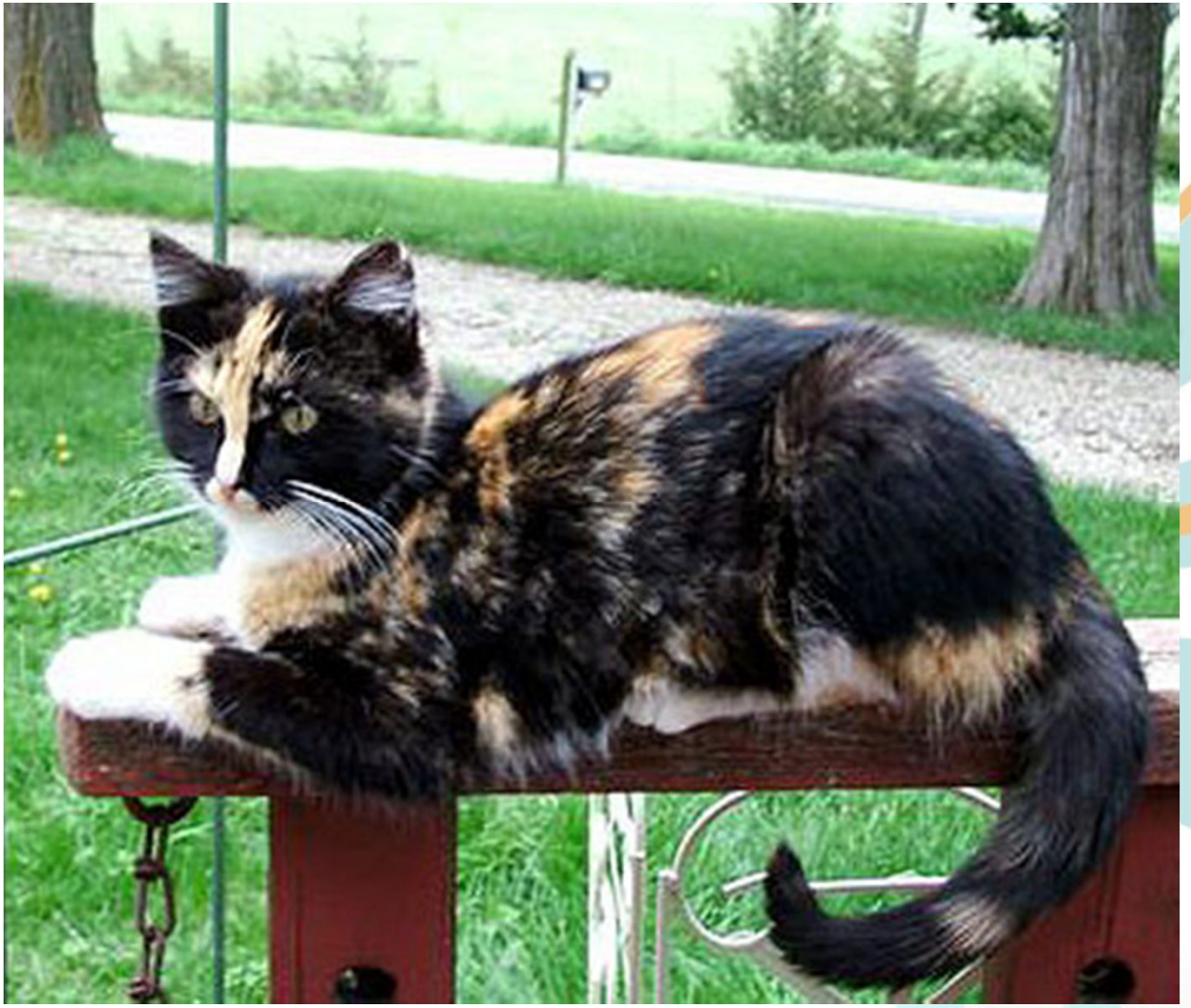
Klinefelter Syndrome (47 XXY)



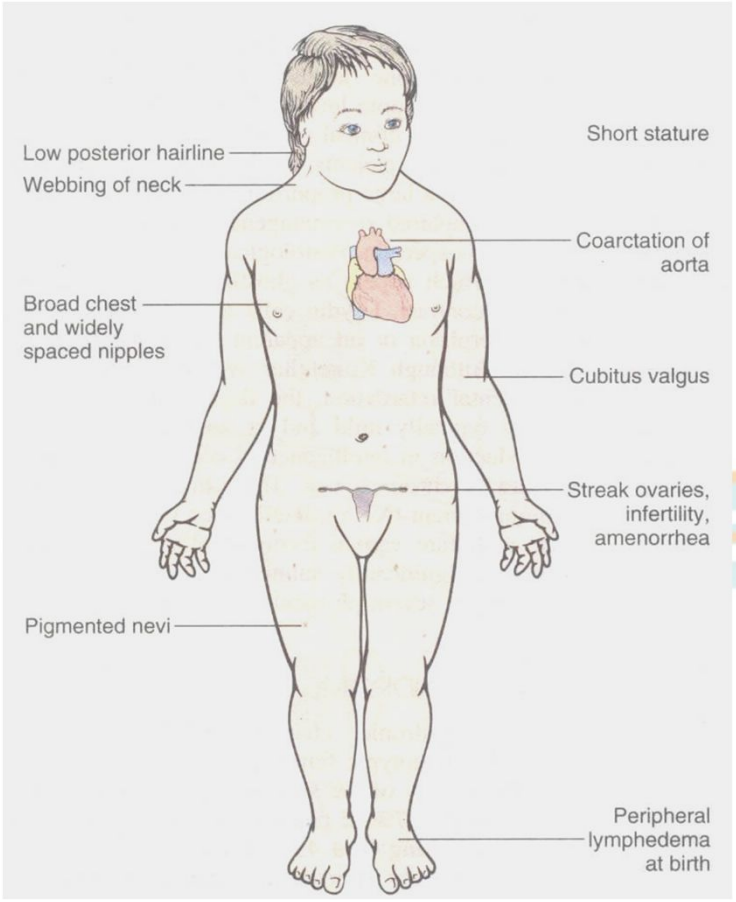
Extra chromosome can be maternal or paternal (advance maternal age or irradiation), 15% patients show mosaicism

Microorchidism, hypogonadism, longer legs, gynecomastia, infertility, may be mildly mentally retarded, reduced facial & pubic hair

Klinefelter Syndrome (47 XXY)



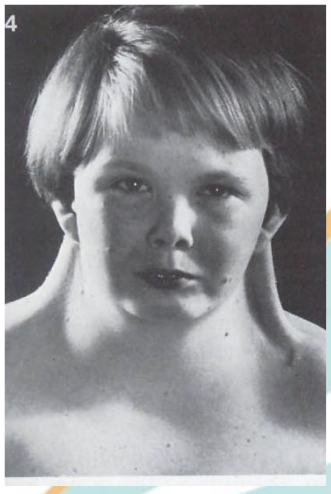
Turner Syndrome

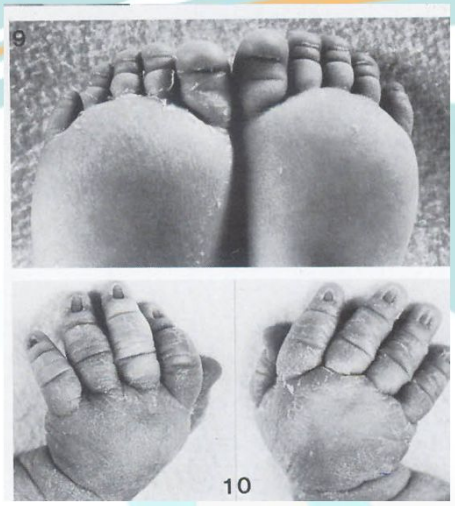


Turner syndrome encompasses several chromosomal abnormalities, of which monosomy X is the most common 57% rest may have i (X)
43% of patients mosaic

Short stature
Lymphoedema
Broad chest
Reproductive sterility
Obesity

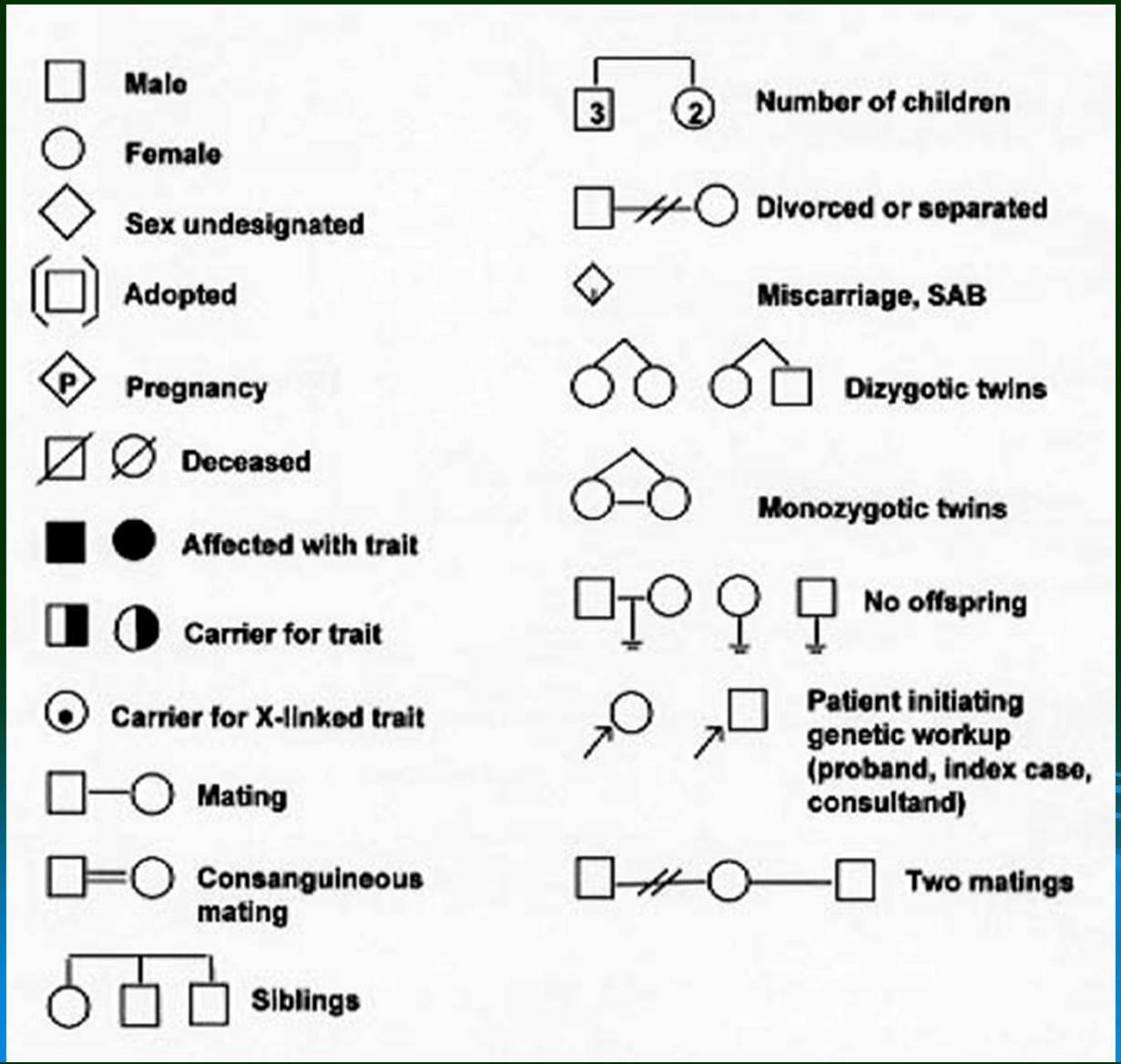
Turner Syndrome

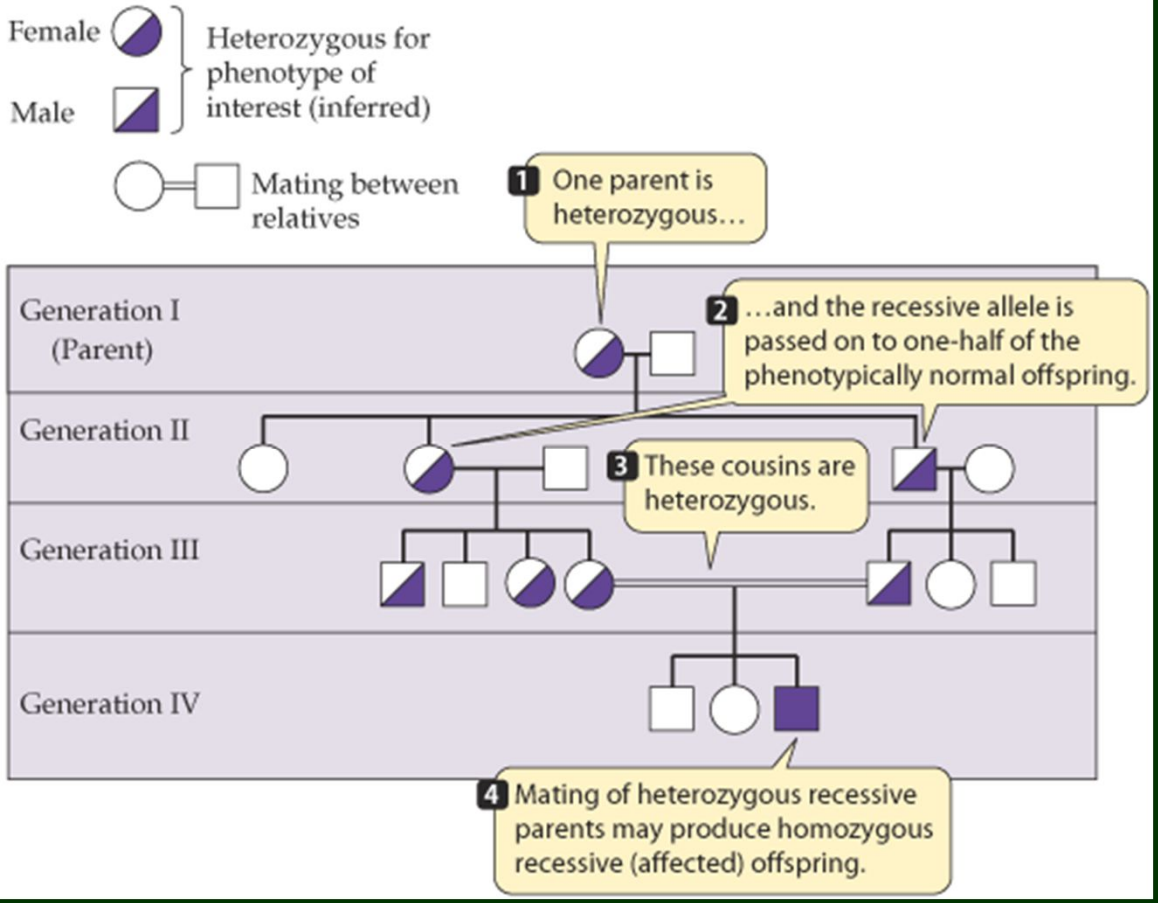




END

Pedigree Symbols




Female
Male
Heterozygous for phenotype of interest (inferred)
Mating between relatives
1 One parent is heterozygous…
2 …and the recessive allele is passed on to one-half of the phenotypically normal offspring.
3 These cousins are heterozygous.
4 Mating of heterozygous recessive parents may produce homozygous recessive (affected) offspring.
Generation I (Parent)
Generation II
Generation III
Generation IV

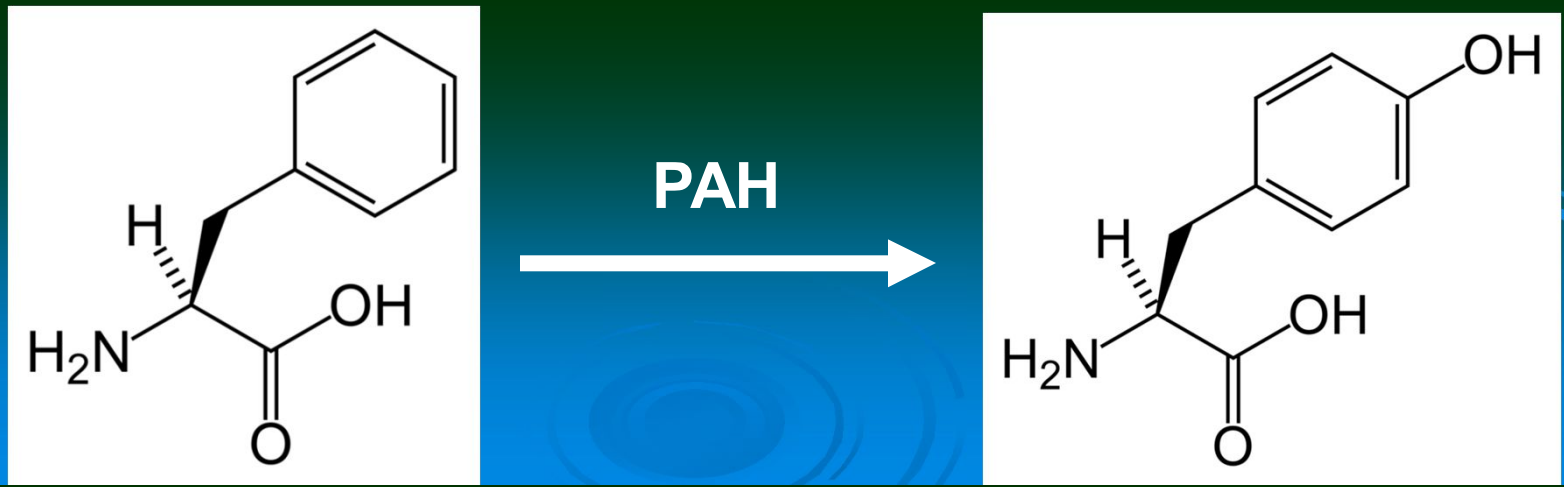
Pedigrees

Phenylketonuria (PKU)

Phenylketonuria (PKU) is an autosomal recessive genetic disorder characterized by a deficiency in the enzyme phenylalanine hydroxylase (PAH)

This enzyme is necessary to metabolize the amino acid phenylalanine to the amino acid tyrosine

When PAH is deficient, phenylalanine accumulates and is converted into phenylketones, which are detected in the urine



PKU

Untreated PKU can cause problems with brain development, leading to progressive mental retardation and seizures

PKU is one of the few genetic diseases that can be controlled by diet

A diet low in phenylalanine and high in tyrosine can bring about a nearly total cure

the disease may present clinically with seizures, albinism (excessively fair hair and skin), and a "musty odor" to the baby's sweat and urine (due to phenylacetate, one of the ketones produced

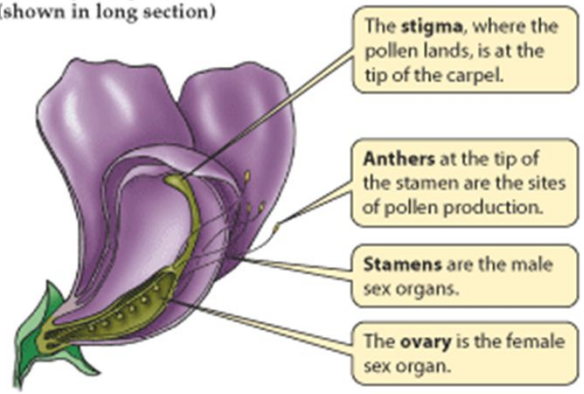
Neurofibromatosis

Neurofibromatosis causes tumors to grow on nerves and is is characterized by spots of increased skin pigmentation(café au lait spots), combined with peripheral nerve tumors and a variety of others dysplastic abnormalities of the skin, nervous system, bones, endocrine organs and blood vessels

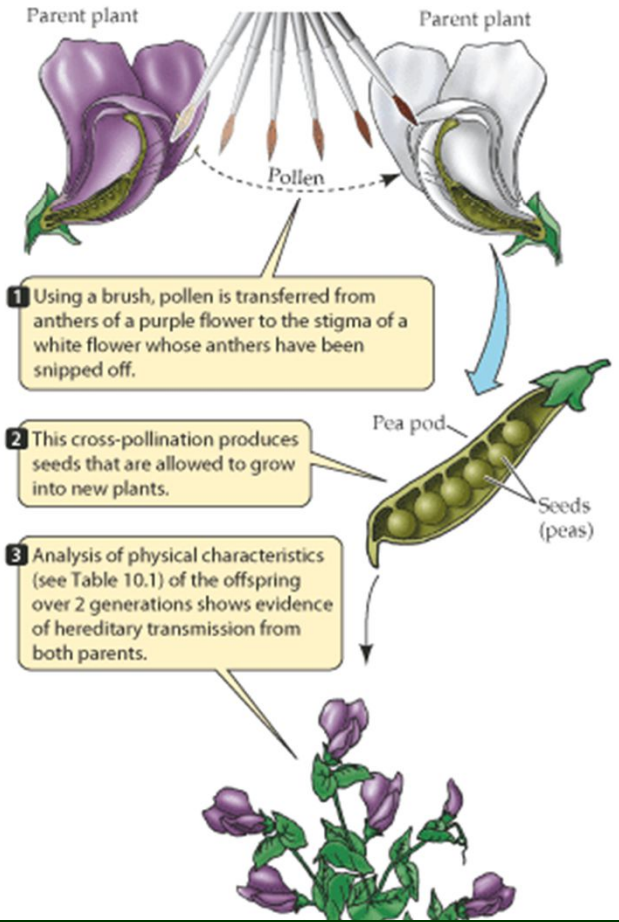
Neurofibromin a negative regulator of RAS (OP) is mutated

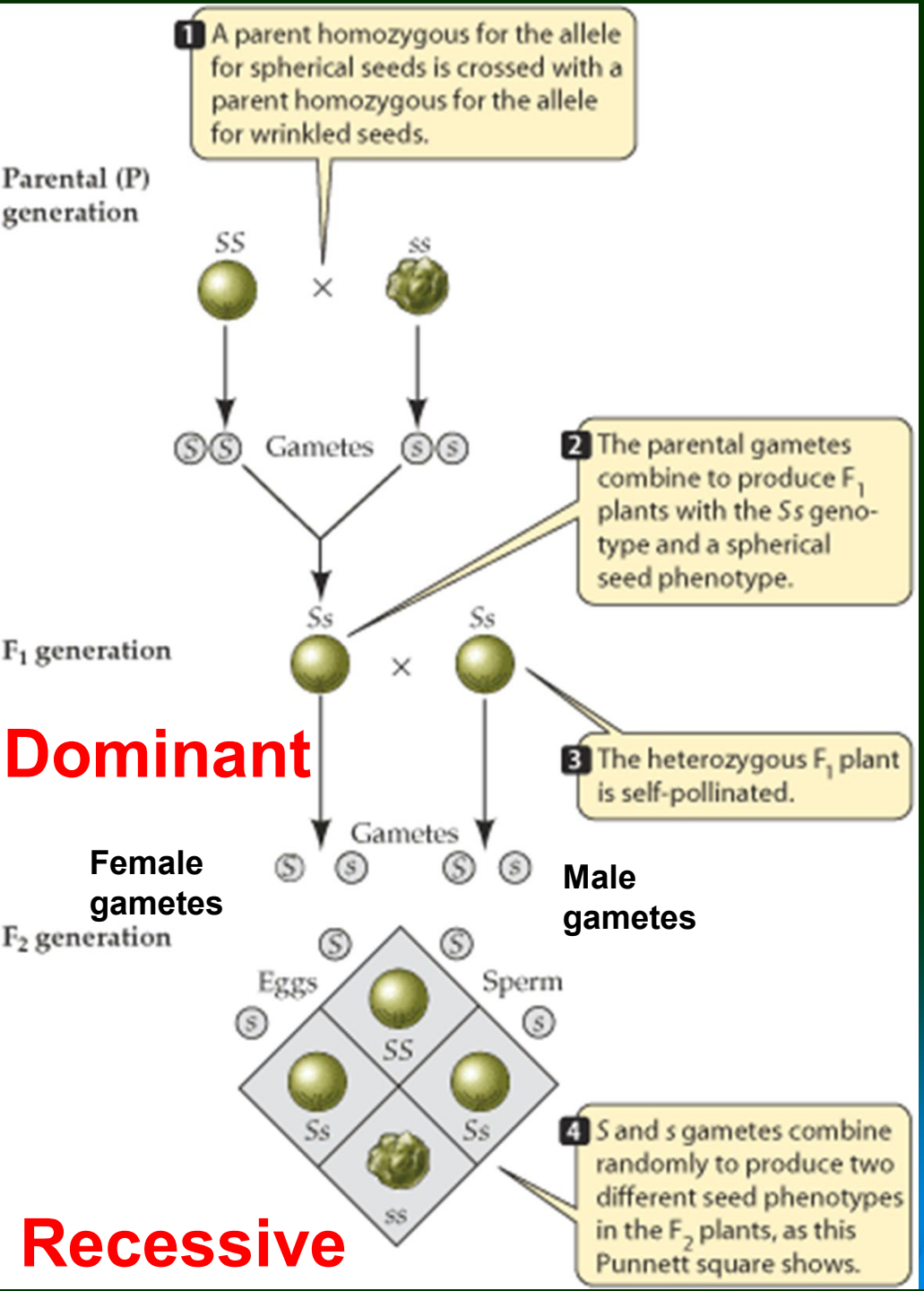
RESEARCH METHOD

Anatomy of a pea flower (shown in long section)



Pea flower cross-pollination





Dominant & Recessive

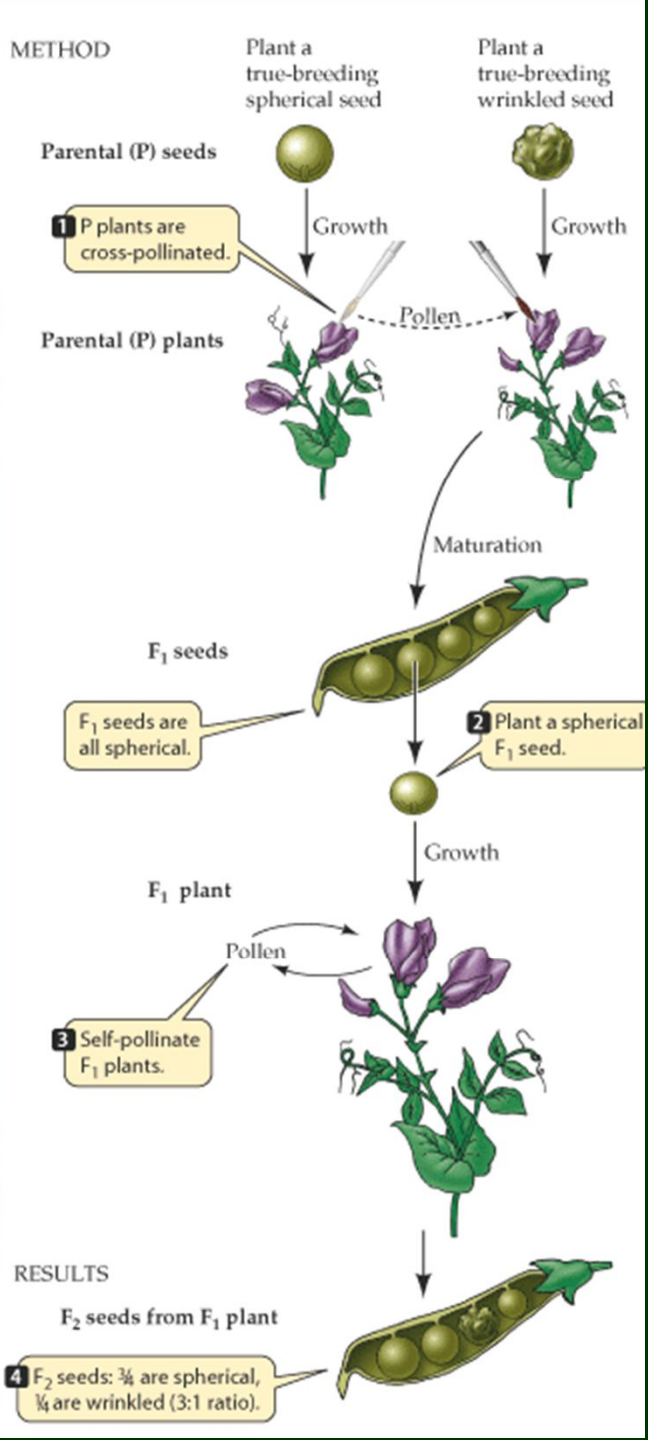
Mendel observed that the wrinkled seed trait was never expressed in the F1 generation, even though it reappeared in the F2 generation

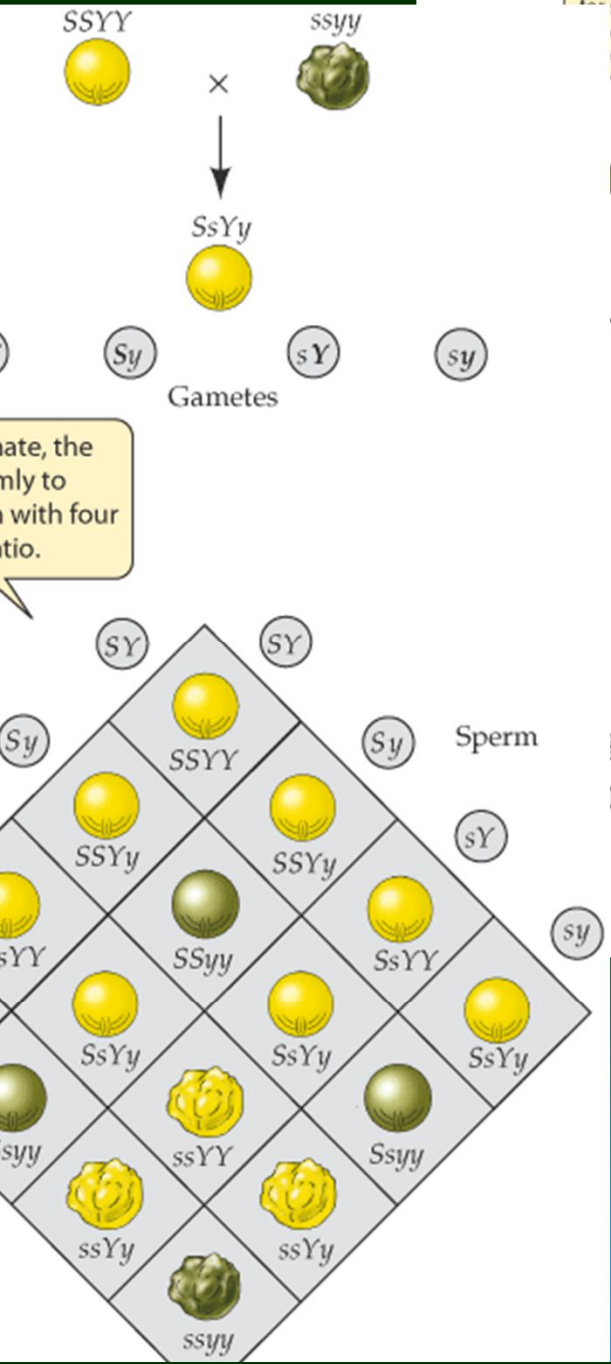
No Blending

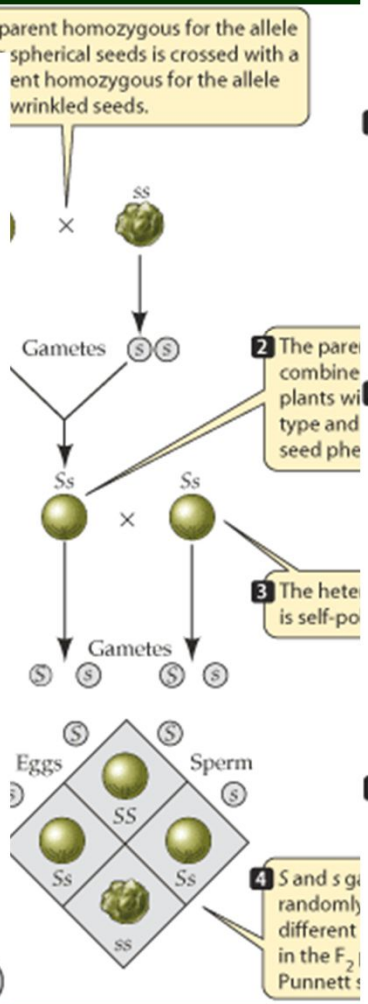
Particulate Theory

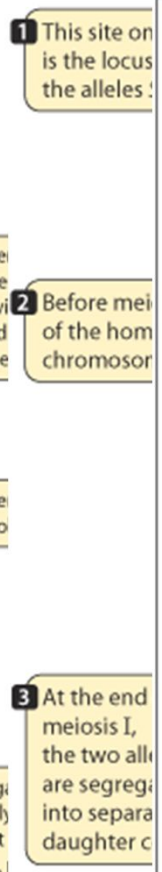
10.1 Mendel's Results from Monohybrid Crosses

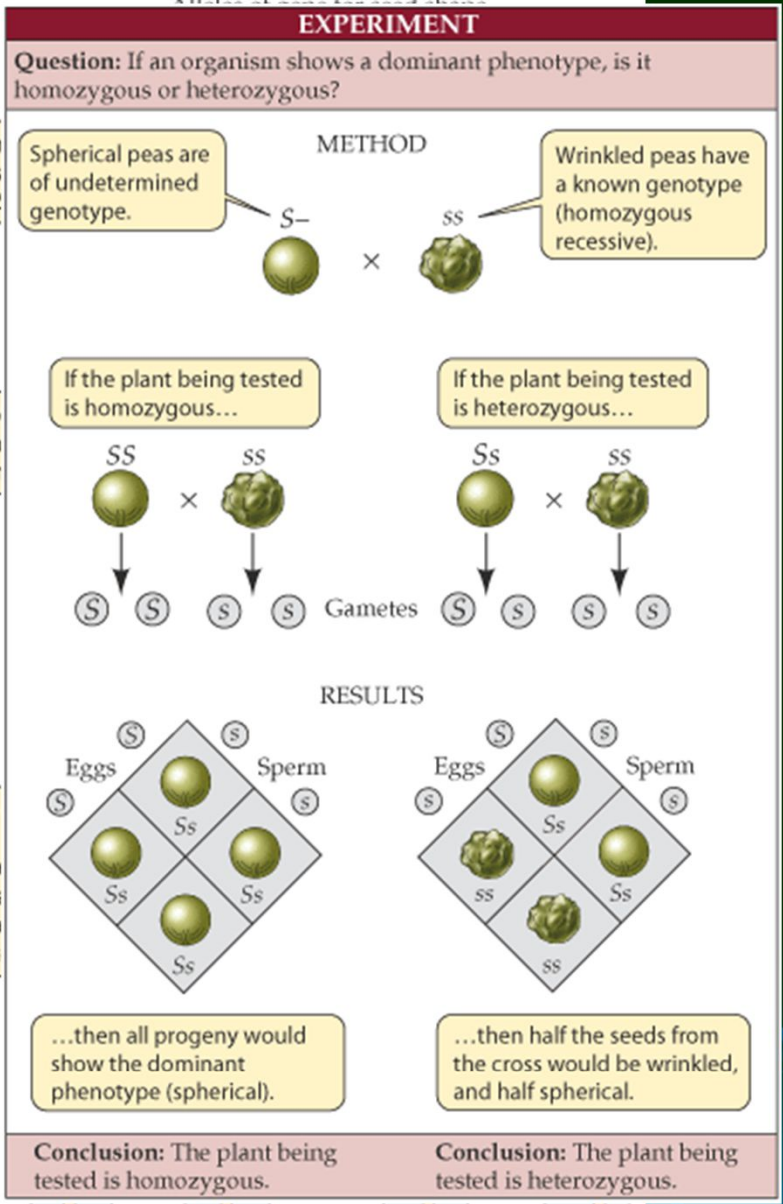
PARENTAL GENERATION PHENOTYPES			F_2 GENERATION PHENOTYP	
	DOMINANT	RECESSIVE	DOMINANT	RECESSI
	Spherical seeds ×	Wrinkled seeds	5,474	1,850
	Yellow seeds ×	Green seeds	6,022	2,001
	Purple flowers ×	White flowers	705	224
	Inflated pods ×	Constricted pods	882	299
	Green pods ×	Yellow pods	428	152
	Axial flowers ×	Terminal flowers	651	207
	Tall stems (1 m) ×	Dwarf stems (0.3 m)	787	277

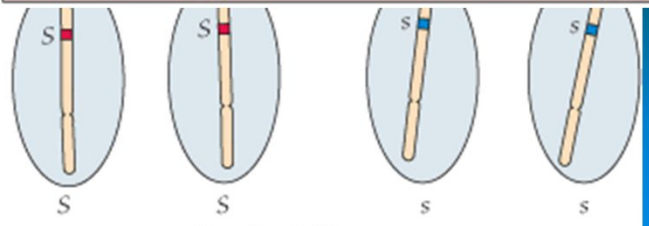


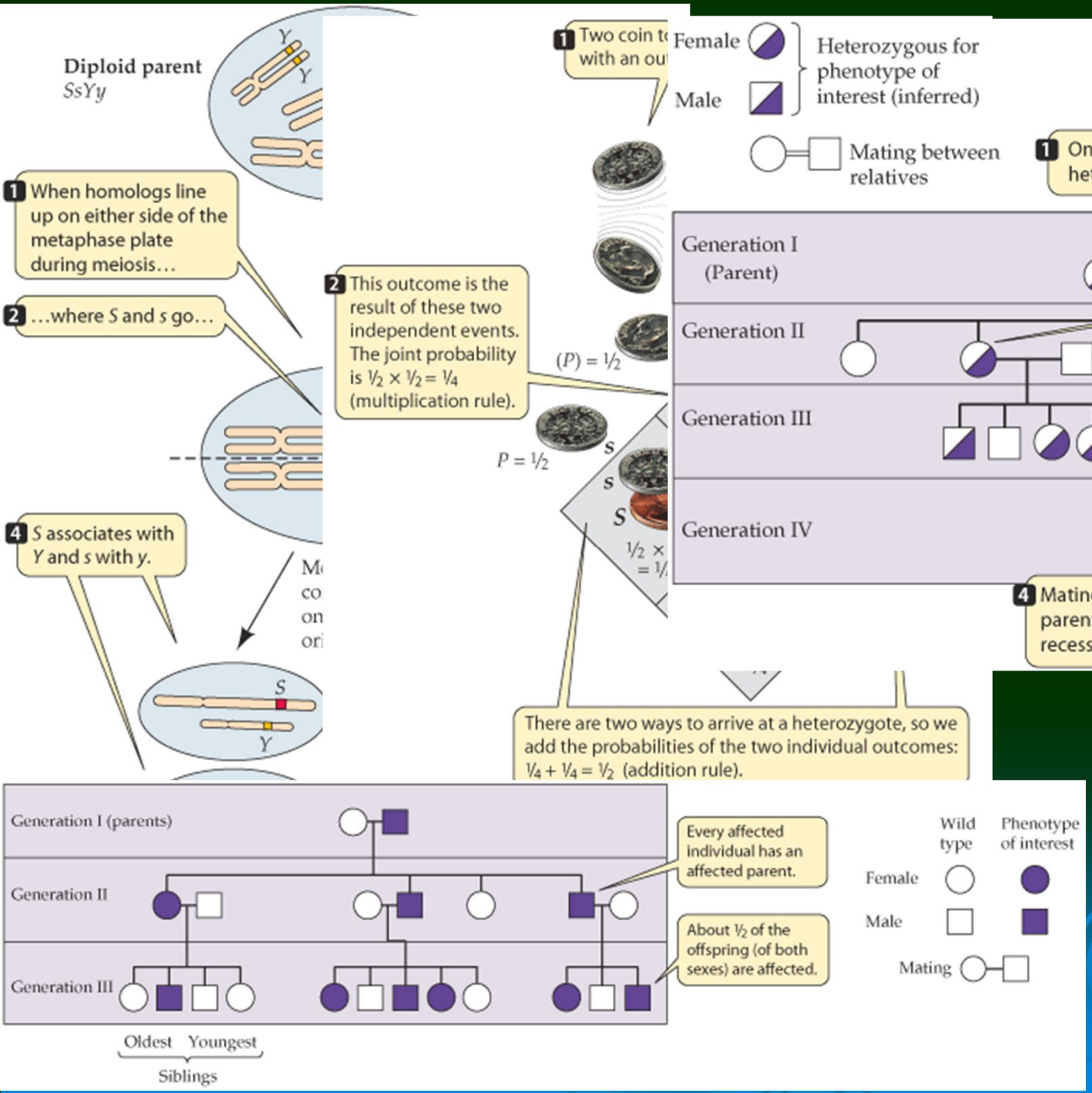

SSYY
ssyy
×
SsYy
Sy
sY
sy
Gametes
ate, the
nly to
with four
tio.
SY
SY
Sy
Sy
Sperm
sY
sy
SSYY
SSYy
SSYy
SSyy
SsYY
SsYy
SsYy
SsYy
ssYY
Ssyy
ssYy
ssYy
ssyy


1 A parent homozygous for the allele
spherical seeds is crossed with a
ent homozygous for the allele
wrinkled seeds.
ss
×
Gametes
S S
2 The pare
combine
plants wi
type and
seed phe
Ss
Ss
×
3 The hete
is self-po
Gametes
S S S S
Eggs
Sperm
S
S
SS
Ss
Ss
ss
4 S and s ga
randomly
different
in the F2
Punnett s

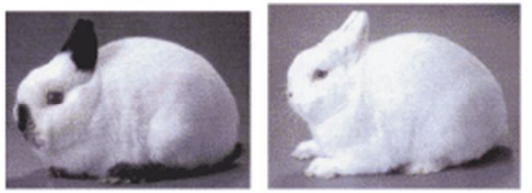

1 This site on
is the locus
the alleles
2 Before mei
of the hom
chromosor
3 At the end
meiosis I,
the two all
are segrega
into separa
daughter c

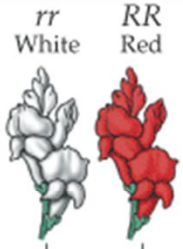

EXPERIMENT
Question: If an organism shows a dominant phenotype, is it homozygous or heterozygous?
METHOD
Spherical peas are of undetermined genotype.
S–
×
ss
Wrinkled peas have a known genotype (homozygous recessive).
If the plant being tested is homozygous…
SS
ss
×
S S
s s
Gametes
If the plant being tested is heterozygous…
Ss
ss
×
S s
s s
Gametes
RESULTS
Eggs
Sperm
S
s
Ss
Ss
Ss
Ss
…then all progeny would show the dominant phenotype (spherical).
Eggs
Sperm
S
s
Ss
Ss
ss
Ss
…then half the seeds from the cross would be wrinkled, and half spherical.
Conclusion: The plant being tested is homozygous.
Conclusion: The plant being tested is heterozygous.


S
S
S
S
s
s
s
s


Diploid parent
SsYy
Y
Y
1 When homologs line up on either side of the metaphase plate during meiosis…
2 …where S and s go…
4 S associates with Y and s with y.
S
Y
1 Two coin t
with an ou
2 This outcome is the result of these two independent events. The joint probability is ½ × ½ = ¼ (multiplication rule).
(P) = ½
P = ½
s
s
S
½ ×
= ½
There are two ways to arrive at a heterozygote, so we add the probabilities of the two individual outcomes: ¼ + ¼ = ½ (addition rule).
Female
Male
Heterozygous for phenotype of interest (inferred)
Mating between relatives
1 One parent is heterozygous…
2 …and the recessive allele is passed on to one-half of the phenotypically normal offspring
Generation I (Parent)
Generation II
Generation III
Generation IV
3 These cousins are heterozygous.
4 Mating of heterozygous recessive parents may produce homozygous recessive (affected) offspring.
Generation I (parents)
Generation II
Generation III
Every affected individual has an affected parent.
About ½ of the offspring (of both sexes) are affected.
Wild type
Phenotype of interest
Female
Male
Mating
Oldest
Youngest
Siblings

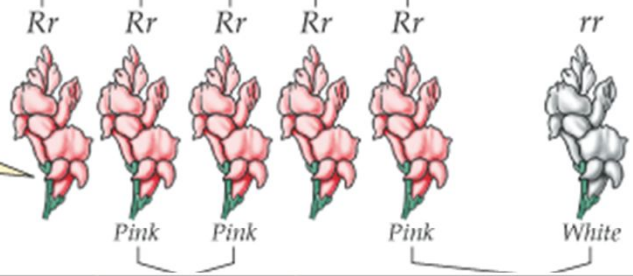
Possible genotypes	CC, Cc^{ch}, Cc^{h}, Cc	$c^{ch}c^{ch}$	$c^{ch}c^{h}$, $c^{ch}c$	$c^{h}c^{h}$, $c^{h}c$	cc
Phenotype	Dark gray	Chinchilla	Light gray	Himalayan	Albino





Parental (P) generation

1 When true-breeding red and white parents are crossed, the F_1 generation are all pink.



F_1 generation

2 Heterozygous snapdragons produce pink flowers—an intermediate phenotype—because the allele for red flowers is incompletely dominant white one

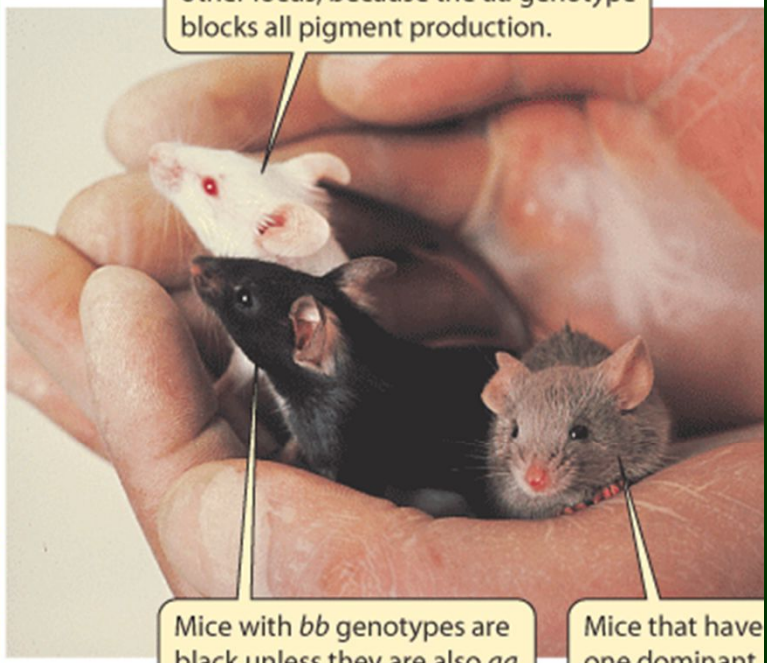
F_2 generat

3 When F_1 pollinate, white, pir offspring

Blood type of cells	Genotype	Antibodies made by body	Reaction to added antibodies	
			Anti-A	Anti-B
A	I^AI^A or I^Ai^O	Anti-B		
B	I^BI^B or I^Bi^O	Anti-A		
AB	I^AI^B	Neither anti-A nor anti-B		
O	i^Oi^O	Both anti-A and anti-B		

Re
dc
ar
ev

Re
th
ar
to
ap



EXPERIMENT

estion: Do alleles for different characteristics always assort independently?

rent (P)

BbVgvg
Wild type (gray body, normal wings)

×

bbvgvg
(Black body, vestigial win

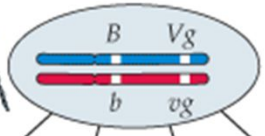
♀ ♂

Gray normal (wild type)

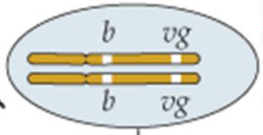
♀

Black vestigial

♂



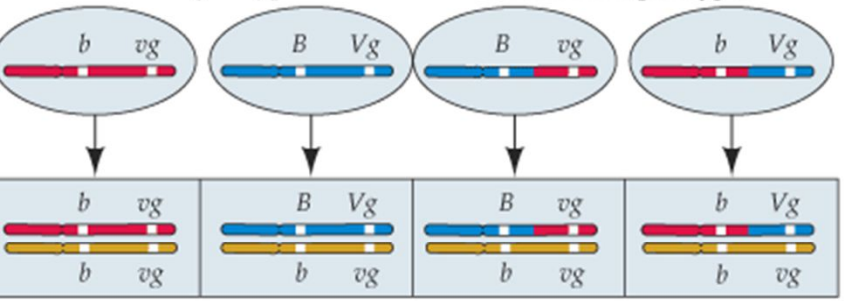
×



Recombination

Parental genotypes

Recombinant genotypes



Black vestigial	Wild type	Gray vestigial	Black normal
944	965	206	185

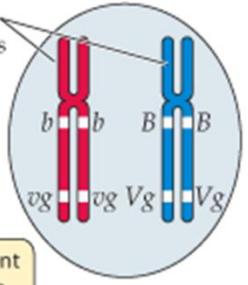
Number of individuals

Parental phenotypes

Recombinant (nonparental) phenotypes

$$\text{Recombinant frequency} = \frac{391 \text{ recombinants}}{2{,}300 \text{ total offspring}} = 0.17$$

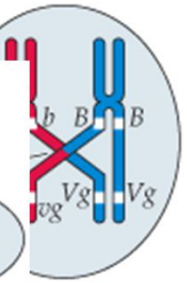
Homologous chromosomes



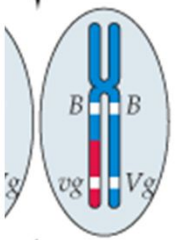
Genes at different loci on the same chromosome can separate and recombine by crossing over.

These are th
from Mendel
(independen

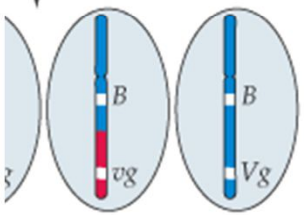
Meiosis I



s

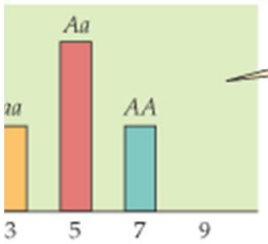


II

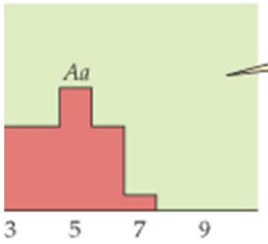


Gametes

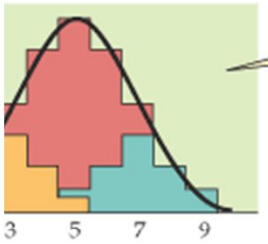
recombinant gametes
t of crossing over.



Quantitativ
3 phenotyp

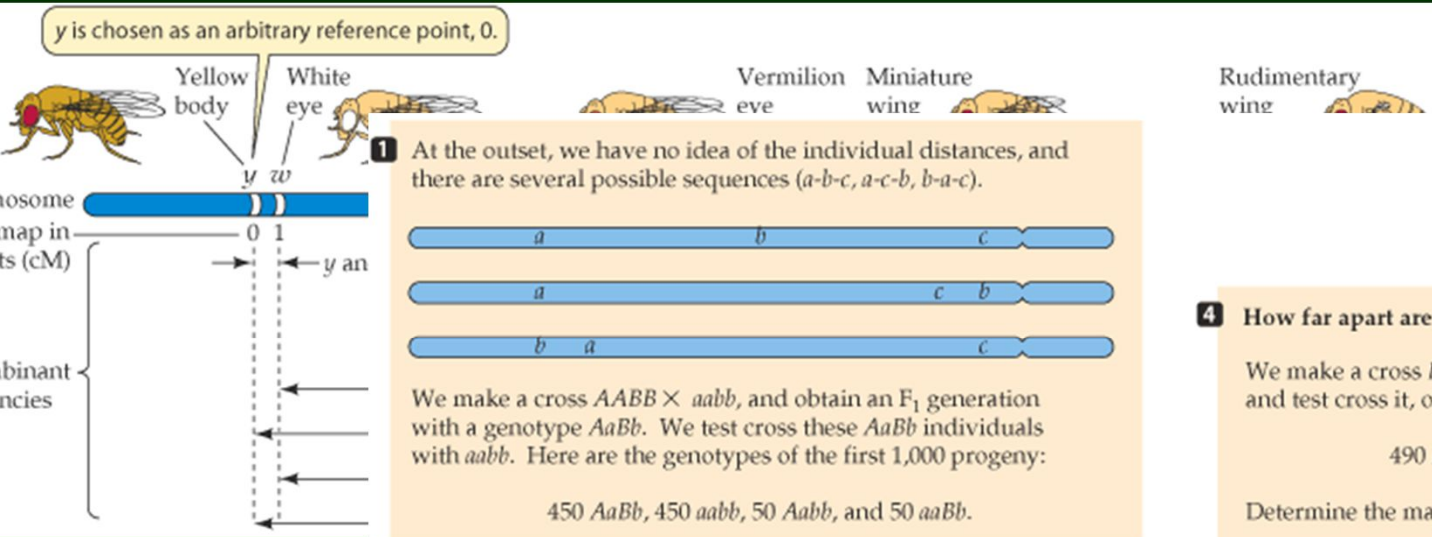


Quantitativ
due to envi



Interaction
environme
continuous

on in phenotype



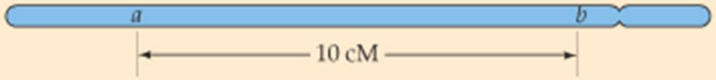
1 At the outset, we have no idea of the individual distances, and there are several possible sequences (*a*-*b*-*c*, *a*-*c*-*b*, *b*-*a*-*c*).

We make a cross *AABB* × *aabb*, and obtain an F_1 generation with a genotype *AaBb*. We test cross these *AaBb* individuals with *aabb*. Here are the genotypes of the first 1,000 progeny:

450 *AaBb*, 450 *aabb*, 50 *Aabb*, and 50 *aaBb*.

2 **How far apart are the *a* and *b* genes?** Well, what is the recombinant frequency? Which are the recombinant types, and which are the parental types?

Recombinant frequency (*a* to *b*) = (50 + 50)/1,000 = 0.1
So the map distance is
Map distance = 100 × recombinant frequency =
100 × 0.1 = 10 cM

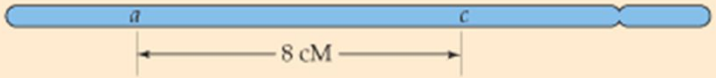


3 Now we make a cross *AACC* × *aacc*, obtain an F_1 generation, and test cross it, obtaining:

460 *AaCc*, 460 *aacc*, 40 *Aacc*, and 40 *aaCc*.

How far apart are the *a* and *c* genes?

Recombinant frequency (*a* to *c*) = (40 + 40)/1,000 = 0.08
Map distance = 100 × recombinant frequency =
100 × 0.08 = 8 cM



4 **How far apart are the *b* and *c* genes?**

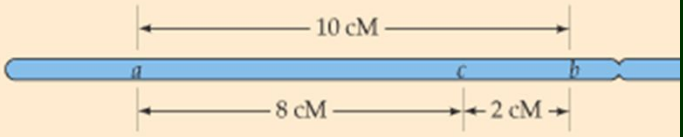
We make a cross *BBCC* × *bbcc*, obtain an F_1 generation, and test cross it, obtaining:

490 *BbCc*, 490 *bbcc*, 10 *Bbcc*, and 10 *bbCc*.

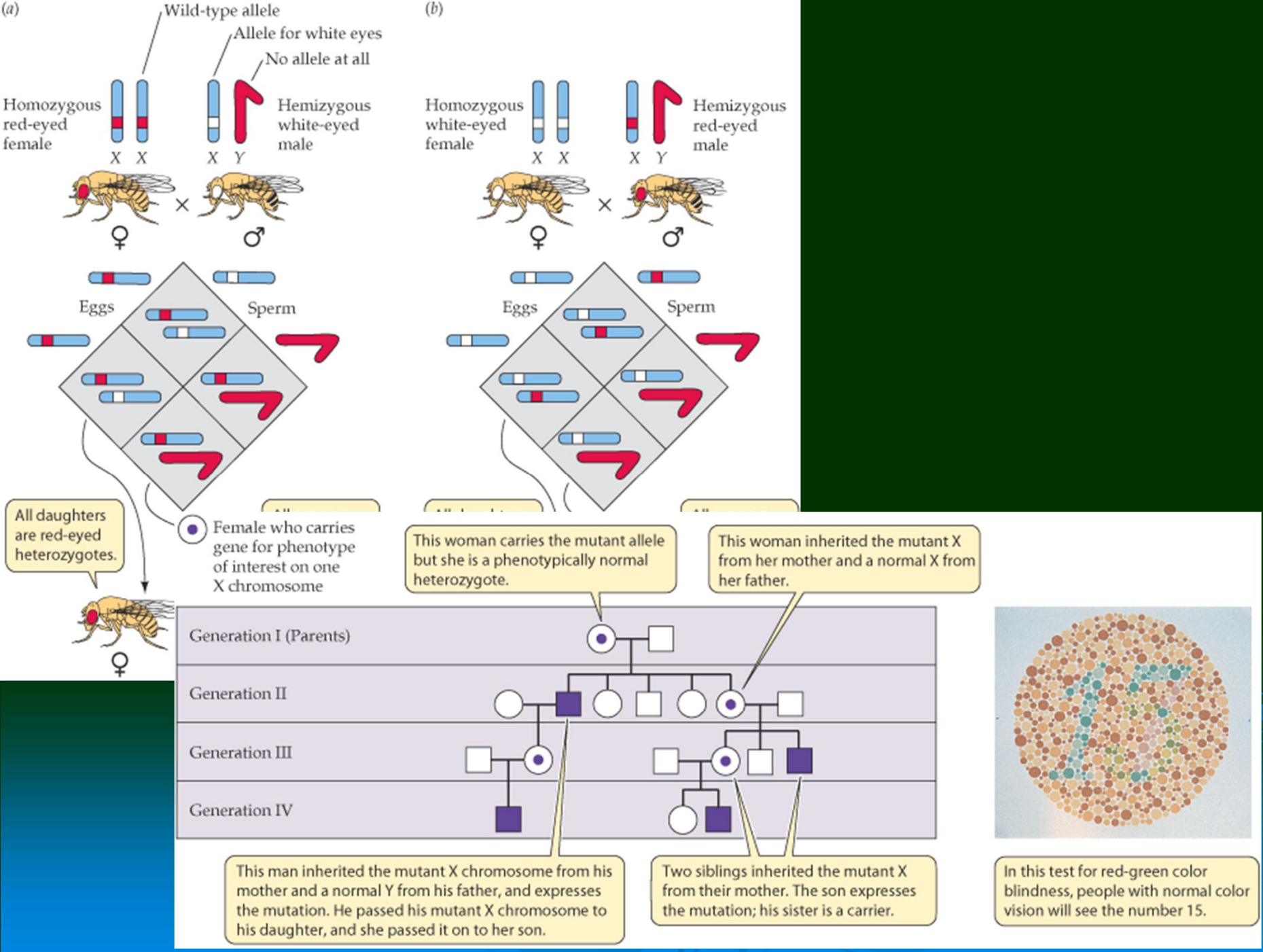
Determine the map distance between *b* and *c*.

Recombinant frequency (*b* to *c*) = (10 + 10)/1,000 = 0.02
Map distance = 100 × recombinant frequency =
100 × 0.02 = 2 cM

5 **Which of the three genes is between the other two?**
Because *a* and *b* are the farthest apart, *c* must be between them.



These numbers add up perfectly, but in most real cases they don't add up perfectly because of multiple crossovers.


(a)
Wild-type allele
Allele for white eyes
No allele at all
Homozygous red-eyed female
Hemizygous white-eyed male
X X
X Y
Eggs
Sperm
All daughters are red-eyed heterozygotes.
Female who carries gene for phenotype of interest on one X chromosome
(b)
Homozygous white-eyed female
Hemizygous red-eyed male
X X
X Y
Eggs
Sperm
This woman carries the mutant allele but she is a phenotypically normal heterozygote.
This woman inherited the mutant X from her mother and a normal X from her father.
Generation I (Parents)
Generation II
Generation III
Generation IV
This man inherited the mutant X chromosome from his mother and a normal Y from his father, and expresses the mutation. He passed his mutant X chromosome to his daughter, and she passed it on to her son.
Two siblings inherited the mutant X from their mother. The son expresses the mutation; his sister is a carrier.
In this test for red-green color blindness, people with normal color vision will see the number 15.

Introduction to Molecular Cell Biology

Hereditary Molecules

Dr. Fridoon Jawad Ahmad

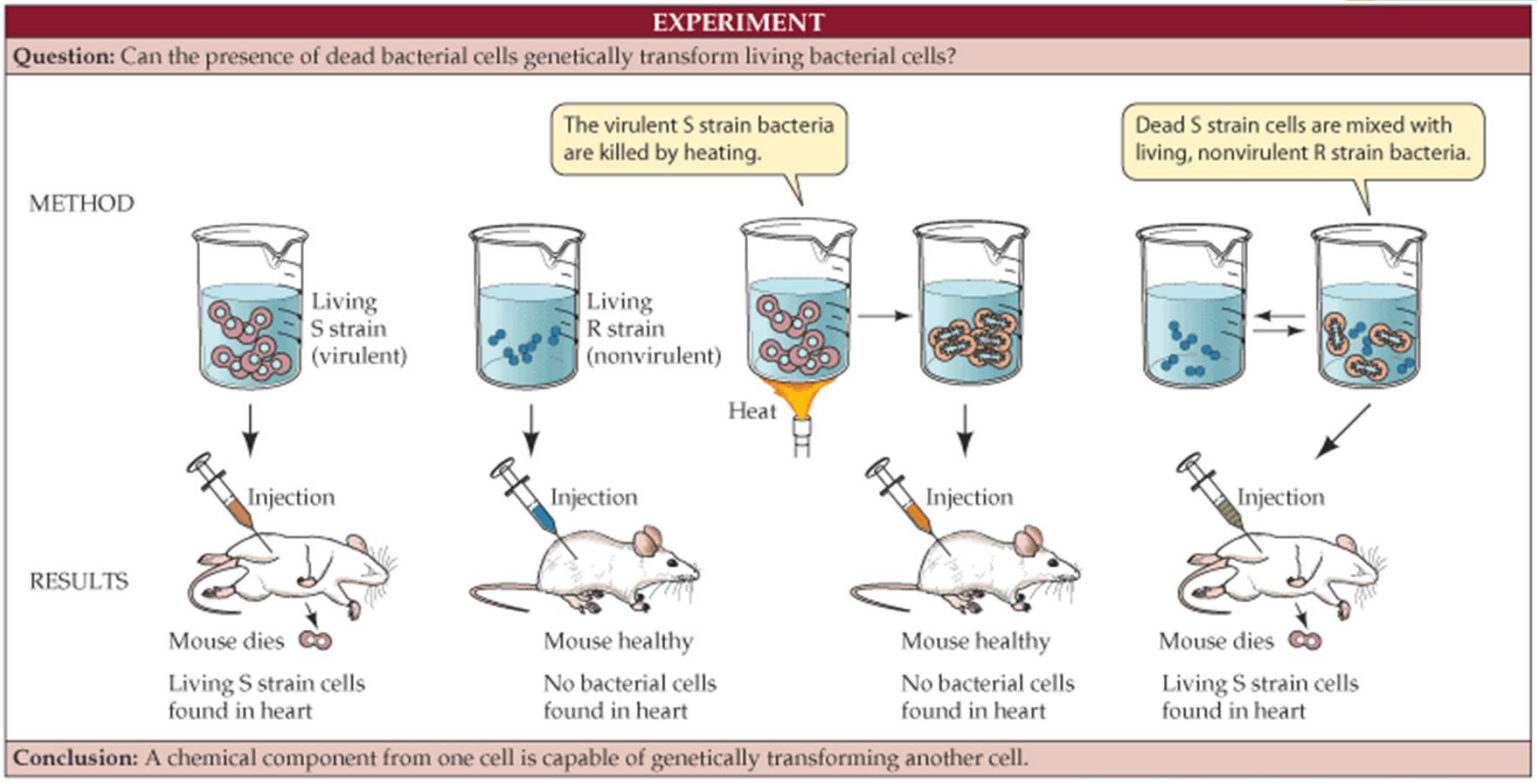
HEC Foreign Professor
King Edward Medical University
Visiting Professor LUMS-SSE

Search for the Heredity Molecule

Search for the Heredity Molecule

Fredrick Griffith Experiment 1920s

Transforming Principle



MacLeaod and MacCarty 1944

The nature of "Transforming Principle"

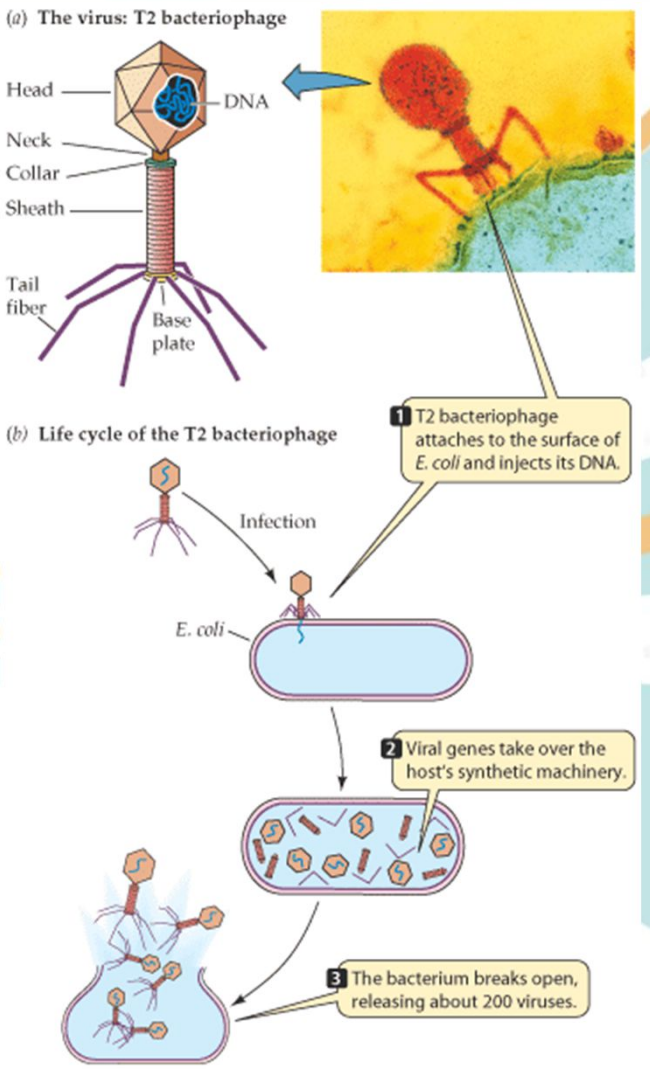
If DNA was destroyed in cell extracts there was no transformation.

Cells are the basic unit of life.

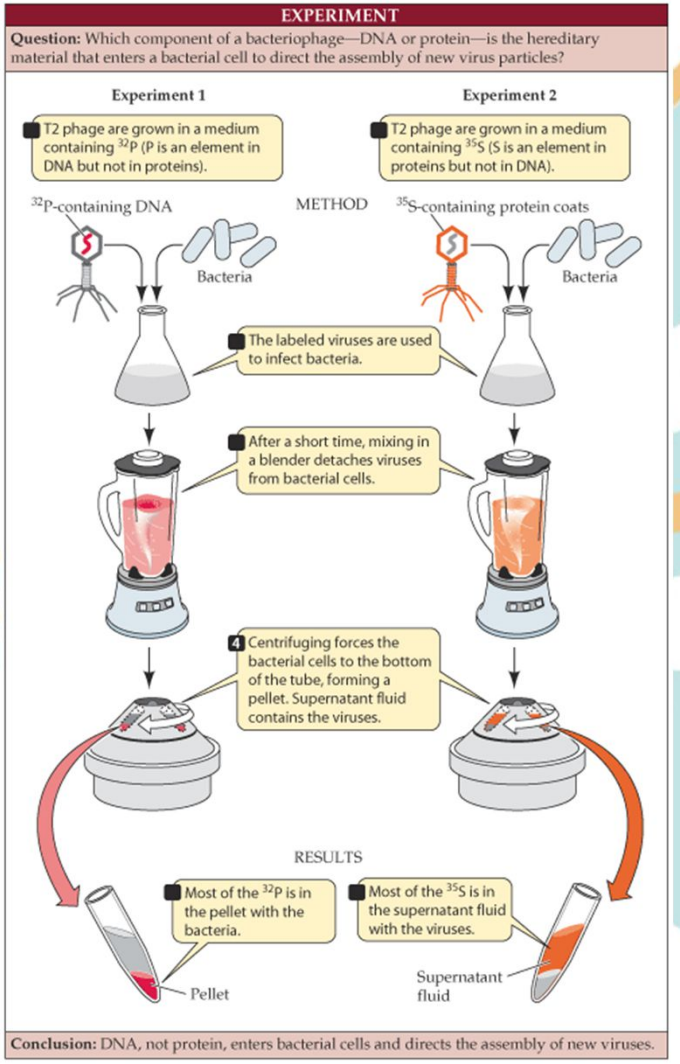
Little impact scientists did not believe DNA could be the genetic material.

Search for the Heredity Molecule 2

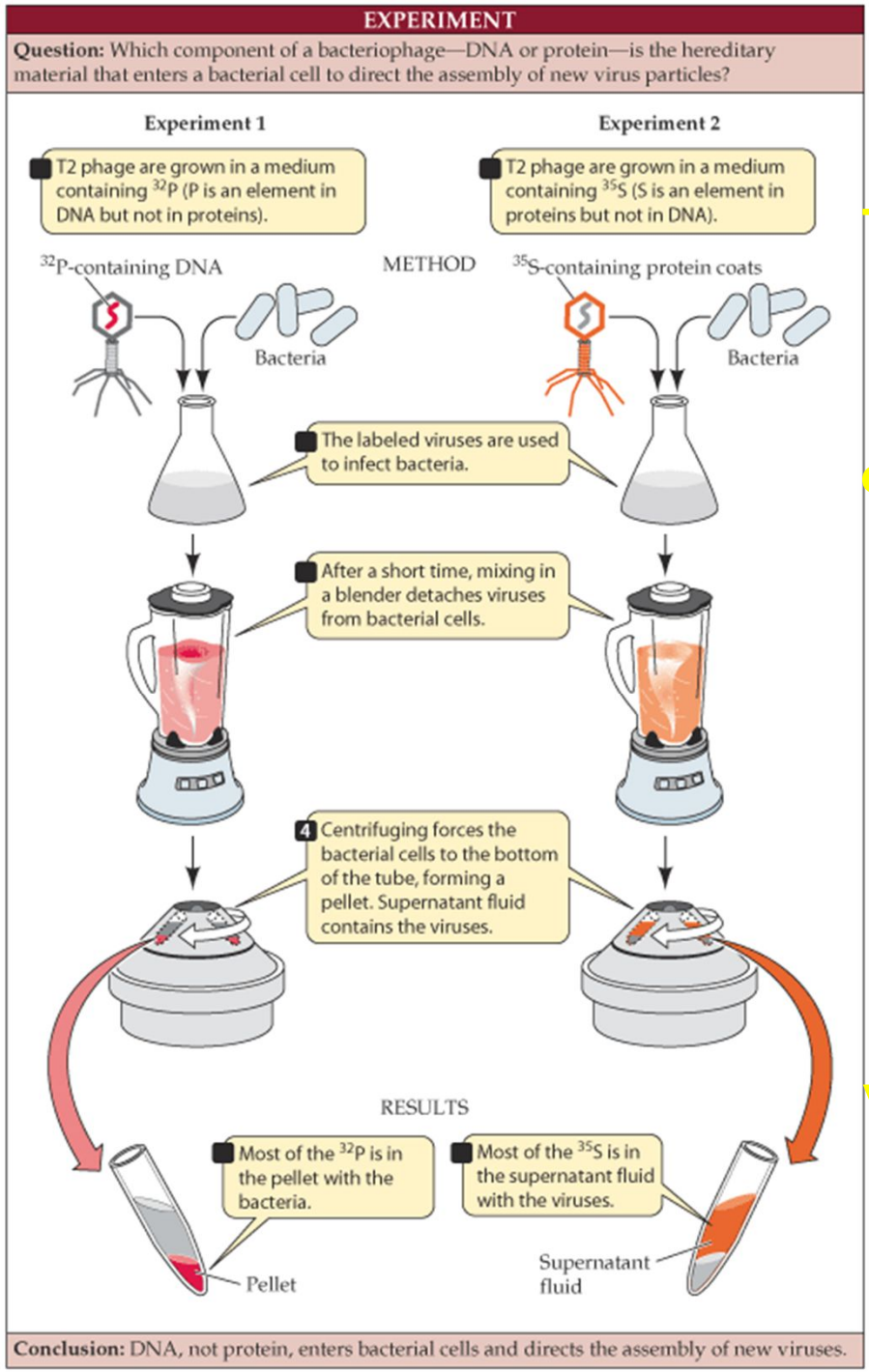
T2 Reproduction Cycle



Hershey & Chase 1944



END



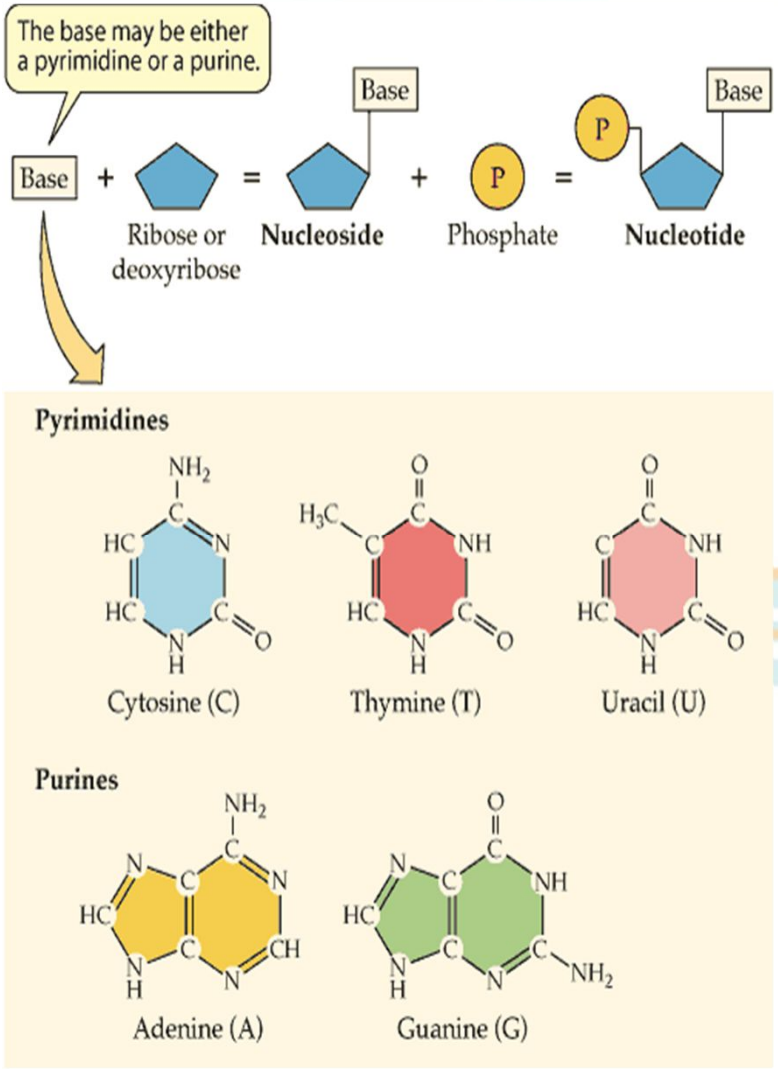
Alfred Hershey & Martha Chase Experiment 1944

This experiment motivated scientists to DNA structure long-range experiments, allowing a progeny (offspring) generation of viruses to be collected. The resulting viruses contained almost none of the original 35S and none of the parental viral protein. However, they contained about one-third of the original 32P—and thus, presumably, one-third of the original DNA. Because DNA was carried over in the virus from generation to generation but protein was not, a logical conclusion was that the hereditary information of the virus is contained in the DNA.

Cell Biology

Investigating:
Heredity Molecule

Investigating: Heredity Molecule



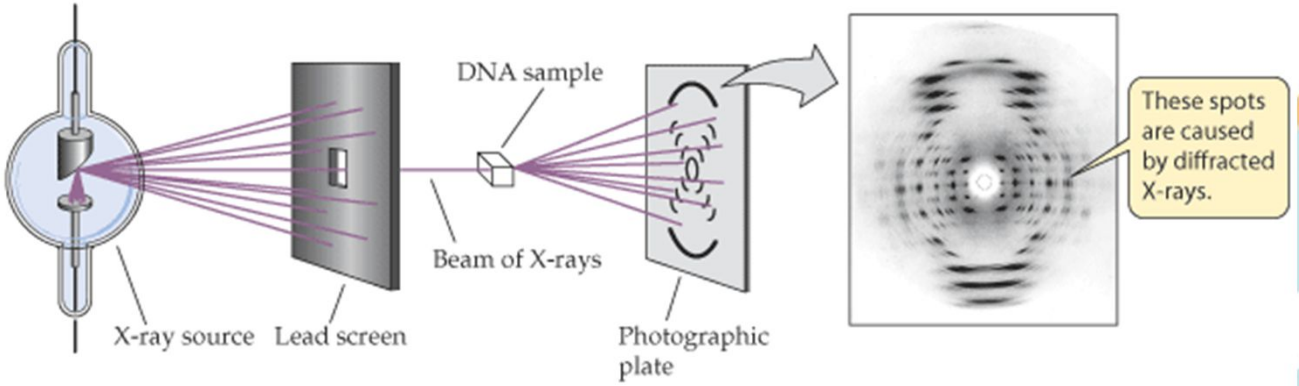
Nucleic Acids

Nucleic acids are polymers made up of nucleotides.

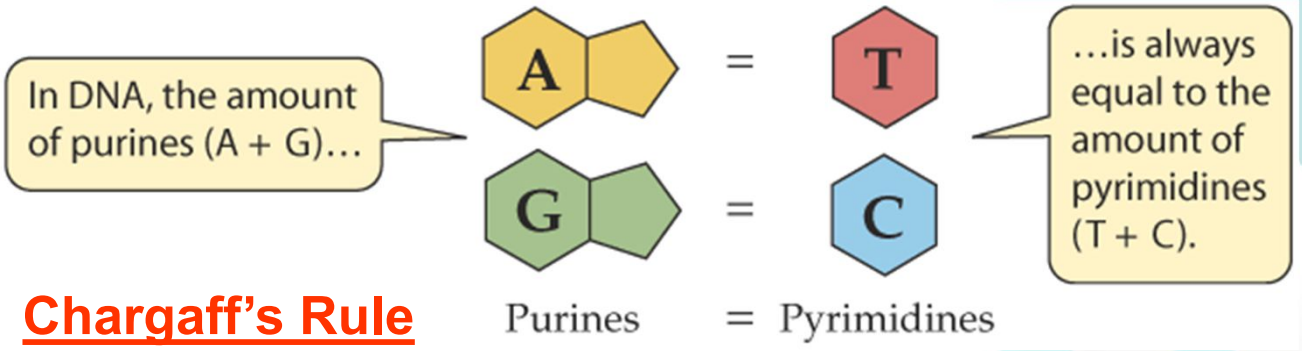
A nucleotide consists of a phosphate group, a sugar (ribose in RNA and deoxyribose in DNA), and a nitrogen-containing base.

Investigating: Heredity Molecule

DNA Structure

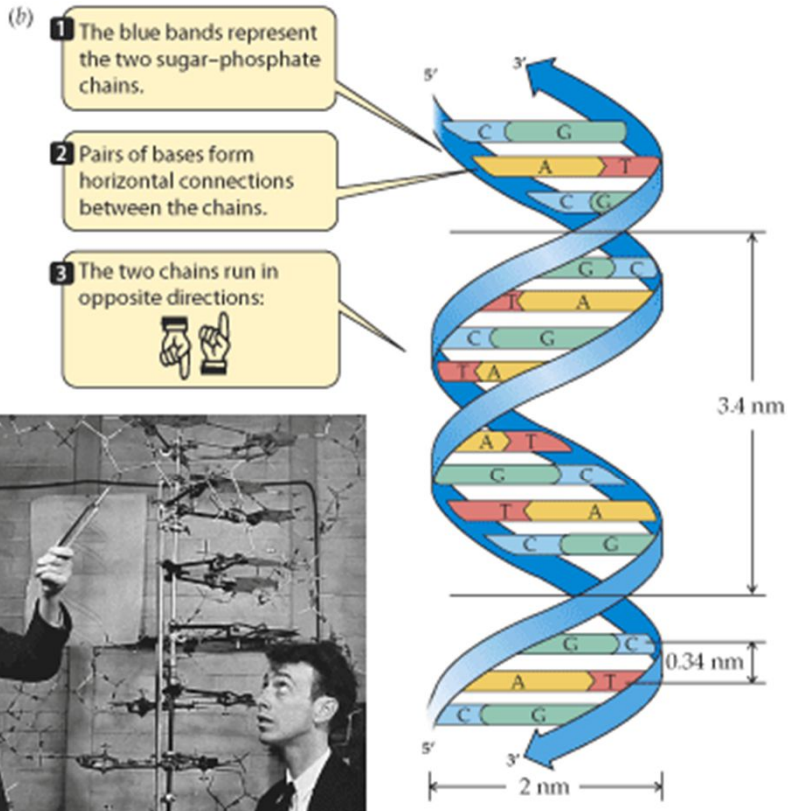


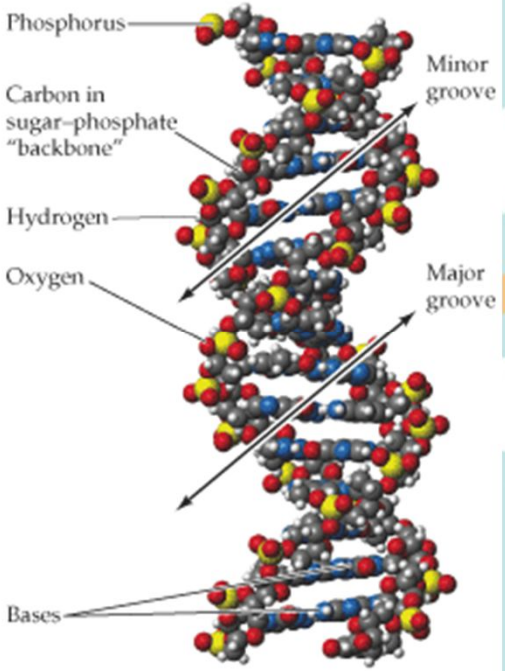
DNA Chemistry



Chargaff's Rule

Investigating: Heredity Molecule





Introduction to Molecular Cell Biology

Hereditary Molecules

Dr. Fridoon Jawad Ahmad

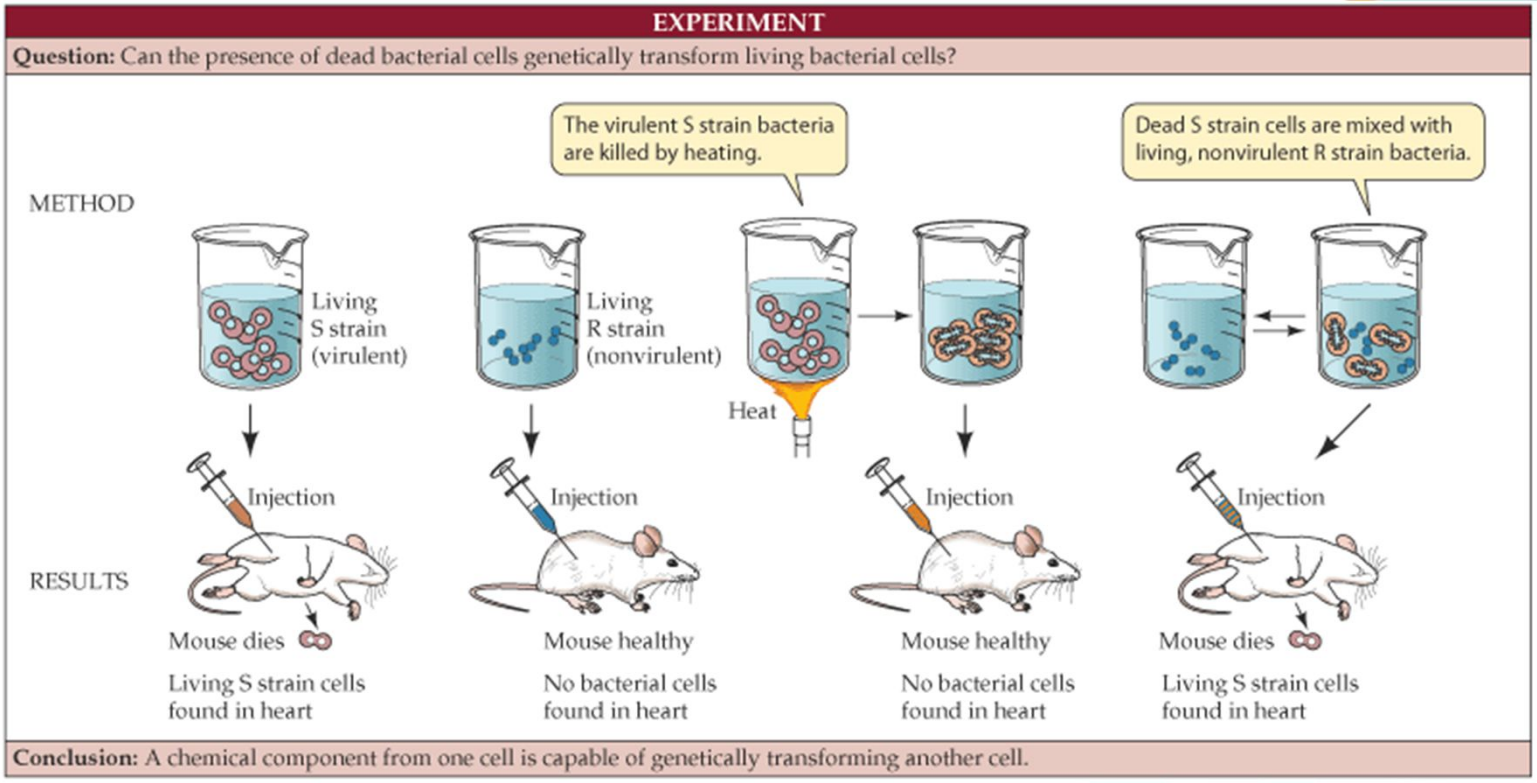
HEC Foreign Professor
King Edward Medical University
Visiting Professor LUMS-SSE

Search for the Heredity Molecule

Search for the Heredity Molecule

Fredrick Griffith Experiment 1920s

Transforming Principle



MacLeaod and MacCarty 1944

The nature of "Transforming Principle"

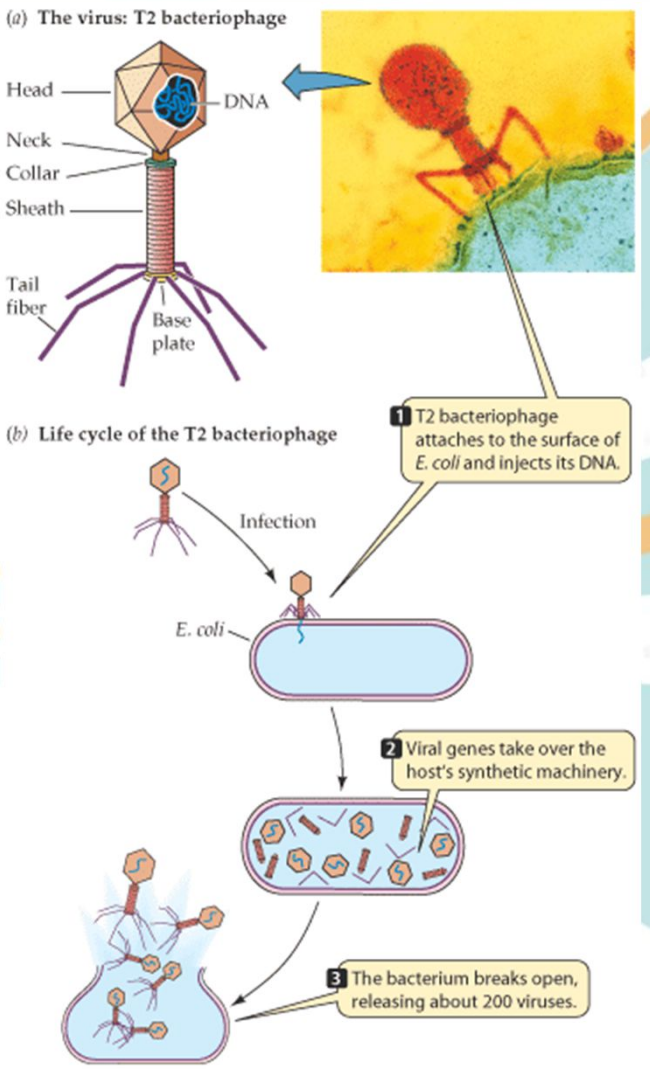
If DNA was destroyed in cell extracts there was no transformation.

Cells are the basic unit of life.

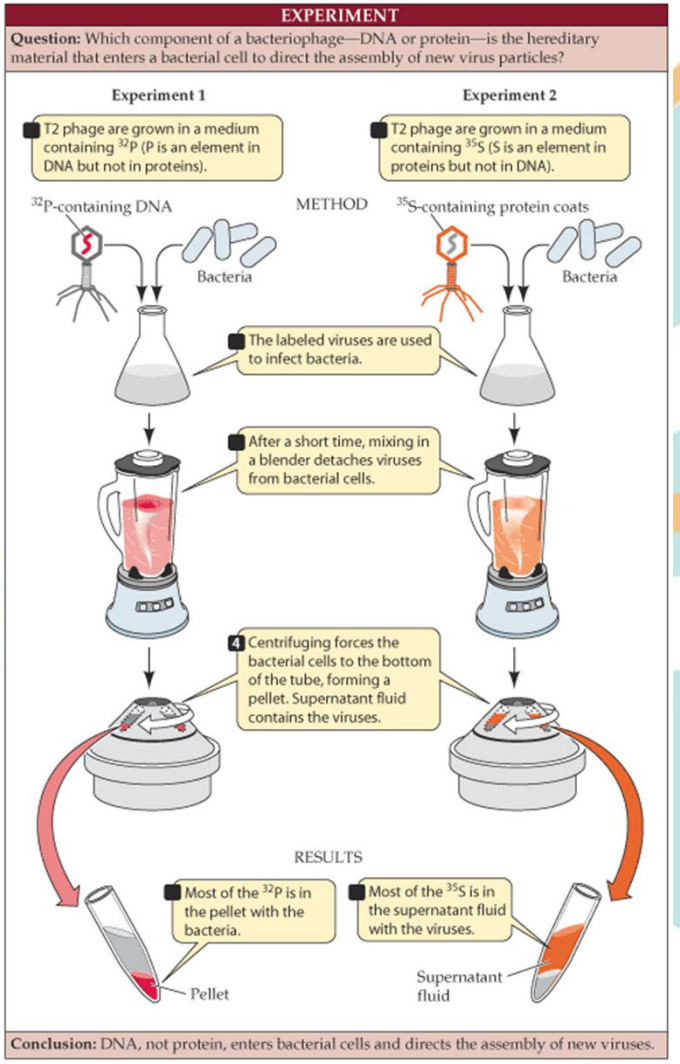
Little impact scientists did not believe DNA could be the genetic material.

Search for the Heredity Molecule 2

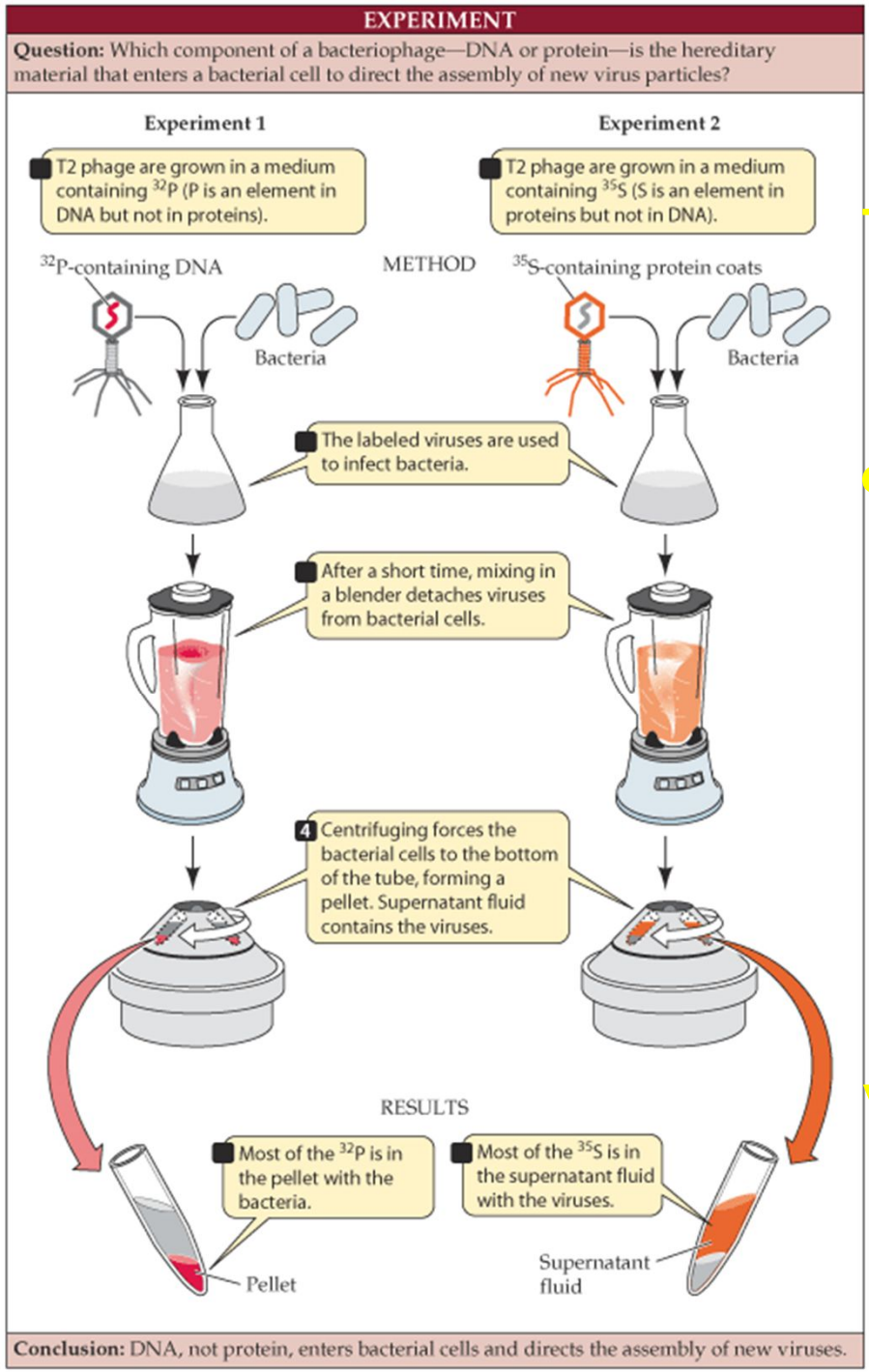
T2 Reproduction Cycle



Hershey & Chase 1944



END



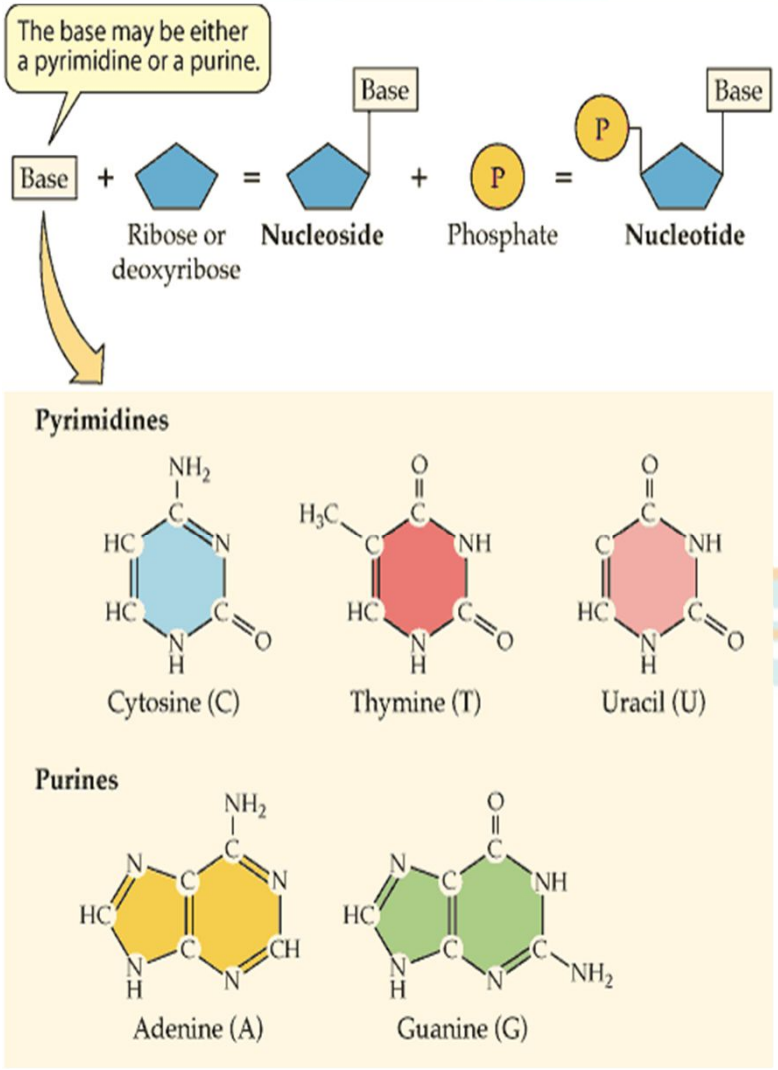
Alfred Hershey & Martha Chase Experiment 1944

This experiment motivated scientists to DNA structure long-range experiments, allowing a progeny (offspring) generation of viruses to be collected. The resulting viruses contained almost none of the original 35S and none of the parental viral protein. However, they contained about one-third of the original 32P—and thus, presumably, one-third of the original DNA. Because DNA was carried over in the virus from generation to generation but protein was not, a logical conclusion was that the hereditary information of the virus is contained in the DNA.

Cell Biology

Investigating:
Heredity Molecule

Investigating: Heredity Molecule



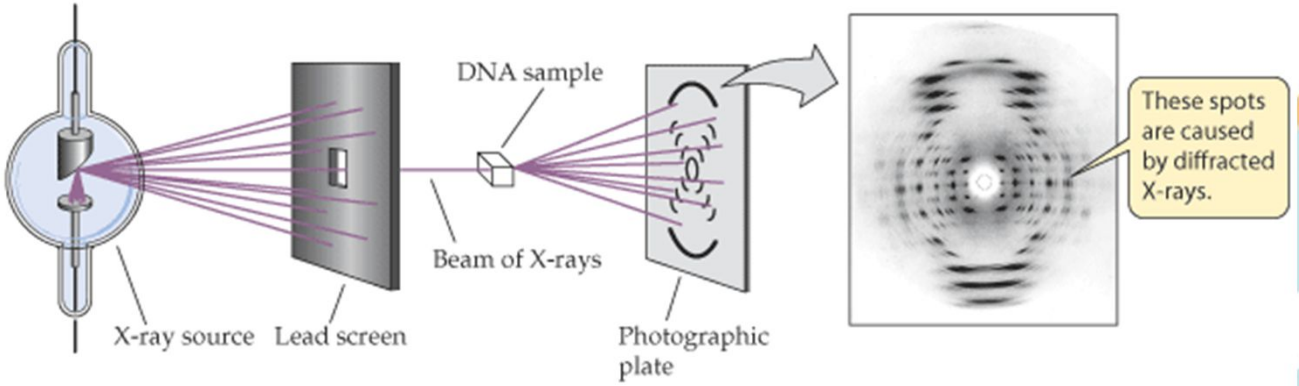
Nucleic Acids

Nucleic acids are polymers made up of nucleotides.

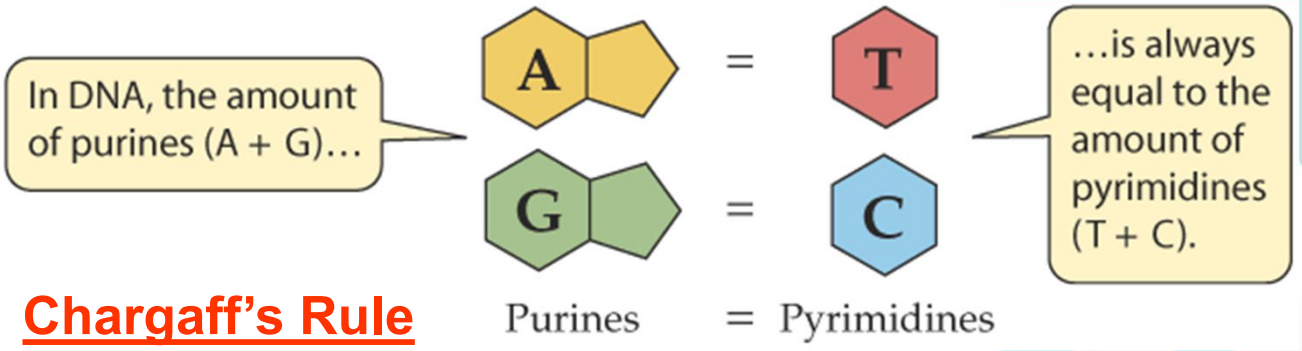
A nucleotide consists of a phosphate group, a sugar (ribose in RNA and deoxyribose in DNA), and a nitrogen-containing base.

Investigating: Heredity Molecule

DNA Structure

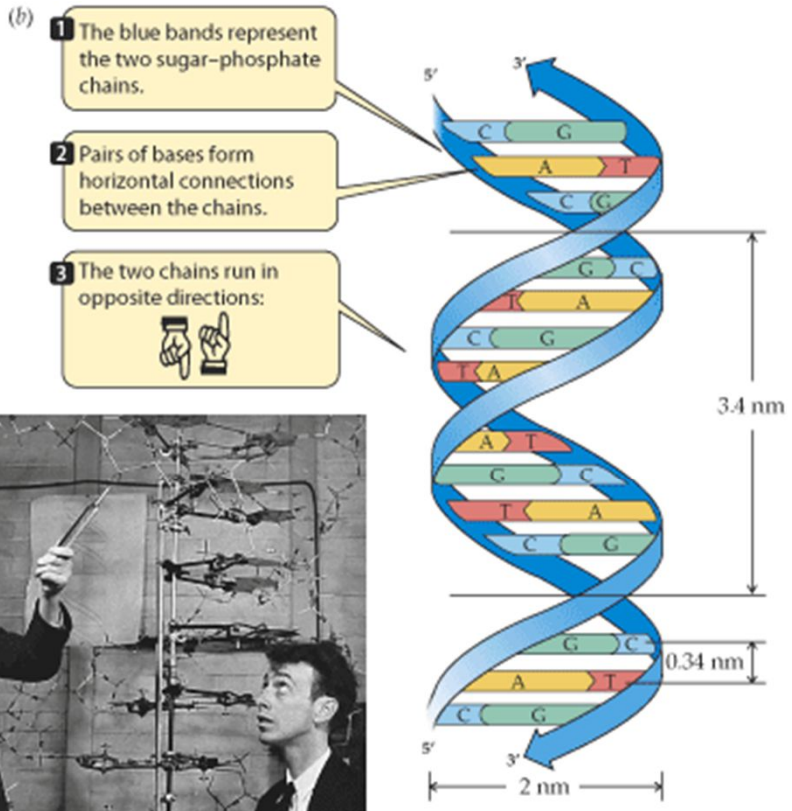


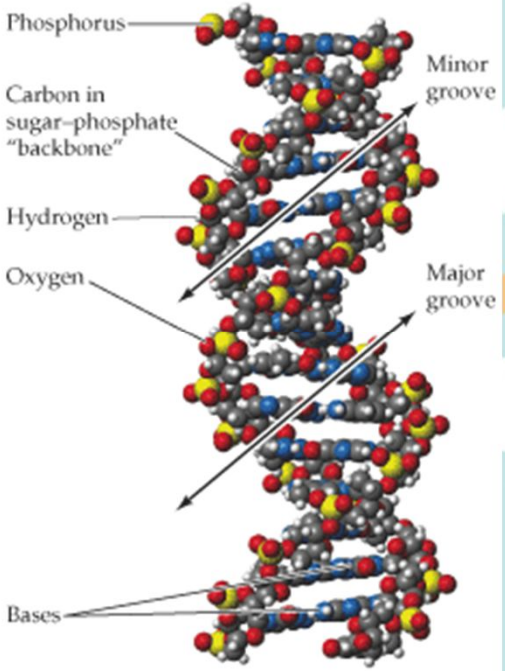
DNA Chemistry



Chargaff's Rule

Investigating: Heredity Molecule



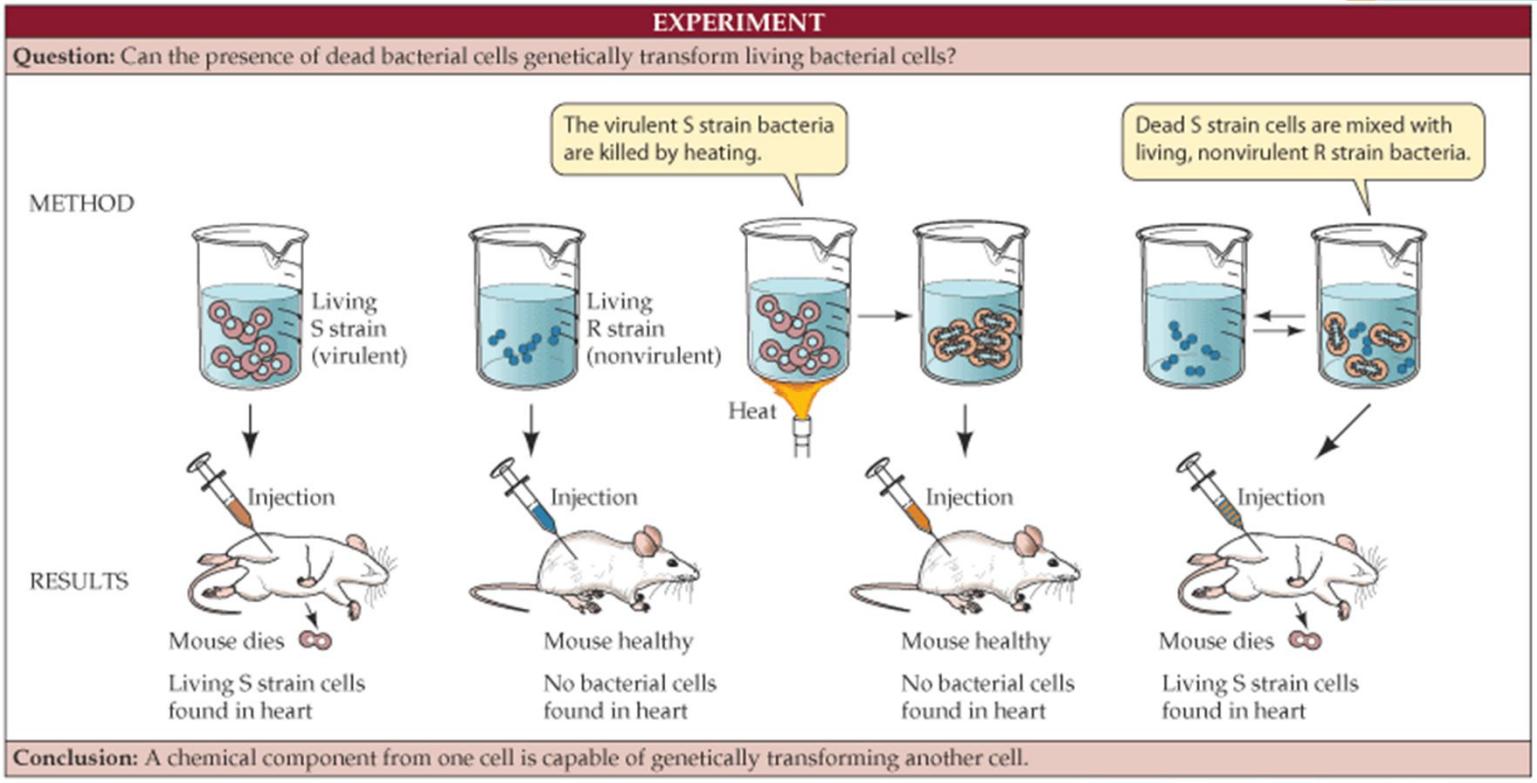


Search for the Heredity Molecule

Search for the Heredity Molecule

Fredrick Griffith Experiment 1920s

Transforming Principle



MacLeaod and MacCarty 1944

The nature of "Transforming Principle"

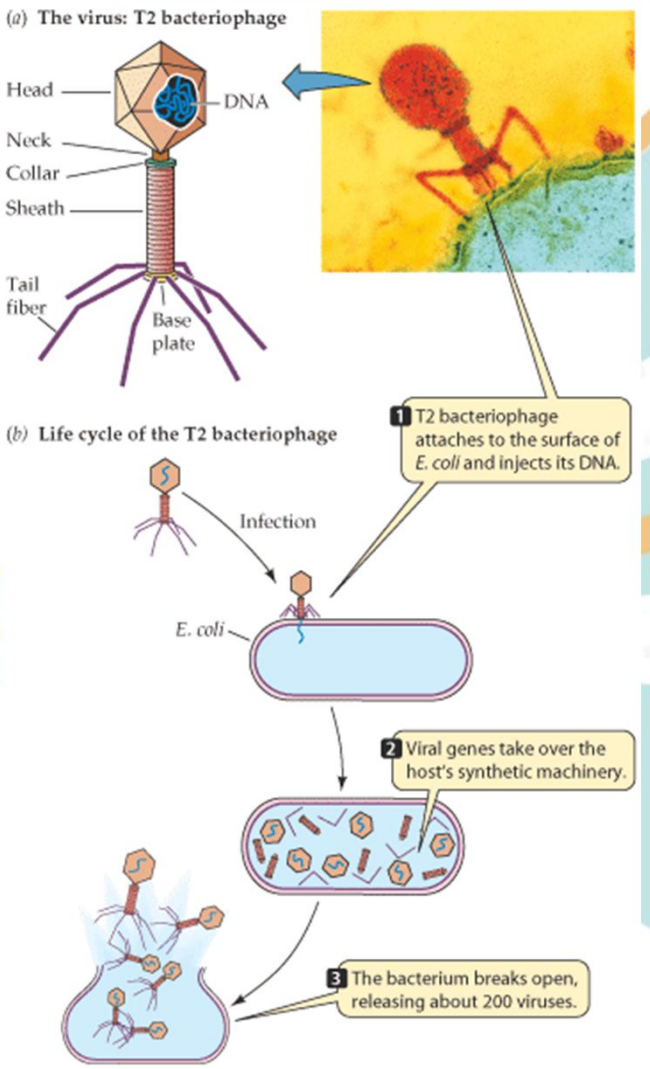
If DNA was destroyed in cell extracts there was no transformation.

Cells are the basic unit of life.

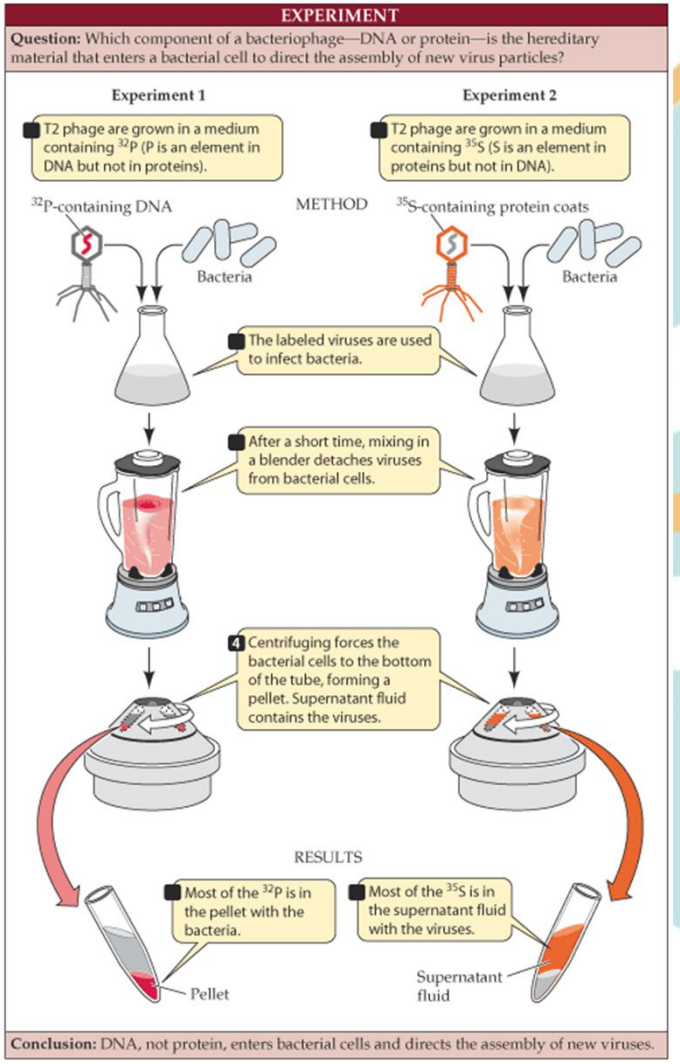
Little impact scientists did not believe DNA could be the genetic material.

Search for the Heredity Molecule 2

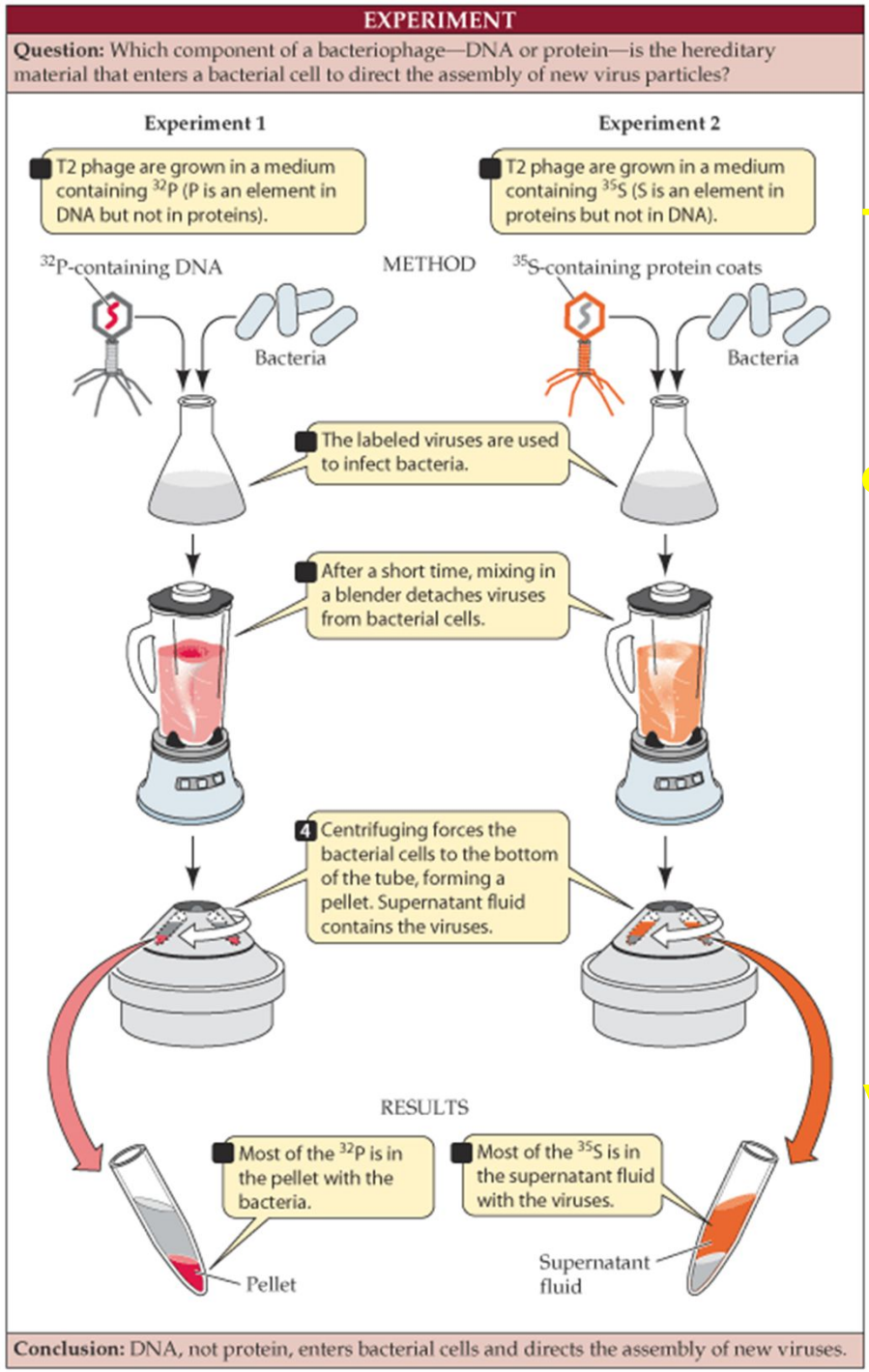
T2 Reproduction Cycle



Hershey & Chase 1944



END



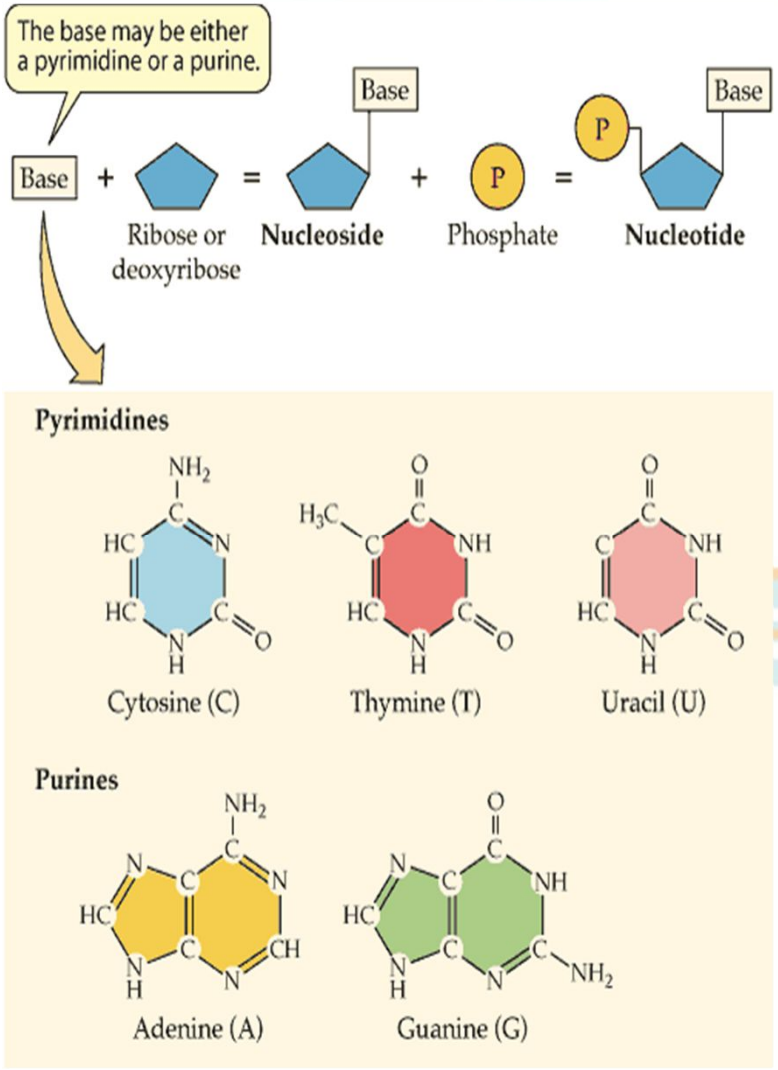
Alfred Hershey & Martha Chase Experiment 1944

This experiment motivated scientists to DNA structure long-range experiments, allowing a progeny (offspring) generation of viruses to be collected. The resulting viruses contained almost none of the original 35S and none of the parental viral protein. However, they contained about one-third of the original 32P—and thus, presumably, one-third of the original DNA. Because DNA was carried over in the virus from generation to generation but protein was not, a logical conclusion was that the hereditary information of the virus is contained in the DNA.

Cell Biology

Investigating:
Heredity Molecule

Investigating: Heredity Molecule



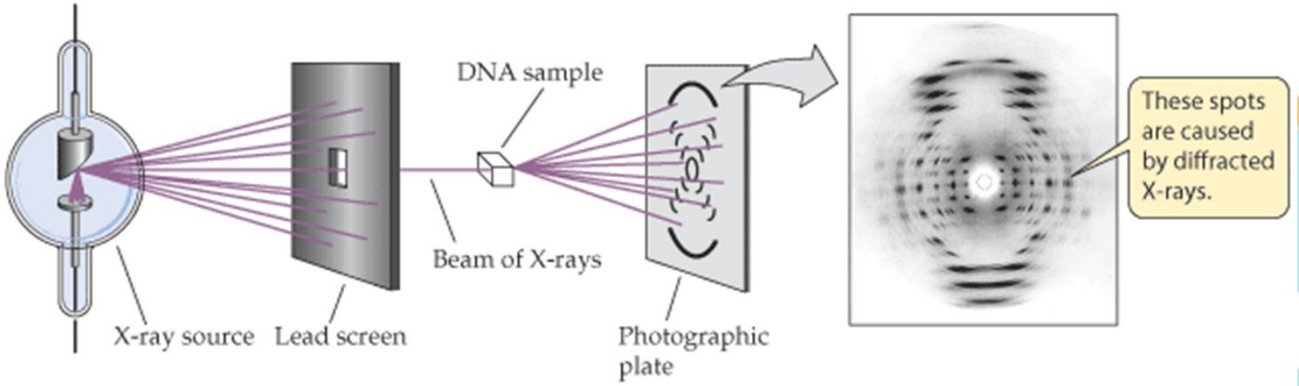
Nucleic Acids

Nucleic acids are polymers made up of nucleotides.

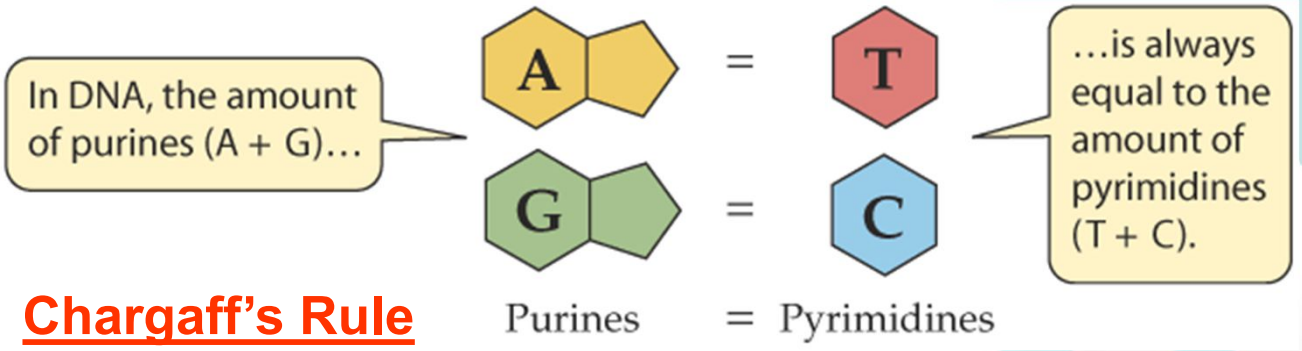
A nucleotide consists of a phosphate group, a sugar (ribose in RNA and deoxyribose in DNA), and a nitrogen-containing base.

Investigating: Heredity Molecule

DNA Structure

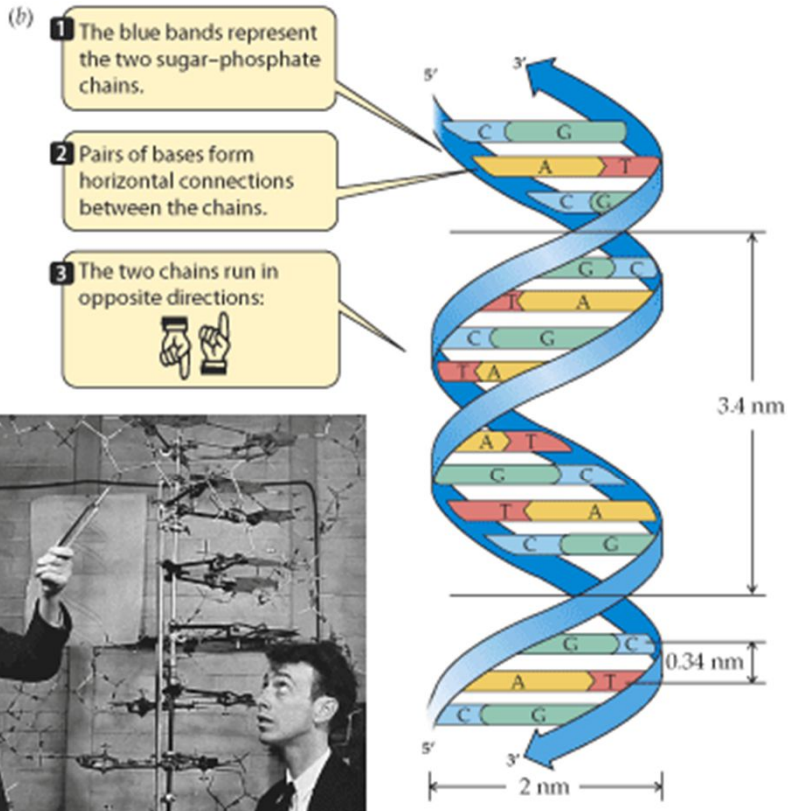


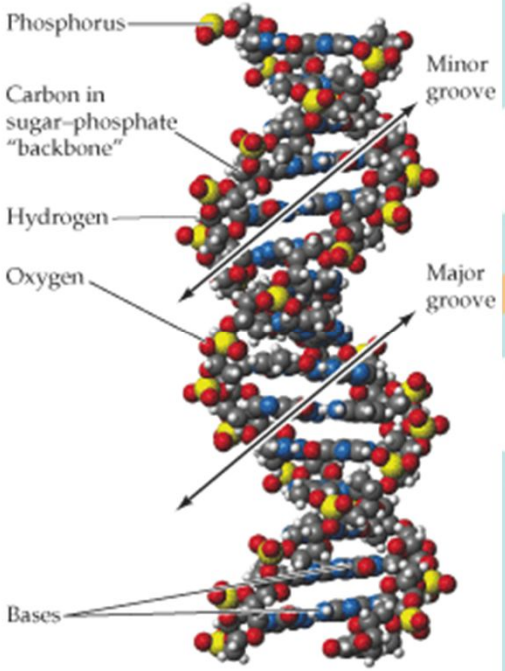
DNA Chemistry



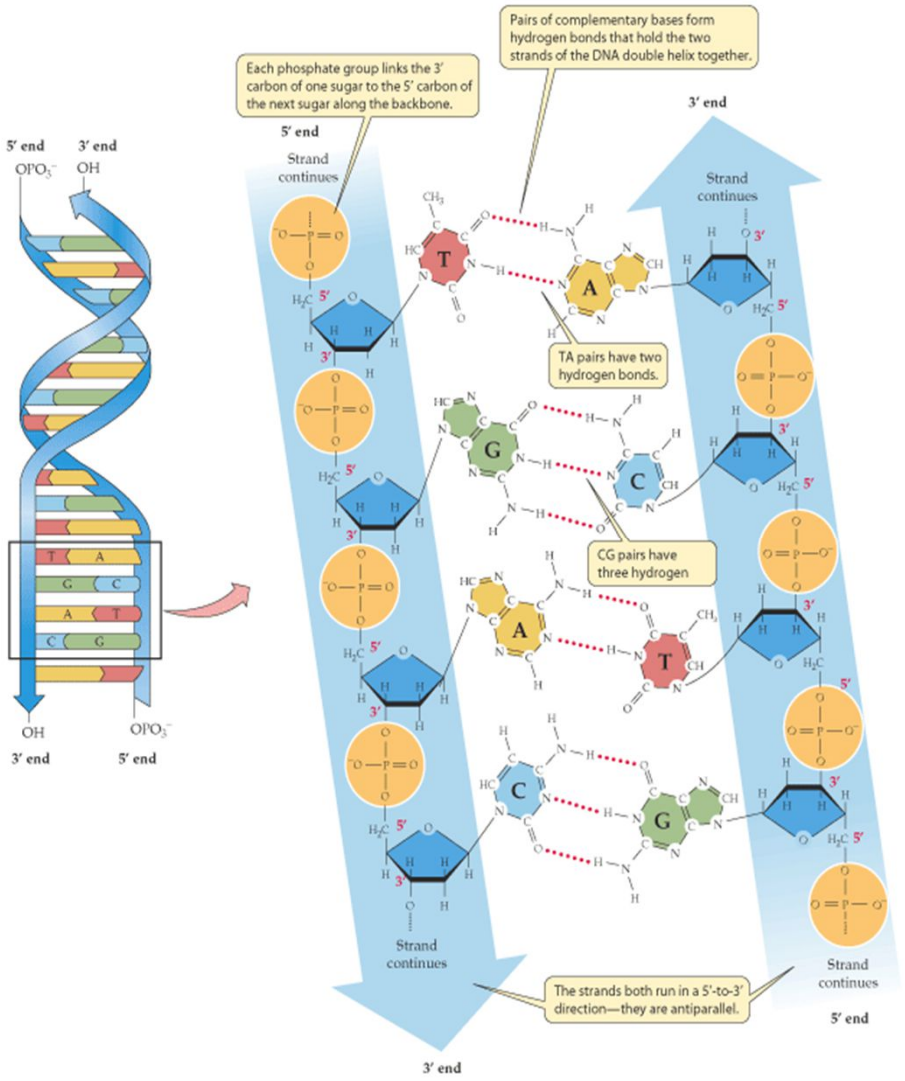
Chargaff's Rule

Investigating: Heredity Molecule





DNA Function From Structure



Double stranded helix

Right handed (threads on a screw)

Antiparellel

Complementary base pairing

A=T

G=C

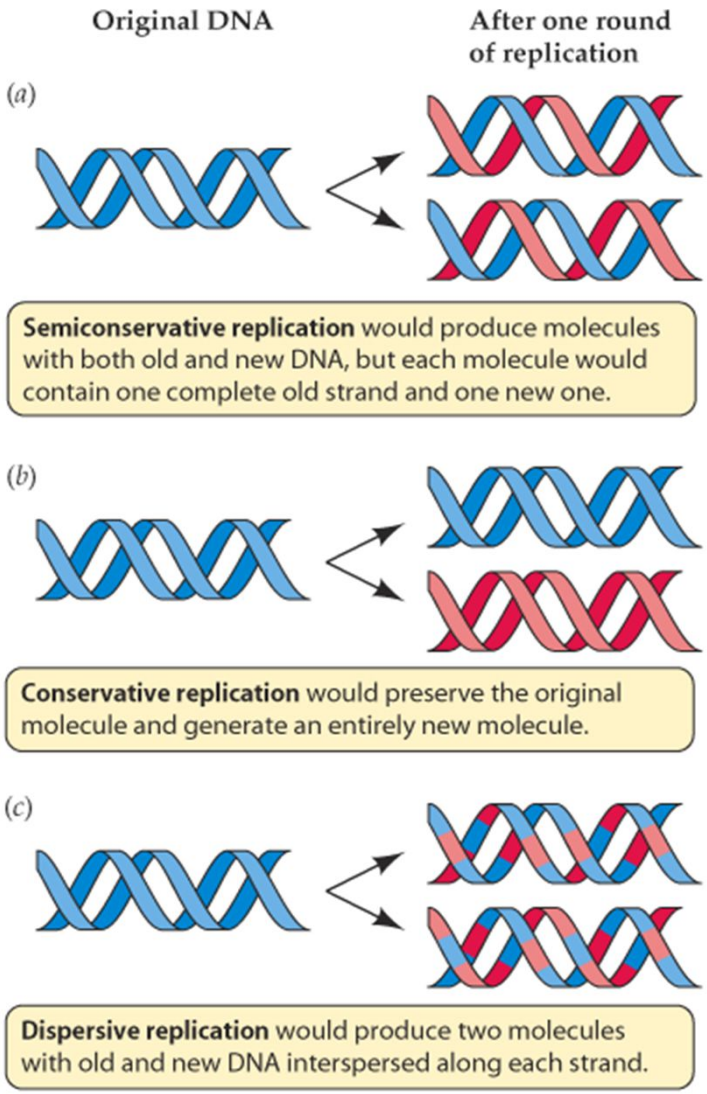
Store an enormous amount of information and could account for species and individual differences

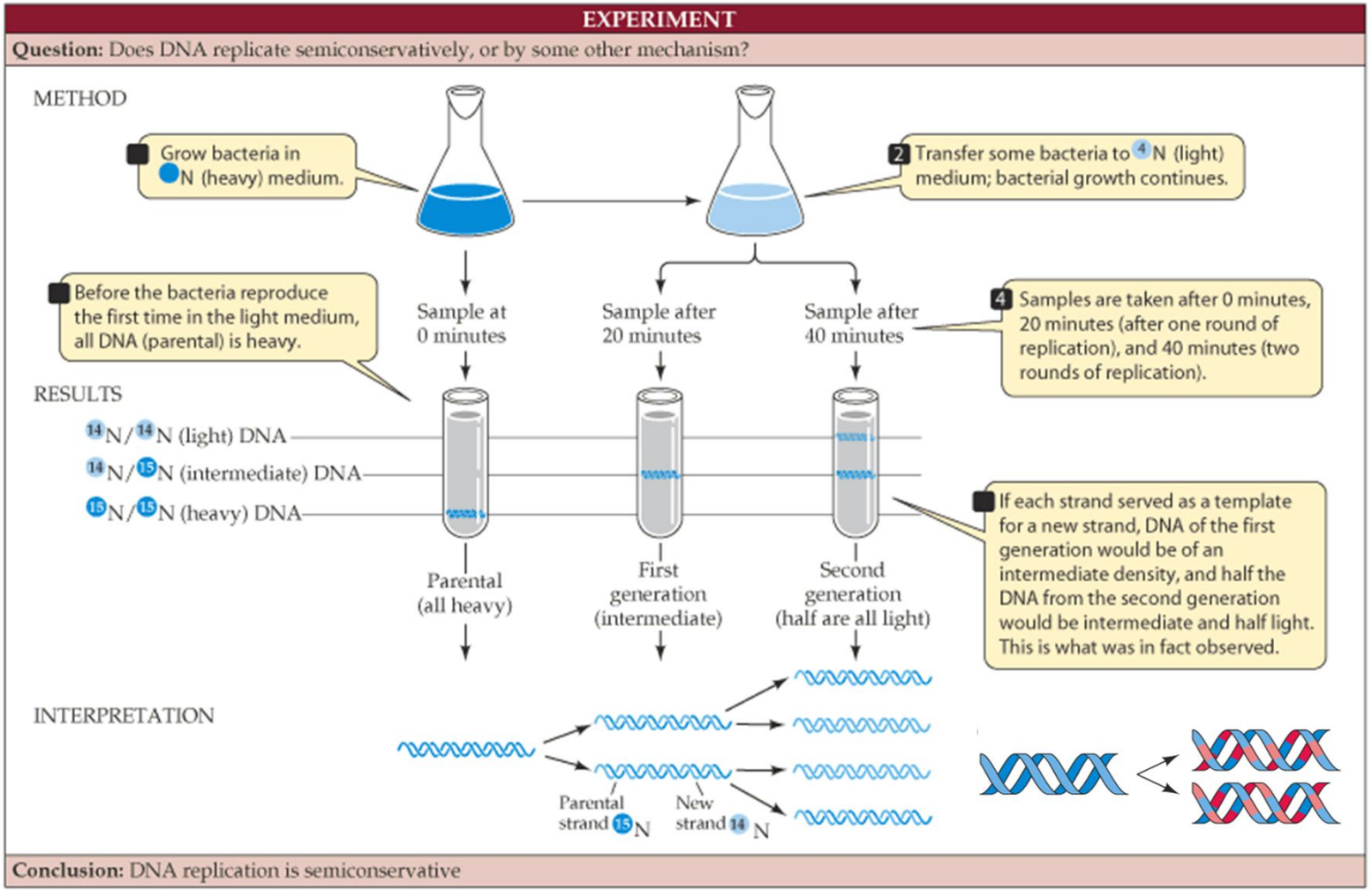
Susceptibility to change (mutate) Change bp

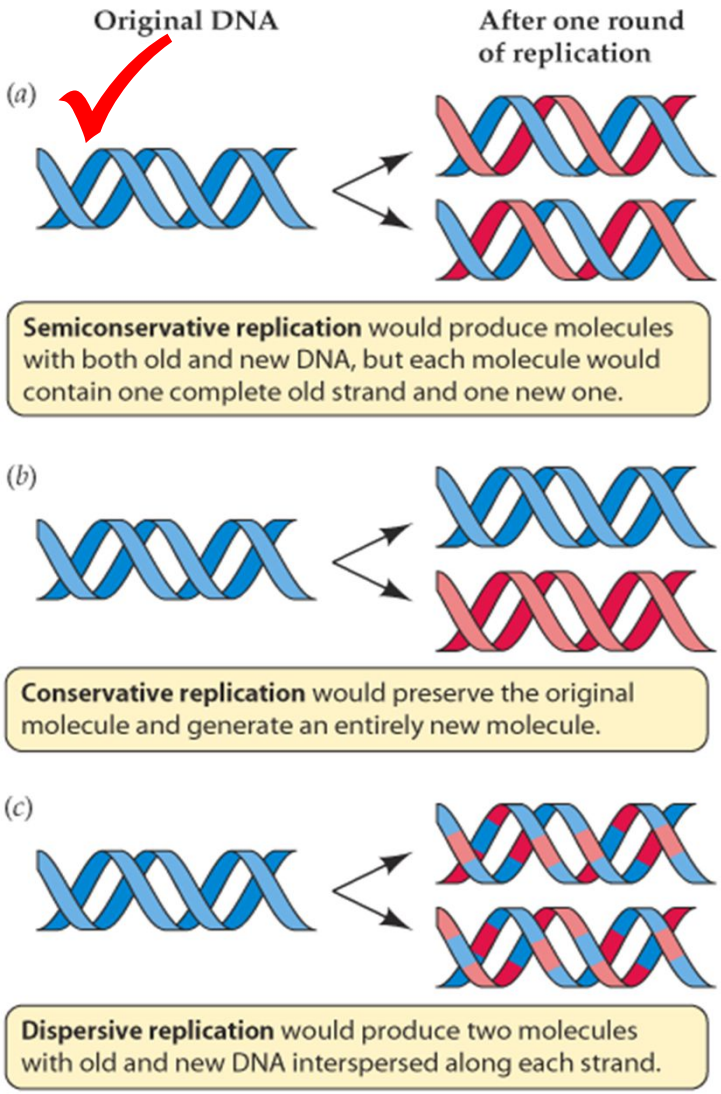
Precise Replication bp is specific

Expression of a phenotype

Replication Models

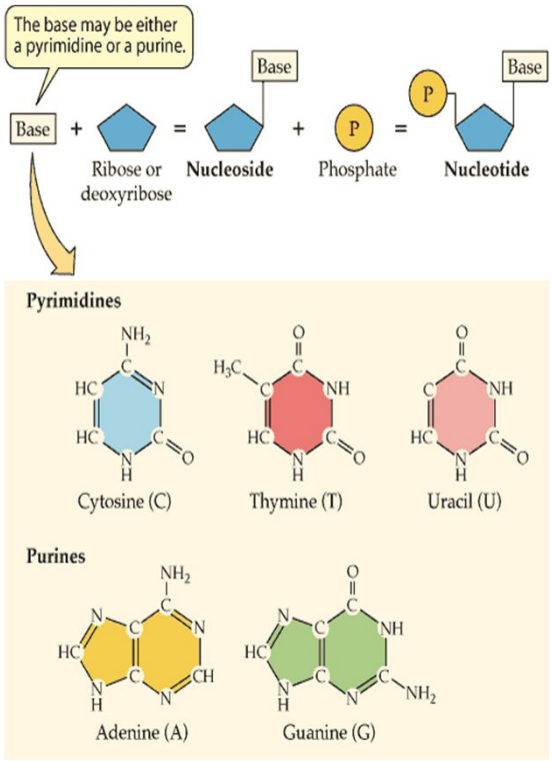

Original DNA
After one round of replication
(a)
Semiconservative replication would produce molecules with both old and new DNA, but each molecule would contain one complete old strand and one new one.
(b)
Conservative replication would preserve the original molecule and generate an entirely new molecule.
(c)
Dispersive replication would produce two molecules with old and new DNA interspersed along each strand.

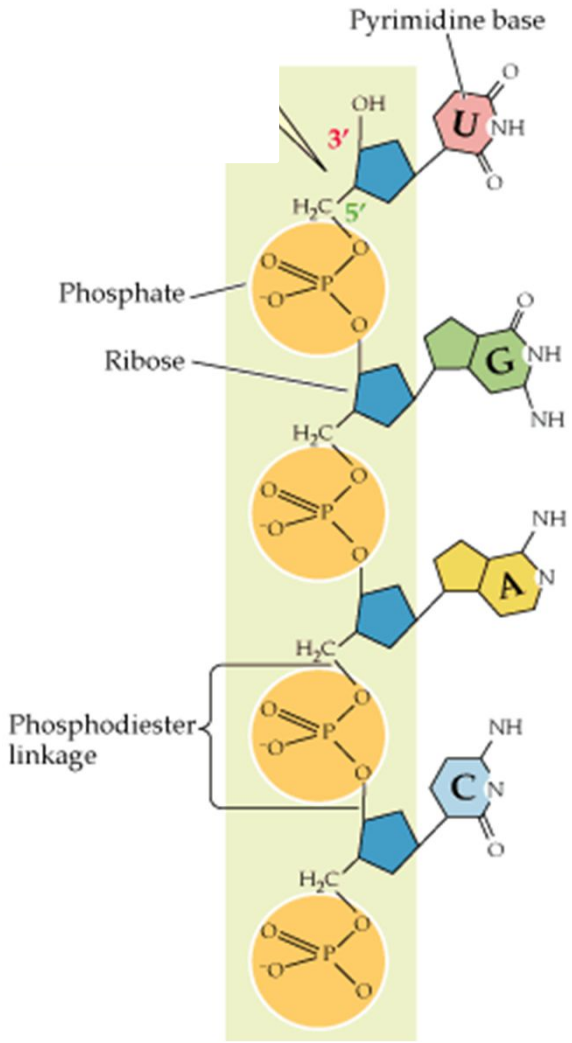

EXPERIMENT
Question: Does DNA replicate semiconservatively, or by some other mechanism?
METHOD
1 Grow bacteria in 15N (heavy) medium.
2 Transfer some bacteria to 14N (light) medium; bacterial growth continues.
3 Before the bacteria reproduce the first time in the light medium, all DNA (parental) is heavy.
Sample at 0 minutes
Sample after 20 minutes
Sample after 40 minutes
4 Samples are taken after 0 minutes, 20 minutes (after one round of replication), and 40 minutes (two rounds of replication).
RESULTS
14N/14N (light) DNA
14N/15N (intermediate) DNA
15N/15N (heavy) DNA
Parental (all heavy)
First generation (intermediate)
Second generation (half are all light)
5 If each strand served as a template for a new strand, DNA of the first generation would be of an intermediate density, and half the DNA from the second generation would be intermediate and half light. This is what was in fact observed.
INTERPRETATION
Parental strand 15N
New strand 14N
Conclusion: DNA replication is semiconservative

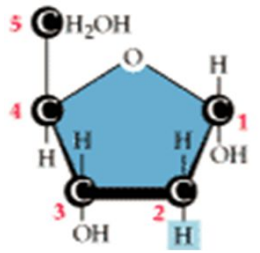

Original DNA
After one round of replication
(a)
Semiconservative replication would produce molecules with both old and new DNA, but each molecule would contain one complete old strand and one new one.
(b)
Conservative replication would preserve the original molecule and generate an entirely new molecule.
(c)
Dispersive replication would produce two molecules with old and new DNA interspersed along each strand.

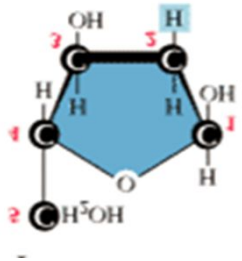
END

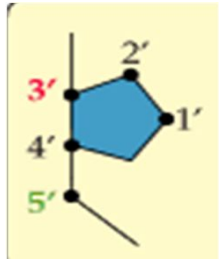
DNA Replication The Molecular View

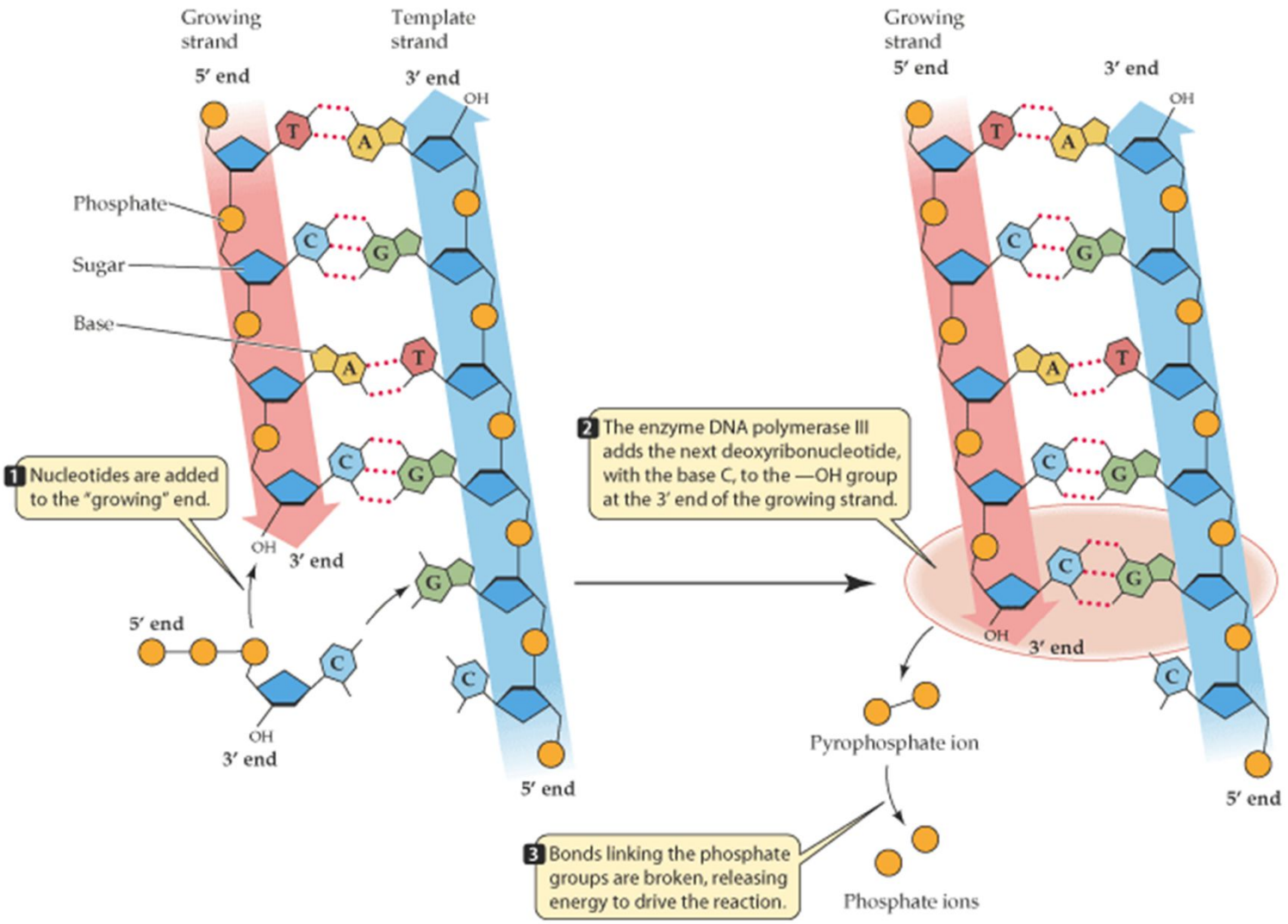

The base may be either a pyrimidine or a purine.
Base
+
Ribose or deoxyribose
=
Base
Nucleoside
+
P
Phosphate
=
P
Base
Nucleotide
Pyrimidines
Cytosine (C)
Thymine (T)
Uracil (U)
Purines
Adenine (A)
Guanine (G)

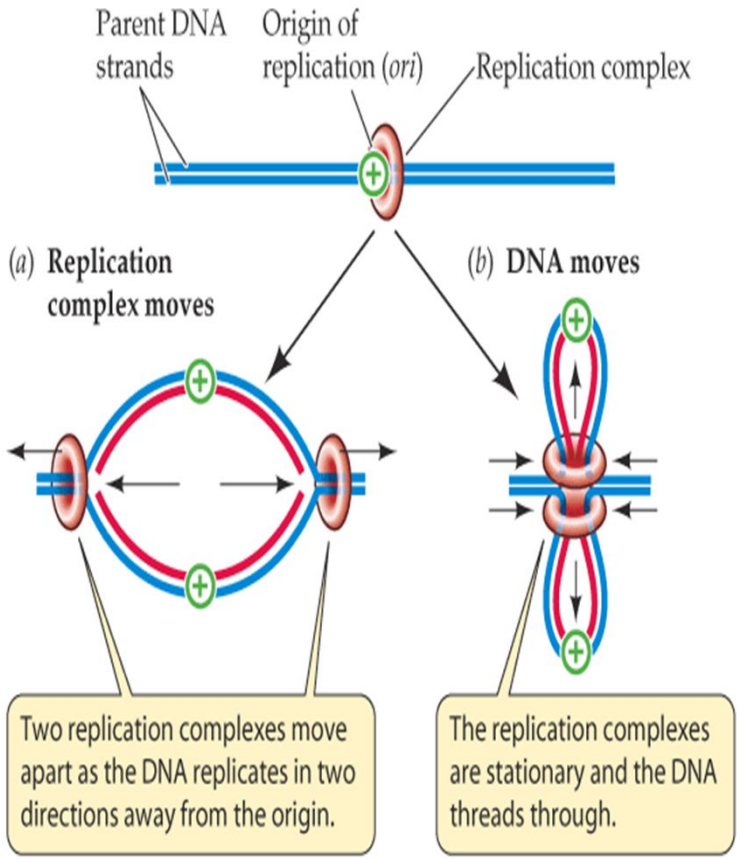

Pyrimidine base
Phosphate
Ribose
Phosphodiester linkage
3′ end
5′ end
U
G
A
C
OH
3′
5′


5
H2OH
O
H
4
1
H
H
H
OH
3
2
OH
H


OH
H
H
H
H
OH
H
O
H
5OH


2′
3′
1′
4′
5′


Growing strand
Template strand
5′ end
3′ end
OH
Phosphate
Sugar
Base
T
A
C
G
1 Nucleotides are added to the "growing" end.
OH
3′ end
5′ end
OH
3′ end
5′ end
2 The enzyme DNA polymerase III adds the next deoxyribonucleotide, with the base C, to the —OH group at the 3′ end of the growing strand.
Growing strand
5′ end
3′ end
OH
OH
3′ end
Pyrophosphate ion
5′ end
3 Bonds linking the phosphate groups are broken, releasing energy to drive the reaction.
Phosphate ions


Parent DNA strands
Origin of replication (ori)
Replication complex
(a) Replication complex moves
(b) DNA moves
Two replication complexes move apart as the DNA replicates in two directions away from the origin.
The replication complexes are stationary and the DNA threads through.

END

DNA Replication The Process

Requirements

1) The Original

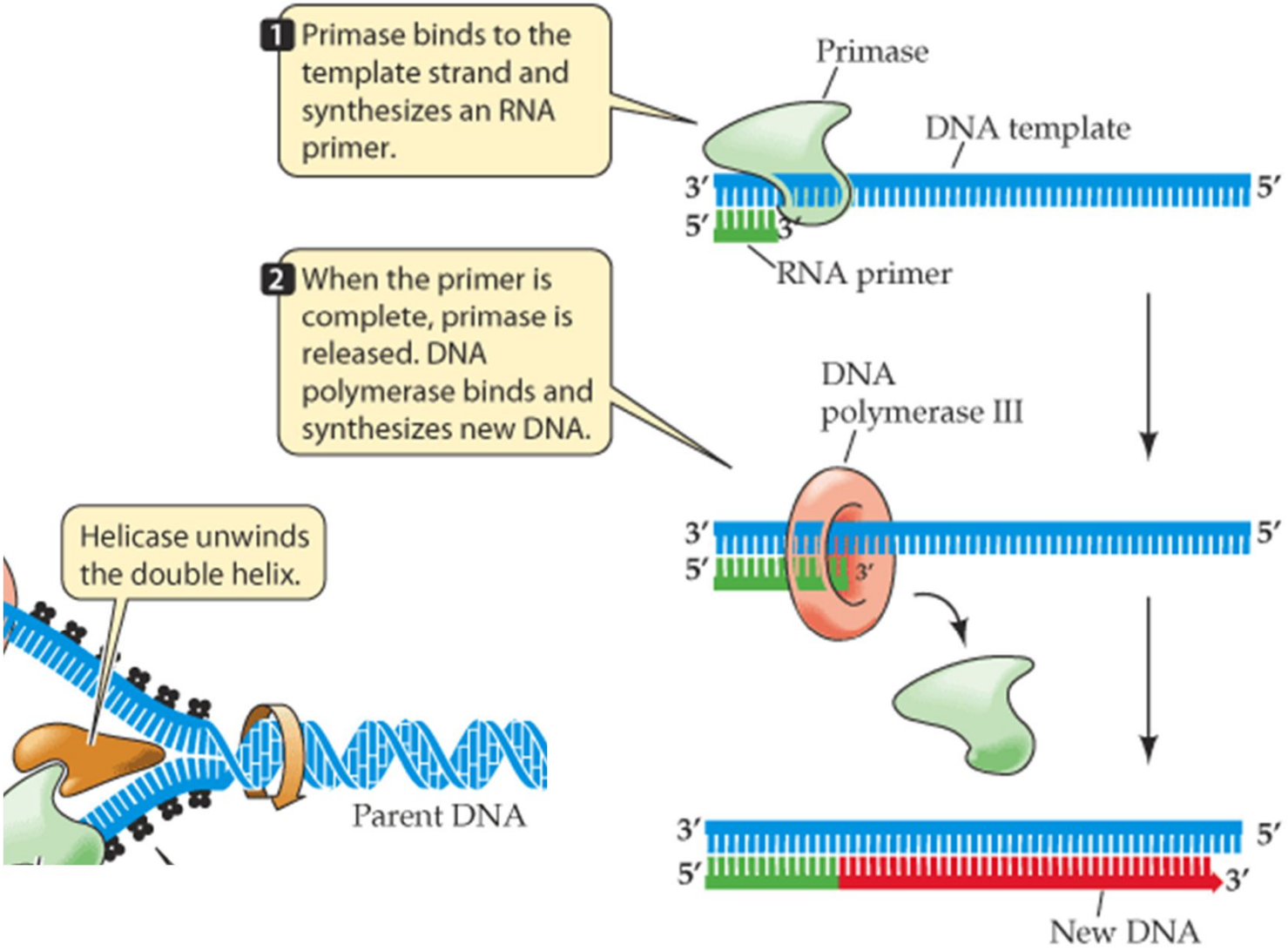
2) The Machinery

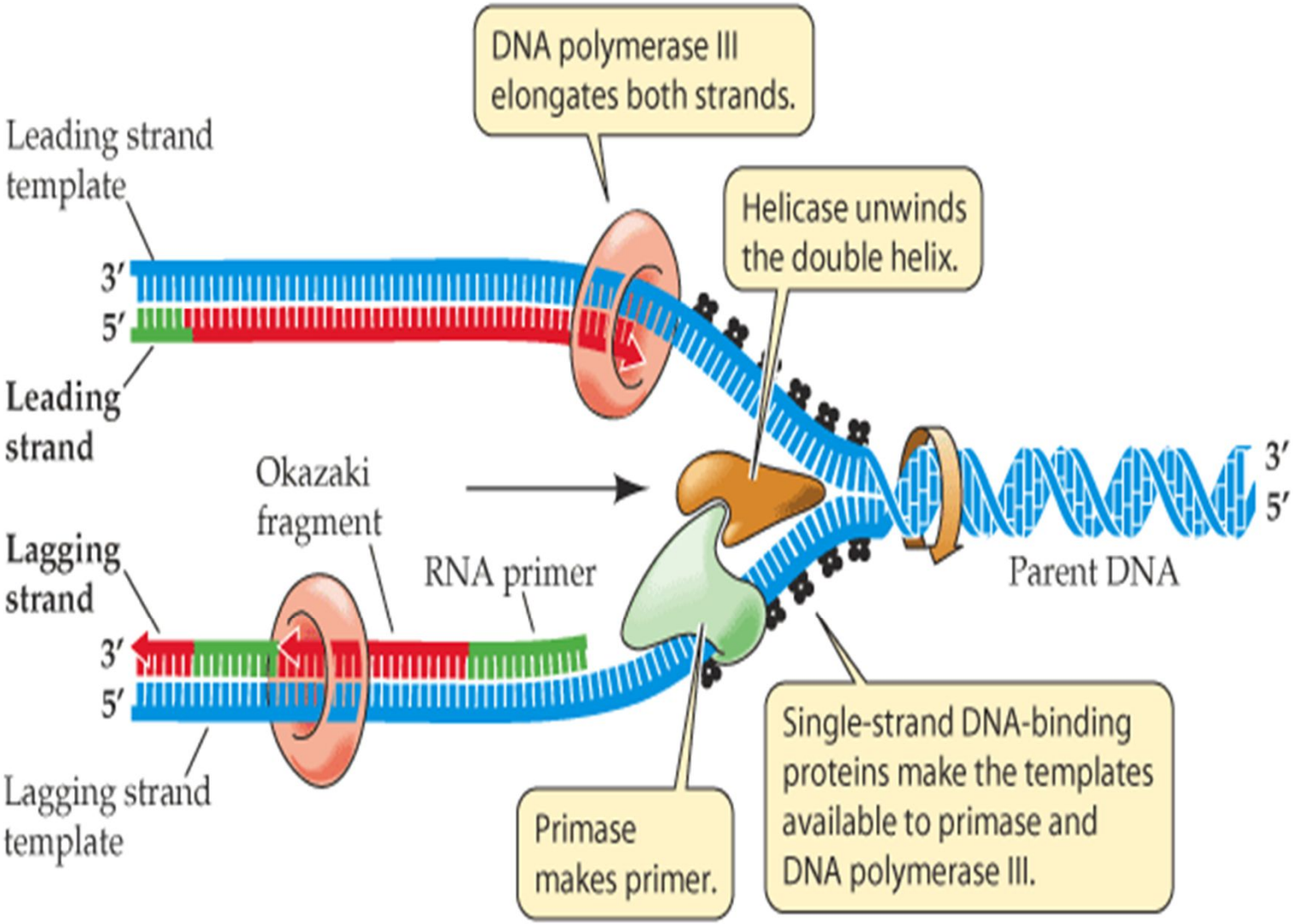
3) The Raw Material

5) Energy

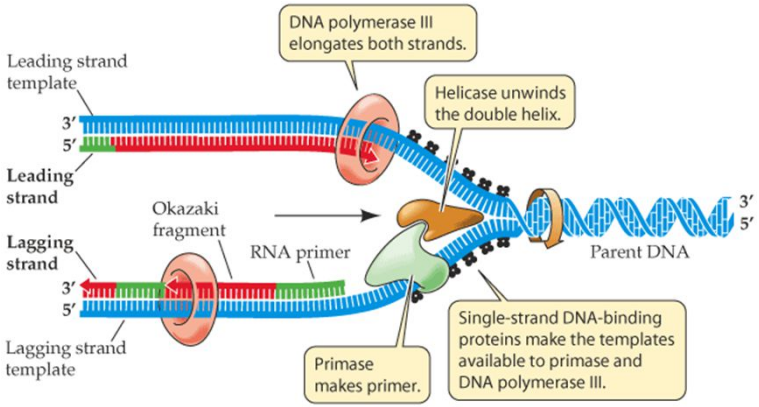
4) Initiator

DNA polymerase
useless without primer


1 Primase binds to the template strand and synthesizes an RNA primer.
Primase
DNA template
3′
5′
5′
3′
RNA primer
2 When the primer is complete, primase is released. DNA polymerase binds and synthesizes new DNA.
DNA polymerase III
3′
5′
5′
3′
Helicase unwinds the double helix.
Parent DNA
3′
5′
5′
3′
New DNA

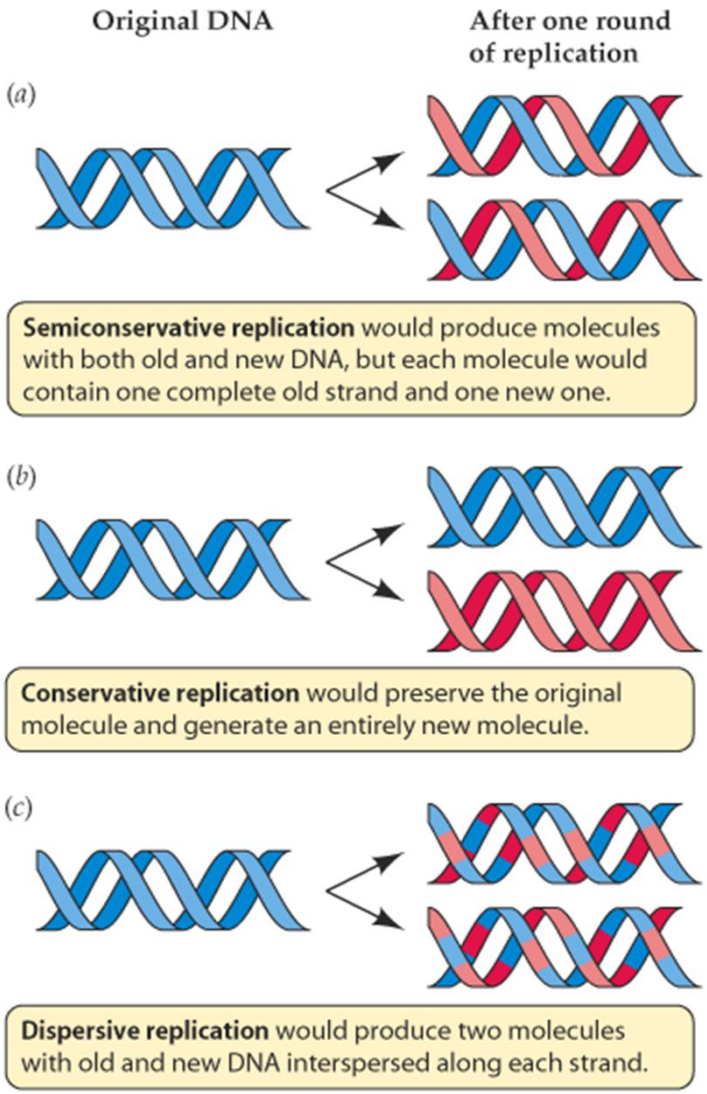

DNA polymerase III elongates both strands.
Helicase unwinds the double helix.
Leading strand template
3′
5′
Leading strand
Okazaki fragment
RNA primer
Lagging strand
3′
5′
Lagging strand template
3′
5′
Parent DNA
Primase makes primer.
Single-strand DNA-binding proteins make the templates available to primase and DNA polymerase III.

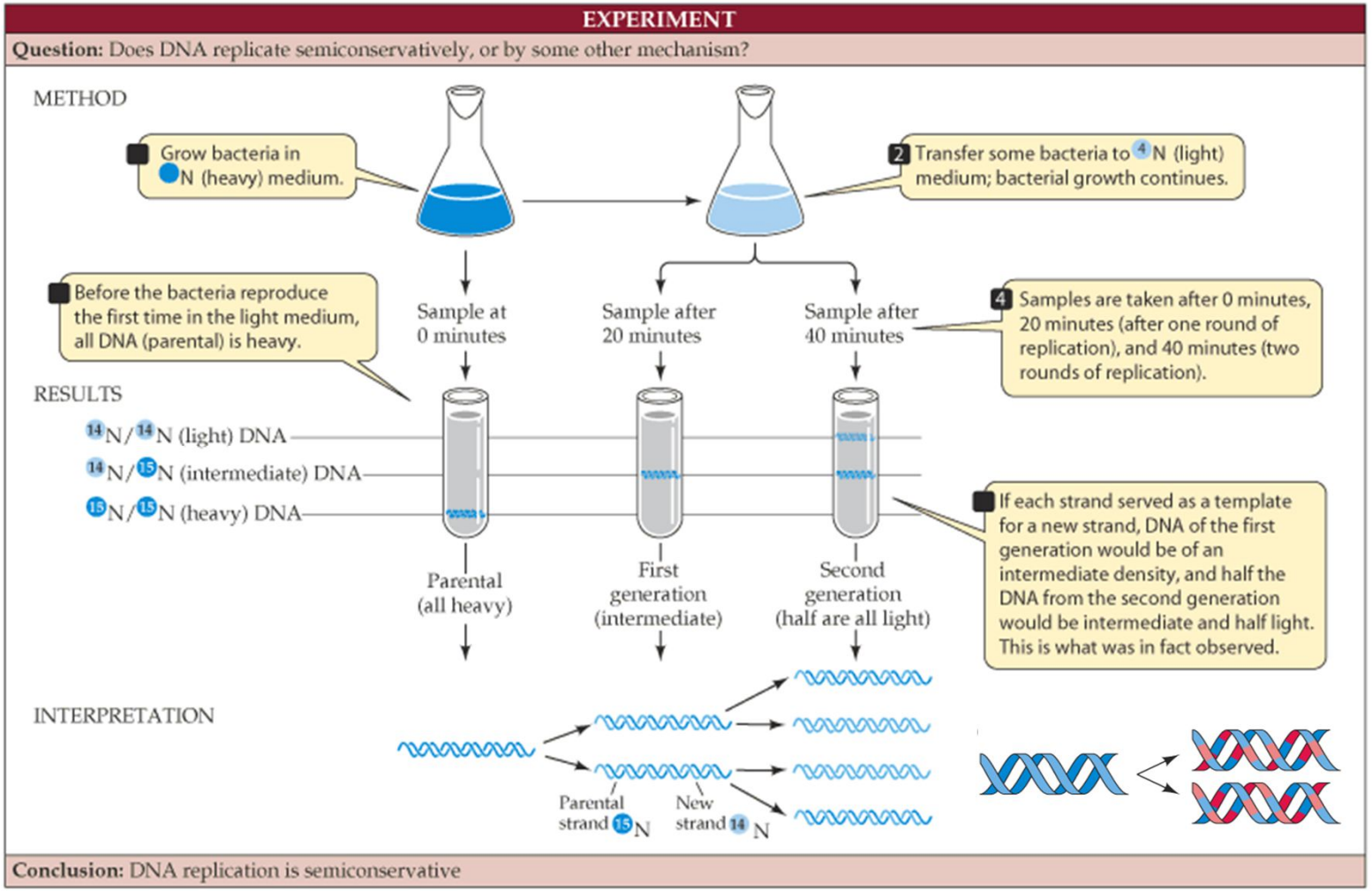
The Three Dilemmas

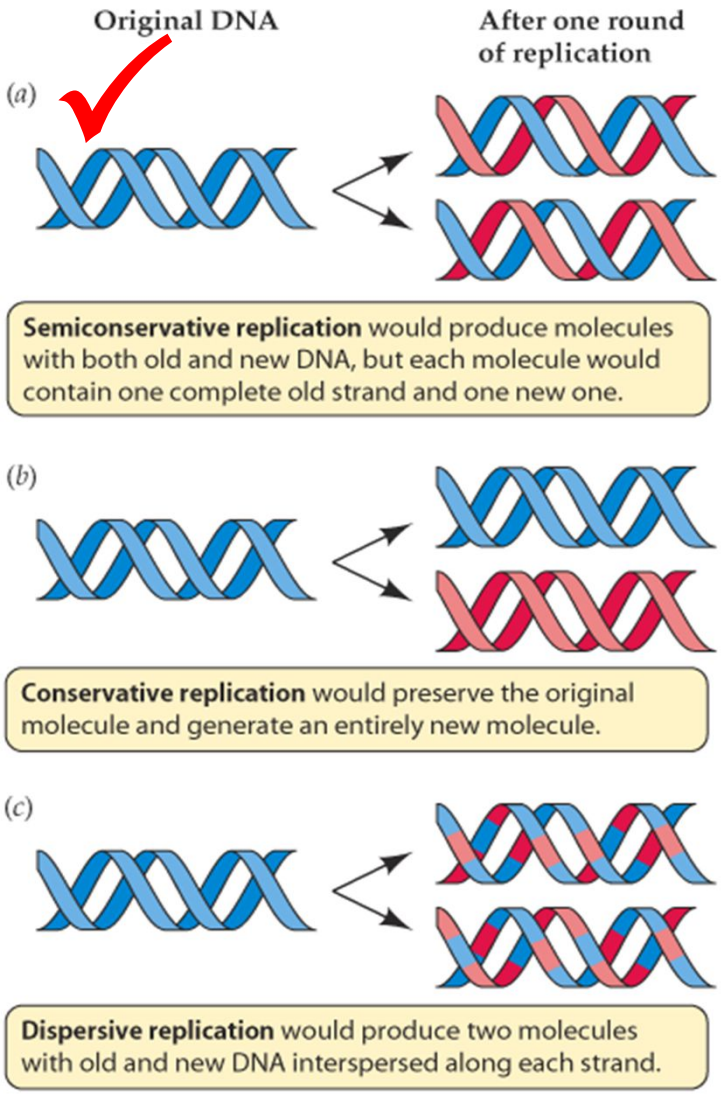


END

Replication Models

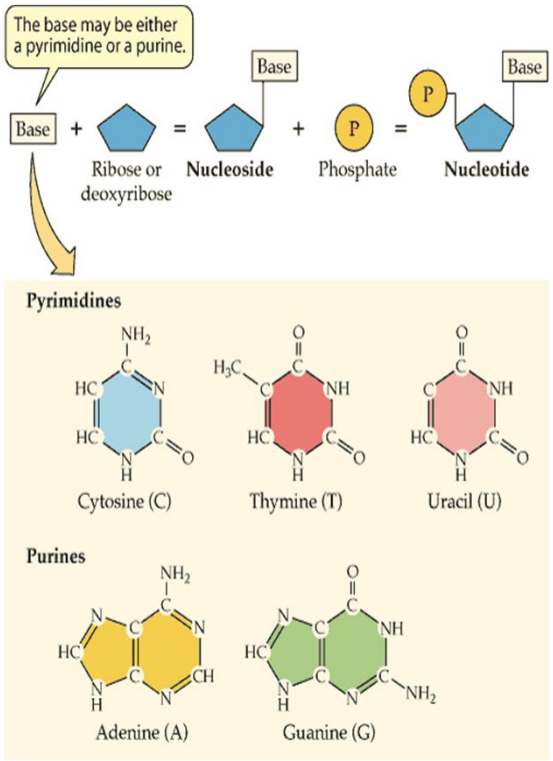

Original DNA
After one round of replication
(a)
Semiconservative replication would produce molecules with both old and new DNA, but each molecule would contain one complete old strand and one new one.
(b)
Conservative replication would preserve the original molecule and generate an entirely new molecule.
(c)
Dispersive replication would produce two molecules with old and new DNA interspersed along each strand.

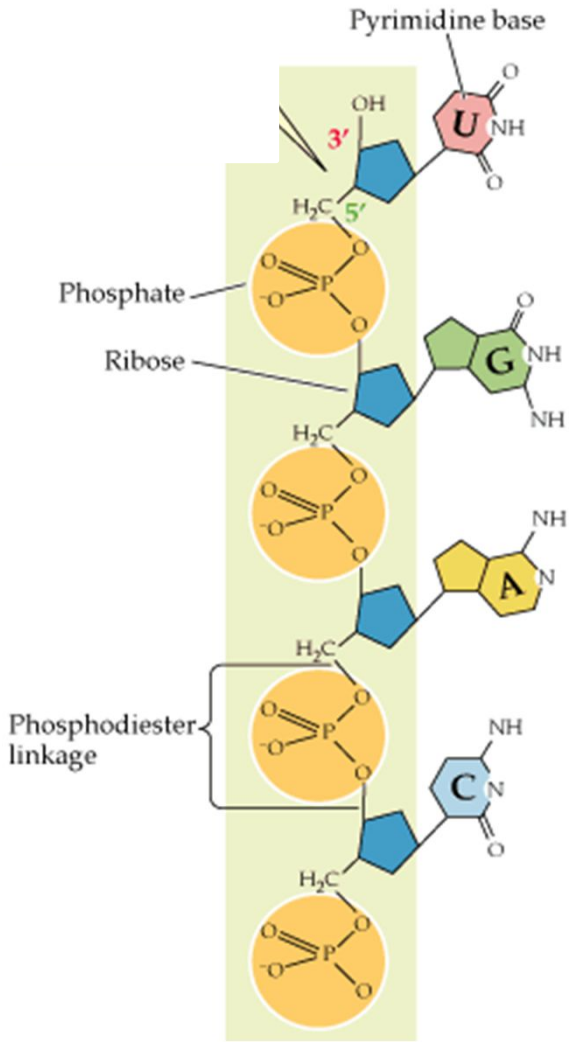

EXPERIMENT
Question: Does DNA replicate semiconservatively, or by some other mechanism?
METHOD
1 Grow bacteria in 15N (heavy) medium.
2 Transfer some bacteria to 14N (light) medium; bacterial growth continues.
3 Before the bacteria reproduce the first time in the light medium, all DNA (parental) is heavy.
Sample at 0 minutes
Sample after 20 minutes
Sample after 40 minutes
4 Samples are taken after 0 minutes, 20 minutes (after one round of replication), and 40 minutes (two rounds of replication).
RESULTS
14N/14N (light) DNA
14N/15N (intermediate) DNA
15N/15N (heavy) DNA
Parental (all heavy)
First generation (intermediate)
Second generation (half are all light)
5 If each strand served as a template for a new strand, DNA of the first generation would be of an intermediate density, and half the DNA from the second generation would be intermediate and half light. This is what was in fact observed.
INTERPRETATION
Parental strand 15N
New strand 14N
Conclusion: DNA replication is semiconservative

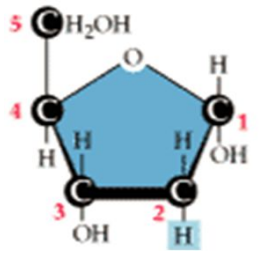

Original DNA
After one round of replication
(a)
Semiconservative replication would produce molecules with both old and new DNA, but each molecule would contain one complete old strand and one new one.
(b)
Conservative replication would preserve the original molecule and generate an entirely new molecule.
(c)
Dispersive replication would produce two molecules with old and new DNA interspersed along each strand.

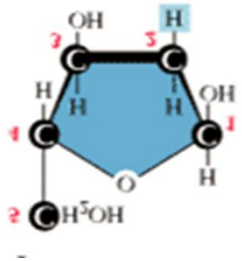
END

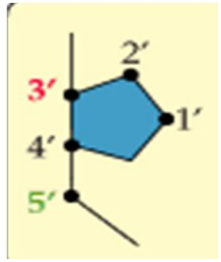
DNA Replication The Molecular View

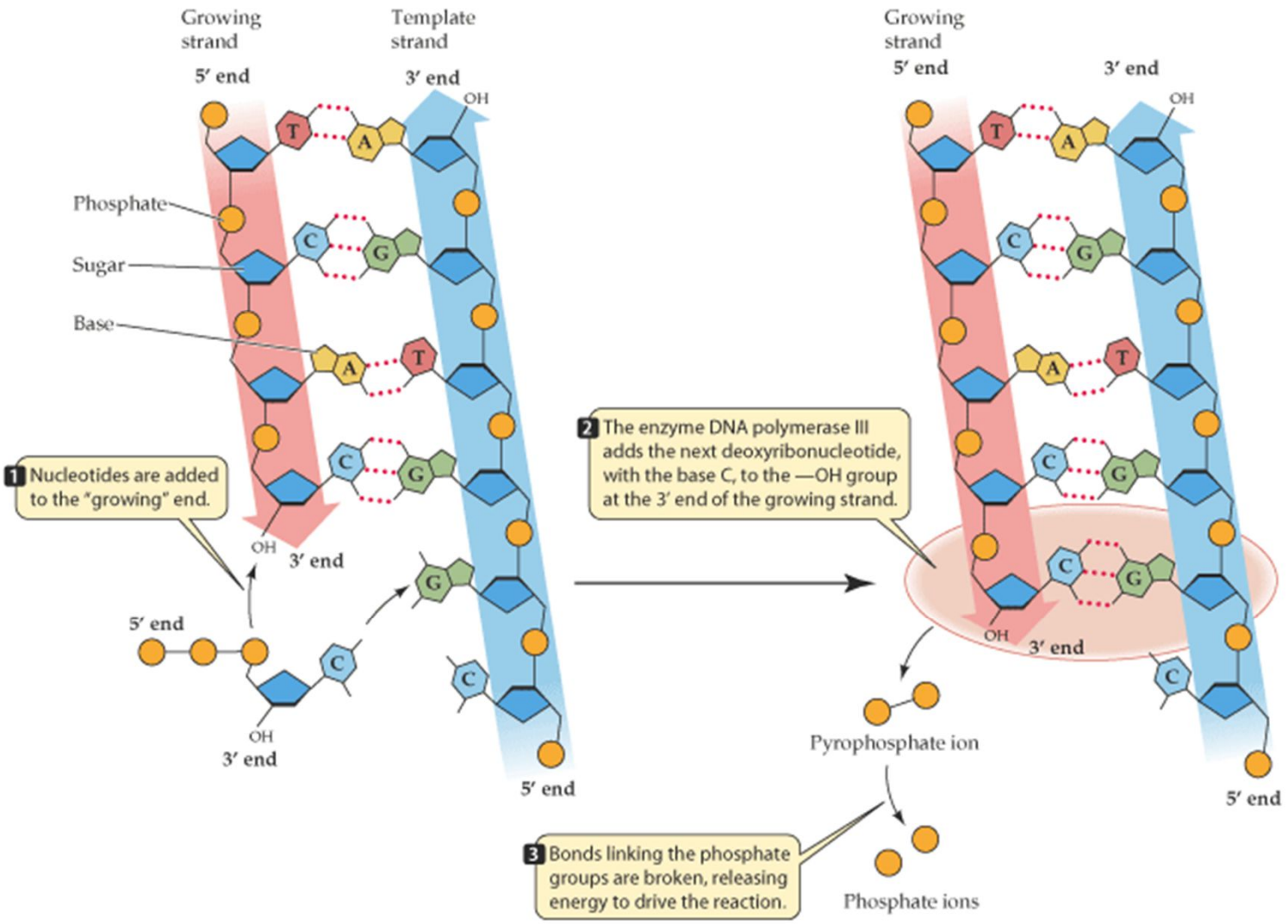

The base may be either a pyrimidine or a purine.
Base
+
Ribose or deoxyribose
=
Base
Nucleoside
+
P
Phosphate
=
P
Base
Nucleotide
Pyrimidines
Cytosine (C)
Thymine (T)
Uracil (U)
Purines
Adenine (A)
Guanine (G)

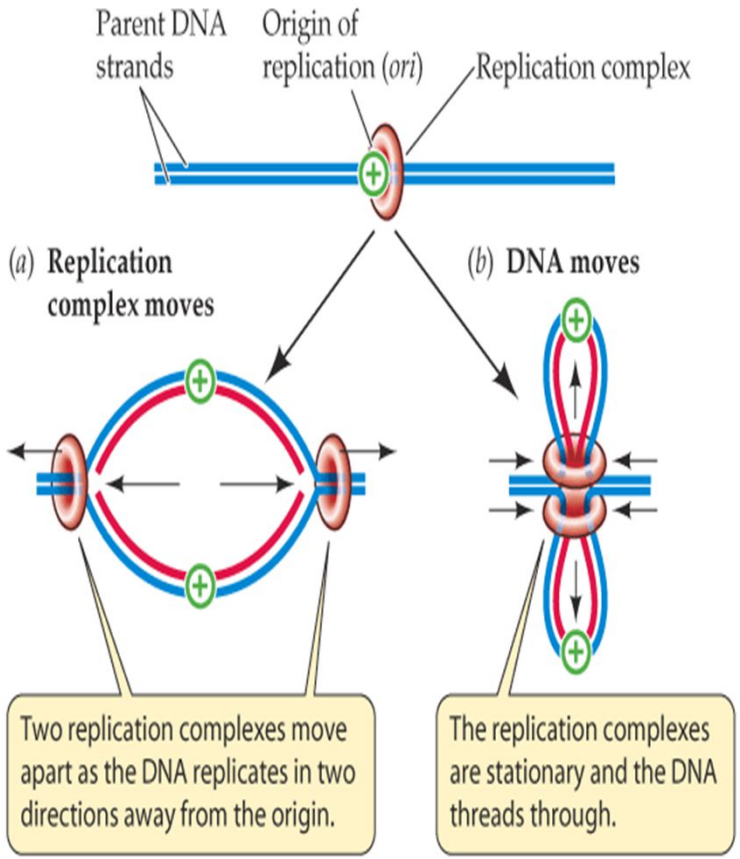

Pyrimidine base
Phosphate
Ribose
Phosphodiester linkage
3′ end
5′ end
U
G
A
C
OH
3′
5′


5
H2OH
O
H
4
1
H
H
H
OH
3
2
OH
H


OH
H
H
H
H
OH
H
O
H
5OH


2′
3′
1′
4′
5′


Growing strand
Template strand
5′ end
3′ end
OH
Phosphate
Sugar
Base
T
A
C
G
1 Nucleotides are added to the "growing" end.
OH
3′ end
5′ end
OH
3′ end
5′ end
2 The enzyme DNA polymerase III adds the next deoxyribonucleotide, with the base C, to the —OH group at the 3′ end of the growing strand.
Growing strand
5′ end
3′ end
OH
OH
3′ end
Pyrophosphate ion
5′ end
3 Bonds linking the phosphate groups are broken, releasing energy to drive the reaction.
Phosphate ions


Parent DNA strands
Origin of replication (ori)
Replication complex
(a) Replication complex moves
(b) DNA moves
Two replication complexes move apart as the DNA replicates in two directions away from the origin.
The replication complexes are stationary and the DNA threads through.

END

DNA Replication The Process

Requirements

1) The Original

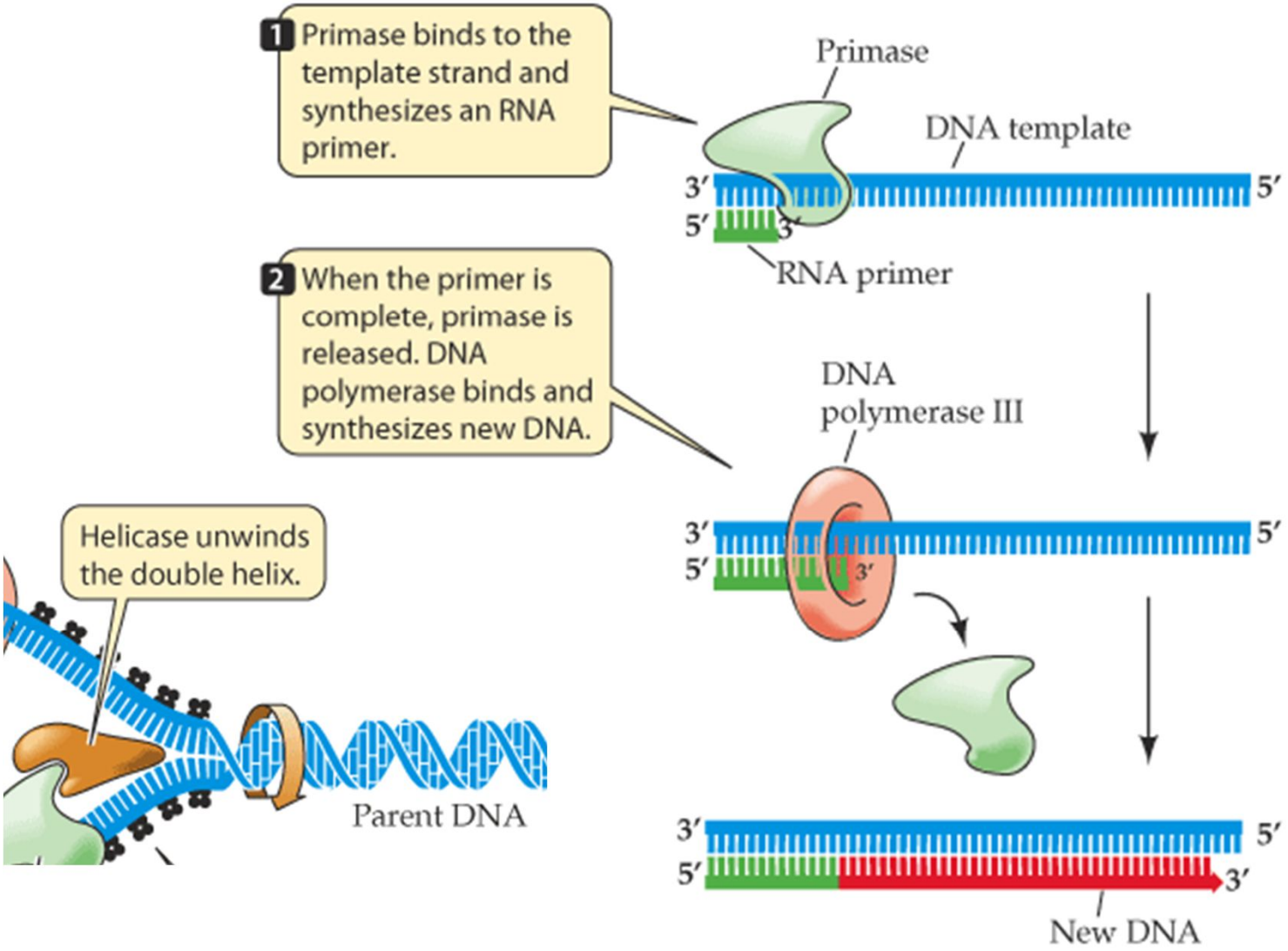
2) The Machinery

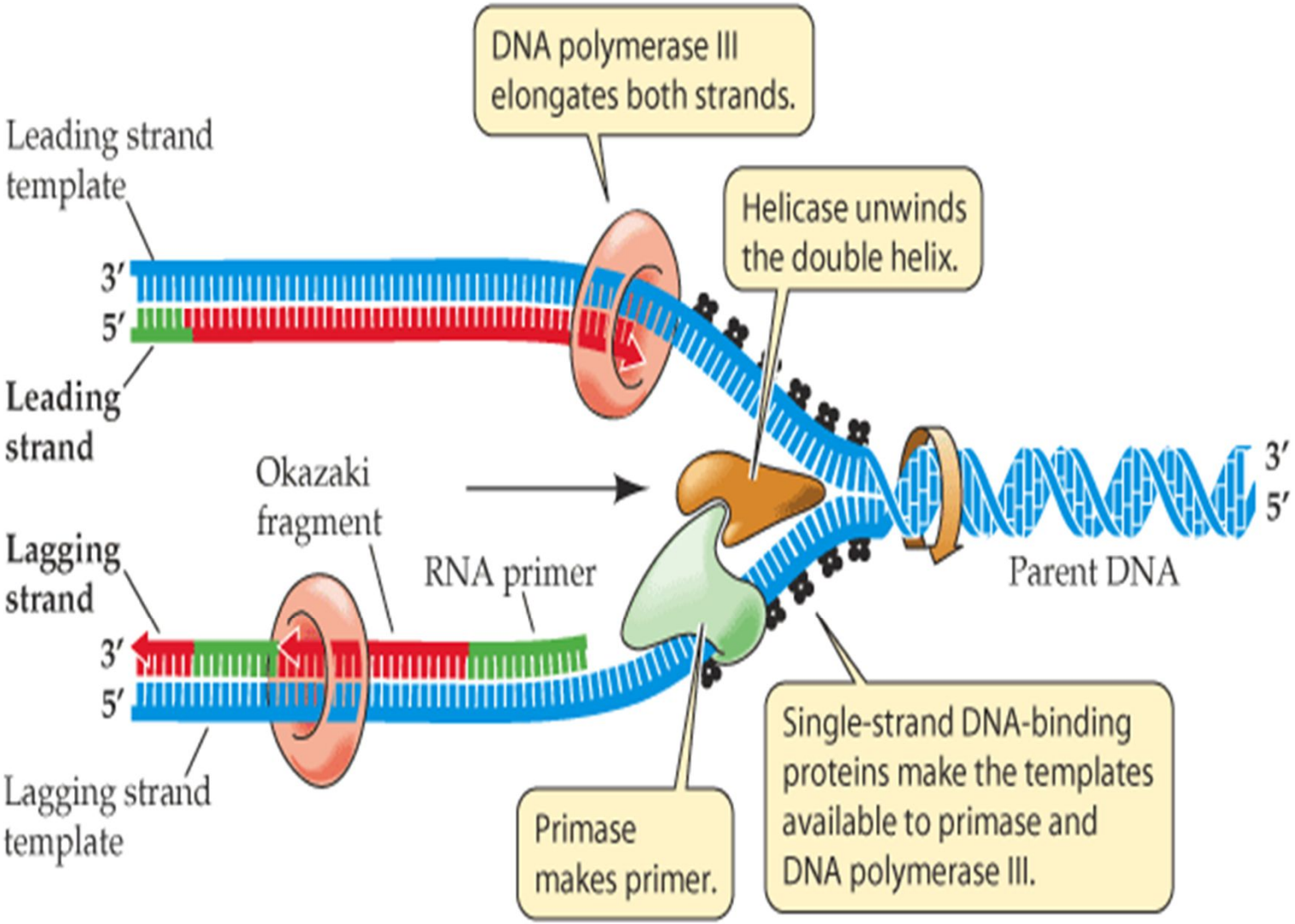
3) The Raw Material

5) Energy

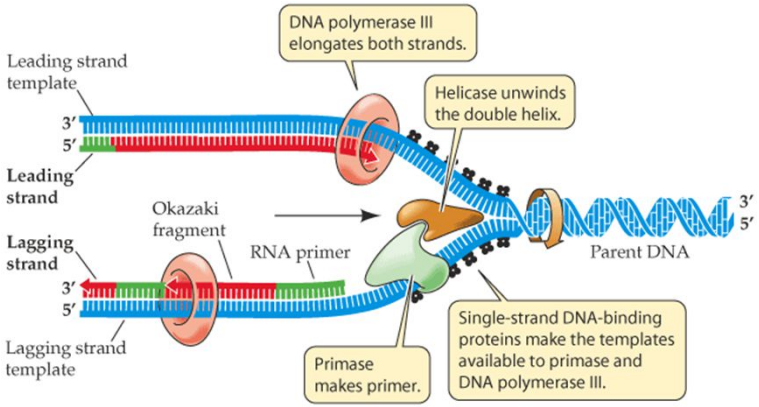
4) Initiator

DNA polymerase
useless without primer


1 Primase binds to the template strand and synthesizes an RNA primer.
Primase
DNA template
3′
5′
5′
3′
RNA primer
2 When the primer is complete, primase is released. DNA polymerase binds and synthesizes new DNA.
DNA polymerase III
3′
5′
5′
3′
Helicase unwinds the double helix.
Parent DNA
3′
5′
5′
3′
New DNA

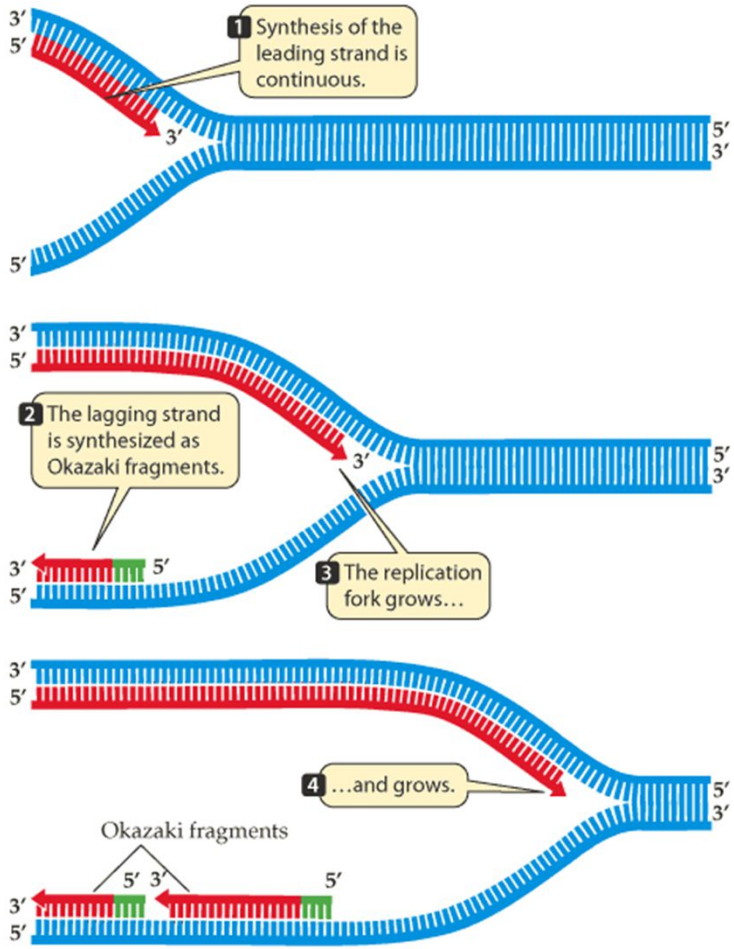

DNA polymerase III elongates both strands.
Helicase unwinds the double helix.
Leading strand template
3′
5′
Leading strand
Okazaki fragment
RNA primer
Lagging strand
3′
5′
Lagging strand template
3′
5′
Parent DNA
Primase makes primer.
Single-strand DNA-binding proteins make the templates available to primase and DNA polymerase III.

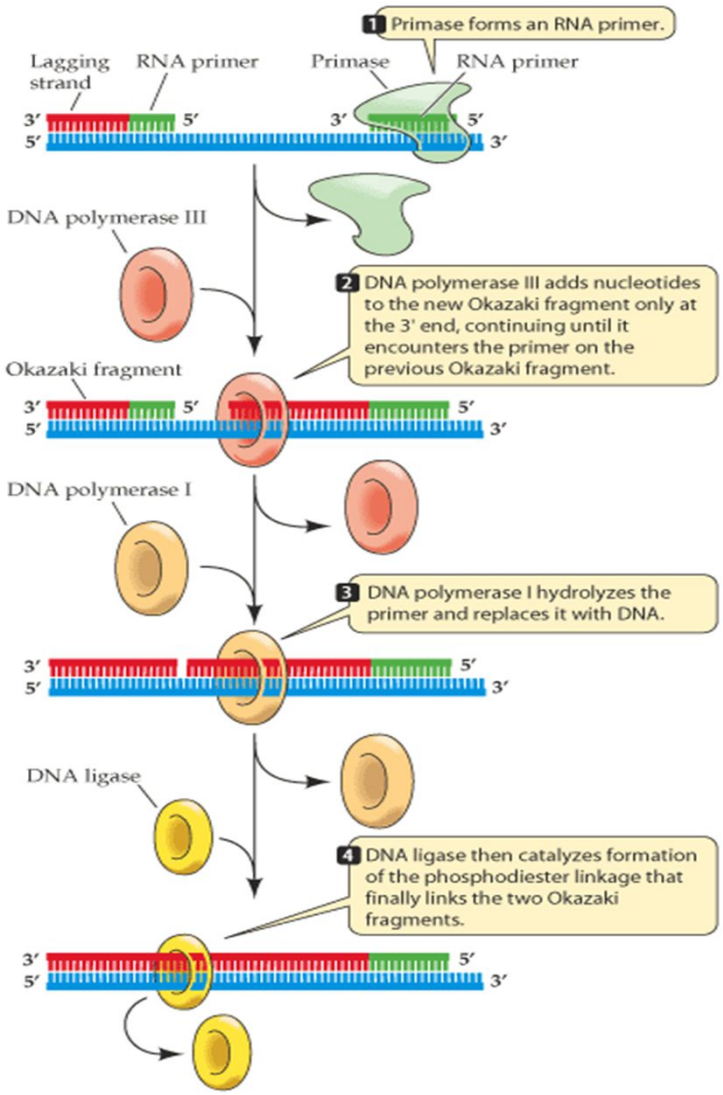
The Three Dilemmas



END

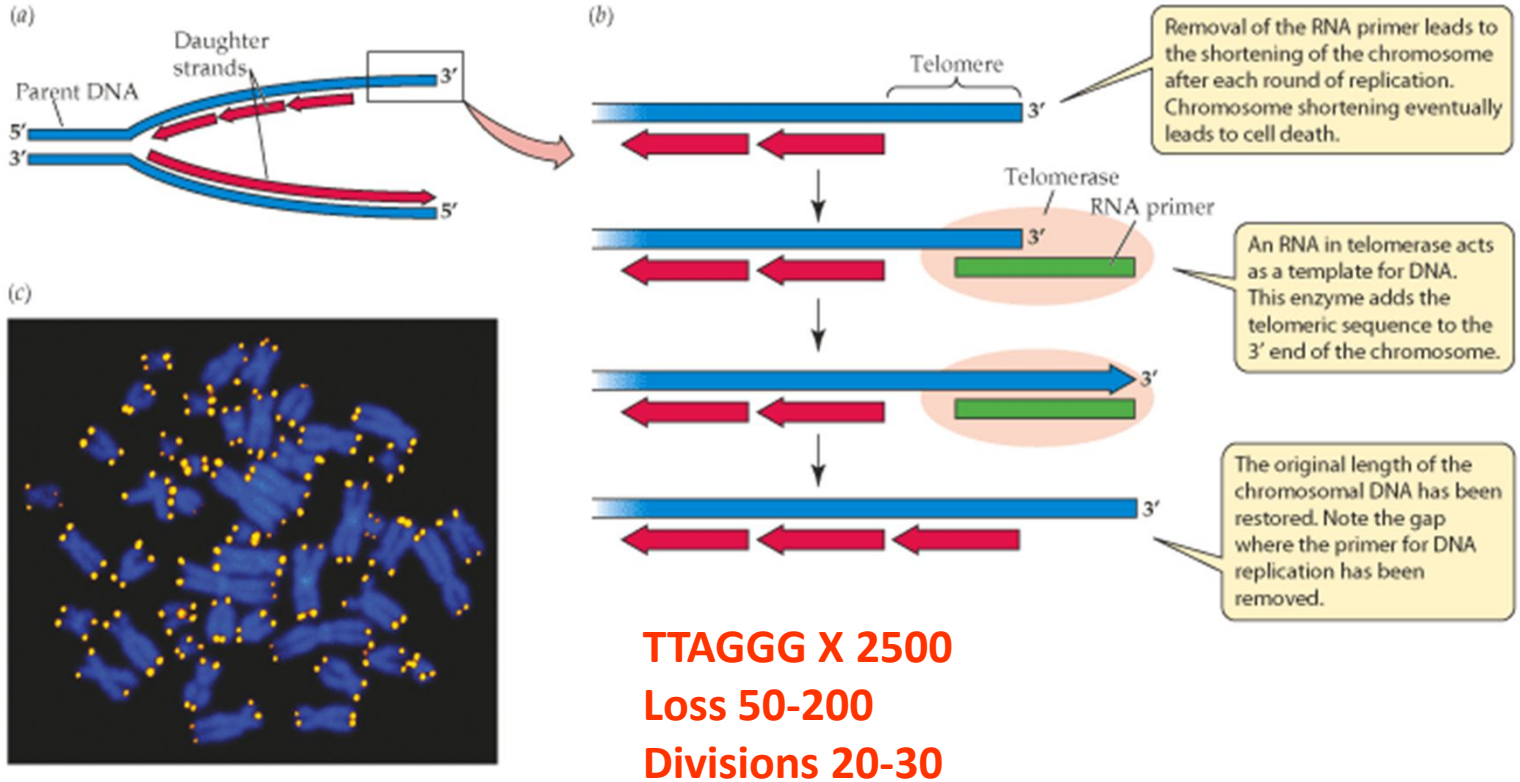
The Dilemma 1


1 Synthesis of the leading strand is continuous.
2 The lagging strand is synthesized as Okazaki fragments.
3 The replication fork grows…
4 …and grows.
Okazaki fragments
3′
5′


1 Primase forms an RNA primer.
Lagging strand
RNA primer
Primase
RNA primer
3′
5′
DNA polymerase III
2 DNA polymerase III adds nucleotides to the new Okazaki fragment only at the 3′ end, continuing until it encounters the primer on the previous Okazaki fragment.
Okazaki fragment
DNA polymerase I
3 DNA polymerase I hydrolyzes the primer and replaces it with DNA.
DNA ligase
4 DNA ligase then catalyzes formation of the phosphodiester linkage that finally links the two Okazaki fragments.

END

The Dilemma 2 & 3

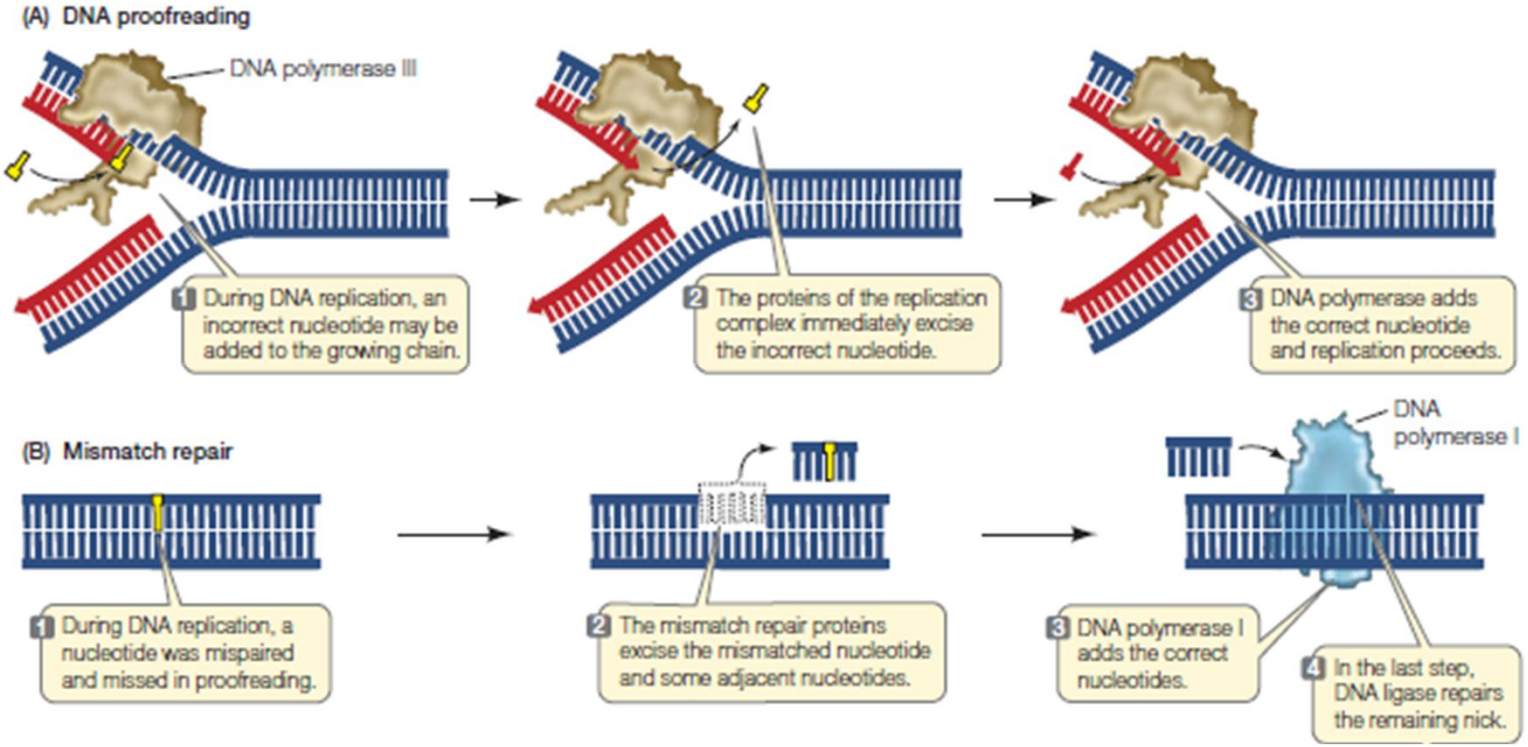


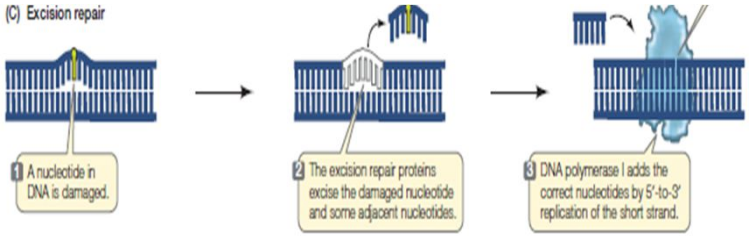
TTAGGG X 2500
Loss 50-200
Divisions 20-30

Error rate for DNA pol 10^6 = 1000 mutations/division

DNA polymerase can recognize mispairing

Efficiency 1 in 10,000 bp Overall error rate = 10^{10}





An excision repair mechanism removes abnormal bases that have formed because of chemical damage and replaces them with functional bases.

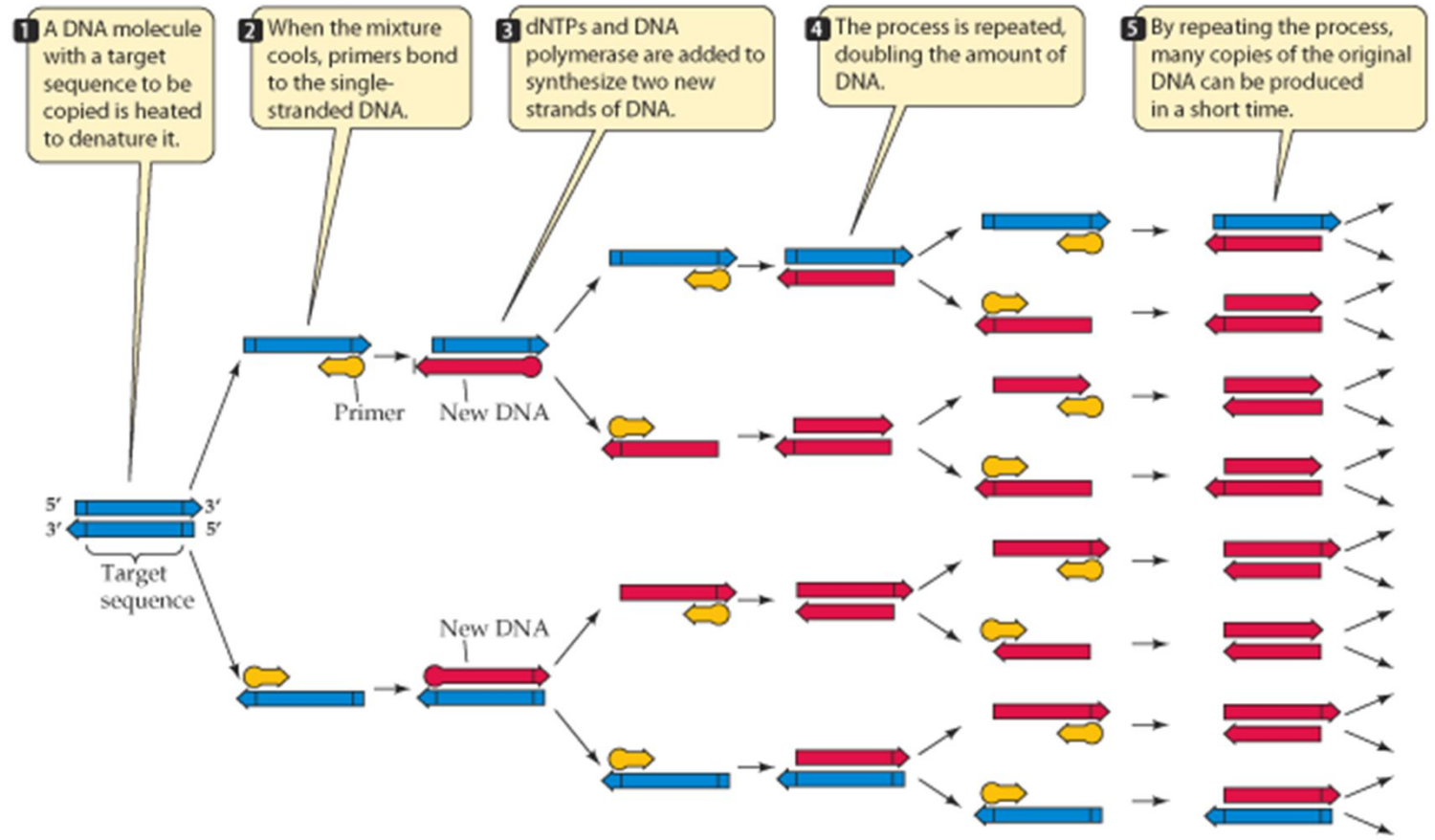
END

The Polymerase Chain Reaction (PCR) Technology

Requirements

1) Target DNA
2) Two oligonucleotide primers
3) DNA polymerase,
4) dNTPs
5) PCR machine

RESEARCH METHOD



Requirements

1) Target DNA
2) Two oligonucleotide primers
3) DNA polymerase,
4) dNTPs
5) PCR machine

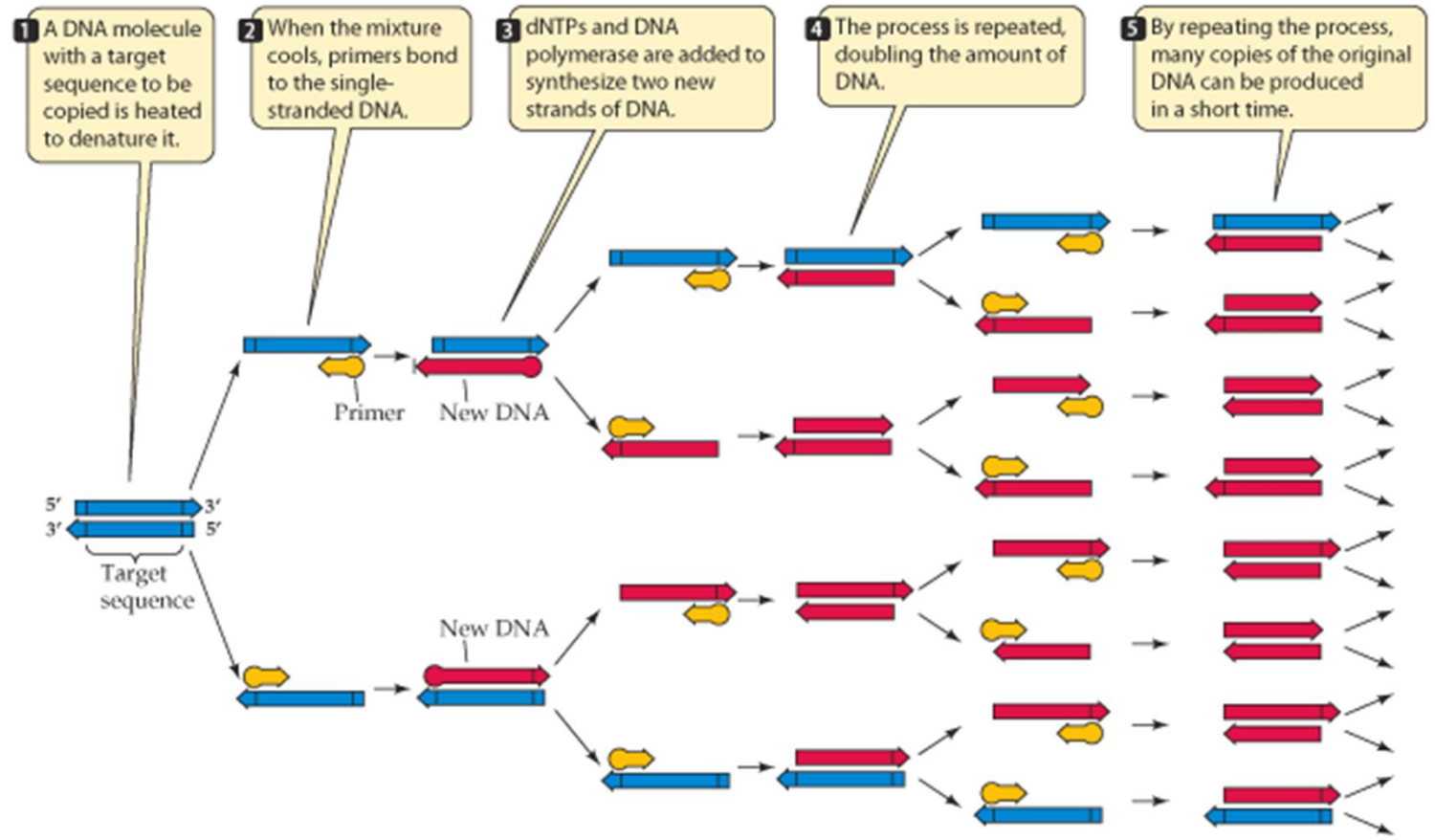
END

The Polymerase Chain Reaction (PCR) Technology

Requirements

1) Target DNA
2) Two oligonucleotide primers
3) DNA polymerase,
4) dNTPs
5) PCR machine

RESEARCH METHOD



Requirements

1) Target DNA
2) Two oligonucleotide primers
3) DNA polymerase,
4) dNTPs
5) PCR machine

END

PCR Animation

PCR Continued

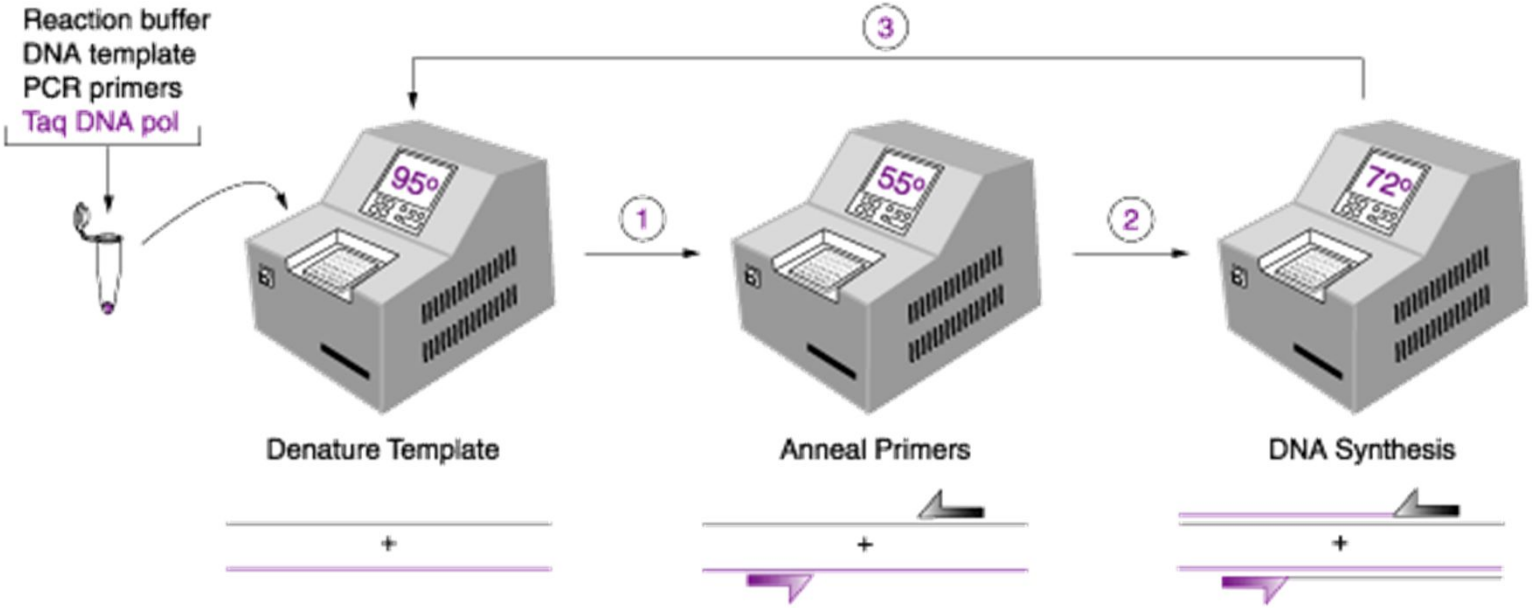
Now

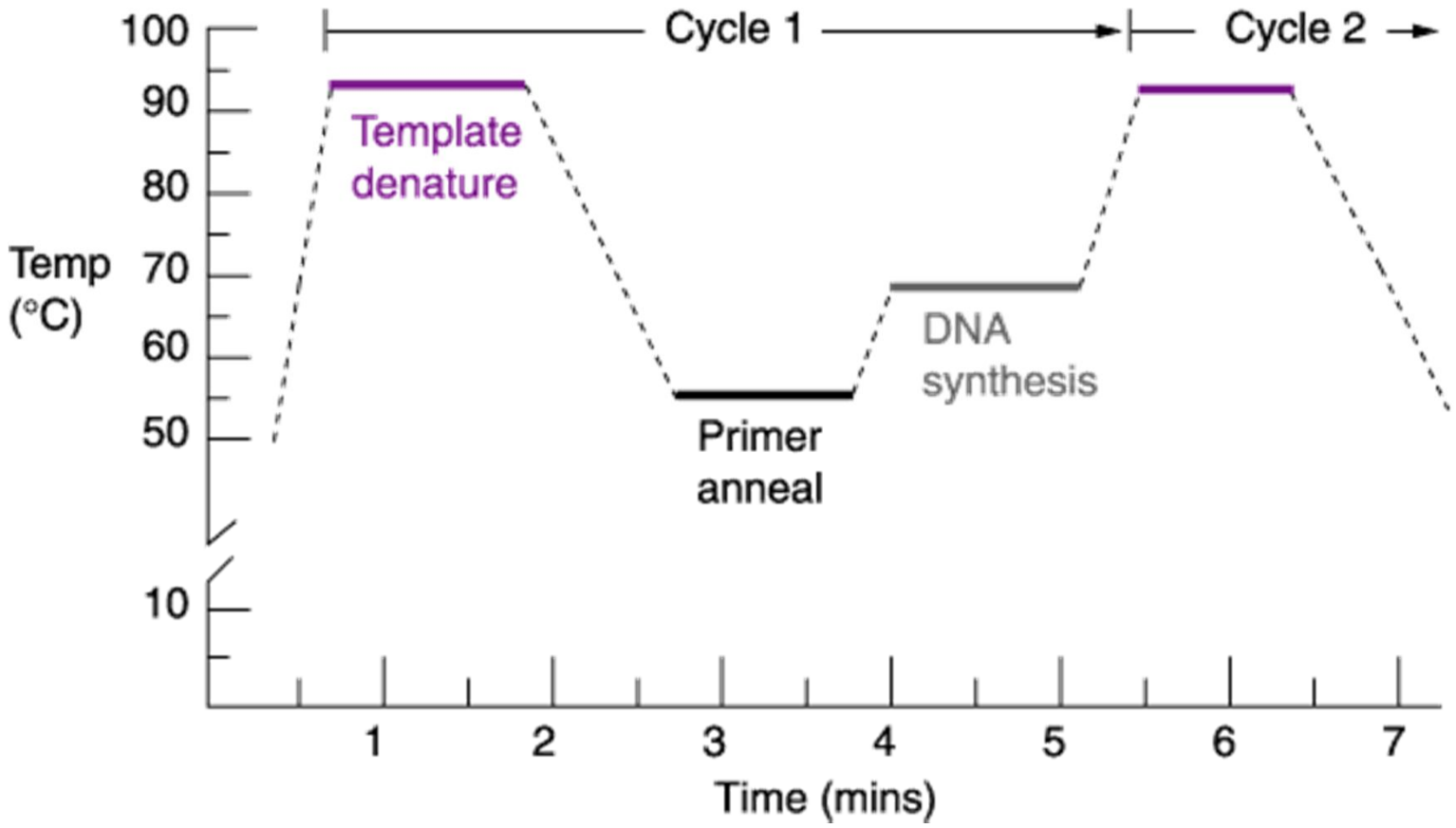
eppendorf 5330 plus
Run
Stop
Caution! Hot Base
eppendorf
Mastercycler 5330
Run
Stop
Prg
Data
Track
Run
Stop
Prg
Data
7 8 9
4 5 6
1 2 3
• 0 Clr

Then

GENE MACHINE
BATH 1
BATH 2
BATH 3
CYCLE CONTROL
INNOVONICS
Please DO NOT leave this machine on overnight.

The Cycles


Reaction buffer
DNA template
PCR primers
Taq DNA pol
95°
55°
72°
3
1
2
Denature Template
Anneal Primers
DNA Synthesis
+
+
+


Cycle 1
Cycle 2
100
90
80
70
60
50
10
Temp (°C)
Template denature
Primer anneal
DNA synthesis
1
2
3
4
5
6
7
Time (mins)

Exponential Amplification

Cycles	Copies
1	2
2	4
4	16
10	1,024
15	32,768
20	1,048,576
25	33,554,432
30	**1,073,741,824**

END

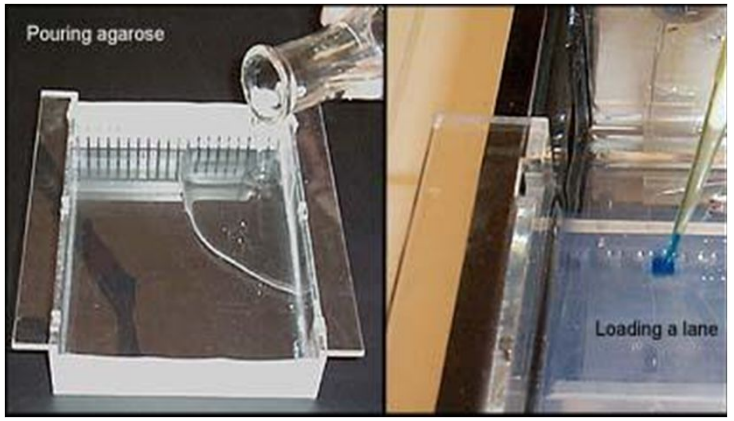
Exponential Amplification

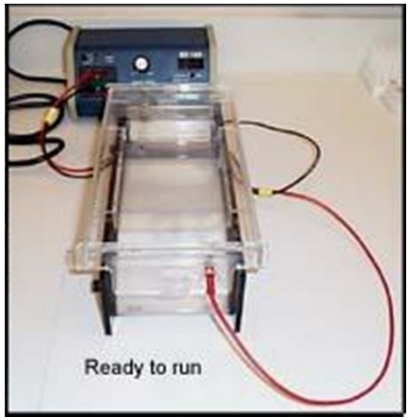
1,073,741,824

END

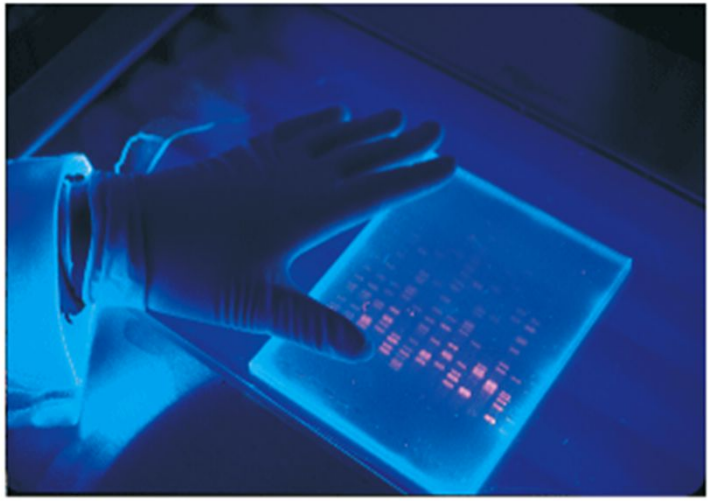
Detecting Amplified DNA

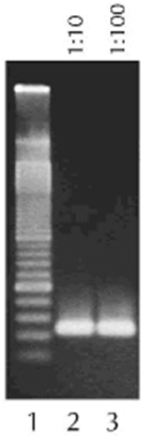
Gel Electrophoresis





UV





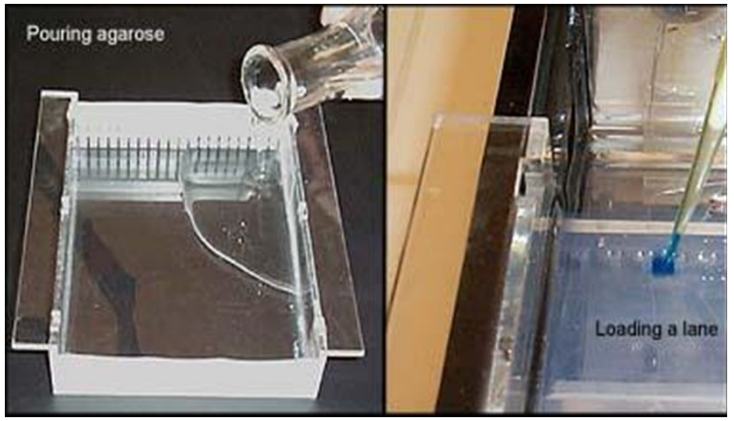
END

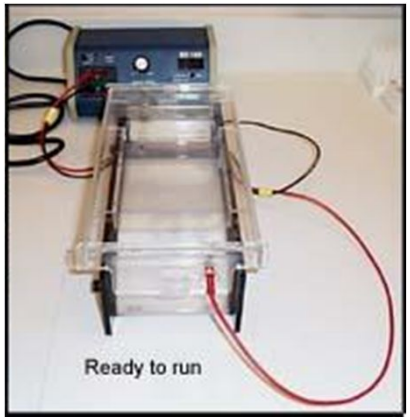
Gel Electrophoresis Animation

Gel Electrophoresis

END

Gel Electrophoresis

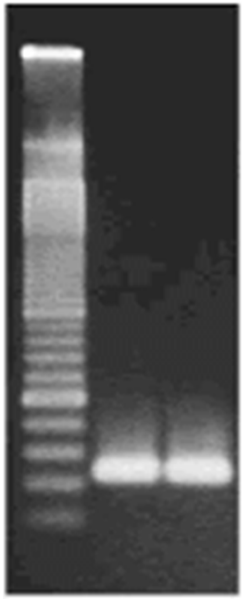




PCR Technology
Applications

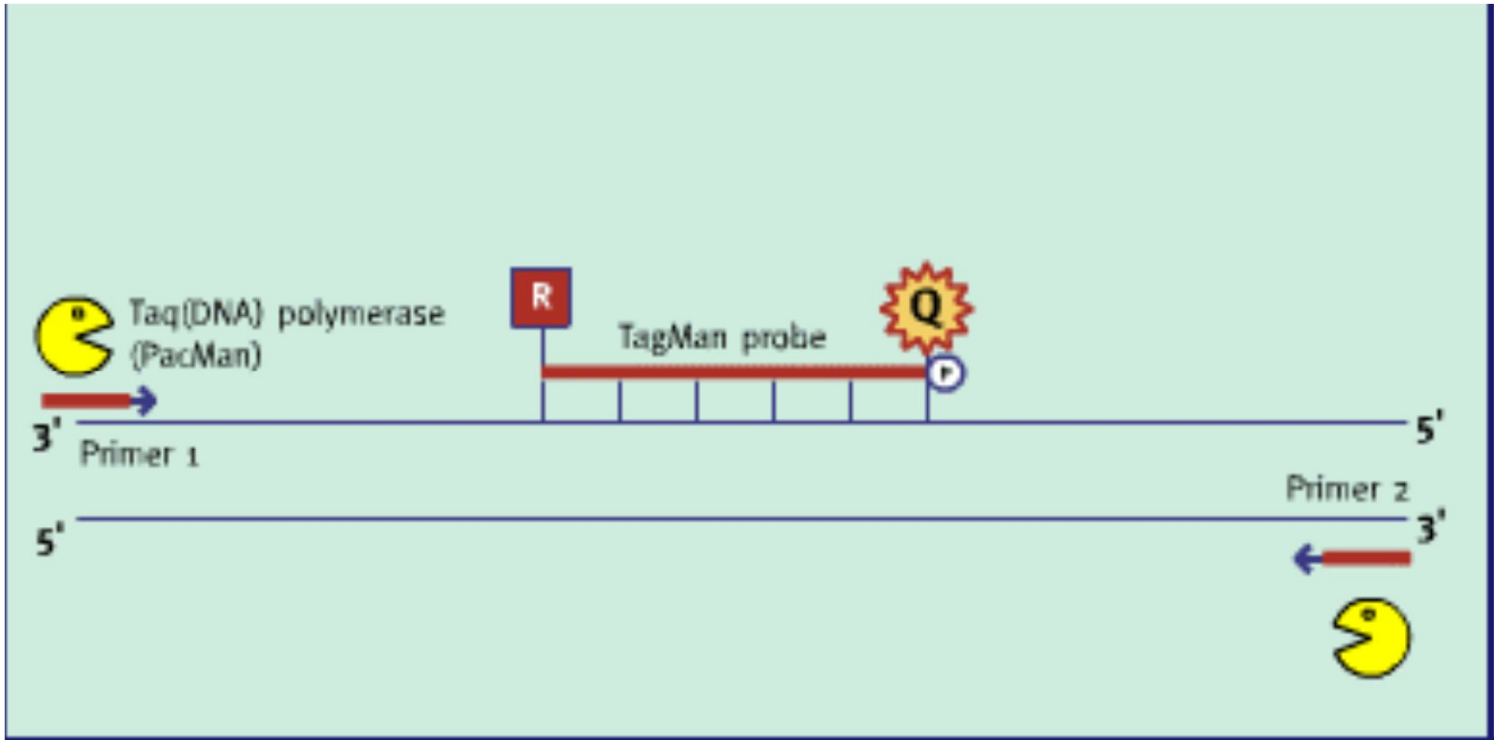
Disease Detection

Many pathogens can be detected from a small sample e.g.

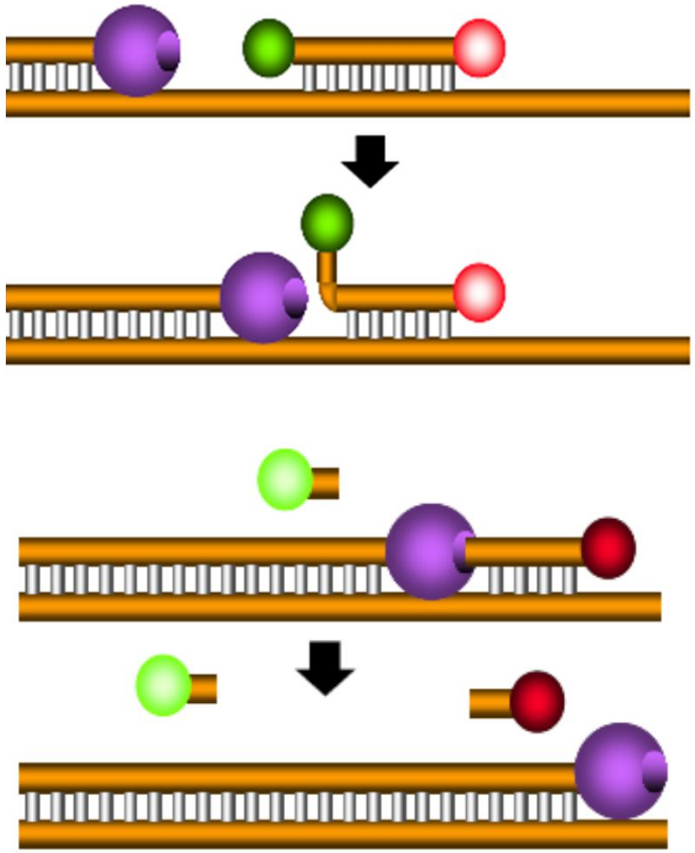


HCV
HBV
HIV
Adenoviruses
Influenza virus
Cytomegalovirus etc

Quantification



Quantification

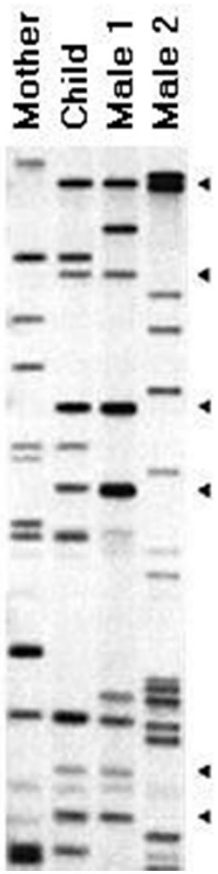


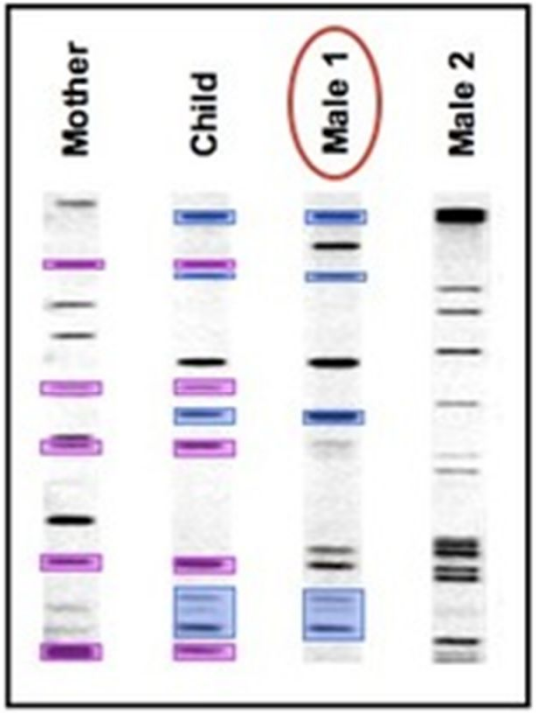
END

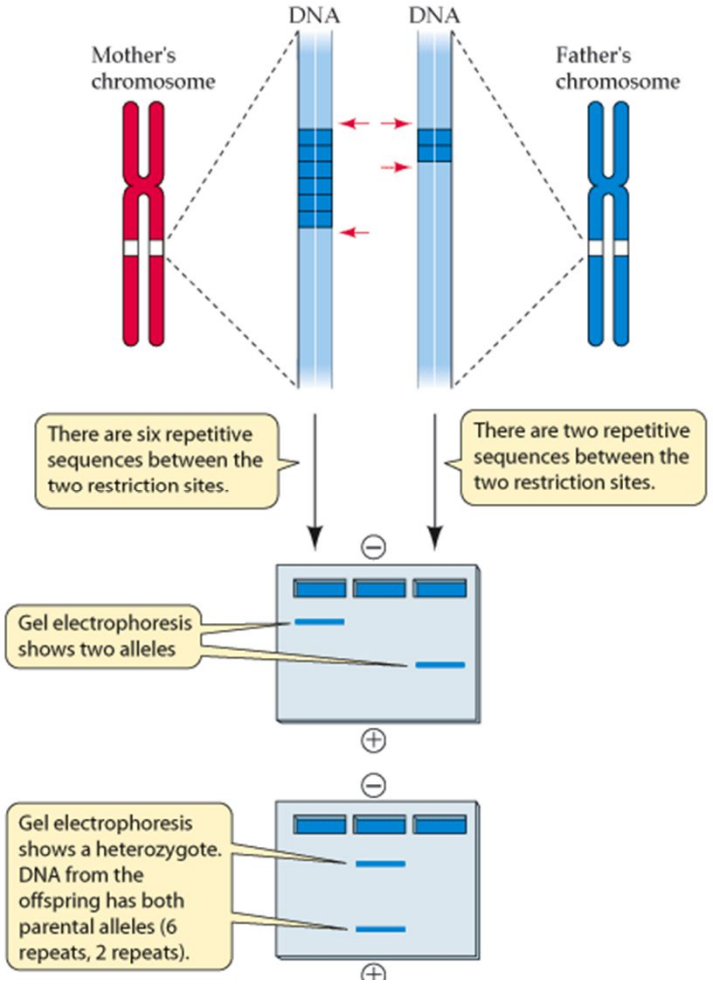
DNA Finger Printing

VNTRs
Variable Number Tandem Repeats

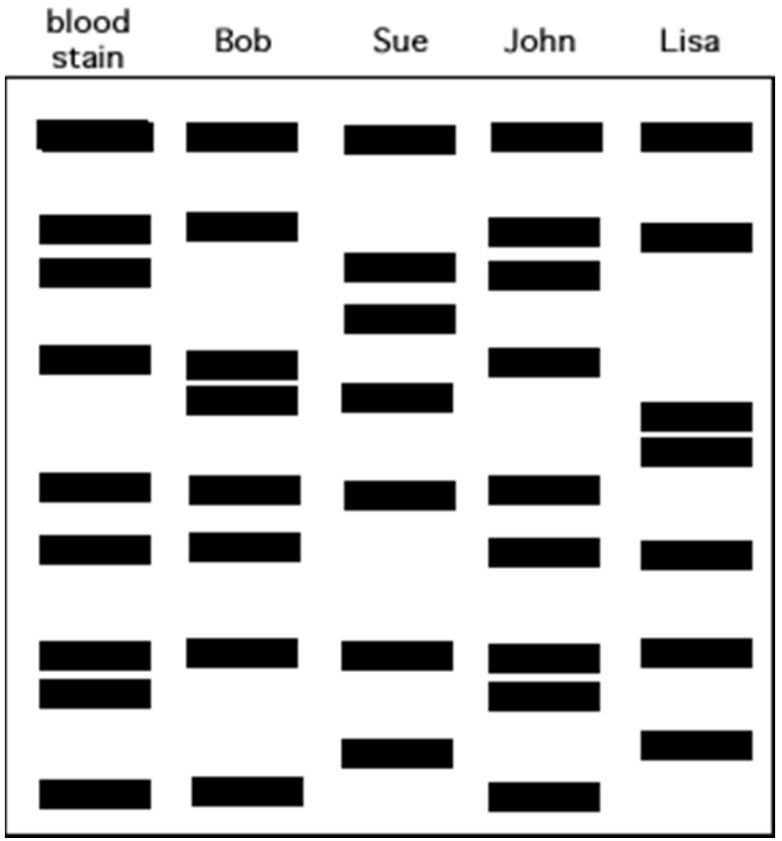
Paternity


Mother
Child
Male 1
Male 2

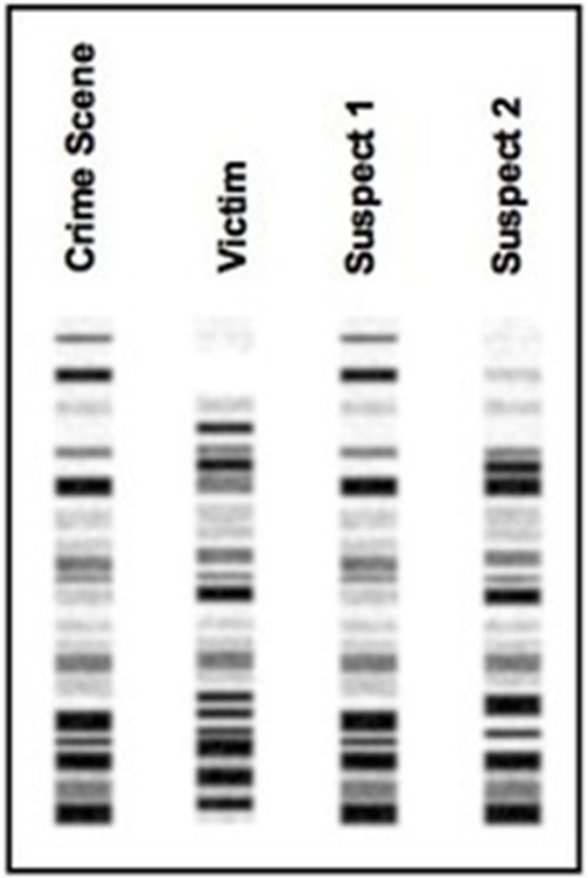

Mother
Child
Male 1
Male 2


DNA
DNA
Mother's chromosome
Father's chromosome
There are six repetitive sequences between the two restriction sites.
There are two repetitive sequences between the two restriction sites.
Gel electrophoresis shows two alleles
Gel electrophoresis shows a heterozygote. DNA from the offspring has both parental alleles (6 repeats, 2 repeats).

Crime Investigation



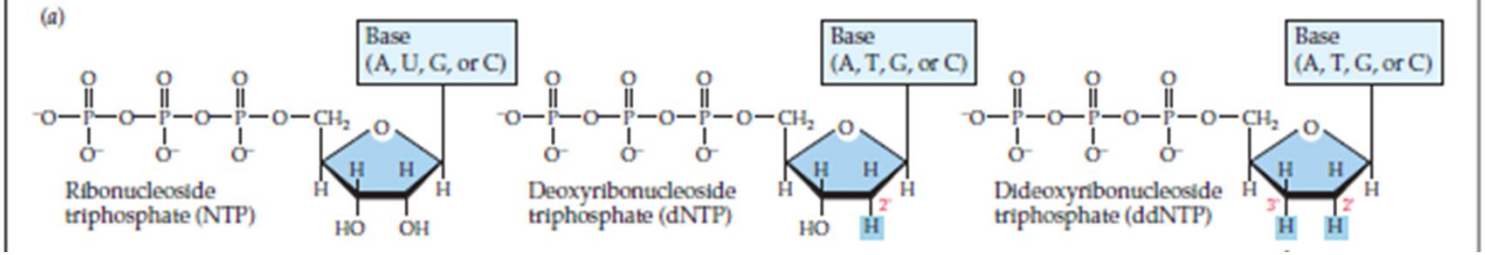
Crime Investigation

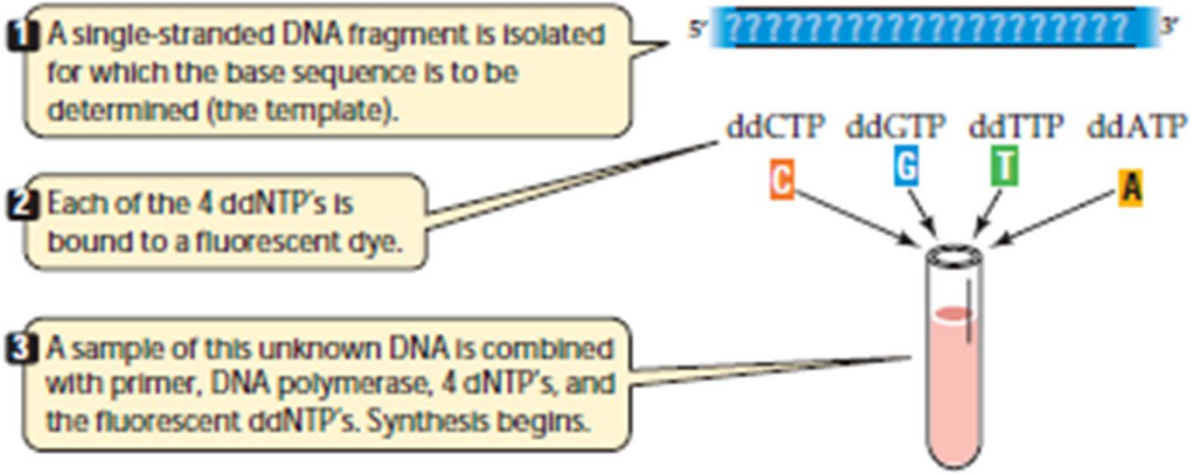


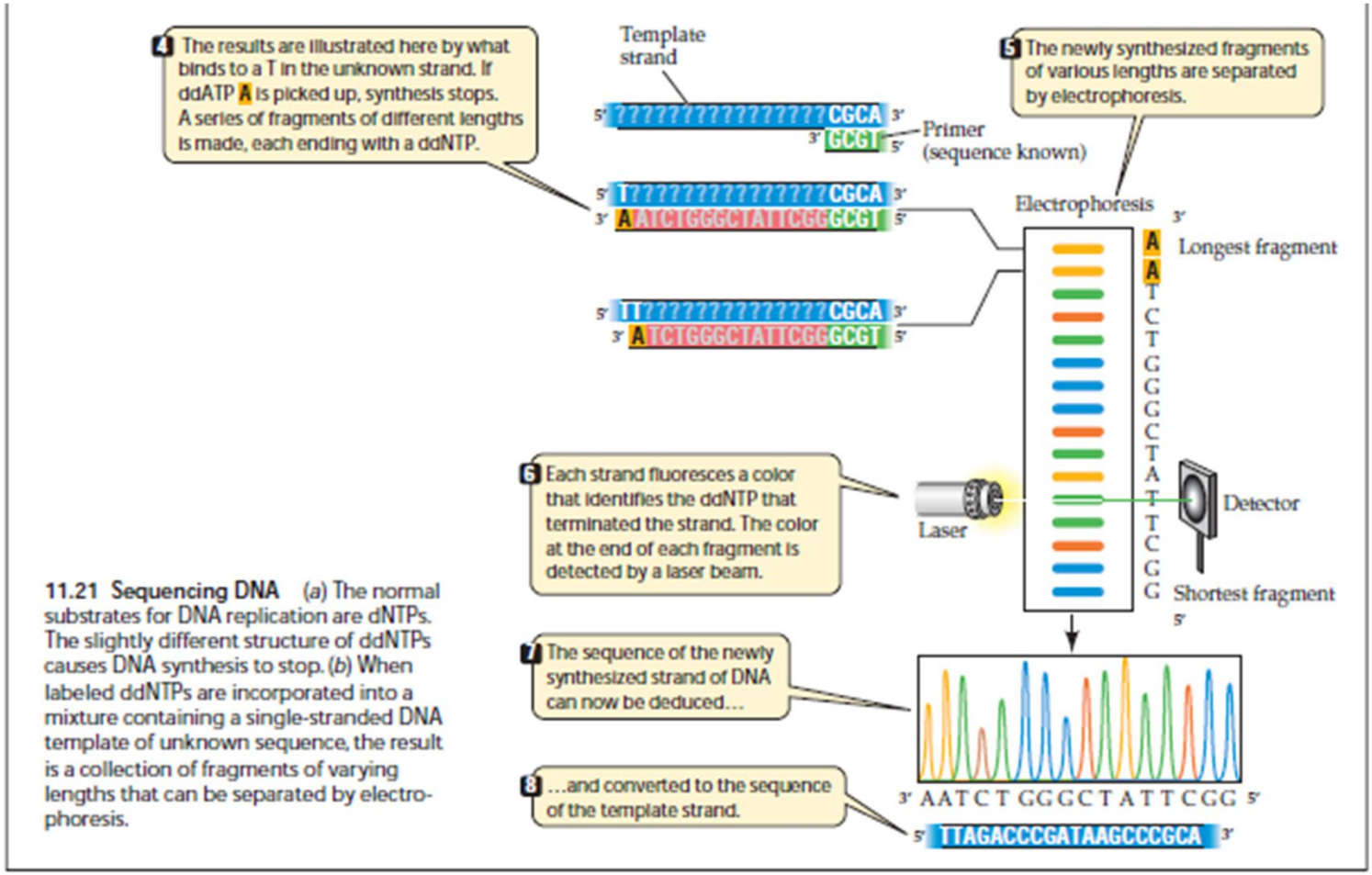
END

DNA Sequencing

RESEARCH METHOD





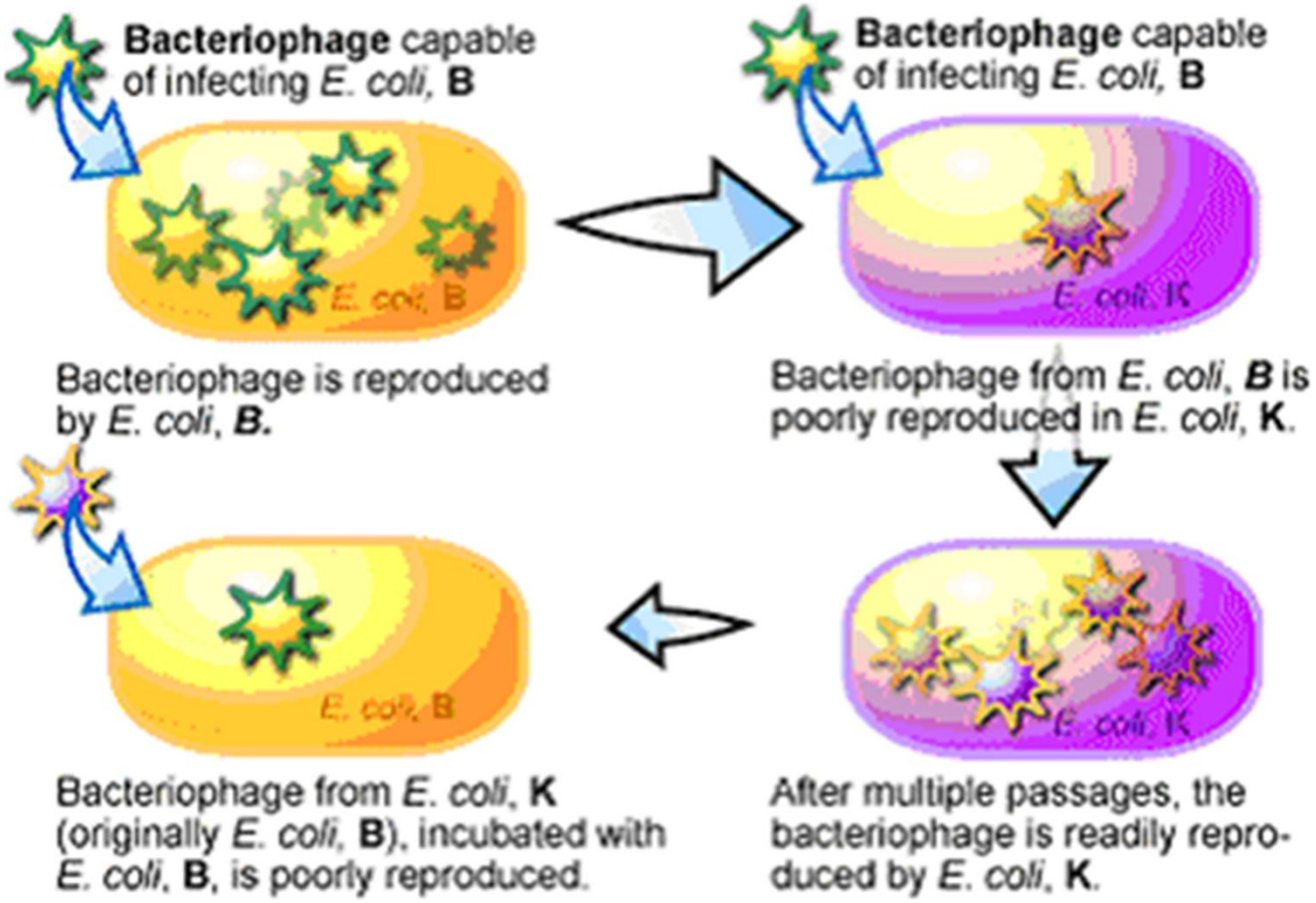


11.21 Sequencing DNA (*a*) The normal substrates for DNA replication are dNTPs. The slightly different structure of ddNTPs causes DNA synthesis to stop. (*b*) When labeled ddNTPs are incorporated into a mixture containing a single-stranded DNA template of unknown sequence, the result is a collection of fragments of varying lengths that can be separated by electrophoresis.

END

Restriction Endonucleases

Arber, Nathans & Smith (1978 Nobel Prize



Special bacterial enzyme can cut double stranded DNA at specific sites in a test tube.

Restriction endonucleases (restriction enzymes) act as a kind of immune system, protecting the cell from the invasion of foreign DNA (virus).

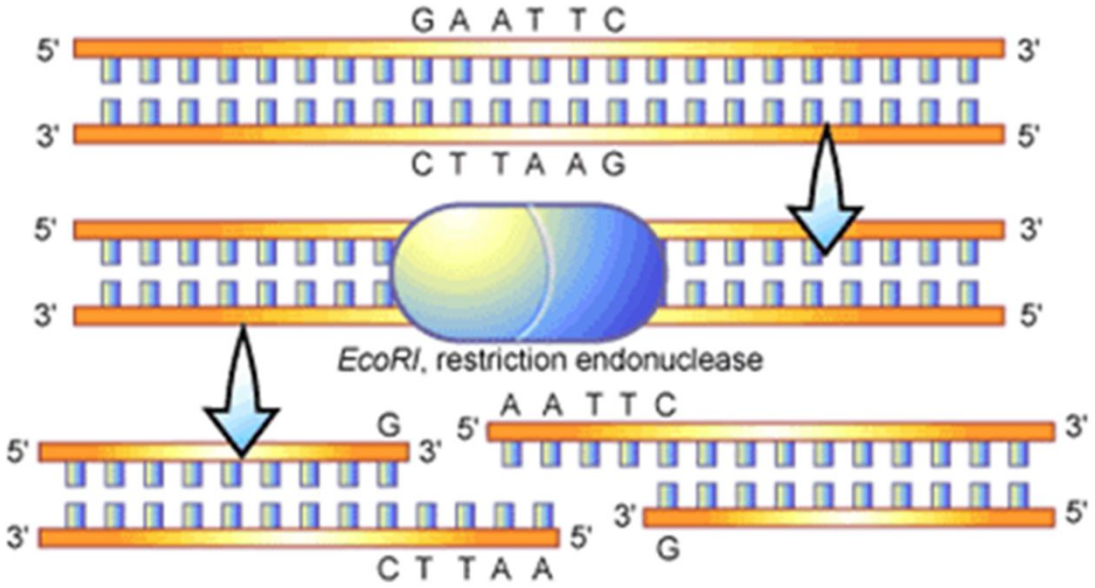
Many recombinant DNA technologies, which the field of biotechnology heavily relies on, are unlikely to have been developed without the discovery of restriction enzymes

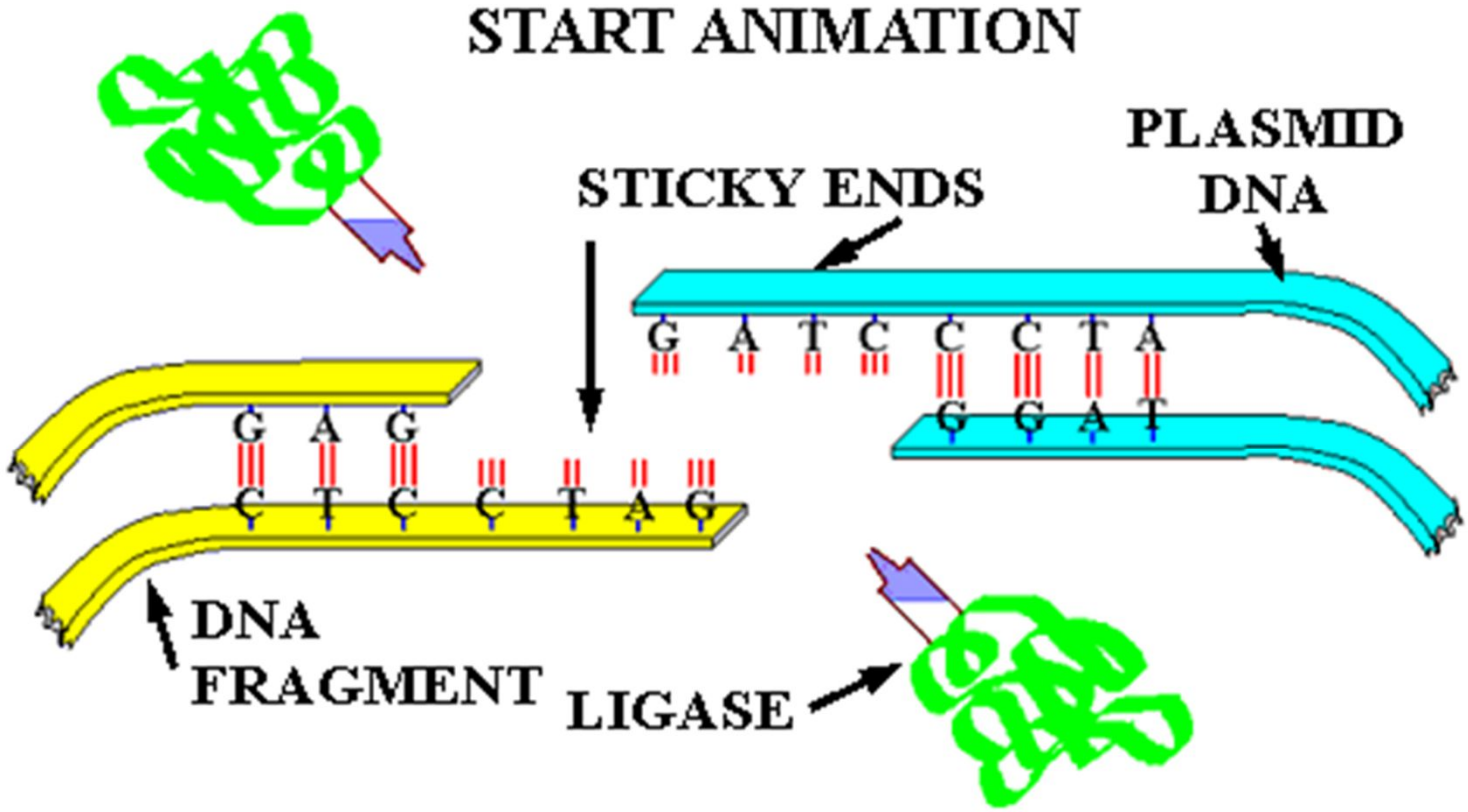
Palindromic Sequences: A palindromic sequence is the same when read in 5′ to 3′ direction on either complementary strand of DNA.

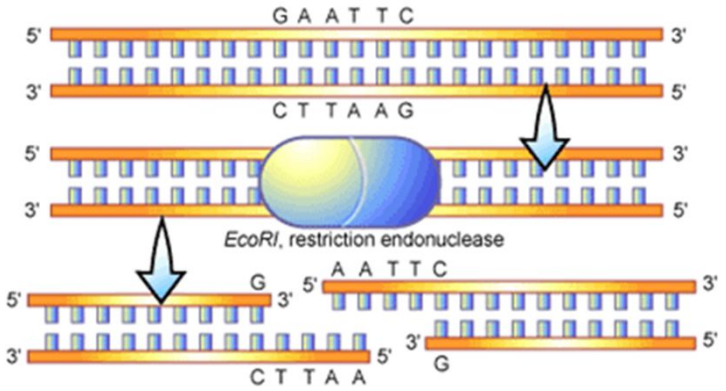
DENNIS SINNED

GTAATG VS GTATAC

.



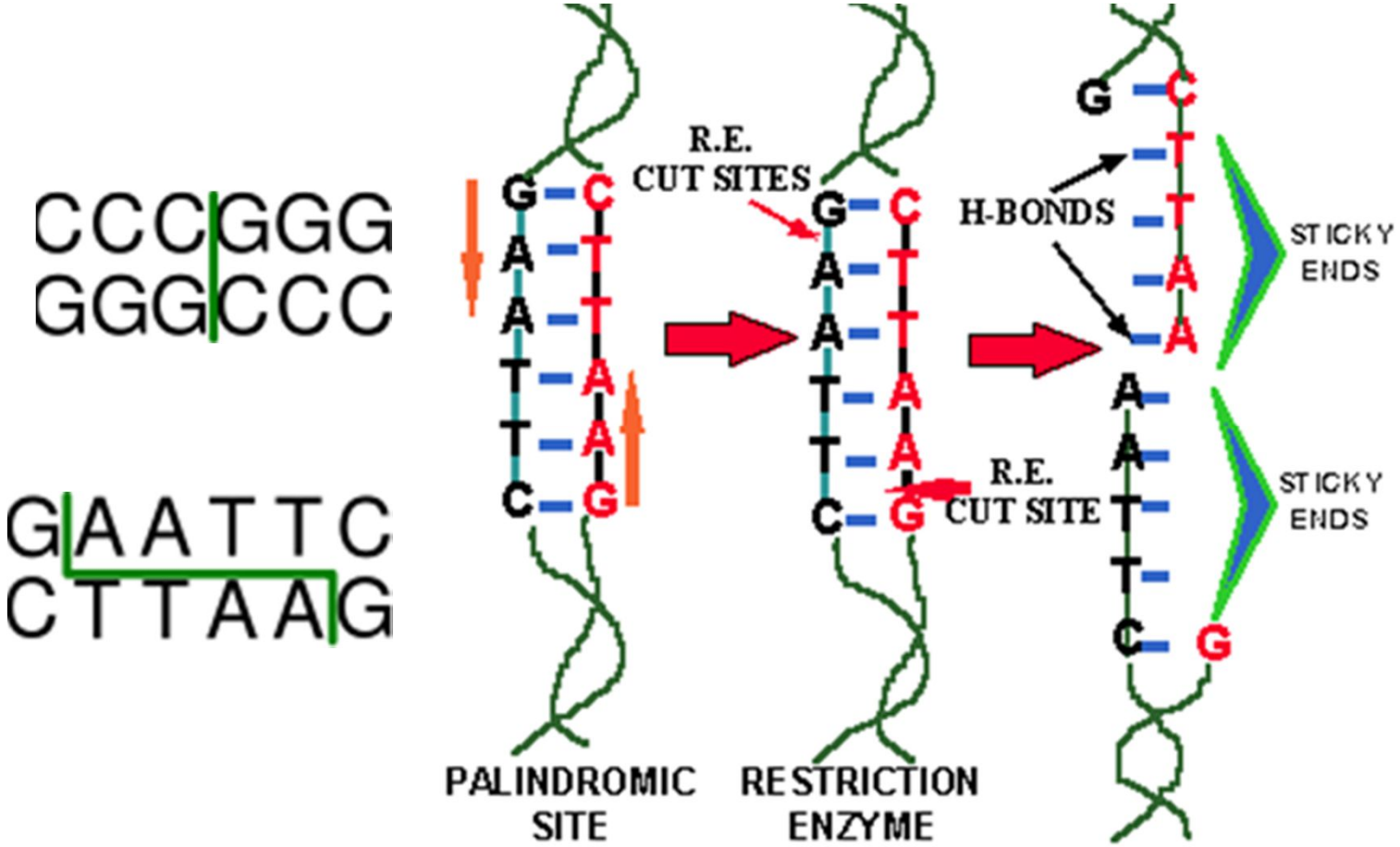

START ANIMATION
PLASMID DNA
STICKY ENDS
G A T C C C T A
G G A T
G A G
C T C C T A G
DNA FRAGMENT
LIGASE

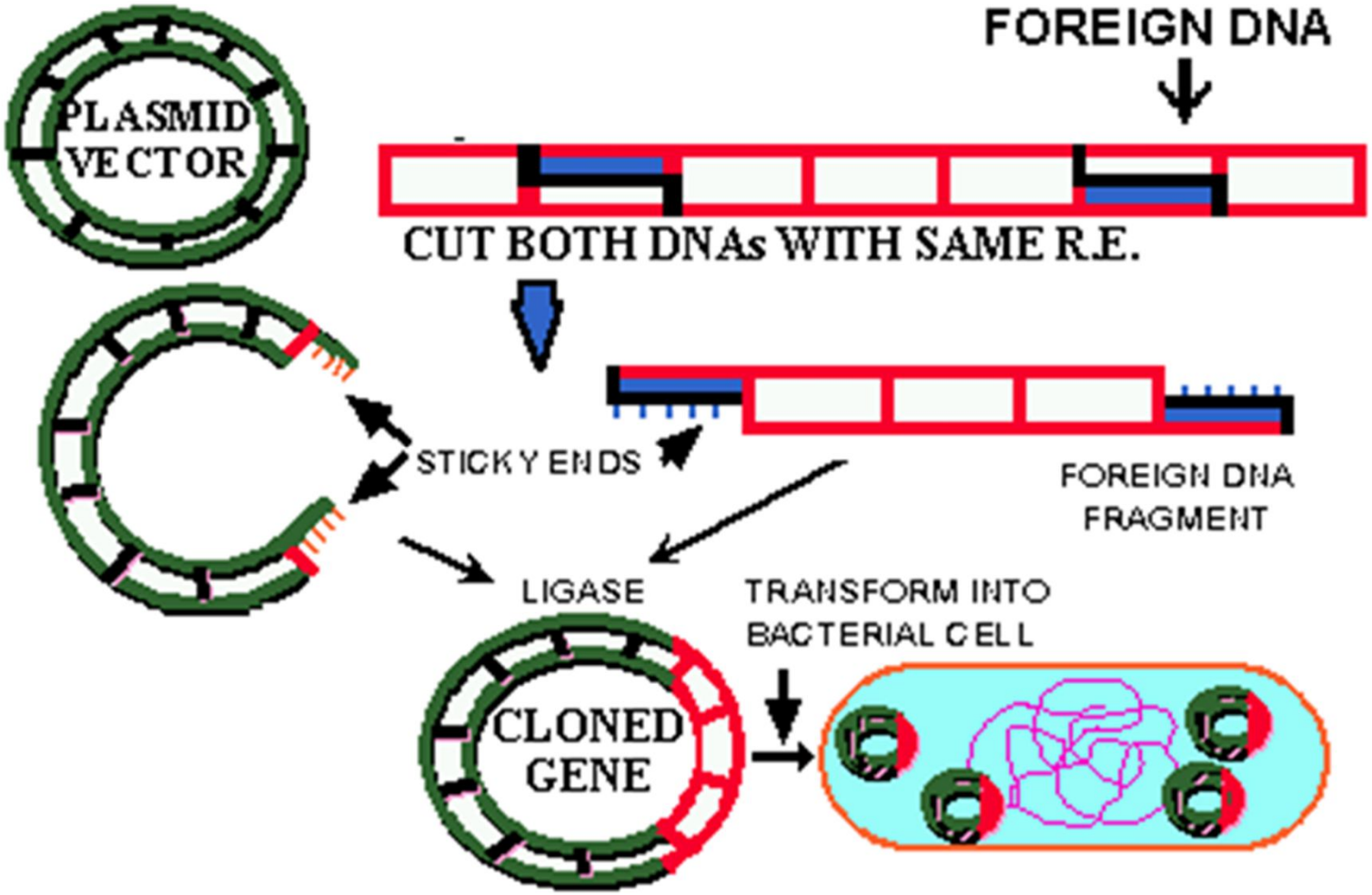

G A A T T C
5'
3'
3'
5'
C T T A A G
5'
3'
3'
5'
5'
3'
EcoRI, restriction endonuclease
G
5'
3'
A A T T C
5'
3'
3'
5'
3'
5'
C T T A A
3'
G

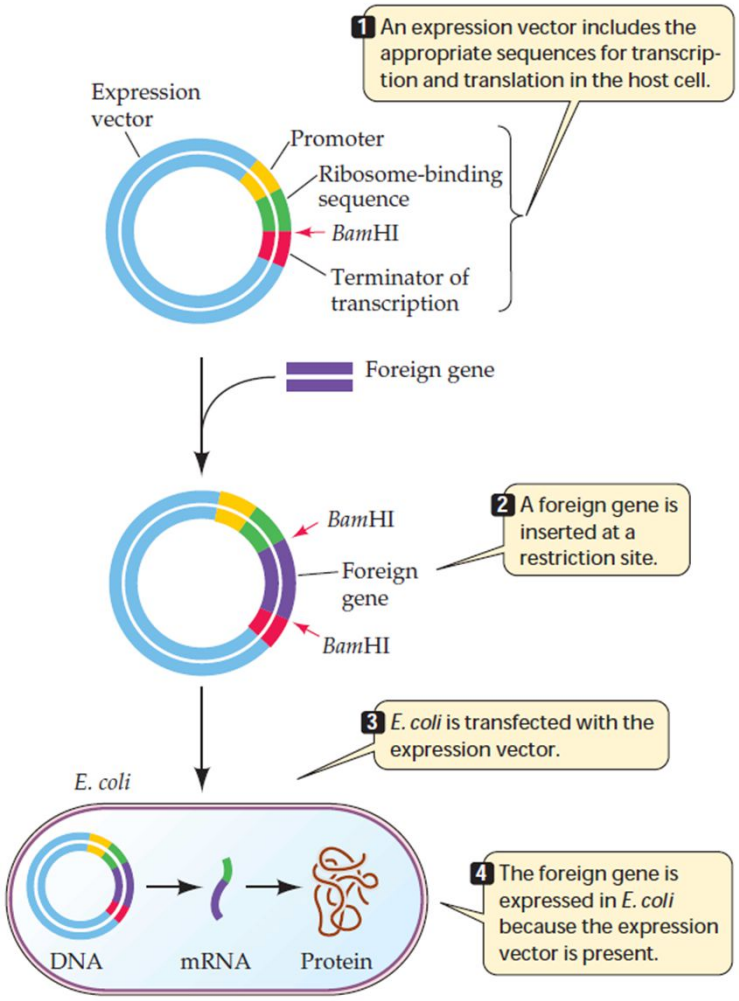
END

Genetic Engineering

Blunt Ends Sticky Ends

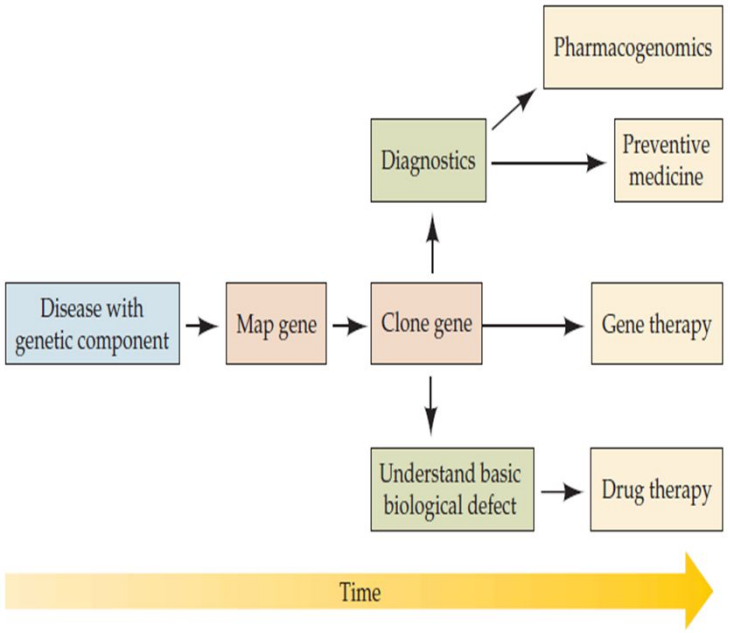



PLASMID VECTOR
FOREIGN DNA
CUT BOTH DNAs WITH SAME R.E.
STICKY ENDS
FOREIGN DNA FRAGMENT
LIGASE
TRANSFORM INTO BACTERIAL CELL
CLONED GENE


1 An expression vector includes the appropriate sequences for transcription and translation in the host cell.
Expression vector
Promoter
Ribosome-binding sequence
BamHI
Terminator of transcription
Foreign gene
BamHI
Foreign gene
BamHI
2 A foreign gene is inserted at a restriction site.
3 E. coli is transfected with the expression vector.
E. coli
DNA
mRNA
Protein
4 The foreign gene is expressed in E. coli because the expression vector is present.

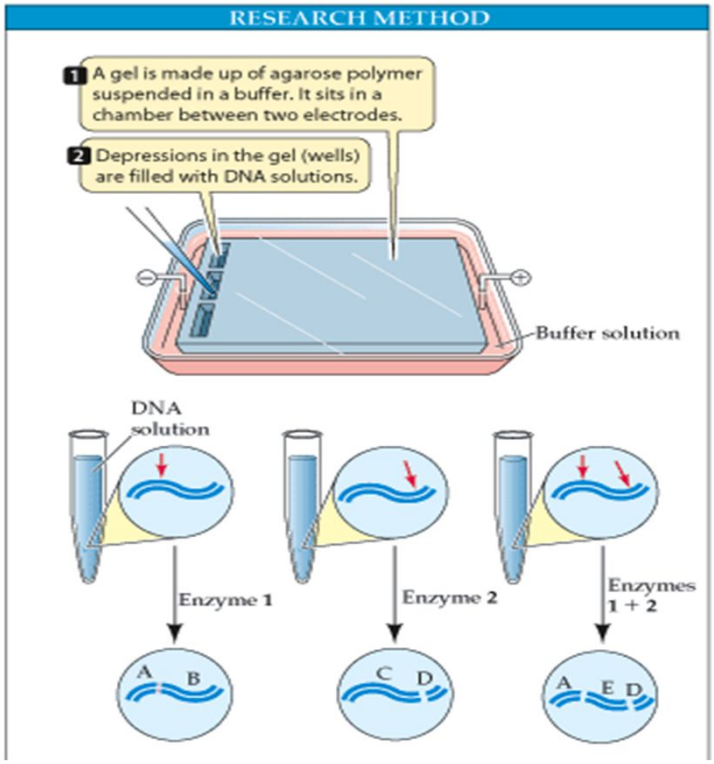
SEQUENCING: Sequencing determines the base pair sequence of a gene. By reading the 3-letter code, sequencing also describes the AMINO ACID SEQUENCE translated from that gene.
MUTATION: It is possible to study the effects of SINGLE AMINO ACID CHANGES on the function of the gene product, which is, after all, the ultimate purpose of the exercise (rev genetics).
MASS PRODUCTION OF PROTEINS: To use the amplified gene to make HUMONGOUS QUANTITIES of the GENE PRODUCT for commercial purposes.

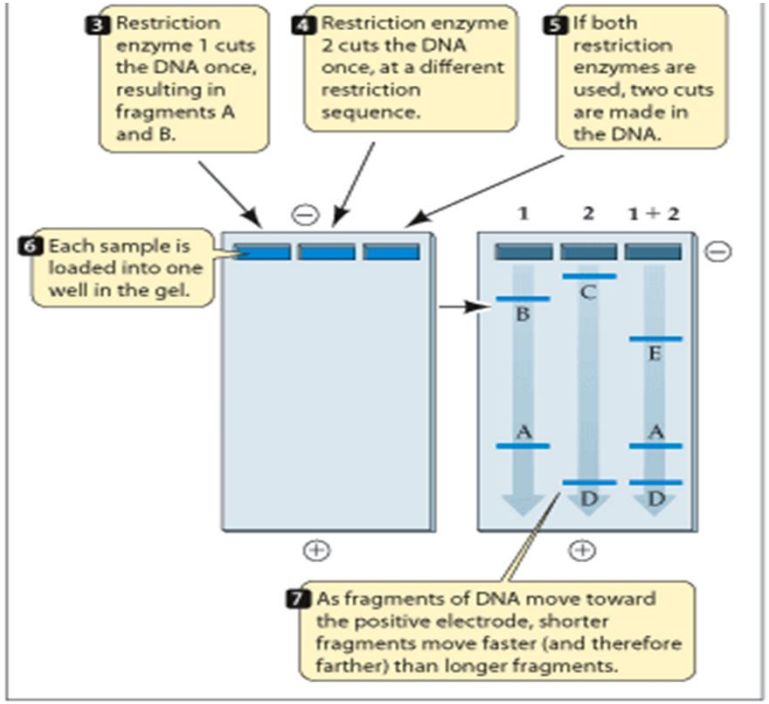
Human Health



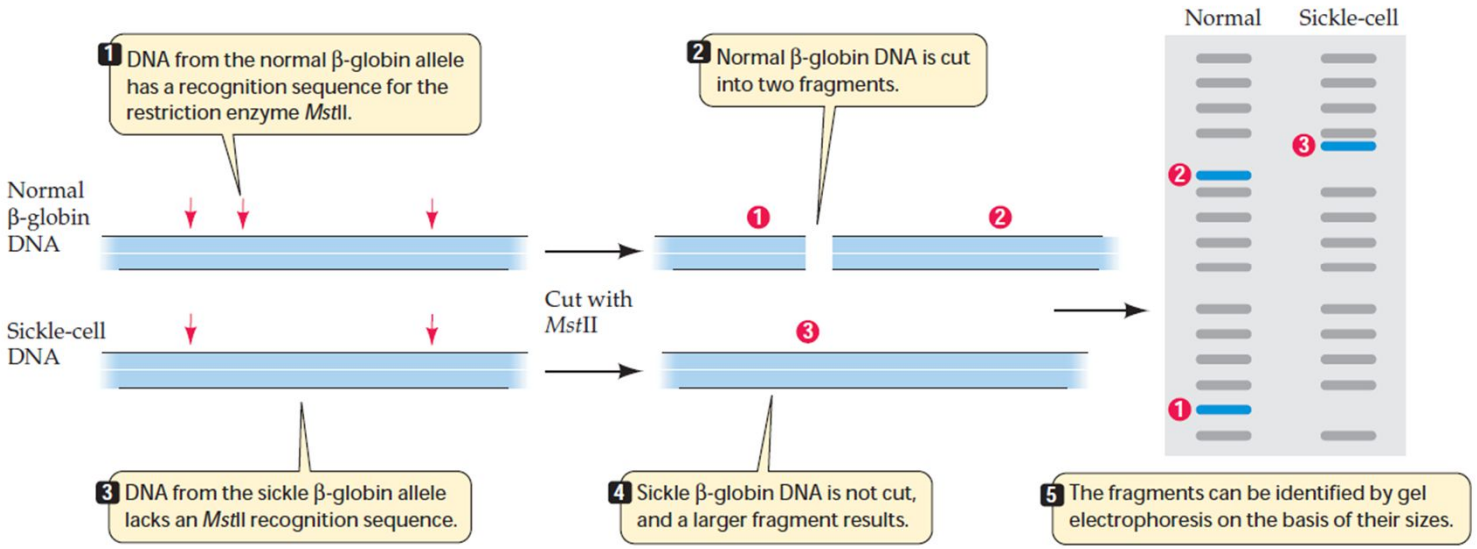
END

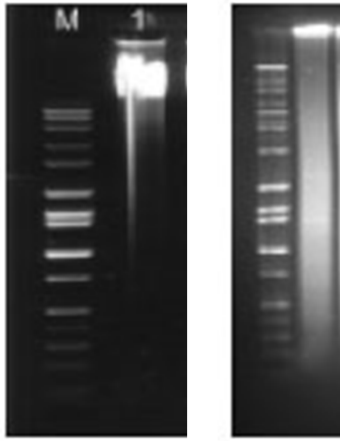
RFLP


RESEARCH METHOD
1 A gel is made up of agarose polymer suspended in a buffer. It sits in a chamber between two electrodes.
2 Depressions in the gel (wells) are filled with DNA solutions.
Buffer solution
DNA solution
Enzyme 1
Enzyme 2
Enzymes 1 + 2
A B
C D
A E D

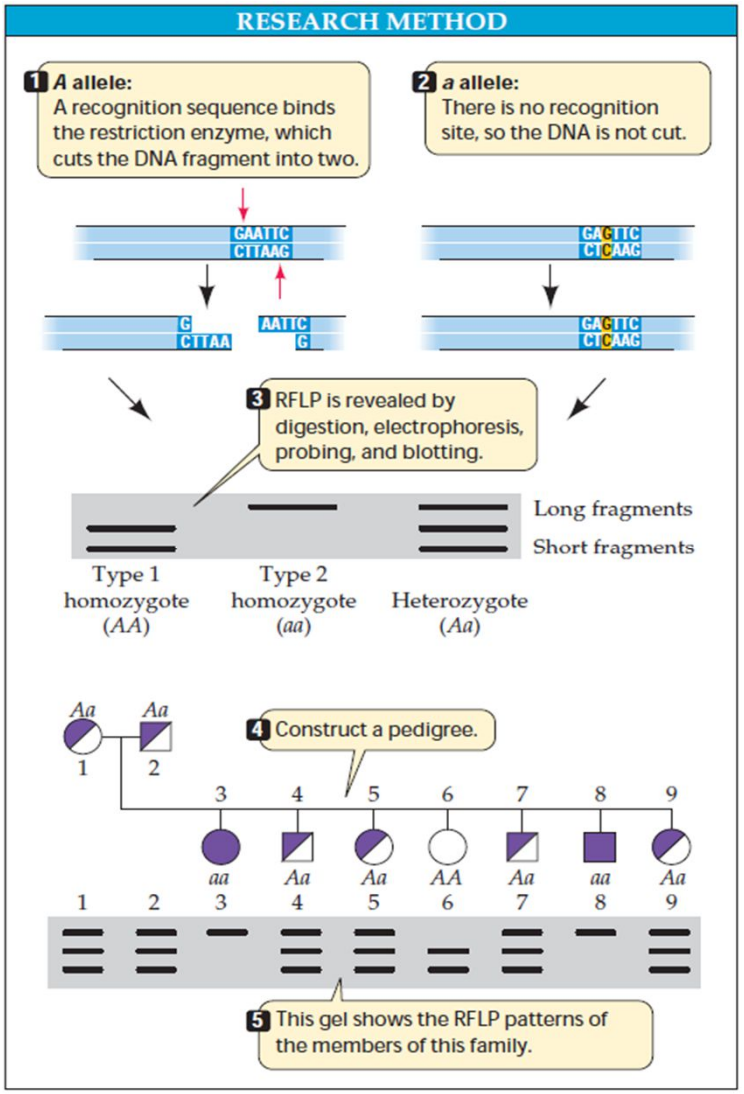

3 Restriction enzyme 1 cuts the DNA once, resulting in fragments A and B.
4 Restriction enzyme 2 cuts the DNA once, at a different restriction sequence.
5 If both restriction enzymes are used, two cuts are made in the DNA.
6 Each sample is loaded into one well in the gel.
1
2
1 + 2
B
C
E
A
A
D
D
7 As fragments of DNA move toward the positive electrode, shorter fragments move faster (and therefore farther) than longer fragments.

RESEARCH METHOD

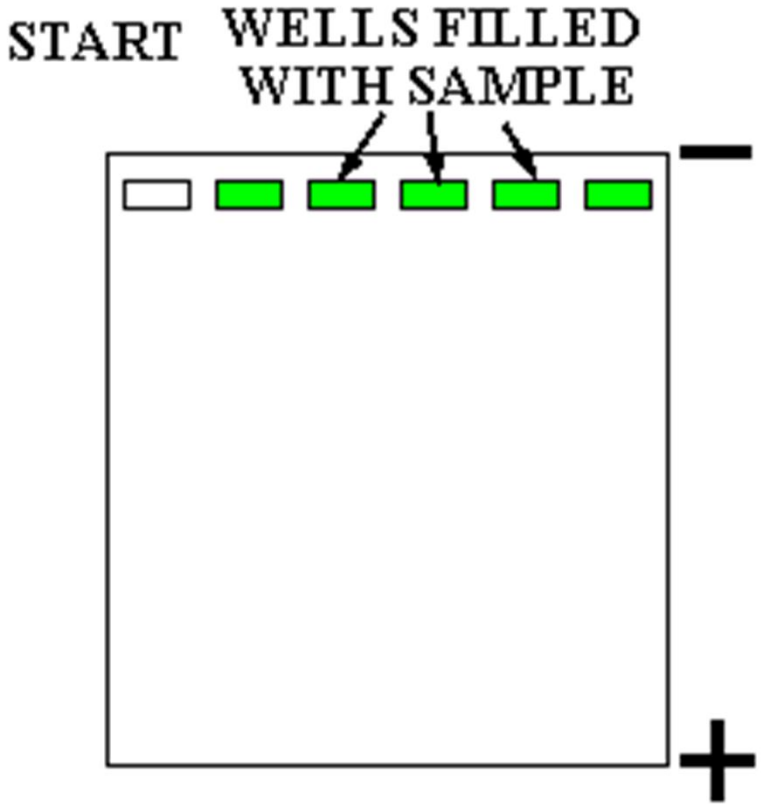




Restriction Fragment
Length polymorphisms (RFLP)
Can reveal mutations in DNA which may indicative of disease

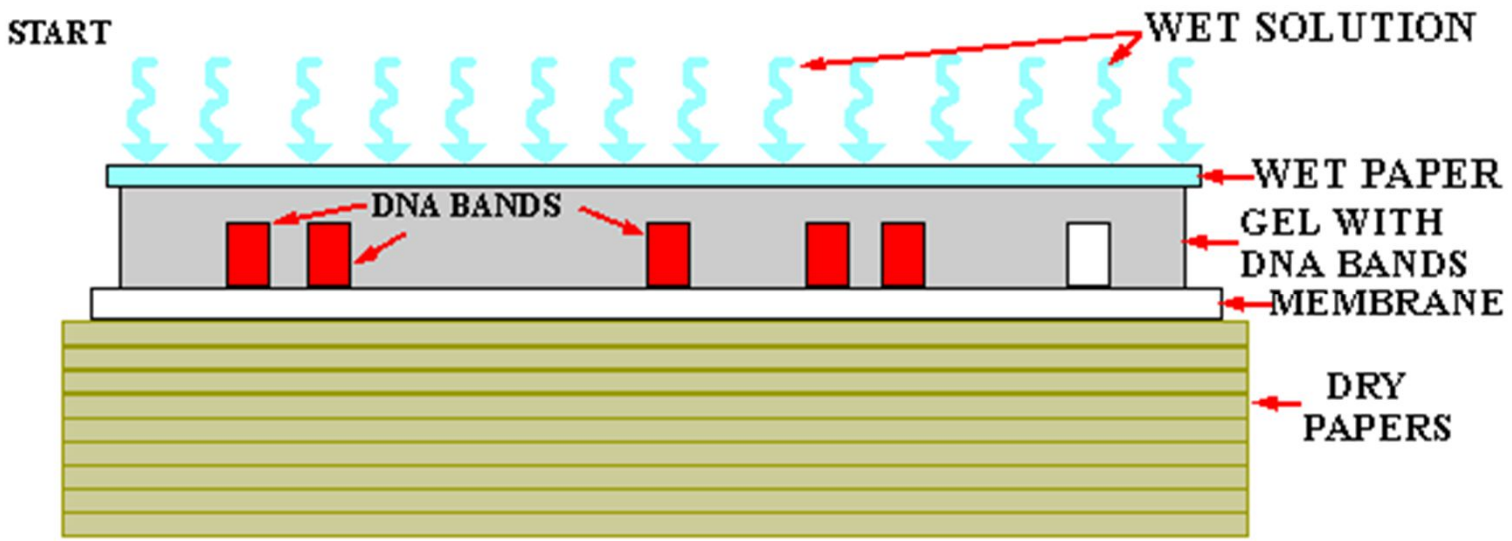


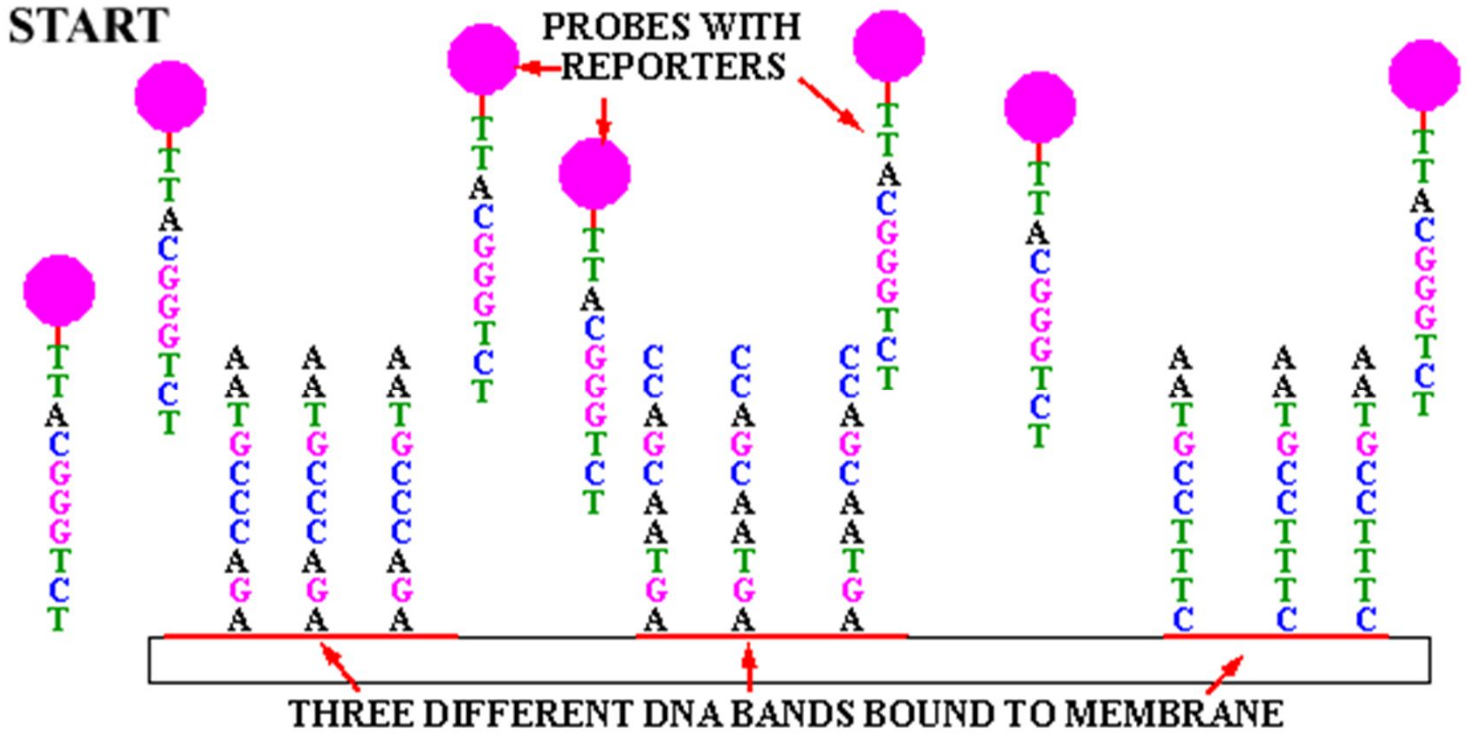
Cut Gel Pic



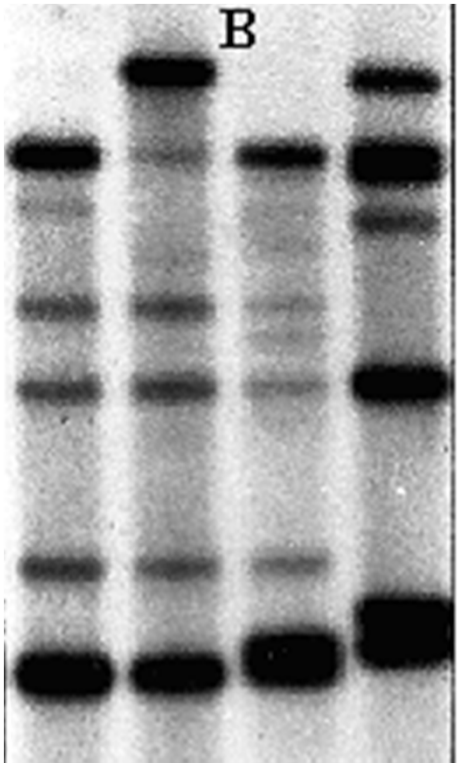
END

DNA Transfer


START
WET SOLUTION
WET PAPER
DNA BANDS
GEL WITH
DNA BANDS
MEMBRANE
DRY
PAPERS


START
PROBES WITH REPORTERS
THREE DIFFERENT DNA BANDS BOUND TO MEMBRANE

DNA Bound to Probe Exposed to X-ray film



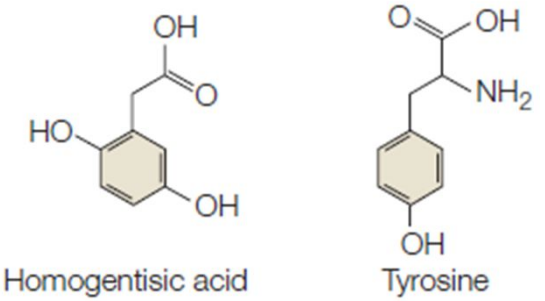
Genetic Information is Converted into Protein

Alkaptonuria ("black urine")

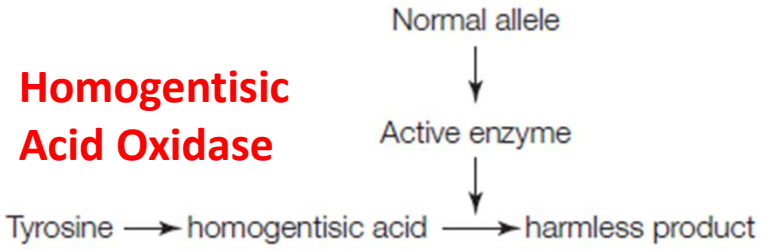
The molecular basis of phenotypes was actually discovered before it was known that DNA was the genetic material.

He linked the biochemical phenotype of the disease to an abnormal gene and a missing enzyme.

Most common in children whose parents were first cousins 12.5%



Homogentisic Acid Oxidase



Neurospora

An altered gene resulted in an altered phenotype, associated with an altered enzyme

Neurospora haploid (n) recessive alleles

X-rays, which act as a mutagen prototrophs (original) converted to auxotrophs (increase)

Some mutant strains could no longer grow on the minimal medium. One group could grow if supplemented with the arginine (*arg* mutants)

Neurospora

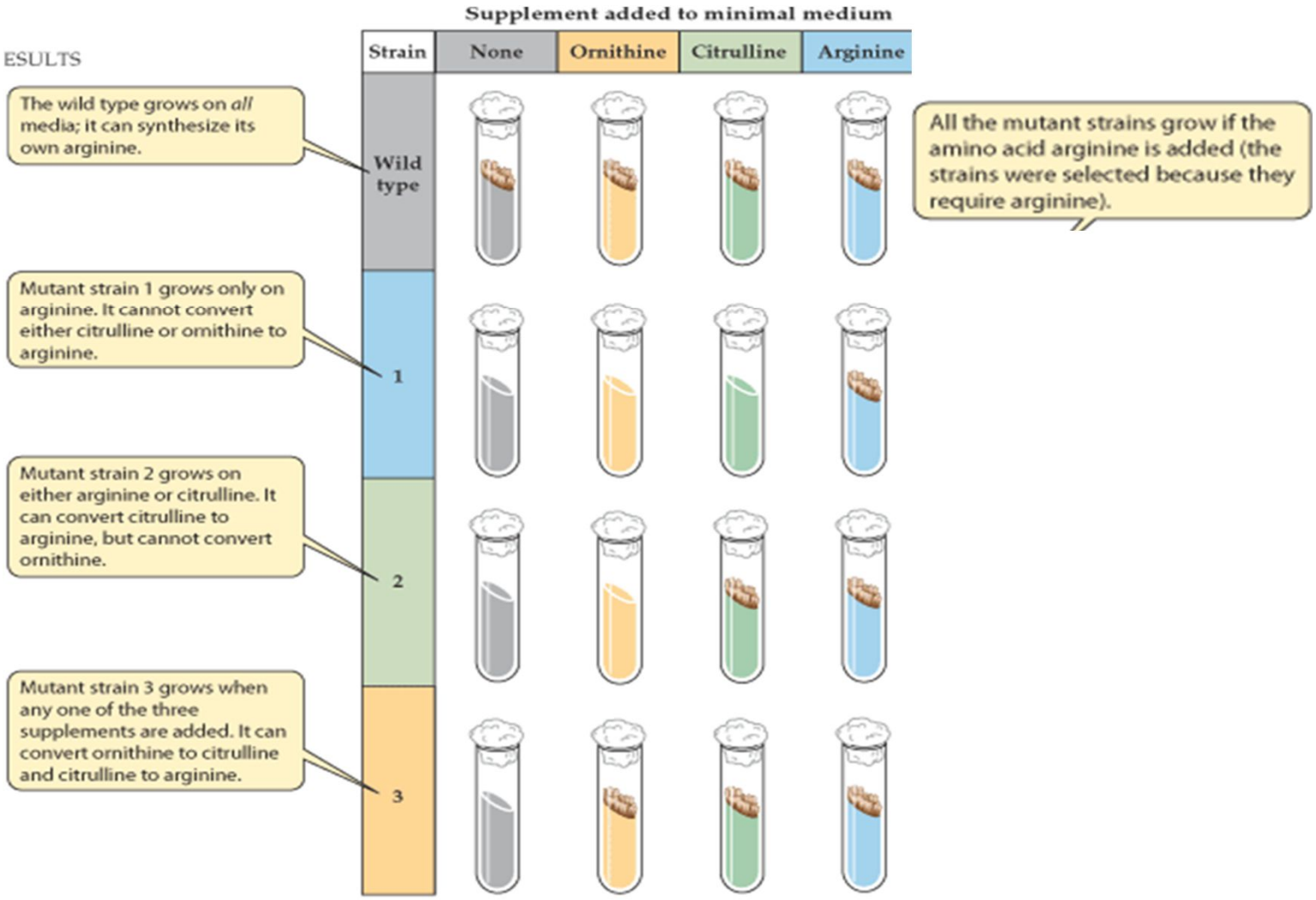
arg mutants were grown in presence of various compounds *suspected intermediates* in the synthetic metabolic pathway for arginine,

B & T classified each mutation as affecting one enzyme or another.

wild-type and mutant cells examined for enzyme activities.

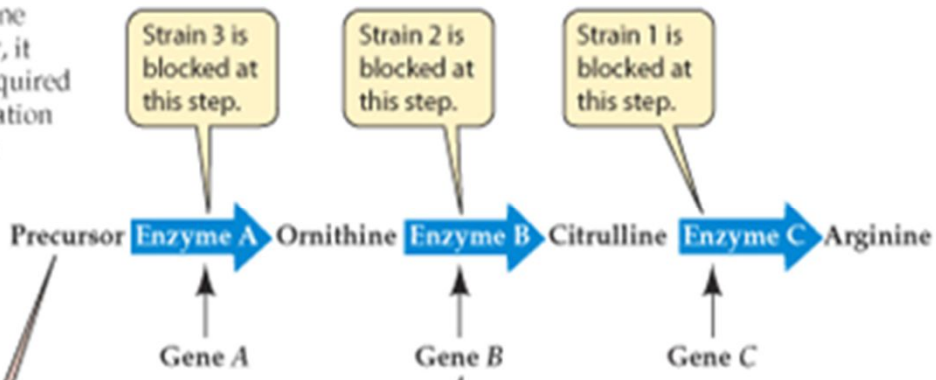
results confirmed: Each mutant strain was indeed missing a single active enzyme in the pathway.

ESULTS

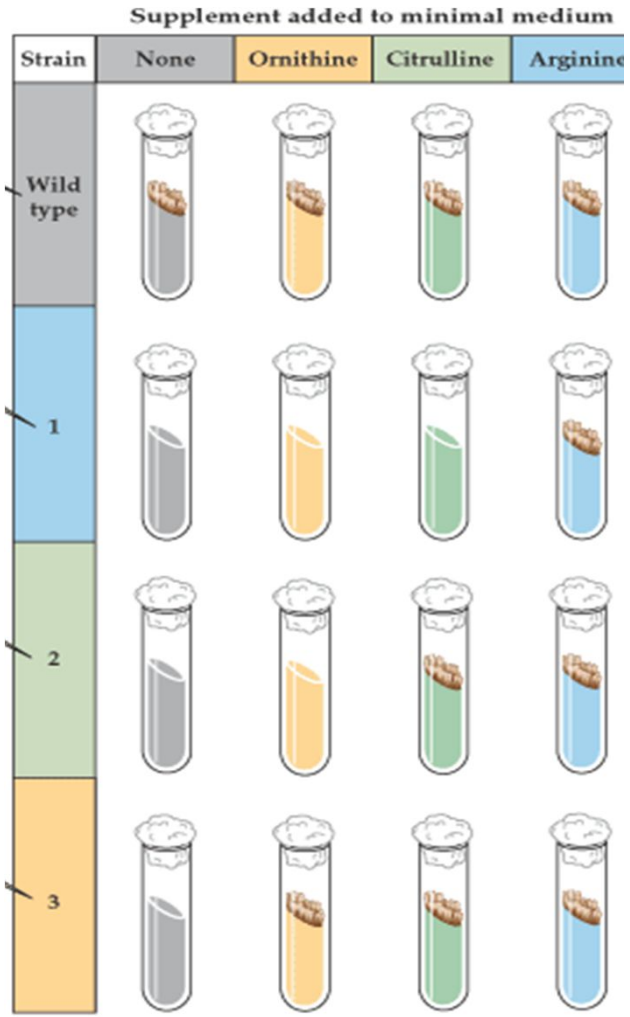

Supplement added to minimal medium
Strain
None
Ornithine
Citrulline
Arginine
Wild type
1
2
3
The wild type grows on *all* media; it can synthesize its own arginine.
Mutant strain 1 grows only on arginine. It cannot convert either citrulline or ornithine to arginine.
Mutant strain 2 grows on either arginine or citrulline. It can convert citrulline to arginine, but cannot convert ornithine.
Mutant strain 3 grows when any one of the three supplements are added. It can convert ornithine to citrulline and citrulline to arginine.
All the mutant strains grow if the amino acid arginine is added (the strains were selected because they require arginine).

INTERPRETATION

If an organism cannot convert one particular compound to another, it presumably lacks an enzyme required for the conversion, and the mutation is in the gene that codes for that enzyme.



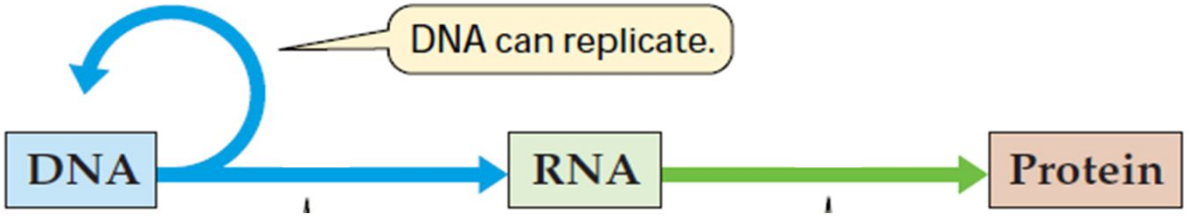
Conclusion: The synthesis of arginine proceeds like this. Each gene specifies a particular enzyme.


Supplement added to minimal medium
Strain
None
Ornithine
Citrulline
Arginine
Wild type
1
2
3

END

The Dogma

Info flows from DNA to RNA to proteins



How does genetic information get from the nucleus to the cytoplasm?

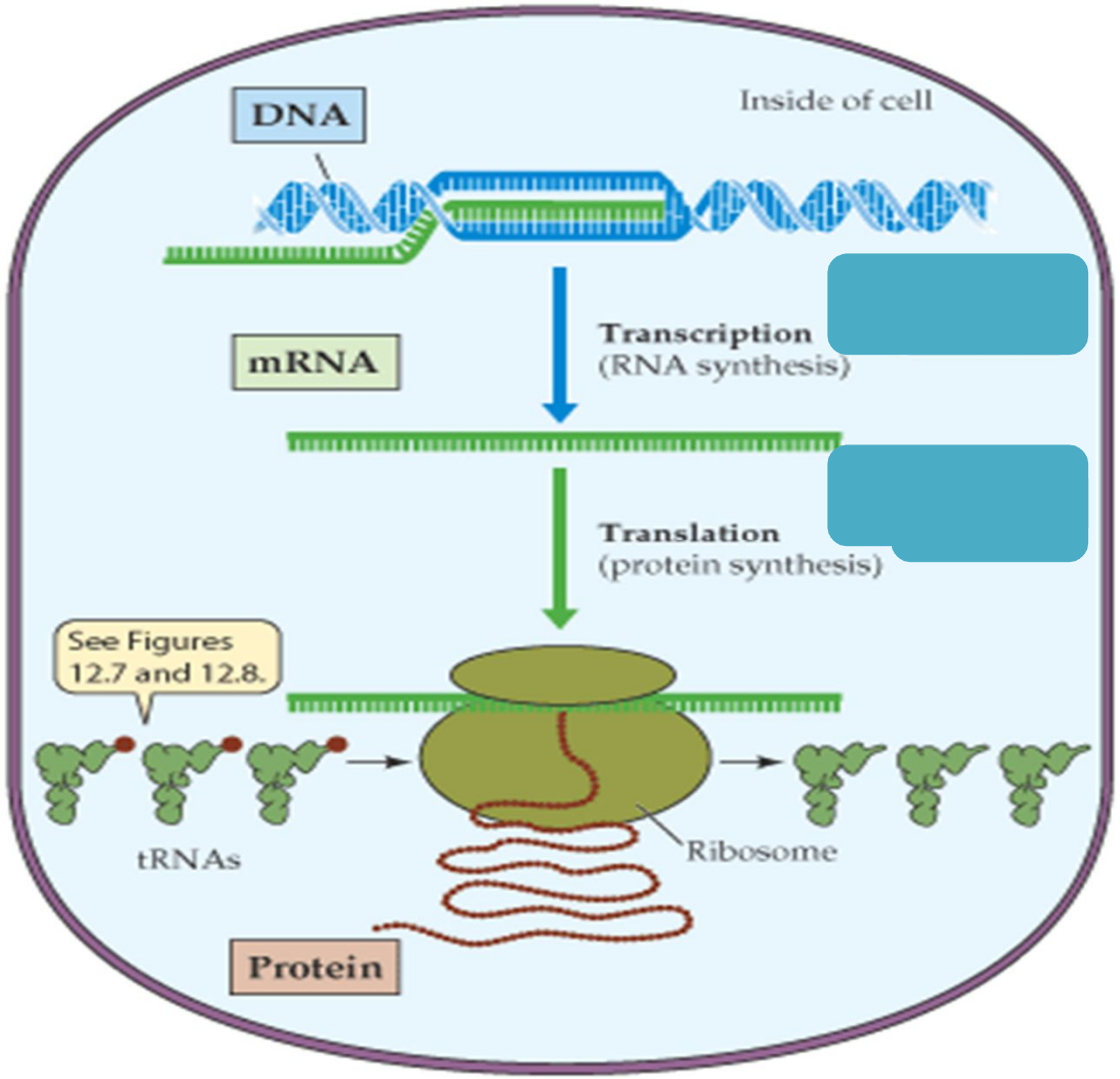
What is the relationship between a specific nucleotide sequence in DNA and a specific amino acid sequence in a protein?)

THE MESSENGER HYPOTHESIS AND TRANSCRIPTION

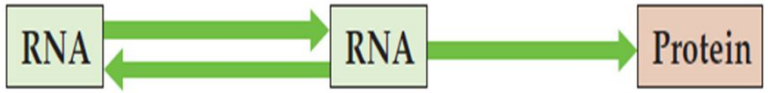
RNA molecule forms complementary copy of one DNA strand, moves to cytoplasm serves as template for protein synthesis.

THE ADAPTER HYPOTHESIS AND TRANSLATION

Adapter molecule binds a specific amino acid with one region and recognize a sequence of nucleotides with another region to translate the language of DNA into the language of proteins

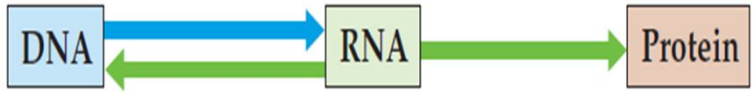

DNA
Inside of cell
mRNA
Transcription
(RNA synthesis)
Translation
(protein synthesis)
See Figures
12.7 and 12.8.
tRNAs
Ribosome
Protein

Influenza virus, & poliovirus



Transcribing from RNA to RNA

HIV

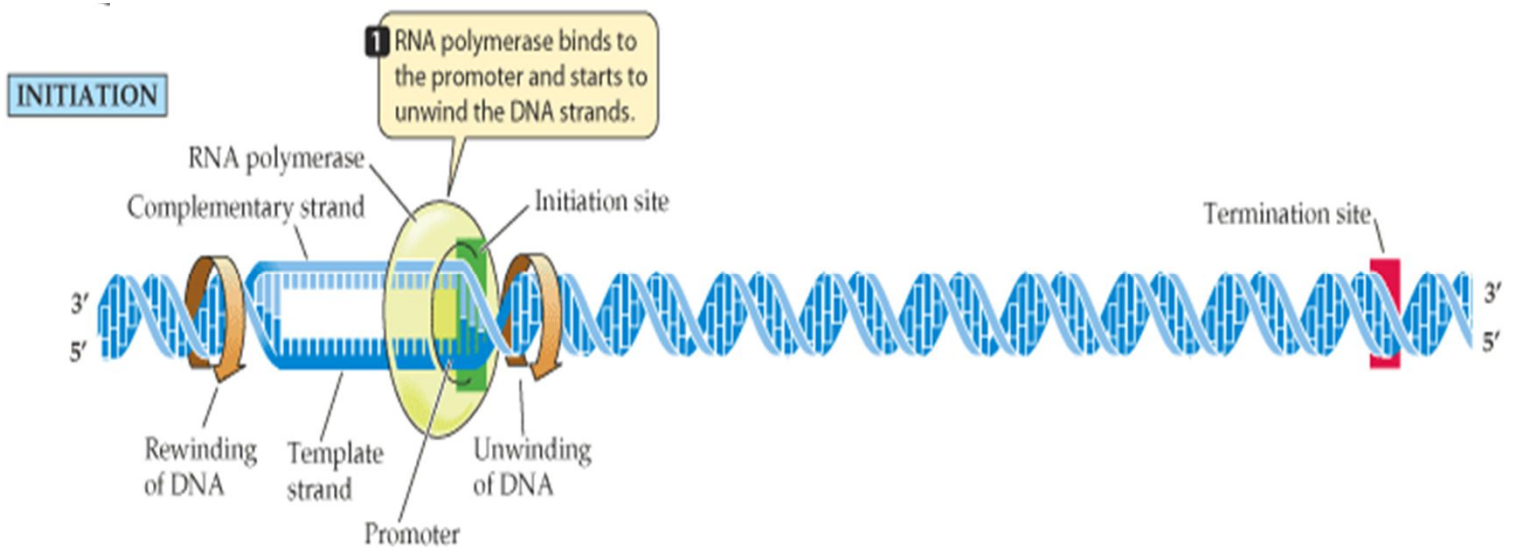


make a DNA copy of their genome

END

Transcription

Initiation begins at promoter -tightly binds RNA Pol

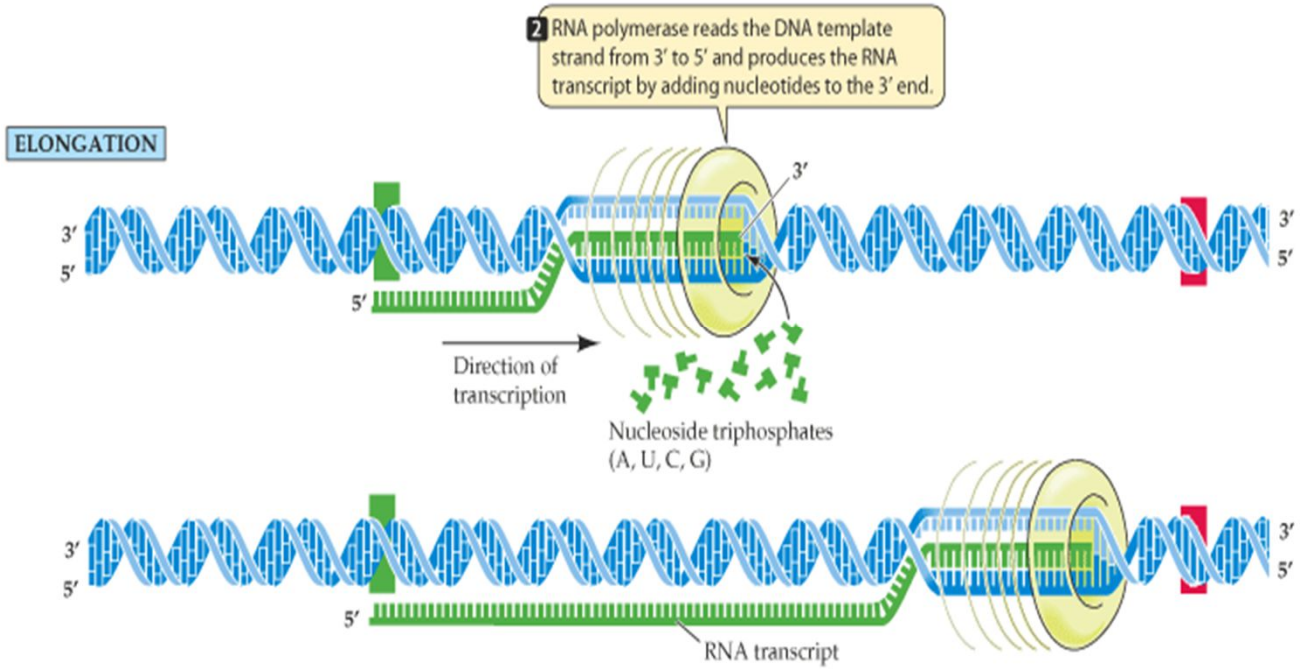


1) Where to start transcription.
2) Which strand to read.
3) The direction to take start from.

Part of promoter is initiation site (transcription start).

RNA Pol moves 3' to 5', synthesis 5' to 3', ~~primer~~

Unwinds 20 bp at a time.

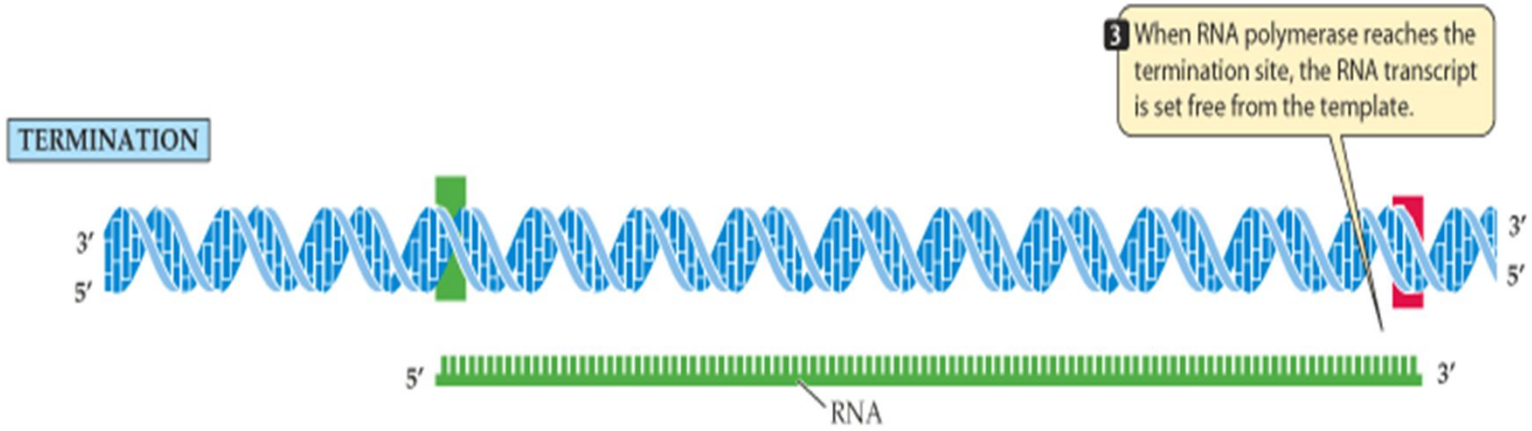


Transcript antiparallel to DNA template strand.

No proof reading error rate 1 in 10,000-100,000

Translation starts before termination in prokaryotes.

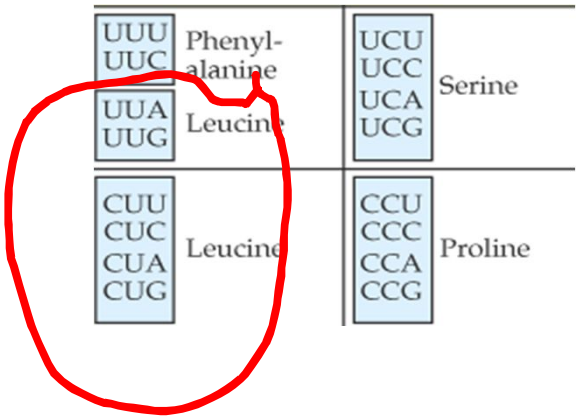
Pre-mRNA 1st product of transcription in eukaryotes.



Promoter
Terminator
5'
3'
3'
5'

Genetic Code

Three mRNA nucleotides form a codon

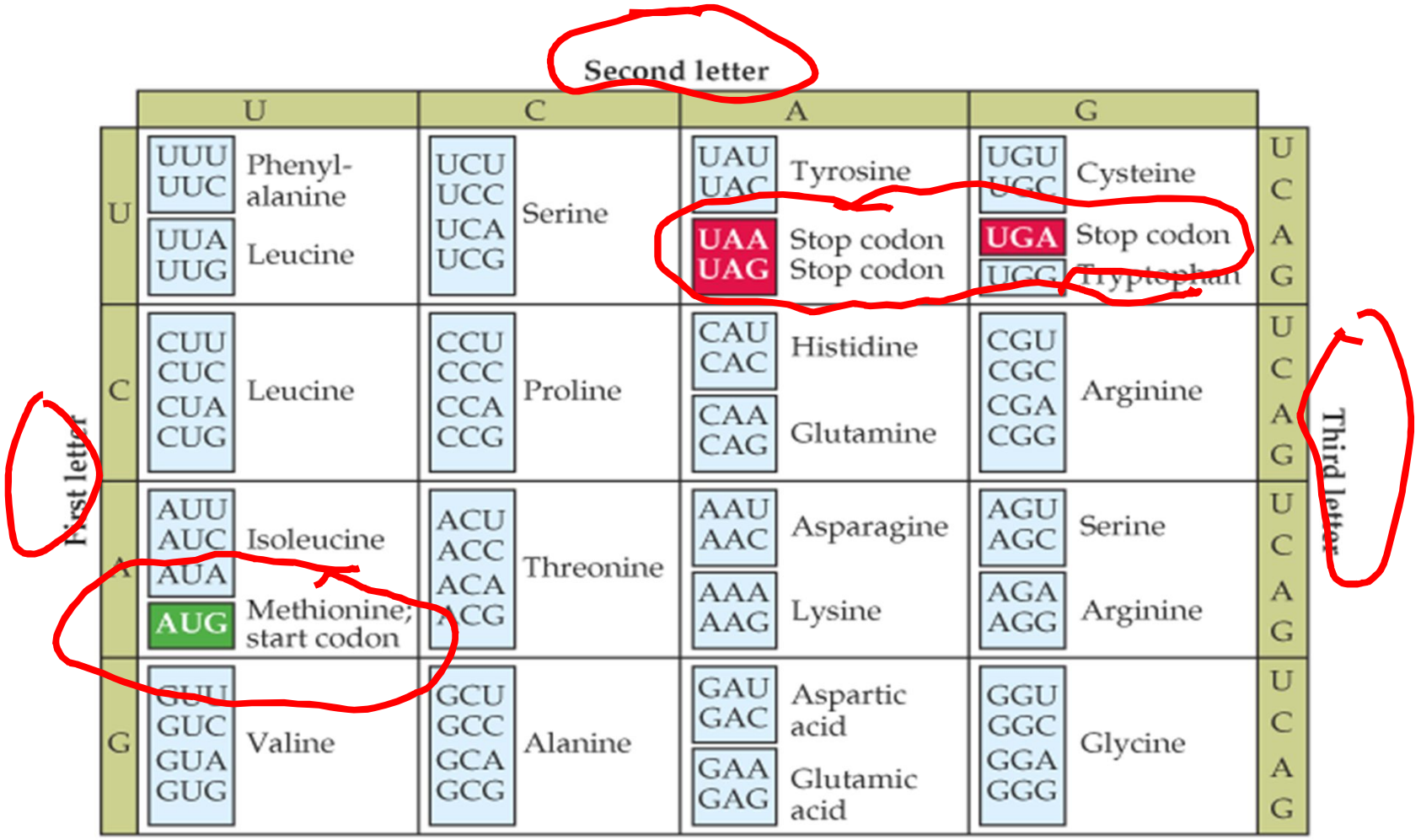


which is complementary & antiparallel to corresponding DNA.

Genetic code relates codons to their specific AA.

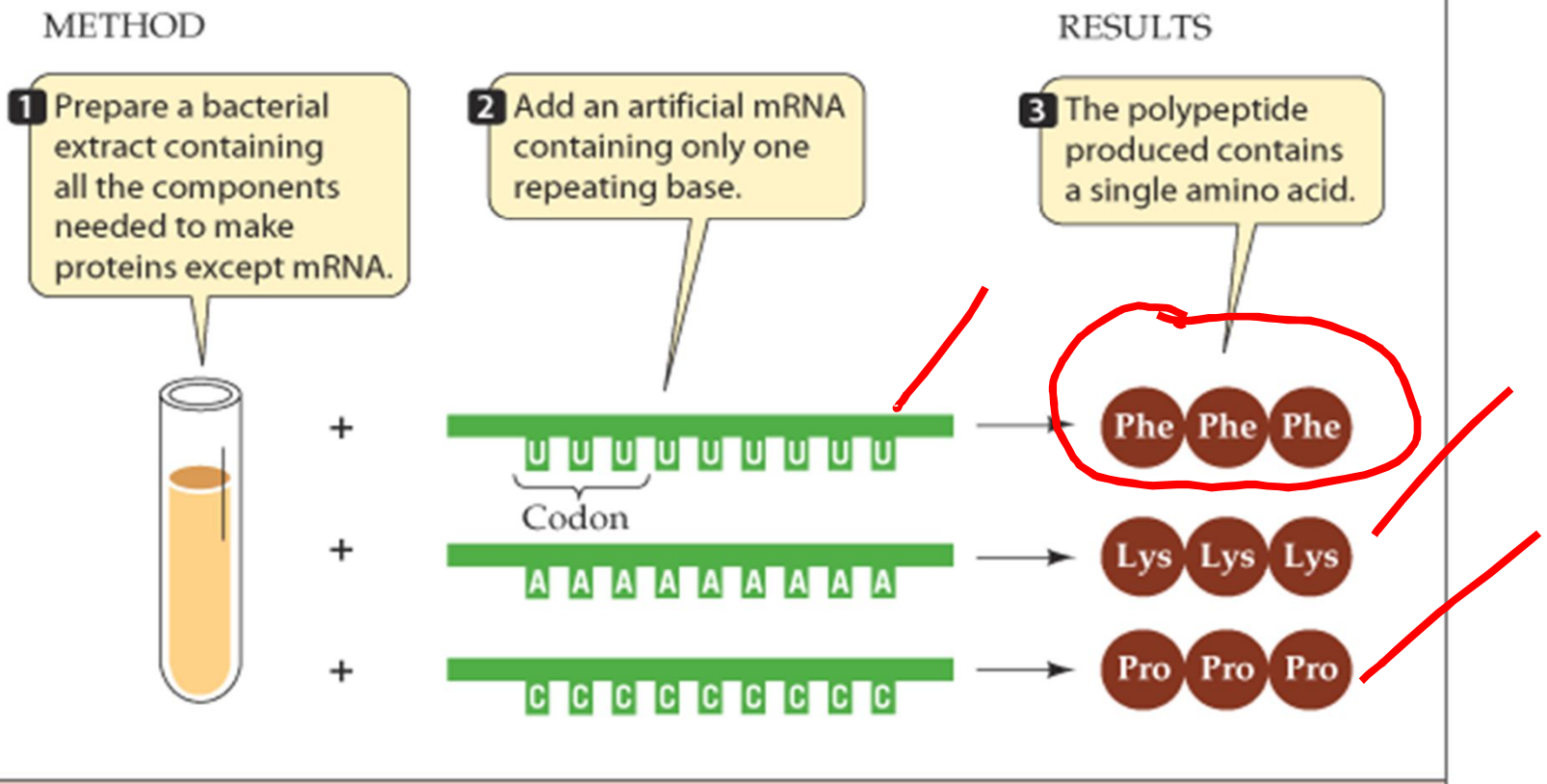
~ Universal & Redundant but not ambiguous some exceptions.

Start & Stop


Second letter
U
C
A
G
First letter
Third letter
U
UUU UUC Phenyl-alanine
UUA UUG Leucine
UCU UCC UCA UCG Serine
UAU UAC Tyrosine
UAA Stop codon
UAG Stop codon
UGU UGC Cysteine
UGA Stop codon
UGG Tryptophan
C
CUU CUC CUA CUG Leucine
CCU CCC CCA CCG Proline
CAU CAC Histidine
CAA CAG Glutamine
CGU CGC CGA CGG Arginine
A
AUU AUC AUA Isoleucine
AUG Methionine; start codon
ACU ACC ACA ACG Threonine
AAU AAC Asparagine
AAA AAG Lysine
AGU AGC Serine
AGA AGG Arginine
G
GUU GUC GUA GUG Valine
GCU GCC GCA GCG Alanine
GAU GAC Aspartic acid
GAA GAG Glutamic acid
GGU GGC GGA GGG Glycine

EXPERIMENT

Question: What are the amino acids specified by the triplet codons UUU, AAA, and CCC?



Conclusion: UUU is an mRNA codon for phenylalanine.
AAA is an mRNA codon for lysine.
CCC is an mRNA codon for proline.

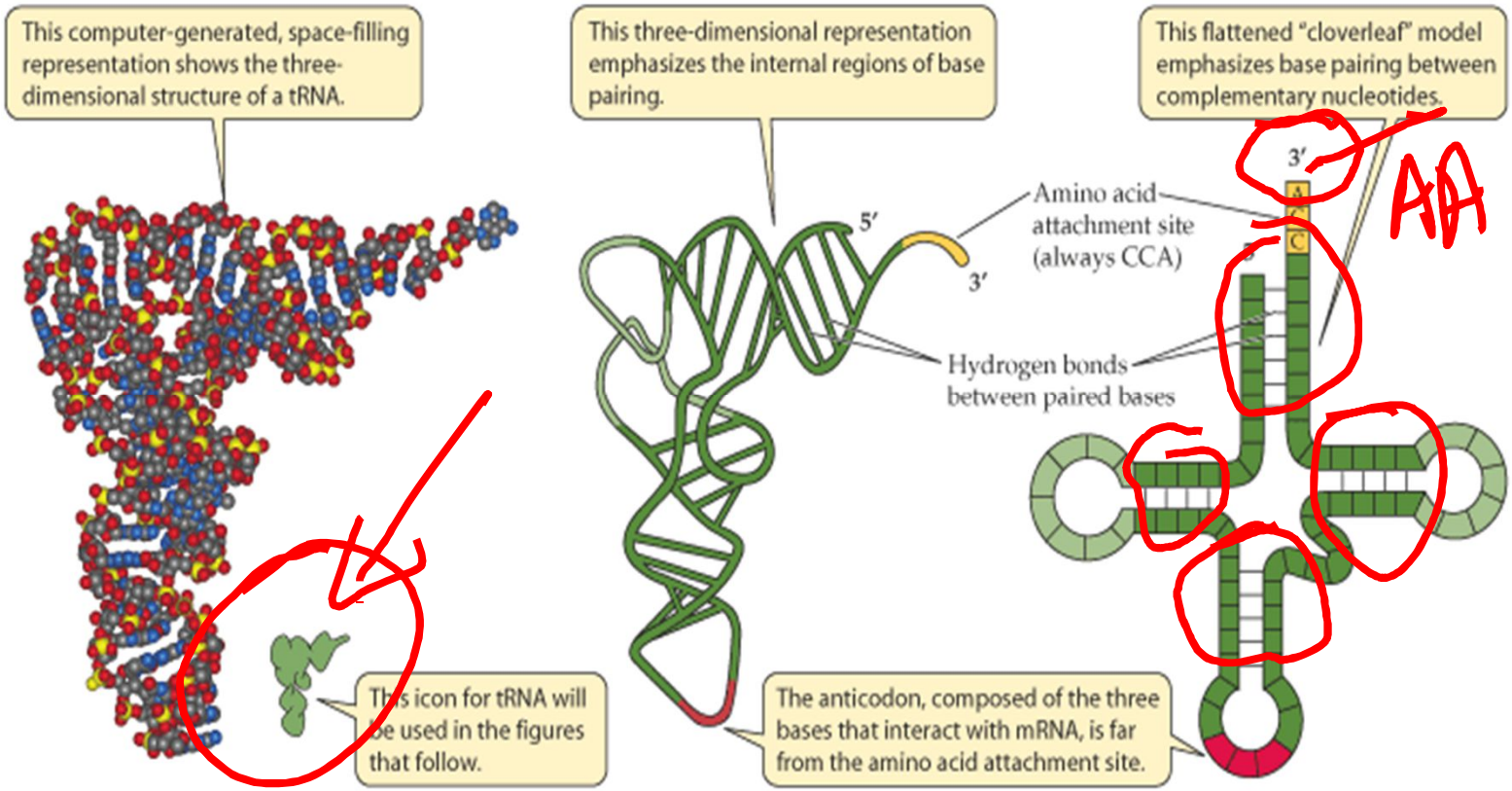
tRNA Translates Nucleotide Language into AA Language

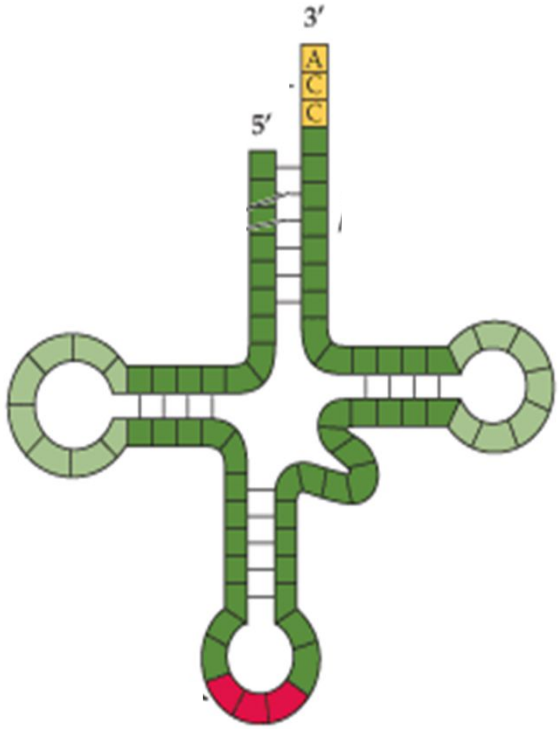
1) Carries AA
2) Associates with mRNA
3) Binds ribosome

tRNA has 75-80 nucleotides with internal H-bonding.

Conformation of tRNA allows it to bind ribosome.

3' end of tRNA (CCA) is attached to AA & mid point has anticodon (antiparallel).


This computer-generated, space-filling representation shows the three-dimensional structure of a tRNA.
This three-dimensional representation emphasizes the internal regions of base pairing.
This flattened "cloverleaf" model emphasizes base pairing between complementary nucleotides.
5′
3′
Amino acid attachment site (always CCA)
3′
A
C
C
Hydrogen bonds between paired bases
This icon for tRNA will be used in the figures that follow.
The anticodon, composed of the three bases that interact with mRNA, is far from the amino acid attachment site.


3′
A
C
C
5′

Charging tRNA

Aminoacyl-tRNA Synthetase

Specific for one AA & its corresponding tRNA.

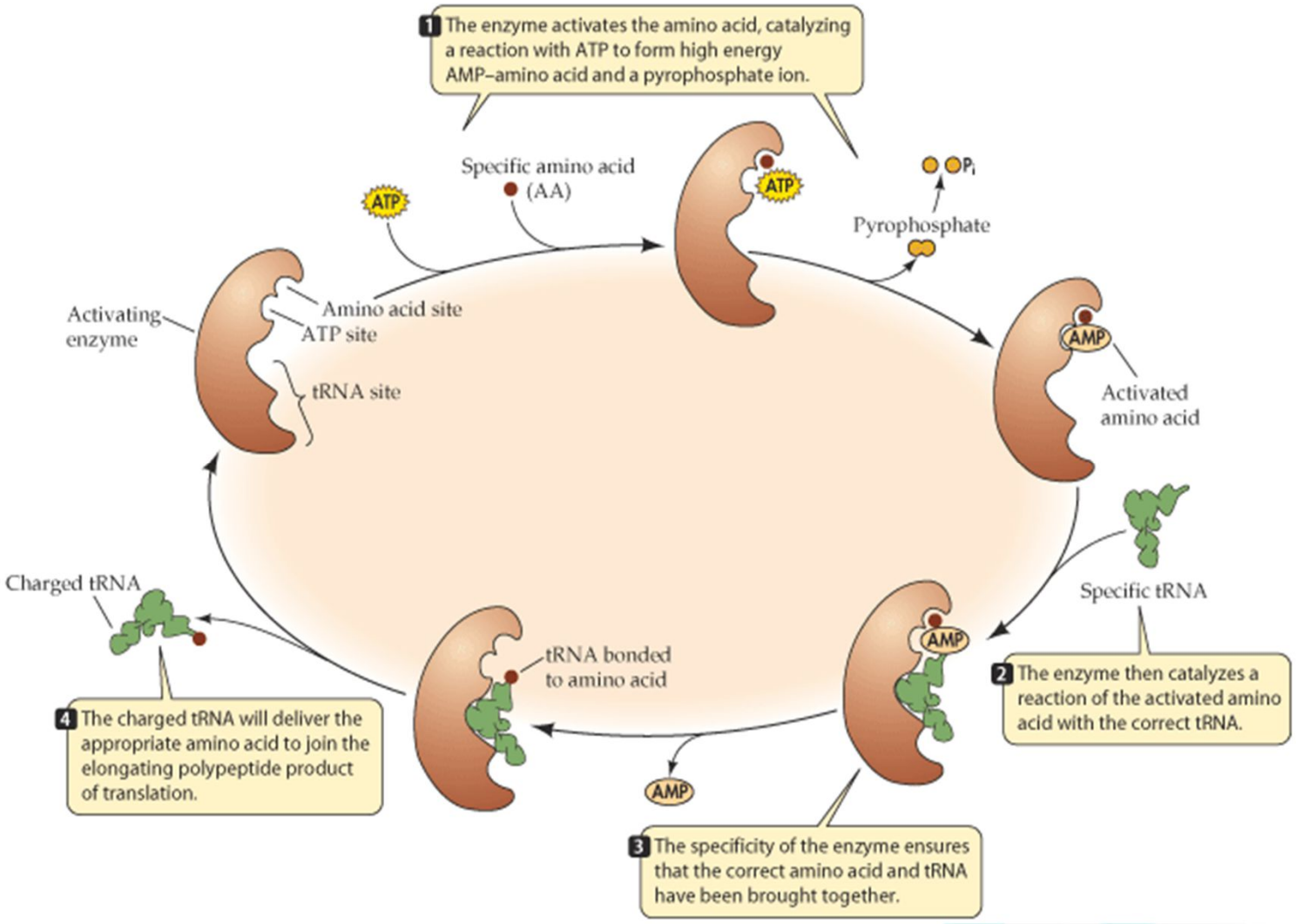
Enzyme has three active sites for three molecules.

amino acid + ATP → aminoacyl-AMP + PPi
aminoacyl-AMP + tRNA → aminoacyl-tRNA + AMP

-COOH of AA is attached to 3' end of tRNA's free OH on ribose.

This high energy bond provides energy for peptide bond synthesis (2nd G code).

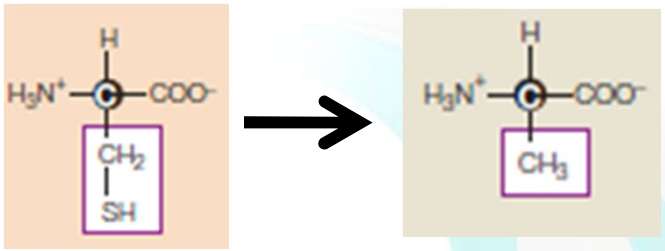
Charging


1 The enzyme activates the amino acid, catalyzing a reaction with ATP to form high energy AMP–amino acid and a pyrophosphate ion.
Specific amino acid (AA)
ATP
Activating enzyme
Amino acid site
ATP site
tRNA site
Pyrophosphate
Pi
AMP
Activated amino acid
Specific tRNA
2 The enzyme then catalyzes a reaction of the activated amino acid with the correct tRNA.
3 The specificity of the enzyme ensures that the correct amino acid and tRNA have been brought together.
tRNA bonded to amino acid
Charged tRNA
4 The charged tRNA will deliver the appropriate amino acid to join the elongating polypeptide product of translation.

Benzer Experiment

Amino acid/tRNA recognized?

1) Took the cys-tRNA molecule and chemically modified the cysteine, converting it into alanine.
2) Hybrid charged tRNA was put into a protein synthesizing system.
3) Everywhere in the synthesized protein where cysteine was supposed to be, alanine appeared instead.
4) Protein synthesis machinery recognizes the anticodon of the charged tRNA, not the amino acid attached to it.



Charging

Charging

Translation Workbench

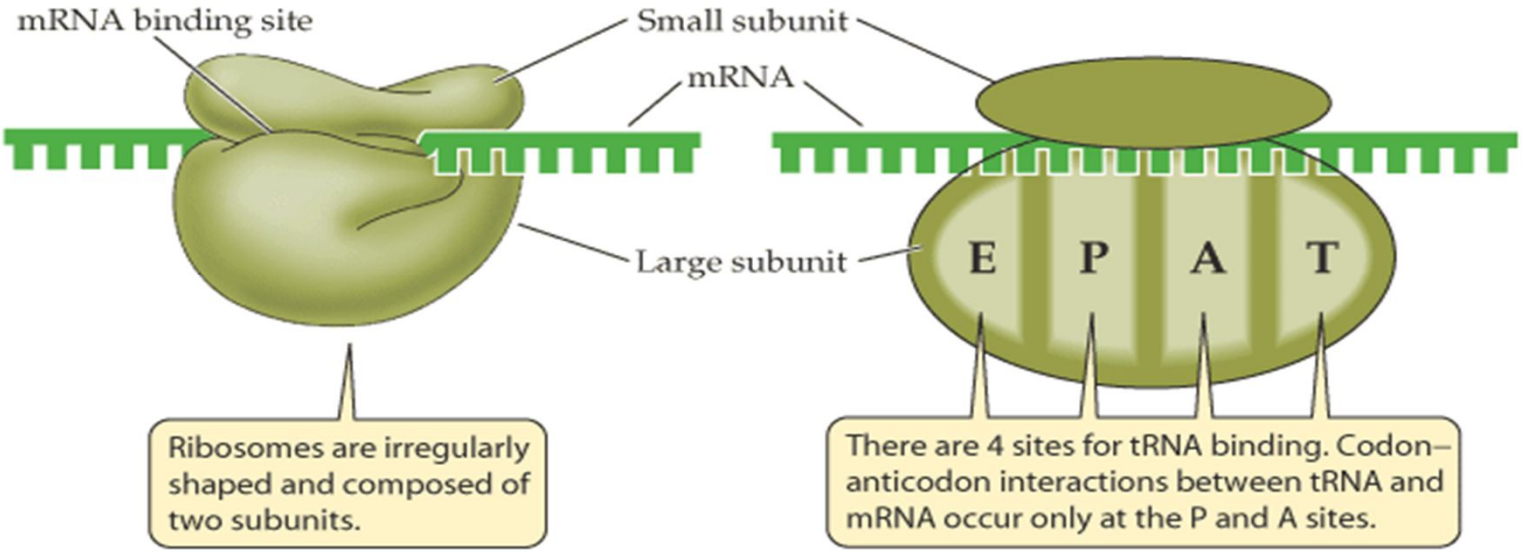
Two Subunits

Large subunit = 3 rRNAs + 45 poly peptides.

Small subunit = 1 rRNA + 33 poly peptides.

Ionic & hydrophobic forces hold complex together.

Complex self assembles



Ribosomes

A ribosome can use any mRNA and all species of charged tRNAs, and thus can be used to make many different polypeptide products.

The ribosomes of prokaryotes Have different ribosomal proteins and RNAs.

Antibiotics that target the prokaryotic ribosome (tetracycline) can be used to kill bacteria without harming a patient's cells.

Mitochondria and chloroplasts also contain ribosomes, some of which are similar to those of prokaryotes

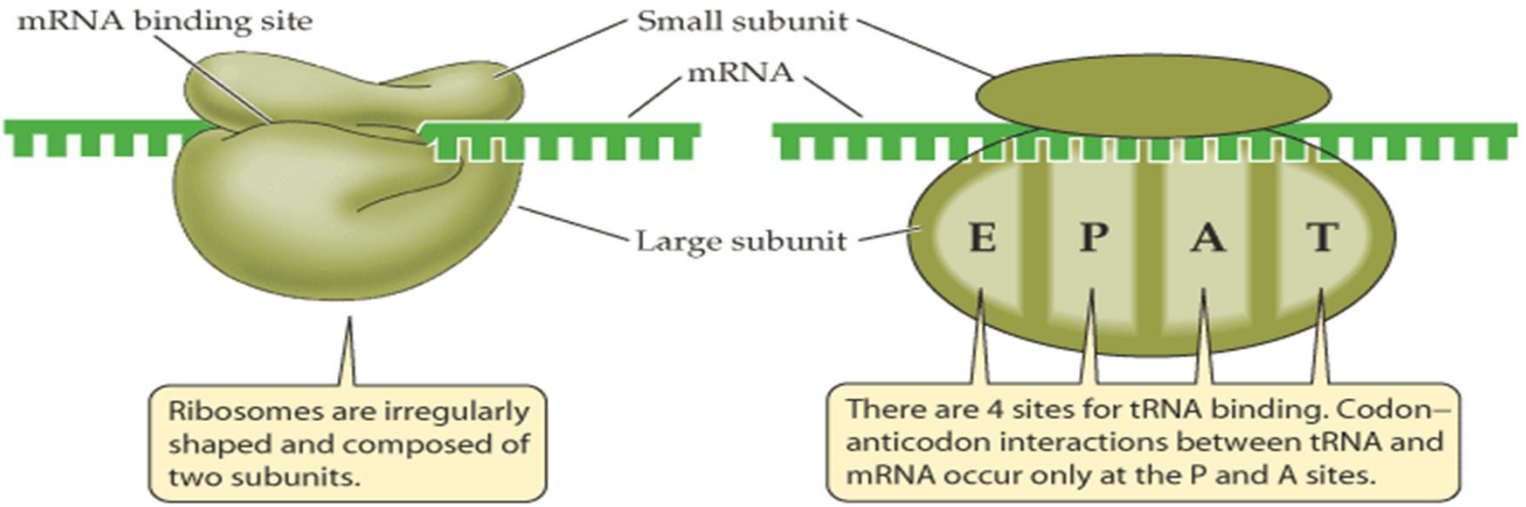
Large Subunit

T (transfer) site: first site where charged tRNA lands with the help of transfer factor

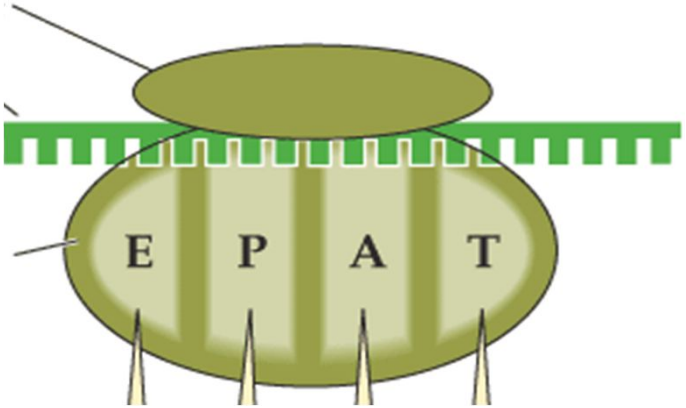
A (AA) site: Anticodon binds to mRNA codon

P (polypeptide) site: tRNA adds AA to polypeptide

E (exit) site: after AA transfer tRNA exits here

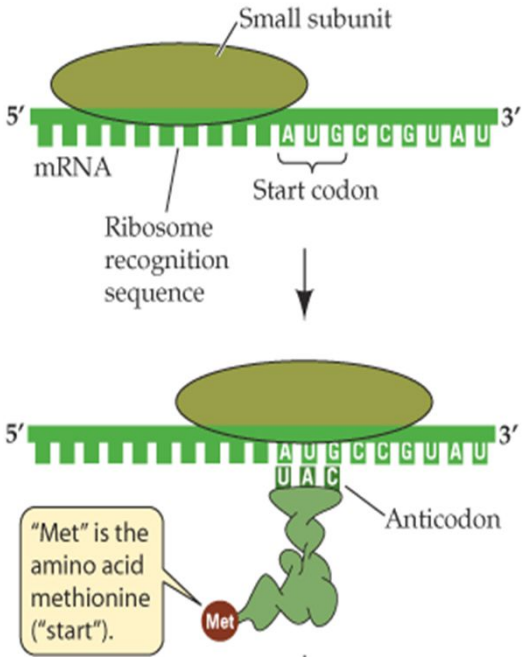


Ribosome



END

Translation

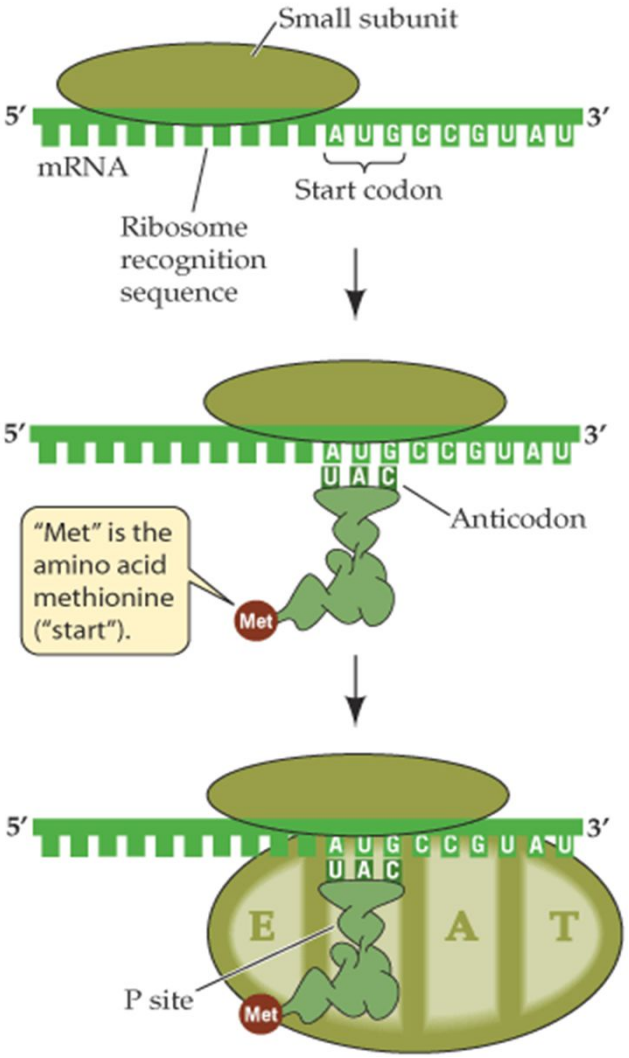


1) In prokaryotes, the rRNA of the small ribosomal subunit first binds to a complementary ribosome binding site upstream start codon (AGGAGG; known as the Shine–Dalgarno sequence) on the mRNA.

2) Anticodon of methionine charged tRNA binds start the codon.

mRNA 5′ . . . A G G A G G. . . (start codon) . . . 3′

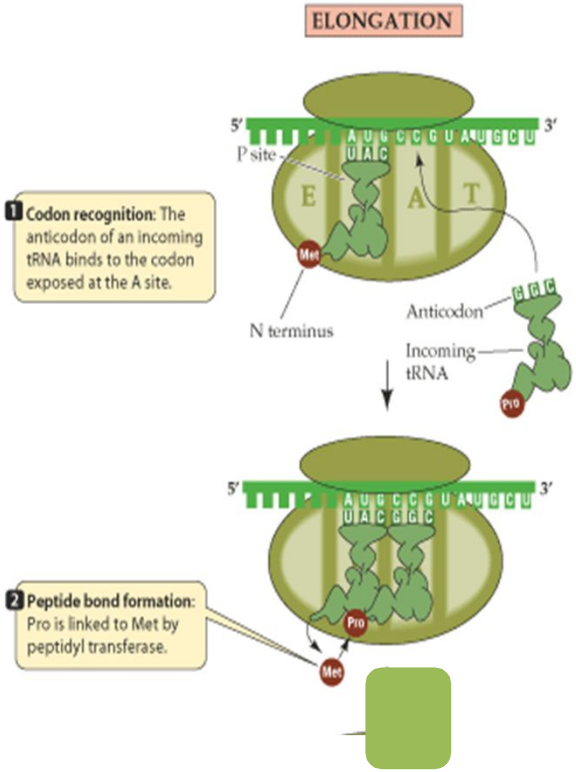
rRNA 3′ . . . U C C U C C. . . (P site) . . . 5′



3) Mehtionine can be removed after translation.

4) rRNA in small subunit stabilizes H-bonding between t & m RNA.

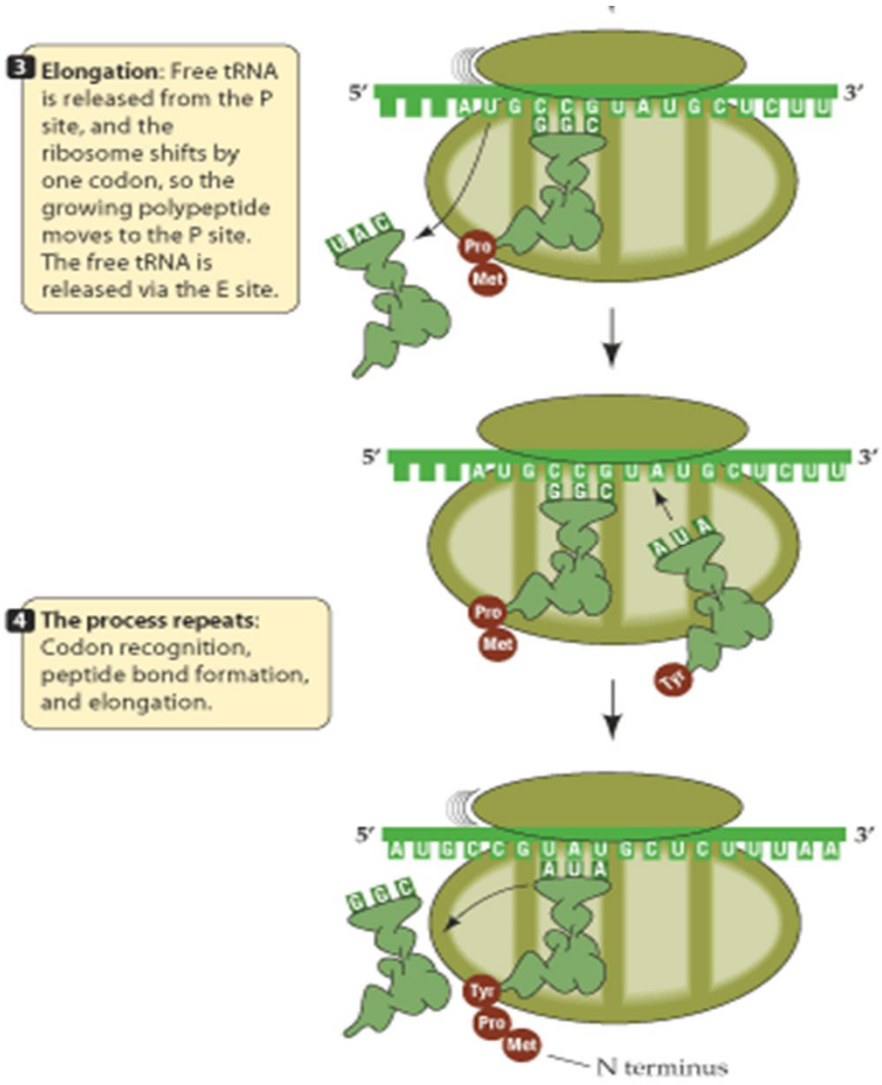
5) Large subunit facilitated by initiation factors binds the complex with methionine charged tRNA at P site.

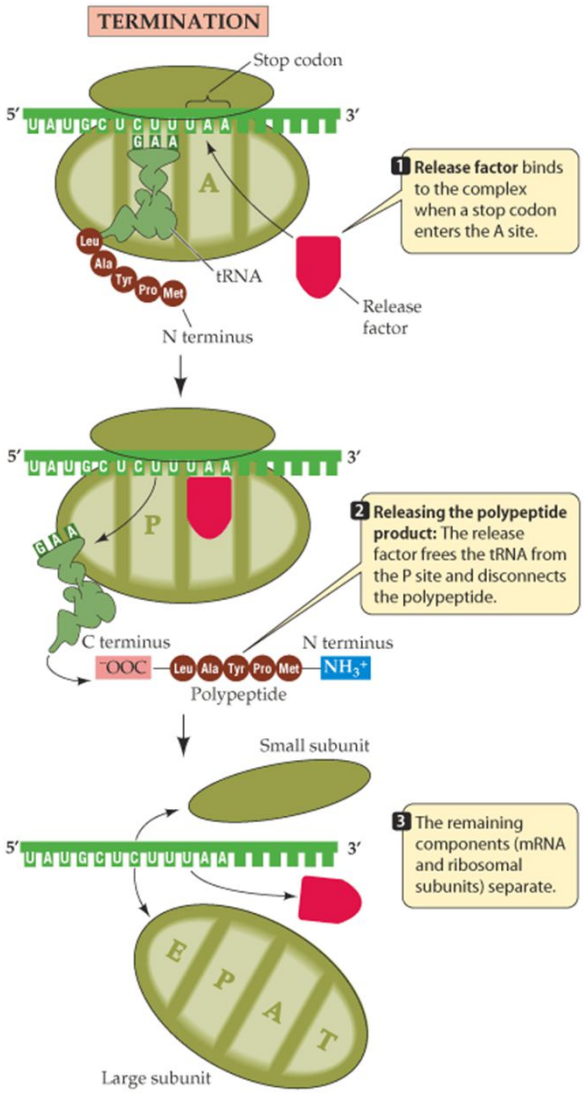


1) A charged tRNA with anticodon to next codon binds A site.

2) Large subunit breaks bond between tRNA & its AA at P site.

3) Assisted by elongation factors large subunit Catalyzes formation of peptide bond between the AA and the one attached to tRNA in A site (peptidyl transferase activity).


3 Elongation: Free tRNA is released from the P site, and the ribosome shifts by one codon, so the growing polypeptide moves to the P site. The free tRNA is released via the E site.
4 The process repeats: Codon recognition, peptide bond formation, and elongation.
5′
3′
U A C
Pro
Met
A U A
Tyr
N terminus

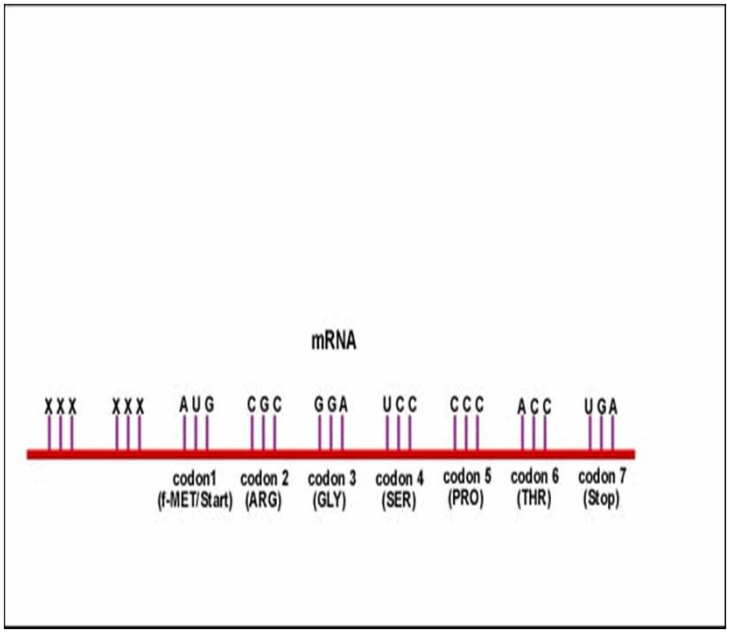


1) When stop codon enters A site elongation is terminates.

2) Stop codons bind release factor, which hydrolysis bond between polypeptide chain and tRNA at P site.

3) Protein folding information is a function of AA sequence.

mRNA
X X X
X X X
A U G
C G C
G G A
U C C
C C C
A C C
U G A
codon1
(f-MET/Start)
codon 2
(ARG)
codon 3
(GLY)
codon 4
(SER)
codon 5
(PRO)
codon 6
(THR)
codon 7
(Stop)

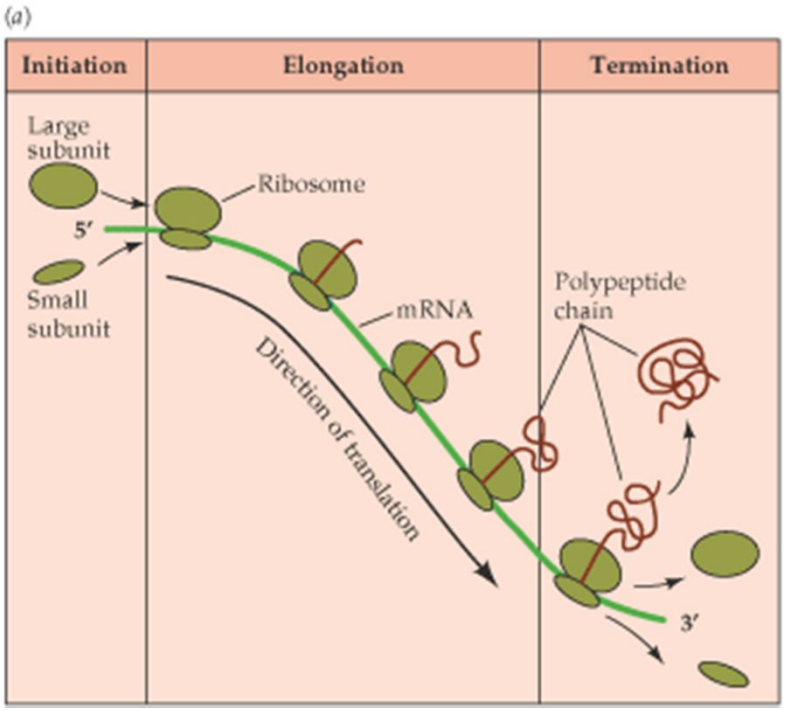

mRNA
XXX
XXX
AUG
CGC
GGA
UCC
CCC
ACC
UGA
codon1 (f-MET/Start)
codon 2 (ARG)
codon 3 (GLY)
codon 4 (SER)
codon 5 (PRO)
codon 6 (THR)
codon 7 (Stop)

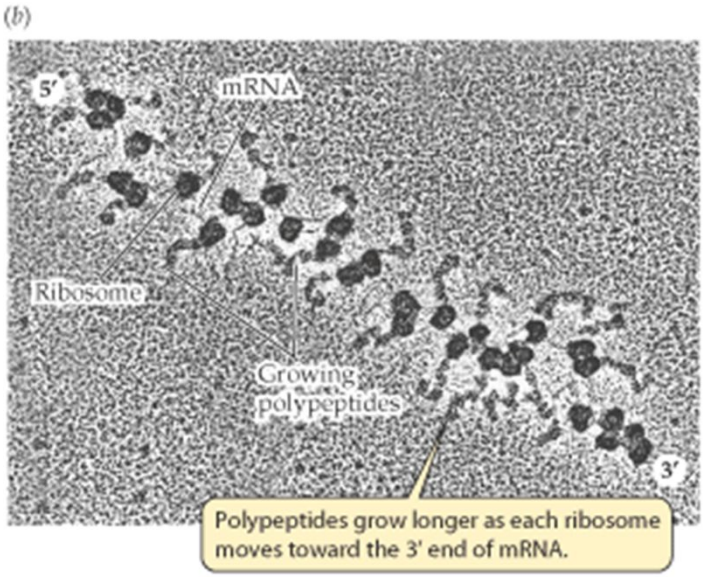
END

Protein Targeting

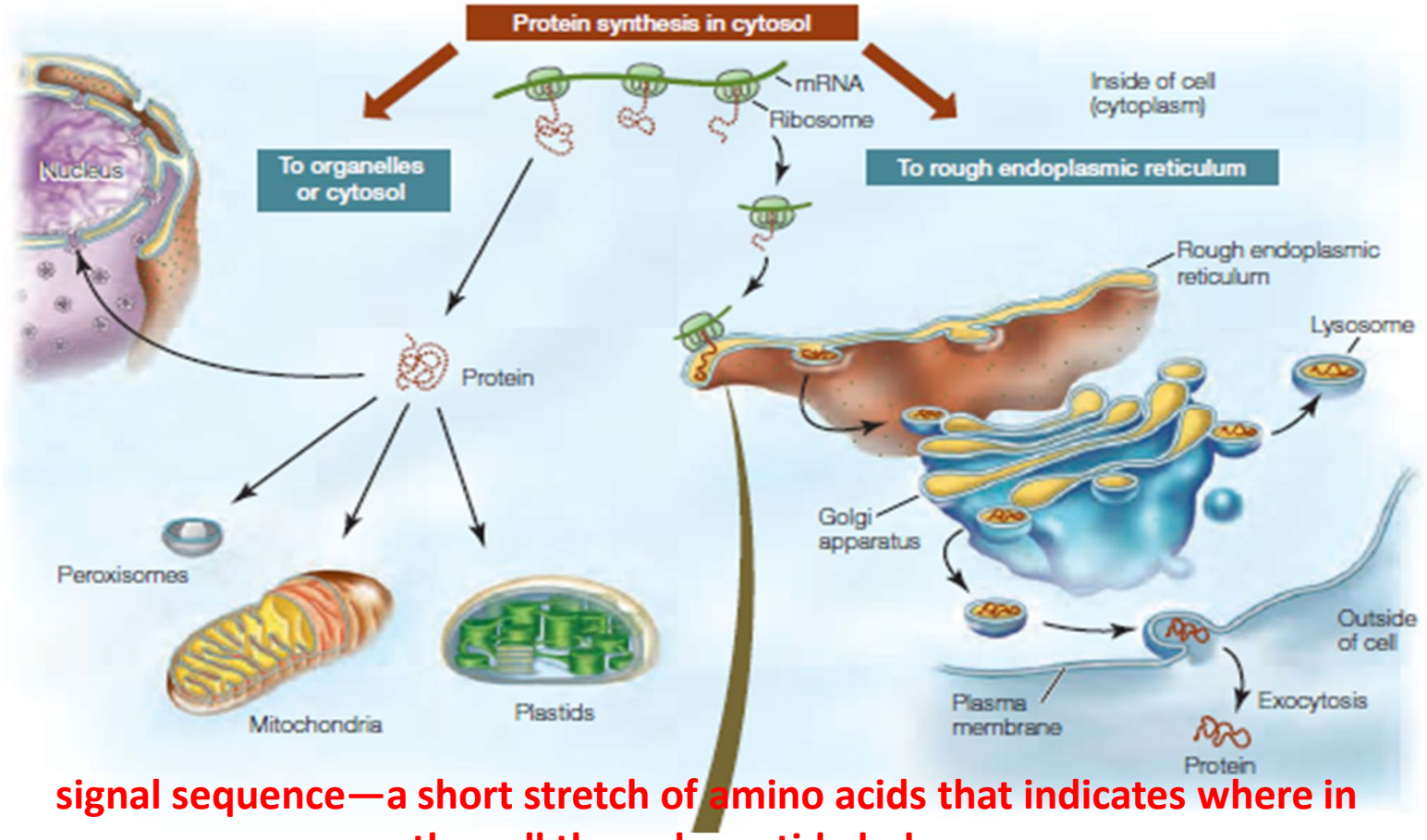
Polyribosome/polysome

Simultaneous Transcription and Translation

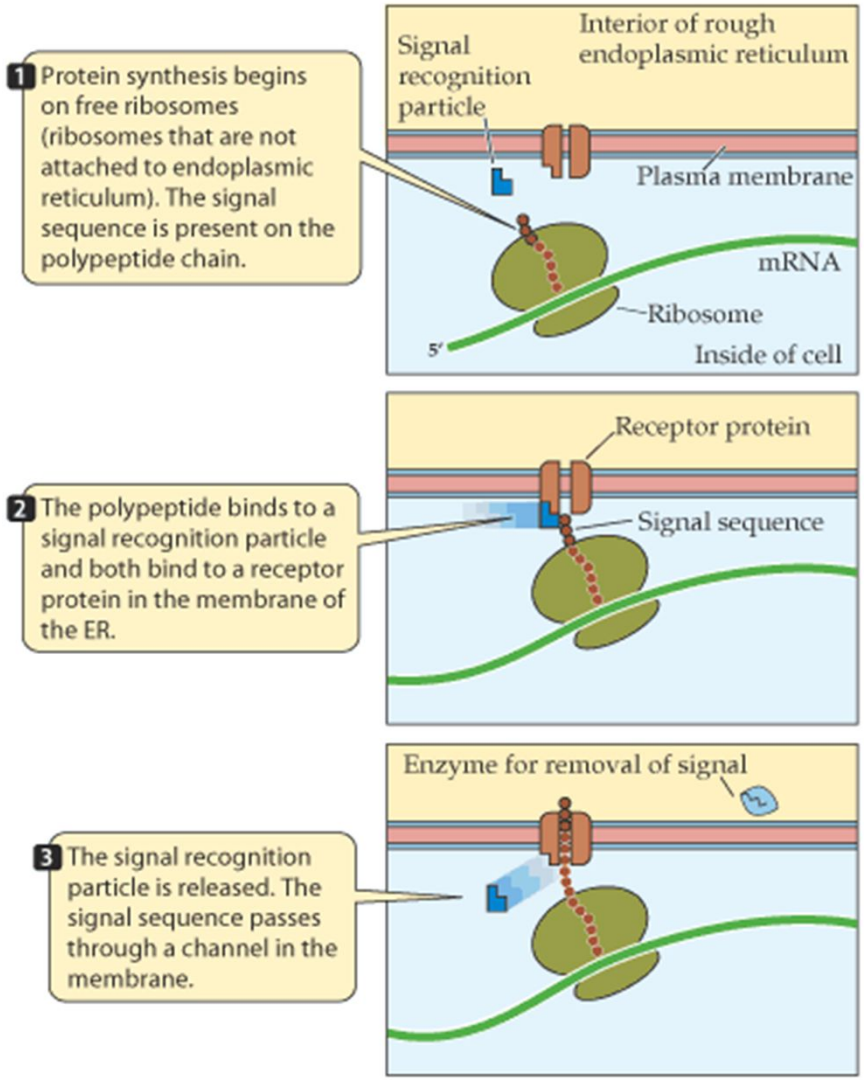




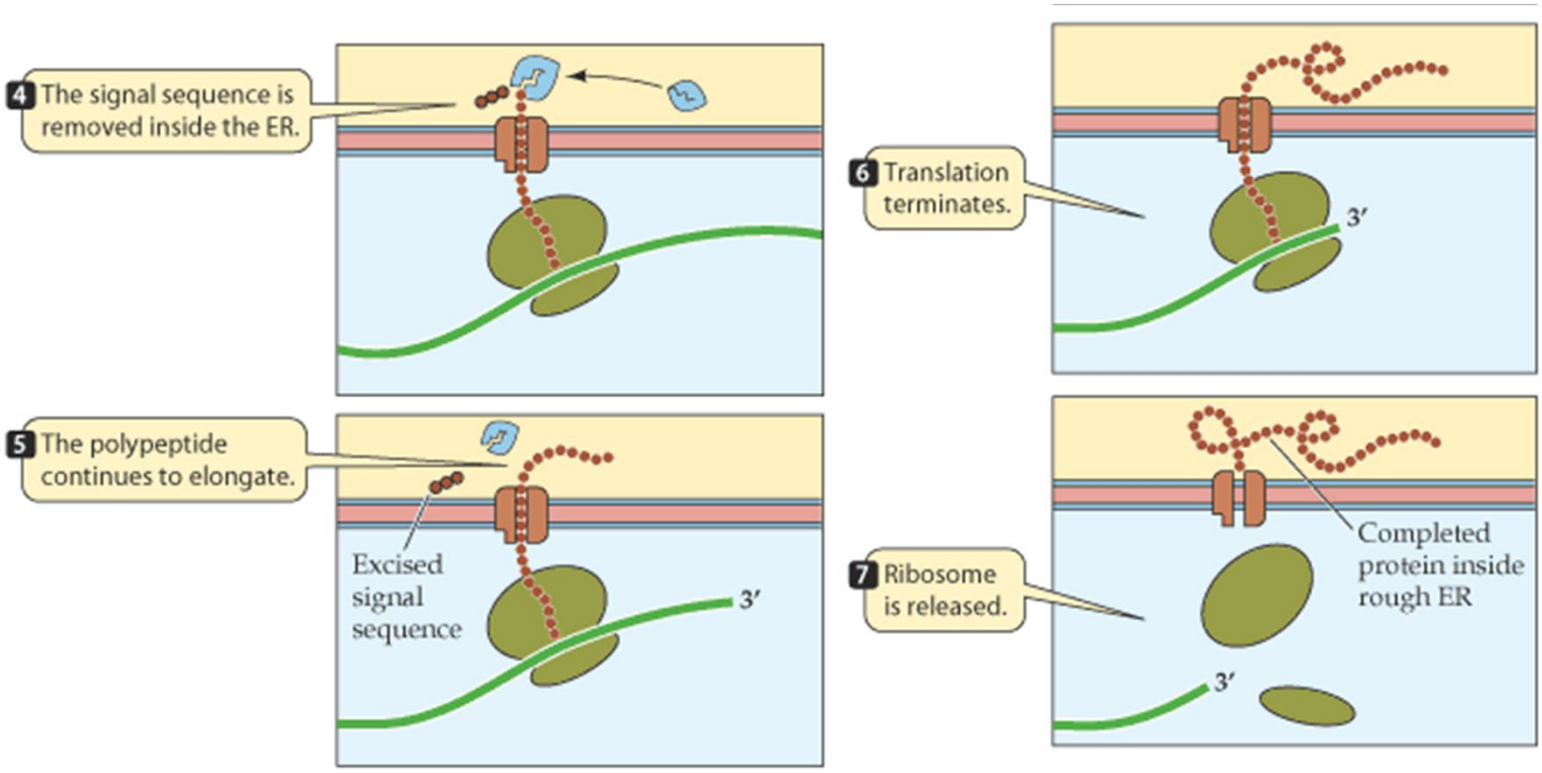
Compartmentalization



signal sequence—a short stretch of amino acids that indicates where in the cell the polypeptide belongs

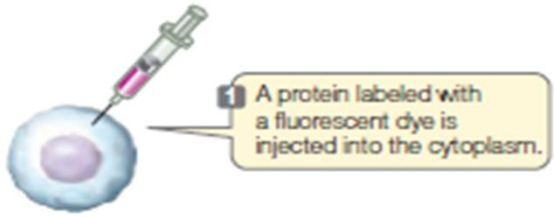


If a polypeptide carries a particular signal of about 20 hydrophobic amino acids at its N terminus, it will be directed to the rough endoplasmic reticulum (ER)

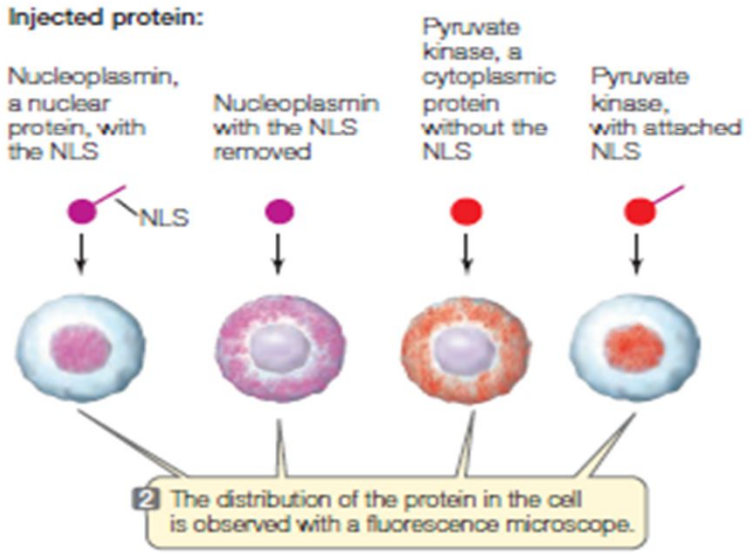

4 The signal sequence is removed inside the ER.
5 The polypeptide continues to elongate.
Excised signal sequence
3′
6 Translation terminates.
3′
7 Ribosome is released.
Completed protein inside rough ER
3′

HYPOTHESIS An NLS is necessary for import of a protein into the nucleus.

Method



Results



CONCLUSION The NLS is essential for nuclear protein import and is sufficient to direct a normally cytoplasmic protein to the nucleus.

Mutations

Heritable Changes in Genetic Information

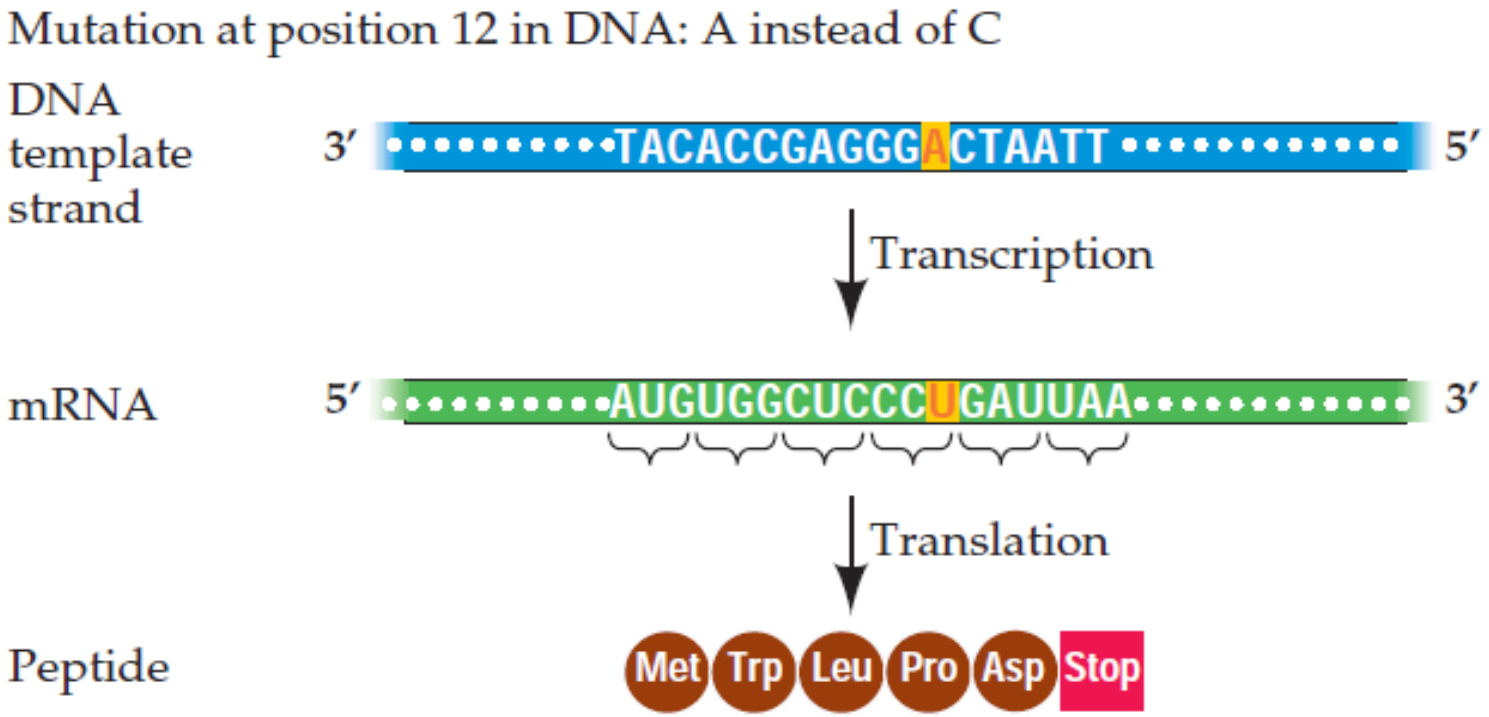
Somatic mutations : Occur in somatic (body) cells & passed on to the daughter cells after mitosis but not passed on to the children.

Germ line mutations: Occur the germ line cells that give rise to gametes. A gamete with the mutation passes it on to embryo after fertilization.

Point mutations: mutations of single base pairs

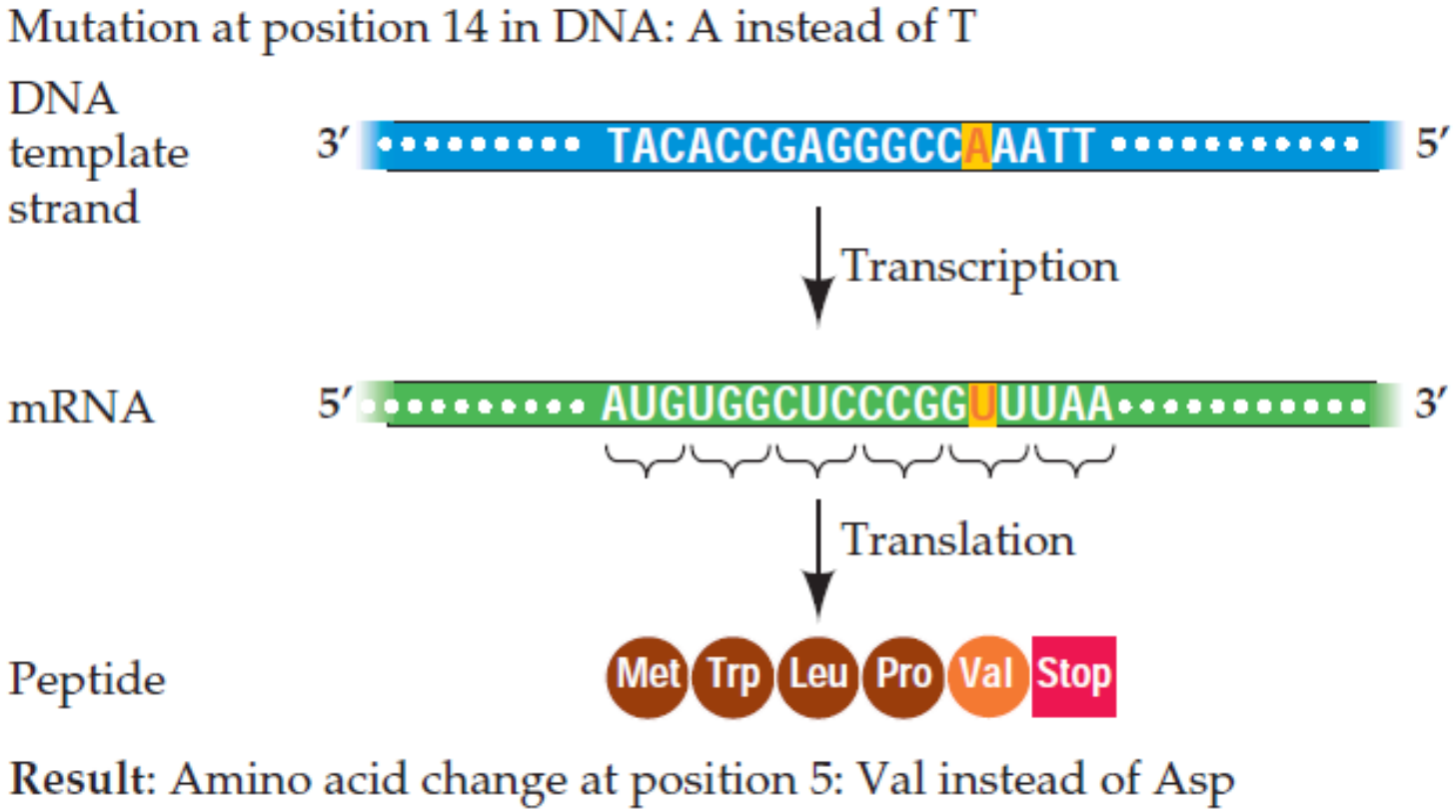
Chromosomal mutations: change the position or orientation of a DNA segment

Silent Mutations: Mutations result in no change in amino acids due to redundancy genetic code



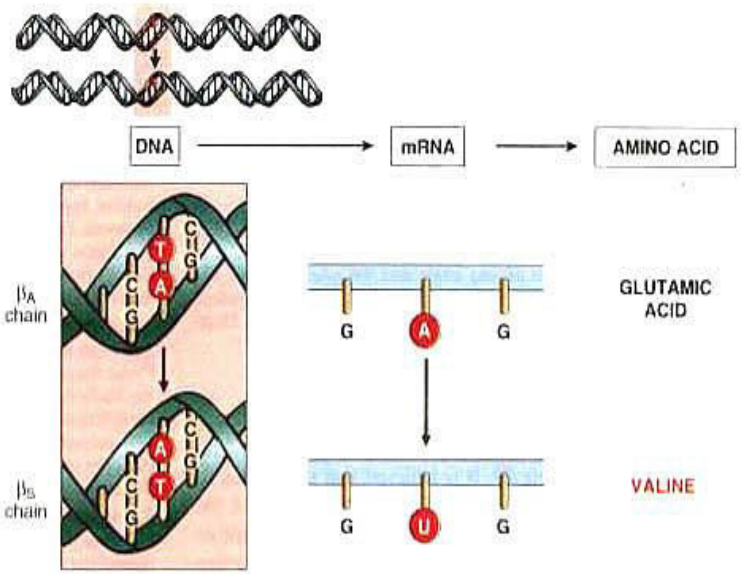
Result: No change in amino acid sequence

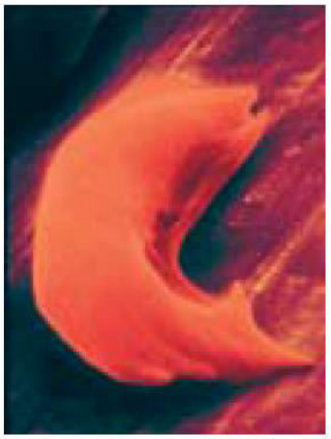
Missense Mutations: Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein.



Result: Amino acid change at position 5: Val instead of Asp

Sickle-cell disease results from a defect in human β-globin gene, a protein in human red blood cells that carries oxygen.

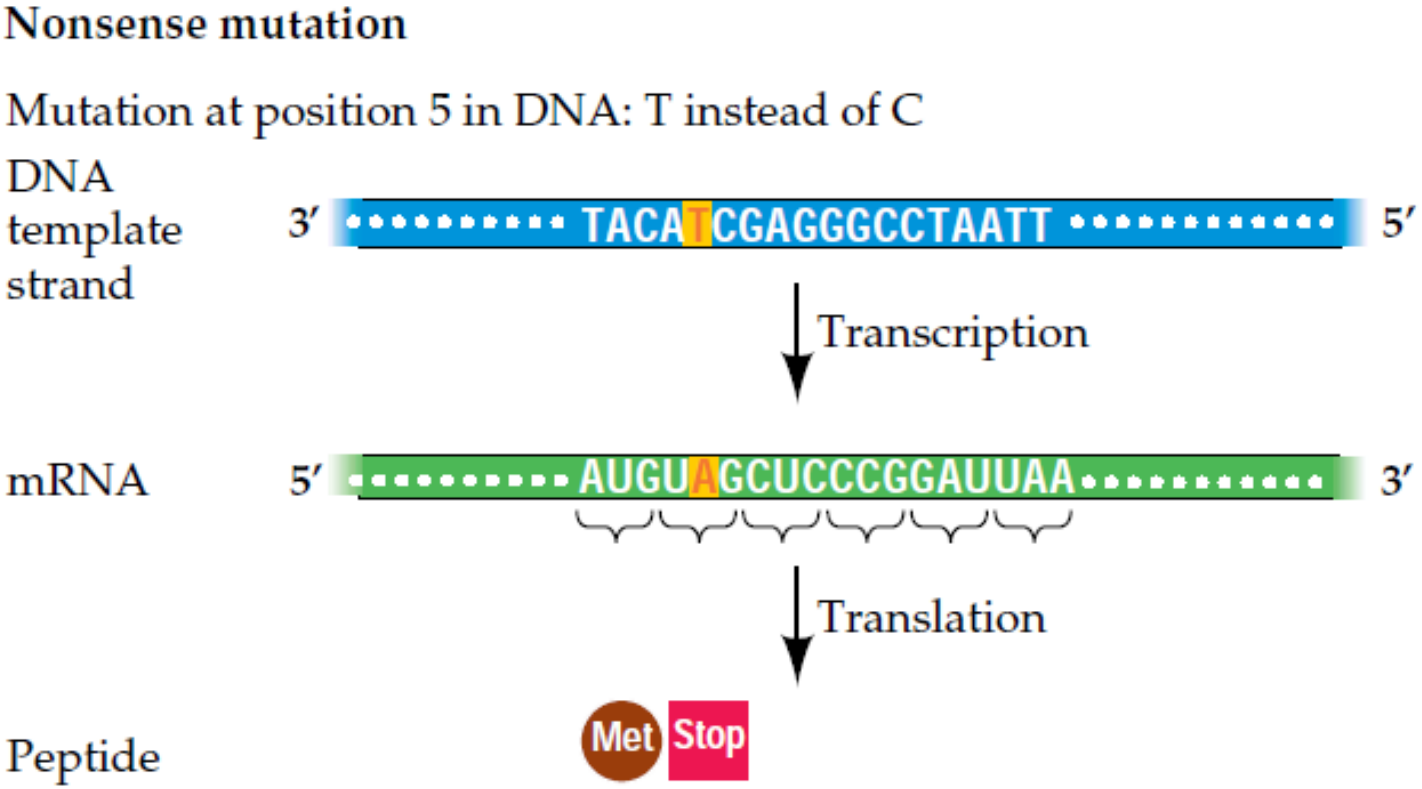




Sickle-cell phenotype

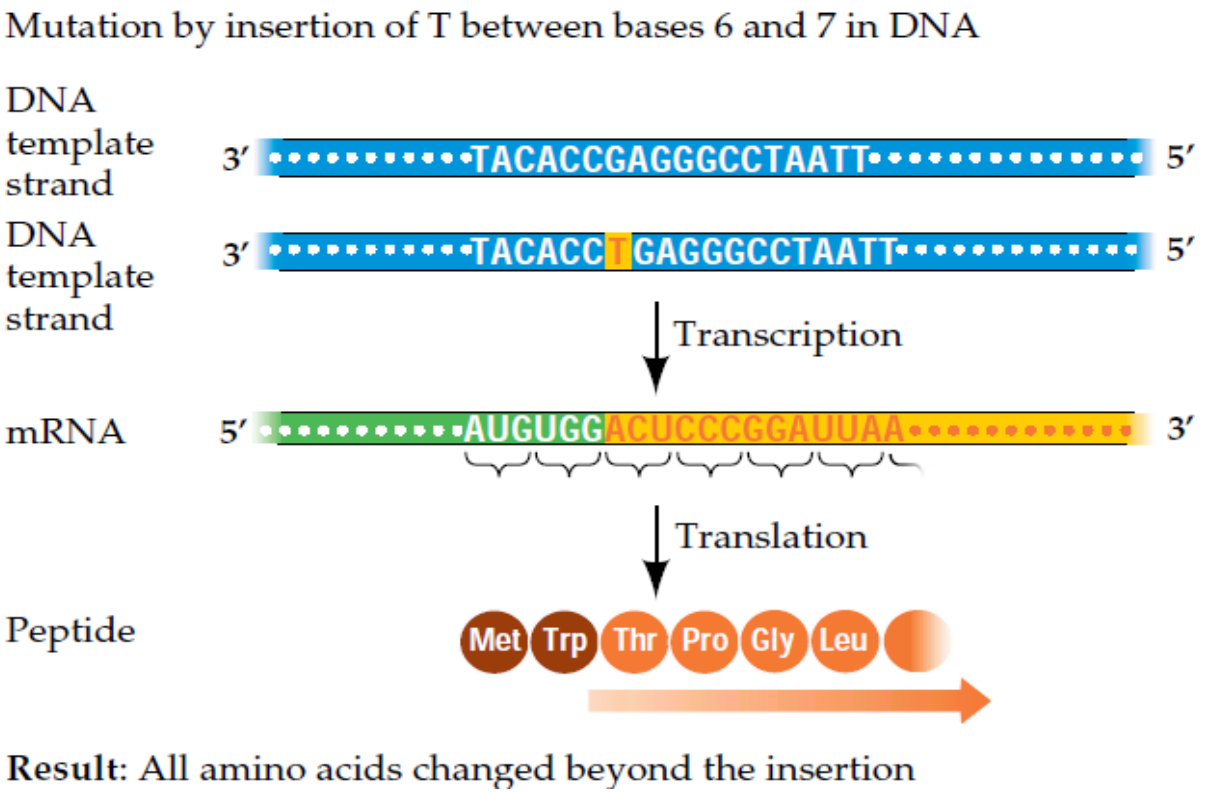
Normal phenotype

Nonsense mutations: Base substitution results in stop codon, (UAG) to form in the mRNA product:



Result: Only one amino acid translated; no protein made

Frame-Shift Mutations: No substitutions. Single base inserted or deleted from DNA. Genetic message out of register "reading frame" changed



Result: All amino acids changed beyond the insertion

Good Mutations

Advantages (βs)

Diversity

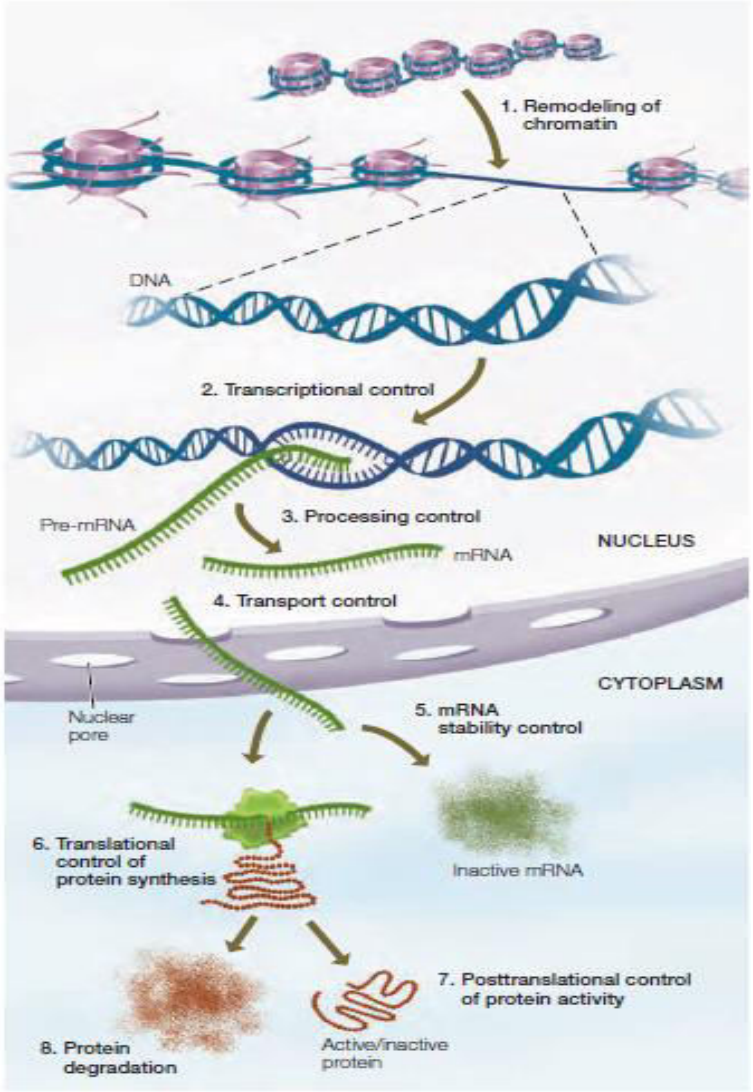
Evolution

END

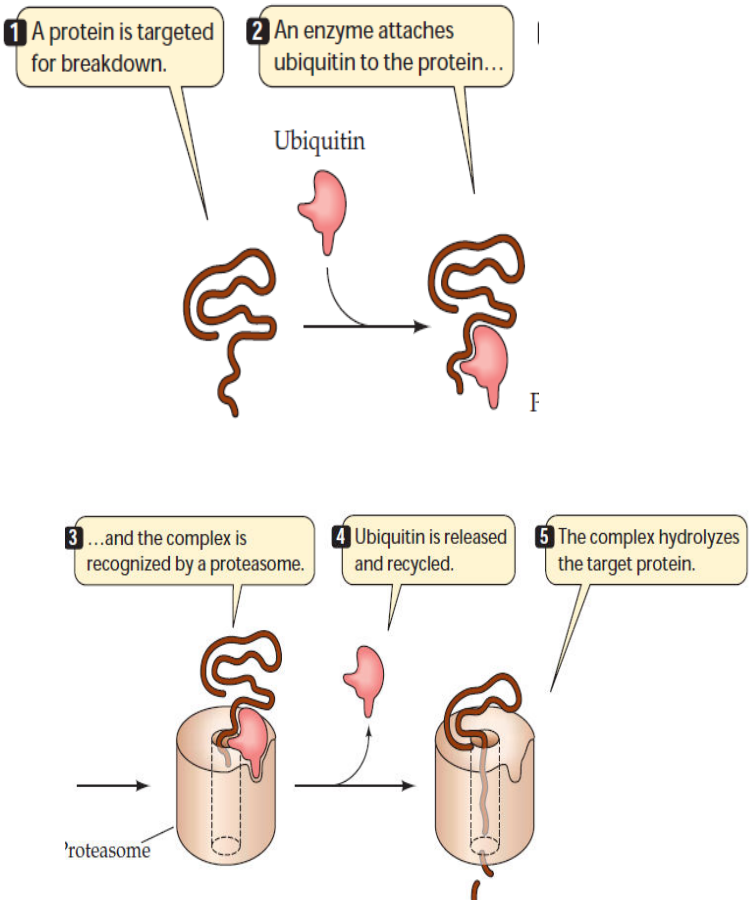
Transcription Regulation

Gene Gategories

- Some of genes are expressed in all cells all the time. Housekeeping genes are responsible for the routine metabolic functions (e.g. respiration).
- Some are expressed as a cell enters a particular pathway of differentiation.
- Some are expressed all the time in only those cells that have differentiated in a particular way.
- Some are expressed only as conditions around and in the cell change. For example, the arrival of a hormone may turn on/off certain genes in that cell.



Epigenetics is the study of heritable changes in gene expression that do not involve changes in the DNA sequence.



14.22 The Proteasome Breaks Down Proteins Proteins targeted for breakdown are bound to ubiquitin, which "leads" them to the proteasome, a complex composed of many polypeptides.

END

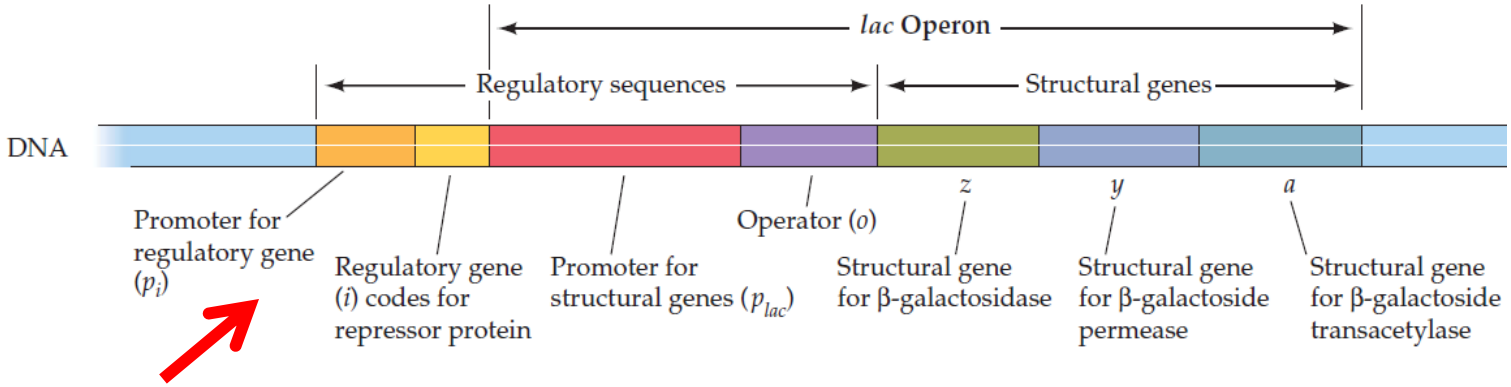
Transcription Regulation in E. Coli

François Jacob & Jacques Monod (Nobel Prize)

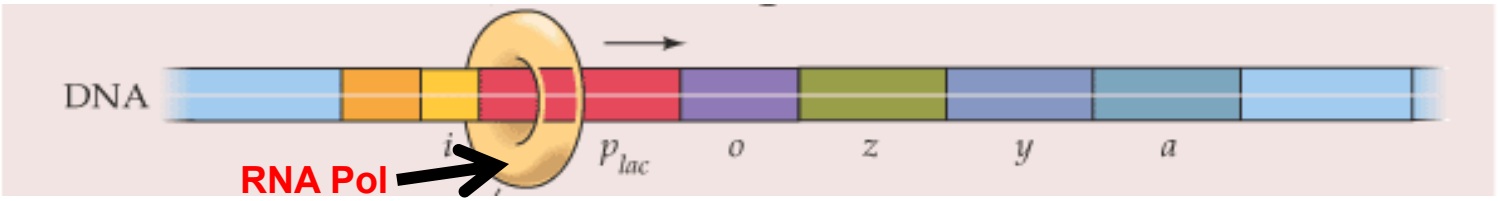
Cluster of genes with 1 promoter is called operon

Operon consists of a promoter, an operator, and two or more structural genes.

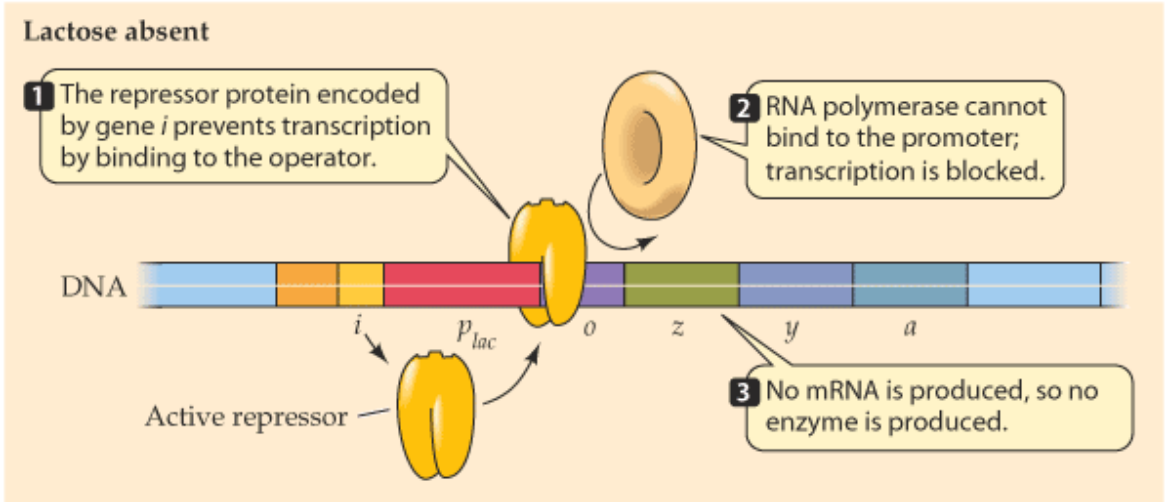
The operator lies between the promoter & structural genes binds regulatory proteins that either activate or repress transcription.

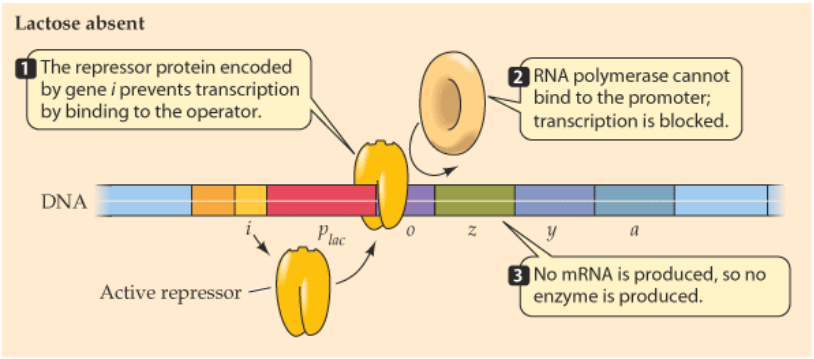


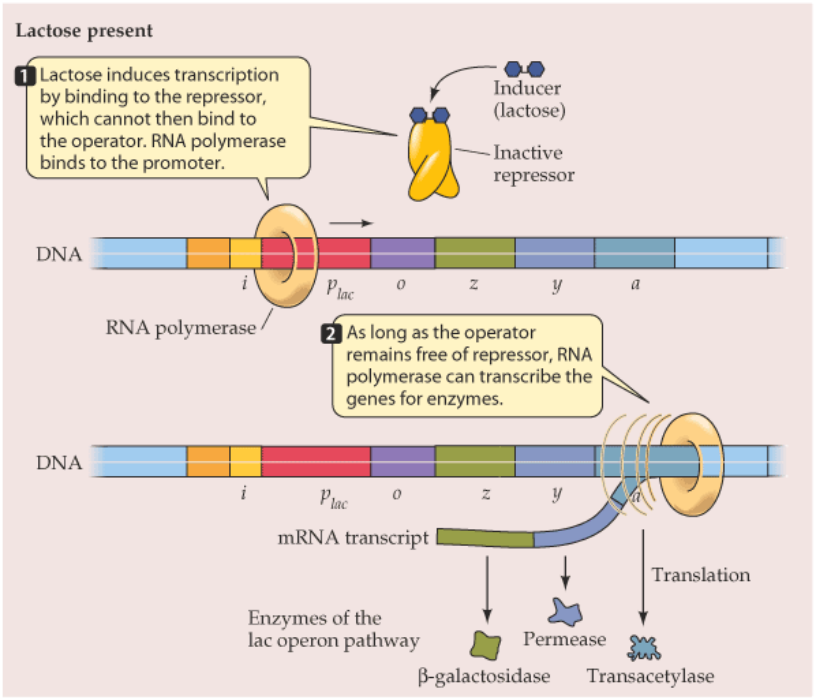
The lac operon contains a promoter, to which RNA polymerase binds to initiate transcription, and an operator, to which a repressor protein can bind.



Repressor is bound, transcription is blocked.



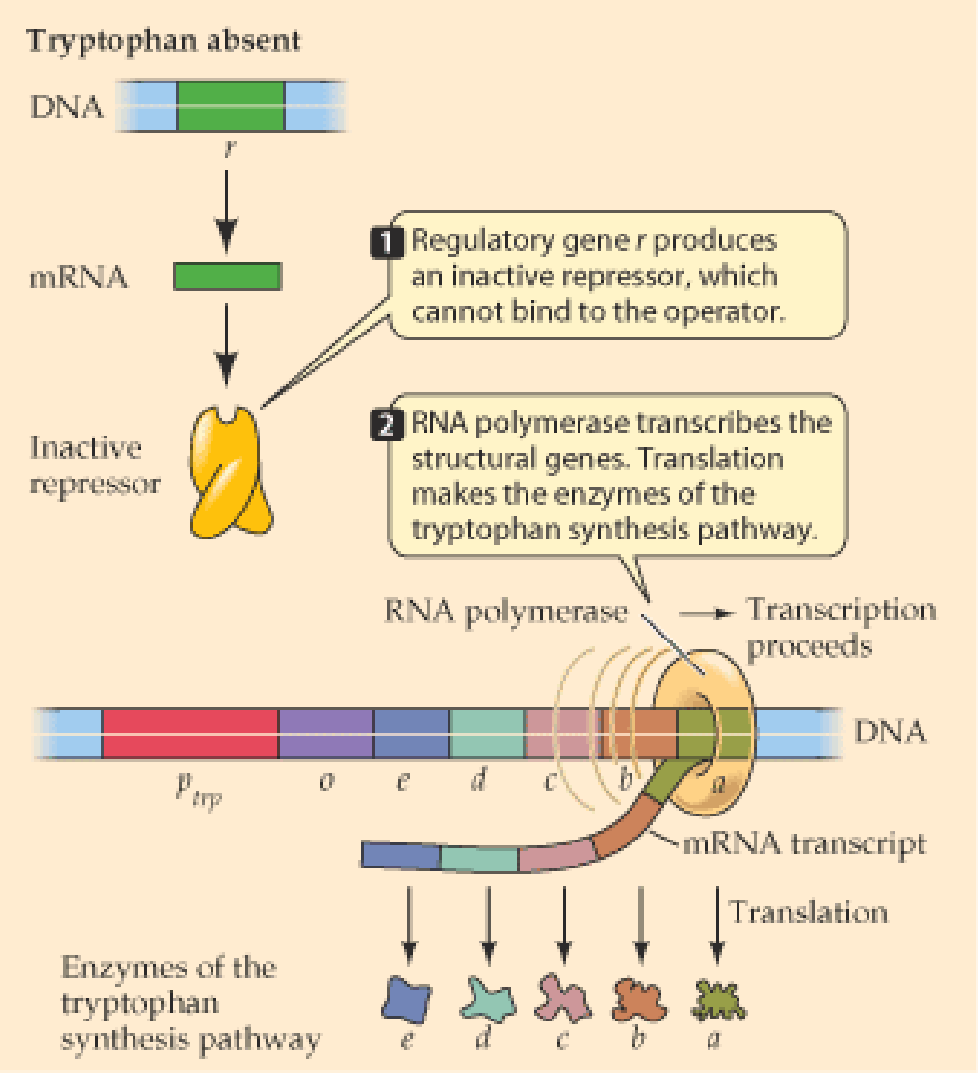


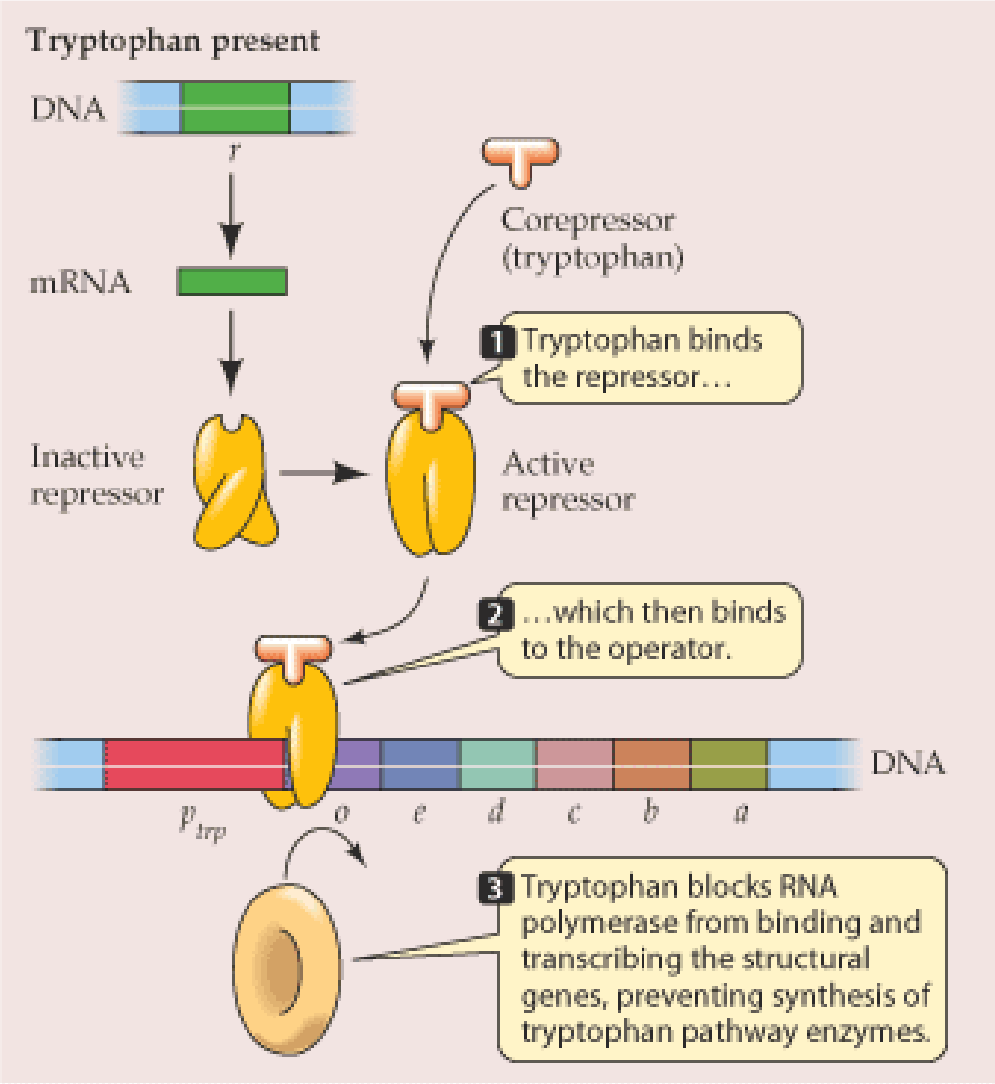


E. Coli

The lac operon contains a promoter, to which RNA polymerase binds to initiate transcription, and an operator, to which a repressor protein can bind.

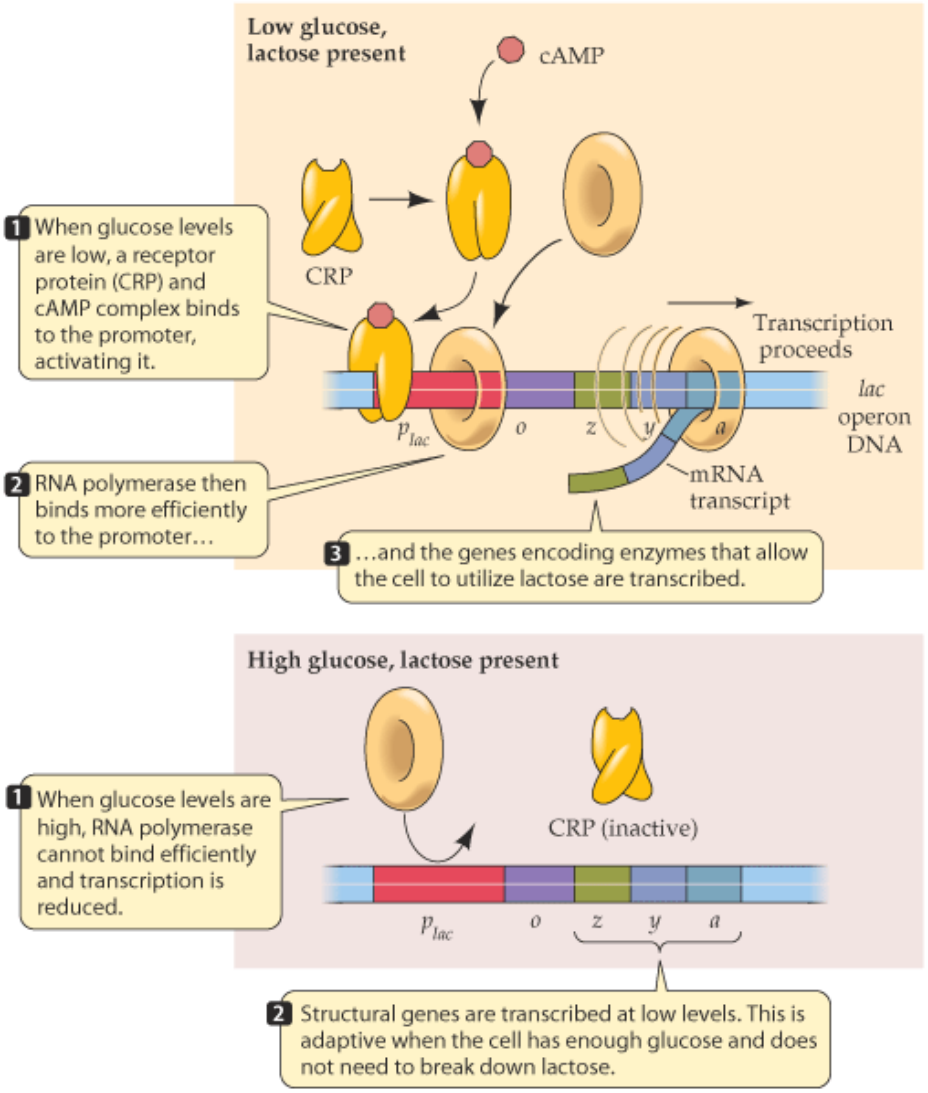
Repressible System





Negative Control of Transcription

The regulatory molecule (the repressor) prevents transcription.



Positive Control of Transcription

the regulatory molecule (the CRP–cAMP complex) enhances transcription.

The Three Systems

- An inducible operon regulated by a repressor protein

- A repressible operon regulated by a repressor protein

- An operon regulated by an activator protein

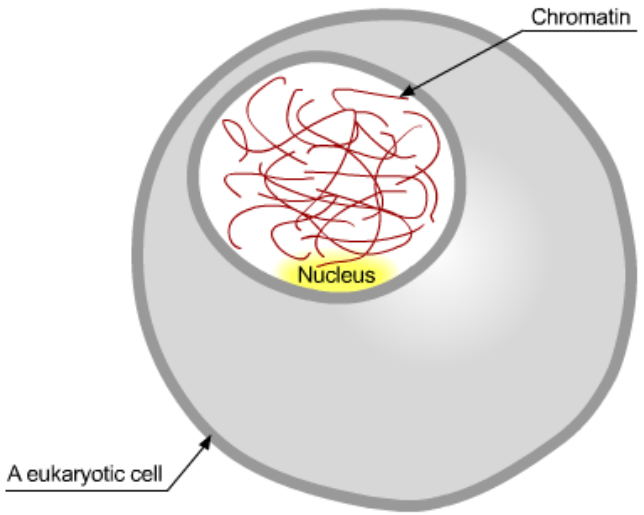
END

Transcription Regulation
In Eukaryotes

The Eukaryotic Dilemma

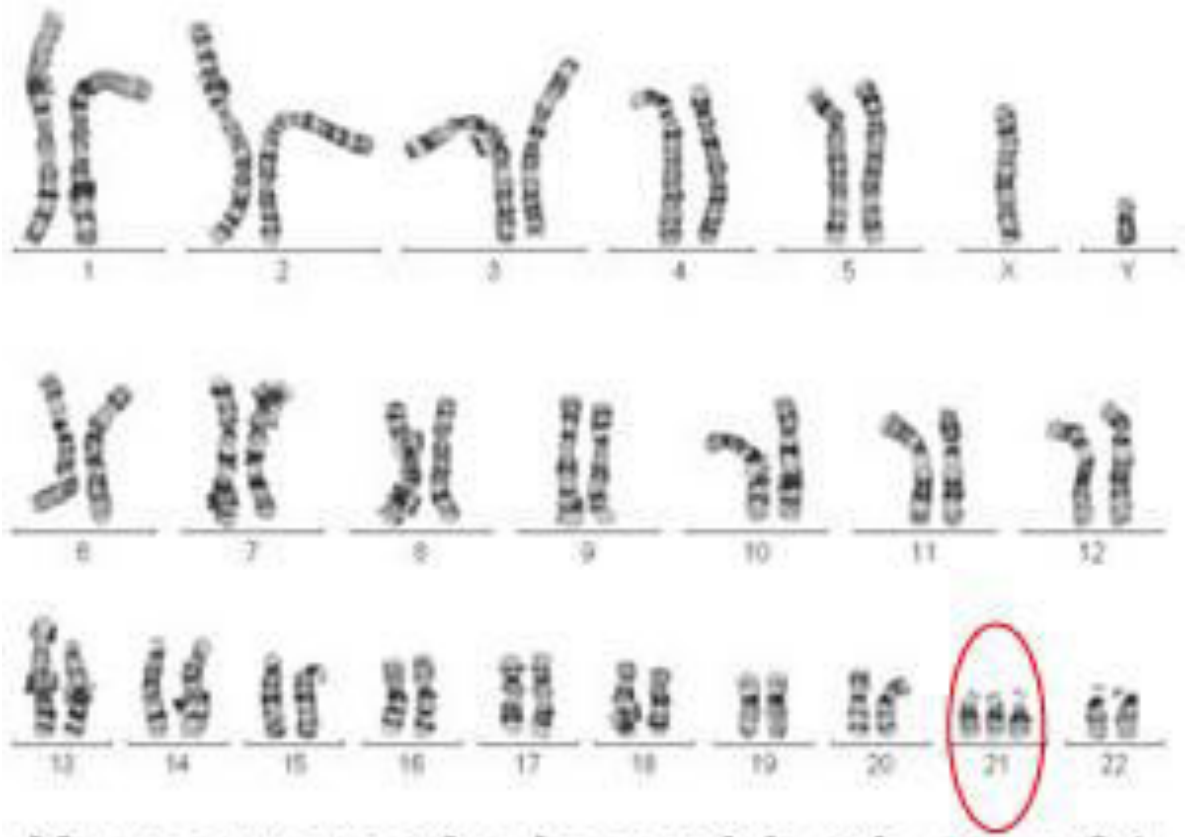
For a cell to develop and function properly it must turn on the right genes and right time.

Genes are located in the nucleus in a tangled chromatin material wrapped around histones.

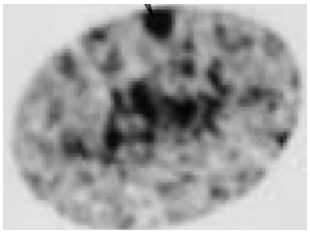


The structure of chromatin dictates whether a gene can be transcribed not.

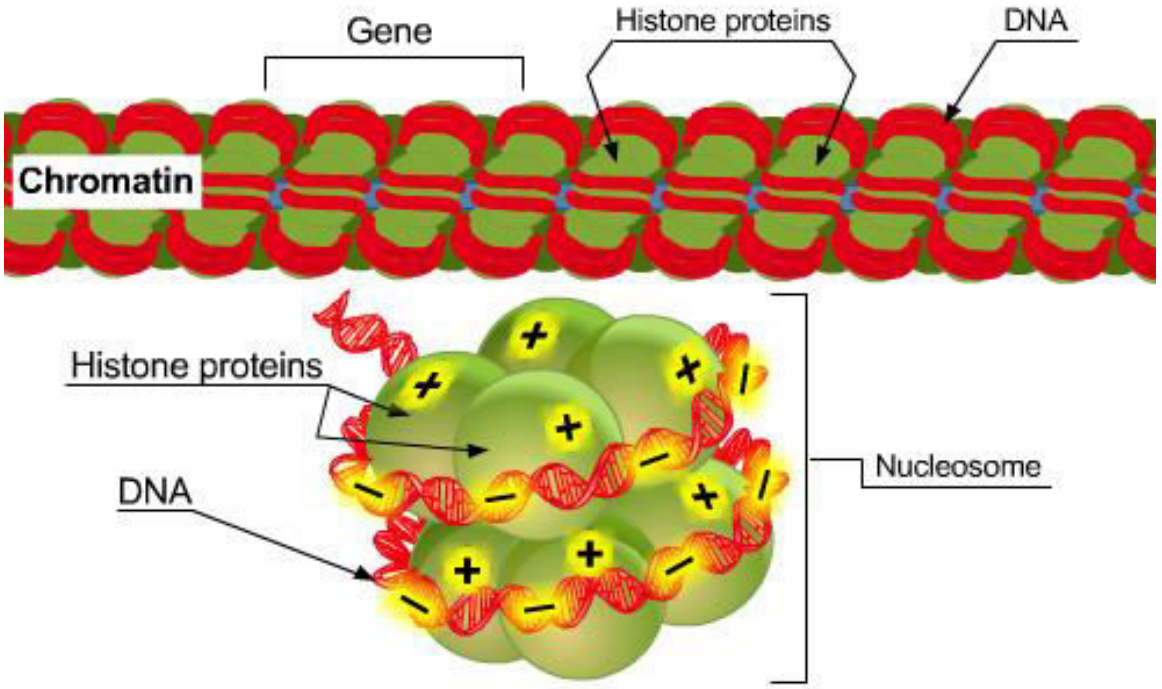
Eu & Hetero Chromatin



Karyogram of a boy with trisomy 21

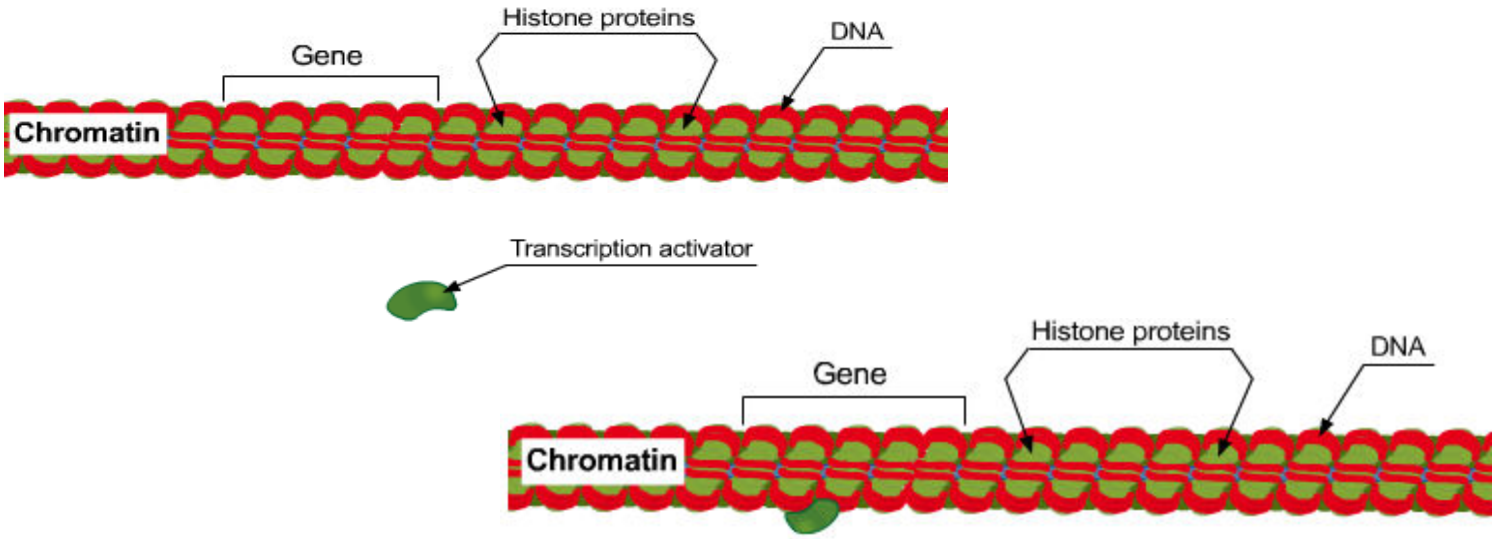


Most of DNA is highly ordered and tightly coiled around positively charged histone proteins and is virtually inaccessible to RNA polymerase.



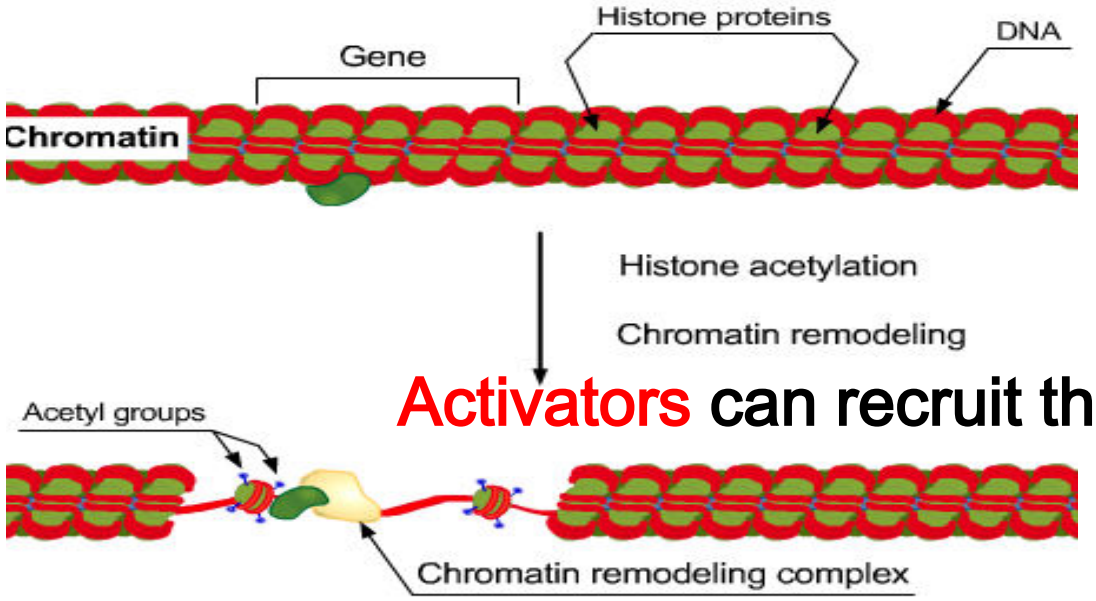
In response to specific signal a cell may produce or activate a transcription activator.

Transcription activator recognizes and binds a specific sequence of a specific gene and work in conjunction with other proteins that can alter chromatin structure

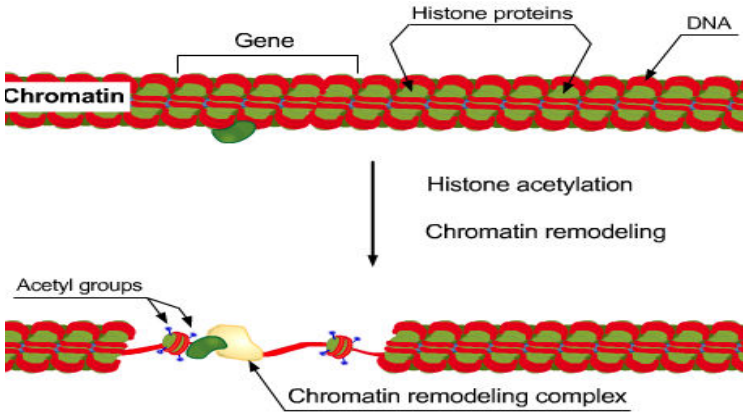


The alteration may occur by addition of acetyl groups to histones which reduce positive charge.

Chromatin remodeling complex can also alter the structure of chromatin to make it accessible to transcription factors that attract RNA Pol.



Activators can recruit the CRC.


Histone proteins
DNA
Gene
Chromatin
Histone acetylation
Chromatin remodeling
Acetyl groups
Chromatin remodeling complex

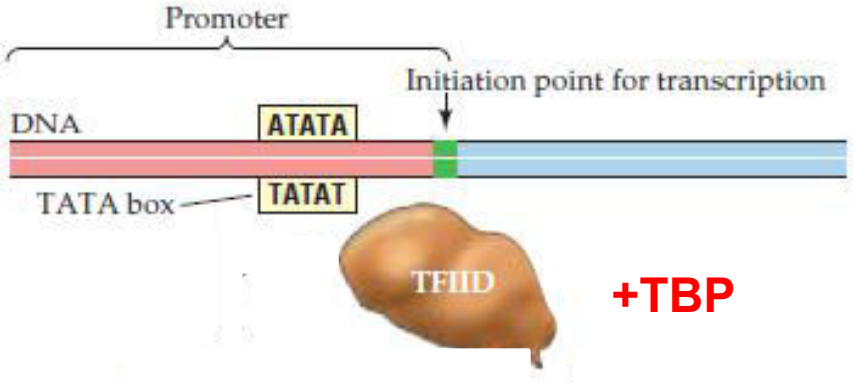
END

Eukaryotic Transcription

Initiation

Promoter has two essential sequences.
1) Recognition sequence (RNA polymerase).
2) TATA box where DNA begins to denature

Eukaryotic RNA polymerase II cannot simply bind to the promoter and initiate transcription. It needs help from transcription factors. First, the protein TFIID

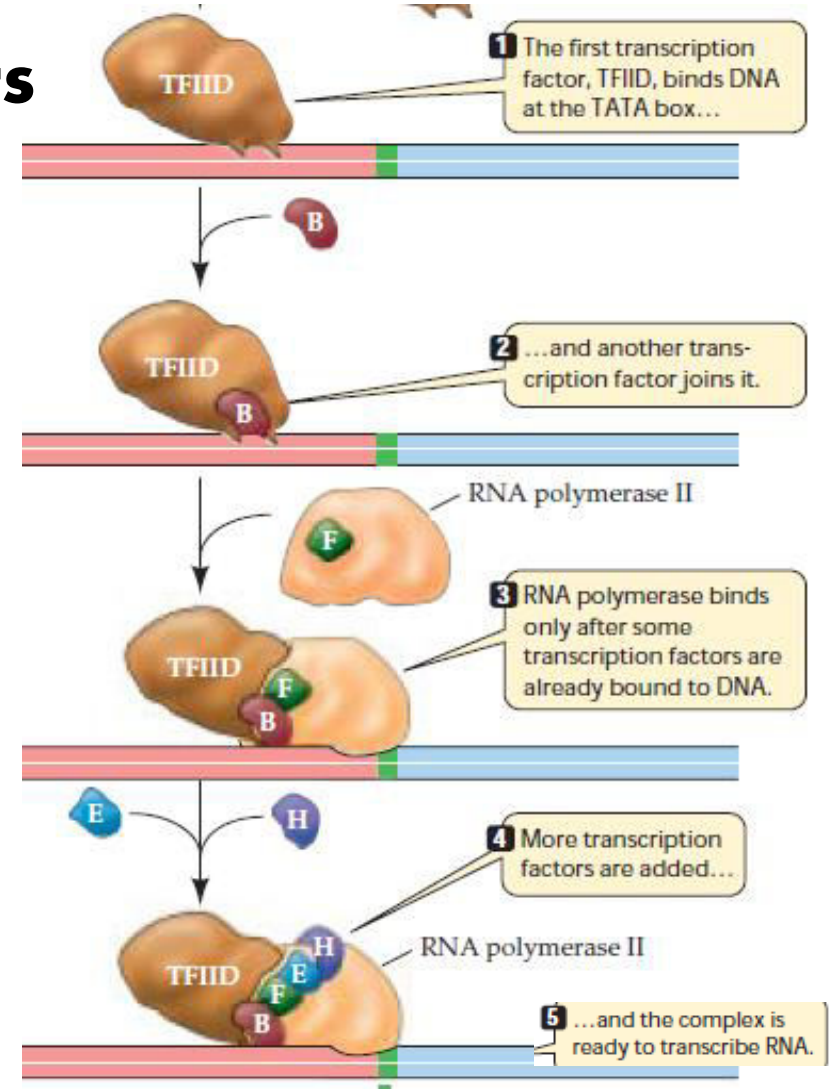


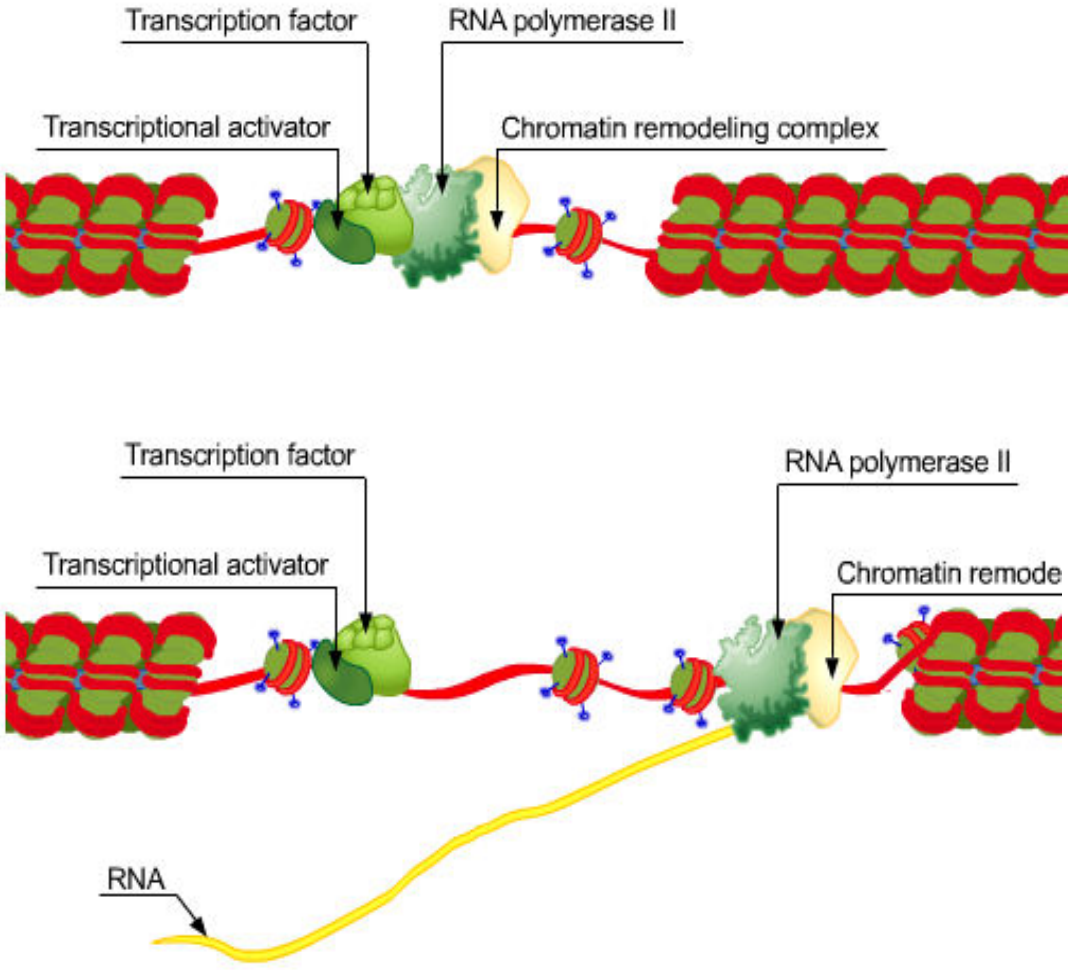
Most eukaryote promoters have TATA box, which is located 25bp upstream of start site.

Transcription factors

TFIID binding changes its own shape and that of the DNA, presenting a new surface that attracts the binding of other transcription factors to form a transcription complex.

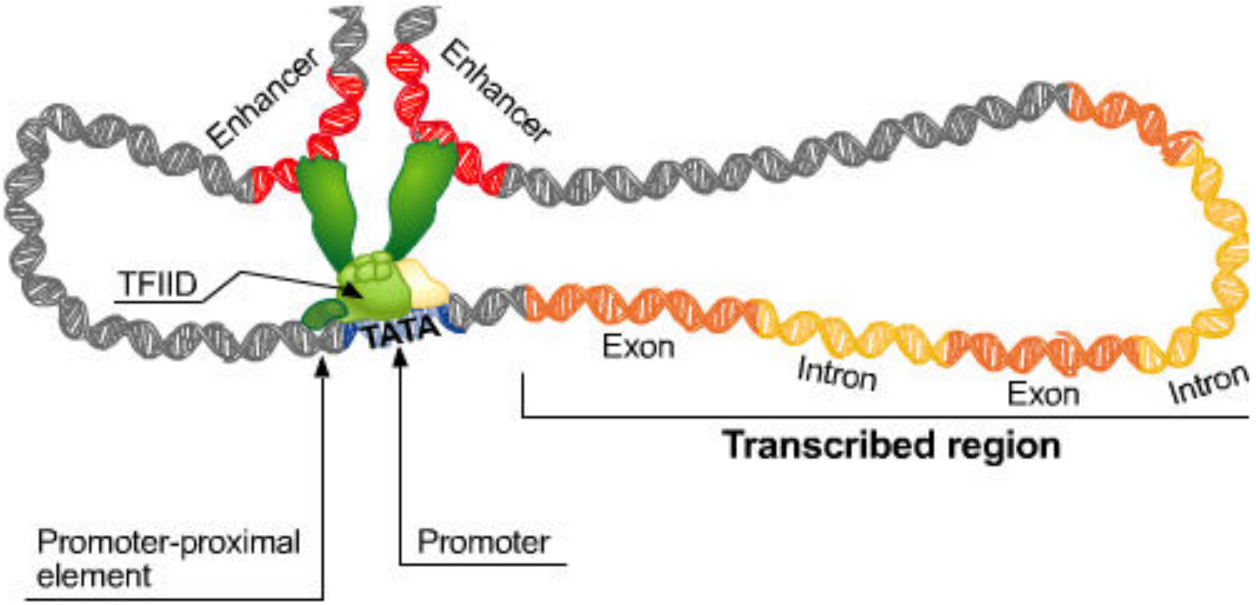
RNA Pol last to join

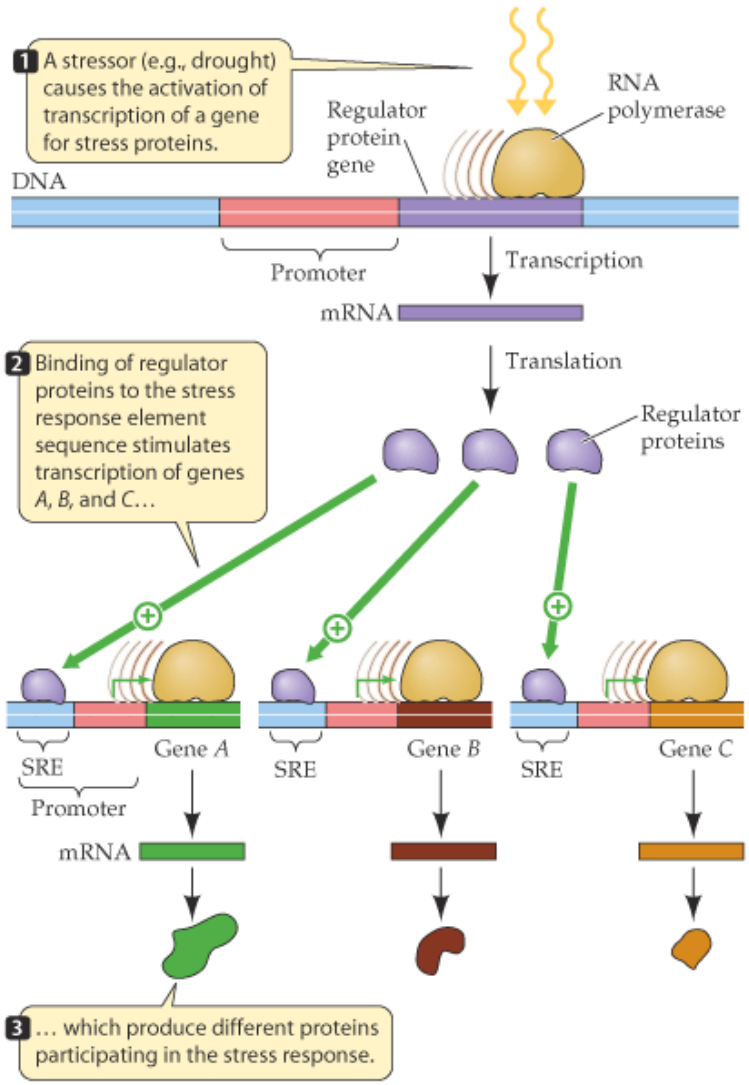



Transcription factor
RNA polymerase II
Transcriptional activator
Chromatin remodeling complex
Transcription factor
RNA polymerase II
Transcriptional activator
Chromatin remode
RNA

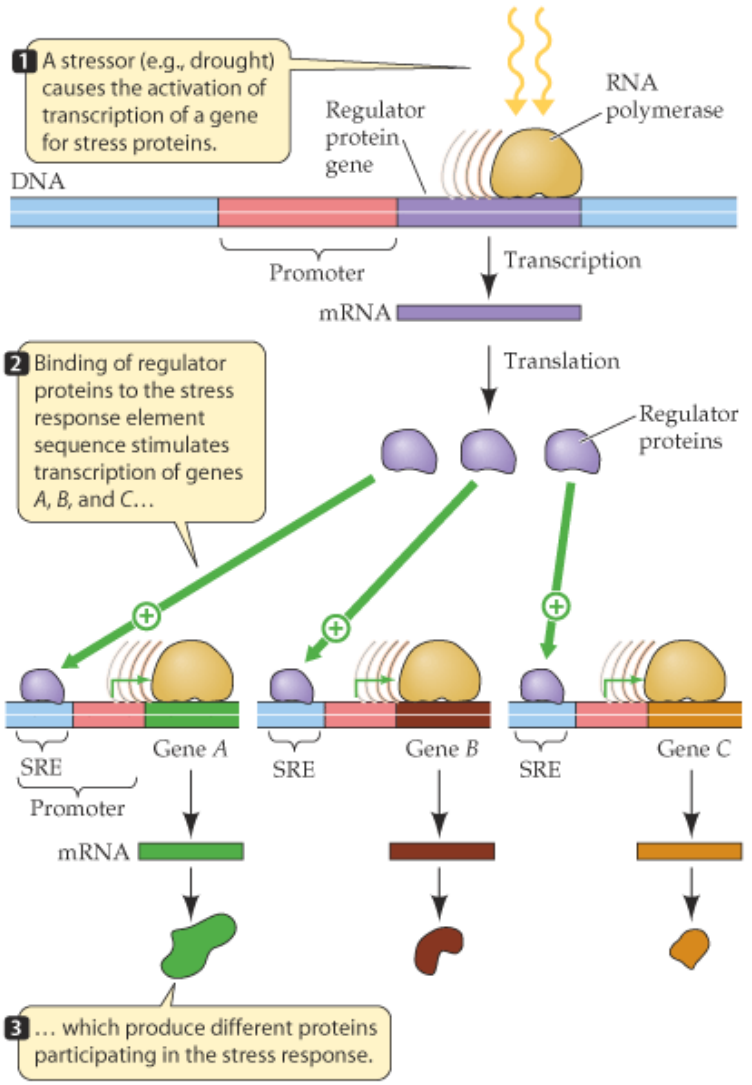
Detailed View

Many genes also have enhancer (1000 nucleotide away) where specific activators only made by certain cells can bind.





The simultaneous regulation of widely separated genes is possible through common sequences in their promoters, to which the same regulatory proteins bind.


1 A stressor (e.g., drought) causes the activation of transcription of a gene for stress proteins.
RNA polymerase
Regulator protein gene
DNA
Promoter
Transcription
mRNA
Translation
2 Binding of regulator proteins to the stress response element sequence stimulates transcription of genes A, B, and C…
Regulator proteins
SRE
Promoter
Gene A
SRE
Gene B
SRE
Gene C
mRNA
3 … which produce different proteins participating in the stress response.

END

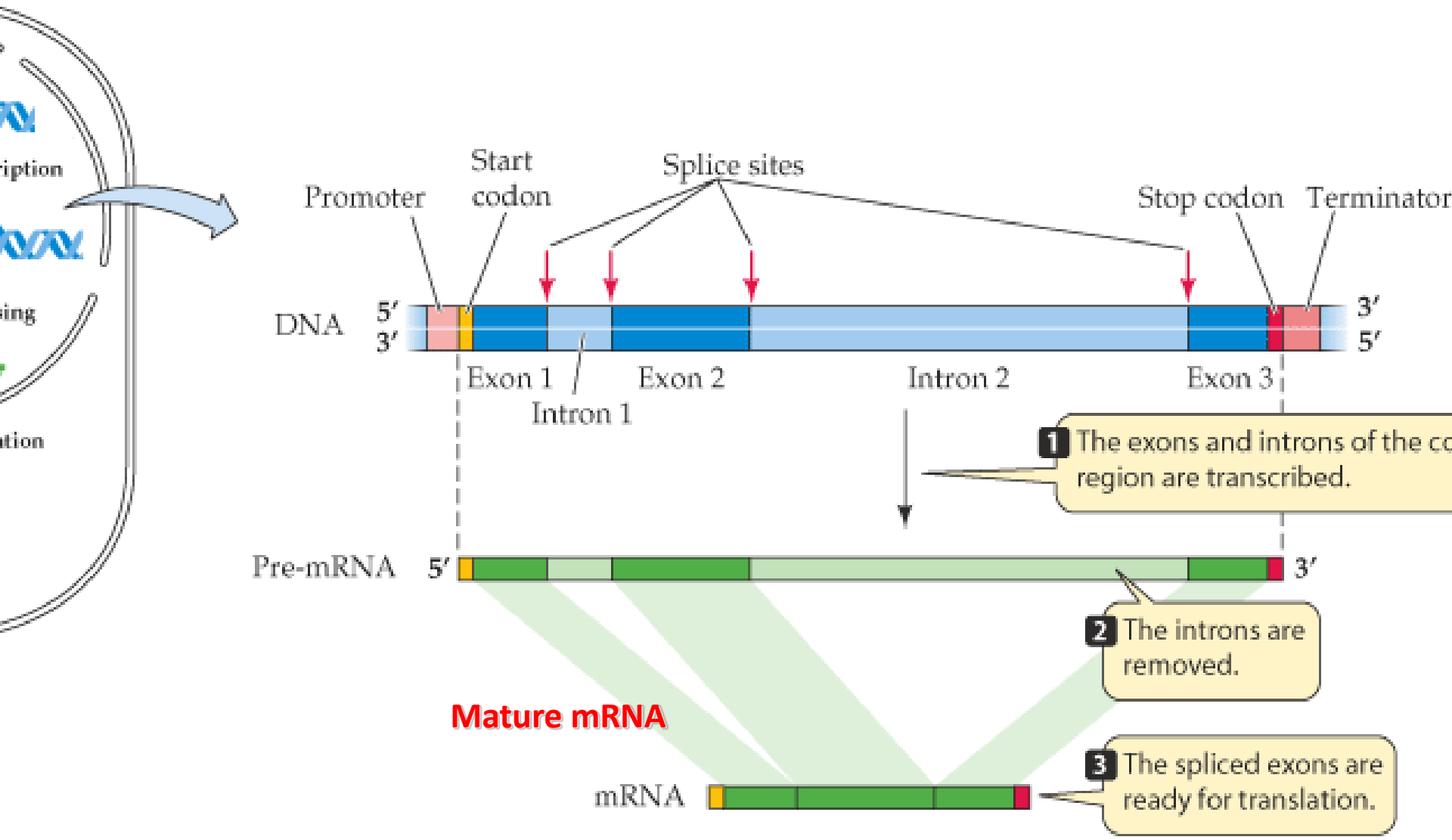
RNA Processing

pre-mRNA (freshly transcribed) is processed to make it functional.

1-Introns are removed during RNA maturation

2) The transcribed pre-mRNA is altered by the addition of a G cap (modified GTP) at the 5′ end to protect RNA and facilitate translation.

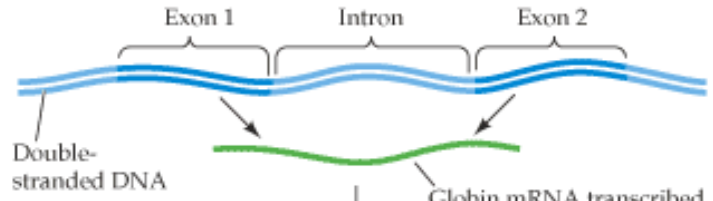
3) After the last codon a poly A tail (100-300) is added at the 3′ end for export and protection

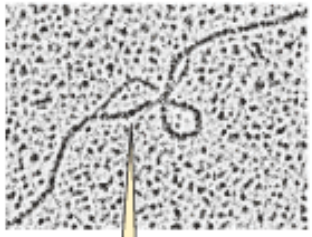

Promoter
Start codon
Splice sites
Stop codon
Terminator
DNA
5′
3′
3′
5′
Exon 1
Intron 1
Exon 2
Intron 2
Exon 3
1 The exons and introns of the co region are transcribed.
Pre-mRNA
5′
3′
2 The introns are removed.
Mature mRNA
mRNA
3 The spliced exons are ready for translation.

EXPERIMENT

Question: Are there regions within the coding sequence of a gene that do not end up in its mRNA?

METHOD



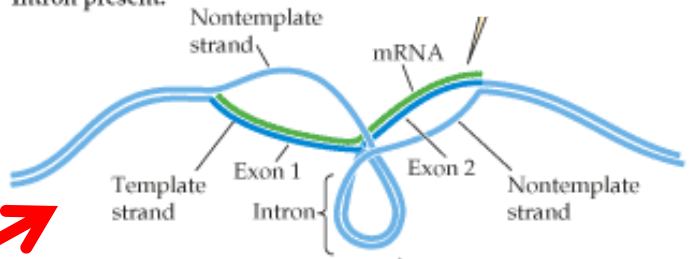


1 Mouse DNA is partially denatured and hybridized to mRNA from a mouse gene.

2 The mRNA hybridizes with the template DNA of its gene, forming thick double strands. The thin loops are formed by the nontemplate strand of DNA that was displaced by the mRNA.

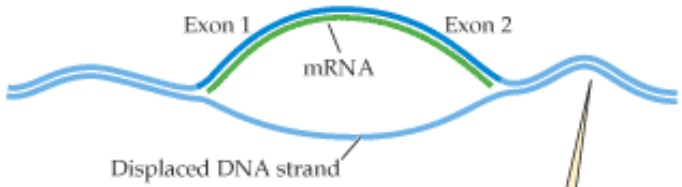
RESULTS

Intron present:



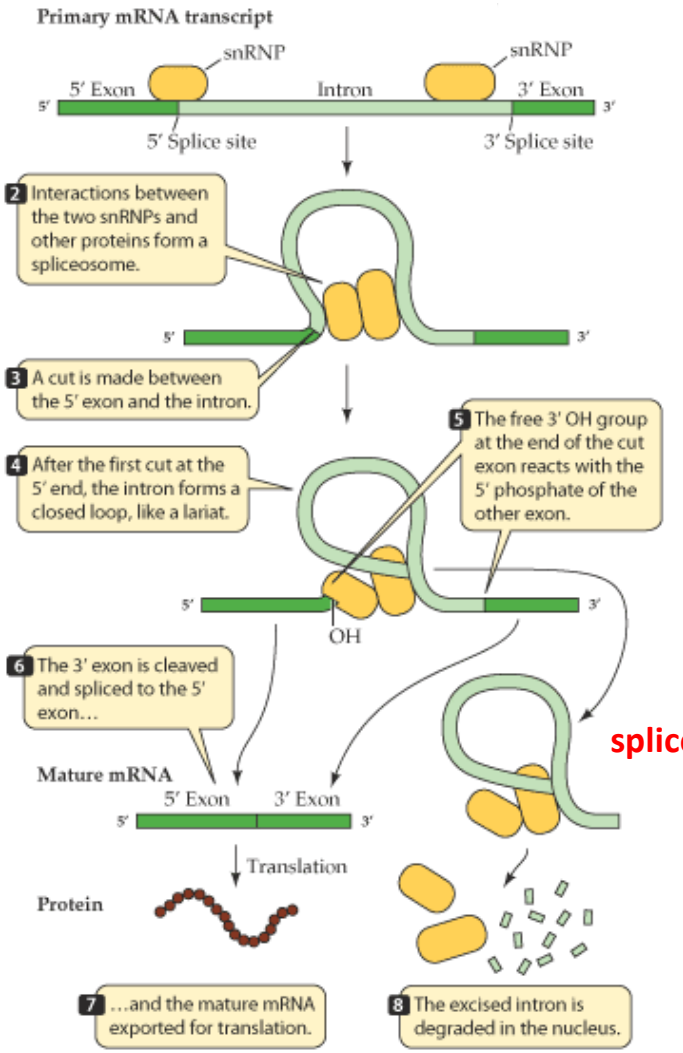
3 The double-stranded intron is forced into a loop by the mRNA, bringing the two exons together.

If there was no intron:



4 If there was no intron, the DNA would hybridize with the mRNA in a continuous strand.

Conclusion: The DNA contains noncoding regions within the genes that are not present in the mature mRNA.



spliceosome

The Mechanism

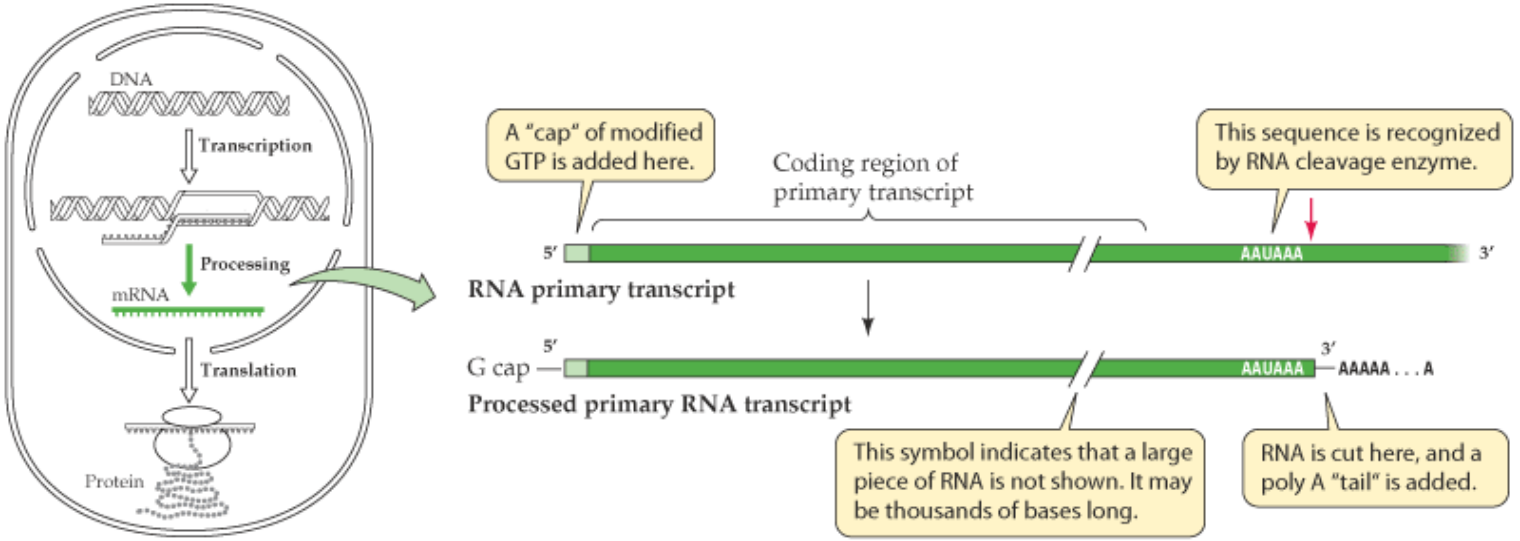
snRNPs bind Intron-exon bounderies consensus sequences

RNA of U1 has complementary bases that bind CS at 5’ exon-intron boundery.

RNA of U2 has complementary bases that bind CS near 3’ intron-exon boundery.

2) The trans ibed pre-mRNA is altered by the addition of a G cap (modified GTP) at the 5′ end to protect RNA and facilitate translation.

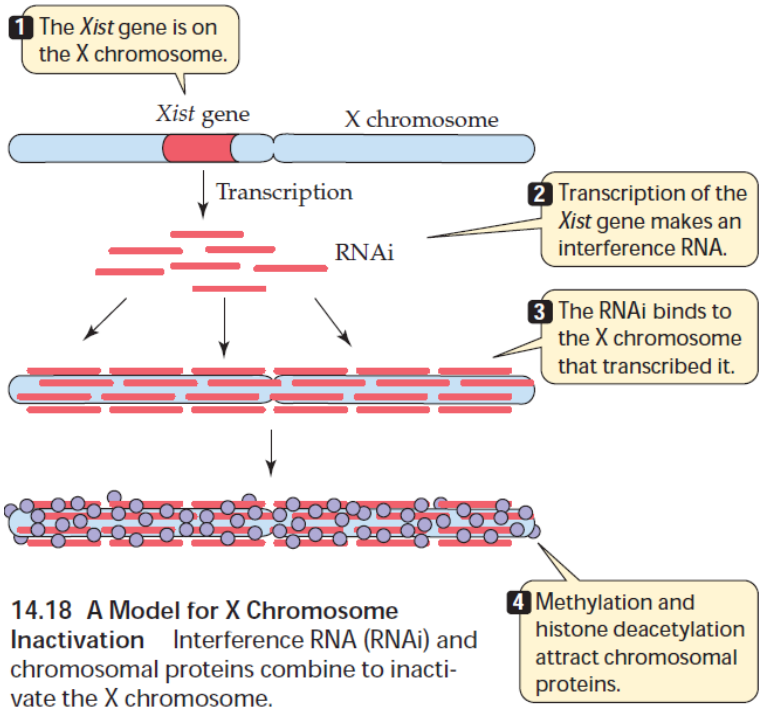
3) After the last codon a poly A tail (100-300) is added at the 3′ end for export and protection



The Mechanism

X chromosomes have a special gene Xist (X inactivation-specific transcript).
RNA transcribed from it does not leave nucleus.
It bind to the X chromosome from which it is transcribed, and this leads to a spreading of inactivation along the chromosome.

The active X makes interference RNA (RNAi) anti-Xist gene, appropriately called Tsix. This gene codes for an RNAi that binds by complementary base pairing to Xist RNA at the active X chromosome.



14.18 A Model for X Chromosome Inactivation Interference RNA (RNAi) and chromosomal proteins combine to inactivate the X chromosome.

END

Cell Signaling & Prehistoric world

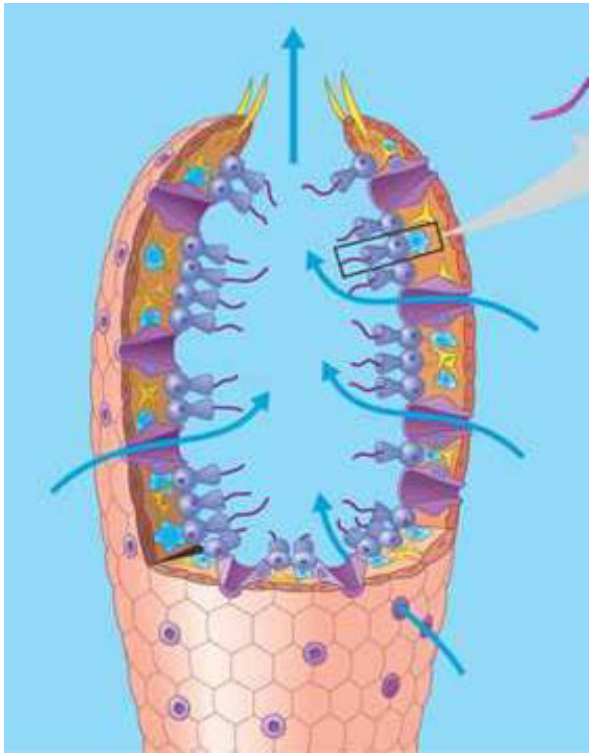
The Prehistoric world

Earths changing atmosphere

Ability to adapt key to survival and continuity

Ability to sense changes quintessential

Nutrition, Safety & Mate



Multicellular = Specialization = Coordination
Ability to sense & respond to external and internal environment

The Necessity

In order to survive even simplest organisms need to sense and respond to their environment.

It is critical that the cells of multicellular organisms communicate to coordinate their efforts (Running).

Multicellular organism cells are specialized and rely on each other for the support (brain sugar).

During development there have to be checks balances on differentiation (analogy society).

Signals

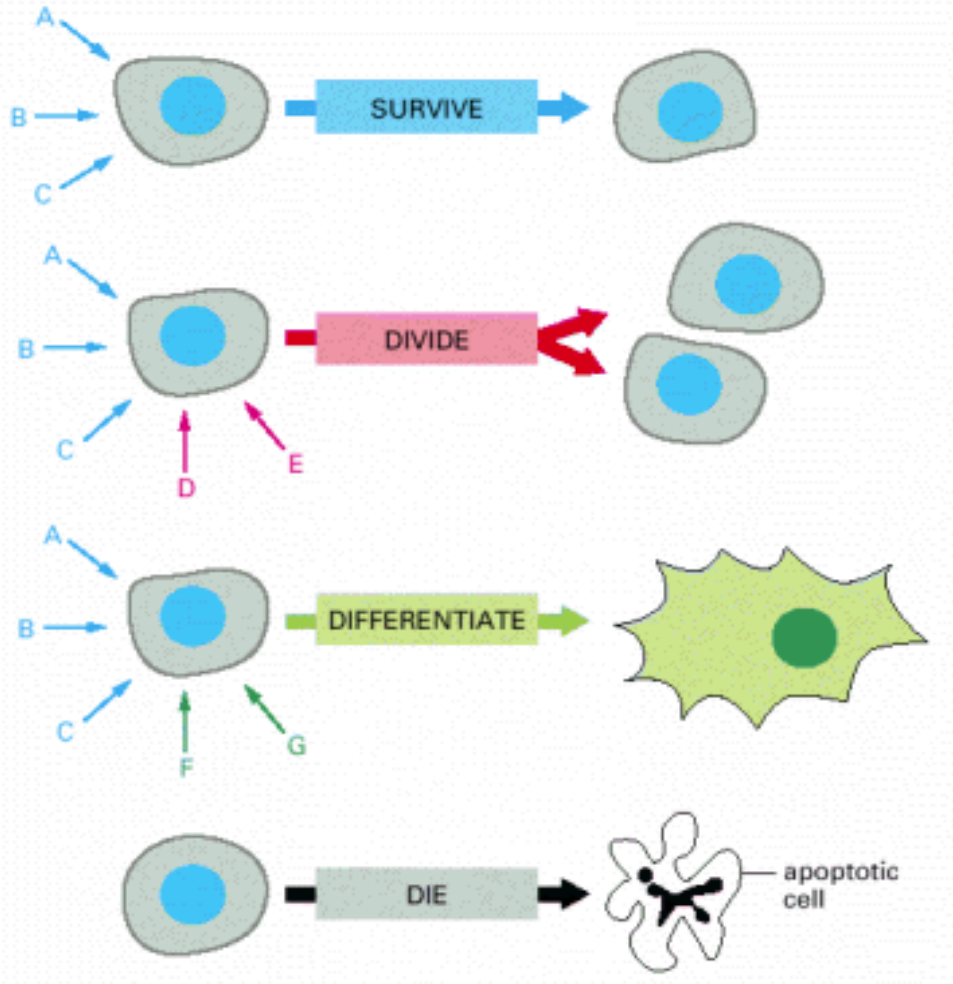
Our sense organs allow us to respond to light, odors and tastes (chemical), temperature, touch, and sound.

Magnetism can be a signal: some bacteria and birds orient themselves to Earth's magnetic poles.

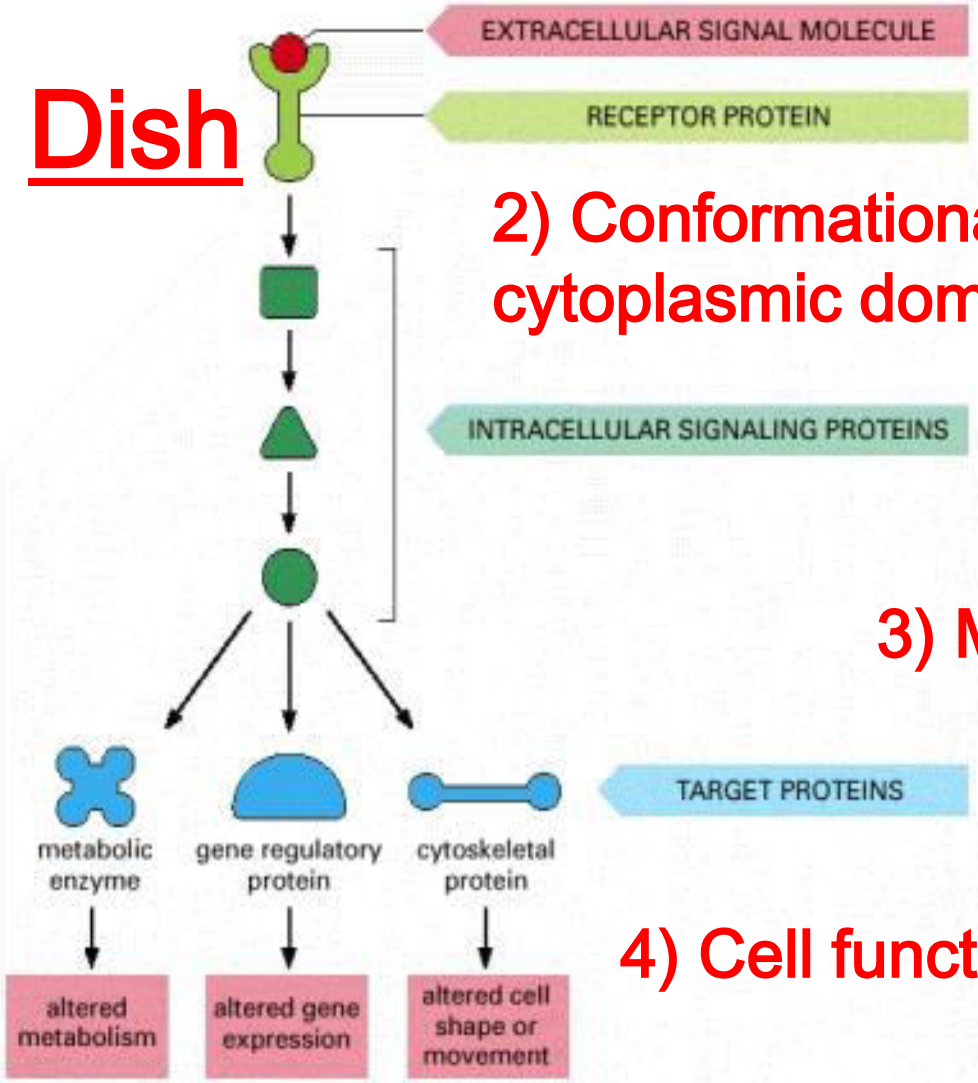
END

The Instructions & The Design

Manipulate Gene expression & Protein Functions



Dish



1) Ligand binding

2) Conformational change cytoplasmic domain

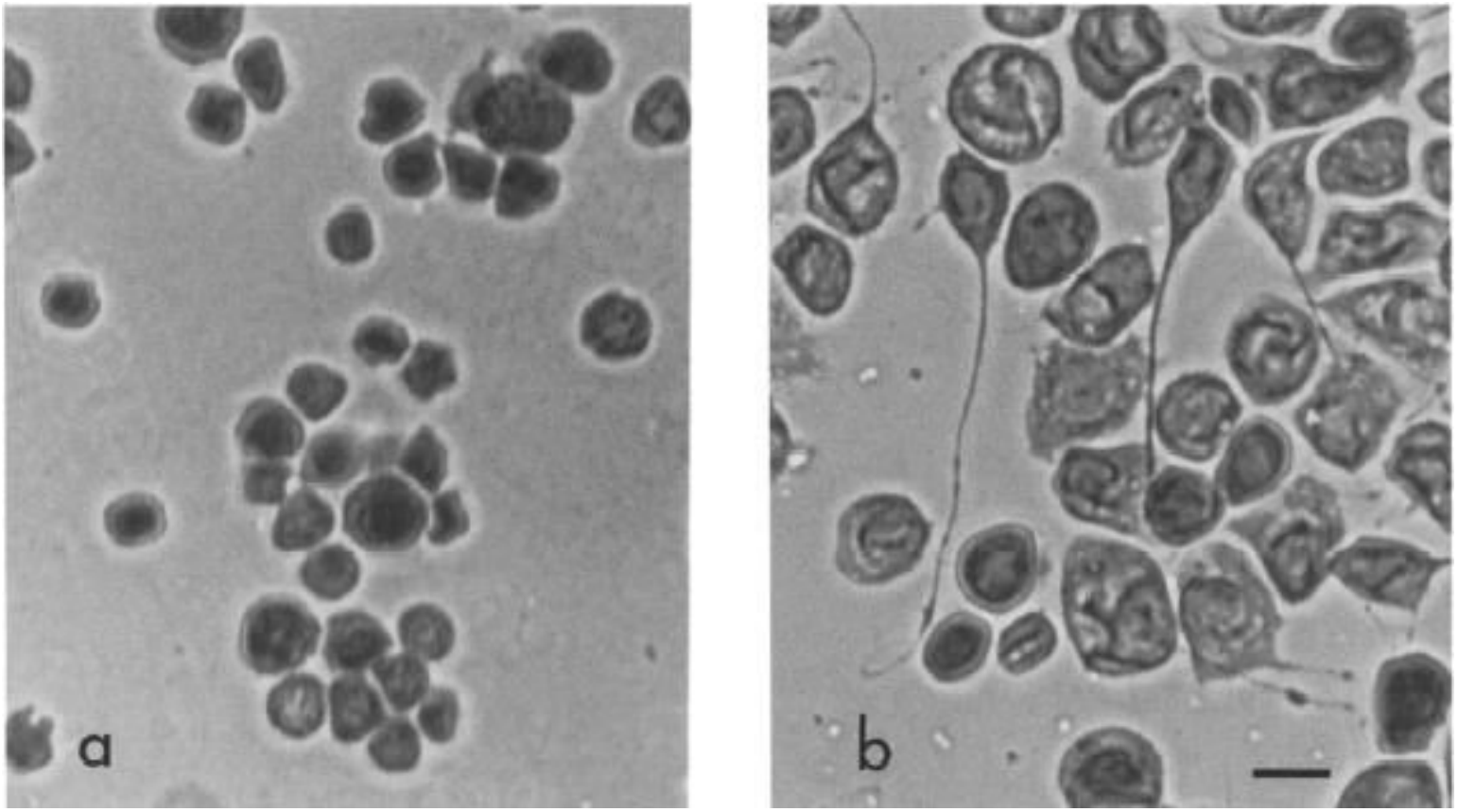
3) Mediators

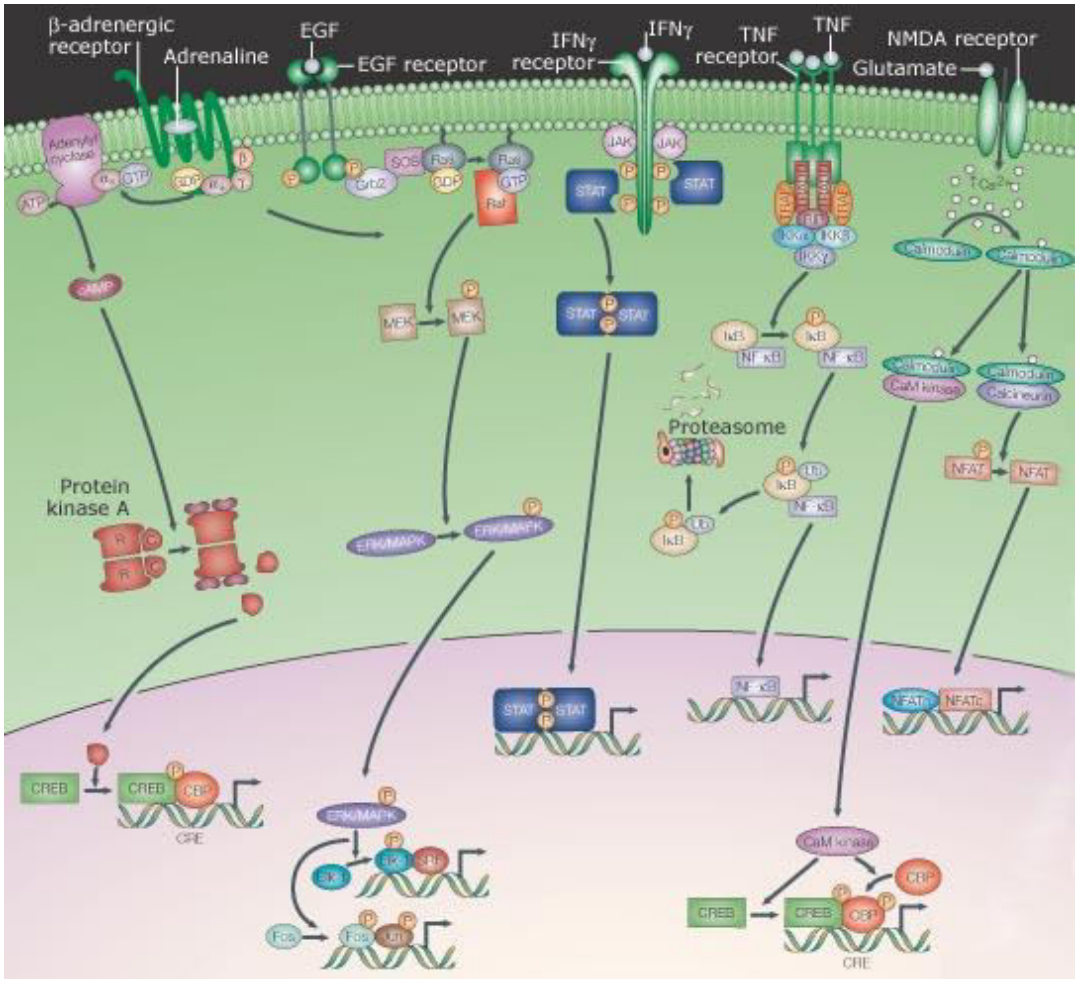
4) Cell function modified

Gene Gategories

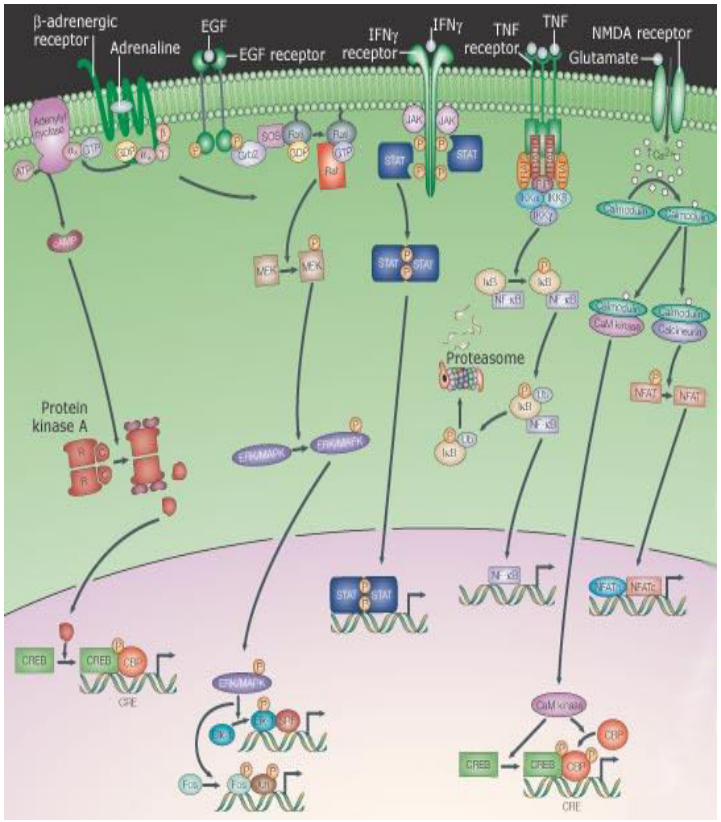
- Some of genes are expressed in all cells all the time. Housekeeping genes are responsible for the routine metabolic functions (e.g. respiration).
- Some are expressed as a cell enters a particular pathway of differentiation.
- Some are expressed all the time in only those cells that have differentiated in a particular way.
- Some are expressed only as conditions around and in the cell change. For example, the arrival of a hormone may turn on/off certain genes in that cell.

Alters Phenotype/Physiology of Cells



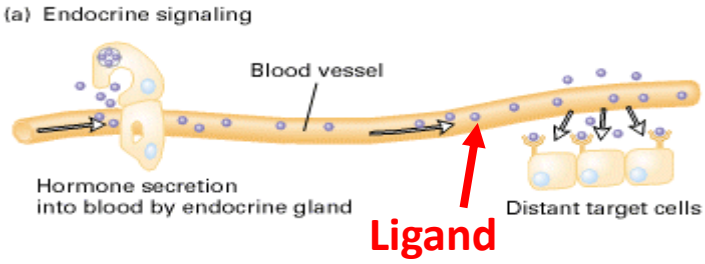


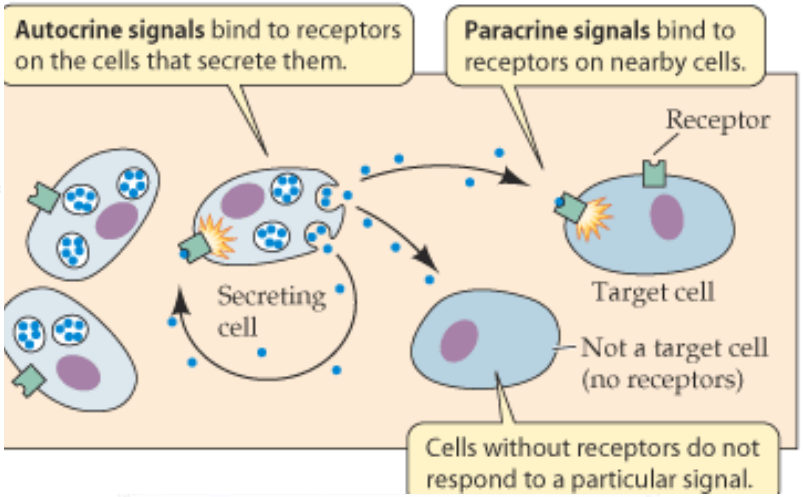
The entire signaling process, from signal detection to final response is called signal transduction pathway/ signaling cascade

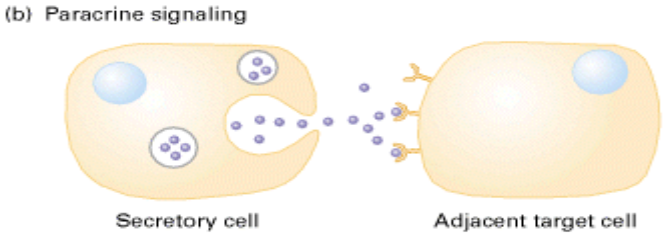

β-adrenergic receptor
Adrenaline
EGF
EGF receptor
IFNγ receptor
IFNγ
TNF receptor
TNF
NMDA receptor
Glutamate
Protein kinase A
Proteasome
CREB
CRE

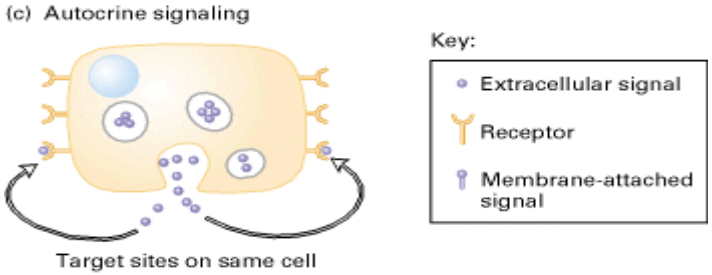
END

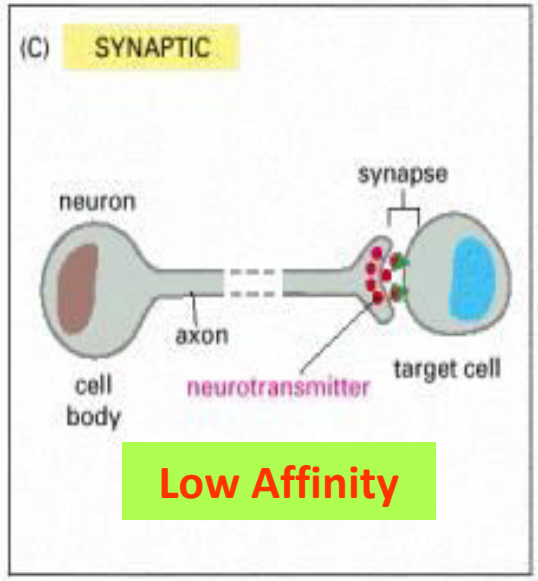
The Modes & Principles

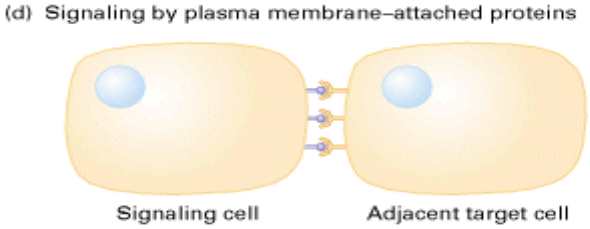

(a) Endocrine signaling
Blood vessel
Hormone secretion into blood by endocrine gland
Distant target cells
Ligand


Autocrine signals bind to receptors on the cells that secrete them.
Paracrine signals bind to receptors on nearby cells.
Receptor
Secreting cell
Target cell
Not a target cell (no receptors)
Cells without receptors do not respond to a particular signal.


(b) Paracrine signaling
Secretory cell
Adjacent target cell

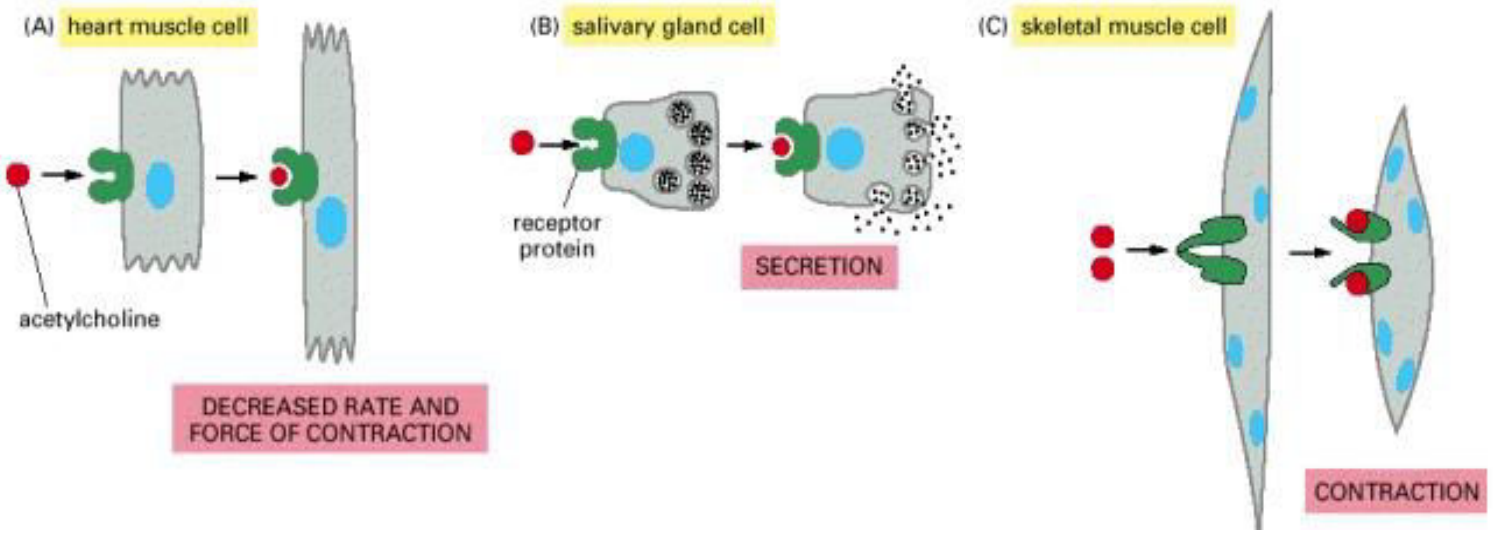

(c) Autocrine signaling
Target sites on same cell
Key:
Extracellular signal
Receptor
Membrane-attached signal


(C) SYNAPTIC
neuron
synapse
axon
cell body
neurotransmitter
target cell
Low Affinity


(d) Signaling by plasma membrane–attached proteins
Signaling cell
Adjacent target cell

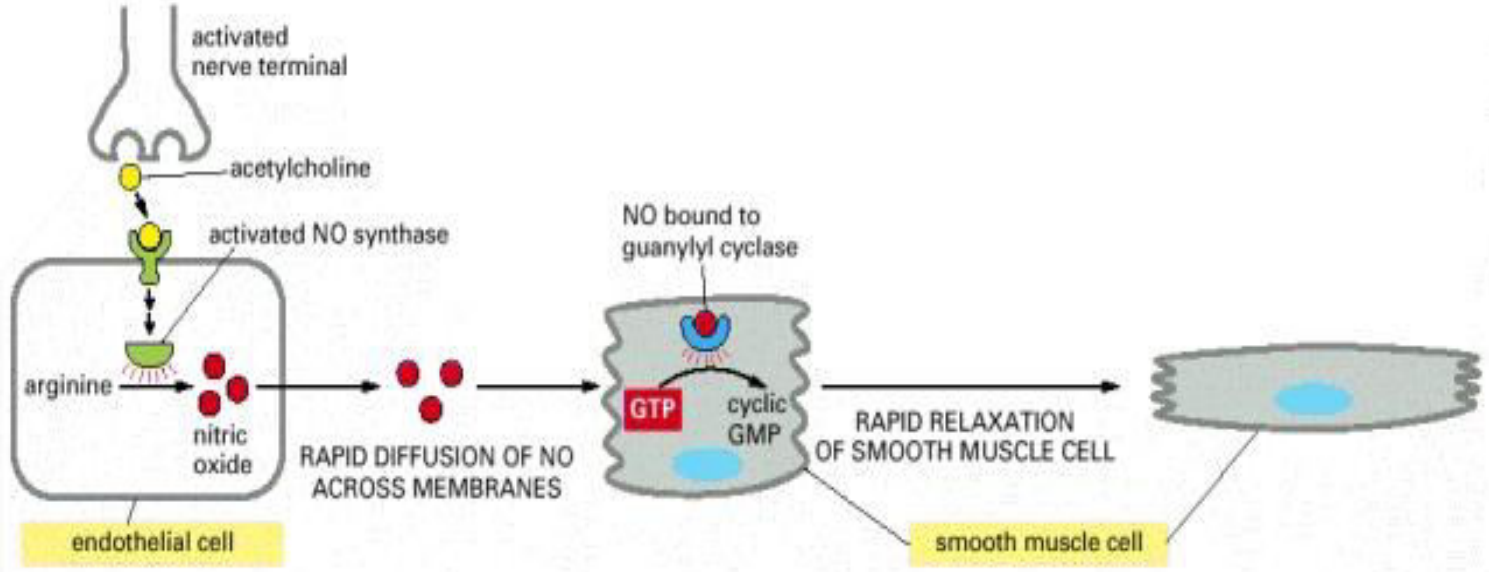
Receptor combinations confer cell behavior in an environment flooded with hundreds of ligands

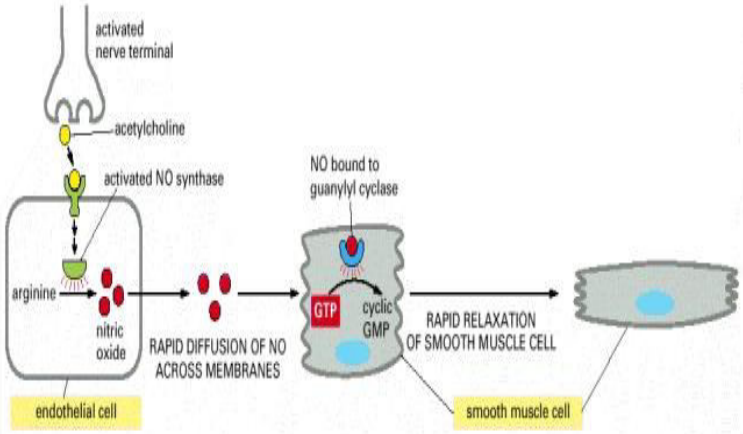
Cellular machinery specifies cell response to a particular ligand



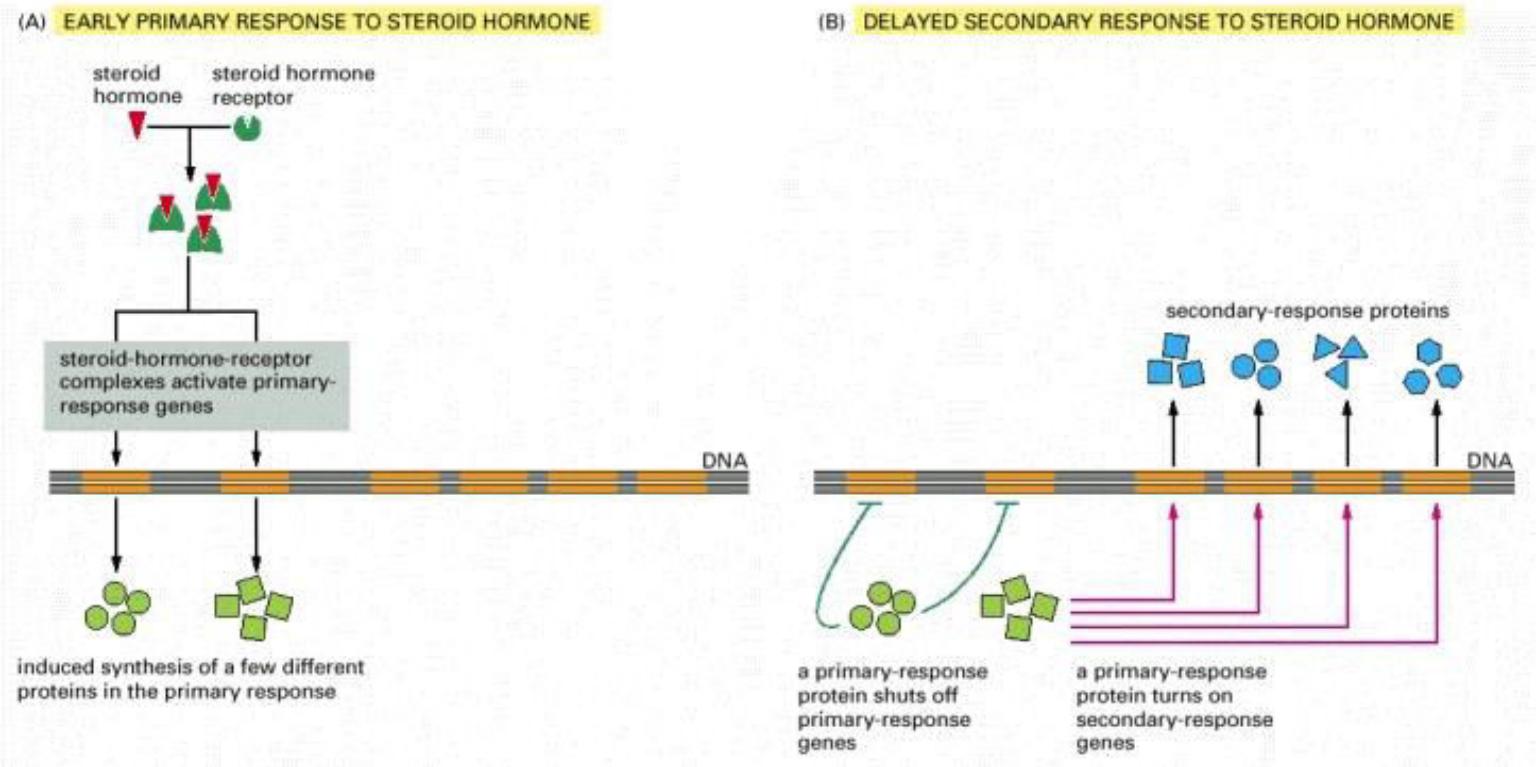
Ach → Receptor-ACh → NO synthase → Deamination
NO → Diffusion → G-cyclase → cGMP → Relaxation

NO half life 5 seconds




activated
nerve terminal
acetylcholine
activated NO synthase
NO bound to
guanylyl cyclase
arginine
nitric
oxide
RAPID DIFFUSION OF NO
ACROSS MEMBRANES
GTP
cyclic
GMP
RAPID RELAXATION
OF SMOOTH MUSCLE CELL
endothelial cell
smooth muscle cell

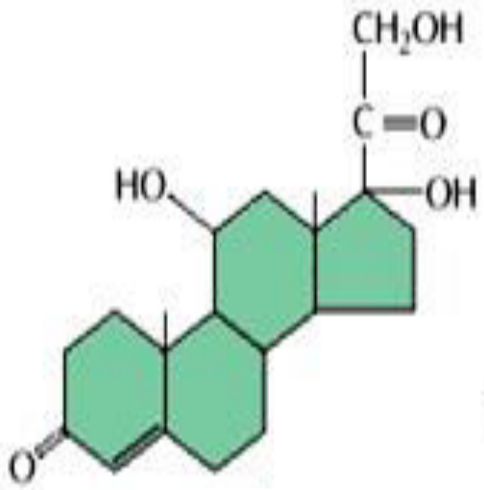
The Modes & Principles II



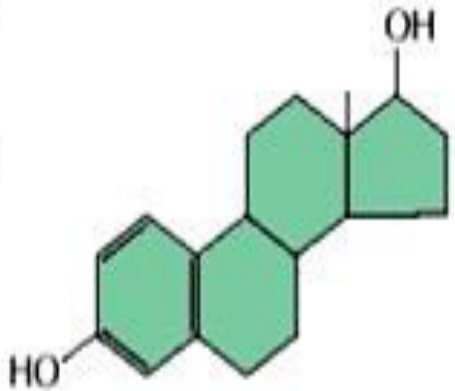
Ligand binding removes inhibitory proteins and facilitates binding of transcription activators

Small Hydrophobic Lipid soluble molecules eg steroid & thyroid hormones, retinoids & Vit D etc

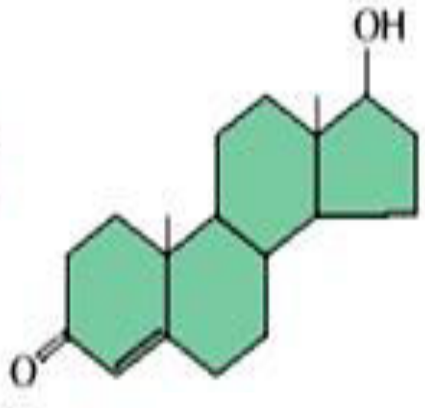
Blood transport via carrier proteins longer life (thy days Ach ms) Carrier left outside



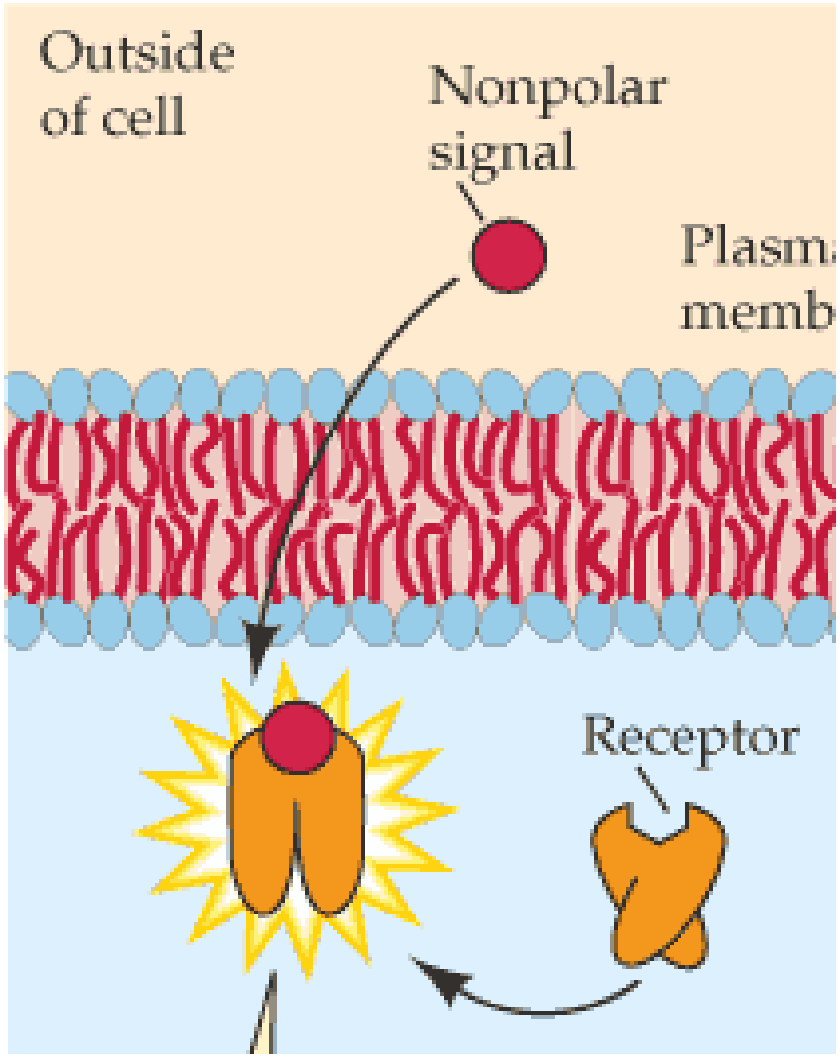
Cortisol



Estradiol

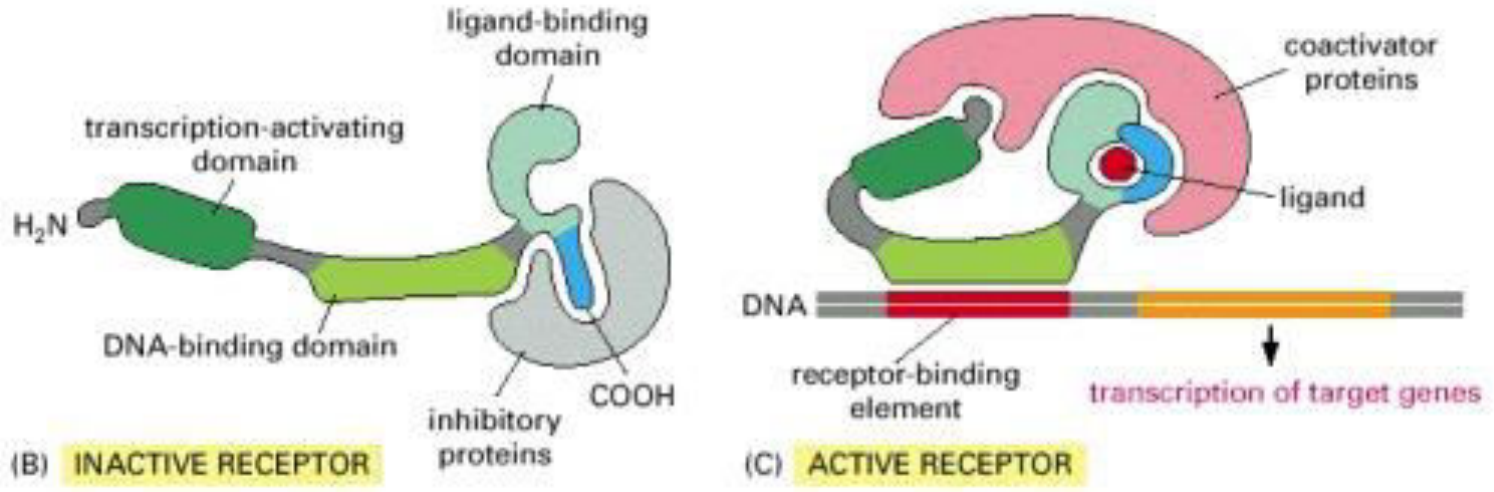


Testosterone



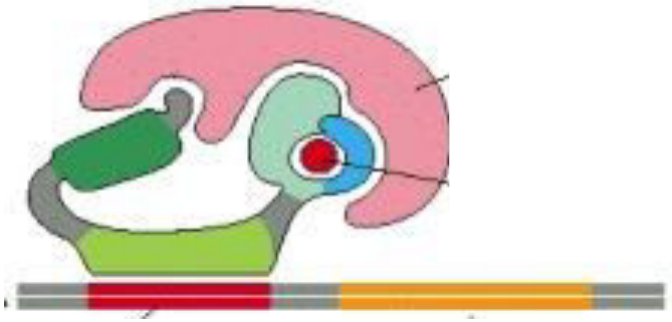
Inactive ICR may be DNA bound or in cytoplasm (NLS nonfunctional)

Activated receptor binds DNA induces gene transcription



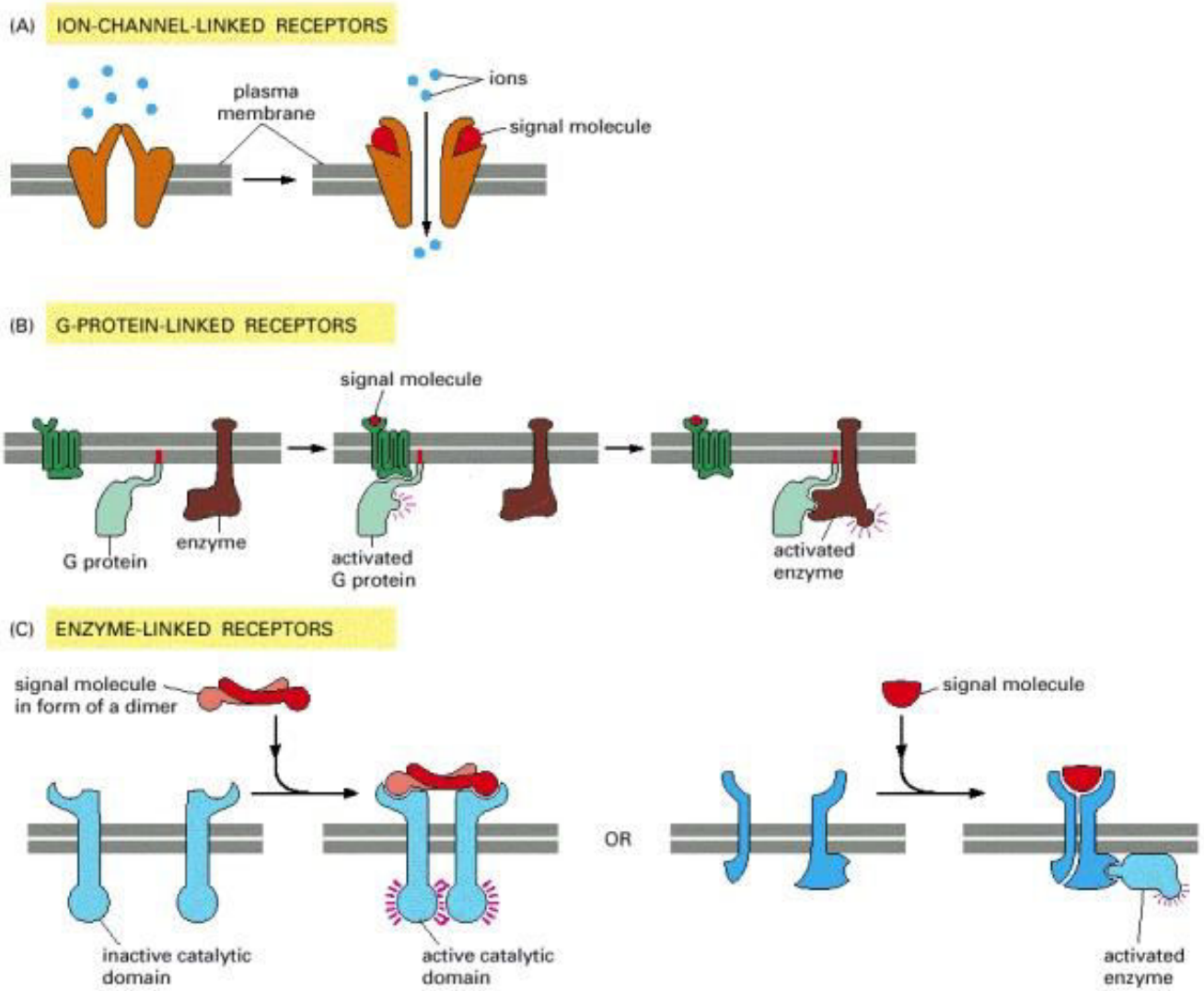
Different cells with identical ICRs regulate different genes due to other cell specific mediators

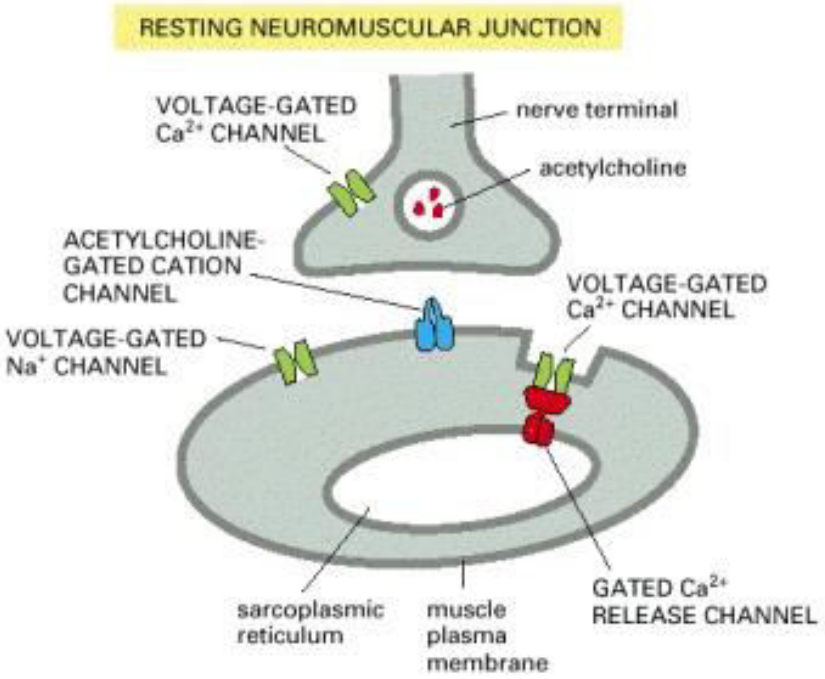
Right combination of co-activators/gene regulators required to transcribe specific genes (testosterone)

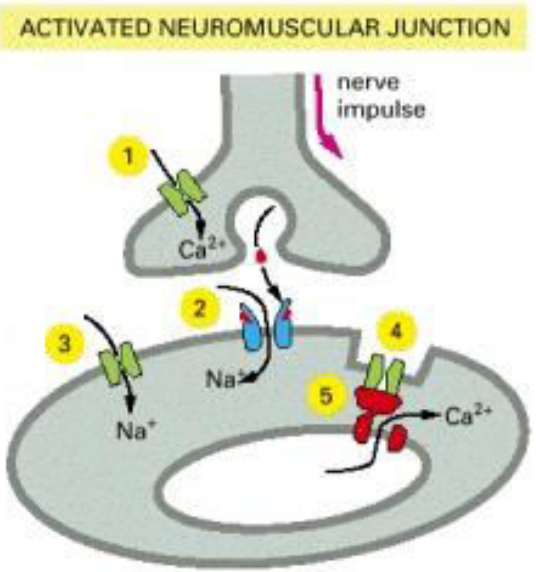


END

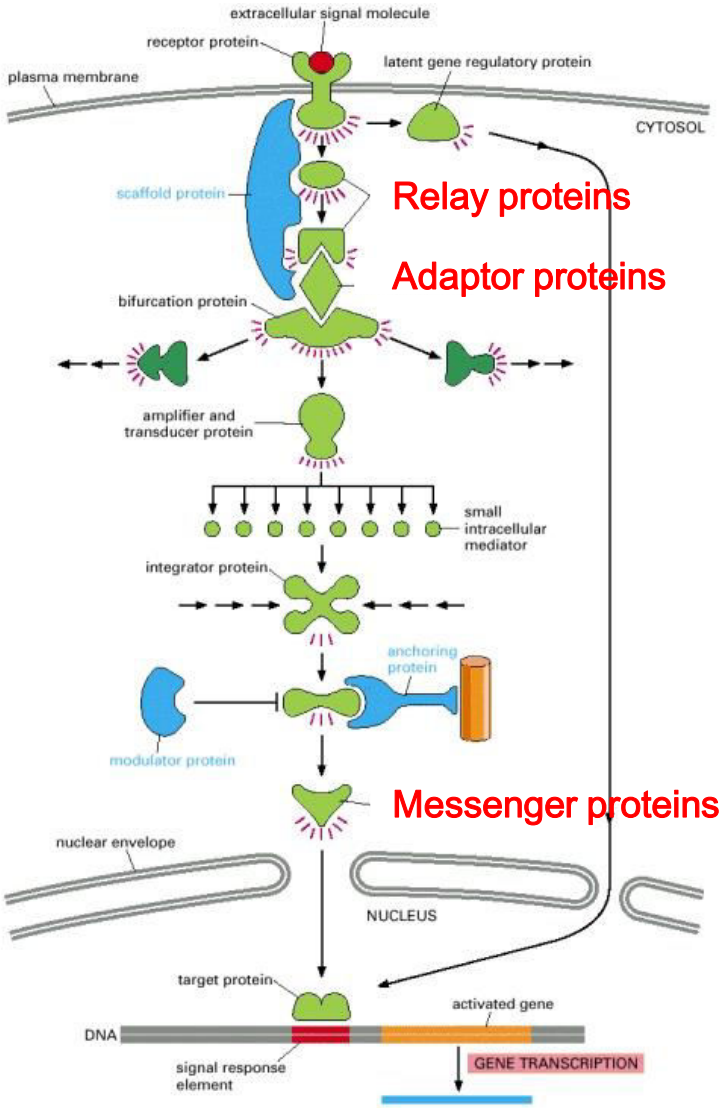
The Modes & Principles III


(A) ION-CHANNEL-LINKED RECEPTORS
plasma membrane
ions
signal molecule
(B) G-PROTEIN-LINKED RECEPTORS
signal molecule
G protein
enzyme
activated G protein
activated enzyme
(C) ENZYME-LINKED RECEPTORS
signal molecule in form of a dimer
inactive catalytic domain
active catalytic domain
OR
signal molecule
activated enzyme





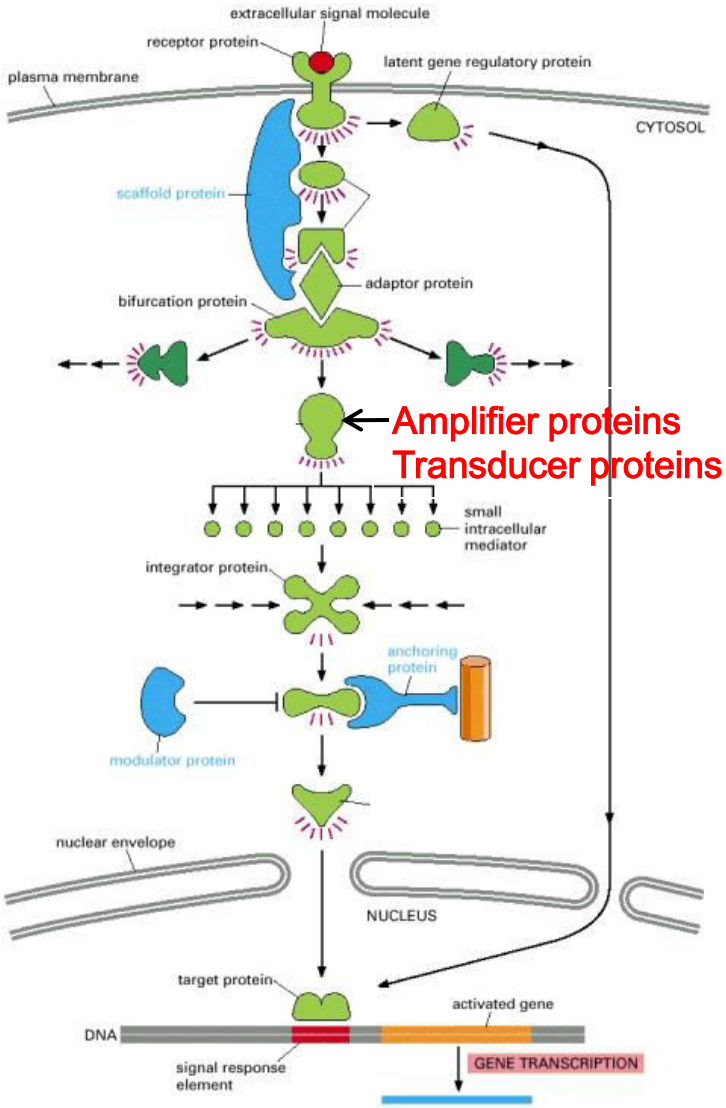
Neurotransmitters produce all or noting response



1) Relay proteins pass the message to the next signaling component in the chain.

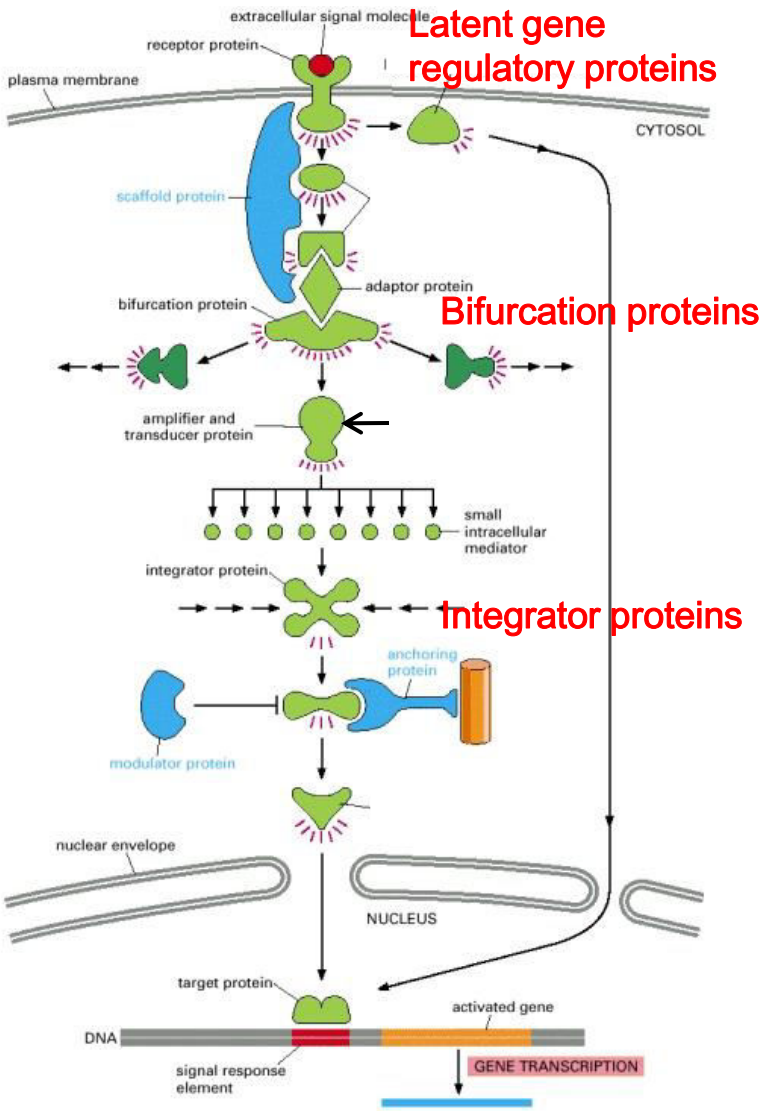
2) Messenger proteins carry the signal from one part of the cell to another (cytosol to nucleus).

3 Adaptor proteins link one signaling protein to another, without themselves conveying a signal.



4) Amplifier proteins either enzymes or ion channels, increase the signal they receive, (Produce large amounts of small intracellular mediators/activate signaling proteins (adenylyl cyclase).

5 Transducer proteins convert the signal into a different form (adenylyl cyclase).



Latent gene regulatory proteins

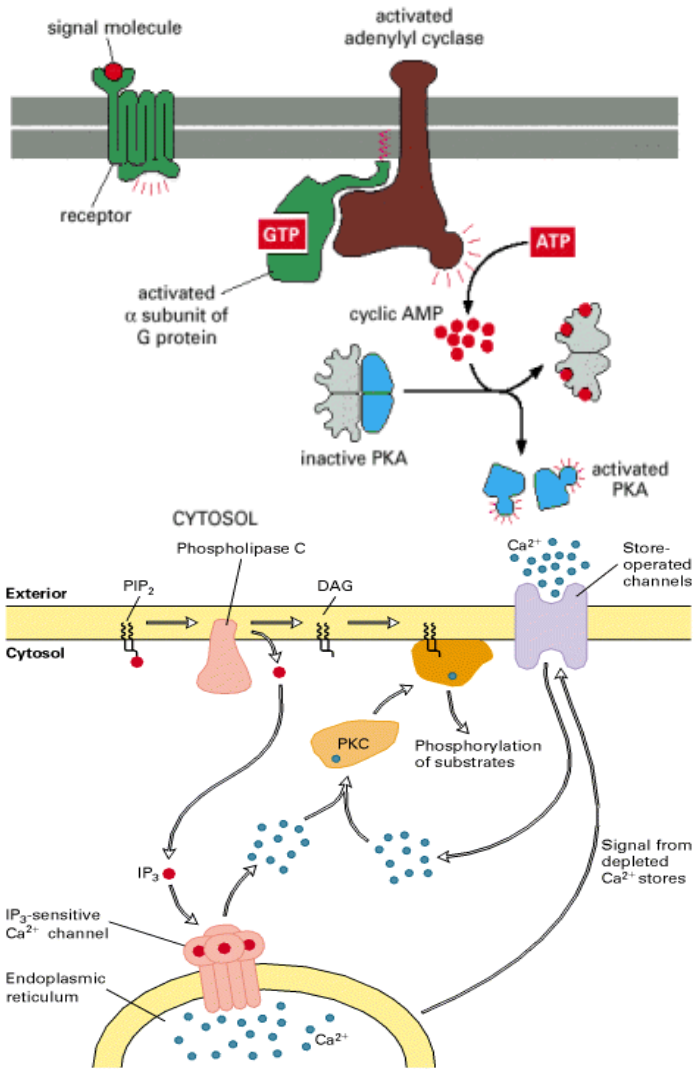
Bifurcation proteins

Integrator proteins

6 Bifurcation proteins spread signal from one pathway to another.

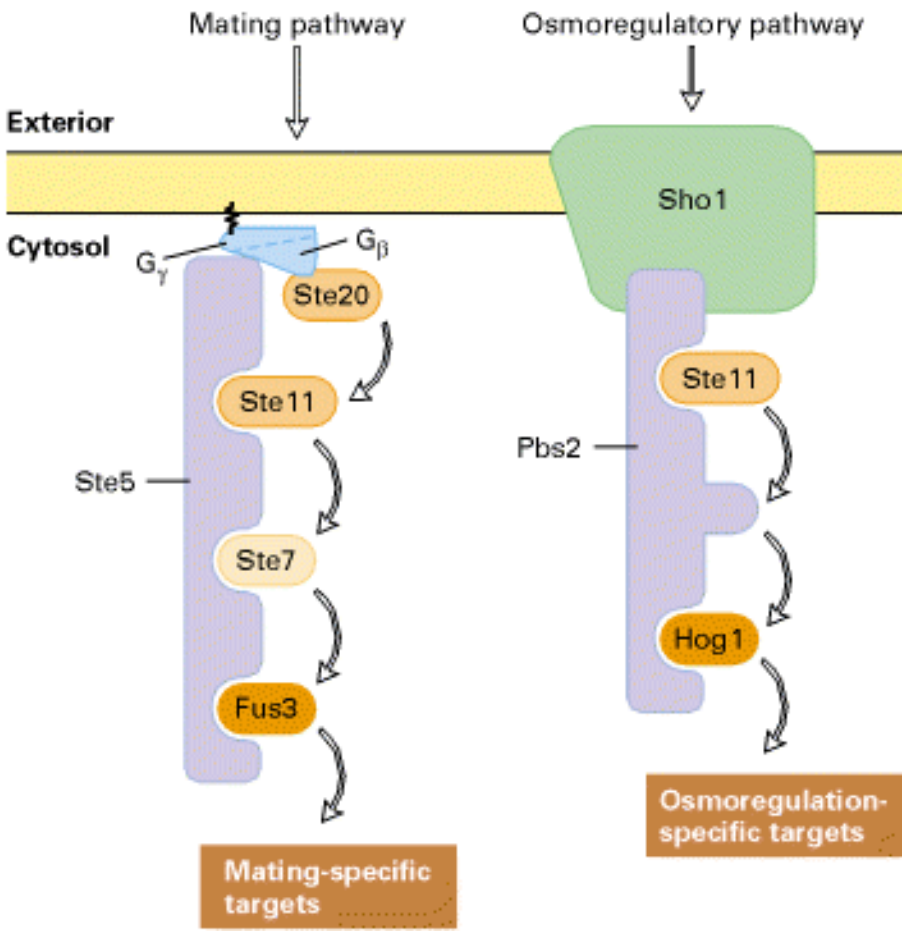
7 Integrator proteins receive signals from 2++ pathways integrate them before relaying a signal onward.

8 Latent gene regulatory proteins activated at the cell surface then migrate to nucleus to stimulate gene transcription.



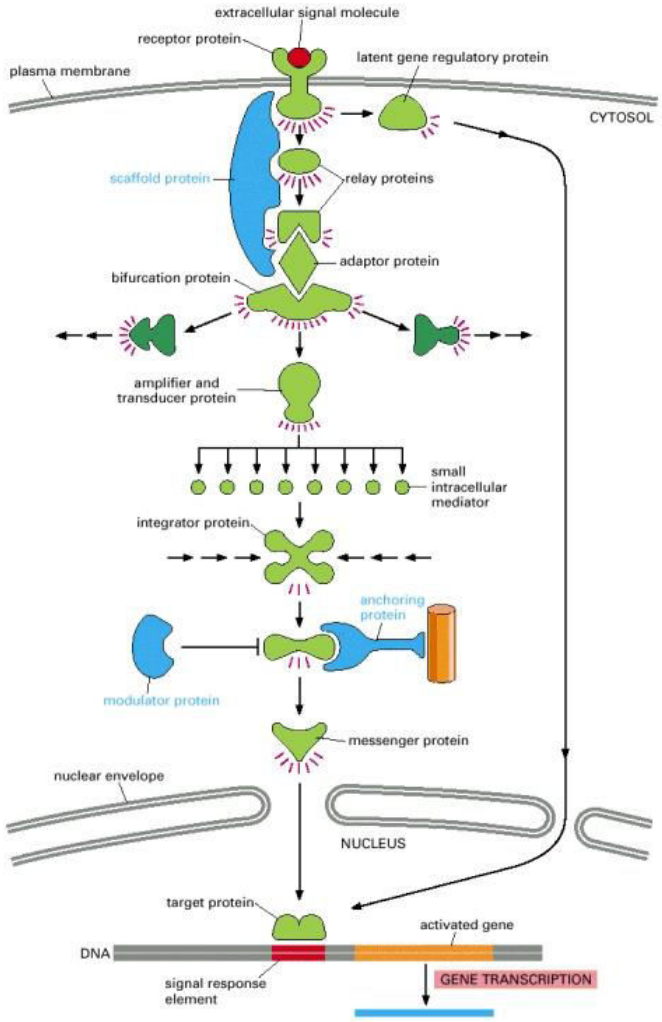
SICMs are produced/released in response to signal received by the receptor

SICMs donot have an enzymatic activity of their own however they modify the function of other molecules



Same ligand-receptor complexes may produce different responses

Many cells use scaffold proteins to enhance speed efficiency & specificity of response


extracellular signal molecule
receptor protein
latent gene regulatory protein
plasma membrane
CYTOSOL
scaffold protein
relay proteins
adaptor protein
bifurcation protein
amplifier and
transducer protein
small
intracellular
mediator
integrator protein
anchoring
protein
modulator protein
messenger protein
nuclear envelope
NUCLEUS
target protein
activated gene
DNA
signal response
element
GENE TRANSCRIPTION

END

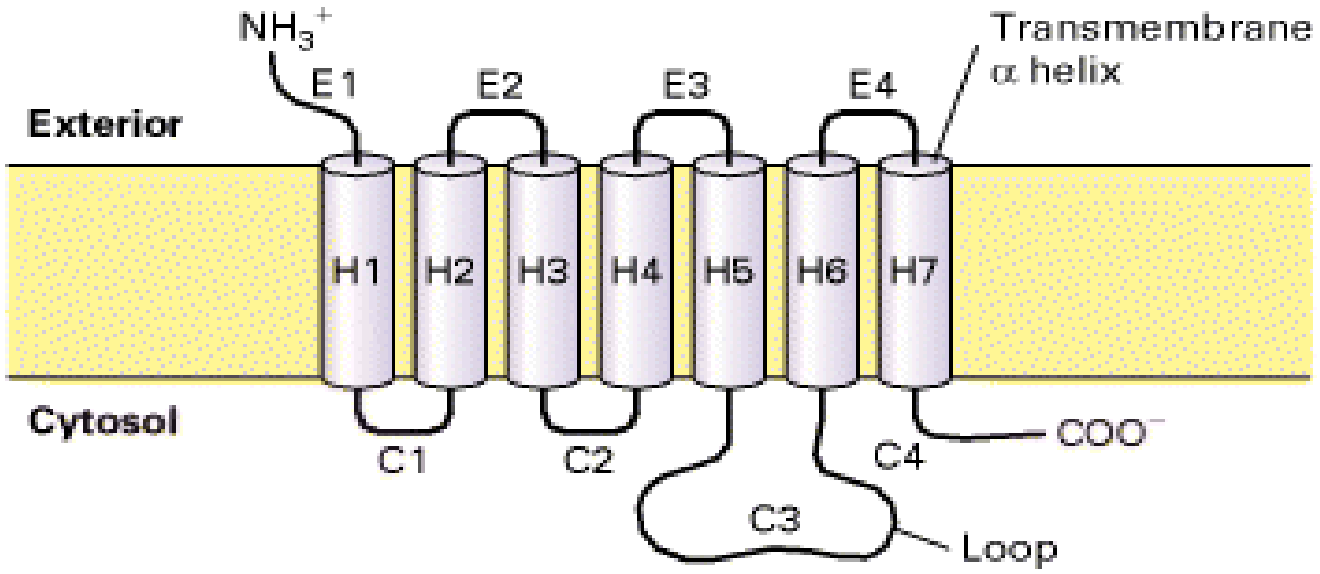
G Protein-Linked Receptors

Helices 3, 5 & 6, that participate in agonist binding

Ligand binding moves helices 5 and 6, closer

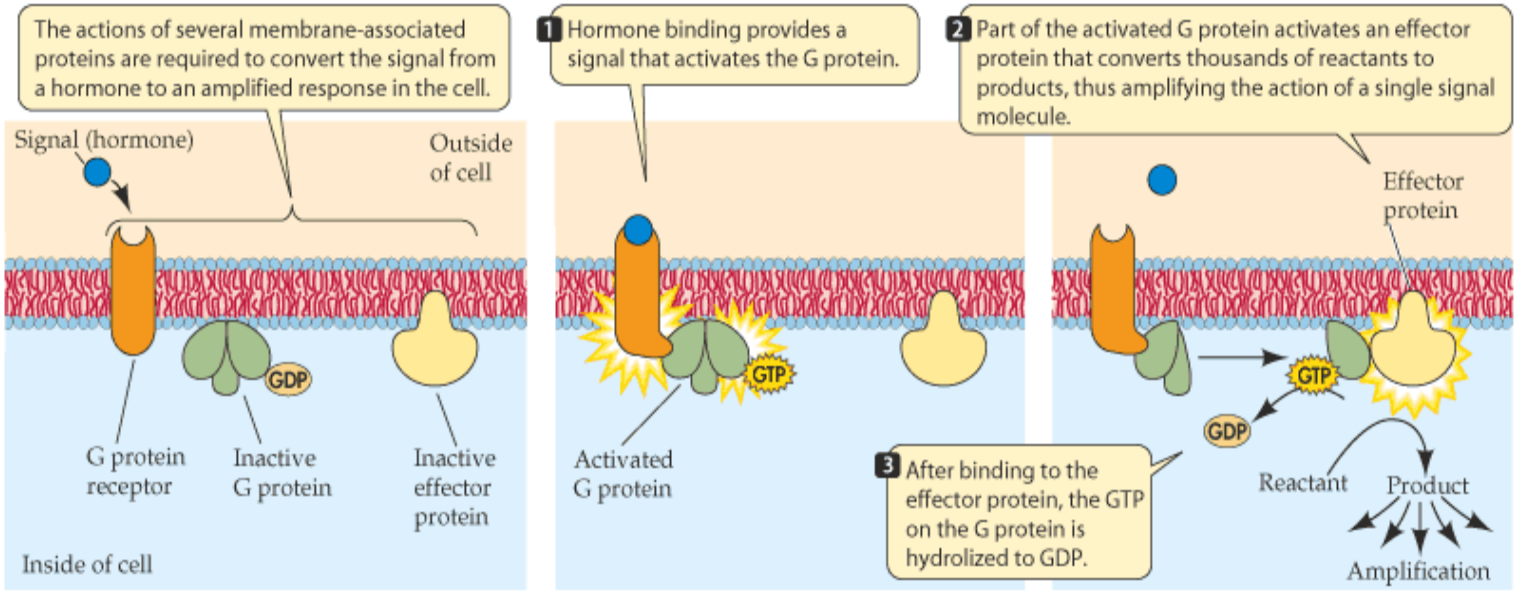
C3 loop conformation & binds/activates G protein

Helix 7 functions in determining ligand specificity



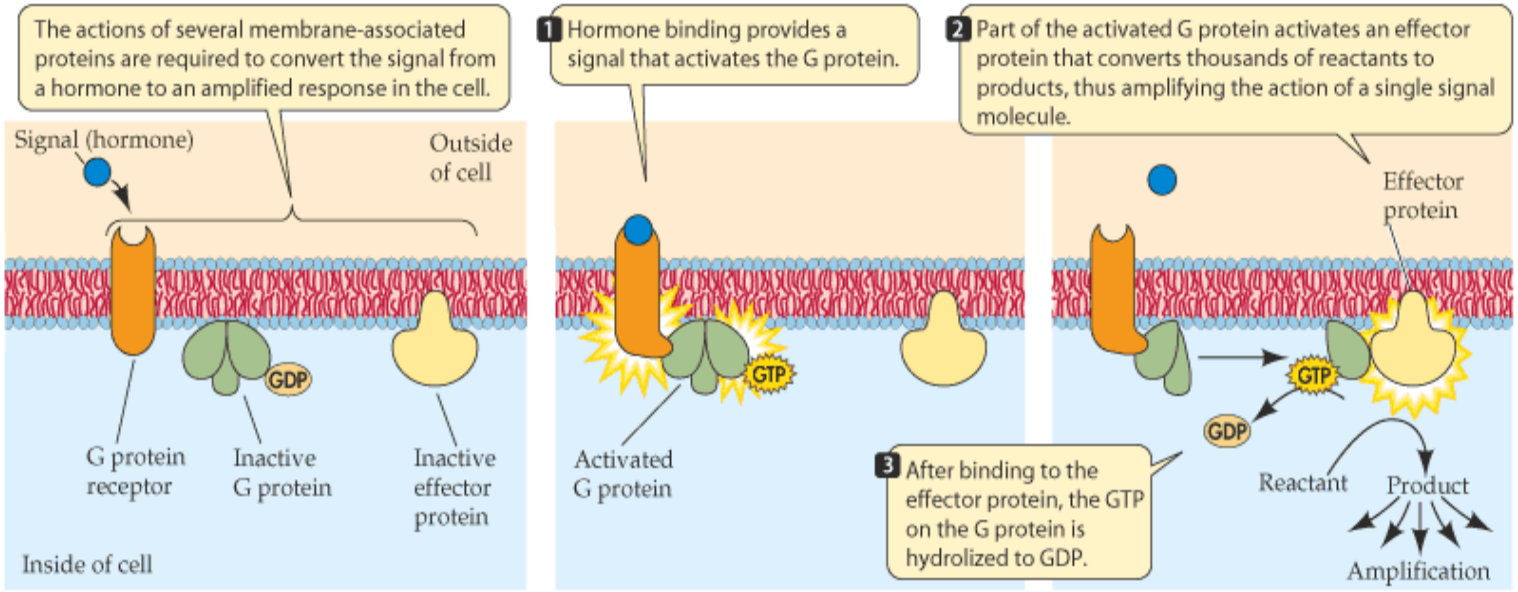
Ligand binding causes a structural change permitting G protein to bind receptor.

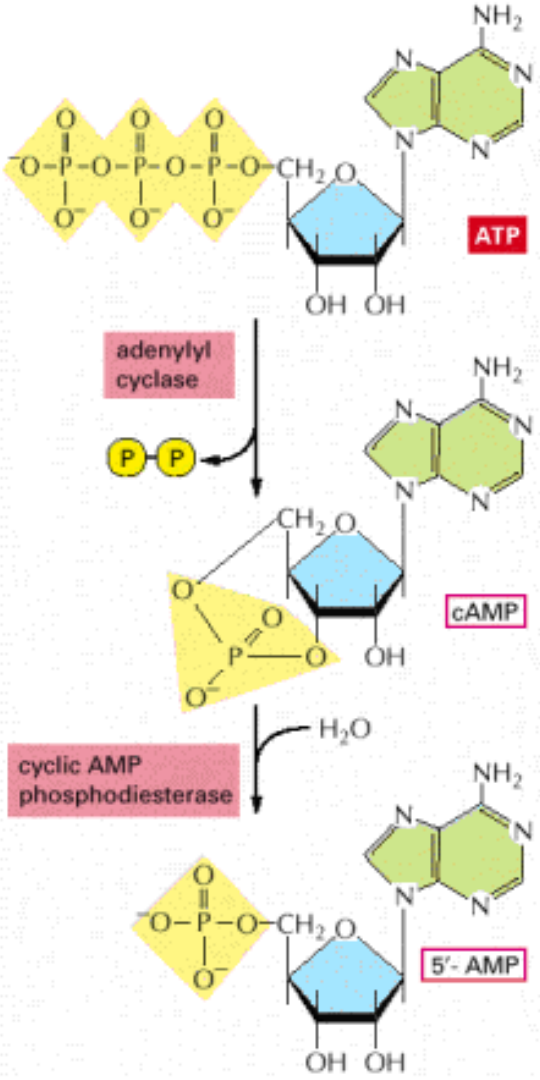
Binding of G protein to activated receptor causes it to exchange GDP for GTP (ligand releases).



Subunit of G protein separates and activates an effector molecule (causing a functional change).

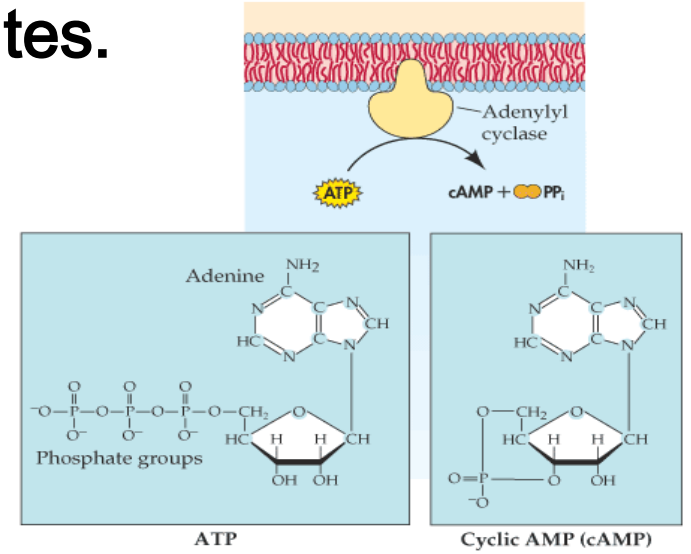
Epinephrine effects different cells differently (heart muscle contracts, intestinal vascular smooth muscle relaxes more nutrients absorbed).

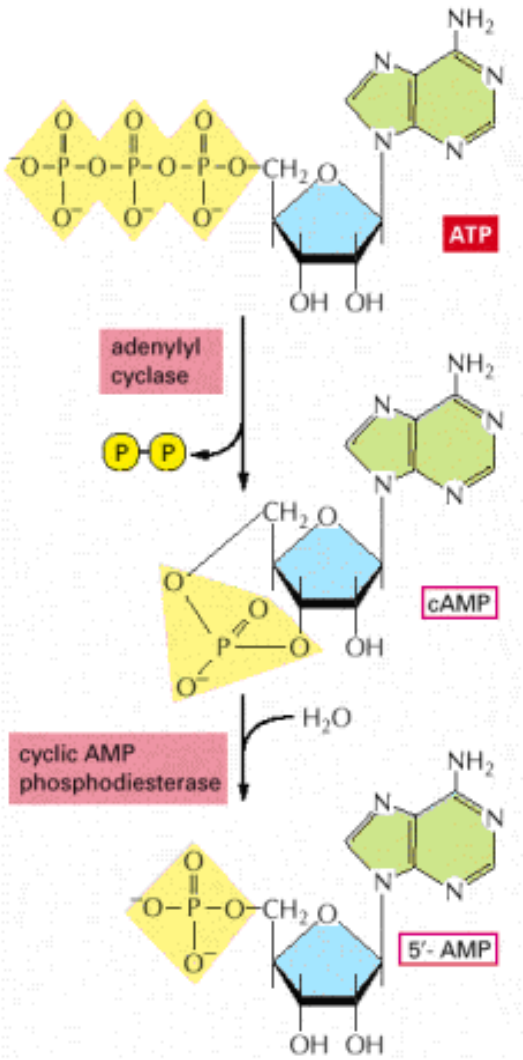




Second messengers are allosteric regulators and do not have enzymatic activity

Cyclic AMP (cAMP) can bind ion channels to open them or bind enzymes to exposing their active sites.





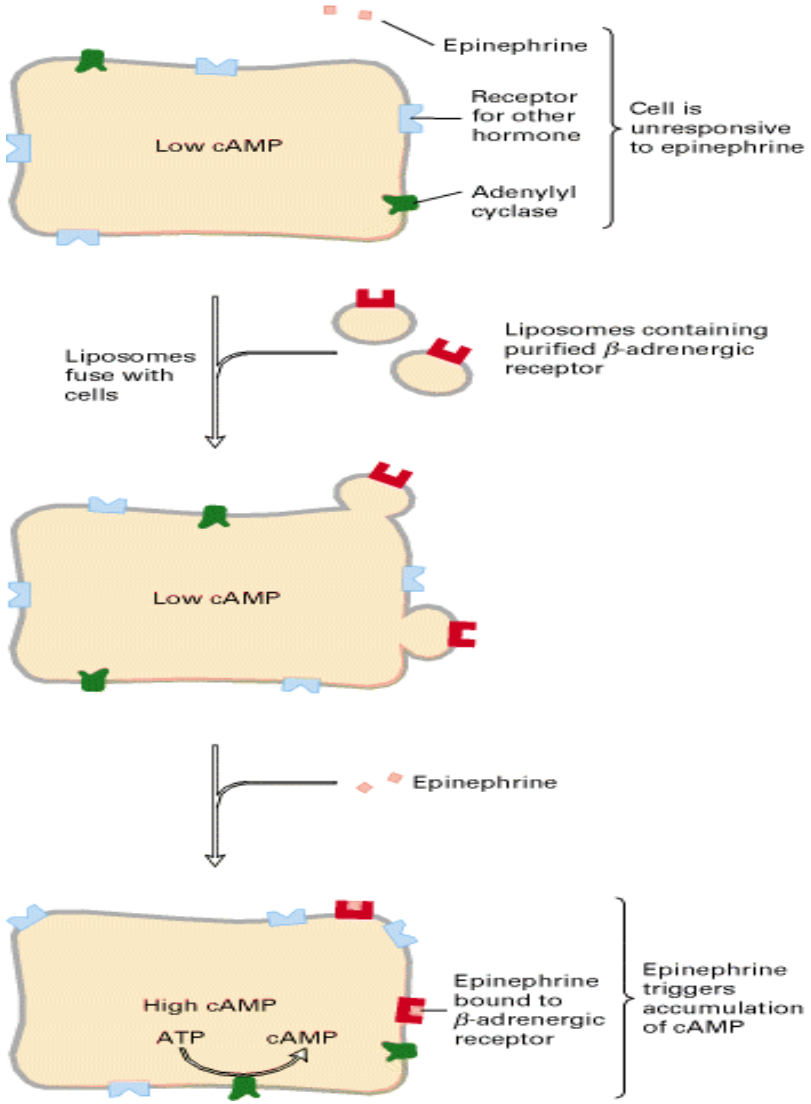
Adenylyl cyclase

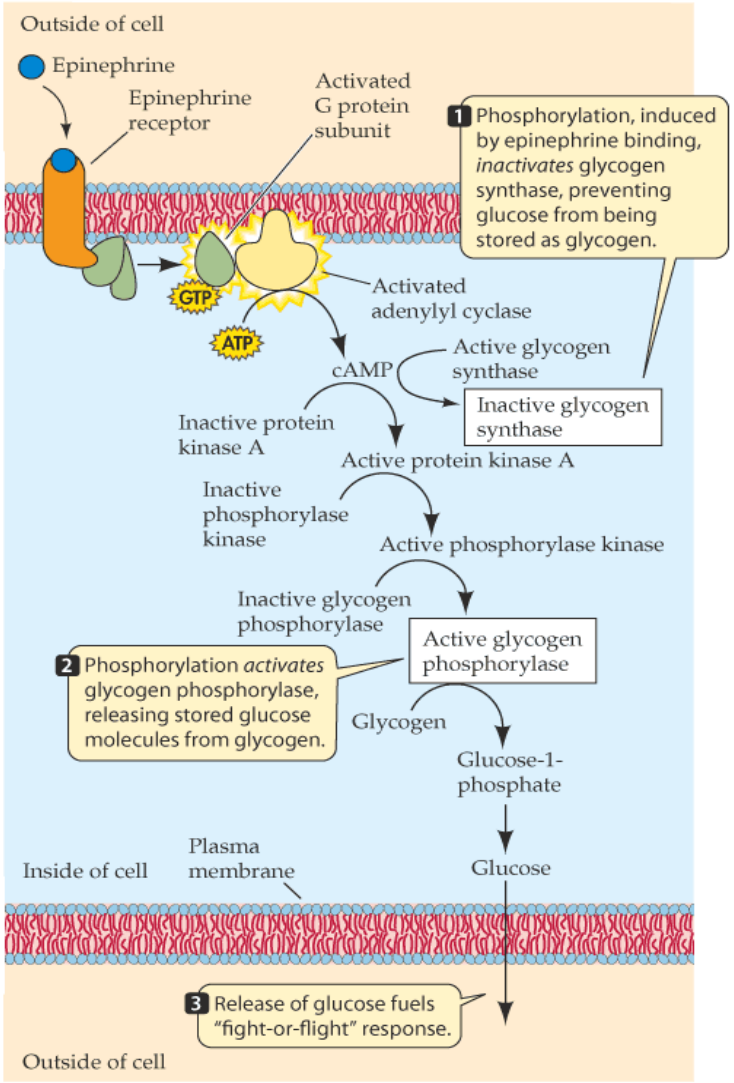
one receptor 100 active Gs

G-GTP, activates a single AC

One Gs bound AC catalyzes synthesis of many cAMP

Several hundred cAMP molecules are produced per receptor-hormone complex


Epinephrine
Receptor for other hormone
Low cAMP
Adenylyl cyclase
Cell is unresponsive to epinephrine
Liposomes fuse with cells
Liposomes containing purified β-adrenergic receptor
Low cAMP
Epinephrine
High cAMP
ATP
cAMP
Epinephrine bound to β-adrenergic receptor
Epinephrine triggers accumulation of cAMP


Outside of cell
Epinephrine
Epinephrine receptor
Activated G protein subunit
1 Phosphorylation, induced by epinephrine binding, *inactivates* glycogen synthase, preventing glucose from being stored as glycogen.
GTP
ATP
Activated adenylyl cyclase
cAMP
Active glycogen synthase
Inactive glycogen synthase
Inactive protein kinase A
Active protein kinase A
Inactive phosphorylase kinase
Active phosphorylase kinase
Inactive glycogen phosphorylase
Active glycogen phosphorylase
2 Phosphorylation *activates* glycogen phosphorylase, releasing stored glucose molecules from glycogen.
Glycogen
Glucose-1-phosphate
Plasma membrane
Inside of cell
Glucose
3 Release of glucose fuels "fight-or-flight" response.
Outside of cell

END

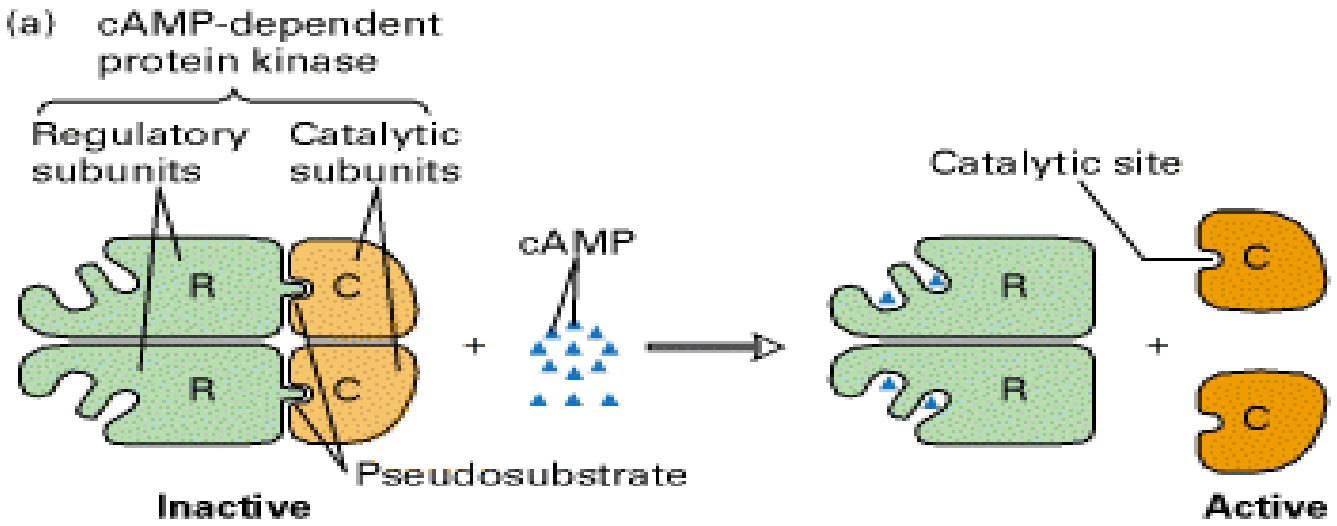
Protein Kinases A

AKA cAMP-dependent protein kinase (cAPK)

Diverse effects of cAMP are mediated through the action of cAMP-dependent protein kinases

Phosphorylate specific serine & threonine residues

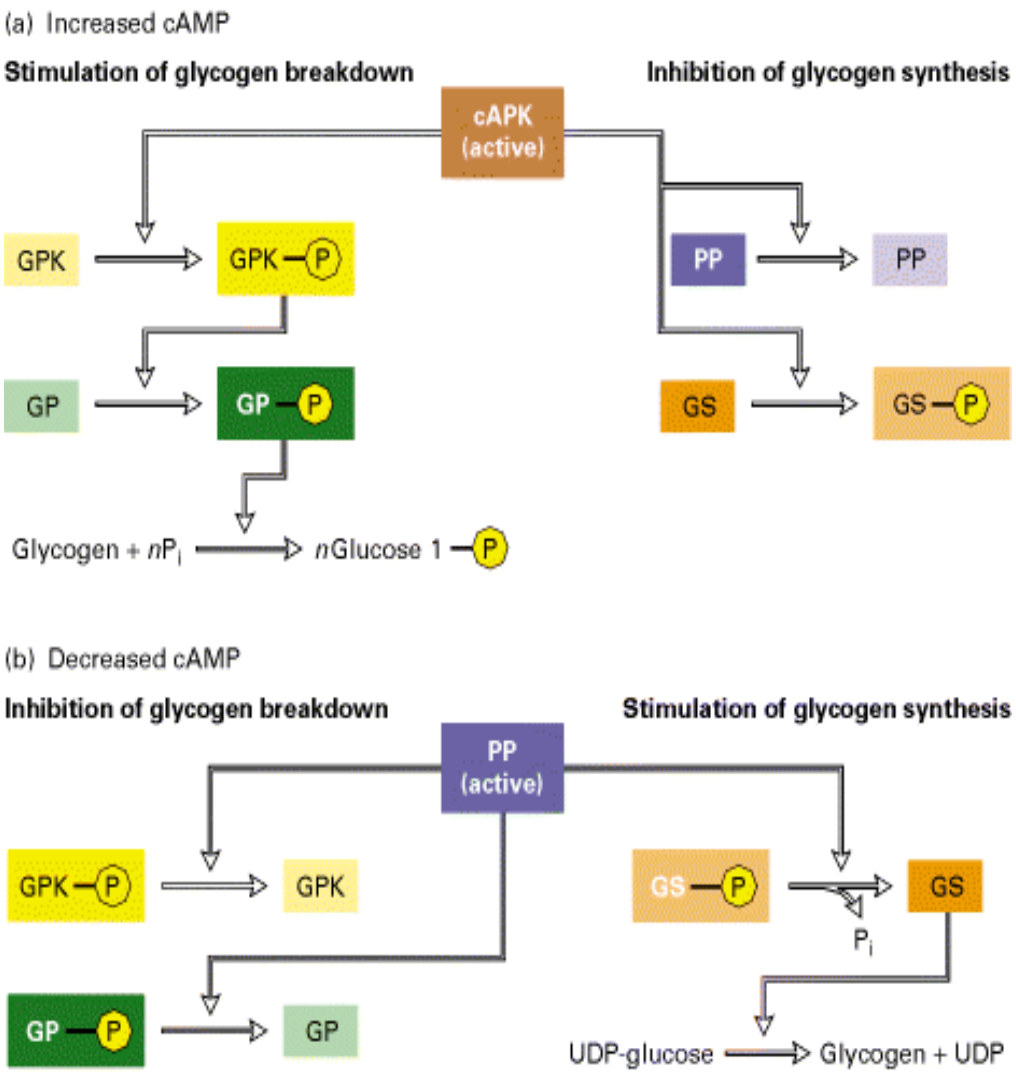
Phosphorylation of many enzymes increases their catalytic activity, whereas phosphorylation of others decreases their activity.



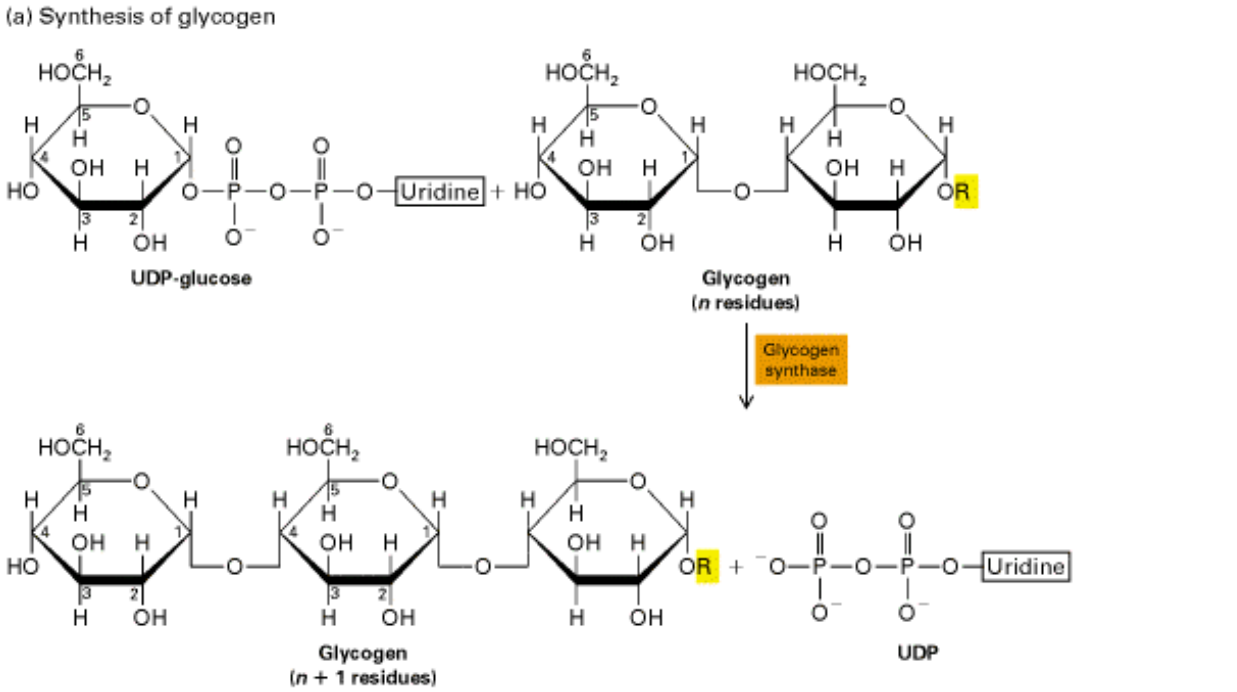
The PKAs are tetramers, two regulatory (R) subunits and two catalytic (C) subunits.

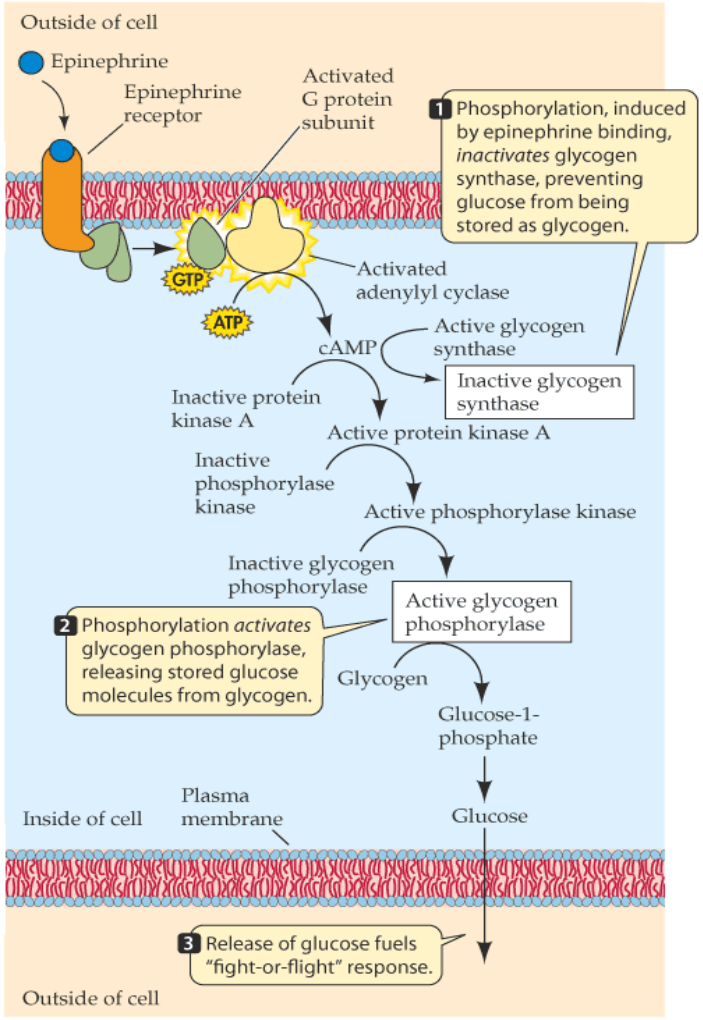
cAMP binding to the R subunits (pseudosubstrate) causes dissociation of the two C subunits

Cooperative binding



glycogen phosphorylase kinase (GPK), glycogen phosphorylase (GP) phosphoprotein phosphatase (PP), glycogen synthase (GS)


(a) Synthesis of glycogen
UDP-glucose
Uridine
Glycogen
(n residues)
Glycogen synthase
Glycogen
(n + 1 residues)
UDP
(b) Degradation of glycogen
Glycogen
(n residues)
Glycogen phosphorylase
Pi
Glucose 1-phosphate
Glycogen
(n − 1 residues)

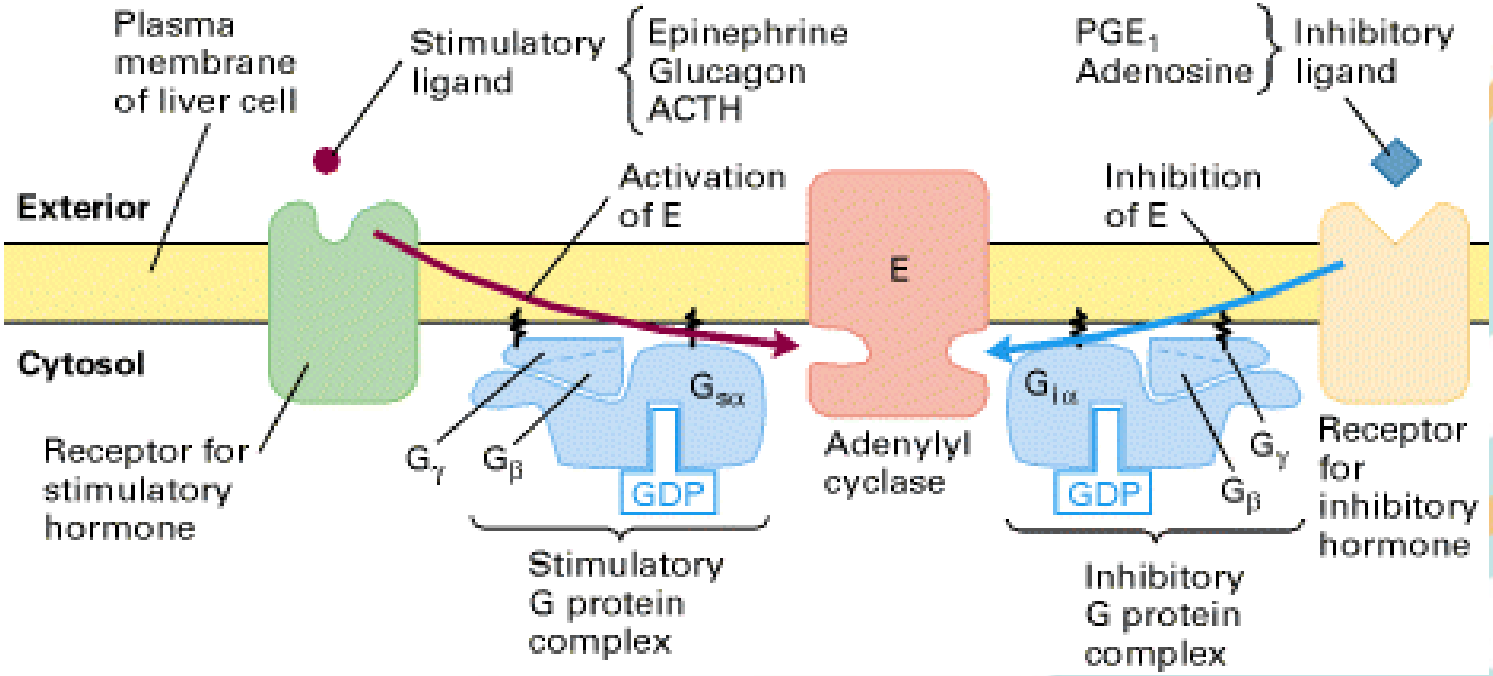

Outside of cell
Epinephrine
Epinephrine receptor
Activated G protein subunit
1 Phosphorylation, induced by epinephrine binding, *inactivates* glycogen synthase, preventing glucose from being stored as glycogen.
GTP
ATP
Activated adenylyl cyclase
cAMP
Active glycogen synthase
Inactive glycogen synthase
Inactive protein kinase A
Active protein kinase A
Inactive phosphorylase kinase
Active phosphorylase kinase
Inactive glycogen phosphorylase
Active glycogen phosphorylase
2 Phosphorylation *activates* glycogen phosphorylase, releasing stored glucose molecules from glycogen.
Glycogen
Glucose-1-phosphate
Glucose
Plasma membrane
Inside of cell
3 Release of glucose fuels "fight-or-flight" response.
Outside of cell

END

Cell Biology

G Proteins

$G_{i\alpha}$



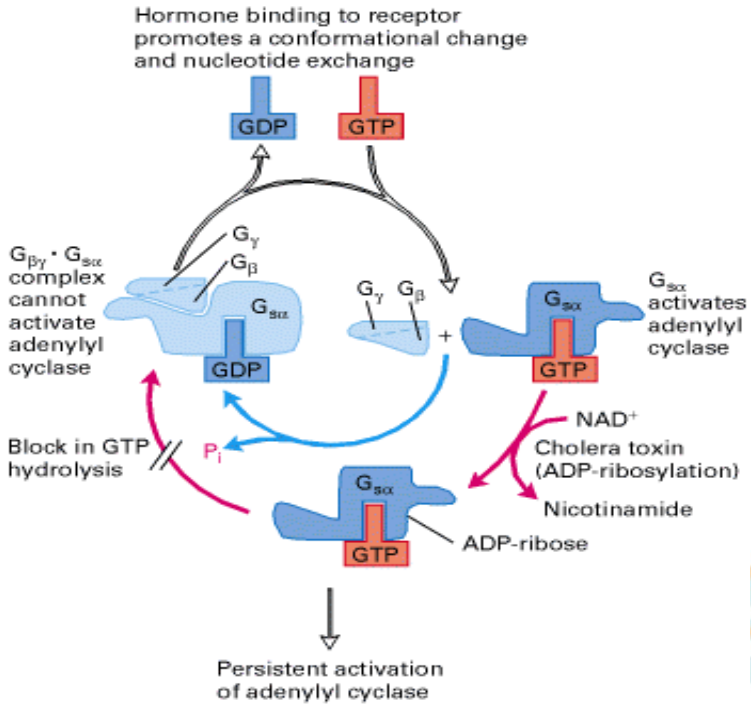
PGE_1/adenosine receptors interact with G_i,

Active $G_{i\alpha}$ (same β and γ different α) inhibits AC

$G_{\beta\gamma}$ also can directly inhibit some isoforms of AC

$G_{\beta\gamma}$ open K channels in heart cells (inh contract)

Cholera Toxin

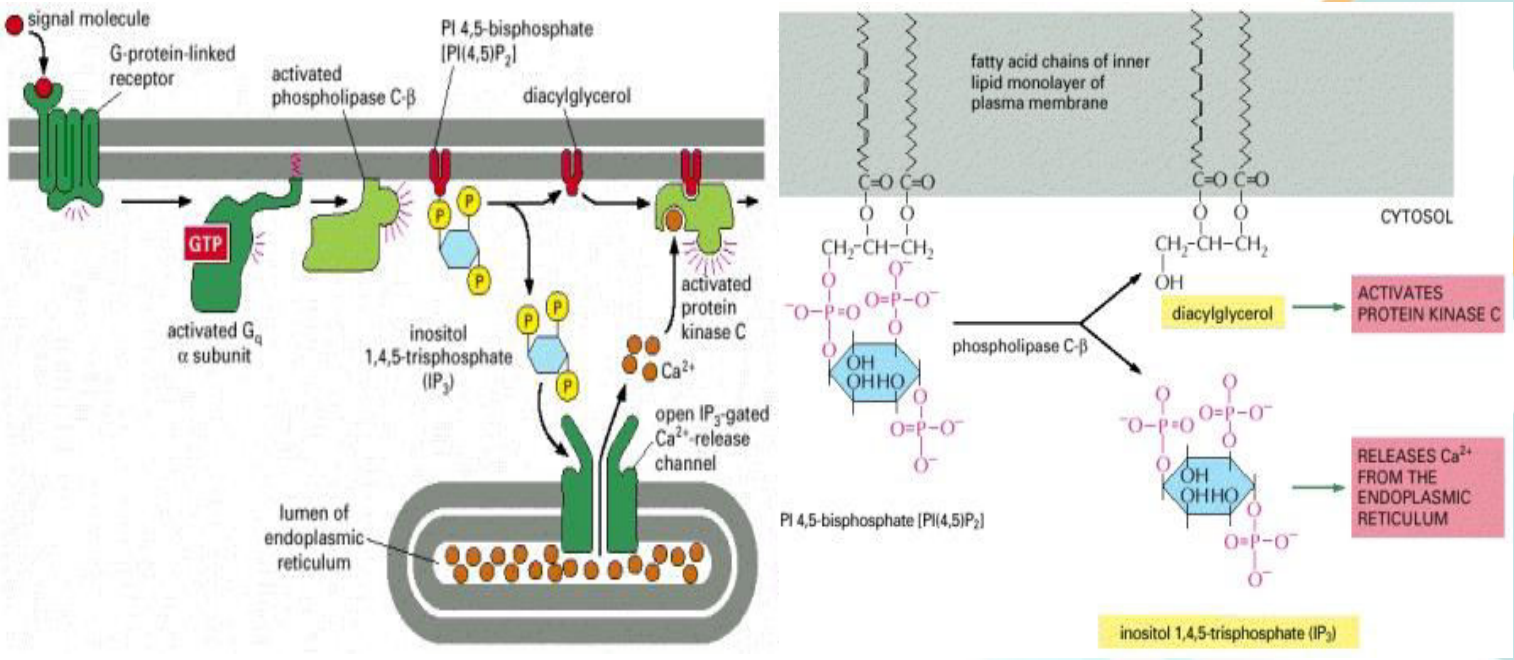


The α subunit of cholera toxin is an enzyme penetrates the plasma membrane prevents bound GTP to GDP hydrolysis.

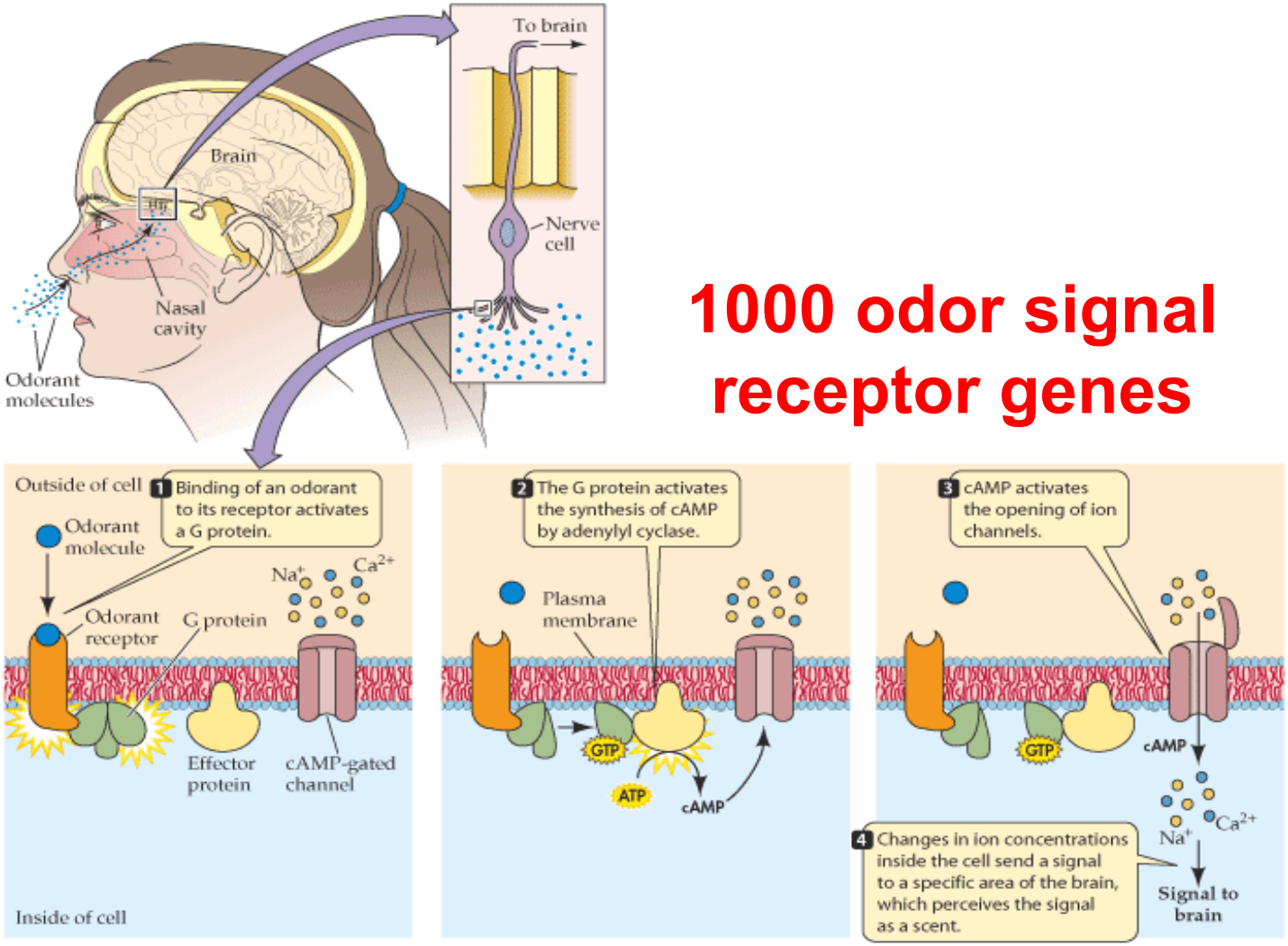
cAMP rise causes membrane proteins to permit a massive flow of water from the blood into the intestinal lumen, causing diarrhea (water loss) resulting in death from dehydration.

$G_{q\alpha}$

DAG→ Arachidonic acid→ Eicosanoids (PG)

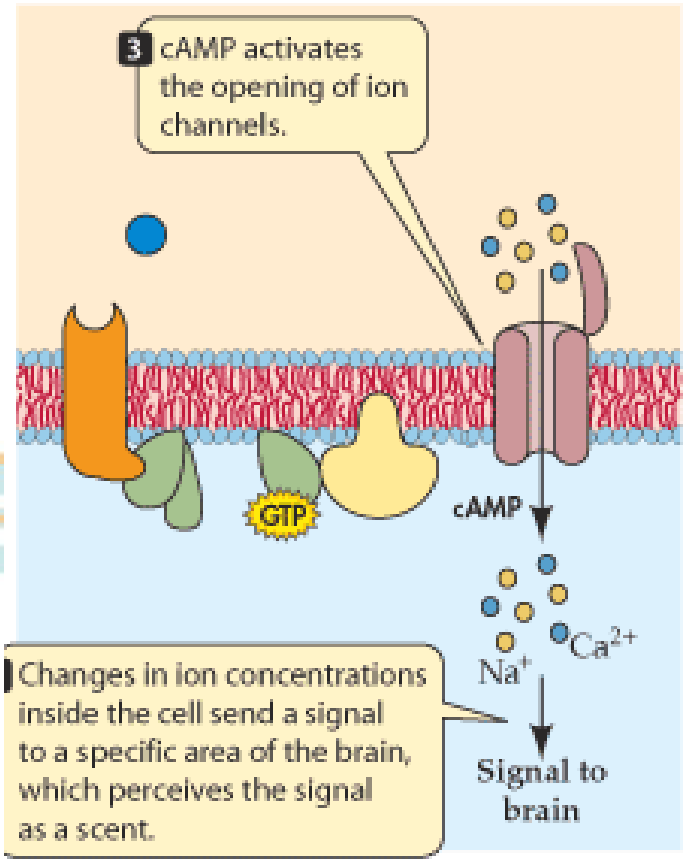


Olfaction



1000 odor signal receptor genes

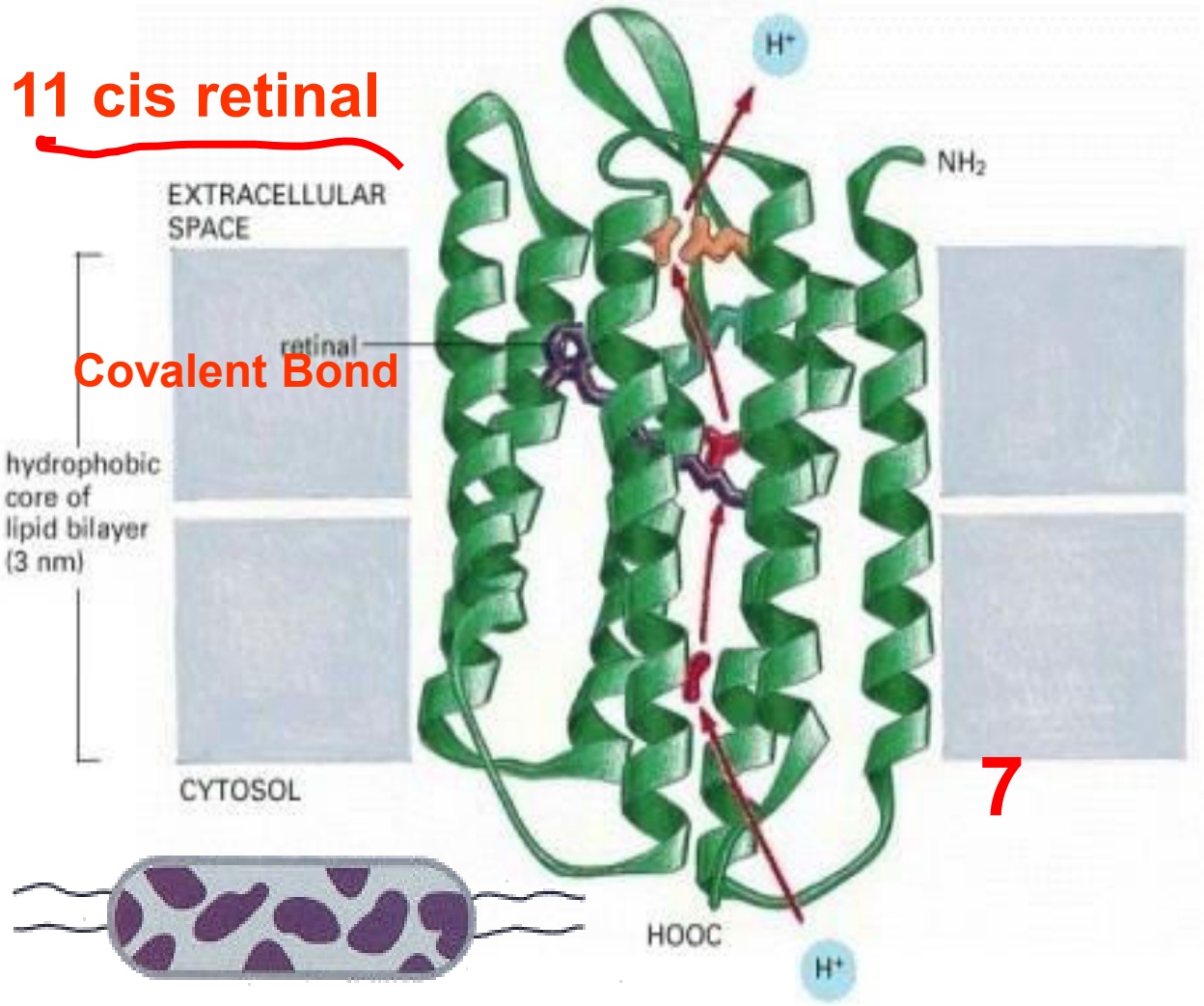
Olfaction


3 cAMP activates the opening of ion channels.
GTP
cAMP
Ca^{2+}
Na^{+}
Changes in ion concentrations inside the cell send a signal to a specific area of the brain, which perceives the signal as a scent.
Signal to brain

Cell Biology

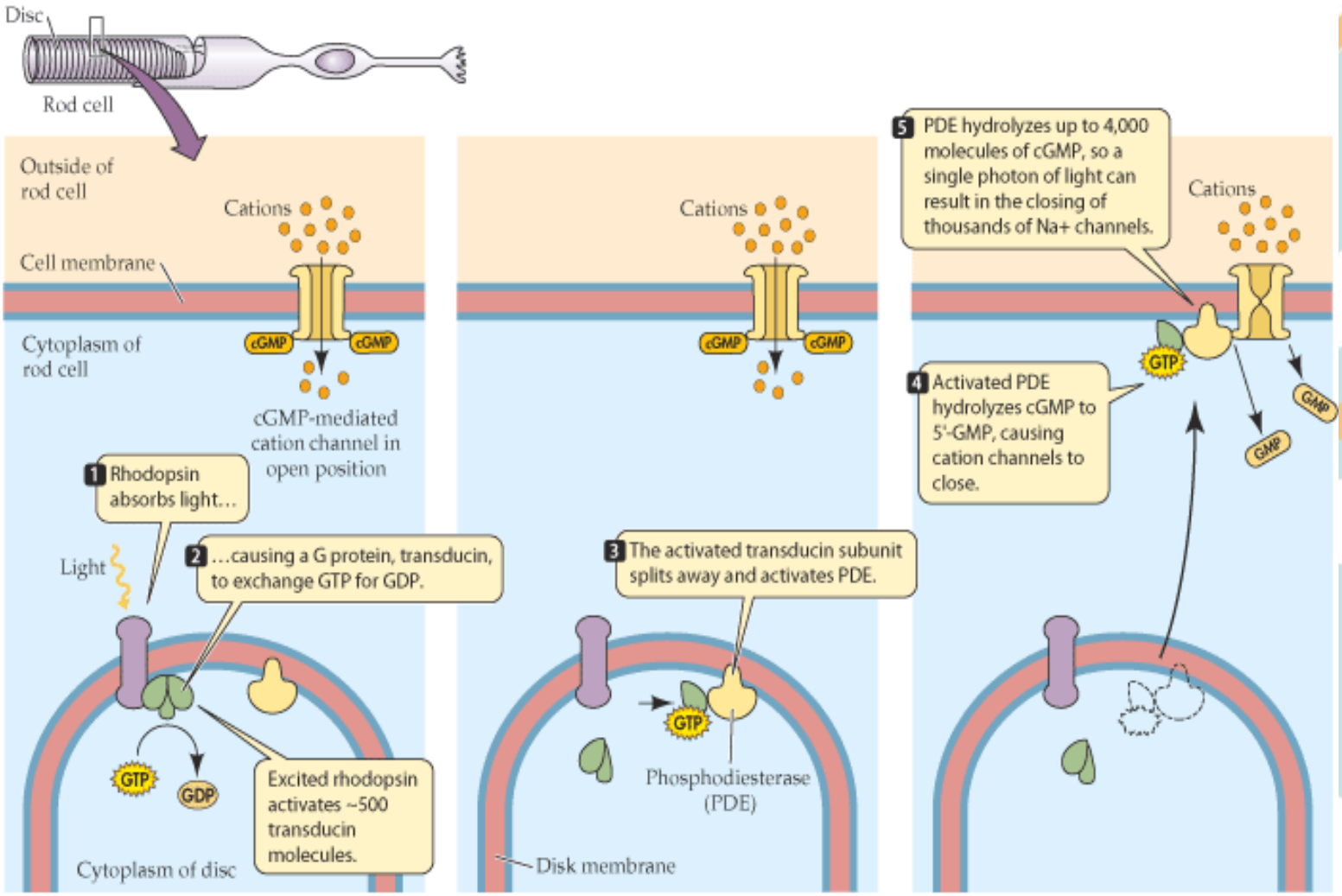
G Proteins Vision

ATP Production & Vision

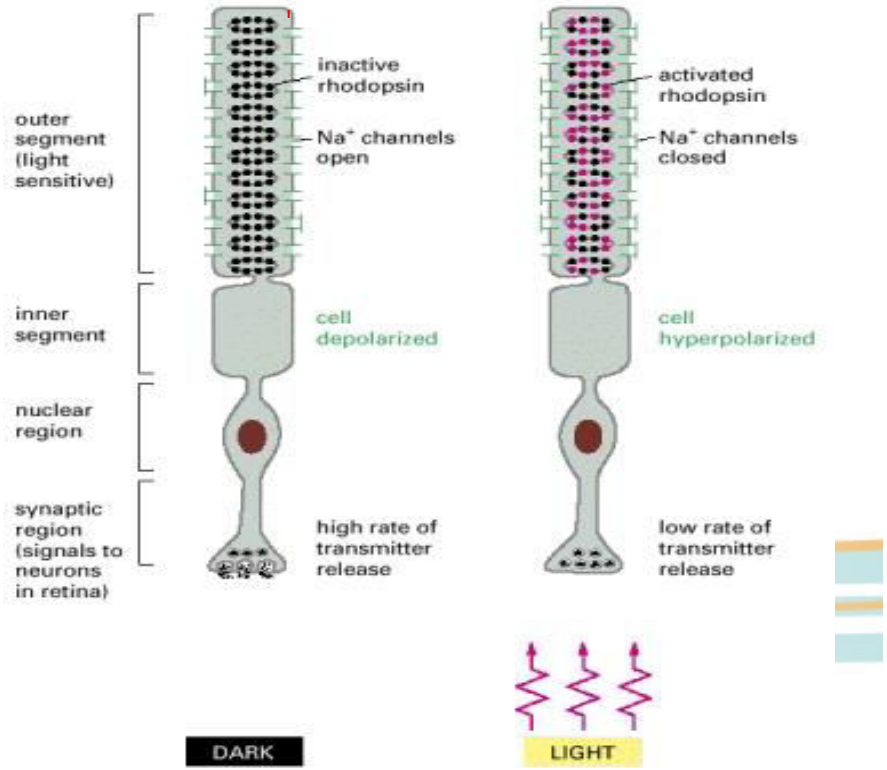


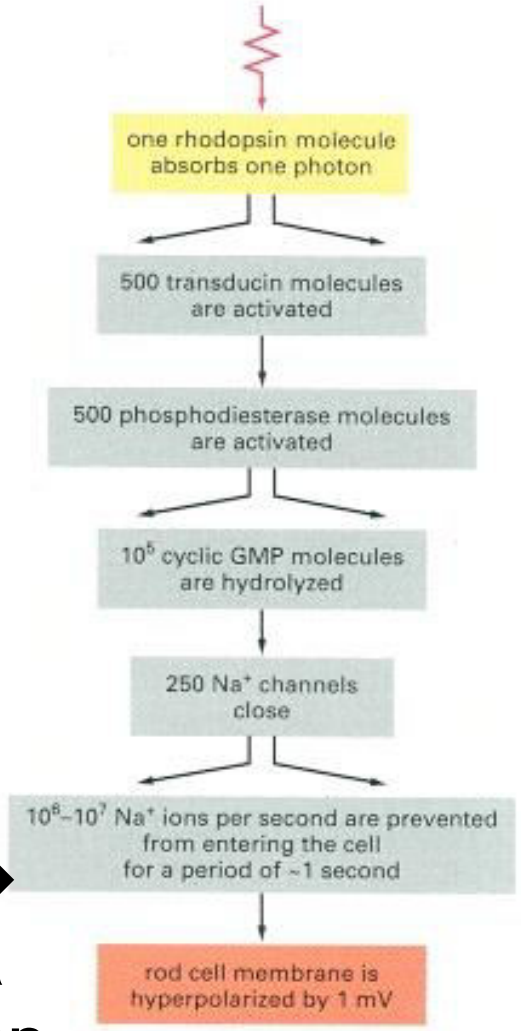
Halobacterium Salinarum

Rod Cells


Disc
Rod cell
Outside of rod cell
Cytoplasm of rod cell
Cell membrane
Cations
cGMP
cGMP
cGMP-mediated cation channel in open position
1 Rhodopsin absorbs light...
Light
2 ...causing a G protein, transducin, to exchange GTP for GDP.
GTP
GDP
Excited rhodopsin activates ~500 transducin molecules.
Cytoplasm of disc
Cations
cGMP
cGMP
3 The activated transducin subunit splits away and activates PDE.
GTP
Phosphodiesterase (PDE)
Disk membrane
5 PDE hydrolyzes up to 4,000 molecules of cGMP, so a single photon of light can result in the closing of thousands of Na+ channels.
Cations
GTP
4 Activated PDE hydrolyzes cGMP to 5'-GMP, causing cation channels to close.
GMP
GMP

Rod Cells



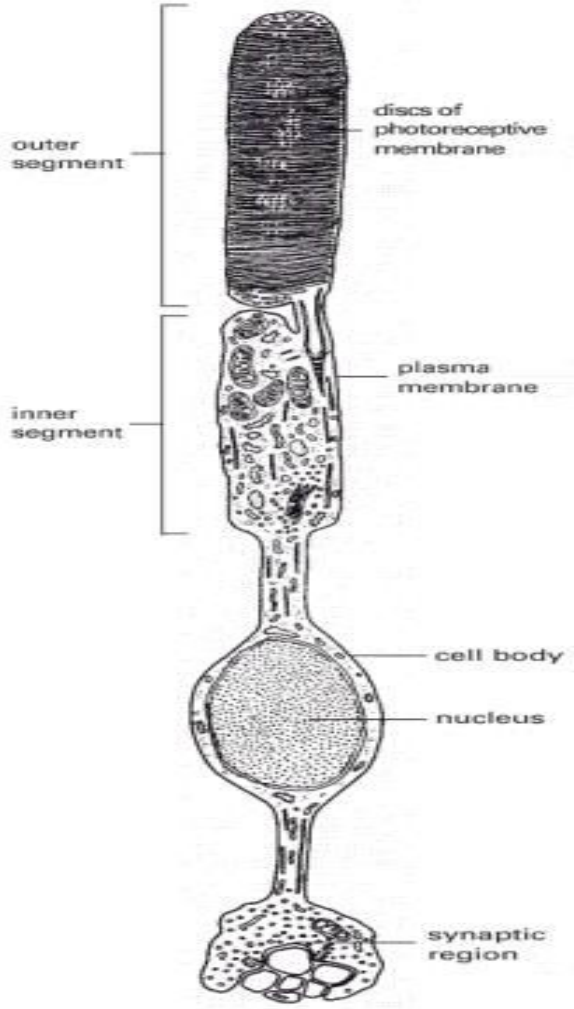


Photon→ trans retinal→ opsin conformation→ $G_{t\alpha}$ (tranducin)→ cGMP phosphodiesterase→ Na channel close→ hyper polarization

G Protiens

SOME FAMILY MEMBERS	ACTION MEDIATED BY	FUNCTIONS
G_s	α	activates adenylyl cyclase; activates Ca^{2+} channels
G_{olf}	α	activates adenylyl cyclase in olfactory sensory neurons
G_i	α	inhibits adenylyl cyclase
	βγ	activates K^+ channels
G_o	βγ	activates K^+ channels; inactivates Ca^{2+} channels
	α and βγ	activates phospholipase C-β
G_t (transducin)	α	activates cyclic GMP phosphodiesterase in vertebrate rod photoreceptors
G_q	α	activates phospholipase C-β

Rod Cells



END

Cell Biology

Cell Signaling & Us

Epinephrine Adrenalin

β1- and β2-adrenergic receptors (Heart & Lungs Respectively) are coupled to G proteins (Gs), which activate adenylyl cyclase.

α1 and α2 receptors are coupled to two other G proteins, Gq and Gi, respectively. adenylyl cyclase Inhibition & phospholipase C activation

Epinephrine bound to α2 receptors causes the arteries to constrict, cutting off circulation to the blood vessels in the intestinal tract, skin, and kidneys (peripheral organs)

Epinephrine Adrenalin

Agonists (terbutaline)

Mimic the function of a hormone by binding to its receptor and causing the normal response

Antagonists (β-blockers)

Bind to the receptor but do not activate hormone induced effects.

An antagonist acts as an inhibitor of the natural hormone (or agonist) by competing for binding sites on the receptor, thereby blocking the physiological activity of the hormone.

Epinephrine Adrenalin

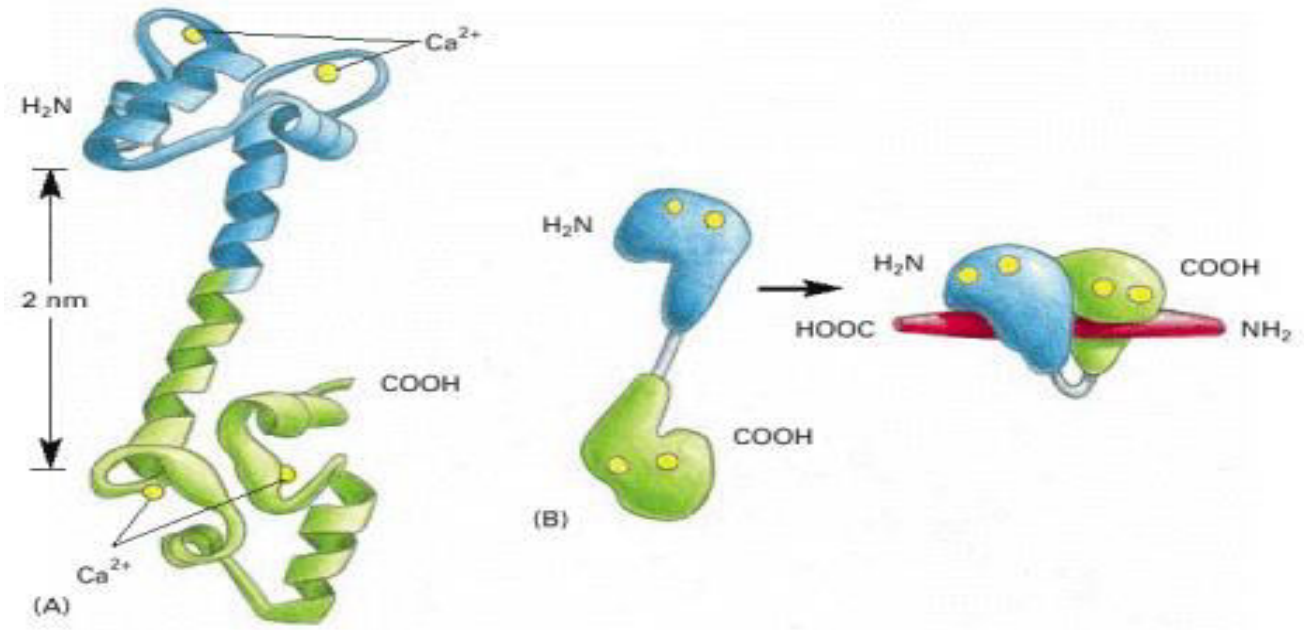
Tamoxifen

END

Cell Biology

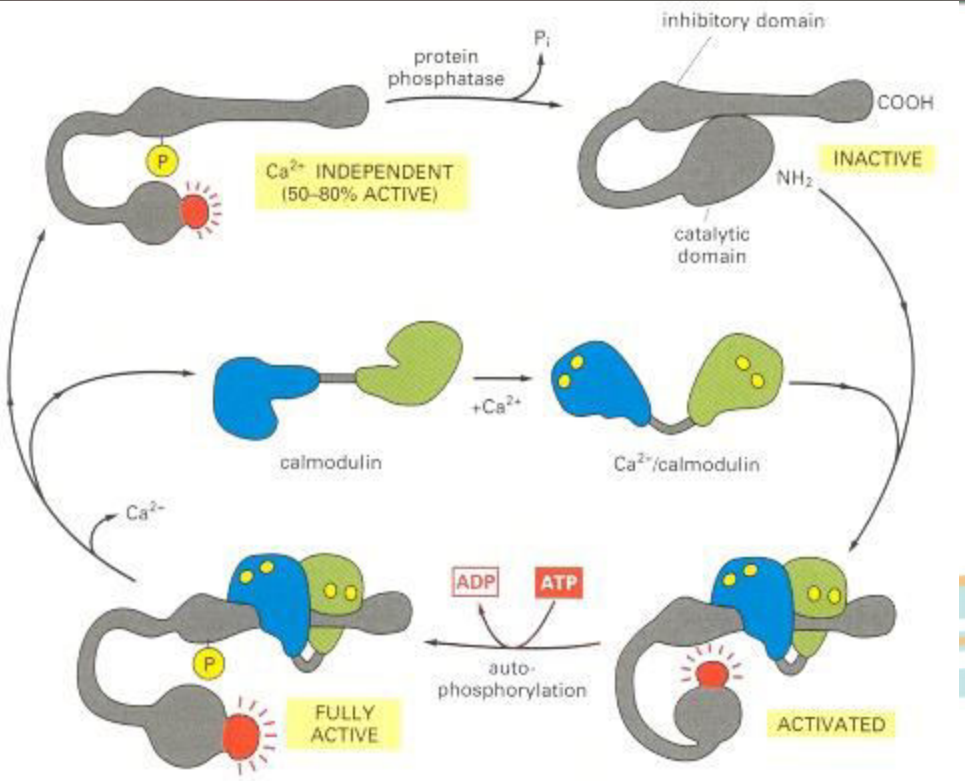
Cell Signaling & Memory

Ca^{2+} Binding Proteins



Troponin in muscle contraction (1st Ca^{2+}binding)
Calmodulin binds 4 Ca^{2+} allosteric (PKA) (1%)
Calmodulin-dependent Protein Kinases (CaM-Kinases) (serine/threonine)
CaM-Kinases phosphorilate myosin LT chain SM

CaM-kinase II (2%) & Memory

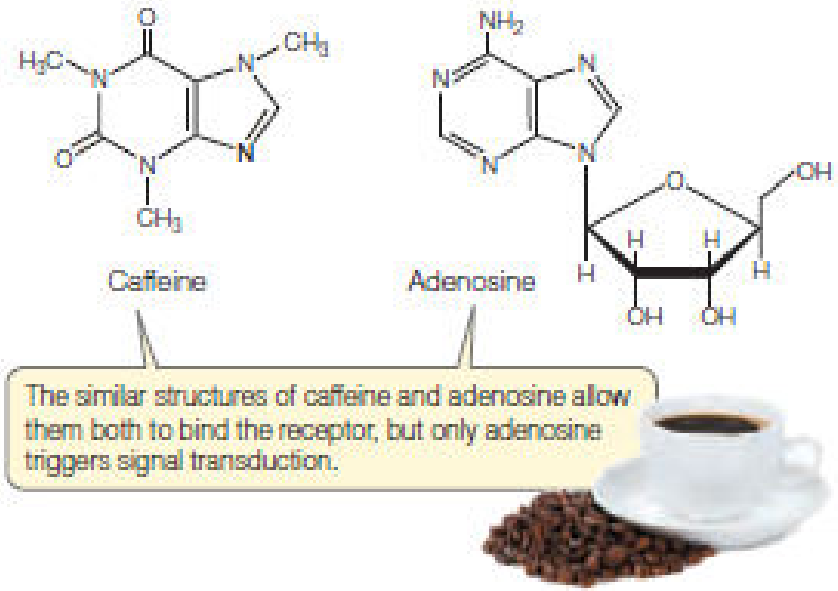


Calmodulin binding → conformation catalytic domain

phosphorylates inhibitory domain prevents disociation

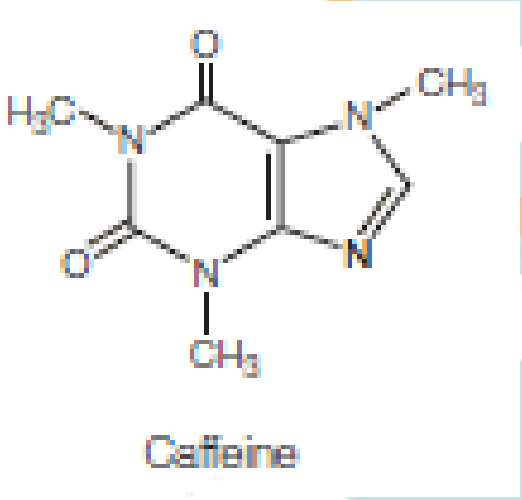
CaM-kinase II active w/o signal mutants poor memory

CaM-kinase II (2%) & Memory



Tired brain produces adenosine that binds specific receptor, decreases brain activity → drowsiness. Caffeine's structure is similar to adenosine, it occupies adenosine receptors without inhibiting brain cell function, and alertness is restored.

CaM-kinase II (2%) & Memory



Caffeine

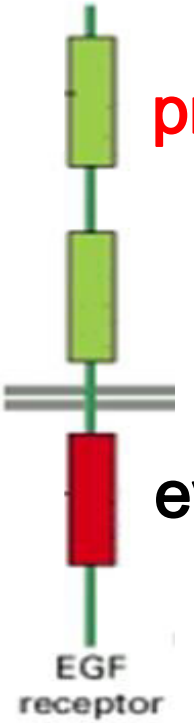
END

Cell Biology

Enzyme Linked Receptors

Enzyme Linked Receptors

Growth Factors



Extracellular signal promote growth, proliferation, differentiation, or survival of cells.

Usually act as local mediators at very low concentrations (10^{-9}-10^{-11} M).

The responses slow (hours) eventually lead to changes in gene expression

Extracellular ligand-binding domainon
1 transmembrane segment

Cytosolic domain either has an intrinsic enzyme activity or associates directly with an enzyme.

Enzyme Linked Receptors

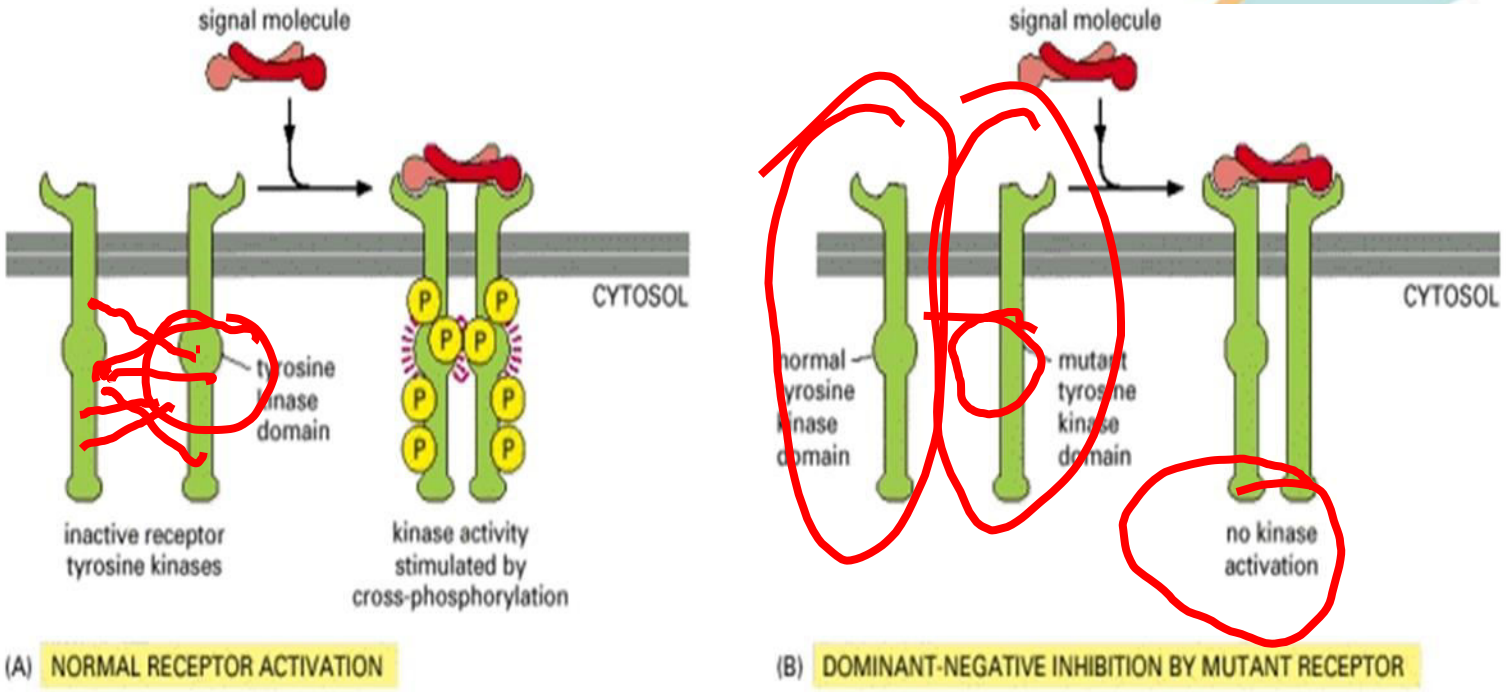
SIGNALING LIGAND	RECEPTORS	SOME RESPONSES
Epidermal growth factor (EGF)	EGF receptor	stimulates proliferation of various cell types
Insulin	insulin receptor	stimulates carbohydrate utilization and protein synthesis
Insulin-like growth factors (IGF-1 and IGF -2)	IGF receptor-1	stimulate cell growth and survival
Nerve growth factor (NGF)	Trk A	stimulates survival and growth of some neurons
Platelet-derived growth factors (PDGF AA, BB, AB)	PDGF receptors (α and β)	stimulate survival, growth, and proliferation of various cell types

Receptor Tyrosine Kinases

2 receptor come together, form dimer

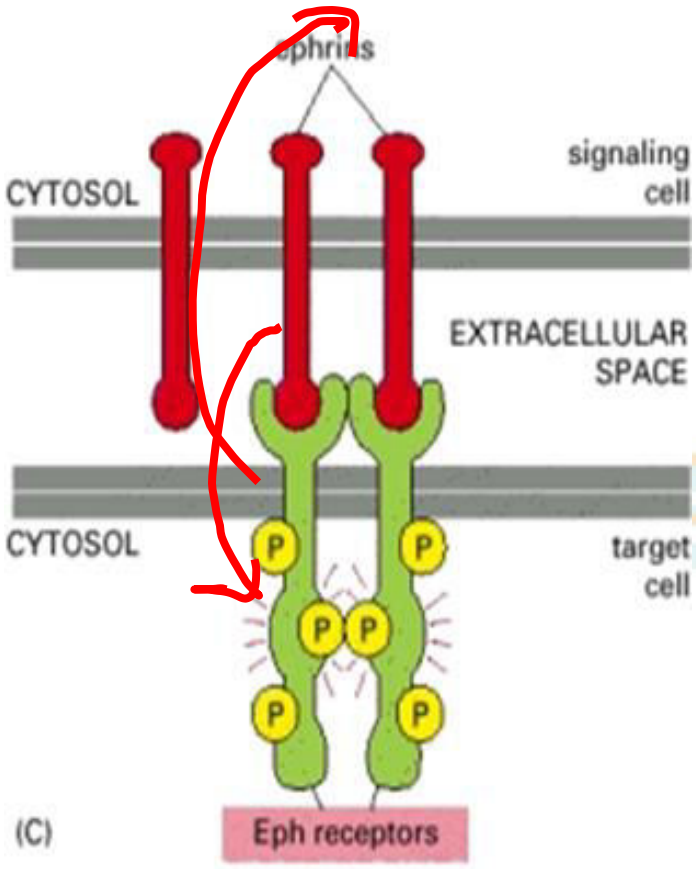
Neighbouring kinase domains cross-phosphorylate each other on multiple tyrosines, autophosphorylation

Mutant receptor that dimerizes normally inactive kinase domain dominant-negative



Receptor Tyrosine Kinases

Ephrins & Eph Receptors



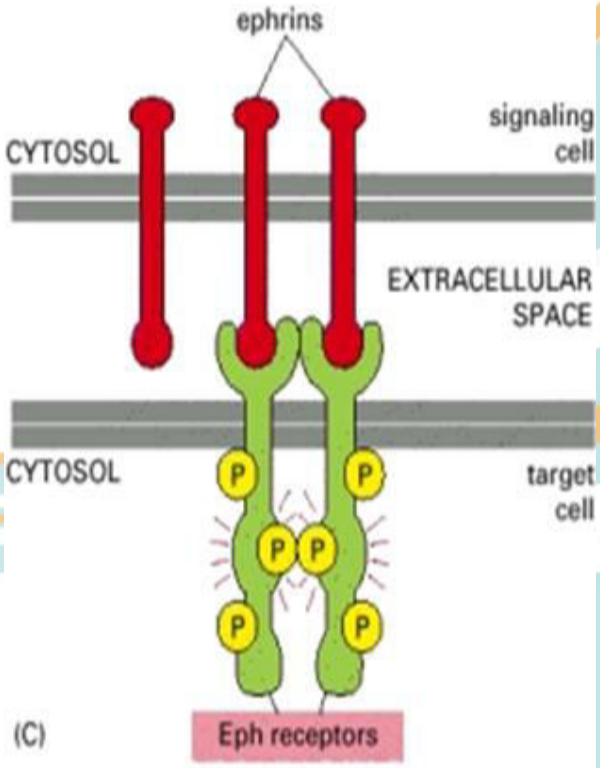
Regulate the cell adhesion & repulsion responses

Migration of cells and axons development

Most numerous receptor tyrosine kinases.

Act as ligand & receptor bidirectional reciprocal signaling

Receptor Tyrosine Kinases



END

Cell Biology

Enzyme Linked Receptors 2

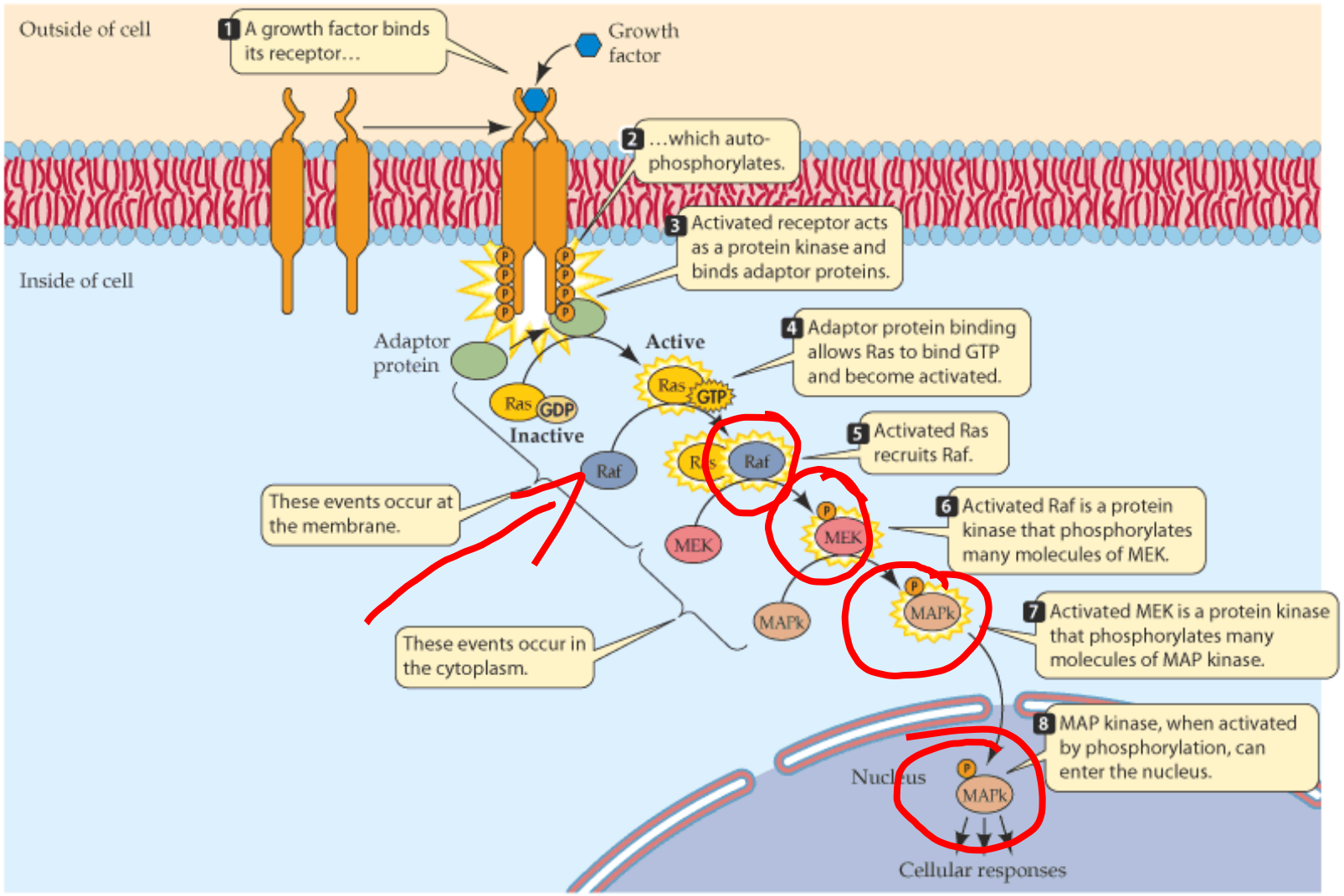
RTK → Ras Activation ?

Fibroblast cells can be induced to proliferate by treatment with a mixture of platelet-derived growth factor (PDGF) and epidermal growth factor (EGF).

Microinjection of anti-Ras antibodies into these cells blocks cell proliferation

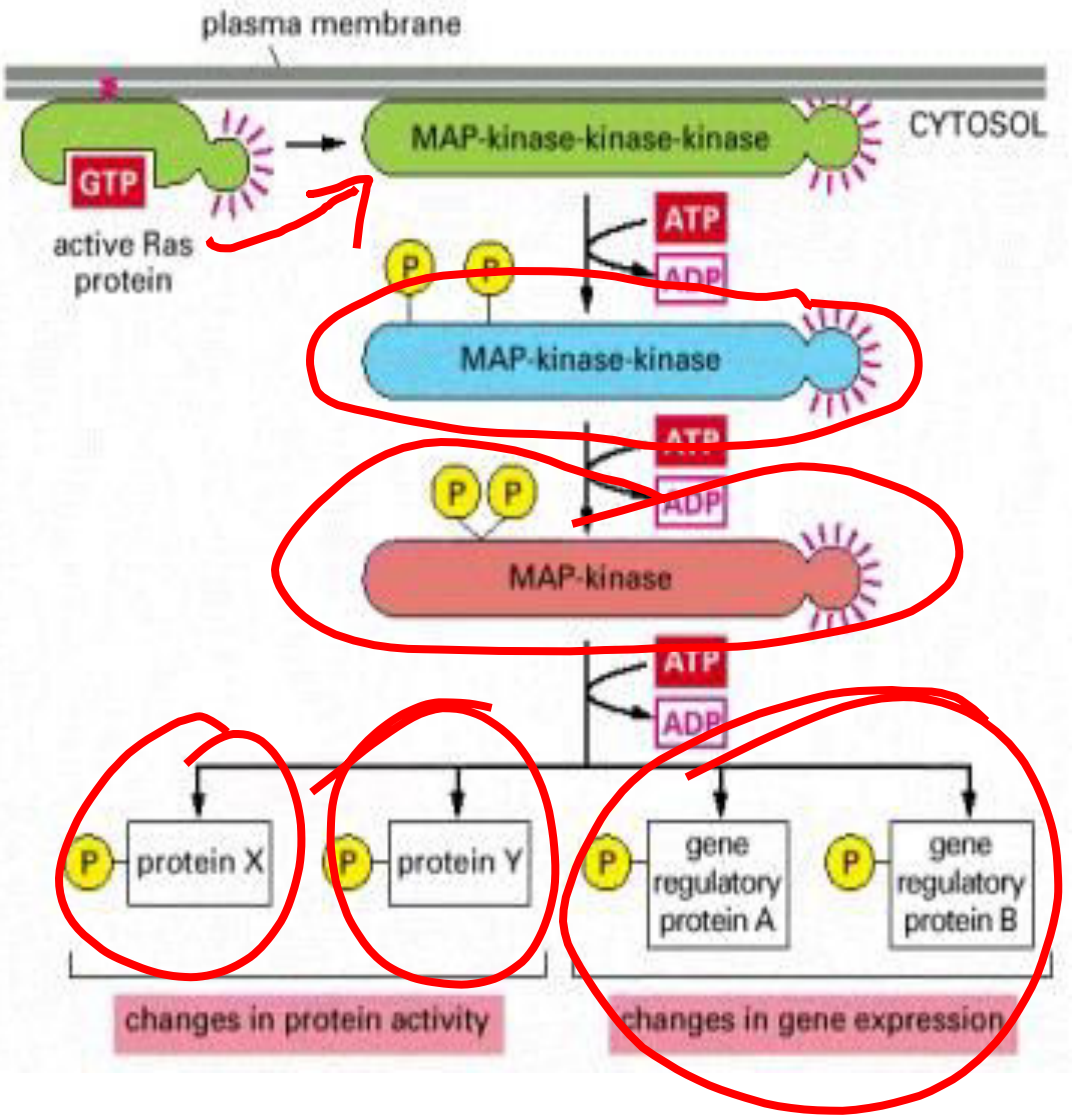
Conversely, injection of a constitutively active mutant Ras protein (i.e., Ras^{D}), which hydrolyzes GTP very inefficiently and thus persists in the active state, caused the cells to proliferate in the absence of the growth factors.

Protein Kinase Cascade



Signal Amplification

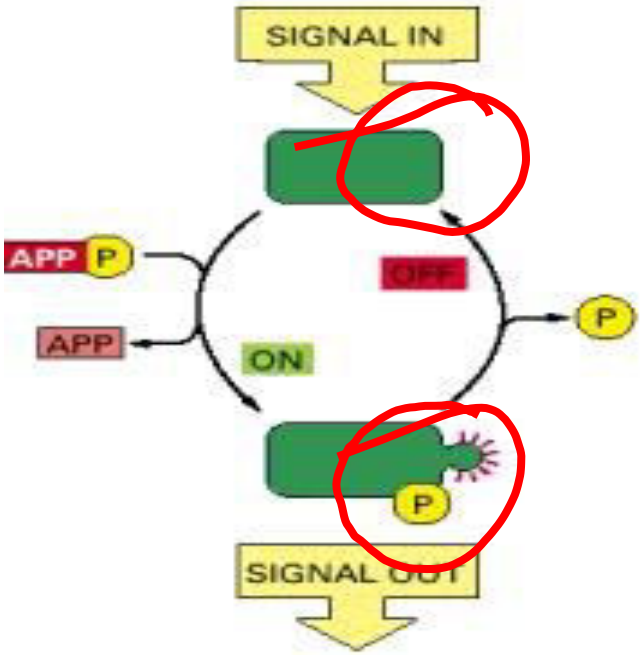
Pathway



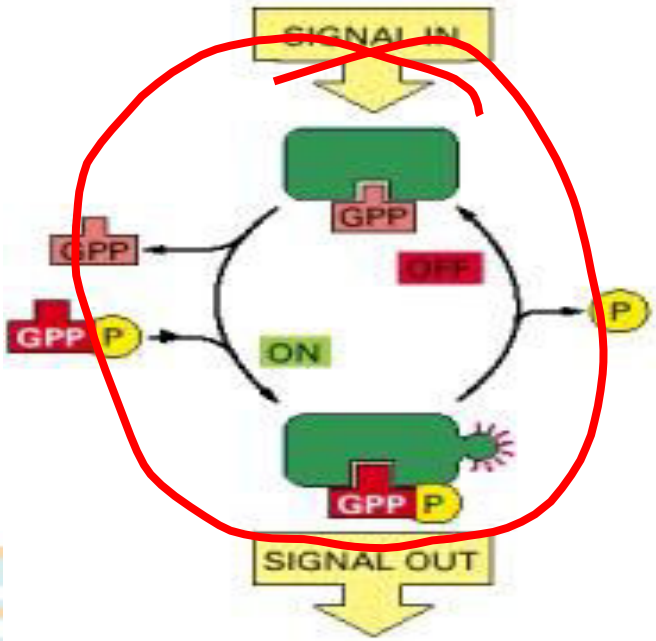
EGF activated MAPK stays active for 5 min, NGF activated MAPK is active for several hours

5MAPK,7MEK & 7 Raf, specificity via scaffolds

Switches



SIGNALING BY PHOSPHORYLATION



SIGNALING BY GTP-BINDING PROTEIN

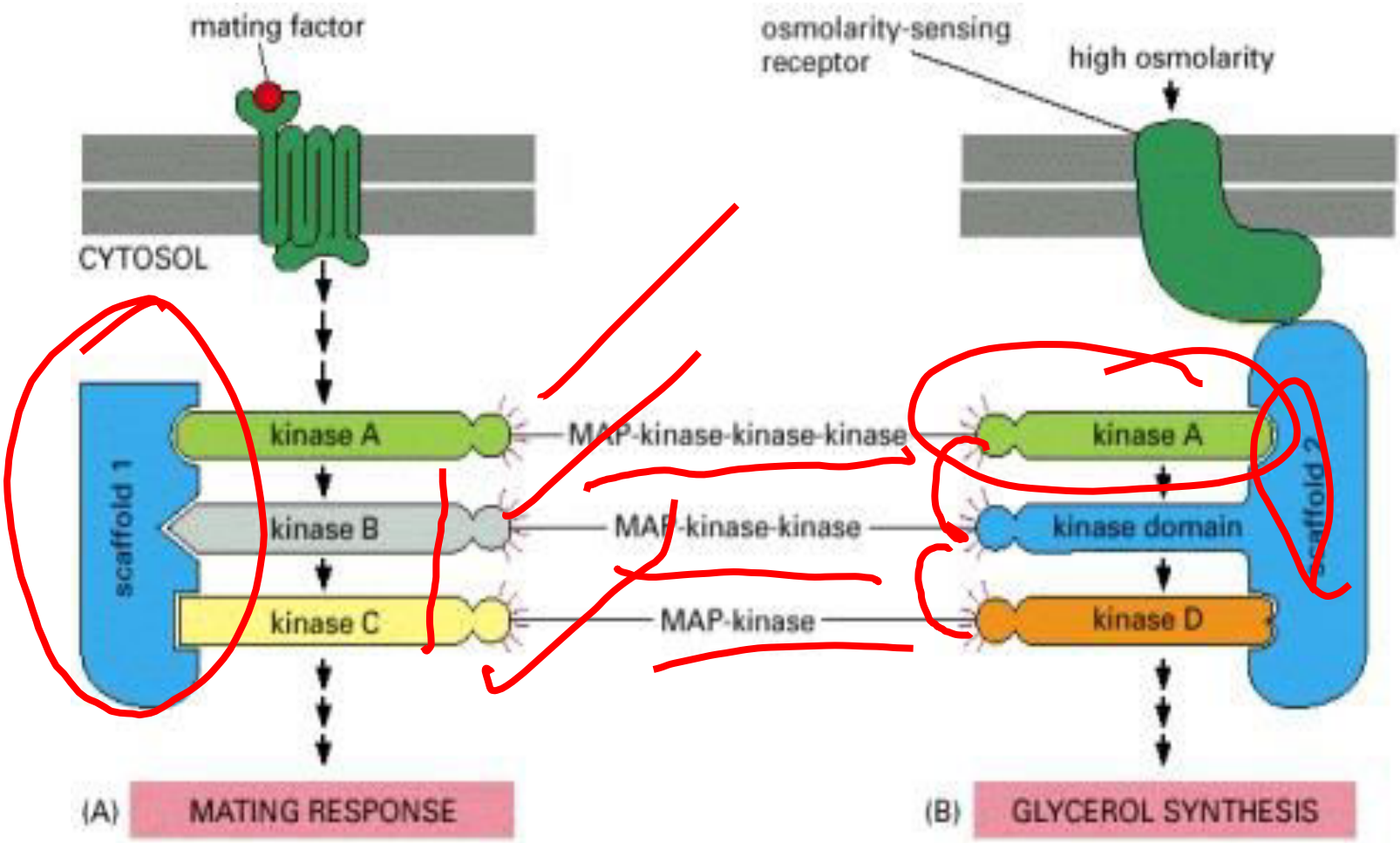
Phosphorilation: Kinases & Phosphatases

1/3 proteins are phosphorilated

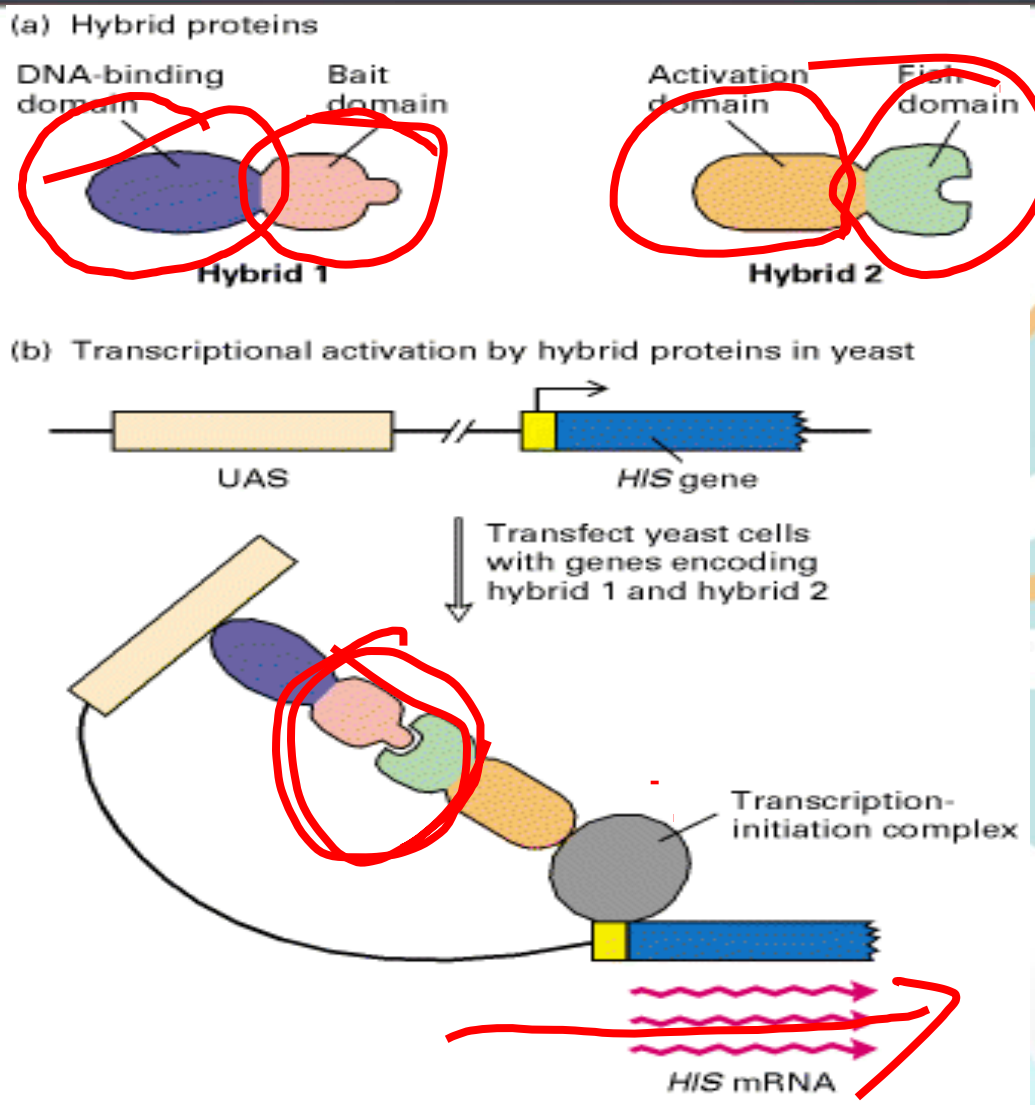
~2% genes encode Kinases

G-Proteins:Trimeric Bind GTP/GDP

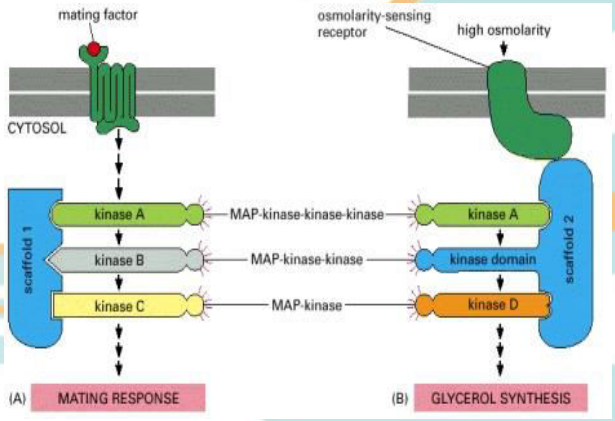
Specificity



Interactions?



Specificity



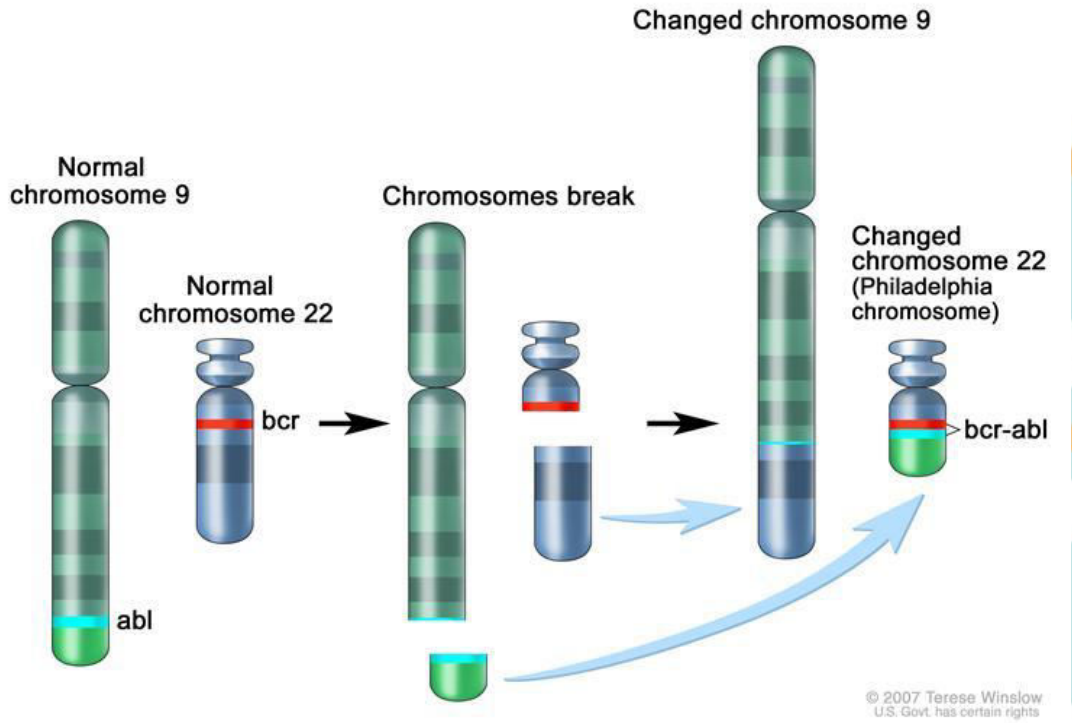
END

Cell Biology

RTK Signaling Gone Wrong

Chronic Myelogenous Leukemia

Philadelphia chromosome



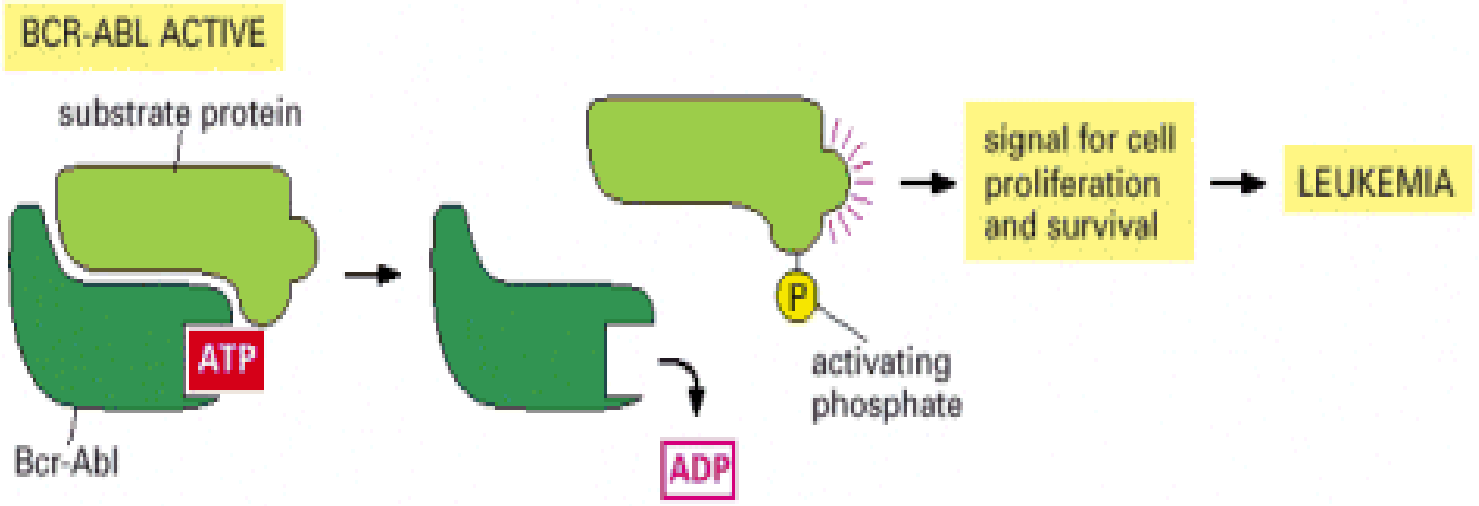
CML results from a chromosomal translocation between chromosome 9 and 22

Abelson (Abl) tyrosine kinase gene

BCR/ABL Fusion Protein

BCR/ABL fusion protein produced by the Philadelphia chromosome activates constitutively the cytosolic tyrosine kinase ABL that normally would be activated only when the cell is stimulated by a growth factor (PDGF).

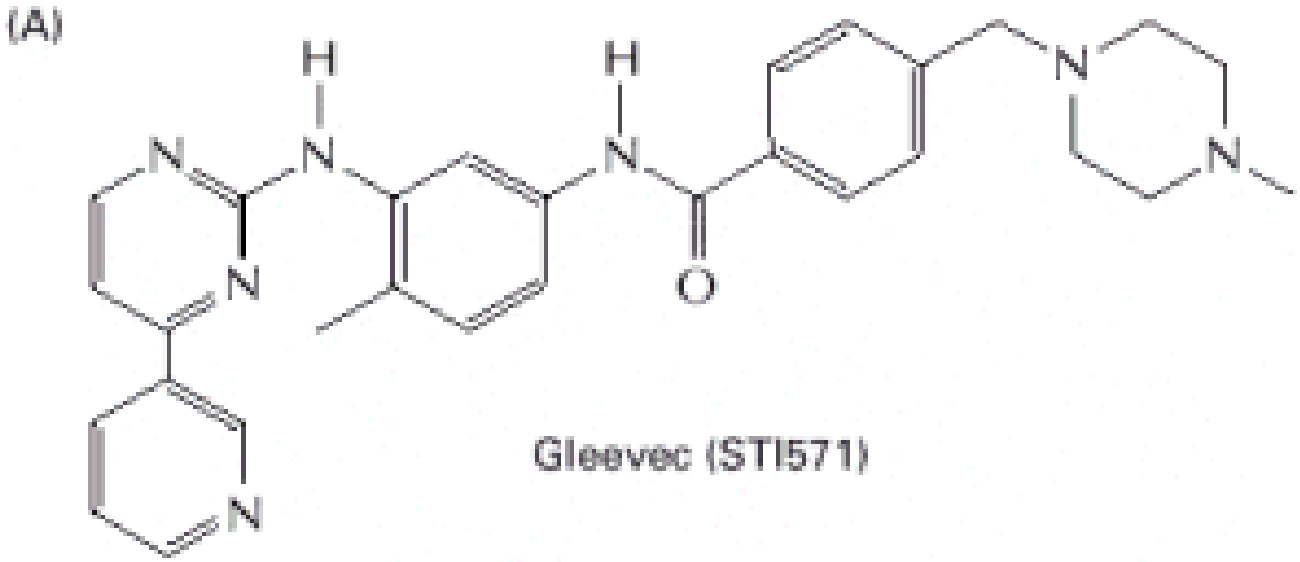
The result: chronic myelogenous leukemia



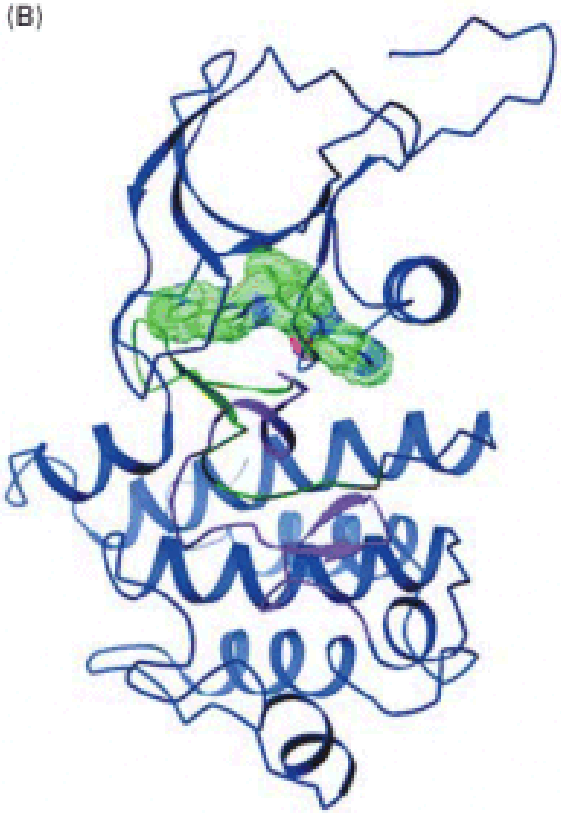
The Cure

Imatinib mesylate (Gleevec® AKA STI571)

The drug can be given orally.



Imatinib/Gleevec

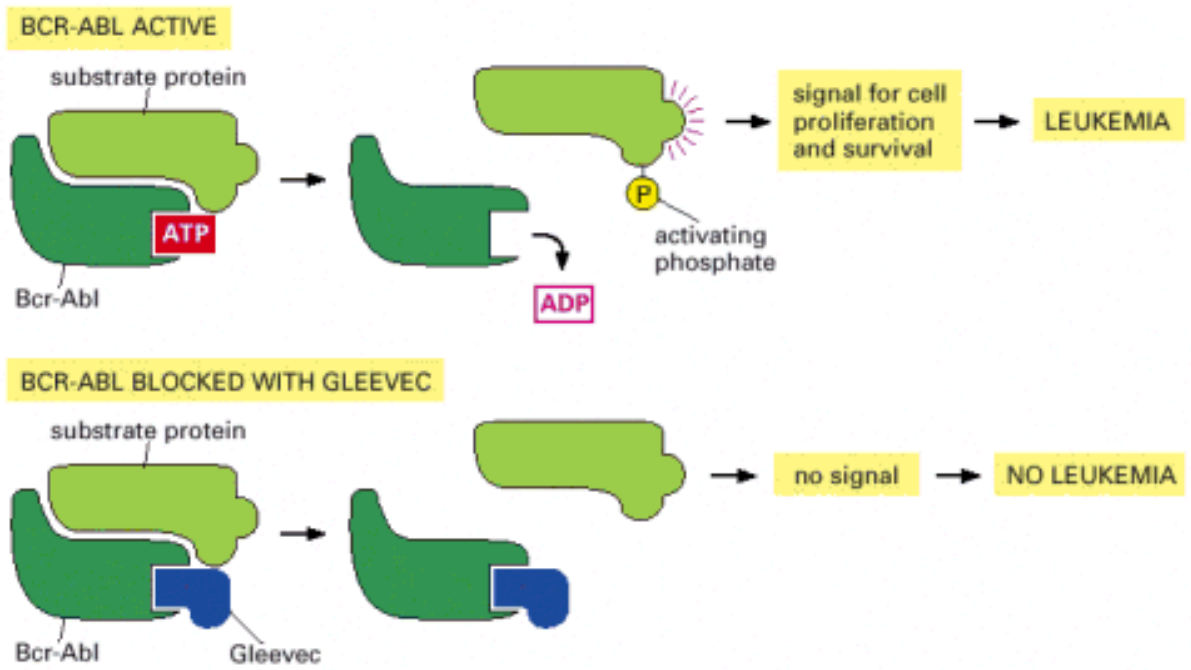


This molecule fits into the active site of the ABL protein preventing ATP from binding .

Without ATP as a phosphate donor, the ABL protein cannot phosphorylate its substrate(s).

Almost 90% of the CML patients treated with the drug showed no further progression of the disease

Imatinib/Gleevec



Gleevec prevents Bcr-Abl from transferring a phosphate group from ATP onto a tyrosine residue in a substrate protein.

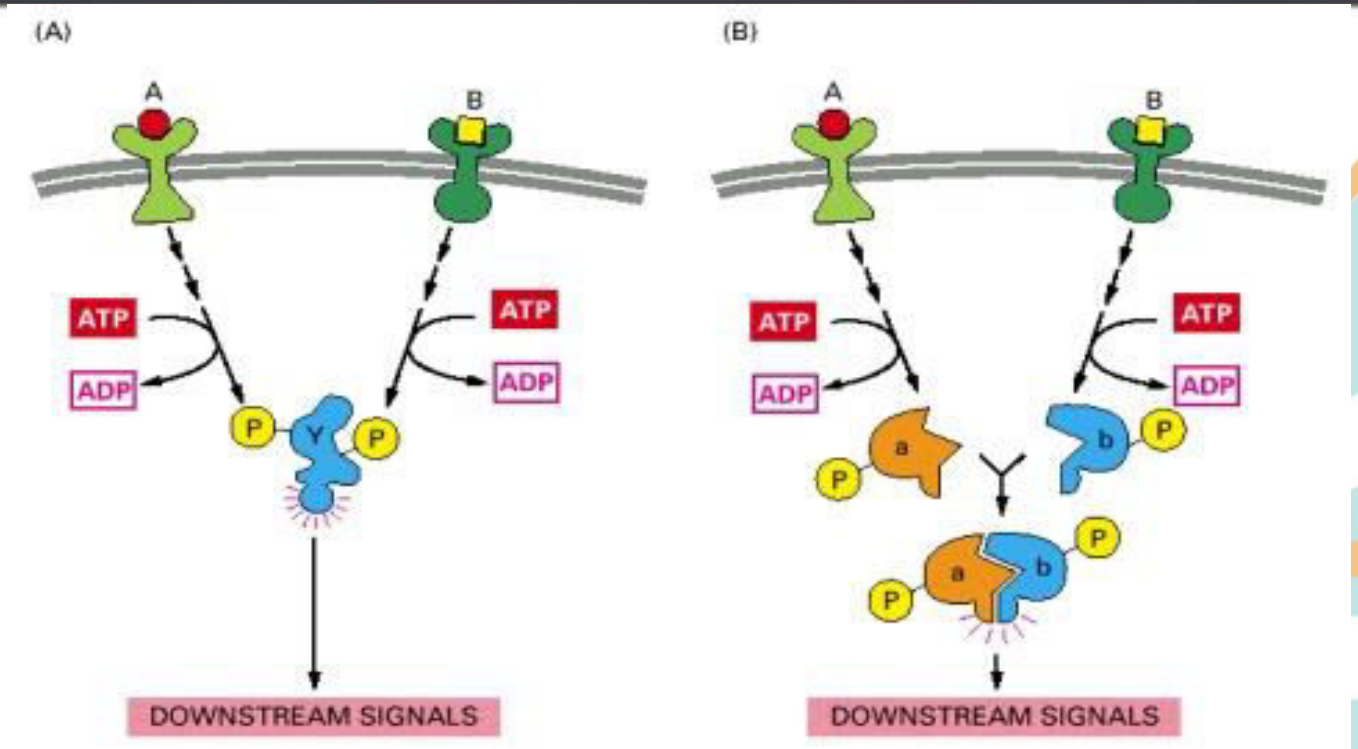
This blocks onward transmission of a signal for cell proliferation and survival.

Neurofibromatosis

Neurofibromatosis causes tumors to grow on nerves abnormalities of the skin, nervous system, bones, endocrine organs and blood vessels

Neurofibromin a negative regulator of RAS (OP) is mutated

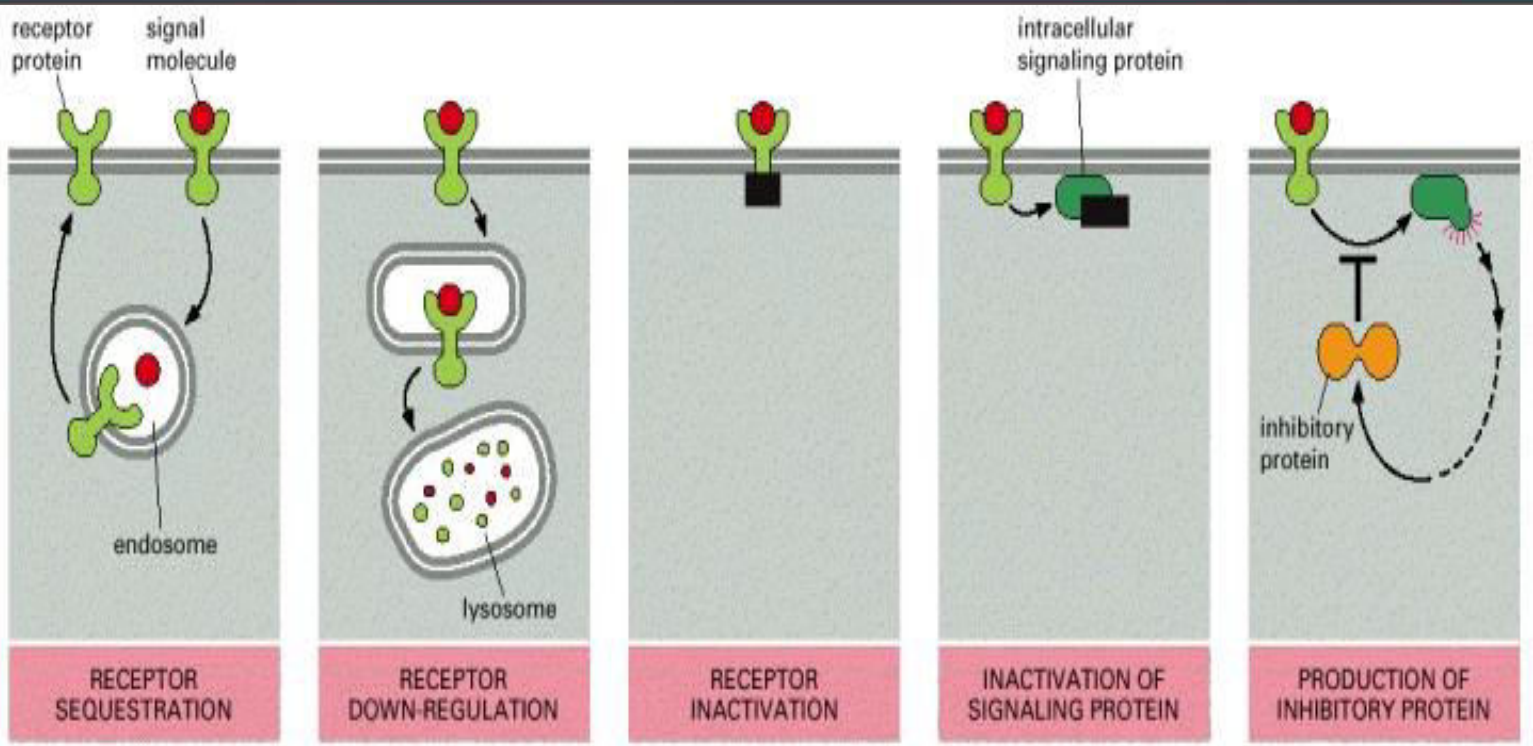
Signal integration



Complex cellular behavior (survival, differentiation & proliferation) result of specific combination of signals

Response dependent on integrator proteins (coincidence)

Desensitization



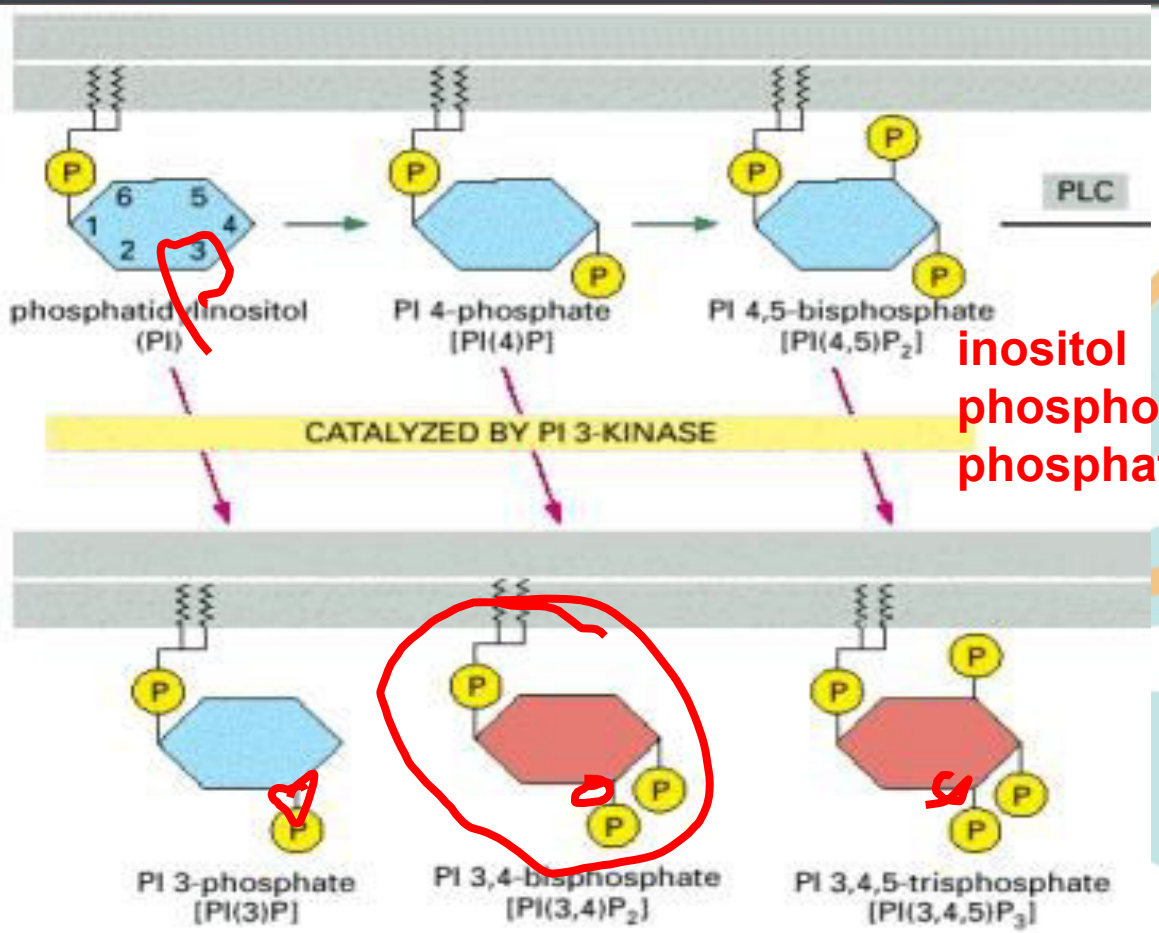
Imatinib/Gleevec

END

Cell Biology

PI3-kinase

PI3-kinase

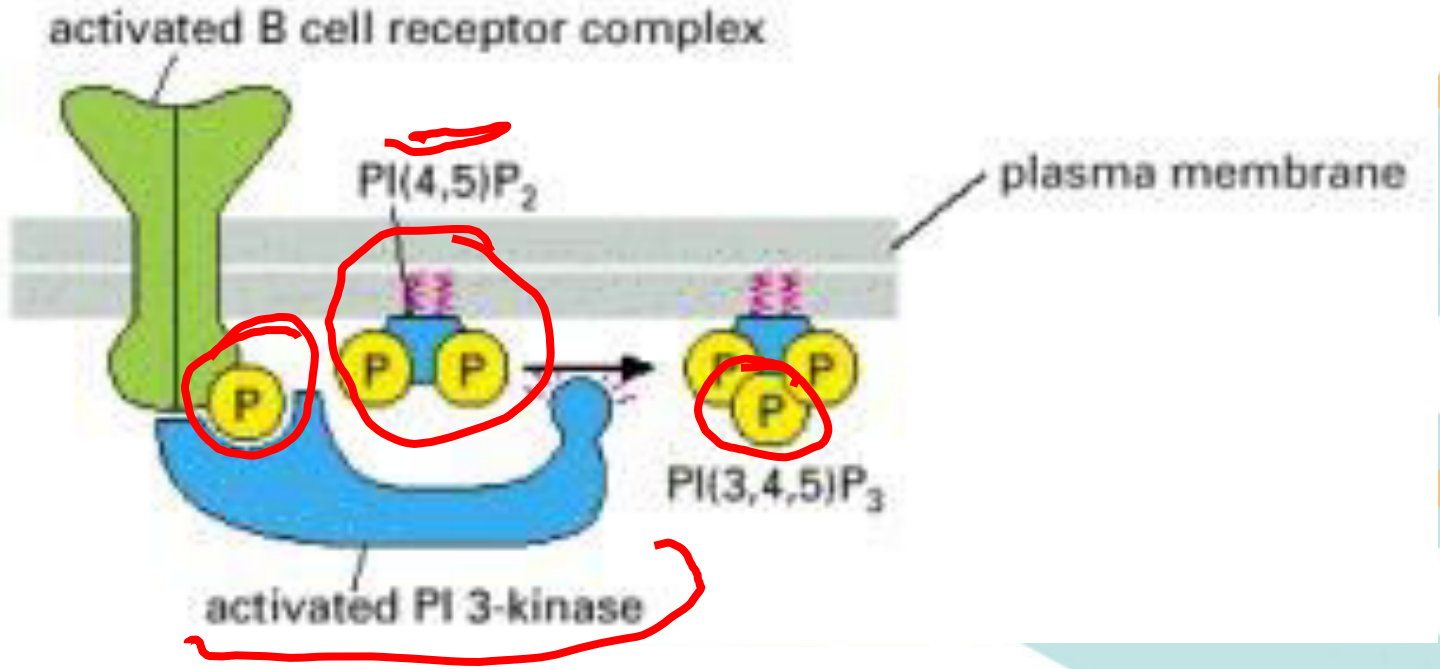


inositol phospholipid phosphatases

Signaling proteins bind PI(3,4)P2 and PI(3,4,5)P3 Pleckstrin homology (PH) domain 2

PI(4,5)P2 is cleaved by PLC-β (gpcr) or PLC-γ (RTK) 200 proteins

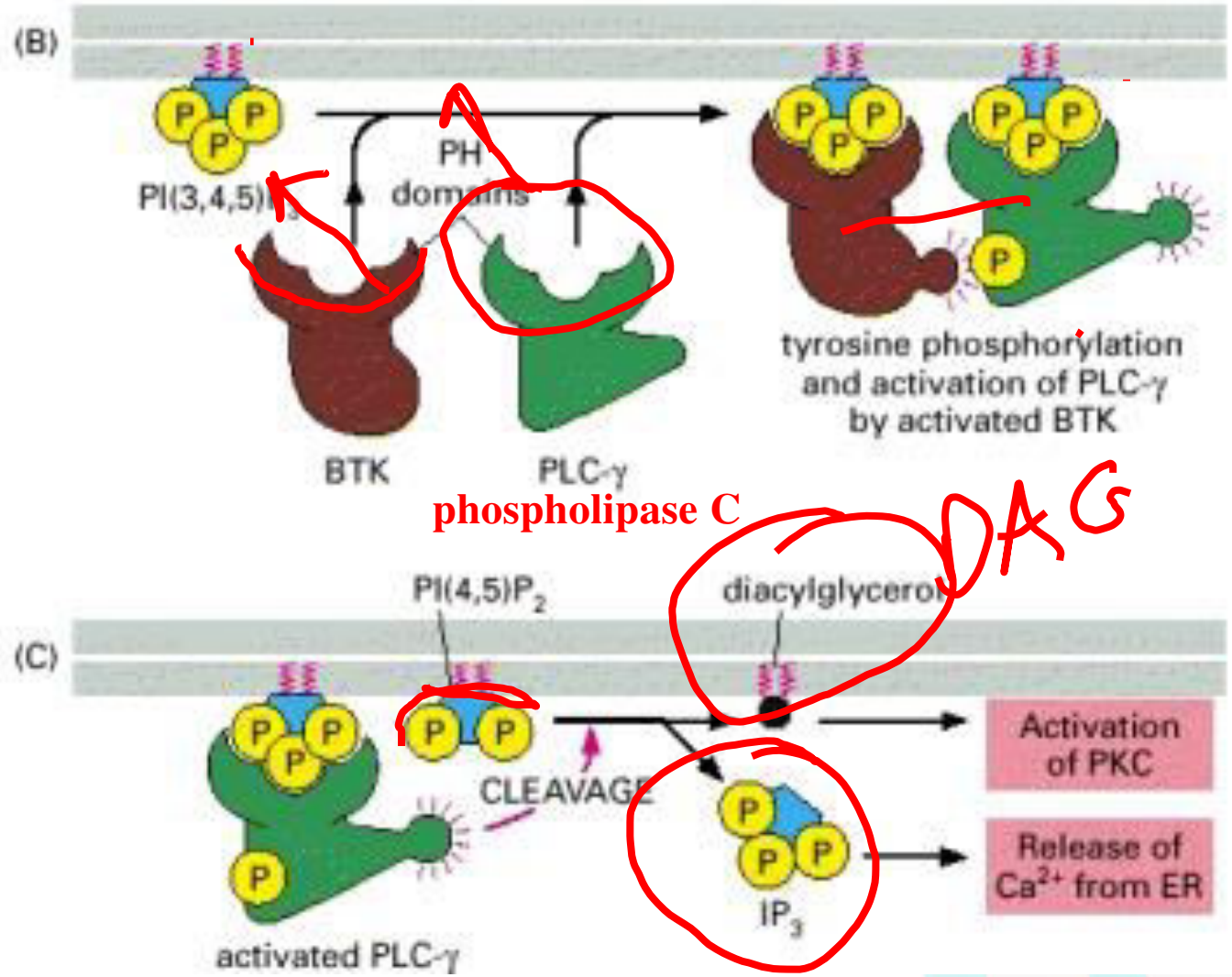
PI3-kinase



Antigen receptors on B lymphocytes (B cells) activate PI 3-kinase, the resulting inositol lipid docking sites recruit both BTK and PLC-γ to the cytoplasmic face of the plasma membrane.

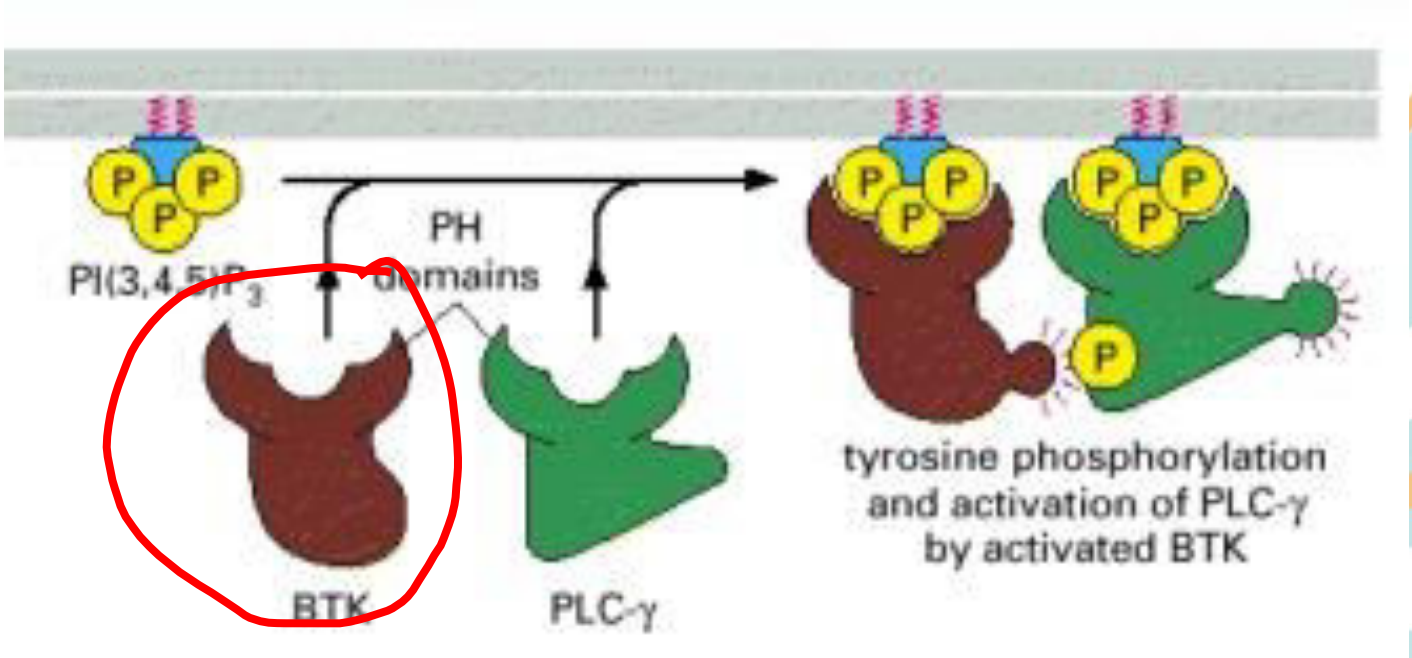
platelet protein Pleckstrin

PI3-kinase



BTK phosphorylates and activates PLC-γ, which then cleaves PIP2 to generate IP3 and diacylglycerol to relay the signal onward

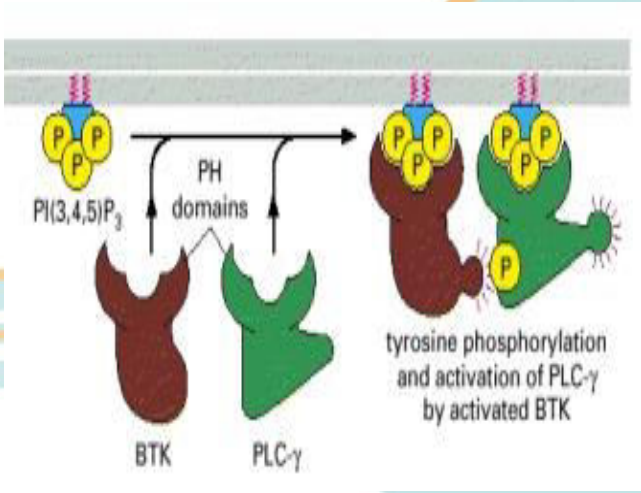
PI3-kinase



Bruton's tyrosine kinase, a protein whose mutation is associated with X-linked agammaglobulinemia

platelet protein Pleckstrin

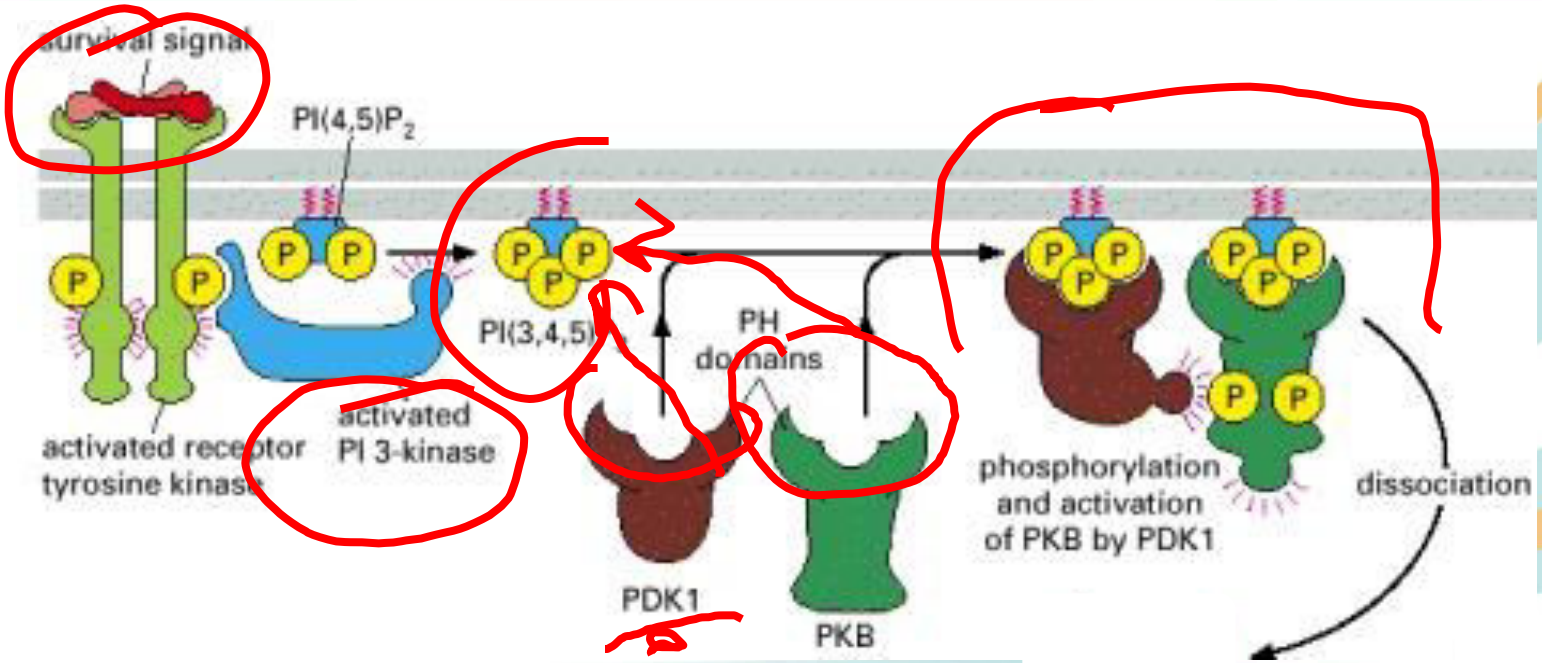
PI3-kinase


PI(3,4,5)P3
PH
domains
BTK
PLC-γ
tyrosine phosphorylation
and activation of PLC-γ
by activated BTK

Cell Biology

PKB Apoptosis
Inhibition

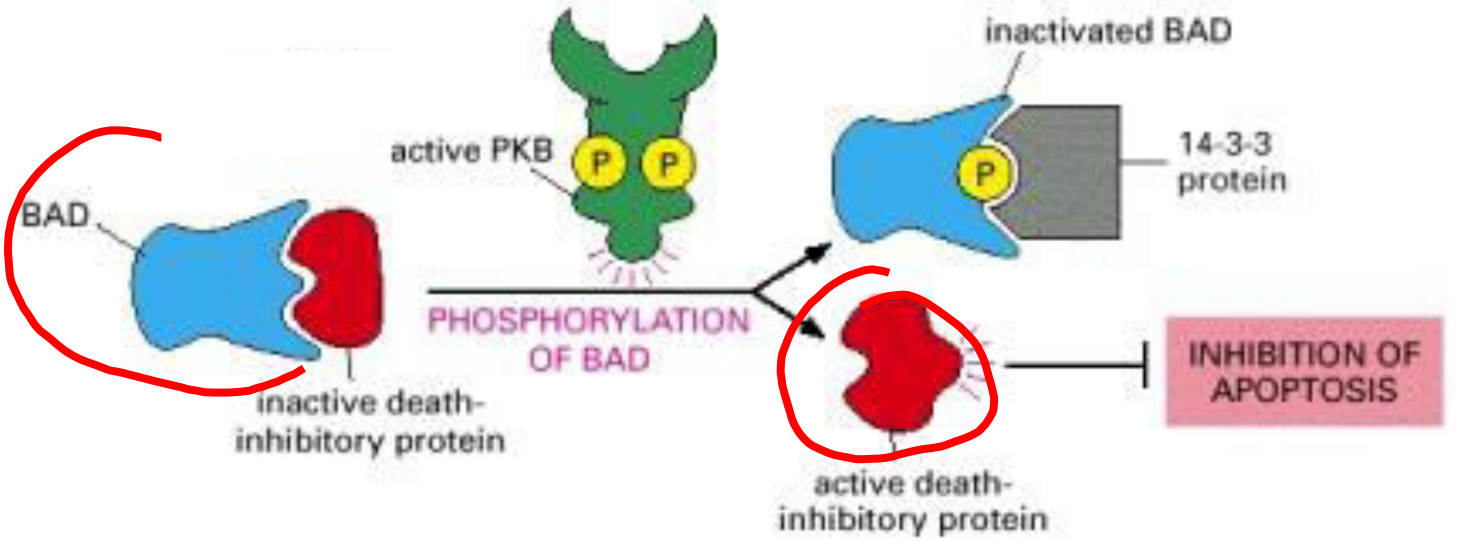
PKB Apoptosis Inhibition



PI 3-kinase signals cells to survive, by indirectly activating protein kinase B (PKB) (Akt). PDK1 PKB PH domains direct them to the plasma membrane where PDK1 phosphorilates PKB

PI Dependent Kinase

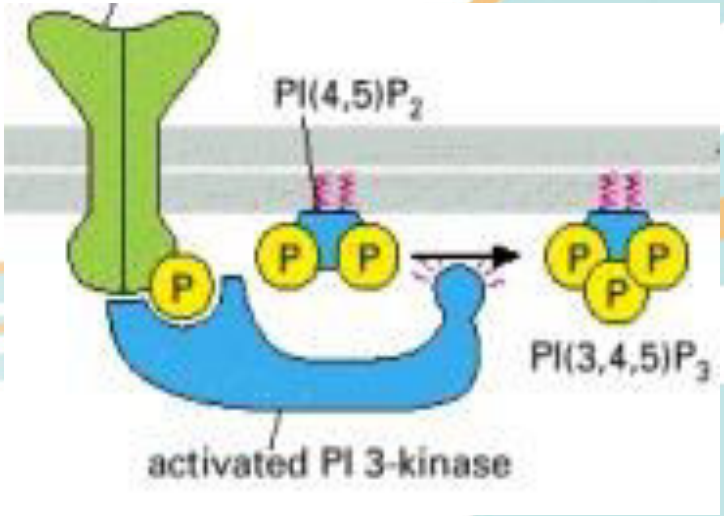
Apoptosis Inhibition



Activated PKB returns to the cytoplasm and phosphorylates BAD. BAD encourages cells to undergo apoptosis, promoting cell survival.

PKB also promotes cell survival by inhibiting other cell death activators, in some cases by inhibiting the transcription of the genes that encode them.

Ab Deficiency



END

Cell Biology

Jak-STAT Signaling

γ-interferon Antiviral Defense

Increase antigen presentation of macrophages

Activate & increase lysosome activity macrophages

Cells to increase expression of MHC class I

Promotes adhesion and binding required for leukocyte migration

Promotes NK cell activity

Activate inducible Nitric Oxide Synthase

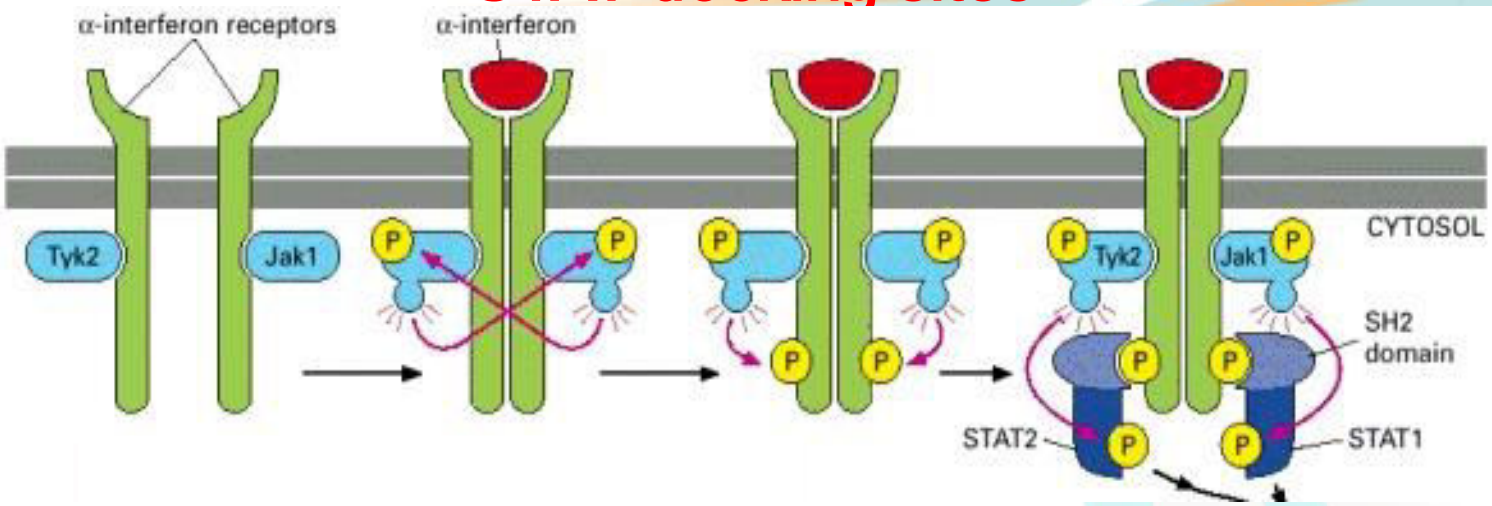
Jak-STAT Signaling

Jak-STAT one of the most direct signaling.

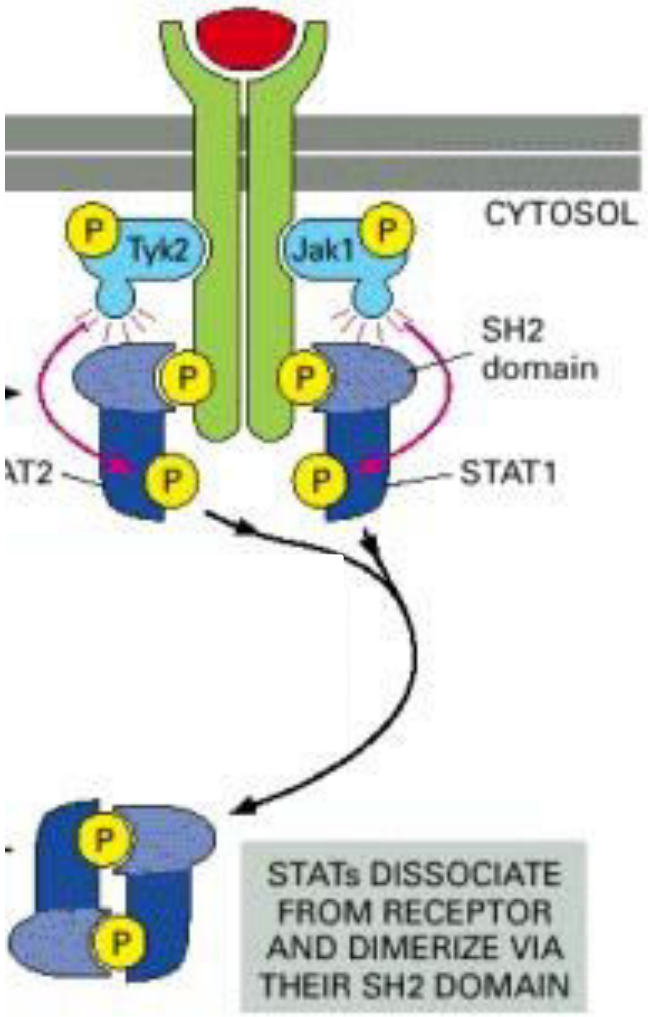
Ligand binding induces receptor to dimerize.

Binding brings Jaks close to cross-phosphorylate each other, increasing kinase activity.

The Jaks then phosphorylate receptors, creating STAT docking sites



SH2 Two Functions

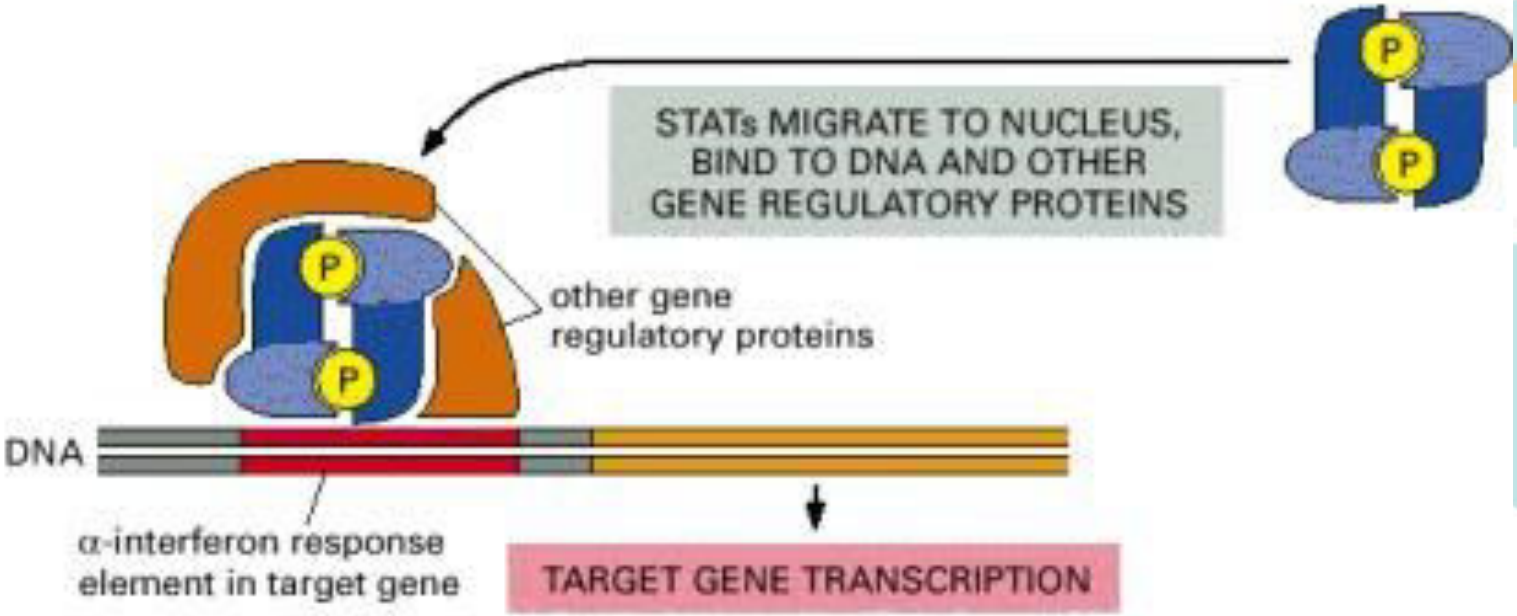


1) STAT binds receptor Jaks phosphorylate STAT forming SH2 domain & dissociation from the receptor.

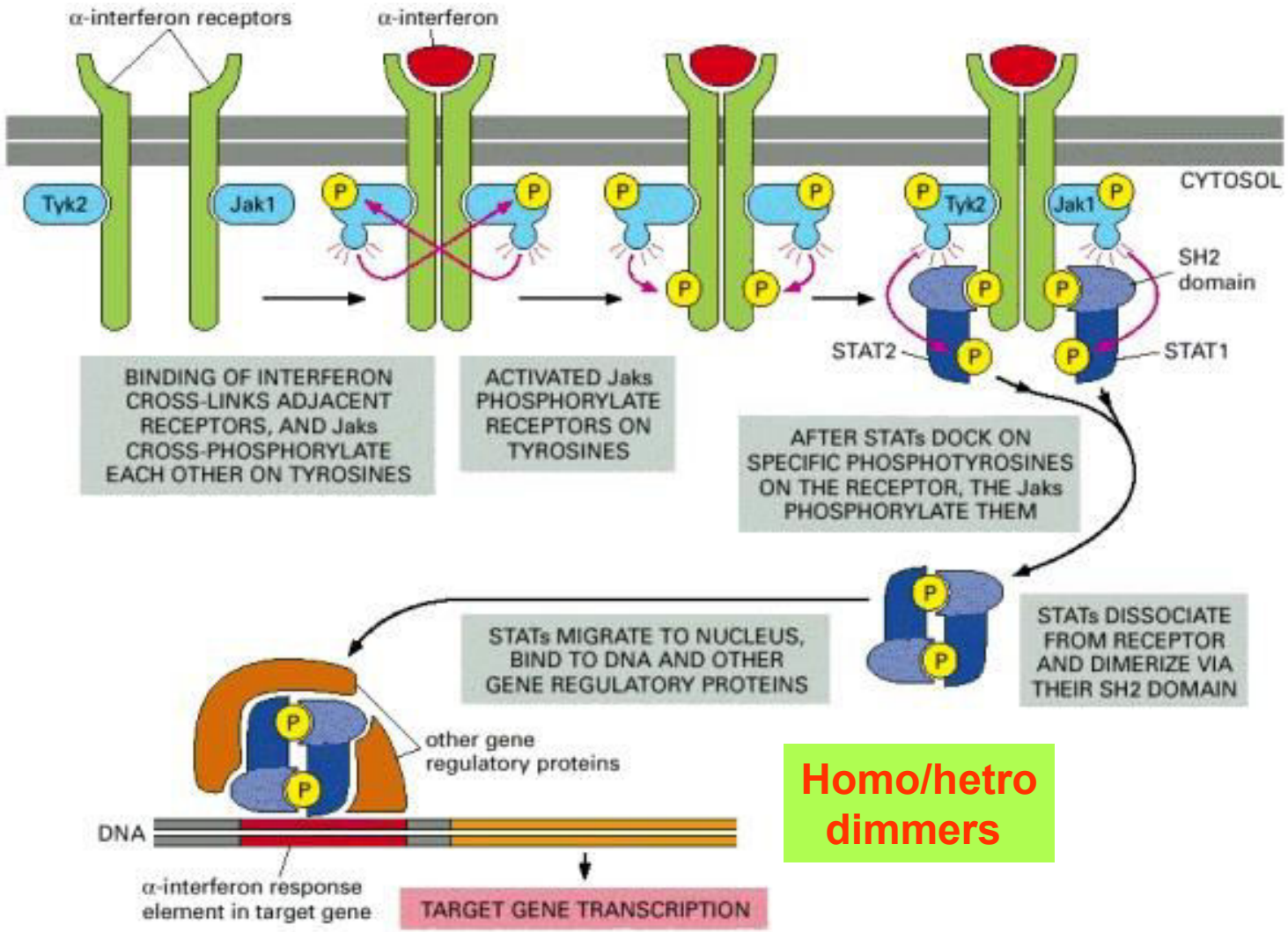
2) STAT SH2 domain mediates binding to another STAT molecule, forming STAT homo/heterodimer

STAT & Transcription

STAT dimer moves to the nucleus, where, in combination with other gene regulatory proteins, it binds to a specific DNA response element in various genes and stimulates their transcription



Jack STAT Signaling



Homo/hetro dimmers

Jack STAT Signaling

Erythropoietin stimulates erythrocyte precursor cells to survive, proliferate, and differentiate, is associated with Jak2.

Jak2-deficient mice erythrocyte development fails, and the mice die early in development.

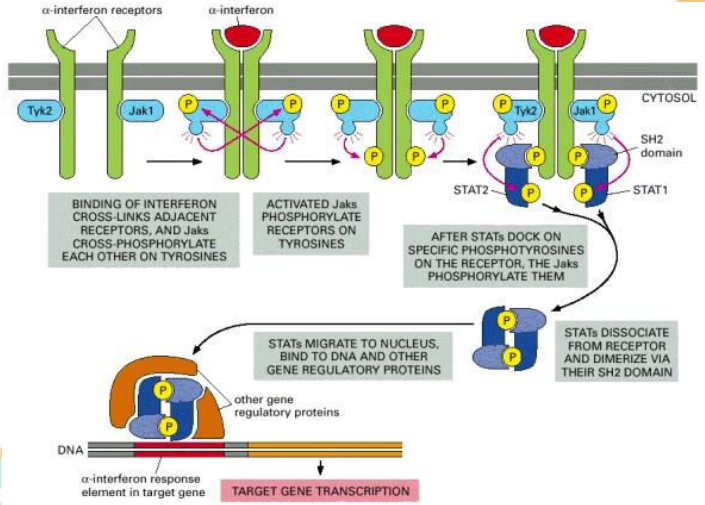
Prolactin stimulates breast cells to produce milk
Activated STAT5 stimulates the transcription of genes that encode milk proteins.

α-interferon, for example, recruit both STAT1 & STAT2, γ-interferon receptors recruit only STAT1

Jack STAT Signaling

SIGNALING LIGAND	RECEPTOR-ASSOCIATED JAKS	STATS ACTIVATED	SOME RESPONSES
γ-interferon	Jak1 and Jak2	STAT1	activates macrophages; increases MHC protein expression
α-interferon	Tyk2 and Jak2	STAT1 and STAT2	increases cell resistance to viral infection
Erythropoietin	Jak2	STAT5	stimulates production of erythrocytes
Prolactin	Jak1 and Jak2	STAT5	stimulates milk production
Growth hormone	Jak2	STAT1 and STAT5	stimulates growth by inducing IGF-1 production
GM-CSF	Jak2	STAT5	stimulates production of granulocytes and macrophages
IL-3	Jak2	STAT5	stimulates early blood cell production

Jack STAT Signaling



END

Cell Biology

TGF-β Receptor Family

TGF-β Receptor Family

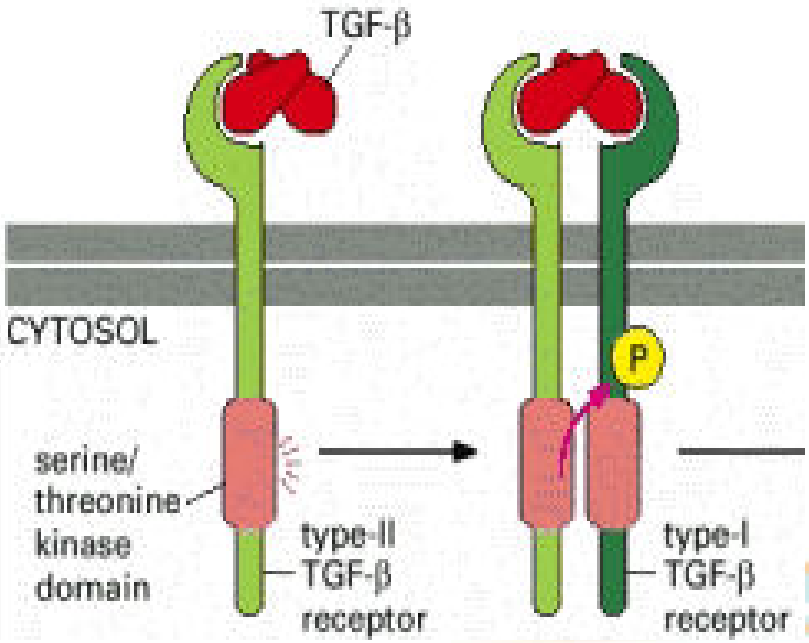
Consists of a large number of structurally related, secreted, dimeric proteins.

During development, they regulate pattern formation and influence various cell behaviors, ie proliferation, differentiation, extracellular matrix production, and cell death.

In adults, tissue repair and in immune regulation.

Act through enzyme-linked receptors that are single-pass transmembrane proteins

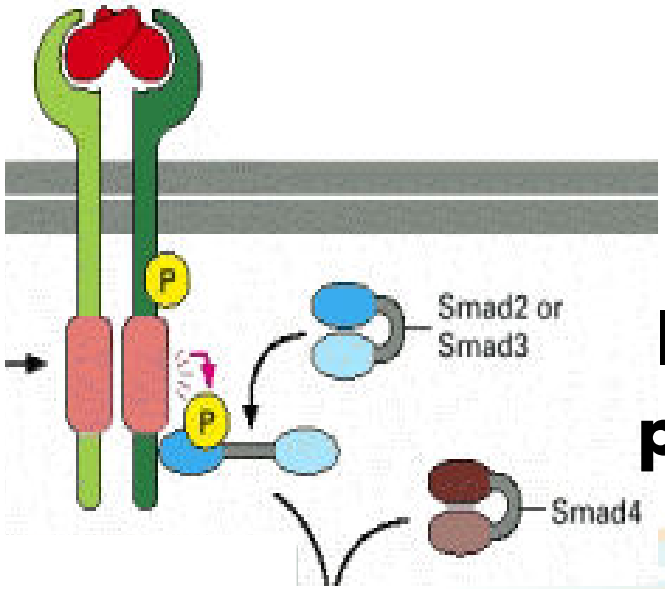
TGF-β Receptor Family



Both type-I and type-II receptors, required for signaling.

Ligand first binds and activates a type-II receptor homodimer, which recruits, phosphorylates, and activates a type-I receptor homodimer, forming an active tetrameric receptor complex

TGF-β Receptor Family

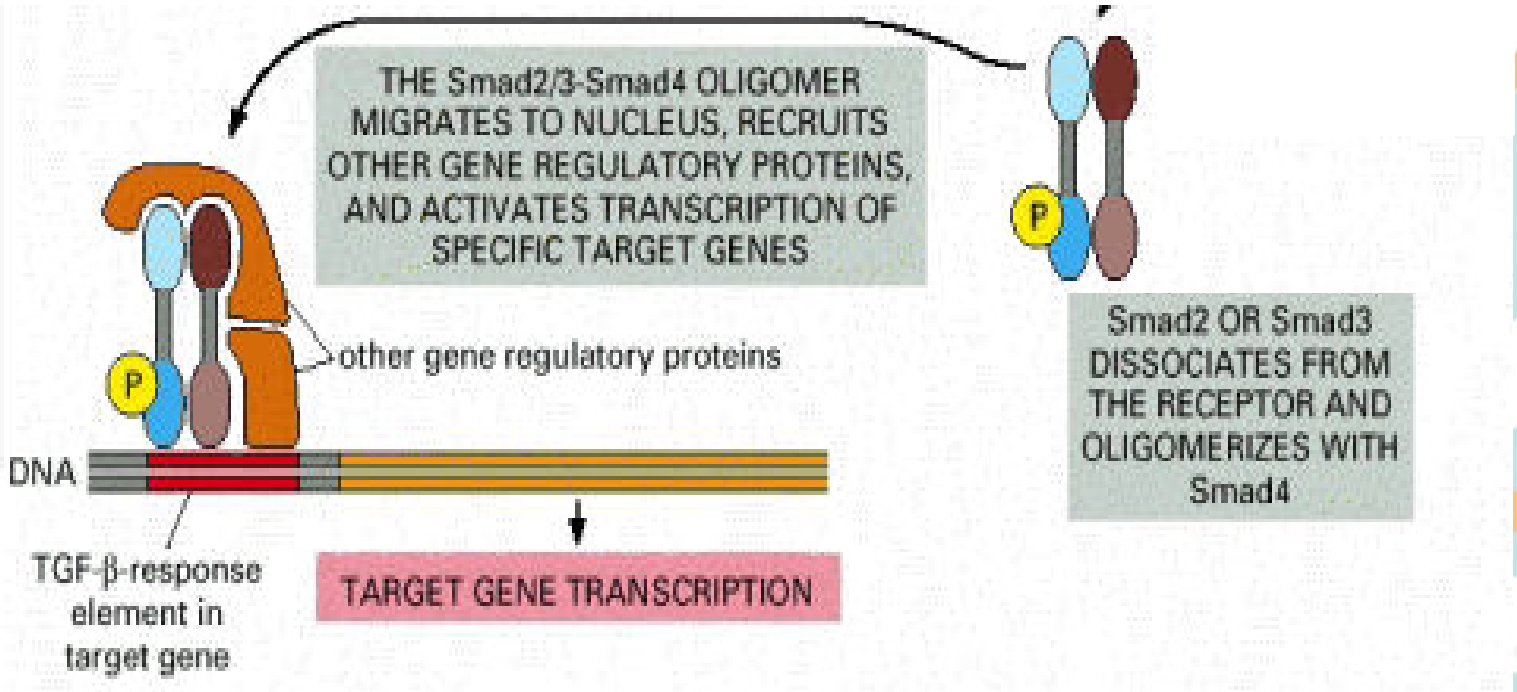


The type-I receptor directly binds and phosphorylates a latent gene regulatory protein of Smad family

Activated TGF-β receptors phosphorylate Smad2 or Smad3, activated BMP receptors phosphorylate Smad1, Smad5, or Smad8.

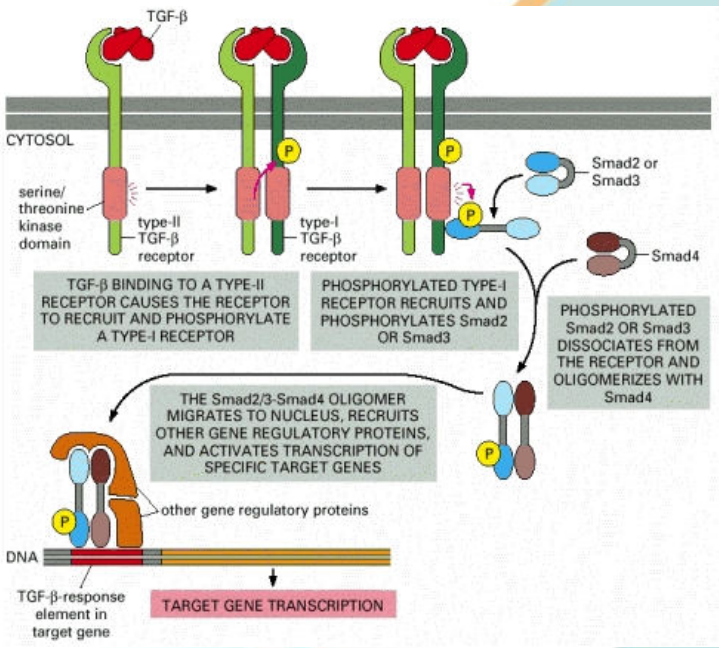
One of these Smads then dissociates from the receptor and binds to Smad4

TGF-β Receptor Family



Smad complex moves into the nucleus, it associates with other gene regulatory proteins, binds to specific sites in DNA, and activates a particular set of target genes.

TGF-β Receptor Family



END

Cell Biology

Proteolysis

Signaling Pathways

Proteolysis Signaling Pathways

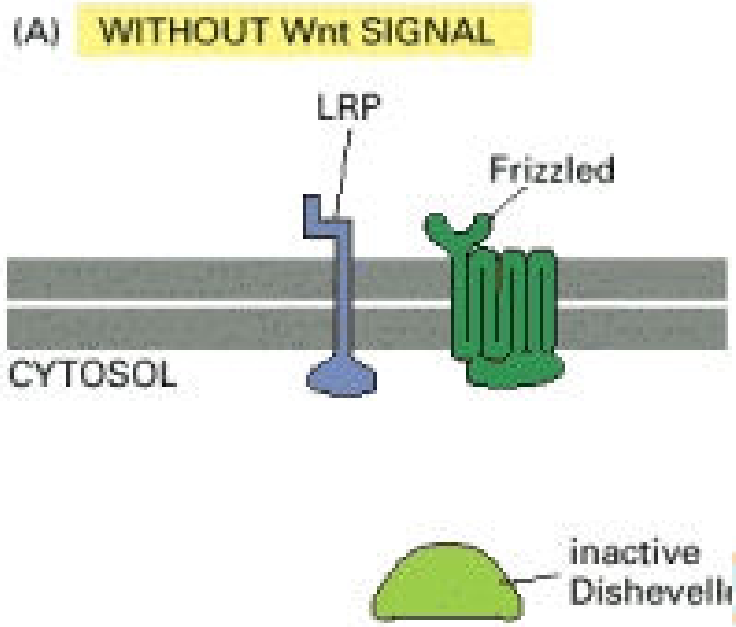
Each cell in the embryo has to be guided along one developmental pathway according to its history, position & the character of its neighbors.

Cells must exchange signals with their neighbors to coordinate theirs behavior with others.

Developmental processes depend on signaling pathways that involve regulated proteolysis.

Defective Notch, Wnt, and Hedgehog signalling results in seriously disturbed development, the mouse dies as an embryo or at birth.

WNT Signaling



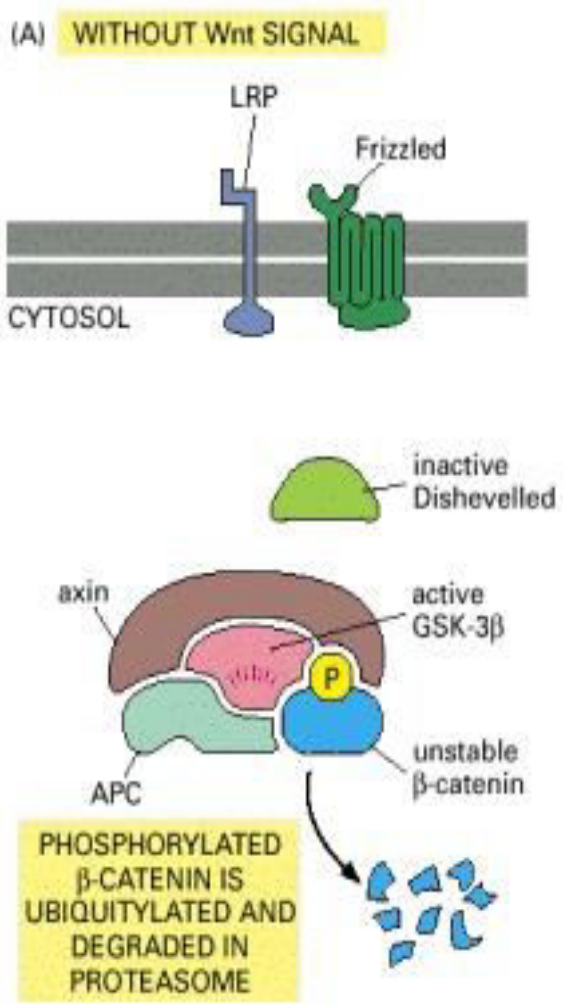
Wnt proteins are secreted signal molecules

Frizzled (receptor) family of seven-pass transmembrane proteins resemble GPCR.

Co-receptor LDL-receptor-related protein (LRP).

Cytoplasmic signaling protein called Dishevelled is activated when Wnt is present.

WNT Signaling



β-catenin functions both in cell-cell adhesion and as a latent gene regulatory protein.

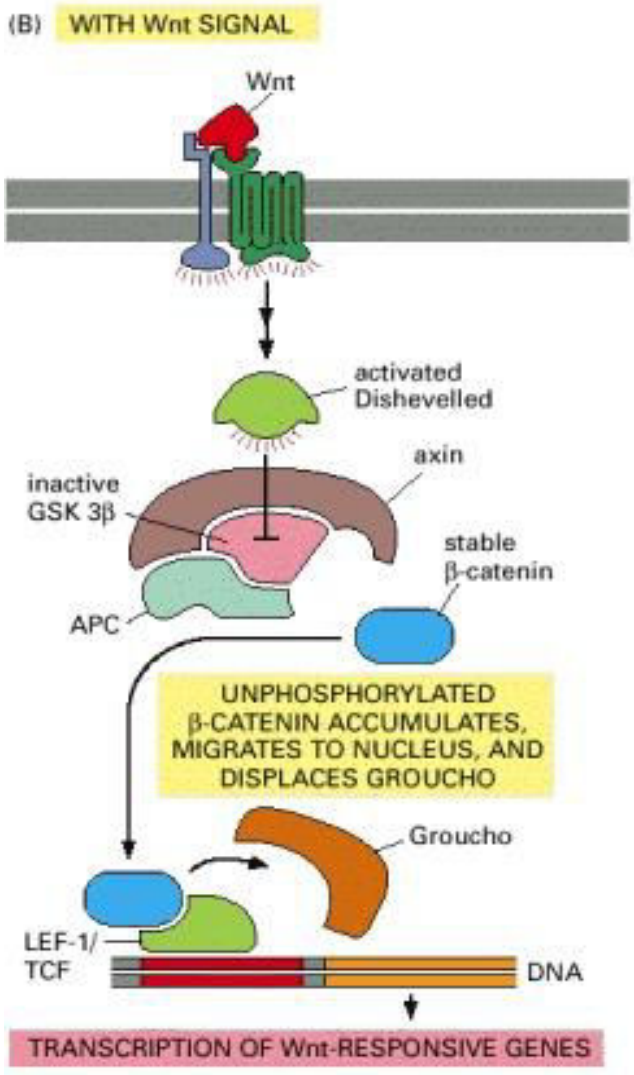
β-catenin links the cadherins to the actin cytoskeleton

β-catenin not associated with cadherins is rapidly degraded

glycogen synthase kinase-3β (GSK-3β) phosphorylates β-catenin, thereby marking the protein for ubiquitylation

adenomatous polyposis coli (APC) adherens junctions scaffold protein called axin

WNT Signaling

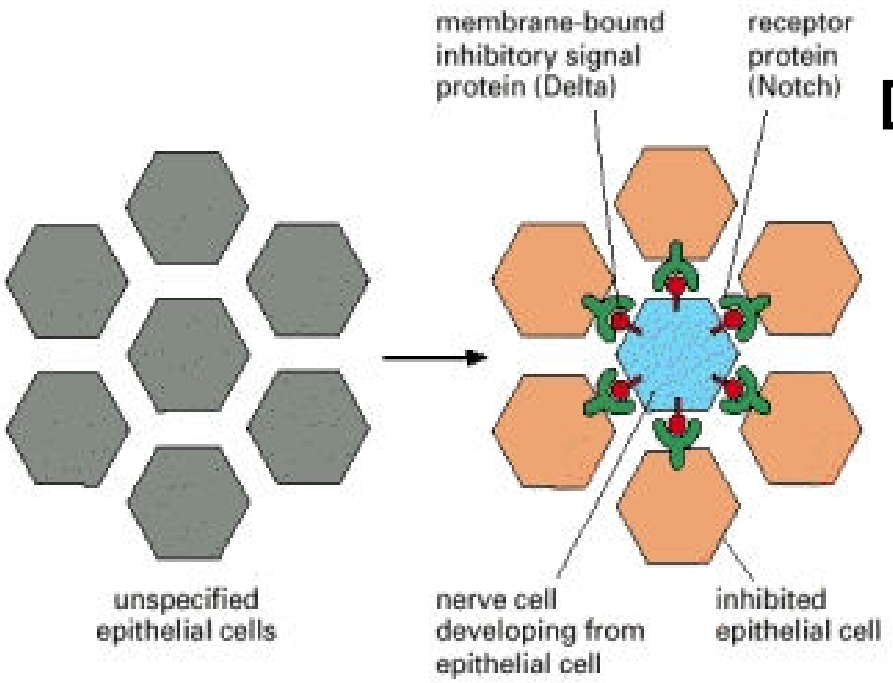


Wnt binding to Frizzled and LRP inhibits of β-catenin phosphorylation/degradation.

β-catenin is a coactivator & induces transcription of Wnt target genes

APC gene mutations occur in 80% of human colon cancers. APC binding to β-catenin, inhibited so that β-catenin accumulates

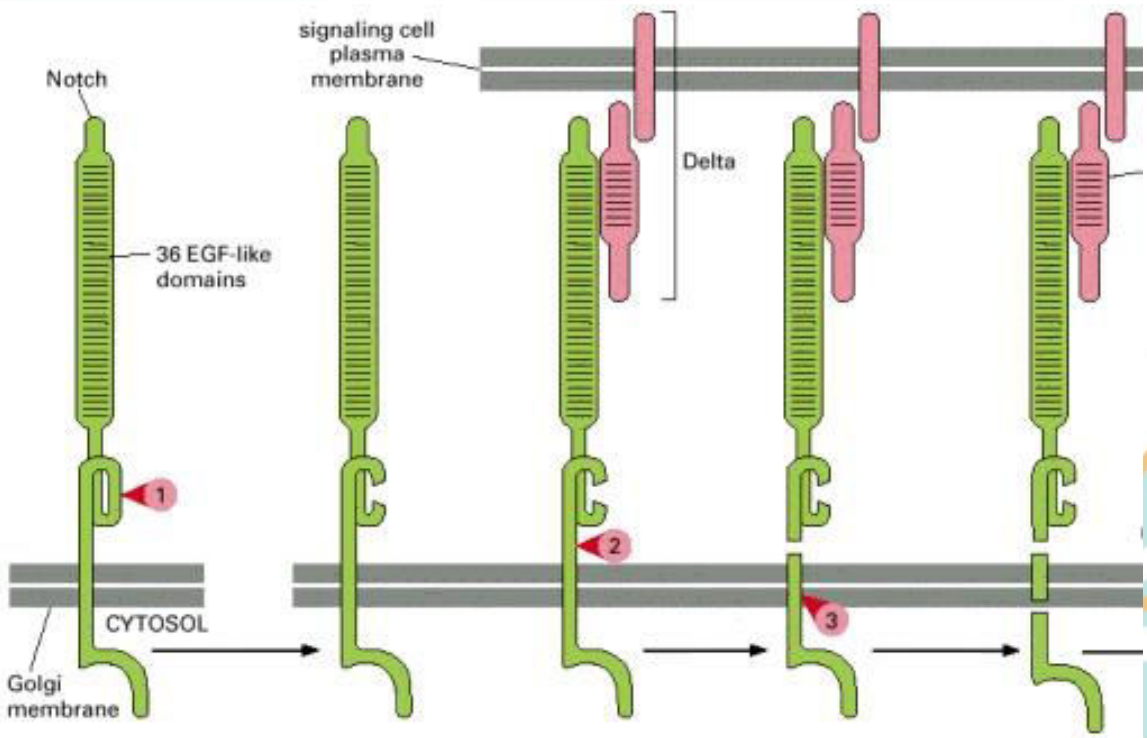
Signaling Pathways Proteolysis



Delta, on the surface of future neural cell binds Notch on a neighboring cell, Delta signals to the neighbor not to become neural

Nerve cells arise as isolated single cells within an epithelial sheet of precursor cells.
Future nerve cell signal its immediate neighbors not to develop in the same way at the same time, (lateral inhibition) contact-dependent signaling.

Signaling Pathways Proteolysis



single-pass trans-membrane proteins

Intracellular protease cleaves off the cytoplasmic tail of Notch when it binds Delta,
Tail moves into the nucleus to activate the transcription of a set of Notch-response genes.

3 proteolytic cleavages last 2 depend on Delta.

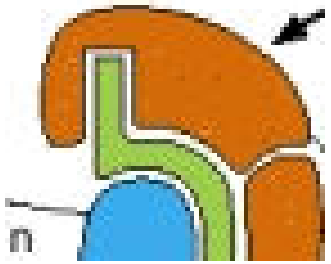
Heterodimer

Signaling Pathways Proteolysis

Intracellular protease cleaves off the cytoplasmic tail of Notch when it binds Delta,.

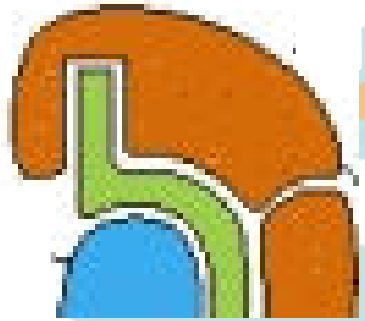
Tail moves into the nucleus to activate the transcription of a set of Notch-response genes.

The products of the main genes directly activated by Notch signaling are themselves gene regulatory proteins, but with an inhibitory action: they block the expression of genes required for neural differentiation



β-amyloid precursor protein (APP) Alzheimer's

Signaling Pathways Proteolysis



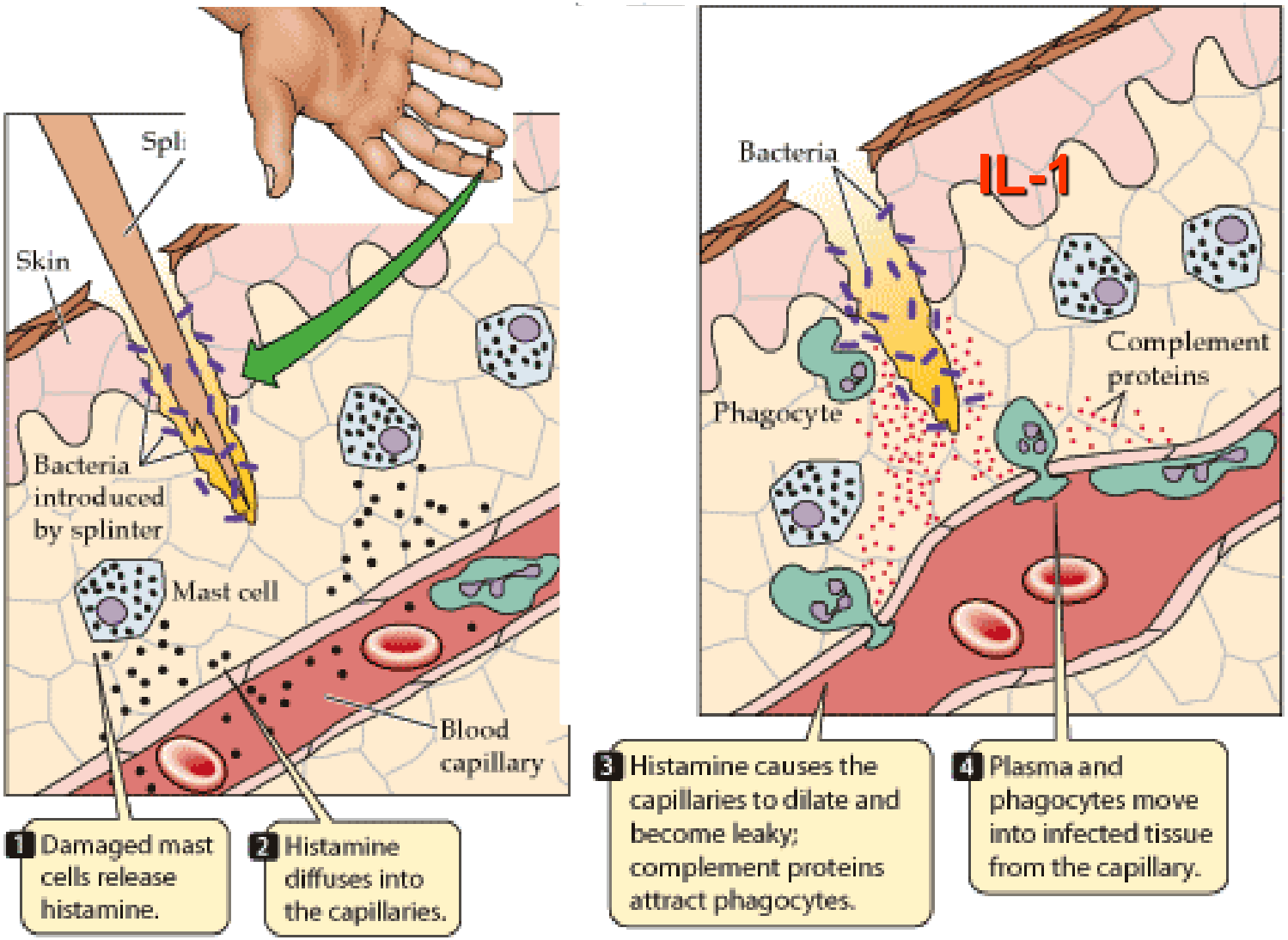
END

Cell Biology

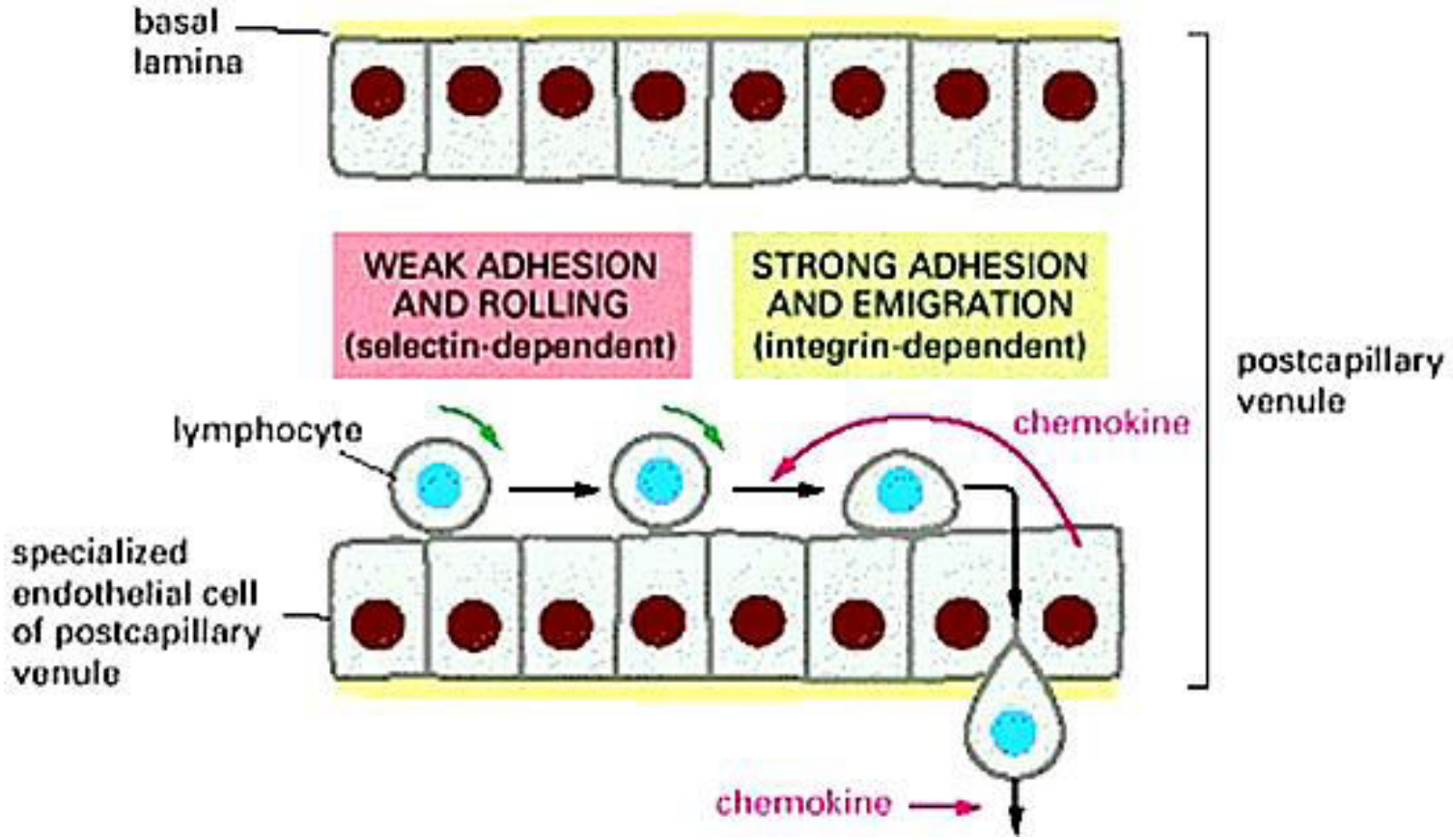
Immune System 2

demonstration

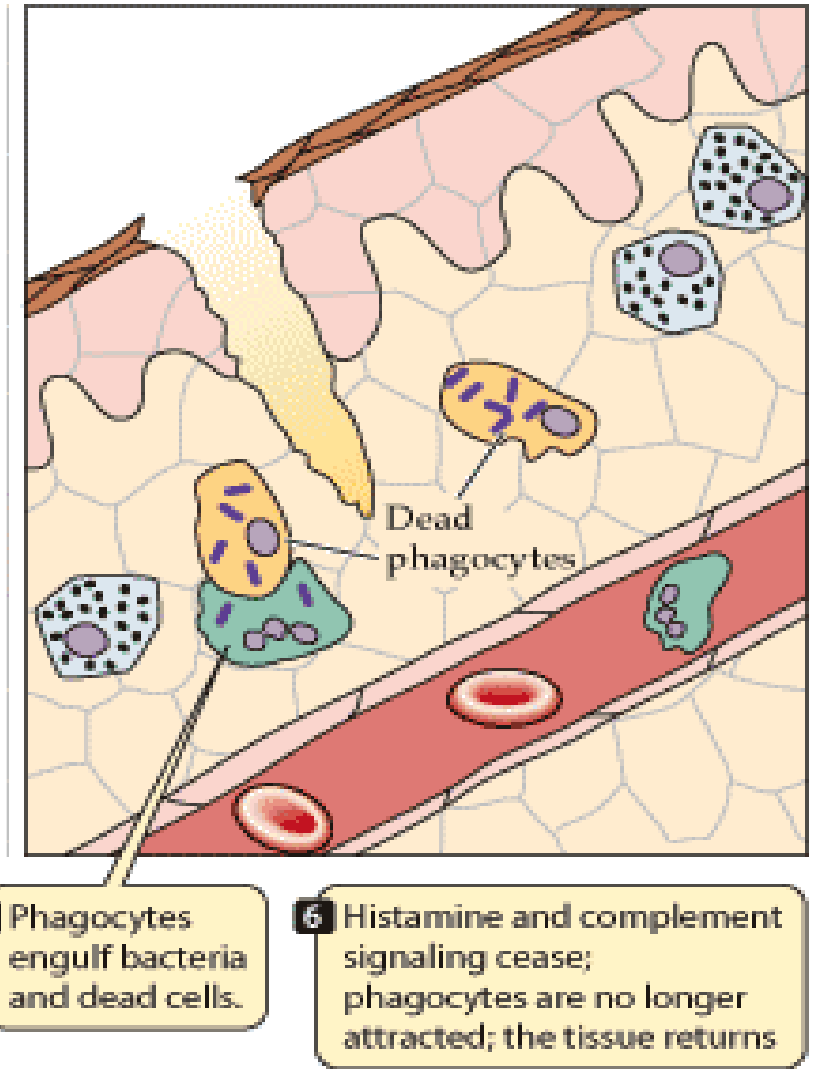
The Attack


Spl
Skin
Bacteria introduced by splinter
Mast cell
Blood capillary
1 Damaged mast cells release histamine.
2 Histamine diffuses into the capillaries.
Bacteria
IL-1
Complement proteins
Phagocyte
3 Histamine causes the capillaries to dilate and become leaky; complement proteins attract phagocytes.
4 Plasma and phagocytes move into infected tissue from the capillary.

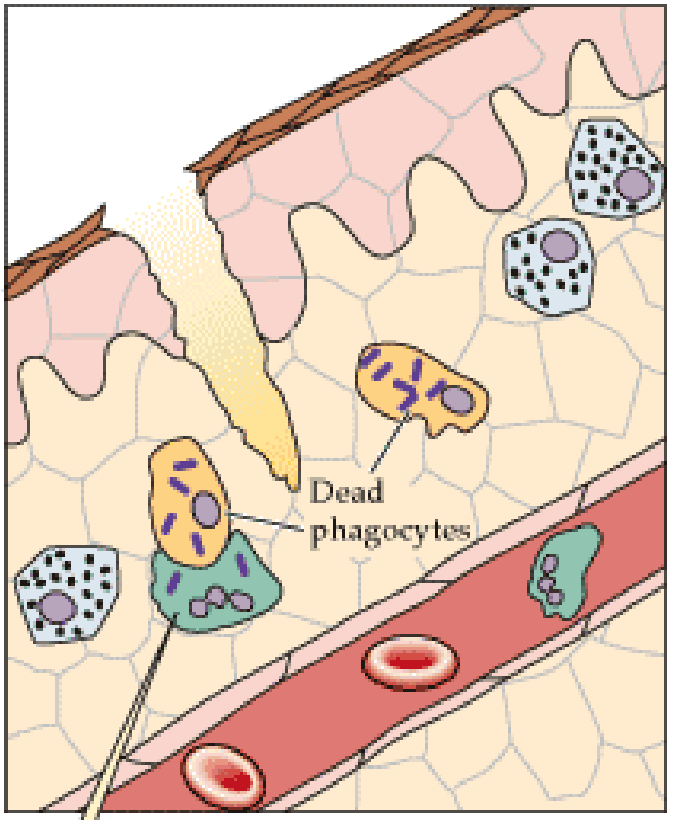
The Attack


basal lamina
WEAK ADHESION AND ROLLING (selectin-dependent)
STRONG ADHESION AND EMIGRATION (integrin-dependent)
postcapillary venule
lymphocyte
chemokine
specialized endothelial cell of postcapillary venule
chemokine

The Attack


Dead
phagocytes
5 Phagocytes engulf bacteria and dead cells.
6 Histamine and complement signaling cease; phagocytes are no longer attracted; the tissue returns

The Attack



5 Phagocytes engulf bacteria and dead cells.

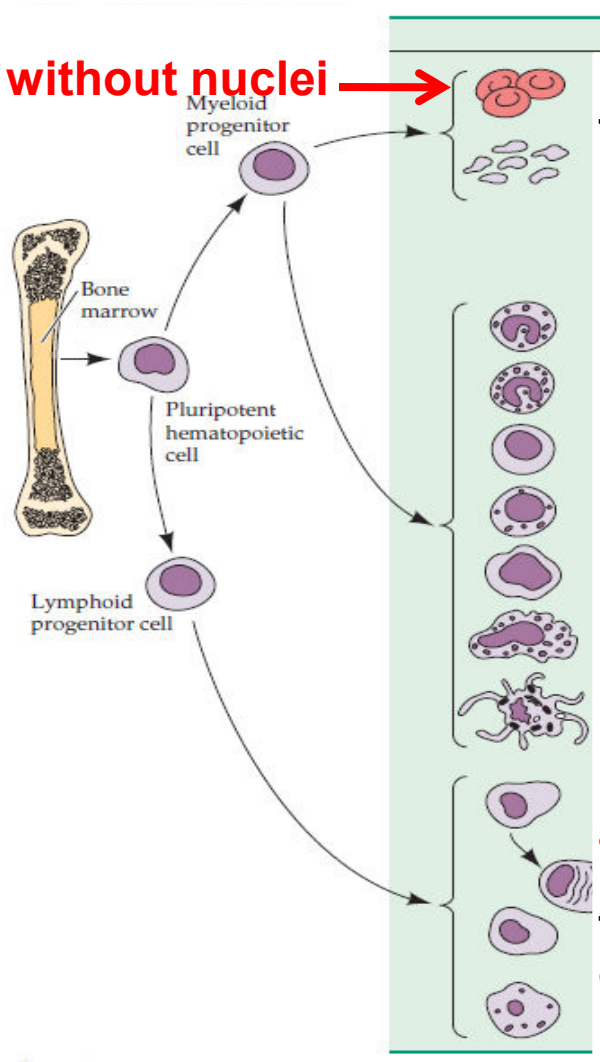
6 Histamine and complement signaling cease; phagocytes are no longer attracted; the tissue returns

END

Cell Biology

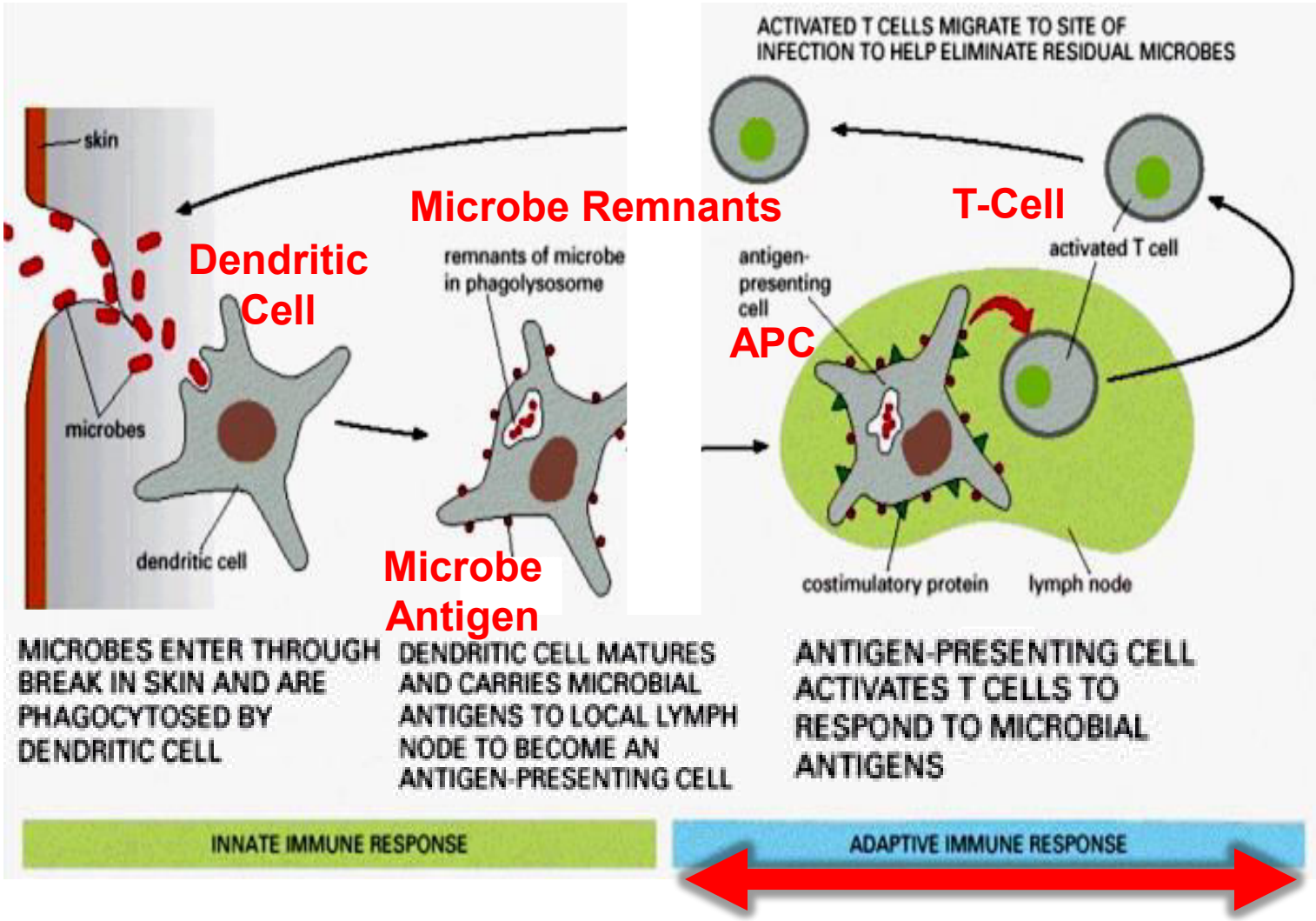
Immune System 3

Immune System

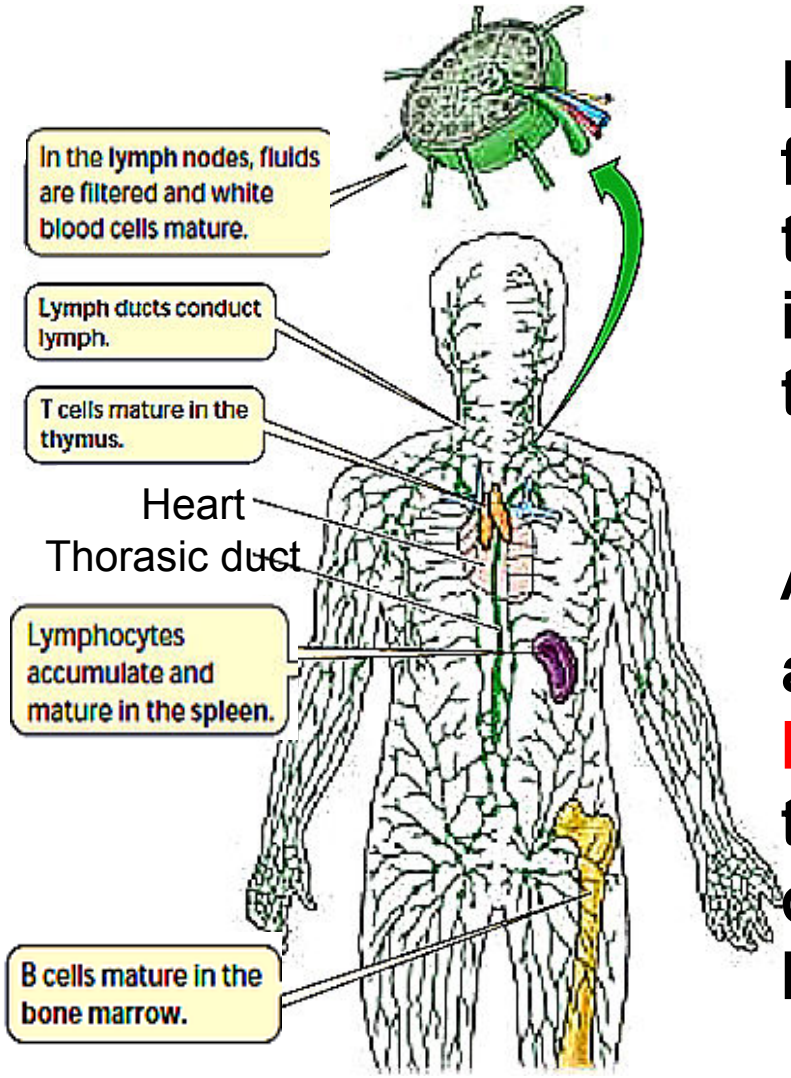


TYPE OF CELL	FUNCTION
without nuclei →	Red blood cells RBC (erythrocytes) Transport oxygen and carbon dioxide
	Platelets (cell fragments Initiate blood clotting)
	Basophils Release histamine
	Eosinophils Kill antibody-coated parasites
	Neutrophils Phagocytose antibody-coated pathogen
	Mast cells Release histamine when damaged
	Monocytes Develop into macrophages
	Macrophages Engulf and digest microorganisms
	Dendritic cells Present antigens to T cells
	B cells Differentiate form antibody-producing cells and memory cells Plasma cells Secrete antibodies
	T cells Kill virus-infected cells; regulate activities of other white blood cells
	Natural killer lyse virus-infected/cancerous cells

Immune System



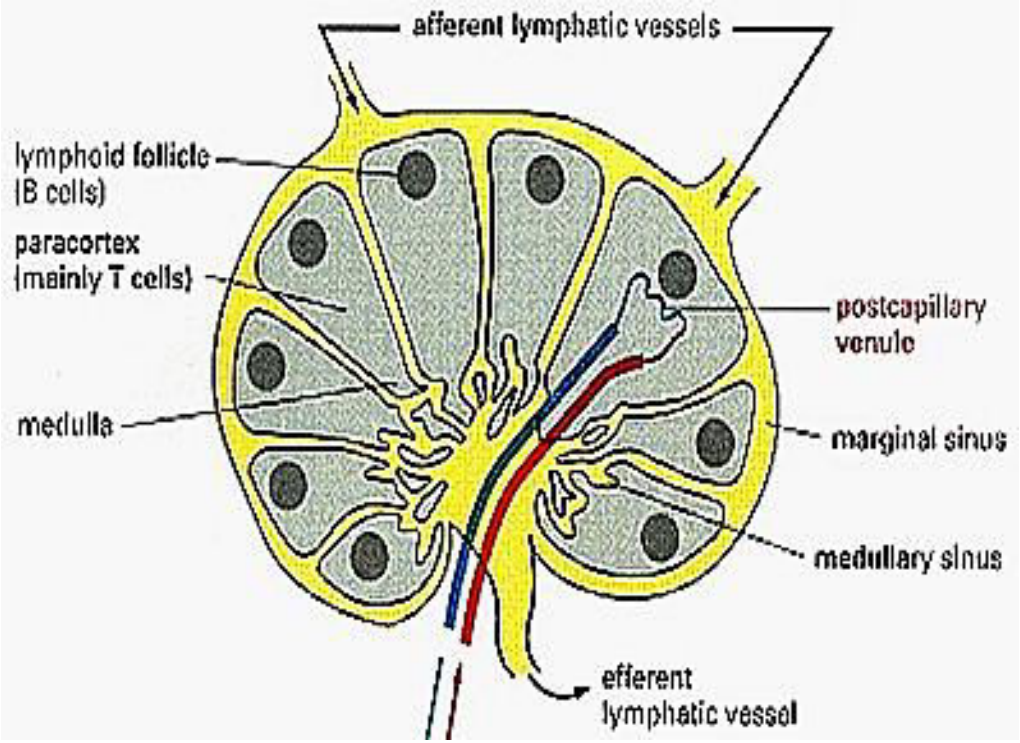
Human Lymphatic System



Lymph is a fluid derived from blood and other tissues that accumulates in intercellular spaces throughout the body.

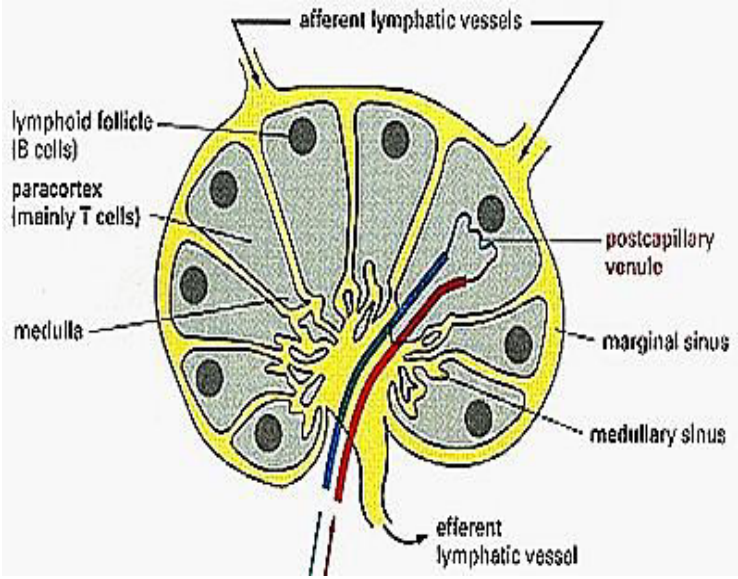
A network of ducts and vessels collects lymph (w/o RBCs) from the body's tissues and carries it toward the heart.

Human Lymphatic System



Lymph Nodes, contain a variety of WBCs. As fluid passes through a node, it is filtered and inspected for non-self materials by these defensive cells.

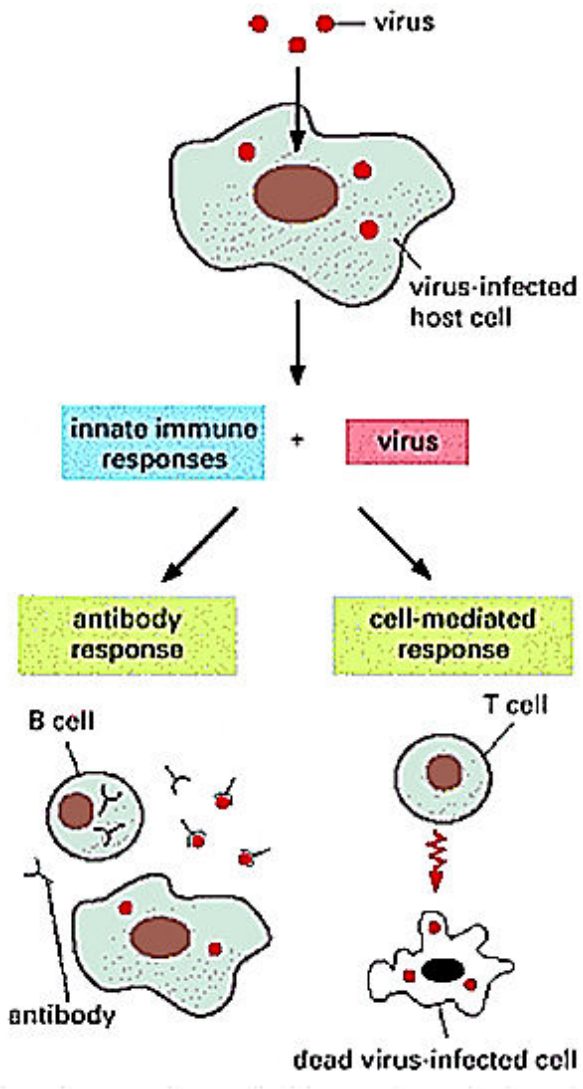
Human Lymphatic System



Cell Biology

AIS

Adaptive Immune System (AIS)



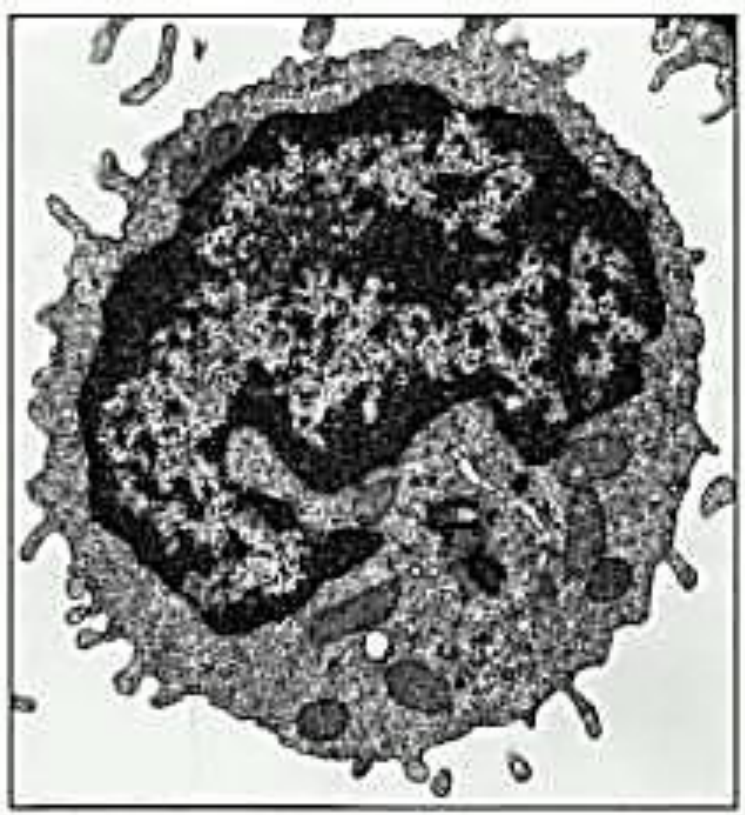
AIS has two components.

Humoral immunity is mediated by soluble antibody proteins produced by B lymphocytes that neutralize and kill pathogens/toxins (20%-30%).

Cellular immunity is mediated by T lymphocytes which target cells infected by a pathogen (60%-70%).

Adaptive Immune System (AIS)

The Bs & the Ts



(A) resting T or B cell

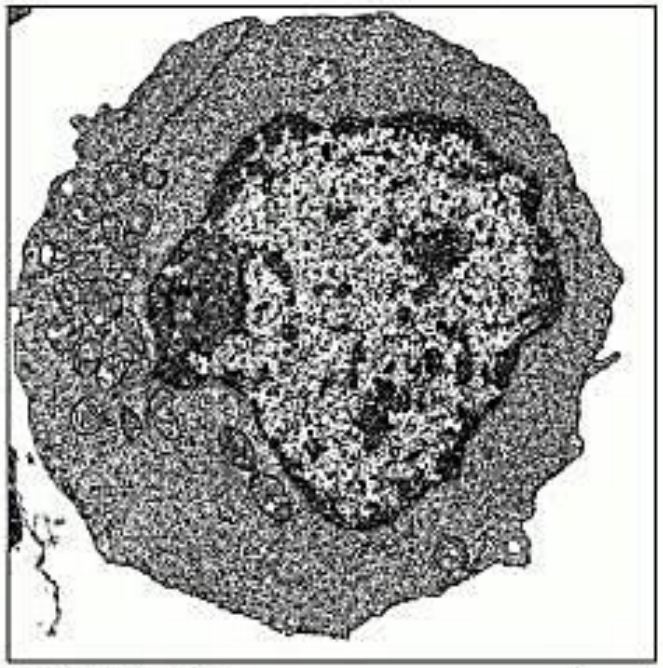
1 µm

Adaptive Immune System (AIS)

The Bs & the Ts

(B) effector B cell (plasma cell)

1 µm



(C) effector T cell

1 µm

AIS Activation

IIS Phagocytic cells have receptors that have affinity for certain pathogen-associated molecules

IIS programmed to be activated by specific pathogen molecules (Toll Receptors)

IIS activates AIS

Adaptive Immune Response AIR is highly specific to the pathogen that induced it.

There are 2×10^{12} lymphocytes in the human body, making the immune system comparable in cell mass to the liver or brain.

AIS Features

IIS is activated at sites of infection, whereas AIS are activated in peripheral lymphoid organs

AIR can distinguish self from non-self

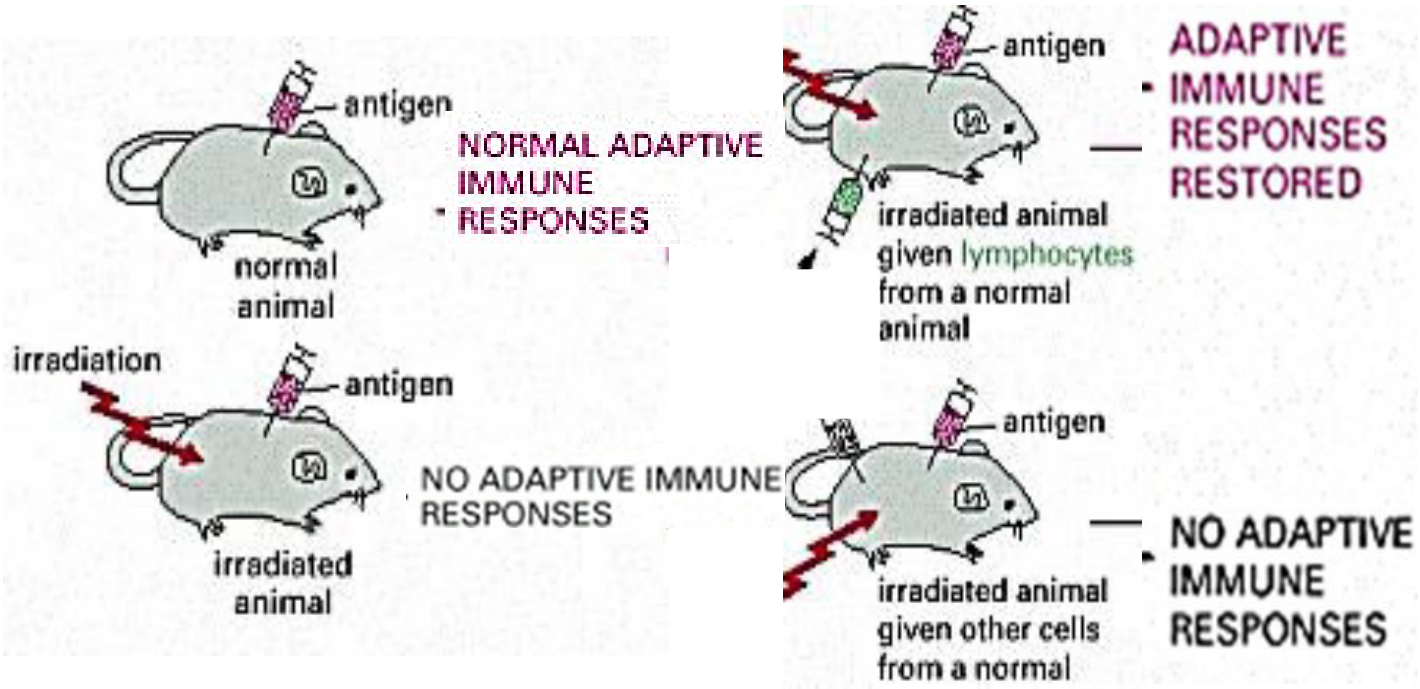
Long term protection (measles)

AIR activated by self is harmful

Substances that activate AIS are called antigens

AIR can distinguish even optical isomers

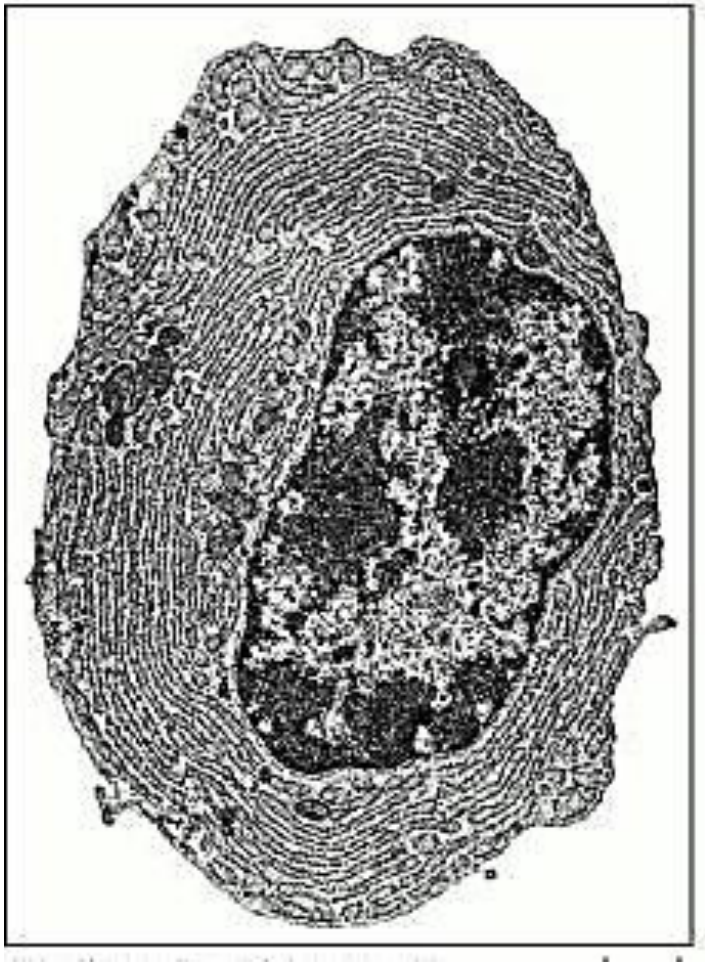
AIS



Rats were heavily irradiated to kill most of their WBCs including lymphocytes. These rats were unable to mount AIR . Various types of cells were transferred into the animals . Only lymphocytes restored the AIR of irradiated animals

Adaptive Immune System (AIS)

The Bs & the Ts

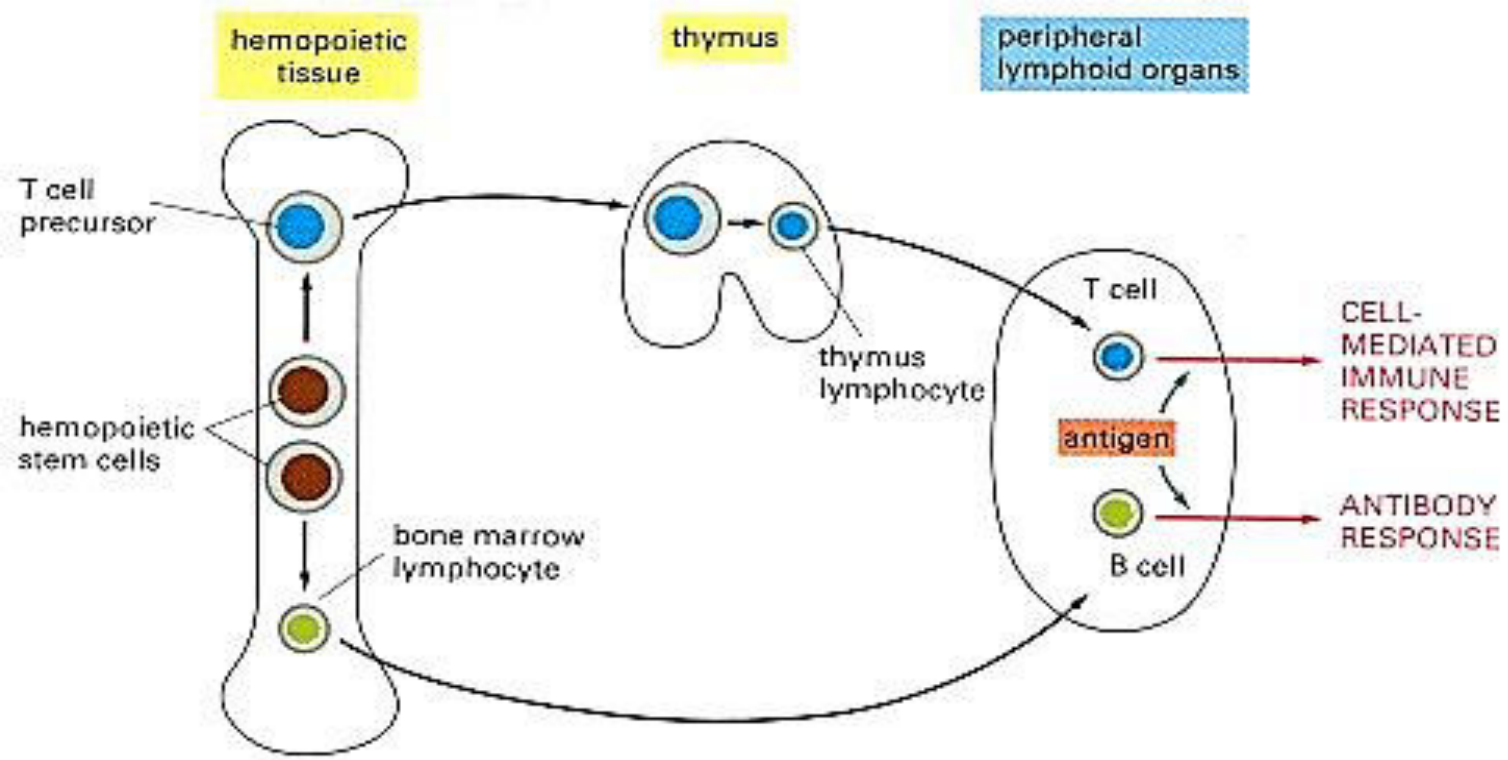


END

Cell Biology

The B-Cells

The B-Cells



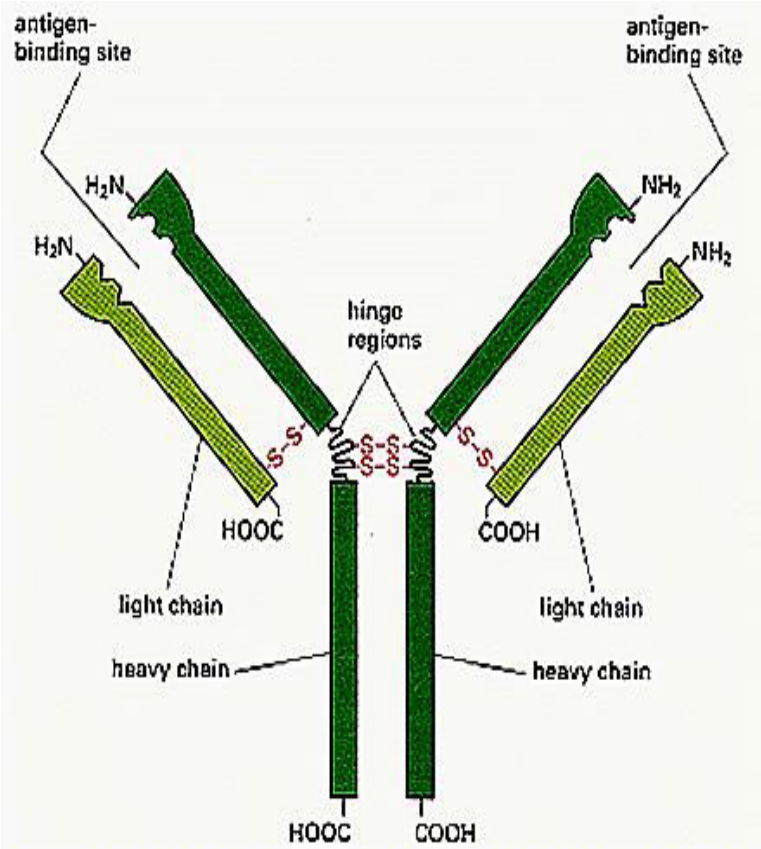
T cells develop in the thymus, and B cells, in mammals, develop in the bone marrow

Most lymphocytes die in central lymphoid organ

B-Cell Maturation Antigen Independent

The B-Cells

Antibody Structure



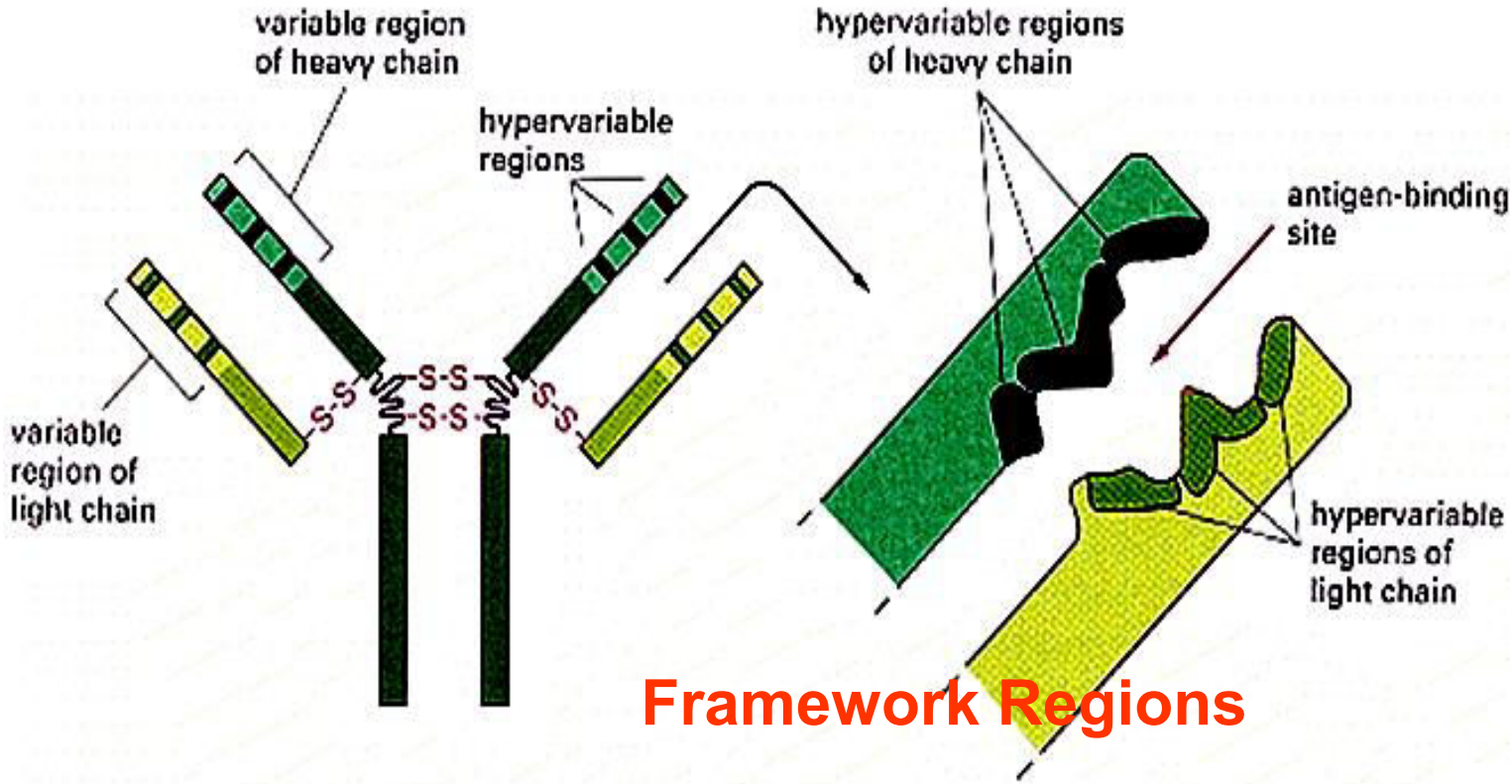
It is composed of four polypeptide chains 2 identical heavy chains & 2 identical light chains.

Tail (Fc) & hinge region are formed by the two heavy chains.

2 identical Binding Sites.

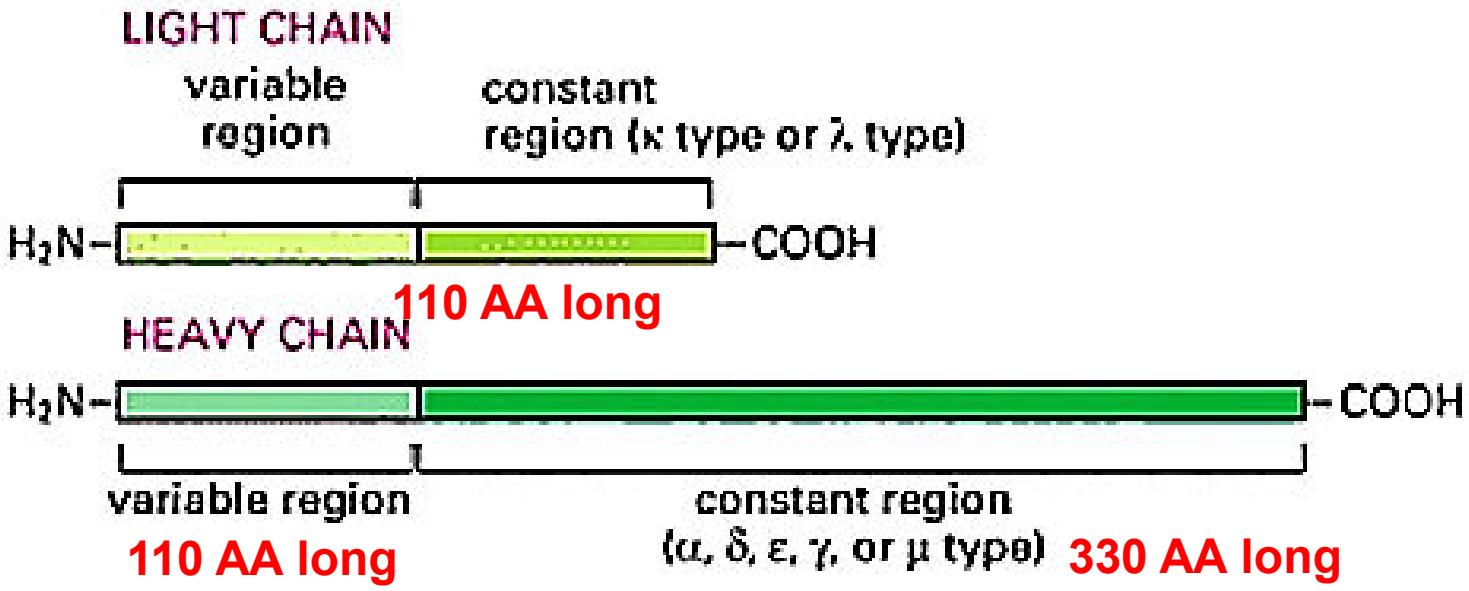
Flexible hinge region improves efficiency of antigen binding and cross-linking.

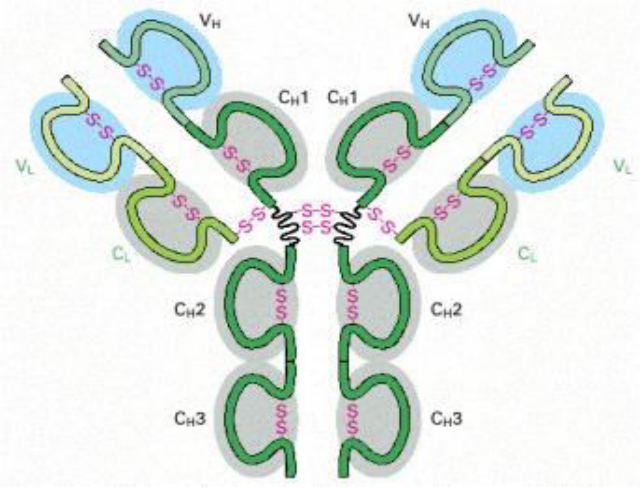
Antibody Structure



The two antigen-binding sites are identical, each formed by the N-terminal region of a light chain and the N-terminal region of a heavy chain.

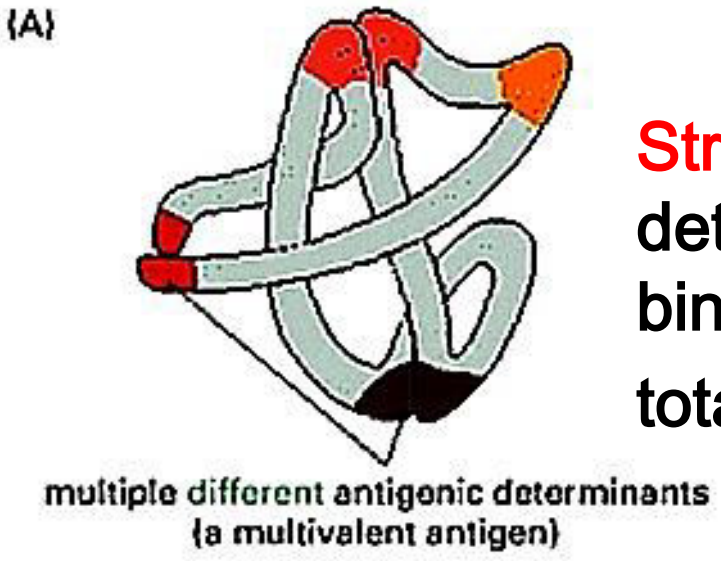
Antibody Structure





Immunoglobin (Ig) domain is 110 AA long and is repeated suggesting gene duplication.

Antigen



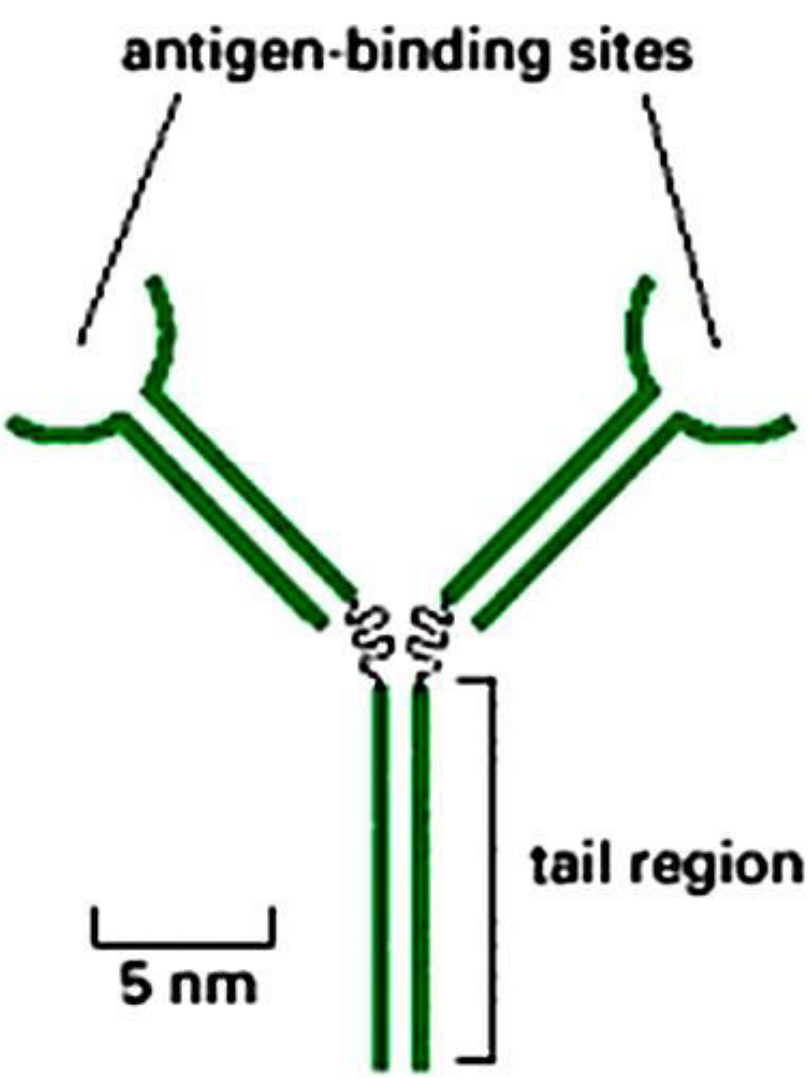
Affinity

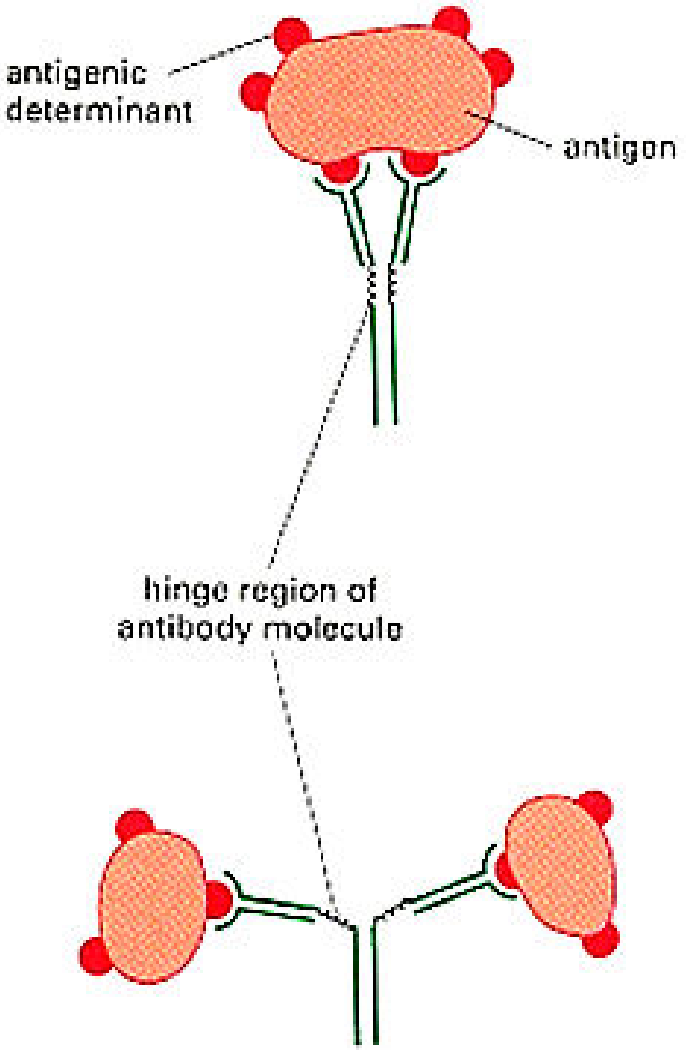
Strength of a single anitgenic determinat with a single Ab binding site (# independent)

total binding strength: Avidity

IgG binds Polyvalent 100 times stronger than monovalant

Mode of Action

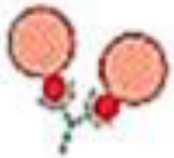


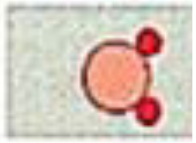


Mode of Action

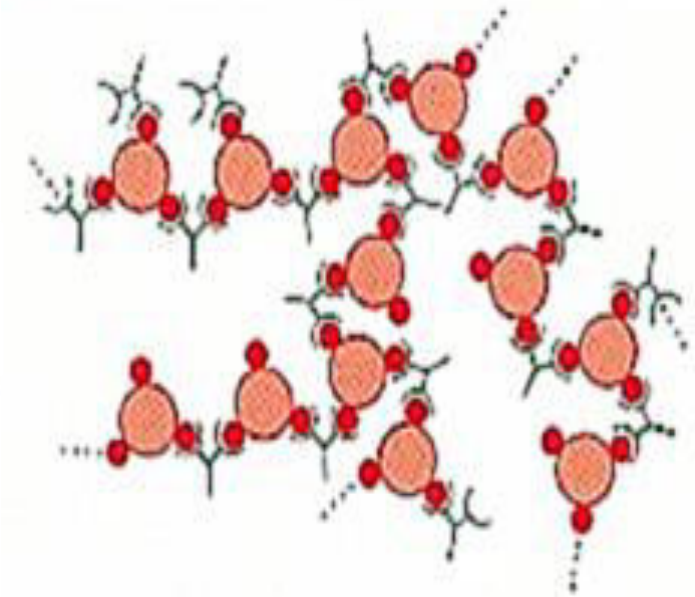
One antigenic determinants

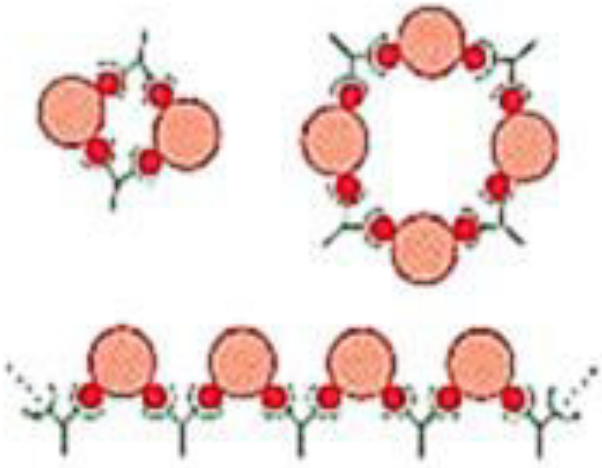
Three or more antigenic determinants



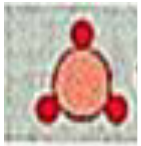


Two antigenic determinants





Mode of Action



Three or more antigenic determinants

END

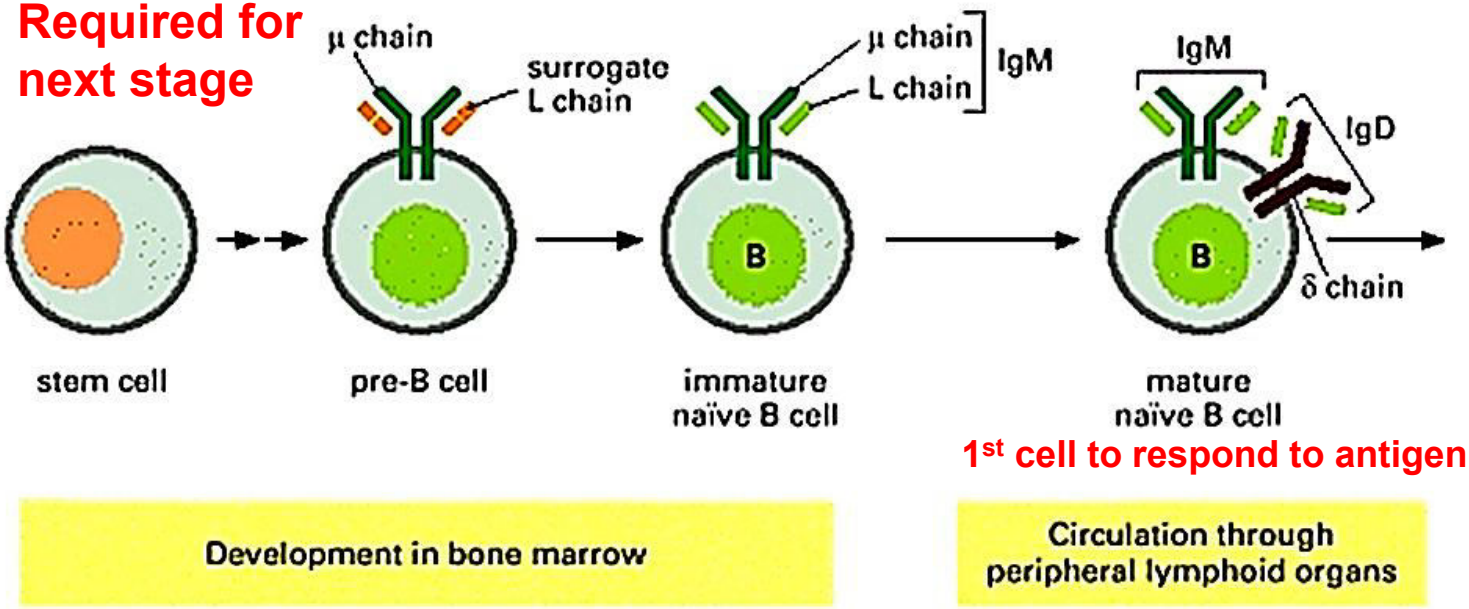
Cell Biology

The B-Cells 2

The B-Cells

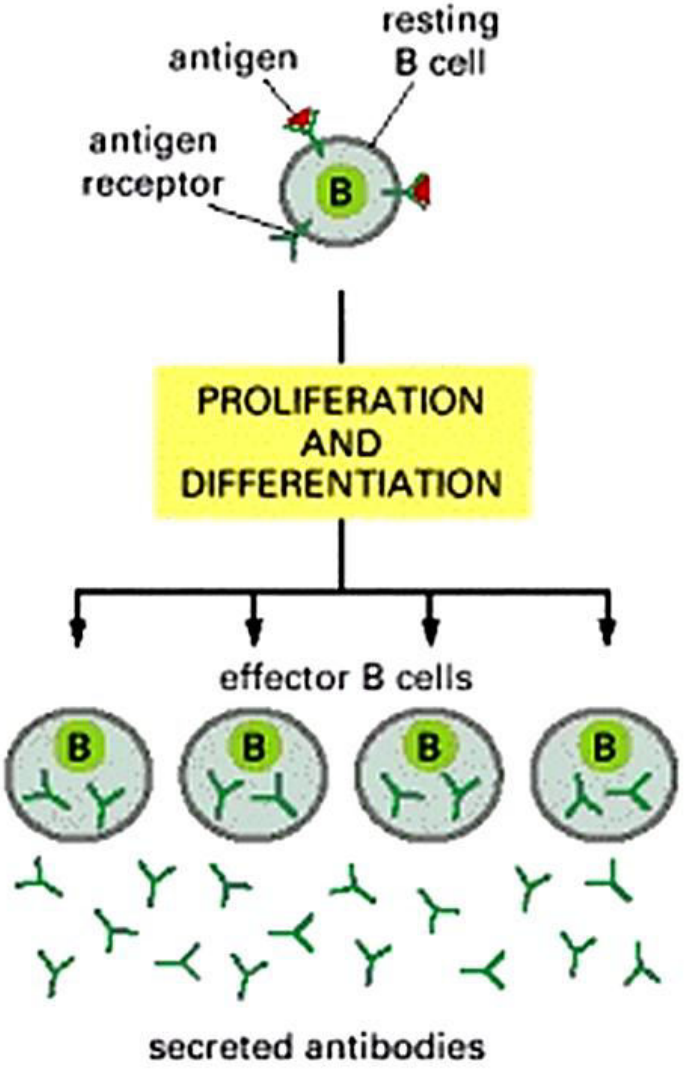
B-Cell Maturation Antigen Independent

Sub for LC
Required for
next stage



IgM is the 1st class of Ab to appear on cell surface and secreted B cells can then switch Ab class

Mode of Action

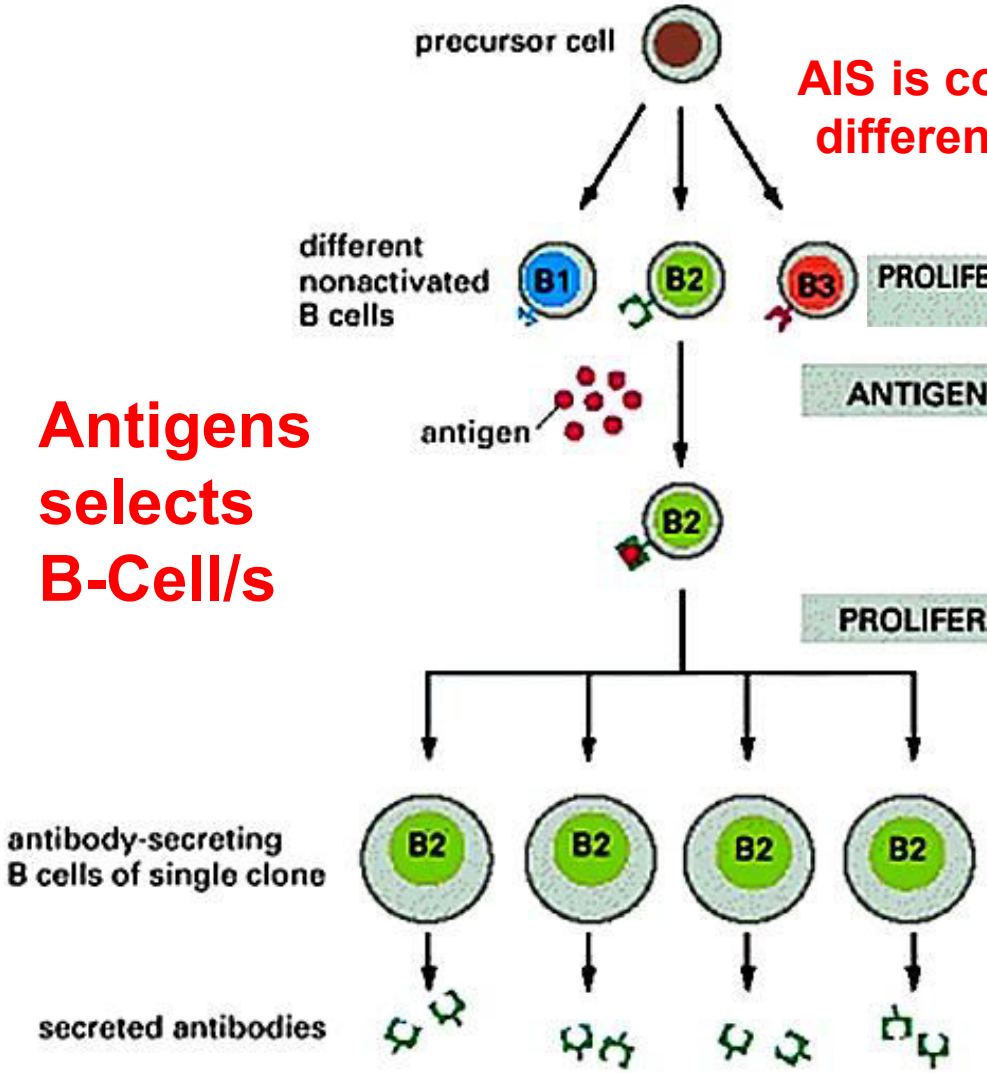


AIS can respond to millions of different foreign antigens in a highly specific way.

Animals randomly generate a vast diversity of lymphocytes

Lymphocytes that react against foreign antigens encountered by animal are selected for action

Clonal Slection

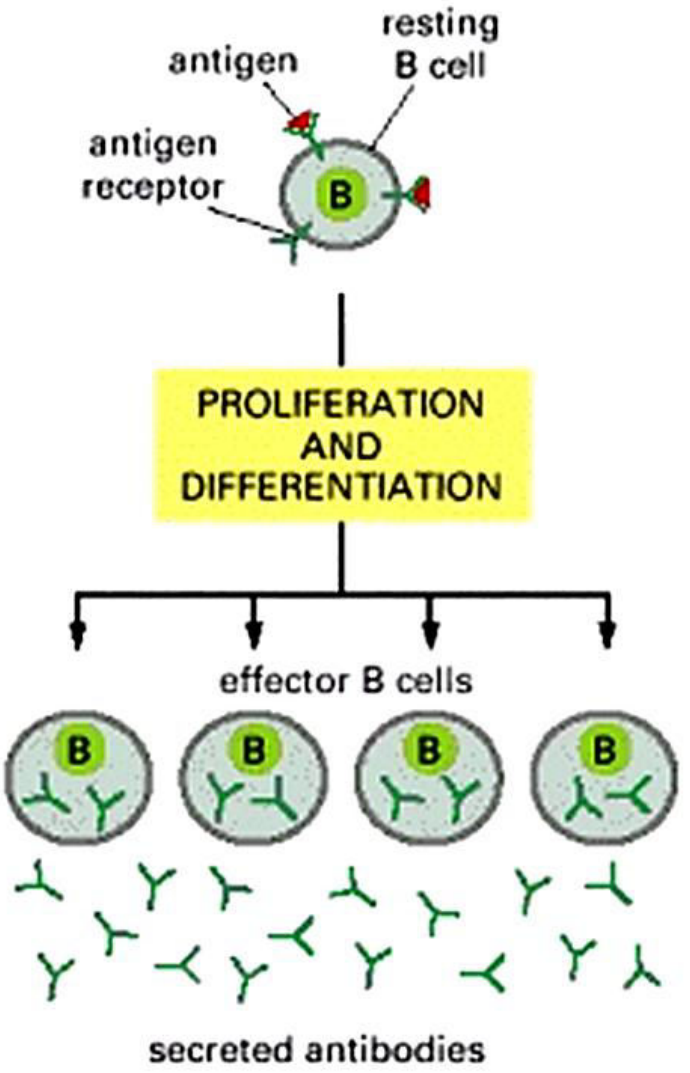


AIS is composed of millions of different lymphocytes clones

Antigens selects B-Cell/s

Ready-Made Not Made-To-Order

Mode of Action



Cell Biology

The Antibodies

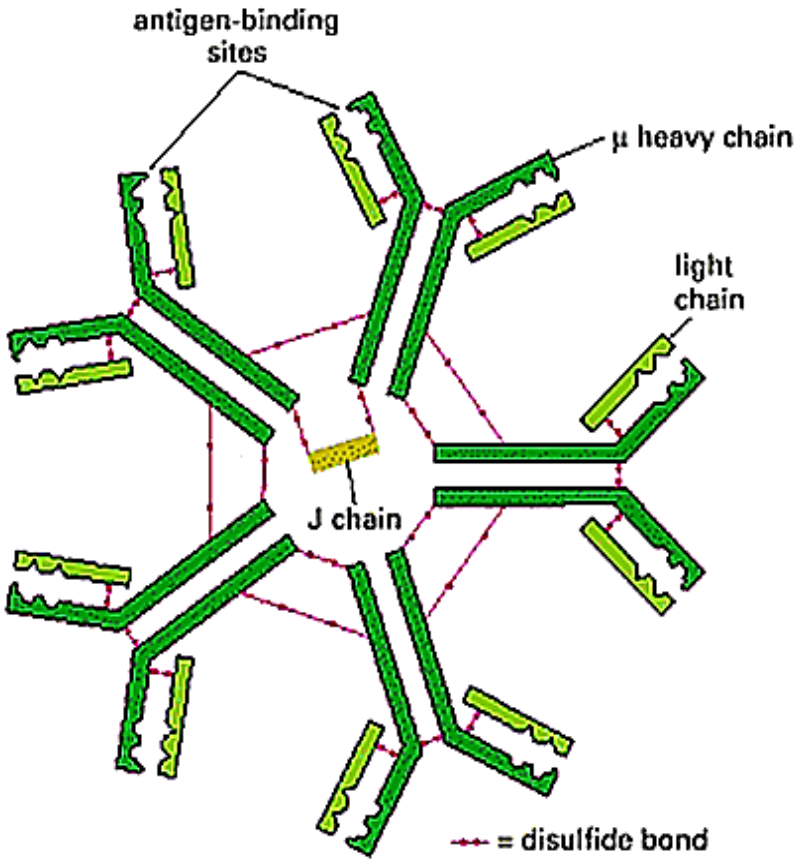
IgM

5 classes of antibodies, IgA, IgD, IgE, IgG, & IgM, class of heavy chain—α, δ, ε, γ, & μ, respectively.

Four human IgG subclasses (IgG1, IgG2, IgG3, and IgG4), having γ1, γ2, γ3, andγ4 heavy chains, respectively.

The various heavy chains give a distinctive conformation to the hinge and tail regions of antibodies, so that each class (and subclass) has characteristic properties of its own.

IgM

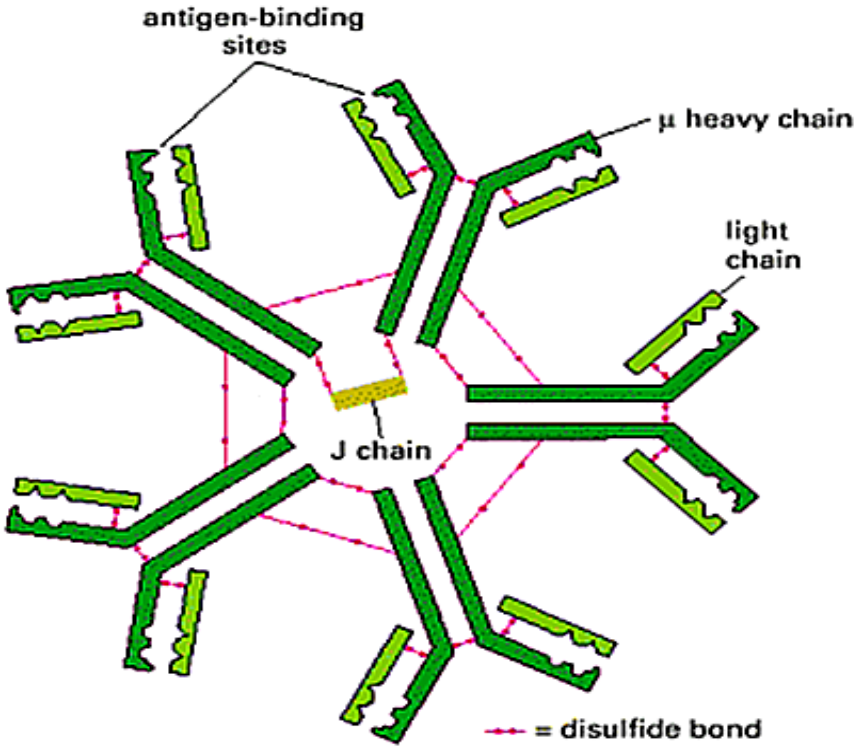


IgM first class of antibody to appear on the surface of a developing B cell.

The major class secreted in the early stages of a primary antibody response.

Secreted form, IgM is a pentamer, J chain is required for pentamer formation.

IgM



The binding of an antigen to a single secreted pentameric IgM molecule can activate the complement system, wich can either mark the pathogen for phagocytosis or kill it directly.

IgG

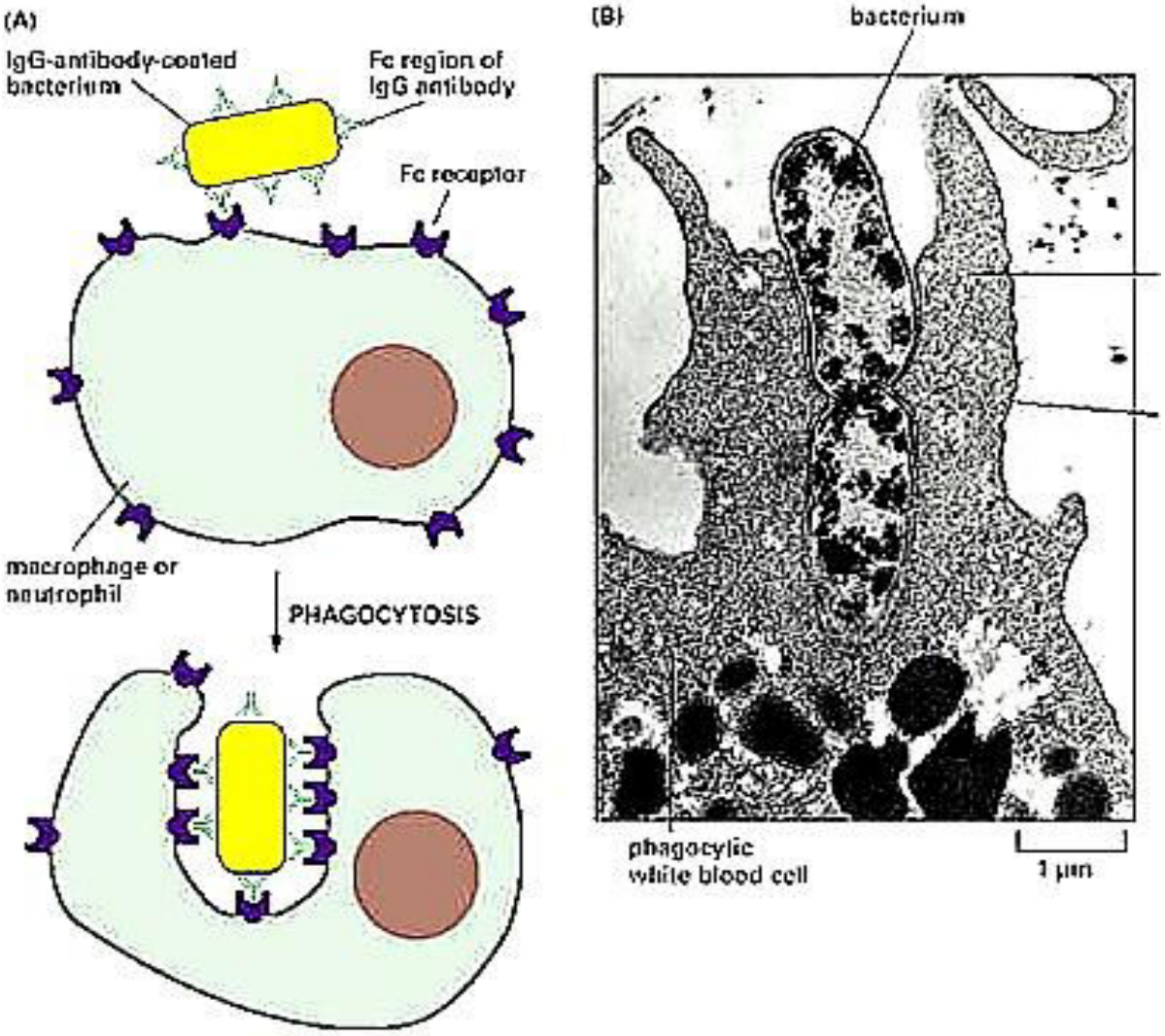
IgG is a monomer

Phagocytic cells bind, ingest, and destroy infecting microorganisms that have become coated with the IgG antibodies produced in response to the infection

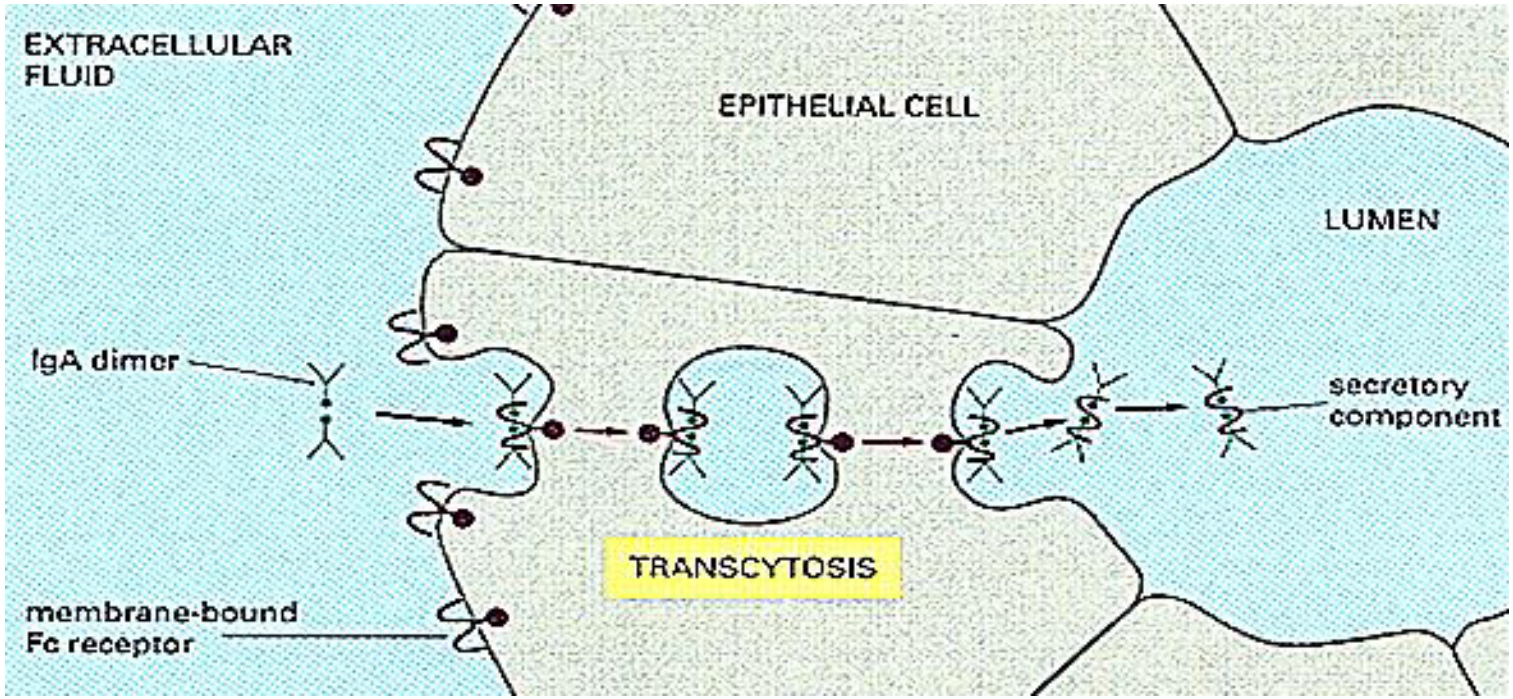
IgG only Ab that can pass from mother to fetus

placenta cells that are in contact with mothers blood have Fc receptors that are use to grab IgG and pass it to the fetus.

IgM


(A)
IgG-antibody-coated bacterium
Fc region of IgG antibody
Fc receptor
macrophage or neutrophil
PHAGOCYTOSIS
(B)
bacterium
phagocytic white blood cell
1 μm

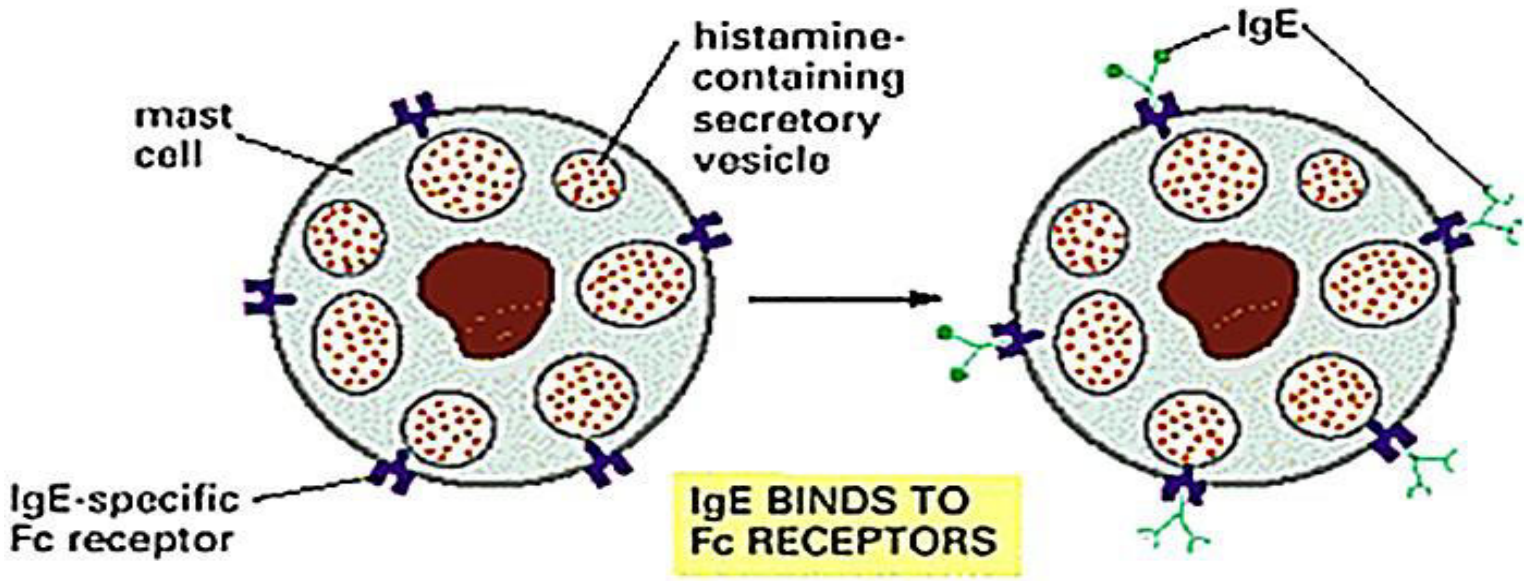
IgA



IgA is transported through secretary epithelial cells from the extracellular fluid into the secreted fluid by another type of Fc receptor that is unique to secretory epithelia.

This Fc receptor can also transport IgM into secretions less efficiently

IgE

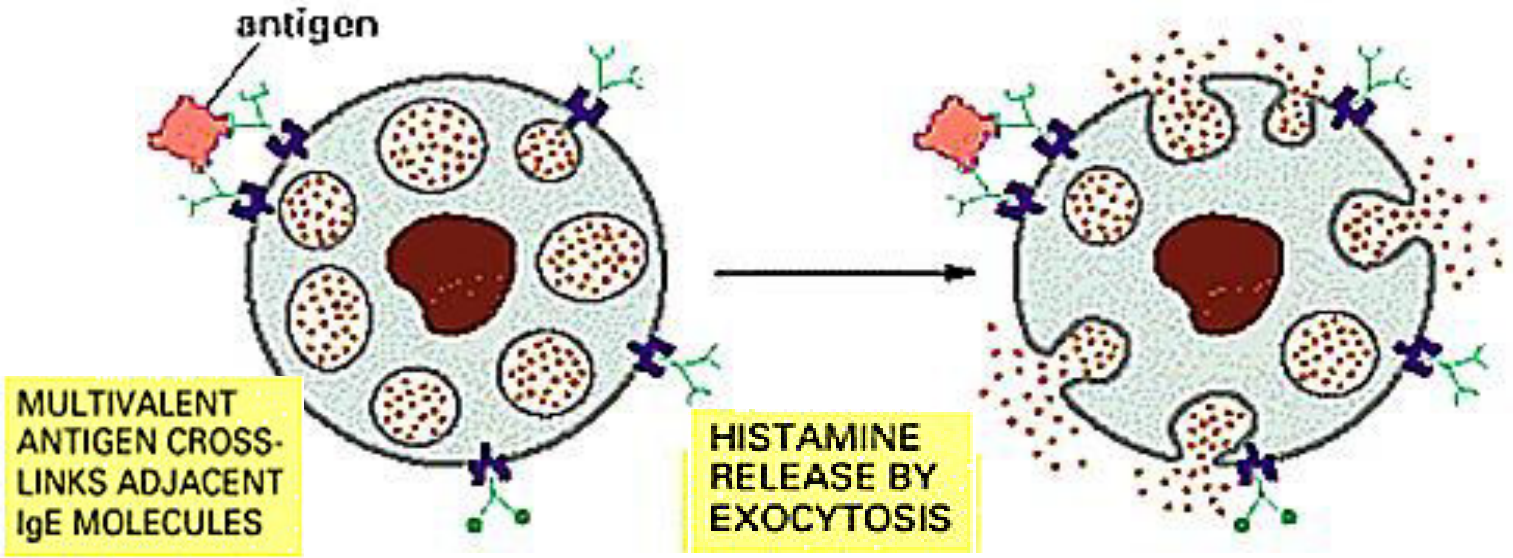


Tail region of IgE molecules four-chain monomers, binds (high affinity) another class of Fc receptors.

These receptors are located on the surface of mast cells in tissues and of basophils in the blood.

The IgE molecules bound to them function as passively acquired receptors for antigen.

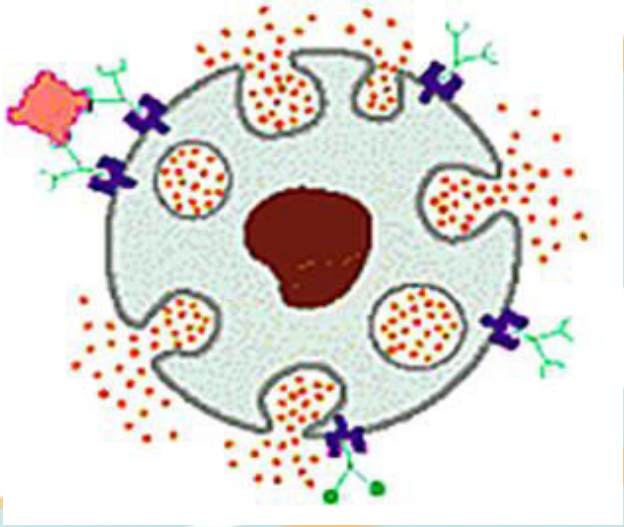
IgE



Antigen binding triggers the mast cell or basophil to secrete a variety of cytokines and biologically active amines, especially histamine.

These molecules cause blood vessels to dilate & become leaky, which helps WBCs, antibodies, complement sys to enter sites of infection.

IgE



These molecules cause allergic reactions as hay fever, asthma, and hives.

END

Cell Biology

Immunological Memory

Immunological Memory

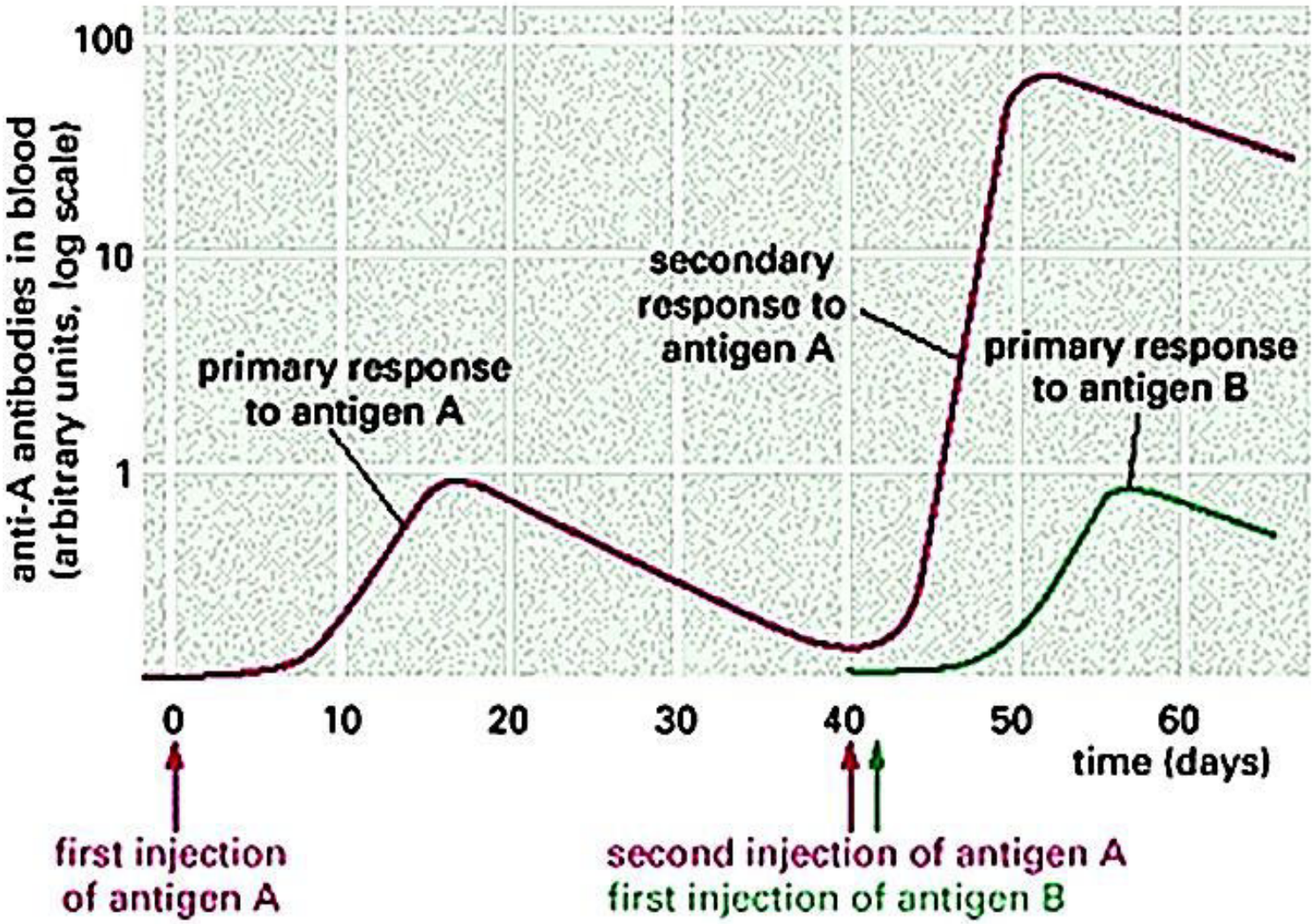
The adaptive immune system, like the nervous system, can remember prior experiences.

First exposure to an antigen A: Immune response appears after several days, rises rapidly and exponentially, and then, more gradually, declines.

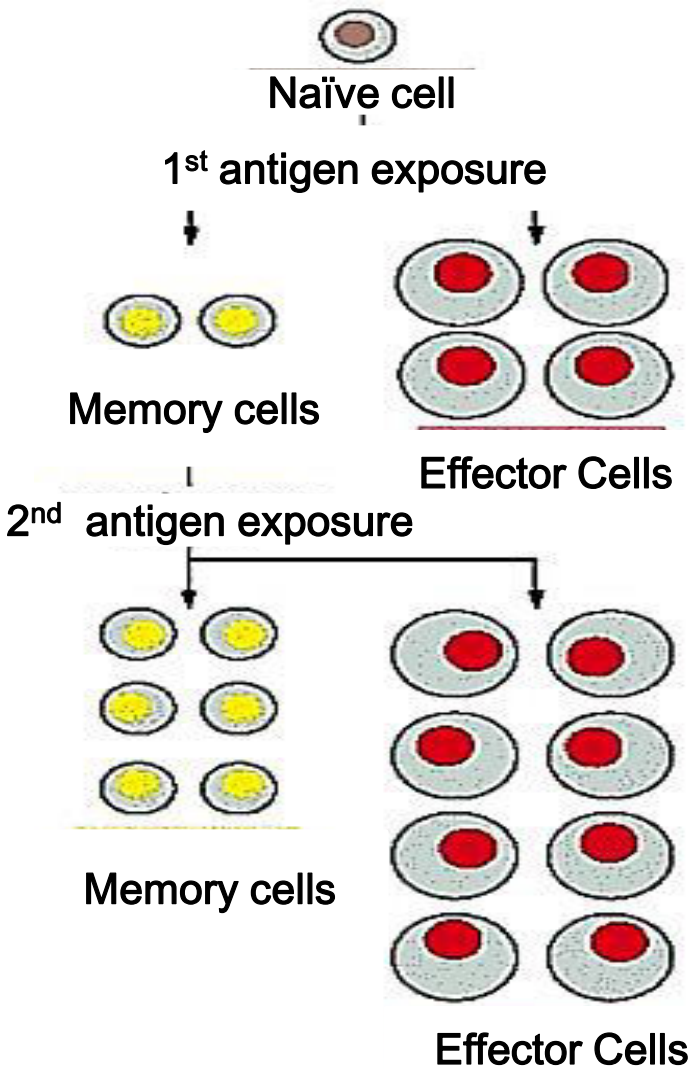
Secondary immune response: Lag period is shorter, and the response is greater indicating animal remembers its first exposure to antigen A.

If animal is given a different antigen B instead of second injection of antigen A the response is typical primary not secondary, immune response

Immunological Memory



Immunological Memory



Cellular Basis of Immunological Memory

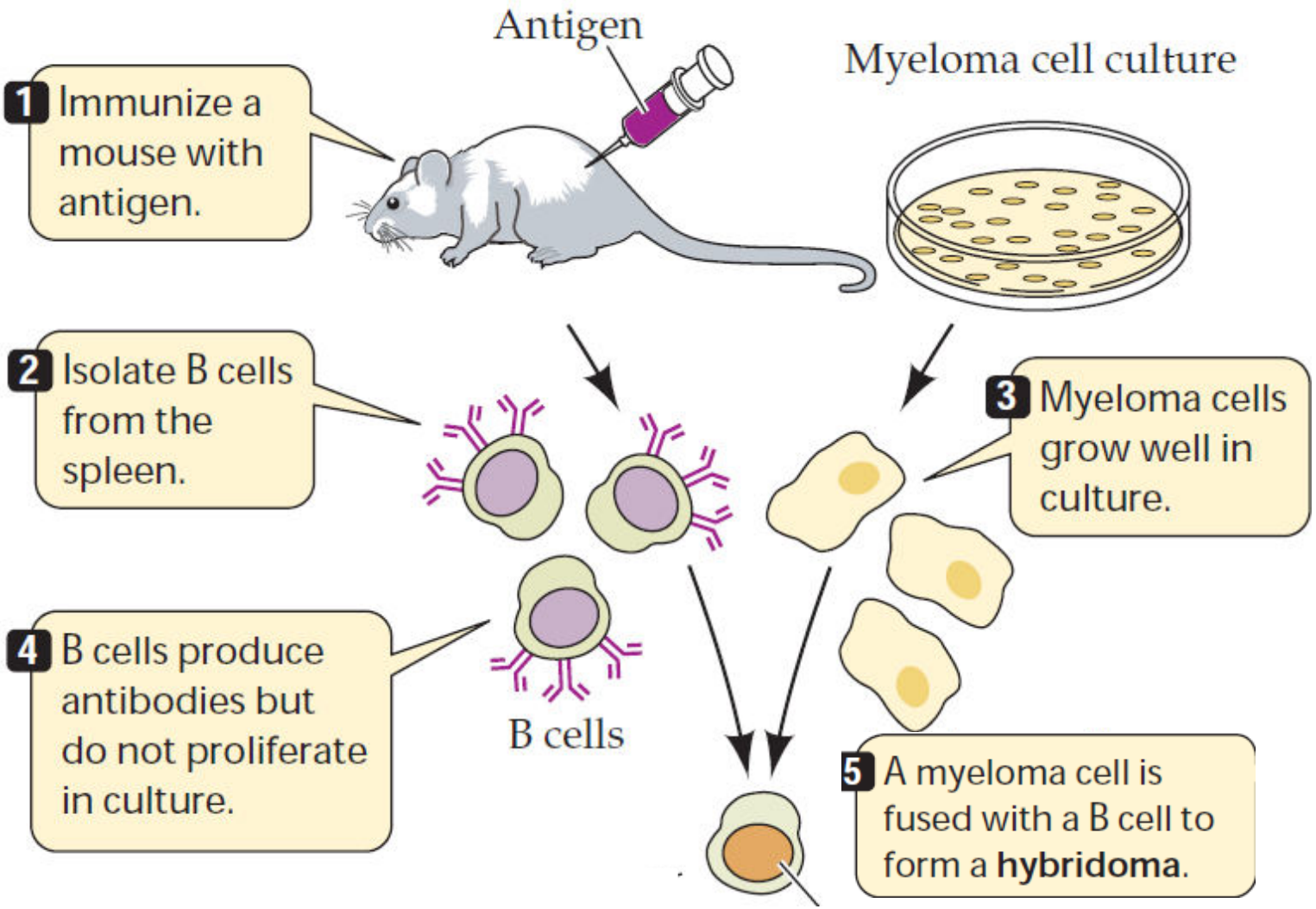
When naïve lymphocytes are stimulated by their specific antigen, they proliferate and differentiate. Most become effector cells which function & then die, while others become long-lived memory cells.

Immunological Memory

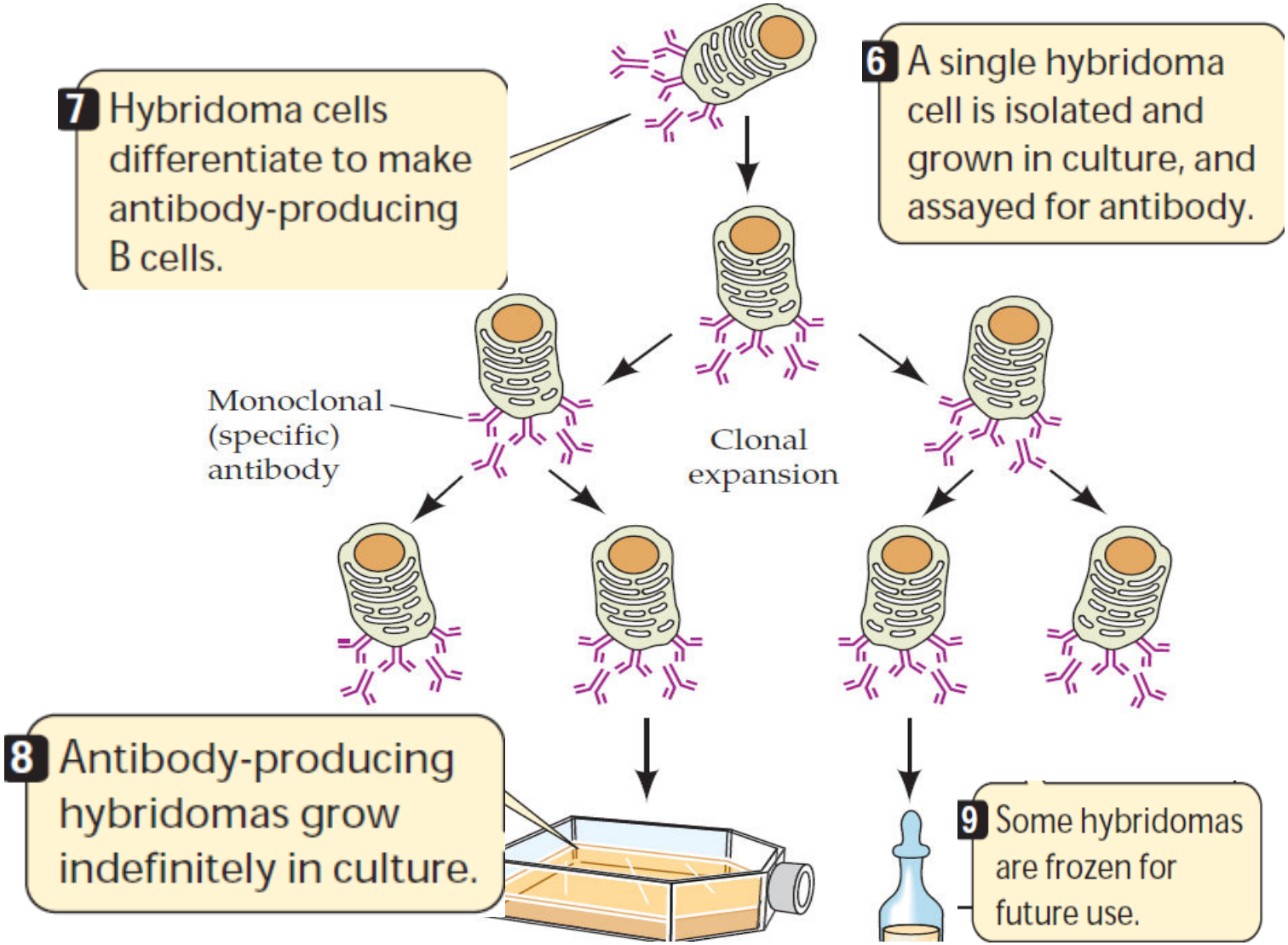
Vaccination

INFECTIOUS AGENT	DISEASE	VACCINATED POPULATION
Bacteria		
Bacillus anthracis	Anthrax	Exposed in biological w
Bordetella pertussis	Whooping cough	Children and adults
Clostridium tetani	Tetanus	Children
Corynebacterium diphtheriae	Diphtheria	Children
Haemophilus influenzae	Meningitis	Children
Mycobacterium tuberculosis	Tuberculosis	All people
Salmonella typhi	Typhoid fever	Areas exposed to agent
Streptococcus pneumoniae	Pneumonia	Elderly

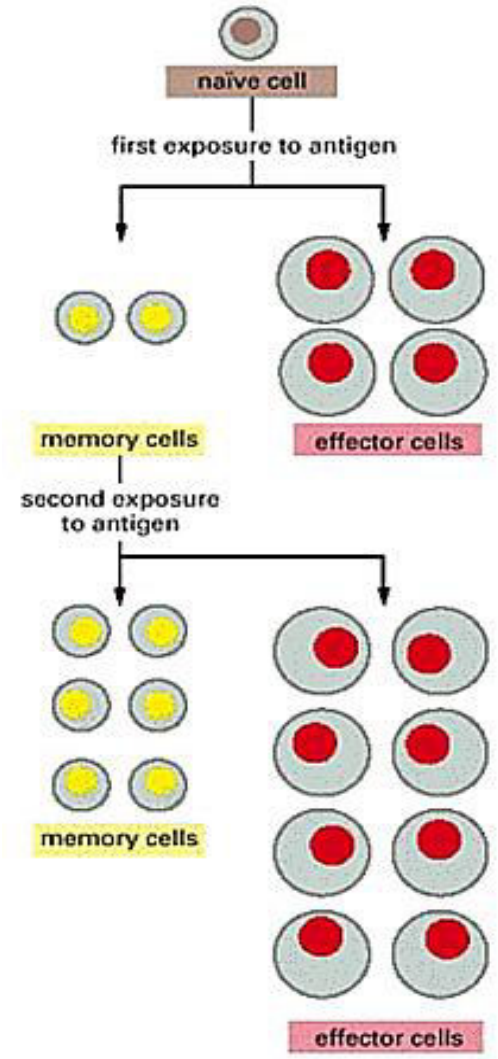
Hybridomas


Antigen
Myeloma cell culture
1 Immunize a mouse with antigen.
2 Isolate B cells from the spleen.
3 Myeloma cells grow well in culture.
4 B cells produce antibodies but do not proliferate in culture.
B cells
5 A myeloma cell is fused with a B cell to form a hybridoma.

Hybridomas


7 Hybridoma cells differentiate to make antibody-producing B cells.
6 A single hybridoma cell is isolated and grown in culture, and assayed for antibody.
Monoclonal (specific) antibody
Clonal expansion
8 Antibody-producing hybridomas grow indefinitely in culture.
9 Some hybridomas are frozen for future use.

Immunological Memory



END

Cell Biology

Antibody Diversity

Antibody Diversity

The Dilemma

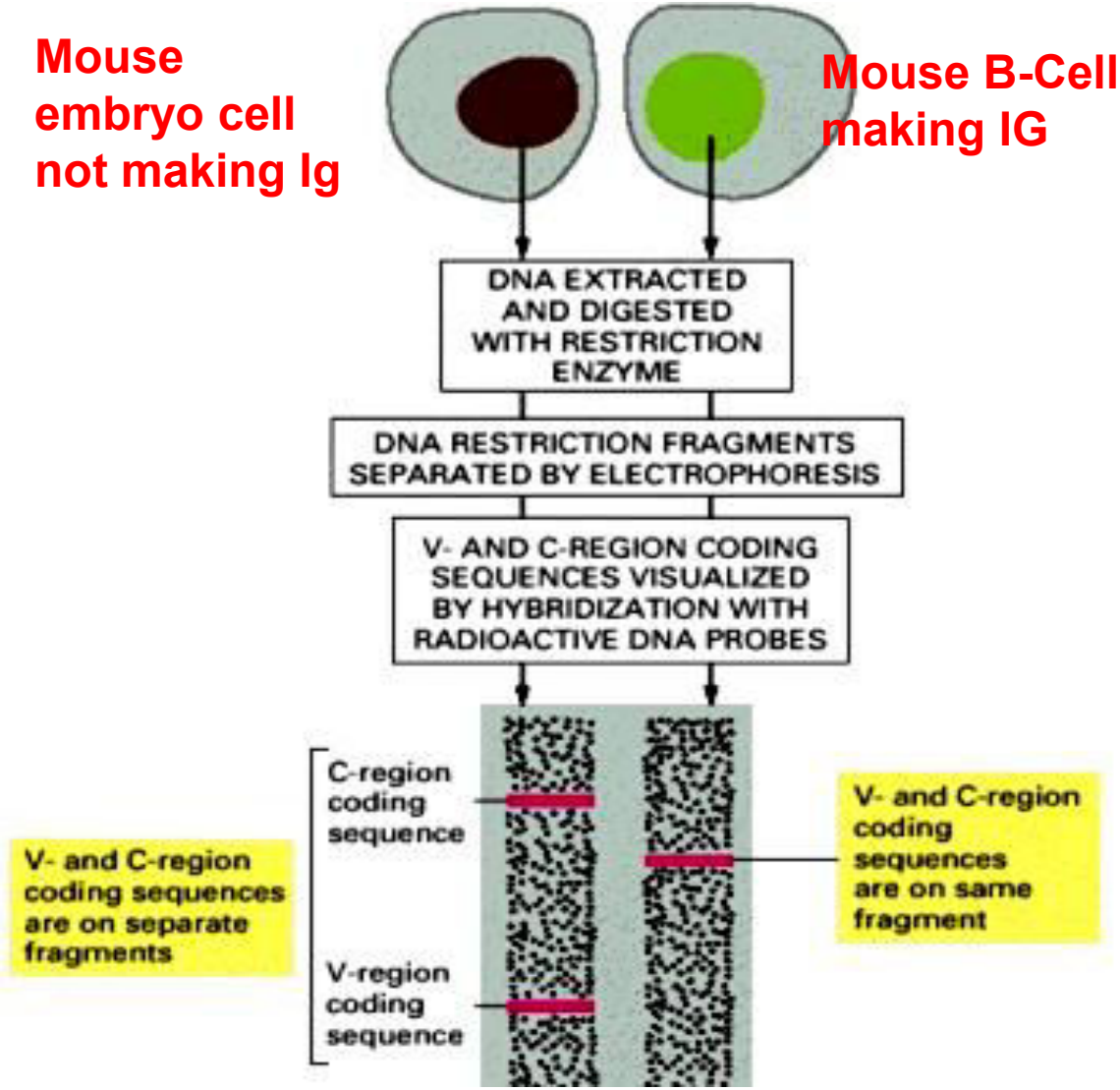
Even in the absence of antigen humans can make more than 10^{12} different antibody molecules.

Estimated number of human genes is 30,000

Mammalian immune system has evolved unique genetic mechanisms that enables it to generate an almost unlimited number of different light & heavy chains in a remarkably economical way, by joining separate gene segments before TRANSCRIPTION.

Birds and fish sheep and rabbits

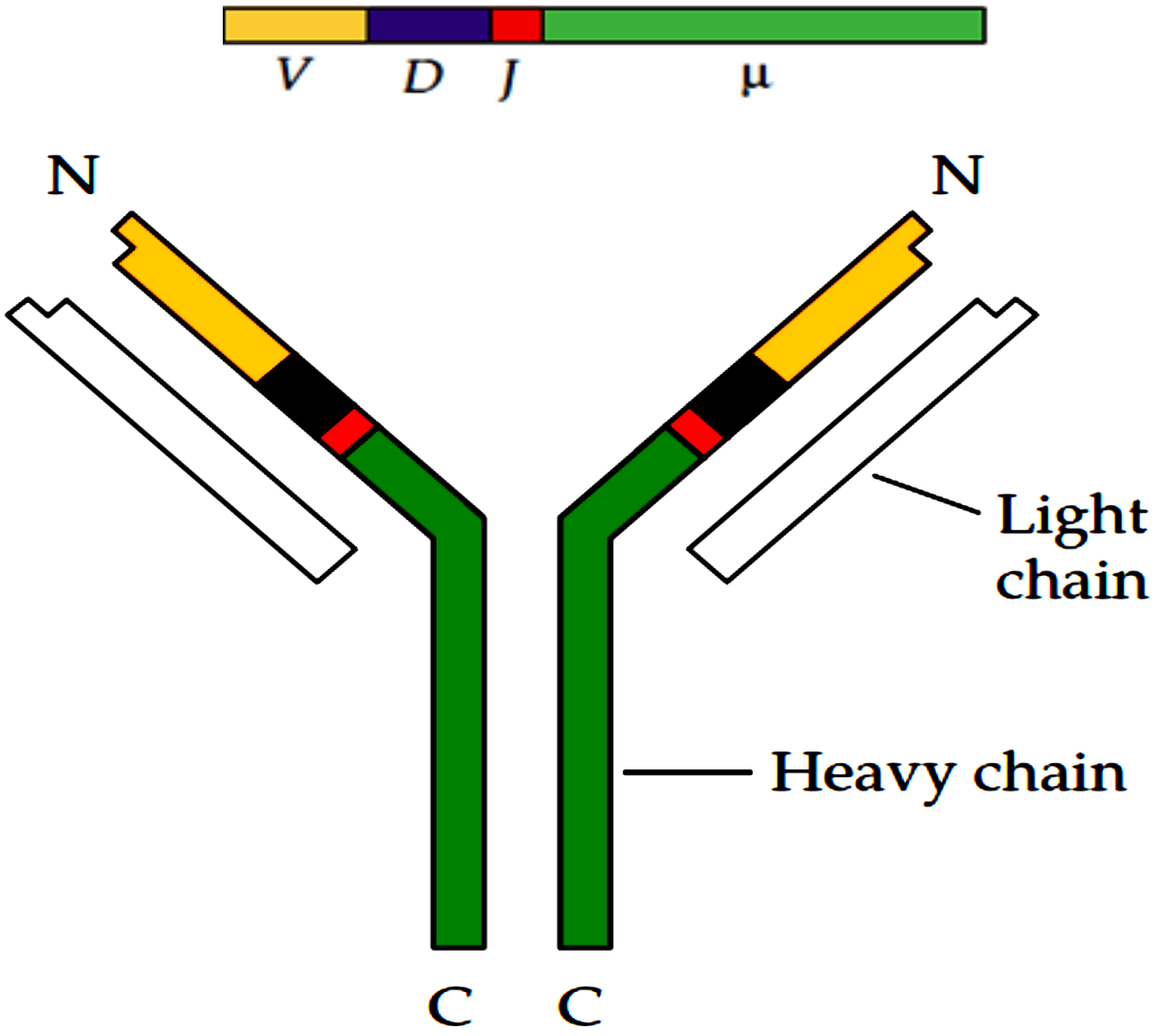
Antibody Genes Assembly



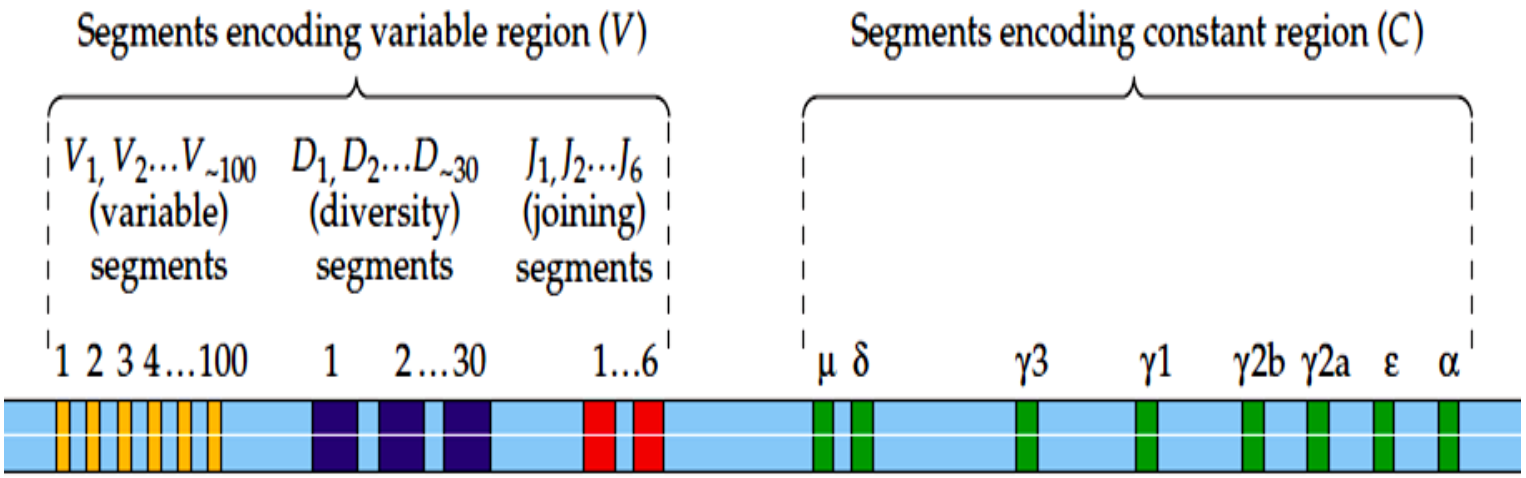
The First Evidence

1970

Antibody Genes Assembly



Antibody Genes Assembly



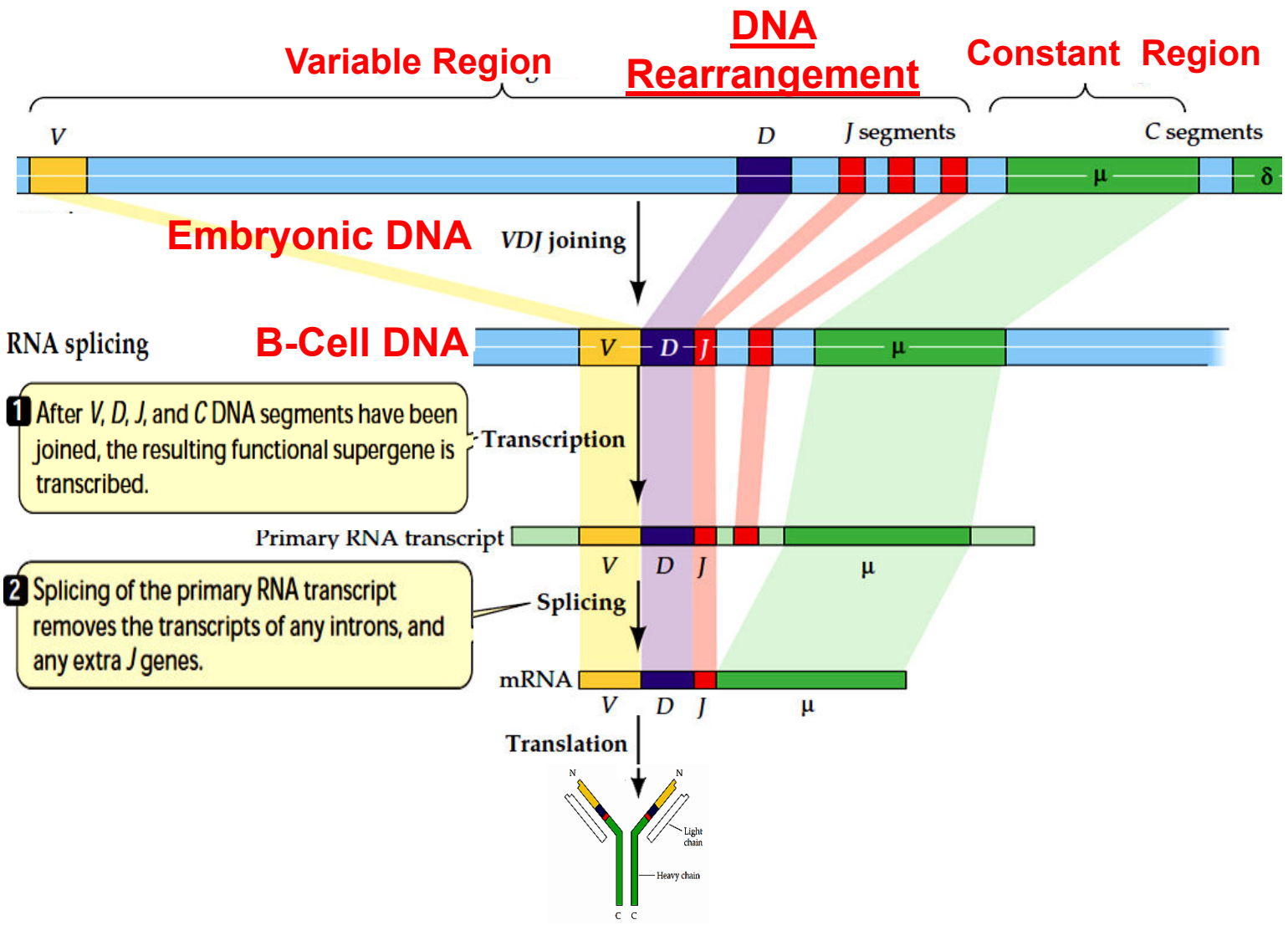
The variable region for the heavy chain of a particular antibody is encoded by one V segment, one D segment, and one J segment. Each of these segments is taken from a pool of like segments.

The constant region is selected from another pool of segments.

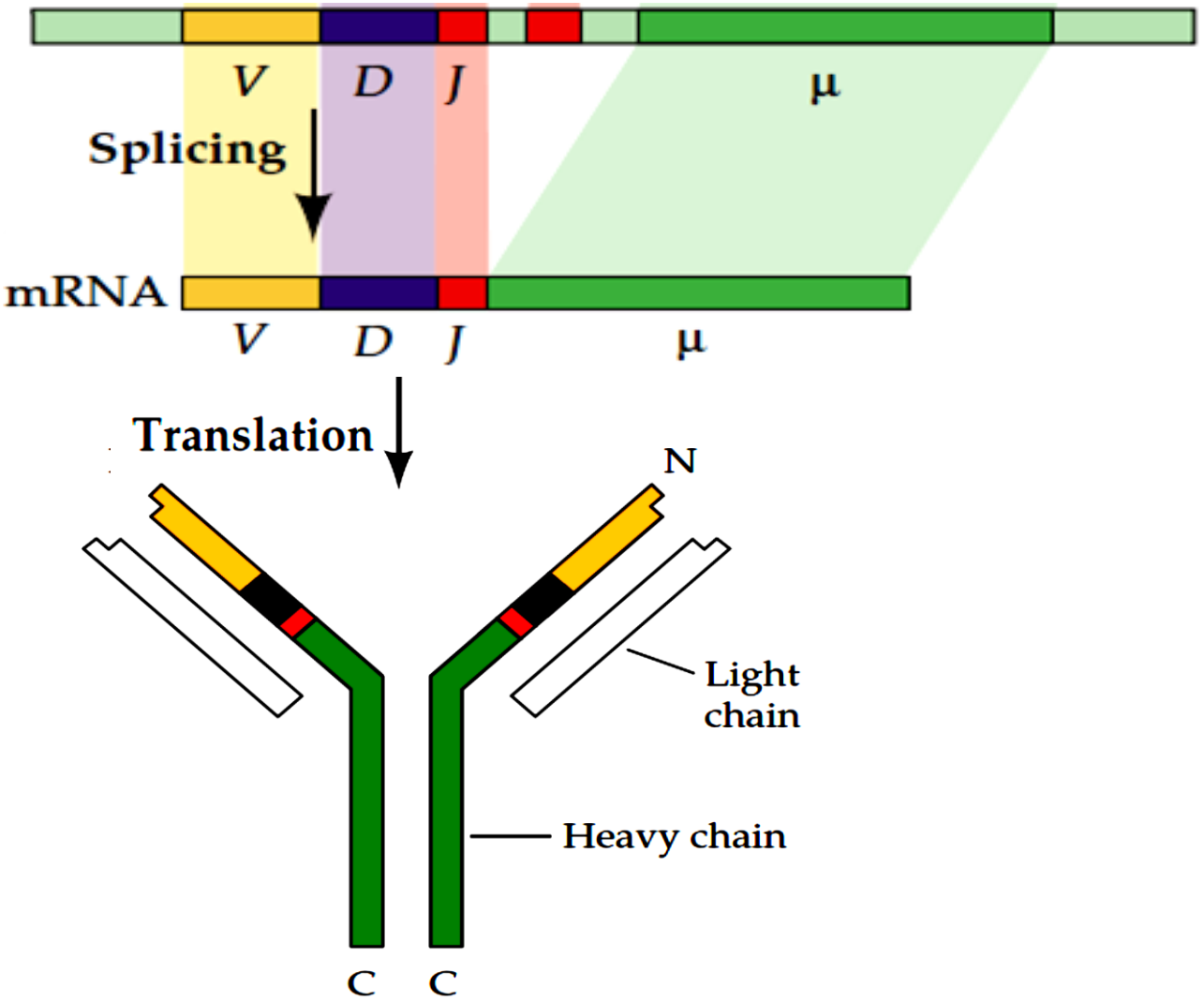
The number of possible combinations to make an immunoglobulin heavy chain from this set of genes is:

(100 *V*)(30 *D*)(6 *J*)(8 *C*) = 144,000

Antibody Genes Assembly



Antibody Genes Assembly



Antibody Diversity

Additional Mechanisms

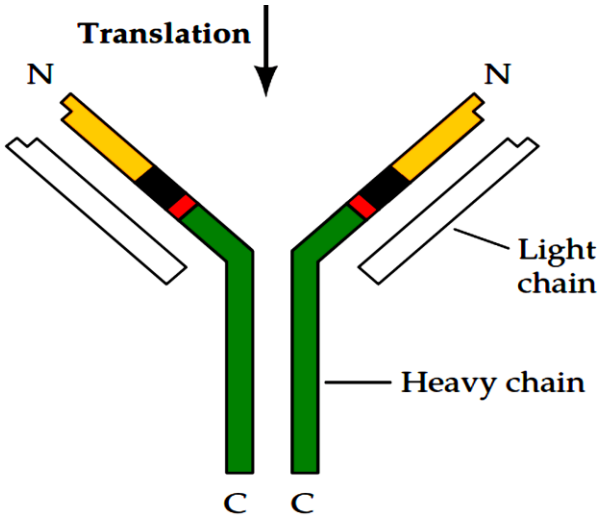
V, J, and C regions are rearranged imprecisely, errors at junctions create new codons resulting in amino acid changes.

After the DNA sequences are cut out and before they are joined, terminal transferase, adds some nucleotides to the free ends of the DNAs.

Immunoglobulin genes have relatively high mutation rate which creates new alleles & adds to antibody diversity.

Birds and fish sheep and rabbits

Immunological Memory



END

Cell Biology

Antibody Diversity

Antibody Diversity

The Dilemma

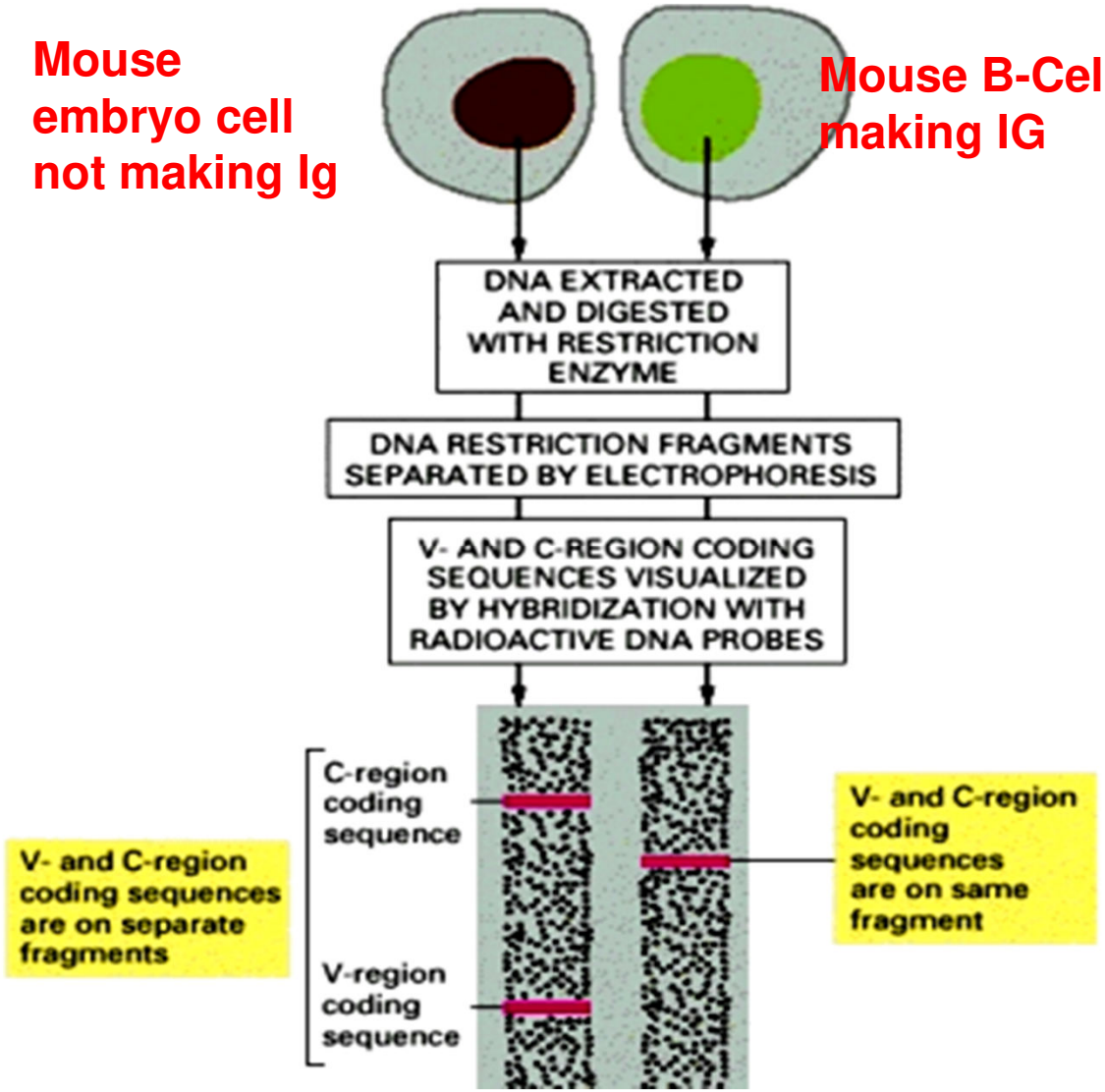
Even in the absence of antigen humans can make more than 10^{12} different antibody molecules.

Estimated number of human genes is 30,000

Mammalian immune system has evolved unique genetic mechanisms that enables it to generate an almost unlimited number of different light & heavy chains in a remarkably economical way, by joining separate gene segments before TRANSCRIPTION.

Birds and fish sheep and rabbits

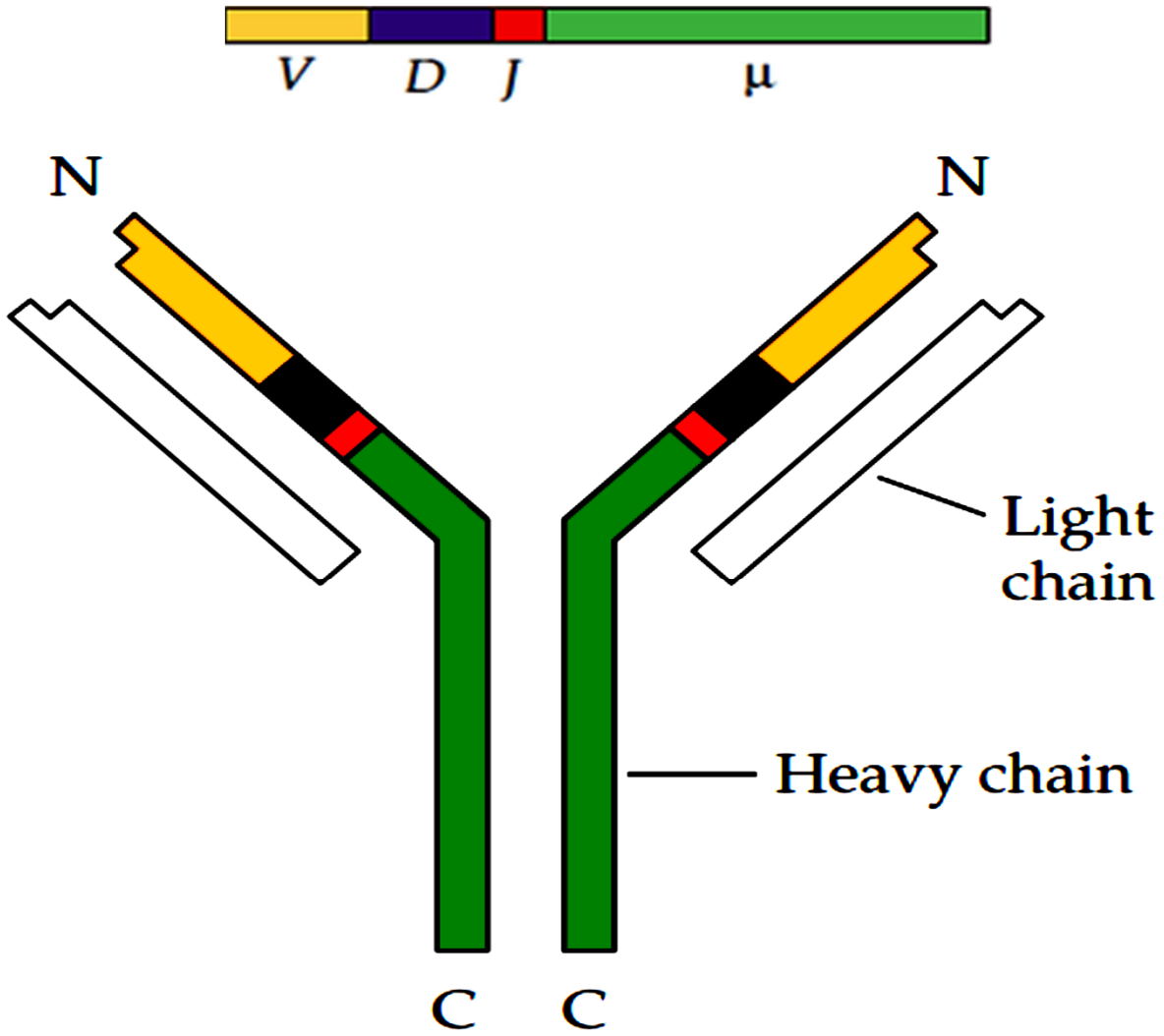
Antibody Genes Assembly



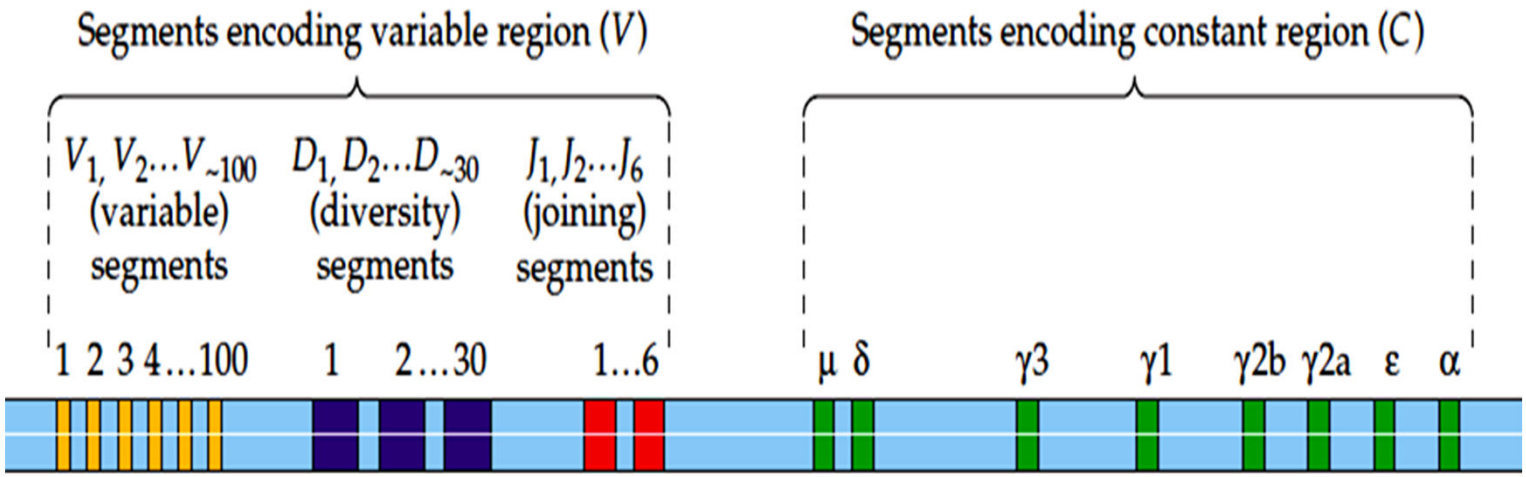
The First Evidence

1970

Antibody Genes Assembly



Antibody Genes Assembly

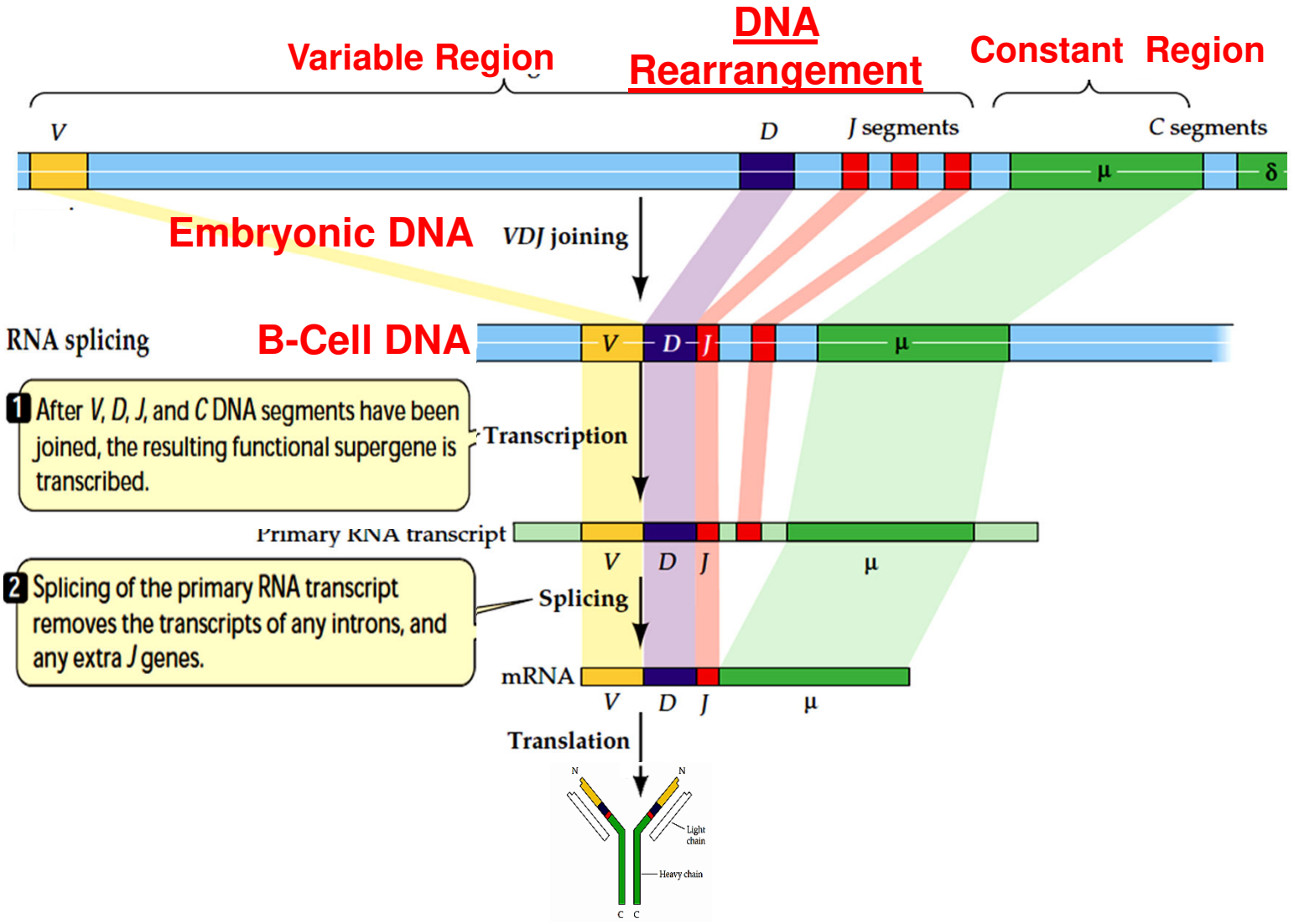


The variable region for the heavy chain of a particular antibody is encoded by one V segment, one D segment, and one J segment. Each of these segments is taken from a pool of like segments.

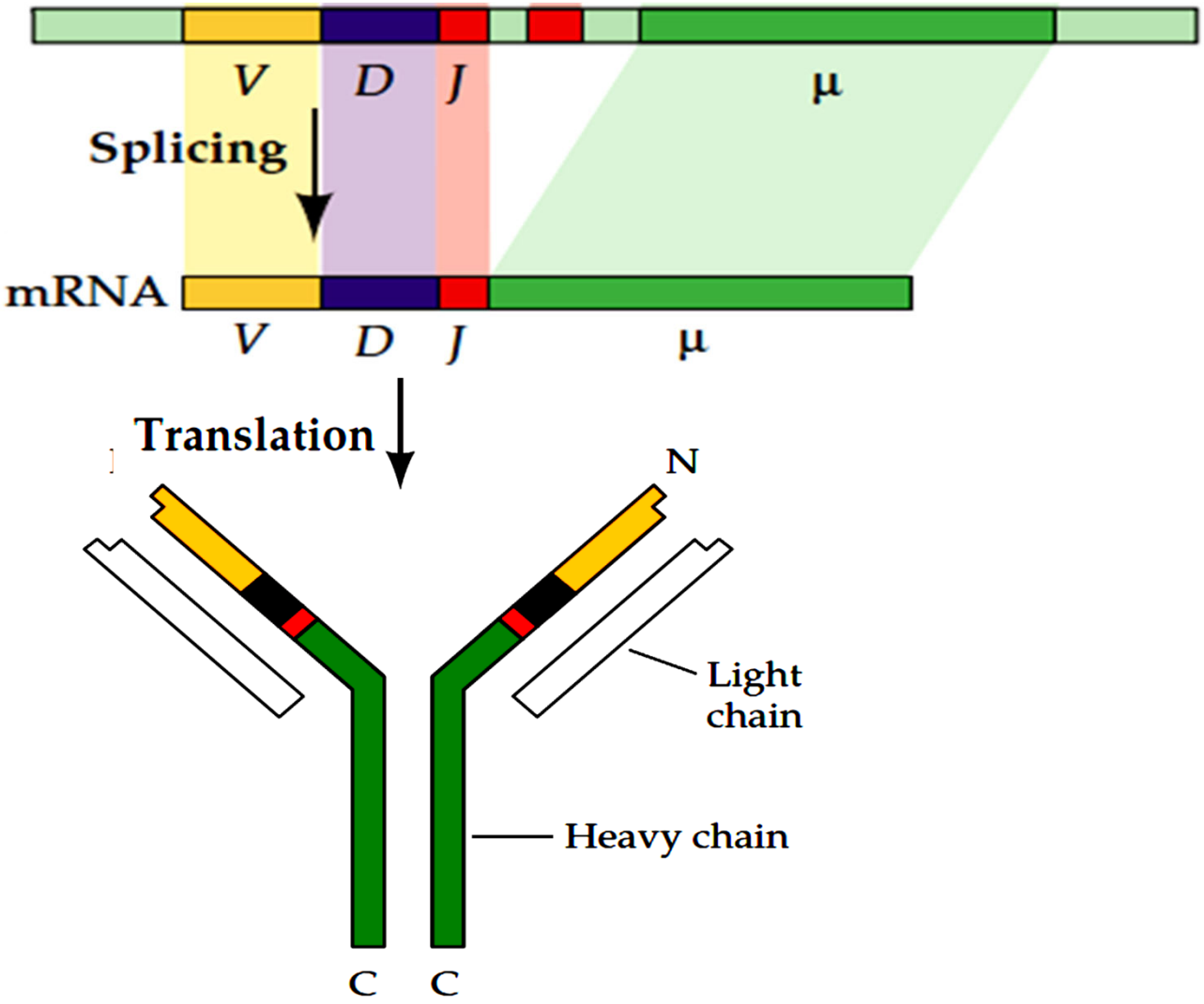
The constant region is selected from another pool of segments.

The number of possible combinations to make an immunoglobulin heavy chain from this set of genes is:
(100 *V*)(30 *D*)(6 *J*)(8 *C*) = 144,000

Antibody Genes Assembly



Antibody Genes Assembly



Antibody Diversity

Additional Mechanisms

V, J, and C regions are rearranged imprecisely, errors at junctions create new codons resulting in amino acid changes.

After the DNA sequences are cut out and before they are joined, terminal transferase, adds some nucleotides to the free ends of the DNAs.

Immunoglobulin genes have relatively high mutation rate which creates new alleles & adds to antibody diversity.

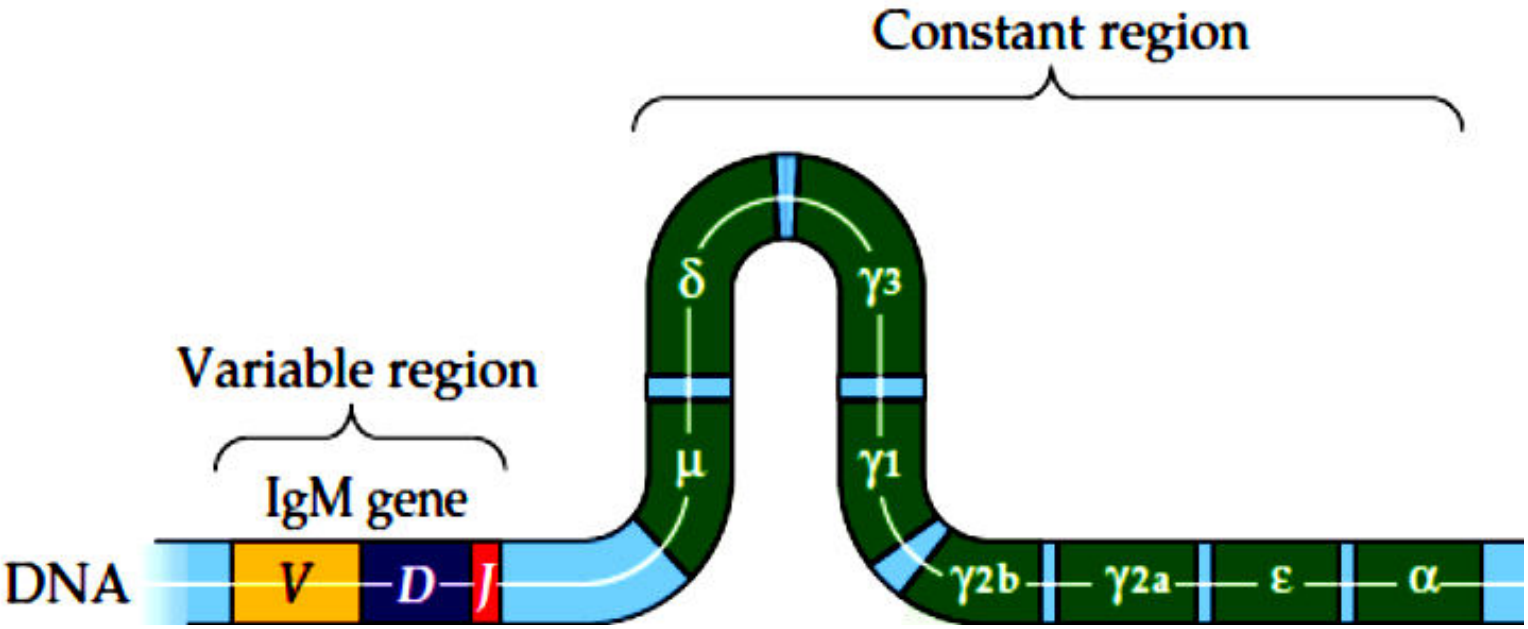
Birds and fish sheep and rabbits

Cell Biology

Class Switching

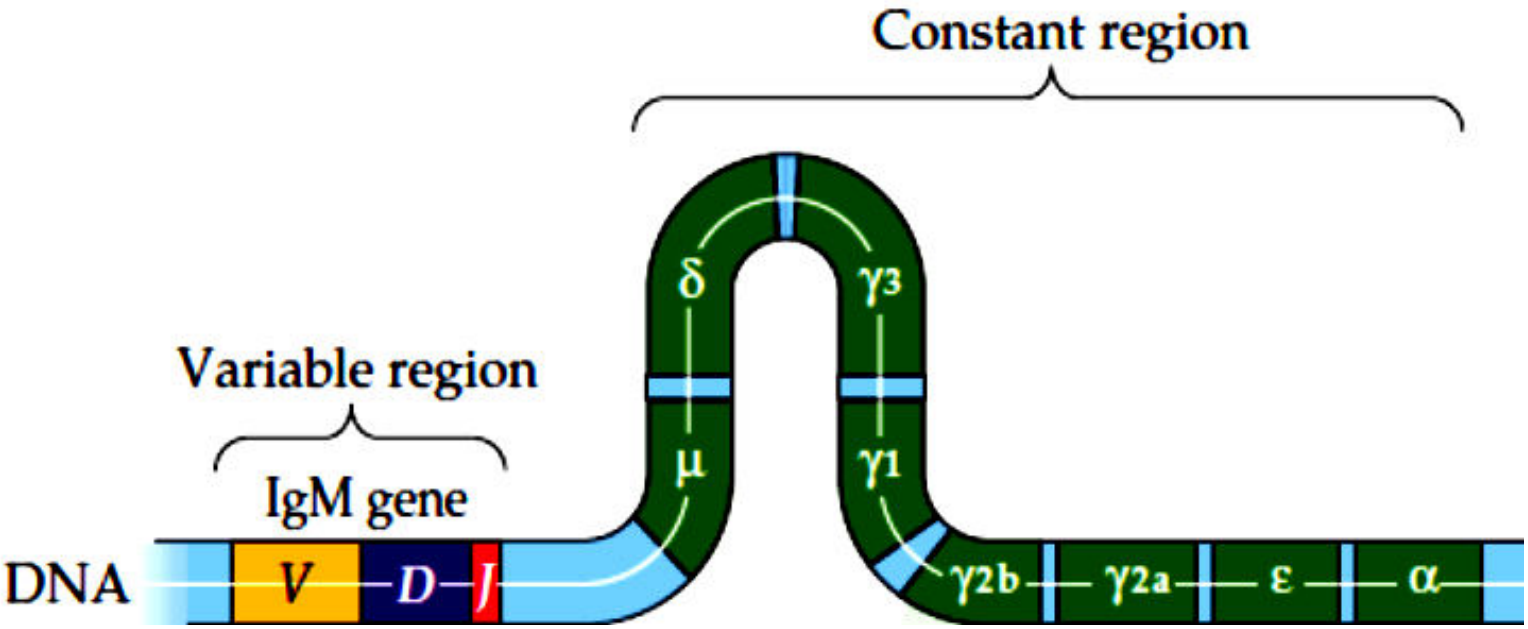
Class Switching

B-Cell initially produces IgM (antibody's heavy chain is encoded by the first constant region gene, the μ segment) molecules, which recognise specific antigenic determinant.

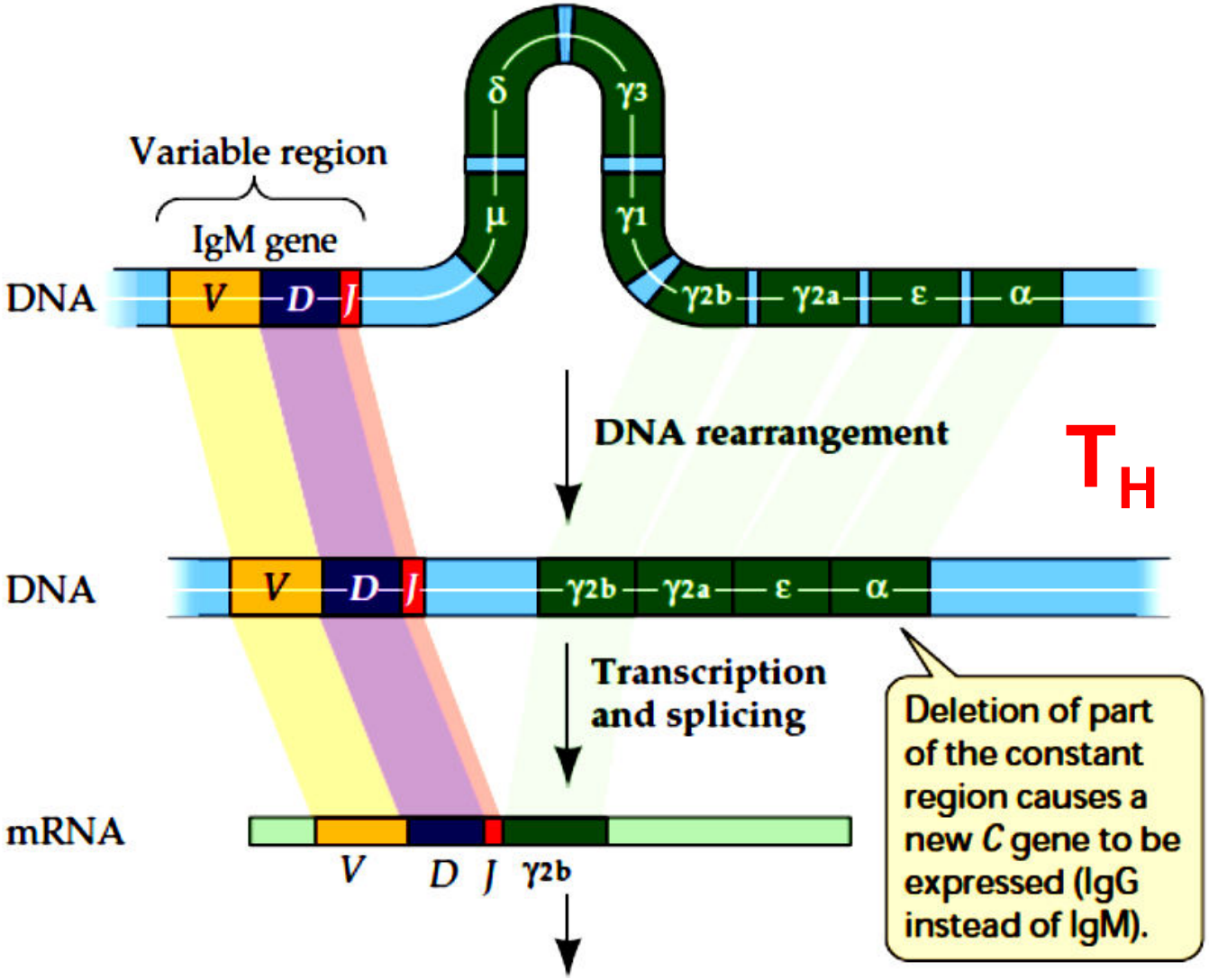


Class Switching

When B cell becomes a plasma cell another deletion occurs in the cell's DNA, which position heavy-chain variable region gene next to a different constant region (γ, ε, or α)

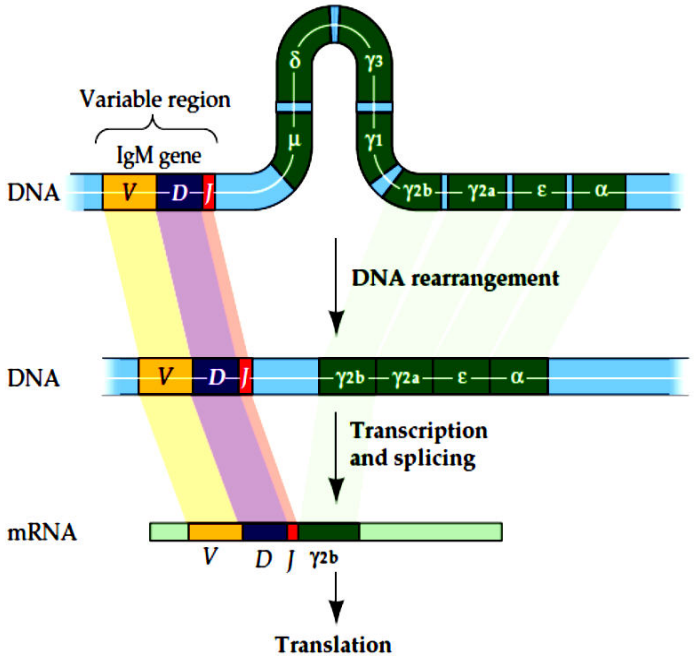


Class Switching



same variable regions of the light and heavy chains, and tantigen specificity

Class Switching



END

Cell Biology

T cells & TCR

Cell-mediated Immune Reactions

2 types : Cytotoxic T cells & Helper T cells.

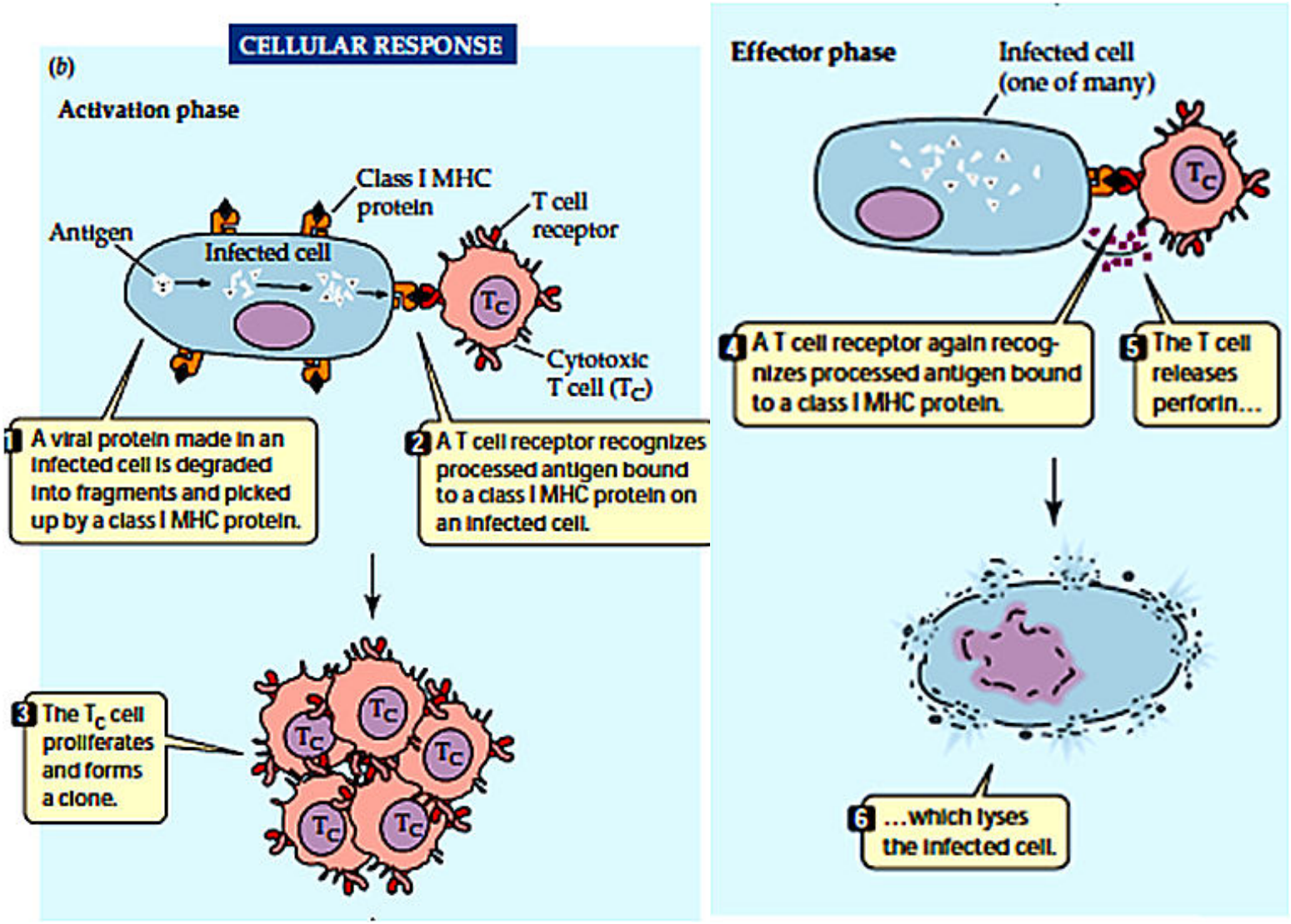
T cells eliminate intracellular pathogens.

T cells can only recognize antigen partly degraded fragments associated with MHC.

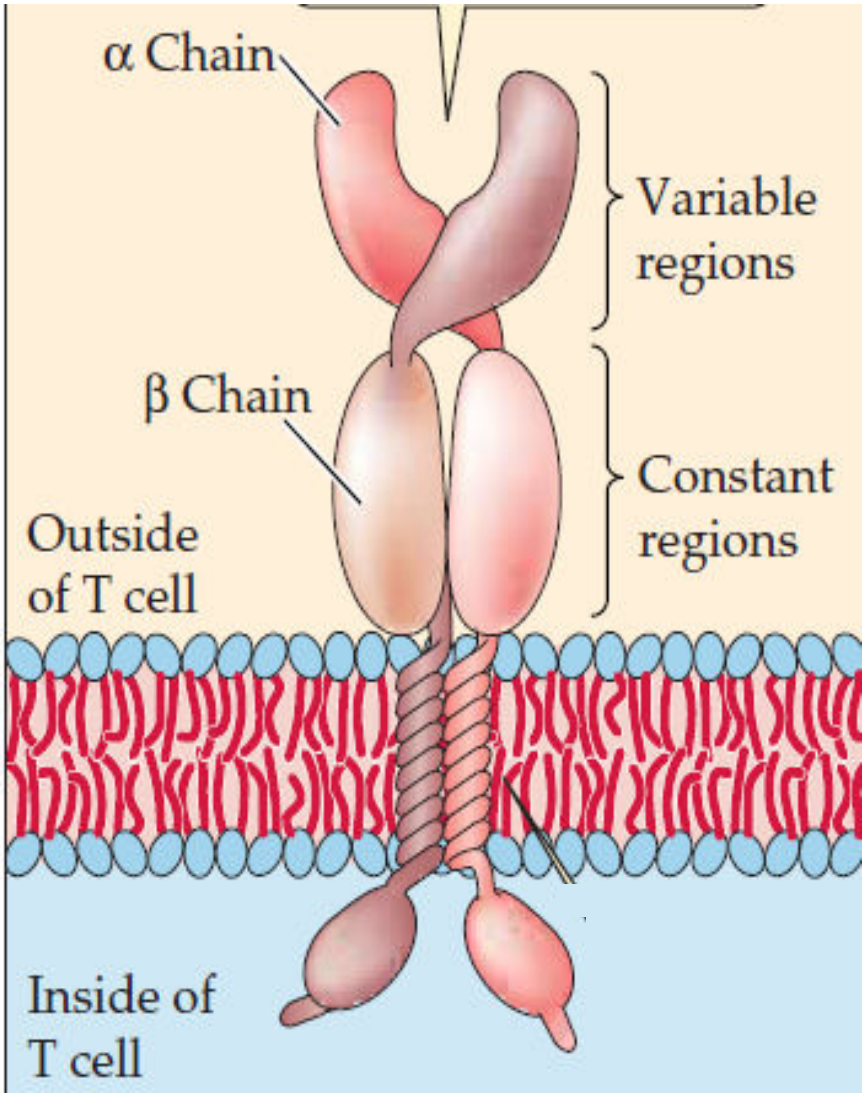
Effector cytotoxic T cells directly kill cells infected with a virus or some other intracellular pathogen.

Effector helper T cells stimulate the responses of other cells phagocitic macrophages, B cells, and cytotoxic T cells.

Cell-mediated Immune Reactions



TCR



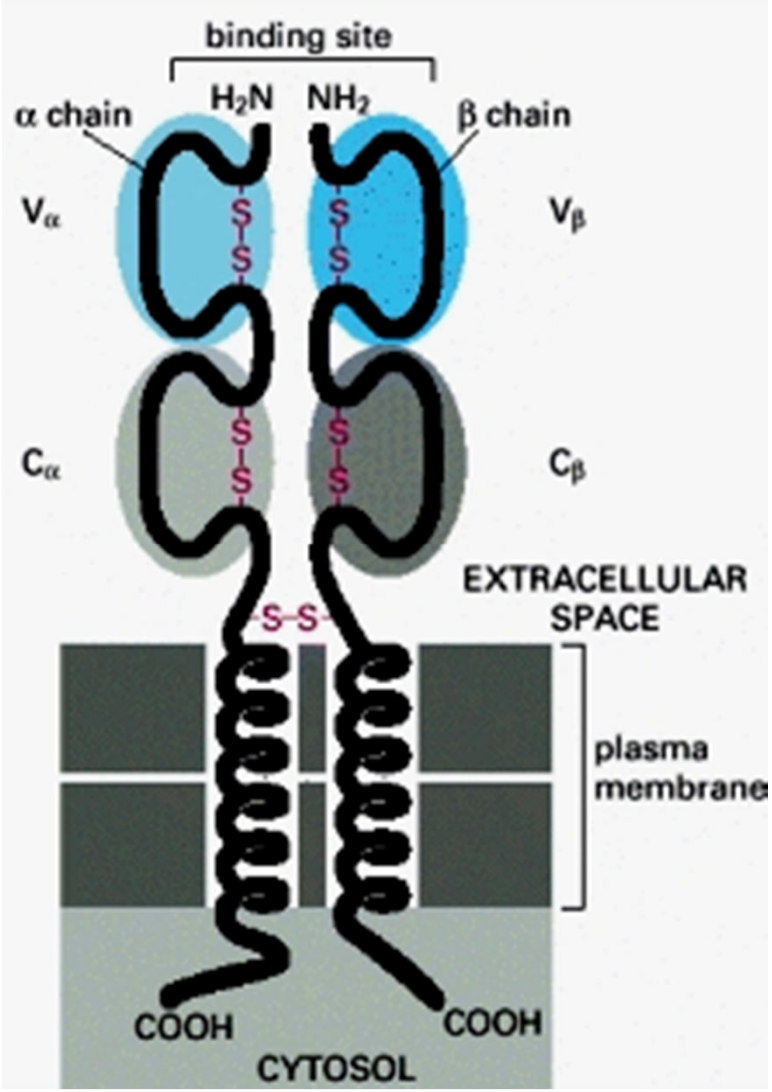
Heterodimer

Glycoproteins not immunoglobulins

Two polypeptide chains, each encoded separate genes, homologous to BCR genes

Source may be evolutionarily more ancient group of genes.

TCR



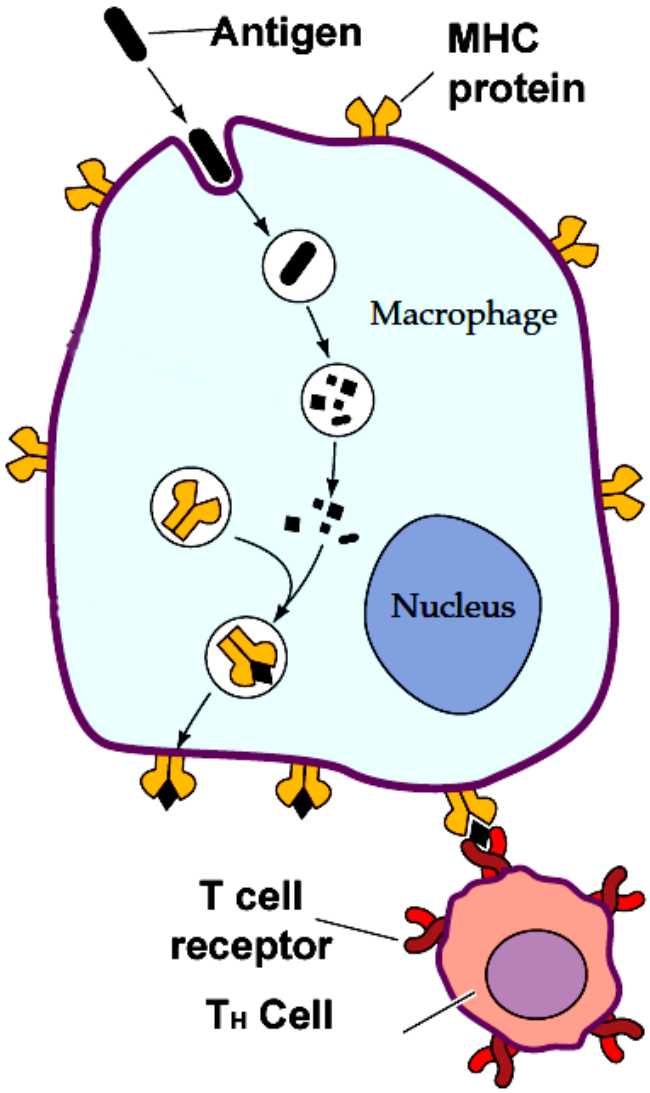
T cell receptor pools contain separate V, D, and J gene segments, which are brought together by site-specific recombination during T cell development in the thymus.

T cells recognize protein fragments of antigen that have been partly degraded inside the antigen-presenting cell.

Cell Biology

APCs Activate
T Cells

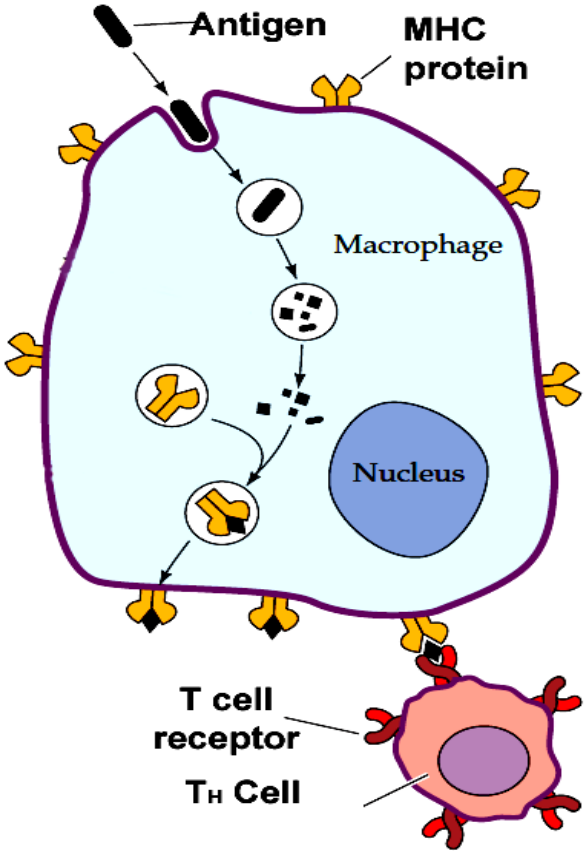
APCs Activate T Cells



T cells are initially activated in peripheral lymphoid organs by recognizing foreign antigen on the surface of an APC (dendritic cell, macrophage & B cell).

Antigen is peptide fragments that are generated by the degradation of foreign protein antigens inside the antigen-presenting cell.

APCs Activate T Cells



The recognition process depends on the presence of MHC proteins on the antigen-presenting cell, MHC bind these fragments, carry them to the cell surface, and present them.

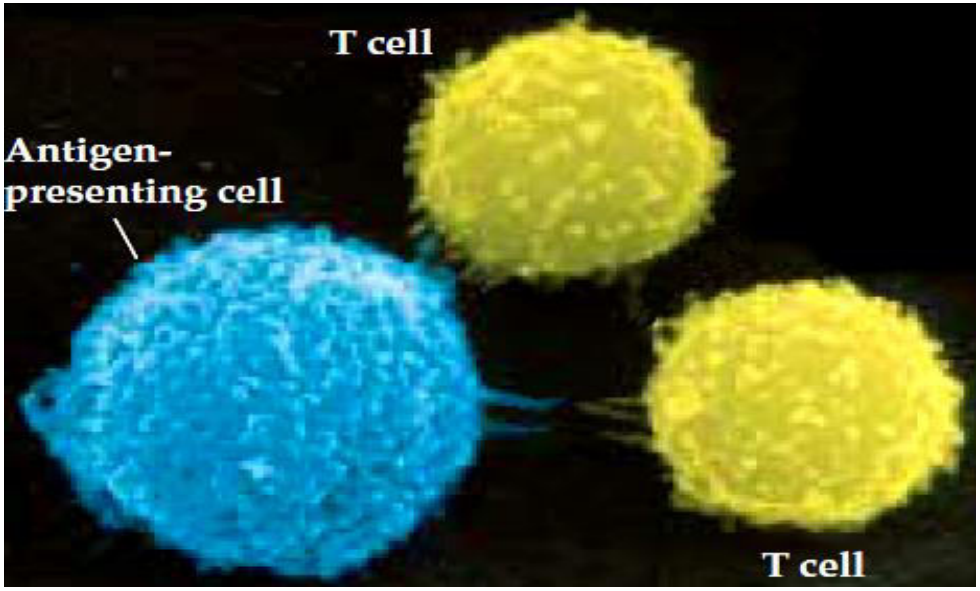
Activated, effector T cells then recognize the same peptide-MHC complex on the surface of the target cell they influence.

B cell, a cytotoxic T cell, or an infected macrophage in the case of a helper T cell, or a virus-infected cell in the case of a cytotoxic T cell.

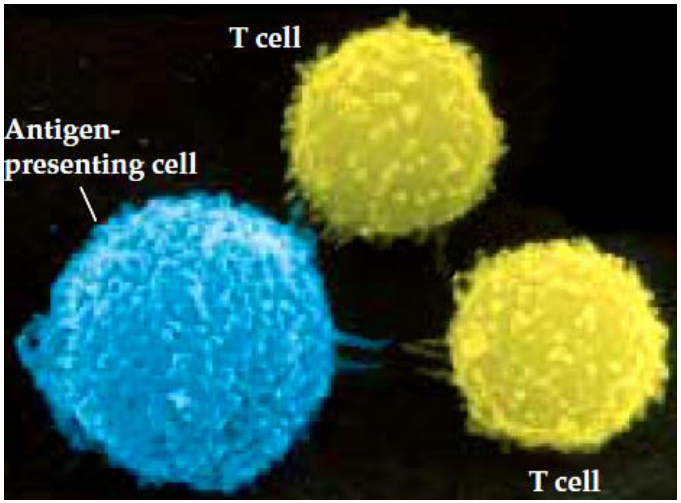
APC & MHC

APC express major histocompatibility complex (MHC) genes which encode proteins that present antigens to the immune system.

There are two types of MHC molecules MHC class I and MHC class II with distinct functions.



APCs & MHC



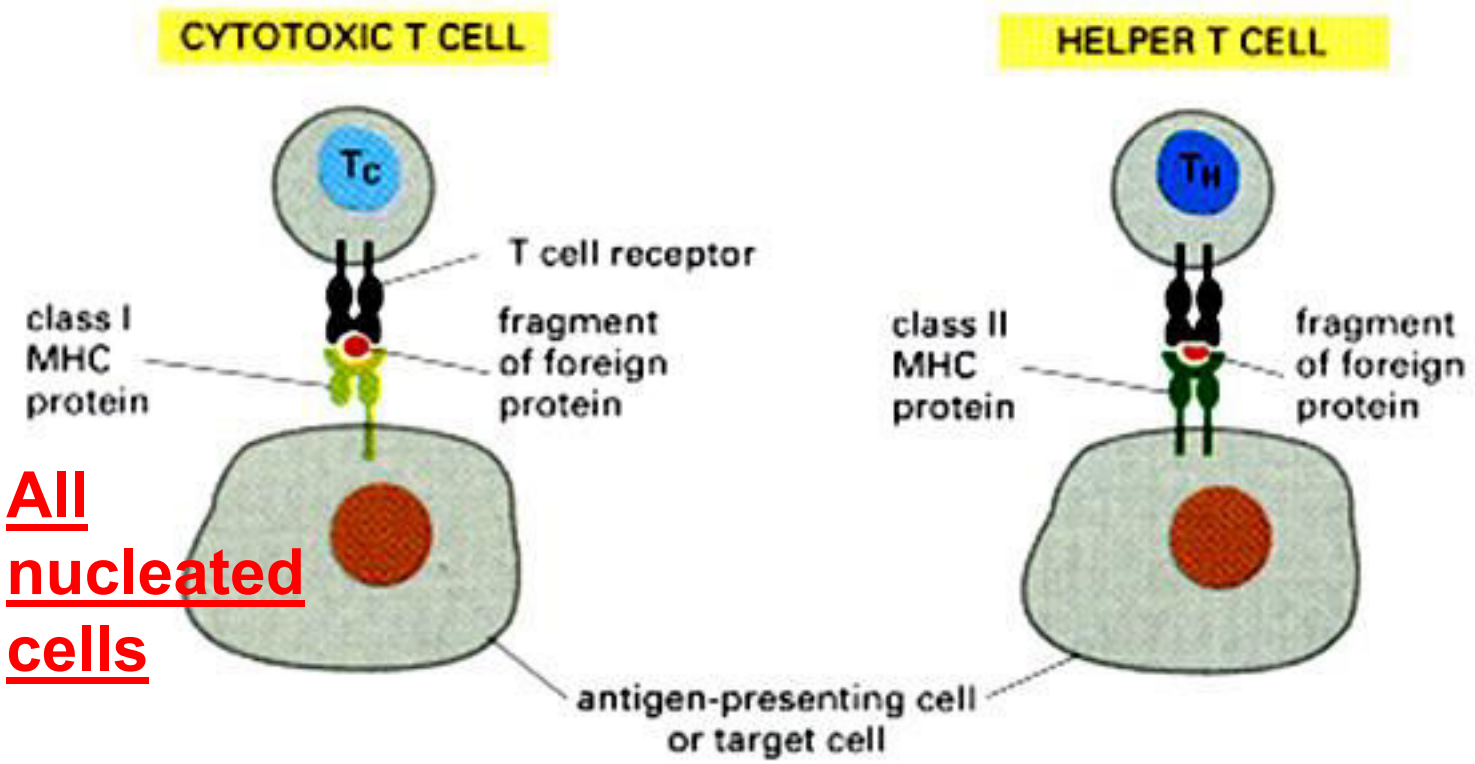
END

Cell Biology

MHC

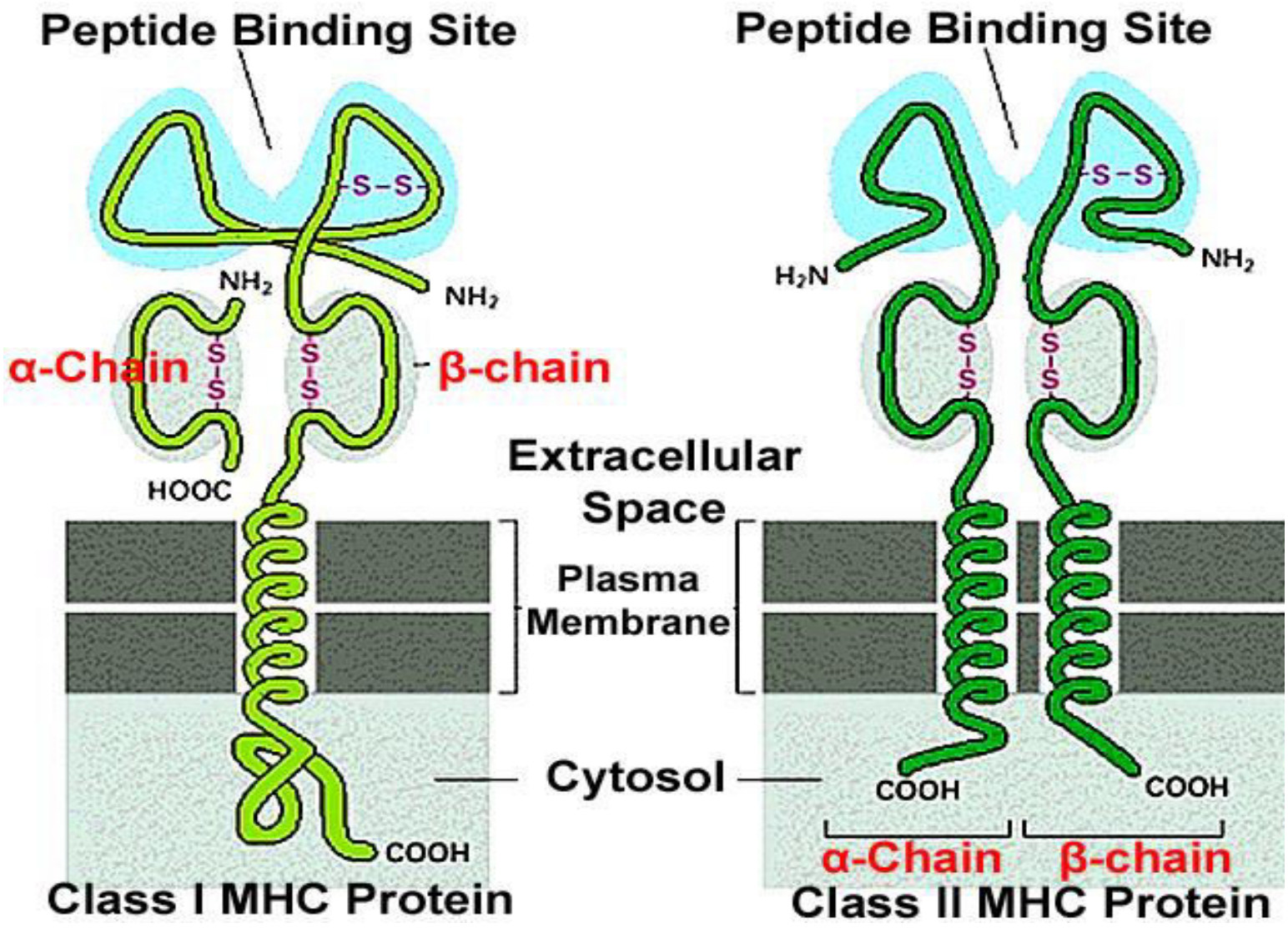
MHC

2 main structurally & functionally distinct classes of MHC proteins: class I MHC proteins, present foreign peptides to cytotoxic T cells, & class II present foreign peptides to helper T cells.



All nucleated cells

MHC Class I & II



both transmembrane heterodimers extracellular N-terminal domains that bind antigen for presentation to T

MHC Class I &II

Genetic Diversity

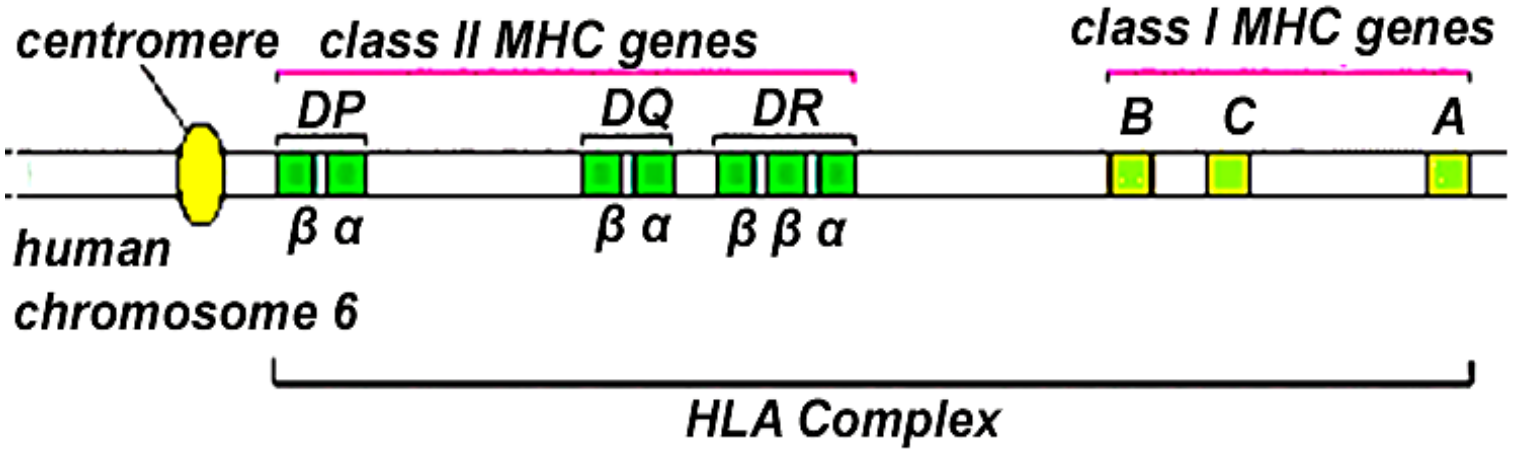
3 genes for class I proteins (HLA-A, B &C)

3 genes for class II proteins (HLA-DP, DQ &DR)

6 genes each for both the classes

Class II can α & β maternal & paternal can pair up

Polymorphism VS Gene Rearrangement



rare for two unrelated individuals to have an identical set of MHC proteins

Class I Not Diverse Enough?

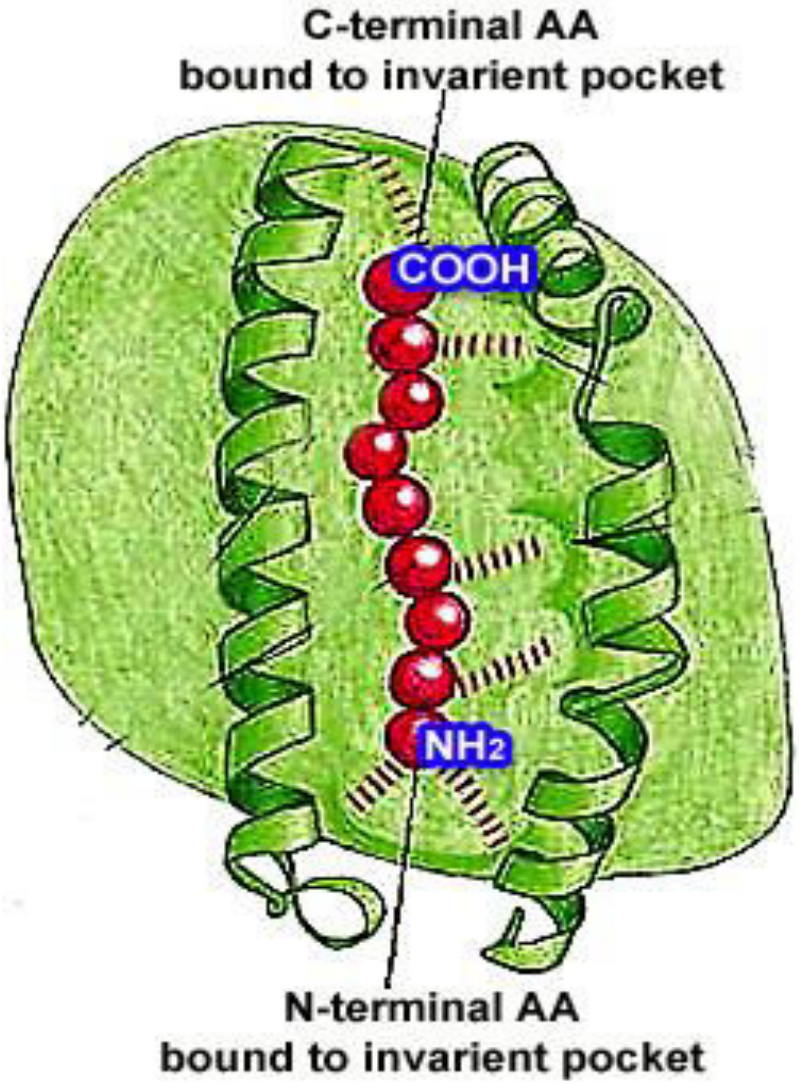
A person makes a small number of MHC proteins MHC should be able to present peptide fragments from almost any foreign protein to T cells

MHC groove can accommodate an extended peptide about 8–10 amino acids long.

In extended conformation terminal amino group binds to an invariant pocket at one end of the groove and its terminal carboxyl group bound to an invariant pocket at the other end of the groove.

Pockets recognize peptide backbone features common to all peptides, each class I MHC protein can bind a peptides of diverse sequence.

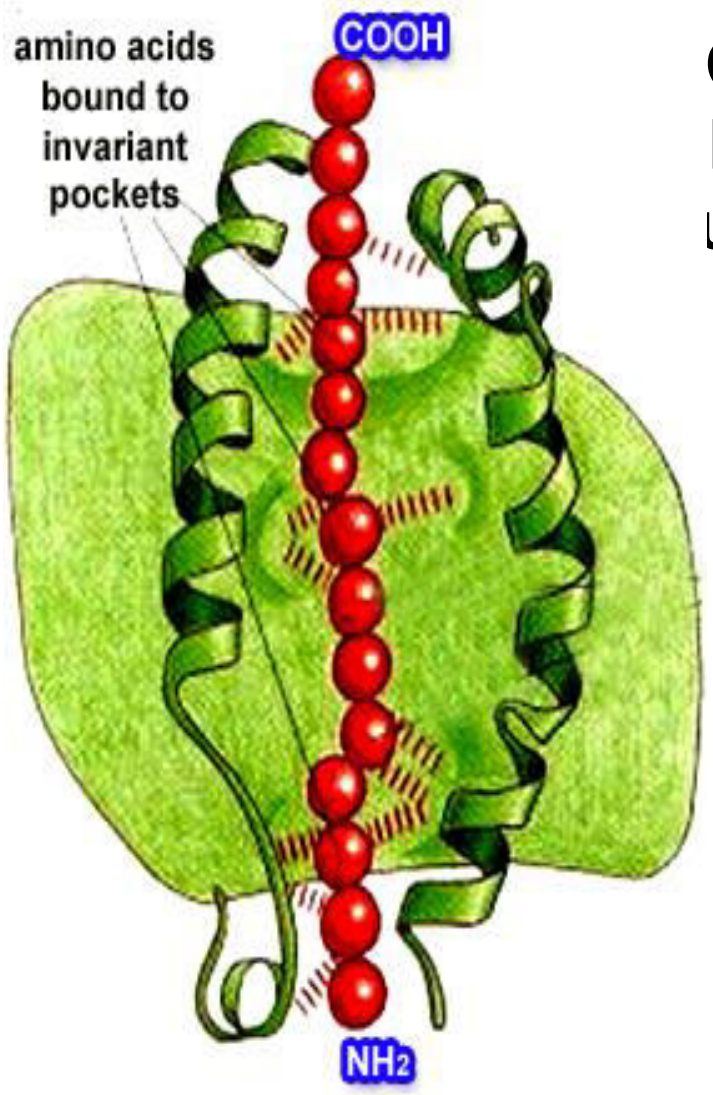
MHC Class I



Antigen Binding

The side chains of other amino acids of the peptide point outward, in a position to be recognized by receptors on cytotoxic T cells.

MHC Class II

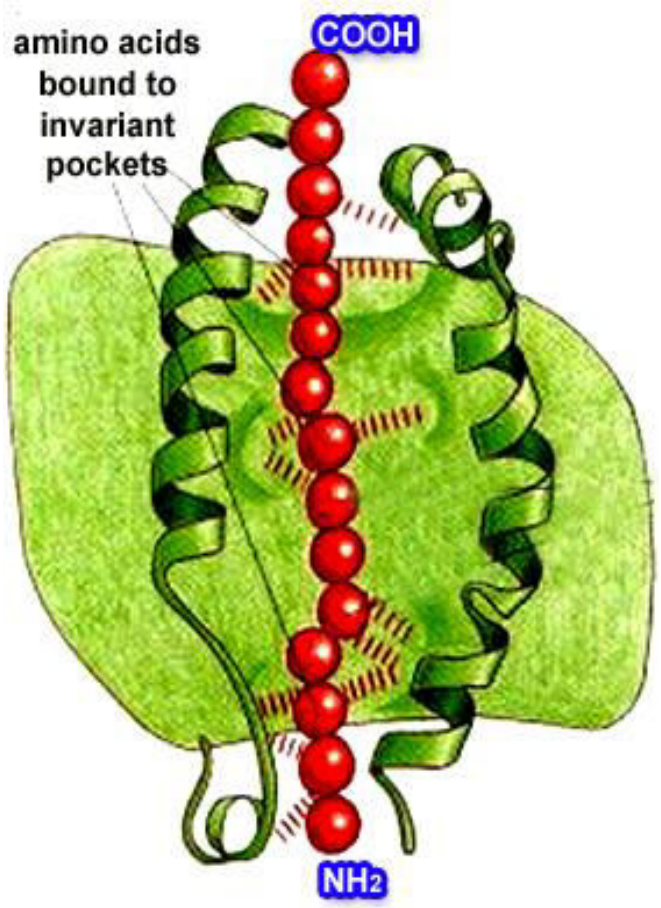


Class II can accommodate longer peptides, which are usually 13–17 amino acids.

Peptide not bound by ends.

It is held in the groove by parts of its peptide backbone that bind to invariant pockets formed by conserved amino acids that line all class II MHC peptide-binding grooves

APCs & MHC

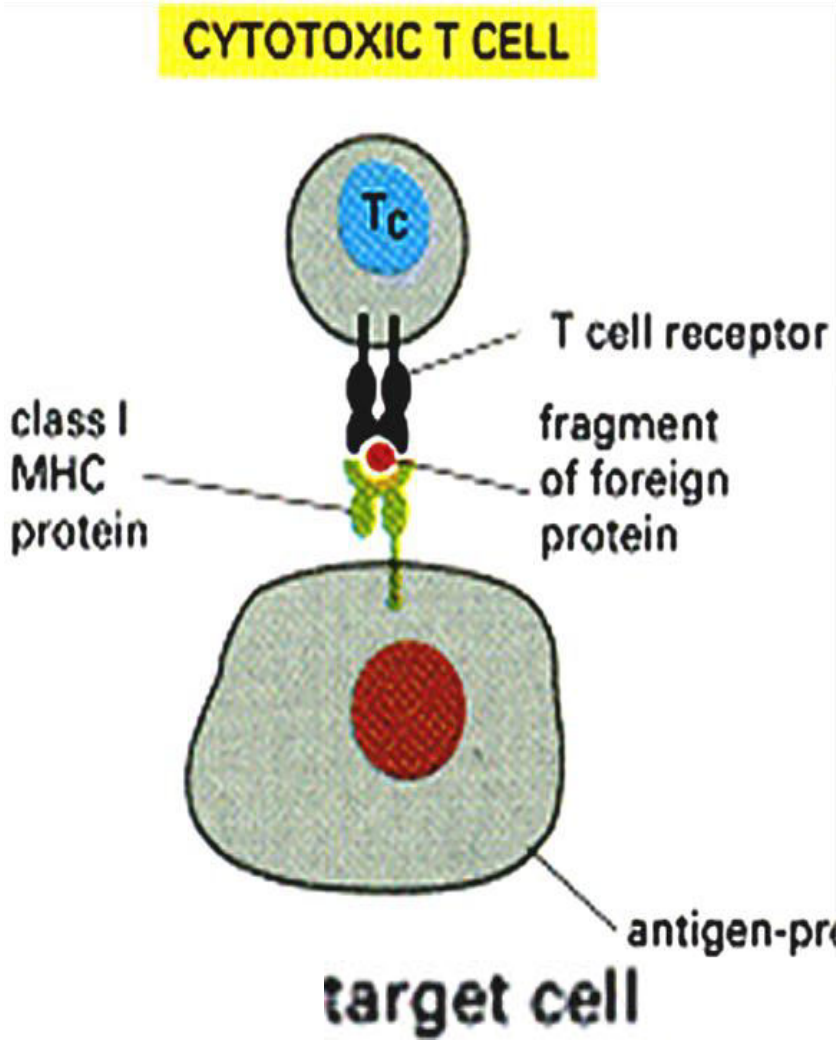


END

Cell Biology

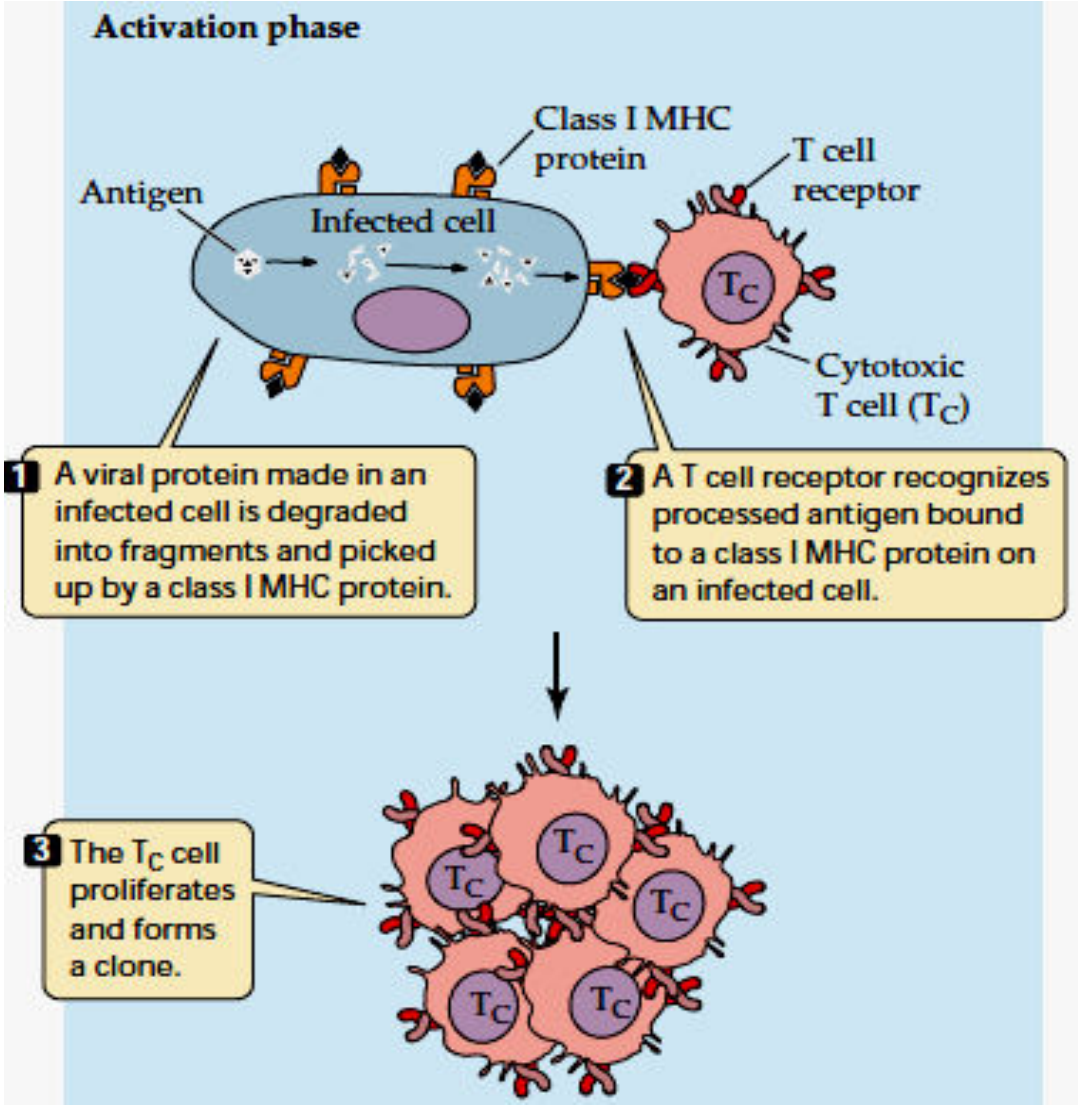
Cytotoxic T cells

Cytotoxic T cells

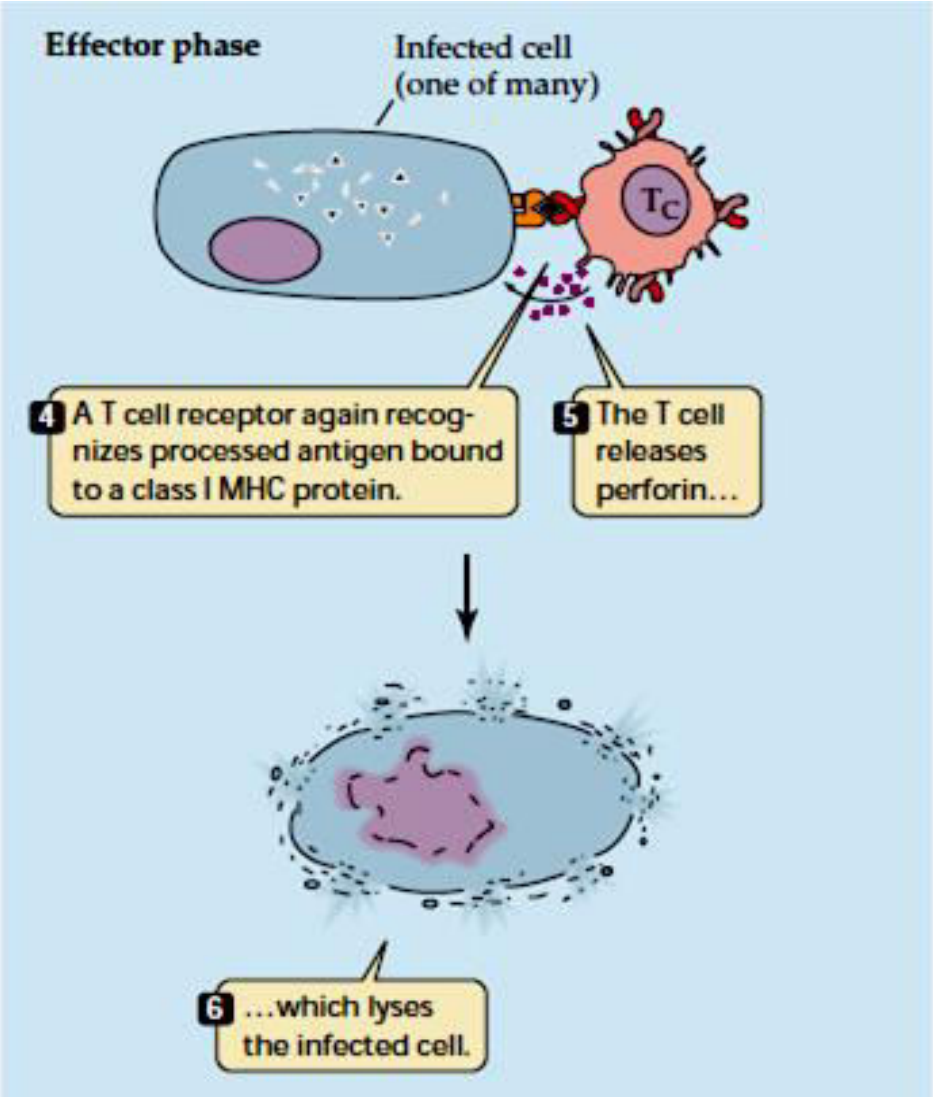


Cytotoxic T cells provide protection against intracellular pathogens such as viruses and some bacteria and parasites that multiply in the host-cell cytoplasm, where they are sheltered from attack by antibodies.

Cytotoxic T cell Activation

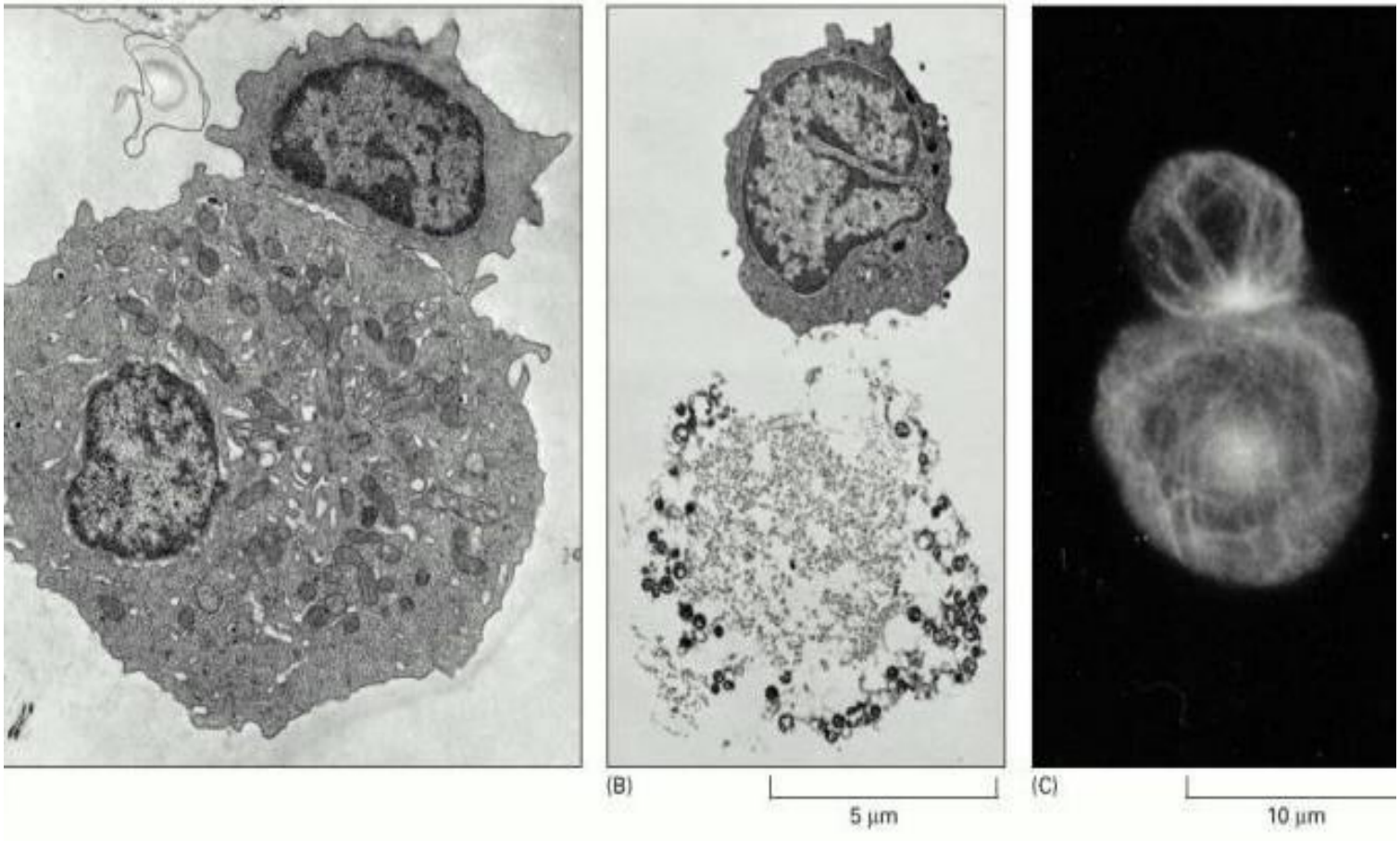


The Effector Phase

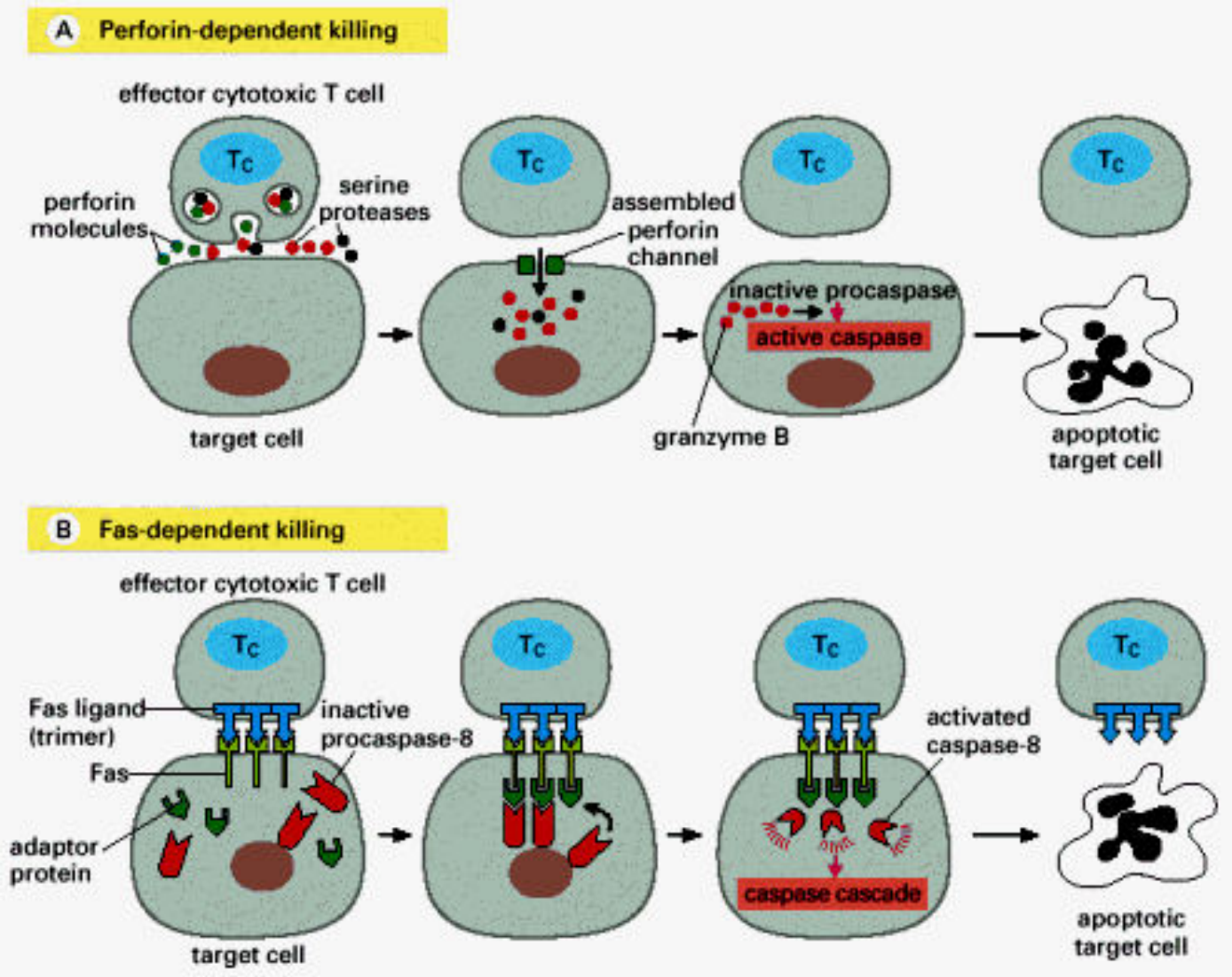


Cytotoxic T cells inaction

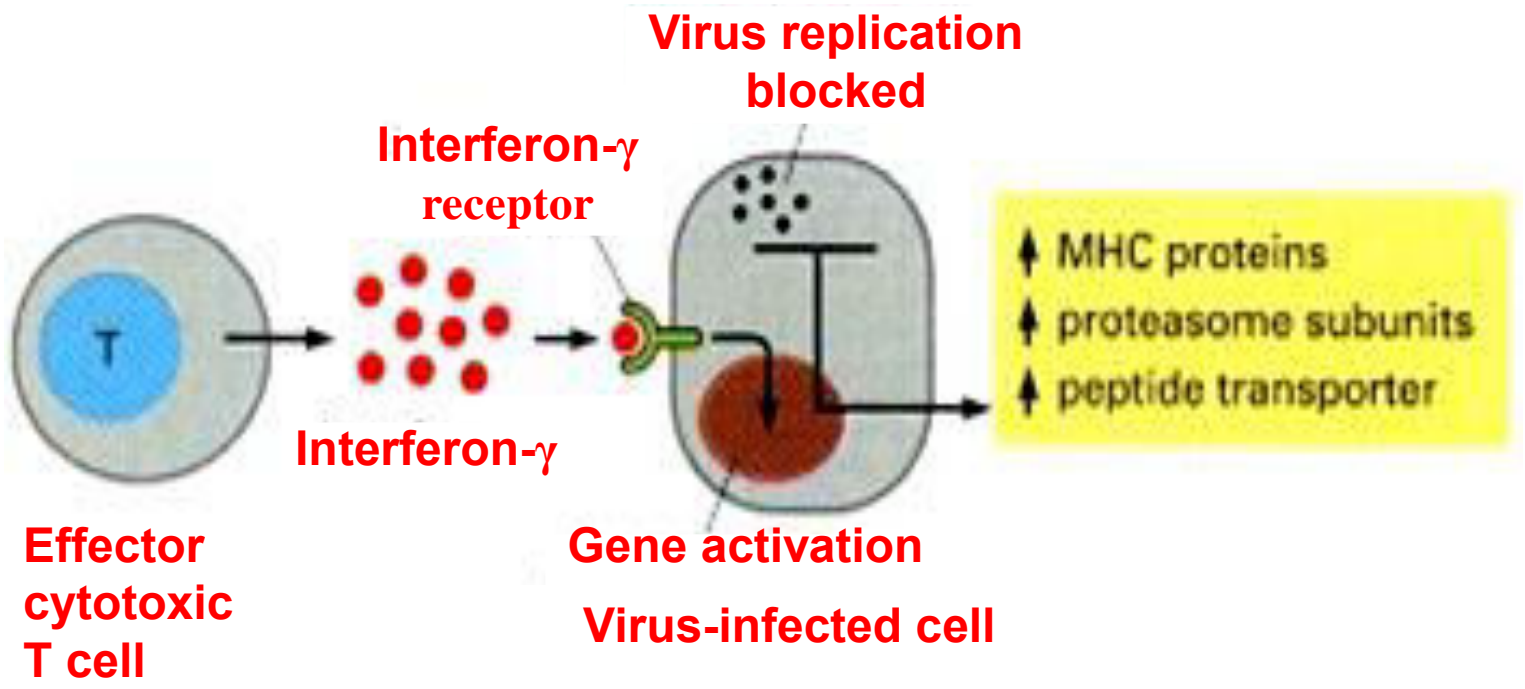
Cytotoxic T cells in action



The Killing Machine



Increased susceptibility to killing



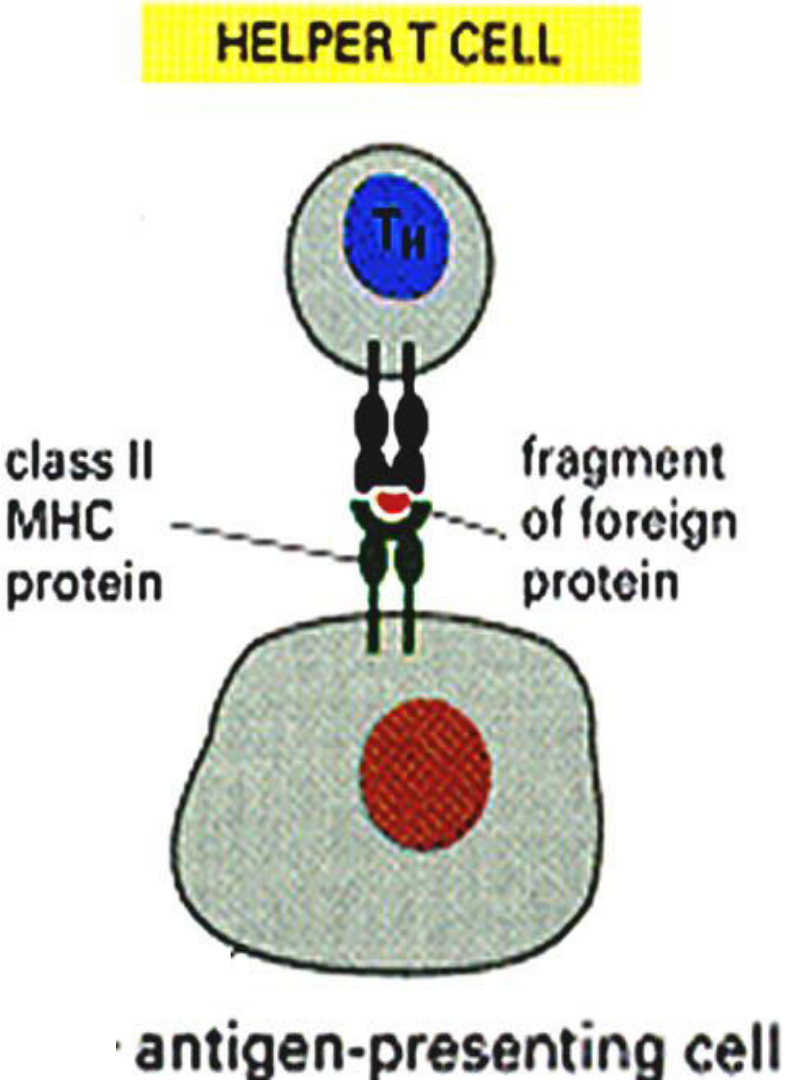
Cytotoxic T cells

END

Cell Biology

Helper T cells

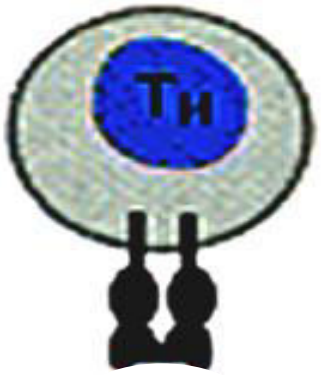
Helper T cells



Helper T cells are crucial for defense against both extracellular and intracellular pathogens.

Once a helper T cell is activated by an antigen-presenting cell to become an effector cell, it then help activate other cells.

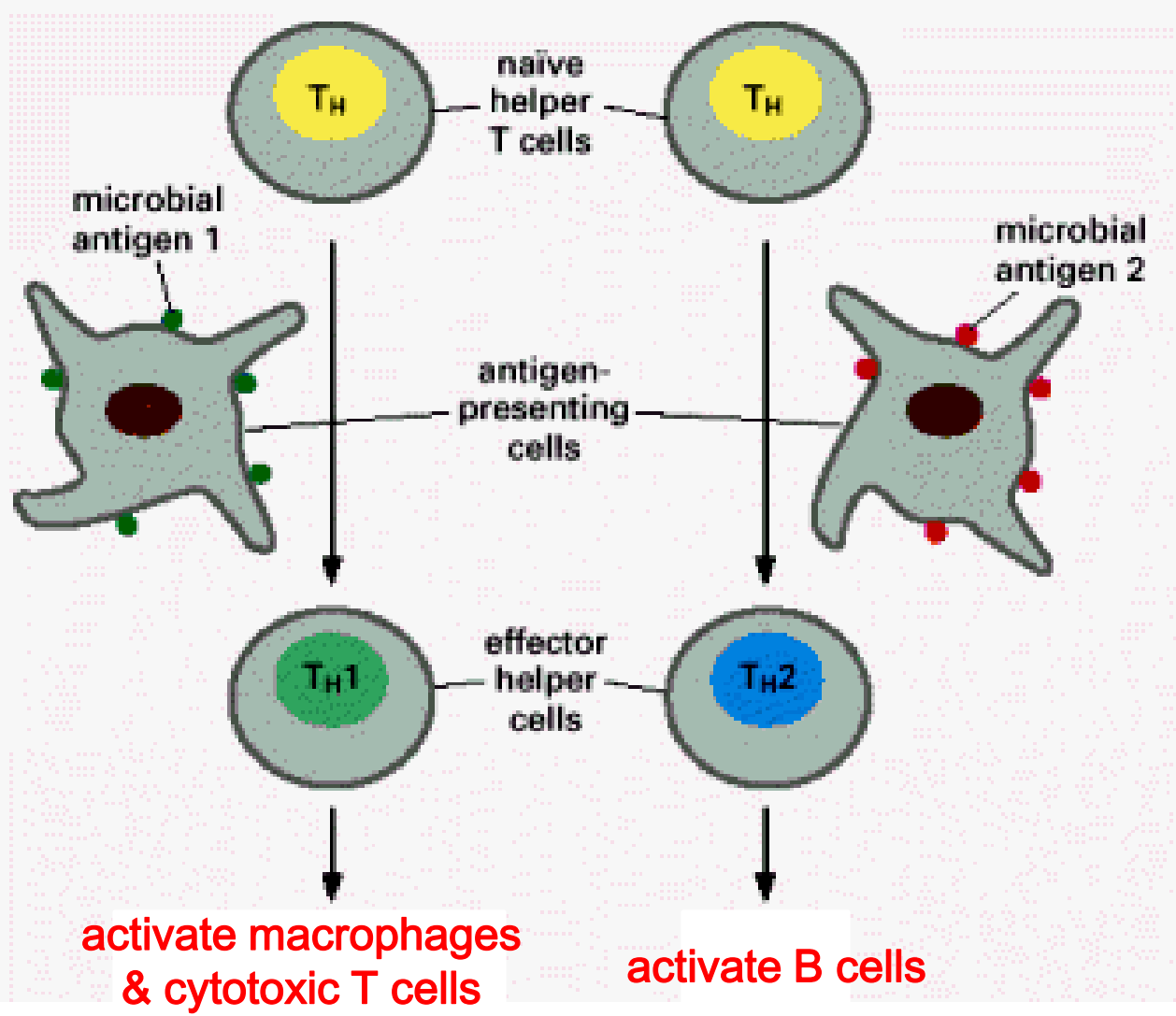
Helper T cells



Helper T cells stimulate B cells to make antibodies that inactivate or eliminate extracellular pathogen & their toxic products.

When activated by an antigen-presenting cell, a naïve helper T cell can differentiate into either of two distinct types of effector helper cell, called TH1 and TH2. TH1 cells mainly help activate macrophages and cytotoxic T cells, whereas TH2 cells mainly help activate B cells

T cell differentiate into TH_1/TH_2



nature of the invading pathogen and the types of innate immune

The Two Signals

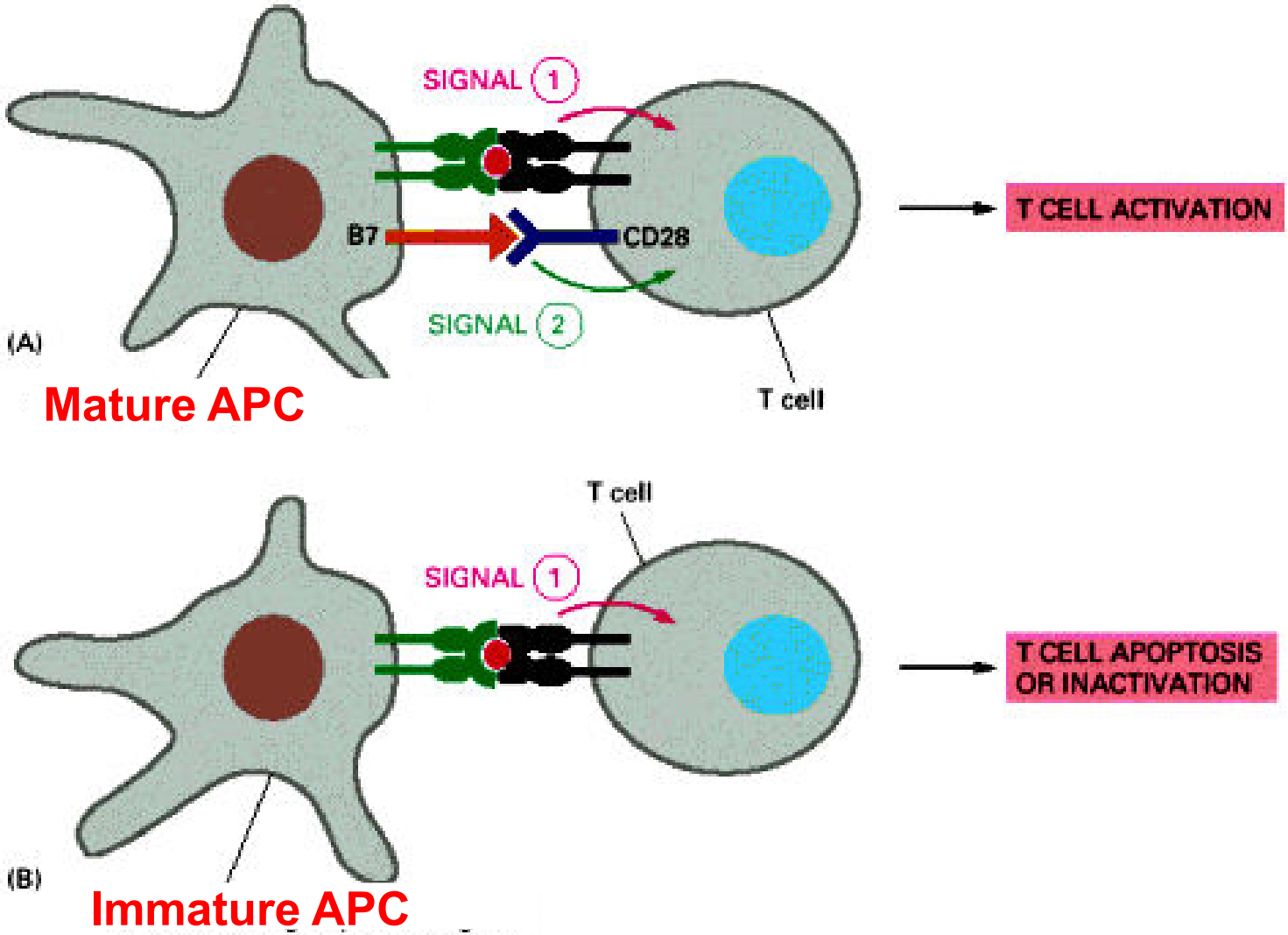
APCs provides two kinds of signals.

Signal 1 is provided by a foreign peptide bound to an MHC protein on the surface of the presenting cell, peptide-MHC complex signals through the T cell receptor and its associated proteins.

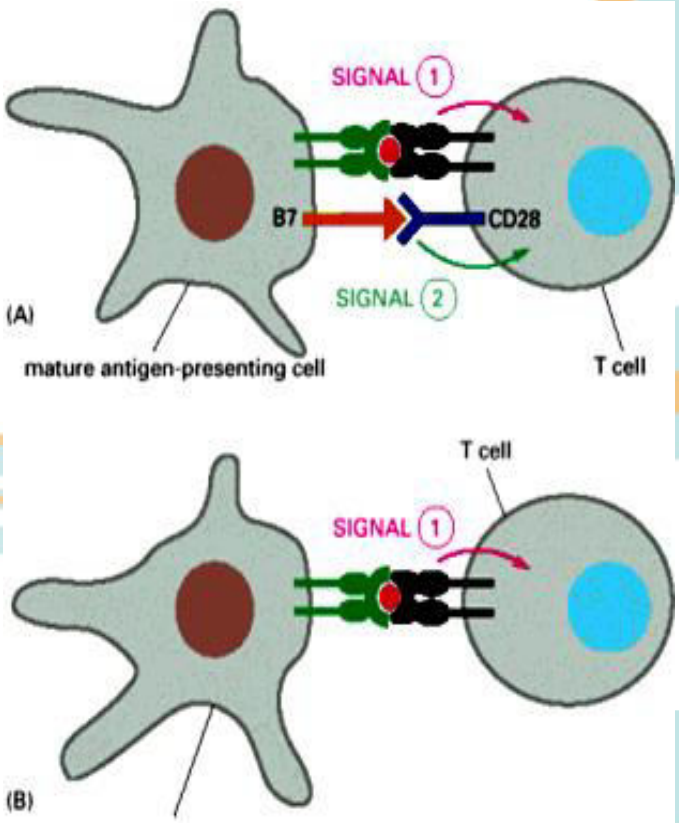
Signal 2 is provided by costimulatory proteins, especially the B7 proteins (CD80 and CD86), which are recognized by the co-receptor protein CD28 on the surface of the T cell.

The expression of B7 proteins on an antigen-presenting cell is induced by pathogens during the innate response to an infection.

The Two Signals



Helper T cells

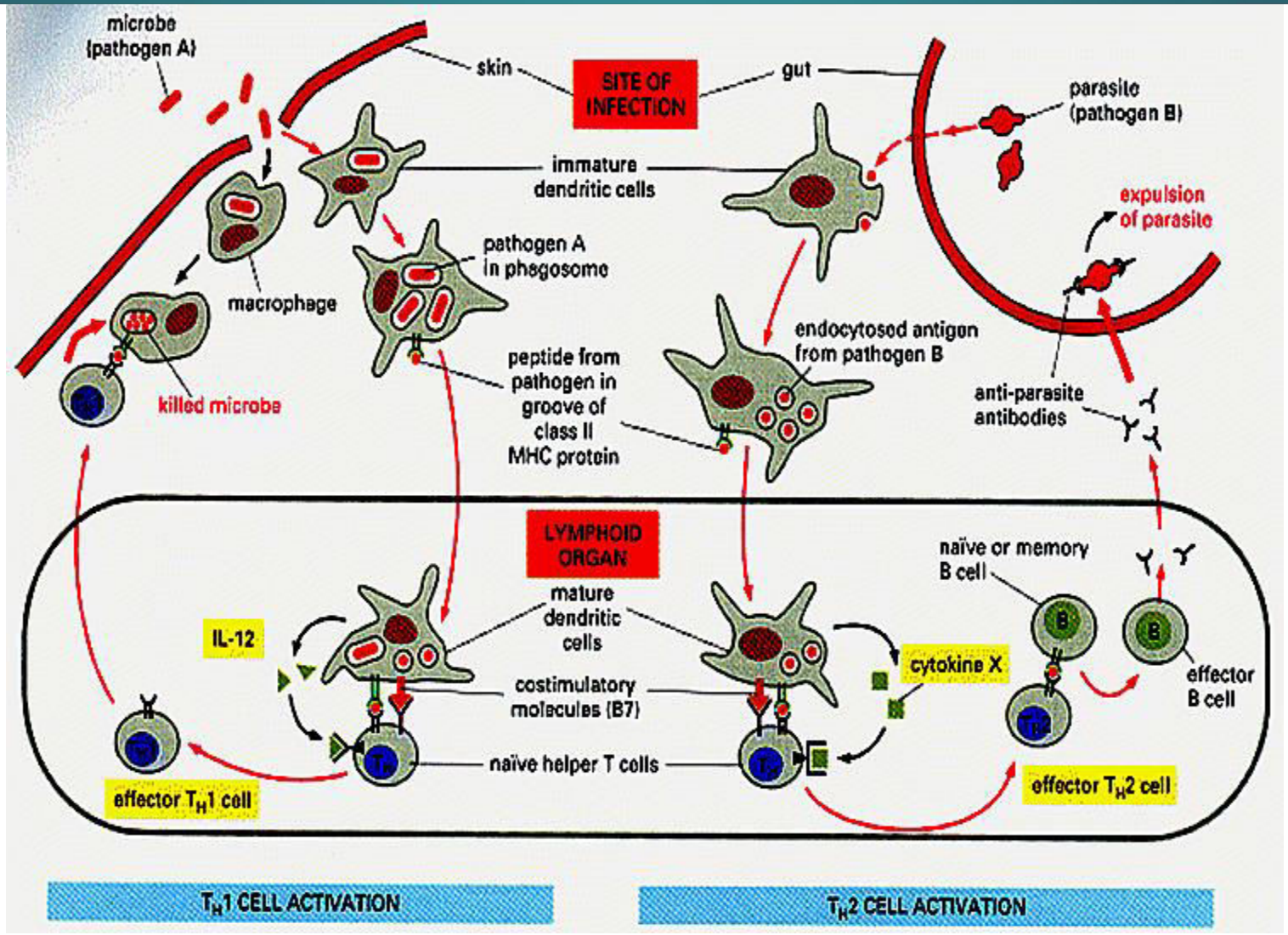


END

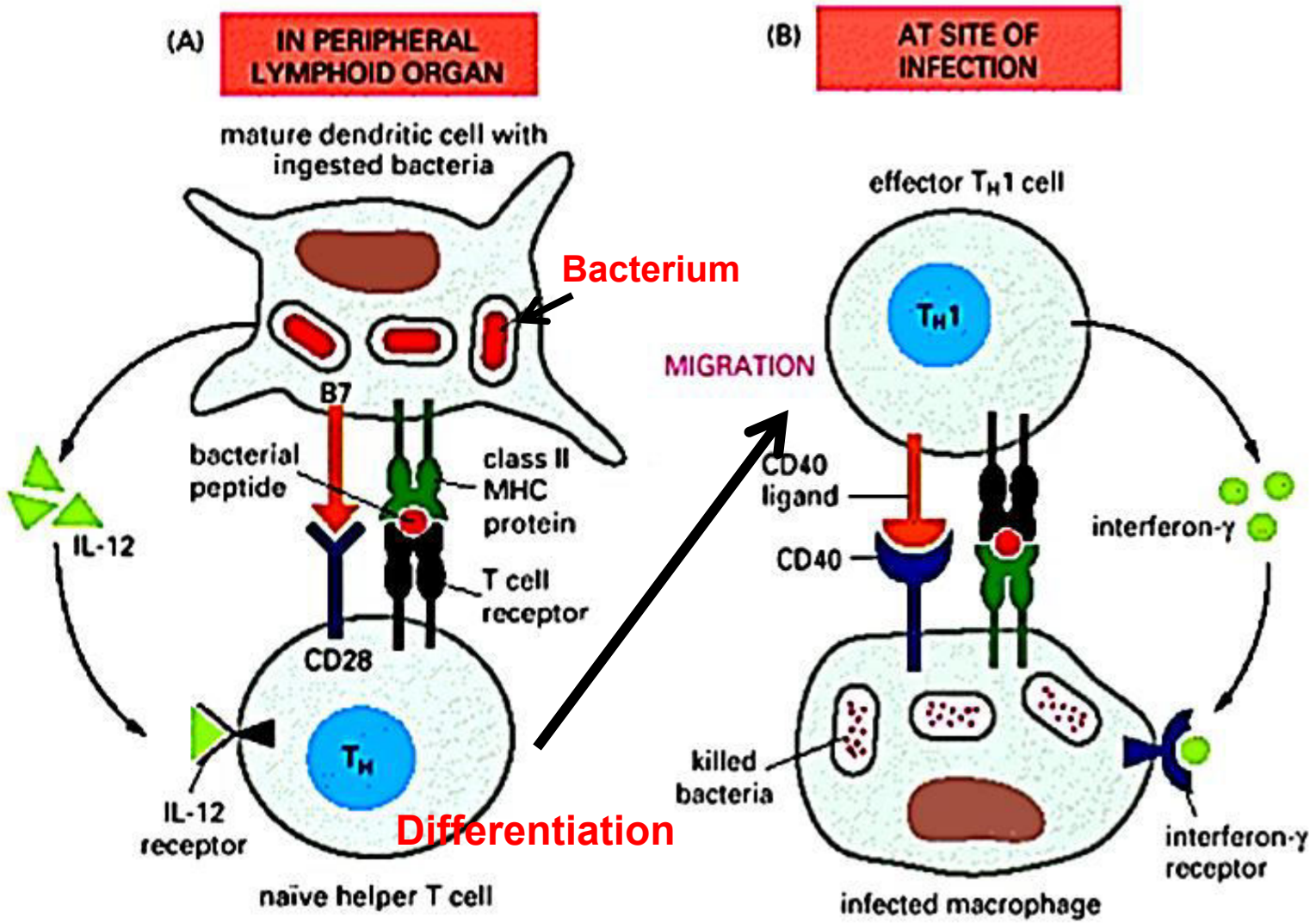
Cell Biology

T_H1 & T_H2

The Two Signals



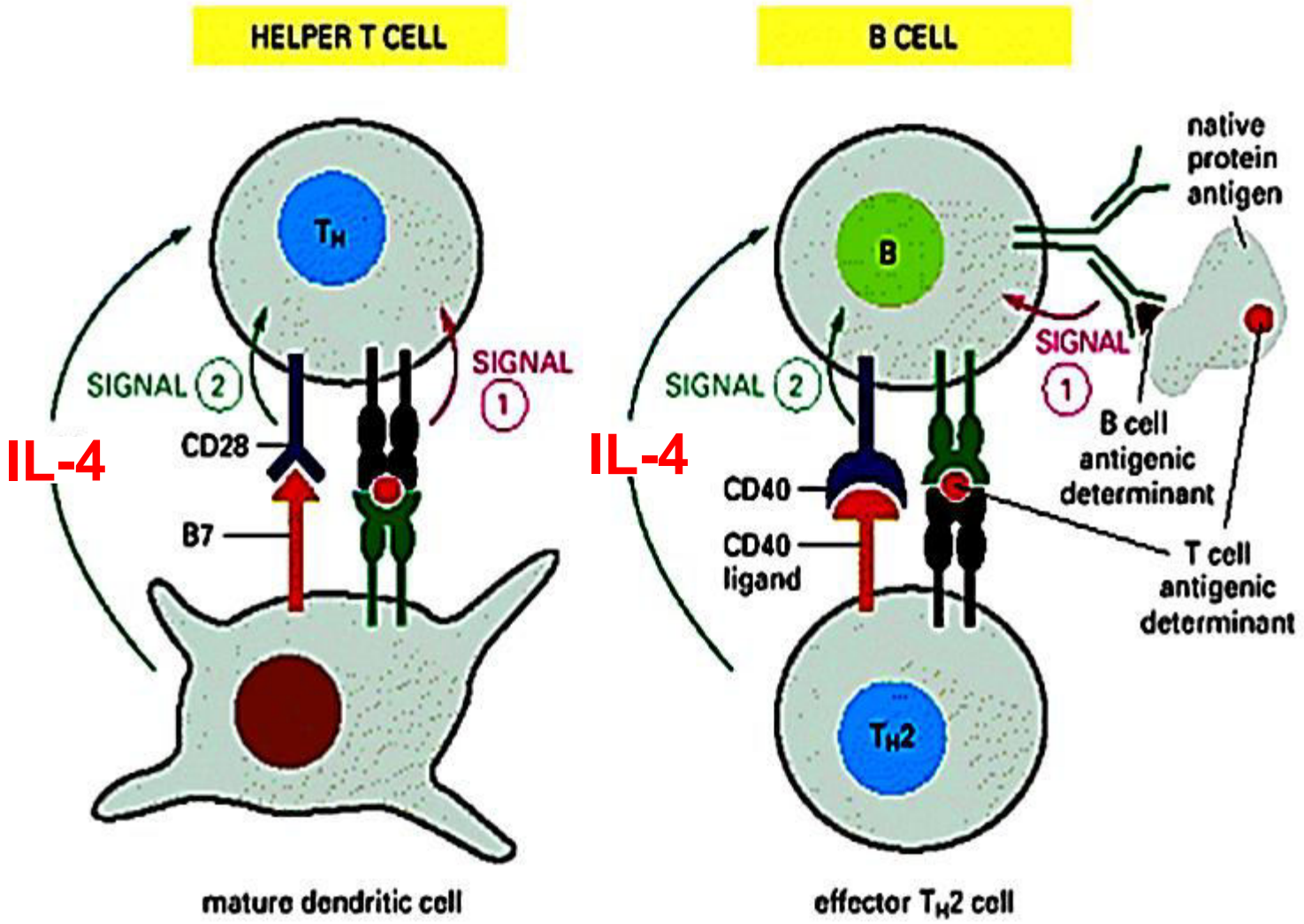
The Two Signals



Dendritic cell Stimulates TH1

Mature TH1 activates Macrophage to kill

The Two Signals



APCs & MHC

Shortcut

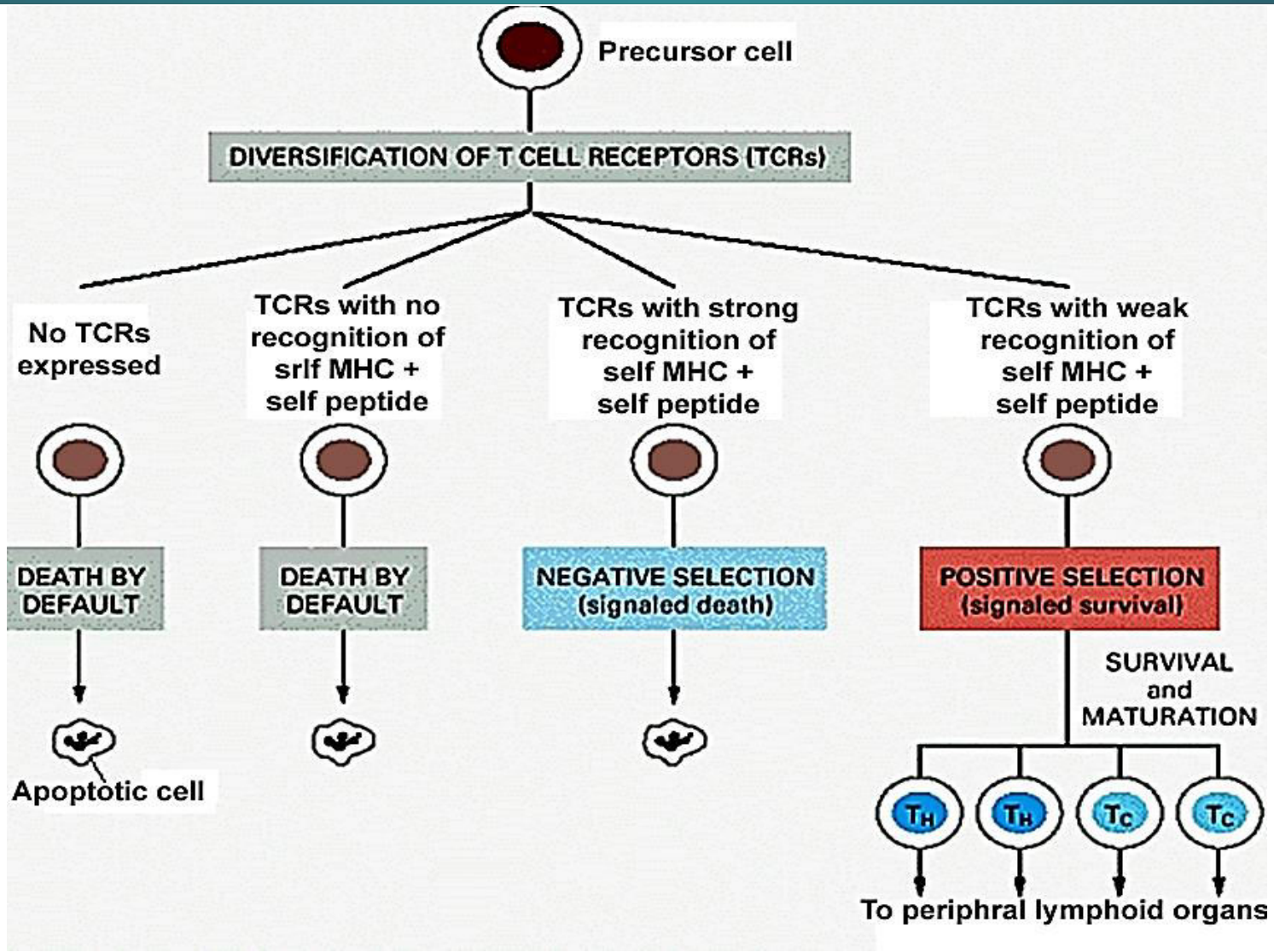
B cell as APC

END

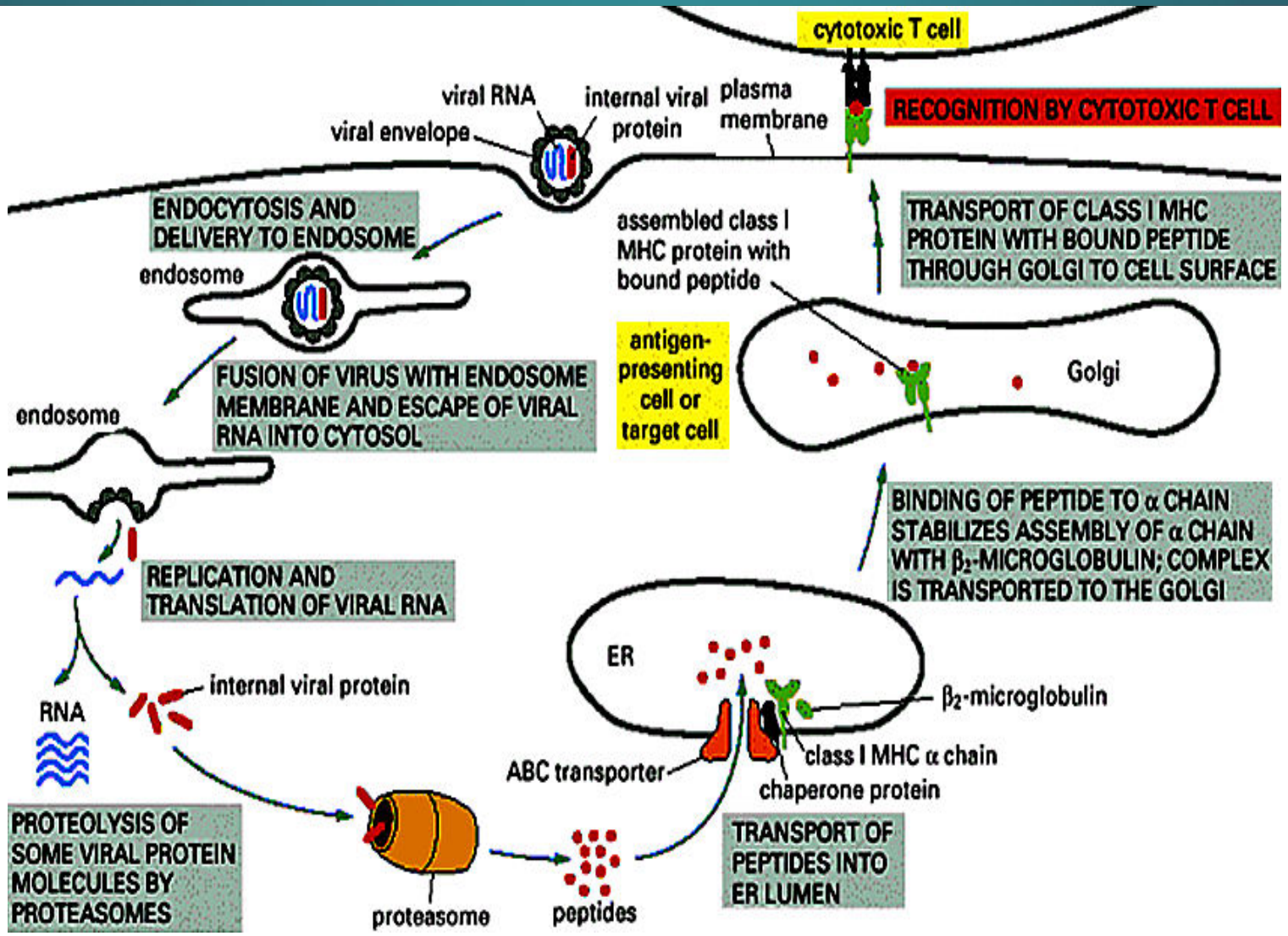
Cell Biology

Checks & Balances

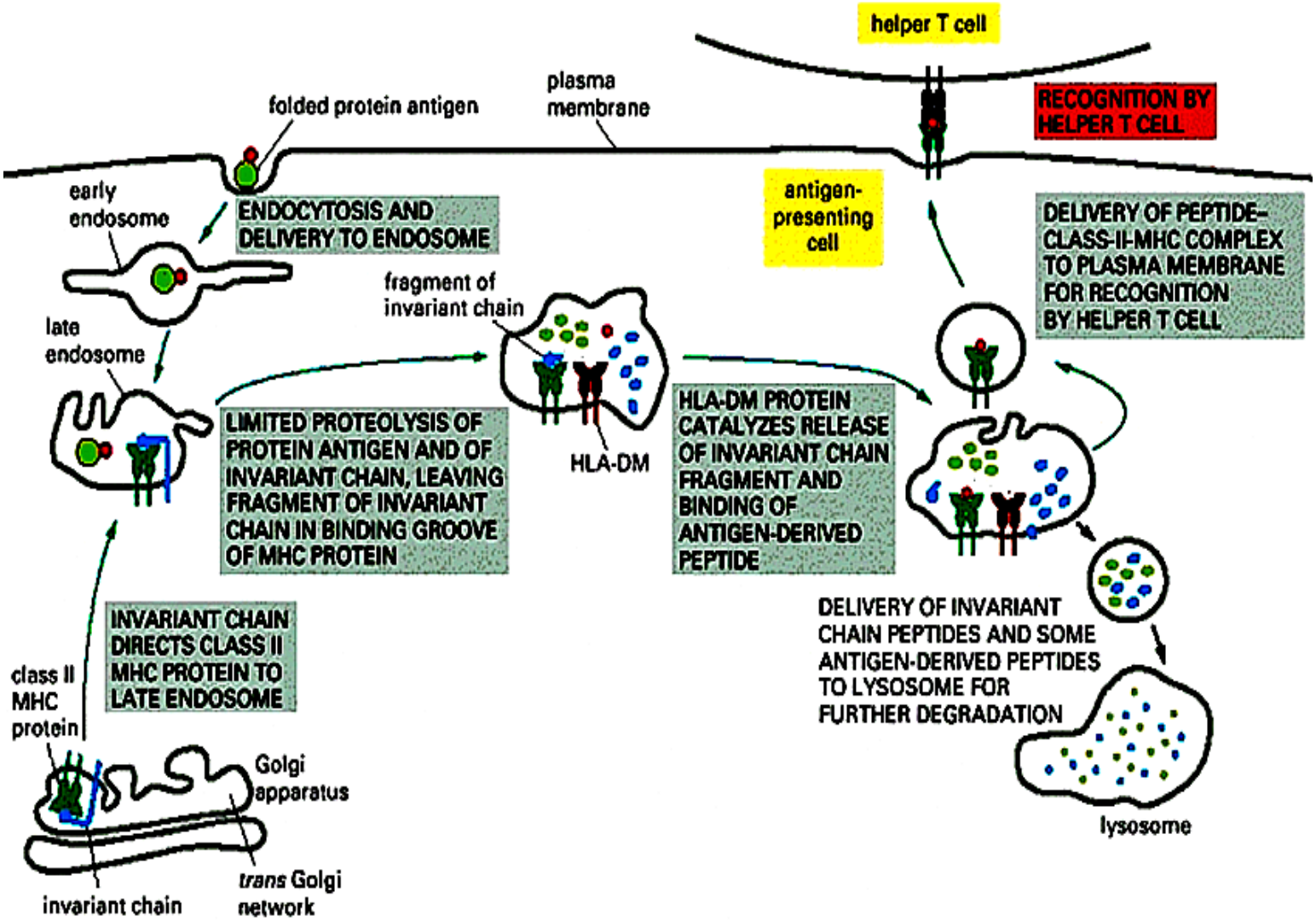
Checks & Balances



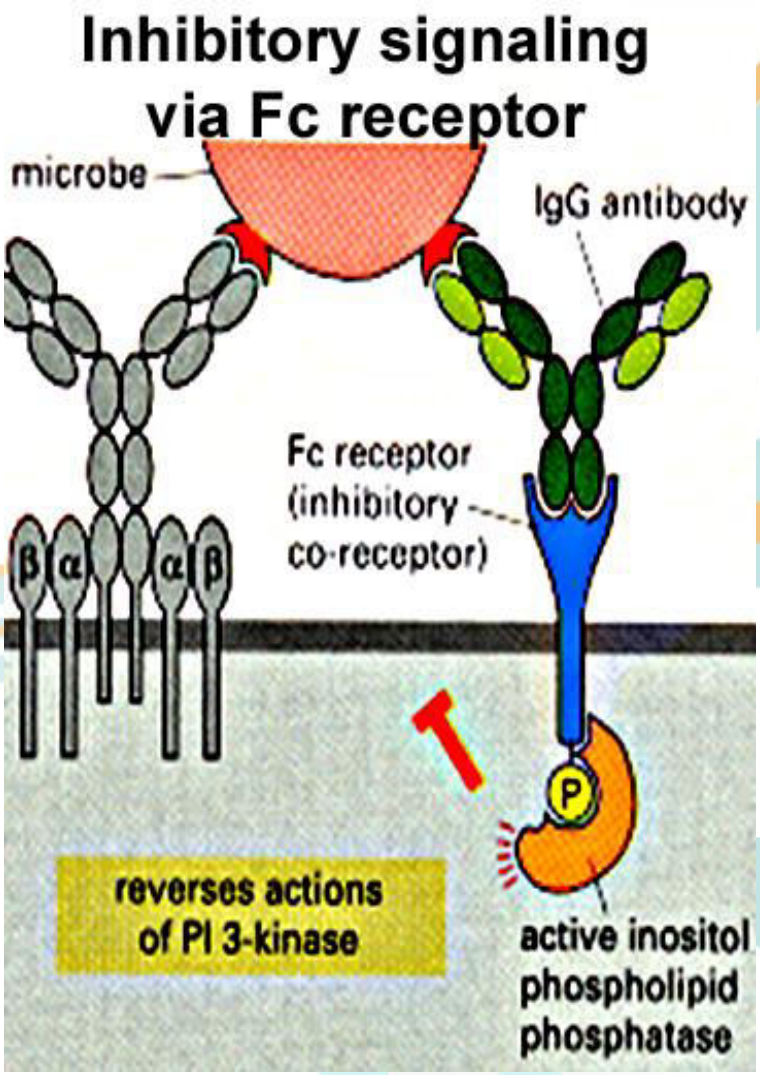
Checks & Balances



Checks & Balances



Helper T cells



END

Cell Biology

Complement System

Complement System

Most of the time, the binding of antibodies to antigen performs no useful function until and unless it can activate an effector mechanism

Complement system serves several effector roles

Complement system provides the actual protection from the response while the interaction of antibodies and antigen provides the specificity of the response

Put another way, antibodies "finger" the target, complement destroys it

Complement System

The complement system consists of some 30 proteins circulating in blood plasma

These proteins are synthesized mainly in the liver, and they account for about 5% of the globulin fraction of blood serum

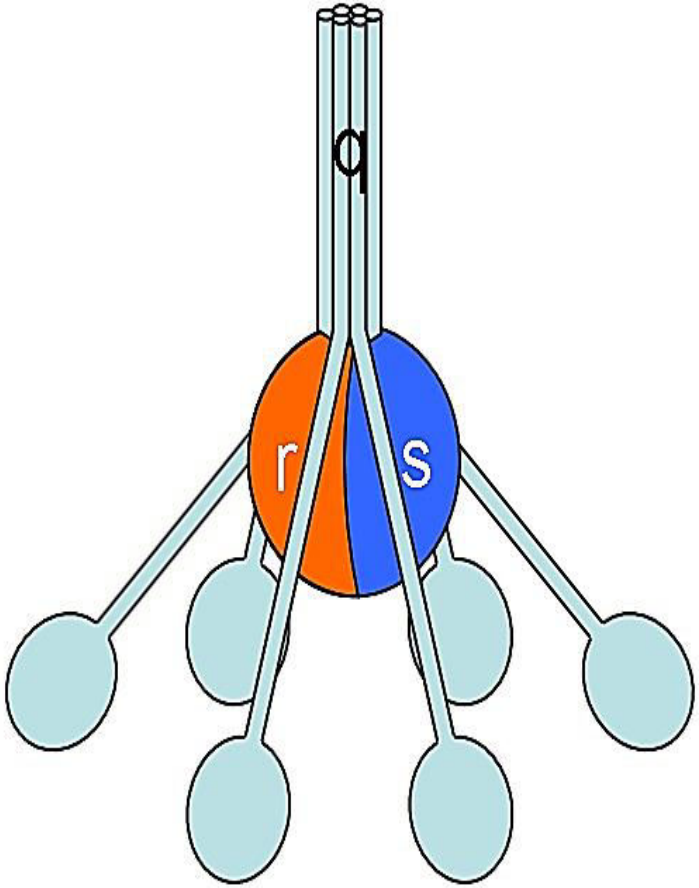
Most of these are inactive until

they are cleaved by a protease which, in turn,

converts them into a protease

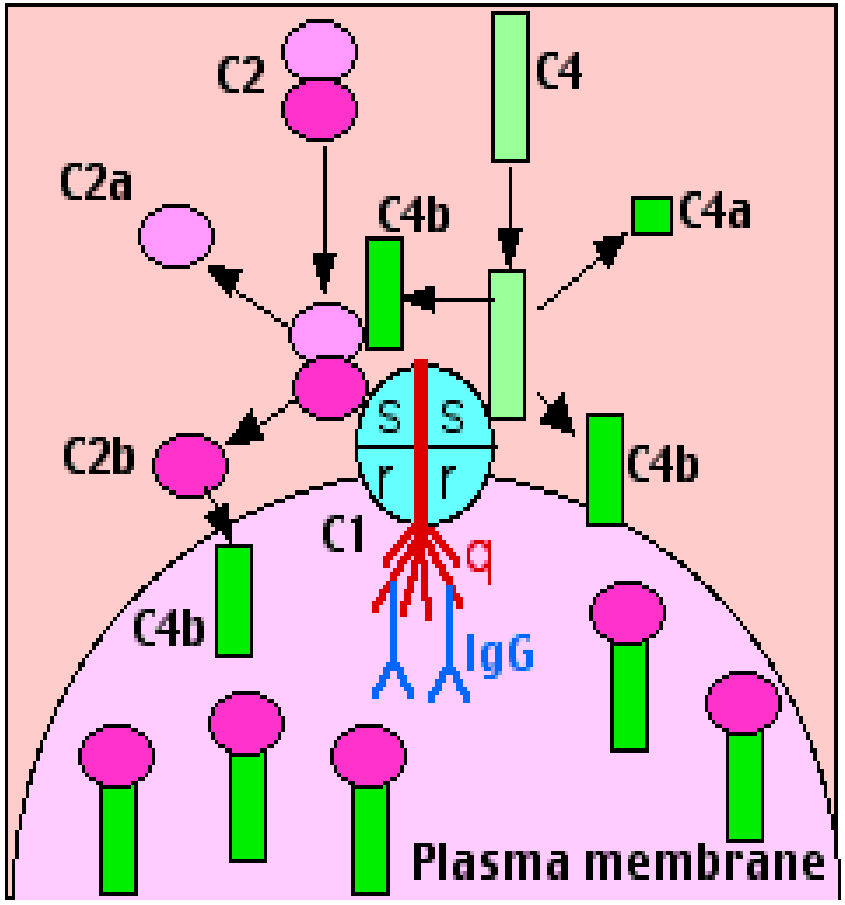
Thus many components of the system serve as the substrate of a prior component and then as an enzyme to activate a subsequent component

Classical complement pathway



The classical pathway is triggered by activation of the C1-complex (which consists of six molecule C1q and two molecules C1r and C1s), either by C1q's binding to antibodies from classes M and G, complexed with antigens, or by its binding C1q to the surface of the pathogen

Classical complement pathway

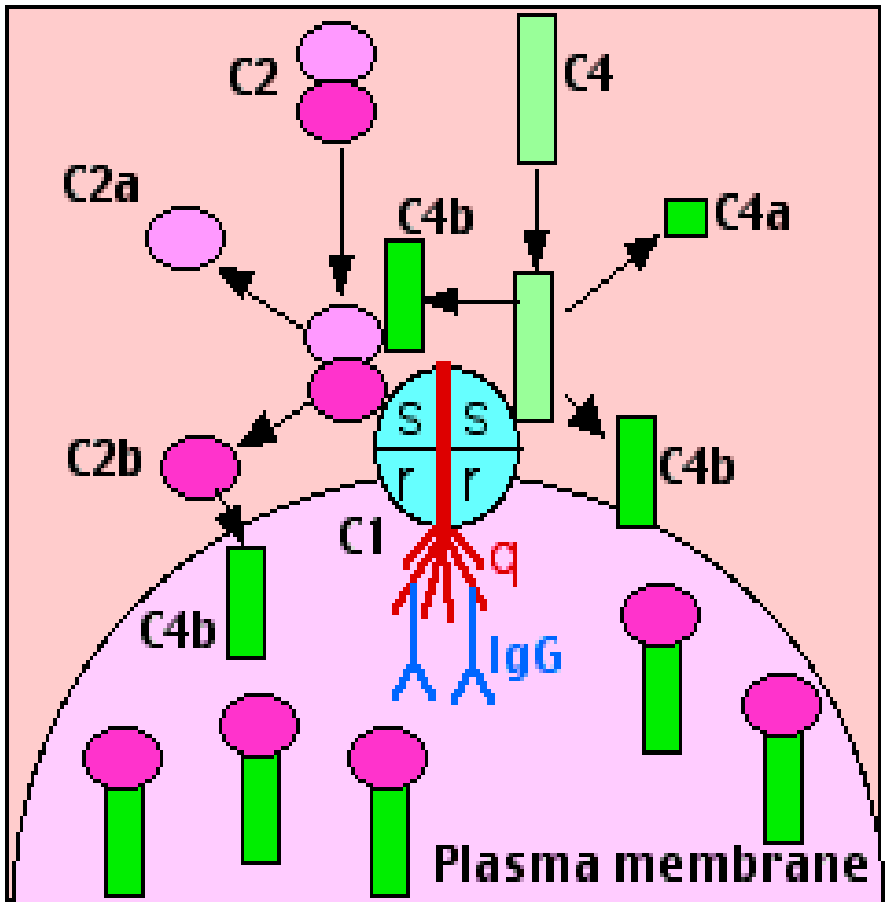


This binding leads to conformational changes in C1q molecule, which leads to the activation of two C1r (serine protease) molecules

Then they cleave C1s (another serine protease)

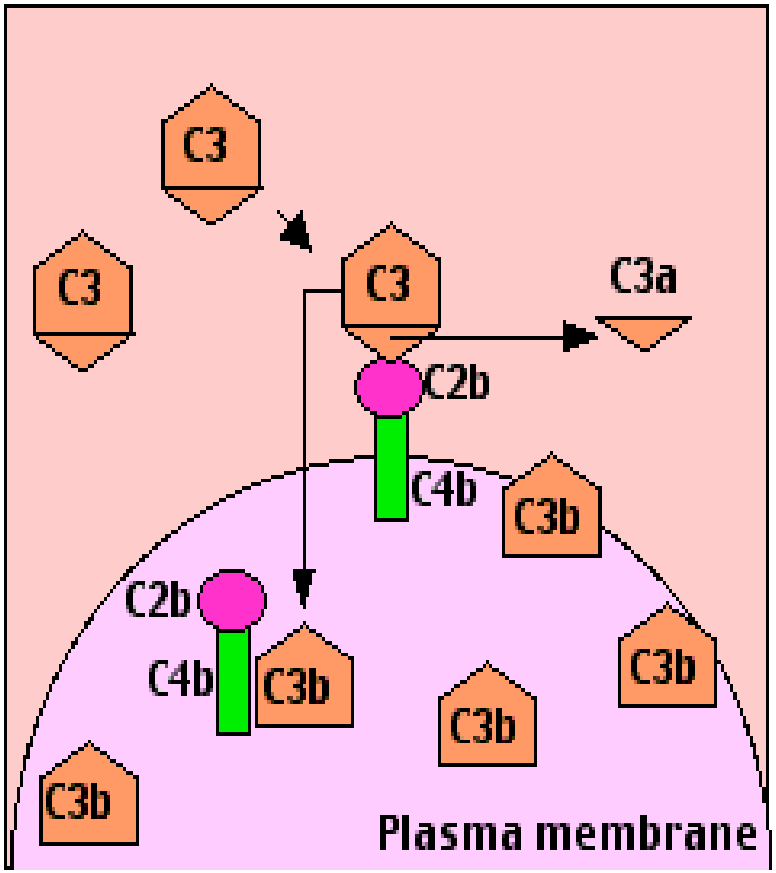
The C1-complex now binds to and splits C2 and C4, producing C2a and C4b

Classical complement pathway



C4b, which binds covalently to sugar residues on cell-surface glycoproteins, and a smaller, inactive, fragment of C4a which diffuses away C2 is cleaved into C2b, which binds noncovalently to a site on C4b, leaving a smaller, inactive, fragment of C2a which diffuses away

Classical complement pathway

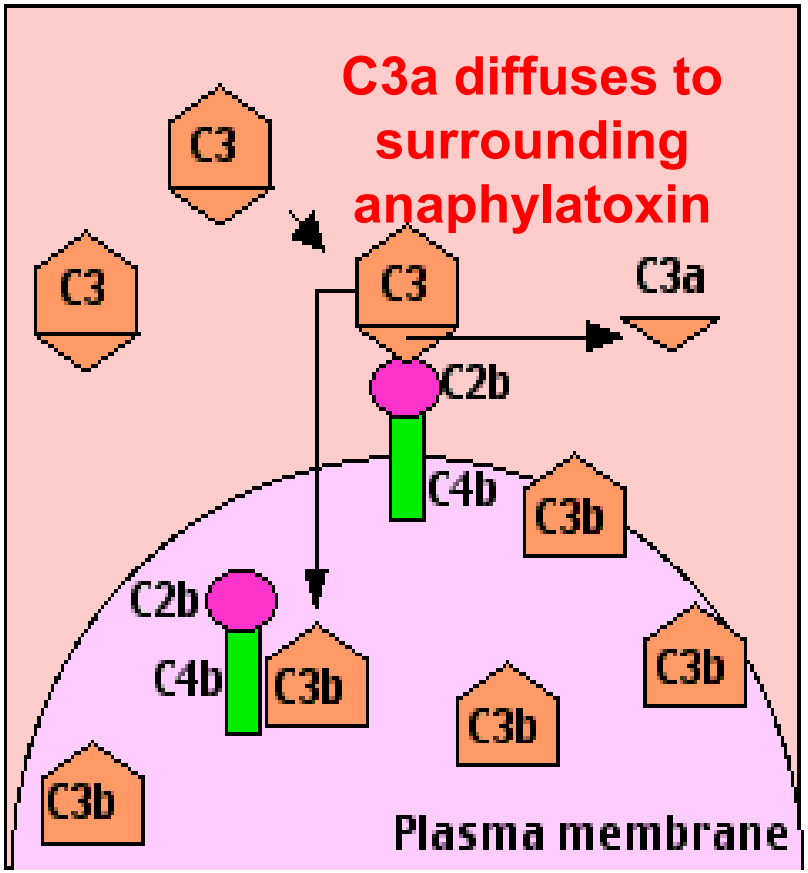


This effect qualifies C3b as an opsonin

The complex of C4b•2b is called "C3 convertase" because it catalyzes the cleavage of C3

C3b binds covalently to glycoproteins scattered across the cell surface. Macrophages and neutrophils have receptors for C3b and can bind the C3b-coated cell or particle

Classical complement pathway



This effect qualifies C3b as an opsonin

The complex of C4b•2b is called "C3 convertase" it catalyzes the cleavage of C3

C3b binds covalently to glycoproteins scattered across the cell surface. Macrophages and neutrophils have receptors for C3b and can bind the C3b-coated cell or particle

Histamine release from mast cells vascular permeability

Classical complement pathway

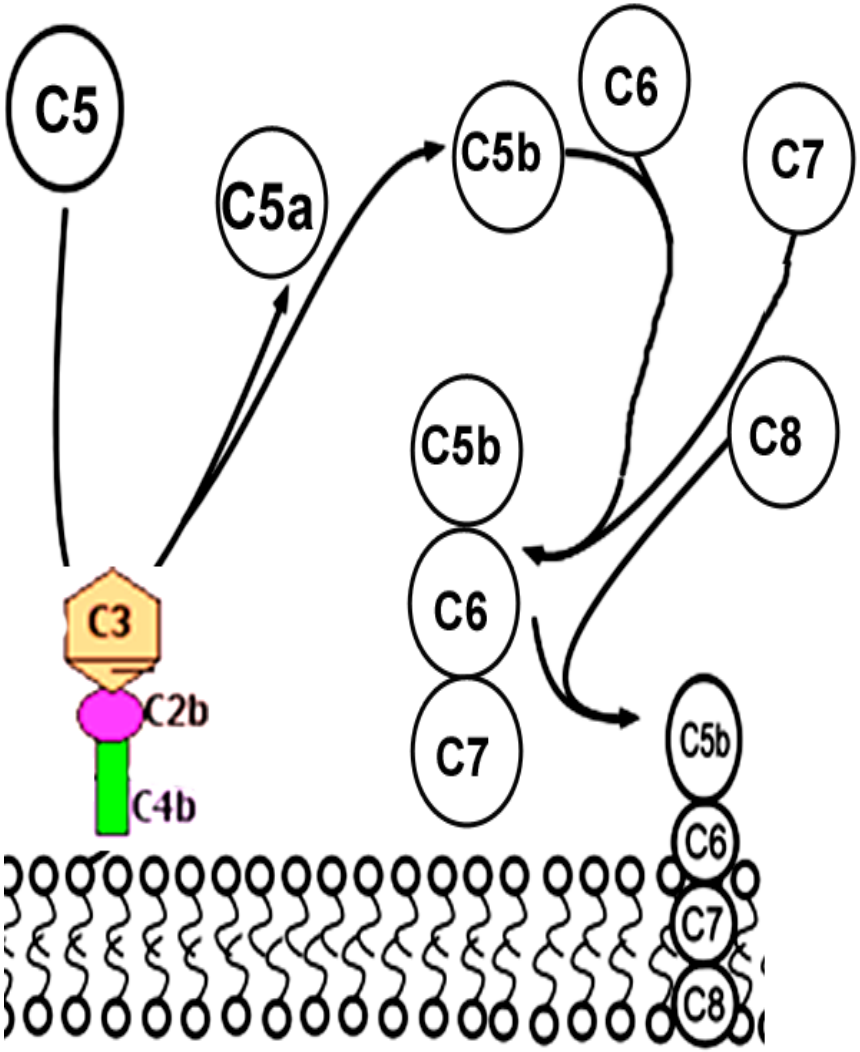
Cleavage of C5

Some of the C3b binds to molecules of C5 creating an allosteric change that exposes them to cleavage by C4b•2b (which is thus a "C3/C5 convertase")

Cleavage of C5 by the C3/C5 convertase initiates the assembly of a set of complement proteins that make up the membrane attack complex.

The membrane attack complex can also be formed by another C5 convertase produce by the "alternative pathway"

Membrane Attack Complex

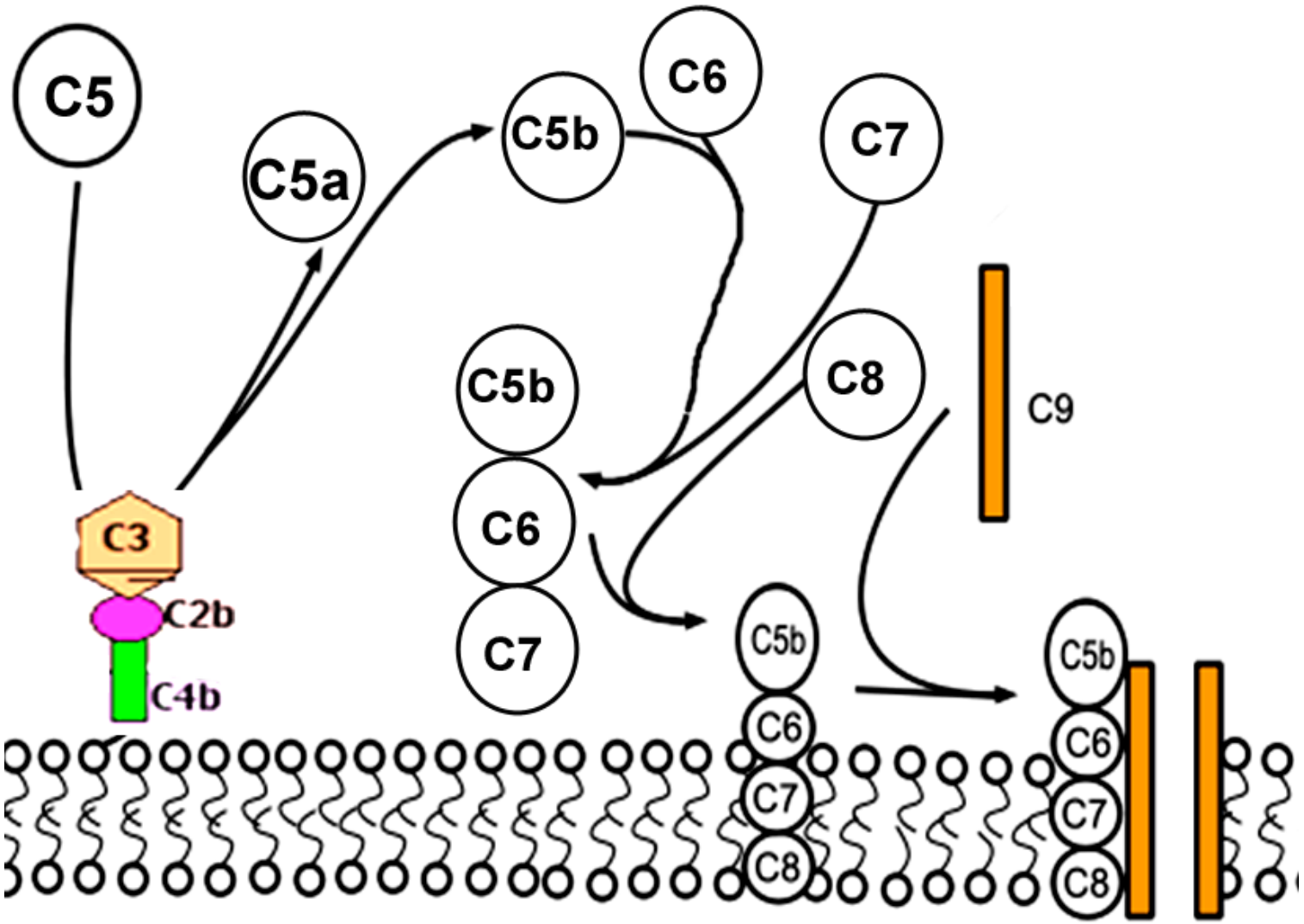


Cleavage of C5 produces:

C5a, which is released into the fluid surroundings where it is a potent anaphylatoxin

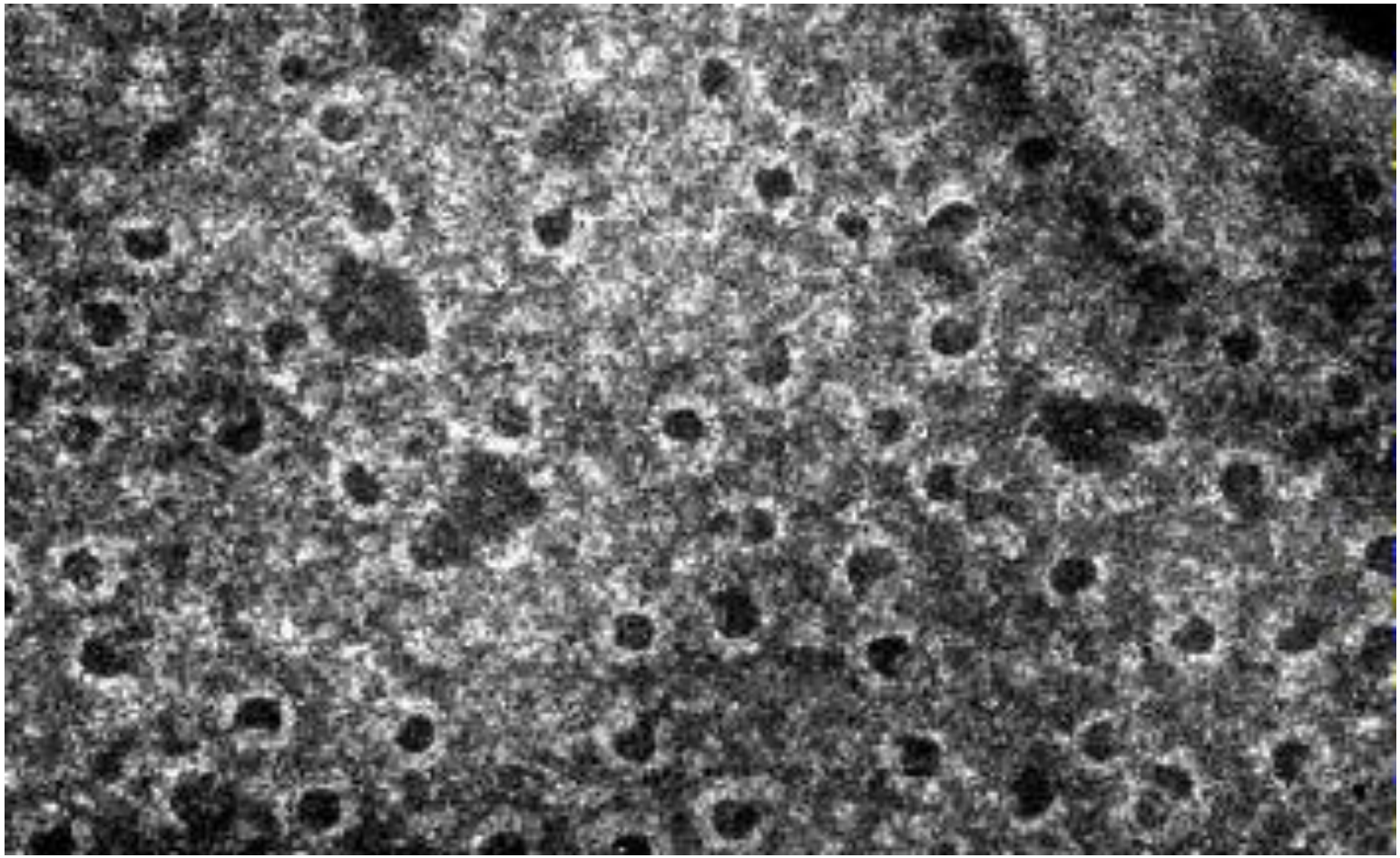
C5b, which serves as the anchor for the assembly of a single molecule each of C6, C7 and C8

Membrane Attack Complex



C5b•6•7•8 guides the polymerization of 18 molecules of C9 inserted into the lipid bilayer passage H2O causing cell lyses.

Membrane Attack Complex



electron micrograph shows holes punched through the cell wall of the bacterium components of the complement system

Cell Biology

Complement System 2

Classical complement pathway

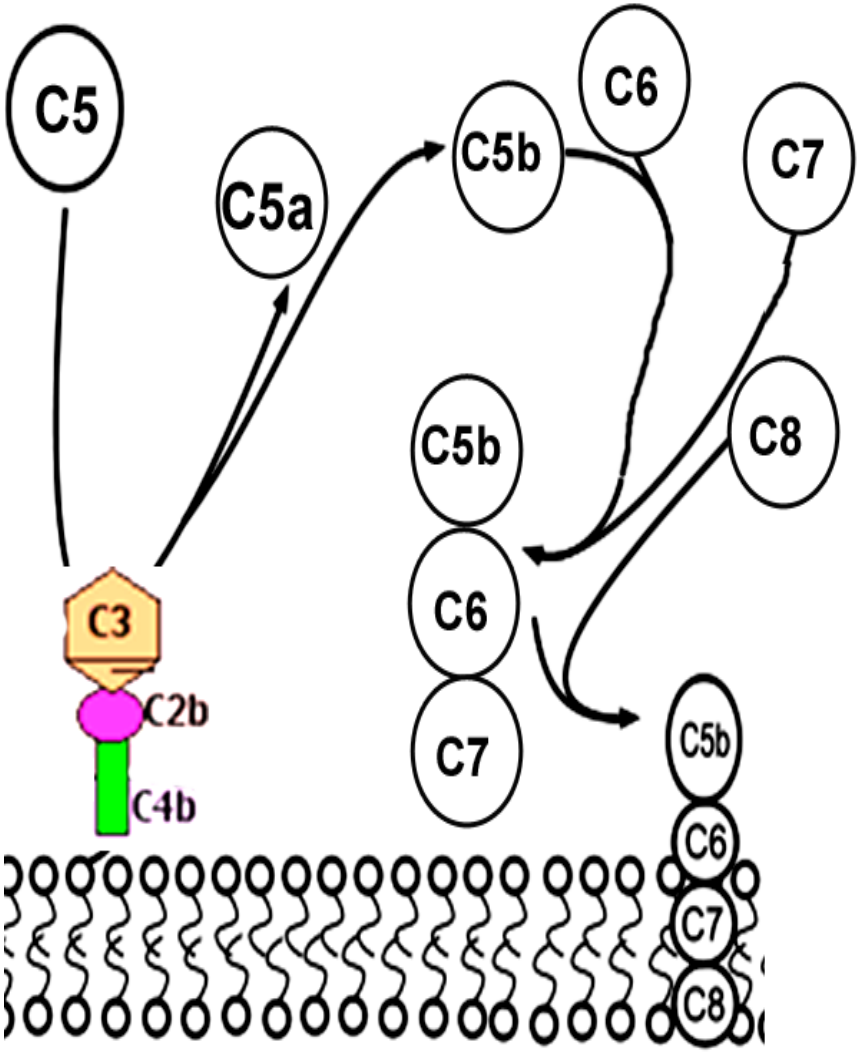
Cleavage of C5

Some of the C3b binds to molecules of C5 creating an allosteric change that exposes them to cleavage by C4b•2b (which is thus a "C3/C5 convertase")

Cleavage of C5 by the C3/C5 convertase initiates the assembly of a set of complement proteins that make up the membrane attack complex.

The membrane attack complex can also be formed by another C5 convertase produce by the "alternative pathway"

Membrane Attack Complex

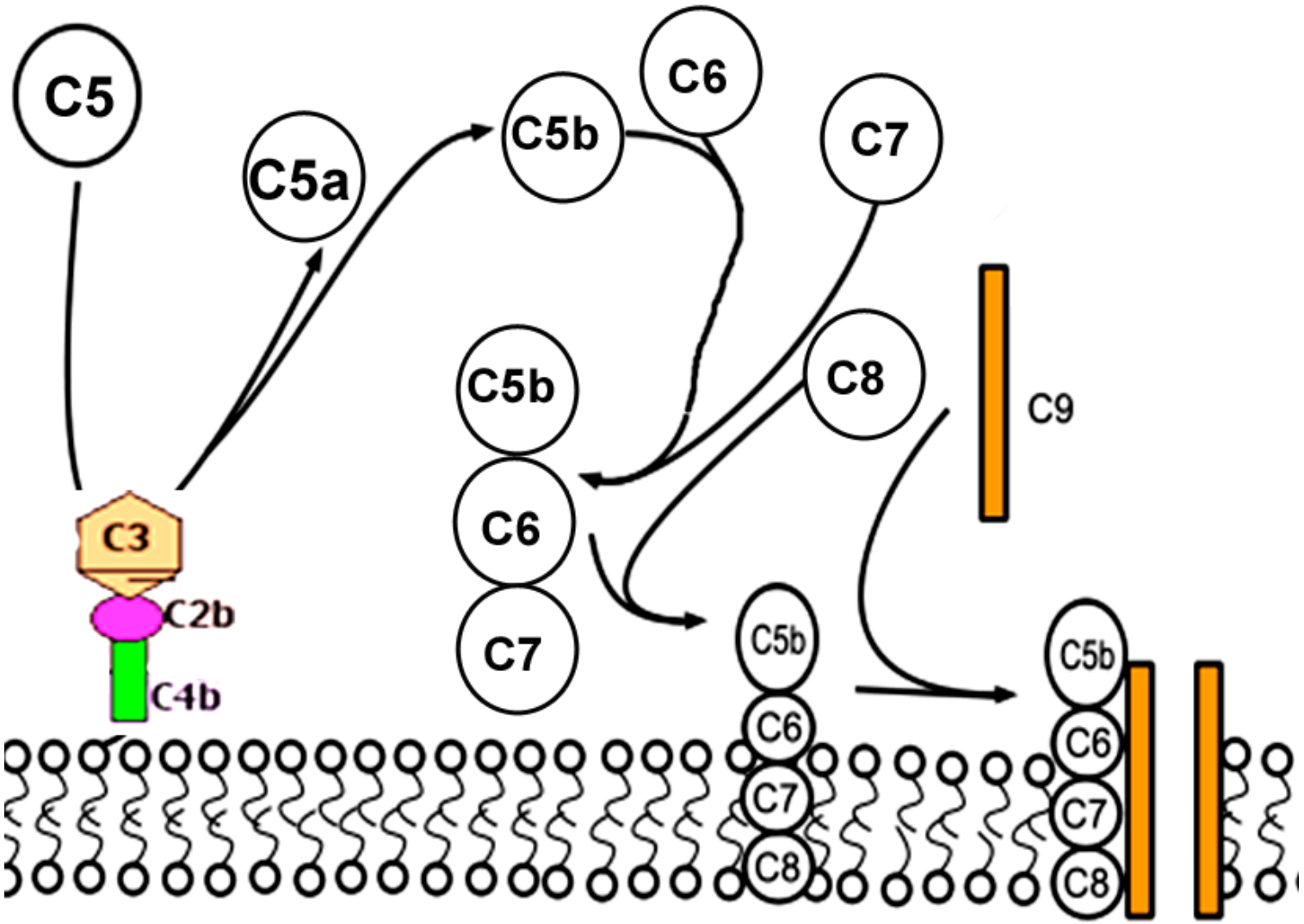


Cleavage of C5 produces:

C5a, which is released into the fluid surroundings where it is a potent anaphylatoxin

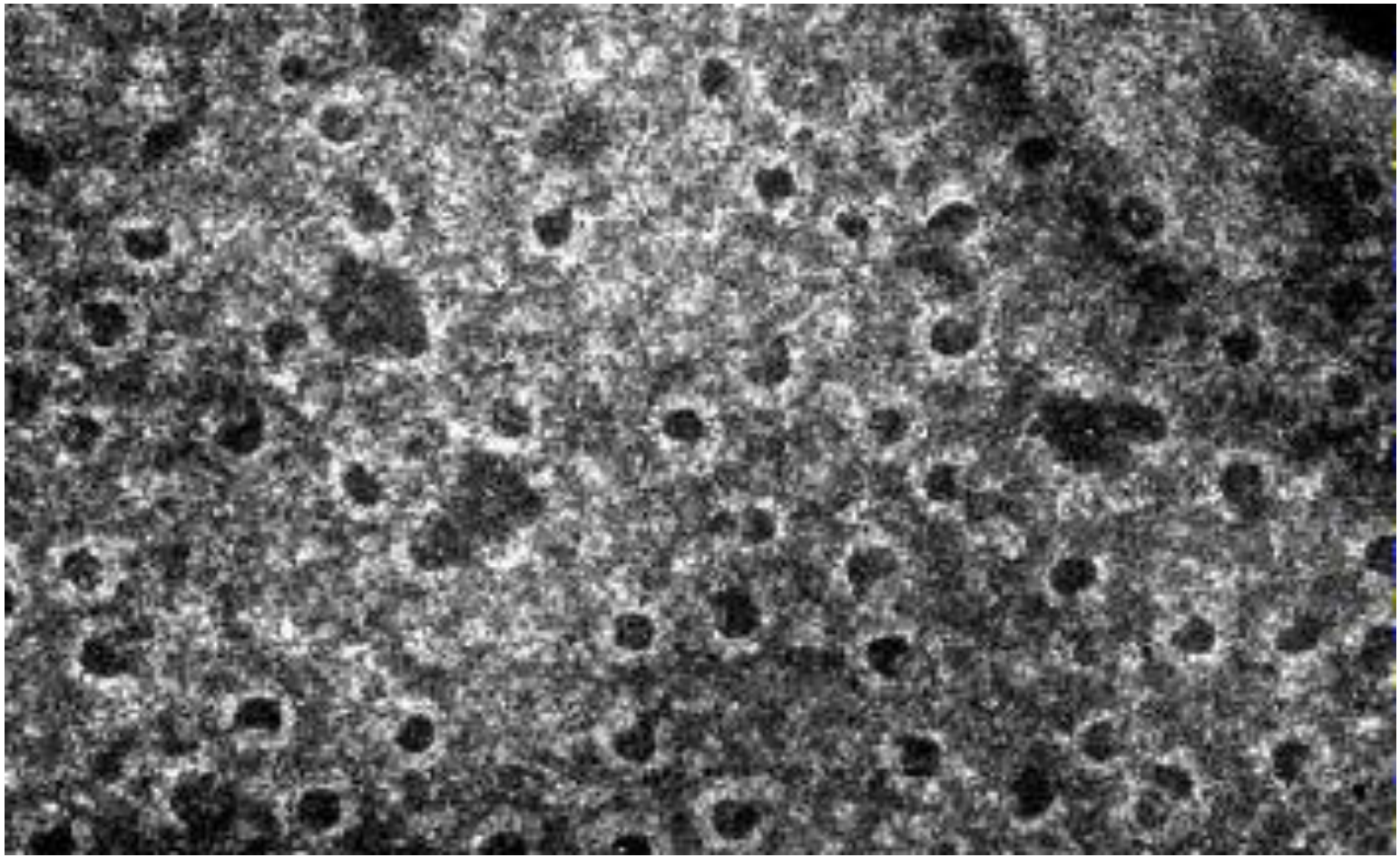
C5b, which serves as the anchor for the assembly of a single molecule each of C6, C7 and C8

Membrane Attack Complex



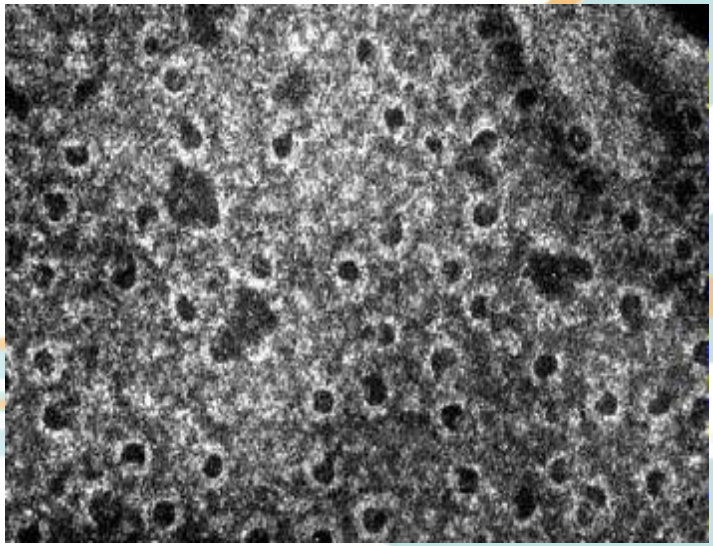
C5b•6•7•8 guides the polymerization of 18 molecules of C9 inserted into the lipid bilayer passage H2O causing cell lyses.

Membrane Attack Complex



electron micrograph shows holes punched through the cell wall of the bacterium components of the complement system

Helper T cells



END

Cell Biology

The Alternative Pathway

The Alternative Pathway

The complement system can also be triggered without antigen-antibody complexes

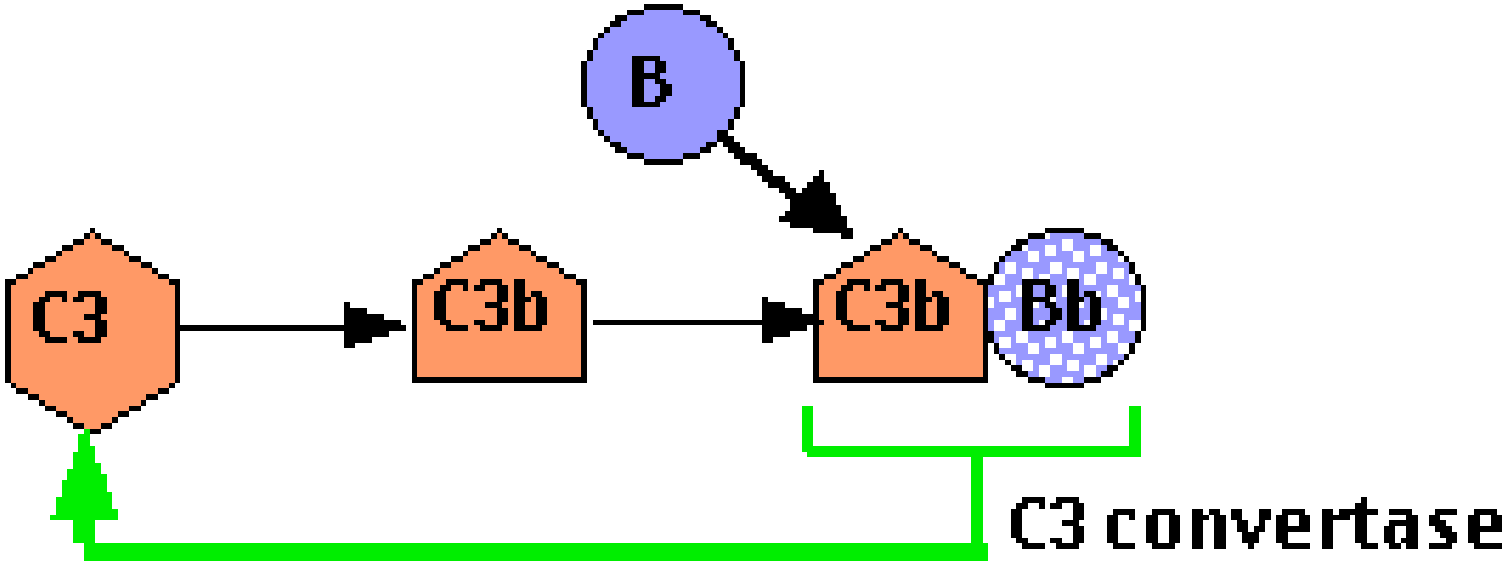
Even in their absence, there is a spontaneous conversion of C3 to C3b

Ordinarily the C3b is quickly inactivated: the C3b binds to inhibitory proteins and sialic acid present on the surface of the body's own cells, and the process is aborted

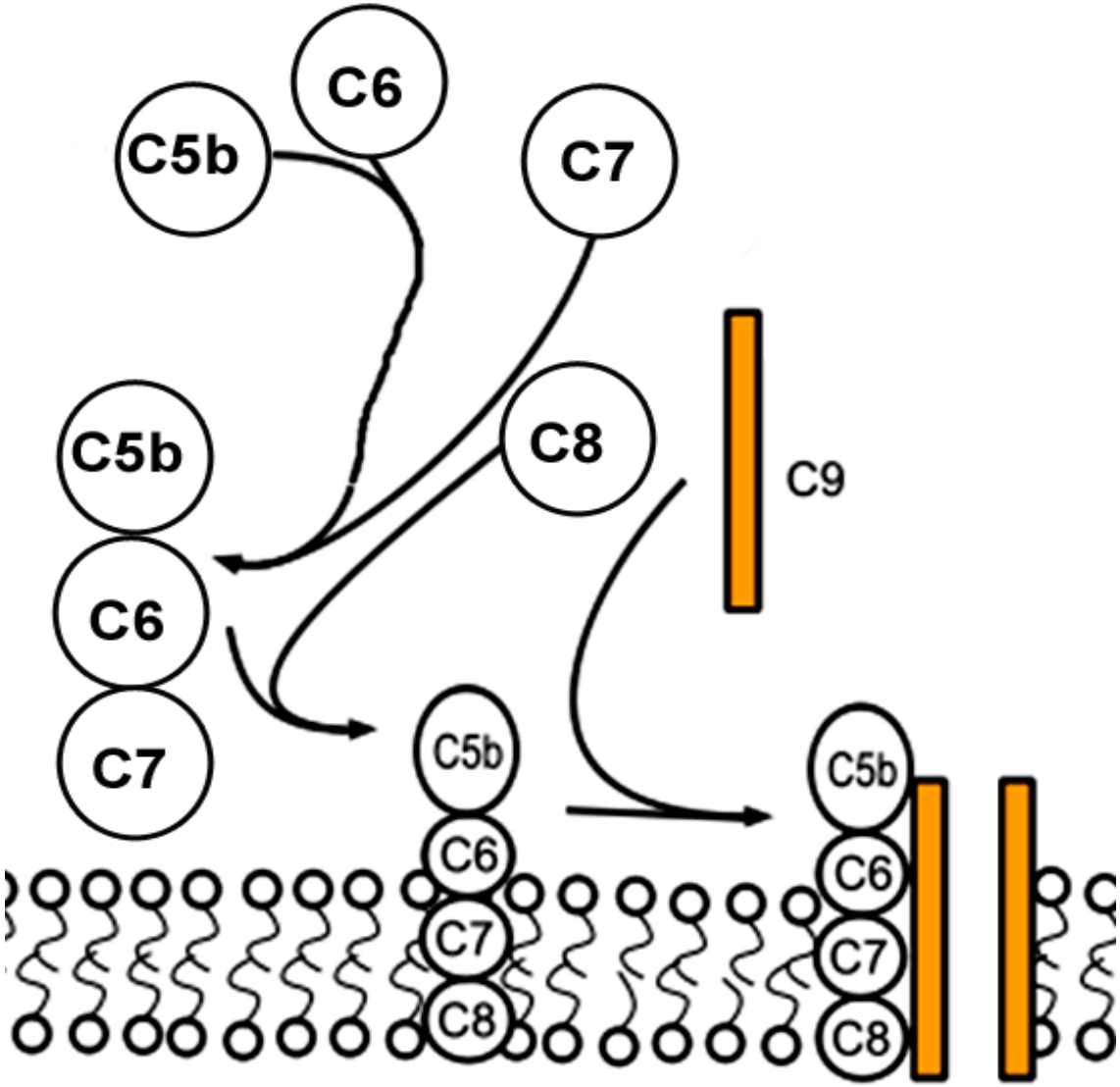
However, bacteria and other foreign materials that may get into the body lack these proteins

The Alternative Pathway

C3b binds a protein called Factor B forming a complex of C3b•Bb which is a C3 convertase, acts on more C3 to form: C3b•Bb•C3b, which is a C5 convertase and can start the assembly of the membrane attack complex & more C3b (amplification massive production of C3b)



The Alternative Pathway



Lectin Pathway

The lectin pathway is homologous to the classical pathway, but with the opsonin, mannan-binding lectin (MBL) and ficolins, instead of C1q

This pathway is activated by binding mannan-binding lectin to mannose residues on the pathogen surface, which activates the MBL-associated serine proteases, MASP-1, MASP-2, MASP-3, which can split C4 into C4a and C4b and C2 into C2a and C2b

C4b and C2a then bind together to form C3-convertase, as in the classical pathway

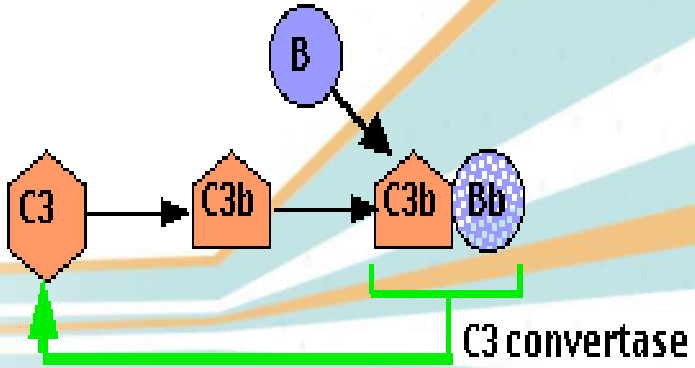
Complement System Regulation

The explosive potential of the complement system requires that it be kept under tight control, at least 12 proteins are known that do this

Factor H removes Bb from the alternative pathway C3 convertase breaking the positive feedback loop. Factor I inactivates C3b

C1 inhibitor (C1INH) binds to sites on activated C1r and C1s shutting down their proteolytic activity, when C1 is activated by antigen-antibody complexes, there is only a brief interval during which it can cleave C4 and C2 before it is deactivated by C1INH.

Helper T cells



END

Cell Biology

Neoplastic Transformations

Neoplastic Transformations

Proto-oncogene

A proto-oncogene is a normal gene that can become an oncogene, either after mutation or increased expression.

Proto-oncogenes code for proteins that help to regulate cell growth and differentiation.

Proto-oncogenes are often involved in signal transduction and execution of mitogenic signals, usually through their protein products.

Upon activation, a proto-oncogene (or its product) becomes a tumor inducing agent, an oncogene.

Neoplastic Transformations

Proto-oncogene

Some oncogenes, usually involved in early stages of cancer development, increase the chance that a normal cell develops into a tumor cell, possibly resulting in cancer

Classes of Oncogenes

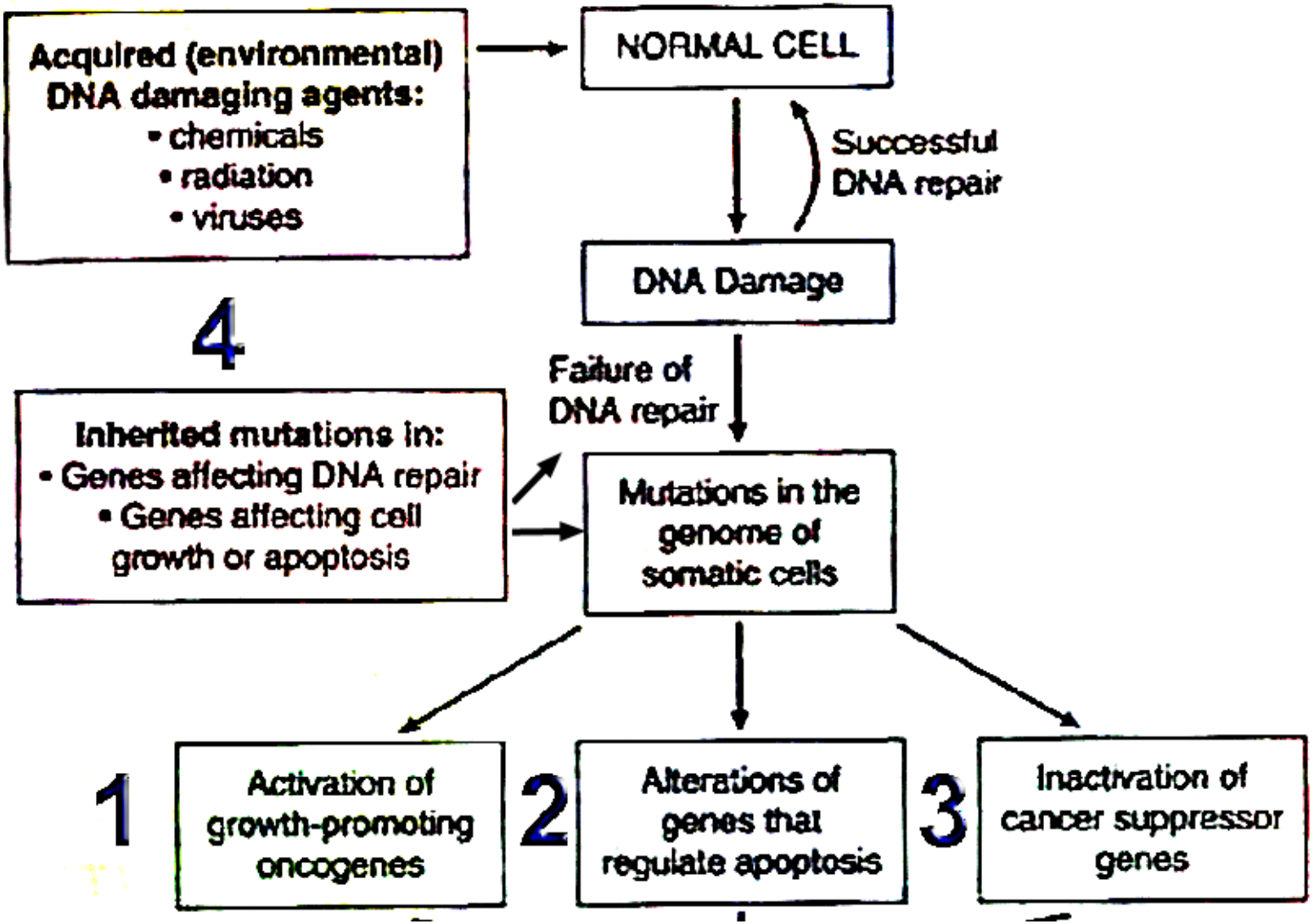
Growth Promoting Oncogenes

Growth-Inhibiting Cancer Suppressor Genes

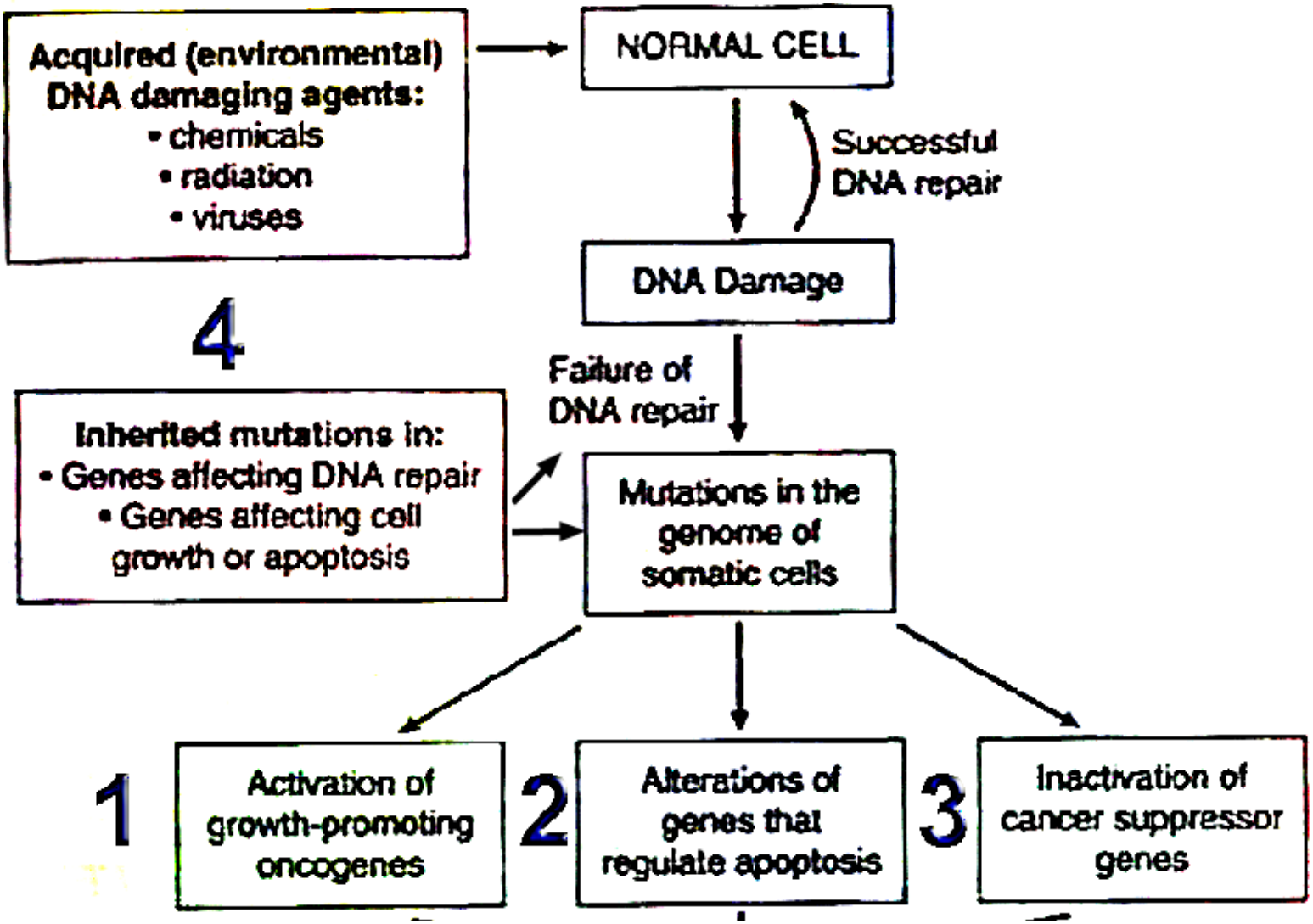
DNA Repairing Genes

Apoptosis Genes

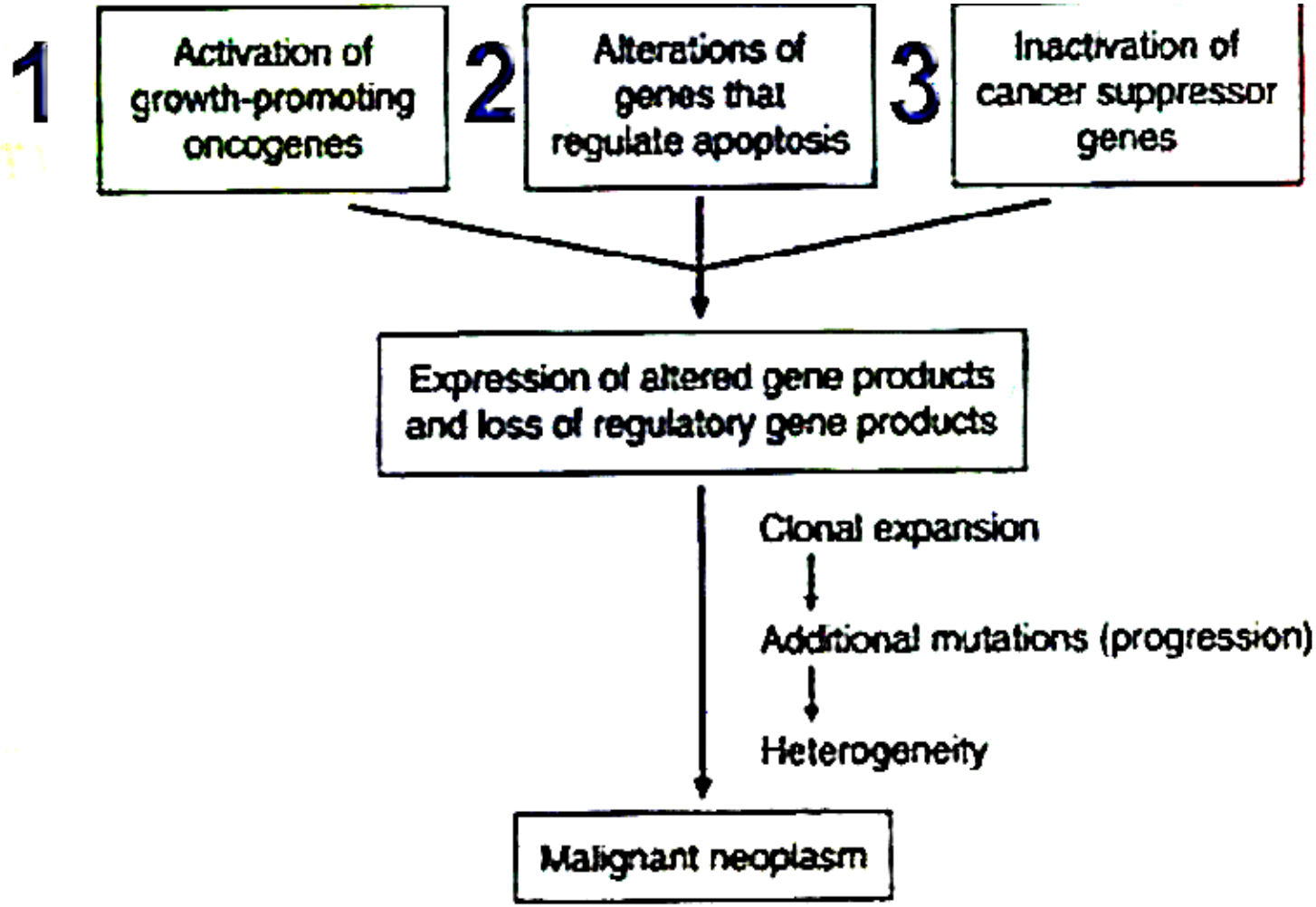
Neoplastic Transformations



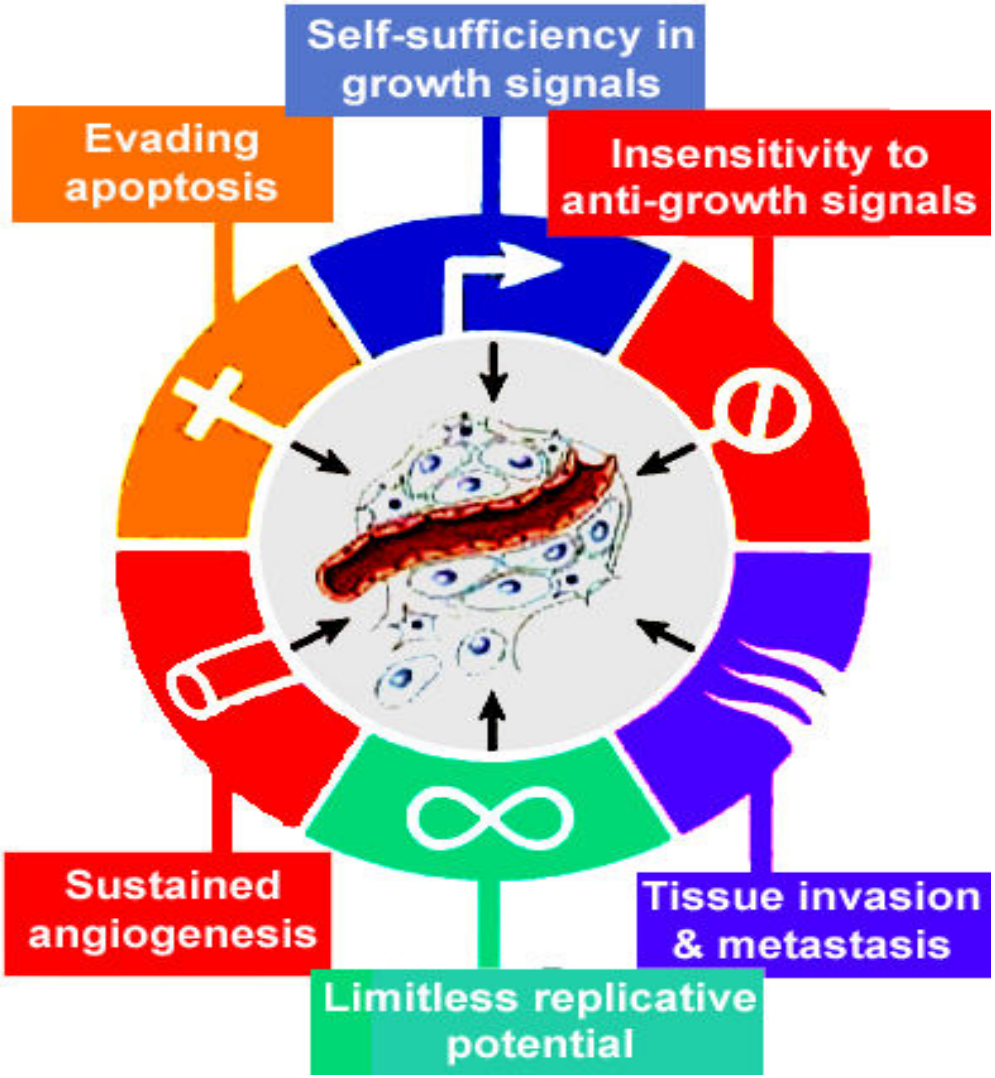
Neoplastic Transformations22



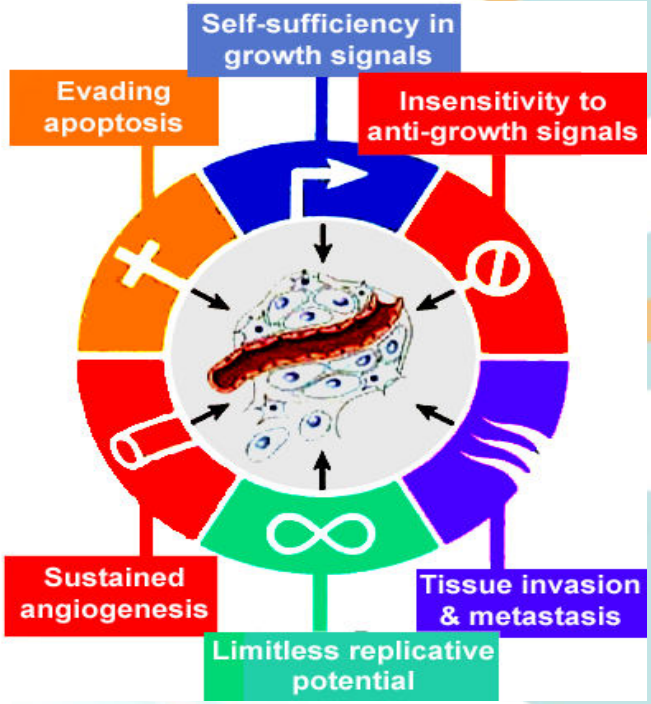
Neoplastic Transformations



Neoplastic Transformations



Neoplastic Transformations



END

Growth Promoting Oncogenes

Growth Promoting Oncogenes

Autonomous Cell Growth In Cancer Cells Is Promoted By Following Elements

a) Growth Factors

b) Growth Factor Receptors

c) Signal-Transducing Proteins (secondary messengers)

d) Nuclear Transcription Factors

e) Cyclins and Cyclin Dependent Kinases (CDKs)

a) Growth Factors

Some cancer cells acquire the ability through mutations to produce growth factors to which they are responsive (autocrine signaling).

Many glioblastomas secrete PDGF and sarcomas make TGF- α. Similar loops are fairly common many types of cancers

b) Growth Factor Receptors

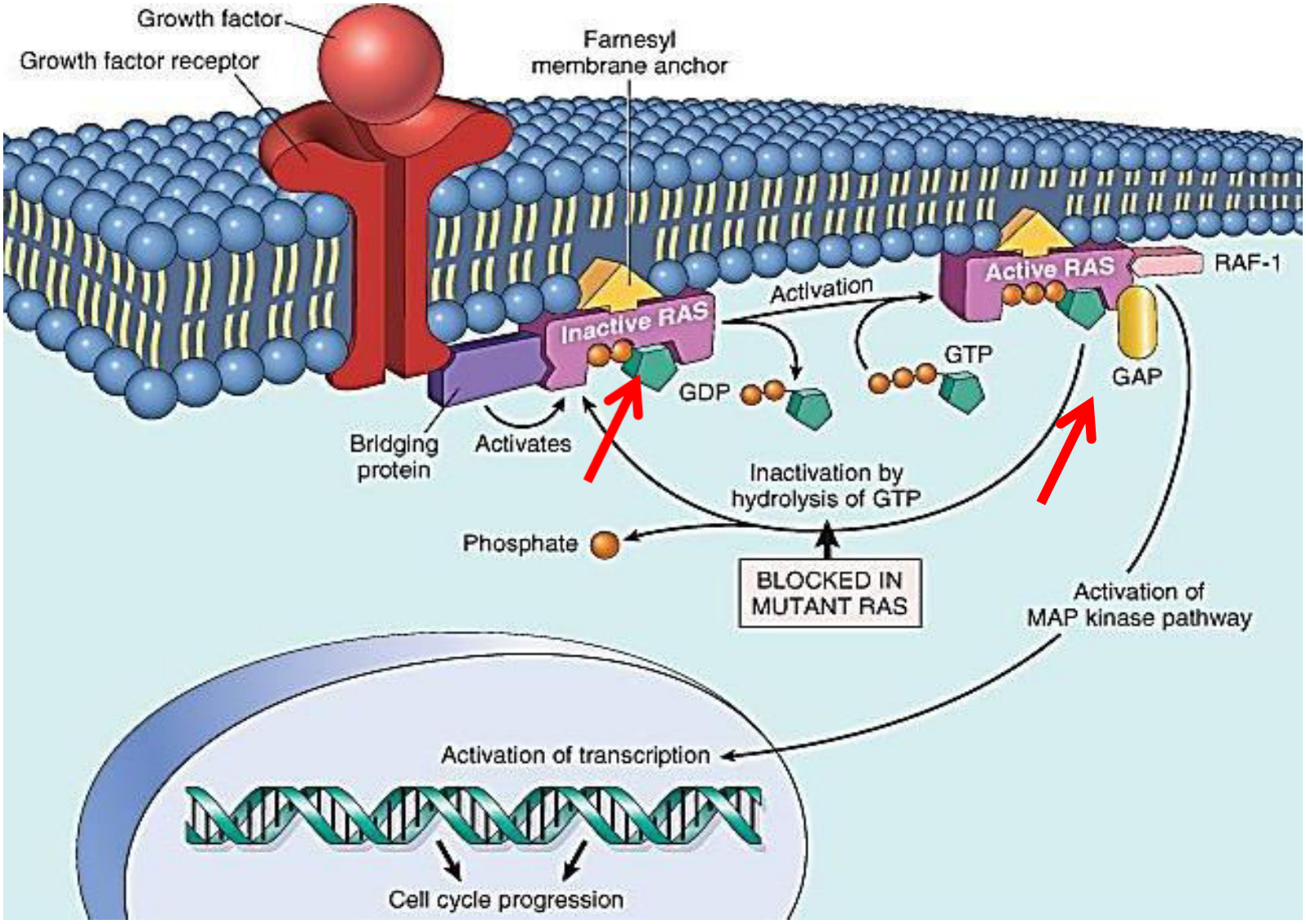
Mutated Growth Factor Receptors continue to deliver continues signals in the absence of the ligand.

EGF receptor ERBB1 is over expressed in 80% of SCC of lung.

ERBB2/HER2 is amplified in 25%-30% of breast cancers and adenocarcinomas of lung , ovary and salivary glands.

These tumours are sensitive to small amounts of growth factors.

c) Signal-Transducing Proteins



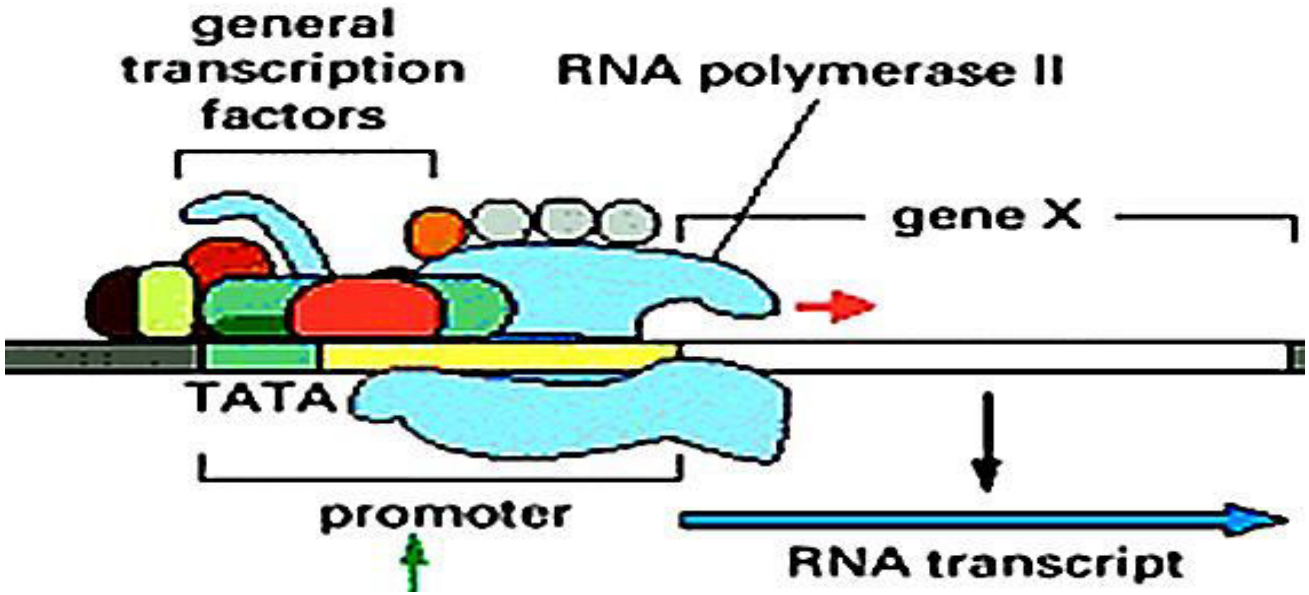
c) Signal-Transducing Proteins

Mutaion of RAS is the most common oncogene abnormality, approximately 30% of all human tumors contain mutated version of RAS gene.

GAP/neurofibramin 1 (a RAS deactivating protein) is mutated in neurofibramatosis type 1.

BCR-ABL hybrid gene activates RAS pathways (like p53 Normal ABL gene also promotes apoptosis via NLS domain).

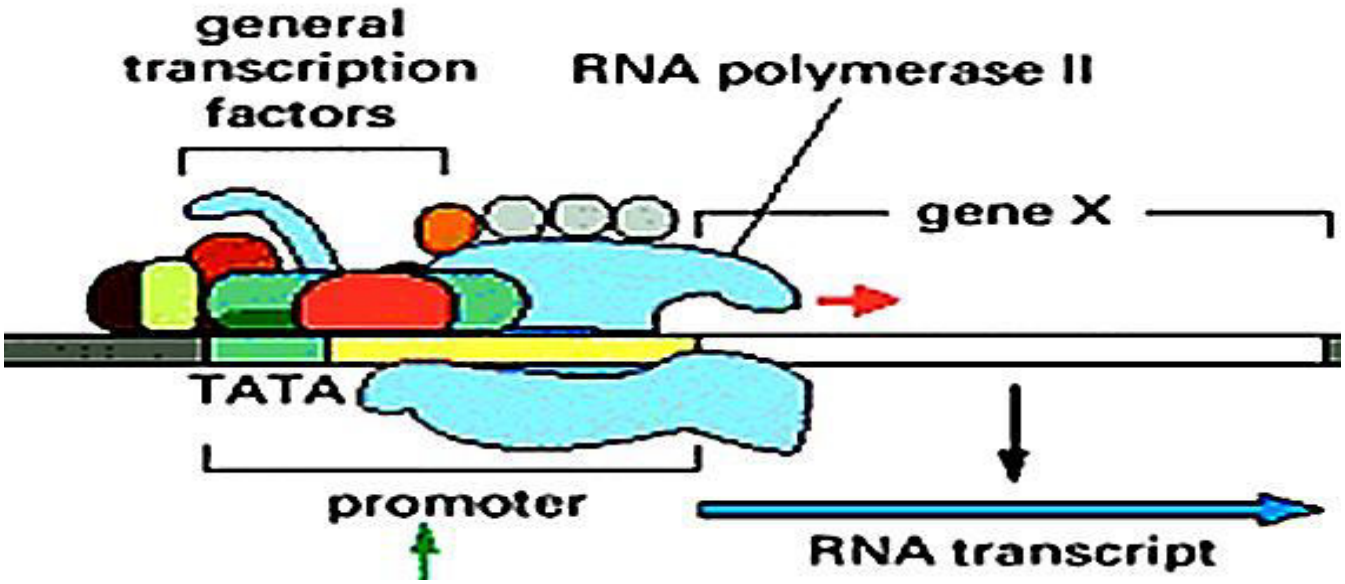
d) Nuclear Transcription Factors



MYC and the polymerase assemble on a specific DNA sequence and drive transcription of genes.

MYC protooncogene is expressed in virtually all the cells, and the MYC protein is rapidly induced when cells receive a signal to divide.

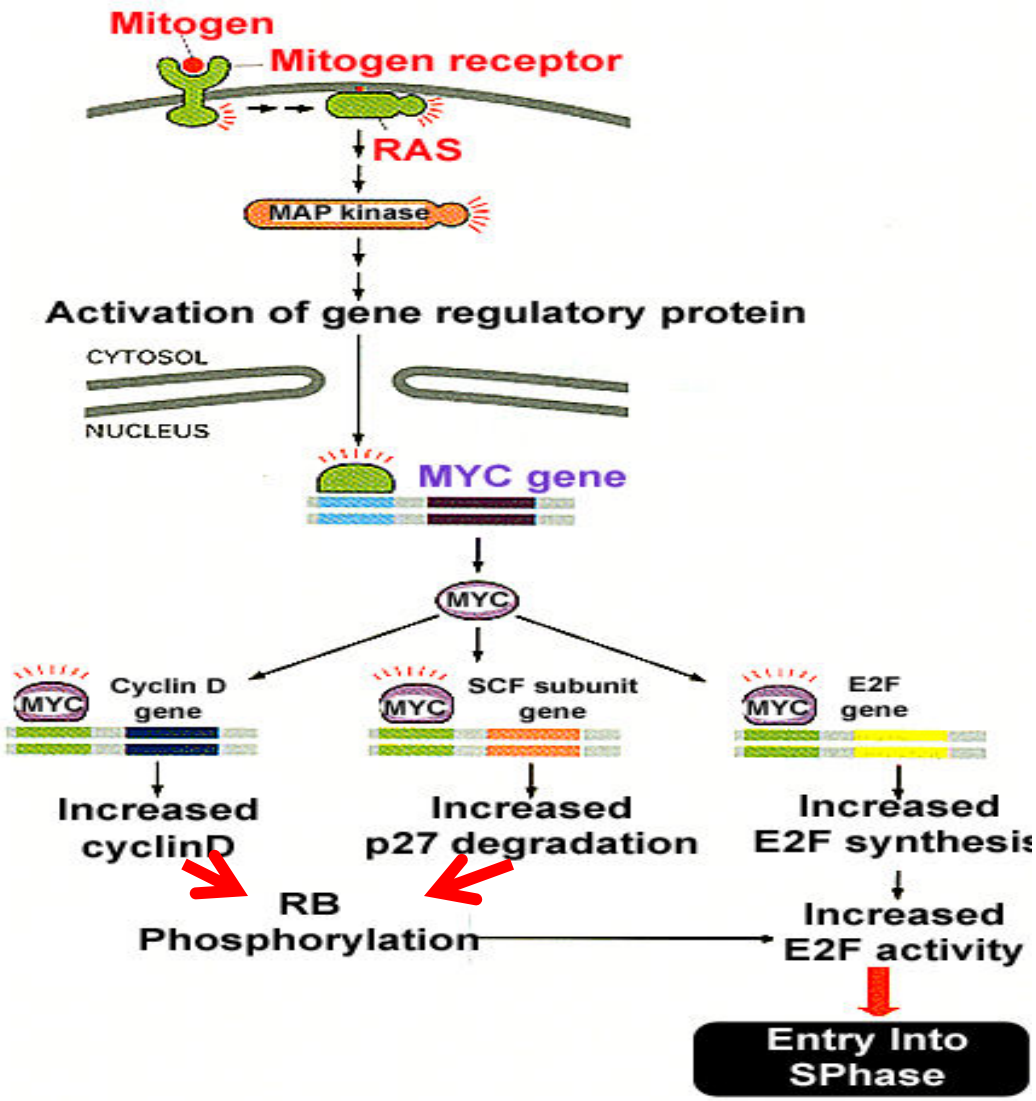
d) Nuclear Transcription Factors



MYC binds DNA and causes transcriptional activation of many growth related proteins.

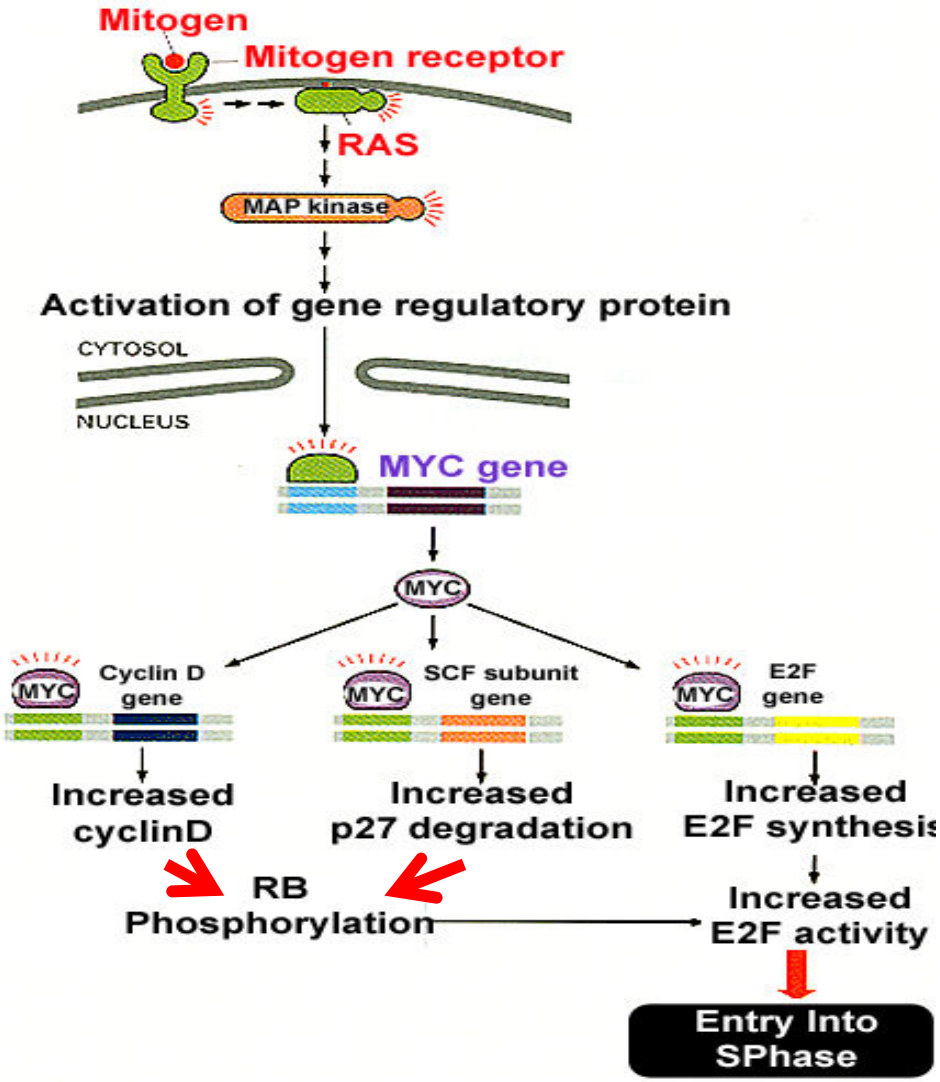
Dysregulation of of MYC gene occurs in Burkitt Lymphoma (B-cell tumor); MYC is also amplified in many other cancers.

MYC Gene



The binding of mitogens to cell-surface receptors leads to the activation of Ras and a MAP kinase cascade.

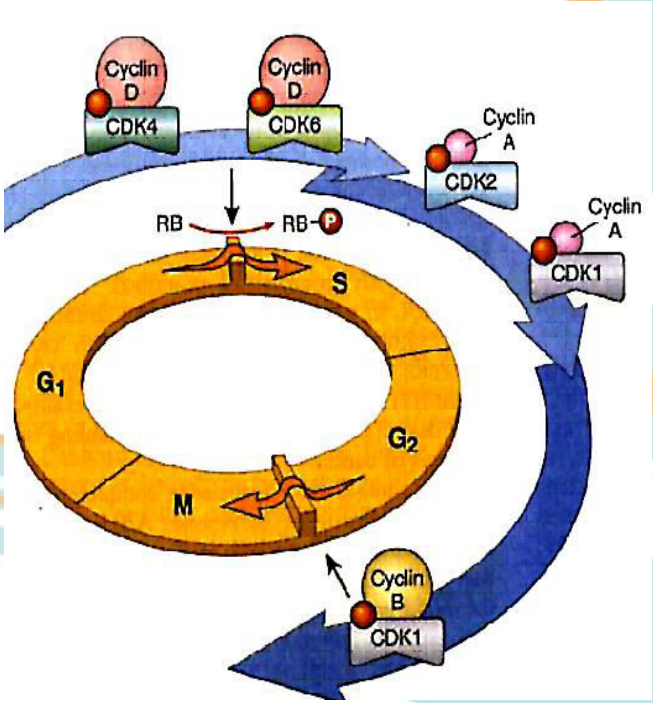
MYC Gene



One effect of this pathway is the increased production of the gene regulatory protein Myc.

Myc increases the transcription of several genes, including the gene encoding cyclin D

Growth Promoting Oncogenes



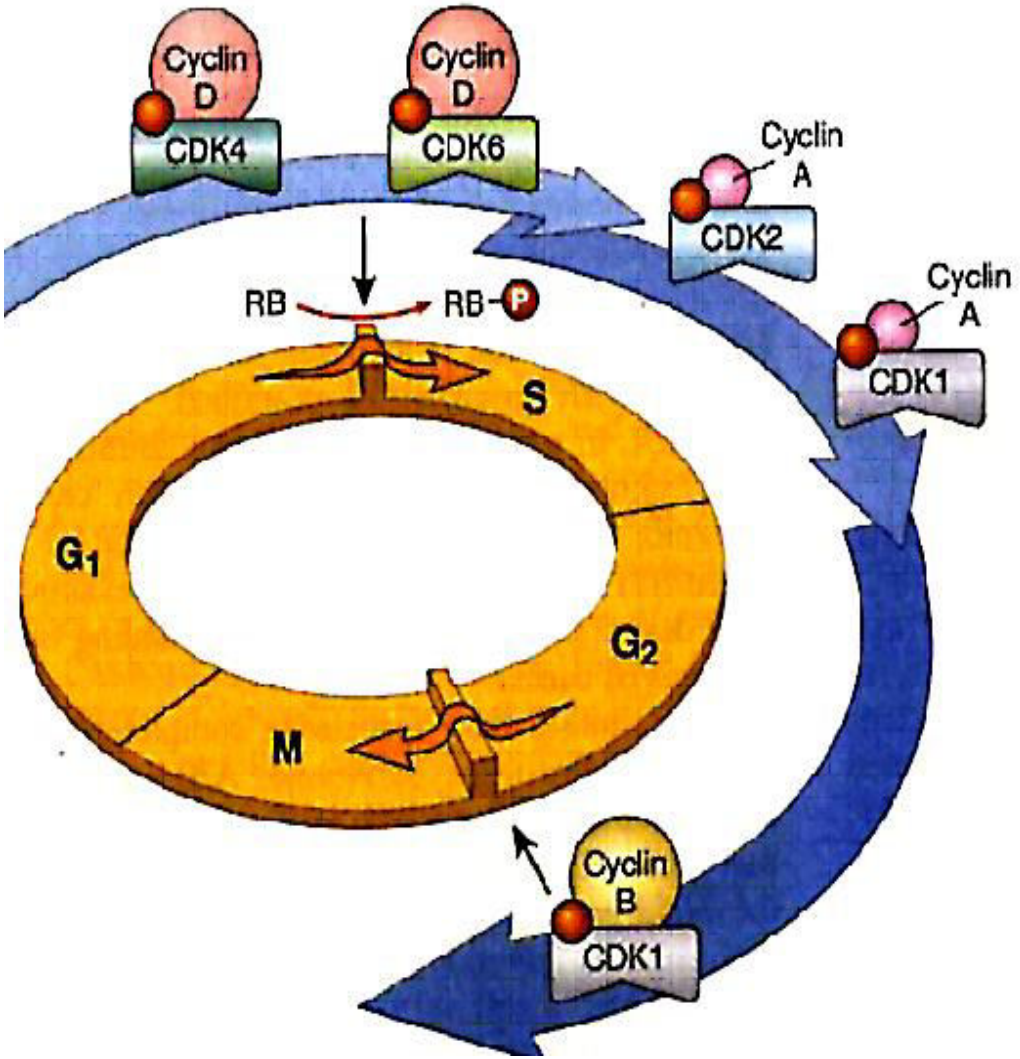
END

Cell Biology

Growth Promoting Oncogenes

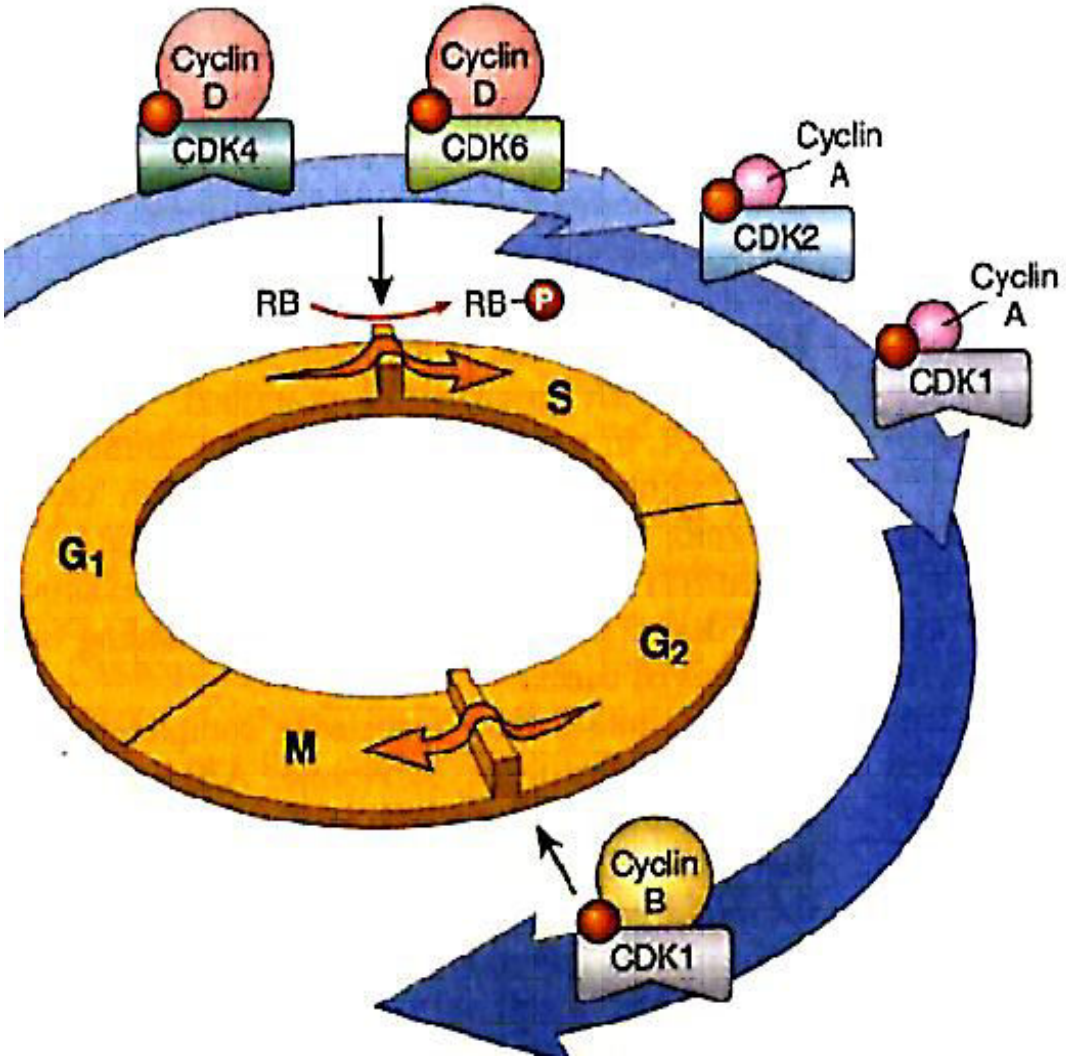
Cyclins & CDKs

e) Cyclins & CDKs



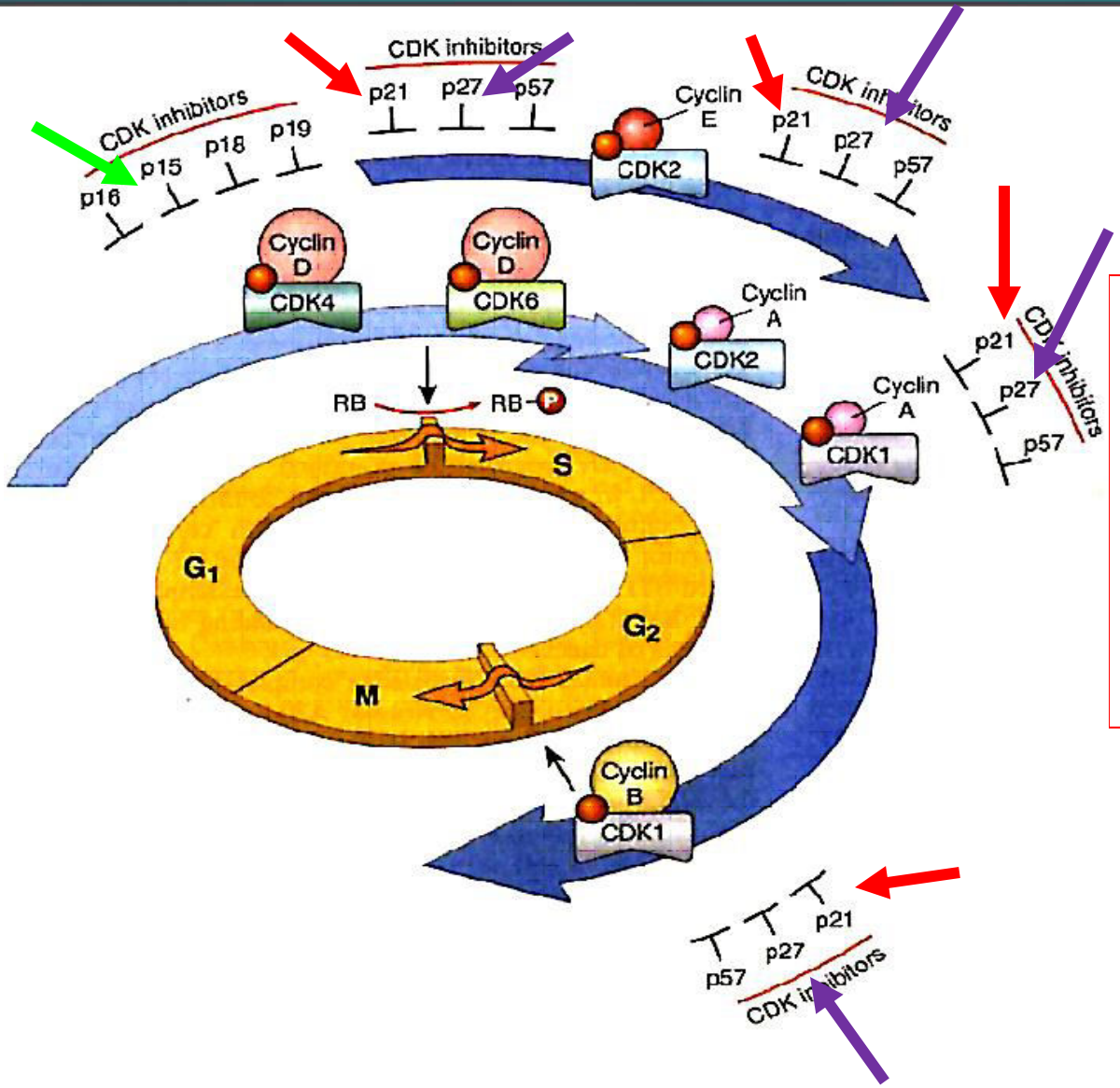
CDKs Phosphorilate crucial **target** proteins & are **expressed constitutively** during cell cycle in **inactive** form however **cyclins** are synthesized during **specific phases** of cell cycle.

e) Cyclins & CDKs



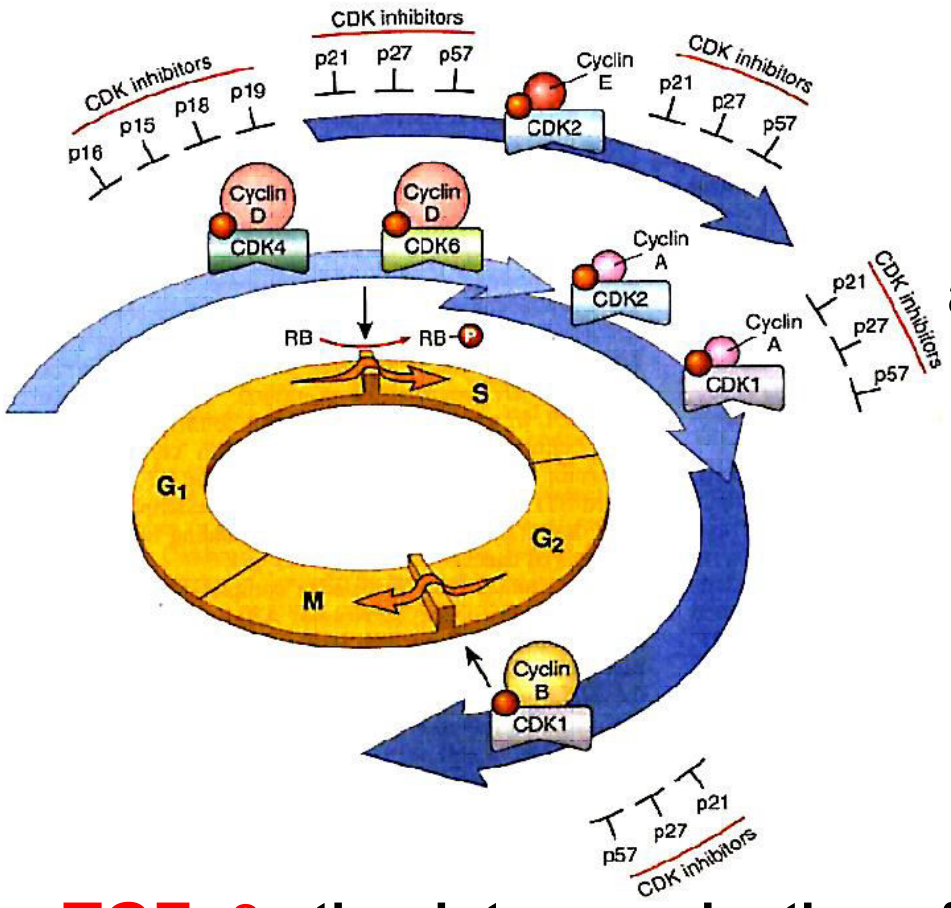
Activated CDK4 and CDK6 phosphorylate RB which over come the hurdle of G1 to S hurdle

e) Cyclins & CDKs



The activity of CDKs is regulated by CDK inhibitors (CDKIs)

e) Cyclins & CDKs

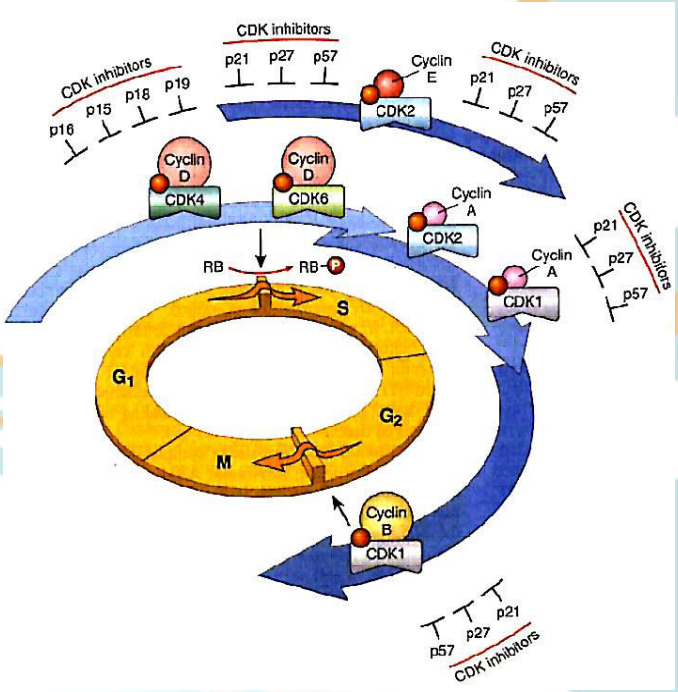


Cyclin D genes are over expressed in many cancers, e.g. affecting the breast, esophagus, liver & subset of lymphomas

Mutations affecting other cyclins and CDKs also occur in some neoplasms.

TGF- β stimulates production of p15 and inhibits the transcription of CDK2, CDK4 and cyclin A

Growth Promoting Oncogenes



END

Cell Biology

DNA Repairing Genes

TP53/ P53

p53 has many anti-cancer mechanisms:

It can activate DNA repair proteins when DNA has sustained damage.

p53 can also arret the cell cycle at the G1/S regulation point on DNA damage recognition (if it holds the cell here for long enough, the DNA repair proteins will have time to fix the damage & the cell will be allowed to continue the cell cycle.)

It can initiate apoptosis, the programmed cell death, if the DNA damage proves to be irreparable.

TP53/ P53

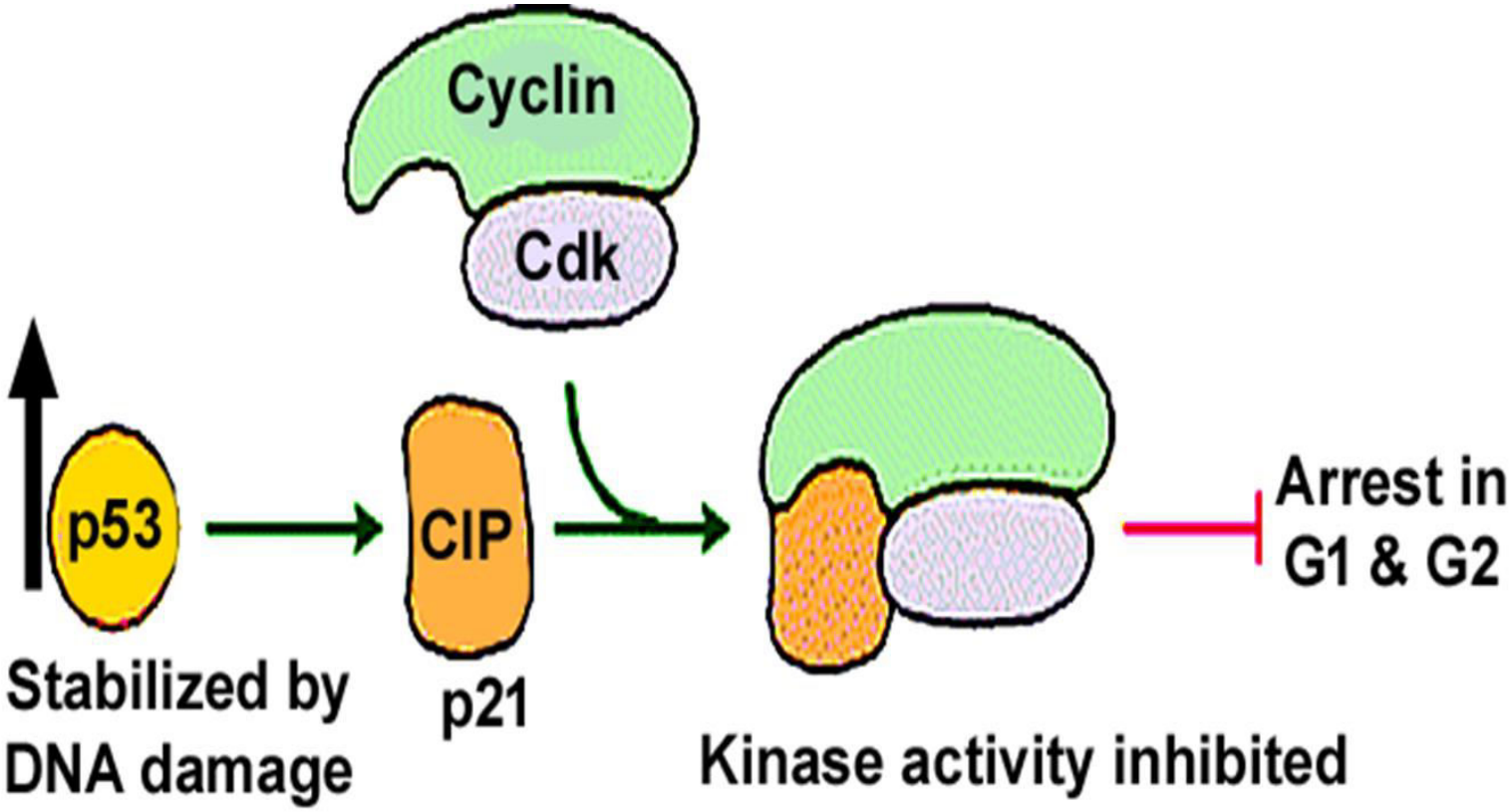
p53 is one of the most commonly mutated genes in human cancers (it can sense DNA damage via other proteins e.g. ATM).

p53 has an anti-proliferative effects and also regulates apoptosis.

The normally unstable p53 protein is stabilized by damaged DNA, so its concentration increases.

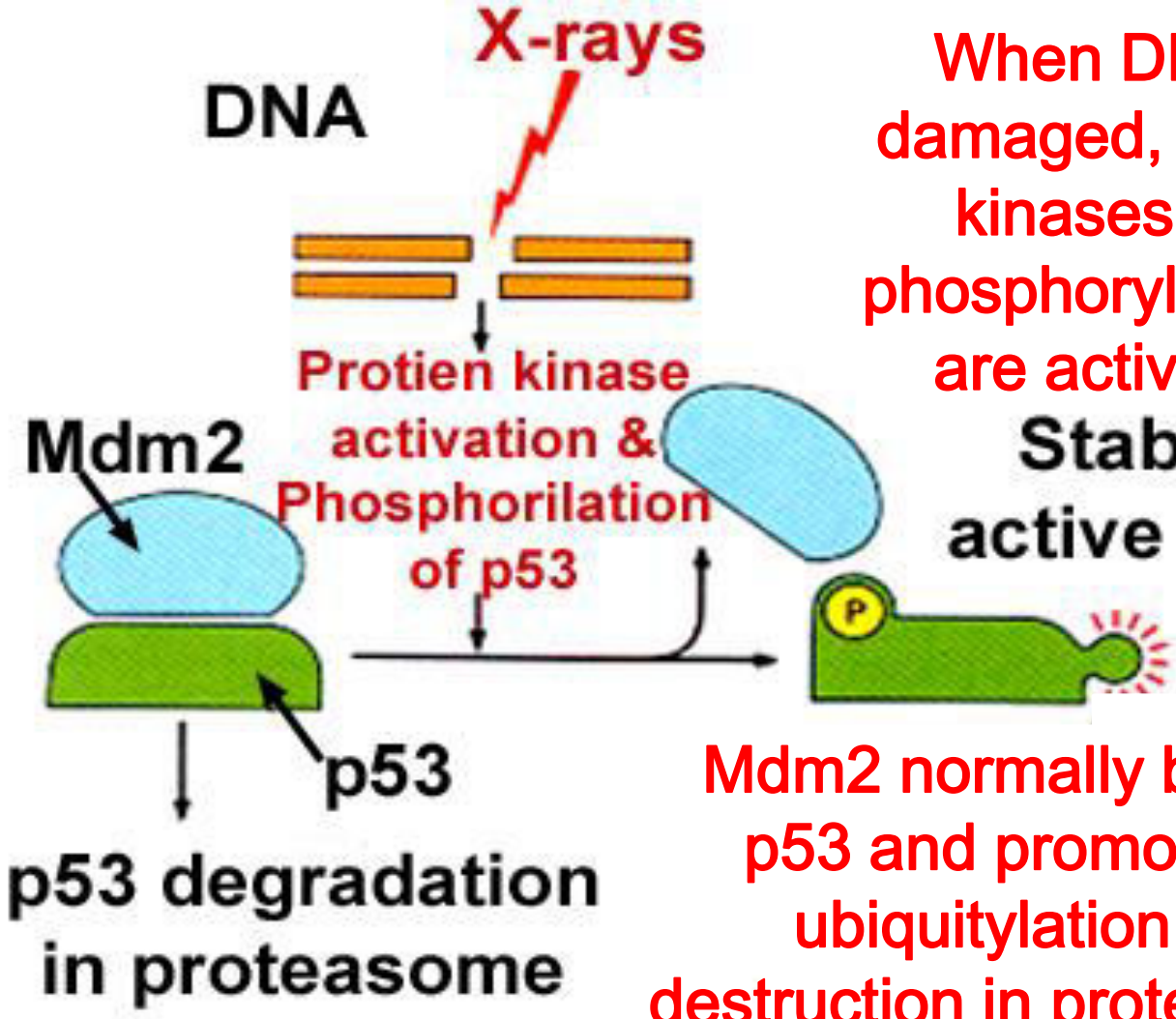
Acting as a transcription factor, p53 induces expression of p21, a cyclin-kinase inhibitor that inhibits all Cdks (Cdk1-, Cdk2-, Cdk4-, and Cdk6-cyclin complexes).

TP53/ P53



Binding of p21 to these Cdk-cyclin complexes leads to cell cycle arrest in G1 and G2.

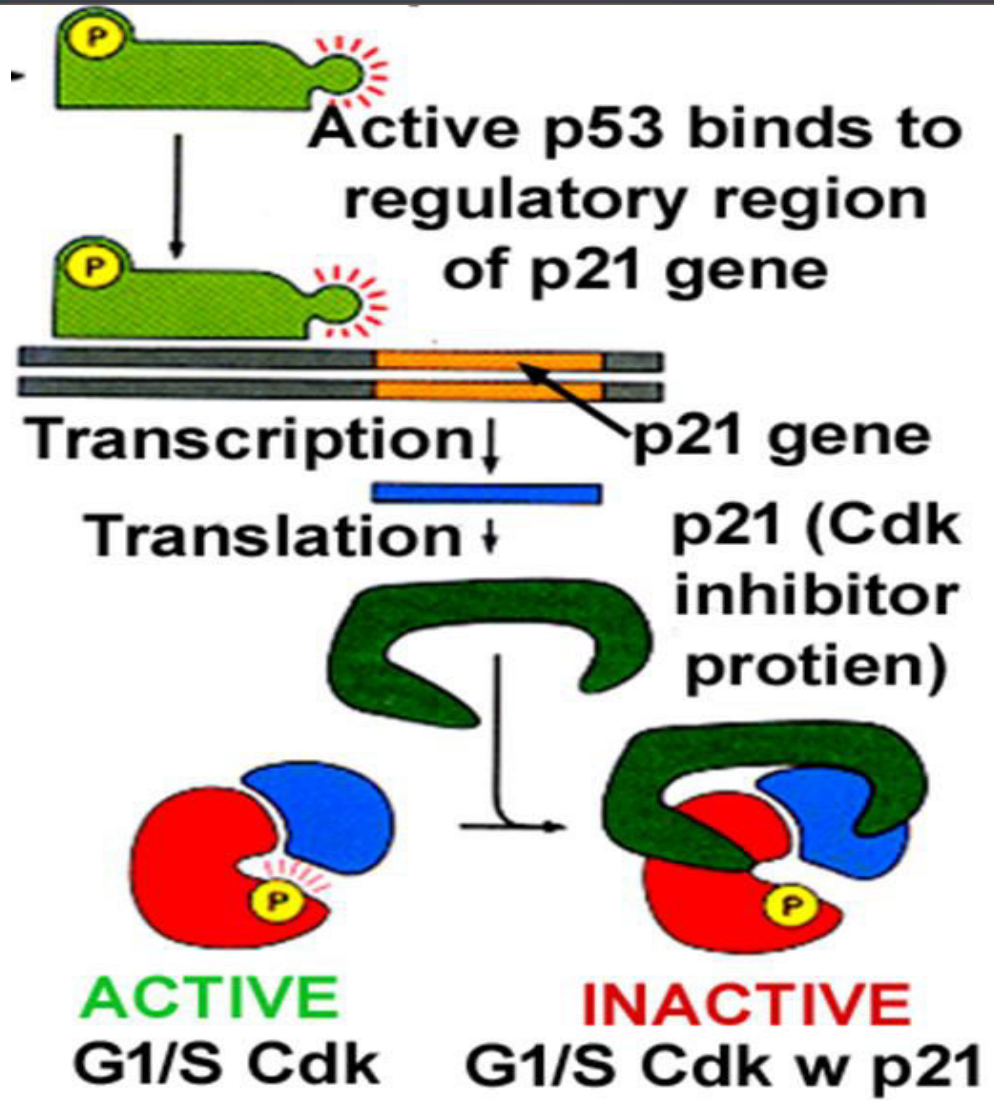
TP53/ P53



When DNA is damaged, protein kinases that phosphorylate p53 are activated.

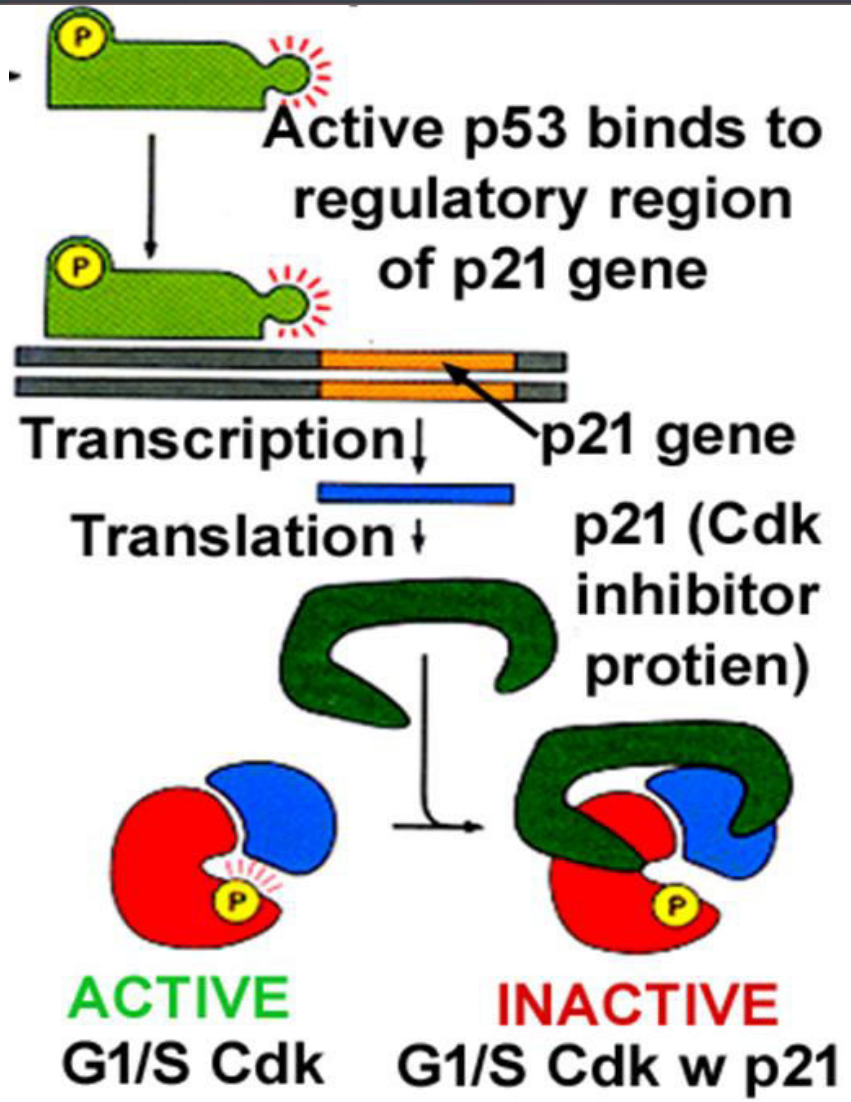
Mdm2 normally binds to p53 and promotes its ubiquitylation and destruction in proteasomes.

TP53/ P53



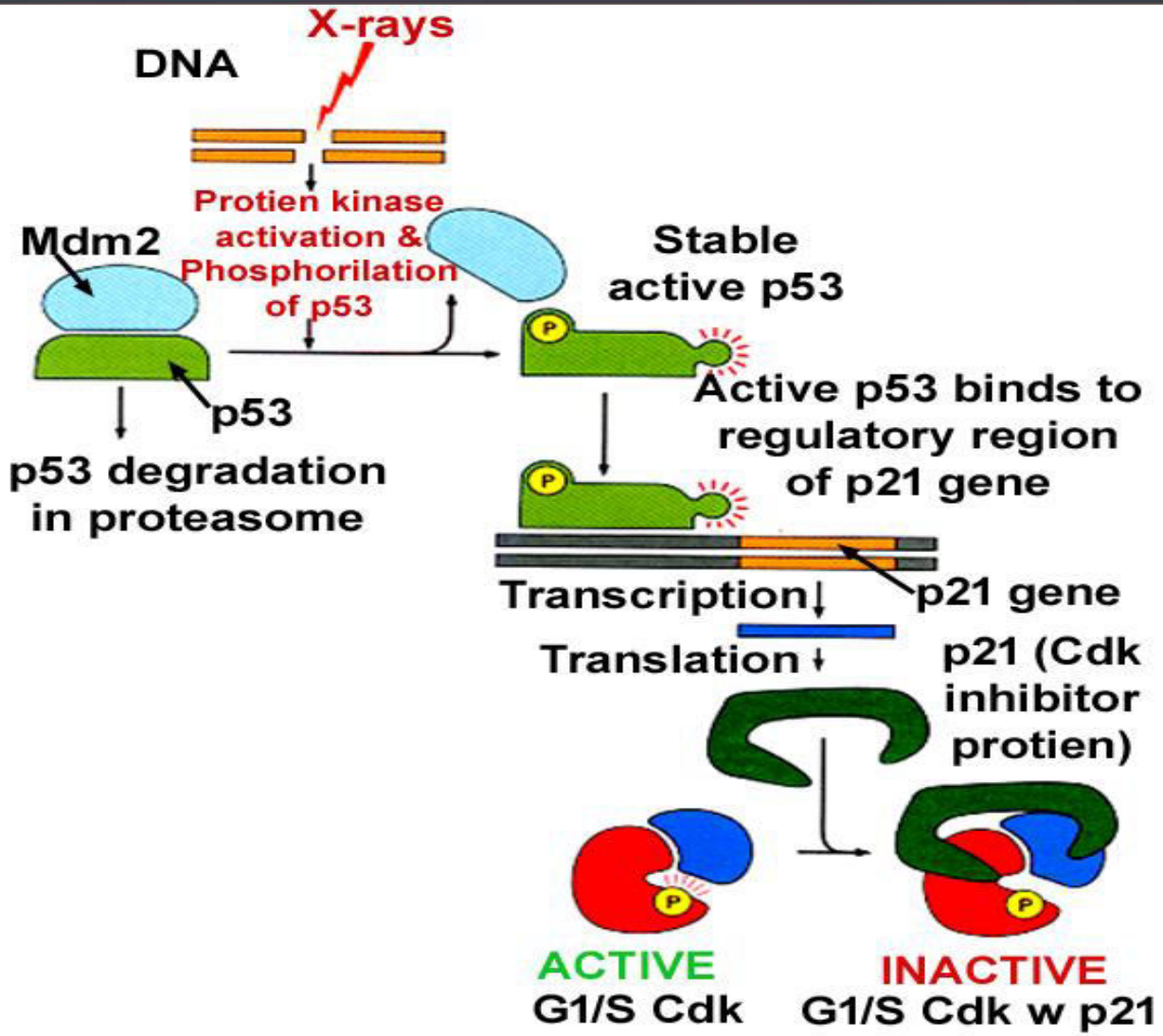
Phosphorylation of p53 blocks its binding to Mdm2; as a result, p53 accumulates to high levels and stimulates transcription of the gene that encodes the CKI protein p21.

TP53/ P53

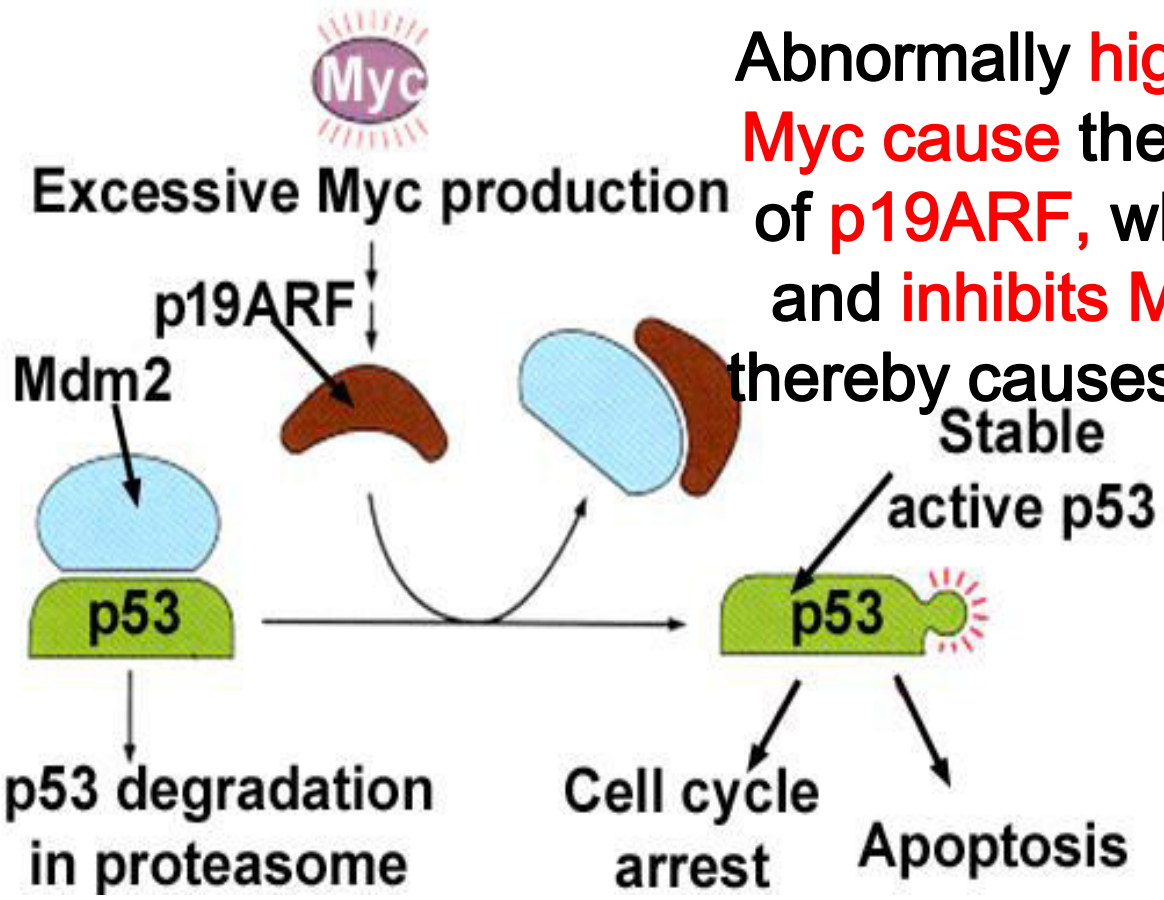


The p21 binds and inactivates G1/S-Cdk and S-Cdk complexes, arresting the cell in G1. In some cases, DNA damage also induces either the phosphorylation of Mdm2 or a decrease in Mdm2 production, which causes an increase in p53 .

TP53/ P53


X-rays
DNA
Protien kinase activation & Phosphorilation of p53
Mdm2
Stable active p53
p53
p53 degradation in proteasome
Active p53 binds to regulatory region of p21 gene
Transcription
p21 gene
Translation
p21 (Cdk inhibitor protien)
ACTIVE
G1/S Cdk
INACTIVE
G1/S Cdk w p21

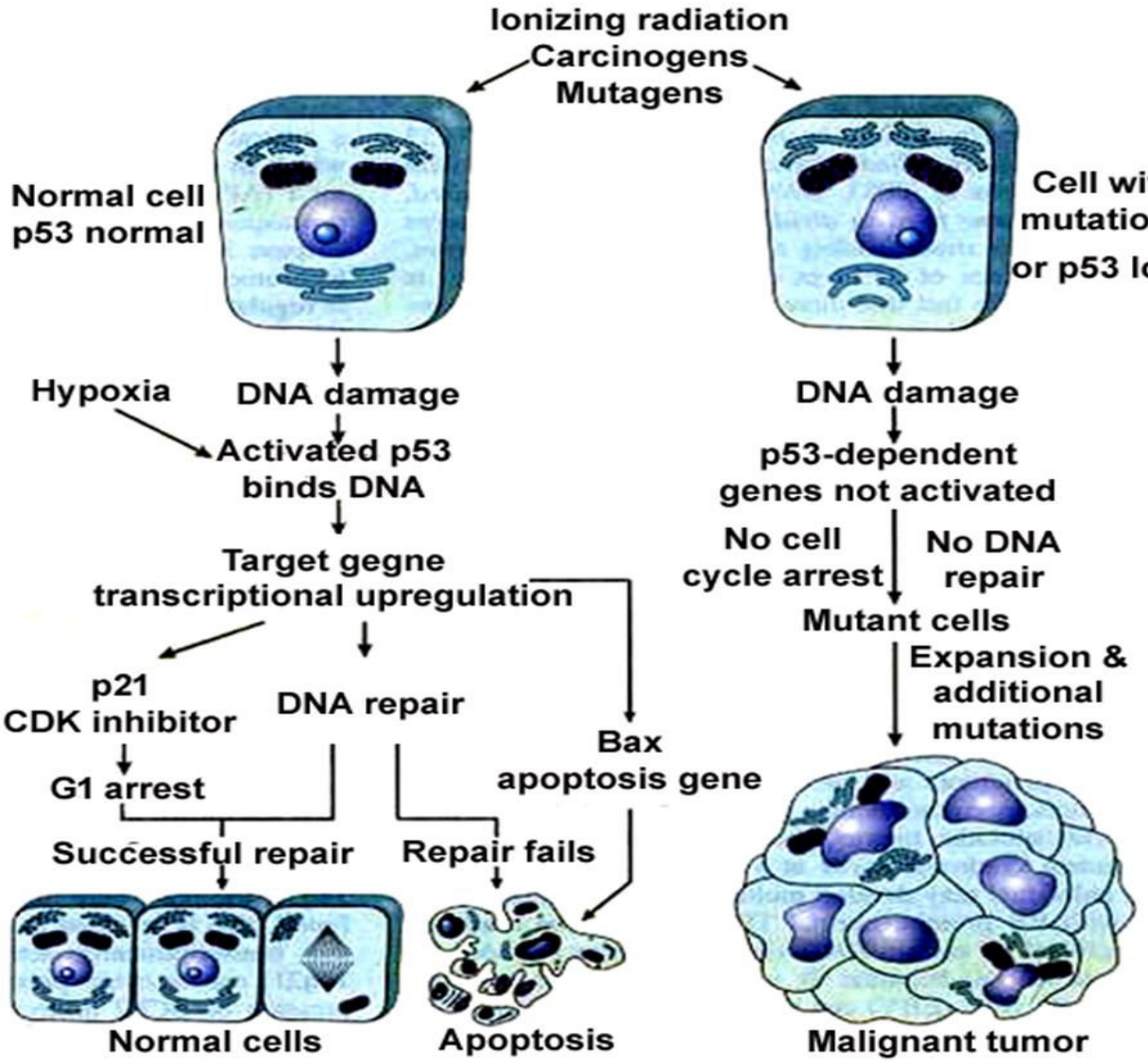
TP53/ P53



Abnormally high levels of Myc cause the activation of p19ARF, which binds and inhibits Mdm2 and thereby causes increased p53 levels.

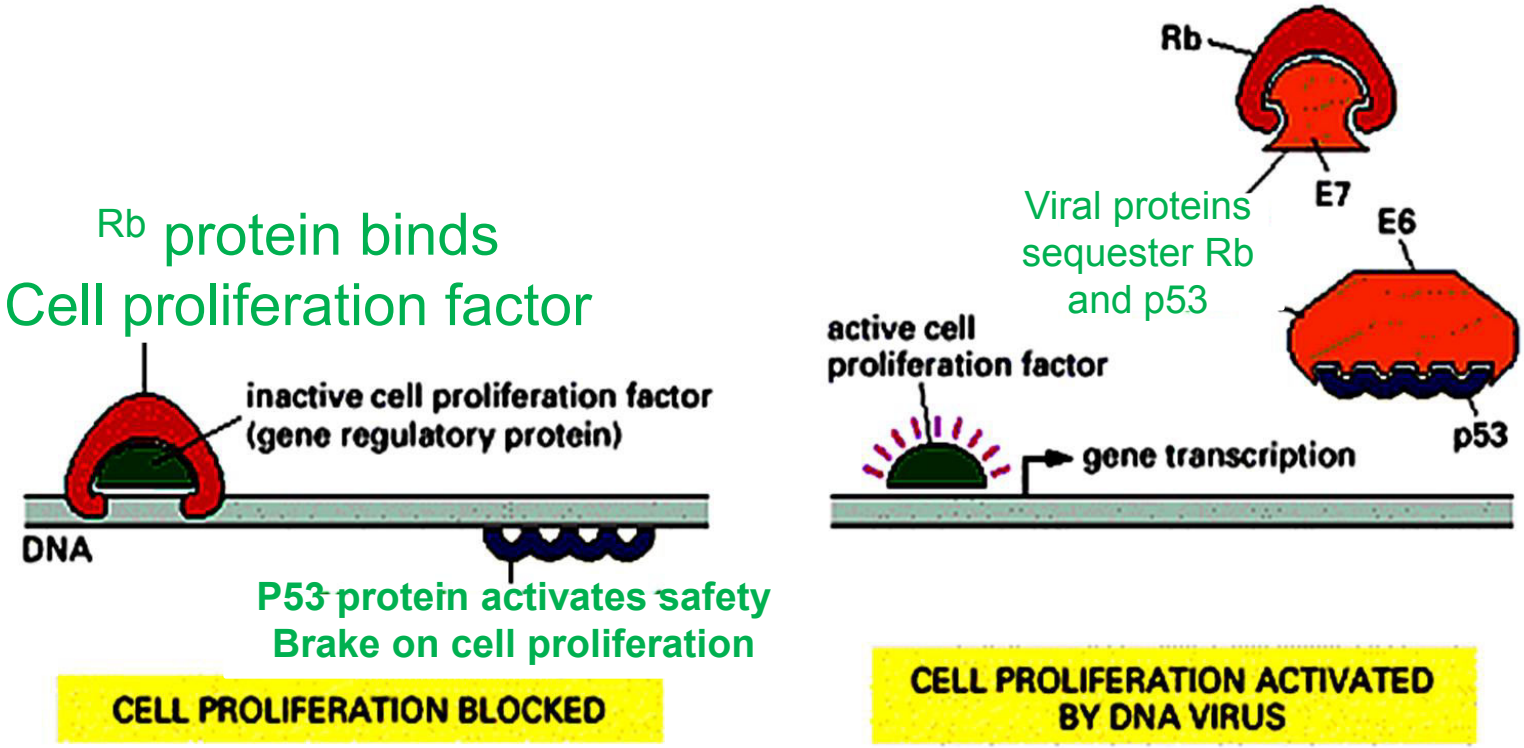
Depending on the cell type p53 then causes either cell-cycle arrest or apoptosis.

Li-Fraumeni syndrome



One of the P53 alleles is mutated which causes 25 fold increased chance of eveloping a malignant tumor by age 50

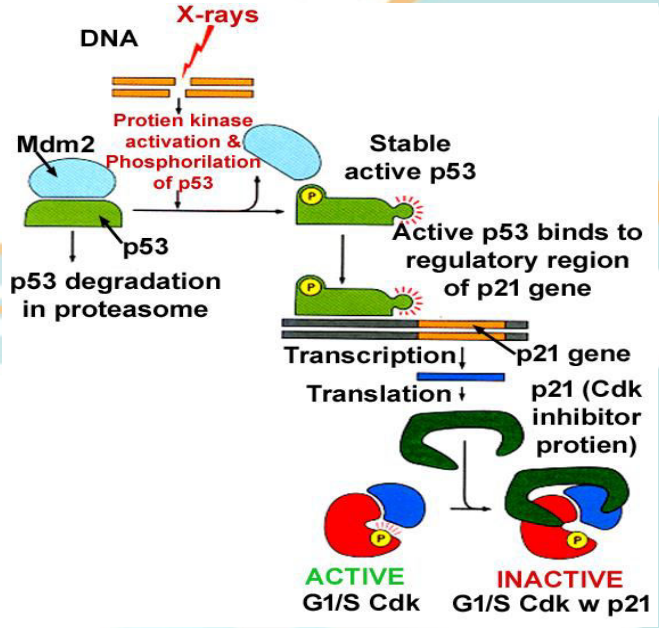
The Viral Protein Targets



Papillomavirus uses two viral proteins, E6 and E7, to sequester the host cell's p53 and Rb. The SV40 virus uses a single dual-purpose protein called large T antigen, for the same purpose.

(a related virus that infects monkeys)

Growth Promoting Oncogenes



END

Cell Biology

Protein Sorting

Protein Sorting

Protein Sorting Essential for Sub-cellular Specialization

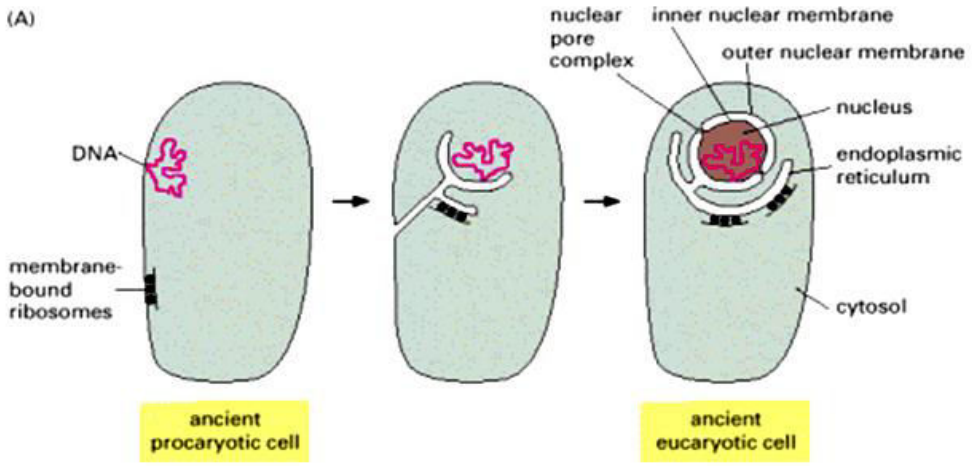
Cytoplasm elaborately subdivided into functionally distinct, membrane-enclosed compartments.

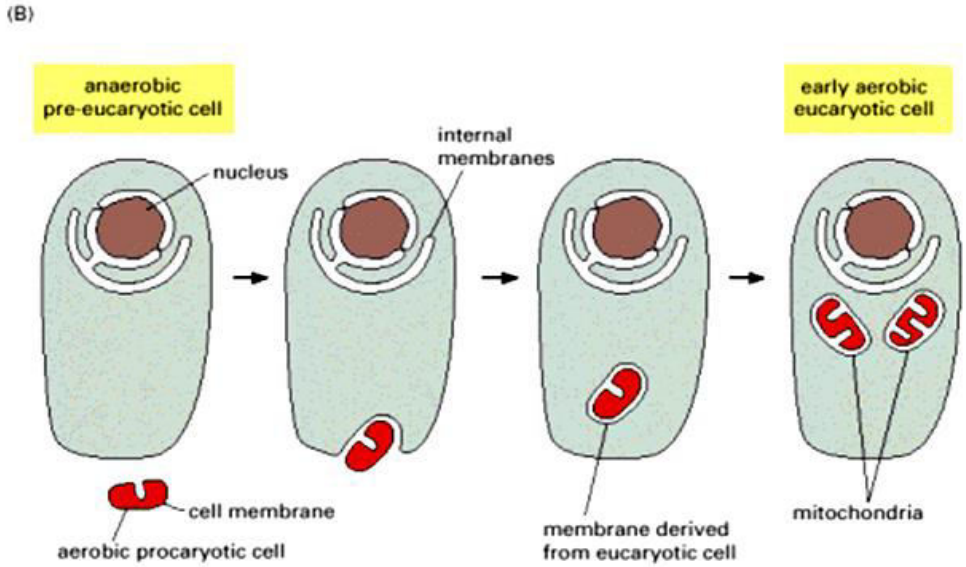
All most all the organelles are membrane enclosed & membrane-enclosed organelles are present in all eukaryotes.

Each organelle, contains its own characteristic set of enzymes and other specialized molecules.

Proteins confer upon each compartment its characteristic structural & functional properties.

Evolution of Organelles



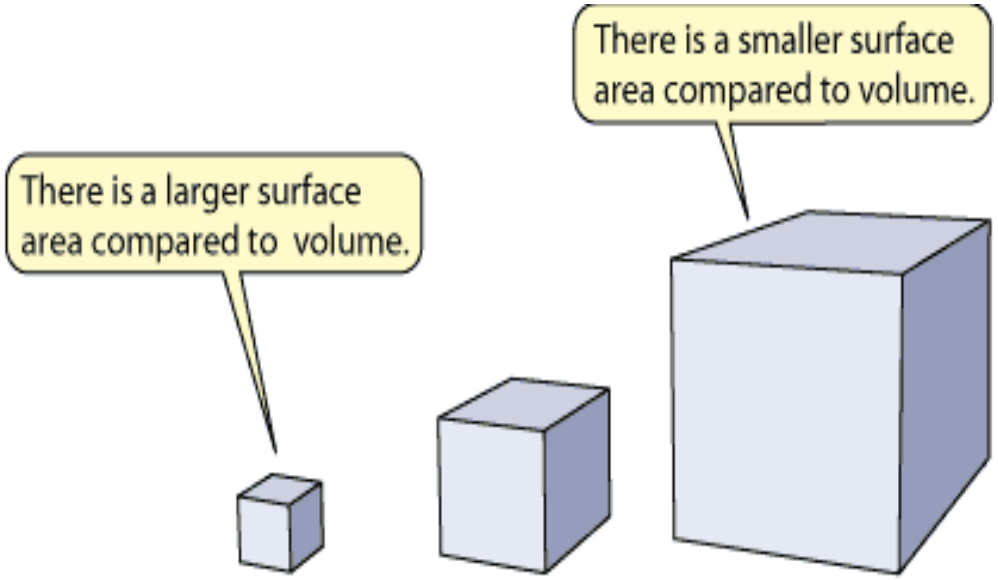


Specialized Compartments

Division of Labor

Vital biochemical processes take place in or on membrane

Membrane Scaffolds



	1-mm cube	2-mm cube	4-mm cube
Surface area	6 sides × 1^2 = 6 mm^2	6 sides × 2^2 = 24 mm^2	6 sides × 4^2 = 96 mm^2
Volume	1^3 = 1 mm^3	2^3 = 8 mm^3	4^3 = 64 mm^3
Surface area-to-volume ratio	6/1	3/1	1.5/1

Greater Surface Area for Chemical Reactions

Allows Cells to Have Bigger Size

Protein Sorting

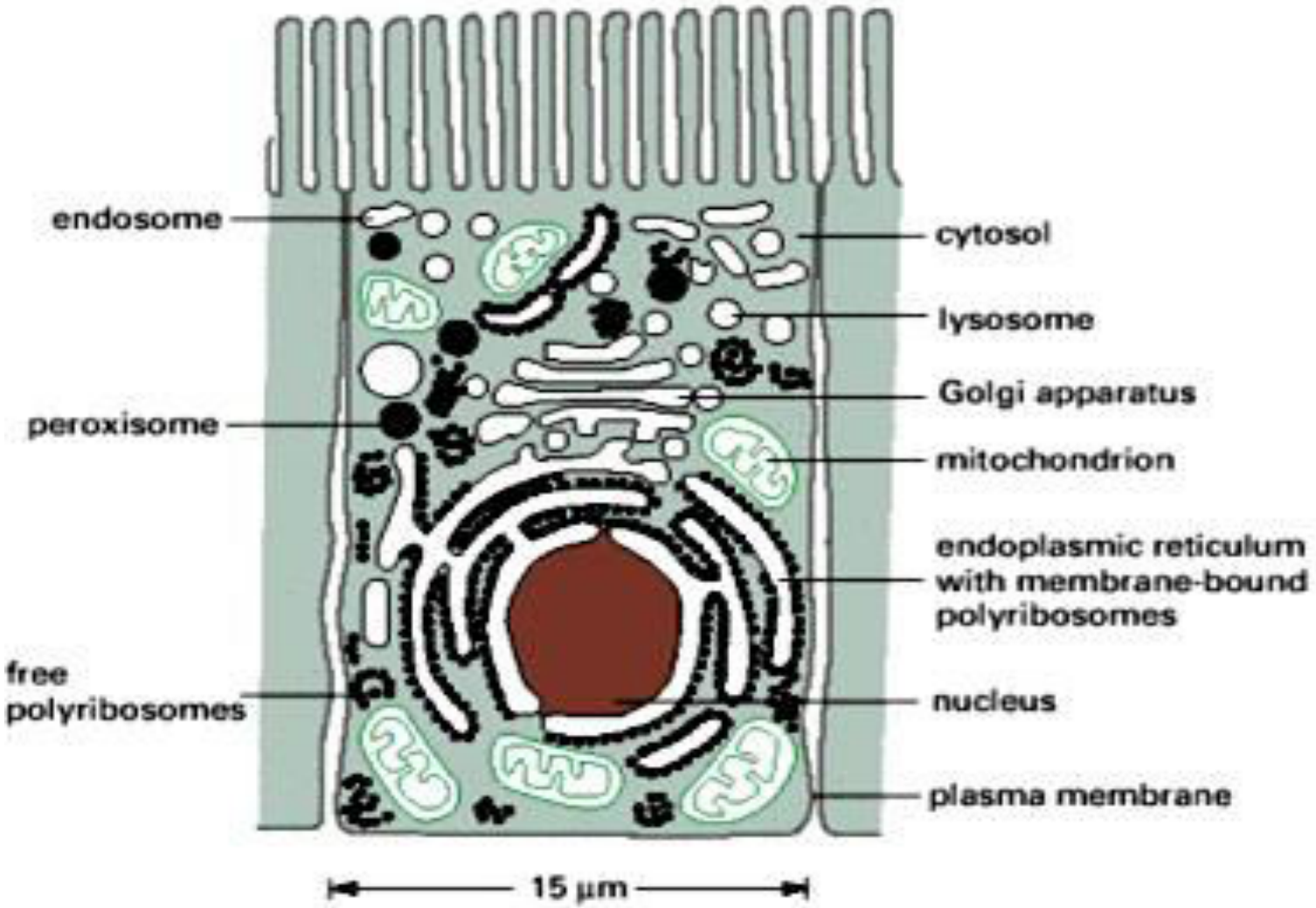
To understand the eucaryotic cell, it is essential to know how molecules move between them, and how the compartments are maintained.

An animal cell contains about 10 billion (10^{10}) protein molecules 10,000–20,000 kinds, synthesis of almost all of begins in cytosol.

Each newly synthesized protein is delivered specifically to compartment that requires it.

THE CHALLENGE: Lipid bilayer of organelle membranes is impermeable to most hydrophilic molecules

Protein Sorting



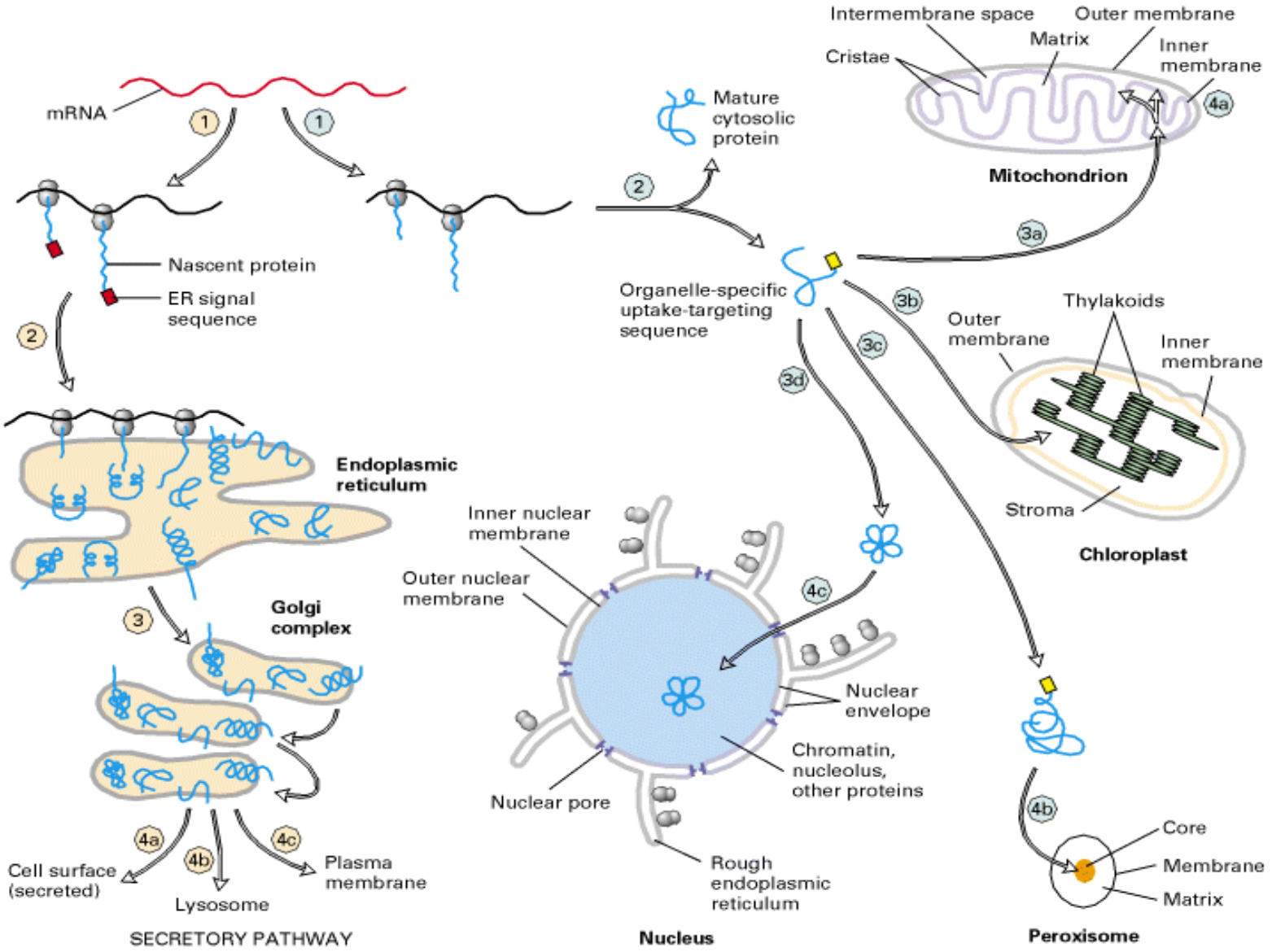
Organelles and Membrane

Membrane of each organelle contains membrane transport proteins that are responsible for the import and export of specific metabolites

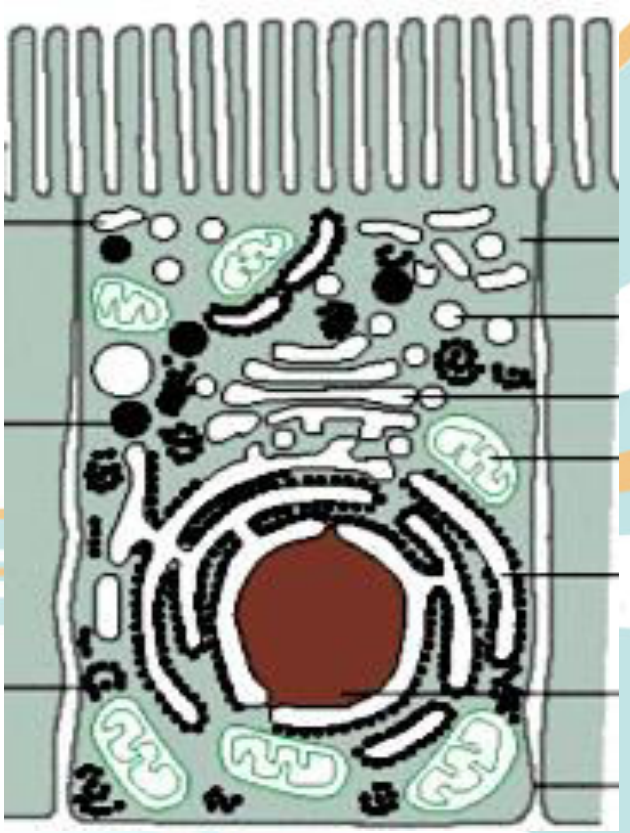
Each organelle membrane has a mechanism for importing, & incorporating specific proteins that makes the organelle unique.

All proteins synthesis initiates in the cytosol, (mitochondria & plastids), their subsequent fate depends on their amino acid sequence, which can contain sorting signals that direct their delivery to locations outside the cytosol, no sorting signal cytosol permanent residents.

Organelles and Membrane



Organelles and Membrane



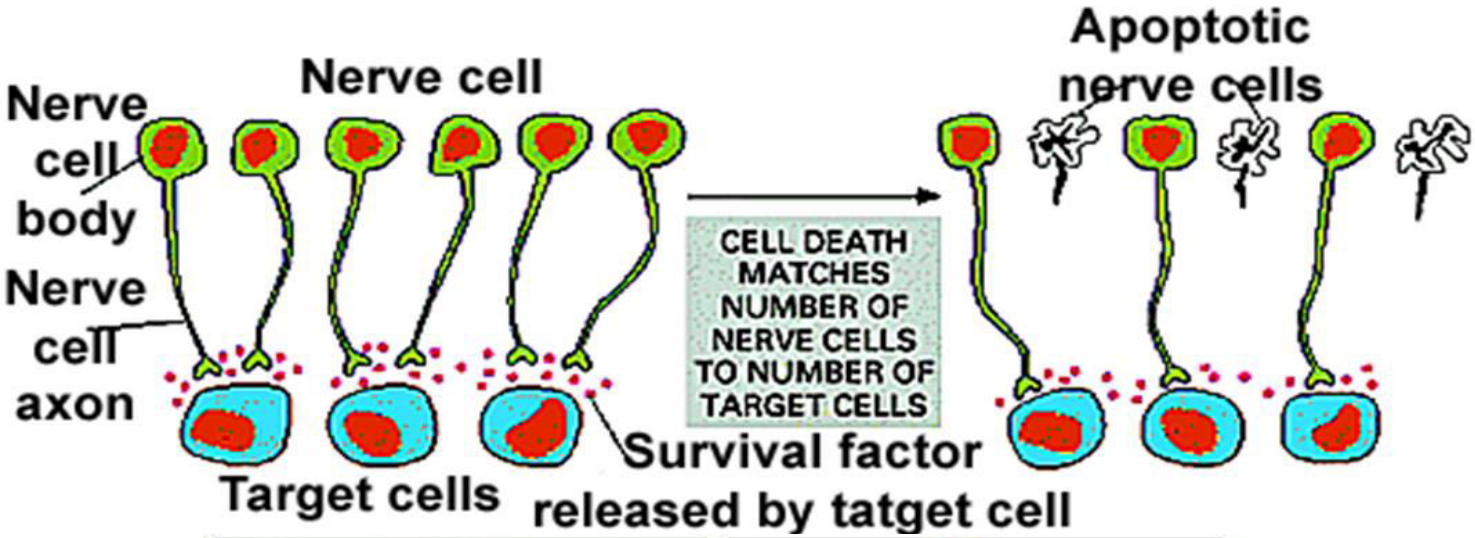
END

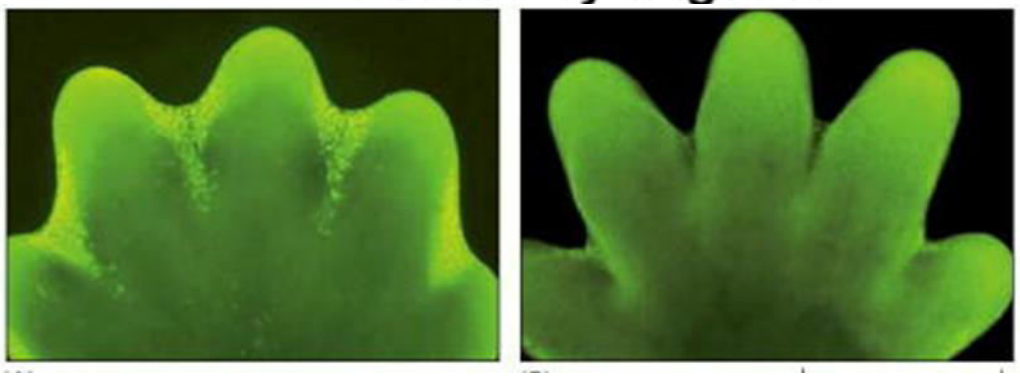
Cell Biology

Apoptosis Genes

Apoptosis Introduction

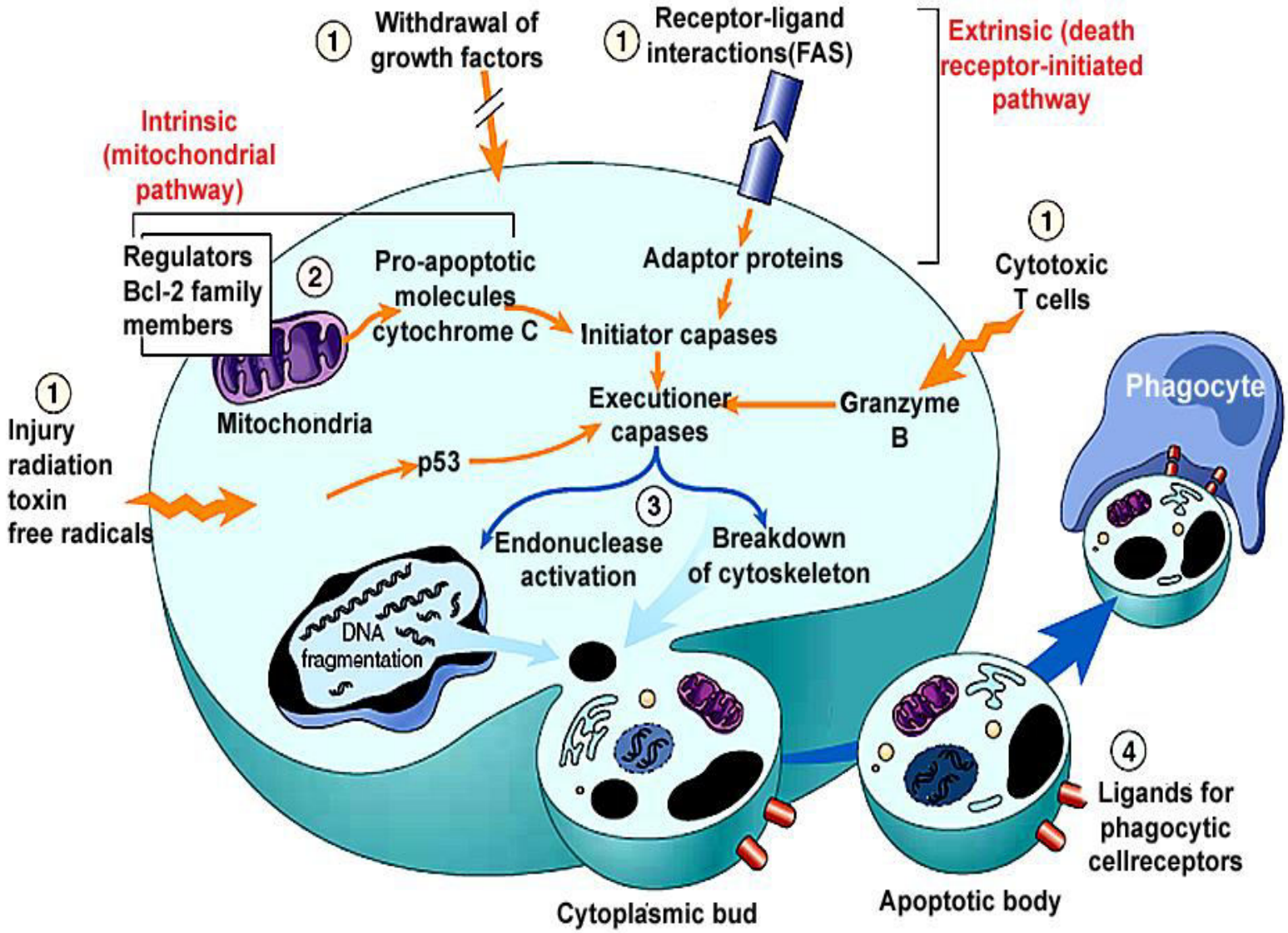
Billions of bone cells die in a healthy adult human
In adults, cell death exactly balances cell division



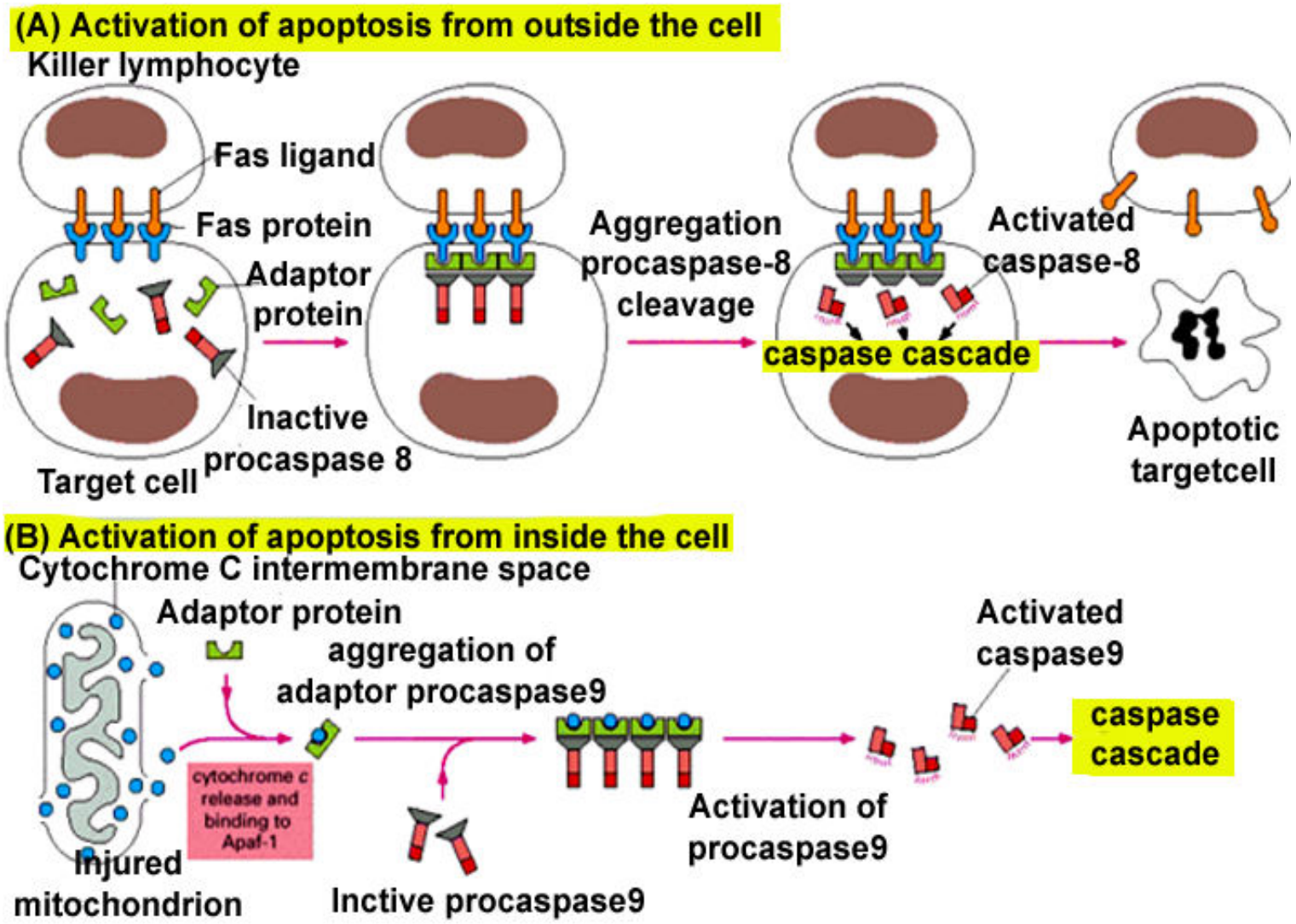


Sculpting the digits in the developing mouse paw by apoptosis)

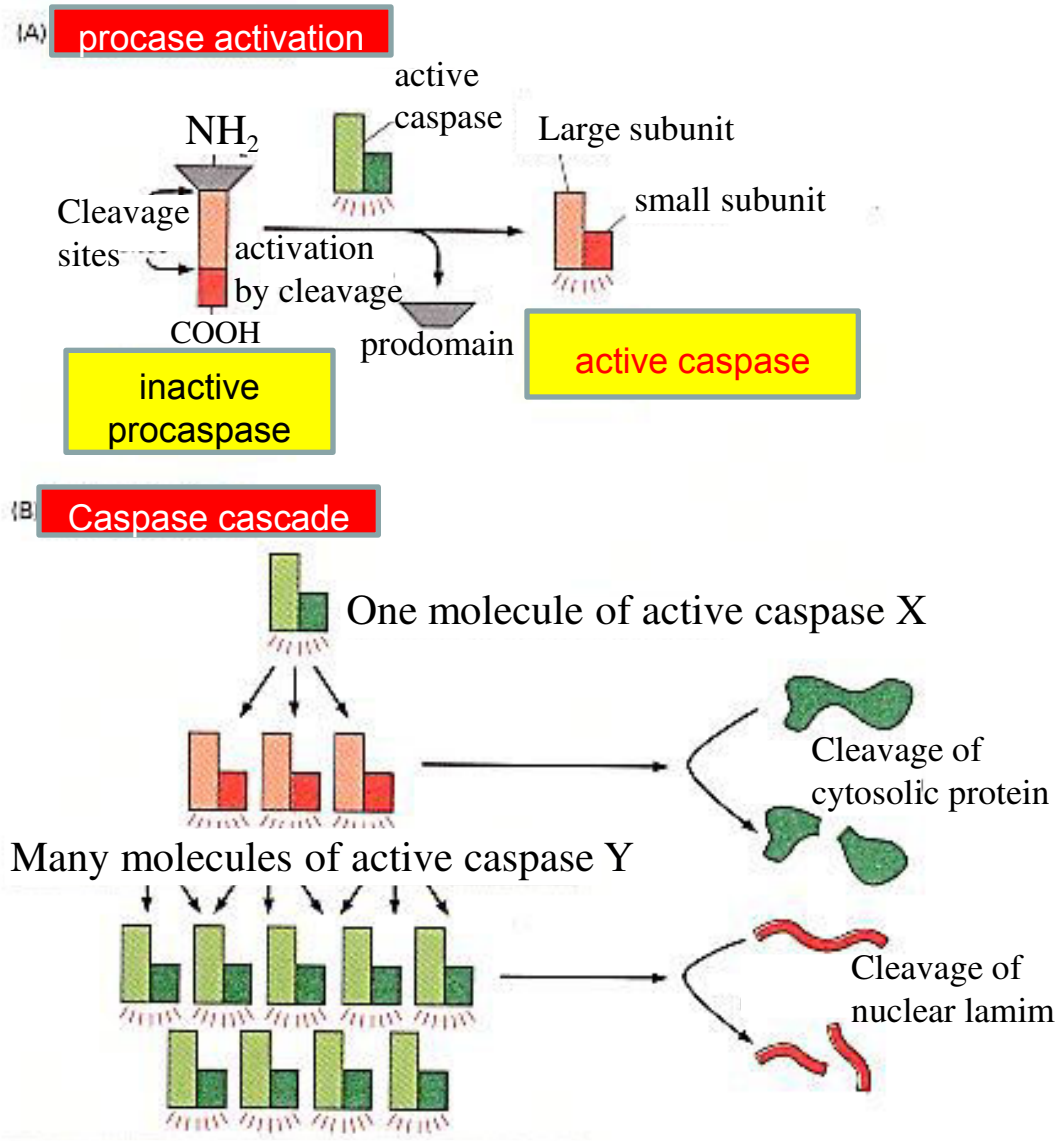
Apoptosis Overview



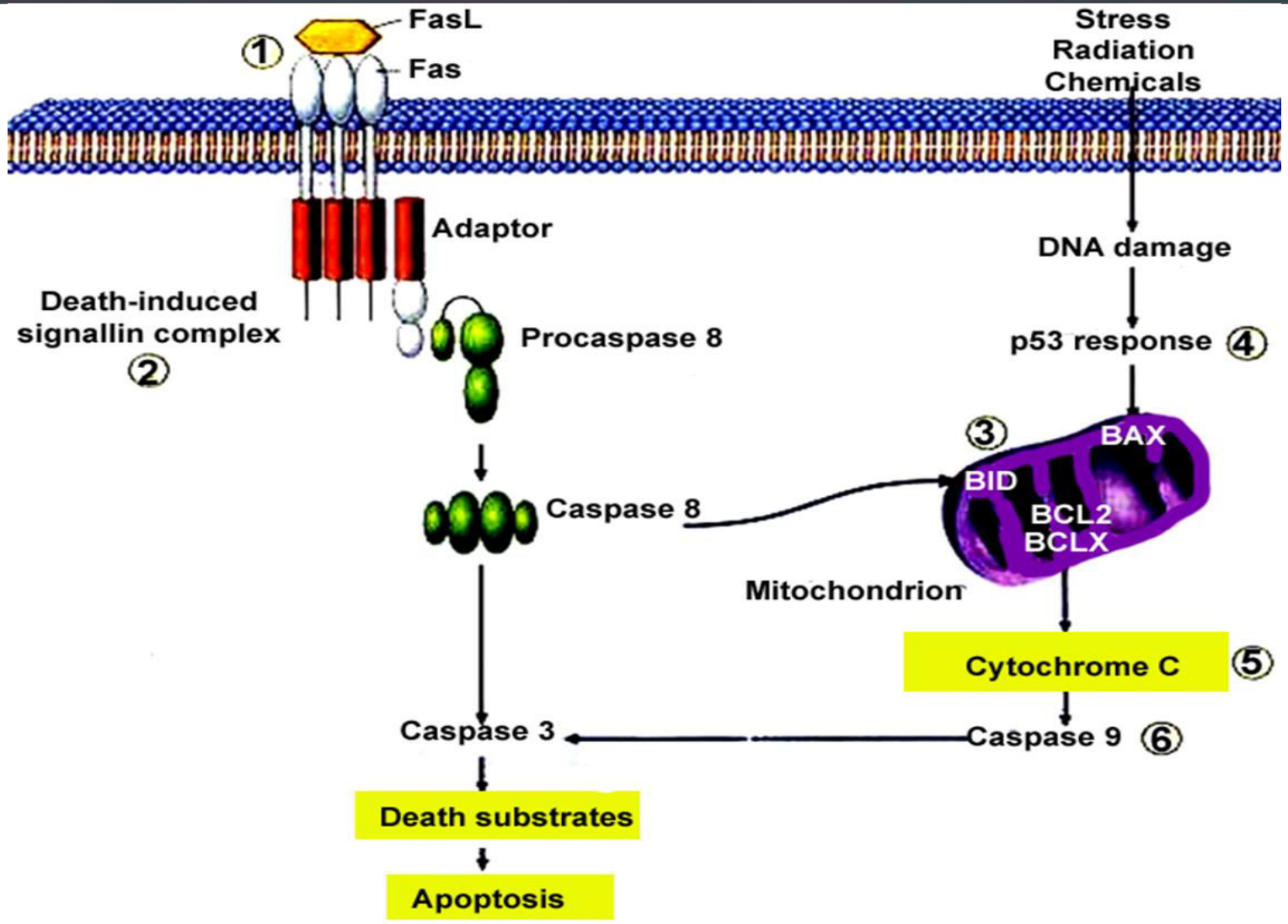
The Two Mechanisms



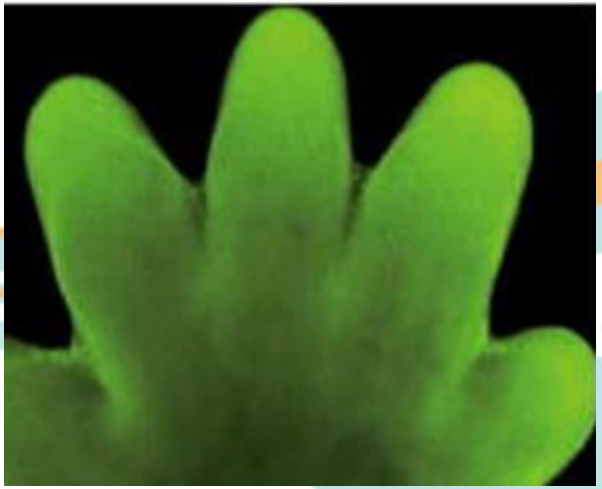
Apoptosis Overview



The Two Mechanisms



Apoptosis

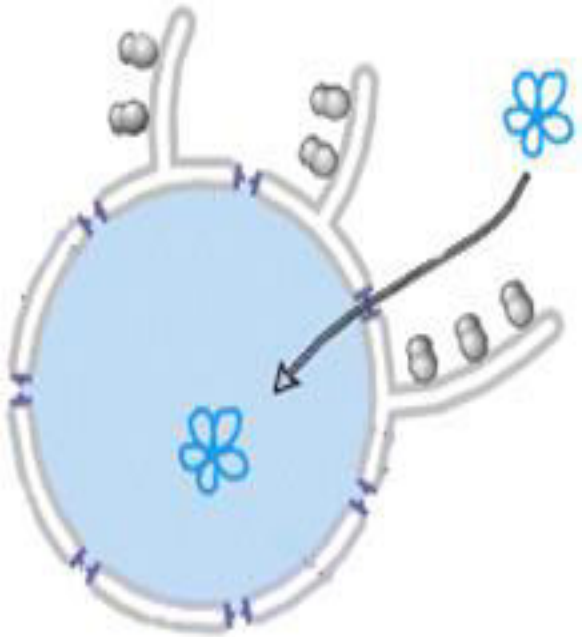


END Tail

Cell Biology

Protein Sorting Mechanisms

Protein Sorting Mechanisms



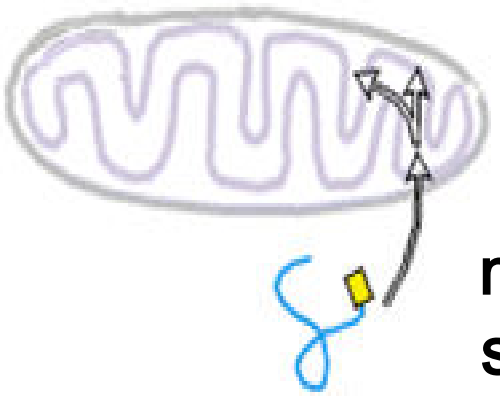
Gated Transport

The protein traffic between the cytosol and nucleus occurs between topologically equivalent spaces, which are in continuity through the nuclear pore complexes

The nuclear pore complexes function as selective gates that actively transport specific macromolecules and macromolecular assemblies, although they also allow free diffusion of smaller molecules 5000 daltons or <

Protein Sorting Mechanisms

Transmembrane Transport

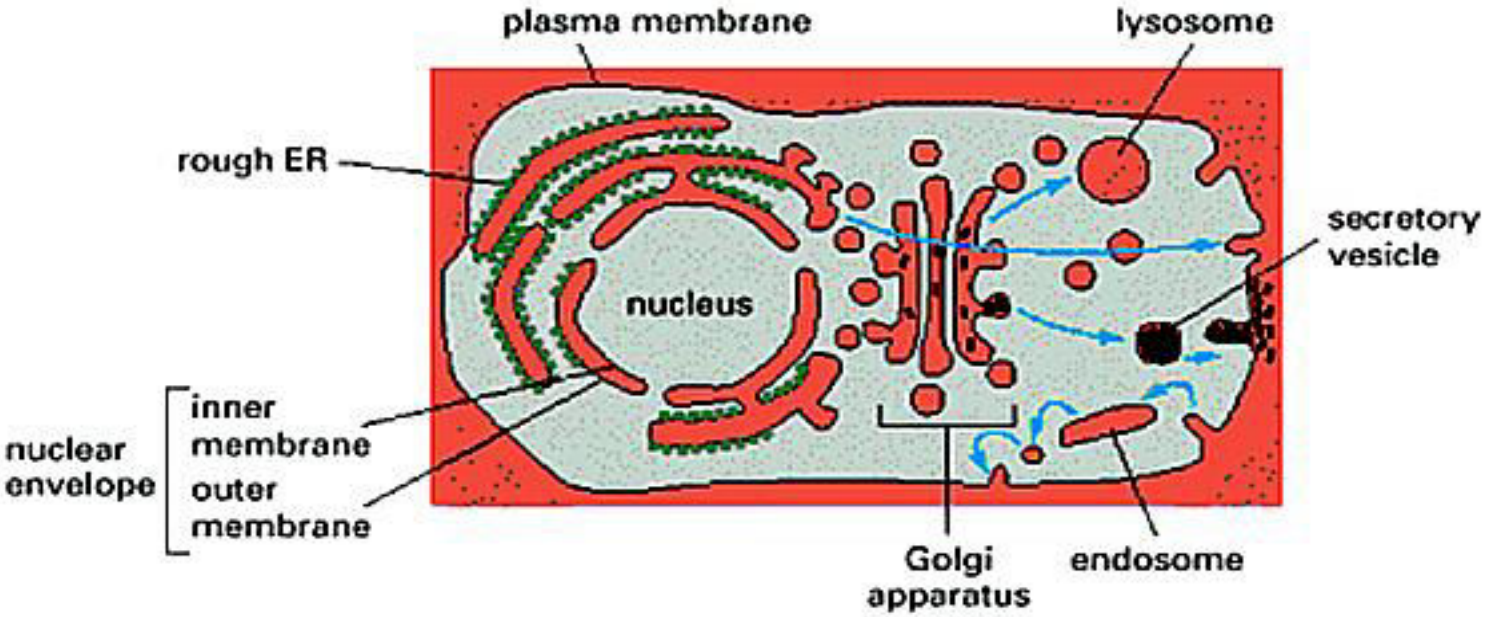


Membrane-bound protein translocators directly transport specific proteins across a membrane from the cytosol into a space that is topologically distinct

The transported protein molecule usually must unfold to snake through the translocator (initial transport of selected proteins from the cytosol into the ER lumen or from the cytosol into mitochondria, for example, occurs in this way)

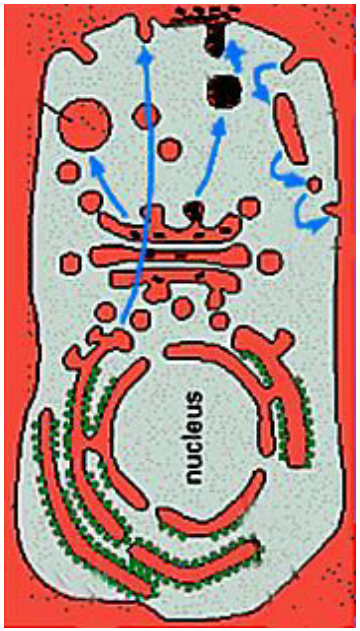
Protein Sorting Mechanisms

Vesicular Transport



Membrane-enclosed transport intermediates—may be small, spherical transport vesicles or larger, irregularly shaped organelle fragments—ferry proteins from one compartment to another.

Protein Sorting Mechanisms

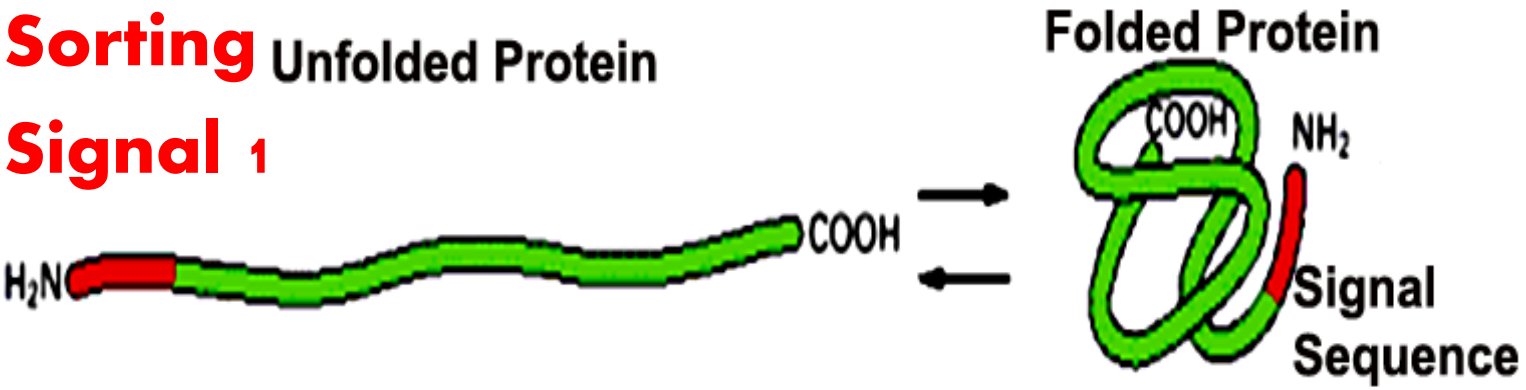


Vesicular Transport

The transport vesicles become loaded with a cargo of molecules derived from the lumen of one compartment as they pinch off from its membrane; they discharge their cargo into a second compartment by fusing with that compartment.

The transfer of soluble proteins from the ER to the Golgi apparatus, transported proteins do not cross a membrane, vesicular transport moves proteins between topologically equivalent compartments

Two Types of Sorting Signals

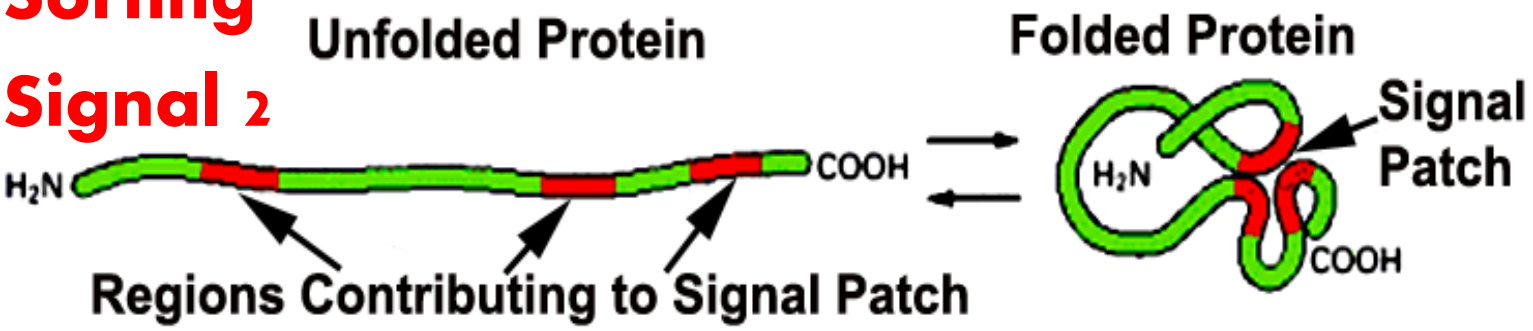


Sorting signal1 (SS) : Continuous stretch of amino acid sequence, 15–60 residues. Some SS removed from finished protein by specialized signal peptidases after sorting is complete.

SS1 direct proteins from the cytosol into the ER, mitochondria, chloroplasts, and peroxisomes.

Two Types of Sorting Signals

Sorting Signal 2



Sorting signal 2 (SS) : Consists of a specific three-dimensional arrangement of atoms on the protein's surface that forms when protein folds.

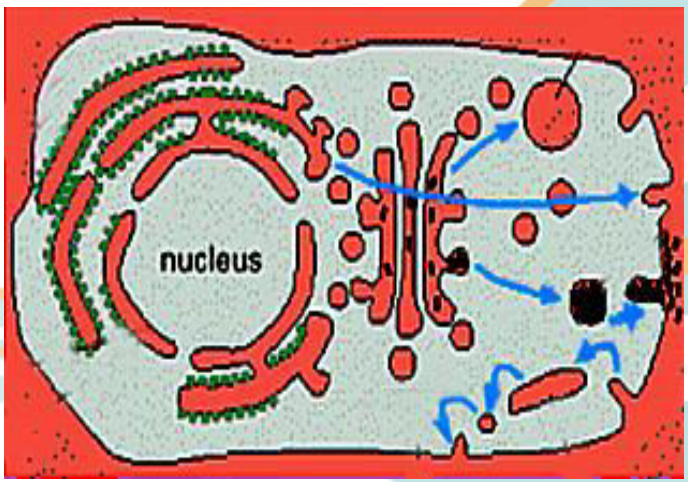
The amino acid residues that comprise this signal patch can be distant from one another in the linear amino acid sequence, and they generally persist in the finished protein

The Sorting Signals

FUNCTION OF SIGNAL SEQUENCE	EXAMPLE OF SIGNAL SEQUENCE
Import into nucleus	-Pro-Pro-Lys-Lys-Lys-Arg-Lys-Val-
Export from nucleus	-Leu-Ala-Leu-Lys-Leu-Ala-Gly-Leu-Asp-Ile-
Import into mitochondria	$^{+}H_3N$-Met-Leu-Ser-Leu-Arg-Gln-Ser-Ile-Arg-Phe-Phe-Lys-Pro-Ala-Thr-Arg-Thr-Leu-Cys-Ser-Ser-Arg-Tyr-Leu-Leu-
Import into plastid	$^{+}H_3N$-Met-Val-Ala-Met-Ala-Met-Ala-Ser-Leu-Gln-Ser-Ser-Met-Ser-Ser-Leu-Ser-Leu-Ser-Ser-Asn-Ser-Phe-Leu-Gly-Gln-Pro-Leu-Ser-Pro-Ile-Thr-Leu-Ser-Pro-Phe-Leu-Gln-Gly-
Import into peroxisomes	-Ser-Lys-Leu-COO^{-}
Import into ER	$^{+}H_3N$-Met-Met-Ser-Phe-Val-Ser-Leu-Leu-Leu-Val-Gly-Ile-Leu-Phe-Trp-Ala-Thr-Glu-Ala-Glu-Gln-Leu-Thr-Lys-Cys-Glu-Val-Phe-Gln-
Return to ER	-Lys-Asp-Glu-Leu-COO^{-}

Some characteristic features of the different classes of signal sequences are highlighted in color. Where they are known to be important for the function of the signal sequence, positively charged amino acids are shown in *red* and negatively charged amino acids are shown in *green*. Similarly, important hydrophobic amino acids are shown in *yellow* and hydroxylated amino acids are shown in *blue*. $^{+}H_3N$ indicates the N-terminus of a protein; COO^{-} indicates the C-terminus.

Protein Sorting Mechanisms

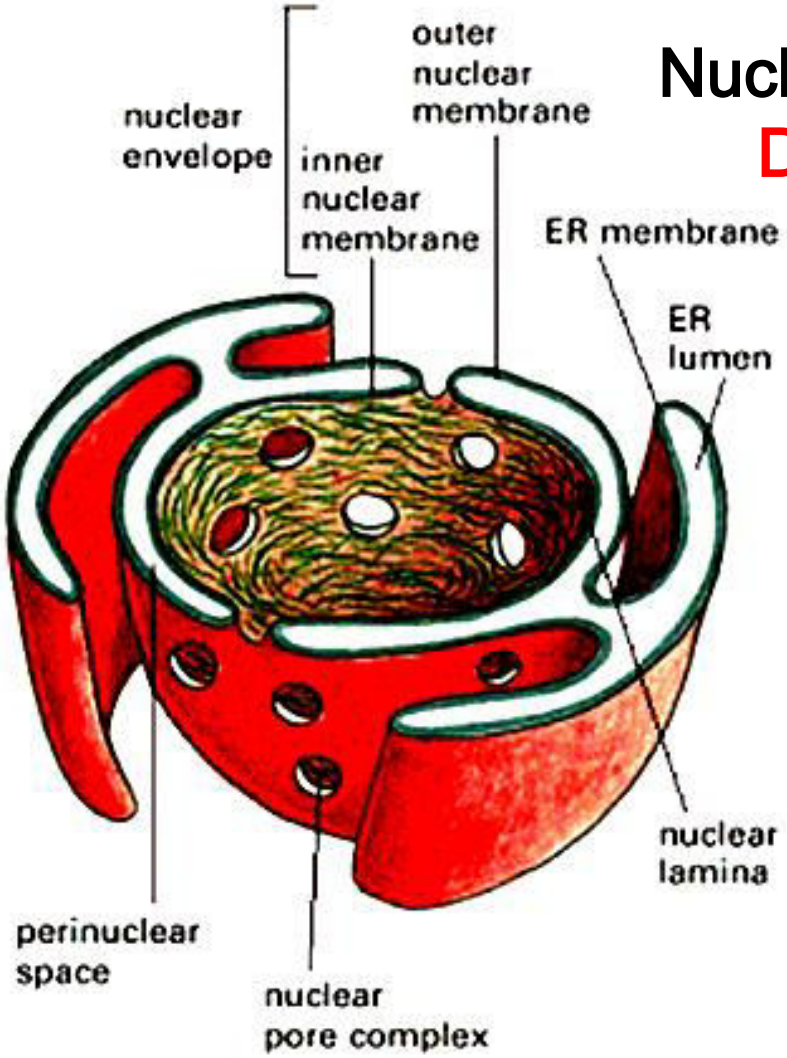


END

Cell Biology

Nuclear Transport

Nuclear Transport



Nuclear envelope encloses DNA & defines nuclear compartment

Envelope consists of 2 concentric membranes that are penetrated by nuclear pore complexes

The inner & outer membranes are continuous & maintain distinct protein compositions.

Nuclear Transport

The inner nuclear membrane contains specific proteins that act as binding sites for chromatin & protein meshwork of the nuclear lamina that provides structural support.

The inner membrane is surrounded by the outer nuclear membrane, which is continuous with the membrane of the ER

The outer nuclear membrane is studded with ribosomes engaged in protein synthesis

Nuclear Transport

Bidirectional traffic between the cytosol and the nucleus

Many proteins that function in nucleus—including histones, DNA and RNA polymerases, gene regulatory proteins, & RNA-processing proteins—are selectively imported into the nuclear compartment from the cytosol,

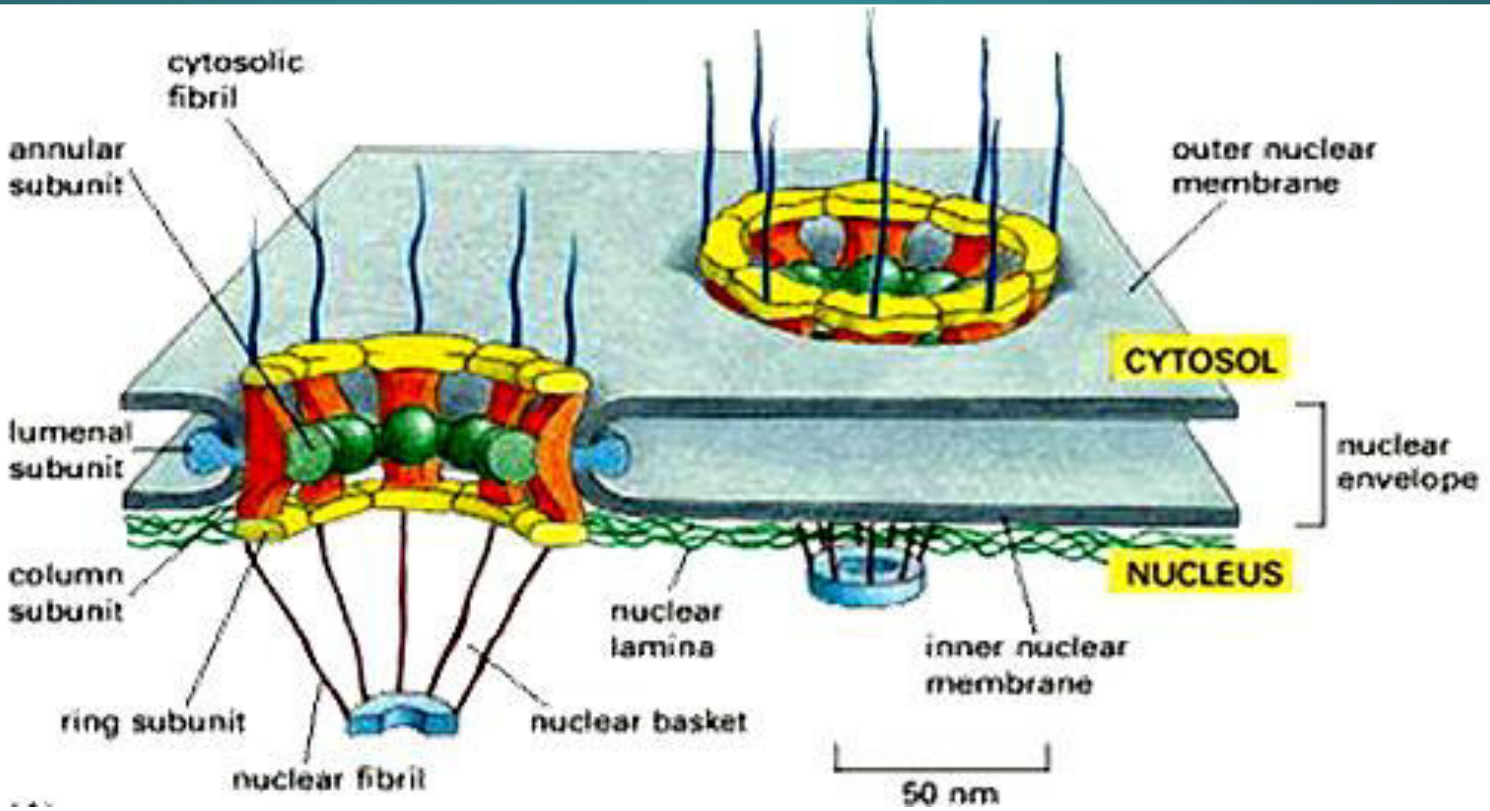
At the same time, tRNAs and mRNAs are synthesized in the nuclear compartment and then exported to cytosol

Nuclear Transport

Like the import process, the export process is selective; mRNAs, for example, are exported only after they have been properly modified by RNA-processing reactions in the nucleus

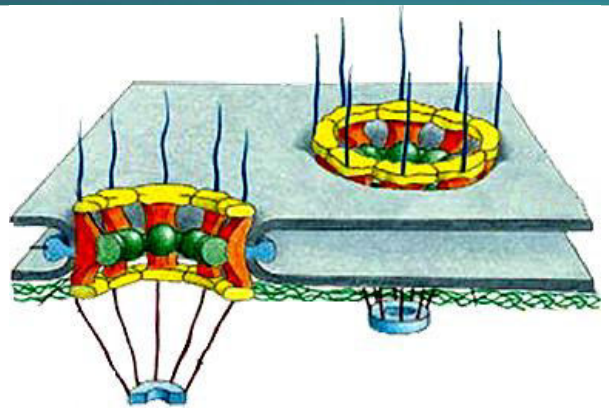
Ribosomal proteins, are made in the cytosol, imported into the nucleus—where they assemble with newly made ribosomal RNA into particles—and are then exported again to the cytosol as part of a ribosomal subunit

Nuclear Pore Complexes



Eukaryotic nuclear envelope is perforated by large, elaborate structures called nuclear pore complexes which selectively facilitate nuclear transport

Nuclear Pore Complexes



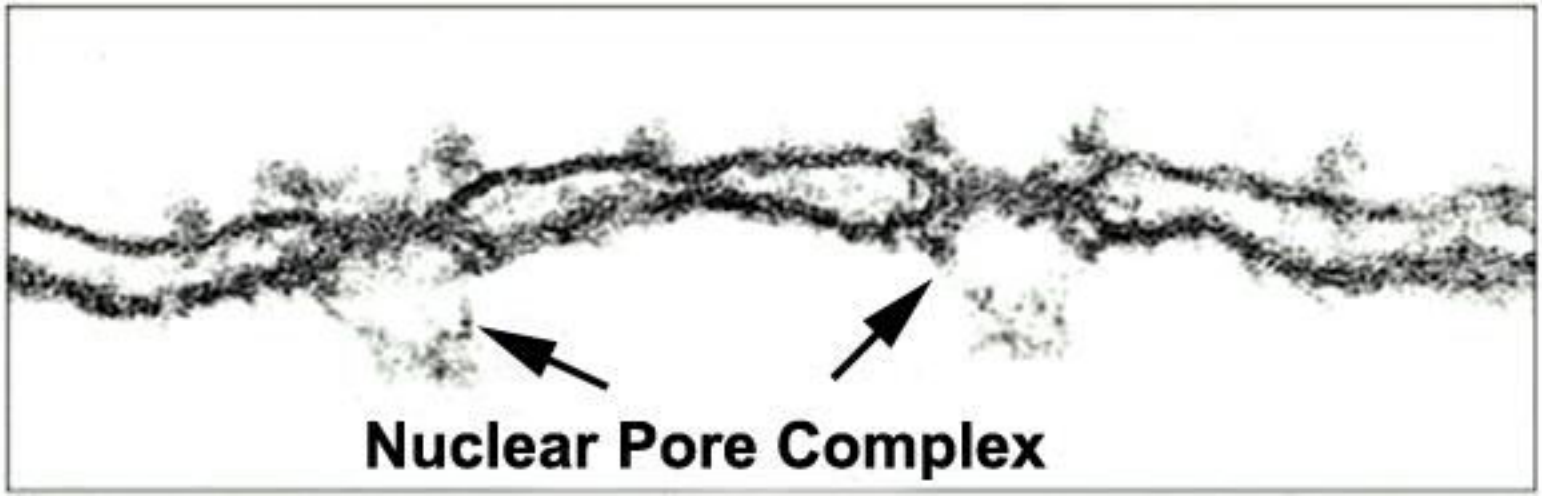
Each pore complex contains one or more open aqueous channels through which small water-soluble molecules can passively diffuse

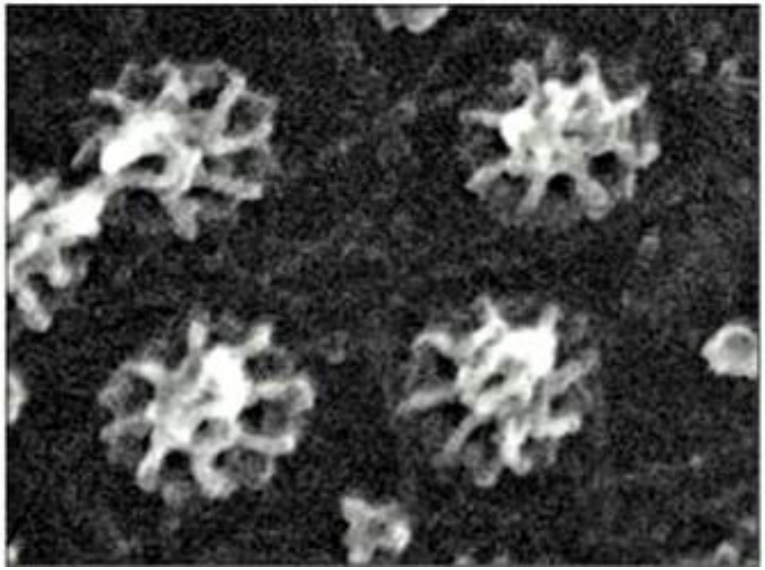
Each complex has a molecular mass of about 125 million & is composed of aprox 50 different proteins, called nucleoporins, arranged octagonal symmetry

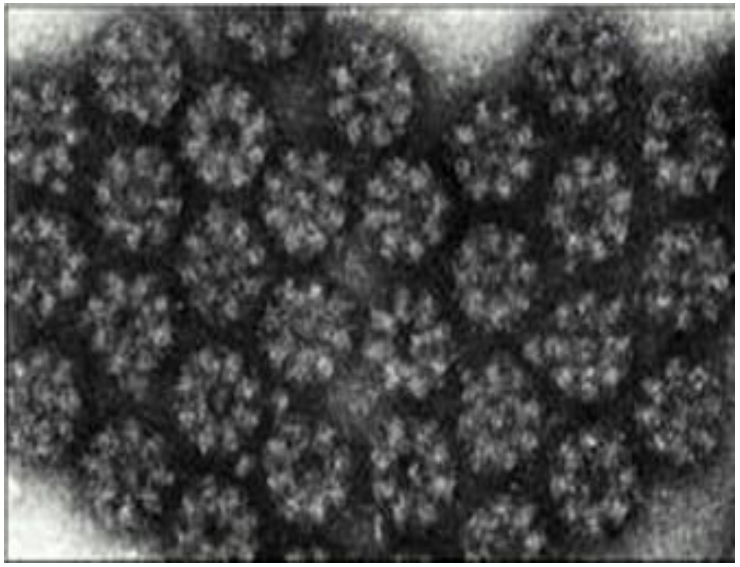
Small molecules (5000 daltons or less) diffuse in so fast that the nuclear envelope can be considered to be freely permeable to them, a protein of 17,000 daltons takes 2 minutes to proteins larger than 60,000 daltons are unable to enter the nucleus.

(injecting labeled water-soluble molecules)

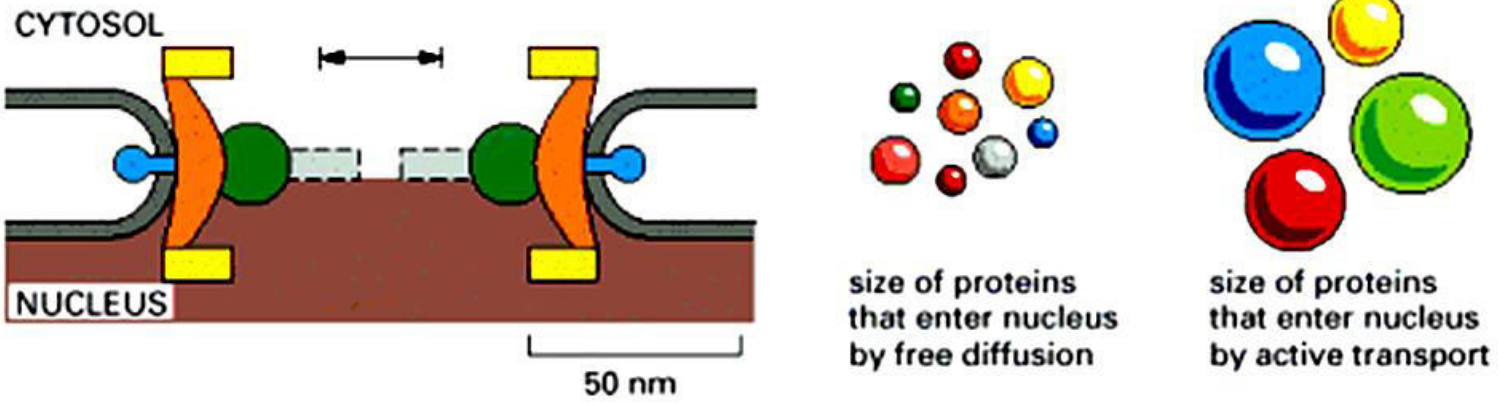
Nuclear Pore Complexes







Nuclear Pore Complexes



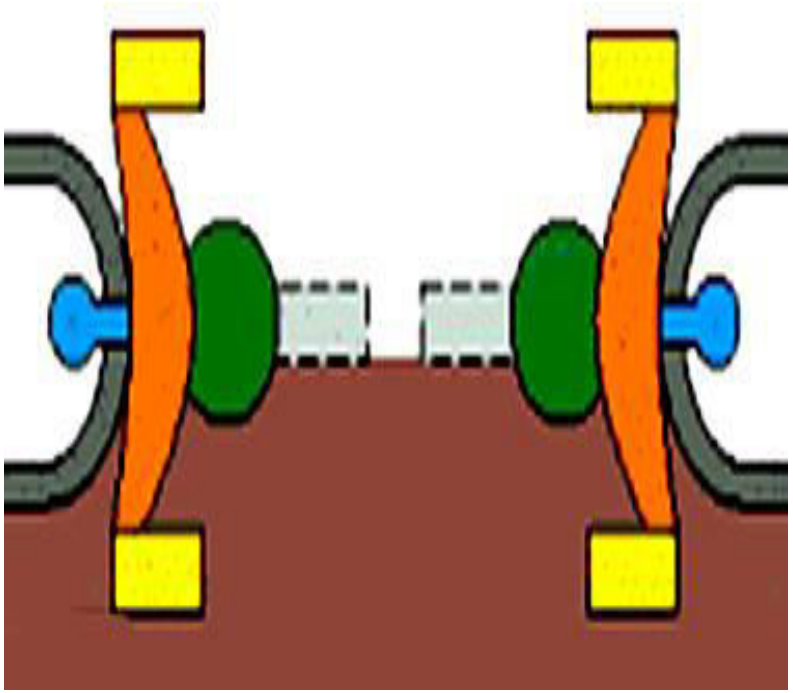
Nuclear pore contains a pathway for free diffusion 9 nm in diameter & 15 nm long, this channel occupies a small fraction of total volume of the pore complex.

The more active the nucleus is in transcription, the greater the number of pore complexes

The nuclear envelope of a typical mammalian cell contains 3000–4000 pore complexes

equivalent to a water-filled cylindrical channel

Nuclear Pore Complexes



Cells synthesizing DNA, need to import about 10^6 histone molecules from the cytosol every 3 minutes to package DNA into chromatin, ie each pore complex needs to transport about 100 histone molecules/minute

Rapidly growing cell need to transport about 6 large & small ribosomal subunits/minute from nucleus, where they are produced, to the cytosol.

Nuclear Transport

END

Cell Biology

Nuclear Localization Signals (NLSs) & Transport

NLS

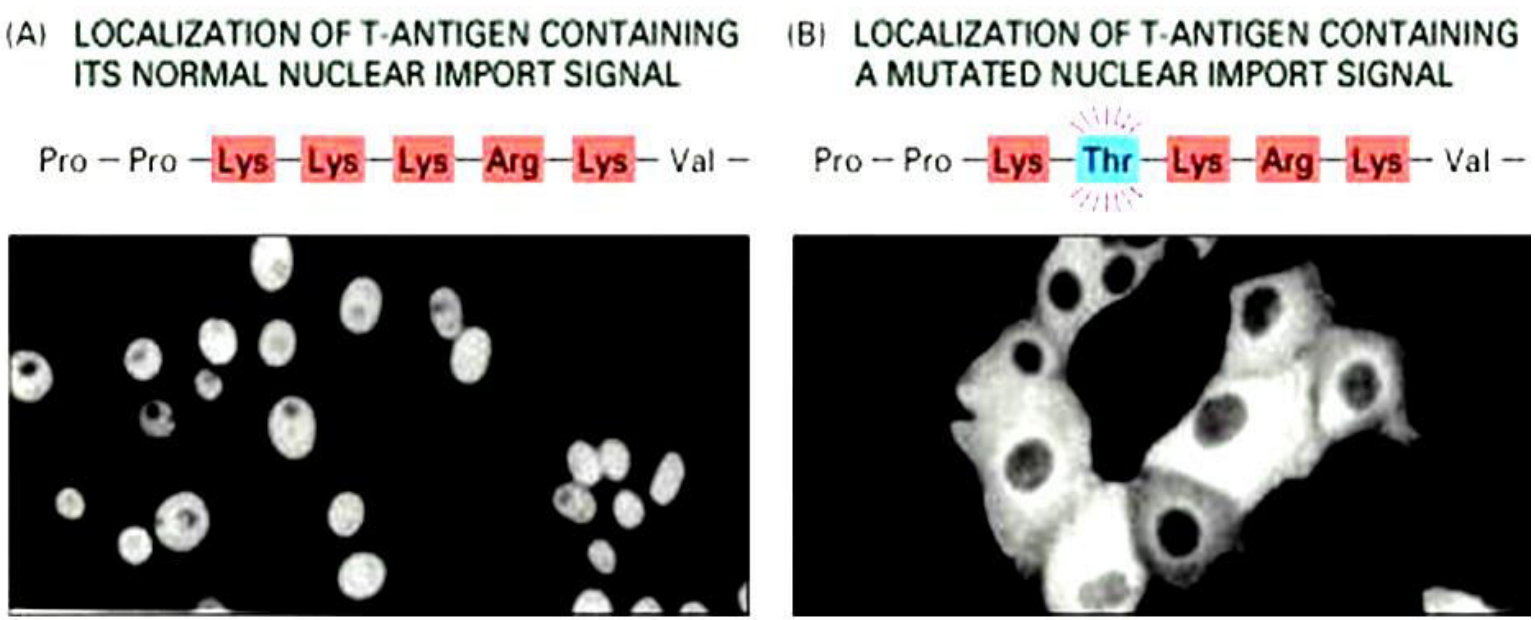
Import into nucleus -Pro-Pro-Lys-Lys-Lys-Arg-Lys-Val-

When proteins are experimentally extracted from the nucleus and reintroduced into the cytosol even the very large ones re-accumulate efficiently in the nucleus

The selectivity of this nuclear import process resides in nuclear localization signals (NLSs), which are present only in nuclear proteins

The signals have been precisely defined in numerous nuclear proteins by using recombinant DNA technology

NLS Experiment



Immunofluorescence micrographs showing the cellular location of SV40 virus T-antigen containing or lacking a short peptide that serves as a nuclear localization signal

Transport Across Nuclear Pore

The mechanism of macromolecular transport across nuclear pore complexes is different from the transport mechanisms involved in protein transfer across the membranes of other organelles, because it occurs through a large aqueous pore rather than through a protein transporter spanning one or more lipid bilayers.

Nuclear proteins can be transported through a pore complex while they are in a fully folded conformation

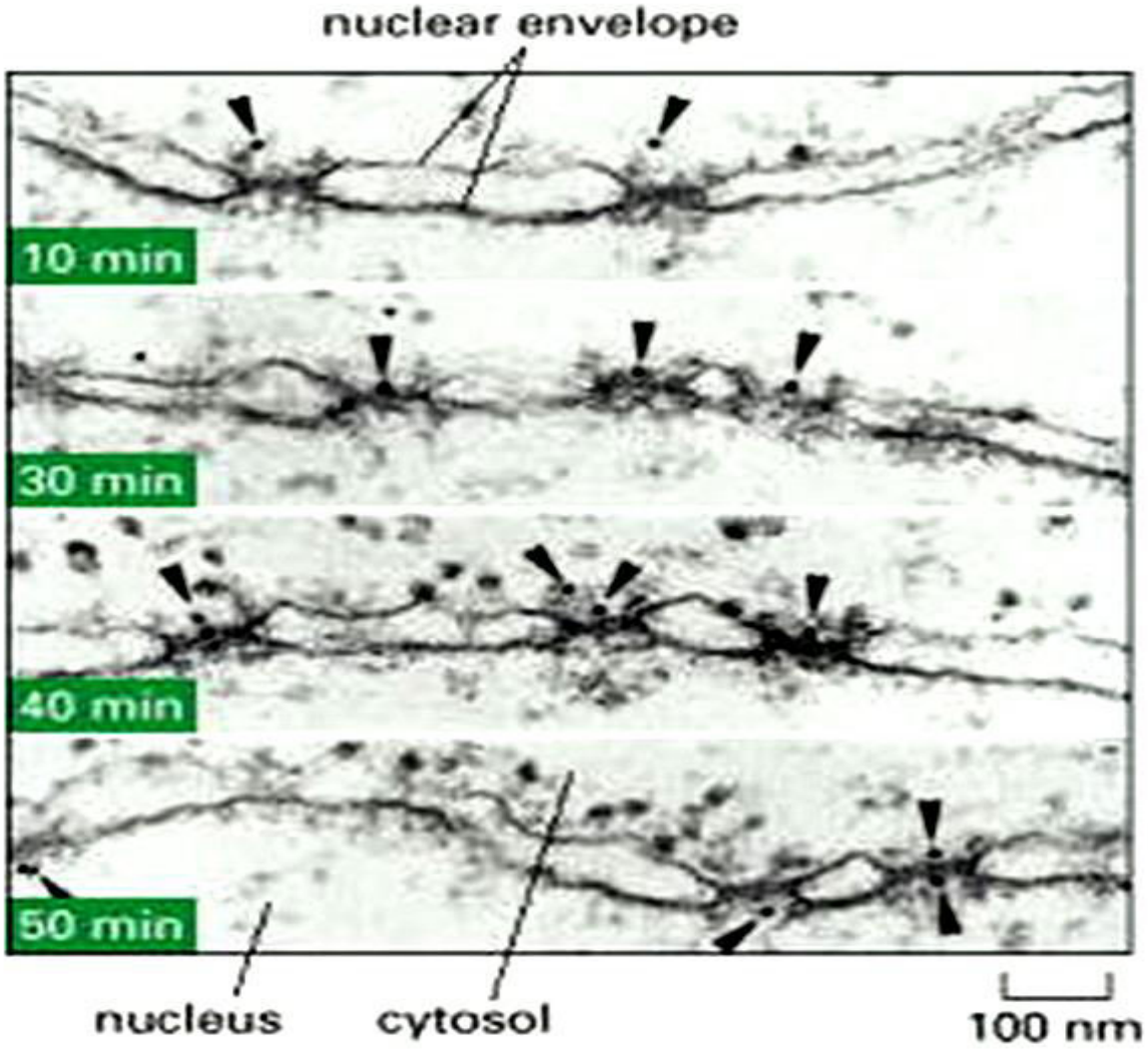
A newly formed ribosomal subunit is transported out of the nucleus as an assembled particle

Transport Across Nuclear Pore

By contrast, proteins have to be extensively unfolded during their transport into most other organelles

Very large particles traversing the pore seem to constrict as they squeeze through the nuclear pore complex, indicating that at least some of them must undergo restructuring during transport e.g. very large mRNAs

Transport Across Nuclear Pore



Nuclear Transport

-Pro-Pro-Lys-Lys-Lys-Arg-Lys-Val-

END

Cell Biology

Nuclear Receptors

Nuclear Import Receptors

To initiate nuclear import, NLS must be recognized by nuclear import receptors, encoded by family of related genes

Each family member encodes a receptor protein that is specialized for the transport of a group of nuclear proteins sharing structurally similar nuclear localization signals

Nuclear Import Receptors

FG-repeats line the path through the nuclear pore complexes taken by the import receptors and their bound cargo proteins

These protein complexes move along the path by repeatedly binding, dissociating, and then re-binding to adjacent repeat sequences

Once in the nucleus, the import receptors dissociate from their cargo and are returned to the cytosol

Nuclear Import Receptors

Nuclear import receptors do not always bind to nuclear proteins directly

Additional adaptor proteins are sometimes used that bridge between the import receptors and the nuclear localization signals on the proteins to be transported

The combined use of import receptors and adaptors allows a cell to recognize the broad repertoire of nuclear localization signals that are displayed on nuclear proteins

Nuclear Export

Nuclear export like nuclear import, but in reverse

Nuclear export of large molecules, e.g. ribosome subunits & RNAoccurs through nuclear pore & depends on a selective transport system

Nuclear export signals are present on molecules to be exported, & on nuclear export receptors

These receptors bind both the export signal and nucleoporins to guide their cargo through the pore complex to the cytosol

Export & import receptors are structurally related encoded by the same gene family karyopherins

Ran GTPase

Ran GTPase Drives Directional Transport

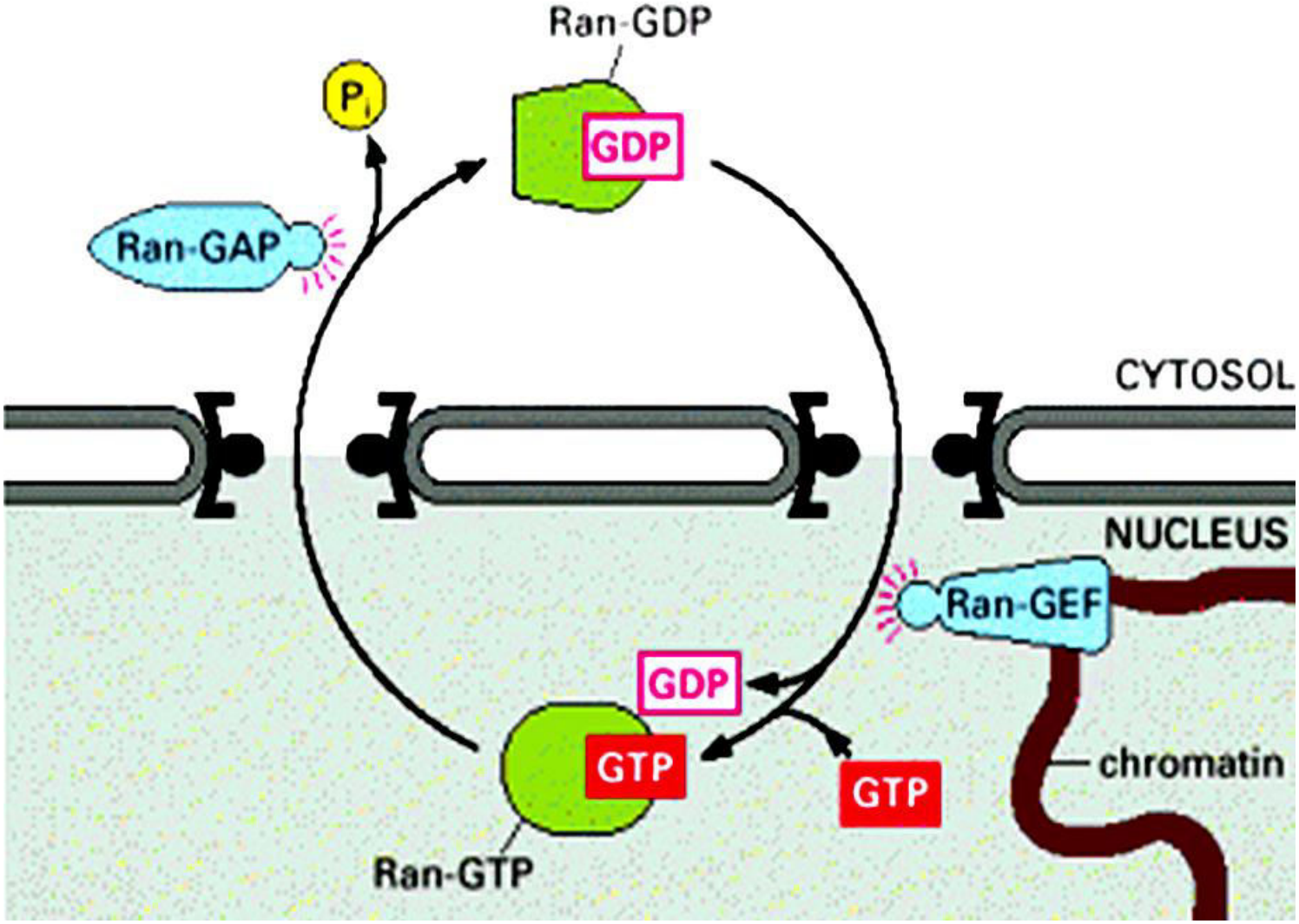
The import of nuclear proteins through the pore complex concentrates specific proteins in the nucleus, thereby increasing order (decreasing entropy) in the cell, which consumes energy

The energy is provided by the hydrolysis of GTP by the monomeric GTPase Ran

Ran is found in both the cytosol and the nucleus, and it is required for nuclear import & export

Ran is a molecular switch that can exist in 2 states, depending on whether GDP or GTP is bound

Ran GTPase


Ran-GDP
P_i
GDP
Ran-GAP
CYTOSOL
NUCLEUS
Ran-GEF
GDP
GTP
GTP
chromatin
Ran-GTP

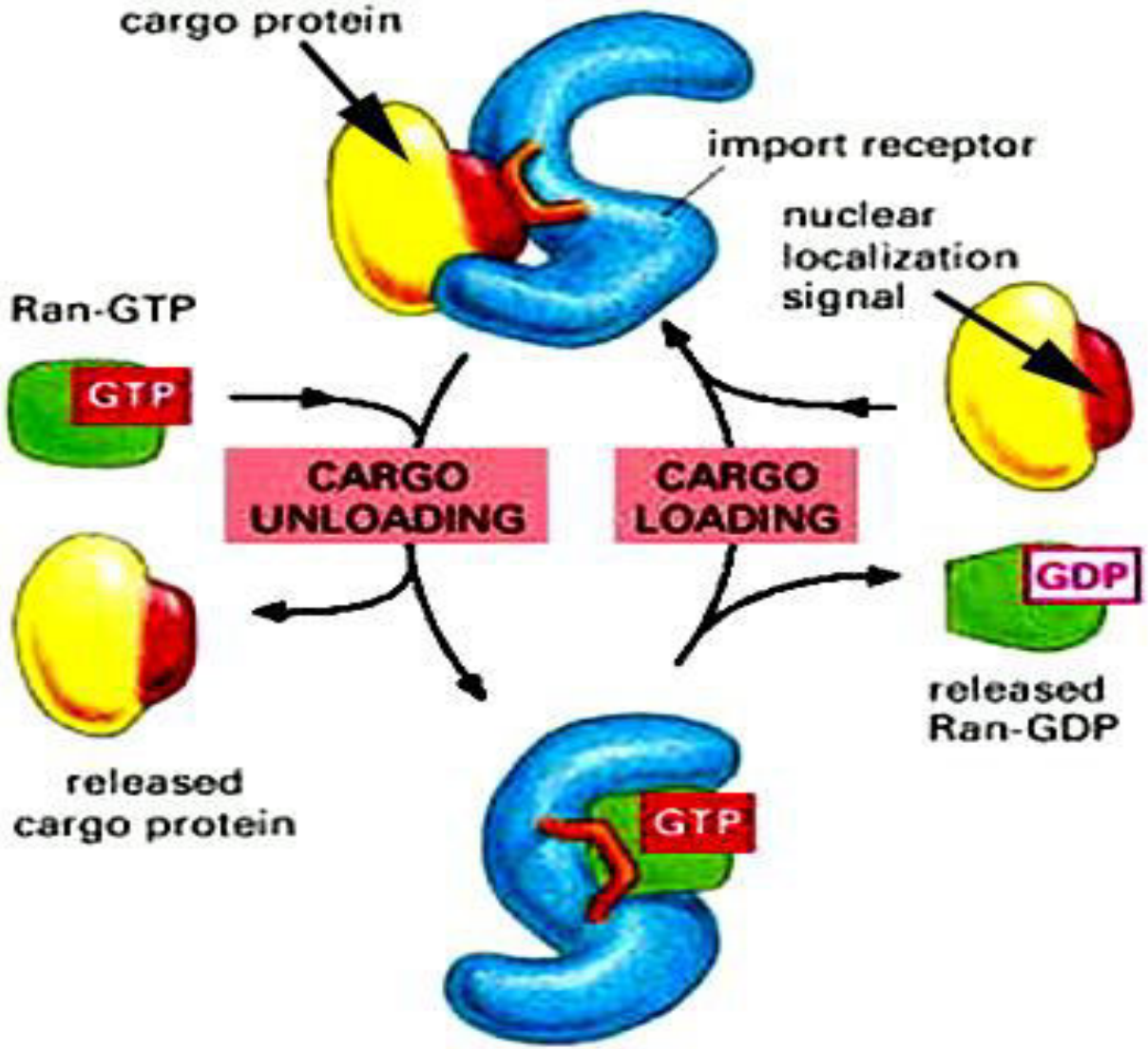
Ran GTPase

Conversion between the two states is triggered by two Ran-specific regulatory proteins: a cytosolic GTPase-activating protein (GAP) that triggers GTP hydrolysis and thus converts Ran-GTP to Ran-GDP, and a nuclear guanine exchange factor (GEF) that promotes the exchange of GDP for GTP converts Ran-GDP to Ran-GTP

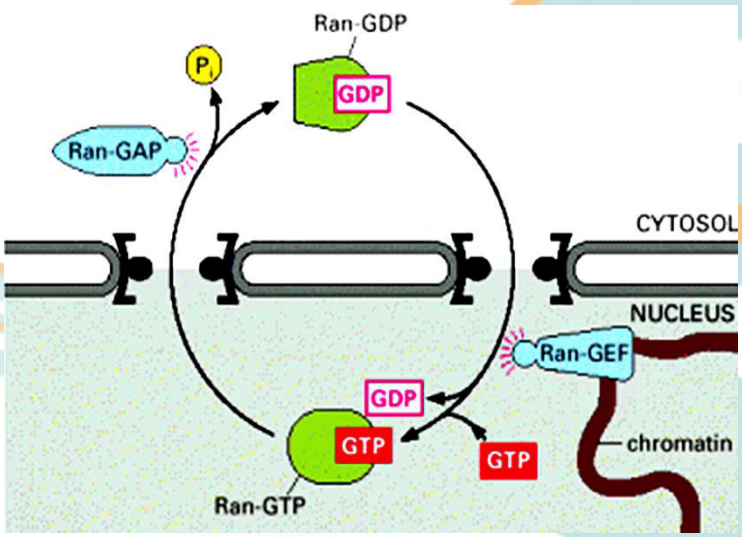
Ran-GAP is located in the cytosol and Ran-GEF is located in the nucleus, the cytosol primarily contains Ran-GDP, & the nucleus Ran-GTP

This gradient of the 2 conformational forms of Ran drives nuclear transport in appropriate direction

Ran GTPase


cargo protein
import receptor
nuclear localization signal
Ran-GTP
GTP
CARGO UNLOADING
CARGO LOADING
GDP
released Ran-GDP
released cargo protein
GTP

Nuclear Transport



END

Cell Biology

Control Of Import And Export

Control Of Import And Export

The activity of some gene regulatory proteins is controlled by keeping them out of the nuclear compartment until they are needed there

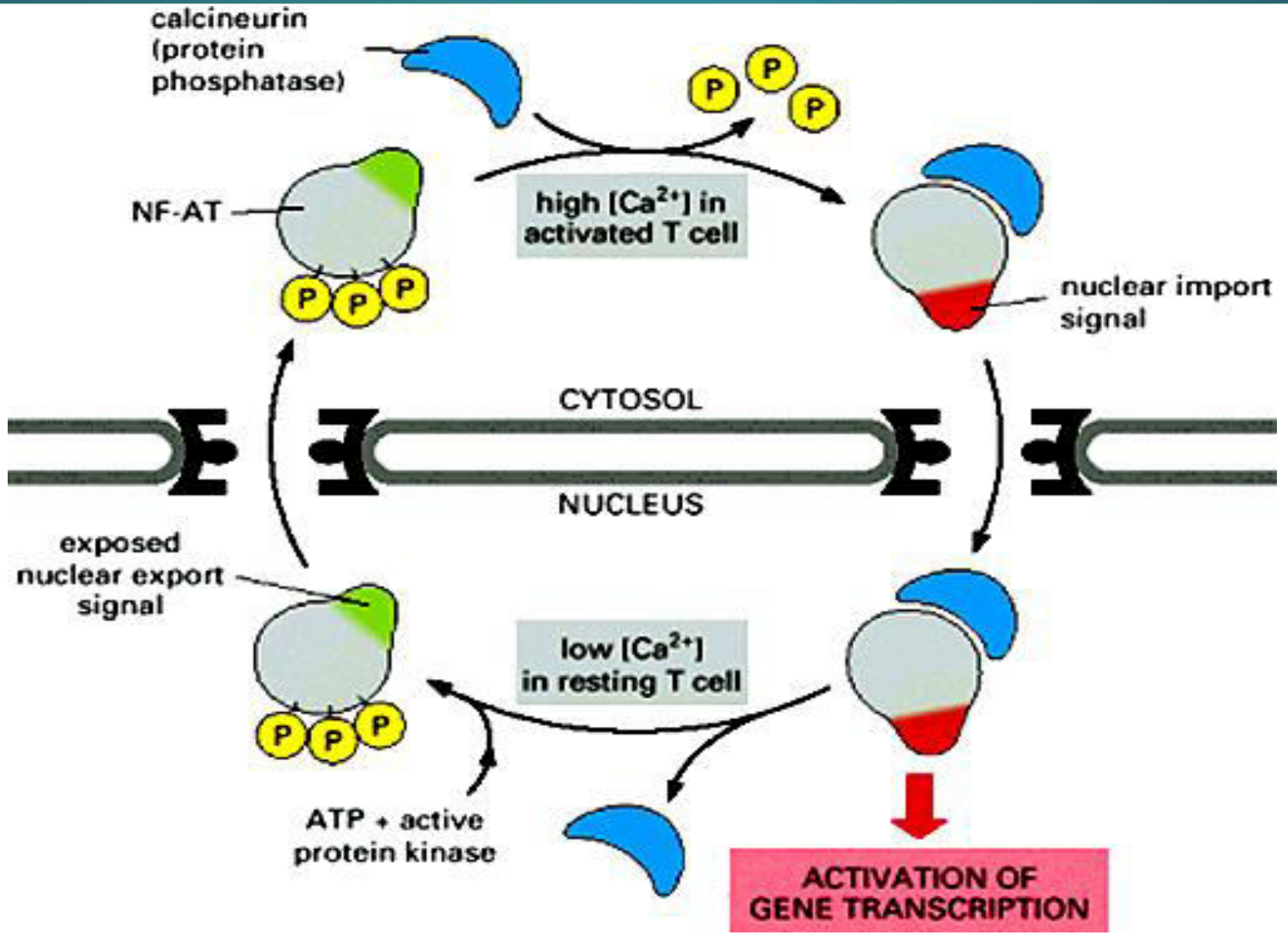
In many cases, this control depends on the regulation of nuclear localization and export signals; these can be turned on or off, often by phosphorylation of adjacent amino acids

Gene regulatory proteins are bound to inhibitory cytosolic proteins that either anchor them in the cytosol or mask their NLSs so that they are unable to interact with nuclear import receptors

When the cell receives a signal the gene regulatory protein is released & transported into the nucleus

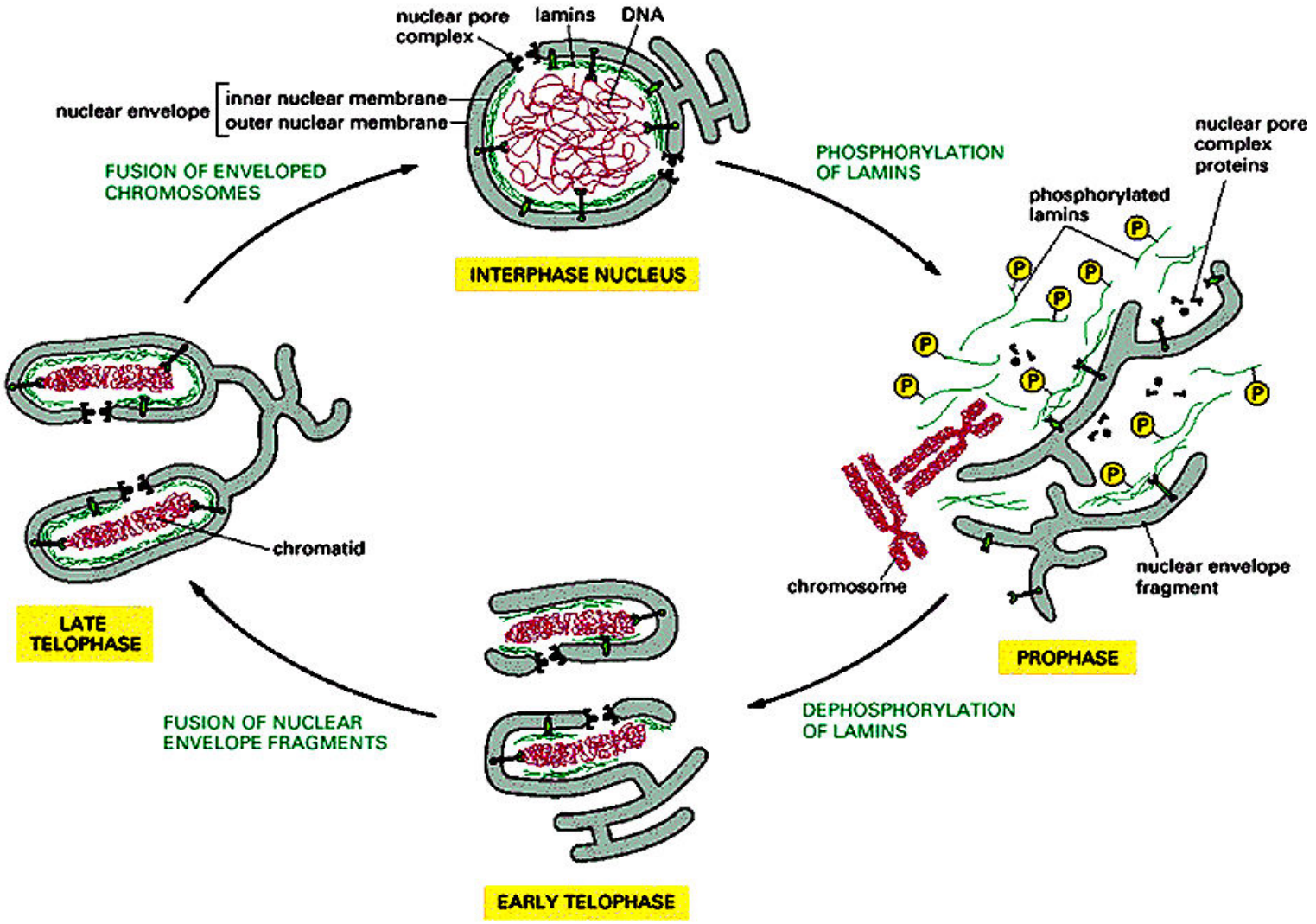
The transport is stringently controlled

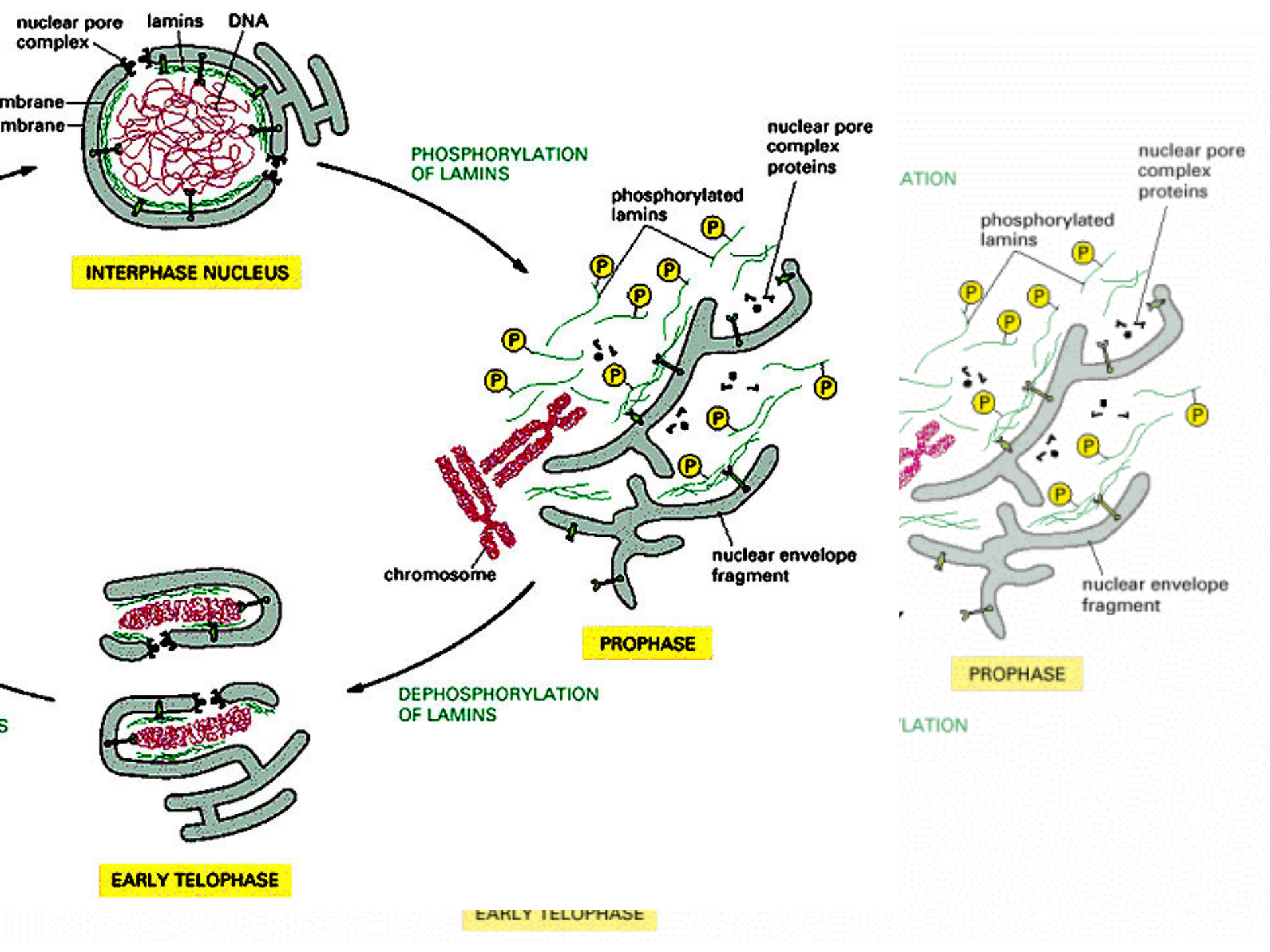
Import Control & T-cell activation



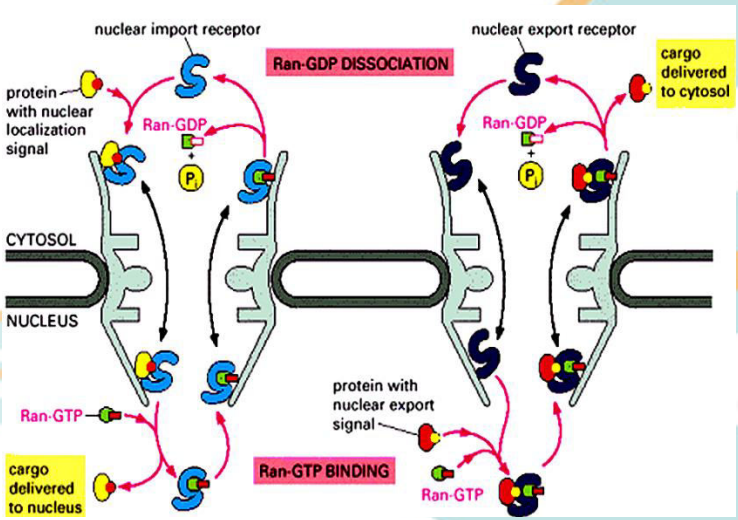
The activity of some gene regulatory proteins is controlled by keeping them out of the nucleus

Cell Division & Import Control




nuclear pore complex
lamins
DNA
mbrane
mbrane
INTERPHASE NUCLEUS
PHOSPHORYLATION OF LAMINS
nuclear pore complex proteins
phosphorylated lamins
P
chromosome
nuclear envelope fragment
PROPHASE
ATION
nuclear pore complex proteins
phosphorylated lamins
nuclear envelope fragment
PROPHASE
'LATION
DEPHOSPHORYLATION OF LAMINS
S
EARLY TELOPHASE
EARLY TELOPHASE

Nuclear Transport

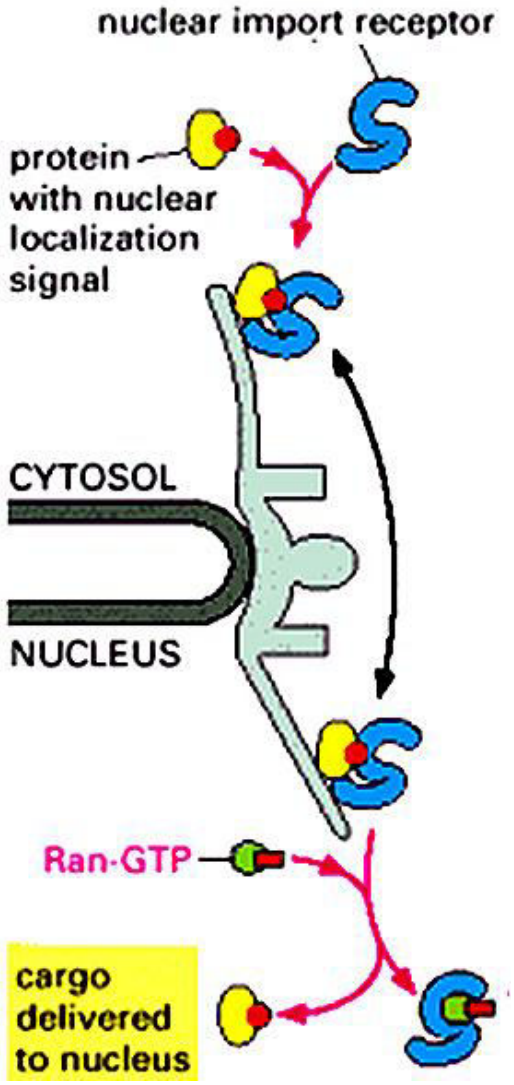


END

Cell Biology

The Ran Cycle

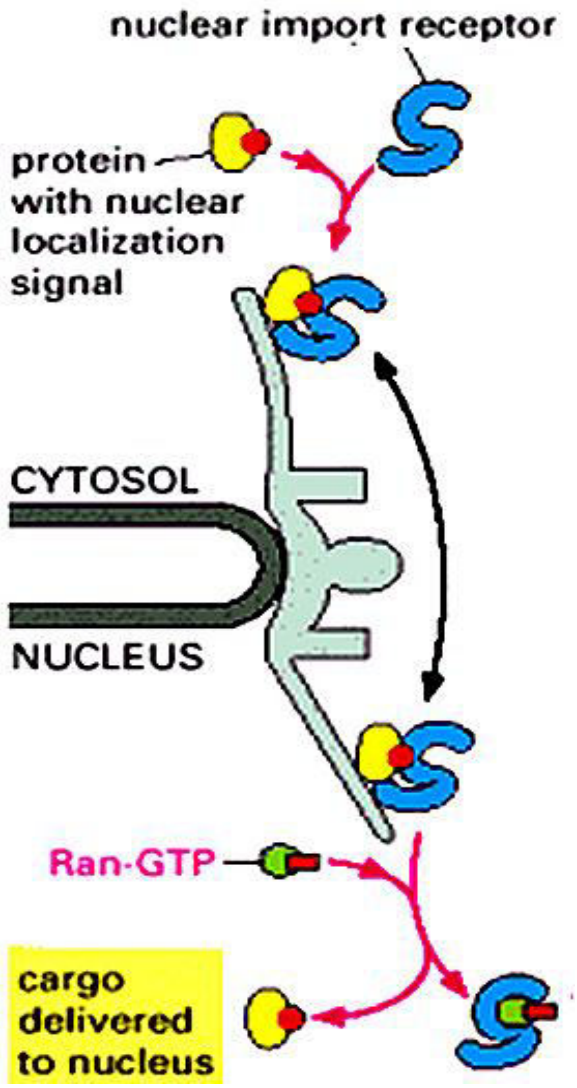
Nuclear Import Receptors



Docking of nuclear import receptors to FG-repeats on the cytosolic side of the nuclear pore complex, occurs only when these receptors are loaded with an appropriate cargo.

Import receptors with their bound cargo then move along tracks lined by FG-repeat sequences until they reach the nuclear side of the pore complex, where Ran-GTP binding causes the import receptors to release their cargo .

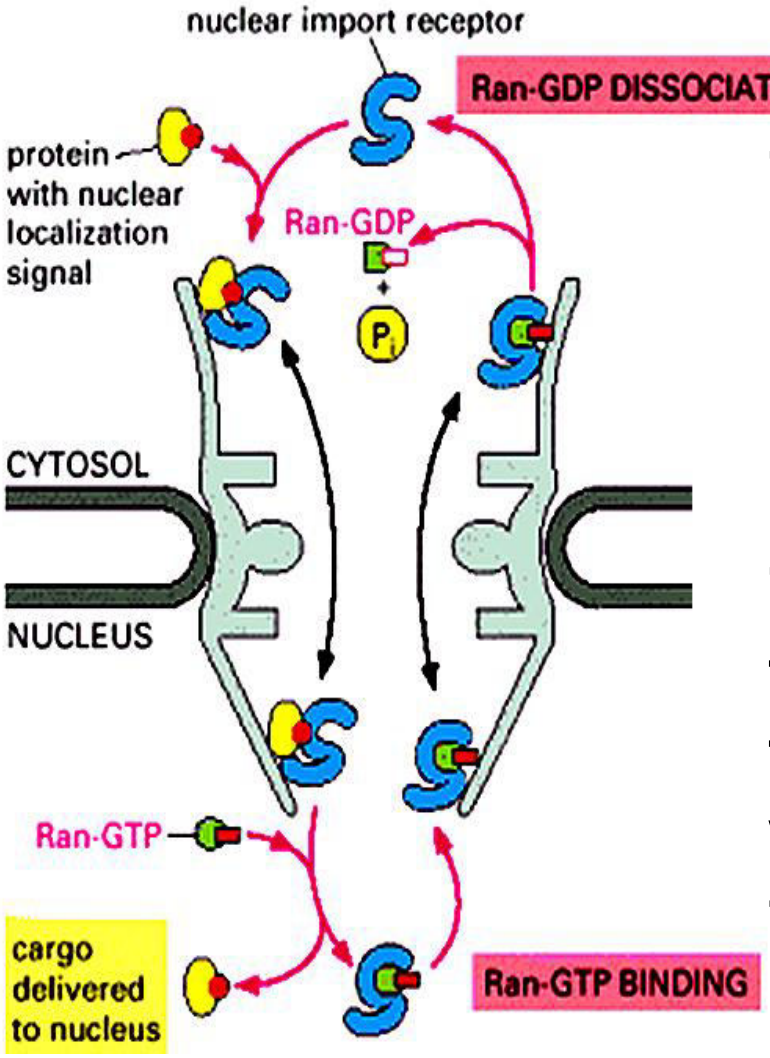
Nuclear Import Receptors



By favoring cargo-dependent loading of import receptors onto the FG-repeat track in the cytosol and Ran-GTP-dependent cargo release in the nucleus, the nuclear localization of Ran-GTP imposes directionality.

Having discharged its cargo in the nucleus, the empty import receptor with Ran-GTP bound is transported back through the pore complex to the cytosol

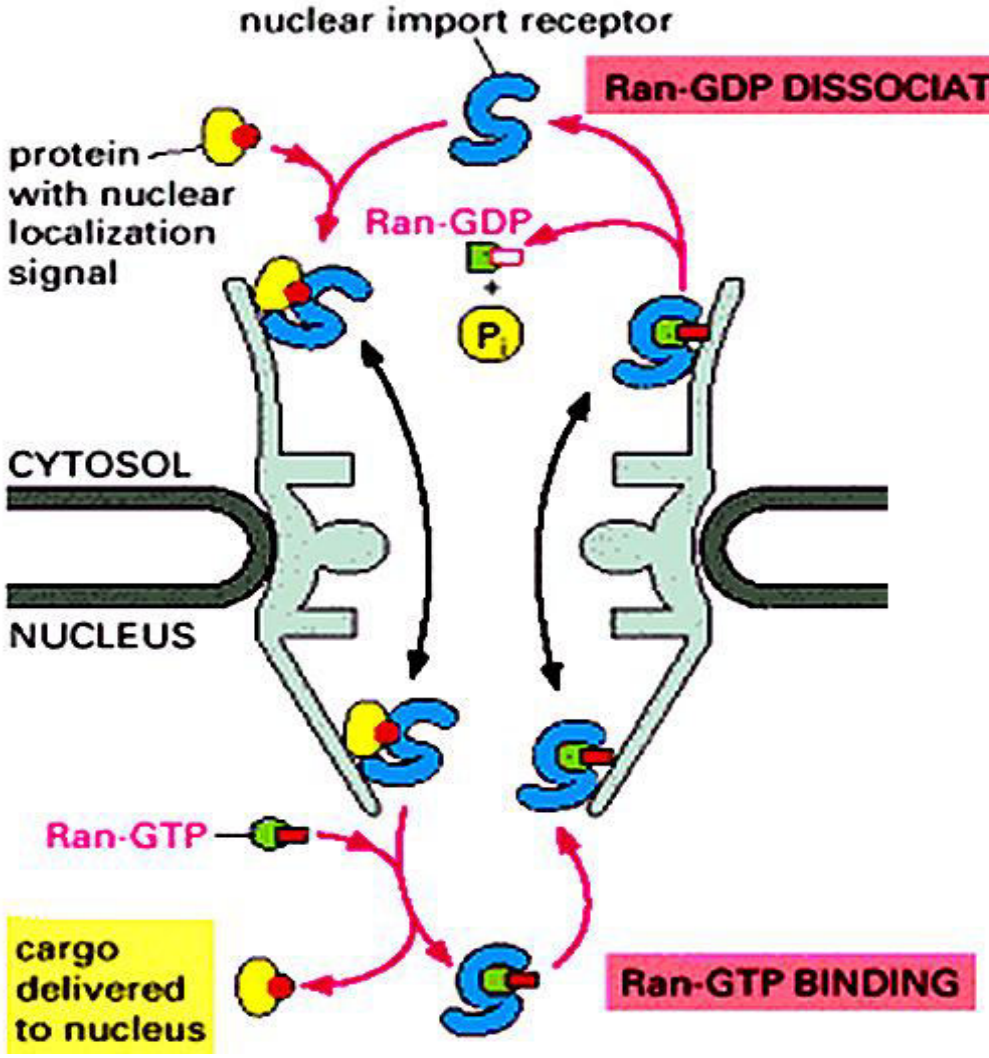
Nuclear Import Receptors



There, two cytosolic proteins, Ran Binding Protein and Ran-GAP collaborate to convert Ran-GTP to Ran-GDP.

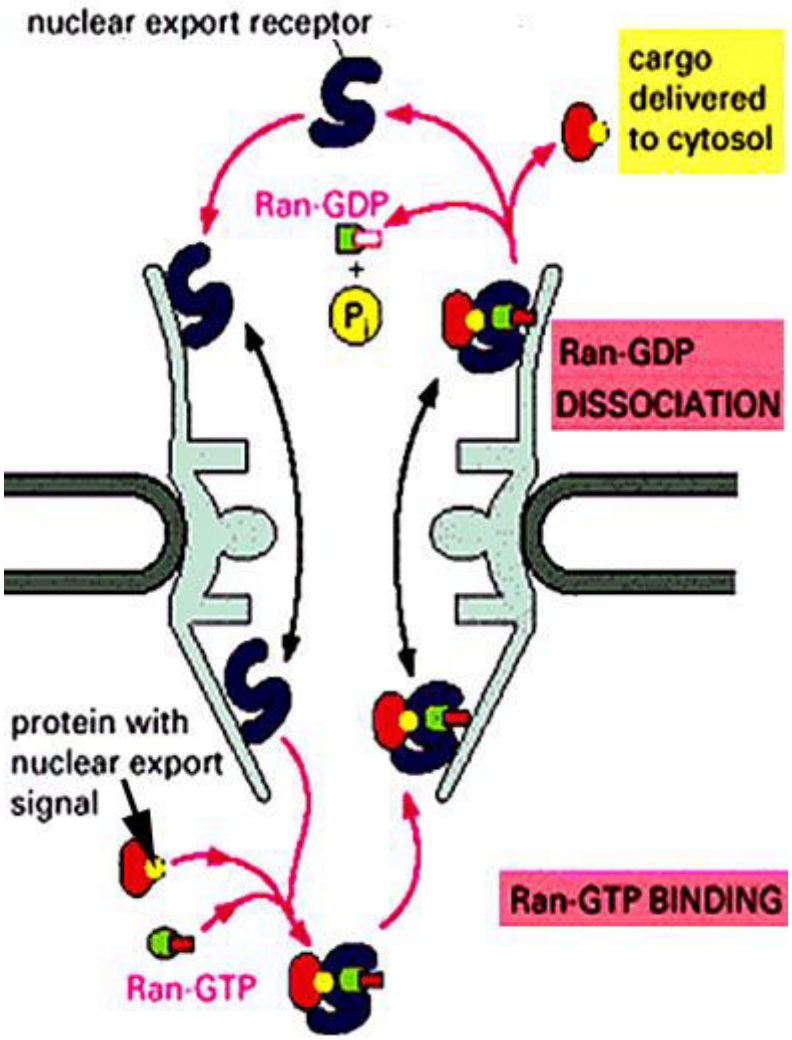
The Ran Binding Protein first displaces Ran-GTP from the import receptor, which allows Ran-GAP to trigger Ran to hydrolyze its bound GTP

Nuclear Import Receptors



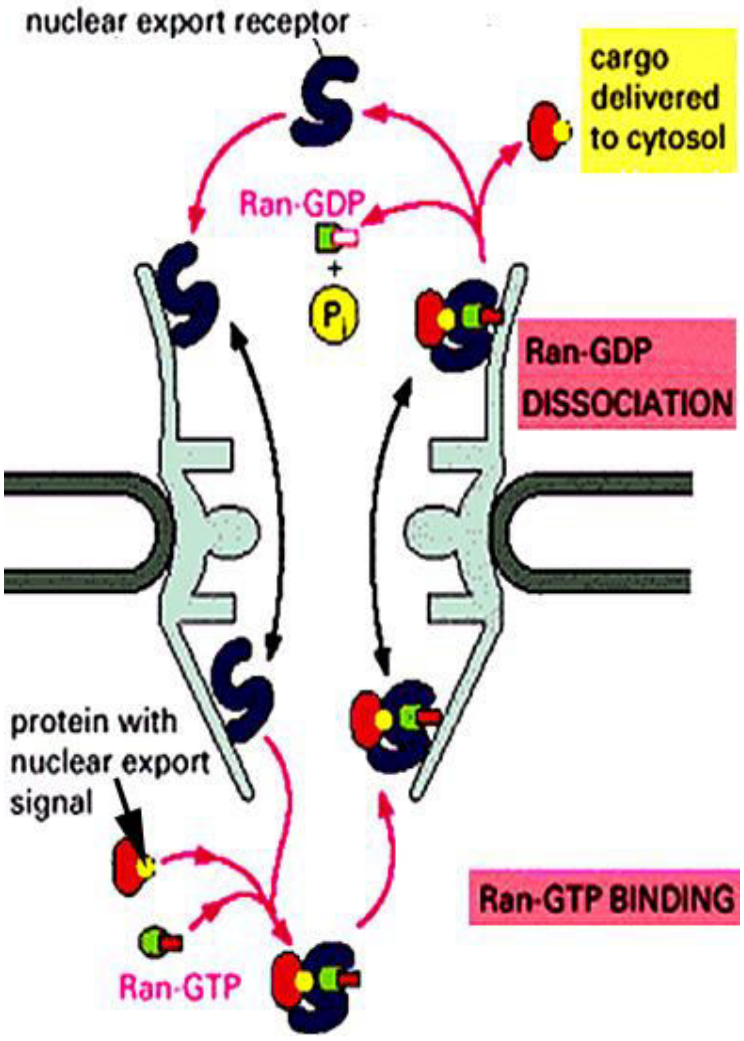
The Ran-GDP then dissociates from the Ran Binding Protein and is reimported into the nucleus, completing the cycle.

Nuclear Export Receptors



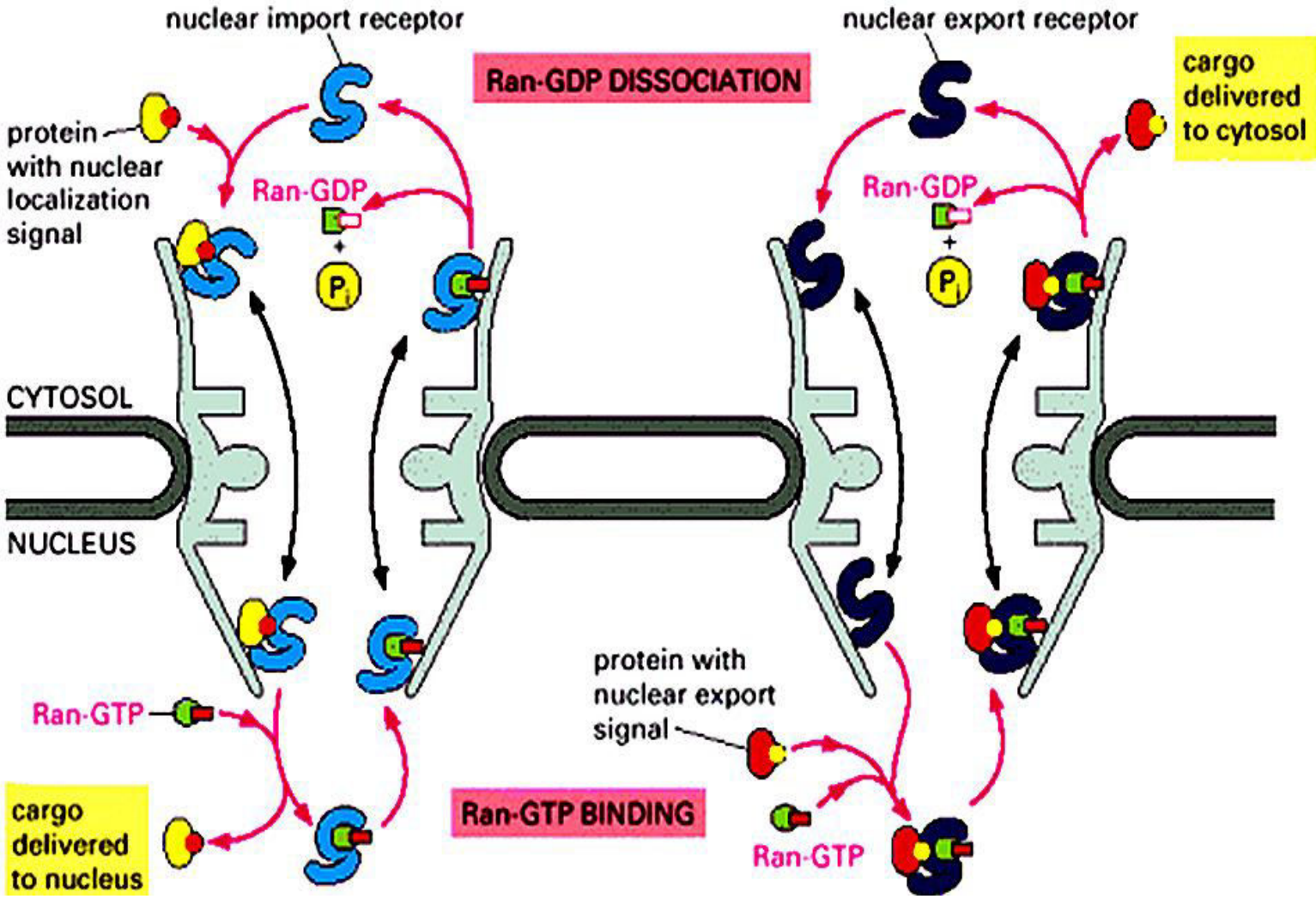
Nuclear export occurs by a similar mechanism, except that Ran-GTP in the nucleus promotes cargo binding to the export receptor and the binding of the loaded receptor to the nuclear side of the pore complex

Nuclear Export Receptors

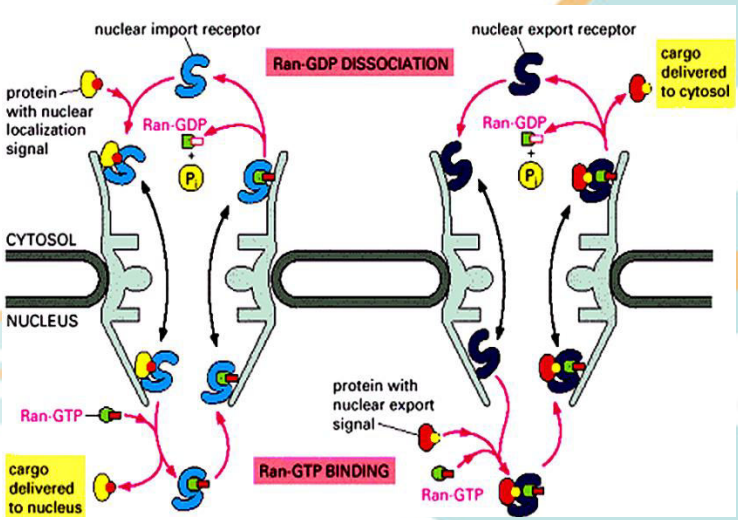


Once in the cytosol, Ran encounters Ran-GAP and Ran Binding Protein and hydrolyses its bound GTP. The export receptor then releases both its cargo and Ran-GDP in the cytosol and dissociates from the pore complex, and free export receptors are returned to the nucleus to complete the cycle

Nuclear Export Receptors



Nuclear Transport

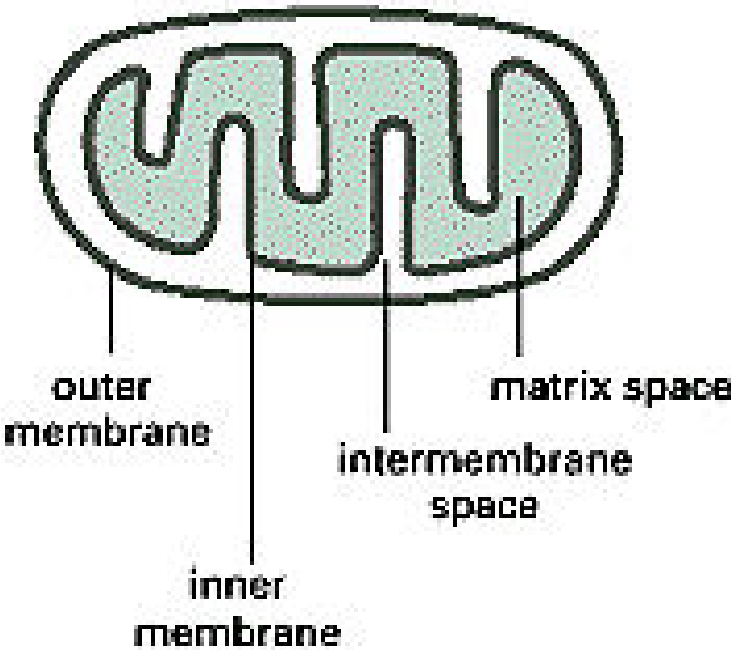


END

Cell Biology

Protein Transport into Mitochondria

Mitochondria Structure



Mitochondria is double-membrane-enclosed organelle

It contains its own DNA, ribosomes, and other components required for protein synthesis, most of its proteins are encoded in the cell nucleus and imported from the cytosol

There are two subcompartments in mitochondria: the internal matrix space and the intermembrane space

Mitochondrial Import

Proteins imported into the matrix of mitochondria are usually taken up from the cytosol within seconds or minutes of their release from ribosomes.

Mitochondrial proteins are first fully synthesized as precursor proteins in the cytosol & then translocated into mitochondria by a posttranslational mechanism

Mitochondrial precursor proteins have a signal sequence at their N terminus that is rapidly removed after import by a protease (the signal peptidase) in the mitochondrial matrix

The signal sequences are both necessary and sufficient for import of the proteins that contain them

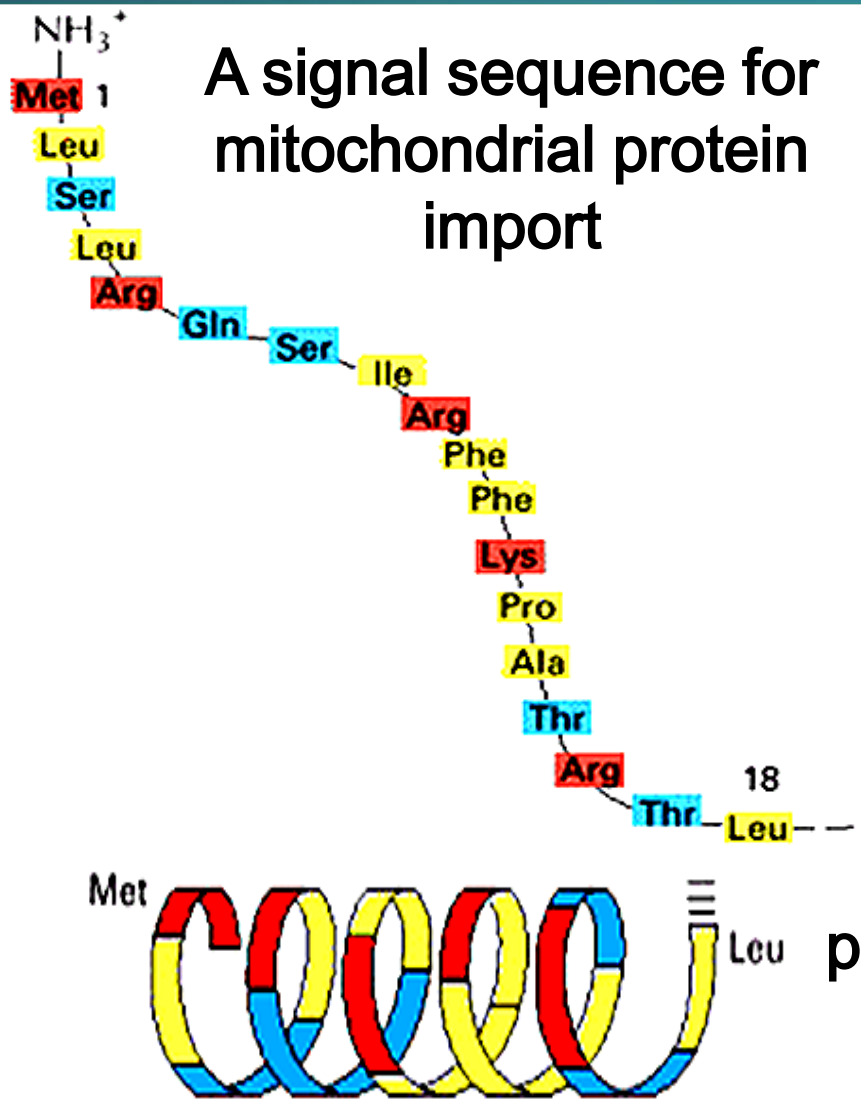
genetic engineering signals linked to cytosolic protein to direct the protein into the mitochondrial matrix

Targeting Sequence

Matrix signal sequences have propensity to fold into an amphipathic α helix, in which positively charged residues are clustered on one side of the helix, while uncharged hydrophobic residues are clustered on the opposite side

This configuration—rather than a precise amino acid sequence—is recognized by specific receptor proteins that initiate protein translocation

Signal Sequence



A signal sequence for mitochondrial protein import

When the signal sequence is folded as an α helix, the positively charged residues (red) are seen clustered on one face of helix, while the nonpolar residues (yellow) are clustered on opposite face

amphipathic α helix, is recognized by receptor proteins on mitochondrial surface

Signal Sequence

Mitochondrial precursor proteins do not fold into their native structures after they are synthesized; instead, they remain unfolded through interactions with other proteins in the cytosol.

chaperone proteins belonging to the hsp70 family

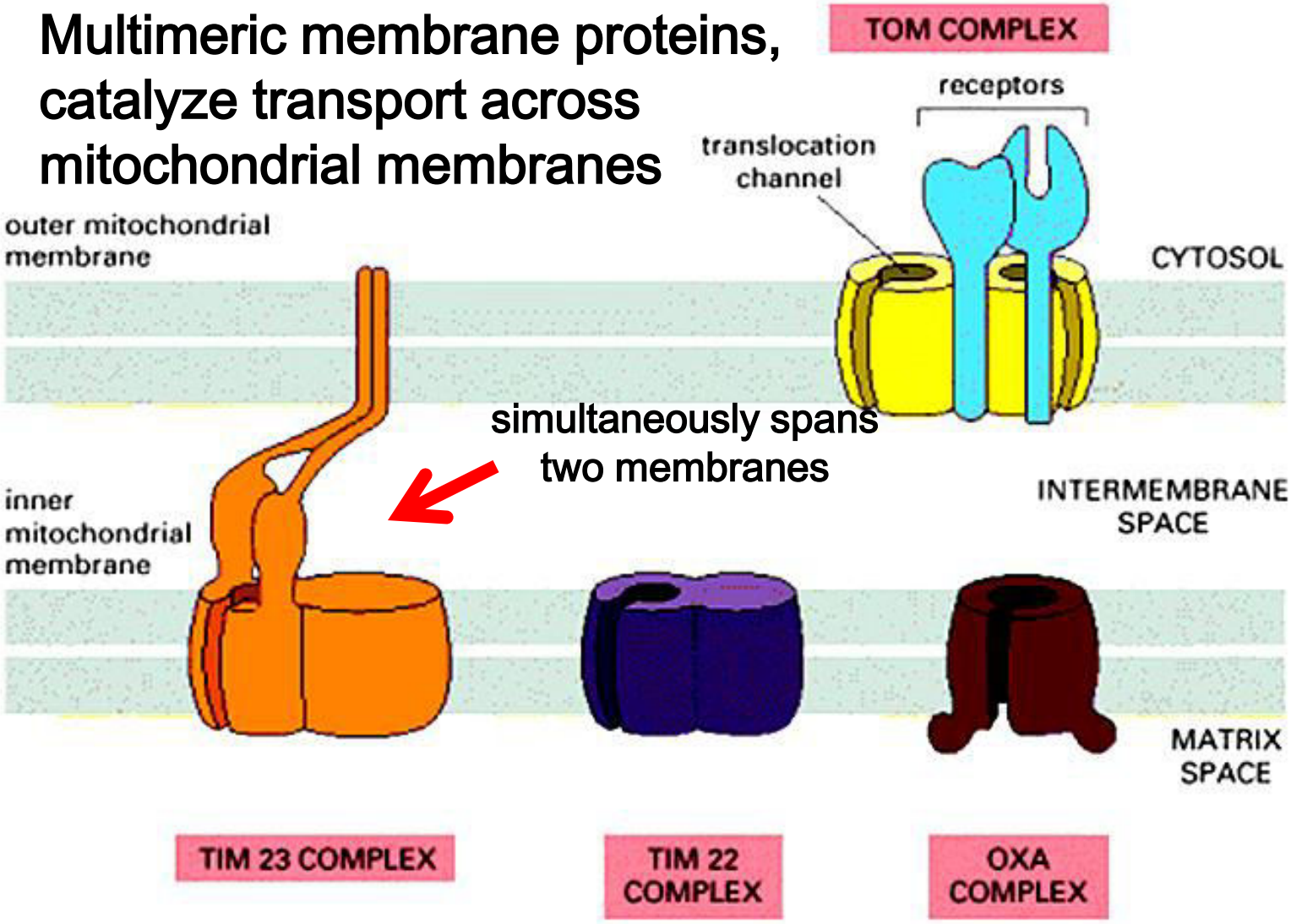
END

Cell Biology

The Translocators

The Translocators

Multimeric membrane proteins, catalyze transport across mitochondrial membranes



The Translocators

TOM and TIM complexes functions are coupled to allow protein transport across both membranes simultaneously, both translocators can work independently.

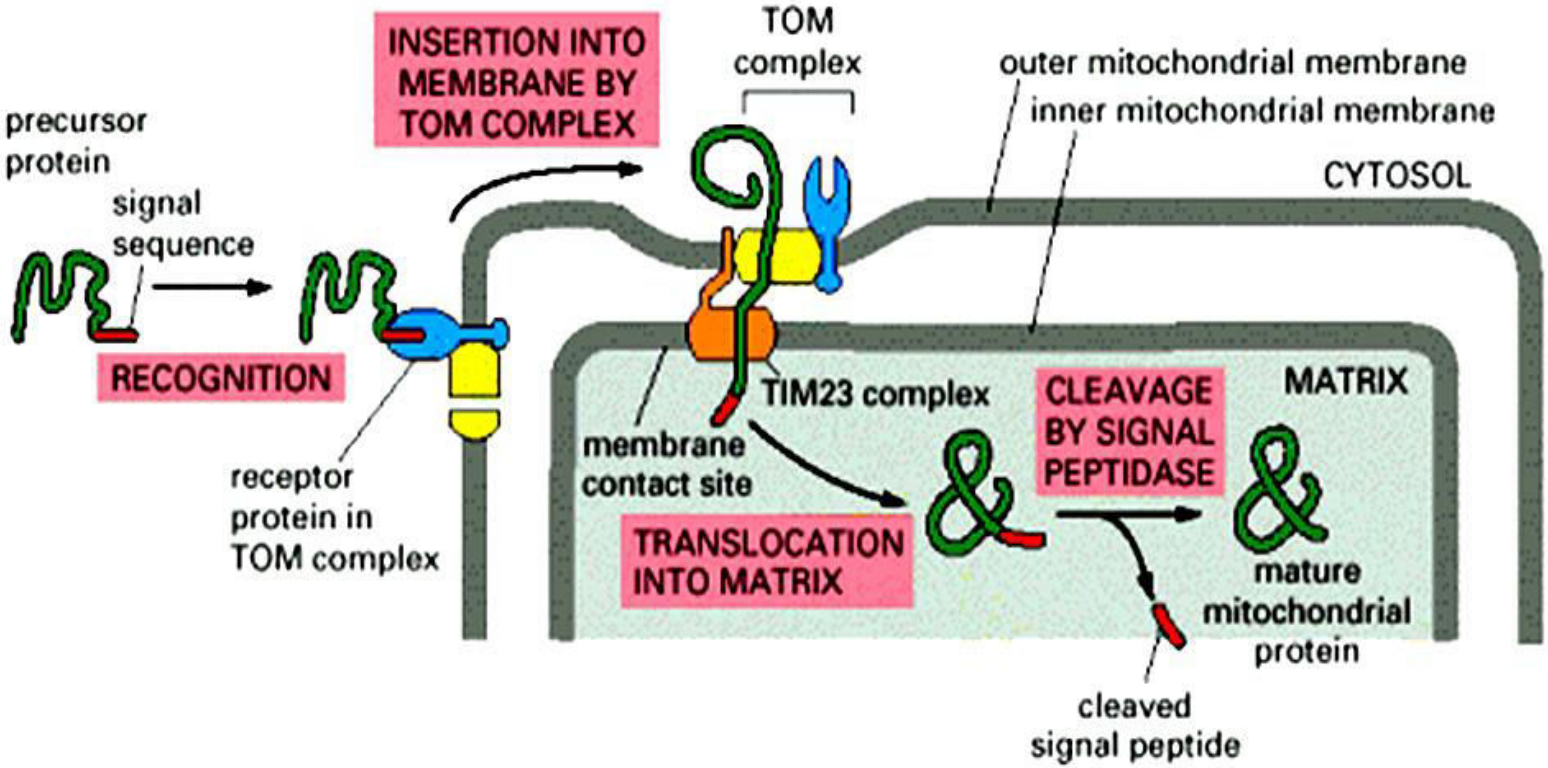
TIM22 complex mediates the insertion inner membrane proteins.

OXA complex, mediates insertion of inner membrane proteins synthesized within mitochondria.

The N-terminal signal sequence of precursor protein is recognized by receptors of the TOM complex

The protein is translocated across both mitochondrial membranes at or near contact sites

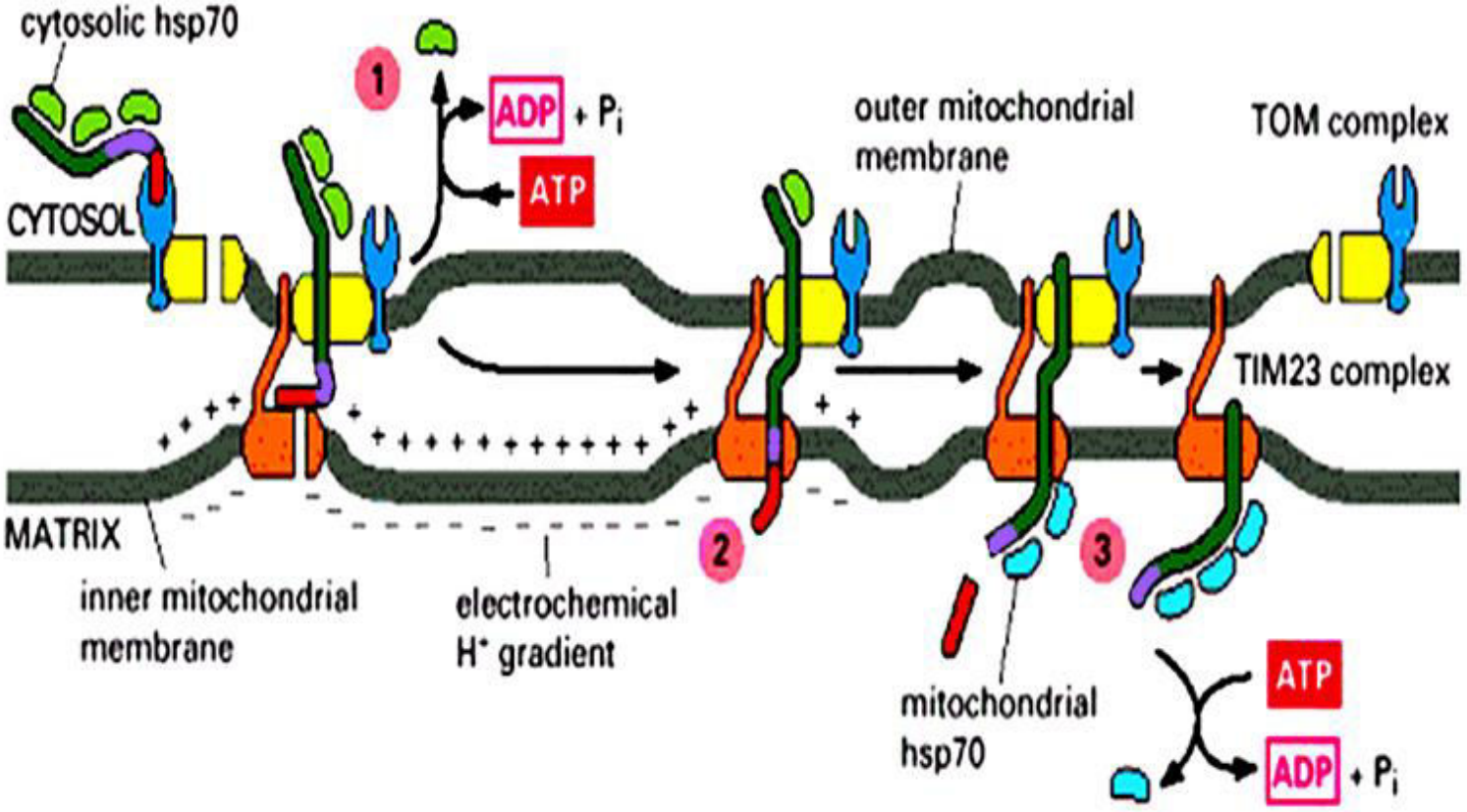
The Translocators



The N-terminal signal sequence of precursor protein is recognized by receptors of the TOM complex

The protein is translocated across both mitochondrial membranes at or near contact sites.

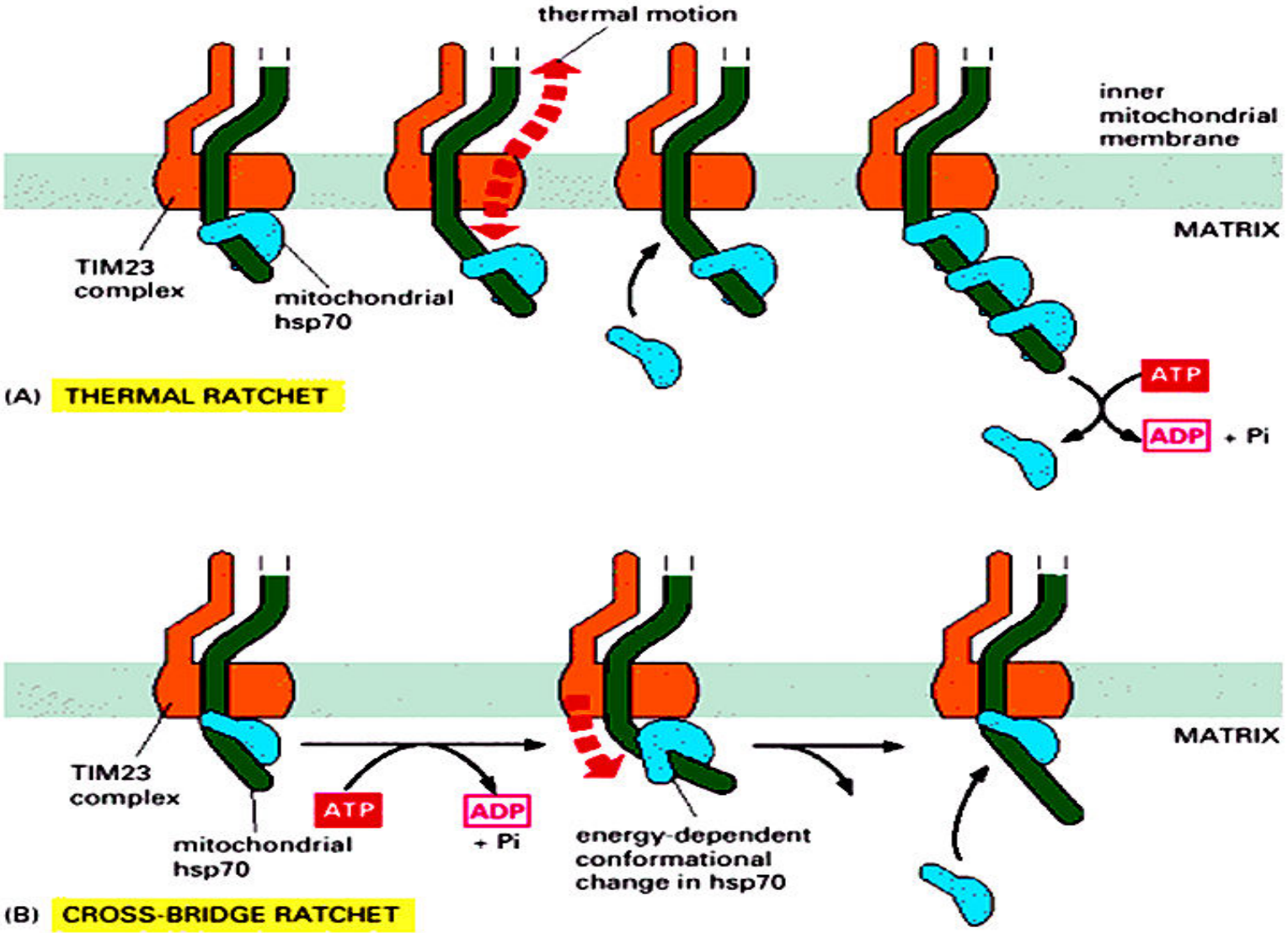
Directional transport requires energy



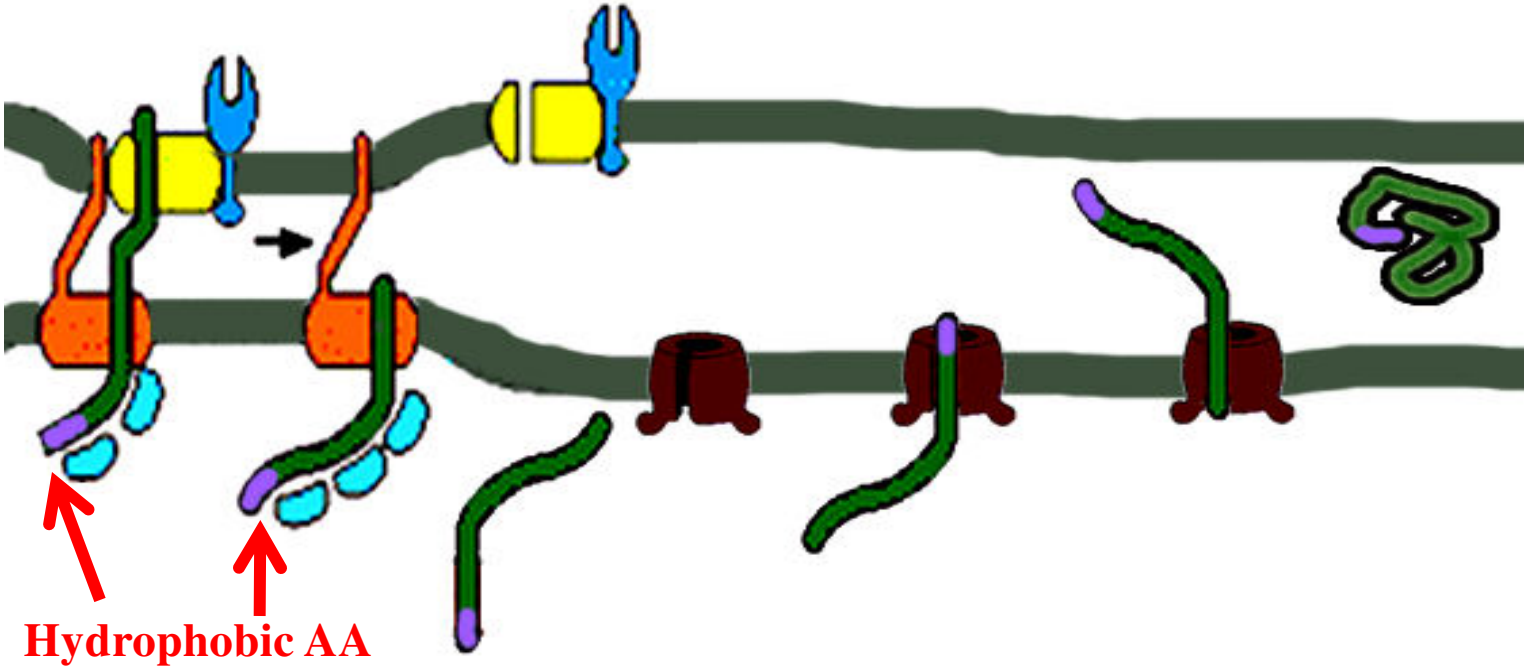
Protein import is fueled by ATP hydrolysis at 2 sites, (1) outside mitochondria & (2) in the matrix + (3) H^+ gradient across the inner mitochondrial membrane.

Electrochemical --- Unfolded proteins with SS can be translocated by TOM without energy

Directional transport requires energy



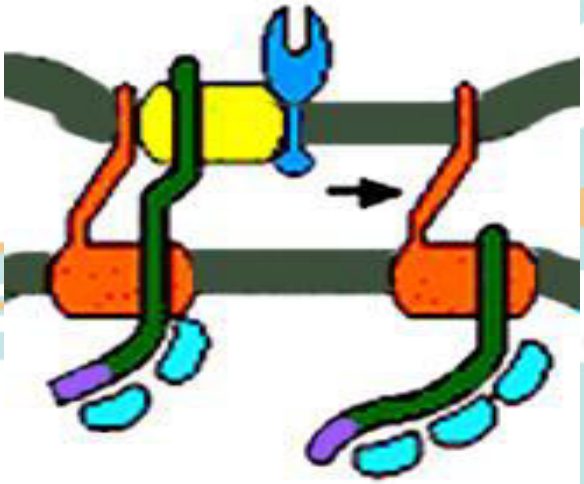
Directional transport requires energy



Protein Transport into the Intermembrane Space Requires Two Signal Sequences.

OXA complex is also used to insert proteins encoded in the mitochondrion into the inner membrane

The Translocators

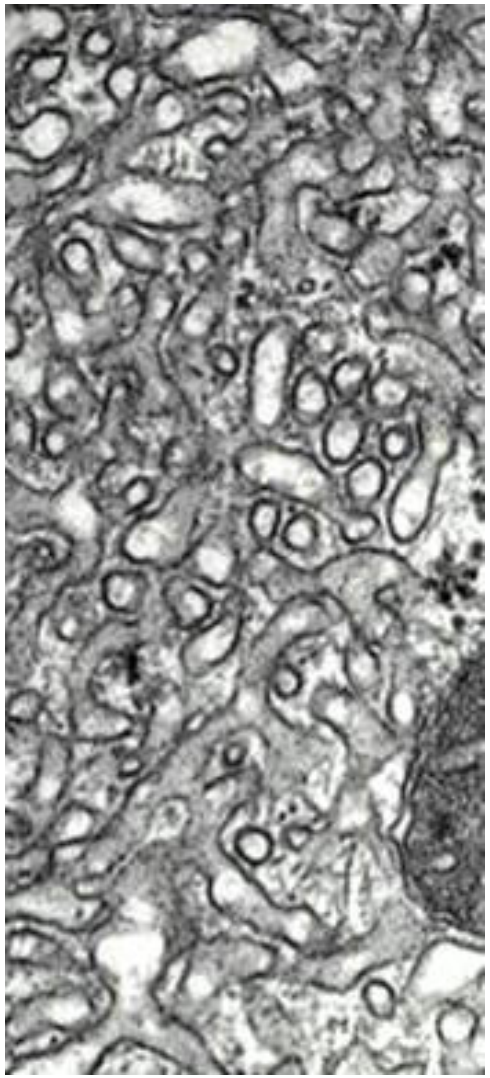


END

Cell Biology

ER Transport Introduction

The Endoplasmic Reticulum

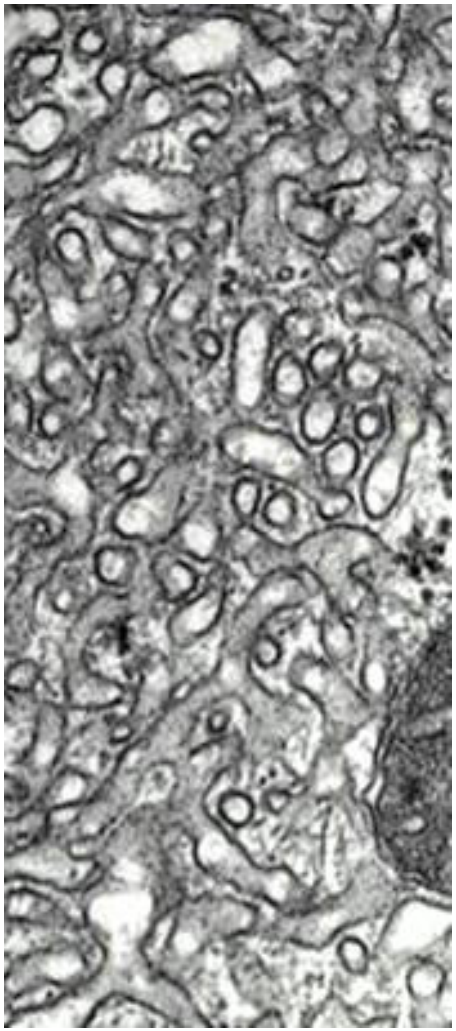


Smooth ER

Regions of ER that lack bound ribosomes are called smooth endoplasmic reticulum, or smooth ER in the great majority of cells, such regions are scanty and are often partly smooth and partly rough

Smooth ER contains ER exit sites from which transport vesicles carrying newly synthesized proteins and lipids bud off for transport to the Golgi apparatus

The Endoplasmic Reticulum

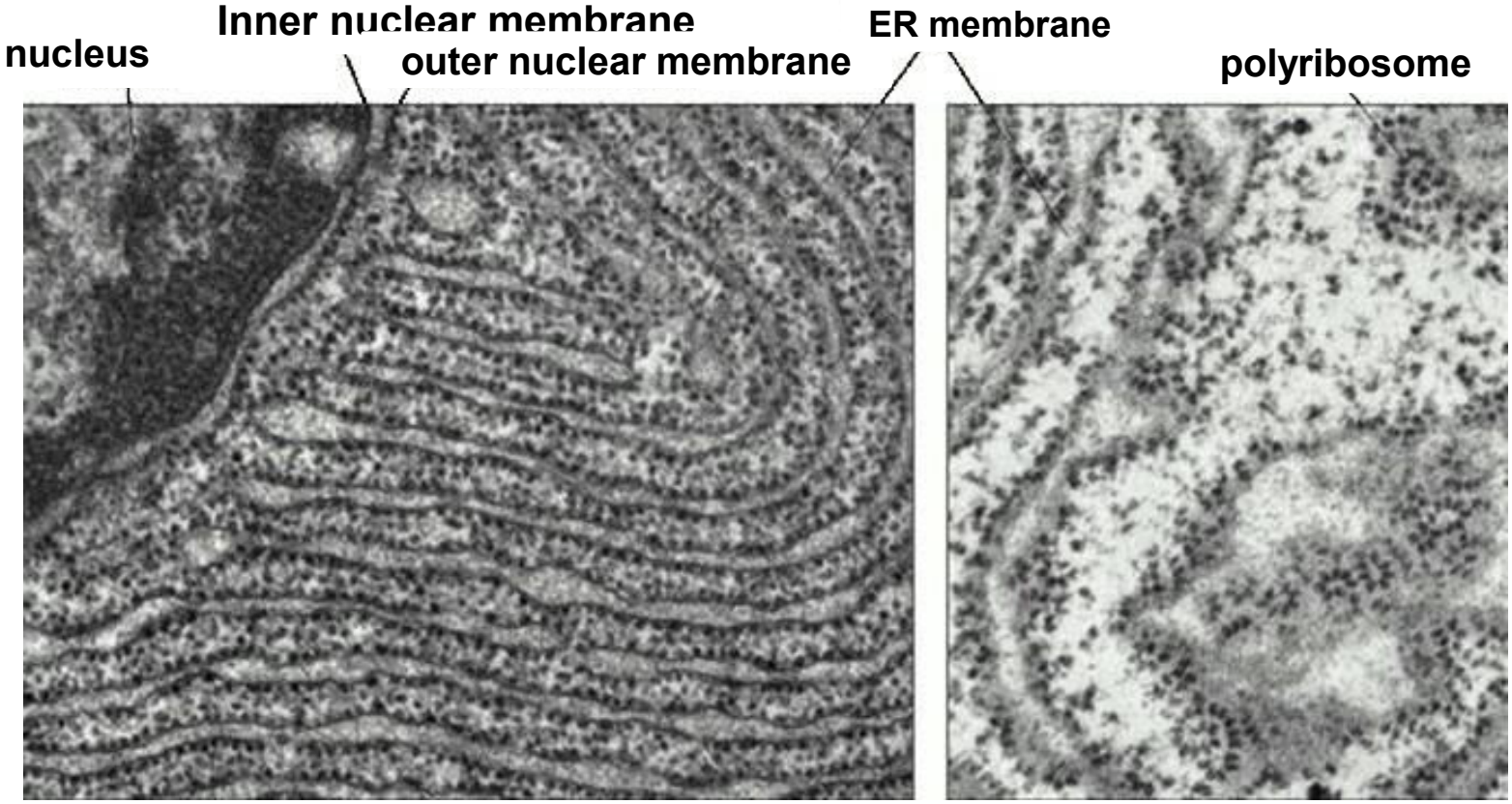


Smooth ER

In hepatocytes smooth ER doubles in surface area during detoxification of both lipid-soluble drugs and various harmful compounds produced by metabolism e.g. phenobarbital

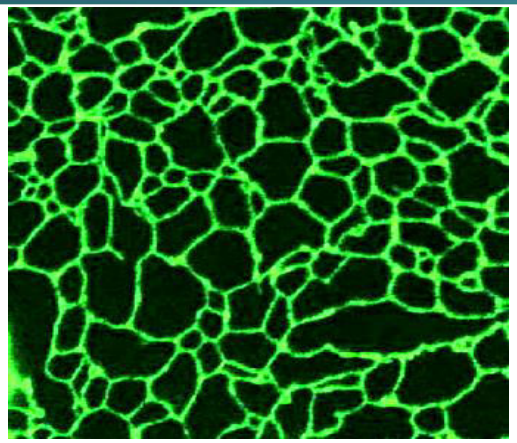
ER in most eucaryotic cells stores $Ca2^{+}$ e.g. Muscle cells, have an abundant specialized smooth ER, "sarcoplasmic reticulum", which sequesters Ca2+ from the cytosol

The Endoplasmic Reticulum



The ribosome that is synthesizing the protein is directly attached to the ER membrane creating regions termed rough endoplasmic reticulum

The Endoplasmic Reticulum



All eucaryotic cells have an endoplasmic reticulum (ER) which is organized into a netlike branching tubules and flattened sacs throughout the cytosol

The tubules and sacs are interconnected the ER membrane encloses a single internal space

This highly convoluted space is called the ER lumen or the ER cisternal space.

The ER membrane separates the ER lumen from the cytosol, and it mediates the selective transfer of molecules between these two compartments

The Endoplasmic Reticulum

The ER captures selected proteins from the cytosol as they are being synthesized

Two types: 1)transmembrane proteins, partly translocated across the ER membrane and become embedded in it, 2)water-soluble proteins, which are fully translocated across the ER membrane and are released into the ER lumen

Some of the transmembrane proteins function in the ER, but many are destined to reside in the plasma membrane or the membrane of another organelle, water-soluble proteins are destined either for the lumen of an organelle or for secretion.

The Endoplasmic Reticulum

All of these proteins, have same kind of signal sequence & are translocated by similar mechanisms

ER protein import is a co-translational process the import of proteins into mitochondria, chloroplasts, nuclei, and peroxisomes, is posttranslational processes

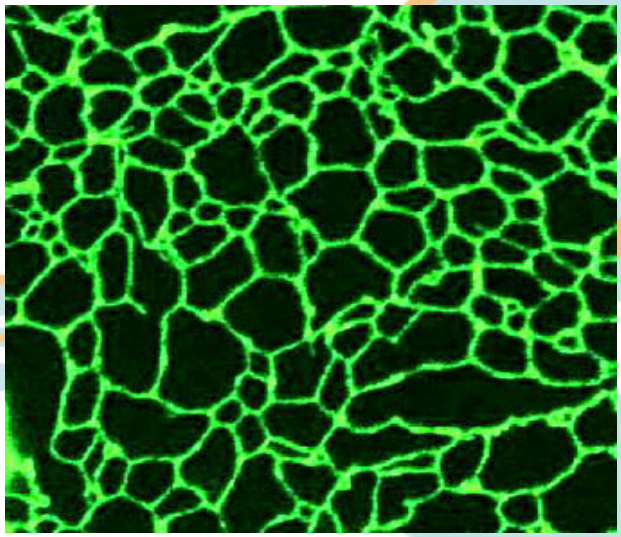
One end of the protein is usually translocated into the ER as the rest of the polypeptide chain is being made, the protein is never released into the cytosol and therefore is never in danger of folding up before reaching the translocator in the ER membrane

The Endoplasmic Reticulum

Animation

133-203

The Endoplasmic Reticulum



END

Cell Biology

ER Transport

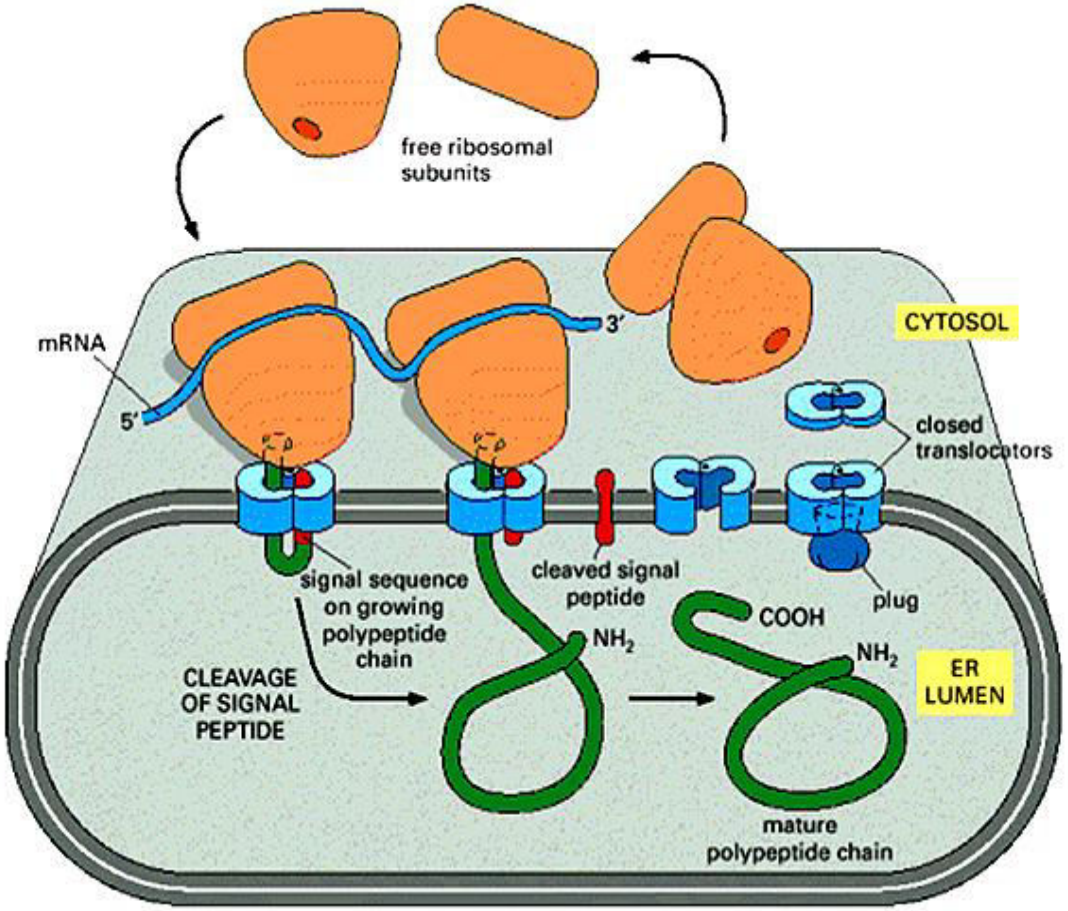
ER Transport

Import into ER

$^{+}H_3N$-Met-Met-Ser-Phe-Val-Ser-Leu-Leu-Leu-Val-Gly-Ile-Leu-Phe-Trp-Ala-Thr-Glu-Ala-Glu-Gln-Leu-Thr-Lys-Cys-Glu-Val-Phe-Gln-

Return to ER

-Lys-Asp-Glu-Leu-COO^{-}

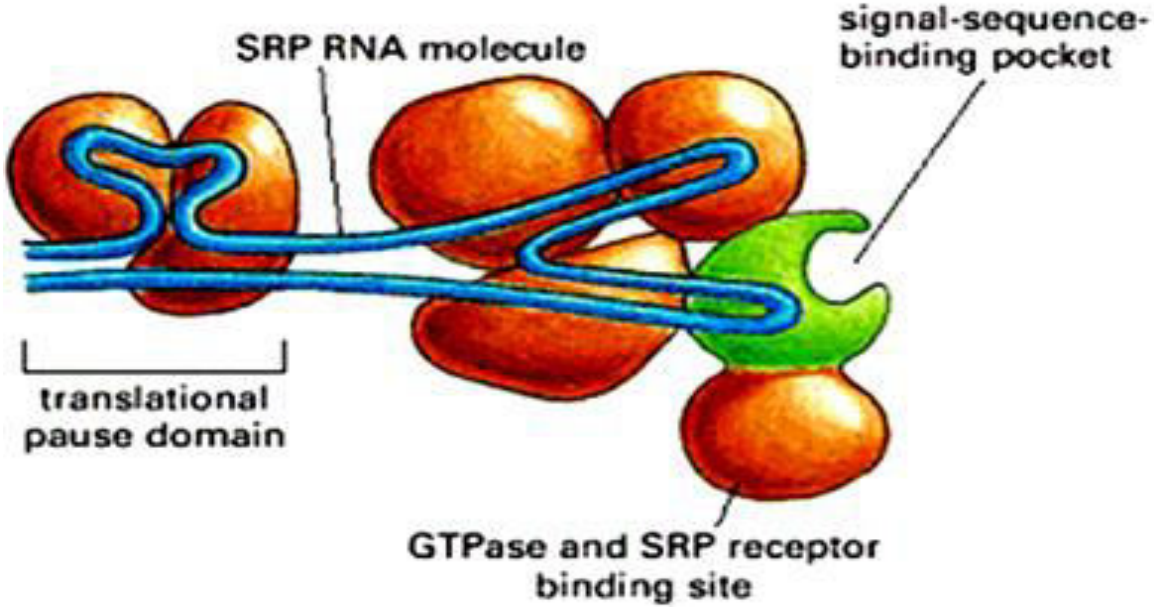


Signal Sequences Were First Discovered in Proteins Imported into the Rough ER

Signal-Recognition Particle (SRP)

The ER signal sequence is guided to the ER by 2 components:
1) SRP which cycles between the ER membrane and the cytosol & binds to the signal sequence, & 2) SRP receptor in the ER membrane

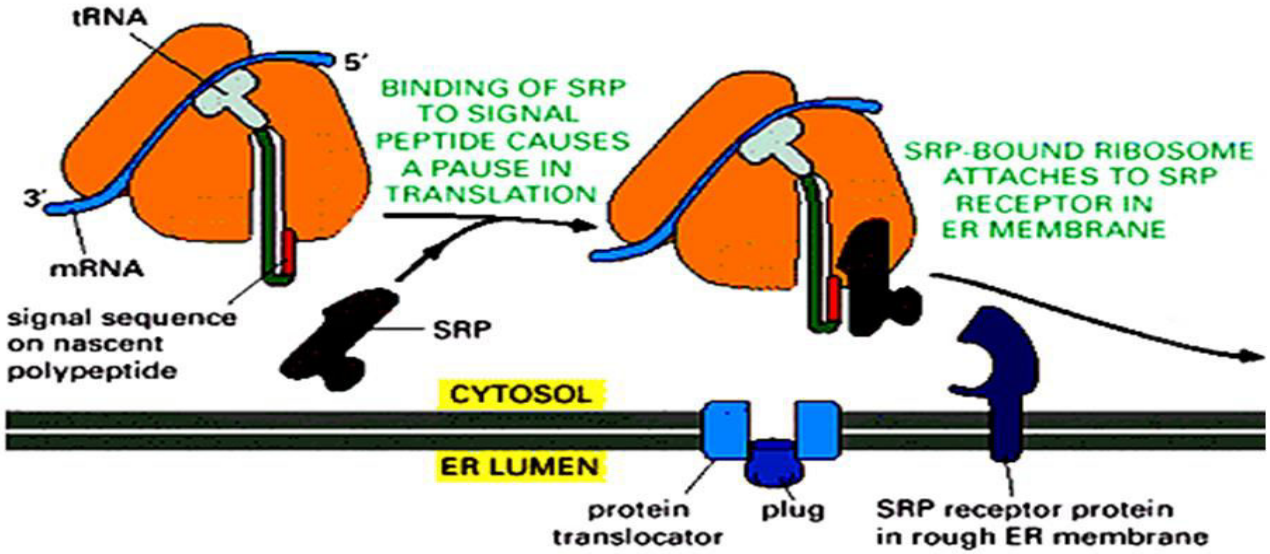
The SRP consisting of six different polypeptide chains bound to a single small RNA molecule



SRP

The SRP binds to the ER signal sequence as soon as the peptide has emerged from the ribosome causes a pause in protein synthesis, ensuring that the protein is not released into the cytosol

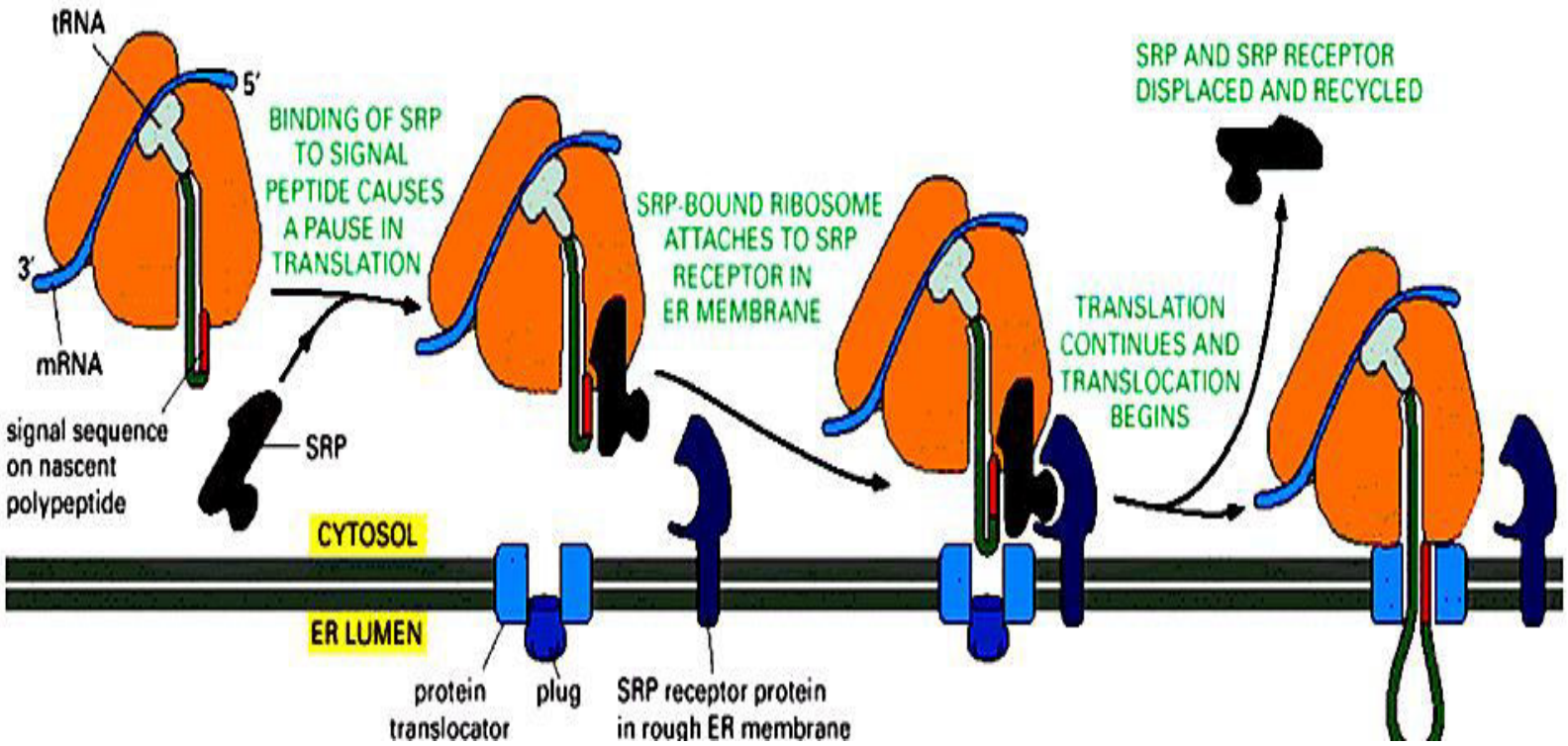
This is safety mechanisim especially for lysosomal hydrolases that could wreak havoc in the cytosol



the pause presumably gives the ribosome enough time to bind to the ER

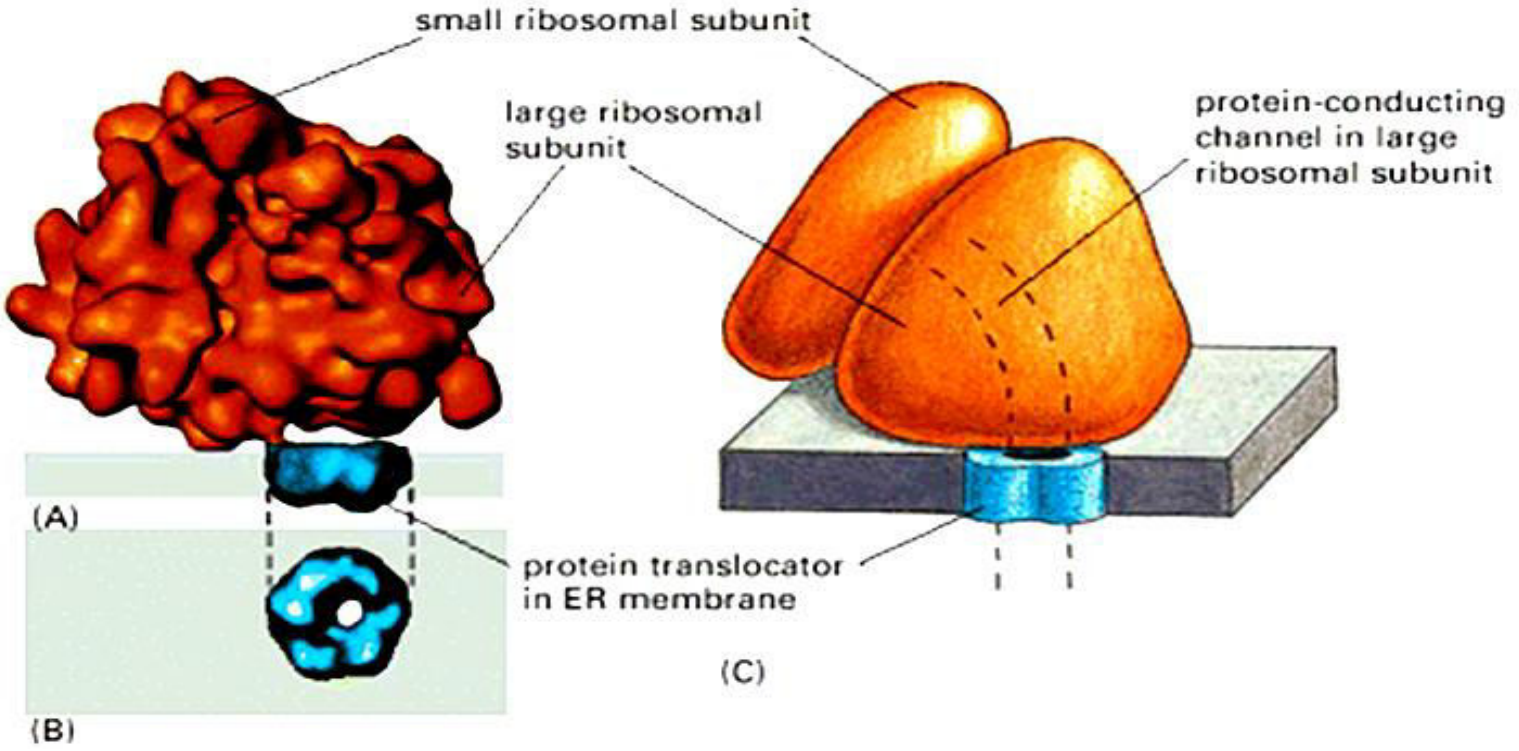
ER Transport

SRP-ribosome complex binds to the SRP receptor, this interaction brings the SRP-ribosome complex to a protein translocator SRP & SRP receptor are then released, and the growing polypeptide chain is transferred across the membrane



The Translocator

The translocator Sec61 complex forms a water-filled pore in the membrane through which the polypeptide chain traverses the membrane, it consists of 4 protein which assemble into a donut like structure



The Translocator

When a ribosome binds, the central hole in the translocator lines up with a tunnel in the large ribosomal subunit through which the growing polypeptide chain exits from the ribosome

The bound ribosome forms a tight seal with the translocator, such that the space inside the ribosome is continuous with the lumen of the ER and no molecules can escape from the ER

The pore in the translocator cannot be open permanently, however; if it were, Ca2+ would leak out of the ER when the ribosome detaches

The Translocator

A lumenal ER protein serves as a plug or that the translocator itself can rearrange to close the pore when no ribosome is bound

Thus, the pore is a dynamic structure that opens only transiently when a ribosome with a growing polypeptide chain attaches to the ER membrane

The signal sequence in the growing polypeptide chain triggers the opening of the pore: after the signal sequence is released from the SRP and the growing chain has reached a sufficient length, the signal sequence binds to a specific site inside the pore itself, thereby opening the pore

The Translocator

An ER signal sequence is therefore recognized twice: first, by an SRP in the cytosol, and then by a binding site in the ER protein translocator

This may help to ensure that only appropriate proteins enter the lumen of the ER

Some proteins, however, are imported into the ER after their synthesis has been completed, demonstrating that translocation does not always require ongoing translation

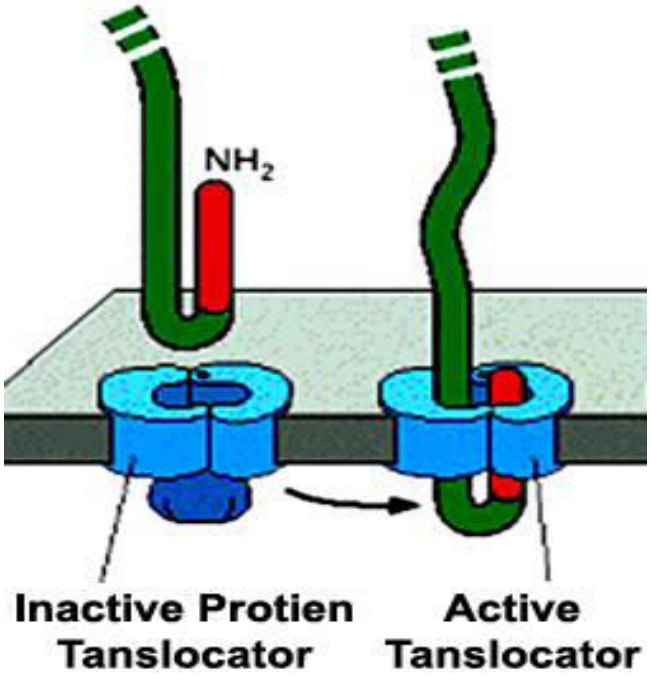
The Translocator

END

Cell Biology

Translocation

The N-terminal ER signal

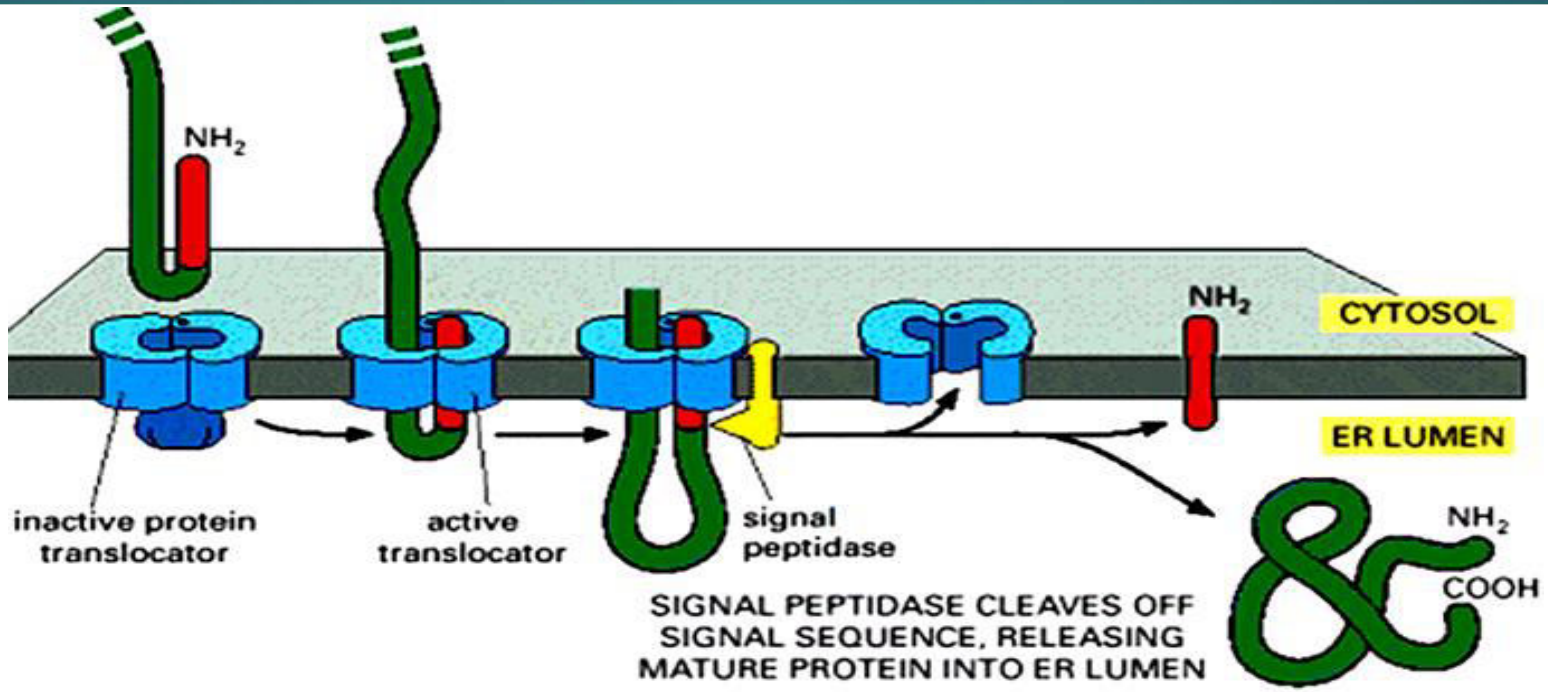


The N-terminal ER signal sequence of a soluble protein has two signaling functions

1) It directs the protein to the ER membrane 2) It serves as a start-transfer signal (or start-transfer peptide) that opens the pore.

Even after it is cleaved off by signal peptidase, the signal sequence remains bound to the translocator while the rest of the protein is threaded continuously through the membrane as a large loop

The N-terminal ER signal



Once the C-terminus of the protein has passed through the membrane, the translocated protein is released into the ER lumen the signal sequence is released from the pore and rapidly degraded to amino acids by other proteases in the ER

signal sequences contact with Sec61 &hydrophobic lipid core of the membrane

Single-Pass Proteins

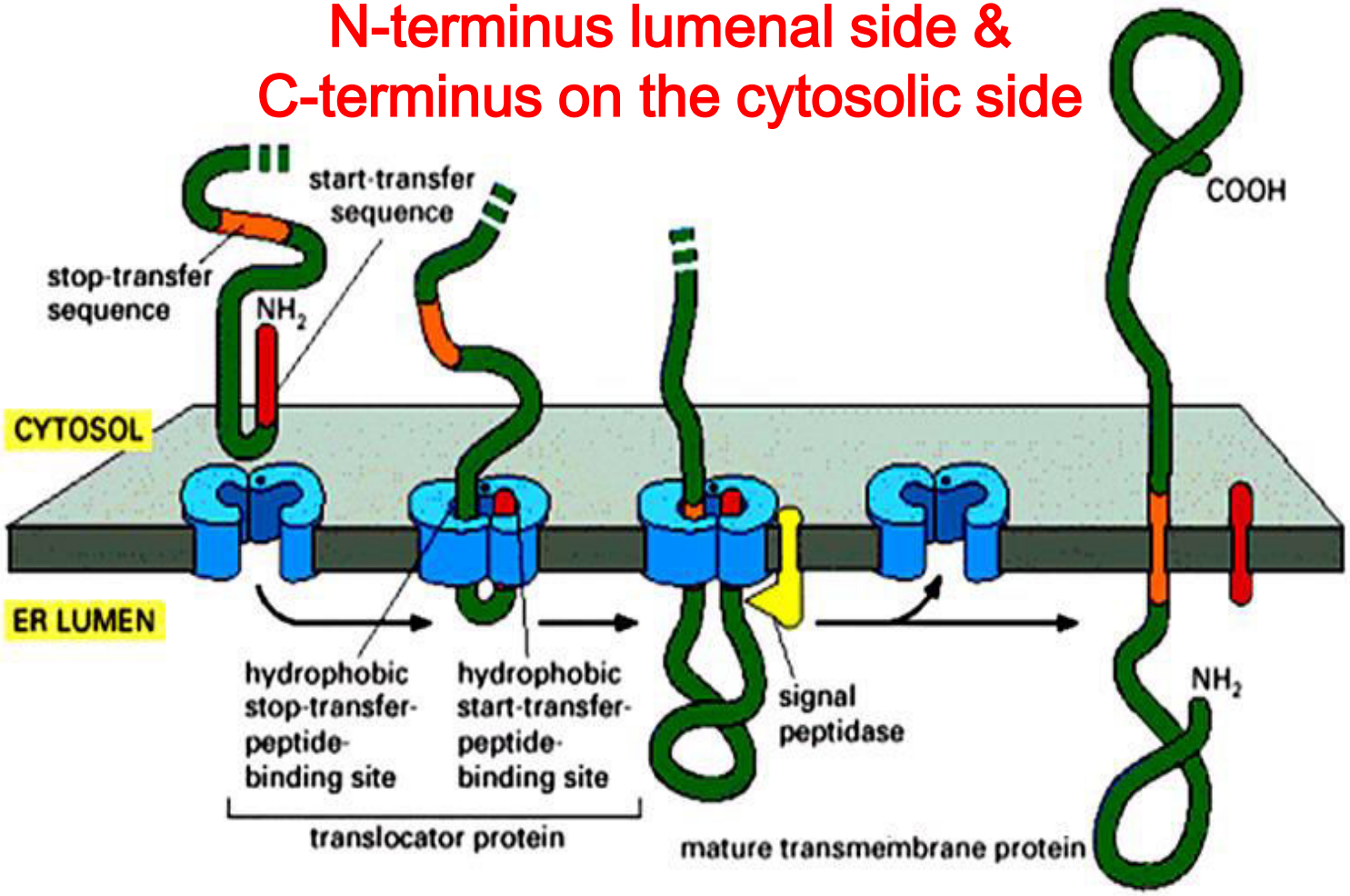
An N-terminal signal sequence initiates translocation, just as for a soluble protein, but an additional hydrophobic segment in the polypeptide chain stops the transfer process before the entire polypeptide chain is translocated

This stop-transfer signal anchors the protein in the membrane after the ER signal sequence (the start-transfer signal) has been released from the translocator and has been cleaved off

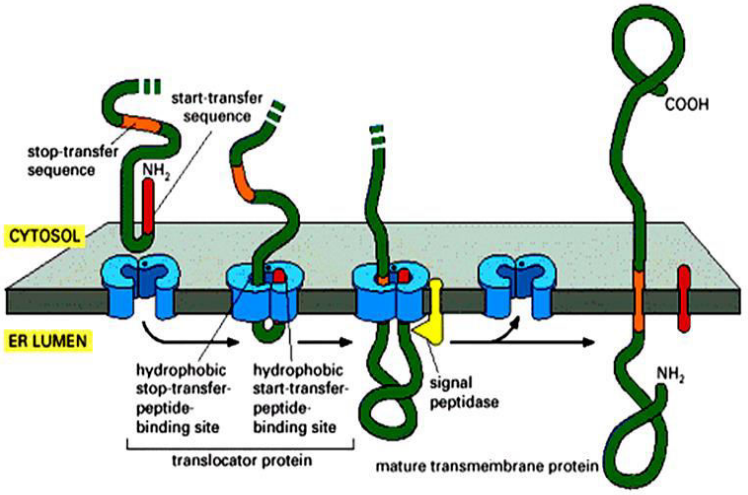
The stop-transfer sequence is transferred into the bilayer by the lateral gating mechanism, and it remains there as a single α-helical membrane-spanning segment.

The N-terminal ER signal

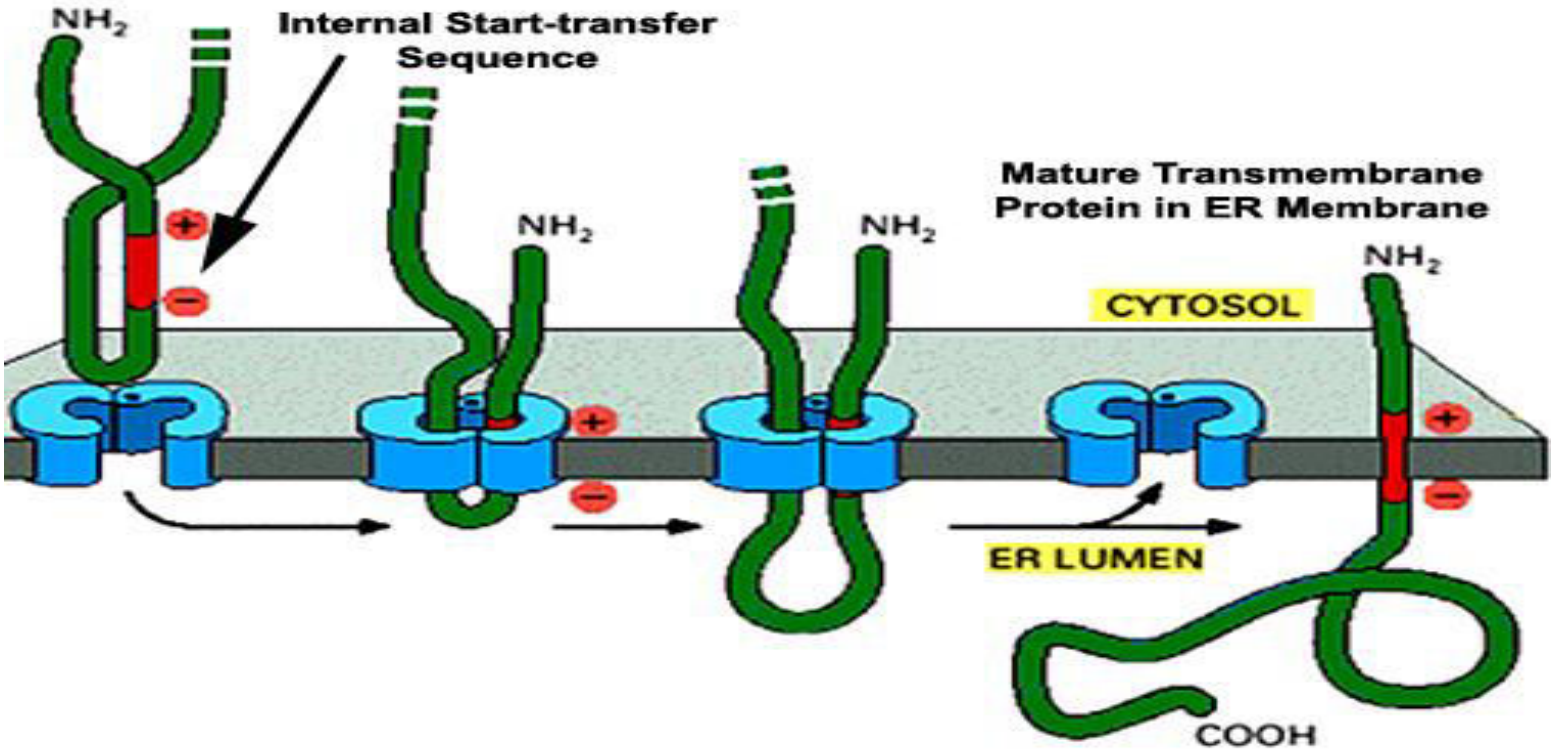
N-terminus lumenal side &
C-terminus on the cytosolic side



The N-terminal ER signal



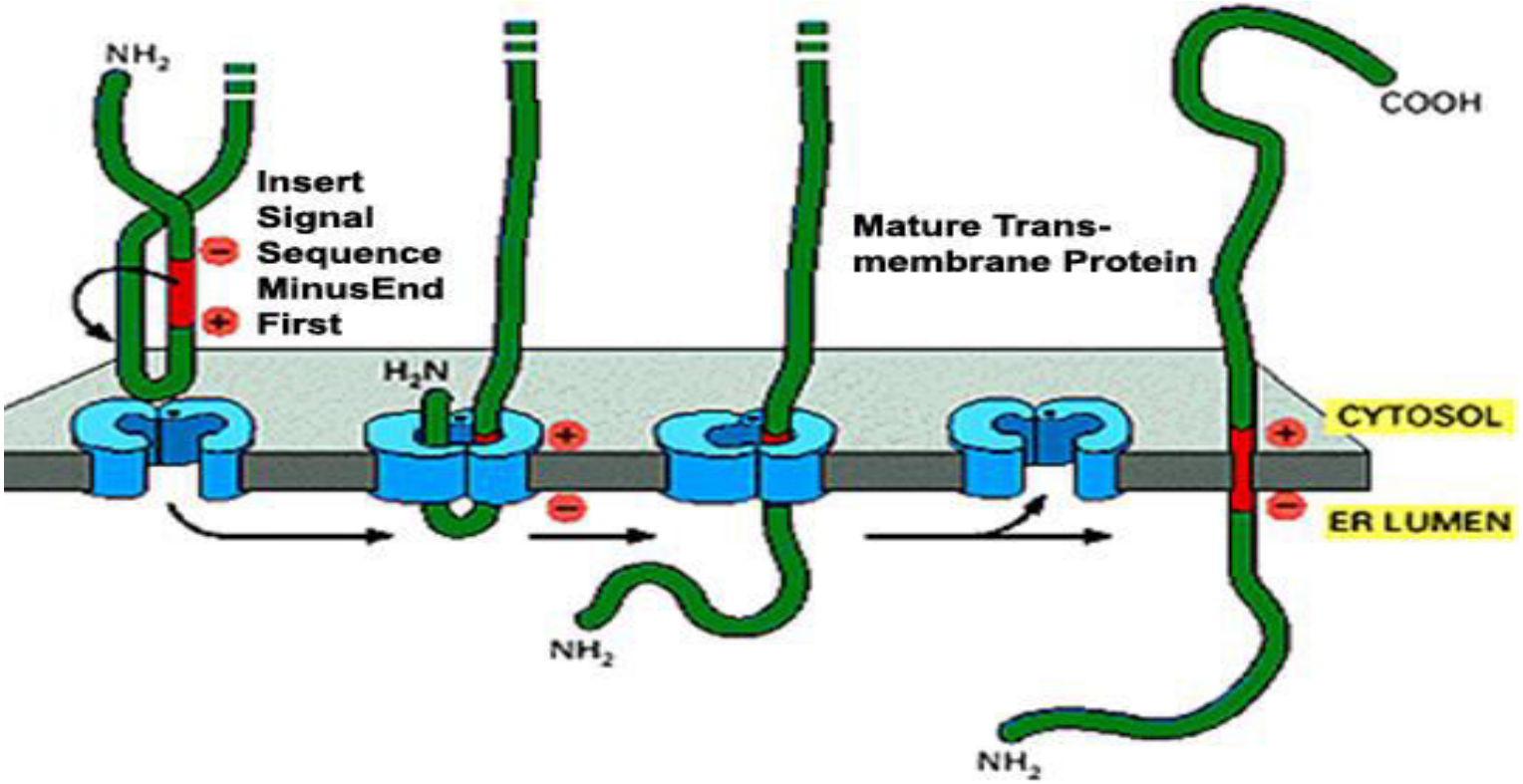
N out Vs C out



If there are more +ly charged AAs immediately preceding the hydrophobic core of start-transfer sequence (STS) than there are following it, the STS is inserted in the orientation shown here.

C-terminal to the start-transfer sequence will therefore be passed across the membrane

N out Vs C out



If there are more +ly charged amino acids immediately following the hydrophobic core of the STS than there are preceding it, the start-transfer sequence is inserted into the translocator in the orientation shown here

N-terminal to the start-transfer sequence will therefore be passed across the membrane

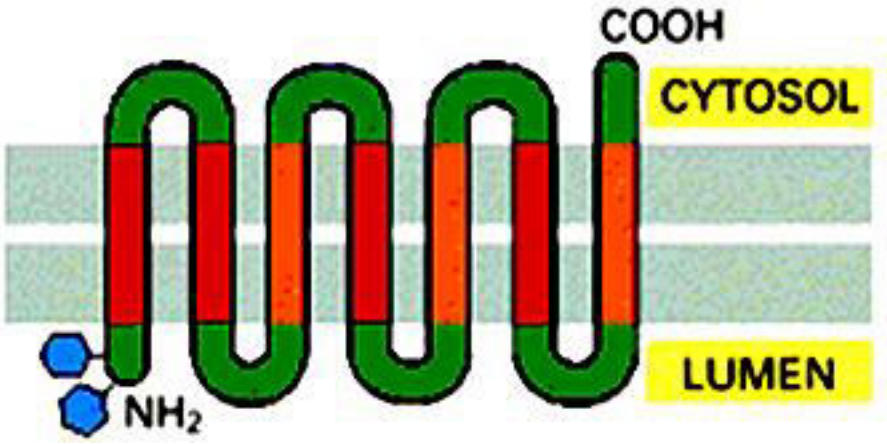
Multipass Transmembrane Proteins

In multipass transmembrane proteins, the polypeptide chain passes back and forth repeatedly across the lipid bilayer an internal signal sequence serves as a start-transfer signal in these proteins to initiate translocation, which continues until a stop-transfer sequence is reached

In double-pass transmembrane proteins, for example, the polypeptide can then be released into the bilayer

Multipass Transmembrane Proteins

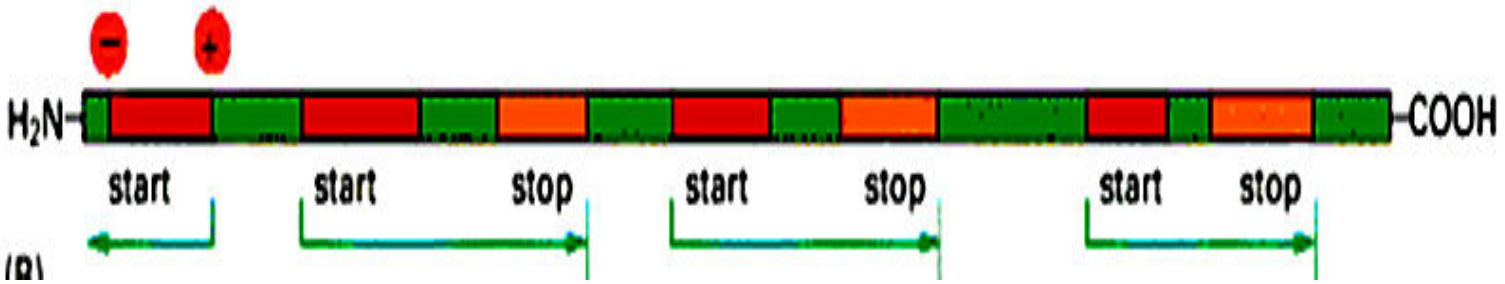
In more complex multipass proteins, in which many hydrophobic α helices span the bilayer, a second start-transfer sequence reinitiates translocation further down the polypeptide chain until the next stop-transfer sequence causes polypeptide release, and so on for subsequent start-transfer and stop-transfer sequences



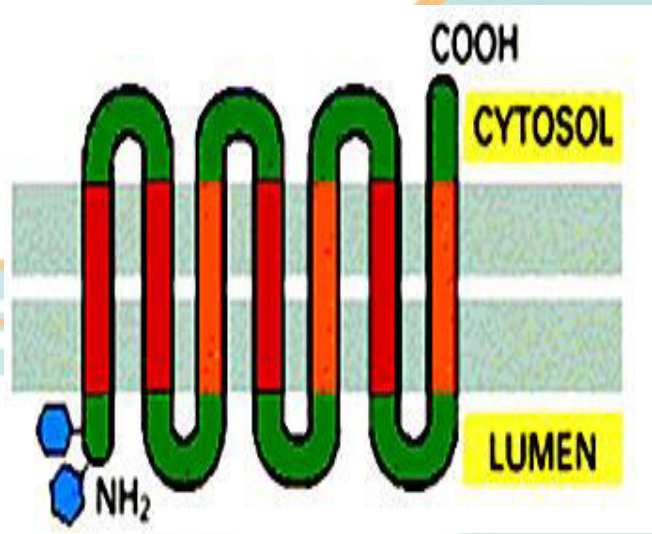
Multipass Transmembrane Proteins

Whether a given hydrophobic signal sequence functions as a start-transfer or stop-transfer sequence depends on its location in a polypeptide chain, since its function can be switched by changing its location in the protein using recombinant DNA techniques

The distinction between start-transfer and stop-transfer sequences results mostly from their relative order in the growing polypeptide chain



ER signal



END

Cell Biology

Vesicle Transport

Golgi Apparatus

The Golgi complex is a major site of carbohydrate synthesis, as well as a sorting and dispatching station for the products of the ER.

Many of the cell's polysaccharides are made in the Golgi, including the pectin and hemicellulose of the cell wall in plants and most of the glycosamino-glycans of the extracellular matrix in animals

Golgi apparatus also lies on the exit route from the ER, and a large proportion of the carbohydrates that it makes are attached as oligosaccharide side chains to the many proteins & lipids that ER sends

Golgi Apparatus

A subset of these oligosaccharide groups serve as tags to direct specific proteins into vesicles that then transport them to lysosomes

Most proteins and lipids, after acquiring their appropriate oligosaccharides in the Golgi, are recognized in other ways for targeting into the transport vesicles going to other destinations

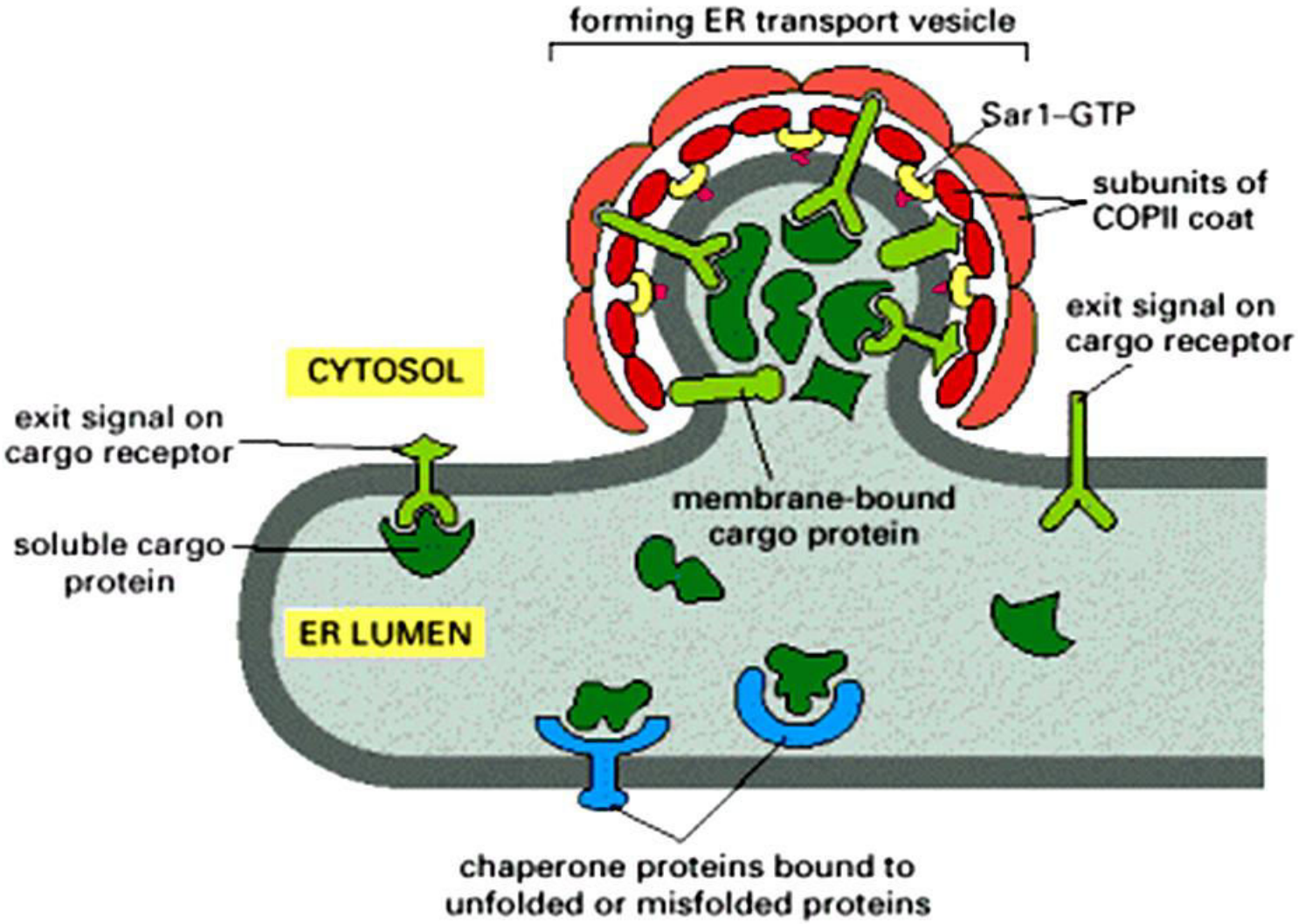
Proteins that enter the ER and are destined for the Golgi apparatus are packaged into small COPII-coated transport vesicles which, bud from specialized regions of the ER called ER exit sites whose membrane lacks bound ribosomes

Golgi Apparatus

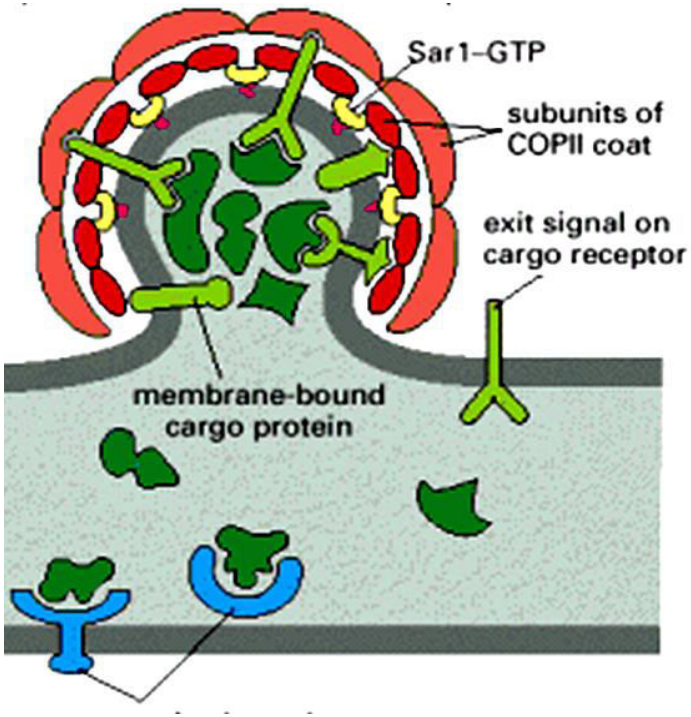
Some cargo proteins are actively recruited into such vesicles, where they become concentrated, these cargo proteins display exit (transport) signals on their surface that are recognized by complementary receptor proteins that become trapped in the budding vesicle by interacting with components of the COPII coat

Cargo proteins display exit (transport) signals on their surface that are recognized by complementary receptor proteins that become trapped in the budding vesicle by interacting with components of the COPII coat, some ER resident proteins without such exit signals can also get packaged in vesicles .

Golgi Apparatus



Golgi Apparatus



The ERGIC53 protein

To exit from the ER, proteins must be properly folded and completely assembled

Misfolded or incompletely assembled are retained in the ER, where they are bound to chaperone proteins

The ERGIC53 protein serves as a receptor for packaging some secretory proteins into COPII-coated vesicles

Its role in protein transport was identified because humans who lack it owing to an inherited mutation have lowered serum levels of two secreted blood-clotting factors (Factor V and Factor VIII) and therefore bleed excessively

The ERGIC53 protein

The ERGIC53 protein is a lectin that binds mannose on Factor V and Factor VIII proteins, thereby packaging the proteins into transport vesicles in the ER

END

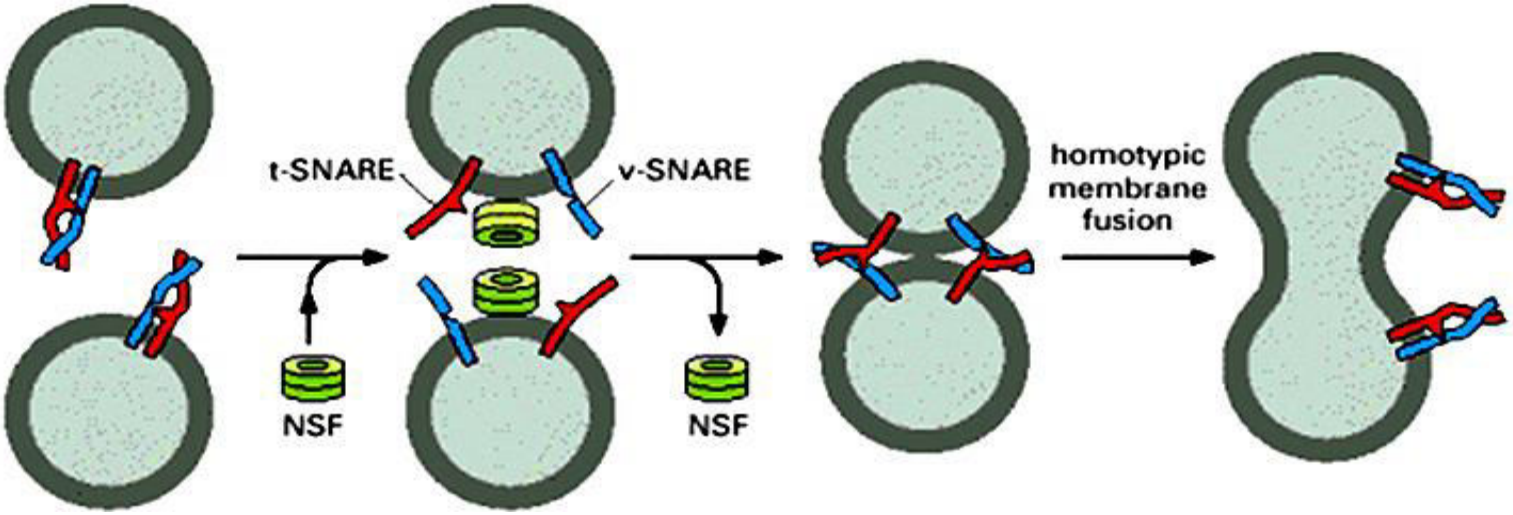
Cell Biology

COPI

Snares

After transport vesicles have budded from an ER exit site they shed their coat, and then they begin to fuse with one another

This fusion of membranes from the same compartment is called homotypic fusion, to distinguish it from heterotypic fusion, in which a membrane from one compartment fuses with the membrane of a different compartment



Vesicular tubular clusters

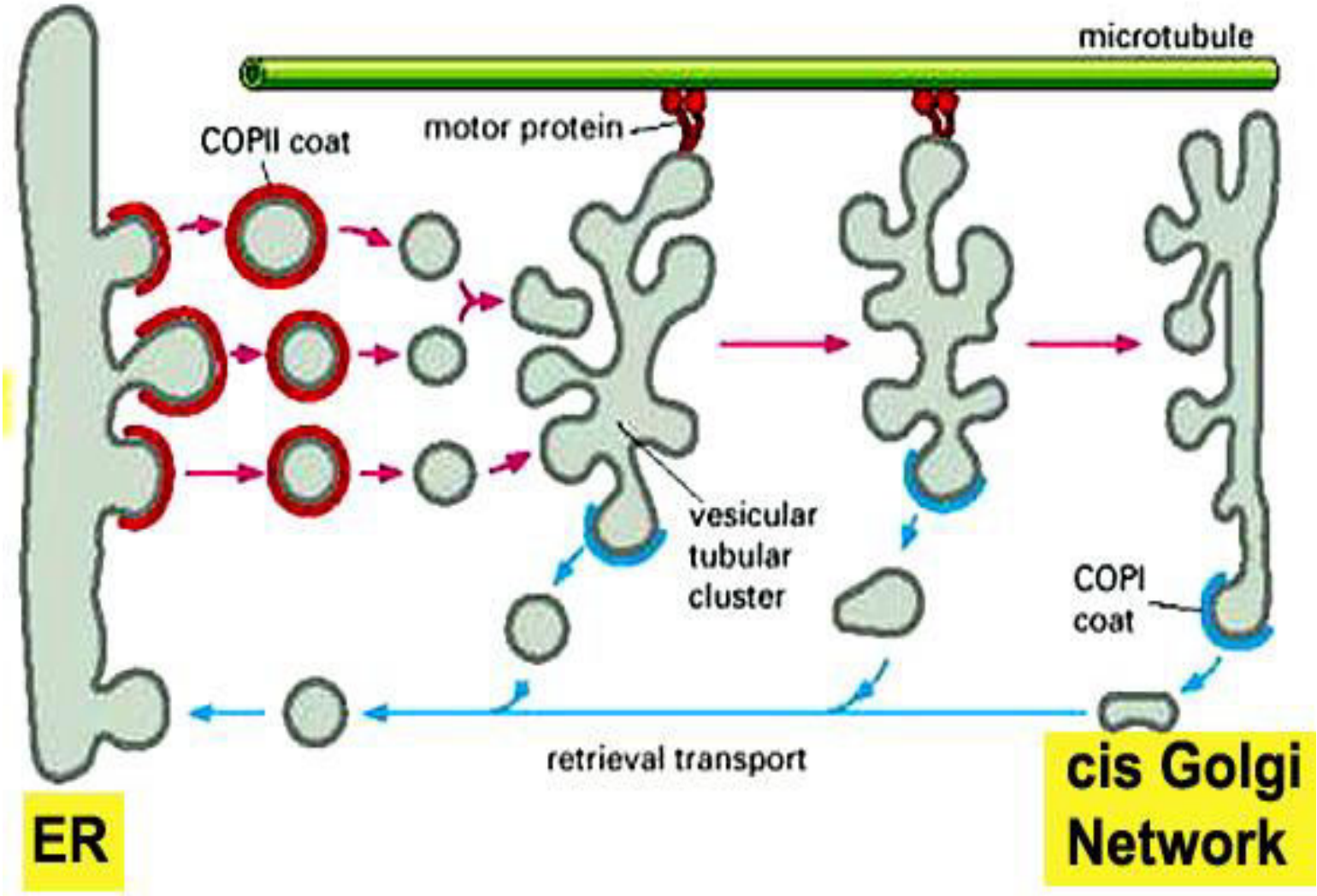
As with heterotypic fusion, homotypic fusion requires a set of matching SNAREs

The interaction is symmetrical, with v-SNAREs and t-SNAREs contributed by both membranes

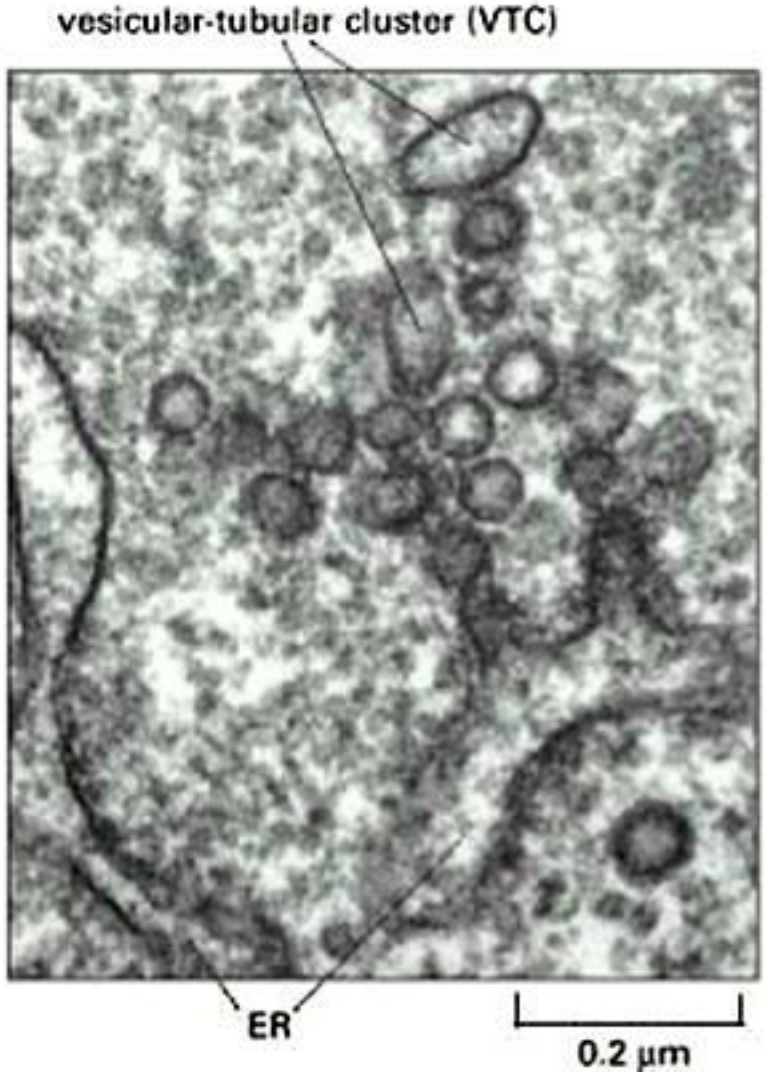
The structures formed when ER-derived vesicles fuse with one another are called vesicular tubular clusters

These clusters constitute a new compartment that is separate from the ER and lacks many of the proteins that function in the ER

Vesicular tubular clusters



Vesicular tubular clusters



COPI The Retrieval Process

As soon as vesicular tubular clusters form, they begin budding off vesicles of their own

Unlike the COPII-coated vesicles that bud from the ER, these vesicles are COPI-coated

They carry back the ER resident proteins that have escaped, as well as proteins that participated in the ER budding reaction and are being returned

This retrieval process demonstrates the exquisite control mechanisms that regulate coat assembly

The COPI coat assembly begins only seconds after the COPII coats have been shed

COPI The Retrieval Process

The clusters continuously mature, gradually changing their composition as selected proteins are returned to the ER a similar retrieval process continues from the Golgi apparatus, after the vesicular tubular clusters have delivered their cargo

Resident ER membrane proteins, for example, contain signals that bind directly to COPI coats and are thus packaged into COPI-coated transport vesicles for retrograde delivery to the ER

Signal of this type consists of two lysines, followed by any two other amino acids, at the extreme C-terminal end of the ER membrane protein

COPI The Retrieval Process

It is called a KKXX sequence, based on the AA code

Soluble ER resident proteins, such as BiP, also contain a short retrieval signal at their C-terminal end, but it is different: it consists of a Lys-Asp-Glu-Leu or similar sequence

If this signal (called the KDEL sequence) is removed from BiP by genetic engineering, the protein is slowly secreted from the cell

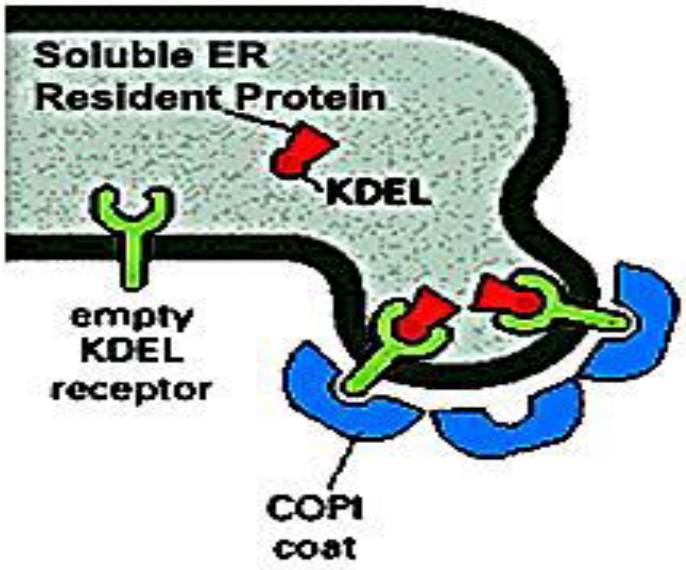
If the signal is transferred to a protein that is normally secreted, the protein is now efficiently returned to the ER, where it accumulates

COPI The Retrieval Process

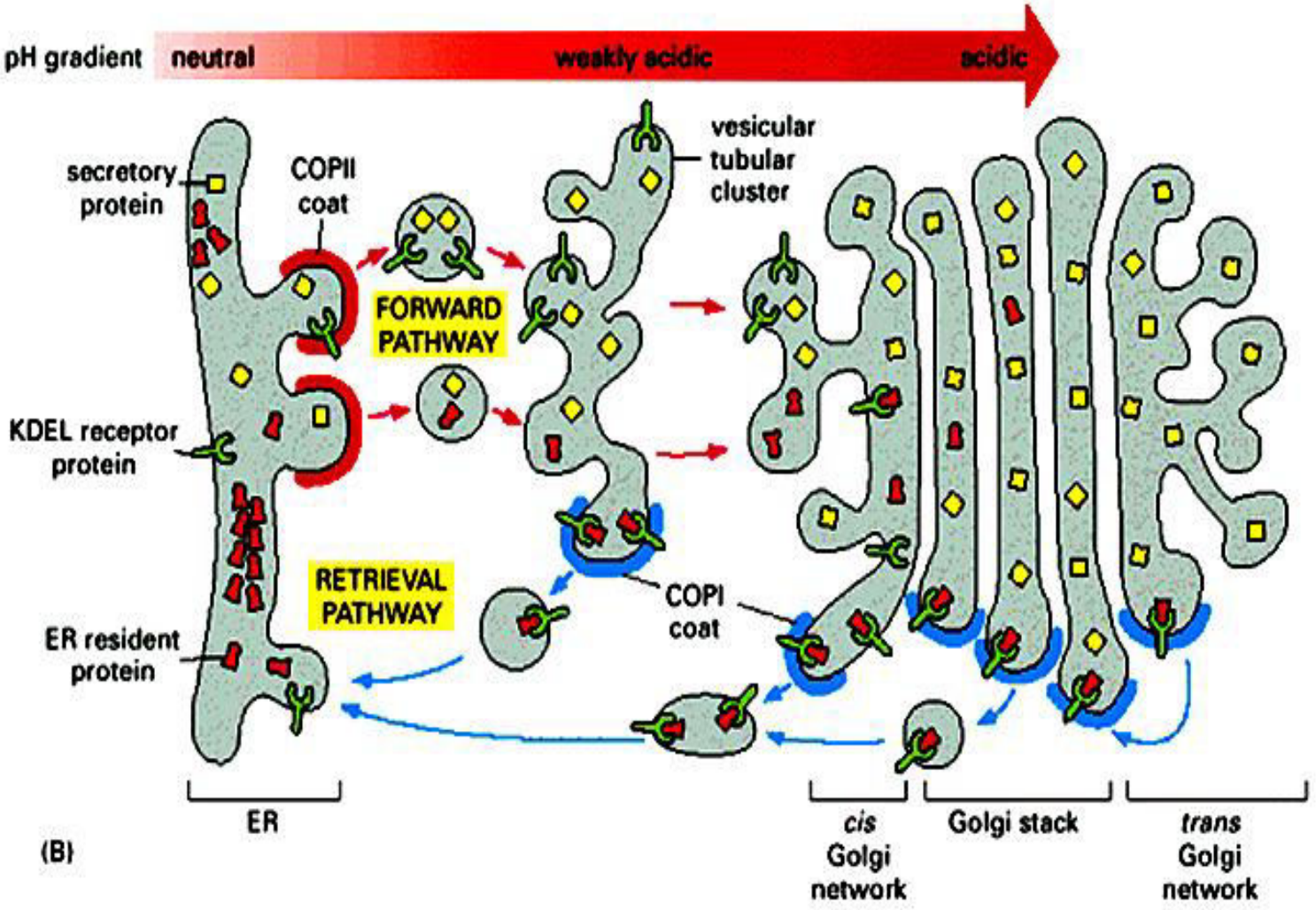
ER resident proteins must bind to specialized receptor proteins such as the KDEL receptor—a multipass transmembrane protein that binds to the KDEL sequence and packages any protein displaying it into COPI-coated retrograde transport vesicles

ER resident proteins bind to one another, thus forming complexes that are too big to enter transport vesicles Aggregation of proteins that function in the same compartment—called kin recognition—is a general mechanism that compartments use to organize and retain their resident proteins

COPI The Retrieval Process



COPI The Retrieval Process



Cell Biology

Lysosomal Transport

Lysosomal Transpot

Lysosomes are membrane-enclosed compartments filled with hydrolytic enzymes that are used for the controlled intracellular digestion of macromolecules

They contain 40 types of hydrolytic enzymes, including proteases, nucleases, glycosidases, lipases, phospholipases, phosphatases, and sulfatases they are all are acid hydrolases

For optimal activity they require an acid environment, and the lysosome provides this by maintaining a pH of about 5.0 in its interior

The contents of the cytosol are doubly protected against attack by the cell's own digestive system

Lysosomal Transpot

Lysosomal hydrolases and membrane proteins are synthesized in the rough ER and transported through the Golgi apparatus to the lysosome.

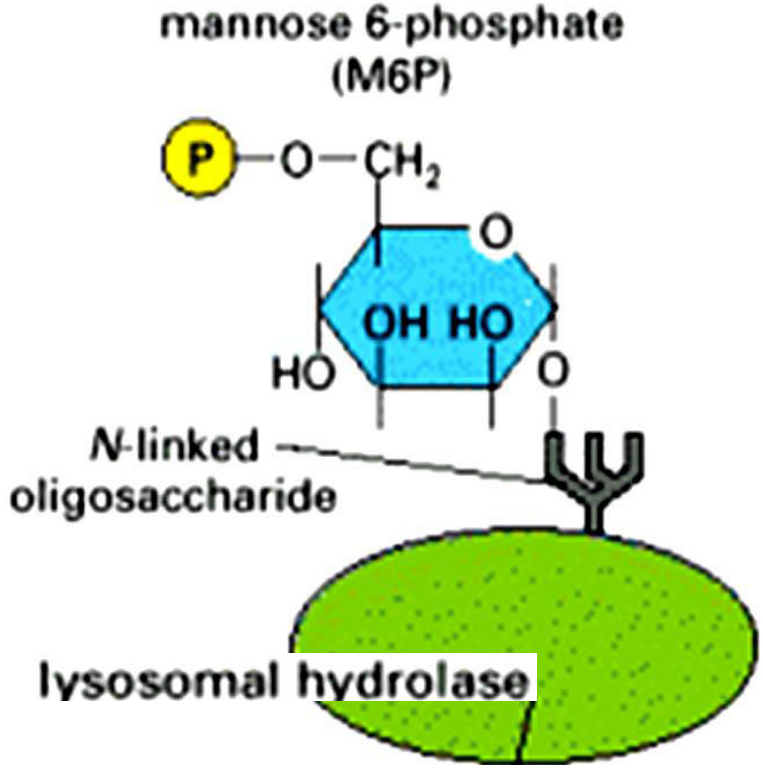
The transport vesicles that deliver these proteins to lysosomes bud from the trans Golgi network

These vesicles incorporate the lysosomal proteins & exclude other proteins being packaged into different transport vesicles for delivery elsewhere

Lysosomal proteins are recognized and selected in the trans Golgi network with the required accuracy

Lysosomal hydrolases carry a unique marker in the form of mannose 6-phosphate (M6P) groups,.

Lysosomal Transpot

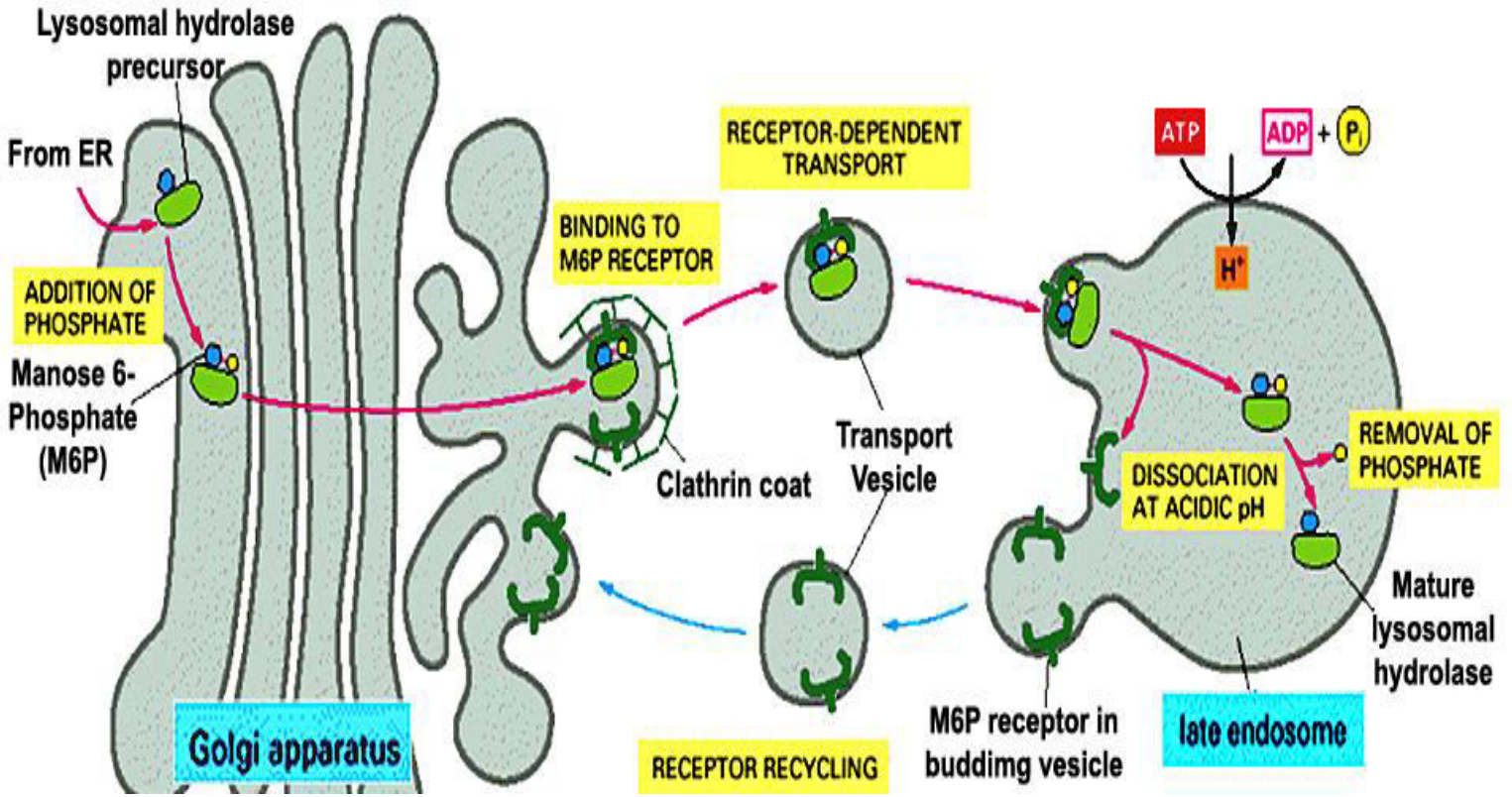


M6P is added to N-linked oligosaccharides of the soluble lysosomal enzymes in the lumen of the C-Golgi

The M6P groups are recognized by trans-membrane M6P receptor proteins, which are present in the T-Golgi

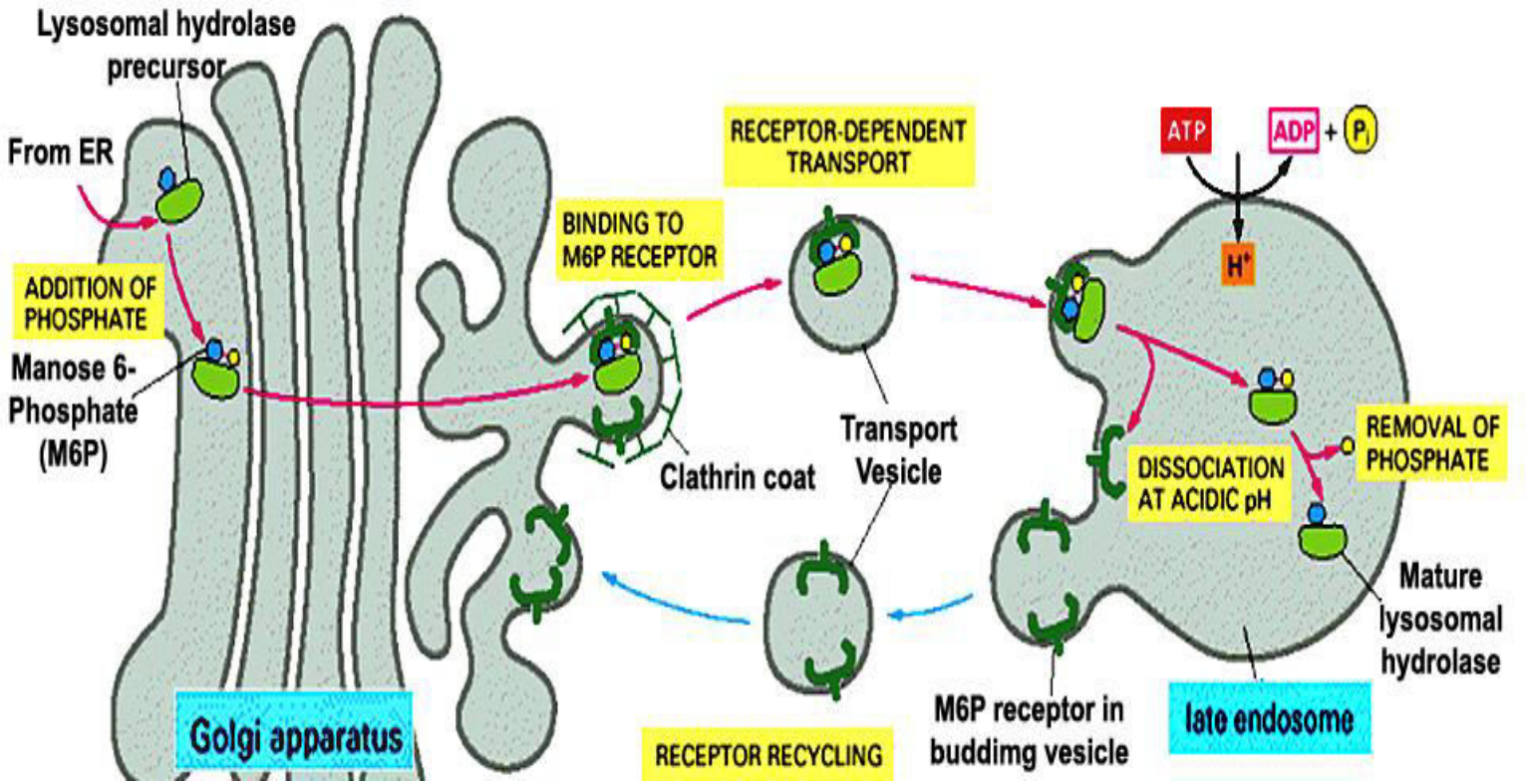
Receptors bind to lysosomal hydrolases on the lumenal side of the membrane and to adaptins to assemble clathrin coats on the cytosolic side

Lysosomal Transpot



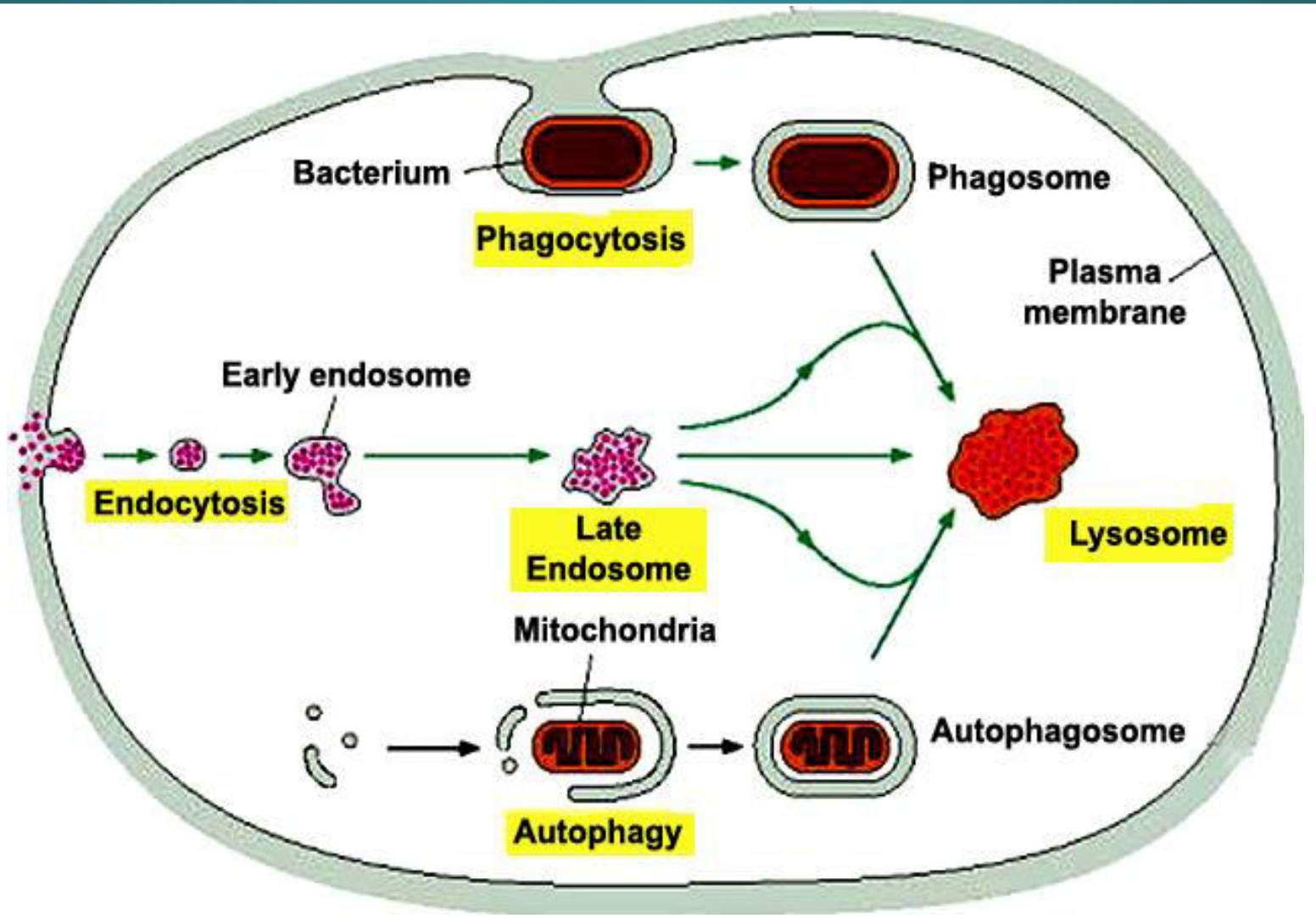
Receptors help package the hydrolases into clathrin-coated vesicles that bud from the T-Golgi network
Vesicles contents are subsequently delivered to Lysosomes

Lysosomal Transpot



M6P receptor binds its specific oligosaccharide at pH 6.5 in the T-Golgi network & releases it at pH 6, late endosomes M6P receptors are retrieved into transport vesicles from late endosomes the receptors are returned to the T-Golgi for reuse.

Lysosome Function



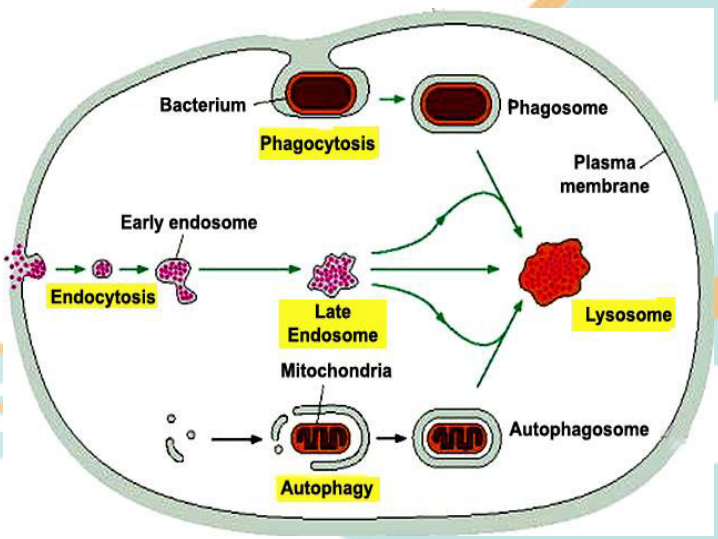
Inclusion-cell (I-cell) Disease

In this disease almost all of the hydrolytic enzymes are missing from the lysosomes of fibroblasts, and their undigested substrates accumulate in lysosomes, which consequently form large "inclusions" in the cells

In I-cell disease patients, all the hydrolases missing from lysosomes are found in the blood

Because they fail to be sorted properly in the Golgi apparatus, the hydrolases are secreted rather than transported to lysosomes.

The Retrieval Process



END

Cell Biology

Endocytosis

Endocytosis

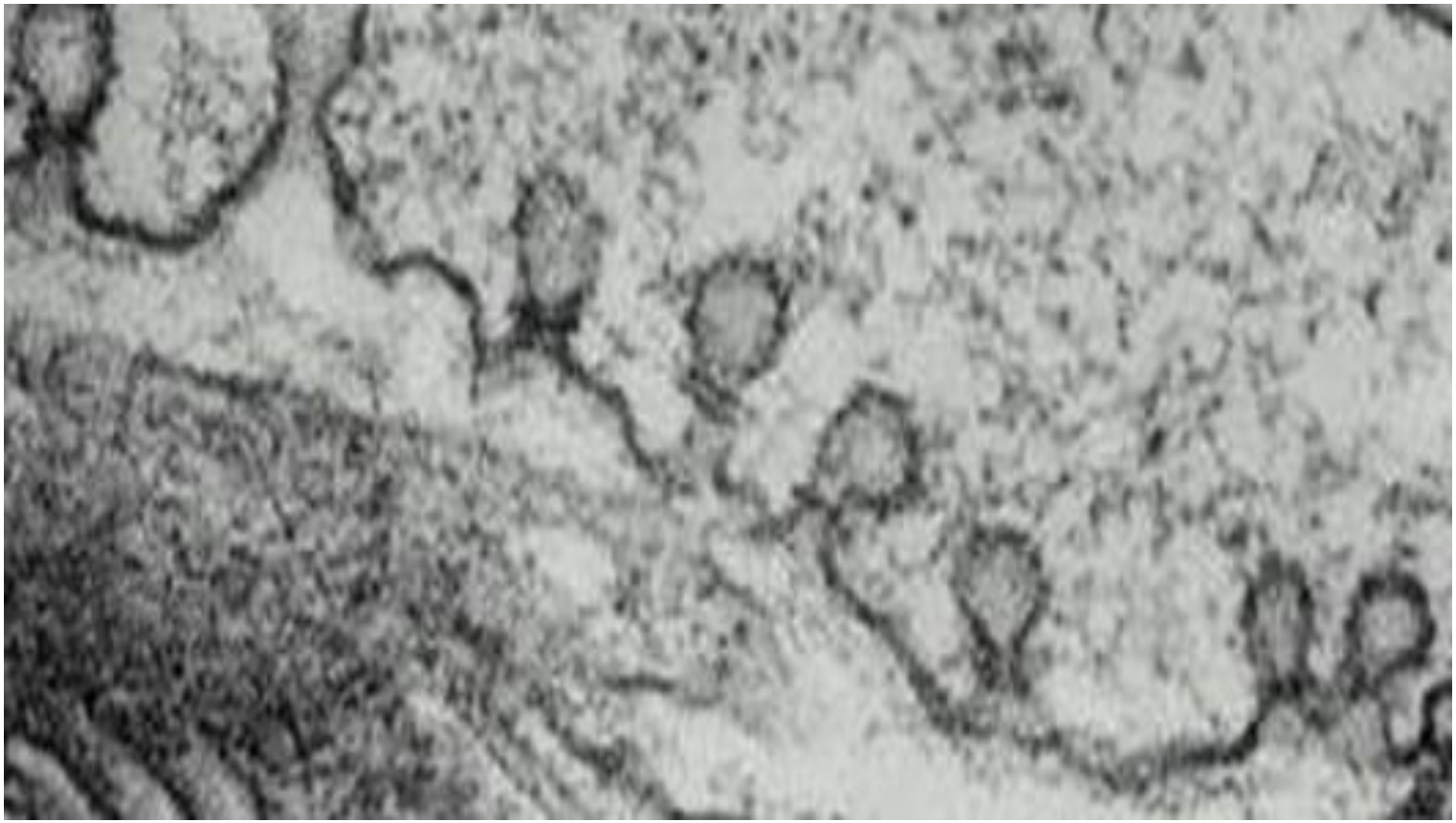
The material to be ingested is progressively enclosed by a small portion of the plasma membrane, which first invaginates and then pinches off to form an endocytic vesicle containing the ingested substance or particle

Phagocytosis involves the ingestion of large particles microorganisms or dead cells via large vesicles called phagosomes (>250 nm in diameter)

Pinocytosis involves ingestion of fluid & solutes via small pinocytic vesicles (about 100 nm in diameter)

Most eucaryotic cells are continually ingesting fluid and solutes by pinocytosis; large particles are most efficiently ingested by specialized phagocytic cells

Endocytosis



Endocytosis

In mammals, three classes of white blood cells act as professional phagocytes macrophages, neutrophils, and dendritic cells

These cells develop from hemopoietic stem cells and they defend us against infection by ingesting invading microorganisms.

Macrophages also scavenge senescent cells and cells that have died by apoptosis.

In quantitative terms, the latter function is by far the most important: our macrophages phagocytose more than 10^{11} senescent red blood cells every day.

Endocytosis

Particles first bind to the surface of the phagocyte.

Phagocytes have a variety of specialized surface receptors that are functionally linked to the phagocytic machinery of the cell.

Phagocytosis is a triggered process, requiring that receptors be activated that signal the cell interior and initiate the response

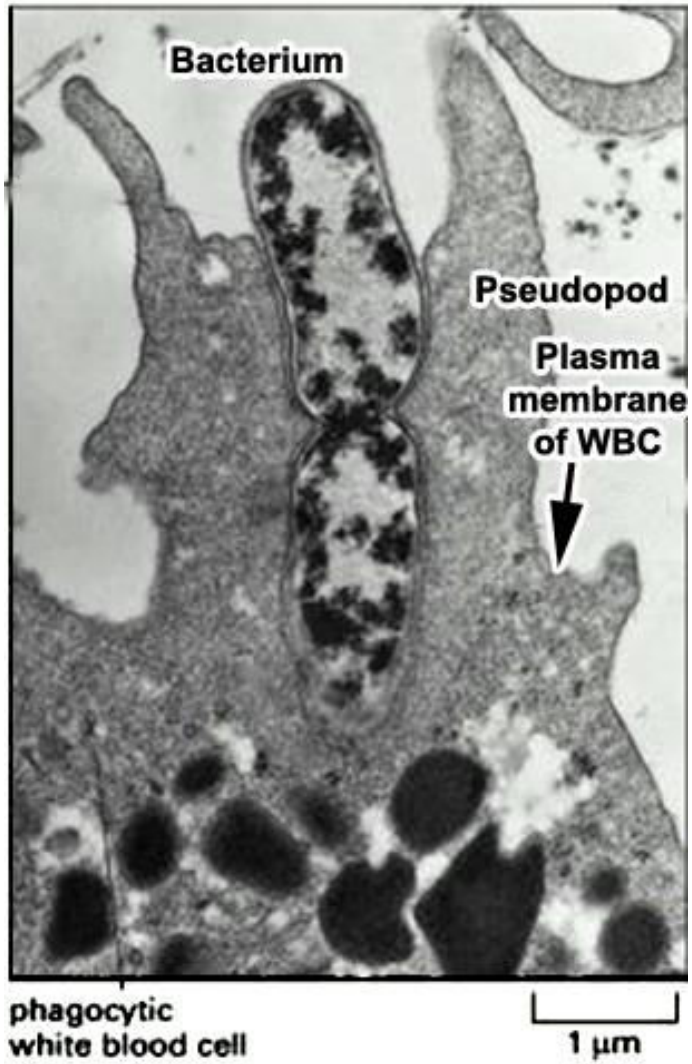
Best-characterized triggers are antibodies, which protect us by binding to the surface of infectious microorganisms to form a coat in which the tail region of each antibody molecule, called the Fc region, is exposed on the exterior

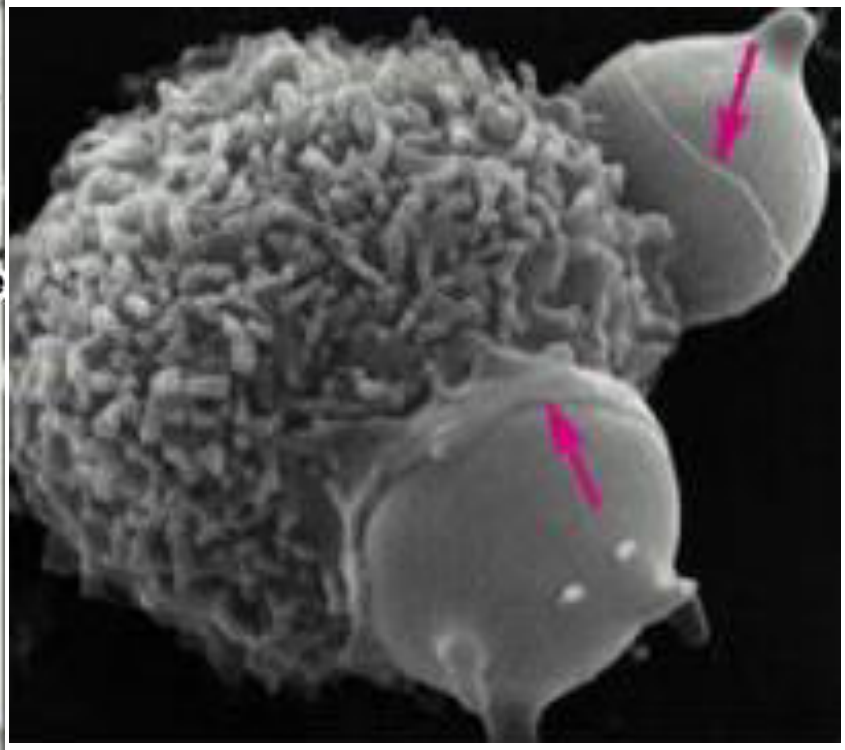
Endocytosis

This antibody coat is recognized by specific Fc receptors on the surface of macrophages and neutrophils, whose binding induces the phagocytic cell to extend pseudopods that engulf the particle and fuse at their tips to form a phagosome.

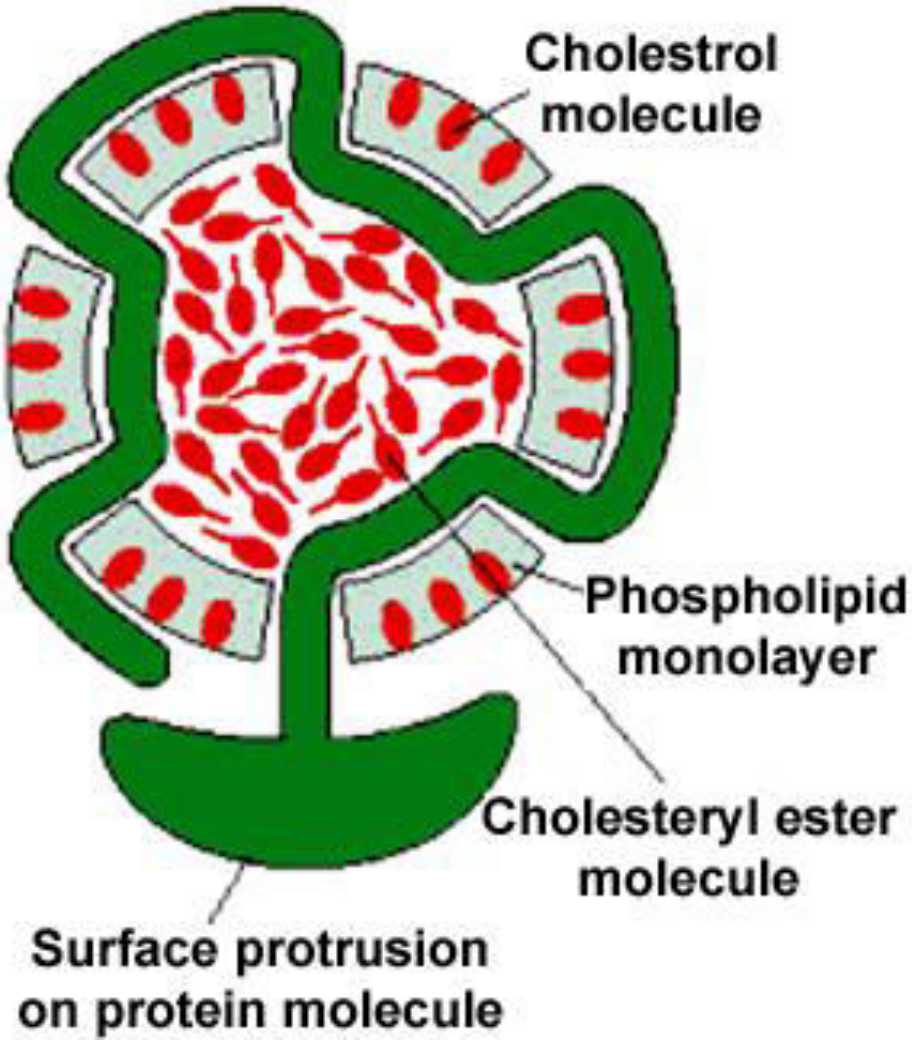
Apoptotic cells lose the asymmetric distribution of phospholipids in their plasma membrane as a consequence, negatively charged phosphatidylserine, which is normally confined to the cytosolic leaflet of the lipid bilayer, is now exposed on the outside of the cell, where it triggers the phagocytosis of the dead cell

Endocytosis



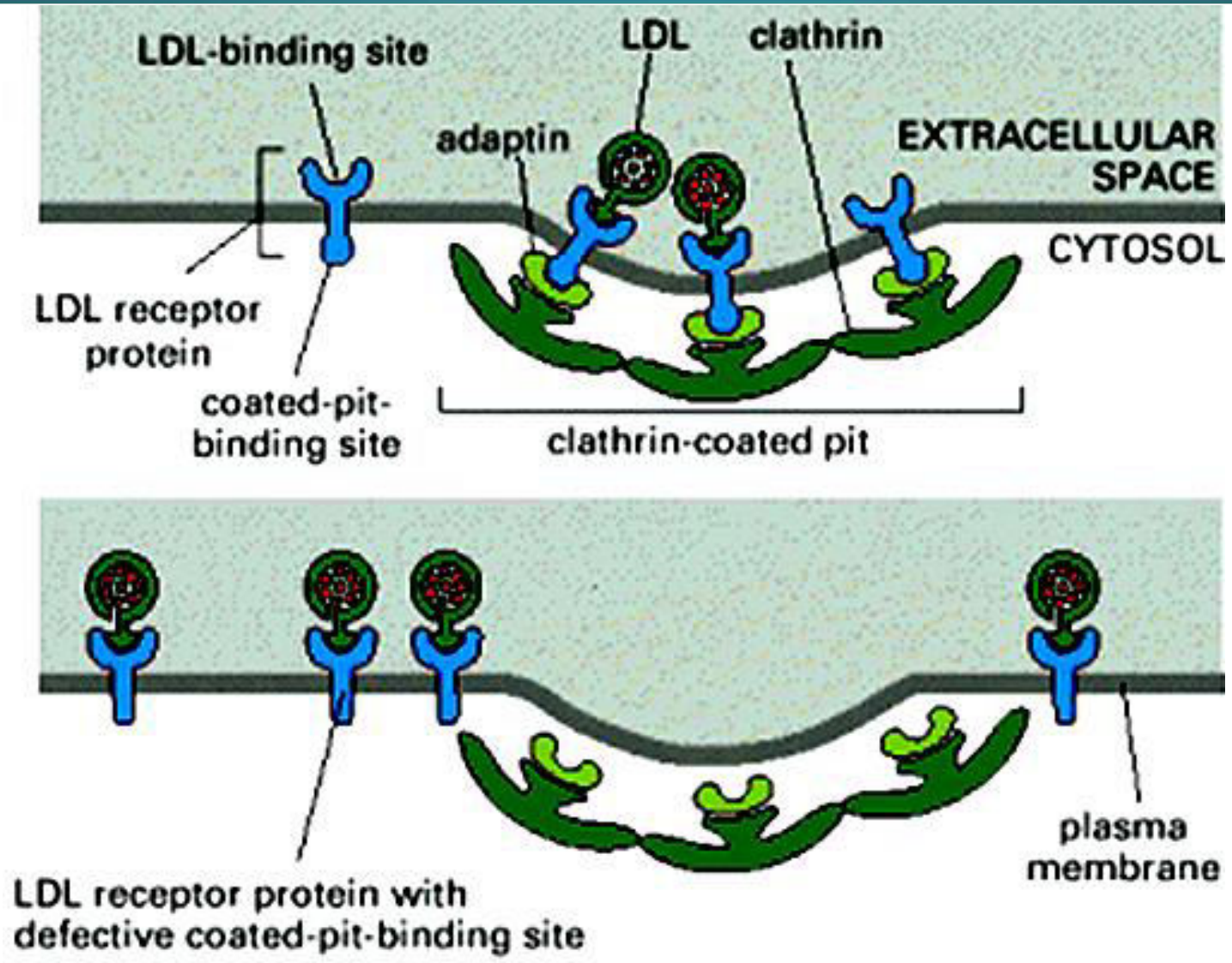


Low-density lipoprotein (LDL) particle



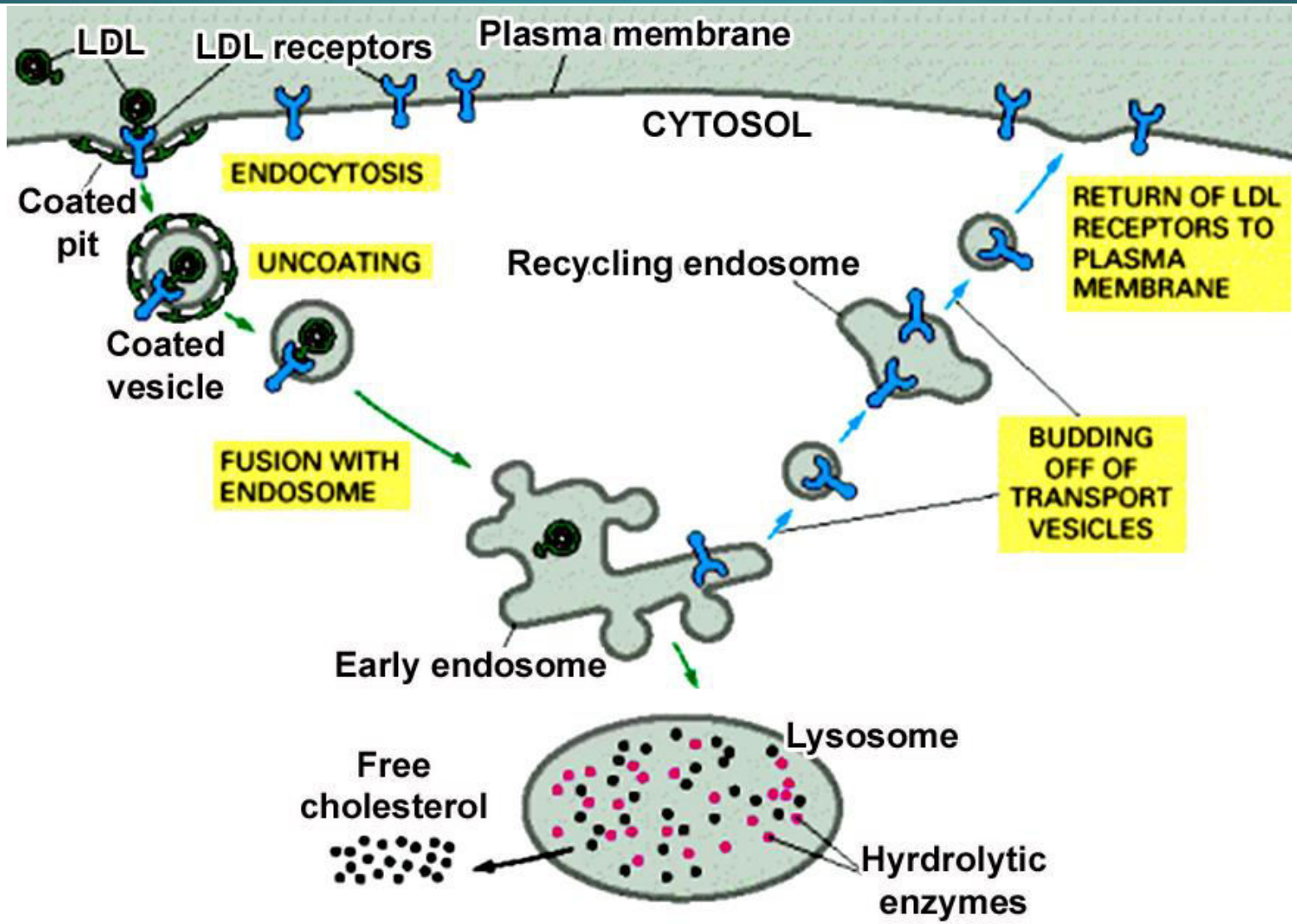
Each particle contains a core of about 1500 cholesterol molecules esterified to long-chain fatty acids

Low-density lipoprotein (LDL) particle



Low-density lipoprotein (LDL) particle

Low-density lipoprotein (LDL) particle



Cell Biology

Tissue Replacement

Tissue Replacement

A Natural Phenomenon

Bones regenerate every 10 years

Skin regenerates every two weeks

Macrophages scavenge senescent RBCs (10^{11}/day)

Tissue repair in response to minor injury

All organs have limited ability to regenerate (1 cm)

Tissue Replacement

Animation

Tissue Replacement

Every 30 Seconds a Person Dies from a Disease that can be Treated with

Tissue Replacement

Human Intervention History

Blood Transfusion

- 1492: Pope Innocent VIII
- 1901: Human blood groups discovery Austrian Karl Landsteiner (1939 Rh)
- 1925: First transfusion center Alexander Bogdanov Moscow

Human Intervention History

Solid Tissue

- 1905: Cornea transplant Eduard Zirm
- 1954: Kidney transplant Joseph Murray
- 1998: Hand transplant France
- 2005: Partial face transplant France

Current Approaches

Organ Repair via Stem Cells

Cytokine Induce Stem Cell Proliferation & Differentiation

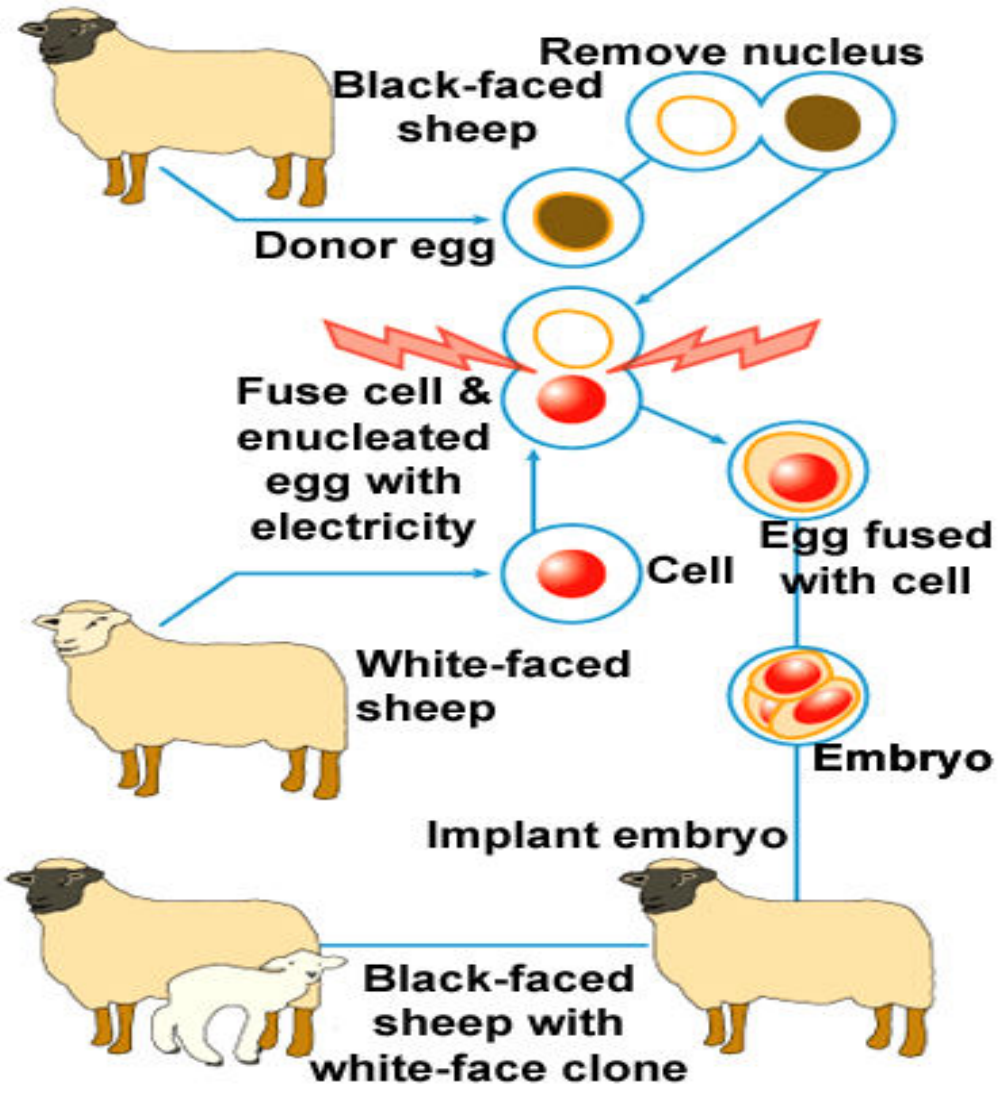
In-vitro Organ/Tissue Fabrication

Cell Biology

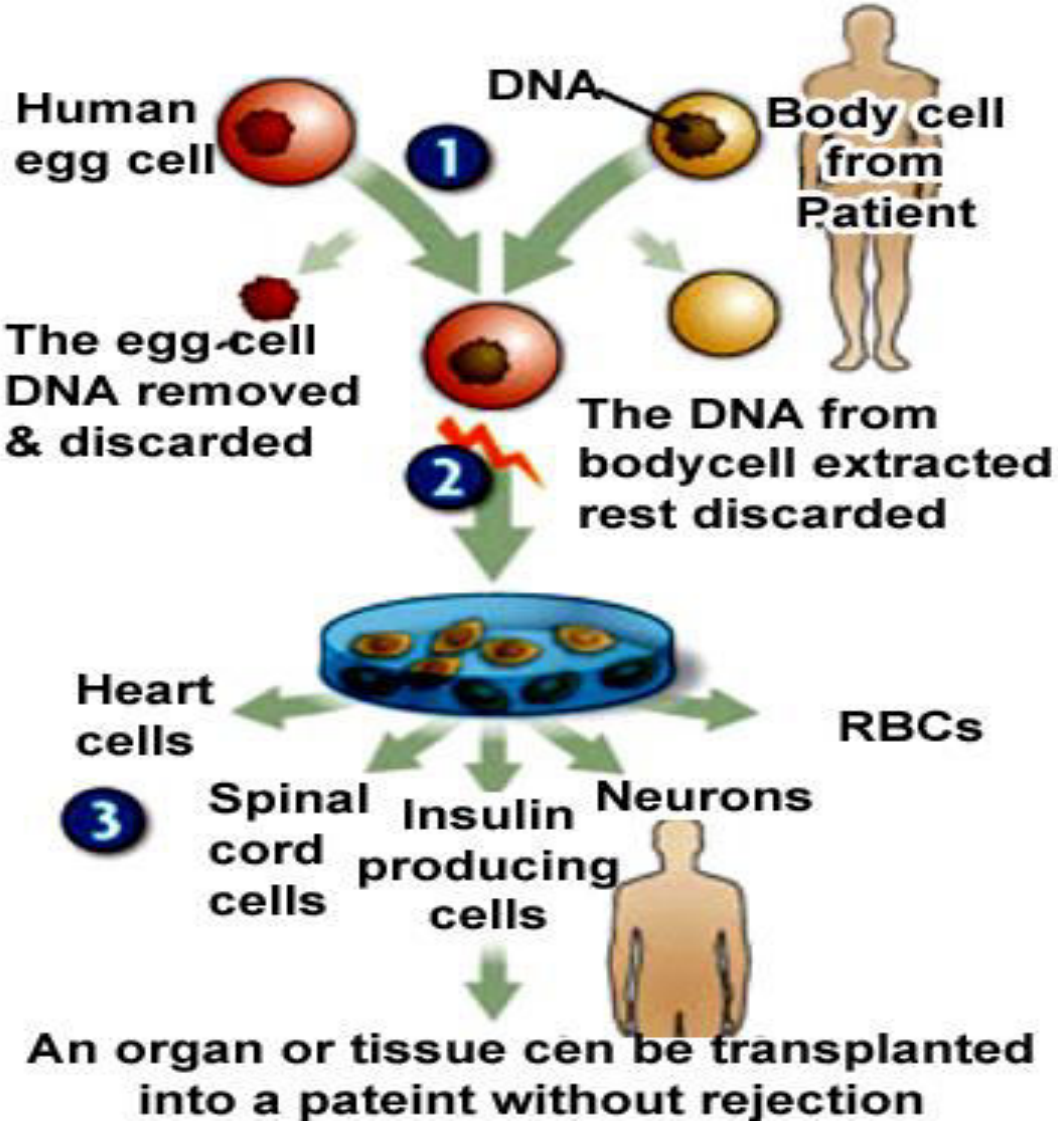
Customized

Stem Cells

Stem Cells


Remove nucleus
Black-faced sheep
Donor egg
Fuse cell & enucleated egg with electricity
Egg fused with cell
Cell
White-faced sheep
Embryo
Implant embryo
Black-faced sheep with white-face clone

Stem Cells


Human egg cell
DNA
Body cell from Patient
1
The egg cell DNA removed & discarded
The DNA from bodycell extracted rest discarded
2
Heart cells
RBCs
3
Spinal cord cells
Insulin producing cells
Neurons
An organ or tissue cen be transplanted into a pateint without rejection

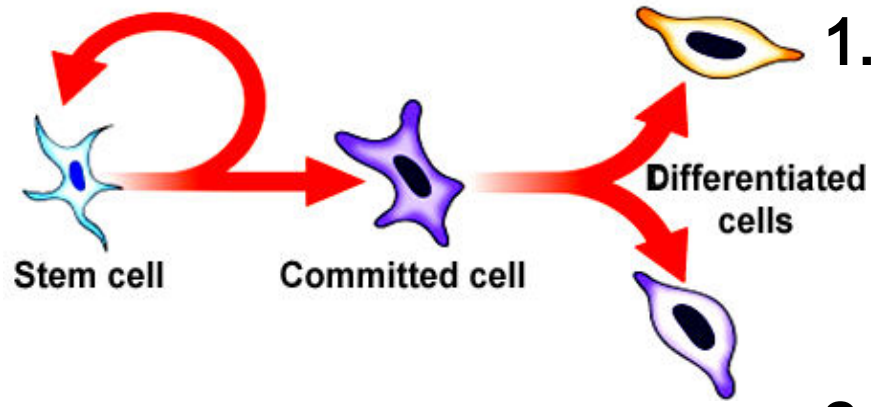
Tissue Replacement

Animation

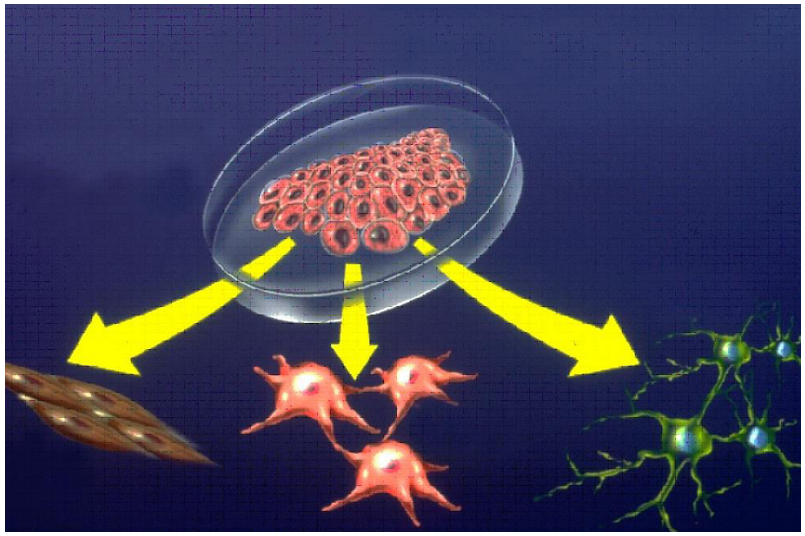
Cell Biology

Organ Repair via Stem Cells

Stem Cells



1. Stem cells are unspecialized cells that renew themselves.



2. Under certain physiologic or experimental conditions, Stem cells can be induced to become cells with special functions.

Stem Cells

Embryonic Stem Cells

Are derived from the embryos (ethical dilemma)
BB Option

Have the ability to differentiate into any cell type in the body

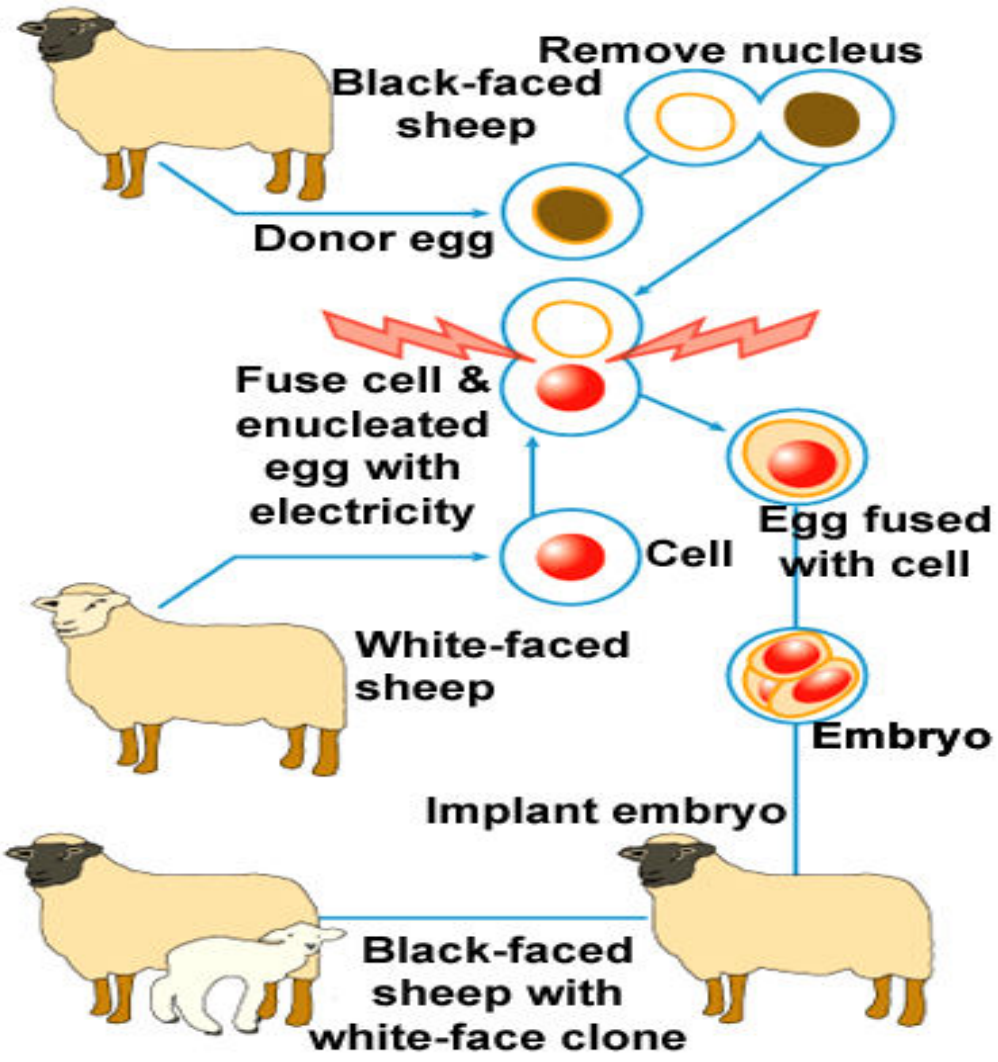
Have unlimited ability to renew themselves

Being foreign in origin, when transplanted can cause immune rejection

Tissue Replacement

Animation

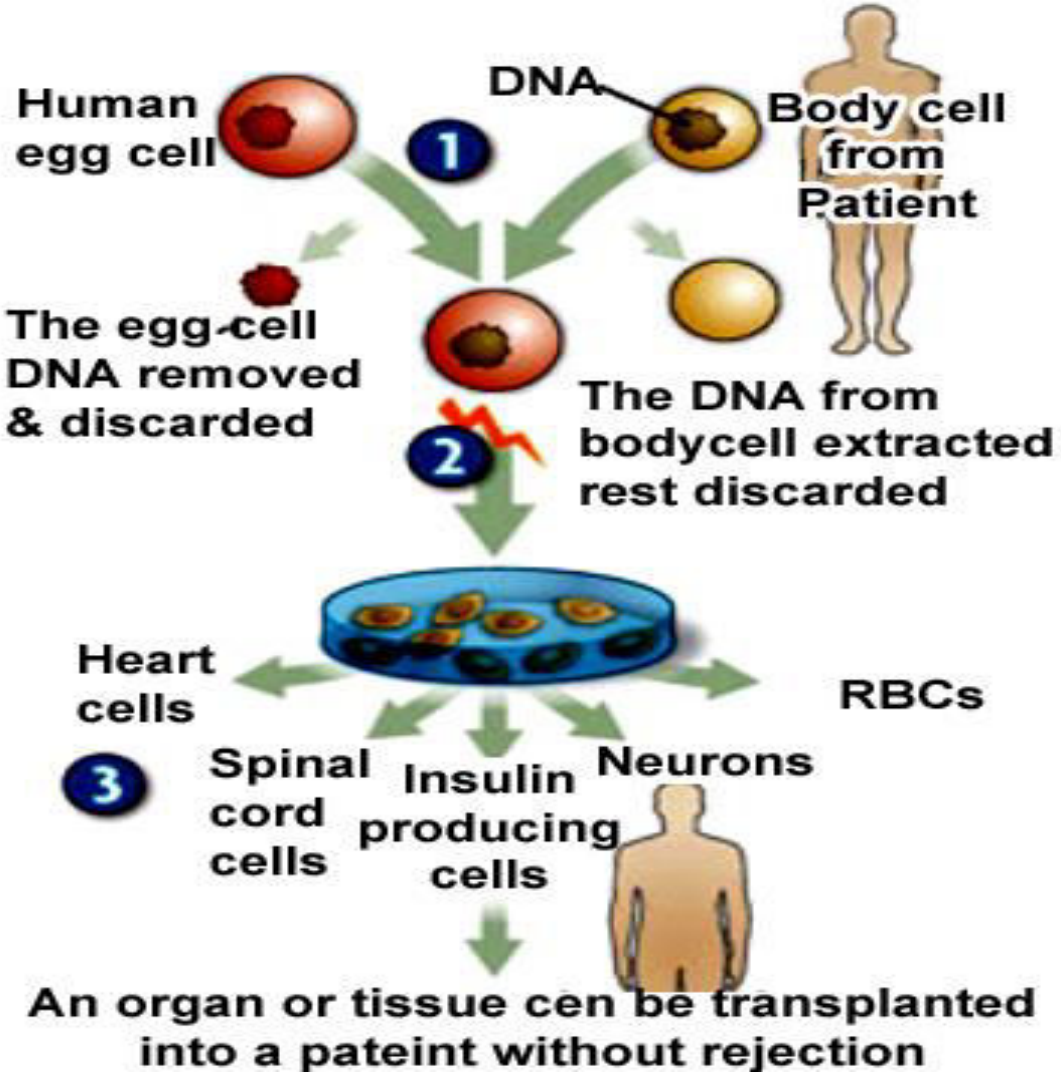
Stem Cells


Remove nucleus
Black-faced sheep
Donor egg
Fuse cell & enucleated egg with electricity
Egg fused with cell
Cell
White-faced sheep
Embryo
Implant embryo
Black-faced sheep with white-face clone

Tissue Replacement

Animation

Stem Cells


Human egg cell
DNA
Body cell from Patient
1
The egg cell DNA removed & discarded
2
The DNA from bodycell extracted rest discarded
Heart cells
RBCs
3
Spinal cord cells
Insulin producing cells
Neurons
An organ or tissue cen be transplanted into a pateint without rejection

Stem Cells

Adult Stem cells

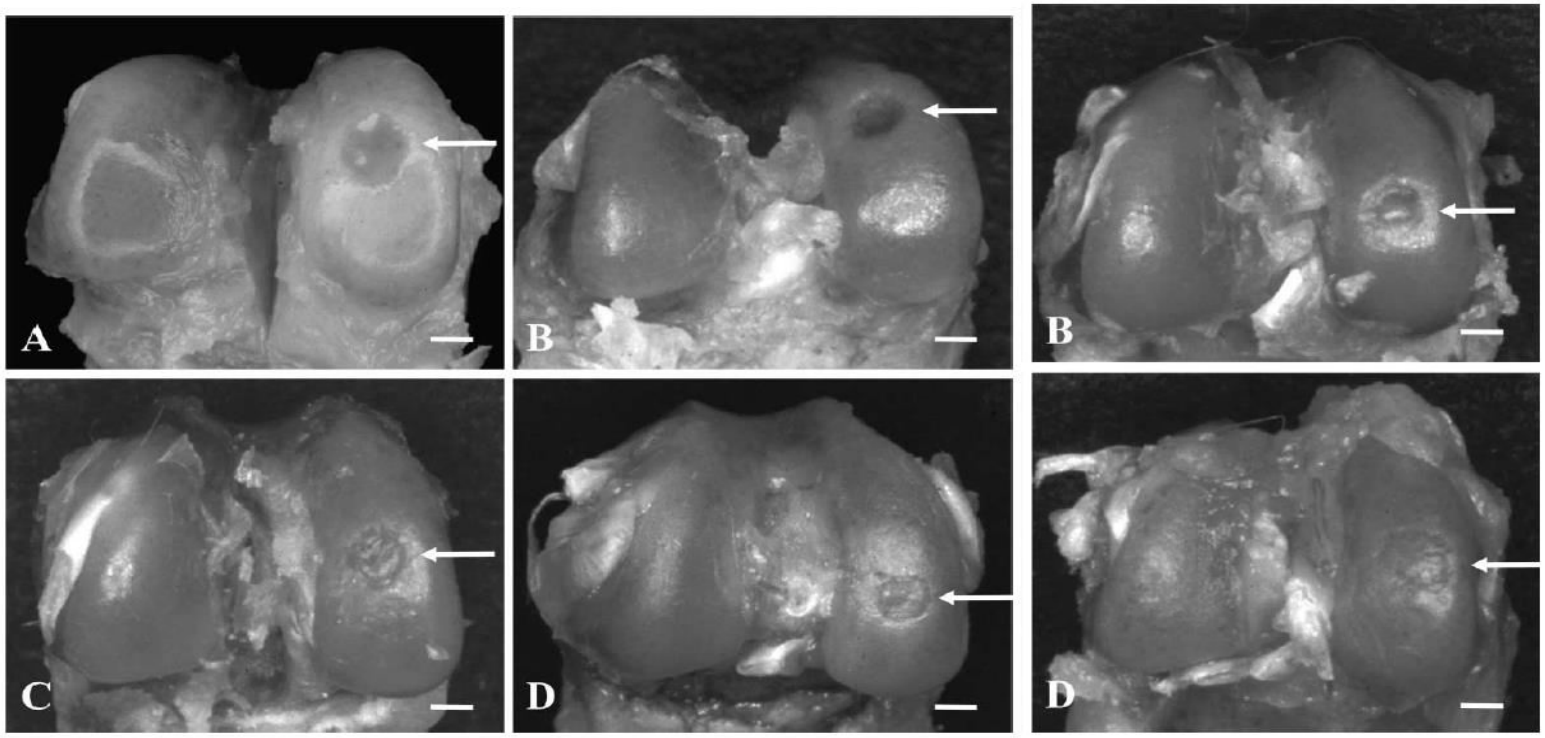
Are harvest from there niches from a (postnatal) individual

No ethical dilemma for their use in research

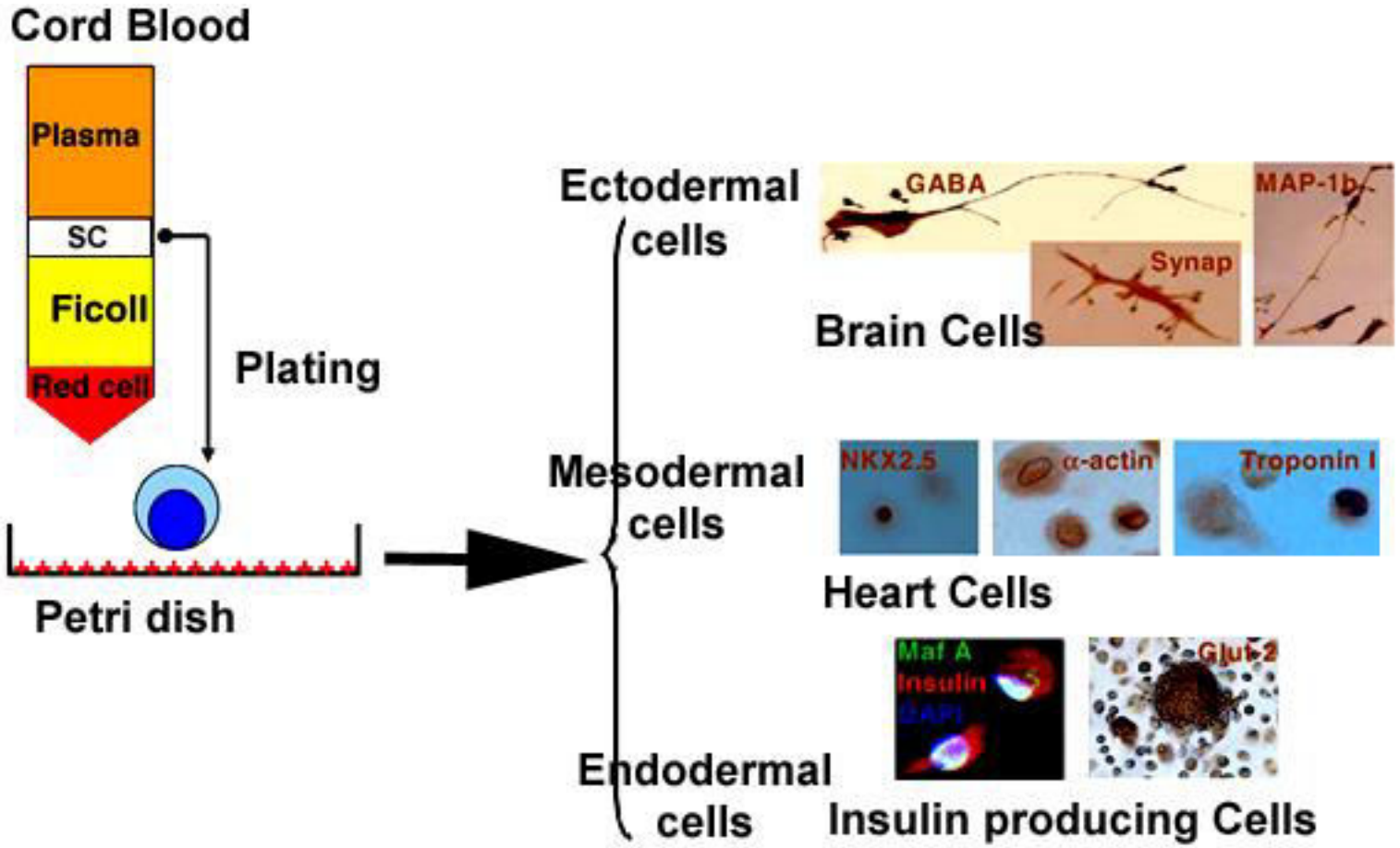
Have a limited ability to renew themselves and differentiate in to different cell types

Do not cause an immune rejection

Stem Cells

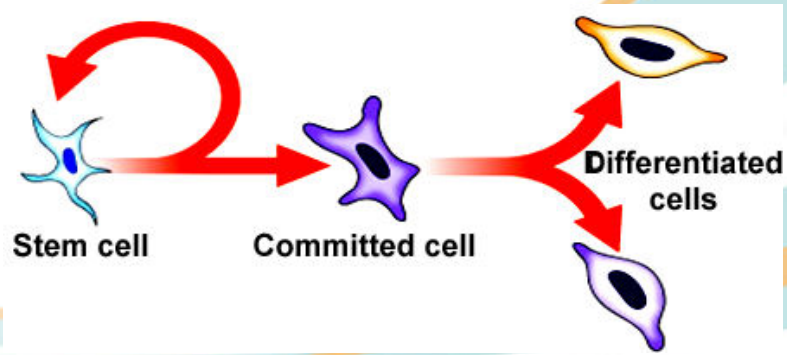


Stem Cells


Cord Blood
Plasma
SC
Ficoll
Red cell
Plating
Petri dish
Ectodermal cells
GABA
Synap
MAP-1b
Brain Cells
Mesodermal cells
NKX2.5
α-actin
Troponin I
Heart Cells
Endodermal cells
Maf A
Insulin
DAPI
Glut-2
Insulin producing Cells

Heart

Tissue Replacement



END

Cell Biology

Adult

Stem Cells

Stem Cells

Adult Stem cells

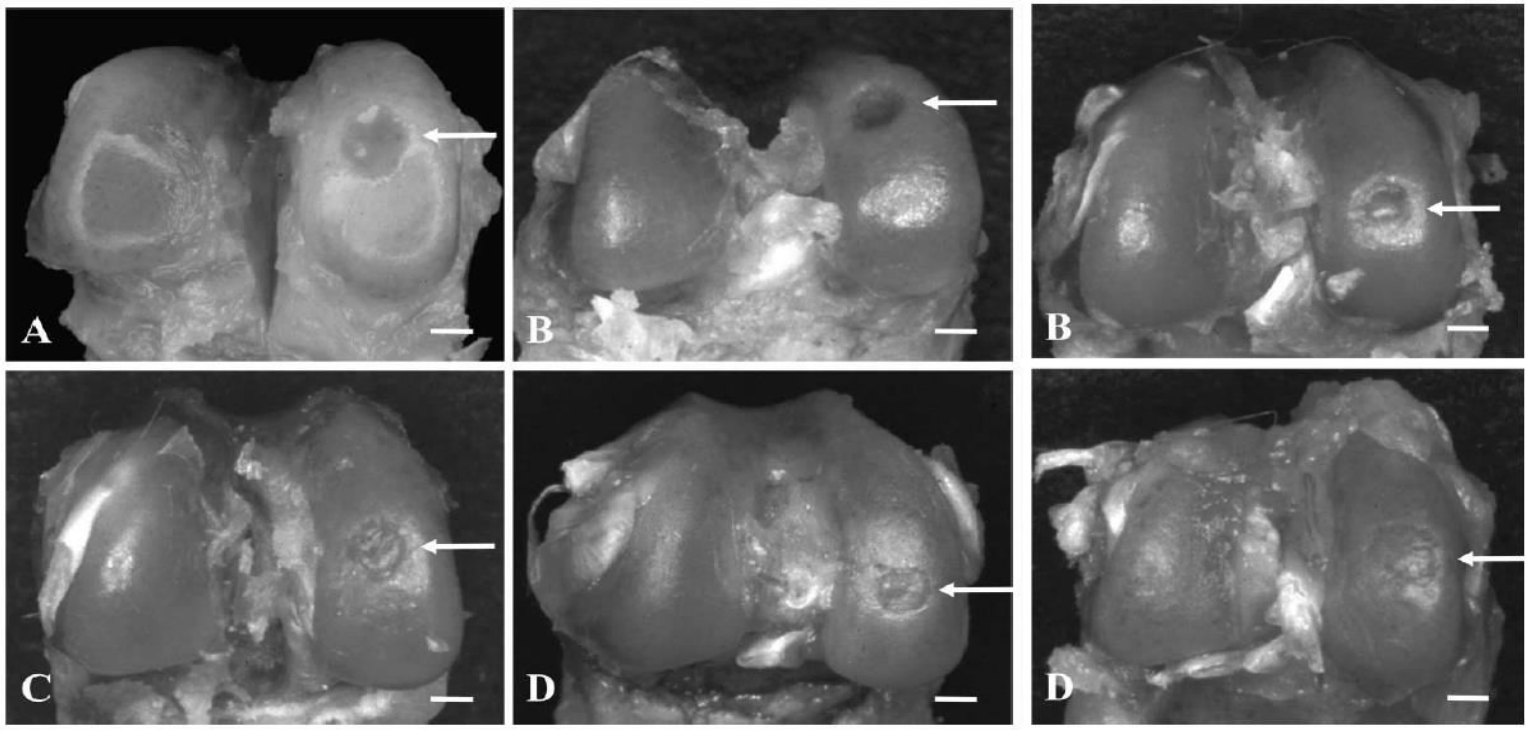
Are harvest from there niches from a (postnatal) individual

No ethical dilemma for their use in research

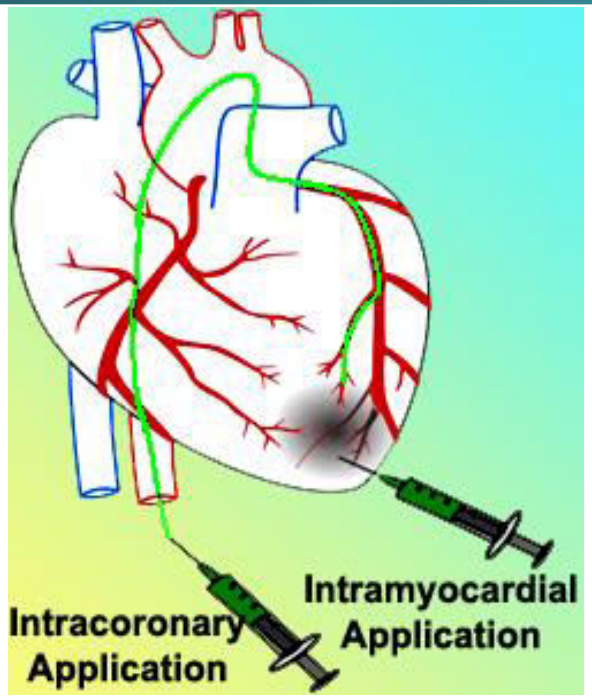
Have a limited ability to renew themselves and differentiate in to different cell types

Do not cause an immune rejection

Stem Cells



Heart

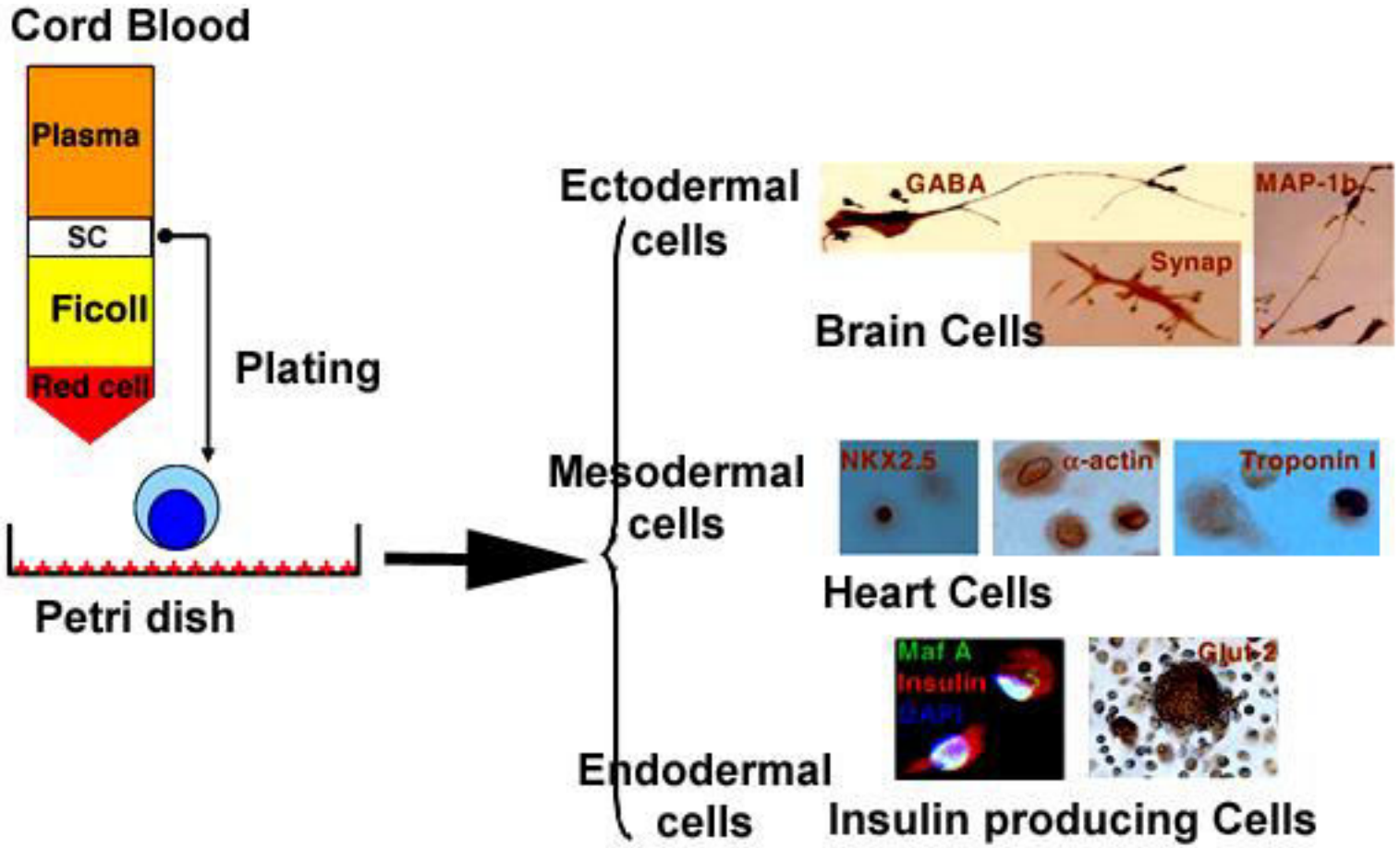


Düsseldorf group performed clinical stem cell application for the first time in 2001, treating acute myocardial infarction by intracoronary cell transfer.

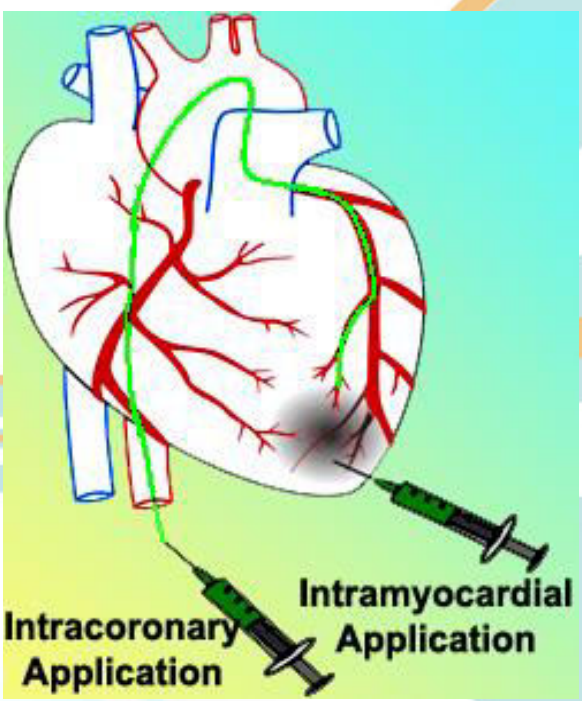
Currently, the optimal time point for cardiac stem cell application in a ischemic setting seems to lay within the first month after myocardial infarction

"There is growing evidence that bone marrow-derived stem/progenitor cells improve cardiac function" (Leri 2009).

Stem Cells


Cord Blood
Plasma
SC
Ficoll
Red cell
Plating
Petri dish
Ectodermal cells
GABA
Synap
MAP-1b
Brain Cells
Mesodermal cells
NKX2.5
α-actin
Troponin I
Heart Cells
Endodermal cells
Maf A
Insulin
DAPI
Glut-2
Insulin producing Cells

Tissue Replacement

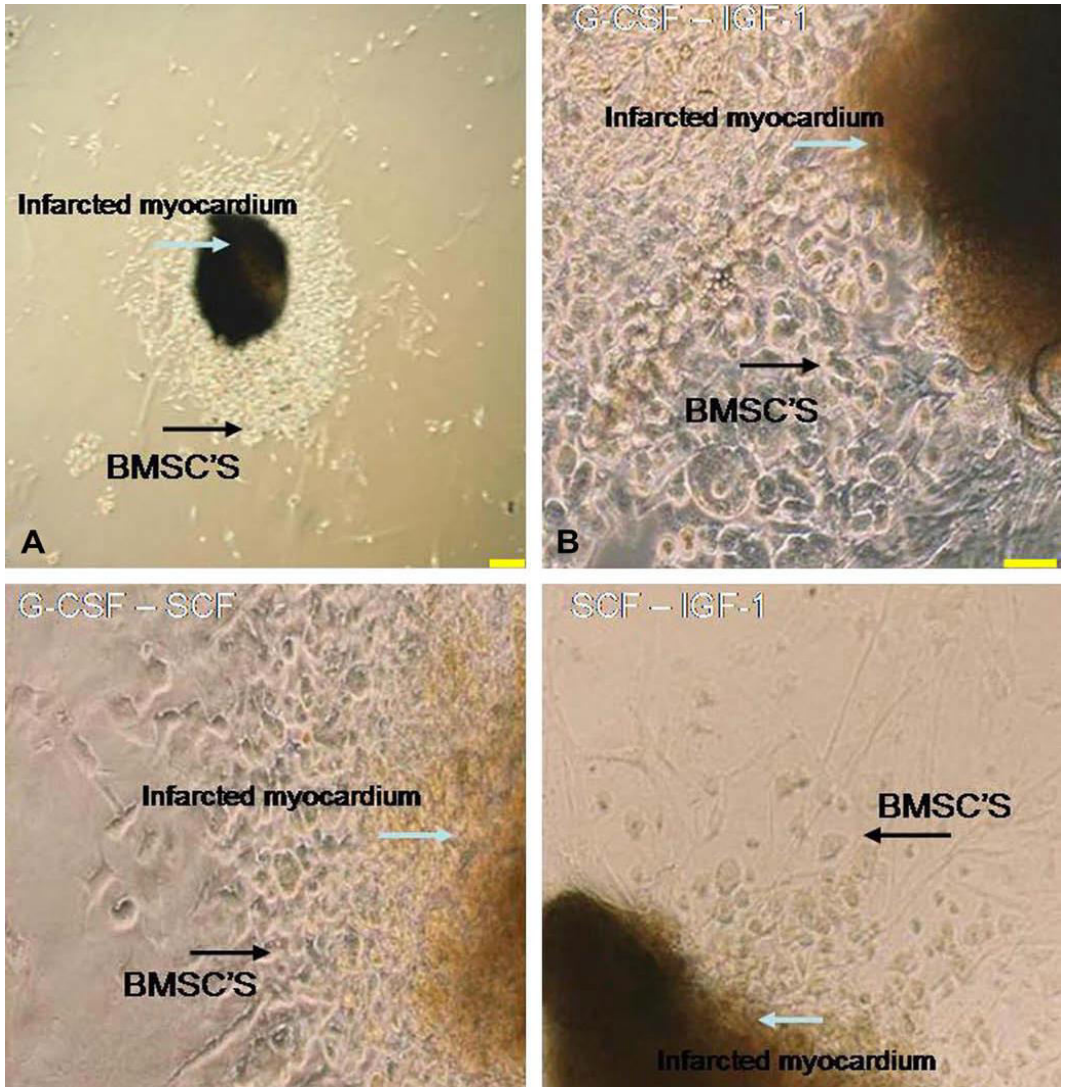


END

Cell Biology

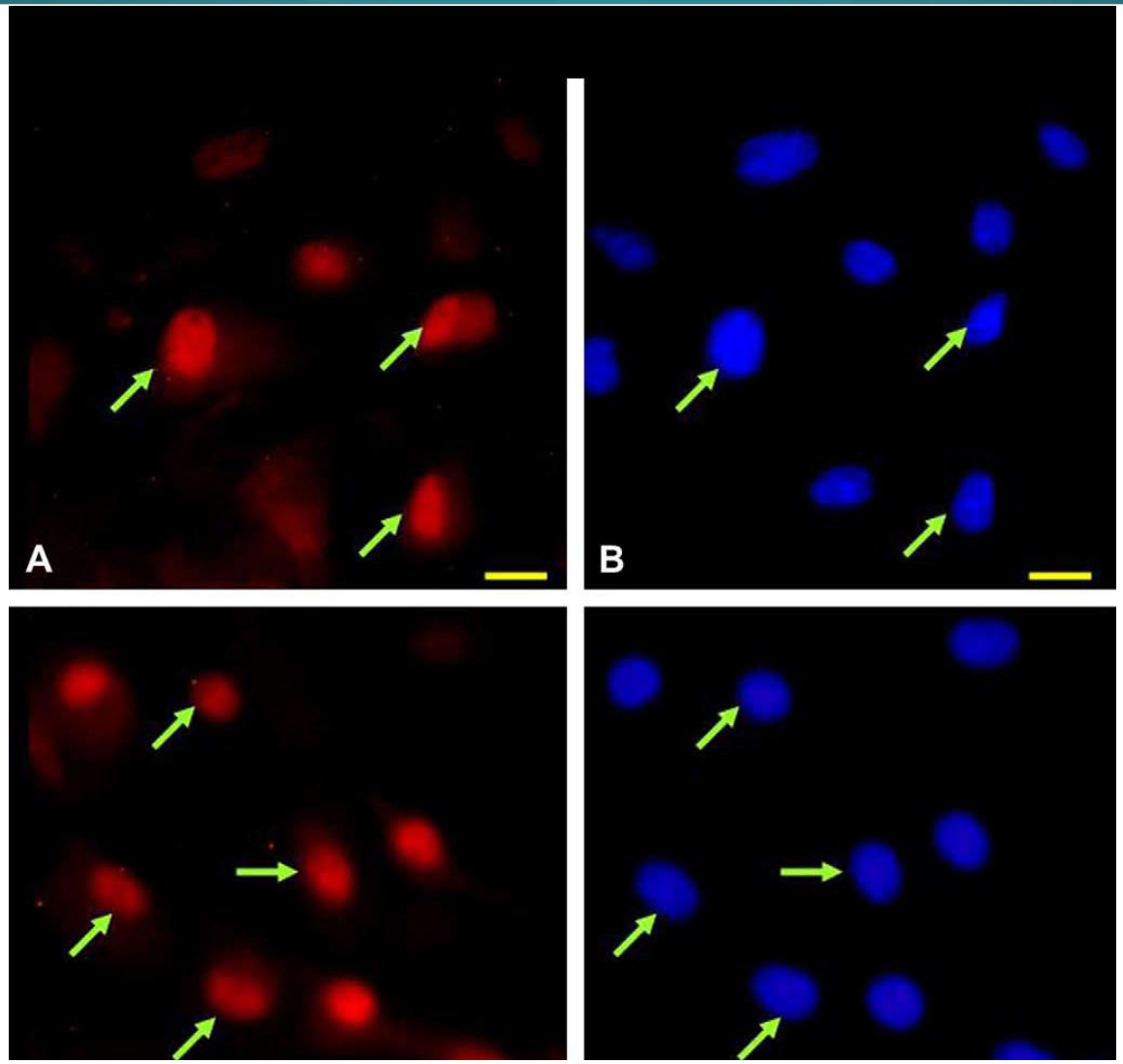
Cytokine Induced Stem Cell Proliferation & Differentiation

Infarcted Myocardium



IGF-1 & G-CSF → BMSC migration infarcted myocardium

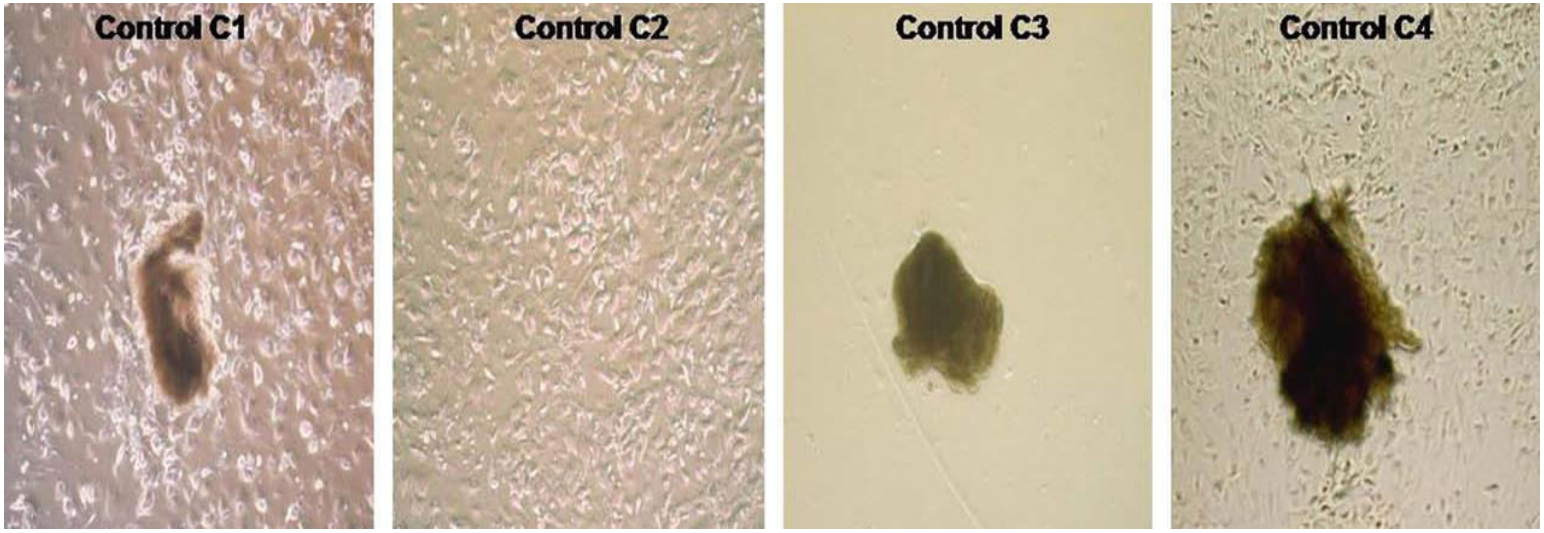
MEF-2 Expression



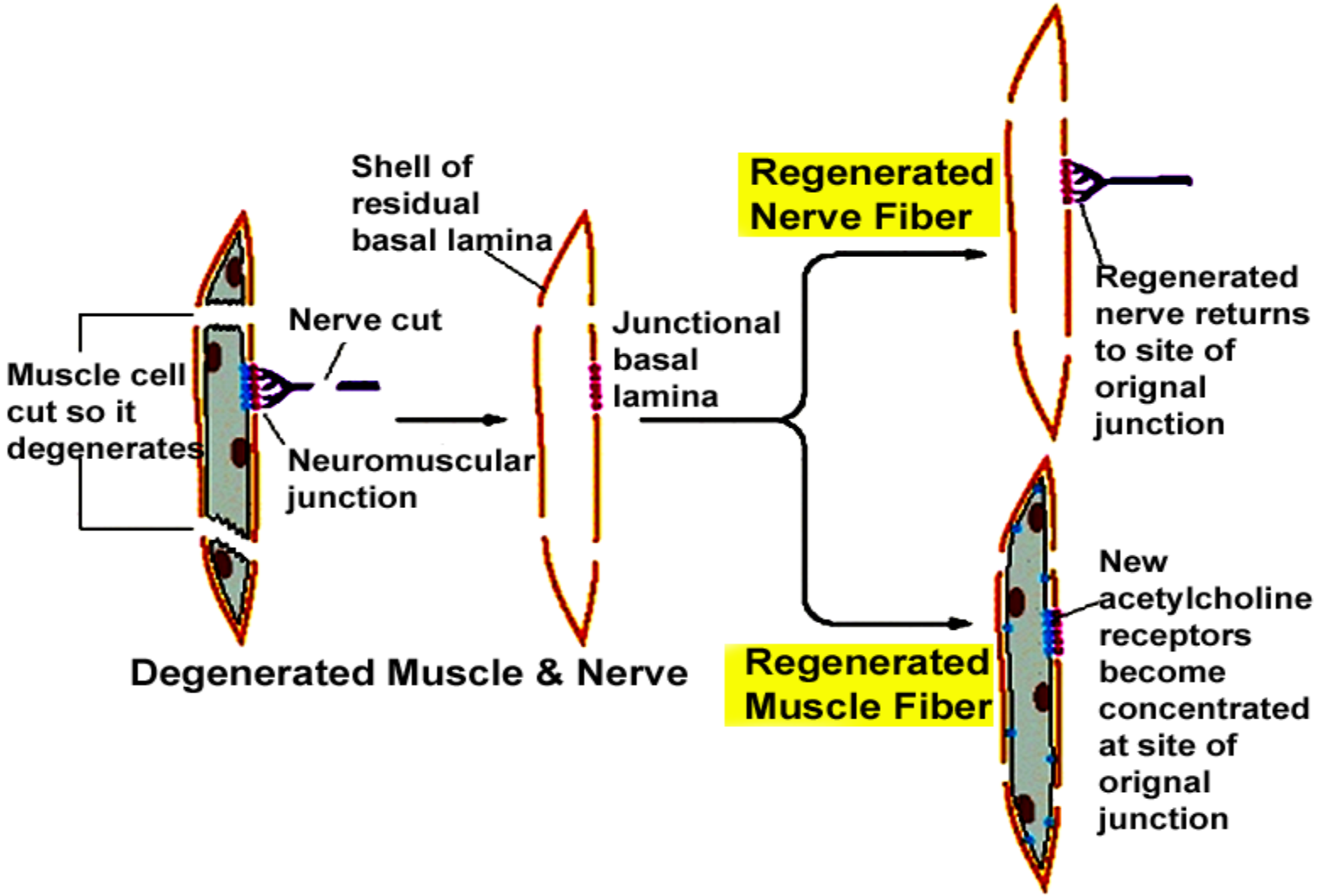
Myocyte enhancer factor 2 transcription factors regulates muscle development

Infarcted Myocardium

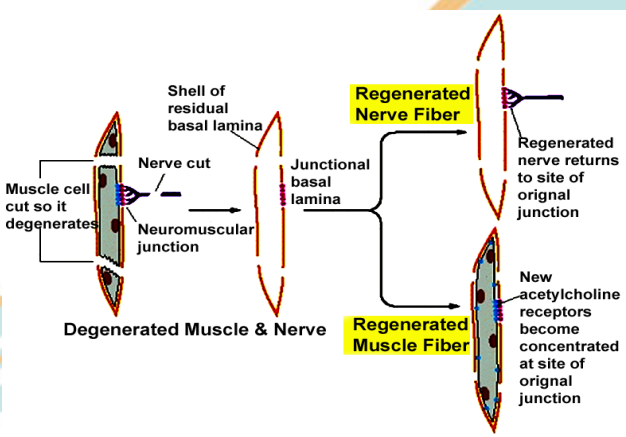
Controls



ECM Memory


Shell of residual basal lamina
Regenerated Nerve Fiber
Regenerated nerve returns to site of orignal junction
Nerve cut
Junctional basal lamina
Muscle cell cut so it degenerates
Neuromuscular junction
New acetylcholine receptors become concentrated at site of orignal junction
Degenerated Muscle & Nerve
Regenerated Muscle Fiber

Tissue Replacement

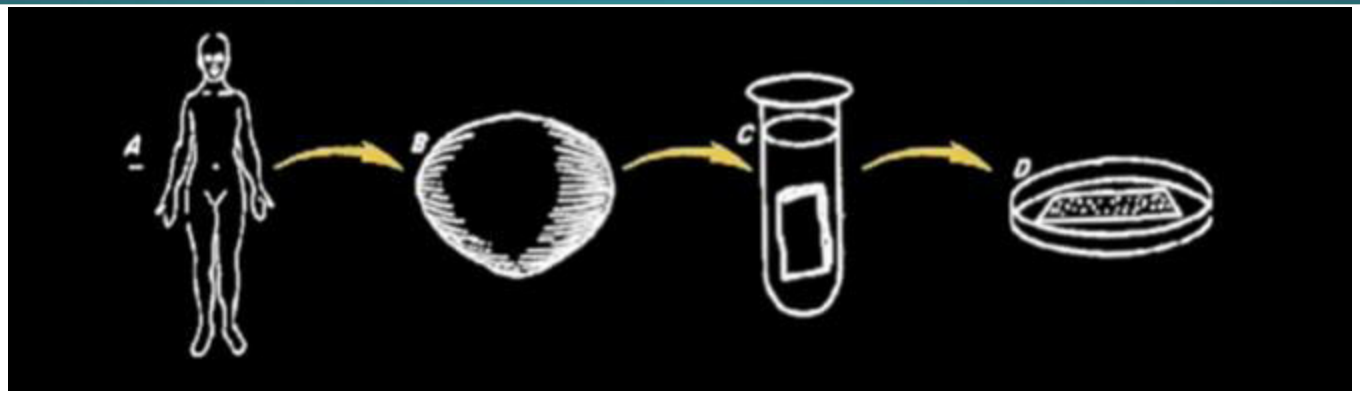


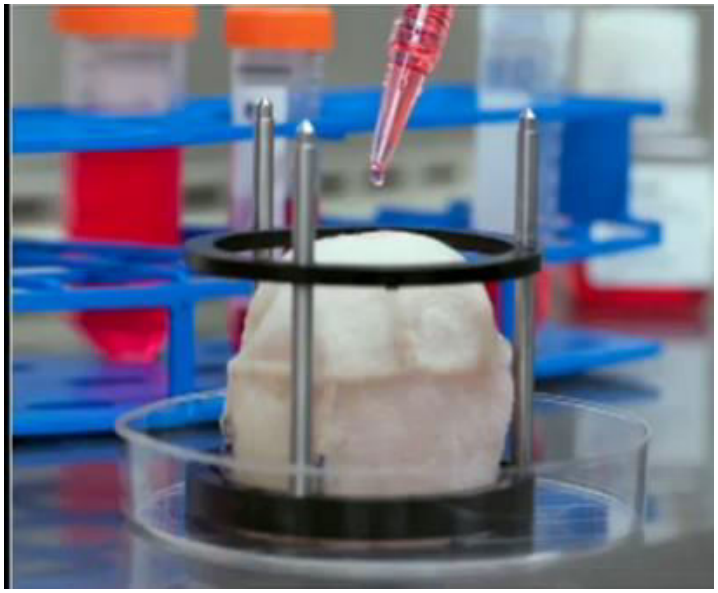
END

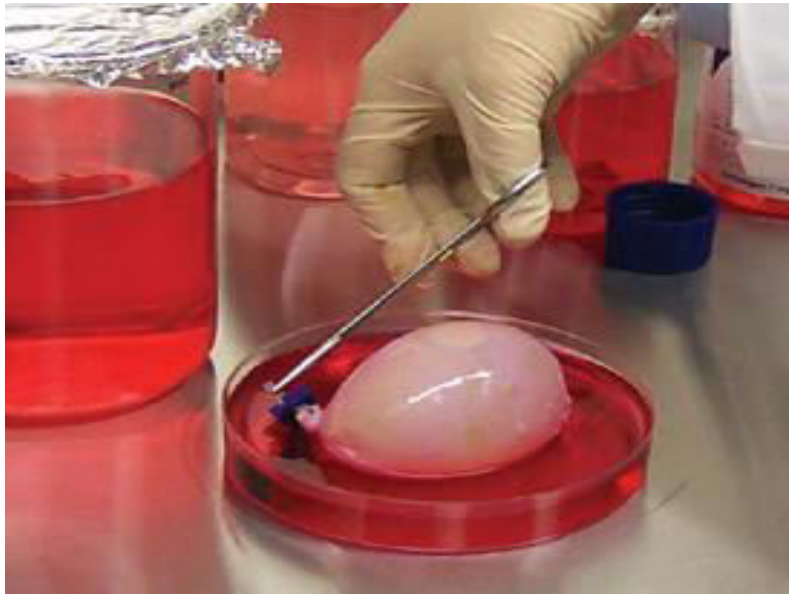
Cell Biology

In-vitro
Organ/Tissue
Fabrication

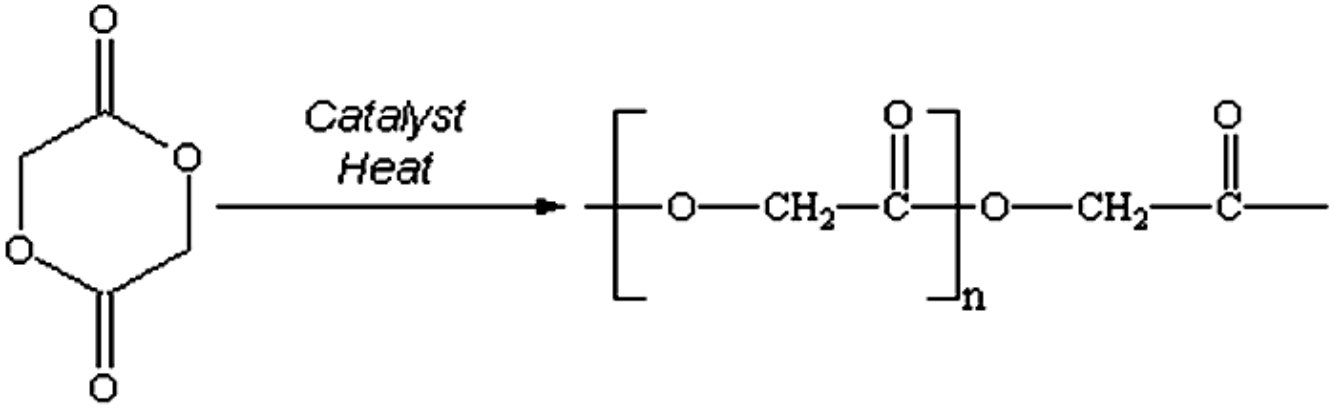
Unipotents + Artificial Scaffolds







Unipotents + Artificial Scaffolds

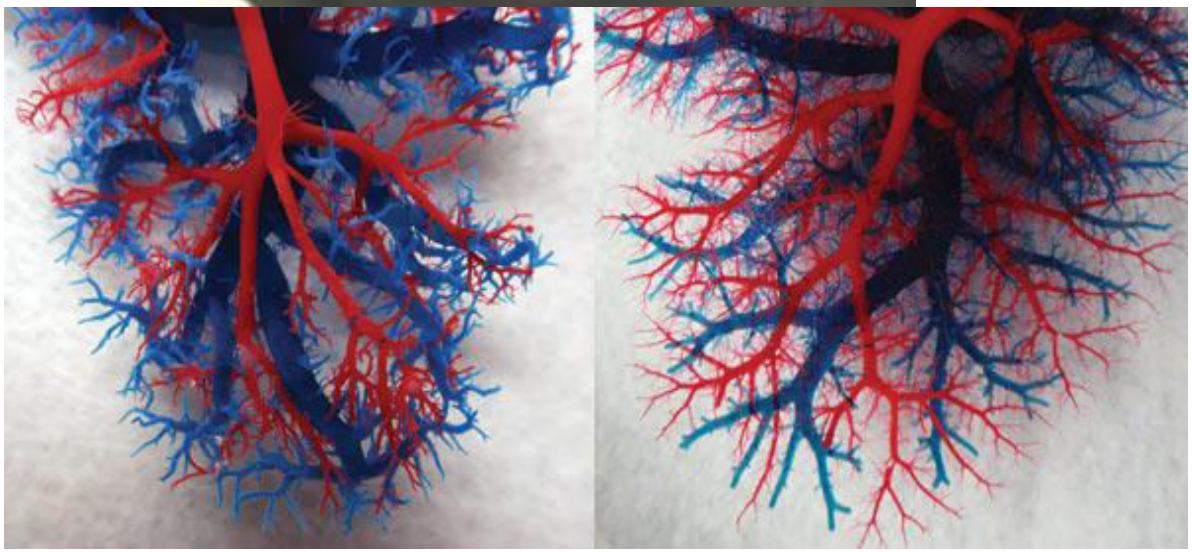


Glycolide **Polyglycolide**

PGA

Bio-artificial Tissue

Bio-artificial Tissue



Tissue Replacement

END

In 1924, Bogdanov started his blood transfusion experiments, apparently hoping to achieve eternal youth or at least partial rejuvenation. Lenin's sister Maria Ulianova was among many who volunteered to take part in Bogdanov's experiments. After undergoing 11 blood transfusions, he remarked with satisfaction on the improvement of his eyesight, suspension of balding, and other positive symptoms. The fellow revolutionary Leonid Krasin wrote to his wife that "Bogdanov seems to have become 7, no, 10 years younger after the operation". In 1925-1926, Bogdanov founded the Institute for Haemotology and Blood Transfusions, which was later named after him. But a later transfusion cost him life, when he took the blood of a student suffering from malaria and tuberculosis. (Bogdanov died, but the student injected with his blood made a complete recovery.) Some scholars (e.g. Loren Graham) have speculated that his death may have been a suicide, because Bogdanov wrote a highly nervous political letter shortly beforehand. Others, however, attribute his death to blood type incompatibility, which was poorly understood at the time.[16] In May 2014 the media reported on a blood transfusion that contains a protein called GDF11 to possibly cure Alzheimer's disease and heart disease.[17]

Rules & Cell Culturing Equipment

The Golden Rules

Laboratory coats should be worn and fastened at all times when working with chemicals or biological agents.

Wear Safety Glasses when working with liquids or other material that could get into your eyes.

Footwear that properly protects the foot from chemical splashes should be worn at all times in the laboratory.

Working in laboratory is fun way to learn but first and foremost is your safety to ensure follow instructions

The Golden Rules

- Special gloves should be worn when handling chemicals or samples.

- Do not touch door and/or fridge handles etc. while wearing protective gloves.

- Do not pipette by mouth.

- Do not eat, drink or smoke anywhere in the laboratory space.

Laminar Flow Hood

Provides sterile working area where cells can be handled without worrying about contamination.

Changing media Air curtain Invisible barrier

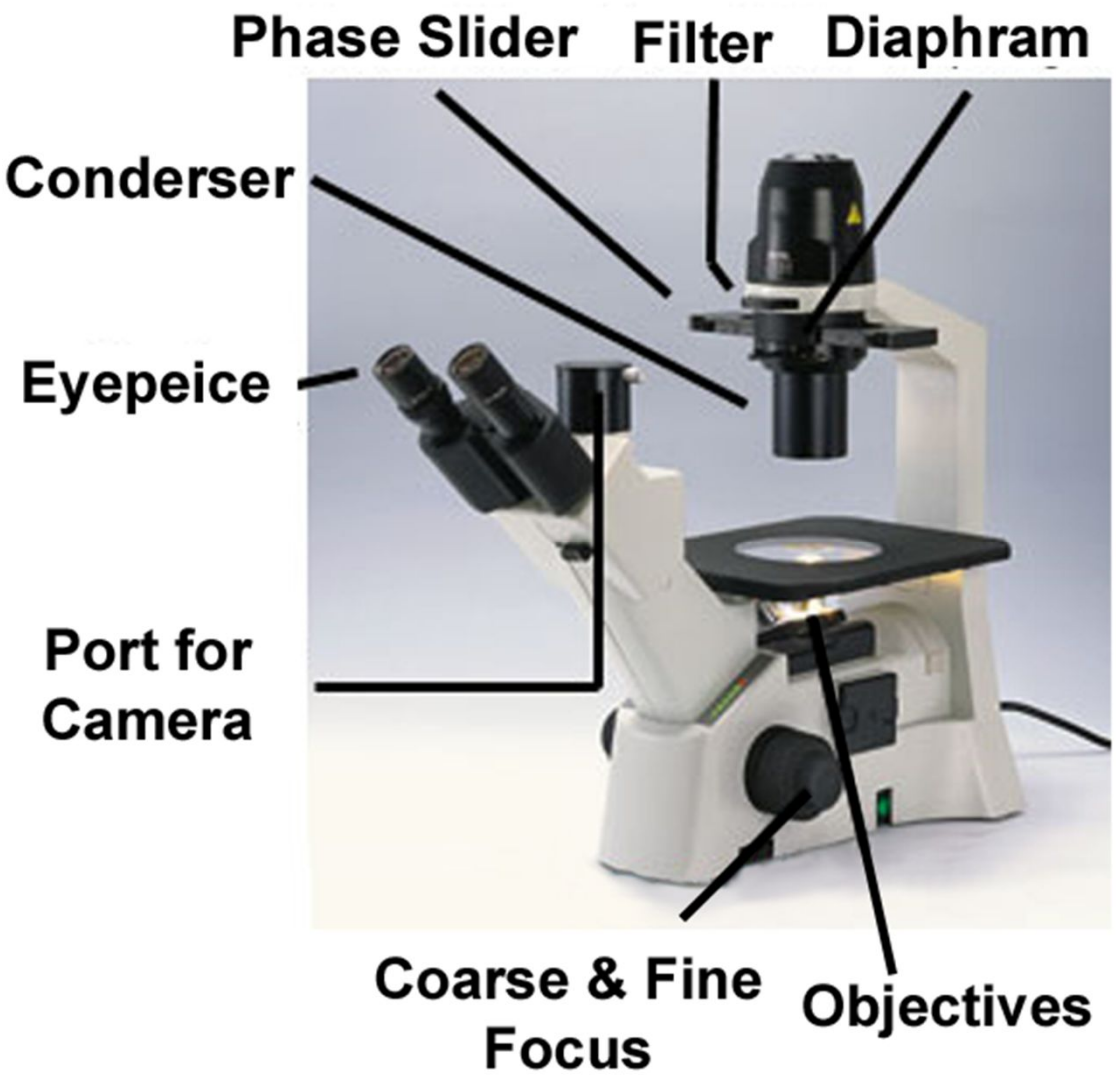
Incubator

Maintains optimal temperature & humidity for cell survival and proliferation.

Double door.

CO_2 to maintain pH .

Increasing atmospheric CO2 decreases the pH of the medium and vice versa.



Microscope

Separating cells from the media washing

Centrifuge

Rotates sample around a fix axis simulating high gravity causing denser substances to sink to the bottom of the sample tube.

Separating cells from the media washing

Water bath

Maintains a fix temperature.

Used to thaw samples.

Incubate samples for enzymatic digestion

END

Rules & Cell Culturing Equipment

The Golden Rules

Laboratory coats should be worn and fastened at all times when working with chemicals or biological agents.

Wear Safety Glasses when working with liquids or other material that could get into your eyes.

Footwear that properly protects the foot from chemical splashes should be worn at all times in the laboratory.

Working in laboratory is fun way to learn but first and foremost is your safety to ensure follow instructions

The Golden Rules

- Special gloves should be worn when handling chemicals or samples.

- Do not touch door and/or fridge handles etc. while wearing protective gloves.

- Do not pipette by mouth.

- Do not eat, drink or smoke anywhere in the laboratory space.

Laminar Flow Hood

Provides sterile working area where cells can be handled without worrying about contamination.

Changing media Air curtain Invisible barrier

Incubator

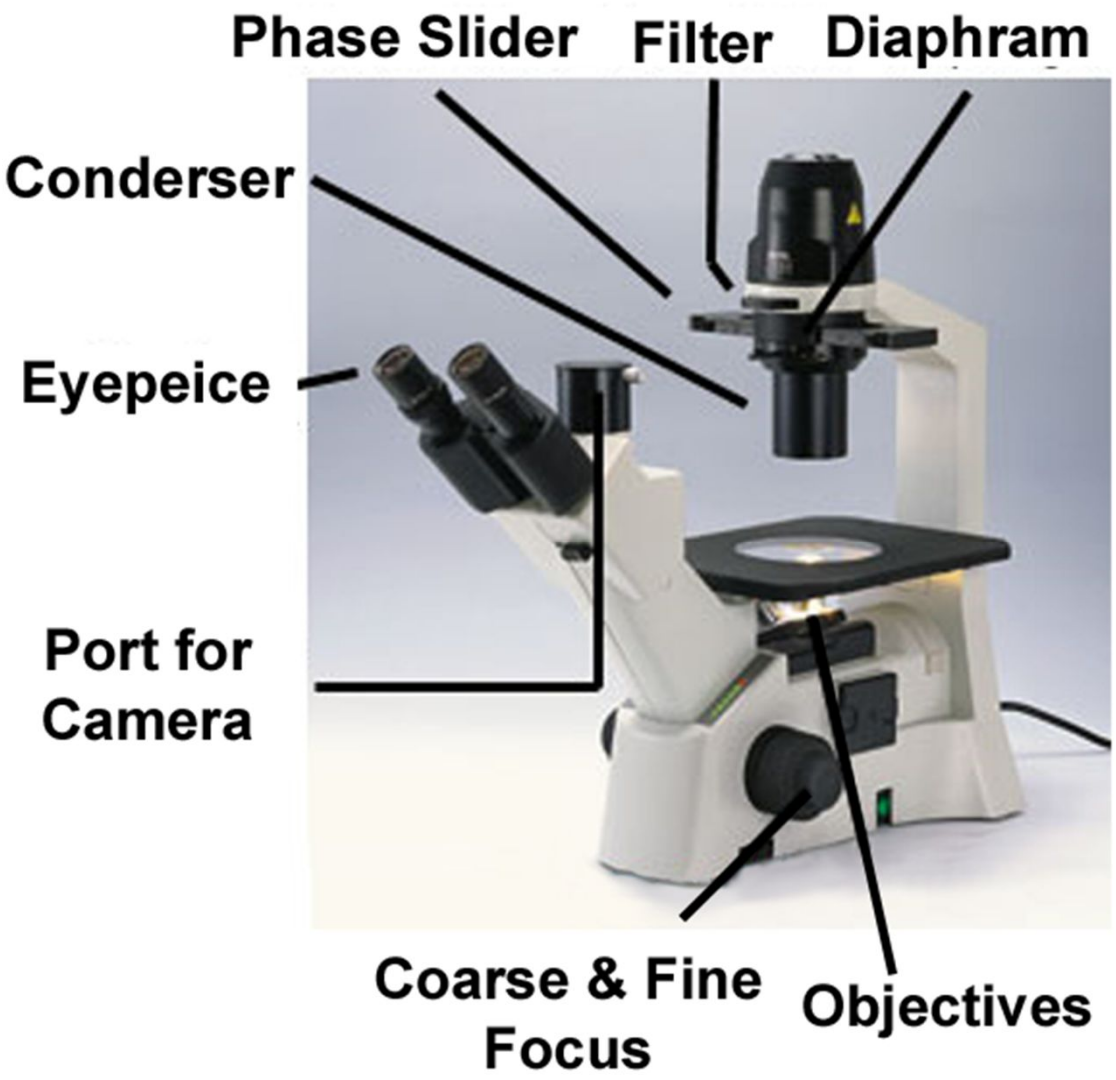
Maintains optimal temperature & humidity for cell survival and proliferation.

Double door.

CO_2 to maintain pH

.

Increasing atmospheric CO2 decreases the pH of the medium and vice versa.



Microscope

Separating cells from the media washing

Centrifuge

Rotates sample around a fix axis simulating high gravity causing denser substances to sink to the bottom of the sample tube.

Separating cells from the media washing

Water bath

Maintains a fix temperature.

Used to thaw samples.

Incubate samples for enzymatic digestion

END

Getting Started

Check Equipment Settings before Usage

- Check the incubator Settings before you place your cells most incubators are set at 37°C .
- 100% humid environment.
- Cells in Bicarbonate media should be place in CO2 incubators& 5-10% CO_2
- CO2 + H2O yields H+ HCO3- buffer system
- Bicarbonate Maintains PH of the media
- Some cells are grown in Bicarbonate free media

Sterilize Reagents and Equipment.

- UV
- 70% ethanol
- Bake
- Autoclave
- Flame
- Filter

Cross linking: permeability: Heat: Heat & pressure: Filter: oxidation not heat

Filter Types

Filters come in two pore sizes 0.2 um and 0.45 um

0.2 um filter to remove bacteria from the tissue culture media

Smaller particles i.e. Mycoplasma and viruses can pass through 0.2 um filter

0.45 is primarily used to remove particulates from solutions

Use an appropriate filter size and type Cellulose, polycarbonate

Media Should be filtered (prevent deactivation)

Tissue culture hood should be cleaned with 70% ethanol and disinfected with UV

Plastic can be Autoclaved, UV treated or sprayed with 70% ethanol

Glass can be Autoclaved, UV treated, sprayed with 70% ethanol or Baked

Metal can be Autoclaved, UV treated, sprayed with 70% ethanol or Baked

Flame the mouth of bottles when after and before closing

Cross linking: permeability: Heat: Heat & pressure: Filter: oxidation not heat

Sterilize the Tissue culture hood (UV + Ethanol)

All manipulations must be carried out in a sterile cabinet

Turn the UV light off (Should ordinarily be on)

Bring materials into the hood

Light up the flame or gas

Begin your work

Flame all caps and lids

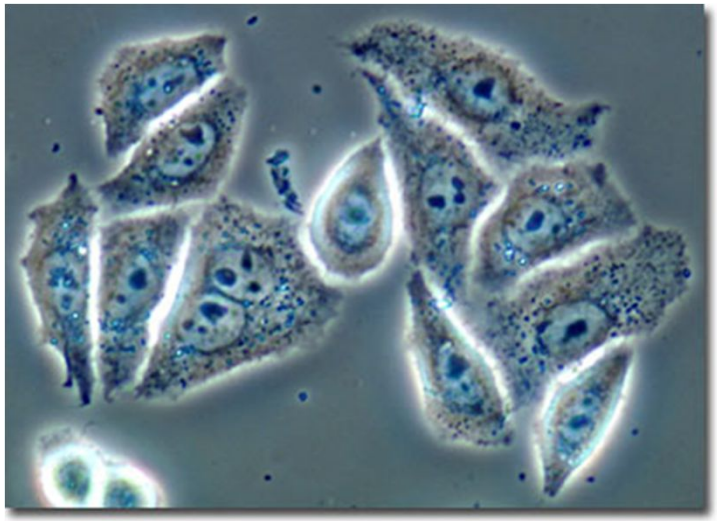
END

Mammalian Cell Culture

What is cell culture?

In vitro cultivation of cells from different tissues is referred to as cell culture.

A vertebrate can have more than 100 different cell types, each cell type requires specific culturing conditions.



Why do we need to culture cells in vitro?

To study or manipulate cells in controlled environment.

Clinical use of cell culture:

Fertility treatment, (IVF and embryo transplant)

Chromosomal studies, amniocentesis, CVS,
Blood bone Marrow Tissue grafts, (stem cell)

Cytotoxicity testing of therapeutic drugs
tissue/organ engineering etc.

Use of cell culture in Research

To study cell structure and function.

To study behaviour of cells (developmental stages, response).

Generation of monoclonal antibodies.

Generation of recombinant proteins.

Culturing viruses for research.

Types of cell culture:

Primary cells are cells straight from the tissue with no passages. A primary cell culture may be composed of mixtures of cell types.

Cell lines have at least one passage. With each subsequent subculture. There are several means of obtaining immortalized cell lines. Some cells, such as tumor cells, are inherently immortal.

Pure cultures: Use of selective media to suppress growth of unwanted cells and promoting growth of desired cells.

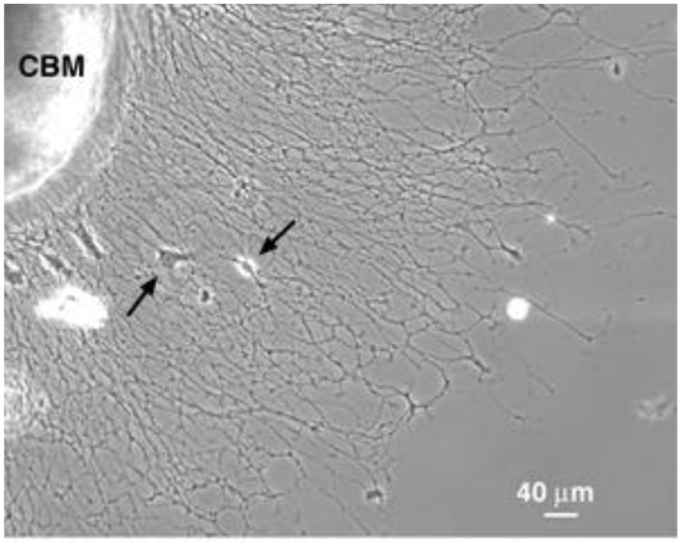
Adherent Cell Culture	Suspension Cell Culture
Appropriate for most cell types, including primary cultures	Appropriate for cells adapted to suspension culture and a few other cell lines that are nonadhesive
Requires periodic passaging, but allows easy visual inspection under inverted microscope	Easier to passage, but requires daily cell counts and viability determination to follow growth patterns;
Cells are dissociated enzymatically or mechanically	Does not require enzymatic or mechanical dissociation

Explant Culture:

A piece of a tissue is placed in an appropriately coated dish containing media.

Cells typically move out o f the tissue and start to proliferate in the dish.

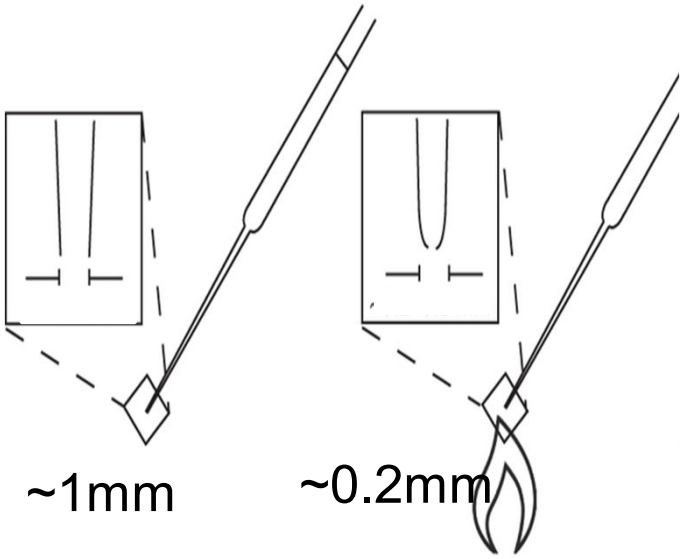
This is useful for sensitive tissue, which may be prone to damage by proteases or other enzymes.



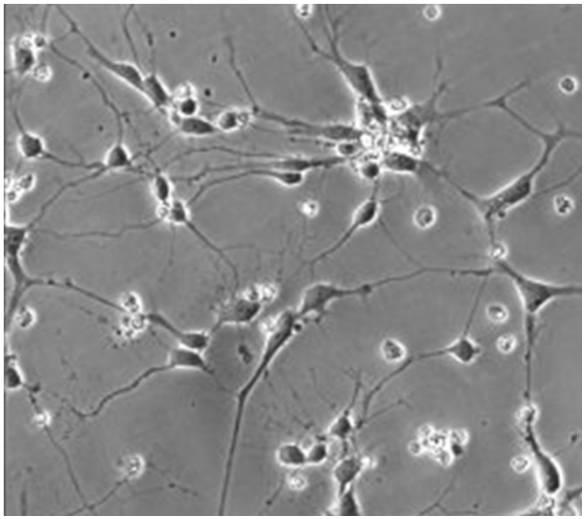
In case of neuronal tissue explants provide a dense area of axons with uniform polarity

Dissociated **Culture**:

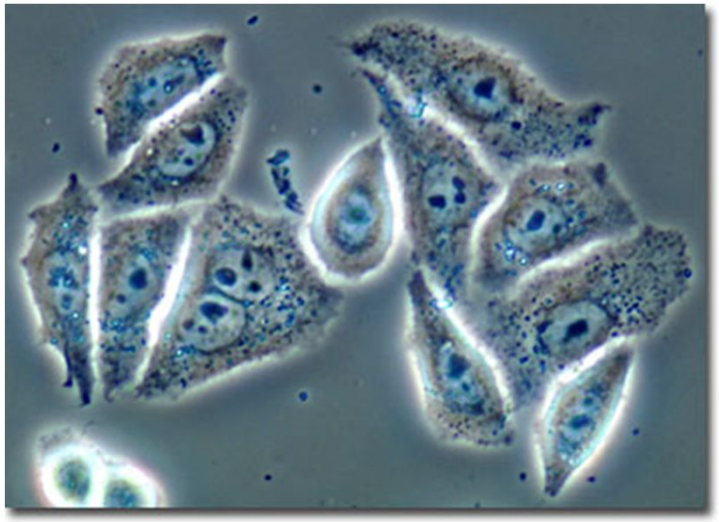
Tissue sample is dissociated, either mechanically (titration) or enzymatically, producing a cell suspension.



Cells are plated on coated tissue culture dish, cells attach to dish and start dividing.



The next day media is changed, which also removes dead cells



END

Culturing Cells

Procedure: Primary culture

Tissue is obtained from the animal under aseptic conditions.

It is usually necessary to treat the tissue with enzymes e.g. trypsin collagenase.

Serum to bock the effect of these enzymes

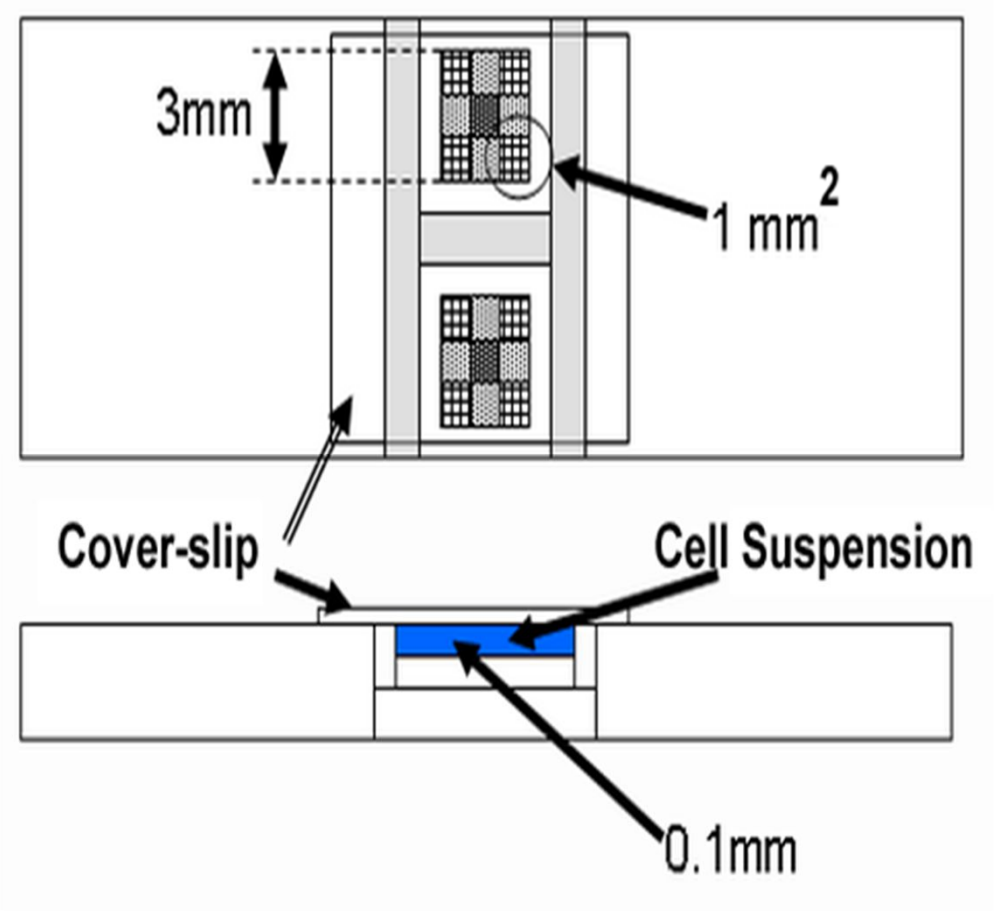
Trituration.

Evaluate the quality and quantity of cells.

Plating cells on appropriately coated dish.

Separating cells from the media washing

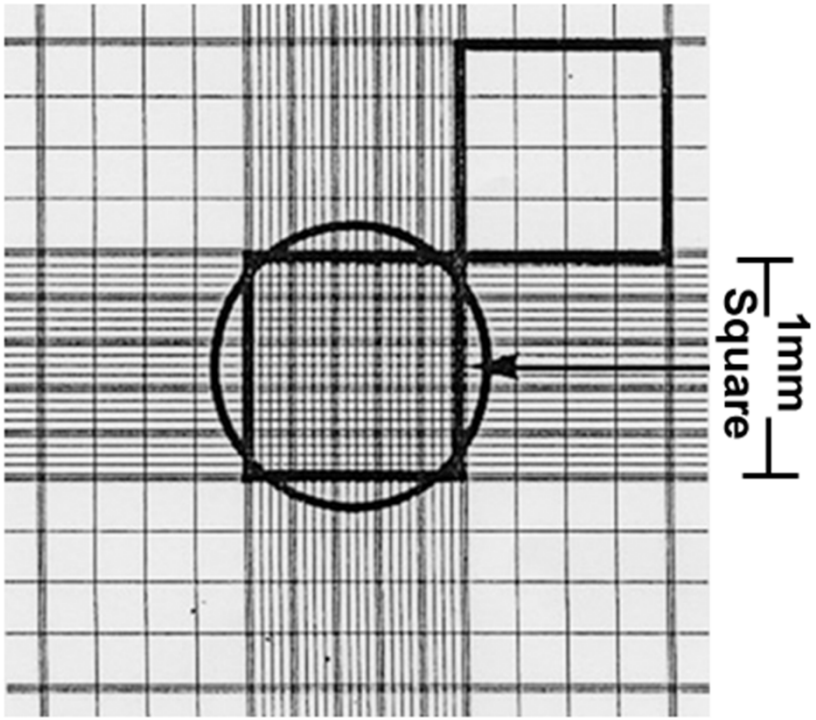
Use of the Neubauer Haemocytometer



The haemocytometer is a specialised microscope slide

The central region is 0.1 mm lower than the rest of the slide.

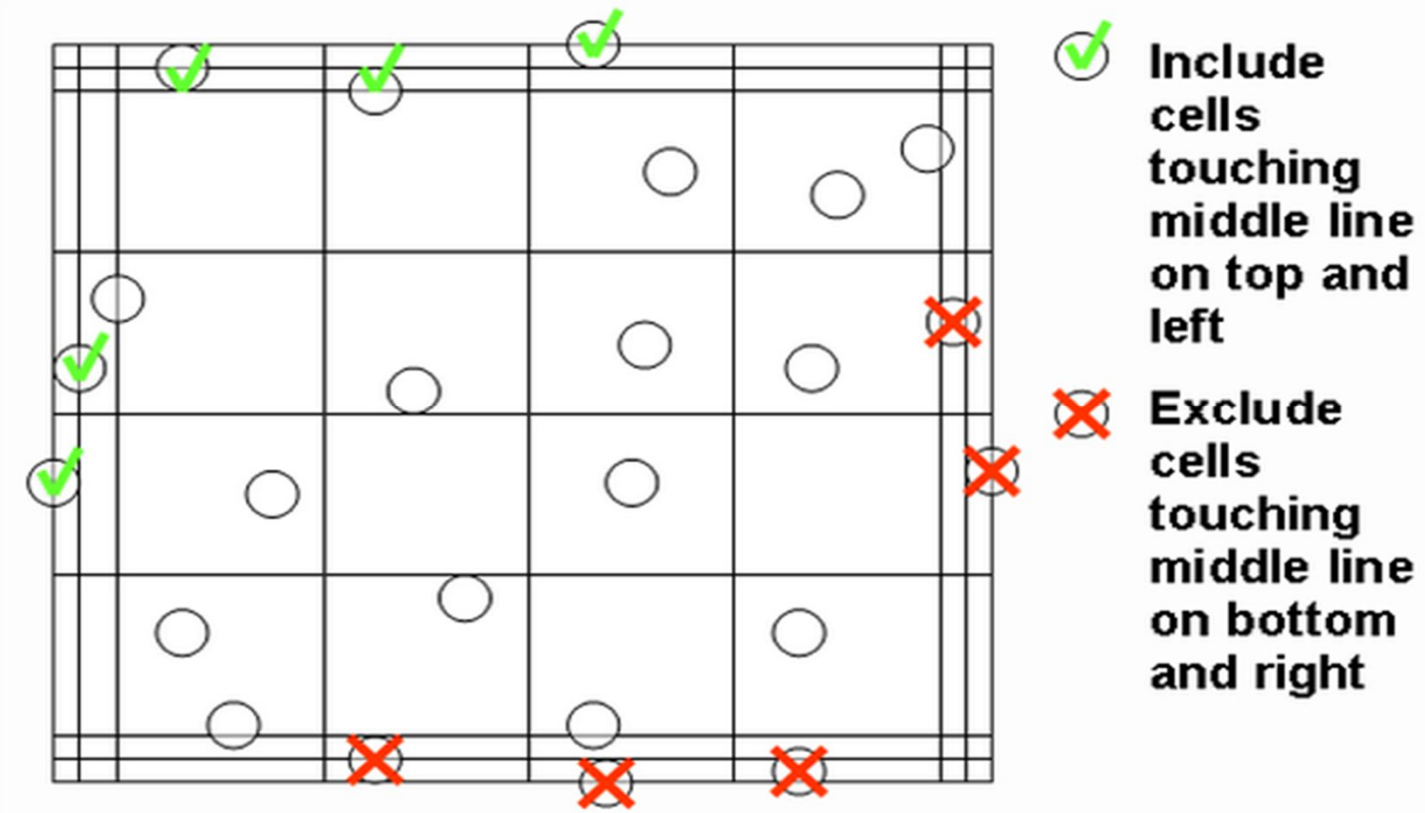
Use of the Neubauer Haemocytometer



Each grid comprises 25 large squares.

This creates a region of known volume (0.1 mm^3 = 0.1 x1x1= 0.1 microliter = $1x10^{-4}$ ml) when a special coverslip is placed over the central region.

Ten µl of culture are then pipetted under the cover slip



Cells counted in a proportion of the grid squares, multiplying the total number of cells in the entire grid by 10^4 gives the number of cells per ml.

$C = \tilde{N} \times 10^4$ where

C = cells per ml $\tilde{N}$= average of cells counted

10^4 = volume conversion factor for 1mm2

Total Yield = C x V (V= total volume of cells (ml))

Example: -

Count1 = 182 cells/1mm2

Count 2 = 175 cells/1mm2

Volume of Cells = 55 ml

Average cells counted =(Count1 + Count)/2 =

(182 + 175)/2 = 178.5

$C = 178.5 \times 10^4 = 1{,}785{,}000$ cells/ml

Total yield = C x V

1,785,000 x 55 = 98,175,000 cells

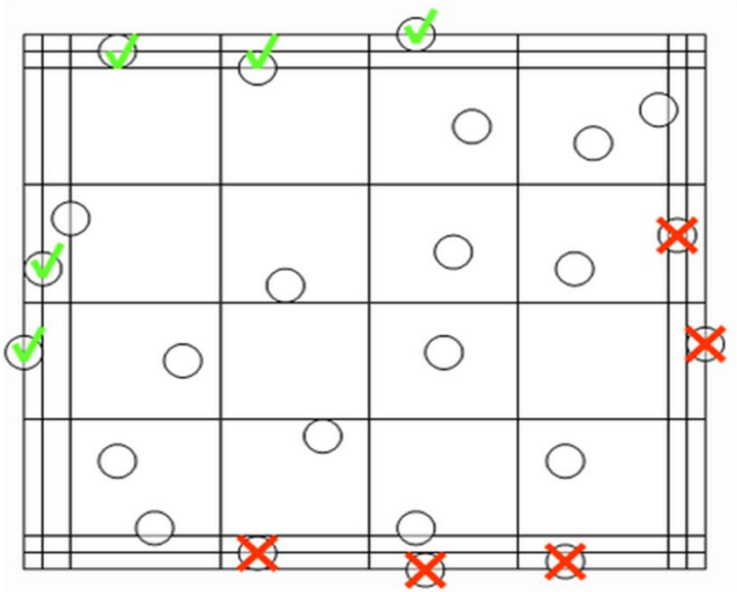
Pipet out appropriate volume of cell suspension & place it in a tissue culture dish or flask.

Make sure the tissue culture dish or flask are coated with appropriate substrate for the cells e.g. Collagen, Fibronectin, Laminin, Poly-L-lysine etc.

Most tissue culture dishes are pre-coated.

Next day change media to get rid of dead cells

Dead cells may release toxins, which may be injurious to healthy cells



END

Sub-culturing & Cell Banking

Cells should be sub-cultured just before they approach confluency.

A confluent monolayer is an adherent cell culture (dish, plate or flask) in which the cells have formed a single layer over the entire surface area available for growth.

Once the cells have started to form clusters above the first layer or have started to lift up from the surface, the culture is past confluency stage.

Cells that are repeatedly passaged at densities past confluency display decreased doubling times, decreased viabilities, & decreased ability to attach.

The culture is considered to be unhealthy

To increase the sample and keep cells healthy.

Suspension cultures should be passaged before they reach a density of 2.0 to 2.5 x 106 cells/ml and diluted back to 0.7 to 1.0 x 106 cells/ml

An average cell can divide 50 times before it gets unhealthy

Cells are treated with Trypsin-EDTA

EDTA is a Ca^{++} chelator EDTA binds Ca^{++}

This lowers the Ca^{++} concentration in the media causes cell rounding

Trypsin is highly active, relatively non-specific, protease, always make fresh trypsin.

Cleaves proteins on the cell surface and extracellular matrix.

This causes cells to detach from the surface .

Over trypsinization can damage cells.

Presence of EDTA allows us to use lower concentration of Trypsin.

Always block Trypsin with Serum to prevent it from damaging cells.

Storing cells at low temperatures

Advantages

1. Cells don't have to be maintained
2. Cells can be readily plated
3. Cells can be transported long distances

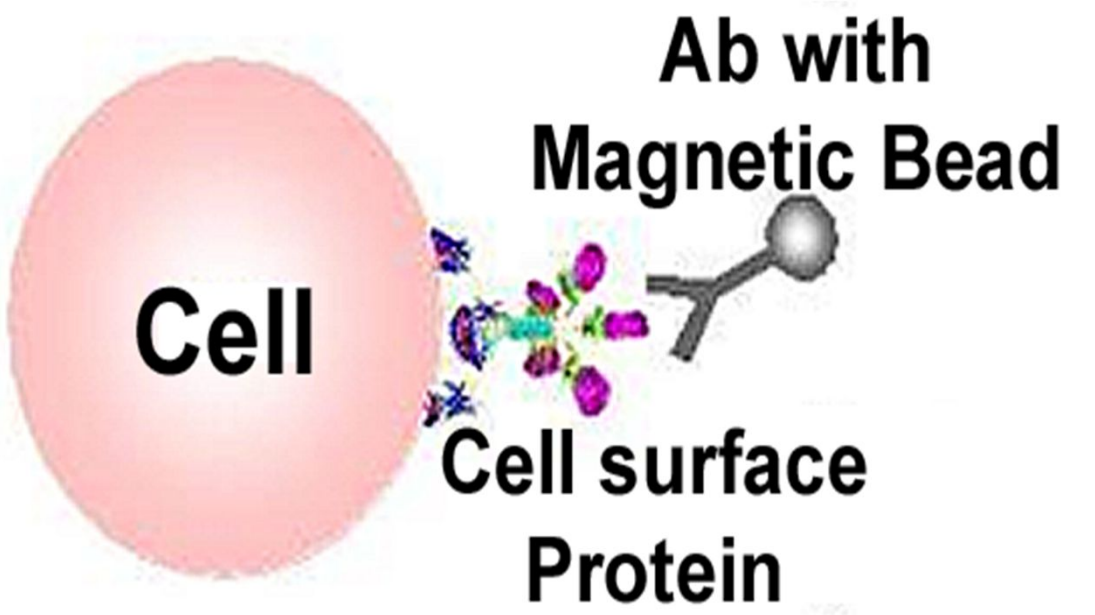
Cryoprotectants

DMSO or Glycerol

DMSO is more penetrating and is usually the agent of choice for larger, more complex cells

Procedure

1. Trypsinize the cells
2. Remove cells from the container
3. Spin the cell remove trypsin
4. Block trypsin with Media + 10% serum
5. Spin the cell remove blocking media
6. Suspend cells in 5%-10% DMSO+ media+20%serum
7. Freeze cells slowly in tubes inside Styrofoam
8. Store cells in LN



Cells have surface proteins that are unique to them.

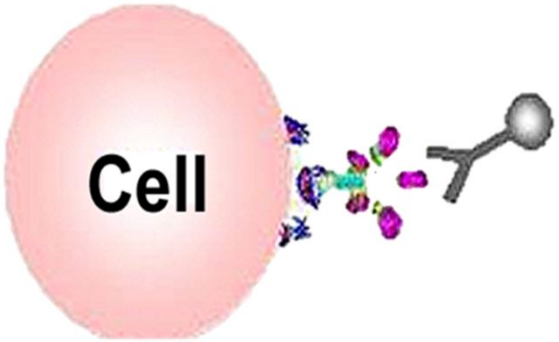
Cells are incubated with an antibody that recognizes an antigen present on the surface of cell of interest .

Antibodies are conjugated to magnetic beads

Cells are passed through a magnetic column cells with magnetic beads are retained others cells pass through

Magnet is remove and the column is subsequently washed

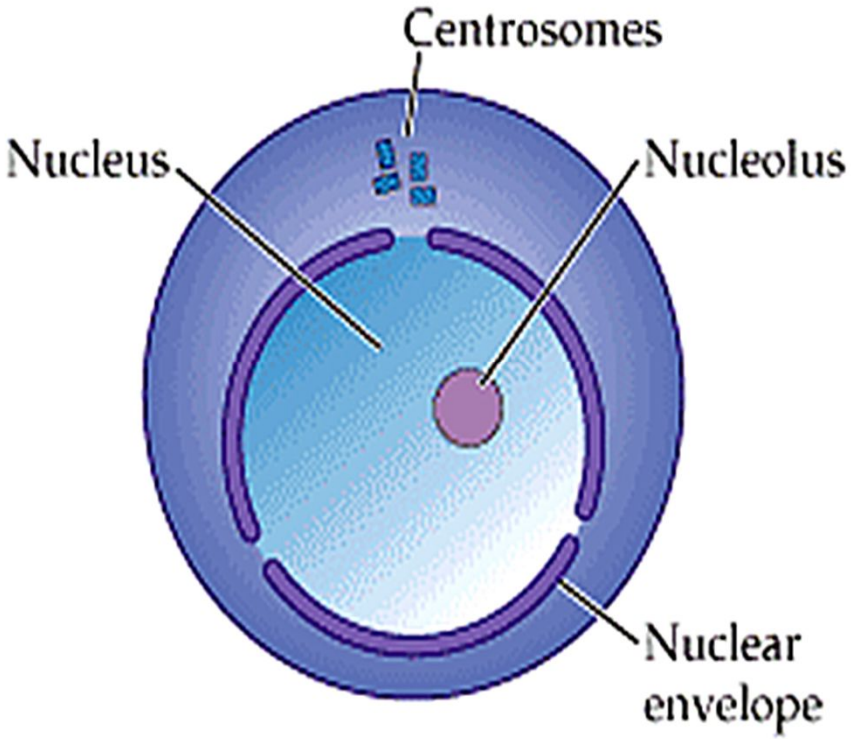
Cells are collected from the second wash will be homogenous

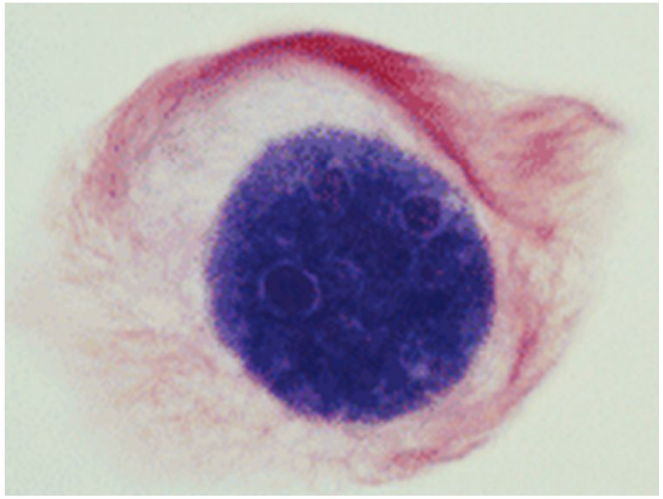


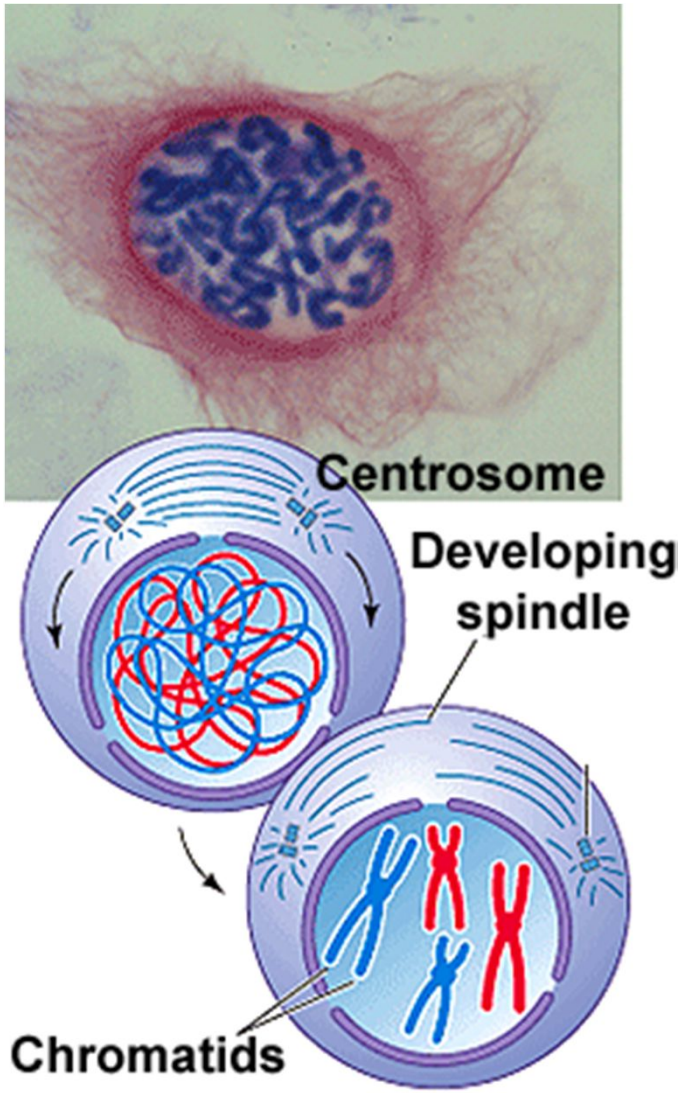
Pure cultures isolating stem cells from blood bone marrow

Identifying different stages of cell cycle

1 During the S phase of interphase, the nucleus replicates its DNA and centrosomes.



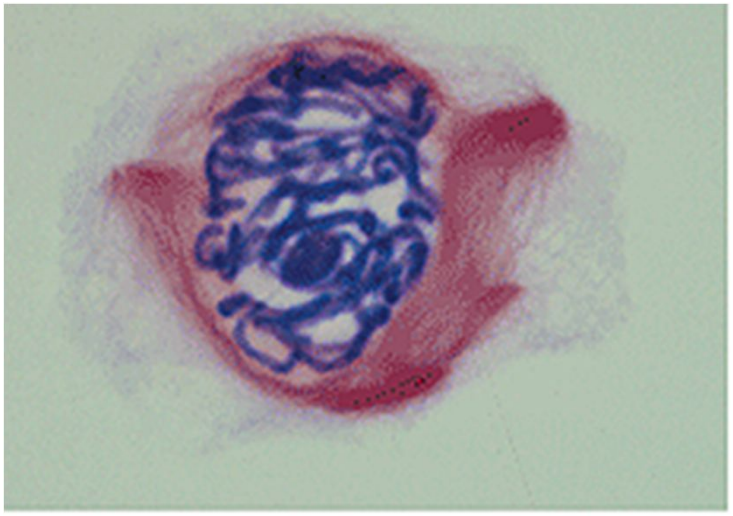


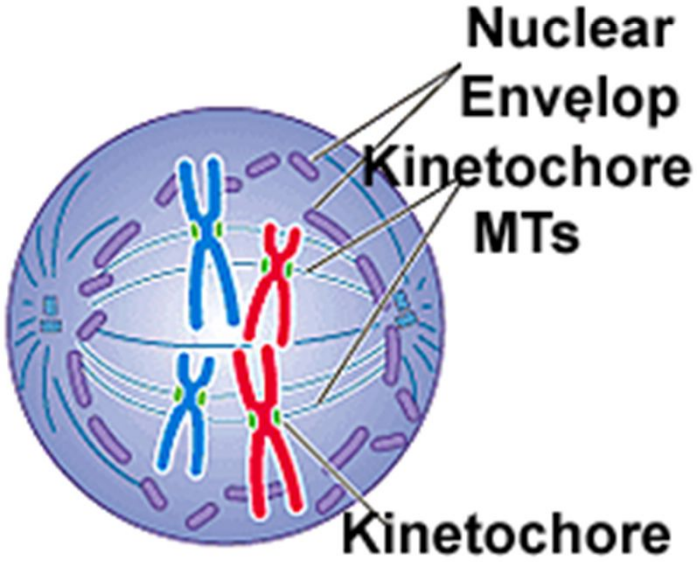


Prophase

The beginning of mitosis. so the individual chromatids become visible.

Microtubules form between each pole and the chromosomes to make up a spindle.



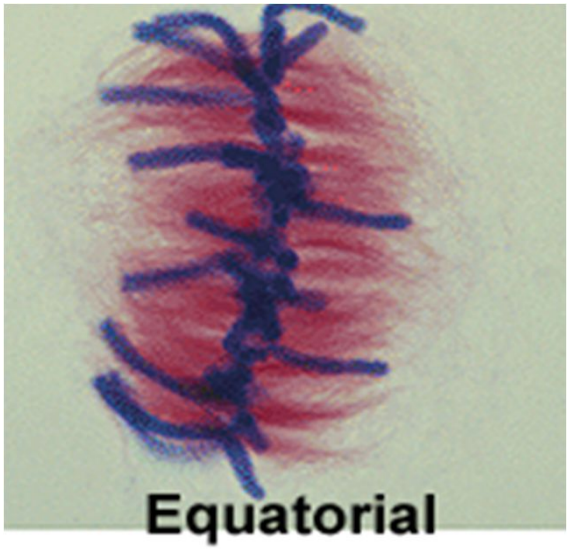


Prometaphase

Prometaphase is marked by the disappearance of the nuclear envelope.

Chromosomes begin to move toward the poles.

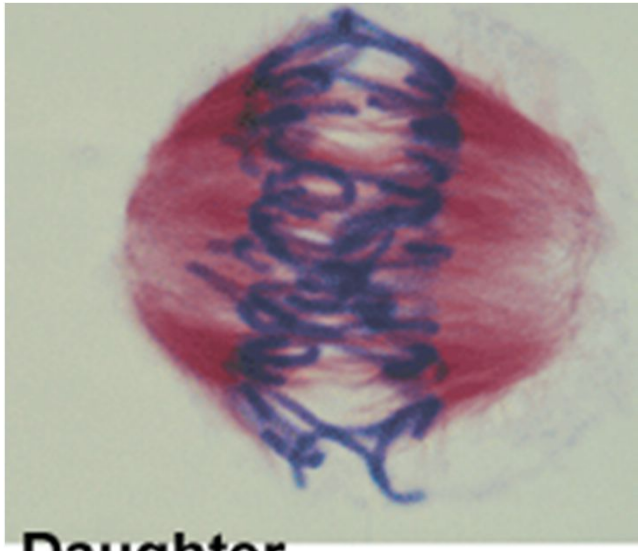
Stem Cell tissue Banking in the west advantages



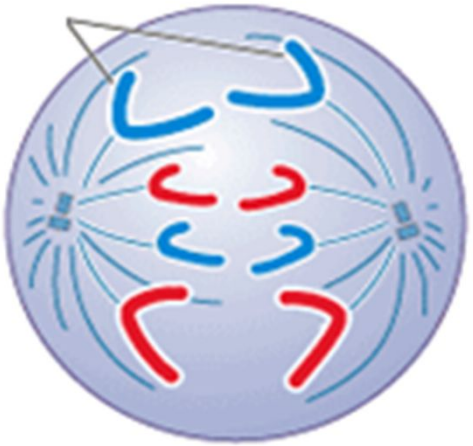
Metaphase

All the centromeres arrive at the equatorial plate.

Metaphase is the best time to see the sizes and shapes of chromosomes because they are maximally condensed.



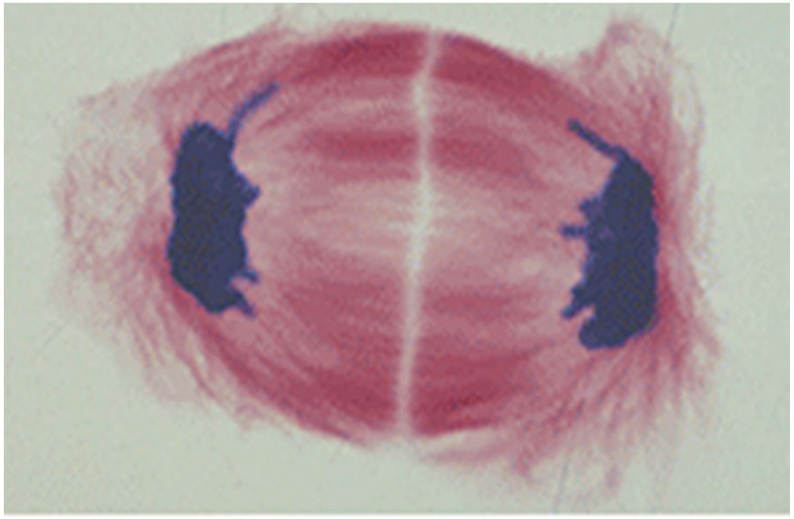
Daughter chromosomes

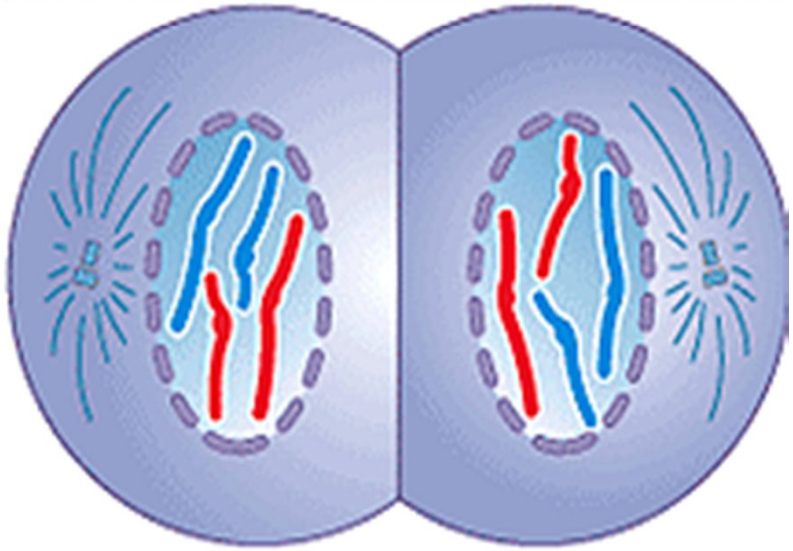


Anaphase

Separation of the chromatids marks the beginning of anaphase, during which the two sister chromatids move to opposite ends of the spindle.

Each chromatid con-tains one double-stranded DNA molecule and is now referred to as a daughter chromosome.



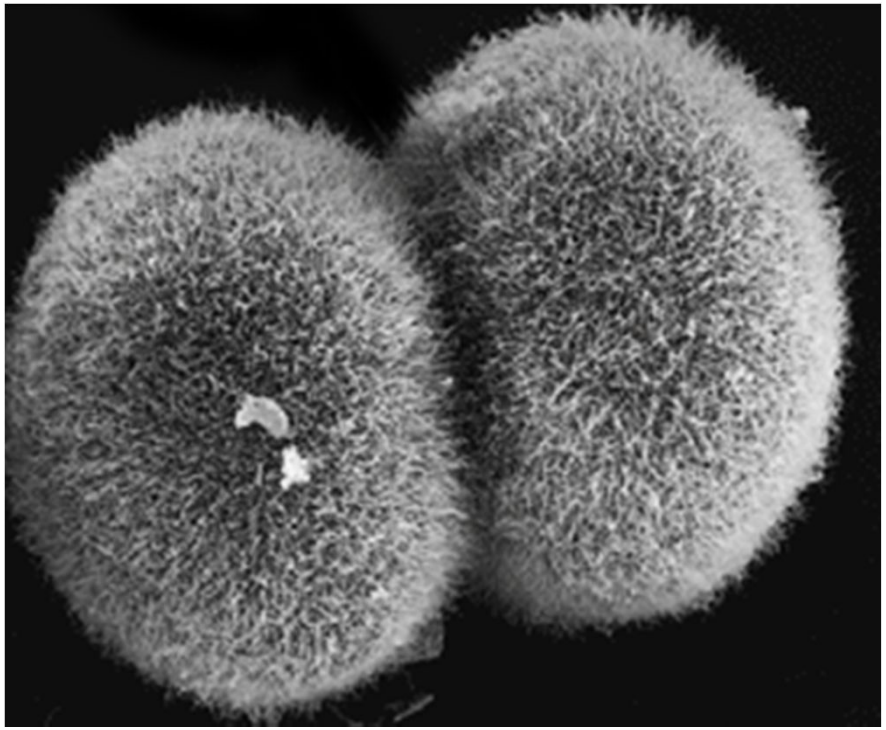


Telophase

Chromosomes stop moving at the end of anaphase, the cell enters telophase.

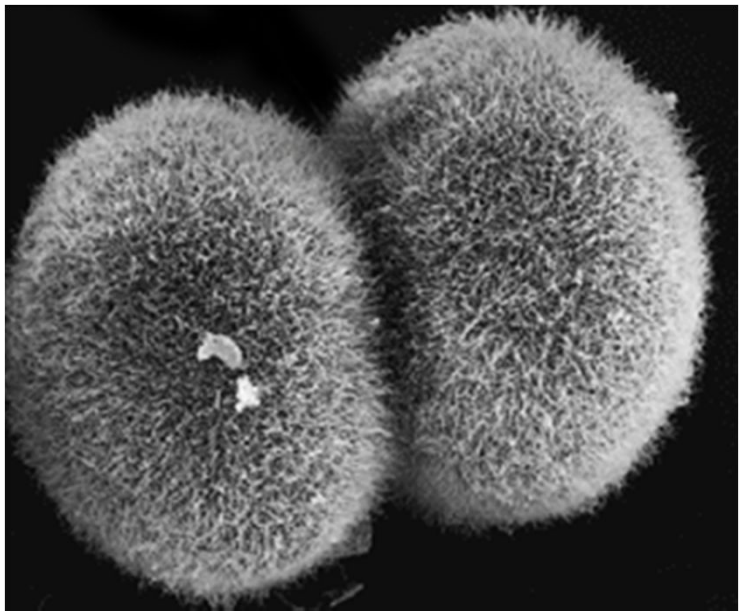
The chromosomes begin to uncoil, the nuclear envelopes and nucleoli, reform their respective structures.

Division of the Cytoplasm



Begin before telophase ends.

Animal cells usually divide by a furrowing of the plasma membrane, as if an invisible thread were tightening between the two poles.



END